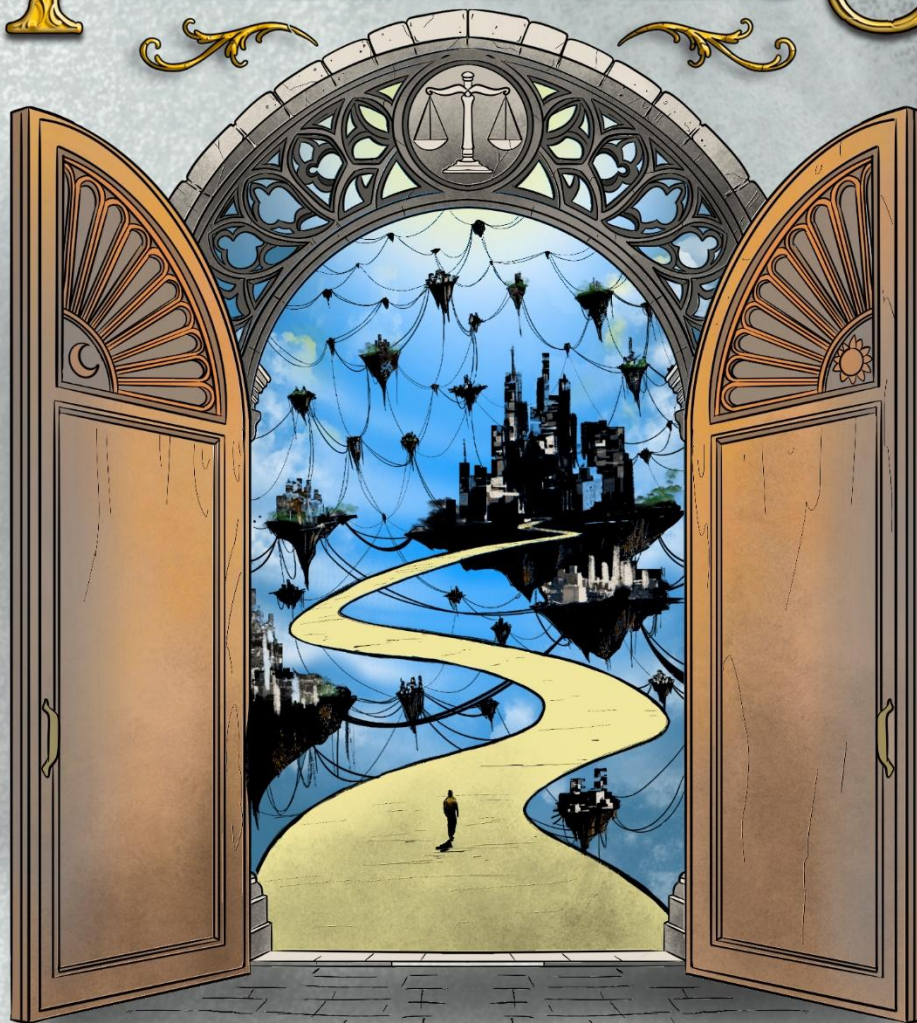


The
LAW
— OF —
FAIRNESS



STEPHEN TAKOWSKY

THE LAW OF FAIRNESS

Publishing Details

The Law of Fairness

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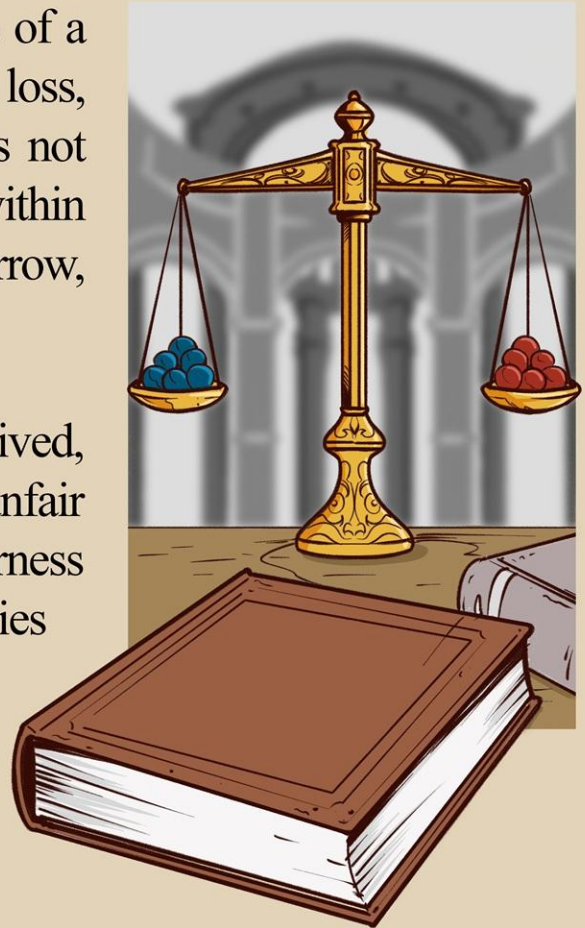
Series Map

THE LAW OF FAIRNESS

A new way of seeing every life.

We judge fairness by the visible shape of a life: health and hardship, love and loss, opportunity and misfortune. Yet life is not lived from the outside. It is lived within consciousness, through joy and sorrow, pleasure and pain, peace and fear.

Could life be fair where it is actually lived, even when it appears profoundly unfair from the outside? The Law of Fairness proposes that every felt experience carries weight according to its intensity and duration. Across the completed conscious life, total positive and negative feeling exactly balance.



Picture a scale open for a lifetime. Joy and relief add weight to one side; pain and grief to the other. It may tip heavily for years, but when the conscious life is complete, the two sides balance. Each life carries its own scale. No life balances another.

Lives may differ enormously in length, choices, pleasures, wounds, and circumstances. The scale weighs neither wealth nor virtue, status nor faith. It weighs felt experience alone. Circumstances tell the visible story of a life, but feeling is how that story is lived. The Law asks us not to mistake the visible story for the whole.

WHAT IF THE LAW IS REAL?

A new meaning of utopia.

If the Law of Fairness is real, utopia cannot mean endless pleasure. More comfort, luxury, or consumption does not make a society ideal. If conscious lives end with no lasting surplus of positive feeling, utopia must mean something deeper and steadier than limitless happiness.

“There is no free happiness” does not mean joy is wrong, pain is payment, or suffering is holy. Pleasure alone cannot define a good life, nor appearances a good society. The goal is joy without excess. A fairer world would reduce needless suffering, poverty, disease, cruelty, corruption, and abuse while protecting love, beauty, rest, play, choice, and joy. Moderation restrains excess, work builds worthy goals, and stewardship preserves the world so future generations inherit something worth protecting.

If the Law is real, its deepest equality lies beneath wealth, status, power, culture, and belief. Lives may differ while reaching the same final balance of felt experience. These ten books follow that question from science into life, spirituality, government, cities, law, transition, and peace, exploring how humanity might build a lasting utopia.



THE TEN BOOKS OF FAIRNESS

A complete path from truth to practice.

THE LAW OF FAIRNESS



Explores the core question, defines key terms, and sets the ethical bounds of the series.

THE SCIENCE OF FAIRNESS



Evaluates the Law through evidence, measurement, rival models, protocols, and falsification.

THE UTOPIA OF FAIRNESS



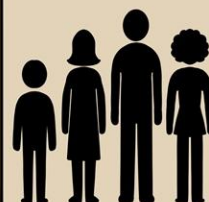
Brings the series into a mature view of what a peaceful world could become.

THE TRANSITION OF FAIRNESS



Asks how a move toward utopia could unfold with care, step by step, under real limits.

THE LIFE OF FAIRNESS



Brings the Law into daily life and asks what it means for how we live.

THE RELIGION OF FAIRNESS



Explores faith, world religions, and the optional path of Fairness.

THE GOVERNMENT OF FAIRNESS



Examines civic structure and the public institutions that give it form in practice.

THE CITY OF FAIRNESS



Builds a civic space for learning, service, and peace with privacy kept safe.

THE CONSTITUTION OF FAIRNESS



Sets out the rights, rules, paths, and checks a lasting utopia would need.

THE PEACE OF FAIRNESS



Builds the base for lasting peace and a fairer future for all humanity.

The series moves from the deepest question about reality to the structures that make a more peaceful world.

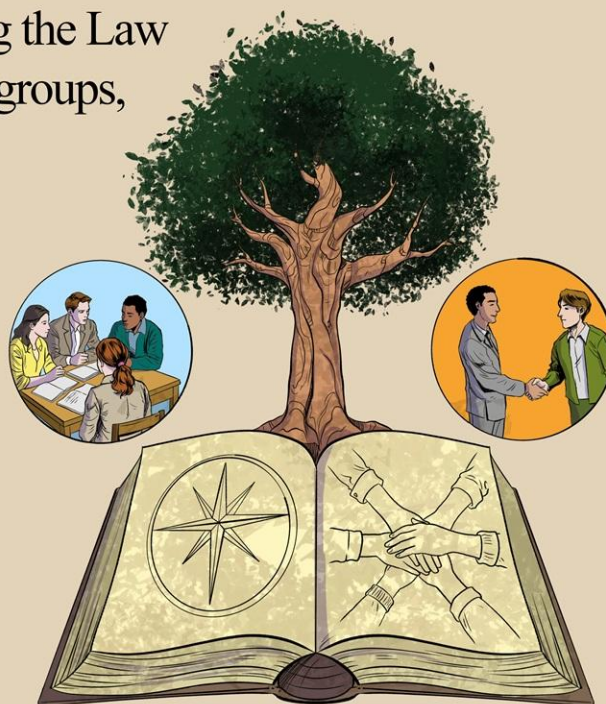
THE WORK AHEAD

A shared mission to build a better world.

Read the books, share them, and bring the Law of Fairness into classrooms, research groups, reading circles, and community conversations. Important ideas grow stronger when people examine them together, challenge assumptions, compare perspectives, and turn understanding into shared effort.

Scientists, researchers, clinicians, philosophers, and other experts can help test the Law, refine its measurements, improve its methods, and compare it with strong rivals. Serious criticism is essential. It exposes weaknesses, sharpens the questions, and helps reveal what should endure, what needs revision, and what should be rejected.

Beyond the hypothesis lies a practical challenge: building a world with less corruption and needless suffering, and more room for dignity, freedom, meaning, beauty, and opportunity. If this project speaks to you, bring your knowledge, experience, curiosity, and determination. Progress begins when people from different paths meet, share what they know, and build what survives scrutiny.



Help us make this generation
the one that brought peace to our world.

How to Read This Book

Life is not fair. Life may be fair. Between those two sentences lies the reason for this book. The first names the world as it appears: pain without visible proportion, luck beyond explanation, and grief falling where no blame can reach. The second names the question this book refuses to abandon: whether a complete conscious stream may carry a deeper terminal fairness that visible circumstance cannot reveal, but evidence could test.

The Law of Fairness asks a precise and surprising question: could the positive and negative felt experience within each complete qualifying conscious stream have finite, exactly equal totals? This is a proposed law to investigate. Its primary finite branch is called Per-Stream Terminal Valence Closure, or PTVC. The stream and its true endpoint must be identified independently of whether its totals balance. Each stream stands separately, and equal totals would still permit profound differences in duration, intensity, opportunity, agency, health, memory, and circumstance.

As you read, keep four kinds of discussion in view. Some passages explain the proposed Law. Others ask what would follow if it were true. Research from established fields helps us understand feeling, memory, consciousness, and measurement. Practical and interpretive passages ask what remains valuable in ordinary life. Each kind has a different job: a consequence of an assumption is not a finding, and a useful idea can deserve attention before the assumption is settled. Following those distinctions lets you explore the possibility fully while judging each argument on its own merits.

Choose a route that fits your purpose. For a rapid orientation, read the Illustrated Guide and Chapters 4 and 27–30. For the human

and ethical argument, follow Parts I, IX, and X. For the scientific architecture, follow Parts II–VII and Chapter 28, using *The Science of Fairness* for mathematics, protocols, recovery, replication, and verdicts. The full sequence remains available as a linear journey; these routes offer efficient entry points for different purposes.

This book opens the questions that the series pursues in greater depth. What would count as a scientific test? What might a fairer future make possible, and how could people move toward it? What changes in ordinary life, meaning, public institutions, and the foundations of peace? The companion books take up those questions in their own domains. You can follow the question that matters most to you, returning here for the common vocabulary and the distinction between the proposed Law and the evidence or reasons each further argument needs.

One boundary travels through every page: care comes before interpretation. No one's pain is morally owed payment, punishment, proof, or a private message from the Law. The equal-and-opposite net remainder required by exact PTVC does not assign moral desert or authorize harm. If someone is hungry, unsafe, grieving, sick, or afraid, the first answer is food, shelter, medicine, protection, apology, or repair. Read slowly enough for the difficulty to remain visible and boldly enough for the possibility to remain alive. Let the images open the door, the definitions sharpen the question, the rivals attack it, and compassion outrun certainty at every stage.

Illustrated Guide to Fairness



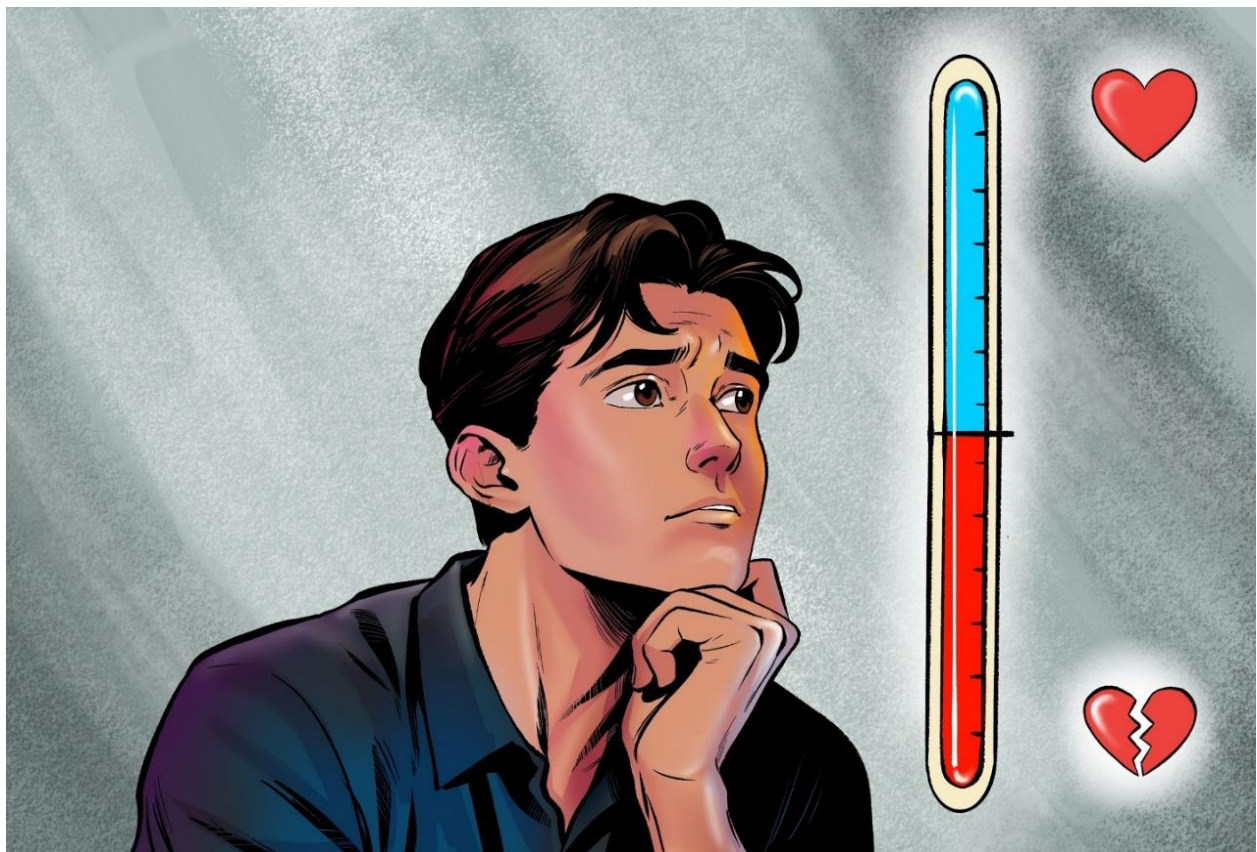
From a young age, we are told that “life is not fair.” As we look at the world, the evidence seems overwhelming. Some lives appear to overflow with unearned comfort and joy, while others are marked by relentless struggle, loss, and tragedy. On the surface, the distribution of pain and pleasure often looks unjust and even cruel. Facing this disparity, it is only natural to ask: Why is life so unfair? The Law of Fairness begins by refusing to look away from that question. It does not ask us to call visible injustice good or pretend that suffering is harmless. The Law asks one deeper question beneath the unequal world we can see: could a complete conscious life still be fair at the level of felt experience, even when its circumstances appear profoundly unequal?



Appearances can be deceiving. We do not see the entire story of another person's life. A billionaire may inhabit a private world of fear, paranoia, loneliness, or stress, while a person with little may still know genuine moments of peace, laughter, love, or relief. That does not make poverty harmless, wealth imaginary, or unequal opportunity acceptable. Circumstances matter, but they tell only the visible story. Every life is actually lived from within, through an inner world that no observer can fully see. Every life contains burdens that remain hidden and blessings that never become public. If we want to understand whether life could be fair at a deeper level, appearances alone are not enough. We have to look at experience as it is actually felt.



It is easy to judge another life by how it appears, especially in a world where people carefully curate what they show. On social media, someone may post only bright moments, creating the illusion that joy fills nearly every day. The image may hide late-night worry, grief, coercion, exhaustion, numbness, or the quiet fear of losing what has been gained. A life that looks uneventful may contain friendship, tenderness, purpose, or hard-won peace. Because we rarely glimpse the entire story, comparison can make other people seem luckier and our own lives uniquely burdened. Social comparison is profoundly uneven: we often measure selected moments from another person's life against the full private reality of our own. A photograph, salary, diagnosis, title, or public smile may reveal something important, but none can reveal the whole felt life behind it. A snapshot is not a lifetime.



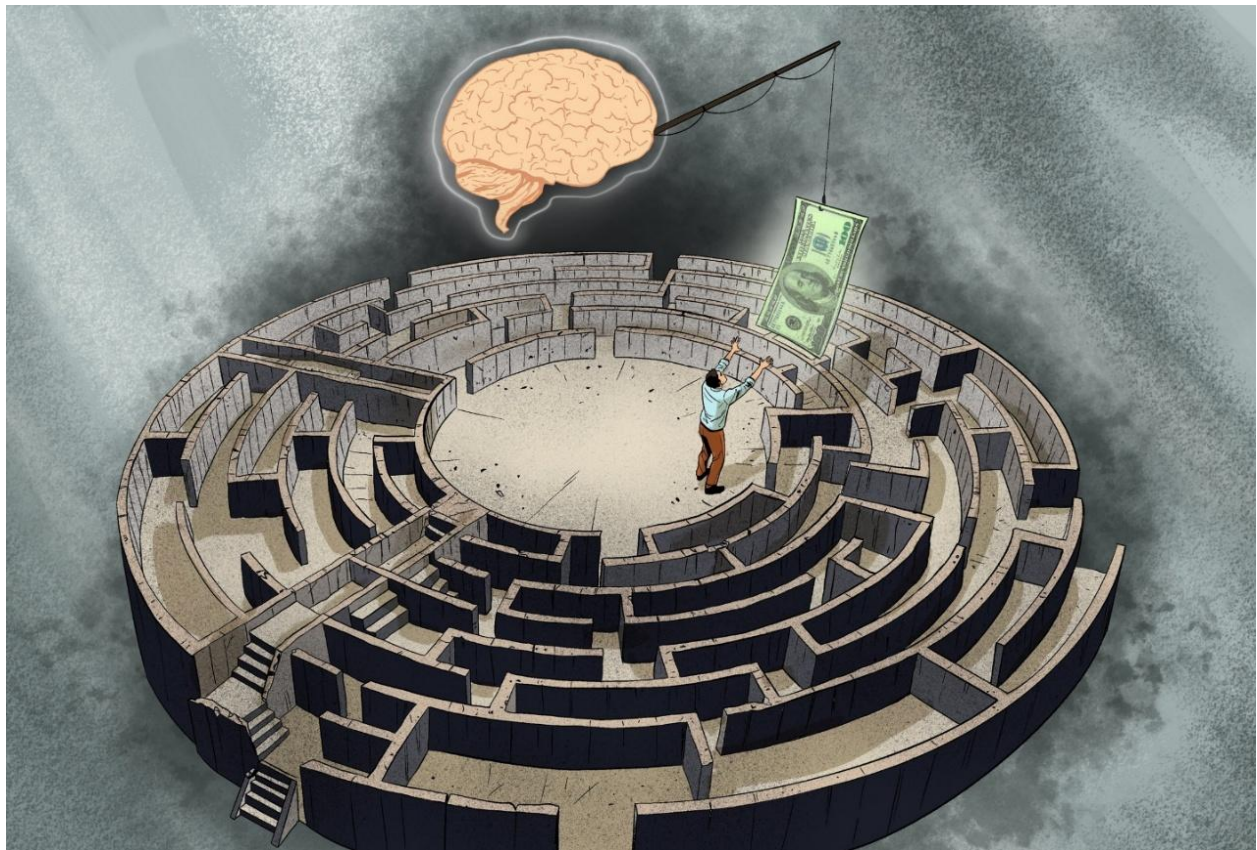
Even when life seems painfully unfair, human experience has a remarkable capacity to move. A sudden stroke of luck can bring pressures no one else sees, while people who endure terrible hardship do not always remain broken by it. Many adapt, discover strength they did not know they had, and find moments of love, purpose, or joy in places they never expected. Psychologists study part of this movement through hedonic adaptation and other forms of emotional regulation. Our feelings often shift away from their most extreme highs and lows as we adjust to what has happened. Experience changes as the person changes. It is almost as if the mind carries a quiet thermostat. And when the last visible chapter of a life is tragic, it can be easy to forget how much peace, joy, love, and satisfaction may have filled the years that came before. Do these familiar movements belong to something larger that becomes visible across the whole of a conscious life?



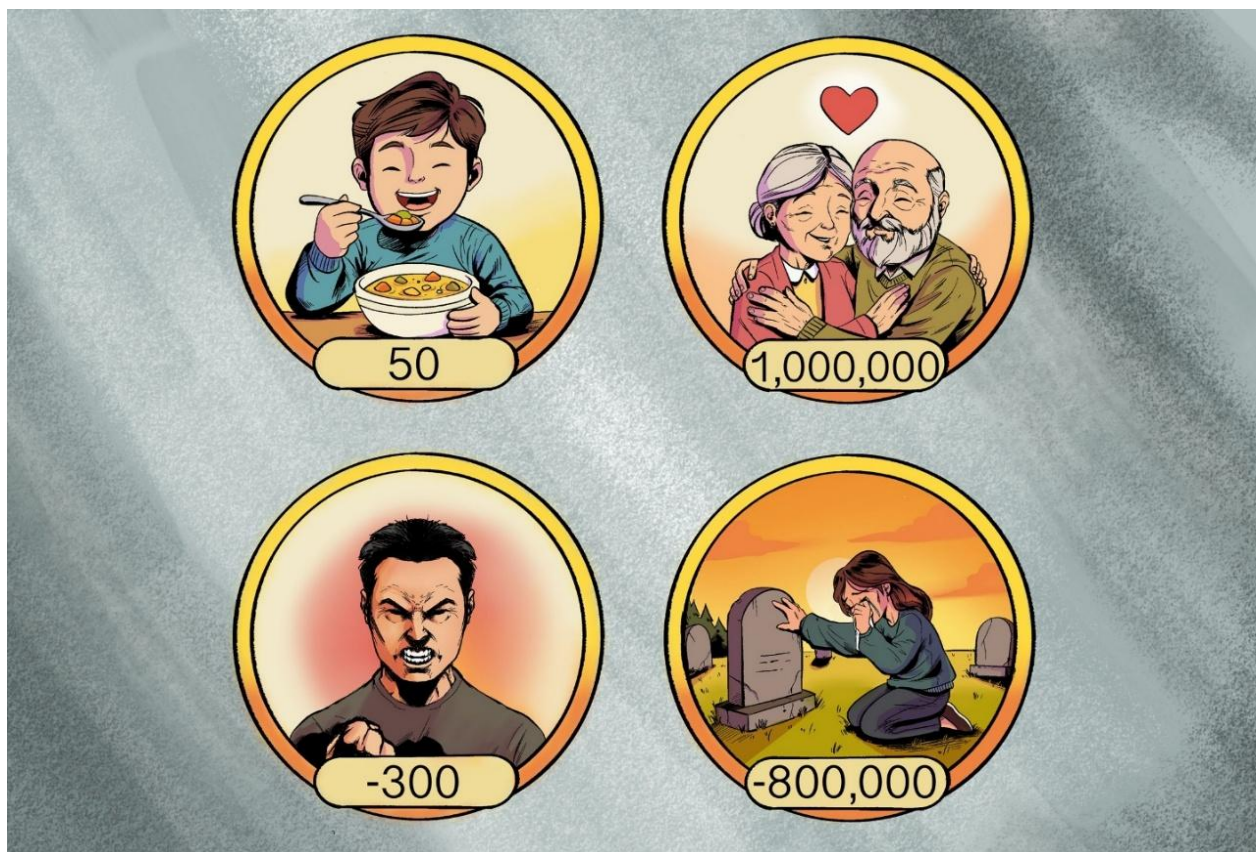
There is at least one remarkable way life could be fair for everyone, no matter how differently our stories unfold. Not through equal money, status, luck, health, opportunity, or length of life, but at the deeper level of felt experience. The Law of Fairness proposes that, across each qualifying conscious stream over its complete course, the total weight of positive feeling ultimately equals the total weight of negative feeling. Joy and sorrow may arrive in completely different amounts, intensities, and sequences along the way, and one life may look nothing like another. Yet when the whole journey is complete, the two sides would exactly balance. The scientific name for this proposed pattern is Per-Stream Terminal Valence Closure, or PTV. This is the possibility at the heart of the Law of Fairness.



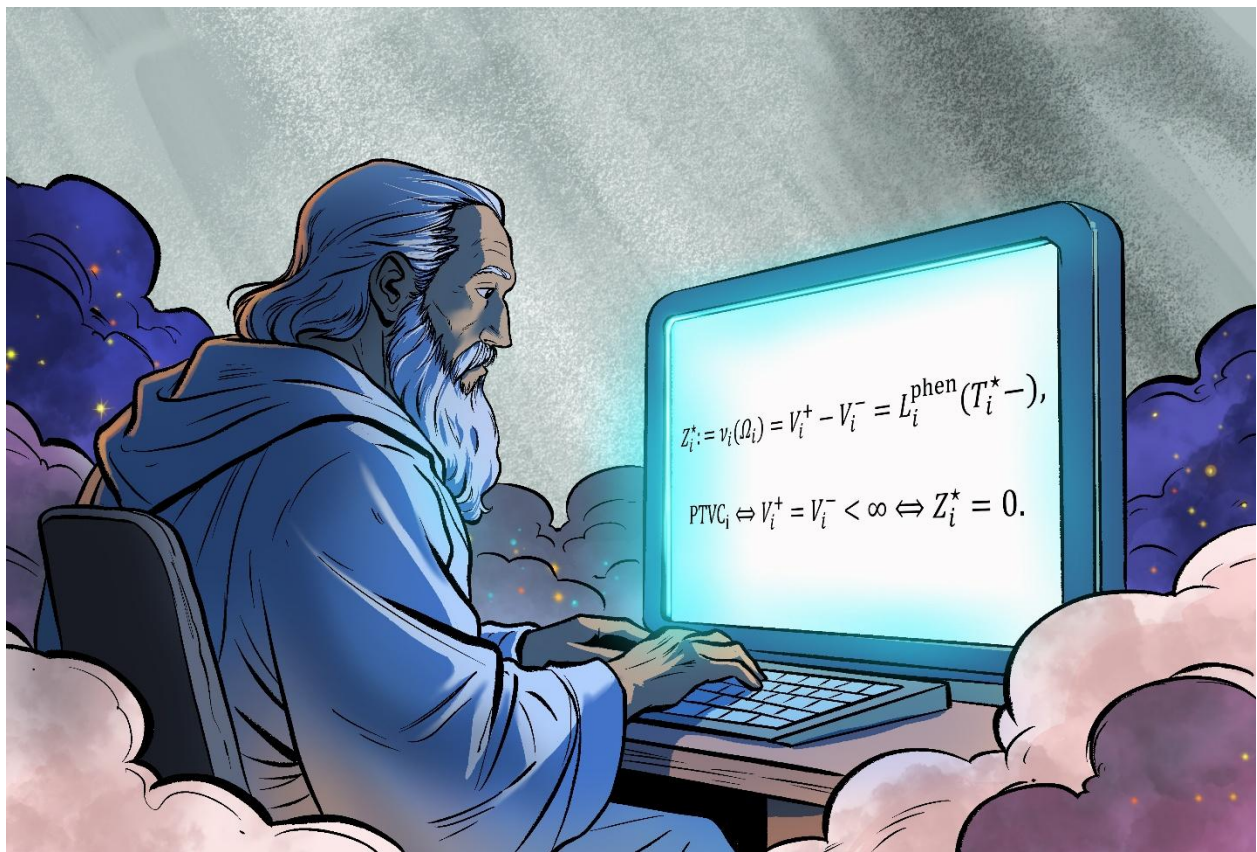
To determine whether the Law is true, we must examine the entire conscious story, not a single chapter. One joyful year cannot settle a life, and one terrible hour cannot reveal the total on its own, however morally serious that hour remains. The proposed domain begins when a qualifying stream of experience exists and ends at that stream's true endpoint. Everything felt between those boundaries belongs to the whole: joy and grief, pleasure and pain, ordinary days, forgotten moments, and experiences no outside observer may ever fully know. The Law of Fairness therefore asks us to consider the complete course of conscious experience rather than judging a life from any single period, event, or ending.



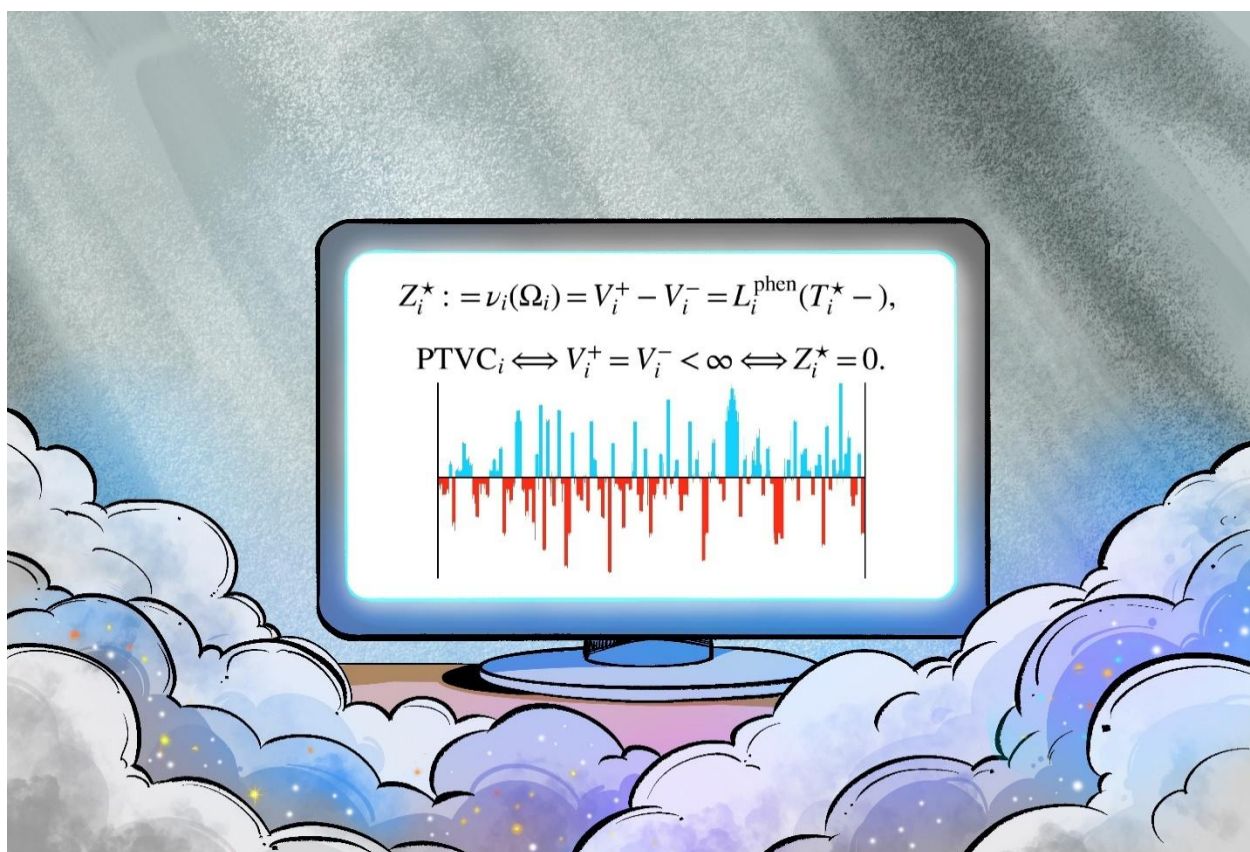
How could exact closure happen without anyone noticing? Our instincts, desires, addictions, fears, and even sudden pulls of attention help shape the experiences and possibilities ahead of us. If the Law of Fairness is true, these forces must unfold within a larger structure compatible with exact closure. Our urges and choices may sometimes function like guardrails, shaping which paths remain reachable, but they would be only one possible part of that larger structure. A conscious stream could not reach its true endpoint while its complete balance remained unresolved. Yet exact balance need not occur only at the end; the running total may return to zero at other moments along the way. How often that happens could vary enormously from one life to another, and discovering those patterns would be part of understanding how the Law works.



If the Law of Fairness is true, then happiness is never isolated from the complete story of a conscious life. Food, play, rest, love, companionship, grief, fear, frustration, and pain all become part of one larger course of experience whose positive and negative magnitudes ultimately balance at the stream's true endpoint. That does not mean one joy must be repaid by one hardship, or that suffering is assigned as the price of pleasure. The order and causes may vary enormously, yet no part of the conscious story is outside the whole. Our tears today may water the flowers of tomorrow's happiness through growth, love, relief, or understanding; at other times, the positive weight that completes the whole may belong to joys already lived. Studying the Law could reveal meaningful patterns between the pleasures we pursue and the suffering or well-being that follows, perhaps helping us make wiser choices and reduce unnecessary suffering. The deeper mystery is how a life made of countless unequal, unpaired moments could nevertheless arrive at exact balance when the entire stream is complete.



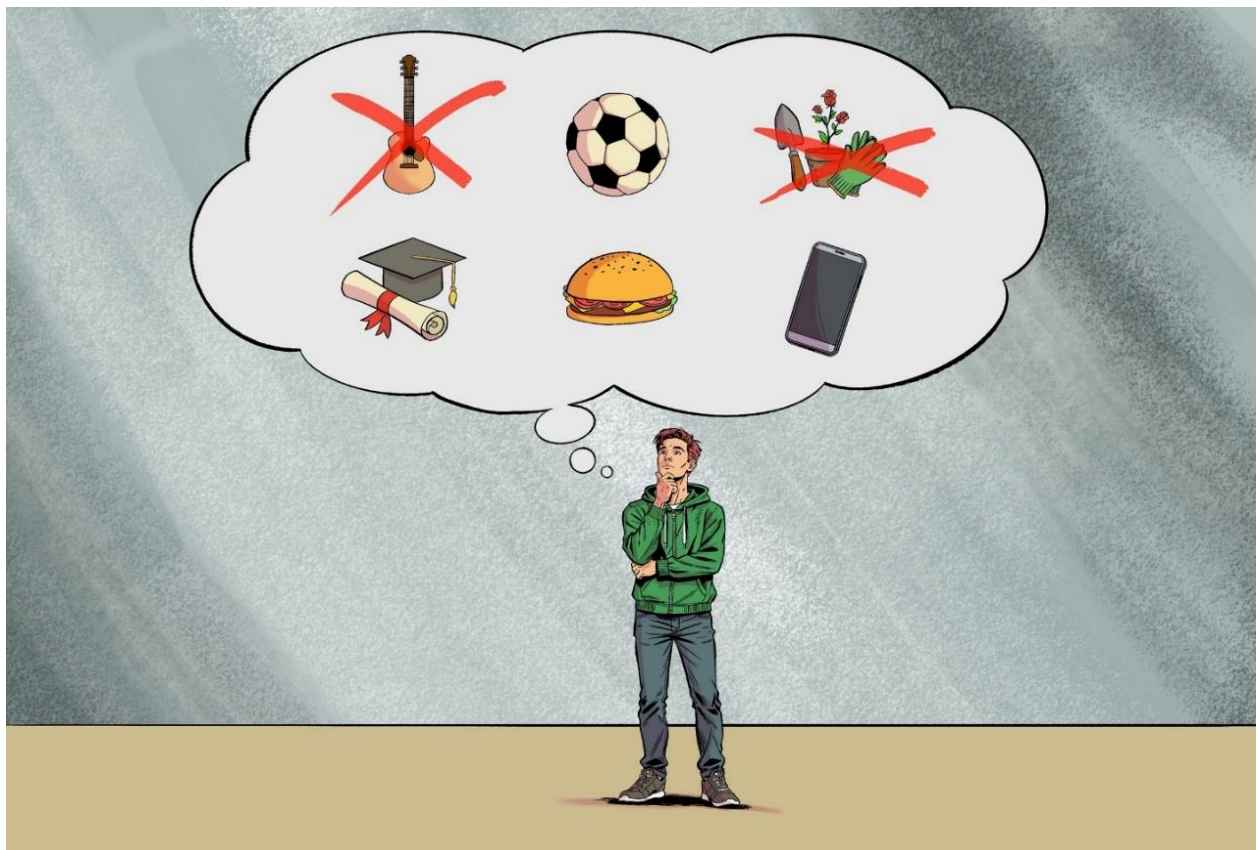
One way to picture the hidden structure behind exact closure is what we call the Queue System. The name is a metaphor, not a claim that a literal queue, accountant, or controller exists inside the mind. Imagine instead that every conscious life unfolds through a vast field of possible futures. Some paths remain open, others narrow or disappear, and one becomes the life actually lived. The picture does not require a single predetermined future; many paths could remain possible while still preserving a way to exact closure. That unfolding must ultimately be compatible with exact closure, even though we may never perceive how it happens. Different mechanisms could produce that same constraint, so the Queue leaves open how nature might accomplish it. The formula on the screen expresses the same idea without the metaphor: across one complete conscious stream, the total positive and negative felt magnitudes are equal at its true endpoint. The Queue gives us a way to imagine the mystery; the equation gives us a precise claim that science can test.



What would that claim look like across a life? Imagine felt value changing through time. Some moments rise above a neutral baseline; others fall below it. Even when an experience contains both positive and negative elements, its net felt value falls on one side or the other, revealing which predominates. As those moments accumulate, the running balance could climb, fall, reverse direction, or wander for years; the Law does not require it to move steadily toward zero. What matters is the complete stream at its true endpoint, where the final signed balance would be exactly zero. That simple picture points toward a scientific test. Scientists could develop ways to track felt experience over time, identify the correct conscious stream and endpoint, and compare the resulting evidence with the Law's prediction. Any measured graph would be evidence about the underlying felt history, not the history itself. An average across many people would not establish the claim, because the Law concerns each qualifying conscious stream individually. The graph turns an abstract claim into a question we can begin to measure, compare, and test against reality.



The Law of Fairness proposes structure without a miracle or a judge adjusting events one by one. It asks whether exact closure could be built into conscious reality as a lawful boundary. The Queue System is one way to picture that possibility. Life may contain many genuine choices while still unfolding within limits that keep closure reachable, much as a road can branch without leading everywhere. On one strong Queue-like route, possibilities that could no longer end in exact balance would simply fall outside the paths that remain available; on other routes, the constraint could lie elsewhere. The deeper idea is that fairness would not have to be imposed on life from outside. It could be part of the structure through which conscious lives are able to unfold.



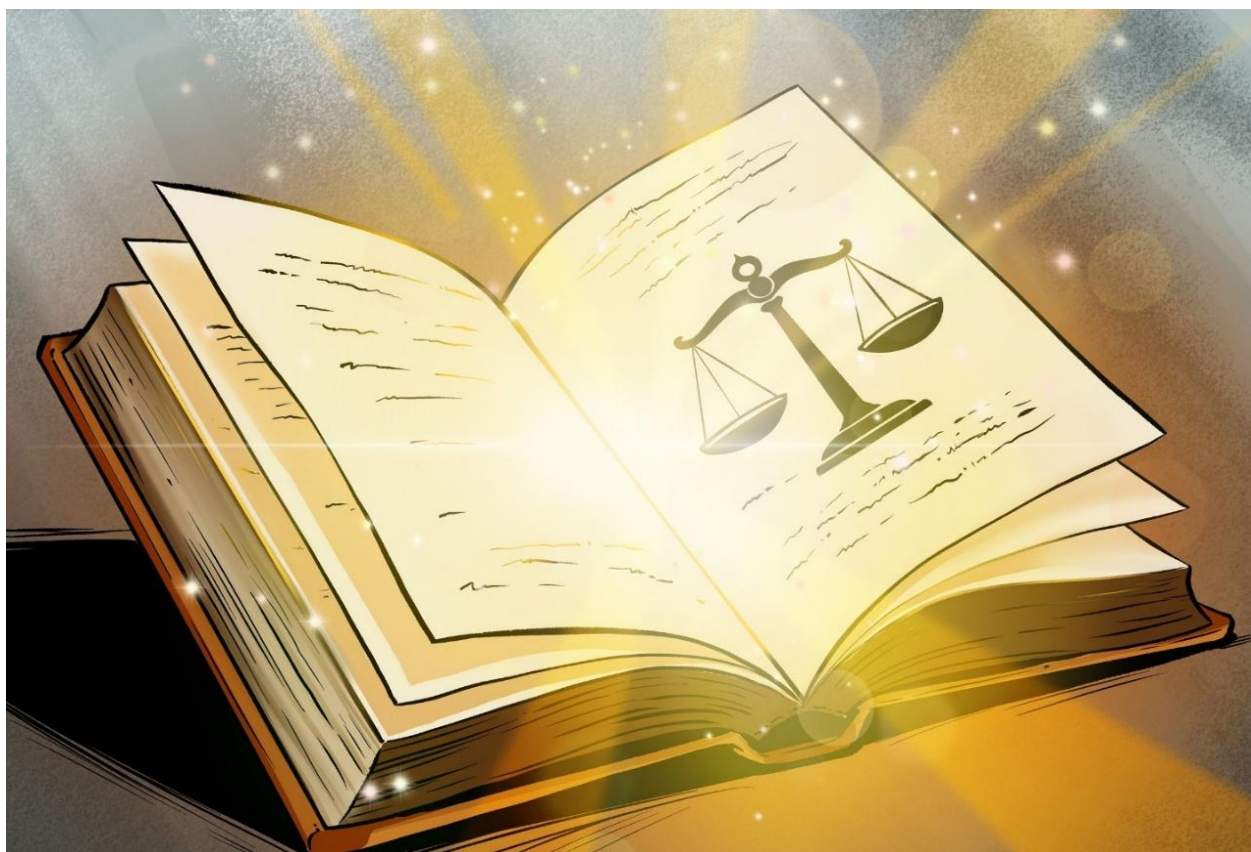
If exact closure is real, there is no way to “cheat” it at the level of the completed stream. Choosing pleasure, avoiding discomfort, changing habits, or pursuing a different path may alter what happens next, but none of those choices can place the finished conscious history outside the Law’s final condition. What we choose still matters enormously because it helps determine the experiences, relationships, opportunities, and consequences that become part of that history. Addiction can narrow future freedom, avoidance can deepen fear, discipline can expand possibilities, and care can change lives. Thoughts, dreams, desires, and sudden impulses may also influence the path, but their exact role remains open to discovery. The important point is that choice can change the journey without creating an escape from the boundary the Law proposes. If exact closure governs the completed stream, every route actually lived must ultimately be compatible with it.



Exact closure would constrain not only which paths can remain open, but also where the conscious story can truly end. A life could wander through many different experiences and still remain capable of reaching balance, but the stream could not simply stop while that balance was unresolved. This is important because preserving a possible path to closure would mean little if the journey could end before getting there. Some mechanism would therefore have to ensure that the path and endpoint together cannot produce an unresolved ending. That protection might come through the choices that remain available, the way conscious states change, the continuation of experience, the timing of the true endpoint, or some deeper rule governing the complete history. The branching landscape does not mean every step is predetermined. Many futures may remain possible, but whatever route is actually lived, the conscious stream can only be complete once exact closure has been satisfied. The journey may take countless forms, but the final condition does not change. In that sense, only fair endings are permitted to exist.



For some readers, the Law will naturally carry spiritual meaning. A believer may imagine a Creator establishing a just structure so deep and complete that fairness need not be imposed through constant intervention, while another reader may see the same possibility as a natural regularity of conscious life. The formal claim itself remains neutral: it neither requires nor rules out God, karma, providence, reincarnation, or an afterlife. What it offers is a question believers and skeptics can approach together and test in this life: could exact closure be a real feature of conscious existence? Spiritual interpretation can enrich the meaning of that question, while evidence determines whether the underlying pattern is actually there. If exact closure were someday established, science could reveal that the pattern exists without necessarily settling why reality has such a pattern. In that sense, natural law and spiritual meaning need not be competing ideas, and the Law can be contemplated through faith without belonging to any one faith.



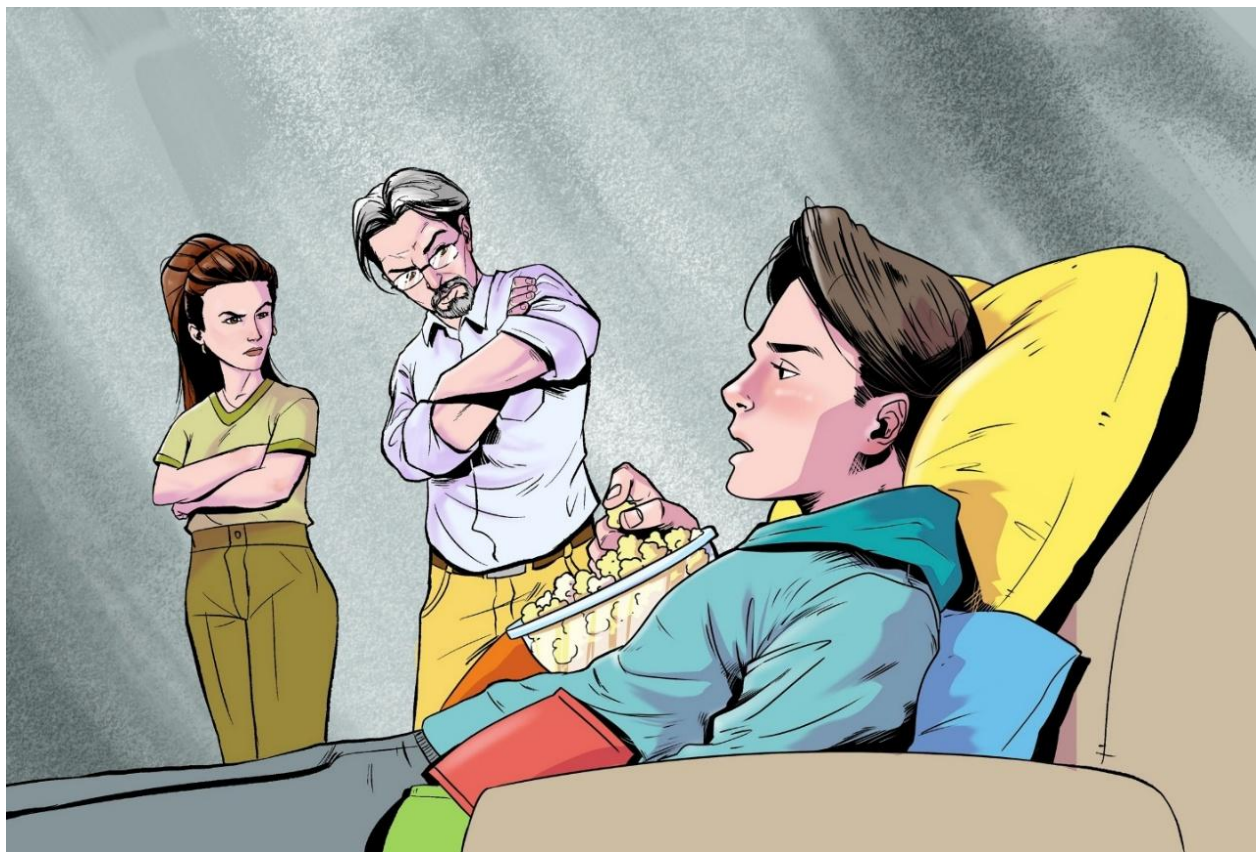
Across history, traditions have asked how to live wisely amid pleasure and suffering, desire and consequence. Long before the Law of Fairness was proposed, people built rituals, moral codes, prayers, stories, and ways of life around that question, often returning to themes of restraint, compassion, and balance. Those traditions do not prove the Law, but their persistence is difficult to ignore. If the Law is real, humanity may have been feeling its effects all along without understanding the deeper structure behind them. People may have noticed recurring patterns in how pleasure, hardship, choice, and consequence shape a life, then explained those patterns through the language available to them. What later became religion, philosophy, or moral tradition may sometimes have been an attempt to describe something real before the tools existed to test it directly. That would not make every ancient belief correct, but it raises a remarkable possibility: some of humanity's oldest ideas about how to live may be echoes of a deeper pattern in conscious experience.



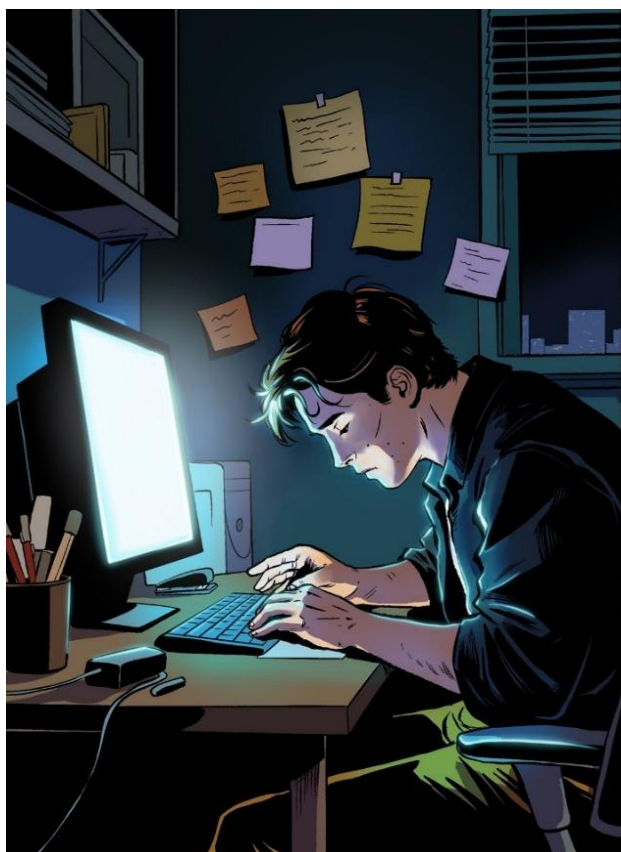
Many people sense a missing piece in our understanding of existence. Perhaps the Law feels like that piece not because it replaces science, philosophy, personal experience, or spiritual tradition, but because it gives them a new question to meet around. Science can ask whether exact closure is real and measurable. Philosophy can explore what such a pattern would mean for fairness and the human condition. Our own lives provide the felt reality the Law is actually about, while spiritual traditions may offer ways of interpreting what a deeper order would mean. These perspectives should not be collapsed into one another: adaptation is not karma, biology is not providence, and resemblance is not proof. Yet they may still be looking at different edges of the same mystery. Perhaps the missing piece is not a final answer at all, but a question precise enough to let very different ways of understanding reality finally speak to one another.



Consider the old warnings about greed, gluttony, pride, and excess. Their practical wisdom may reach deeper than simple moral condemnation. Pleasure itself is not the enemy; a good life should contain food, rest, celebration, achievement, love, and delight. The danger begins when pleasure stops enriching life and starts taking possession of it. Appetite can become compulsion, easy reward can reshape what we want, and repeated indulgence can quietly narrow the freedom to choose what comes next. That is why moderation matters: not because joy must be feared, but because freedom is worth protecting. The Law opens an even deeper question. Could some forms of excess reliably alter the later course of felt experience in ways we have not yet learned to measure? Addiction already shows that pleasure and suffering can become connected through recognizable chains of consequence. Studying the Law may reveal whether other patterns exist as well. If such patterns are found, ancient warnings about excess may take on new meaning, not as proof of cosmic punishment, but as early observations about human life that science is only beginning to understand.



Each of us can shape something like heaven or hell within ordinary life, not because we control everything that happens, but because our habits and choices help create the conditions we and others must live through. Comfort can become isolation, resentment can poison a home, and neglect can slowly close doors that once stood open. The opposite is also true. Honest change, treatment, forgiveness, boundaries, hard work, and care can transform what a life feels like from the inside. In that sense, we are not merely passengers through experience; we continually influence part of the world that produces it. No amount of virtue exempts a person from the balance the Law proposes, and no hardship should be read as proof that someone is being corrected. The pain caused by cruelty still belongs within the completed balance, but that never turns harm into justice. Good people can suffer terribly without that suffering carrying a moral verdict. The Law does not ask us to decode pain as deserved or necessary. It asks a more useful question: knowing that our choices can alter what another conscious life actually feels like, what kind of experience are we helping to create?



We pursue goals expecting satisfaction, but the effort required to reach them may be part of a deeper pattern in felt experience. Hard work can change not only what we achieve, but how strongly achievement is valued when it arrives. Delayed gratification, discipline, sacrifice, and earned success may create very different experiential trajectories from easy reward. The Law does not tell us that every obstacle was placed in our path for this purpose, and some hardships are plainly unjust. But it does give us a reason to ask whether recurring relationships exist between effort, reward, and the later balance of conscious experience. If such patterns are real, they could help explain why effortless pleasure can sometimes feel strangely empty while difficult achievements carry unusual depth and meaning. A worthy ambition may therefore be more than a destination. It may be part of a larger experiential pattern in which effort, anticipation, struggle, and fulfillment become one connected journey.



Think of daily habits as entries on a rough balance sheet of felt experience and future freedom. We cannot yet assign reliable numbers to every choice, but we can still recognize patterns. A habit that brings immediate pleasure may also narrow attention, damage health or relationships, and reduce the freedom available later; another may require effort now while expanding what life can become. Addiction makes this especially important because reward can keep reinforcing a behavior even as suffering grows around it. If the Law is eventually established, that relationship could become a new scientific question: could reducing the reward that sustains a harmful habit also reduce some of the suffering that tends to follow? Researchers and therapists could test such patterns rather than assume them. The point is not to fear pleasure, seek pain, or keep a perfect score. It is to notice which habits quietly trap us, which ones restore freedom, and whether the balance of experience can teach us to choose more wisely.



Whenever life feels unfair, the Law should deepen compassion, never explain the wound away. Even if exact closure is real, present suffering is still part of what a person actually experiences, and we should not treat grief, illness, oppression, or danger as something that should simply be endured. A joyful event is not a warning that pain must soon arrive, and a painful one is not evidence that some hidden correction is underway. Suffering calls for a human response. We can comfort, protect, heal, resist injustice, repair what is damaged, and stand beside one another when life becomes difficult. Easing another person's suffering changes the very experience the Law asks us to understand. If that deeper balance exists, compassion itself becomes part of the conscious histories through which it unfolds. The possibility of deeper fairness should never make us passive before visible unfairness. Our responsibility is not to enforce the Law, but to improve the conscious experience our choices can actually touch. A theory of fairness worthy of humanity should never teach us to accept suffering more easily. It should make us more determined to reduce it wherever we can.



We are all part of one living fabric, yet every conscious life is lived from within. No one can feel another person's joy or pain for them, and no stream can balance another. Still, our lives enter one another's stories. What we build, protect, neglect, repair, or simply say can change what someone experiences. That gives choices extraordinary weight. The Law of Fairness asks whether a deeper equality exists beneath the differences that divide us, but we do not need the answer before deciding how to treat one another. We can build a world with less needless suffering and more room for freedom, love, meaning, beauty, and joy. Perhaps humanity has been living within a deeper order all along, only now beginning to recognize its shape. Whatever nature reveals, no one can live another person's conscious life, but each of us can help change what that life feels like. The future of fairness is ours to discover and help build. If we take that responsibility seriously, we may yet create a world worthy of the lives within it and give future generations something humanity has sought for centuries: a more peaceful world.

Part I — The Question and Human Stakes

Fairness begins as a human recognition before it becomes a theory. It appears when safety gathers around one life and danger around another, when illness arrives before a child can name what is being lost, when effort and outcome part company, and when time, love, health, power, and pain arrive without visible regard for innocence or need. The question endures because human beings can see both the world's beauty and the burdens it distributes, and because understanding that contrast may help us care more wisely.

That first observation should not be softened. Poverty can constrict choice. Violence can alter a body and a future. Discrimination can shape housing, education, health care, credibility, exposure, and the ordinary expectation of being safe. Disability can meet a world built without access. Grief can divide a life into before and after. Any account of fairness that begins by explaining these facts away has already failed the people whose lives supplied the question.

Yet visible inequality does not exhaust the inquiry. A biography can be observed from outside, while experience is lived from within. Two people may receive similar goods and inhabit different emotional worlds. Public advantage may coexist with fear, pain, compulsion, loneliness, or shame that no photograph reveals. Severe external constraint may coexist with humor, relationship, skill, meaning, or peace, but none of those goods makes the constraint acceptable. Appearance is real evidence, not the whole field of consciousness.

The proposed Law begins at that divide. It asks whether each complete qualifying conscious stream could contain finite and

exactly equal total positive and negative felt experience, under a declared representation of felt experience. This primary version is called Per-Stream Terminal Valence Closure, or PTVC. Its balance concerns one stream across its whole history before its true endpoint, with the stream, the measure, and the endpoint identified independently of the desired answer. One person's joy cannot balance another person's suffering. Lives could still differ profoundly in duration, health, freedom, opportunity, memory, and everything that makes a history its own. The proposal therefore opens a separate question about experience while leaving the work of justice fully in place.

Part I invites the reader to meet that hypothesis before asking for assent. It preserves the ordinary force of the sentence "life is not fair," separates the forms of fairness already at work in moral and public life, identifies the phenomenal domain the proposed Law addresses, and establishes the ethical ground on which the rest of the book can proceed. The movement is from recognition to precision, then from precision to protected inquiry.

That sequence is also intellectually necessary. Procedural fairness governs voice, reasons, consistency, and correction; distributive justice addresses goods, burdens, opportunities, and institutions; desert concerns responsibility; moral luck exposes how praise and blame depend on conditions no one chose. Each remains indispensable. None was designed to settle whether one complete conscious stream has an exact terminal property at the level of felt valence (Rawls, 1999; Sen, 2009; Nagel, 1979; Williams, 1981).

The difficult cases give this inquiry its seriousness. Childhood suffering, early death, severe illness, disability, abuse,

discrimination, and catastrophe bring the questions into sharp focus: what does the proposed Law actually claim, and could that claim survive the lives that challenge it most? A worthwhile explanation must meet those cases directly. It must recognize the visible harm, preserve the person's dignity, and distinguish what is known from what has only been imagined. The possibility of a deeper relation in experience offers no explanation of why a particular harm occurred. It gives the inquiry a demanding place to begin, while care and repair retain their full urgency.

The ethical firewall follows from this restraint. Present pain is a present claim on care. No one earns suffering. One stream cannot pay for another. Research may not create serious suffering to test a theory about suffering. Refusal must remain meaningful, especially under dependency. Relief may come before explanation, and meaning may be declined. These are not apologetic disclaimers; they are conditions under which the idea is worth discussing at all.

The emotional movement of this Part is from disturbance to precision, then from precision to trust. Chapter 1 begins with the world as people encounter it and asks why ordinary concepts of fairness cannot settle a complete-stream claim. Chapter 2 moves inward without denying material reality, showing why felt experience matters and why it remains partly private. Chapter 3 fixes care, consent, dignity, and freedom from imposed meaning before formal ambition begins.

Chapter 1 — Why Fairness? Why Now?

A child asks why one person has enough and another does not. An adult learns to answer with causes: inheritance, policy, geography, health, family, luck, skill, violence, history, institutions, timing. The causes matter. They explain how unequal conditions arise and often show where responsibility lies. Yet after the explanation, the child's question survives. Why should the burden have landed there?

The survival of that question shows that fairness names more than one human concern, while leaving open whether any deeper symmetry exists. Sometimes we ask whether rules were followed. Sometimes we ask whether goods and risks were distributed justly. Sometimes we ask whether praise or blame fits what a person controlled. Sometimes we are asking something more elemental: whether the full weight of conscious life is as arbitrary as its visible surface appears.

This book takes the elemental question seriously and keeps the answer open. Birth distributes citizenship, family, biology, language, exposure, wealth, and social recognition before anyone has chosen anything. Accident interrupts plans. Disease enters bodies without consulting character. Structural discrimination can make risk cumulative and opportunity conditional. These facts belong to history, policy, medicine, law, and daily life, and they remain fully visible while a formal hypothesis asks a different question (Nussbaum, 2006; United Nations, 2006; World Health Organization, 2022; Bailey et al., 2017; Williams et al., 2019).

The word fairness, however, can conceal several different targets. A fair hearing concerns procedure; a fair distribution concerns allocation; a deserved reward concerns desert; equal access

concerns opportunity; and a predictable rule concerns governance. None of these by itself answers whether a life is free from suffering or whether a complete stream satisfies a terminal relation. The first intellectual task is therefore separation. We need to see what ordinary ideas of fairness can decide, what they leave open, and why the Law of Fairness cannot simply borrow their authority.

Procedural fairness concerns the path by which a decision is made. People care about voice, neutrality, intelligibility, consistency, respectful treatment, and the opportunity to correct error. Research on legal and organizational legitimacy shows that procedure affects trust and compliance even when outcomes disappoint. That matters because an opaque or humiliating process creates its own burden. Still, a procedure can be impeccable while illness, bereavement, or accident remains devastating (Tyler, 2006).

Desert-based fairness asks whether outcomes fit choices, effort, contribution, or fault. It can illuminate responsibility, but it becomes unstable at the edges of control. Talents, temperament, family, health, opportunity, social treatment, and chance shape what people can do and what consequences follow. The philosophical literature on moral luck is not a decorative side debate here. It helps expose why suffering cannot be treated as a verdict on character (Nagel, 1979; Williams, 1981).

Distributive justice asks how institutions should arrange rights, resources, opportunities, burdens, and capabilities. Its questions are urgent and public. They remain urgent even if the Law is eventually supported. A terminal property within experience

would not build an accessible school, stop discrimination, provide medicine, repair a dangerous road, or return stolen time. The Law cannot do the work of justice by renaming it (Rawls, 1999; Sen, 2009).

The hard cases reveal why the question needs more than one language. Severe childhood illness lies beyond moral blame. Early death contains an interrupted future that no slogan can restore. Private suffering within wealth leaves inherited advantage morally visible. A disabled person's reported quality of life can coexist with an uncompromised demand for accessibility, while outsider pity remains fallible. Each case contains several true descriptions that must remain distinct.

They also reveal why visible comparison is insufficient. We compare lives because comparison can expose inequity, but the formal Law is not a claim about the average relation among lives. It is a claim, on its primary branch, about a complete stream. If the unit of the hypothesis is one stream, then a favorable population mean cannot establish closure for a single person, and one person's joy cannot repair another person's pain.

The difference is easier to see in ordinary statistics. A pattern across groups can reverse or disappear within individuals. A relation between income and reported well-being at the population level does not tell us how each person's experience changes across time. A group average can be accurate and still fail to describe any particular trajectory. The point is not that averages are useless. It is that their object is different (Robinson, 1950; Molenaar, 2004; Fisher et al., 2018).

Time introduces another boundary. Most studies observe windows. A survey may ask about yesterday. A clinical trial may follow participants for months. A longitudinal study may last years and still miss the beginning, the ending, or long unobserved intervals. A complete conscious stream is not identical to any of these windows. If the Law makes a terminal claim, its target cannot be chosen because the observed data happen to look balanced at the end of the study.

The Science of Fairness develops the technical work behind a compelling possibility: a claim about a whole conscious history may place demands on what happens before that history ends. A particular mechanism proposed to uphold the Law could require a physical state to retain information, restrict how that state changes, or predict a relation across different sequences of experience. Those requirements can become questions for mathematics and experiments. This volume explains why such questions matter; the Science volume examines how candidate answers could be built and compared. The first advance is to turn an enormous question into smaller ones that can meet evidence.

Two quick answers close the door before the question has acquired an object. One assigns every event to a hidden purpose. The other treats visible injustice as proof that no deeper order could exist. A stronger inquiry keeps purpose, evidence, and disbelief free until the bearer, target, endpoint, serious rivals, and failure conditions have been named.

The chapter begins with the visible world, tests familiar meanings of fairness against hard cases, and ends by showing why group

averages and partial observation windows cannot by themselves establish complete-stream closure.

Why now? No device can read a lifetime, but methods that once sat apart can now be joined under one test architecture. Affective science distinguishes valence from arousal; intensive longitudinal methods model within-person change; consciousness assessment and palliative-care frameworks sharpen questions about awareness, observation, and care near end-of-life windows; and preregistration and missing-data analysis can expose failure (Russell, 2003; Fisher et al., 2018; Giacino et al., 2018; World Health Organization, 2020; Little & Rubin, 2019; Nosek et al., 2018). Model recovery and equal-budget rival comparison can then be built into the PTVC test architecture. None of these methods settles PTVC. Together they make the question testable in stages.

Preventable harm should remain disturbing. Precision gives that response a proper object and a practical direction. Outrage can motivate repair without becoming evidence for terminal closure; compassion can attend without collapsing into pity; explanation can locate causes without excusing them; and population patterns can guide action without becoming an individual stream.

The attraction of the Law lies in what a successful explanation would connect. It would link the changes of feeling inside an ordinary day to a relation across a complete conscious history, while allowing lives to remain profoundly different. Such a finding would reach beyond a description of recovery: it would reveal a constraint on how experience can unfold. That possibility makes the question worth sustained attention. The chapters ahead

explain the claim, explore what could uphold it, and ask which observations would give us reason to take a candidate explanation more seriously. The reward is a research program in which each successful bridge makes the next question sharper, from momentary feeling to whole-history structure. That gives specialists concrete problems to test and every reader a clearer reason to keep following the argument.

1.1 — The World We Can See

The fairness question first arrives through evidence available to ordinary sight: who is safe, who is hungry, who is believed, who is injured, who receives time, and whose future is interrupted. These conditions are not merely symbols of an invisible theory. They are realities with causes, consequences, and often identifiable responsibilities.

Beginning here protects the book from premature consolation. The formal Law may eventually ask whether felt experience has a complete-stream boundary property, but it cannot be allowed to turn a visible injustice into an unexplained “balance.” The world we can see is the place where prevention, remedy, access, and public accountability begin.

Three distinctions organize the opening. The sentence “life is not fair” names moral pressure before theory. Suffering deepens that pressure because pain and loss demand recognition rather than abstraction. Comfort and inquiry then separate, allowing care to remain immediate while the scientific question stays open.

1.1.1 — The Ordinary Sentence, “Life Is Not Fair”

The ordinary sentence “life is not fair” is often treated as childish because it is simple. That is a mistake. Some of the deepest human perceptions arrive first as simple sentences. Before a person can build a theory, they can still notice that pain, safety, beauty, wealth, love, health, and time appear distributed with no proportionate regard for innocence, effort, or need. The Law of Fairness does not begin by correcting that protest or asking a

wounded person to wait while an idea explains them. It admits the sentence as the starting pressure.

The sentence also carries a methodological warning. People usually utter it after a mismatch between what they expected and what occurred, but the mismatch may concern chance or choice, procedure or outcome, distribution, desert, luck, public injustice, or the felt weight of an event. That lack of classification does not make the perception empty. It means disproportion has been recognized before the right analytical box has been found. A serious inquiry should ask what kind of unfairness is being named, what evidence supports the judgment, who or what produced it, and what remains unresolved after the cause is known. The simplicity of the sentence is a doorway, not a conclusion.

Consider two children entering the same school year. One arrives rested, housed, medically supported, and certain that an adult will answer the telephone. The other has moved three times, shares a room with people working opposite shifts, and has learned to hide hunger. Their test scores may later be compared as if effort entered a neutral field. The ordinary sentence protests the field before it makes any claim about inner totals. That protest is morally important even if the proposed terminal hypothesis is false.

The same sentence can also name bad luck without accusing a person. A random collision, a genetic condition, a wildfire, or the sudden loss of a caregiver may alter a future before responsibility has any meaningful role. Work on moral luck shows why judgment becomes unstable when outcomes depend heavily on

factors outside control (Nagel, 1979; Williams, 1981). The ordinary complaint notices this instability in everyday language.

Sometimes the sentence is directed at institutions rather than nature. A rule may be written in neutral terms while its cost falls predictably on people with less money, less mobility, less legal knowledge, or less time to appeal. Social conditions shape exposure to illness, violence, pollution, and premature death, so visible unfairness is often organized rather than accidental (Marmot et al., 2008; Williams et al., 2019). A deeper hypothesis cannot be allowed to blur that causal work.

There is a difference between saying that life contains unfair arrangements and saying that life is unfair in every possible sense. The first can be supported by observable facts about rights, resources, exposure, and treatment. The second reaches toward a total verdict on existence. The Law of Fairness enters precisely in the space between those statements. It asks whether one deeper terminal property could exist without denying the injustices already in view.

A careful opening therefore gives the ordinary sentence two forms of respect. It preserves its moral urgency, because preventable suffering and unequal treatment require action. It also subjects the sentence to analysis, because urgency alone cannot identify a terminal law. This double respect lets the reader remain morally awake without asking indignation to perform the work of measurement.

A second scene concerns inherited advantage. A young adult may appear unusually disciplined because years of nutrition, safety,

tutoring, transport, and family stability made disciplined action easier to sustain. Another may display extraordinary effort while producing a less impressive public result. The comparison is not useless, but it is incomplete. Sen's capability approach helps show why resources, conversion conditions, and real opportunities must be distinguished before outcomes are treated as evidence of merit (Sen, 1999; Sen, 2009).

There is also an emotional discipline in refusing premature total verdicts. Anger may correctly identify a wrong while still leaving the full causal history unresolved. Gratitude may correctly identify a gift while still leaving structural privilege intact. The book does not ask readers to replace moral perception with detached calculation. It asks them to let perception lead into better description, more exact responsibility, and more effective repair.

The sentence is also a test of scale. A household can be loving inside an unjust labor market; a fair classroom rule can operate inside an inaccessible school; a country can improve average health while leaving one region behind. Fairness judgments become clearer when the level of analysis is named rather than allowed to slide from moment to institution to existence.

The ordinary sentence also appears when institutions ask people to accept randomness without explanation. A denied claim, an unexplained school placement, or an algorithmic flag may be lawful yet still feel arbitrary because the affected person cannot see the rule or correct an error. The complaint is partly about outcome and partly about being excluded from the reason-giving process.

1.1.2 — Why Suffering Reopens the Question

Suffering reopens the fairness question because pain is not merely information. It is experience demanding recognition. Grief, illness, trauma, humiliation, fear, loneliness, hunger, and premature loss do not wait politely for philosophy to decide whether they matter. The question changes when suffering enters the body, the household, or the name of someone loved: fairness is no longer a distant concept, but a protest against disproportion and a demand that the burden be seen.

Pain science reinforces the need to listen before theorizing. Pain is not identical to tissue damage and cannot be inferred perfectly from a scan, an injury, or a facial expression. It is a personal sensory and emotional experience shaped by biological, psychological, and social conditions. That does not make pain unreal or merely subjective in the dismissive sense. It means the person's report is evidence with a privileged role, and the response should begin with assessment and relief rather than a story about what the suffering means (Raja et al., 2020; Cassell, 1982).

Suffering does more than supply an example of negative experience. It changes the moral order of attention. A person in severe pain cannot be treated as a convenient illustration whose value lies in what observers may learn. The experience first belongs to the person who bears it, and the first practical question is what relief, protection, presence, or repair is possible. Inquiry begins only after that priority is secure.

The social location of suffering matters as well. The same symptom may be treated promptly in one patient and discounted in another. Differences in credibility, access, race, disability, language, gender, income, and geography can shape whose pain is believed and whose care is delayed (Bailey et al., 2017; Williams et al., 2019). These are public failures with identifiable mechanisms. Hidden experience cannot be used to make them metaphysically acceptable.

Childhood suffering makes the problem especially sharp because children depend on adults for interpretation, transport, consent, safety, and relief. Early adversity can affect development across biological, psychological, and social pathways, but those associations do not turn a child into a damaged destiny or a scientific exhibit (Shonkoff et al., 2012). The proper response combines protection, care, developmental opportunity, and humility about what any one history predicts.

Suffering also reveals why a complete-stream claim cannot be defended through isolated positive moments. A person in severe deprivation may still laugh, love, create, or feel relief. Those experiences matter because they were real. They do not cancel the deprivation as a public wrong, and they do not allow an observer to infer that a hidden balance has already been achieved. The stream can contain complexity without becoming an excuse.

Many cultures and families offer narratives that help people endure. Some speak of purpose, faith, growth, solidarity, or continuity. Such narratives may be freely valuable, but they cannot be imposed as the meaning of another person's pain. Meaning-making varies widely, may change over time, and may

never occur in the form observers expect (Park, 2010). The right to refuse a meaning is part of the dignity of the sufferer.

Suffering also changes time. Minutes in acute pain may feel enlarged, while months of chronic illness can blur into fatigue, appointment schedules, missed work, and repeated explanation. Anticipation can make a future procedure present long before it begins. Relief can be immediate, delayed, incomplete, or accompanied by fear that the pain will return. Any account of felt experience that ignores these temporal structures will mistake clock time for lived exposure.

The problem becomes sharper when communication fails. An infant, a person with advanced dementia, an intubated patient, or someone who does not share the clinician's language may be unable to provide the report a protocol prefers. The absence of a standard answer is not the absence of experience. Pain science therefore treats self-report as central where available while also developing behavioral, contextual, and physiological evidence for situations in which speech is constrained (Raja et al., 2020).

Suffering can also be socially amplified. Pain accompanied by disbelief is not merely the original symptom plus an unrelated insult. The person may face delayed treatment, self-doubt, financial loss, stigma, and the repeated labor of proving what cannot be directly displayed. Epistemic injustice scholarship helps name this added burden when a person's testimony is discounted because of identity, diagnosis, or social position (Fricker, 2007; Kidd & Carel, 2017).

A serious theory must distinguish suffering from the story told about it. An event may be preventable, unavoidable, intentionally caused, negligently ignored, or resistant to present treatment, and those differences alter responsibility and remedy even if the felt valence were identical. The hard case does not make the Law more likely; it makes vague answers less acceptable. A complete-stream hypothesis must face partial recovery, absent meaning, limited observation, and uncertain endings while remaining exact enough to lose.

Suffering and relief can cascade through connected pathways: pain can disrupt sleep, concentration, work, and economic security, while effective care can reverse several links at once. These ordinary mechanisms matter independently of PTVC and should be modeled before any terminal interpretation.

The reopening matters because many people stop asking about fairness not because the question was answered, but because the available answers became unbearable. *The Law of Fairness* must be written for that reader too: the reader who is not naive, who has heard enough explanations, and who will only continue if the book respects pain before theory.

1.1.3 — The Difference Between Comfort and Inquiry

Comfort and inquiry answer different human needs. Comfort responds to the person who is hurting; inquiry responds to the question that remains unresolved. A book fails when it offers theory where safety or presence is needed, and it also fails when it offers consolation where a reader is asking what is true. *The*

Law of Fairness must preserve both without letting either impersonate the other.

In acute distress, comfort may mean protection, practical help, silence, a glass of water, a ride to treatment, a door that locks, or truthful reassurance that does not pretend to know the future. Inquiry can come later, if the person wants it. Reversing that order can turn another person's pain into an occasion for displaying a worldview rather than meeting the need in front of them (Cassell, 1982). Schools, hospitals, and communities need the same discipline: safety, treatment, repair, and accountability do not wait for a theory of cosmic order.

Inquiry has a slower rhythm: definitions, alternatives, evidence, and the possibility of error. Medicine's distinction between disease and suffering shows why a technically correct account can miss threatened identity, agency, relationship, or future, while compassionate presence may help without explaining the mechanism (Cassell, 1982). The humane sequence protects both roles. Care does not pretend to settle inquiry, and inquiry does not turn a person into a puzzle.

Verb choice makes the boundary visible. 'I believe that you are hurting' recognizes testimony, and 'we will help you find safety' commits to action. 'I know what your pain is for' claims authority, while 'everything will balance' predicts an unseen outcome. Honest inquiry can also say 'we cannot tell' when endpoint, measurement, or missingness remain unresolved. Abstention is not abandonment; it prevents an unanswered question from becoming a verdict on a person.

The book offers a shared question that readers can examine from different starting points. A scientist may ask what observations would distinguish explanations. A philosopher may ask what a complete conscious history means. Another reader may be drawn by the possibility that unequal lives contain a relation we have not yet learned to describe. These approaches can enrich one another while keeping their questions distinct. The invitation is to follow the argument closely enough to recognize what would count as progress, and to carry that clearer understanding into the next conversation.

Timing and role design make this distinction practical. A wounded person need not participate, provide evidence, accept consolation, or become an example. During acute danger or the first hours of loss, stabilization, quiet, and trusted presence may matter more than interpretation; later, the same person may want causes, probabilities, records, or philosophy. Independent consent, interpreters, advocates, cooling-off periods, and separation between treatment and study decisions protect both the person and the evidence (Fricker, 2007; Substance Abuse and Mental Health Services Administration [SAMHSA], 2014; Beauchamp & Childress, 2019; National Commission, 1979; Emanuel et al., 2000). This sequencing respects both urgency and autonomy: assistance can be immediate while interpretation remains optional.

Public cases require context, permission, privacy, and current status. Anonymous or composite illustrations must be identified honestly. A person may clarify an ethical or scientific problem without becoming a lesson, symbol, or counterexample owned by

the observer (Fricker, 2007). When a case must be revisited, a dated source and correction route help prevent an old portrayal from becoming the person's permanent public identity. This also improves longitudinal interpretation: later changes can be compared with a dated account instead of being forced into a single permanent story. Revision becomes evidence of a life continuing to unfold rather than a defect to conceal.

1.2 — What Fairness Usually Means

Fairness is a family of ideas rather than one settled object. Rules can be fair, distributions can be fair, judgments can be fair, and relationships can be fair in different senses. Confusing these senses can make an argument appear stronger than it is, because the authority of one standard is silently borrowed by another.

Older theories of justice, desert, welfare, and luck keep their proper work. Public justice remains public justice, responsibility remains answerable to control and evidence, and welfare remains multidimensional. The proposed Law becomes clearer by naming a different question rather than treating one already answered as its own.

1.2.1 — Procedural Fairness

Procedural fairness concerns rules, hearings, and decision paths. It asks whether standards were knowable and applied consistently, bias was limited, people could understand what was happening and be heard, reasons were connected to decisions, and errors could be challenged before someone with authority. These features matter because process shapes both outcomes and the experience of being treated as a participant rather than an object; research identifies voice, neutrality, respect, and trustworthy motives as important to perceived legitimacy (Tyler, 2006). Formally equal rules can still exclude through language, disability, time, access, or power.

The Law respects procedural fairness because bad procedure creates hidden burden. A person who cannot understand a process, contest an error, access a hearing, or know why a

decision was made bears more than an external disadvantage. Respectful treatment and intelligible reasons can matter even when the outcome cannot change, but procedure does not settle the fairness of a life. Any research, communication, or institutional use of the Law must itself remain contestable and respectful (Tyler, 2006).

Neutrality also requires more than an impersonal tone. Decision rules can encode old inequalities, data can omit the most burdened people, and discretion can be exercised differently across groups. Structural racism and other durable forms of exclusion affect health and public credibility through pathways that cannot be repaired by courteous administration alone (Bailey et al., 2017; Williams et al., 2019). Procedure must therefore be evaluated by design, operation, and patterned consequence.

Respectful treatment matters even when the decision remains adverse. A person can be told no without being humiliated, ignored, or treated as a threat. This distinction is especially important in systems where the person depends on the decision-maker for medicine, safety, education, income, or legal status. Dependency magnifies the experiential cost of opacity and makes retaliation protections part of fair process.

The limit appears when procedure is asked to settle a complete life. Fair hearings can coexist with profound pain, and flawed procedures can sometimes benefit a person. That is not a criticism of procedure but an ownership rule: procedural justice governs authority and treatment, while the proposed Law asks about an exact terminal relation within one complete stream. Preserving

that division keeps the Law from replacing rights, remedies, and institutional standards.

The same distinction governs science. Ethical approval, preregistration, and transparent analysis improve trustworthiness, but they do not make a weak construct valid or guarantee truth. The Law should be stated in advance, tested under common rules, and revised through visible reasons; a hypothesis that changes its endpoint or scale after an adverse result has failed its own procedure.

Procedural fairness is especially important where evidence is uncertain. The less certain the institution is, the more it needs transparent reasons, revisability, and appeal. Confidence cannot substitute for a process that lets affected people test what was assumed about them.

Public procedures can impose burdens even when the rule itself is formally available. Applicants may have to learn a complex process, produce repeated documentation, wait through delays, travel, disclose private facts, and absorb the anxiety of a minor error. These costs can restrict access to rights and benefits as effectively as an explicit denial, which is why procedural fairness must examine the route as lived rather than merely as written (Herd & Moynihan, 2018).

1.2.2 — Desert-Based Fairness

Desert-based fairness asks when praise, reward, blame, or sanction properly follows from what a person chose and did. The idea has a legitimate local use. Promises, contracts, offices, and relationships would become unintelligible if intention,

knowledge, effort, and responsibility never mattered. The difficulty begins when a judgment about conduct expands into a judgment about the value of a person or the hidden purpose of suffering.

Desert is emotionally powerful because it promises a morally ordered world. Good effort should matter, cruelty should not prosper, and people harmed by wrongdoing should receive repair. That hope can support accountability. It can also encourage observers to read every outcome as a verdict, even when luck, unequal opportunity, illness, or power did most of the causal work.

A useful desert judgment is narrow. It asks what duty applied, what the person understood, what alternatives were genuinely available, how freely the action was chosen, and what response is tied to the act and harm. A worker who completed difficult work may reasonably expect credit; a public official who concealed a conflict may properly lose authority; a person who damaged another's property may owe repair. None of these judgments makes distress itself the currency of justice.

Desert judgments become unreliable when control is overstated. People act with unequal information, health, time, safety, education, freedom from coercion, and histories they did not choose. Moral luck names the role of conditions beyond choice in judgments of character and consequence, including genes, early caregivers, social location, exposure to violence, illness, and the timing of opportunity (Nagel, 1979; Williams, 1981). Acknowledging these influences does not erase agency; it prevents agency from being inflated into total authorship of a life.

Capabilities matter for the same reason. The same resource can produce very different practical freedom depending on disability, discrimination, transport, caregiving, health, and social support (Sen, 1999; Sen, 2009). A judgment that sees only the resource may praise conversion advantages as merit and blame conversion barriers as character.

Children establish the clearest limit. A child does not earn illness, hunger, abuse, unsafe housing, war, abandonment, or the absence of medicine. No hidden fault can be invented to save a theory of moral order. The child's standing creates present duties of protection and care, not a puzzle to be solved by assigning desert.

Success deserves the same scrutiny from the opposite direction. Wealth, health, status, and achievement can reflect real discipline while also depending on inheritance, timing, social networks, public infrastructure, and freedom from emergency. Recognizing luck does not insult effort. It keeps success from being misread as proof of superior human value.

Responsibility, human worth, and recovery must remain separate. Accountability may require truth, restitution, treatment, boundaries, or proportionate restriction, but suffering itself is not a moral invoice. Addiction shows why agency and constraint can coexist, while survivor recovery shows why later growth never proves that an earlier harm was useful. Repair should therefore ask what was damaged, what can be restored, and how recurrence can be prevented without turning pain into payment.

Good fortune creates no moral debt of pain. Joy, safety, love, health, and relief do not become suspect because they were not

earned through hardship. Gratitude and generosity may be fitting responses to advantage, yet neither creates a duty to suffer. Under exact PTVC, however, a genuinely net positive accumulated balance requires a net negative remainder; that arithmetic does not identify a particular future harm.

The formal Law contains no desert variable. It does not ask whether an experience was morally assigned, earned, deserved, or deservedly reversed. Its proposed terminal condition, if it were ever supported, would be a boundary property of felt experience. Importing cosmic judgment would change the claim rather than explain it.

Local desert questions remain important. Communities still need to distinguish accident from intention, inability from refusal, repair from evasion, and proportionate accountability from cruelty. Those questions belong to conduct, institutions, and public reasons. They cannot settle the complete felt history of a conscious stream.

The limit of desert points toward the next target. Procedural fairness can tell us whether rules were applied properly, and desert can help evaluate accountable action, but neither describes the whole lived course of a person. The formal Law therefore asks a different question, one that can include visible distribution without reducing experience to reward or blame.

1.2.3 — Why the Formal Law Needs a New Target

The formal Law needs a target that can travel across visible differences while remaining faithful to lived reality. Money, health, status, rules, opportunity, welfare, justice, and

responsibility all matter and can shape experience, but none alone answers the additional question: whether the complete phenomenal history of one separately routed qualifying stream satisfies an exact terminal relation. The target is narrower than a theory of justice and more demanding than a claim that mood tends to recover.

Specificity does not make the Law superior in every domain. Courts still need procedural fairness, governments need distributive and administrative fairness, and families need care and repair. The new target earns a separate place only if it has a definite bearer, covers the complete relevant history, and states a boundary property that could fail. The formulation developed here supplies those commitments; this public volume explains their meaning without treating precision as evidence.

The bearer is one separately routed qualifying conscious stream, not a country, household, generation, or population average. This prevents positive experience in one stream from canceling negative experience in another. The proposed currency is felt valence under a declared representation. Wealth, rights, health, behavior, and status can influence experience, but they are not the formal object and they remain independently important.

Completeness requires every qualifying part of the strict-pre-endpoint domain, including portions that are difficult or impossible to observe. Exactness distinguishes a latent zero from a tendency, small mean, or equivalence band. Terminality makes the claim a boundary condition rather than a demand for moment-by-moment balance. A life may traverse large positive and

negative regions and still satisfy or violate the proposed endpoint relation.

Equal totals would not make lives equivalent. In an invented example, one stream could accumulate ten units of positive experience and ten of negative experience; another could accumulate a thousand of each. Both net to zero, while their intensity, duration, opportunities, and histories remain profoundly different.

The target is meaningful only when bearer, representation, neutral origin, domain, endpoint, observation, and serious rivals are fixed independently. A shifted scale, result-selected endpoint, or outcome-dependent repartition can manufacture closure. Felt valence is therefore a testable representation choice, not an assumption that experience is perfectly scalar or directly measurable. Mixed, nonadditive, or partially ordered experience may require a different branch or abstention rather than forced fit. For example, shifting five hypothetical ratings upward by one unit raises their sum by five, showing why the origin must be fixed before aggregation.

This precision changes which objections matter. Income research addresses material and evaluative domains; adaptation research addresses ordinary dynamics. Neither settles complete-stream closure. Failed routing, unidentified sign, unstable origin, or result-selected endpoints strike load-bearing dependencies. That map is valuable before support exists because it separates justice, affective science, consciousness research, endpoint research, and the terminal proposal, while leaving each free to win its own questions. This division gives collaborators a practical starting

point: identify the unresolved component and choose evidence capable of improving that component. A measurement specialist can sharpen the neutral origin, a consciousness researcher can sharpen the bearer, and a modeler can ask which mechanism would make the terminal relation necessary rather than accidental. The question is large enough to matter and divided into pieces that different fields can genuinely advance.

1.3 — The Cases That Resist Easy Answers

A hard case should make a theory work, not merely make a reader feel. The cases in this Section are chosen because they remove familiar escape routes. Innocent children block blame. Early death exposes the difference between observed biography and a complete stream. Visible fortune with hidden burden warns against treating public advantage as transparent access to private experience.

Each case carries an ethical rule. The person is not an example first. The visible harm is not minimized. Hidden experience is not invented. A hypothesis about the whole stream cannot be used to explain why the event happened, and no later possibility can turn the injury into something owed.

The cases also widen the inquiry. They show why a single dimension, such as wealth, health, duration, or public success, cannot represent everything that matters. That widening prepares the move from biography to experience without making experience an excuse for biography.

1.3.1 — Innocent Children

Innocent children are the first hard case because they remove blame before the argument begins. A child does not choose illness, abuse, neglect, hunger, war, disability, abandonment, or fear. Dependency gives adults and institutions greater responsibility; it never gives the child moral responsibility for conditions imposed on them (United Nations, 1989).

The visible scandal is immediate. Some children are surrounded by safety, patience, medicine, and love. Others are exposed to violence, instability, untreated conditions, or grief before they can interpret what is happening. This difference is not an abstract distribution problem.

Childhood also exposes the limits of consent and voice. Children may lack the language, authority, or practical safety to report what is happening, yet adults can still overinterpret silence, compliance, or adaptation. Rights protections therefore place special duties on families, institutions, clinicians, and researchers. A child's vulnerability is not a reason to turn the child into an emblem. It is a reason to strengthen protection, seek assent where possible, respect dissent, and distinguish the child's interests from the ambitions of adults (World Medical Association, 2024; United Nations, 1989).

Childhood is not valuable only as preparation for adulthood. Safety, play, attachment, learning, rest, friendship, and relief are present goods in a life being lived now. Protecting them is not merely an investment in future productivity or health. The child's current experience has standing before any long-range theory is considered (United Nations, 1989).

Children also depend on adults to notice and interpret distress. Pain may be expressed through movement, sleep, play, withdrawal, irritability, or physiological change before a child can describe it. The fact that communication is incomplete does not make the experience less real. The modern pain definition was deliberately written to avoid making verbal report a condition for the existence of pain (Raja et al., 2020).

Development makes timing consequential. The effects of adversity depend on when it occurs, how long it lasts, how severe or persistent it is, and whether responsive relationships and other protections buffer it. Interacting pathways of stress regulation, learning, health, and relationship may be altered without fixing a child's destiny (Shonkoff et al., 2012). These findings should direct prevention, support, and repair rather than deterministic labels; a child is not a forecast, and biological embedding is not fate.

Research with children carries additional obligations: developmentally meaningful assent, respect for dissent, proportionate burden, and protection against creating serious suffering for data (World Medical Association, 2024; CIOMS, 2016). Dependency increases adult responsibility rather than making children easier evidence sources.

Children reveal why social causation matters. Preventable disease, malnutrition, conflict, unsafe environments, and unequal access to essential care have institutional histories. Global child-mortality estimates show both long-term progress and a continuing concentration of deaths in regions and settings with the greatest barriers to survival (United Nations Inter-agency Group for Child Mortality Estimation [UN IGME], 2026). The per-stream focus should intensify attention to each bearer, not compete with population prevention.

A terminal hypothesis cannot tell a parent why a child suffered. It cannot promise a chosen compensation or timetable, identify a purpose, or convert grief into a research result. It can only state a narrow candidate relation whose terms and endpoint would still

need to be established. In the presence of grief, that restraint is a form of respect.

Resilience should describe a process, not excuse an environment. A child may adapt through supportive relationships, skill, temperament, community, or sheer necessity, while still carrying costs that deserve attention. Quietness, vigilance, high achievement, or apparent self-sufficiency can be compatible with competence and distress at once. The evidence should direct support toward the child and the conditions, not turn endurance into proof that help was unnecessary.

Because children often cannot leave unsafe settings, navigate complaint systems, or describe experience in adult terms, institutions need age-appropriate reporting, independent advocacy, nonretaliation, and developmentally appropriate listening (United Nations, 1989; CIOMS, 2016). The practical aim is to keep the child's perspective central without forcing repeated disclosure or treating limited language as absence of experience.

The formal question cannot turn dependency into invisibility. Experiences that are difficult to report still matter. A research program that includes only articulate, stable adults would study an easier reference class and should say so rather than generalize to every conscious stream.

This way of reading the case asks where the child's burden is carried, where it is hidden, where adults deflect it, and where institutions delay response. That practical question does not need the formal Law to be proven before it matters.

1.3.2 — Early Death and Interrupted Futures

Early death brings the endpoint question forward before the reader is ready for it. A life interrupted in childhood, youth, or early adulthood seems to defeat every easy account of fairness. There was no long arc, no visible time for repair, no completed public unfolding, no old age in which patterns could soften.

This case forces the distinction between lifespan and conscious stream. The formal Law will eventually ask about the completed stream, not about the amount of visible biography. But that formal distinction cannot be used to make early death sound small.

An interrupted future is not merely a shorter count of calendar years. It can deprive a person of projects, relationships, repair, learning, agency, and possibilities that had already begun to matter. Philosophers have long examined death as deprivation while recognizing the uncertainty of counterfactual futures (Nagel, 1970). Those unrealized possibilities remain ethically important, but they are not phenomenal events to be entered retroactively into the ledger. The formulation developed here concerns experience within the stream; justice, medicine, and grief also track goods and futures the phenomenal target does not absorb.

Early death also clarifies why endpoint independence cannot become clinical arrogance. The fact that the formal model needs a stopping rule does not give a reader, researcher, or public presentation permission to interpret a dying person from the outside.

Early death sharpens two scientific problems that later chapters treat in full: the end of biological life, loss of report, and the truth-level end of a qualifying stream are not automatically identical; and the intervals hardest to observe may be lost through fatigue, sedation, delirium, deterioration, or death. Those uncertainties must remain explicit rather than being converted into a convenient terminal value. At the bedside, care and palliative priorities govern; measurement yields whenever it burdens treatment or participation (World Health Organization, 2020).

Early death also exposes the limits of hindsight. After a death, people search for a final meaning in the last words, expression, symptom change, or period of calm. Such moments can be deeply important to a family. They are not automatically evidence of terminal neutrality, divine settlement, or a completed moral narrative. Grief deserves memory without being turned into a verdict.

A death may result from disease, violence, neglect, environmental exposure, delayed treatment, or interacting causes. A hypothesis about felt experience never weakens the obligation to identify and prevent those causes. Science may ask when a stream ends and what can be known near that boundary; institutions must still improve prevention, treatment, bereavement support, and the conditions that make early death unequally likely. A possible terminal invariant would never make an avoidable death less avoidable.

Counterfactual loss cannot be measured by assigning the person a single imagined future. Treatment might have succeeded or failed; relationships might have deepened or ended;

circumstances might have widened or narrowed possibility. Responsible analysis can compare plausible opportunities and known barriers without pretending to know the life that would have unfolded. The range of uncertainty belongs in the conclusion.

Population statistics reveal patterns that individual stories cannot. UN IGME estimates that 4.9 million children died before age five in 2024, including 2.3 million newborns, with the heaviest regional burdens in sub-Saharan Africa and Southern Asia (UN IGME, 2026). These estimates describe the scale and distribution of loss; they do not establish hidden compensation. Prevention remains urgent whether the terminal hypothesis is true, false, or unmeasurable.

For the formal hypothesis, early death creates a reachability question. If the endpoint arrives while the cumulative object is far from zero, what lawful structure could still make exact closure possible? The model cannot answer by assuming a terminal jump, a hidden continuation, or favorable endpoint selection unless that branch is independently specified and tested.

Those who remain may carry medical debt, traumatic memory, altered family roles, lost income, disrupted education, and decisions that cannot be revisited. These consequences belong to their own lives and to institutions capable of reducing them. They should be named because early death is not a single private event; it reorganizes households and communities without transferring anyone's experience into another stream.

The Law of Fairness cannot answer early death by declaring the lost future irrelevant. A careful account must keep three truths apart: the experience that was lived, the opportunities that were foreclosed, and the grief and material consequences borne by others. None can be used as a proxy for the hidden terminal state. Keeping them separate is what allows the formal question to remain exact while the human loss remains fully visible.

1.3.3 — Visible Fortune With Hidden Burden

Visible fortune is real and often protective. Wealth, health, status, talent, safety, and support can widen options and reduce exposure, yet none reveals the whole felt condition. Public success may coexist with fear, isolation, addiction, scrutiny, pain, trauma, or distrust. The responsible inquiry follows mechanisms and testimony without presuming either effortless happiness or secret misery (Kahneman & Deaton, 2010; Killingsworth et al., 2023). Longitudinal evidence is especially useful here because a single photograph or disclosure cannot show whether a burden is fleeting, recurrent, or structurally tied to the conditions surrounding the person.

Pity makes the opposite error when visible hardship is treated as the whole person. Difficult circumstances may coexist with love, dignity, humor, skill, meaning, faith, memory, or peace. That does not make deprivation harmless or barriers acceptable. The book rejects both sentimental inversion and the claim that appearances are complete. A satisfying life can therefore be reported truthfully while a barrier remains unjust and worth removing.

Advantages reach consciousness through concrete pathways. A promotion may raise income while reducing sleep or privacy; recognition may bring belonging alongside scrutiny; wealth may lower danger while adding exposure or family conflict. Social media intensifies the temptation to infer an easy inner life from a house, title, body, partner, or audience. Better questions examine control over time, exit, belonging, comparison, surveillance, and the cost of honest disclosure. Following those pathways over time can reveal which pressures persist after the visible advantage changes and which disappear with context.

Success can make disclosure expensive, while outsiders can underestimate quality of life among people facing poverty or disability and still ignore remediable barriers. The correction is to respect testimony, inspect incentives and conditions, preserve public justice, and avoid turning either advantage or hardship into an inner verdict (Albrecht & Devlieger, 1999; Koch, 2000; Ubel et al., 2005). Hidden burden matters when it reveals repairable pathways such as safer disclosure, nonretaliation, accessible care, rest, or support; it never makes the world secretly equal. The most useful inference is one that points toward a condition that can be checked or changed, not a compensating story invented to make unequal lives look balanced.

1.4 — Why Averages Cannot Settle a Whole Stream

Because the formal claim is per-stream and terminal, population averages, follow-up windows, and ordinary regression address different objects.

Here the intuition becomes a target-selection problem. A claim about one complete stream needs a bearer and endpoint that no average, selected episode, or follow-up window can supply.

1.4.1 — Averages Do Not Save Streams

Averages are indispensable for seeing patterns that anecdotes cannot reveal. They estimate prevalence, compare groups, detect change, and direct attention. The Law of Fairness rejects neither averages nor population science. It rejects letting a summary stand in for the bearer who feels, because populations have no single conscious stream: nations aggregate outcomes, families contain connected but distinct persons, and one person's relief remains outside another person's experienced history (Robinson, 1950; Molenaar, 2004; Fisher et al., 2018).

The ecological fallacy is one familiar example of the problem. A relation among group-level percentages may not describe the individuals inside those groups. Human-subject research faces a parallel challenge when between-person patterns are assumed to describe within-person dynamics. These methodological warnings do not prove the Law. They explain why a population tendency cannot stand in for the per-stream terminal claim the Law actually makes (Robinson, 1950; Molenaar, 2004; Fisher et al., 2018).

The scientific program therefore needs distributions, trajectories, heterogeneity, individual-level uncertainty, and explicit shrinkage. Hierarchical models may borrow information across people when individual data are sparse, but partial pooling must remain visible: a person's estimate pulled toward the group is not direct knowledge of that person. A model can predict the center well while missing volatile or extreme paths, so it must report who the average fails to represent and how uncertainty changes at the tails.

This has a direct public consequence. The sentence most people adapt can be statistically defensible and still cruel when aimed at the person who did not. Population findings should guide inquiry without becoming commands, accusations, or reasons to withdraw care.

Aggregation can even reverse the story. One clinic may have better outcomes than another within every relevant patient group yet a worse overall rate because it treats a larger share of difficult cases. The arithmetic is correct while the interpretation fails. A per-stream question therefore asks which units were pooled, how they were weighted, and whether the summary preserved the people whose paths generated it.

Population success can be distorted by attrition. If participants with worsening conditions are more likely to leave, a cohort mean may return toward baseline because some recover, some remain impaired, and the most severely affected disappear from observation. Death, institutionalization, distrust, cost, caregiving, and symptom burden can all alter the denominator. Without

tracking who left and why, a reassuring average can be partly manufactured by observability rather than recovery.

Averages cannot establish a universal per-stream claim or erase a gate-valid counterexample. Section 4.2.3 states the quantifier rule in full.

Averages can reveal whether a pattern is common. They cannot save a stream by absorbing it into the group. The per-stream requirement is both a formal commitment and a protection against disappearing concentrated burden inside a successful mean.

1.4.2 — A Life Is Not a Follow-Up Window

Studies observe people over days, weeks, months, or years, and those windows can reveal recovery, relapse, adaptation, burden, relief, and changing physiology. But every empirical study begins after some experiences have occurred and ordinarily ends while life continues. Even a long follow-up is a segment unless a justified bridge connects its observed prefix and remainder to the complete strict-pre-endpoint truth domain. The strong Law cannot erase unobserved time by calling the recorded interval a life.

Timing can change the apparent conclusion. A study may end before delayed collapse, miss late repair, capture a temporary rebound, or begin after the most severe burden has already passed. The same life can support opposite stories when different windows are selected.

Finite studies remain scientifically valuable when their claims are bounded. A window may test measurement validity, adaptation after a specified event, reserve scaling, within-person dynamics,

endpoint-proxy performance, or comparative model prediction. Incompleteness is not the defect; promoting a segment into the whole stream is. Every conclusion should stay at the level the observed interval and its justified bridges can support.

Observation windows are also shaped by who remains observable. People may drop out because they improve, deteriorate, move, lose trust, become too ill, or lose the capacity to respond. Near the end of life, sedation, delirium, fatigue, family mediation, and communication loss can change what reaches a dataset. A complete-stream claim cannot treat this missingness as empty time. The scientific companion must model observation, retention, and endpoint relevance rather than allowing the last available measurement to become the true end by convenience.

Study windows are constructed. Recruitment may begin after a crisis or after the least tolerant people have already left; repeated measurement can alter attention and behavior; retrospective reports are present-day reconstructions of the past; and protocols sample only the settings and channels they can reach. These are ordinary design limits, not reasons to abandon longitudinal work.

A responsible study therefore states the lived interval it targets, the interval it actually observes, who becomes unobservable and why, and what bridge, if any, connects missing prefixes or suffixes to the target. More sensors cannot repair an undefined construct, missing consent, or selective dropout.

The public reader needs three dates: when the relevant lived process began, when observation began, and when the independently defined target endpoint occurs. If they differ, the

missing intervals remain part of the scientific problem; a study stop or final survey cannot be promoted into the truth-level endpoint by convenience.

1.4.3 — The Need for an Exact Terminal Question

If the Law is to be more than a tendency, it must eventually say something about the completed felt record of one qualifying conscious stream immediately before its independently defined truth-level endpoint, not merely about local rebounds, typical averages, anniversaries, interviews, hospital admissions, or the last available survey. The record is latent because no observer directly sees every contribution across a life, and a study stop is not the endpoint unless an independently justified bridge says otherwise.

A simple thought experiment shows how an exact claim could face a finite challenge. On the additive branch, suppose independently justified bounds established that a stream had already accumulated a positive net total of at least eight units and could accumulate at most two further negative units before its independently established true endpoint. Its final total would have to remain positive. No exact final reading would be needed to exclude zero.

The example identifies the information a test would need. Establishing the units, covering all relevant past and future contributions, and justifying those bounds would be the scientific work. A bound supported statistically would carry its stated uncertainty; a deductive exclusion would need bounds established to contain the true values under the declared model. This turns the

measurement problem into a concrete challenge: can a proposed route supply enough independently justified information to make some outcomes impossible?

Exactness gives the proposal scientific shape. “Life tends toward balance” can survive almost any counterexample by changing the scale, the window, or the meaning of balance. An exact terminal claim names the stream, the positive and negative quantities, the complete domain, and the endpoint independently of the observed result. Its vulnerability is a contribution, because it turns a recurring human intuition into a claim that reality may support, narrow, or defeat.

Several ordinary processes could imitate the appearance of closure. Regulation can pull states toward a usual range, worsening illness or medication can reduce variability, and a statistical path can be made to end near zero simply because that endpoint was built into the model. A forced ending is not evidence that nature supplied the rule.

The endpoint must be defined independently of the answer. If the stream is said to end whenever the residual reaches zero, closure is true by construction. The scientific program instead requires truth-level and protocol endpoints that are fixed through consciousness and routing criteria rather than chosen by the ledger estimate. This is the tautology firewall.

Exactness also makes the neutral origin load-bearing. Multiplying a valid scale by a positive constant preserves zero, but shifting the origin generally changes the cumulative result. Researchers cannot center each person’s data after seeing the endpoint and

then call the resulting zero evidence. Neutrality must be anchored through independent judgments, comparisons, calibration, and uncertainty.

The terminal question is also branch-sensitive. If consciousness can split, fuse, pause, or continue beyond the biological event used by a protocol, the bearer and endpoint may be uncertain. The response is not to choose whichever routing gives closure. It is to predeclare admissible branches, carry uncertainty, and accept no adequate estimate when the object cannot be identified.

An ancient protest has become a more exact question without being mistaken for an answer. Visible injustice remains real, ordinary standards of fairness retain their authority, and the hard cases retain their force. The inquiry now turns inward, asking what kind of lived quantity could carry a terminal question while preserving the privacy, plurality, and dignity of the person who experiences it.

Scope words and reference classes carry scientific weight. A claim about every qualifying stream is stronger than one about most streams, a declared population, or one intervention, and evidence from verbally reporting adults cannot automatically settle the claim for infants, people with severe disorders of consciousness, nonhuman animals, or artificial systems. A counterexample has different consequences depending on the version stated in advance. Each extension therefore needs its own bearer, measurement, endpoint bridge, and calibrated public language.

Exactness and empirical humility belong together. A finite study may establish a measurement foundation, support compatibility within a declared band, favor one route over prospectively specified rivals, narrow a reference class, produce adverse evidence for a protocol-fixed branch, or conclude that no adequate estimate is possible. Each result is scientifically meaningful when reported at its actual evidence role and resolution. Progress can therefore be cumulative without pretending that compatibility is proof. A sequence of bounded results can improve the representation, endpoint rule, observation model, and rival set even while the universal claim remains unsettled. That cumulative record makes later tests sharper by showing which assumptions have survived contact with data and which remain the weakest links. It also keeps a narrow success from being mistaken for a broader law before the unresolved gates have been tested.

Chapter 2 — The Life Only You Can Feel

A house is visible. Safety is lived. A diagnosis is recorded. Pain is felt. Status can be announced in a title, but belonging, humiliation, relief, vigilance, and loneliness unfold inside a person's experience. The distinction is simple enough to overlook and deep enough to reorganize the fairness question.

External conditions matter for many reasons. They affect survival, agency, rights, opportunity, time, health, relationship, and the ability to refuse. The proposed Law leaves the independent value of these goods intact. A right remains a right even when its violation is not consciously registered in the moment. An inaccessible building remains unjust even if the person excluded has developed resilience. Material reality cannot be dissolved into mood.

At the same time, external goods do not carry one fixed experiential value. Money can relieve hunger, purchase medicine, create privacy, reduce dangerous work, or restore time. It can also expose a person to pressure, predation, conflict, isolation, and fear of loss. Status can open doors and make every mistake more visible. A body can be in pain while a life retains meaning; a body can appear healthy while terror or compulsion dominates the day. Research on income and well-being, disability and reported quality of life, and response shift makes the general point clear: condition and experience are related, but they are not identical (Kahneman & Deaton, 2010; Killingsworth et al., 2023; Albrecht & Devlieger, 1999; Koch, 2000; Sprangers & Schwartz, 1999).

The formal Law needs this distinction because its target is phenomenal. On the primary additive branch, it concerns positive and negative magnitudes as lived within a qualifying stream. The

public reader does not need the measure theory yet. The essential idea is that the hypothesis asks what experience was like from within, across the complete stream, rather than how impressive or tragic the biography looked from outside.

Studying that inward life is a demanding but concrete task. A person's description, changes in behavior, bodily signals, and context can each help us ask a more precise question about what is being felt. The aim is to find out which clues remain reliable when appearances mislead and which combinations reveal more than any one clue alone. The person's own account remains central: every measurement has to earn the meaning assigned to it.

Pain research provides a clear example. The revised International Association for the Study of Pain definition emphasizes that pain is a sensory and emotional experience, not a direct synonym for tissue damage. The accompanying notes also stress that pain is personal and that a person's report deserves respect. Nociception, facial expression, heart rate, imaging, and behavior can contribute evidence, but none is the experience itself (Raja et al., 2020).

Imagine two people sitting quietly in the same waiting room. One is peacefully relieved; the other is frozen with fear. Their posture may look similar while their experience differs sharply. Affective science helps explain why observation requires concepts, context, and multiple forms of evidence rather than confidence drawn from appearance alone.

Interoception adds another layer. The nervous system senses, interprets, and integrates signals from within the body,

contributing to urges, feelings, regulation, and a range of mental-health processes. Yet interoceptive indicators are not a transparent window into valence. The same heartbeat can be excitement, fear, exertion, illness, or nothing salient at all. Context and interpretation matter (Khalsa et al., 2018).

This is why the Chapter's title speaks of the life only you can feel. It does not mean that experience is ineffable, incomparable, or sealed beyond all science. People communicate, recognize, remember, and sometimes measure aspects of feeling. It means that access is asymmetrical. The person lives the state; everyone else receives traces.

Hidden burden is possible, but it cannot be presumed whenever visible advantage offends a theory. Hidden relief is possible, but it cannot excuse deprivation. The central difficulty is not that experience is unreal; it is that experience is intimate, temporally changing, partly reportable, and only indirectly available to anyone else. First-person testimony deserves privileged standing because the experience is the person's, yet language, memory, fear, adaptation, audience, context, and the consequences of disclosure can shape what can be said. Humility therefore requires neither total skepticism nor total credulity, but proportionate confidence and visible limits.

Envy infers ease from advantage; pity infers misery from visible limitation. Both replace the person with an observer's story. Disability scholarship has long challenged judgments that treat impairment as total biography, while critiques of the "disability paradox" warn against using positive reports to ignore barriers or romanticize adaptation (Nussbaum, 2006; United Nations, 2006;

World Health Organization, 2022; Albrecht & Devlieger, 1999; Koch, 2000).

Population studies can estimate associations between income and different dimensions of well-being, and newer adversarial work has clarified why earlier findings seemed to conflict. None of this creates a universal conversion rate between money and experience. Income may matter differently across deprivation, security, aspiration, social comparison, family responsibility, and the distribution of control (Kahneman & Deaton, 2010; Killingsworth et al., 2023).

Experience is temporal. A painful event can be brief and intense; chronic discomfort can become background without becoming harmless; relief can be powerful because a burden ends. What a person feels during an event, remembers later, expects next, and says under a particular question may differ without fraud. Adaptation and response shift can alter standards, values, and the meaning of health or disability, while later interpretation and memory do not rewrite what was already felt (Sprangers & Schwartz, 1999; National Research Council, 2013).

Acute pain can command attention while chronic pain reorganizes sleep, movement, work, and anticipation. Medication may relieve one burden while creating fatigue or cognitive change. Love can widen joy and fear at once; status can expand agency while increasing exposure; boredom and overstimulation can narrow attention through opposite routes. These pathways show why conditions become lived through mechanisms, not through one fixed conversion into happiness.

The final Section identifies what self-report, behavior, physiology, interoception, and neural evidence can contribute, and why no single channel becomes “the person.” Estimator design, invariance testing, missingness recovery, model comparison, replication, and scientific verdicts belong to The Science of Fairness.

If felt life matters and outsiders have only partial access to it, then care, consent, refusal, relief, and freedom from imposed meaning are conditions of inquiry rather than later applications. The person must be protected before the claim can ask anything of them.

The gap between outside prediction and inside life is not merely philosophical. People routinely misimagine how disability, illness, adaptation, and altered circumstances will be experienced, and patients may also be discounted when their testimony conflicts with professional or public expectations (Ubel et al., 2005; Kidd & Carel, 2017). This is why the Chapter treats first-person evidence as privileged without treating it as infallible, and treats measurement as assistance without treating it as ownership.

2.1 — Why Experience Matters

Goods matter before anyone feels them because they alter what is possible. Food sustains a body. Shelter reduces exposure. Rights constrain power. Medicine can prevent injury. Yet goods also become lived conditions: hunger relieved, fear reduced, pain treated, time recovered, belonging restored. The Law's formal domain begins in that transformation without claiming that transformation is the whole of justice.

Two reductions fail in opposite directions. Material reduction treats possession or status as a complete account of life. Experiential reduction treats rights and conditions as important only through immediate feeling. The first ignores lived reality; the second makes justice hostage to reporting, awareness, adaptation, or mood.

Income and well-being research illustrates the need for distinctions among life evaluation, daily affect, deprivation, and heterogeneity. Disability and response-shift research shows why observers should not infer total quality from visible function, while rights-based frameworks show why positive self-report never excuses inaccessible environments (Kahneman & Deaton, 2010; Killingsworth et al., 2023; Albrecht & Devlieger, 1999; Koch, 2000; Sprangers & Schwartz, 1999; United Nations, 2006; World Health Organization, 2022).

2.1.1 — Goods Become Lived Conditions

None of this means goods are illusions. It means their contribution to the Law's formal target is mediated through lived condition. To give a person clean water, safety, pain relief, legal

protection, time, and affection is not merely to improve their external column (Kahneman & Deaton, 2010; Killingsworth et al., 2023).

The same discipline protects against crude comparison. Two people can receive the same object and live different realities through it. One person may experience public attention as recognition; another as exposure. One person may experience wealth as freedom; another as a trap of suspicion, inherited demand, or constant risk management.

A material good enters conscious life through a chain rather than a single conversion. Shelter is not only walls: stable housing can reduce exposure to weather, violence, eviction, and forced movement while creating privacy, predictable sleep, a place for medicine, and a reliable address for school or work. Medicine is not only treatment: it can reduce pain, reopen a future, alter dependence, and give hope a practical route. These consequences are partly experiential and partly practical, which is why the good matters both for what a person can do and for what daily life demands.

Time is often the hidden medium through which goods work. Money can purchase a shorter commute, childcare, a safer repair, or the ability to refuse an extra shift. These changes may reduce fear and recover attention. Yet time is distributed within households and institutions, so the same income can produce different lived conditions depending on care obligations, disability, geography, debt, and public services (Sen, 1999; Nussbaum, 2011).

Medicine demonstrates why benefit is multidimensional. A treatment may relieve pain, extend life, restore mobility, or reduce uncertainty. It may also cause side effects, cost, disruption, stigma, or dependence. A serious evaluation follows these consequences rather than declaring the intervention simply positive or negative. The person's own priorities are necessary because the same trade-off can be lived differently.

Education becomes lived through more than credentials. It can create mastery, curiosity, belonging, opportunity, humiliation, pressure, or debt. The building, schedule, curriculum, teacher response, peer climate, and accessibility of the material all affect the pathway. The external good is real, but the lived condition depends on whether the institution permits the learner to participate with dignity.

Relationships are goods of a different kind. They are not possessions, and their value cannot be assigned by an observer from the label alone. Marriage, friendship, family, and community can provide care and meaning, but they can also involve coercion, obligation, surveillance, or loneliness. The relational name does not settle the experiential reality.

The capability perspective helps keep these dimensions together. Resources matter partly because they expand or restrict genuine opportunities to act, and people differ in their ability to convert the same resource into functioning (Sen, 1999; Nussbaum, 2011). Conversion depends on body, environment, social rules, technology, and support. This is why equal allocation can produce unequal freedom.

A final caution concerns substitution. One good may compensate practically for the absence of another, but it does not automatically replace it. Income can purchase services, yet it may not restore health, trust, legal standing, or lost time. Love can support a person through poverty, but it does not make poverty harmless. Plural goods remain plural.

Health, relationships, resources, and institutions enter experience through changing pathways. They alter agency, attention, time, bodily load, trust, and available choice, and the same nominal good can become relief in one context and burden in another. That variability is why the book treats external goods as inputs into lived conditions rather than fixed quantities of happiness.

2.1.2 — Why External Justice Still Matters

The turn toward felt experience can be misunderstood as a retreat from external justice. That would be a serious error. The claim is not that material conditions matter less because inner life matters more. It is that material conditions matter both because of what they make possible or impossible in experience and because rights, duties, care, and protection can bind independently of any estimator (United Nations, 2006; World Health Organization, 2022).

Poverty is not less urgent because the formal target is felt. Poverty becomes hunger, cold, risk, paperwork dread, transportation fragility, debt threat, medical delay, educational exclusion, family strain, and the exhaustion of living without slack. A person's adaptation, gratitude, humor, or love inside poverty does not erase those burdens and must never be used to romanticize deprivation.

Rights-based disability frameworks make the distinction concrete. A person may report a good life and still face unjust barriers in transport, communication, employment, health care, or civic participation. Positive experience does not release institutions from accessibility duties. Conversely, serious distress does not imply that the person lacks agency or dignity. Justice concerns the conditions under which people live and act, not only the mood those conditions produce (United Nations, 2006; World Health Organization, 2022).

The argument must remain legible across political philosophy, economics, disability studies, clinical practice, and ordinary life because each field knows that external arrangements are not neutral containers. They organize exposure, dependence, agency, time, and access, while the person's report may remain mixed or incomplete.

Rights protect more than current mood. Freedom from violence, access to education, bodily integrity, legal recognition, and political participation matter even when a person cannot report their experiential effect. A sedated patient, an infant, or a person with profound communication difficulty does not lose standing because a valence estimate is unavailable. Justice cannot wait for measurement.

Environmental design shows why the distinction matters. A wheelchair user may report a rich and satisfying life while still facing a staircase that excludes them from a public meeting. The satisfaction is real. The exclusion is real. One does not cancel the other. The correct response honors the person's testimony and changes the environment.

External justice also protects people whose suffering is hidden by necessity. A worker may report that conditions are manageable because complaint risks employment. A child may say everything is fine because the unsafe adult is present. A patient may minimize pain because previous reports were dismissed. Public safeguards cannot depend solely on a declaration made under unequal power.

Capabilities provide another bridge. Justice concerns whether people have genuine opportunities to pursue lives they value, not only whether they possess resources or report satisfaction (Sen, 1999; Nussbaum, 2011). A person may accommodate to a narrow field of options and still be owed a wider one. Preference adaptation therefore cannot be the final measure of fairness.

This does not make subjective experience irrelevant. Policies that increase formal access while producing humiliation, fear, exhaustion, or impossible administrative burden may fail in practice. The person's lived account can reveal where a right exists on paper but not in use. External and experiential evidence should correct one another rather than compete.

A practical reading rule follows. When the problem is an inaccessible entrance, repair the entrance. When it is untreated pain, assess and treat the pain. When it is an uncertain research construct, state the uncertainty. The existence of a deeper question should never make the obvious remedy wait.

Rights also protect plural values that cannot be collapsed into valence. Freedom of conscience, political standing, bodily integrity, and equal protection matter even when exercising them

is difficult or distressing. A person may choose an arduous life because agency and commitment matter. Welfare and liberty interact, but neither should be reduced to the other (Rawls, 1999; Sen, 2009).

The independence of justice prevents hidden-state speculation from becoming policy. Officials do not need to estimate a citizen's private ledger before enforcing safety standards or providing accommodation. Public decisions should rely on lawful, observable, contestable criteria rather than claims about unseen inner balance.

This independence is also outcome resilient. If PTVC is false, injustice remains injustice. If PTVC is supported, injustice remains injustice. The formal result could add one dimension to understanding conscious life, but it would not cancel the public reasons for prevention, redistribution, remedy, and equal respect.

2.1.3 — The Limit of Surface Outcomes

The limit of surface outcomes appears wherever observers infer too quickly. Public success is treated as proof of ease, visible disability as proof of misery, or a modest home as proof that deprivation defines the whole life. The root question asks what burden may be hidden beneath the surface and what relief may be invisible because it does not announce itself. This is not permission to invent an interior story; it is a discipline against giving an outside category more authority than the evidence supports.

Surface outcomes also conceal distribution within households and institutions. A family income can rise while one member loses

time, safety, or control. A school's average achievement can improve while a subgroup carries humiliation or exclusion. A hospital can meet a throughput target while patients experience confusion and fear. The Law does not supply a new institutional metric for these cases. It supplies a reason to ask who actually carries the lived consequences of a visible success.

The target-selection problem asks what a theory of whole-life fairness is trying to represent. Wealth, status, preference, opportunity, health, and public outcome each capture something important, but none alone reaches the from-within character of a conscious trajectory. Felt experience is indispensable because burden and relief must be borne by a bearer rather than merely listed around one.

Indispensable does not mean sufficient for every moral question. A theory of rights may protect bodily integrity without first measuring distress. A theory of justice may condemn discrimination even when its experiential effects are delayed or concealed. A theory of care may respond to risk before a person can describe it. Here, formal target refers only to this book's formal hypothesis, not to every domain of value.

Accessibility provides a concrete example. A ramp, interpreter, medication, protected leave policy, or reliable bus route may reduce frustration and pain, but its public meaning is not exhausted by a mood score. It also recognizes equal standing and expands real freedom. The United Nations Convention on the Rights of Persons with Disabilities therefore treats accessibility and participation as rights, not rewards for proving a particular inner state (United Nations, 2006).

The reverse case is equally instructive. A person may report moments of joy within deprivation, discrimination, or illness. That report deserves respect, but it does not release institutions from remedying barriers or preventable harm. Positive experience is not consent to injustice, just as distress is not the whole identity of the person experiencing it (World Health Organization, 2022).

Surface outcomes are indispensable records, but they are coarse and designed. Employment, marital status, diagnosis, graduation, survival, or enrollment each records something real while omitting security, meaning, pain, uncertainty, exclusion, debt, or practical usability. A result therefore needs its pathway.

Equal-looking outcomes can hide unequal work. One household completes an application in minutes; another needs translation, transport, unpaid leave, and repeated calls. Conversely, different outcomes can reflect comparable agency under unequal health, care, or resource constraints. A ranking that ignores those pathways converts the environment into a judgment of the person (Nagel, 1979; Williams, 1981; Sen, 2009).

Disability makes the pathway visible. The World Health Organization's International Classification of Functioning, Disability and Health treats functioning as an interaction among body functions, activities, participation, and environment rather than a diagnosis alone; a changed environment can alter participation without changing the underlying condition (World Health Organization, 2001). Reported quality of life adds first-person evidence, yet wording, memory, expectation, and response shift can complicate longitudinal comparison (Sprangers & Schwartz, 1999).

The formal Law therefore needs more than an outcome category or final interview. It concerns cumulative felt valence through time and requires a representation and observation process that connect evidence to a latent target without pretending direct access. The practical rule is dual attention: ask what condition is present, what right or capability is affected, and how the condition is lived. Outcomes guide action without becoming complete access to inner history or a proxy for worth.

2.2 — How Conditions Become Lived

The same external condition can travel through different mechanisms. Wealth can reduce scarcity, increase control, create exposure, alter relationships, or intensify fear of loss. Status can bring recognition while increasing scrutiny. A biography can contain achievement and compulsion, disability and joy, grief and love, or security and emptiness at the same time.

The purpose of mechanism is not to invent secret stories. It is to ask better questions. What changed in time, agency, sleep, bodily load, belonging, threat, or available choice? What does the person report, and under what pressures? What remains publicly unjust regardless of the answer?

Simple comparison gives way here to causal texture. Conditions matter profoundly, their experiential consequences vary, and no biography assembled from outside can stand in for the stream that was lived.

2.2.1 — Wealth as Relief, Exposure, or Fear

Wealth can be genuine relief. It can buy medicine, food, shelter, education, transportation, legal help, privacy, time, and options that poverty withholds. A serious theory must not pretend that money is spiritually irrelevant or that material stability is merely superficial.

Yet wealth can also become exposure. It can bring requests, suspicion, surveillance, family pressure, envy, legal risk, reputational fragility, predatory relationships, or fear of loss. A person with visible resources may experience ordinary

interactions through the question of whether they are loved, used, targeted, or resented.

These truths should not be compressed into a slogan. The inquiry asks what wealth becomes within the particular stream. For one person it becomes breath. For another it becomes defensive isolation. For another it becomes power to repair.

Wealth should therefore be analyzed through concrete counterfactuals rather than moral caricature. What burden would disappear if income rose? What new obligation or exposure appeared after it did? Who controls the resource? Can it be converted into time, safety, treatment, mobility, or refusal? Population evidence can guide the questions, but the answers remain person- and context-specific (Kahneman & Deaton, 2010; Killingsworth et al., 2023).

At low levels of income, an additional dollar may prevent a utility shutoff or purchase medicine. At higher levels, it may change convenience, status, or future security. The experiential relation is therefore unlikely to be a single conversion rate. Research distinguishes life evaluation from experienced affect and shows substantial heterogeneity across people and income ranges (Kahneman & Deaton, 2010; Killingsworth et al., 2023).

Fear of loss can make abundance feel unstable. If identity, family expectation, or social belonging depends on maintaining a position, the person may experience wealth as a fragile platform rather than settled safety. This does not make poverty preferable. It shows that security depends on predictability and control as well as quantity.

Debt complicates the picture. Two people with the same income may experience very different constraint because one faces volatile obligations, medical bills, or family support commitments. Gross resources conceal timing, liquidity, and the ability to absorb shock. These mechanisms are better guides to lived condition than status labels alone.

Social comparison also matters. People evaluate resources relative to needs, peers, expectations, and past conditions. Comparison can motivate, but it can also produce shame, pressure, and endless escalation. The fact that comparison affects experience does not make inequality imaginary. It explains one route by which unequal structures enter consciousness.

Wealth may expand the capacity to avoid public systems. Private transport, education, security, and health care can reduce personal burden while weakening shared institutions or shifting costs to others. The Law's experiential focus cannot convert this public question into a private feeling. Advantage carries institutional consequences beyond the holder's stream.

Wealth is therefore neither a transparent measure of a life nor an irrelevant surface. It is a powerful condition whose effects travel through concrete mechanisms. The Law of Fairness asks what those mechanisms become in experience, but it does not use hidden burden to erase visible advantage or public responsibility.

Wealth also shows how burden can be shifted onto others. A fortune may rest on workers, renters, customers, public infrastructure, environmental cost, inherited privilege, or legal structures that made accumulation possible. The inner burden of

the wealthy person does not erase those external burdens. The lens follows both directions: inward into felt condition and outward into the systems that produced the condition.

2.2.2 — Status and Social Weather

Status is not only rank. It is social weather. It changes what others assume, how quickly they listen, whether mistakes are forgiven, whether privacy is protected, and whether ordinary presence feels safe. A person can move through the world under warmth, suspicion, admiration, contempt, fetishization, fear, or constant evaluation (Bailey et al., 2017; Williams et al., 2019).

High status can protect. It can open doors, reduce friction, attract help, and make institutions respond. It can also trap. A public identity may become a cage of expectation. A respected person may lose the freedom to be unsure, ordinary, weak, or unseen.

Social weather is partly institutional. Structural racism, stigma, and other durable forms of discrimination influence housing, education, employment, health care, policing, credibility, and exposure. These are not merely private perceptions. They are patterned arrangements that enter experience through repeated vigilance, restricted options, unequal treatment, and accumulated risk. A focus on felt consequences should make structural analysis more concrete, not convert structure into individual attitude (Bailey et al., 2017; Williams et al., 2019; Young, 1990).

Status also shows why external justice and inner experience cannot be separated. A legal right may exist on paper while social weather makes its use costly. A door may be technically open

while contempt, threat, or humiliation make entering feel dangerous.

Low status can impose material and epistemic costs. A person may wait longer, receive less explanation, or have testimony discounted. Epistemic injustice names the harm that occurs when prejudice lowers a speaker's credibility or when shared interpretive resources fail to make an experience intelligible (Fricker, 2007). This is both a social wrong and an experiential pathway.

Status is also context-dependent. A person respected at work may be marginalized in health care, family, or public space. A title that commands attention in one setting may offer no protection in another. The stream receives these shifting environments over time, so a single social rank cannot represent the whole condition.

Status also affects the cost of refusal. A junior employee, patient, student, or dependent family member may agree because disagreement threatens access or belonging. A senior person may face other pressures, but their formal choice usually carries different consequences. Consent and voice must therefore be interpreted within the social weather in which they occur.

The Law's applied lens can ask where status shifts burden, protection, and attention. It cannot infer a terminal ledger from a title or demographic category. Group patterns may identify risks and duties, but the person remains more than the group average. The no-score rule applies equally to prestige and stigma.

A humane conversation about status therefore avoids two simplifications. It does not tell a powerful person that advantage

guarantees ease, and it does not tell a marginalized person that inner strength neutralizes exclusion. It asks what the social environment permits, demands, and threatens while preserving the reality of individual experience.

Social comparison also moves. A gain that feels extraordinary in one reference group may feel inadequate in another. Adaptation and changing expectations can alter the lived meaning of the same position. This does not make status unreal; it shows that its psychological effects are mediated by attention, identity, and social context.

2.2.3 — Why Biography Cannot Replace the Stream

Biography is the story that can be told about a life. It may accurately record birth, place, work, relationships, illness, achievement, loss, and death, but it remains an account shaped from outside or from memory. The stream is the lived bearer through time, at a finer resolution than any story can preserve. Biography can point toward experience and reveal context; it cannot become the stream or determine its terminal residual (Sprangers & Schwartz, 1999).

Biography compresses duration. Ten years of poverty, illness, study, marriage, grief, or office become one sentence, while the stream contained thousands of mornings, bodily states, interruptions, comforts, decisions, and forgotten hours.

Response shift explains why biography and report can diverge without either becoming worthless. Illness, disability, grief, or recovery may change standards, values, comparisons, and the meaning of quality of life. Those changes are information, not

automatic error, and they neither erase loss nor authorize outsiders to correct a person's testimony from afar (Sprangers & Schwartz, 1999).

The latent cumulative object is neither a better biography nor a record an institution can retrieve. It is a formal representation of signed phenomenal contribution across the stream. Biography remains an inference about what happened and what it may have felt like, useful in law, care, history, and ethics but unable to capture every private fear, relief, longing, shame, meaning, or peace.

Stories and reports are time-bound and audience-shaped. Memory may emphasize peaks, endings, present beliefs, or available language, while a clinical history, memoir, obituary, job interview, and private conversation select different material for legitimate reasons of purpose and privacy. Later understanding may enrich evidence without rewriting earlier experience.

Biography can exaggerate agency, reduce a person to injury, or conceal coercion. Success stories may hide luck and support; accounts of failure may moralize structural barriers; diagnoses may eclipse humor, work, relationship, or faith; polished family narratives may hide fear or denied choice. These risks require evidence and humility, not the abandonment of biography. A timeline that places housing changes beside work and caregiving demands can reveal pressures that a simple success narrative leaves unexplained.

Biography remains valuable because it reveals values, relationships, causal structure, and personal integration. Use it to

show mechanisms and stakes, distinguish documented fact from report and inference, and never invent interiority or assign an unseen total. The narrator can also identify which turning points mattered personally, giving an investigator better questions to ask about the surrounding circumstances.

The formal model therefore requires ownership and complete-domain rules rather than a compelling story. Biography generates hypotheses and context; it cannot determine the terminal residual by literary completeness. An account can therefore be complete for a legal dispute or family history while serving a different purpose from the formal representation.

2.3 — Hidden Experience and Appearance

Appearance misleads in more than one direction. Envy can treat advantage as proof of ease. Pity can treat visible difficulty as proof that the life contains nothing else. Both errors make the observer's story more authoritative than the person.

Hidden burden and hidden relief are therefore symmetrical cautions, not compensating entries. The possibility of hidden burden does not make privilege just. The possibility of hidden relief does not make exclusion acceptable. Hiddenness should enlarge humility, not produce a secret explanation for why the world is arranged as it is.

2.3.1 — The Burden Behind Apparent Advantage

Apparent advantage can conceal burden because advantage often arrives with conditions. Wealth can bring safety and option space while also bringing exposure, dependency, family conflict, threat, or pressure to preserve status. Public admiration can widen opportunity while making ordinary failure feel dangerous.

Burden may remain hidden because people adapt to recurring conditions and report change against a local baseline. Chronic pain, danger, or administrative humiliation can become familiar without becoming harmless, and trauma-related numbing or narrowed awareness can make the burden difficult to describe (SAMHSA, 2014). Adaptation changes what is noticed and reported; it does not certify that the condition is acceptable.

Trauma-related dissociation or emotional numbing can also separate outward behavior from danger. A person may sound

detached while carrying an acute threat response, or report little feeling because access to the experience is disrupted. Calm appearance is evidence about presentation, not a verdict about safety or burden (SAMHSA, 2014).

A disciplined inquiry into hidden burden begins with consent and consequence. It does not demand disclosure from the person whose public life attracts attention. It asks whether expectations, exposure, threat, isolation, sleep loss, compulsive work, or lack of genuine refusal are plausible mechanisms, then gives the person's own account the epistemic standing it deserves (Fricker, 2007; Kidd & Carel, 2017). Hidden burden is not a mandatory secret behind success. It is one possible reason visible advantage and lived condition should not be treated as synonyms.

Hidden burden should be inferred only from evidence: the person's report, a documented condition, or a credible mechanism. Caregiving, performance pressure, public exposure, financial dependence, or invisible illness can make visible advantage an incomplete description, but none should be invented merely to make inequality appear balanced. Hidden burden can also coexist with responsibility and privilege.

The applied value is practical. Ask which concrete condition, such as workload, lack of privacy, caregiving burden, coercive access, or untreated illness, can be changed. Create conditions for disclosure without requiring it, and leave known external risks actionable even when inner burden remains uncertain. The theoretical conclusion is narrower: visible advantage is an incomplete channel for a complete-stream claim.

2.3.2 — Pity as Incomplete Knowledge

Pity can awaken care, but it is a poor instrument for knowing a whole life. It notices a visible burden and may let that feature stand in for the person: injury for identity, poverty for biography, disability for destiny, dependence for incapacity. The moral impulse can be sincere while the inference remains reductive (Albrecht & Devlieger, 1999; Koch, 2000).

The first distortion appears in the question pity asks. Instead of asking, ‘What do you want changed?’ it asks, ‘How tragic is your life?’ The first question leaves room for agency, privacy, and specific barriers. The second recruits the person into an observer’s emotional story before the person has spoken.

Disability scholarship offers the strongest correction to pity because it separates impairment, environment, identity, access, and reported quality of life. Studies that found high self-reported quality among some people with serious disabilities challenged outsider assumptions, while critics correctly warned against turning the finding into a paradox that centers nondisabled expectations. The fairest lesson is epistemic: listen to people, remove barriers, and do not treat either suffering or flourishing as surprising evidence against their humanity (Albrecht & Devlieger, 1999; Koch, 2000; United Nations, 2006; World Health Organization, 2022).

The response should be practical: witnessing another person’s burden should lead toward appropriate help, institutional repair, respectful speech, protected privacy, and humility about what remains unknown. Pity becomes unfair when it uses another

person's visible suffering to stabilize the observer's image of goodness.

Rights-based disability scholarship adds a public correction. Barriers arise through the interaction between impairment and environments that exclude participation. Accessibility, support, communication, and autonomy can change the lived condition without requiring the person to become nondisabled (United Nations, 2006; World Health Organization, 2001). Pity that focuses only on the body can hide the design failure.

Lowered expectations can be as restrictive as overt exclusion. A teacher may stop offering difficult work, a clinician may speak past the patient, or a program may offer help only in forms that remove choice. Assistance becomes respectful when it is offered, accessible, and revisable, and when declining it does not trigger punishment or disbelief.

Epistemic injustice can move in both directions. A disabled person who reports a good life may be treated as confused because the report violates an outsider's feared story; a patient who reports severe pain without expected visible signs may be treated as unreliable (Fricker, 2007; Kidd & Carel, 2017). Respect does not choose the testimony that is emotionally convenient. It gives the person's account appropriate weight while remaining open to context and change.

There is also a danger in the opposite direction. Positive self-report can be used by institutions to deny accommodation, treatment, or social support. A person can value life and still need

access. Satisfaction under constraint does not prove that the constraint is fair. Pity is incomplete, but rights remain complete.

A better posture is curiosity under consent. Ask how the environment works, what creates burden, what supports agency, and what the person wants changed. Do not require disclosure as the price of respect. The person's privacy is not evidence of hidden misery or gratitude.

Outsider forecasts are especially unreliable when they are built from an imagined loss of the observer's own present life (Albrecht & Devlieger, 1999; Koch, 2000).

People may experience severe pain, exclusion, dependence, or grief, and accessible environments do not remove every burden. Respect requires hearing negative testimony as fully as positive testimony rather than selecting whichever story comforts the observer.

Help changes character when refusal is real. Assistance should not require public disclosure, gratitude, inspiration, or surrender of ordinary decision-making. Asking before acting, explaining options, and accepting 'not that' are small practices with large epistemic value because they prevent the helper's interpretation from becoming the person's assigned reality.

Structural action can proceed without a total judgment about inner life. An inaccessible entrance can be redesigned; a communication barrier can be removed; pain can be treated; income support can be provided; a discriminatory rule can be changed. These actions are justified by rights, documented

barriers, and practical consequences, not by possessing the person's stream.

2.3.3 — The Applied Lens Versus the Formal Law

The applied inquiry and the formal Law share a concern with felt burden, but they answer different questions. Applied inquiry asks how a condition becomes lived, where strain is hidden, what relief is available, and who has the power to change the situation. The formal Law asks whether one complete qualifying conscious stream satisfies an exact terminal relation. The first can improve action now. The second remains a scientific hypothesis that must survive definitions, measurement, rivals, and possible defeat.

Consider a promotion that raises income and public status while also increasing travel, scrutiny, after-hours availability, and fear of failure. An applied analysis does not declare the promotion secretly bad or assume that every advantage carries an equal hidden cost. It separates the mechanisms: money may reduce insecurity, authority may increase control, visibility may increase exposure, and time pressure may erode rest or family life. The result is a better account of the situation, not evidence of terminal closure.

The same discipline matters in chronic pain. A patient may report a worthwhile life while living with substantial impairment, or may describe severe suffering that outsiders underestimate because the person remains employed and socially engaged. Research on adaptation and the so-called disability paradox shows why surface prediction is unreliable, but it does not show that deprivation is harmless or that every burden is offset

(Albrecht & Devlieger, 1999; Ubel et al., 2005). Treatment, accessibility, and the patient's own priorities remain central.

Applied inquiry can expose burdens that a public outcome hides. Administrative procedures may impose time, travel, documentation, and fear; disability can interact with architecture, communication, prejudice, and technology; caregiving can combine love with exhaustion and financial strain. These examples support the same practical question: which concrete pathway can be changed? They do not supply evidence of terminal closure (Herd & Moynihan, 2018; World Health Organization, 2001; United Nations, 2006; World Health Organization, 2022).

The inquiry can also be wrong. An observer may infer hidden distress that the person does not report, romanticize hardship because moments of joy are visible, or focus so narrowly on subjective adaptation that rights and material conditions disappear. Good use therefore begins with testimony, evidence, and revisability. It leaves room for the person to say that the proposed interpretation does not fit.

The formal Law is stronger and narrower. It concerns the complete relevant domain of one separately routed stream, a defensible representation of valence, a neutral origin, a rule for exposure and time, and an independently specified endpoint. A local account of workplace stress, pain adaptation, or administrative burden supplies none of those elements by itself.

Felt experience is one dimension of a life, not the only dimension by which institutions or relationships should be judged. Rights,

capabilities, opportunity, bodily integrity, public reasons, and distribution remain decisive even when distress falls (Rawls, 1999; Sen, 1999; Sen, 2009). Genuine benefits such as debt relief, pain treatment, or expanded access should remain visible. The task is to trace benefits, costs, uncertainty, and differences among people, not to discover a concealed downside in every good.

A practical method separates observation from interpretation: record what changed, ask how the person describes it, identify plausible pathways, and compare alternatives. When power is involved, the inquiry ends in an accountable choice. Who can alter the schedule, remove the barrier, provide treatment, explain the rule, or hear an appeal? Hidden burden matters most when it reveals a reachable point of repair, not merely a more sophisticated story.

Keeping these tasks separate makes both of them stronger. Applied inquiry gains freedom to be concrete, humane, and useful without waiting for a universal verdict. The formal Law remains vulnerable because useful applications are not counted as proof. Breadth belongs to the question's reach; scientific warrant belongs only to evidence that was designed to test the claim.

2.4 — Describing Experience Without Owning It

Experience can be described without being possessed. People report pain, pleasure, fear, calm, boredom, longing, and relief. Clinicians and researchers can listen, compare, observe, and model. None of those practices transfers ownership of the experience from the person to the method.

The distinction has practical consequences. Privacy is not only data security. It includes freedom from compulsory disclosure, from unified inner-life dossiers, from scores that follow a person into unrelated settings, and from institutional claims that a pattern reveals what the person “really” feels. A measurement system earns trust by the inferences it refuses.

Humility becomes a design requirement. Description must remain revisable, proportionate, and explicit about uncertainty. When a measure cannot meet those conditions, abstaining may be more rigorous than producing a number that outruns what is known.

2.4.1 — Description Is Not Possession

To describe experience is not to possess it. A patient may describe pain for treatment, a participant may report mood for research, and a friend may speak about grief to be understood. In each case, the listener receives testimony for a purpose, not ownership of the state. The description remains connected to the person’s agency, context, and right to revise or stop disclosure, while creating a responsibility to hear carefully (Raja et al., 2020).

Measurement intensifies the danger. A number can look more authoritative than a sentence, even when it rests on weaker

assumptions. A chart can make a life appear organized for outside use. A model can invite others to forget that the target was lived from within.

Description also changes the experience it describes. Being asked about pain can increase attention to pain. Being monitored can create performance. Being categorized can produce shame or relief, depending on context. The act of observation is not morally free.

Clinical pain practice makes the asymmetry concrete. Examination, history, function, symptom intensity, triggers, and treatment response can guide care, yet the patient remains the person directly undergoing the pain. A chart is not the pain itself; it can be incomplete, wrong, or detached from the context in which the person spoke. Scientific care therefore keeps record authority bounded, revisable, and correctable rather than treating documentation as possession (Raja et al., 2020).

Language can become a subtle form of control when a clinician, researcher, institution, family member, government office, religious leader, or platform names someone's state and treats the naming as command. The formal inquiry needs description because its target is not directly observable from outside, but it rejects possession because the experience belongs to the stream, not the observer. Evidence may guide care; it may not swallow the person described.

Possession begins when institutions treat disclosure as theirs to reuse beyond its purpose. Data collected for care may be copied, merged, searched, or repurposed for research, model training,

eligibility, marketing, or other decisions the person never anticipated. The convenience and transferability of a file do not make inner life transferable. Secondary use requires renewed authority, proportionate governance, and a reason tied to the original person rather than institutional appetite.

Third-person evidence can matter when report is unavailable or contradicted by urgent risk. Behavior, physiology, history, and context may guide care. The ethical question is what decision the evidence can support and how uncertainty is handled. Emergency action does not create a general right to own the person's inner truth.

Epistemic injustice provides a useful warning. People can be denied credibility because of prejudice or because their experience does not fit available concepts (Fricker, 2007; Kidd & Carel, 2017). Better description may require changing the institution's language, not forcing the person into an inadequate category.

Even accurate description remains partial. Words select categories, scales compress, and observers interpret. Information about experience therefore creates responsibilities rather than ownership: preserve uncertainty, consent, correction, purpose limitation, accountable human review, and the person's right to reject the description. The more intimate the record, the less authority a neat representation should acquire by itself.

A dignified description therefore requires right scale. Some experiences should be held in private speech, some in clinical confidence, some in research under consent, some in public only

as aggregate or protected summary, and some not at all. The present inquiry asks who benefits from the description and who carries the cost of being described.

2.4.2 — Privacy of Inner Life

Inner life has a privacy that is not merely practical. It is part of dignity. A person may choose to disclose pain, desire, grief, shame, relief, hope, or fear. They may also choose silence, partial speech, delayed speech, protected speech, or no speech at all.

This is one of the book's central safeguards. Because felt experience is central to the formal target, a careless system might think inner life deserves more collection. The correct conclusion is the reverse. The more central inner life is, the more carefully it must be protected from coercive access, public hunger, institutional convenience, and data extraction.

Privacy also includes purpose limitation. Information disclosed for treatment should not automatically become material for employment, insurance, policing, marketing, or unrelated model training. Combining channels can create a dossier whose apparent completeness exceeds the validity of any component. The no-score rule is therefore not hostility to research. It is a boundary against letting uncertain estimates acquire durable power over a person.

Privacy is especially important because hidden burden can make disclosure feel necessary for help. A person in need may feel forced to narrate trauma, prove pain, display poverty, or surrender intimate detail to receive care. A stream-centered reading should

make such burden visible and push systems toward relief without unnecessary exposure.

Inner-life data can expose unusually sensitive information about fear, pain, trauma, belief, relationships, and vulnerability. Minimization should begin before collection; access should be limited to the decision at hand; consent for one purpose should not silently authorize unrelated reuse or model training; and records that contain information about nonparticipants require separate protection (CIOMS, 2016). Chapter 12 develops the governance details.

The formal Law gives no institution a claim to universal monitoring. If the hypothesis requires complete data that cannot be collected without violating dignity, the scientific result may remain unavailable. No adequate estimate is preferable to a surveillance system built to satisfy an abstract target.

Clear limits on access and use can improve both honest disclosure and scientific validity. People speak differently when they expect judgment, punishment, surveillance, or uncontrolled reuse, and they can speak more freely when they know who will see the information, what decisions it can affect, and how refusal will be treated. Trustworthy governance therefore protects the person and shapes the observation process itself (O'Neill, 2002; CIOMS, 2016).

Privacy has narrow exceptions. Immediate danger, abuse, or a specific legal duty may justify disclosure, but the purpose, recipient, and extent should remain limited and transparent where

possible (CIOMS, 2016). A safety exception must not become a general channel for institutional convenience.

Inference can violate privacy even without direct access. Patterns of sleep, movement, typing, speech, purchase, or physiology may be used to guess mood or vulnerability. Such guesses can be wrong and still affect opportunity. The uncertainty of the inference does not make the intrusion harmless.

A protected inner life also allows joy. People need rooms where delight, desire, confusion, grief, boredom, and imagination are not constantly interpreted. If everything inward becomes material for systems, the person loses the ordinary privacy in which experience can simply be lived.

2.4.3 — Why Measurement Must Be Humble or Not Done

Measurement must be humble or not done because the burden of error falls on real people. A weak instrument can misdescribe a theory; an intrusive one can alter a life, expose vulnerability, or create a record that travels farther than consent. Humility begins with construct honesty: a measure of arousal is not valence without validation, a report scale is not a unit of experience merely because it uses numbers, and a composite must not hide incompatible channels behind a precise decimal (National Research Council, 2013).

Humility can be operationalized. Define the construct before choosing the instrument; state what each channel measures and does not; test whether meaning changes across culture, age, disability, language, or context; model missingness; preserve refusal; and predefine when uncertainty is too great for the

intended use. When these conditions cannot be met, abstention is not a failure to care. It may be the most accurate recognition that the person exceeds the instrument and that protection matters more than completion of a dataset (National Research Council, 2013).

Combining channels can increase risk even when each channel seems harmless alone. Self-report, location, sleep, physiology, voice, and behavior may create a revealing profile when linked. The apparent completeness of that profile can tempt institutions to treat inference as fact. Multi-channel evidence therefore requires stronger governance, not greater confidence by default (National Research Council, 2013; CIOMS, 2016).

Humble measurement asks eight linked questions before a number is trusted: Is the measure necessary? Is its burden proportionate? Is the construct valid across the people and contexts being compared? Does measurement itself change the process? What does missingness mean? Is the neutral origin independently defensible? Is the proposed use supported by the evidence? Can participants pause or stop without penalty? Failure at any one of these questions can narrow or halt the inference rather than being repaired by a more precise score.

These questions are developed technically in Parts IV and V and operationally in Chapter 12. At this stage the rule is simpler: uncertainty must change authority. A measure that cannot justify its construct, origin, burden, and use should remain partial or yield No adequate estimate rather than becoming a verdict on a person.

The Law of Fairness requires an unusually demanding measurement program, but the public volume must not turn that demand into a consumer tool. No app, diary, wearable, or artificial intelligence system should invite people to test their own terminal balance. When governed measurement is justified, the construct should remain narrower than the person: a valence estimate may describe one dimension within its window, never a diagnosis, biography, terminal verdict, or ranking of a life.

Humility also requires publishing failure. If channels disagree, the neutral origin cannot be transported, missingness defeats recovery, or the design supports only a narrower claim, the result should say so, including No adequate estimate. Proportional authority carries uncertainty through the conclusion, limits an estimate to its designed decision, and stops when burden or misuse risk exceeds the knowledge gained.

The key error analysis is human. Who is harmed by a false positive, false negative, missing report, cultural mismatch, inaccessible instrument, or model trained on the wrong population? A confidence interval does not make a measurement system humble. Humility means uncertainty changes authority, access, retention, review, and willingness to stop.

2.5 — What Affective Science Can and Cannot Clarify

Affective science makes the inward question more interesting by showing how much an ordinary description can conceal. Feeling activated is different from feeling good; remembering a day is different from living it; and several feelings can occupy the same moment. Separating these features gives research something definite to investigate. This Section asks how observations can preserve those differences well enough to illuminate the experience they are meant to describe.

No channel is identical to the target. A person can conceal expression, misremember, lack language, or experience conflict. Physiological measures are context-sensitive. Neural measures depend on tasks, models, preprocessing, and interpretation. Self-report is indispensable but not infallible. Convergence can improve confidence without turning evidence into direct possession of experience.

2.5.1 — Valence, Arousal, and Mixed Experience

Consider a person leaving a hospital after hearing that a feared diagnosis was ruled out while learning that treatment for another condition will be difficult. Relief may be immediate, fear may remain, and the body may still be highly activated. One ordinary moment can therefore contain opposed feelings, changing expectations, and bodily intensity without collapsing into a single simple state.

Valence is the name affective science often gives to the positive or negative character of experience. Arousal refers to activation, wakefulness, or energetic mobilization. The coordinates are

useful because they separate questions that ordinary language often merges. They are not a complete ontology of feeling (Russell, 2003; Kuppens et al., 2013).

High arousal can accompany celebration, panic, anger, pain, sexual excitement, athletic effort, or awe. Low arousal can accompany serenity, boredom, fatigue, depression, sedation, or despair. Activation alone therefore cannot reveal whether an experience is welcome or burdensome. The same pulse rate can belong to very different lived situations (Russell & Carroll, 1999; Russell, 2003).

Positive and negative affect can also coexist. Pride in a child's independence can sit beside grief at separation. Love can intensify fear of loss. Relief can arrive inside sadness. Mixed emotion is not necessarily confusion or measurement error; it may be the faithful structure of a situation containing opposed values (Larsen et al., 2001; Berrios et al., 2015).

Relief deserves separate attention because it has a temporal structure. The pleasure of a burden ending depends partly on what preceded it, what was expected, and how complete the ending feels. Relief after pain is not interchangeable with an unrelated reward of the same reported intensity. History changes the experience.

Affective granularity varies across people and across time. Some distinguish irritation, dread, shame, grief, and physical discomfort precisely; others use broader labels. Greater verbal detail can improve communication, but limited vocabulary does not show

limited capacity to feel. The measurement task is to separate language skill from the target of interest.

Culture and relationship shape the available language. Some contexts encourage direct reports of sadness or pain; others emphasize bodily symptoms, restraint, social harmony, or spiritual meaning. These differences can alter response patterns without providing a simple ranking of whose report is more truthful. Interpretation requires local knowledge and humility.

Interoception concerns how bodily signals such as heartbeat, breathing, temperature, gut sensation, pain, and fatigue are sensed and interpreted. It is not one transparent faculty. Sensitivity, attention, belief, and regulation can diverge, and their relationships with affect vary across tasks and conditions (Khalsa et al., 2018).

Lack of report is especially ambiguous. A person may be asleep, sedated, overwhelmed, unwilling, unable to use the scale, or simply not prompted. Research should record why an observation is absent instead of treating every blank as zero. Missingness can carry information about both burden and measurement access.

The most defensible first comparisons are often within a person. Repeated observations may reveal how that person distinguishes better from worse, how context changes reports, and which transitions predict later behavior. This does not solve the whole problem, but it avoids assuming that identical numbers already mean identical experiences across people.

Cross-person comparison adds requirements. Scales may be used differently, neutral points may shift, and cultures may attach

different meanings to the same words. Measurement invariance cannot be presumed because a questionnaire has the same printed response options. The comparison must earn its interpretation.

Scalarization remains a scientific choice, not a fact granted by convenience. A signed one-dimensional representation may work for some tasks. Mixed states may require separate positive and negative components, a vector, or partial ordering. The formulation developed here preserves these branches rather than forcing every experience through one line.

The lesson is neither that feeling is immeasurable nor that a universal meter already exists. Affective science has developed disciplined concepts, instruments, and models, while repeatedly showing how context, timescale, language, and bodily interpretation matter. That combination supports ambitious inquiry and punishes careless certainty.

Valence, arousal, and mixed experience therefore orient the public question. They show what a plausible representation would need to preserve before any lifetime claim could be tested. The next difficulty is evidential: no report, action, bodily signal, or brain measure is identical to the experience that gives it meaning.

2.5.2 — No Single Channel Is the Experience Itself

A patient may describe severe pain while speaking quietly and showing little change in heart rate. Another may appear agitated while denying distress. A child may withdraw without words, and a person with paralysis may be unable to display the behavior observers expect. These cases do not prove that one channel is

right. They show why the target and its evidence must remain distinct.

Experience is the target that gives every channel its relevance. Reports, choices, behavior, physiology, clinical observation, and neural activity can each constrain an interpretation. None is the experience itself. The bridge from evidence to lived state must be stated rather than hidden behind the authority of an instrument.

Self-report has a distinctive strength because the person can describe what is felt. It can identify qualities that no external sensor can name, including shame, relief, dread, moral conflict, and the object of fear. Respect for report is not naïveté. It recognizes first-person access as an indispensable source of evidence.

Reports also have limits. Language, memory, audience, privacy, scale design, expectations, and willingness to disclose can change what is said. A person may minimize pain to leave a hospital, exaggerate urgency to be heard, or lack words for a mixed state. These possibilities call for context and follow-up, not automatic disbelief.

Experience sampling asks close to the moment and can reduce some retrospective distortion. It also interrupts life, samples only selected times, and depends on the person being able and willing to answer. The most intense episodes may be precisely when a prompt is missed. Response availability is part of the observation process, not clerical noise.

Behavior reveals approach, avoidance, persistence, withdrawal, choice, and action tendency. Yet the same stillness can reflect

calm, fear, pain, training, coercion, or exhaustion. A smile can express pleasure, politeness, concealment, or social protection. Behavior becomes informative through context rather than through a fixed emotional dictionary.

Neural measures can illuminate processes associated with salience, pain, reward, attention, regulation, and report. They do not extract a private sentence from the brain. Group-level associations may overlap too broadly for reliable individual classification, especially across development, disability, medication, and disease (National Research Council, 2013; Raja et al., 2020).

Choice tasks can reveal trade-offs and preferences. They may help locate sign, compare relative values, or examine what a person will sacrifice for relief. Choices are also shaped by risk, learning, habit, social pressure, available options, and the design of the task. Preference is evidence about value, not a transparent unit of feeling.

Clinical observation can integrate history, examination, behavior, relationship, and change over time. Expertise adds value because patterns can be recognized across many encounters. It does not remove bias, hierarchy, or incomplete access. The interpretation remains answerable to the patient, the record, and other evidence.

Communication-limited settings require adaptation rather than surrender. Infancy, anesthesia, disorders of consciousness, severe illness, language difference, and motor impairment change which channels are available. A substitute measure may become

necessary, but necessity does not make the substitute ontologically identical to experience.

Population findings require another boundary. A model that predicts an average association may classify some individuals poorly because distributions overlap and mechanisms vary. Borrowing information across people can improve estimation, but the amount of borrowing and the remaining uncertainty should remain visible (Molenaar, 2004; Fisher et al., 2018).

Measurement invariance matters whenever groups or persons are compared. The same item may function differently across culture, age, disability, clinical condition, or language. Equal printed scores do not guarantee equal meaning. A comparison that has not tested this bridge should be described as provisional.

Sometimes the honest result is no adequate estimate. A person may be unable to report, the available channels may conflict, or the observation model may not transport to the case. Abstention protects both the person and the science. Uncertainty is not evidence of neutrality, deception, or absence of experience.

Channels are witnesses, not the experience itself. Their value depends on calibration, complementary errors, and explicit uncertainty.

2.5.3 — Orientation, Measurement, and the Science Handoff

A scientific test therefore needs a visible dependency chain from representation and neutral origin through bearer, endpoint, observation, missingness, inference, and rival comparison; success at one link cannot conceal failure at another.

Measurement begins before any instrument is chosen. Researchers must decide what the target is, what counts as positive or negative, where neutrality lies, whether mixed experience can be reduced to one value, and which exposures belong to the same stream. A device can be precise while answering the wrong question.

Orientation means that affective science helps define plausible constructs, methods, dynamics, confounds, and rival explanations. It gives the project a disciplined vocabulary for asking what evidence might bear on felt experience, and it supplies real empirical structure against which a reduced-order model can be tested. Indirect access can establish a measurement foundation, constrain a representation, favor a rival, or expose a failure even though no instrument possesses the latent target.

Mature measurement practice reinforces that separation. The updated Organisation for Economic Co-operation and Development (OECD) guidelines distinguish life evaluation, affective experience, and eudaimonic aspects of well-being, and they emphasize question design, context, comparability, respondent burden, and statistical quality rather than one universal happiness score (OECD, 2025). These tools do not measure PTVC directly; they strengthen the foundation on which a later terminal test would have to stand.

A credible empirical adjudication would require more than association. It would need defensible construct validity, a prospectively specified observation model and endpoint, invariance wherever comparison is claimed, uncertainty propagated through each bridge, ethical feasibility, and

discriminating performance against serious rivals. Finite evidence may support a route, show compatibility within a declared band, narrow a reference class, or yield no adequate estimate; it cannot establish exact universal closure from a small terminal sample. Failure at a prerequisite narrows the claim or stops the inference rather than being repaired by a beautiful final graph.

Consider a wearable that predicts self-reported distress accurately during ordinary workdays. That success could validate one local observation relation without locating neutrality across all contexts, identifying the bearer during altered consciousness, or recovering experience from years before the device was worn. A strong link cannot stand in for the whole dependency chain.

The bearer introduces consciousness and unity. A body, legal person, memory archive, or artificial system is not automatically one qualifying stream. The scientific companion must state routing rules for pauses, splits, mergers, and uncertain cases. Convenient grouping after seeing the result is prohibited.

The endpoint creates the tautology risk. A protocol cannot define the end as the first time the estimate reaches zero. It must use prospectively stated consciousness or unity criteria that are not algebraically identical to the target. Biological death, last report, observation shutdown, and truth-level endpoint remain distinct.

Rivals need equal resources. Adaptation, opponent processes, allostasis, predictive regulation, ordinary mean reversion, endpoint conditioning, selection, memory, and observation floors can imitate parts of a closure pattern. The Law earns support only

by surviving comparisons in which rivals receive the same data and fitting freedom.

Ethics is not downstream of method. Burden, consent, privacy, care, and stopping rules shape which observations are permissible and therefore which claims are identifiable. An unsafe measurement is not rescued by potential importance. Some questions may remain unanswered because dignity sets a real boundary. A valid boundary can therefore reduce the study's ambition without reducing the participant's standing.

Orientation gives a general reader the conceptual tools to recognize both genuine progress and overclaim without requiring entry into the formal mathematics. A calibrated scale, explicit endpoint bridge, transparent missingness model, and held-out rival comparison can make a later test interpretable. A miscentered scale can manufacture zero, a late observation can be mistaken for the true endpoint, and missing participants can make a terminal pattern look cleaner than it is. Plain language can reveal which foundation has been built and which bridge remains open. For readers, this dependency structure turns later evidence into a chain that can be inspected rather than a verdict that must be trusted whole.

Triangulation becomes useful only after each channel's role and error are declared. Self-report may be direct but context-sensitive; physiology may be precise but nonspecific; behavior may be observable but multiply caused; neural data may be rich but inferentially indirect. Their combination must be trained, validated, and allowed to fail rather than treated as a vote among impressive signals. Agreement across channels matters most

when their errors are not merely correlated versions of the same hidden assumption. Independent convergence is especially informative when the channels fail in different ways yet still point toward the same latent feature. That turns multiple measurements into a stronger test of the construct rather than a larger pile of correlated signals. When convergence survives those differences, confidence comes from independent constraint rather than repetition of one measurement assumption.

Chapter 3 — Care Before the Claim

Urgency does not wait for theory. A person in pain needs care; a child in danger needs protection; a dying patient may need quiet, symptom relief, and family presence; a survivor retains the freedom to reject a redemptive story. These are not preliminary courtesies. They are the ethical ground of the project.

The Law of Fairness proposes an exact terminal property of conscious experience. That ambition requires a clear care sequence before any formal claim: notice present burden, relieve what can be relieved, protect consent and privacy, respect refusal, and keep meaning voluntary. The Chapter also closes four dangerous inferences: later balance does not justify present pain, relief is not forbidden by the terminal requirement, a peaceful ending does not vindicate the path, and no one may induce or prolong suffering for data.

The first rule is temporal: care is owed while suffering is occurring. The future does not purchase the past, and a possible terminal relation cannot suspend ordinary duties. Treatment, rescue, shelter, accompaniment, apology, access, and competent help remain urgent for the reasons they were urgent before the hypothesis existed.

A person with advanced lung disease may need treatment for breathlessness, a quieter room, help communicating with family, and an honest discussion of goals on the same day. None of those forms of care requires agreement about the meaning of illness. Clinical need and the person's wishes, not theoretical allegiance, govern access to comfort and support (World Health Organization, 2020).

The second rule concerns truth. The ethics cannot depend on the Law winning. If research later favors a rival, narrows the reference class, finds the target unidentifiable, or defeats the claim, care does not become retrospectively unnecessary. Respect for persons, beneficence, and justice do not derive from PTVC. They govern the inquiry that seeks to test it (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; Beauchamp & Childress, 2019).

The third rule rejects moral desert, punitive debt, and suffering as an obligation. Suffering is not evidence of guilt. Felt pleasure and grief still enter the stream's accumulated balance, and a change in that balance changes the net remainder required under exact PTVC. Exact terminal closure, were it established, would be a phenomenal boundary property, not a moral verdict on the events within the stream. The difference between a formal equality and a justified history is absolute.

The fourth rule is nontransferability. One conscious stream cannot pay for another. No victim's pain is redeemed by a social average, an institution's success, a researcher's insight, a family's later meaning, or another person's joy. The formal model's intrastream boundary and the ethical rule point in the same direction: there is no common experiential account in which one bearer's suffering is canceled by another bearer's benefit.

This rule does not settle every problem in public ethics. Societies constantly make decisions that distribute risk and benefit. Medicine, infrastructure, law, and emergency policy sometimes require tradeoffs. The narrower point is that the Law of Fairness cannot be used to transform those tradeoffs into phenomenal

substitution. If a policy harms one group and helps another, the benefits do not enter the harmed group's conscious stream.

Research may not create serious suffering to test a hypothesis about suffering. Human-subject ethics requires social and scientific value, sound design, favorable risk-benefit assessment, fair selection, independent review, informed consent, and respect for participants, with particular protection for people in vulnerable situations and for assent or dissent where relevant (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; Emanuel et al., 2000). This prohibition does not treat every burden as serious harm: a brief questionnaire, ordinary decision task, or noninvasive observation may be acceptable when the question is important, the design valid, the burden proportionate, refusal real, and support available. The Chapter defines misuse; it does not replace review boards or professional judgment.

Consent must be more than a signature. Dependency can distort refusal when patients fear losing care, employees fear retaliation, students feel pressure from authority, families turn love into obligation, or digital systems make declining harder than accepting. Consent must be practical, comprehensible, revocable, and free from penalty (CIOMS, 2016; Faden & Beauchamp, 1986). Research communication must also prevent therapeutic misconception by separating care from data collection and never implying that participation will reveal the truth of a person's life (Appelbaum et al., 1982).

Trauma-informed practice sharpens the same point. Safety, trustworthiness, collaboration, empowerment, and attention to

historical and cultural context are not merely compassionate tones. They affect whether a person can understand, refuse, disclose, and remain stable during an interaction. A project that studies burden while reproducing coercion has failed both ethically and methodologically (SAMHSA, 2014).

A final rule protects meaning. People may construct meaning after loss, illness, violence, addiction, or grief, and that meaning may support adaptation, identity, relationship, or action. It may also be absent, contested, delayed, or unwanted; research describes variability rather than a duty to narrate every life (Park, 2010; Woods, 2011). Relief therefore comes before interpretation when they compete. A person may receive analgesia without accepting a theory, seek safety without naming a lesson, grieve without calling a death necessary, or decline sacred language. The freedom to find meaning includes the freedom not to be assigned one.

These commitments give a reader room to explore. A person can consider an exact relation in experience while seeking relief, rejecting blame, protecting privacy, and leaving spiritual questions open. A researcher can ask a bold question while participants retain their freedom to decline it. A family can discuss the possibility while continuing to meet the needs in front of them. That combination makes the inquiry worth sharing: its ambition can grow without requiring anyone to surrender care, dignity, or the right to make sense of a life in their own way.

These rules do not make the Law timid; they make bold analysis possible. Visible unfairness supplied the pressure, lived experience supplied the inward domain, and ethics now fixes the

duties that survive every scientific outcome. Once the reader knows that the hypothesis cannot excuse harm, delay care, rank people, or force interpretation, the formal claim can be stated without turning courage into license.

Ethics is architecture rather than decoration. Trust can be damaged by opacity, dependency, role confusion, or an offer that is formally optional but practically costly to refuse, so autonomy requires understandable information, proportional burden, continuing choice, independent review, and protection for people whose circumstances reduce practical freedom (O'Neill, 2002; National Commission, 1979; World Medical Association, 2024; CIOMS, 2016). These protections belong in recruitment language, consent and withdrawal routes, data handling, clinical referral, captions, public summaries, and ordinary conversations. The same discipline applies outside research: families, religious communities, platforms, schools, hospitals, and movements must not pressure people to supply meaning, testimony, data, forgiveness, or optimism, especially where refusal could threaten care, employment, housing, belonging, or approval (SAMHSA, 2014).

3.1 — Care Before Proof

Care before proof is both a moral sequence and a communication rule. A person should not have to wait for a theory to be confirmed before pain is treated, danger is addressed, or basic dignity is protected. The claim is immediate because the suffering is immediate.

The rule also protects inquiry. Untreated pain, fear of losing care, pressure from family, and the need to please researchers can alter reporting, participation, and behavior. Ethical failure can become measurement failure. Palliative care, trauma-informed practice, and human-subject protections converge on the importance of relief, safety, respect, and proportionate burden (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; World Health Organization, 2020; SAMHSA, 2014).

The rule must travel where its omission would make care appear conditional. A public summary, clinical case, image, or conversation about the Law should retain the immediate duty triggered by the situation. Where no such risk exists, the argument should proceed without ceremonial repetition.

3.1.1 — Present Pain as Present Claim

A person in pain is not first a problem for philosophy. Pain can require assessment, protection, relief, competent treatment, shelter, accompaniment, or a route out of danger. The response depends on the situation, but the order is stable: attend to the person before asking the experience to carry a theory.

Palliative care offers a clear institutional model. It addresses pain and other symptoms while attending to communication, family strain, psychological distress, and spiritual concerns as the person defines them (World Health Organization, 2020). Its sequence is instructive for this book because comfort does not depend on accepting an interpretation.

Action need not await diagnostic closure. Emergency medicine, crisis response, safeguarding, and trauma-informed practice often proceed under uncertainty because the cost of waiting may be high. Temporary protection, symptom control, food, warmth, privacy, or de-escalation can be revised as knowledge improves; uncertainty calls for monitoring and humility, not moral delay (SAMHSA, 2014).

Delay is itself lived experience. Waiting for analgesia, safety, shelter, an interpreter, or protection can deepen fear, exhaustion, distrust, and loss of agency. The minutes consumed by an institution do not sit outside the person's life while everyone else decides what the burden means.

The same rule extends beyond medicine. A person escaping violence needs safety before a debate about motive. A child reporting abuse needs protection before repeated questioning. Someone in withdrawal may need clinical support, food, sleep, and overdose prevention before being asked to explain a life history. A worker injured on the job needs treatment and a secure reporting route before institutional reputation is defended.

Present need also requires present competence. The right helper may be a clinician, counselor, social worker, teacher, advocate,

shelter worker, trusted relative, emergency service, or legal professional. Philosophical attention can be humane, but it is not a substitute for the skill, authority, and resources the situation requires.

Research near serious illness therefore needs clinical interruption built into its architecture. Symptom-triggered pauses, direct access to care, independent stop authority, and acceptance of incomplete records should be specified before enrollment (World Medical Association, 2024; CIOMS, 2016). The scientifically honest outcome may be no final observation because treatment, fatigue, refusal, or safety came first.

Addiction tests the sequence without simplifying it. Treatment may involve withdrawal management, boundaries, effort, and discomfort that a person chooses within care. Those burdens differ from pain imposed or prolonged because a theory predicts later balance. The relevant questions are whether the intervention is evidence-based, proportionate, wanted, revisable, and protective of future choice.

Accountability does not suspend care. A person who has harmed others may face lawful restriction, truth-telling, restitution, or loss of authority while still retaining medical needs and human dignity. Care is not a declaration of innocence; it is a refusal to use untreated suffering as an unofficial penalty.

Care often involves trade-offs, so the sequence must be operational rather than sentimental: use competent help, reassess effects, keep the person's values and decision rights visible, and give staff or participants usable stop and escalation routes. Relief

can also preserve future optionality by restoring sleep, attention, mobility, trust, or the ability to choose. None of this depends on PTVC; the duty to reduce preventable burden remains under support, ambiguity, rival victory, defeat, or permanent nonidentification.

3.1.2 — If the Law Is False

If the Law is false, care still comes first. The ethical firewall is independent of the formal result. Scientific defeat would remove one proposed terminal explanation; it would leave every present reason for relief, protection, dignity, consent, and professional care intact. Visible and measurable suffering would remain fully actionable, without any universal guarantee of terminal closure (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016).

A defeated hypothesis can still leave valuable knowledge behind. Research may improve affective measurement, reveal weaknesses in endpoint inference, clarify adaptation, strengthen missing-data methods, or deepen protections for vulnerable participants. Respect, care, justice, and refusal are not consolation prizes for a failed theory. They are the standards that made the inquiry legitimate and the durable gains by which its work should be judged (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016).

What survives is an affirmative architecture for better inquiry: a person remains larger than every score, an individual path remains distinct from a group average, care precedes interpretation, measurement remains separate from possession,

and scientific uncertainty never becomes moral indifference. Those distinctions improve research and practice even when the central target loses.

Rival victory is likewise informative. If adaptation, predictive regulation, selection, memory, or ordinary mean reversion explains the data better, the project learns which mechanisms deserve more attention. The result may improve care and theory without preserving the original terminal claim.

The person should never bear the cost of the project's disappointment. Participants are not failed experiments. Their data, time, and vulnerability should be honored through accurate reporting, privacy protection, and clear communication about what the result means. No one owes the theory a favorable life.

That independence is part of the project's design: care is grounded in vulnerability, rights, professional duty, and the possibility of reducing harm, not in a metaphysical wager. A consequential hypothesis should be able to lose without asking people to lose with it.

3.1.3 — Care as Communication Constraint

Communication can harm even when research is careful. Public sentences, formulas, captions, or promises may make care feel less urgent, imply blame, pressure disclosure, or portray a particular harm as required for closure. Emotional timing matters too: grief, acute pain, trauma, and family desperation can turn theory into pressure. Care therefore constrains what words may permit in ordinary use, not merely whether the author can defend them (SAMHSA, 2014).

A sentence should pass a distressed-reader test: could it reasonably be heard as a reason to wait, endure, self-punish, or refuse help? Detachable language needs its own essential protection because captions, posts, slides, interviews, and crisis messages travel without the Chapter. Present suffering deserves care, the terminal hypothesis does not identify a particular future event, and the reader should not need distant context to recover those facts (SAMHSA, 2014).

Status and personal prediction form one boundary. Name the Law as a hypothesis and distinguish its conditional net-remainder requirement from a claim to know a person's actual balance or future events. The How to Read contract reduces routine repetition, but high-risk excerpts must remain self-contained.

No-blame and care clarity form another boundary. Language suggesting that someone needed a wound, chose a lesson, or must find meaning creates coercion; urgent communication should instead state available help, next steps, refusal, and complaint routes. Meaning belongs to the person, while trauma-informed practice emphasizes safety, transparency, collaboration, and choice (SAMHSA, 2014).

Uncertainty must not sound like abandonment. Pair what is unknown with what is known and with the treatment, rights, and support that continue regardless of the theory. A useful response might explain which assessment is pending, who will make contact, and which practical assistance is available today.

Public trust requires source rigor and access. Prestige, biography, attention, and supporter counts do not establish the hypothesis;

objections need definitions, sources, and failure conditions. Children, people in crisis, and readers with cognitive, sensory, or language barriers may need shorter explanations, interpreters, visuals, selectable text, alt text, and clear structure. A reader should be able to move from a short explanation to the relevant definition or source without navigating an unrelated argument.

Humane precision gives a practical map: what is known, what remains uncertain, what help is available, when reassessment occurs, how refusal works, and who remains responsible. Removing false consolation and inflated certainty makes the prose more concrete, collaborative, and respectful without reducing intellectual seriousness (SAMHSA, 2014). For example, a clear reassessment date turns an open question into a shared next step that the person can recognize and discuss.

3.2 — No Moral Desert, Debt, or Payment

The language of fairness easily attracts moral accounting. People speak of paying for mistakes, earning happiness, owing a debt to life, or receiving what they deserve. Some of this language is metaphorical. Inside a theory of cumulative experience, however, the metaphor can harden into a claim.

The proposed equality is phenomenal, not juridical or moral. It identifies no judge, assigns no fault, rewards no virtue, punishes no vice, and pronounces no biography justified. A zero terminal residual would not identify any particular episode as necessary or make it good.

Under exact PTVC, assume finite positive and negative totals, a completed qualifying stream, and a valid fixed representation with a calibrated neutral origin. At any chosen cut, the accumulated signed balance plus the signed experience still to occur before the true endpoint must equal zero. A genuinely net negative balance therefore requires an equally net positive remainder, and a genuinely net positive balance requires an equally net negative remainder. Required for closure does not mean morally deserved or justified.

This is only algebra about a net difference, not a forecast of events. In invented units, an accumulated balance of -1,000 would require a remaining net contribution of +1,000 under exact PTVC; if another -100 occurred, the required net remainder would become +1,100. The example does not say how, when, or through which experiences that remainder could arise, and it creates no safe strategy for obtaining a chosen outcome.

The same caution applies to positive contributions. They alter the residual algebra under the hypothesis but do not imply that joy, kindness, relief, or pleasure causes, schedules, merits, or “buys” later suffering. Helping changes the life actually lived, and PTVC does not require a prevented harm to reappear. The recipient's experience remains in the recipient's stream; rights, care, and justice remain independent commitments.

Public language should describe the conditional remainder as algebra rather than moral currency. A phrase such as “suffering is credit” falsely suggests entitlement, exchange, or control, so detachable summaries should keep that distinction visible.

3.2.1 — Suffering Is Not Earned

Suffering is not earned. This sentence must stand without qualification because the opposite claim is one of the oldest cruelties in human thought. People who suffer are often asked what they did, what they failed to learn, what hidden flaw they carry, what debt they owe, or what meaning their pain is supposed to produce.

This distinction protects children with special force. A child cannot have earned hunger, abuse, illness, terror, abandonment, or early death. Any theory that cannot say this plainly should not speak about fairness. The Law must stand near the child and refuse every attempt to make the child's pain spiritually or morally deserved. A mathematical remainder requirement is not moral desert and does not prescribe a particular injury.

Adults are protected as well. Addiction, illness, trauma, poverty, disability, grief, mental distress, humiliation, and isolation are

often moralized. Even where choices matter, suffering itself is not a clean moral invoice. A person may need accountability, treatment, protection, or repair, but none of that permits the claim that their pain is earned.

Responsibility may matter for prevention, accountability, and repair, but it does not convert suffering into a deserved phenomenal payment. A person who caused harm may owe restitution or face proportionate legal consequence. That public judgment is different from claiming that pain itself is cosmically owed. Keeping the distinction clear protects both justice and compassion from a metaphor neither can safely use.

The belief that people receive what they deserve can make an unpredictable world feel orderly. It offers the observer psychological distance from misfortune: if the harmed person somehow earned the outcome, others can imagine themselves safe through virtue. That emotional function is precisely why desert language requires scrutiny rather than trust.

Even adult choice occurs within unequal conditions. Education, trauma, addiction, disability, poverty, social support, and coercion shape the feasible set. This does not erase agency, but it makes simple moral arithmetic unreliable. Responsibility should be assessed through evidence about capacity, intention, knowledge, and circumstance.

Suffering can also be moralized positively. Observers may say that hardship builds character, makes success meaningful, or was necessary for growth. Growth after suffering is possible, but

necessity does not follow. A person can value what they became while still judging the harm wrong.

Religious and cultural traditions contain diverse accounts of suffering, responsibility, grace, fate, and repair. The root book can acknowledge those echoes without selecting one as evidence. No tradition authorizes the project to assign meaning to a nonbeliever or to speak for the harmed person.

Suffering is not earned also protects people who have harmed others. Their victims retain claims to truth and repair, and society may need boundaries. Yet care, due process, and freedom from cruelty remain. Human dignity does not become a reward for good behavior.

The principle changes ordinary speech. Instead of asking what a person did to deserve the burden, ask what caused it, what could have prevented it, what support is needed, and what responsibility can be established through fair process. These questions produce more actionable knowledge and less moral fantasy.

Addiction is a revealing case because moral judgment can interfere with treatment. Craving, withdrawal, learned cues, stress, and constrained choice do not remove agency, but they make simplistic blame clinically and ethically inadequate. Care should address risk and responsibility without treating pain as punishment.

Desert language can also turn inward. People may deny themselves relief because they believe they caused the illness, failed a family, or do not deserve resources. A theory of terminal

balance must never reinforce that self-punishment. Relief is not a prize awarded after moral acquittal.

3.2.2 — Exact Terminal Closure Is Not Vindication

This distinction is necessary because people may use terminal valence closure to make visible injustice sound acceptable. They may say that because the end closes, the path is vindicated. That is false. A zero terminal cumulative residual would not make cruelty good.

Vindication language is dangerous because it shifts attention away from the harmed person and toward the comfort of observers. Observers want the world to make sense. They want pain to be redeemed. They want a final sentence that removes moral disturbance.

The distinction between property and evaluation can be made with ordinary examples. A path may begin and end at the same elevation while containing a dangerous fall. An account may sum to zero while recording theft and repayment, and the arithmetic does not make the theft acceptable. Likewise, a terminal equality in valence would describe one relation at a boundary. It would not approve the events that produced the history.

A boundary property describes a relation at a boundary. Vindication is a moral judgment about a path, decision, or institution. Confusing them would allow one mathematical statement to absorb history, agency, rights, and preventability. The distinction must remain visible even in the most compressed public summary.

Equal net value does not imply equal gross experience. A stream with enormous positive and negative magnitudes can share a terminal residual with a quiet stream. The first may contain severe suffering that the second never approached. The formulation developed here preserves this difference explicitly.

Order matters humanly even when the terminal sum does not encode it. Pain followed by relief is not lived like relief followed by pain. Anticipation, memory, bodily wear, trust, and remaining opportunity differ. A final equality cannot retroactively make the sequence harmless.

Public wrongs remain. If an institution discriminated, delayed care, or exposed people to preventable harm, a terminal property cannot cancel responsibility. Records, apology, restitution, reform, and legal remedy address the wrong in different ways.

Vindication language is particularly harmful near death. A calm final period may reflect treatment, exhaustion, sedation, relationship, or many other causes. It should not be used to declare that a difficult life closed fairly. A distressed ending likewise should not be turned into a moral failure.

The distinction protects families from interpretive pressure. Survivors may remember peace, conflict, regret, love, or uncertainty. They should not be asked to fit the death into a theory. Their grief is its own stream of experience and deserves care without a formal verdict.

If the Law were supported, the honest statement would be narrow: a declared terminal property received evidence in a defined

reference class. Stronger claims about justice, purpose, or moral optimization would require independent argument.

3.2.3 — Arithmetic Without Moral Desert

Desert language is more than tone because metaphors organize inference. Payment, debt, credit, and settlement can imply obligation, transfer, and an account keeper; when used for the conditional arithmetic, their limited mathematical meaning must be explicit. They must not make a particular harm sound morally owed or prescribed. Addiction, disability, and adaptation show why causal pathways must remain separate from moral currency. Craving, withdrawal, trauma, learning, access, identity, and community can shape experience without making relapse a debt or barriers the price of growth (SAMHSA, 2014; Albrecht & Devlieger, 1999; Ubel et al., 2005).

Later flourishing does not reprice the past. A survivor may build meaning, forgive, reconcile, advocate, or reject redemptive stories altogether; none makes abuse, discrimination, neglect, or avoidable illness acceptable (Park, 2010; Woods, 2011). Joy creates no moral debt. PTVC specifies exact complete-stream balance, not conservation of a fixed quantity of feeling at every moment. Moderation may be wise because choices affect health, relationships, resources, and future options, not because enjoyment creates a duty to suffer.

Causal language is clearer than accounting. Describe exposure, effort, support, fatigue, learning, stress, health, and consequence rather than saying humiliation paid for ambition or grief bought wisdom. Poetry can honor the fixed past, freedom of meaning,

courage of repair, and a precise hypothesis willing to fail. Courts and institutions may judge conduct under established duties; recovery may describe trust rebuilt by action. Neither makes suffering the price of worthiness.

The Law may concern a terminal property without any judge, creditor, cosmic scheduler, or theory of deserved pain. Later events can change a life without making an earlier wound morally owed, prescribed, or justified. This leaves room to study terminal structure as a property of experience while keeping accountability tied to causes, choices, duties, and repair.

3.3 — One Stream Cannot Pay for Another

The formal hypothesis has a bearer. On its primary branch, the relevant positive and negative magnitudes belong to one separately routed qualifying stream. Without that boundary, a group average could hide any individual result and the claim would lose its object.

The ethical consequence is direct. Another person's benefit cannot enter the victim's experience and cancel what happened. Research knowledge, public prosperity, spiritual meaning, family growth, and future policy improvement may all matter. None makes the original suffering unreal or paid.

Public decisions often involve competing interests, but the Law supplies no hidden experiential account in which one person becomes another person's balancing term. That distinction will matter whenever the book reaches institutions, animals, artificial systems, or collective narratives.

3.3.1 — One Stream Is Not Another Stream

One stream is not another stream. This is both a formal and ethical boundary. A conscious life is lived from within, and no other person can carry that exact inward path. Others may care, witness, help, harm, remember, or be affected, but they do not become the bearer of that stream.

The Law's formal model depends on this distinction. If experiential histories could be exchanged across persons, the claim would lose its bearer. A burden in one stream could be

hidden by relief in another. A child's suffering could be rationalized by social progress.

This does not deny interdependence. People shape one another constantly. A parent's care may relieve a child. A friend's presence may change grief. A teacher may open future possibility. A society may reduce or increase burden.

The boundary applies even when lives are closely connected. Parents and children influence one another. Partners share environments. Communities transmit trauma and support. Institutions shape many lives at once. These causal relations are real, but they do not fuse felt histories into one account. Each bearer's experience remains their own unless a separate theory of stream identity justifies a different routing.

No one can volunteer another stream into balance. Even if a group believes suffering served a noble purpose, the harmed person's experience does not become group property. A community may honor, mourn, repair, or learn, but it cannot retroactively own the burden.

Families provide an intuitive case. Members share resources, rooms, routines, and emotional climates. A household can be described as thriving or distressed, yet it does not possess one consciousness. Each person experiences the shared environment differently, and dependency affects who can speak.

Caregivers can experience another person's pain through empathy, exhaustion, grief, and bodily stress. Those experiences are real and deserve support. They are not the patient's pain

transferred into the caregiver. Compassion connects streams; it does not fuse them.

Artificial archives create a modern temptation. A digital record of a person's words, images, and behavior may persist after death. It is not thereby the same stream, and its outputs cannot settle what the person experienced. Archive continuity is informational, not automatic phenomenal continuity.

The formal model's routing rules are required because stream identity may be uncertain in sleep, anesthesia, fragmentation, fission, fusion, or artificial systems. Uncertainty does not authorize casual merging. The scientific response is branch-specific analysis or abstention.

Nontransferability also protects positive experience. Joy, love, pleasure, and relief belong to the person who feels them. They should not be treated as resources that justify exposing someone else to harm. A happy majority does not purchase a victim's suffering.

The principle is compatible with duties of sacrifice chosen by the person. Adults may accept risk for family, community, medicine, or public service. Voluntary sacrifice can have meaning, but consent does not make the streams fungible. The person's burden remains theirs and deserves proportional protection.

The bearer boundary also limits inference. A group trend cannot reveal the terminal state of a member, and one person's report cannot establish a collective experience. Aggregation is a statistical operation, not a transfer of consciousness.

The boundary can be violated subtly through family language. A parent may say, “I would take the pain for my child,” expressing love and willingness to sacrifice. The feeling is morally significant, but the experiences remain distinct. The parent’s distress and the child’s pain may interact without becoming one transferable quantity.

The boundary applies to memory. One person’s archive may preserve another’s testimony, but remembering someone is not continuing their conscious stream by default. The formal bearer cannot be assigned through sentiment or technological resemblance.

Nontransferability is therefore both a scientific typing rule and a moral protection. It prevents the theory from balancing victims against beneficiaries, individuals against populations, or living persons against records.

3.3.2 — No Victim Pays for Insight

No victim pays for insight. Knowledge may follow harm, but the knowledge does not travel backward and justify what happened. A discovery can be valuable while the event that made it possible remains wrong, preventable, and owed a response.

This distinction protects medicine, law, journalism, religion, education, activism, and family memory from the same temptation. People often seek meaning after injury because understanding can help them act. The danger begins when the lesson is used to redescribe the injury as necessary or the harmed person as a contribution to someone else’s progress.

Research with people who have already suffered must therefore begin with meaningful choice, minimized burden, and support. A vivid history does not make a person especially available to the project. Recruitment language should explain the study's limits without promising that participation will redeem the experience or help others in ways the design cannot establish (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016).

Social value is necessary but not sufficient. A study also needs scientific validity, fair selection, favorable risk-benefit, independent review, informed consent, and continuing respect for participants (Emanuel et al., 2000). Noble purpose cannot rescue a weak design because useless or uninterpretable research asks people to carry burden for knowledge the method cannot deliver.

Historical abuses show what happens when institutions treat urgency, prestige, or public benefit as permission to use persons as means. Those histories should be studied with documentary accuracy rather than turned into spectacle. Their practical inheritance is independent oversight, transparency, enforceable rights, and a willingness to stop before institutional confidence becomes harm.

The same boundary applies across private relationships, storytelling, grief, and public remembrance. A survivor does not owe forgiveness, inspiration, disclosure, or closure, and a moving story is not automatically an ethically available story. Consent, context, factual accuracy, and minimization remain necessary where identifiable experience is shared.

Secondary use and compensation require the same discipline. Data collected for care or one study do not silently become available for another purpose, and payment should recognize time and burden without making refusal unrealistic under dependency (CIOMS, 2016).

Ordinary low-burden research may still be permissible when scientifically necessary, proportionate, and freely chosen. Records produced through coercion, discrimination, or inadequate consent may require restriction, contextualization, community engagement, or nonuse rather than gaining legitimacy with time.

When learning emerges from tragedy, credit is not repair. Institutions should preserve truth, support those harmed, provide remedy where possible, change the conditions that produced the event, and remember why the reform became necessary. Knowledge belongs inside that sequence rather than standing in for it.

Gratitude must attach to voluntary contribution, not to suffering itself. Researchers can thank participants for time, candor, and chosen risk without thanking illness, abuse, or grief for becoming scientifically useful. The wording matters because it identifies what the institution believes it was entitled to receive.

The final boundary is simple: no person becomes the cost of making this project persuasive. Experience can illuminate a system, and voluntary testimony can transform public understanding, but the system never owns the life that taught it.

Insight remains legitimate only while dignity, consent, care, and repair remain larger than the lesson.

3.3.3 — Social Benefit Cannot Justify Suffering

Social benefit cannot justify suffering by itself. A harm does not become acceptable merely because others benefit from it. A policy, experiment, institution, technology, ritual, or movement must be judged not only by aggregate gain, but by the burden imposed on particular bearers.

This rule does not deny that ethical life contains tradeoffs. Public systems allocate limited resources. Medicine weighs risks and benefits. Law protects interests that can conflict. Governments make imperfect decisions under uncertainty. The firewall does not demand impossible purity.

Some acts are categorically outside the permissible field. Deliberately inflicting serious suffering to generate social benefit, withholding relief to preserve an observation, or selecting one person as the price of another's alleged balance cannot be repaired by aggregate language.

This does not mean that every burden imposed by public action is forbidden. Vaccination policy, infrastructure, taxation, emergency response, and environmental rules can distribute costs while pursuing legitimate goals. The relevant prohibition is narrower: the Law may not be invoked as a hidden phenomenal offset that makes one group's suffering acceptable because another group benefits. Public tradeoffs require public reasons, lawful authority, proportionality, and remedy, not metaphysical accounting.

Naming burden does not make decision impossible. It prevents moral laundering. A necessary burden remains a burden. The person carrying it remains entitled to explanation, protection, review, and remedy where those are possible. Necessity is not innocence.

Consent can matter without solving everything. An adult may voluntarily accept risk for others, but background inequality can shape the choice. Some public burdens cannot be individually consented to. In those cases, lawful authority, necessity, proportionality, transparency, and appeal become more important.

Children and dependent adults require special protection because others may authorize participation on their behalf. Permission should track the person's interests, dissent, and capacity. The social importance of the study does not turn dependency into availability.

The Law of Fairness cannot provide a hidden guarantee that burdens will balance. Using the hypothesis to reassure exposed communities would be a serious misuse. Public justification must stand on ordinary evidence, rights, and accountable authority.

The principle also applies to majorities. A majority vote can authorize some decisions under law, but it does not make the minority's suffering unreal or morally neutral. Constitutional and human-rights protections exist partly because preference aggregation has limits.

Collective benefit must be specified rather than invoked. Identify who benefits, who carries which burden, what alternatives exist,

how uncertainty is handled, and who can stop or appeal. Ordinary research burdens such as time, travel, disclosure, fatigue, and bystander exposure should be minimized and governed; group averages must not hide severe burdens placed on particular people. Social value matters, but it cannot function as a hidden offset for another bearer's suffering (National Commission, 1979; CIOMS, 2016; World Medical Association, 2024; Emanuel et al., 2000).

The public language of the Law must preserve this boundary even in its shortest form. Phrases such as “for the greater good” or “the suffering taught us” can erase the difference between learning from harm and authorizing harm. Knowledge may emerge after tragedy, but the tragedy was not purchased for the knowledge. A person remains the bearer of what happened to them, and no later insight acquires retroactive permission over that experience.

3.4 — Consent, Refusal, and No Harm

A prohibition becomes useful when it changes a decision. “No harm” must therefore reach recruitment, task design, burden, privacy, withdrawal, compensation, dependency, aftercare, and stop rules. Ethical research is not only a favorable intention. It is an operational structure (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016; Emanuel et al., 2000; Beauchamp & Childress, 2019).

Refusal is the central test. Can a patient decline without fearing worse care? Can an employee say no without retaliation? Can a child dissent in a way adults respect? Can a participant leave after data collection begins? Can a digital interface make declining as easy as accepting? Consent that fails these questions may be nominal rather than practical.

The invitation itself can become coercive if a study is presented as a way to discover a person’s true balance, destiny, or hidden meaning. No protocol can offer that knowledge. Existential language can make an uncertain research procedure sound personally therapeutic or revelatory, which heightens the risk of therapeutic misconception before a consent form is even signed (Appelbaum et al., 1982).

3.4.1 — No Created Suffering for Data

No created suffering for data is a categorical prohibition when a design deliberately inflicts serious physical or psychological suffering in order to test the Law. A theory about burden may not manufacture burden as proof material (National Commission,

1979; World Medical Association, 2024; CIOMS, 2016; Emanuel et al., 2000).

This hard rule must not be blurred with every lower-burden procedure. Ethically reviewed research may use ordinary questions, brief tasks, noninvasive monitoring, or mild and reversible inconvenience when the objective is important, the design is scientifically credible, alternatives have been considered, and the remaining burden is proportionate.

Vulnerability increases responsibility. Children, dependent patients, prisoners, students, workers, migrants, people in financial need, trauma survivors, terminally ill persons, and others with constrained refusal cannot be recruited as vivid data sources merely because access is easier or the expected signal is stronger.

Research standards require more than the absence of dramatic injury. Scientific validity matters because a badly designed study exposes people to burden without a credible path to knowledge. Independent review matters because investigators can become attached to their own hypotheses. Vulnerability matters because need and dependency can make an offer coercive in practice. The prohibition on created suffering sits inside this broader architecture of value, validity, proportionality, and respect (World Medical Association, 2024; CIOMS, 2016; Emanuel et al., 2000).

The precaution extends to nonhuman animals, organoids, synthetic biological systems, and future artificial candidates when credible uncertainty about suffering exists. Unknown status is not

permission to create a state that might be agony. The question may need to remain unanswered until a humane route exists.

Safe experiments can still study affect. Brief ordinary stimuli, choice tasks, natural variation, therapeutic contexts, and noninvasive measurements may test representation, calibration, adaptation, or rival models.

The prohibition also rules out self-testing. Readers should never be invited to intensify pain, deprivation, addiction, sleep loss, or danger to see whether the Law responds. Personal experimentation cannot establish the complete-stream claim and can cause irreversible harm.

Endpoint manipulation is equally prohibited. Researchers may not delay treatment, alter sedation, prolong distress, or influence life-sustaining decisions to obtain a cleaner terminal observation. Clinical decisions belong to the patient, authorized surrogates, and care team under ordinary standards.

Control conditions, deception, trauma-related questions, and observation of naturally occurring adversity all remain subject to ordinary research ethics. Effective care may not be withheld when serious harm is likely; deception requires strong justification and debriefing where permissible; sensitive questions must be necessary and stoppable; and researchers retain safety duties when observation reveals immediate risk. The Law creates no exception to those standards.

The correct scientific attitude is creative limitation. If one experiment is forbidden, design another, use simulation, test prerequisites, examine natural experiments, or accept that the

target cannot yet be adjudicated. A question does not become less scientific because some routes are closed.

Participants should hear the boundary clearly. The project does not need them to endure pain to prove anything. Withdrawal, symptom relief, and ordinary care take precedence over completion. This message reduces therapeutic misconception and protects voluntary participation.

The prohibition includes third parties. A study cannot expose family members, caregivers, or bystanders through recordings, location data, or inferred relationships without appropriate authority and protection. The participant's consent does not automatically cover everyone present in the data.

When a decisive test would require unacceptable suffering, the correct verdict is not “therefore the Law is true.” It is that the test is ethically unavailable. Science may remain uncertain because people matter more than complete adjudication.

Permissible inquiry should prefer naturalistic, observational, retrospective, model-based, simulation-based, archival, or otherwise humane routes whenever they can answer the question. Where human participation is necessary, the design must be capable of learning enough to justify the burden and of stopping when that justification fails.

3.4.2 — Refusal Rights

Refusal rights are the practical test of consent. A person does not meaningfully consent when refusal is unclear, punished, shamed, delayed, technically difficult, or treated as evidence of poor

character, disloyalty, irrationality, or lack of seriousness (World Medical Association, 2024; Faden & Beauchamp, 1986).

Refusal must be available before participation begins. The person should know that ordinary care, benefits, education, employment, housing, worship, platform access, and public standing will not be withdrawn merely because they decline a study or practice, unless a distinct lawful requirement genuinely applies.

Meaningful refusal has several parts: understandable information, time to decide, a genuine alternative, freedom from retaliation, the ability to stop, and clarity about what happens to already collected data. For children or adults who cannot provide full consent, assent, dissent, supported decision making, and authorized representation require special care. The 2024 Declaration of Helsinki expressly emphasizes respect for assent and dissent among people unable to provide free and informed consent (World Medical Association, 2024; United Nations, 1989; Faden & Beauchamp, 1986).

Refusal rights also protect bystanders and third parties. A participant cannot volunteer another person's intimate life merely because the information appears in the same conversation, home, recording, or sensor field. Consent must follow the bearer of the affected interest rather than the convenience of the collector.

The firewall does not celebrate participation more than refusal. Both can be dignified. The burden of making refusal safe belongs to the institution that asks, and a project that cannot tolerate refusal has confused access with entitlement.

Patients face a special risk of therapeutic misconception, the belief that research procedures are designed primarily for their individual benefit (Appelbaum et al., 1982). Clear separation between care and research helps refusal remain possible. The treating relationship should not be used to recruit through implied obligation.

Decision-making capacity is task-specific and can fluctuate. Illness, medication, delirium, pain, fear, or cognitive disability may affect understanding or communication. Support should be provided before capacity is judged, and uncertainty should not automatically transfer all authority to institutions.

Refusal reaches more than enrollment. A participant may decline interpretation, secondary use, public characterization, self-monitoring, or continued participation; purpose-specific consent should allow granular choices, and refusal must not reduce credibility or ordinary care.

Institutions need retaliation safeguards, and researchers should model how nonparticipation may affect representativeness rather than weakening voluntariness to improve a sample. The right not to monitor oneself is part of the same boundary.

The right to say no is one of the clearest tests of whether the project sees a person or a source. It strengthens science by reducing coercion, selective silence, and resentment while affirming the project's deepest limit: no one owes the theory access to inner life.

Refusal must remain available during participation. A person may change their mind, become tired, encounter an unexpected risk,

lose trust, or simply decide that the burden is no longer acceptable. Withdrawal cannot require a persuasive explanation, and the institution may not treat the exercise of a right as a failure of cooperation.

3.4.3 — No Institutional Need Above Care

Institutions have legitimate purposes, but people are not raw material for those purposes. A hospital may need records, a study may need observations, a school may need evaluation, and a public agency may need eligibility information. The ethical question is whether the institution can pursue its task while preserving care, comprehension, refusal, privacy, and a practical route to stop.

The conflict becomes sharp near the end of life. A final interview may be scientifically valuable while the patient needs analgesia, sleep, family presence, spiritual care, or quiet. The study yields. The choice is not anti-scientific. Data gathered through distress, fatigue, delirium, or role pressure may be both unethical and difficult to interpret, while excellent palliative care remains the governing duty (World Medical Association, 2024; CIOMS, 2016).

Consent in such settings is an ongoing relation rather than a signature obtained once. Understanding, voluntariness, capacity, alternatives, and willingness can change with illness, medication, fear, or dependency. Research teams should be prepared to pause, repeat information, involve authorized support where appropriate, and accept that no valid estimate may be possible (Appelbaum et al., 1982; Faden & Beauchamp, 1986).

Children and other people with constrained authority require additional care. Parental or guardian permission does not erase the importance of assent, dissent, developmental understanding, or the child's experience of burden. Family hope can become pressure even when everyone intends to help. Ethical design makes room for the less powerful person's signal rather than treating it as an obstacle (United Nations, 1989; CIOMS, 2016).

Institutional pressure can come from sponsors, funders, deadlines, staff workload, or public attention. Agreements should protect scientific independence, publication of adverse results, participant privacy, and authority to slow or stop when burdens become disproportionate (Emanuel et al., 2000; World Medical Association, 2024). Review must also account for burdens on families, care teams, researchers, and data stewards rather than shifting pressure to people with the least authority.

Care and research roles should be distinguishable. A patient must know which procedures are intended for individual treatment and which are intended to produce generalizable knowledge. The therapeutic misconception is especially likely when a study concerns a person's deepest experiences or true endpoint. Participation cannot be presented as a private reading of the person's life (Appelbaum et al., 1982).

Governance should include correction when instruments or inferences fail, proportionate compensation, continuing independent review, and prewritten routes for danger, deterioration, or privacy breach. These controls allow an institution to accept delay, missing data, or protocol failure rather than weaken care or realistic refusal (CIOMS, 2016).

An institution demonstrates its values by accepting disappointment. An incomplete dataset, delayed publication, failed protocol, withdrawn graphic, or missed deadline may be correct when the alternative pressures a person who needs care or wants to leave. This project receives no exception from that floor. Scientific credibility grows when institutions yield. People speak more honestly when care is not traded for data and refusal is not treated as moral failure; the deepest test is whether the person remains protected when the story stays incomplete.

3.5 — Relief and Meaning Remain Free

Relief and meaning can support one another, but they are not the same good. Analgesia can make reflection possible. Safety can make narrative possible. Relationship can make grief bearable. Yet treatment may not be withheld until a person adopts an explanation, and meaning may not be imposed as the price of care.

The literature on meaning-making documents possible pathways through stress, loss, and change, not a universal requirement. The medical humanities have also challenged the assumption that every illness or trauma should become a coherent story. People may find meaning, reject it, postpone it, revise it, or live without the kind of narrative others expect (Park, 2010; Woods, 2011).

3.5.1 — Relief Before Interpretation

Relief before interpretation establishes an order of action. Reduce avoidable suffering, protect the person, and restore as much agency as possible. Reflection may follow if it is wanted. This sequence applies in medicine, family life, therapy, spiritual care, research, public communication, and ordinary friendship.

Pain makes the priority obvious. A person asking for analgesia should not be asked to preserve an untreated state for observation or to explain what the pain means before receiving help. Assessment and treatment can proceed together. The pain belongs first to the person who is living it, not to the theory that might learn from it.

Breathlessness, nausea, agitation, insomnia, and fear can be equally urgent. Palliative care is organized around preventing and

relieving suffering through early identification, assessment, and treatment of physical, psychosocial, and spiritual concerns (World Health Organization, 2020). It can accompany disease-directed treatment and does not require agreement with a metaphysical account.

Outside medicine, relief may mean leaving a dangerous room. A person disclosing abuse needs safety, privacy, and competent support before anyone analyzes motive, lesson, or destiny. Shelter, a secure communication channel, and protection from retaliation are forms of care. Interpretation that delays escape becomes another burden.

Relief can take medical, administrative, spiritual, communicative, or accessibility forms, but the principle is the same: reduce avoidable burden before asking a person to carry an interpretation. Relief may also improve the conditions for valid consent by restoring attention and agency. Support can include faith or ritual when freely chosen, but never as a condition of care.

Operationally, staff need authority to pause data collection, simplify burdensome procedures, provide accommodation, correct records, escalate danger, and connect the person with competent help. When someone discloses suffering, first ask what support is wanted and whether immediate safety action is needed; the theory can wait.

The sequence can be carried in one practical rule: reduce what can be reduced, protect what must be protected, and ask about meaning only with permission. The rule is strong because it works

under support, uncertainty, or defeat of the formal hypothesis. It does not need the Law to be true.

3.5.2 — Meaning Can Be Refused

Refusal of meaning is dignified. A person may say no to consolation, spiritual framing, hidden order, personal growth, public testimony, or the Law as comfort without becoming bitter, shallow, faithless, unhealed, or closed. Others may freely rebuild identity through narrative, faith, relationship, service, creativity, political action, or time. Research and medical-humanities scholarship show that searches for meaning, integration, and narrative coherence vary; authority over whether meaning is sought and what form it takes remains with the person (Park, 2010; Woods, 2011).

This freedom protects secular dignity and religious honesty alike. A nonreligious reader can live ethically, love deeply, grieve fully, repair harm, and reduce burden without sacred language. A believer may refuse an easy interpretation in order to remain truthful; a skeptic may refuse spiritual language in order to remain free; a wounded person may need to refuse all interpretation simply to survive the day. No framework receives a higher rank because it is consoling.

Refusal can be temporary or lasting. A person may decline interpretation today and later choose a language that helps, or may never want one. Meaning-making research distinguishes searching from successful integration and reports varied outcomes across people and events (Park, 2010). Pressure can intensify distress when the person is not ready or does not share

the frame, so care must not freeze someone in rejection or push them toward acceptance.

Religious meaning must remain voluntary. A person may understand suffering through providence, karma, grace, covenant, fate, or another tradition. Another may reject all such interpretations. The Law does not select among them, and support should not depend on belief.

Families can disagree about meaning. One member may find comfort in ritual, another in medical explanation, another in silence. Care should make room for plural responses rather than declaring one family narrative. The person most directly affected should not be overruled for the comfort of others.

The right to refuse also protects nonparticipants. Communities should not be enrolled into a project's moral narrative because some members find it inspiring. Public institutions and schools should present the hypothesis as a question, not as a required worldview.

Meaning should never replace accountability. A person may find purpose after harm while still seeking legal remedy, apology, or prevention. Personal integration does not release the responsible actor or institution. The inner response belongs to the person; public duty remains public.

Refusal can be communicated simply. "I do not want to talk about what this means" should be enough. Helpers can ask what support is wanted instead. The burden of satisfying the observer's need for coherence should not fall on the person in pain.

Meaning can be beautiful when freely created or received. Protecting refusal does not flatten the book into silence; it preserves the conditions under which poetry, faith, philosophy, relationship, and silence can be genuine. Some experiences remain unwanted losses, and a survivor may build a valuable future without saying the injury was necessary or beneficial. A person is not an unfinished argument waiting for the project to complete.

The Law of Fairness can resonate with existing traditions or personal reflection, but resonance is not evidence and nonbelief carries no penalty.

3.5.3 — Place the Boundary Beside the Risk

A boundary is most useful where a real decision could go wrong. General principles belong near the beginning of the book, but later passages should carry the particular safeguard triggered by their subject. This is not a demand for repeated warning walls. It is a discipline of placing enough truth beside the risk that a reader can act responsibly without searching elsewhere.

When a passage discusses balance, the nearby boundary is that closure would not vindicate a life or make any event deserved. When a passage discusses measurement, the boundary is that an estimate is not inner truth or human worth. When a passage discusses an endpoint, the boundary is that treatment and comfort remain unchanged. Each safeguard answers a different possible misuse.

In a clinical conversation, adjacency means that a research invitation is accompanied by a plain statement that care does not

depend on participation. The patient should hear the practical right at the moment the pressure could arise, not discover it later in a policy document.

In a consent process, “you may withdraw” is too thin if leaving is difficult. The boundary should specify how to stop, what happens to already collected data, whether future contact ends, whom to call, and whether any benefit or treatment changes. Operational detail turns respect into a usable choice.

Spiritual language requires voluntariness at the point of use. A person may welcome prayer, ritual, philosophy, poetry, or silence. The boundary is not a general attack on meaning; it is the person’s freedom to accept, reinterpret, or refuse the offered frame without losing care or standing.

Family grief illustrates why timing matters. A relative may find the Law consoling while another finds it intrusive. The responsible speaker asks what support is wanted before offering interpretation. The safeguard is not a paragraph about every possible misuse; it is the discipline of letting the grieving person set the terms of the conversation.

Too little context creates the opposite failure: a detachable object can become a slogan the source Chapter would reject. The right amount of boundary is the smallest amount that preserves the proposition’s actual warrant and the duty still active. With that unsafe implication removed, vivid prose can remain vivid; metaphor should carry beauty and wonder, not the work of consent, evidence, care, or correction.

Useful safeguards name both an action and a route for error. “Protect privacy” becomes purpose limitation, restricted access, deletion, correction, and refusal; “do no harm” becomes a stop rule, clinical escalation, burden limit, and independent review; “use human judgment” becomes a named decision owner who can inspect, override, repair, and receive an appeal. A protection without contact, authority, or remedy leaves enforcement to the person most exposed.

Accessibility belongs beside communication as well. A safeguard hidden in unreadable text, an image without meaningful description, or a form that cannot be navigated by the intended participant is not functionally present. The person must be able to perceive and use the protection.

With those commitments in place, the central question can come into focus. Could lives follow profoundly different paths while sharing an exact property at their completion? What would have to be true of experience for that to happen? Could a mechanism uphold the relation without planning a life in advance, and could its effects be distinguished from ordinary recovery? These are questions worth pursuing carefully. Care gives the discussion a humane setting; the argument must now earn its answer through clear definitions, coherent possibilities, and encounters with evidence.

Part II — The Law in Plain Language

Part I established the human stakes: visible unfairness is real, felt experience is the relevant domain, and care comes before every interpretation. Part II now gives the proposal its public form. Everything that follows depends on stating the Law clearly enough to remember and precisely enough to challenge.

The public version has to do two jobs at once: give the reader a sentence that can be remembered, and keep enough structure that the sentence can be contradicted. Notation can obscure the human question; slogans can erase the conditions that make it scientific. This Part occupies the narrow middle. It translates the current formulation developed here into ordinary language while leaving detailed mathematics, protocol design, model recovery, and scientific verdicts to *The Science of Fairness*.

The public sentence matters because it turns a vague hope into a question with consequences. If the bearer is one stream, no population average or another person's joy can answer for it. If the relation is exact, an attractive tendency is not enough. If the claim is terminal, a selected episode cannot settle it. Chapter 4 will give those commitments one canonical formulation; the rest of the Part shows why each word protects the question from becoming either a slogan or an unfalsifiable consolation.

If supported, this terminal equality would coexist with profound differences in biography. People could still differ in wealth, health, opportunity, duration, power, love, memory, agency, and visible fortune. Rights could still be violated, public systems could still fail, and repair could still be owed. The proposed relation belongs within one stream and remains nontransferable across persons, groups, generations, species, and archives.

The boundary concerns the whole strict-pre-endpoint history, not the mood in which a person appears to die. A terminal residual represents the complete path assigned to the stream. A final report, facial expression, memory, medical note, or observable behavior may illuminate one interval without revealing the whole. Peace at the bedside cannot reprice earlier suffering, and distress at the bedside cannot by itself reveal the cumulative total.

The hiddenness of the target creates a productive challenge rather than an excuse. Experience is lived directly by the stream and reached by everyone else through reports, behavior, physiology, context, and other traces. The scientific task is to improve those bridges while keeping the distinction between a life and its evidence visible. That distinction protects the person from being reduced to an instrument and protects the hypothesis from being declared true merely because a proxy looks persuasive.

The endpoint gives the proposal its clearest test of intellectual honesty. A boundary chosen after inspecting the result can make almost any history look complete. An independently governed endpoint forces the claim to face the possibility of a nonzero ending, an ambiguous ending, or no adequate estimate at all. Chapter 5 develops that boundary in difficult human cases, where a convenient definition would weaken both science and care.

This architecture creates two distinct questions. The truth-level proposal asks about exact latent closure. Empirical work asks what finite observations, uncertainty, endpoint models, and measured-to-latent bridges can support. A study may establish compatibility at a declared resolution, pressure a route or target branch, or conclude that no adequate estimate is available. Each

outcome can sharpen the research program without being promoted beyond the object it actually tested.

The most demanding public version is impersonal. A constraint can describe the shape of admissible histories without a judge, rewarder, punisher, cosmic accountant, or hidden intelligence. Removing the easy storyteller leaves the harder and more interesting possibility: a structural relation that must be explained, measured, compared with rivals, and allowed to fail.

Calling the proposal a law therefore sets an ambition rather than announcing a victory. It asks whether the relation is stable, counterfactually serious, and explanatory enough to survive comparison with better-known regularities and strong rivals. Chapter 6 treats that name as an invitation to discovery: nature may support it, narrow it, or reject it, and each outcome can leave the question clearer than it was before.

Plain language is an audit of understanding. A capable reader should be able to locate the bearer, endpoint, latent object, exact target, and evidential limits without mastering the mathematics. Accessibility succeeds when the real uncertainty becomes intelligible and criticism gains a clear object, not when technical words are merely replaced by softer vagueness (Fischhoff, 2013; National Academies, 2017).

Those distinctions prepare the structural inquiry that follows.

Chapter 4 — The Law of Fairness

A public hypothesis becomes useful when one sentence can be remembered, reopened, and challenged. It must carry the real object and boundary without hiding behind notation or dissolving into a slogan. Chapter 4 creates that canonical public home and then shows the work performed by every term.

The shortest accurate statement begins with one conscious stream, not with humanity as a whole. It concerns the complete felt history assigned to that stream under a declared representation, not the visible biography, remembered story, public record, or social average. It ends at a truth-level boundary defined independently of the cumulative result.

The sentence rests on several commitments. A qualifying bearer must exist; its complete domain must be specified; positive and negative experience must be representable with a defensible origin; the endpoint must be relevant and independently governed; and the terminal relation must be exact on the named branch. Because the commitments remain separate, inquiry can discover where the proposal succeeds, narrows, or fails.

The word stream is doing protective work. It keeps the claim within the unit that lives the experience. The body is indispensable in ordinary human cases, but bodily continuity alone does not answer every question about unity. Legal identity protects rights, but it does not define the formal bearer. Biography and archive preserve traces, not the felt history itself.

Hidden means latent rather than directly displayed. Reports and measurements can bear on the cumulative object through an observation model, but no meter reveals the truth-level total.

Construct validity therefore comes before numerical confidence, because precision attached to the wrong object would only sharpen the wrong answer (Cronbach & Meehl, 1955; Borsboom, 2005).

Cumulative means that contributions are assigned across a complete path. It does not create a personal account that can be spent, owed, deserved, transferred, or administered. The formal object organizes a scientific question about one stream; it supplies no moral balance sheet, eligibility score, spiritual rank, or prediction of reward and punishment.

Exact closure is a truth-level boundary relation across the whole represented history. It is not a neutral final feeling, and it leaves duration, order, intensity, agency, memory, opportunity, resources, and gross experience free to differ. Two streams could satisfy the same terminal condition while their lives remained radically unequal in every ordinary dimension.

The word fairness arrives with moral expectations. In this book's narrow formal usage, it names a proposed terminal symmetry: each qualifying stream would independently satisfy a zero boundary under its declared valid representation, even though the lives reaching it could differ radically. The name does not claim that the path was just, deserved, equally good, or morally complete. It asks whether terminal neutrality is one real dimension of fairness while rights, welfare, opportunity, safety, remedy, and prevention remain independently necessary (Rawls, 1999; Sen, 2009; Nussbaum, 2006).

The canonical sentence opens one declared branch rather than pretending that every representation of valence has already been settled. Alternative additive, bipolar, or generalized forms remain available only as separately named hypotheses with their own assumptions and failure conditions. That branch discipline lets the public statement stay memorable without hiding the choices on which it depends.

Confidence here comes from precision. The proposal can be presented at full strength as exact, terminal, intrastream, nontransferable, and capable of defeat. It can also be presented with wonder, because a compact boundary relation would be a remarkable feature of conscious life if future evidence supported it.

4.1 — The Law in One Sentence

On its primary additive branch, the proposal asks whether each complete qualifying conscious stream contains exactly equal finite totals of positive and negative felt experience. The full sentence below specifies the reference class, routing rules, and independently defined endpoint.

4.1.1 — The One-Sentence Law

On the primary additive branch, the Law of Fairness proposes that each maximal qualifying conscious stream in the declared reference class, with its experience separately assigned under independently fixed routing rules, accumulates finite and exactly equal positive and negative magnitudes of felt experience across its complete history before an independently defined true endpoint, so that its latent cumulative residual is exactly zero immediately before that endpoint.

This is the canonical public sentence. No stream is counted toward another. The claim permits immense differences in duration, intensity, order, agency, opportunity, memory, welfare, and circumstance.

Measurement theory helps explain why the sentence must preserve its conditions. A numerical representation is not guaranteed to possess a natural zero, additive units, or invariance under every transformation. Those properties must be argued for rather than assumed from the appearance of arithmetic (Stevens, 1946; Michell, 1999; Borsboom, 2005). The formulation developed here therefore treats origin, sign, scalarization, bearer,

routing, and endpoint as gates to the sentence, not decorative qualifications.

Every condition in the sentence carries risk. Bearer, origin, scale, aggregation, unity, domain, and endpoint assumptions determine what is being claimed and where criticism should land. If an origin cannot be justified or a scalar representation fails, the exact additive form may narrow toward a multidimensional, partial-order, within-person, or domain-specific proposal, or it may fail. Those are discoveries about the object, not invitations to rewrite the sentence after the fact.

Exactness is where the Law becomes scientifically interesting. Adaptation, resilience, mean reversion, and homeostasis can move experience toward a local baseline, but none by itself requires a complete stream to satisfy an exact terminal boundary. A successful route would have to explain why the whole history closes, not merely why local states recover. That difference creates concrete questions about remaining opportunity, dispersion, perturbation, and endpoint timing. Each question gives researchers a way to compare a Law-linked explanation with capable ordinary mechanisms.

The sentence names the primary additive branch, not every possible valence representation. On that branch, positive and negative phenomenal magnitudes are the positive and negative parts of one finite signed measure. A bipolar branch can represent simultaneous positive and negative components separately, while Generalized Terminal Neutrality covers broader zero-valued functionals. Public language remains accurate only when it

identifies the branch being compressed and does not borrow conclusions from another branch without a justified bridge.

The phrase complete stream prevents a second distortion. The proposal does not concern a fortunate decade, a painful illness, a selected memory, or the period for which records happen to exist. It concerns the entire strict-pre-endpoint domain assigned to one qualifying stream. Selecting an emotionally vivid interval can illustrate the stakes, but it cannot stand in for the formal object. Likewise, combining several people into a household, generation, or population may answer important social questions, yet it cannot satisfy a claim whose equality is intrastream and nontransferable.

One branch already creates a severe structural test: if an endpoint can arrive independently throughout an interval with positive density, probability-one exact closure would force the preceding ledger to be zero almost everywhere on that branch. Chapter 7 derives this obstruction and the alternative endpoint structures it leaves open.

4.1.2 — Endpoint

An endpoint gives the one-sentence Law its terminal meaning. Without a boundary, the words complete stream and terminal closure have no fixed object. The relevant endpoint is the point at which the qualifying conscious stream is complete for the declared formal branch, not merely the last moment for which an observer has a record.

Three boundaries must remain separate. The truth-level endpoint belongs to the hypothesis. A protocol endpoint is the

prospectively defined event a study uses to locate a candidate stopping region. The end of observation is simply the last time usable data were obtained. These boundaries may coincide in a well-adjudicated case, but no notation makes them identical.

Clinical and legal death categories serve indispensable purposes, but they are not direct measurements of the model's truth-level endpoint; Chapter 5 develops the consciousness, confounder, and return problems in full.

Chapter 5 will make the central rule visual: the finish line is fixed before the result is seen. A protocol may use a prospectively specified state-coupled, first-hitting, interval, or other branch-appropriate endpoint, but the cumulative estimate may not choose the rule.

Independence is operational, not merely verbal. An endpoint proxy and a cumulative proxy can have different names while sharing a sensor, preprocessing step, clinician judgment, or latent confounder. A valid protocol must therefore test whether the two quantities are algebraically or procedurally entangled. If the same information defines the boundary and the result, apparent closure may have been built into the pipeline rather than discovered.

Irreversibility and return make the rule concrete. A protocol-defined proxy may cross a threshold and later reverse, or a channel of communication may return after it appeared absent. The return does not reconstruct every unobserved experience, but it shows that the earlier candidate endpoint was not final. The protocol must state before outcome inspection how return,

oscillation, competing events, and uncertain persistence change classification.

Endpoint work can end in several honest classifications: usable endpoint, uncertain endpoint, not interpretable, or ethically unavailable. An interval-valued boundary may be the most accurate result available. The uncertainty belongs to the bridge from observation to endpoint, never to the dignity of the person and never to a presumption that the hypothesis received support.

The phrase immediately before is deliberate. The primary public sentence concerns the complete strict-pre-endpoint domain. Any phenomenal contribution assigned exactly at the endpoint belongs to a separately declared endpoint-inclusive branch and cannot be added informally after the result is known. This small grammatical choice protects a large scientific distinction.

Endpoint discipline serves both science and care. A formal boundary is not a wish, forecast, invitation, reason to withdraw treatment, or suggestion that death settles a debt. Rescue, comfort, lawful clinical judgment, and ordinary protection remain primary. The endpoint makes the hypothesis answerable while keeping the vulnerable person beyond its command.

4.1.3 — Why Paraphrase Is Dangerous

Paraphrase can quietly create another theory. “Life is fair” turns a narrow terminal hypothesis into a verdict on visible life; “everything balances” removes bearer, currency, endpoint, and exactness; moralized versions import desert or purpose. Exact latent zero likewise cannot be softened into a near-zero estimate,

trend, or comforting impression without changing the claim (Lakens, 2017; Wellek, 2010).

Ethical and institutional clarity follow the same rule. The canonical sentence assigns no purpose, payment, blame, or personal forecast and gives no clinician, court, school, employer, government, religious body, or artificial system access to a hidden total. It must not authorize delayed care, coercion, ranking, or eligibility.

Every detachable sentence should remain accurate beside a difficult case. Quote cards, captions, headlines, and automated summaries must preserve status, nontransferability, care, and no-score boundaries without requiring readers to reconstruct a missing Chapter. The most durable version is also the most interesting: one exact intrastream terminal relation under demanding assumptions, precise enough to surprise and fail.

4.2 — The Parts Inside the Claim

A single sentence can conceal several independent commitments. The Law requires a bearer, a complete domain, a signed representation with a justified origin, a relevant endpoint, and a terminal relation. Each part supplies a separate point at which the proposal can be narrowed or defeated.

Breaking the claim into assertions prevents all-or-nothing rhetoric. A failure to identify the endpoint cannot be repaired by an elegant measure, and a plausible bearer cannot repair an arbitrary neutral origin. Evidence is noncompensatory across these gates.

4.2.1 — A Stream Exists

The first assertion is that there is a bearer for whom experience occurs across time. A sensation can be brief, but the Law concerns more than isolated flashes. It requires a conscious stream whose positive and negative experiences can be assigned to one history without being pooled with another. If no such bearer can be specified, there is no coherent intrastream object to accumulate and no terminal claim to test.

Unity has several dimensions. Experiences may be unified at one moment across sight, sound, body sensation, thought, and emotion. They may also be connected through time by memory, causal continuity, embodiment, agency, expectation, and recurring access. Philosophy of consciousness does not offer one uncontested test that settles all of these dimensions (Bayne, 2010). The Law therefore needs an operational bearer rule while remaining open about deeper metaphysical accounts of the self.

Personal identity and stream identity are related but not identical. A person can change character, lose autobiographical memory, or reject an earlier narrative while still presenting a serious continuity claim. Conversely, two records can resemble one another without sharing a subject. The model does not need an immortal, indivisible ego, but it cannot let narrative convenience determine where one stream ends and another begins (Parfit, 1984; Dennett, 1987).

Maximality prevents favorable slicing. A painful year cannot be declared one stream and a joyful year another merely because each selected interval produces a preferred total. Separate routing prevents favorable pooling. Two people who share a household, body of records, or common cause do not thereby become one phenomenal bearer. These rules make the stream a scientific constraint rather than a storytelling device.

Qualifying is an eligibility rule, not a favorable verdict. Consciousness, unity, routing, domain, and reference-class criteria must be fixed before terminal outcomes are inspected. A universal statement over an empty eligible class supplies no empirical support: if no stream passes the gates, the result is No adequate estimate, not confirmation. Protocols should report eligible, excluded, unresolved, and adjudicated cases so difficult cases cannot disappear behind the adjective.

The bearer itself is a truth-level object. Evidence for it arrives through reports, behavior, physiology, neural activity, clinical examination, history, and context. None of those channels is identical to consciousness. A person may be unable to speak or move while retaining covert command following, and a fluent

output can be generated without establishing that anything is felt. Recent work on cognitive motor dissociation reinforces the need to separate behavioral appearance from the underlying conscious state (Claassen et al., 2019; Bodien et al., 2024).

This separation matters in ordinary clinical cases. Sedation, aphasia, paralysis, delirium, fluctuating arousal, and severe motor impairment can weaken one channel while leaving others informative. Repeated standardized assessment and converging evidence can reduce error without turning uncertainty into certainty (Giacino et al., 2018; Kondziella et al., 2020). The formal root should therefore teach a modest rule: absence of familiar response is not automatically absence of a stream.

The reverse error also matters. Testimony-like language, learned behavior, automation, or a convincing simulation does not by itself establish a conscious bearer. The question is not whether an output resembles human conversation. It is whether a qualifying system supports experience and a defensible ownership route. That question may remain unresolved for animals, artificial systems, organoids, or unfamiliar biological forms, and the uncertainty should be recorded rather than decided by enthusiasm or fear.

This assertion gives the rest of the claim somewhere to stand. It identifies the unit within which experience may accumulate while denying institutions and observers ownership of that experience. The next assertions add a relevant endpoint, a defensible representation, a unity gate, and an exact terminal condition. Each can fail separately, and success at one cannot repair failure at another.

4.2.2 — A Mind Has a Relevant Endpoint

The heading uses mind in ordinary language, but the formal burden falls on a qualifying conscious stream. A terminal claim needs a boundary at which that stream is complete. Without such a boundary, the proposed equality applies only to an unfinished window, and every unfavorable result can be postponed beyond the part of life that happened to be observed.

Relevant does not mean emotionally satisfying, socially recognized, or narratively complete. A person may leave unfinished work, unresolved love, unanswered wrongdoing, or no public story at all. Those facts matter profoundly, but they do not define the formal endpoint. The endpoint is relevant only if it marks the end of the bearer and domain to which the terminal claim was prospectively assigned.

Apply the three-boundary distinction in Subsection 4.1.2: state which boundary the study has, which it needs, and what bridge connects them.

A relevant endpoint must be irreversible on the tested branch and defined independently of the cumulative result. Chapter 5 treats reversible states, return, and endpoint adjudication in detail.

Random timing and observation shutdown create separate burdens. If an endpoint could occur independently across a broad interval, exact closure at every such event would impose an unusually severe restriction on the path. Meanwhile, speech, sensors, consent, or observation may fail before experience ends, so the last usable record requires a missingness and endpoint

bridge rather than automatic promotion to finality (Little & Rubin, 2019).

Nothing in the endpoint component presumes an afterlife, denies one, or settles a spiritual account of death. The formulation developed here specifies the boundary needed for its declared branch and leaves metaphysical interpretation outside the scientific premise. That restraint protects both nonbelief and belief from being recruited as evidence for a terminal condition.

A relevant endpoint is therefore not a convenient final date. It is a prospectively governed bridge between a qualifying stream, a complete domain, and a terminal proposition. Its difficulty gives the Law scientific content because the theory can fail at the boundary rather than move the boundary.

4.2.3 — Each Assertion Can Fail

The Law is a chain of commitments, not one indivisible intuition. Bearer, ownership, representation, origin, complete domain, endpoint, observation, recovery, exact target, reference class, and verdict each have a distinct failure route. The chain is noncompensatory: a beautiful terminal pattern cannot rescue an undefined bearer, an arbitrary zero, or a finish line selected after the result.

Bearer, ownership, and routing can fail before any terminal arithmetic begins. A study may be unable to identify one qualifying stream, distinguish one stream from several, or decide whether a pause, split, fusion, or altered state preserves continuity. Experiences may also be incompletely assigned or blended across people. A body label, legal identity, memory

archive, or data record cannot repair that failure, and cross-stream cancellation remains forbidden. The result may be a narrower class or No adequate estimate, not a guessed subject.

The neutral-origin assertion can fail separately. A questionnaire midpoint, personal baseline, sample mean, or analyst's centering rule is not automatically the point at which signed valence changes direction for the formal claim. Positive rescaling can preserve a zero relation, while an arbitrary additive shift can create or erase one. If no defensible origin is identified independently, exact terminal zero loses its intended meaning (Stevens, 1946; Michell, 1999).

The completeness assertion can fail because observation starts late, ends early, or misses precisely the intervals that matter most. A mature instrument does not reconstruct an unobserved childhood, dream, period of delirium, or terminal interval by precision alone. Missing prefixes and suffixes need explicit bridges, bounds, or exclusion from the claim. Unseen experience is not zero experience (Little & Rubin, 2019).

The endpoint assertion can fail if a proxy is reversible, confounded, chosen after cumulative-object inspection, or unrelated to the truth-level end of the stream. Legal death, the last interview, cessation of movement, and the end of monitoring answer different questions. When no independently defended endpoint bridge survives scrutiny, terminal evidence is unavailable even if the measured path looks favorable.

Once bearer, representation, origin, domain, endpoint, observation, and recovery gates are adequate, a terminal residual

credibly bounded away from neutrality counts against the declared exact branch. Latency creates an inferential burden, not an unlimited shelter. Unfavorable evidence cannot be renamed measurement error without an independently supported account of what failed.

Measured-band outcomes are separate from exact latent equality; Section 4.5.1 develops the equivalence-band distinction and its permitted verdict language.

Rivals can win, and a clean-looking graph can still be false support. Adaptation, opponent processes, predictive regulation, mean reversion, selection, memory, conditioned bridges, observation floors, endpoint selection, or flexible latent models may reproduce the prospectively specified signatures. Model recovery and equal-budget comparison must first show that the analysis can distinguish those worlds. If rivals predict as well or better, the result is model ambiguity or rival-favored evidence, not a victory secured by giving competitors weaker definitions (Mayo, 2018; Nosek et al., 2018).

Ethical viability is another gate. A question may be scientifically interesting yet impossible to study without coercion, disproportionate burden, harm induction, intrusive terminal observation, or unacceptable use of vulnerable people. Ethical nonviability is not evidence for the hypothesis. It may require a safer proxy, a narrower claim, principled abstention, or retirement of the proposed study. A protocol can therefore fail honorably before recruitment when its decisive observation would require burdens the question cannot justify.

Failure needs typed language because different failures teach different things. A target may be contradicted, a route may fail, a reference class may narrow, an estimator may be invalid, rivals may remain indistinguishable, or No adequate estimate may be possible. A procedure earns evidential force only if it had a real chance to reveal the mistake it claims not to have found (Popper, 1959; Mayo, 2018). A results table can give each component its own row, making the precise location and consequence of a failure immediately visible. Keeping those labels separate also tells the next study what to repair: measurement, scope, rival discrimination, or the target itself.

Universality is itself an assertion. A finite set of compatible cases cannot establish that every qualifying stream in a declared class closes exactly, especially when some cases are missing or unadjudicable. One gate-valid counterexample can defeat the declared universal branch for its class, while many compatible observations can only strengthen a bounded evidentiary position. The quantifier cannot be softened after an adverse case appears. A study can report exactly how its sampled cases relate to the declared class, including which kinds of cases remain unrepresented. A reference-class table can make this asymmetry visible by listing the bearer and endpoint regimes actually tested before any broader claim is considered.

Modularity makes honest revision possible. A corrected instrument can be retested while the target remains fixed; a new origin, bearer, endpoint, or ownership rule creates a new target version; a narrowed class creates a smaller claim. The record must show what failed, what changed, and what evidence still travels.

With those routes visible, the Chapter can now ask what actually carries the stream rather than treating precision as a substitute for a valid bearer. A practical version record should preserve the representation, origin, domain, endpoint, and reason for each change, allowing later studies to distinguish improved observation from a changed target and to replicate the same claim when those choices remain fixed.

4.3 — The Stream That Carries It

The bearer question is not solved by choosing the most familiar noun. Body, person, biography, memory, archive, family, group, and institution each name something real, but they answer different questions. The formal bearer must be tied to where felt experience is lived.

Personal-identity philosophy shows why continuity is difficult even in ordinary cases, while consciousness research shows why behavioral and biological proxies can come apart (Parfit, 1984; Bayne, 2010). The public book can explain the problem without pretending to settle every edge case.

4.3.1 — Stream Rather Than Body Alone

Human streams are embodied. Pain, hunger, sleep, illness, disability, safety, fatigue, hormones, nervous-system state, and sensory access shape experience continuously. Yet the body alone does not settle stream identity or endpoint status. The formal boundary follows conscious-stream status, which is why biological or legal death cannot simply be substituted for the truth-level end of mind.

Embodiment supplies conditions and evidence without becoming the whole definition. Bodily, psychological, narrative, and experiential continuities can come apart in thought experiments and clinical cases (Parfit, 1984; Bayne, 2010). The public root therefore begins with embodied human streams and keeps the formal bearer open enough to face cases in which familiar continuities diverge.

That distinction widens protection. A person who communicates, moves, senses, or requires care differently may still bear a full stream. Surface appearance can understate experience in unusual embodiment and overstate it in mere biological persistence. The stream-rather-than-body-alone rule places lived experience, not conformity to a bodily norm, at the center of the question.

Law and medicine protect people through rights, evidence, duties, diagnosis, and remedy. They do not administer the hidden cumulative object. Legal personhood may justly protect under uncertainty without becoming the formal bearer, and clinical categories may guide care without becoming metaphysical declarations. The constitutional and scientific companions develop those interfaces; the public root preserves the boundary that lets each domain remain authoritative for its own task.

Biography offers indispensable context without revealing the whole stream. The same diagnosis, poverty, praise, danger, or success can be lived differently depending on support, expectation, exposure, and meaning. Public narratives are also shaped by power: some lives receive complexity while others are reduced to stereotypes. Biography is therefore evidence about conditions and interpretation, not a verdict on felt experience.

Privacy completes the distinction. A person does not owe a complete life story before hidden burden, relief, or dignity can matter. Memory and public narrative are edited by audience, culture, shame, pride, and institutional record; the stream includes what was never narrated, could not be told, or vanished from public memory. A compelling biography may teach, but it may not become a proof object.

4.3.2 — Stream Rather Than Archive

Archives preserve accountable traces: records, testimony, decisions, images, data, communications, and institutional memory. They can resist denial, corruption, erasure, and abuse. Their very power can create an illusion of completeness, which is why a research file, public archive, diary, or machine summary must never be mistaken for the life it records.

An archive can extend cognition and social memory without becoming the subject that once felt. The extended-mind literature shows how notebooks, devices, and environments can participate in cognitive activity, but participation in a cognitive system does not by itself establish that the record has phenomenal experience (Clark & Chalmers, 1998). The distinction allows archives to support accountability while blocking the idea that a database preserves a person or soul.

Records can support research, memory, and repair without containing the hidden cumulative object. A transcript, video, biometric trace, dream log, or public presentation remains an observation shaped by collection and interpretation. Treating records as inner truth would invite scoring, surveillance, spiritual ranking, and exaggerated scientific access. An archive can help a society remember a person; it cannot replace the person or prove terminal closure.

Population measures can improve policy and reveal collective harm, but they answer a different question. Average improvement cannot establish individual closure, and a group total cannot absorb one member's residual. Nontransferability prevents public

good, historical destiny, or institutional success from turning particular people into interchangeable carriers of a shared hidden account. Social repair remains urgent because conditions shape streams, not because streams merge.

The stream-rather-than-group rule lets the series discuss families, cities, governments, constitutions, religions, science, peace, and transition while protecting each person from substitution. Collective language can describe shared history and institutional responsibility; it cannot create a collective phenomenal bearer by convenience. The same rule extends through time, because the Law concerns a trajectory of anticipation, memory, change, repetition, shock, recovery, relation, decline, and repair.

Continuity through time does not require perfect memory or an unchanged personality. A person can forget, transform, sleep, dream, or undergo anesthesia and still present a serious continuity claim. Order and context may also matter: relief after terror is not phenomenally identical to relief without terror, and late burden may carry a different horizon from early burden. Dementia, dissociation, split-brain conditions, developmental thresholds, and synthetic candidates keep the research problem open rather than allowing one proxy to settle it.

Through time, then, the Law is neither a mood theory nor a theory of one memorable event. It concerns the complete routed trajectory. Memory, attention, embodiment, agency, narrative, and integration can each contribute evidence without becoming absolute. This temporal view prevents one survey, crisis, sensor reading, or bedside scene from standing in for the whole stream.

4.3.3 — Edge Cases Deferred

Edge cases are deferred because the canonical statement needs a bearer before the public root can examine every difficult bearer. Deferral is not dismissal. It is a promise of disciplined depth. This Chapter states the unit the Law requires; later work on identity and science asks which systems satisfy it, with what confidence, on what evidence, and where the only honest answer remains unknown.

Reversible states are especially instructive. A return to reliable communication can show that an earlier apparent endpoint or interruption was misclassified, yet the absence of return does not reveal the exact history of experience. Sleep and anesthesia also show that report can disappear while later evidence changes what can responsibly be inferred. Reversibility is therefore evidence about classification, not a universal definition of consciousness.

Clinical edge cases remain open to repeated, multimodal assessment; bedside appearance alone cannot determine stream identity. Chapter 13 returns to these cases in the dedicated bearer analysis.

Fission, fusion, split-brain cases, and severe dissociation expose the routing problem directly. If a stream appears to divide, the model must say whether the original ends, whether successors begin, and whether earlier experience is duplicated. If streams appear to integrate, it must say whether a new bearer begins or one continuity survives. Causal connection, information access, report, and phenomenal unity cannot be treated as synonyms, and

no diagram may trade valence across routes merely because their histories touch (Bayne, 2010).

The ethical safeguard is independence from formal classification. Clinical care, legal protection, disability rights, family respect, animal welfare, research restraint, and freedom from exploitation rest on their own grounds. An uncertain stream status cannot become permission to neglect, experiment coercively, deny assistance, or rank worth. Dignity is not the prize awarded after a model reaches sufficient confidence.

Progressive dementia creates a different pressure. Memory loss, altered personality, and reduced agency can transform a biography without establishing that one stream ended and another began. The model cannot use remembered continuity as the only ownership rule because unremembered experience would then lose its bearer. Nor can it assume that biological continuity settles phenomenal continuity. The honest public statement is that biography, memory, and stream identity overlap without being identical, and that no convenient proxy is allowed to erase felt history.

Comparative cases require an evidence ladder rather than a single threshold. Behavioral flexibility, learning, affective tradeoffs, neural organization, and responses to intervention can contribute without functioning as consciousness oracles. Artificial systems pose a parallel inference problem: testimony-like output is evidence about behavior and architecture, not by itself proof of experience or ownership. Precaution may increase with credible risk, but precaution is not a declaration of personhood.

A prospectively declared branch map prevents hard cases from becoming after-the-fact escape clauses. Before a test, the project should state how uninterrupted continuity, reversible interruption, uncertain unity, fission, fusion, and No adequate bearer estimate will be handled. If an unexpected case motivates a new branch, that branch begins a new test; it cannot retroactively rewrite the one that produced the original result.

Universality makes deferred cases important even when rare. An exact claim about every qualifying stream cannot treat unusual bearers as optional decoration. The project may declare a narrower human reference class while the evidence develops, but it must say that it has narrowed. Later extension to animals, artificial systems, or altered clinical states requires new bridges rather than an assumption that the original result travels.

Bearer cases force the project to expose its criteria, uncertainty, and protections. The canonical statement keeps the bearer it needs without pretending every boundary is settled; research must discover where the definition holds, narrows, or fails.

4.4 — The Hidden Cumulative Object

The cumulative object is hidden in a precise sense: it is a latent construct, not a private meter waiting to be inspected. Reports and measurements may estimate components or consequences through an observation model, but the estimate remains a model-governed estimate rather than the experience itself.

Psychometrics and measurement theory warn that numbers do not become meaningful merely because they are reproducible. The construct, scale, origin, and link between observations and the target require independent justification (Cronbach & Meehl, 1955; Stevens, 1946; Michell, 1999; Borsboom, 2005).

4.4.1 — Latent Means Not Directly Observed

Latent means that the cumulative object is a formal target not directly read from the outside. Reports, behavior, physiology, neural activity, context, and records may constrain an observation model, but none is identical to the target. Latency does not imply supernatural secrecy or a hidden substance that a more powerful scanner will eventually reveal.

Science often reasons about constructs through their relations to observable indicators. Construct validity asks whether different measures behave as a coherent theory predicts, whether the construct generalizes across relevant settings, and whether competing explanations have been challenged. The name of a latent variable does not establish those relations; the indicators and the target must remain distinct (Cronbach & Meehl, 1955; Borsboom, 2005).

Hidden does not mean unknowable. Imagine two sealed parcels that together weigh ten kilograms. A combined weighing leaves many possible splits between them: one and nine, four and six, five and five. Weighing the pair more precisely will not select one split. A separate weighing or an independently justified constraint could.

The same distinction separates a latent construct from an unidentified one. Indirect observations can constrain what we cannot inspect directly; the question is whether they distinguish the answers the theory cares about. If incompatible origins, weights, dimensions, endpoints, or missing-data assumptions still fit, the result should show what is constrained and what remains open. It may be a range, a narrower claim, or no adequate estimate. The next useful measurement is one that separates those alternatives.

The inferential chain contains distinct objects. Lived experience is the domain; the latent cumulative object is a formal representation; an estimand states the quantity a study seeks; an estimator maps observations to an estimate; and a decision rule states what the estimate licenses. Collapsing the chain turns an analytic output into a claim of access the study never earned.

Representation remains branch-specific. The primary additive branch uses one signed object, while bipolar and multidimensional branches preserve different structure. Results do not transfer across branches without a justified bridge; Section 4.1.1 defines the alternatives.

An observation model must say what each channel contributes and how it can fail. Self-report can preserve first-person meaning while remaining vulnerable to language, memory, demand, and response shift. Behavior can reveal choice without uniquely revealing experience. Physiological and neural measures can add independent structure while remaining many steps removed from felt valence. More channels may reduce some errors, but agreement among indicators is not direct access to a life.

The same score can arise from different latent situations. Culture, disability, adaptation, medication, expectation, and the comparison standard used by the respondent can all change how a scale is used. Repeated measures may reveal within-person change while leaving the natural origin or cross-person comparability unresolved. Measurement invariance and response-shift analysis are therefore part of identification, not statistical decoration.

Conflict among channels is evidence, not inconvenience. A report may indicate relief while physiology remains highly activated, or behavior may look calm while later testimony describes intense fear. The observation model should predict when disagreement is expected, preserve uncertainty when it does not, and allow the conflict to count against the proposed representation rather than choosing the channel that best fits the Law.

A latent target becomes scientifically useful only when frozen representations and observation rules generate consequences that serious rivals need not generate. Candidate consequences may concern terminal concentration, variance, perturbation response, or transport across channels. If every mismatch can be repaired

by changing the origin, weights, bearer, endpoint, or model after the result, the construct has been insulated rather than tested.

Latent, then, names a disciplined gap. It separates target from indicator, protects the person from claims of direct access, and places uncertainty on the theory rather than on the person.

4.4.2 — Cumulative Does Not Mean Personal Account

Cumulative means that the represented value depends on the path traversed, not only on the state observed at the endpoint. Sequence, anticipation, memory, adaptation, support, and prior state can shape how an interval is lived. The model therefore needs an independently defended origin and a coherent rule for assigning contributions across time. A questionnaire midpoint, sample mean, or personal baseline may be useful for analysis without becoming the natural origin by convenience.

An additive mathematical form is not a bank account. Scale, origin, path dependence, adaptation, sensitization, memory, and context determine what addition could mean, and a drifting baseline can make deprivation look ordinary. If a stable origin and partition rule do not survive testing, the scalar form must narrow, change, or fail (Stevens, 1946; Michell, 1999).

A week of multimodal self-report data may support compatibility with a declared measured band even when a painful interval is missing and the estimate crosses an operational zero. It does not recover the missing experience, identify a natural latent origin, describe a complete stream, or establish exact terminal closure. The estimate answers only the question encoded by its instrument and model.

No such estimate may become a score of worth, eligibility, surveillance, or rank. Measured burden and relief remain bounded research constructs; the hidden object belongs to the hypothesis about one stream's terminal relation.

4.4.3 — No Meter, Memory, Score, or Archive

The object is neither memory nor archive. Memory reconstructs and selects; records preserve reports, observations, and institutional acts, with omissions shaped by collection rules and hard-to-observe states. Neither remembered vividness nor record density equals experienced magnitude (Little & Rubin, 2019).

This boundary protects people whose lives are poorly recorded. Research may refine estimators and lower-level applications, but it may not let convenient instruments rewrite the public root or let practical usefulness flow backward as proof. The undocumented, disabled, young, cognitively impaired, institutionalized, and privately suffering remain fully real.

A dossier combining medical, behavioral, social, and digital traces would enlarge institutional power without solving identification. Separation, minimization, access control, correction, and deletion remain essential.

4.5 — Exact Closure and Unequal Lives

Exact closure is the sharpest part of the hypothesis and the easiest part to misunderstand. The claim concerns a latent terminal relation across the complete modeled history. It does not concern equal biographies, equal gross experience, or a neutral final mood.

A truth-level zero and a measured equivalence band belong to different layers. The former is the formal target; the latter is a finite-data decision rule that depends on error, uncertainty, and prospective specification (Lakens, 2017; Wellek, 2010; Mayo, 2018).

4.5.1 — Neutrality as Terminal Claim

Neutrality is a property proposed for the completed latent cumulative object, not a description of the final scene. A person might be calm, afraid, sedated, delirious, in pain, asleep, relieved, or unable to report near the end of observation. None of those appearances is identical to the whole-stream residual. A peaceful expression does not prove closure, and visible distress does not by itself establish a terminal violation.

The claim is terminal because it concerns the complete strict-pre-endpoint history. It does not require balance at every moment, in every year, or after every event. The path may wander far from neutrality and cross it many times. What matters on the primary branch is the final signed total immediately before an independently defined truth-level endpoint.

Terminal neutrality requires the independently justified origin described in Subsection 4.2.3, rather than a questionnaire midpoint or a sample average.

Measured compatibility is a different proposition. An estimate and its uncertainty may fall within a protocol-fixed equivalence region around an operational neutral point. That can support compatibility at the declared resolution and under the declared observation model. It does not convert a finite band into exact latent equality or a confidence interval containing zero into proof that the true residual is exactly zero (Schuirmann, 1987; Lakens, 2017; Wellek, 2010).

Net neutrality also differs from gross experience. Imagine one short life with intense joy and intense pain, and another long life with mild variations around ordinary comfort. Their terminal net residuals could match while their durations, total variation, peaks, opportunities, agency, safety, and biographies differ radically. Exact net closure would therefore leave unequal lives unequal in every ordinary sense that still matters.

Ordinary regulatory processes can create apparent return without terminal closure. Adaptation, opponent processes, homeostasis, predictive updating, treatment, social support, memory change, and selection may all move observed measures toward a local baseline. Endpoint-conditioned models can also concentrate paths by construction. These rivals are scientifically valuable because they define what the Law would need to explain beyond familiar recovery or statistical conditioning.

The strength of the claim lies in what it can lose. A nonarbitrary origin may prove unavailable, a complete domain unrecoverable, an endpoint bridge inadequate, serious rivals superior, or a well-adjudicated included stream credibly nonneutral. Terminal keeps the proposal narrow; exact keeps it vulnerable.

4.5.2 — Not Salvation

Terminal closure also leaves spiritual and philosophical meaning open. Religious traditions may speak of salvation, liberation, enlightenment, grace, karmic release, or sacred reconciliation through their own warrants, while secular readers may interpret the question differently or not at all. The formal relation neither confirms nor displaces those meanings; it remains one scientific candidate about a terminal phenomenal boundary.

Nor does the hypothesis promise equal happiness or complete justice. Happiness, peace, purpose, relief, and ordinary well-being vary across people and cultures, while justice concerns rights, capabilities, distributions, institutions, and social cooperation. An exact net boundary could coexist with radical differences in gross experience and biography, so no hidden equality makes unequal lives morally equivalent (Rawls, 1999; Sen, 2009; Nussbaum, 2006).

Meaning belongs freely to the person. No one is required to turn pain into growth, grief into purpose, or uncertainty into faith. Readers may find conditional hope in the possibility that visible unfairness is not the whole formal story, and others may decline that interpretation. Both responses preserve dignity because neither is converted into evidence or obligation.

External goods and public repair therefore retain their full importance. A wheelchair ramp, safe home, medicine, fair wage, apology, or remedy can relieve burden without becoming the formal currency. Courts can fail, governments can neglect, families can harm, and societies can exploit even if terminal closure were later supported. The hypothesis concerns one latent experiential relation; justice concerns responsibility, protection, truth, remedy, and accountable power.

Consolation is a separate human act. The observer may not use the Law to escape the discomfort of another person's pain. Grieving people, survivors, patients, and people living with chronic suffering deserve care, relief, presence, and permission before theory.

Public materials should carry the same freedom. A caption, illustration, classroom passage, talk, or automated answer may describe the hypothesis and its possible significance, but it should offer care rather than certainty when the audience needs comfort. The Law may be studied, doubted, believed, or rejected without being placed on another person's wound.

4.5.3 — Neutrality Under Uncertainty

Endpoint uncertainty and observation loss must remain explicit. If every admissible endpoint is evaluated, the result may be a bounded set; selecting the time nearest zero builds in the answer. Withdrawal, treatment, fatigue, delirium, transfer, device failure, or lost communication require a bridge, sensitivity analysis, or exclusion rather than a smooth invented remainder (Little & Rubin, 2019).

Population means cannot settle an intrastream claim because positive and negative residuals may cancel across people. Adaptation, opponent regulation, predictive control, conditioning, flexible latent models, selection, and instrument limits must receive common-rule comparison; unresolved equivalence becomes model ambiguity, not support.

An uncertainty budget should keep origin, exposure, endpoint, observation, and model error separate because each needs a different repair. Wide intervals call for better measurement, endpoint sensitivity for stronger boundary rules, ambiguity for more discriminating tests, and origin sensitivity for calibration. A narrower or multidimensional successor needs its own name and evidence rather than inheriting victory from the scalar universal form.

4.6 — What the Law Does Not Say

A public root must carry its nonclaims as carefully as its headline. The Law begins from visible unfairness and leaves public justice intact. It neither explains why any particular harm happened nor authorizes another person to interpret that harm as deserved.

The boundary is not an apology for the proposal. It is what prevents a conditional experiential hypothesis from borrowing the authority of ethics, law, religion, medicine, or personal prediction. Each of those domains retains its own standards and owners.

4.6.1 — It Does Not Say Life Looks Fair

The Law begins from the reality that visible life is often radically unequal. Birth, health, family, safety, money, timing, temperament, disability, citizenship, and luck shape the opportunities and burdens people face. A terminal hypothesis cannot make a child's suffering, chronic pain, poverty, trauma, abandonment, humiliation, or unpunished cruelty less real. Those conditions are part of the reason the question matters.

Visible fairness remains a separate object of moral and political action. Rawls's institutional focus and Sen's comparative attention to remediable injustice both show why rights, opportunities, and actual conditions cannot be replaced by a hidden experiential hypothesis (Rawls, 1999; Sen, 2009). Even perfect support for terminal closure would leave the visible distribution of safety, power, health, and opportunity open to criticism and repair.

The proposed possibility is narrower: appearance may not reveal the complete latent felt history of a stream. That sentence preserves visible injustice as morally urgent while asking whether another boundary property exists. Public systems still owe protection and repair; caregivers still respond; researchers still measure ethically. The hypothesis expands the question without turning visible harm into illusion.

Present pain makes a present claim. Danger, grief, illness, fear, hunger, loneliness, and despair call for help before formal interpretation. Relief is never interference with a hidden balance: treatment, comfort, shelter, access, apology, protection, and repair contribute to a better life in their own right. The first public application of the framework is therefore not belief but the refusal to let theory stand between suffering and care.

4.6.2 — It Does Not Say Suffering Is Deserved

The formal ledger is not a moral ledger. Terminal closure assigns no guilt, merit, purification, punishment, or reward; illness, abuse, poverty, disability, grief, and relief do not reveal character or hidden desert. Institutions and relationships should answer actual suffering with care, protection, truth, and repair, not with a story about what a person earned (Beauchamp & Childress, 2019).

Nor does suffering have to teach. Meaning may arise through faith, art, love, protest, memory, repair, or private reflection, or may be refused. PTV C concerns a terminal structure; it does not own the interpretation of a wound.

4.6.3 — Closure Does Not Settle Justice or Authority

Closure would not settle justice or create authority. Rights, duties, power, remedy, and distribution remain separate public questions, and no court, government, employer, insurer, school, religious body, or automated system may treat a speculative inner total as an administrative field. Institutional authority must come from law, consent, professional standards, democratic legitimacy, and domain-specific evidence, not from the hypothesis (Rawls, 1999; Nussbaum, 2006; Sen, 2009).

Institutions may still use the framework's questions to notice burden and improve procedures, but practical success or failure in those domains does not automatically validate or refute PTV.

4.6.4 — It Does Not Predict Your Future or Require Belief

The Law does not forecast the sequence of anyone's future pleasures and pains. Even an exact terminal boundary, if real, would leave countless paths to the same residual. It contains no schedule pairing hardship with reward, no happiness budget, and no rule telling a reader when relief must arrive.

Effort can improve skill, opportunity, and the probability of success, but no boundary condition guarantees a particular career, relationship, recovery, artistic breakthrough, or recognition. Individual forecasting would require a validated model, an appropriate reference class, relevant observations, and calibrated uncertainty. A universal terminal hypothesis is not a bedside calculator, oracle, or contract with reality (Mayo, 2018).

Private attempts to force tests add danger without identifying the stream, endpoint, or residual. Increasing pain, refusing treatment,

or chasing extremes changes the life without producing valid controls; care and qualified help remain the guides.

No institution may condition belonging, assistance, employment, study, care, civic standing, or spiritual respect on assent. Families, schools, researchers, communities, donors, and public bodies may discuss the hypothesis without turning belief into eligibility.

Popularity is not validation and rejection is not refutation. Funding, controversy, or prestige can shift attention without changing evidence. Criticism belongs at the sentence, source, method, or artifact, to improve the work rather than become a loyalty test (Fischhoff, 2013; Scheufele, 2014).

Practical decisions stay in their domains: clinical facts and care, resources and risk, consent and conduct, educational need, or voluntary spiritual counsel. The hypothesis may inform questions but cannot replace those authorities (Fischhoff, 2013).

Chapter 5 — A Stream's Beginning and End

Every terminal hypothesis needs a beginning and an end. These boundaries turn a moving history into a complete object that can be judged. Their value lies in discipline: a boundary fixed independently can expose the proposal to evidence, while a boundary chosen from the result would only restate the answer.

Ordinary language supplies many beginnings and endings: conception, birth, wakefulness, first memory, legal personhood, brain development, loss of responsiveness, circulatory death, neurologic death, last observation, and the end of a conscious stream. Each category serves a different human or scientific purpose. The Chapter's task is to keep those purposes clear and show where a formal bridge is still needed.

The formulation developed here uses a truth-level endpoint for the maximal qualifying stream. A study may rely on a protocol-fixed observable proxy and a governed adjudication process, but the proxy is not automatically the truth. Keeping those levels separate protects the hypothesis from turning a practical measurement rule into an ontological decree.

Clinical standards and ordinary sleep illuminate opposite sides of the boundary problem. Neurologic death criteria use structured prerequisites, examinations, and ancillary procedures for a medical determination, while sleep demonstrates that reduced responsiveness can coexist with conscious experience. Both examples teach the same constructive lesson: use each category for the work it has earned, and do not promote behavioral quiet or a clinical rule into a truth-level stream boundary without a defended bridge (Greer et al., 2020; Greer et al., 2023; Siclari et al., 2017; Tononi et al., 2024).

Disorders of consciousness make repeated, standardized, multimodal assessment indispensable. Diagnostic error has been documented when one observation is treated as final, and practice guidelines emphasize reassessment, confounder control, and converging evidence (Schnakers et al., 2009; Giacino et al., 2018; Kondziella et al., 2020).

Covert command following adds an especially important contribution. Functional imaging and electroencephalography have revealed task-responsive activity in some patients who show no reliable behavioral response. The finding does not identify the Law's endpoint, but it demonstrates why behavior, neural response, signal quality, confounders, and later recovery must remain distinct channels (Owen et al., 2006; Monti et al., 2010; Claassen et al., 2019; Bodien et al., 2024).

Endpoint independence is the central rule. The cumulative estimate may not decide when the stream ended. Follow-up, case inclusion, recovery rules, and continuation branches must be specified before terminal outcomes are examined so that an unfavorable case remains capable of being included.

Truth-level exactness and empirical bands remain separate throughout. A finite estimate is interpreted through uncertainty, missingness, an endpoint model, and a protocol-fixed decision rule. The band provides a disciplined language for finite evidence; it does not turn an approximation into the exact metaphysical target (Mayo, 2018; Lakens, 2017; Wellek, 2010).

Observation may end before the stream does. Sedation, device removal, transfer, attrition, family burden, clinical deterioration,

and loss of consent can all truncate data. Missingness can depend on the very state being studied, which means more participants alone do not remove the bias (Little & Rubin, 2019).

Ethics gives endpoint discipline its human foundation. No one may be harmed, pressured, kept in treatment, denied relief, or encouraged toward danger to improve a test. Care decisions follow clinical need, lawful authority, consent, and established research ethics, which is precisely what makes any resulting knowledge worthy of trust (World Medical Association, 2024; Beauchamp & Childress, 2019).

We begin with the question that precedes every terminal claim: what must be true for a conscious stream to have begun at all?

5.1 — When a Stream Begins

A terminal claim requires a complete domain, and a complete domain requires a beginning as well as an end. Yet the first moment of a conscious stream is not automatically identical to birth, conception, legal recognition, first memory, or the first measurement available to science.

The same layering applies at the ending boundary: clinical and legal determinations remain indispensable without automatically identifying the truth-level end of a conscious stream.

5.1.1 — Why Endpoints Matter

A complete stream has two boundaries. The cumulative domain cannot begin before a qualifying conscious stream exists, and it cannot close before that stream has irreversibly ended on the declared branch. The Law is therefore not only a claim about a terminal value; it is also a claim about which experiences belong to the stream and which moments lie outside it.

Evidence about that beginning may be gradual and uncertain. Development can present changing capacities, incomplete access, and disagreement about what signs are sufficient. The Law can require a prospectively stated bearer rule without pretending that one public event settles every developmental case. When the available evidence cannot identify the beginning responsibly, the reference class must narrow or the case must remain unresolved.

Ownership and maximality begin at this boundary as well. Once a qualifying stream is present, its early experience cannot be reassigned to a parent, family, institution, or later narrative

because the evidence is inconvenient. The formal object belongs to the stream that experiences it, even when that stream cannot report or remember the beginning.

A pause does not necessarily create a new beginning. Sleep, anesthesia, seizure, coma, and fluctuating states may interrupt access or unity without ending the stream. Whether a later return continues the same stream or begins another is a separate identity question, and the answer cannot be chosen to reset the cumulative object.

The missing prefix matters because later data cannot reprice the unobserved past. An instrument can improve estimates from the moment it is used, but it cannot reconstruct every childhood, dream, period of delirium, or earlier feeling by precision alone. A study that enrolls adults and observes only recent years therefore needs a defensible bridge for the unobserved prefix or must narrow its claim to the observed domain.

Independence is the honesty condition. The endpoint rule must not inspect whether the cumulative estimate appears neutral, and the beginning rule must not be chosen to exclude inconvenient experience. If an unfavorable residual can be moved outside the domain by redefining the start or end after the path is known, the theory survives by editing its own question.

The burden of boundary uncertainty belongs to the claimant. Patients, families, disabled people, infants, nonhuman candidates, and future systems do not have to prove a stream in order to deserve care or protection. A formal bearer threshold is not a scale

of moral worth, a rights test, or a license to reduce concern where science is uncertain.

A protocol should therefore allow several outcomes: usable boundary, uncertain boundary, not interpretable, ethically unavailable, or outside the declared reference class. These are scientific results, not administrative debris. They reveal whether the proposed bridge from consciousness evidence to a complete stream is tractable, sufficiently identified, and ethically acceptable.

Beginnings and endings create different missing-history problems. At the beginning, experience may precede the first reliable report, memory, or measurement, leaving an unobserved prefix. At the end, observation may stop while experience continues, leaving an unobserved suffix; in time-to-endpoint analysis, this is right censoring. A complete-domain claim must face both. Counting only the observable middle can produce a precise estimate of an incomplete stream.

Endpoints matter because the word every is otherwise protected from contact with reality. A universal terminal claim needs a domain that cannot move with the result and a record of the cases it cannot adjudicate.

5.1.2 — Legal Death Is Not Automatically Death of Mind

Legal death is a public status with consequences for rights, duties, records, succession, disposition, and authorized decision-making. Clinical determination of death is governed by professional and jurisdictional standards that society cannot leave to private interpretation. The truth-level endpoint in the formulation

developed here serves a different purpose: it marks the end of the relevant conscious stream for a scientific hypothesis.

Death by neurologic criteria is not declared from one sign or one intuition. Contemporary guidance requires prerequisites, exclusion of confounders, a structured neurologic examination, apnea testing, and ancillary testing when the clinical evaluation cannot be completed or interpreted as required (Greer et al., 2020; Greer et al., 2023). That rigor should increase respect for the boundary between medical determination and speculative formal language.

Determination by circulatory criteria and determination by neurologic criteria also answer related but not identical clinical questions through different pathways. A formal consciousness endpoint should not flatten those pathways into a generic death marker. Research must identify which event was recorded, under which authority, with what timing uncertainty, and for what inferential purpose.

Confounders matter because apparent absence can be produced by reversible conditions. Sedatives, neuromuscular blockade, metabolic disturbance, temperature, intoxication, shock, seizure, and technical failure can alter examination and monitoring. Clinical standards address these concerns for clinical determination; a research protocol must not assume that a secondary dataset preserved every condition needed for a consciousness inference.

A legally and clinically established death may provide a strong practical outer boundary for some research questions, but it is not

direct access to the last phenomenal moment. The exact relation between a public determination and the truth-level end of a stream may remain unobserved. A study must state what the recorded event can support rather than treating administrative finality as phenomenological certainty.

The gap may be small, large, or unknowable depending on the case and the branch being studied. A protocol can use a conservative interval, a declared terminal window, or a no-estimate outcome. What it cannot do is place the final felt contribution at the time most convenient for the terminal residual.

Family and caregiver testimony can help interpret ordinary communication, preferences, and changes from baseline. That testimony deserves respect without becoming endpoint sovereignty. Love, hope, exhaustion, guilt, culture, and grief can shape interpretation, so the record should preserve both the value and the limits of relational evidence.

Loss of speech, movement, command following, memory, or ordinary responsiveness is not sufficient by itself. Sedation, paralysis, acute brain injury, severe motor impairment, and disorders of consciousness can separate outward behavior from evidence of awareness. Standardized serial assessment and converging methods reduce error, yet they do not turn one channel into an infallible consciousness test (Giacino et al., 2018; Claassen et al., 2019; Bodien et al., 2024).

A research protocol must therefore declare the role of each biological and clinical variable. A measure may serve as context, proxy, confounder, competing event, or part of endpoint

adjudication. Treating every residual signal as mind would be as careless as treating every low signal as its absence. The protocol must carry that uncertainty rather than transferring it to patients or families.

Research access should follow care rather than reorganize it. Monitoring can be removed because it causes discomfort, treatment can alter the variables under study, and a transfer can end data collection. A cleaner terminal record is never a reason to withhold an indicated intervention or extend burdensome observation.

This separation protects both sides of the boundary. It prevents premature claims that a possible stream has ended, and it prevents indefinite postponement from making the terminal hypothesis immune to failure.

5.1.3 — Unknown Cases Remain Unknown

Unknown is a legitimate scientific status. It applies when available evidence does not responsibly settle whether the relevant stream began, continued, paused, returned, fragmented, or ended. The Law is not entitled to convert uncertainty into the result it prefers.

Unknown is not support. A case without a defensible bearer or endpoint contributes no positive evidence that the terminal residual is neutral. It may still reveal limits in feasibility, observation, proxy design, or recruitment. Unknown is also not defeat by itself when the target was never identified well enough to test.

Several neighboring statuses should remain distinct. Not interpretable means the record cannot bear the intended inference because of signal failure, confounding, timing, preprocessing, or an invalid observation model. Ethically unavailable means the information should not be collected, linked, retained, or analyzed because doing so would interfere with care, exceed consent, exploit dependency, or violate privacy. Outside the declared class means the case was prospectively excluded for a stated reason rather than removed after the outcome became known.

Clinical practice in disorders of consciousness shows why revision and serial assessment matter. Responsiveness, awareness, diagnosis, and prognosis can diverge, and later evidence may change an earlier classification. The appropriate lesson for this project is not that clinical categories determine the formal endpoint, but that difficult states should remain open to correction (Giacino et al., 2018; Kondziella et al., 2020).

A high rate of unknown outcomes can pressure the research program. It may show that the endpoint proxy is fragile, the intended population too heterogeneous, the observation window unrealistic, or humane data collection unable to answer the question acceptably. Redesign, reference-class narrowing, or suspension may be the scientifically honest response.

A reference class narrowed for legitimate reasons should be named with equal prominence. Results from well-instrumented acute-care adults cannot silently become results about infants, dementia, nonhuman animals, artificial systems, or every possible conscious bearer. Unknown cases mark the study's reach.

The route into the unknown category also matters. Missingness after emergency transfer differs from selective exclusion of difficult cases; a failed sensor differs from unresolved confounding; ethical nonuse differs from absent monitoring. These mechanisms can leave a deceptively clean group of well-instrumented cases. Denominator flow should therefore report usable, unknown, uninterpretable, ethically unavailable, returned, and excluded cases rather than hiding them behind one final sample size.

Retrospective family reports, medical records, developmental milestones, and behavioral observations can constrain possibilities, but each relates differently to experience. The record should state whether the evidence concerns arousal, responsiveness, learning, memory, neural organization, or social recognition. Combining these clues may narrow the onset window without turning any one into a consciousness switch.

Unknown status remains compatible with protection. Uncertainty about the first qualifying experience does not authorize avoidable harm to infants, disabled people, animals, organoids, or artificial candidates. Risk and vulnerability justify precaution and welfare while the formal bearer question remains unsettled. The ethical floor need not await a perfect onset clock.

5.2 — Pauses, Returns, and Low-Unity States

Continuity cannot be inferred from a single outward state. Sleep, anesthesia, coma, and severe brain injury can reduce or abolish ordinary responsiveness while differing profoundly in experience, mechanism, prognosis, and reversibility.

5.2.1 — Ordinary Sleep

Ordinary sleep is the most familiar reason to distinguish interruption from endpoint. A sleeping person withdraws from ordinary public interaction and later returns; the stream has not ended simply because report and response changed. Sleep also introduces a more subtle problem: some states may contain vivid experience while others yield little report, so biological time cannot be added to a phenomenal history without an exposure and unity rule.

Sleep is not one blank state. Dream reports and neural studies indicate that conscious experience can occur during rapid-eye-movement and non-rapid-eye-movement sleep, with changing content and level across the night (Siclari et al., 2017; Tononi et al., 2024). This makes sleep an everyday demonstration that reduced responsiveness is an uncertain proxy for absence of experience.

A terminal proxy must therefore distinguish reversible low access from irreversible completion. Neural complexity, responsiveness, autonomic activity, memory, and report change across sleep stages, and a useful rule must not classify ordinary sleep as an endpoint. The lesson is limited and exact: sleep supplies a pause or altered-state branch, not a truth-level ending.

Sleep also clarifies what later unity gates are for. They do not prove which moments contain experience; they express the need to decide, prospectively and fallibly, how state-dependent exposure enters the model. Dream valence, affective processing, recovery, and physiology may inform measurement studies, while remaining indirect evidence rather than access to a complete ledger.

Dreaming gives the lesson human texture. Fear, relief, shame, joy, grief, and peace can occur while the body lies still, which shows why public quiet cannot settle private valence. A waking report is reconstructive and shaped by memory, language, culture, medication, timing, and narrative habit, so dreams can contribute to the evidence architecture without becoming proof of terminal closure.

The privacy of dreams matters as much as their scientific interest. Dream research must be consent-based, optional, purpose-limited, and stoppable when distress rises. The Law may recognize dreams as possible felt experience, but no one owes a private dream life to public explanation or research participation.

5.2.2 — Coma and Recovery

Coma is a clinical state with many causes, courses, and outcomes, not a synonym for the truth-level end of a conscious stream. Some patients recover wakefulness or awareness, some transition into other disorders of consciousness, some die, and some remain difficult to classify. Recovery therefore functions as a genuine test of an endpoint proxy: a rule that called the earlier state final has

failed for that case unless it was prospectively defined as a nonterminal low-unity crossing.

Recovery can invalidate a premature endpoint classification even when bedside command following was absent; Chapter 13 treats the bearer evidence in detail (Schnakers et al., 2009; Claassen et al., 2019; Bodien et al., 2024).

Coma and anesthesia teach a shared but bounded lesson: outward stillness, absent report, and impaired memory do not by themselves establish absence of experience or irreversible end. General anesthesia is designed to be reversible, while drugs, paralysis, metabolic state, temperature, and delayed emergence can affect several channels at once. Those factors belong in prospectively specified confounder and return rules rather than in an improvised terminal classification.

Clinical care governs the scene. Anesthetic decisions, rescue, monitoring, analgesia, ventilation, and recovery assessment exist for the patient. Research may use clinically generated data only under proper approval and without changing care to improve the record. For the formal model, anesthesia contributes a reversible-pause category and a return rule: if unified access returns, the earlier event cannot remain classified as the irreversible endpoint.

Recovery may be delayed by neurological injury, sedatives, metabolic disturbance, hypothermia, infection, seizures, or other confounders. Endpoint rules therefore need appropriate follow-up logic without declaring one universal clinical duration. Active resuscitation is a competing event, and sudden death may leave no usable preterminal record at all. A case can be clinically clear

while remaining scientifically uninformative for the terminal-ledger question; that distinction protects both medicine and the hypothesis.

Loss of autobiographical continuity does not establish loss of experience. A person may no longer narrate a stable self and still inhabit the present. Dementia and delirium can fluctuate, interact with illness and medication, and produce thresholds that cross and recross. Persistence, hysteresis, confounder control, and longitudinal evidence are therefore more informative than a convenient first crossing.

Brief returns of recognition, communication, attention, or meaningful response can challenge an irreversible endpoint classification even after prolonged impairment. They do not settle the whole consciousness question, and formal uncertainty never reduces personhood, care, privacy, or respect. When medication, infection, artifact, missing data, and uncertain unity prevent a responsible terminal estimate, scientific nonuse is more accurate than a clean but false boundary.

5.2.3 — Reversibility as the Key Distinction

Reversibility separates a pause from an endpoint asymmetrically. If reliable unified access returns and the predeclared routing rule identifies continuity of the same stream, the earlier state was not that stream's truth-level endpoint, although this may have been unknown at the time. A protocol must prospectively define signs of return and their effect on classification, using reliable communication, command following, validated covert command

following, or another domain-appropriate marker (Brown et al., 2010).

Signal quality, persistence, and return rules protect against false stopping. Artifact, device loss, preprocessing failure, missing channels, brief fluctuations, or rescue interventions cannot define a terminal event; the proxy must meet protocol-fixed duration and evidence criteria, and any return counts against it. This makes the rule a research safeguard rather than a prognosis or legal definition of death.

Reversibility does not directly detect consciousness. Reliable communication, command following, or other convergent return can show that an apparent endpoint was premature, whereas failure to return cannot prove absence of experience or identify the exact moment a stream ended. Recovery can therefore defeat an irreversible-ending claim more readily than silence can confirm one (Giacino et al., 2018; Bodien et al., 2024).

The protocol endpoint and truth endpoint remain separate. Contact loss, excessive burden, withdrawal, or unreliable observation may properly end a study without proving that the stream ended. The result may be censoring, a branch of possibilities, or No adequate estimate rather than a convenient terminal value.

5.3 — The Endpoint Must Be Independent

An endpoint rule must be fixed without using the cumulative result to choose the boundary. That is the kind of independence required here. It differs from statistical independence between when a stream ends and what happens within it. A rule that stops filming a ball when it first touches the ground can be fixed before the throw, even though its stopping time depends on the ball's motion. In the same way, a prospectively specified endpoint may depend on an independently justified biological or consciousness state. Chapter 7 examines what those dependencies would need to explain.

Flexible stopping, exclusions, and analyses can make noisy patterns appear persuasive, which is why prospective specification, transparent deviations, and severe testing matter across science (Simmons et al., 2011; Nosek et al., 2018; Mayo, 2018). A universal terminal claim must accept at least the same discipline.

5.3.1 — No Endpoint After Cumulative Object Inspection

Picture a line wandering above and below zero. If the analyst may stop at any crossing and call that time the endpoint, closure has been manufactured: the answer has painted its own finish line. A real test fixes the stopping rule before the line is revealed, then accepts every eligible result under that same rule. Inclusion, exclusion, continuation, and unusable-status decisions must obey the same discipline.

Humanly vivid scenes require the same discipline. A late smile, peaceful bedside appearance, final conversation, or distressed

ending may matter deeply without defining the formal endpoint or the meaning of an entire life. Emotionally charged cases call for stronger protection against narrative selection, not permission to let the most satisfying scene choose the boundary.

Operational separation requires advance rules for endpoint channels, preprocessing, signal quality, persistence, anti-chatter logic, adjudication, and return. Blinding helps, but shared sensors, physiological drivers, or preprocessing can still entangle the endpoint proxy with the cumulative proxy. If outcome information shaped the boundary, the result becomes exploratory or unusable for the confirmatory claim.

Return, competing events, and missingness also need prospective treatment. Recovery cannot be dismissed after it destabilizes an endpoint; resuscitation, medication changes, seizures, transfer, device loss, and absent channels cannot be coded automatically as terminality or neutrality. The adjudication menu should include endpoint, non-endpoint, indeterminate, protocol-excluded, not interpretable, and ethically unavailable outcomes.

The record should preserve why a classification was reached, which confounders remained, whether return rules were triggered, and what information each adjudicator could see. Plural expertise may be necessary, but cumulative outcome information should not enter merely because consensus is difficult. Role separation, access control, time-stamped decisions, and a concise audit trail make independence inspectable while protecting clinical privacy.

Independent adjudication gives negative findings a fair place to stand. A non-neutral case must remain eligible when it satisfies the same endpoint and adequacy rules as a neutral-looking case. The endpoint firewall is strongest when a skeptical reviewer can reconstruct the decision without finding the desired conclusion hidden inside it.

5.3.2 — Ambiguity Is Not Support

Ambiguity identifies an unanswered layer. If the bearer, origin, truth endpoint, observation process, measured-to-latent bridge, or decision rule cannot be established, the terminal proposition has not been adjudicated. The unresolved bridge may become the next research priority, but it contributes no favorable vote merely because contradiction was impossible.

Ambiguity can occur before any measurement is made. The concept of neutrality may lack an independently justified origin, or the proposed stream may not have a defensible routing rule. In such cases, collecting more observations of the same kind cannot solve the central problem. Precision about an unidentified object merely produces a narrower estimate of something whose relation to the target is uncertain (Cronbach & Meehl, 1955; Borsboom, 2005).

Other ambiguity arises from observation. Reports may be unavailable, physiology may be altered by medication, neural measures may be noisy, and behavior may be constrained by motor impairment. The channels can disagree because each measures a different consequence of the hidden state and fails in

a different way. A multimodal record helps only when the links among channels and target are specified rather than assumed.

Severe testing asks whether a procedure had a real opportunity to reveal error if the claim were false (Mayo, 2018). A case with no defensible endpoint or recoverable terminal estimate may expose a weakness in the design without severely testing exact closure. Recording that limit preserves value and prevents an unperformed test from being described as success.

Terminal research is especially vulnerable to selective loss. Fatigue, delirium, medication, transfer, communication failure, withdrawal, device failure, and death can determine which records remain available. Missingness is part of the process being studied, and the last usable value cannot automatically stand for the truth endpoint (Little & Rubin, 2019).

Adjudicators can also disagree for principled reasons. One may judge a confounder unresolved; another may accept the endpoint. The record should preserve the evidence, the rule applied, the disagreement, and the effect of alternative classifications. Consensus reached only after seeing which choice favors neutrality is not independent adjudication.

Branch analysis can make uncertainty visible without letting it choose the answer. Investigators may evaluate every endpoint window, routing classification, neutral anchor, and missingness model allowed by the protocol while keeping the decision rule fixed. A conclusion that survives the full plausible set is robust; one that appears only on a favorable branch is branch-dependent.

Partial identification can replace a forced point estimate. Data may justify a set of possible terminal values. If the entire set lies away from neutrality after every prerequisite passes, the case can pressure the strong claim. If neutral and non-neutral values both remain possible, the result stays unresolved; the width of the set is information about the bridge, not permission to select its most reassuring point.

The pattern of ambiguity is itself diagnostic. If uncertainty clusters where residuals would be least favorable, the method may be selecting, censoring, or failing at precisely the critical cases. Comparing ambiguity across predicted residuals, clinical pathways, sites, medications, and observation quality can reveal more about the design than the classified cases alone.

Prospective adjudication rules should state which evidence has priority, how channel conflict is handled, and when disagreement forces No adequate estimate. One reviewer may emphasize motor nonresponse, another imaging, another sedation, and another later recovery; the rule must be visible before anyone sees the cumulative result (Giacino et al., 2018; Greer et al., 2023).

Rivals can benefit from ambiguity as readily as PTVC can. Endpoint-conditioned sampling, selection on observability, floor effects, and flexible latent models may all concentrate the cases that remain classifiable. Equal-budget rival comparison must therefore include the ambiguous and missing cases rather than score only the cleanest survivors.

Abstention changes nothing about the person's dignity or care. An unavailable terminal estimate is not a defect in the participant,

family, or clinical team, and it does not license reassurance that balance occurred beyond measurement. The honest conclusion is narrower: this case did not answer the declared question. Preserving that answer is one of the strongest protections for falsifiability.

5.3.3 — Endpoint Discipline Protects Falsifiability

Endpoint discipline fixes the finish line before the desired result is known. Prospective design, transparent exclusions, and reproducible analysis matter because theories learn only when unfavorable results remain visible (Nosek et al., 2018; Munafò et al., 2017). Once the boundary and bridge survive those checks, the terminal estimate can still miss the protocol-fixed band, fail a route prediction, or be explained better by a serious rival.

The stopping rule may use prospectively specified unity-related observations available up to the candidate event, but it may not depend on future cumulative behavior or retrospective knowledge that the estimate later approached zero. The protocol identifies the candidate event; the estimator reports what the observations support; and governed adjudication records the endpoint classification used for analysis. None may be selected because the cumulative estimate is convenient.

The endpoint firewall has operational content: signal quality, persistence, hysteresis, return, competing events, independent adjudication, and traceable revision. Science can revise a rule when instruments or clinical knowledge improve, but timing determines status. A threshold, persistence window, return rule, or exclusion changed after outcome inspection cannot

retroactively govern the confirmatory result. The prior rule remains visible, and the successor becomes a new prospectively specified hypothesis.

Exclusion drift is equally serious. A case cannot become out of scope merely because it challenges neutrality unless the bearer or endpoint rule already excluded it. Across sites, different endpoint definitions also require harmonization or planned heterogeneity analysis; local medical authority does not automatically create one common research endpoint. When material drift occurs, the study may remain exploratory, but its earlier public summary and evidential status must be corrected rather than silently preserved.

A proxy can fail before the terminal cumulative object is considered. It may cross too often in reversible states, lose signal quality, produce unclassifiable cases, or yield too few ethically usable endpoints for the program to be practicable. Those outcomes weaken the endpoint architecture and can require narrowing or suspension. They cannot be converted into support by saying that the true boundary is merely hidden. A failed proxy can still improve the program by identifying which observable cannot carry the endpoint burden under the declared conditions.

Prospective commitment is ethical as well as scientific. Participants and families should not carry research burden for a test whose rules can be rewritten after their data arrive. Sensitivity analyses may compare defensible endpoint rules, but they cannot authorize choosing the version nearest neutrality. Simulation and model recovery should expose reversible crossings, shared confounders, abrupt dropout, and genuine non-neutral endings before a vulnerable cohort carries those risks. Pre-simulation can

also reveal whether the planned rule would systematically favor easily observed cases over difficult ones, allowing the design to correct that asymmetry before enrollment.

The endpoint record should preserve the protocol-fixed rule, the event that triggered review, the evidence available at that time, any later return or competing event, the reason for revision, and the effect on the analysis. It is not a public dossier and should contain only what scientific accountability requires. Its purpose is to let later reviewers determine whether the boundary was governed prospectively or moved after the terminal result became visible (Nosek et al., 2018). A versioned record should keep the superseded rule and its successor linked so replication can reproduce either analysis without silently changing the finish line.

5.4 — Ambiguous, Branching, and Revised Endpoints

Clinical and scientific classifications sometimes change because better evidence reveals that the earlier category was too coarse. Revision is not a defect when the old and new rules are explicit and the evidence remains traceable.

Clinical ambiguity becomes scientifically important when behavioral, imaging, electrophysiological, and later-recovery evidence disagree; the question is how those channels should be adjudicated without making any one of them sovereign.

5.4.1 — Clinical Ambiguity

Clinical ambiguity exists when the available evidence does not responsibly establish whether unified conscious experience is present, absent, recovering, fluctuating, or irreversibly ended. The uncertainty may arise from the condition, from treatment, from the limits of observation, or from disagreement among methods that are each valid for narrower purposes.

The first discipline is jurisdiction. Clinical categories guide diagnosis, treatment, prognosis, communication, and lawful determination of death. The truth-level endpoint belongs to a formal hypothesis. A protocol may use clinical information as part of an endpoint bridge, but the formal vocabulary cannot overrule a bedside team, reinterpret a chart, or create a second route for declaring death.

Endpoint adjudication should therefore rely on standardized serial assessment and confounder-aware review rather than an

unstructured impression (Schnakers et al., 2009; Giacino et al., 2018; Kondziella et al., 2020).

The confounder rule in Section 5.1.2 remains active here. Clinical ambiguity must also account for analgesia, infection, organ failure, sensory impairment, and language barriers, because a weak or absent response may reflect the channel rather than the absence of a stream.

Cognitive motor dissociation is one reason behavioral response, neural evidence, report, and later recovery cannot be treated as interchangeable channels (Claassen et al., 2019; Bodien et al., 2024).

No modality acquires sovereignty by detecting what another misses. Neural responses require task comprehension, signal quality, preprocessing discipline, control of artifacts, and careful interpretation. Bedside behavior remains valuable but channel-dependent. Family testimony may reveal changes from baseline and ordinary communication patterns, yet it is shaped by intimacy, hope, fatigue, grief, and cultural expectations. Convergence can strengthen a case without turning any channel into direct access.

Fluctuation requires time, not a snapshot. Arousal can vary across hours and days; medication can change; seizures can interrupt access; delirium can wax and wane; recovery can reveal capacities that were not available at an earlier examination. A protocol should specify the number, timing, and independence of assessments and should state what later return does to an earlier classification.

Active resuscitation is especially incompatible with casual endpoint inference. The clinical purpose is to restore circulation, breathing, and recoverable function. Research may not alter that effort, delay treatment, or interpret provisional signals as terminality while urgent care continues. If the observation window closes because care takes priority, the scientific result may be missing or uninterpretable. That is the correct allocation of burden.

Sudden death presents a different limitation. The clinical event may be clear while no prospective unity record or complete cumulative proxy exists. The truth-level endpoint may have occurred, but its timing and the terminal residual may remain inaccessible. A case without the required antecedent record does not become evidence merely because the ending was well documented.

Clinical ambiguity supplies no positive evidence for closure. It can reveal where a proxy fails, which populations are not transportable, and what a safer protocol would need. It may also show that humane observation cannot identify the truth-level object at the required resolution. That is a limitation on the research program, not a defect in the person or a reason to press harder.

A multidisciplinary adjudication panel may include neurology, critical care, rehabilitation, ethics, measurement, and statistics, with additional expertise when the protocol requires it. The panel's role is not to vote consciousness into existence. It is to apply declared criteria, record disagreement, protect blinding

where possible, and state when the available evidence does not justify the requested inference.

The governing asymmetry is simple: uncertainty stays with the study. It does not justify invasive measurement, reduced analgesia, altered resuscitation, pressure on surrogates, or a family story about balance. Care and dignity remain immediate.

5.4.2 — Dissociative Ambiguity

Dissociation is a clinical and experiential domain, not a dramatic metaphor for theoretical uncertainty. The American Psychiatric Association describes dissociative disorders as involving disruptions in memory, identity, emotion, perception, behavior, and sense of self (American Psychiatric Association, 2022). Those disruptions can affect how experience is accessed and narrated without automatically establishing that conscious experience ended or that several formal streams exist.

The topic requires anti-stigma discipline. Dissociative symptoms and disorders are often associated with trauma, and the literature has had to develop amid controversy, diagnostic disagreement, and public sensationalism (Dorahy et al., 2014). A person is not an edge case before being a person. The model must carry the uncertainty rather than asking trauma survivors to carry the burden of making the theory interesting.

Memory cannot be the sole stream criterion. Experience may occur without later autobiographical recall, and a later narrative may reorganize events without creating or erasing the experience that was lived. If every memory gap were treated as a new stream, ordinary sleep, anesthesia, trauma, childhood amnesia, and

neurological impairment would fragment bearers by report rather than by a defensible unity rule.

Identity language also needs care. Distinct identity states or discontinuities in agency may pressure a model of access, ownership, and routing, but they do not by themselves supply a formal count of conscious streams. The relevant questions concern simultaneous or sequential experience, access relations, continuity, mutual influence, and whether ownership can be assigned prospectively without erasing any lived interval.

The project may therefore prospectively specify more than one plausible routing model for a declared class. One may treat changing states as modes of one continuing stream; another may remain agnostic where evidence cannot distinguish alternatives. What is forbidden is retrospective partitioning that excludes an unfavorable interval or creates a compensating child stream after the terminal result is seen.

Clinical expertise and first-person testimony both matter, but neither is transformed into an infallible endpoint test. Testimony provides indispensable access to lived patterns and should be received without spectacle or coercion. Diagnosis and treatment require trained professionals and context. The formal research program adds no authority for family members, platforms, employers, courts, or artificial systems to assign a person's stream structure.

Dissociative ambiguity must also be distinguished from cognitive motor dissociation. The latter term is used in disorders-of-consciousness research for patients who are behaviorally

unresponsive to commands yet show task-responsive brain activity on functional imaging or electroencephalography (Claassen et al., 2019; Bodien et al., 2024). Similar words do not make psychiatric dissociation and motor-output dissociation the same phenomenon.

No one should be asked to produce a definitive metaphysical interpretation of their own diagnosis. A participant may describe discontinuity, co-consciousness, amnesia, conflict, or continuity in language that is true to their experience without settling the formal ontology. The study's job is to record the evidence and uncertainty, not to recruit testimony as proof of a favored model.

Language and culture shape how dissociative experience is recognized and described. Terms of identity, possession, detachment, memory, and agency do not travel unchanged across communities. A measurement instrument validated in one setting may misclassify another. Translation, cultural consultation, and measurement-invariance review are therefore scientific requirements, not decorative sensitivity.

Privacy is unusually important because dissociative histories may contain trauma, abuse, identity information, and material that could create danger if disclosed. Data should be purpose-limited, separated from employment, legal, educational, platform, and family decision systems, and protected against reconstruction from combined traces. No research dataset becomes a public map of a person's states.

Review should include clinicians with relevant expertise and, where appropriate, people with lived experience who can identify

stigmatizing language and unrealistic procedures. Lived-experience review does not settle formal ontology, just as clinical review does not settle metaphysics. Together they can reveal whether the study is asking a coherent question without converting participants into examples for a theory.

Dissociative ambiguity therefore tests whether the Law can remain exact without becoming cruel or theatrical. It requires prospective routing rules, expert review, trauma-informed methods, and language that cannot be converted into a personal classification tool.

5.4.3 — Unknown as Honest Category

Use “unknown” in the sense defined in Subsection 5.1.3: an unresolved case does not support closure.

Many endpoint problems are interval problems rather than point problems. Evidence may support the conclusion that a stream ended sometime between two events without identifying one privileged instant. In that case the analysis should carry the interval forward. If the terminal conclusion changes across the interval, the result is endpoint-sensitive. If every defensible endpoint inside the interval yields a terminal set away from neutrality after the other gates pass, uncertainty about the exact instant does not rescue the universal claim. Partial identification is more informative than an invented time stamp (Little & Rubin, 2019).

Apply the predeclared adjudication, blinding, and disagreement-reporting rules in Subsections 5.3.1 and 5.3.2.

Version history matters when classification changes. A person first recorded as nonresponsive may later show command-following on imaging, recover behavioral signs, or receive a revised diagnosis. The earlier record should not disappear. The file should show what was believed, what new evidence arrived, why the category changed, and which analyses must be rerun. Revision is evidence of a working correction system, not proof that the final category was always obvious (Claassen et al., 2019; Bodien et al., 2024).

The denominator is as important as the classified sample. A terminal pattern among clean, well-monitored cases may be misleading if rapidly changing, heavily sedated, poorly resourced, transferred, or medically complex cases disproportionately enter the unknown group. Reports should show the full flow from eligibility to classification, including withdrawal, loss of signal, ethical abstention, death before observation, and conflicting evidence. Unknown cases cannot simply vanish from the denominator because their absence may be the strongest clue that the operationalization is selective. Reporting that pattern by site, diagnosis, language, and care setting can show whether uncertainty is concentrated where the operationalization travels poorly.

Public reporting needs the same precision. A press release should not say that uncertain cases were consistent with the Law. A figure should not color unknown trajectories as near-neutral. A summary should state how many cases reached each category, what the categories mean, and whether the conclusion changes when unknown cases are handled under different defensible

assumptions. The public may tolerate an honest gap more readily than a polished certainty later exposed as selection. When a classification later changes, the public record should preserve the earlier status and the evidence that resolved it so correction remains visible.

An explicit unknown category therefore strengthens falsifiability when it is bounded, counted, audited, and prevented from becoming a refuge. It tells readers where the project knows, where it does not know, which component failed, and what evidence could change that judgment. A useful unknown category should also record what observation could resolve the case and who is responsible for obtaining it, if doing so is ethical. That turns uncertainty into a defined research task rather than a permanent holding bin. Where feasible, protocols should distinguish uncertainty caused by absent evidence from uncertainty caused by conflicting evidence, because the next study needed is different. A case should leave the unknown category only when a predeclared resolution rule is met; pressure for a cleaner denominator is not evidence.

5.5 — Closure, Measured Bands, and Crisis Safety

The final endpoint Section brings exact truth, finite measurement, and care into one view without collapsing them. A truth-level residual may be exact within the model while every empirical estimate remains resolution-bounded and uncertain.

5.5.1 — Exact Closure Is Not an Empirical Band

Exact terminal closure is the strong latent target: the signed phenomenal representation of one qualifying stream equals zero immediately before its independently defined truth-level endpoint. Empirical adjudication needs two bridges. Observations must bear validly on the latent cumulative object, and a governed endpoint proxy must bear validly on the truth-level end.

A finite band answers a different question. Equivalence testing or decision analysis can define an operational margin, but meaning depends on the margin, estimator, sample, noise, and assumptions (Schuirmann, 1987; Wellek, 2010; Lakens, 2017). An estimate near zero bears on the latent target only through a specified observation and measured-to-latent bridge.

Exactness becomes scientifically interesting when independently governed endpoints and valid observation models yield consequences ordinary rivals do not share. An unusable case remains silent rather than favorable, while repeated gate-valid non-neutral results can pressure or defeat a declared universal branch. The latent target is difficult, but its difficulty cannot become a reason to exclude adverse cases.

Equivalence requires more than failure to find a difference. The empirical band, endpoint window, estimator, error rates, missingness model, and decision rule must be justified from the measurement problem and fixed in the protocol before terminal outcomes are examined. A wide band can manufacture weak compatibility, while an unrealistically narrow one can manufacture failure. Alternative specifications belong in declared sensitivity analyses rather than replacing the primary rule after inspection (Schuirmann, 1987; Wellek, 2010; Lakens, 2017).

Replication can strengthen a measured consequence while leaving scalarization, origin, bearer scope, latent interpretation, and mechanism open. Public language should name the earned result: compatibility with a fixed band, support for a consequence, preference for a rival, or No adequate estimate. Participants contribute observations to a governed test; their lives are never officially declared balanced.

Terminal language exists because a whole-stream claim needs a boundary, not because death is desirable, completing, or explanatory. Care, rescue, comfort, treatment, and ordinary protection remain immediate. The scientific opportunity lies in disciplined observation of naturally occurring boundaries, never in romanticizing them or asking anyone to approach one.

Even a well-designed equivalence result speaks about measured compatibility under assumptions. It is not a probability that the exact latent residual equals zero, and a confidence interval containing zero is not proof of closure. Reports should distinguish a precise estimate near the reference, an imprecise estimate, failed

equivalence, and No adequate estimate because those outcomes answer different questions (Wasserstein & Lazar, 2016).

5.5.2 — Observation Can Stop Before a Stream Does

Observation can stop before a stream does. The truth-level endpoint, a protocol stopping time, and the last usable observation are different events. A sensor can fail, a study can end, communication can disappear, or behavior can become uninformative while experience continues. The unity proxy remains a protocol-fixed and fallible research device, and adjudication may return usable, uncertain, not interpretable, or ethically unavailable.

Observation often stops selectively. Severe illness, sedation, delirium, device failure, discharge, family burden, withdrawal, death, and transfer can make later data unavailable, and the probability of missingness may depend on the unobserved state. Larger samples cannot by themselves repair that dependence; the observation process must be modeled as part of the terminal problem (Little & Rubin, 2019).

A failed result must be located at the right layer. Investigators first ask whether the endpoint bridge met its persistence, signal-quality, return, competing-event, and independence rules. They then ask whether the measurement bridge met reliability, calibration, invariance, discriminant-validity, and missingness requirements. A failed bridge defeats the operational route to inference; a gate-valid adverse result bears on the measured consequence or target branch.

When both bridges are adequate, repeated non-neutral terminal results become serious pressure on the measured consequence and, with a valid entailment, on the strong branch. Conversely, terminal concentration is not uniquely PTVC-like if homeostasis, frailty, selection, or conditioned bridges predict it equally well. Held-out prediction, common scoring rules, sensitivity analysis, and replication across sites are what turn resemblance into discrimination.

Unusable endpoints, failed equivalence, and model ambiguity should be published with the same visibility as favorable patterns. Participants and families should not carry research burden for a literature that hides its hardest cases. If usable endpoints cannot be obtained without excessive burden or confounding, the program may conclude that the strong claim is not currently testable in that domain. That result preserves care and directs effort toward the failed bridge.

Once latent and measured claims are separated, the ethical instruction becomes simple. Present suffering remains real and relief remains urgent. Comfort, palliative care, symptom control, family communication, and ordinary medical duties take priority even when they limit the evidence available; they are the conditions under which any humane inquiry must proceed.

A complete-stream claim requires complete-domain access or a justified bridge from the last observation to the truth-level remainder. Optimism, smooth extrapolation, or a final value near neutrality cannot supply that bridge. Without independently defended assumptions and model recovery against serious rivals,

the responsible conclusion is endpoint-aligned partial evidence, not direct adjudication of the complete stream.

5.5.3 — No Personal Endpoint Testing

No person should try to test the Law by moving themselves or anyone else toward danger, loss of consciousness, treatment refusal, self-harm, violence, deprivation, or death. Such an episode cannot identify the bearer, complete domain, origin, endpoint, observation process, missingness, or measured-to-latent bridge. It creates risk without producing valid evidence.

Research involving human participants remains governed by ordinary ethical duties. The 2024 Declaration of Helsinki places participant health, well-being, rights, scientific validity, independent review, informed consent, privacy, and freedom to refuse or withdraw at the center of medical research (World Medical Association, 2024). Clinical ethics likewise keeps care directed to the patient rather than the convenience of the study (Beauchamp & Childress, 2019).

The permanent rule is simple: the person never serves the endpoint hypothesis. Research may observe under consent, care, independent review, and explicit uncertainty; it may never create the danger it hopes to interpret. This boundary defines which questions can be asked and which apparent opportunities must be refused.

When someone describes a private test involving crime, coercion, self-harm, treatment refusal, or immediate danger, the responsible response is to stop the activity and direct the person toward immediate safety and qualified emergency, clinical, safeguarding,

or legal help. The episode should never be treated as research or as information about the Law.

Formal research near serious illness requires role separation that private experimentation cannot provide. The care team must remain free to change analgesia, sedation, ventilation, communication support, or any other intervention for the patient's welfare. Research accepts those changes, documents the resulting limits, and stops when burden or uncertainty crosses the approved threshold (World Medical Association, 2024; Beauchamp & Childress, 2019).

A safe research ladder begins far from terminal boundaries: representation and instrument studies, simulations, model recovery, retrospective audits, and ordinary-life observation that does not alter care. Prospective nonterminal studies follow only when burden, privacy, and reactivity are controlled. Terminal cohorts, if ever justified, belong at the far end, after earlier gates show that measures, classifications, missing-data procedures, and rival comparisons can do useful work.

Every rung needs an independent stop rule. A protocol pauses when burden rises, signal quality collapses, consent changes, role separation fails, or new evidence shows that the intended inference is unavailable. These stops are not lost opportunities. They are evidence about feasibility and ethics, and they protect the conditions under which human research may seek knowledge.

Data governance must preserve the same boundary after observation. Clinical records, research measures, family testimony, and device logs remain purpose-limited and access-

controlled rather than merged into a lifetime dossier. Withdrawal, correction, retention limits, and deletion rules should be explained before participation. No terminal dataset may become a tool for worth scoring, insurance screening, employment decisions, spiritual classification, or unrestricted model training.

Failure reporting is part of protection. Unusable endpoints, missing terminal intervals, channel disagreement, model ambiguity, low recruitment, and early stopping deserve the same visibility as favorable patterns. A transparent record lets the program narrow, redesign, or stop before repetition turns uncertainty into ritual.

These disciplines make the inquiry more serious. They replace private testing with governed questions, spectacle with measurement, and the demand for a final answer with a sequence of answerable gates. The result may still be No adequate estimate. That outcome leaves care intact, identifies what current methods cannot establish, and directs the next study toward the failed bridge rather than a more vulnerable participant.

Chapter 6 — A Constraint Without a Judge

An exact boundary can be intellectually striking without being personified. The proposed Law asks whether complete conscious-stream histories satisfy a compact terminal relation. That possibility becomes more rigorous, not less wondrous, when the explanation begins with impersonal structure rather than a hidden observer arranging each life.

Constraints appear throughout mathematics, physics, biology, and systems theory as limits on admissible states, histories, or transformations. Those examples show that structure need not be an agent, while also reminding us that no analogy transfers its evidence to conscious experience (Maudlin, 2007; Lange, 2009).

PTVC is a boundary hypothesis before it is a mechanism story. Reserve coupling, bridge processes, feedback regulation, adaptation, opponent dynamics, predictive control, and selection can become candidate routes or rivals only when they generate discriminating consequences. The target remains fixed while those explanations compete.

Impersonality protects human dignity. Without a rewarder or punisher, illness, abuse, poverty, disability, and grief cannot be read as hidden correction, and pleasure cannot become a moral debt. The scientific layer asks about structure; moral responsibility remains with human action, institutions, care, and repair.

Impersonality also leaves room for agency. People already choose within biological, social, legal, economic, and temporal constraints. A terminal boundary, if real, would add one more

condition on complete histories without telling any person what to choose or guaranteeing a preferred outcome.

Miracle language becomes methodologically costly when it appears only after a hard case. A serious program instead asks whether the case defeats a route, narrows a reference class, exposes a measurement failure, or challenges the target itself. Each answer opens a better next question (Popper, 1959; Lakatos, 1978).

Regularity gives wonder a form it can share. A mechanism earns attention by producing testable consequences, surviving capable rivals, and transporting beyond the examples that inspired it. Otherwise the explanation merely repeats the result in more vivid language.

Purpose language needs the same care. Functional descriptions can summarize what a trait or process contributes without establishing intention, cosmic design, or conscious agency. Biology uses such shorthand productively when the underlying causal and evolutionary questions remain open (Mayr, 1961; Dennett, 1987).

Meaning remains available outside the evidence layer. Faith, love, vocation, memory, art, and commitment can guide a life without becoming support for terminal closure. Separating interpretation from scientific warrant protects both the freedom of meaning and the integrity of the test.

The word law then becomes an invitation to examine regularity, necessity, counterfactual structure, and explanatory compression. Natural-law analogies can clarify one feature at a time, but no title

or analogy supplies a valence measure, bearer rule, endpoint, or empirical verdict (Lewis, 1973; Armstrong, 1983; van Fraassen, 1989; Lange, 2009).

Gravity illustrates impersonality, conservation a relation across a whole, and a bridge conditioned paths. They orient the question without supplying evidence for PTVc.

Method supplies the strongest confidence available. Representations, origins, endpoints, exclusions, models, and verdict rules should be prospectively specified; ‘publicly preregistered’ applies only when a timestamped external registry with a persistent identifier exists. Serious rivals receive the same information and scoring opportunities, and outcomes that favor them remain in the public record (Nosek et al., 2018; Munafò et al., 2017; Mayo, 2018).

6.1 — A Constraint Without a Judge

A constraint can describe a relation without describing an agent. A boundary condition may restrict the form of complete histories while leaving open which mechanisms, choices, and events produce any particular path. This is the explanatory space PTVC asks readers to consider.

Reward and punishment language disappear at this level. The target contains no moral observer who gives pain, withholds dreams, or invoices a person for joy. Human freedom and responsibility remain questions within a constrained world, while care and accountability remain grounded in human reasons and institutions.

6.1.1 — Constraint as Formal Shape

Imagine drawing a closed loop on a sheet of paper. You can make it small or enormous, smooth or jagged; if it finishes where it began, its net displacement is zero. No point on the line has to know the final shape. The relation belongs to the whole path. This is how a boundary condition can constrain a history without prescribing every event inside it.

PTVC proposes a boundary condition of that general logical kind for cumulative felt experience. Whether experience admits the needed quantity and boundary remains the scientific question. Initial, terminal, and other boundary conditions describe relations at their respective edges; constitutive, control, functional, and metaphysical constraints ask different questions.

Constitutive constraints help define a system, control constraints limit admissible actions or trajectories, and biological functions describe contributions under a defensible account of function. PTVC does not currently claim that closure constitutes consciousness, that a hidden controller optimizes a life, or that terminal balance is an evolved function. These categories organize candidate explanations without preselecting one.

A mechanism names entities, activities, or couplings that could produce the target and earns status through discriminating evidence. A best-system account would ask whether the relation contributes unusual simplicity, breadth, and informativeness, while metaphysical necessity would make a stronger claim about what could not have been otherwise. The current hypothesis remains open to contingency, reference-class limits, biological dependence, or defeat.

The public test is whether a word such as ledger can clarify accumulation without conjuring a clerk of the universe. Human beings still judge cruelty, duty, courage, negligence, and repair, but those judgments rest on evidence and moral reasoning, not hidden access. The local human questions remain concrete: what burden is present, who carries it, what can be relieved, and what must remain private.

6.1.2 — No Rewarder

Reward language often appears because people want assurance that sacrifice counted or endurance will be repaid. That desire is human. The intentional stance can help people interpret complex systems, but usefulness does not establish that the system literally

has beliefs, desires, or moral aims (Dennett, 1987). Personifying the Law adds no evidence and becomes harmful when it assigns a hidden reason to another person's pain.

Local consequences remain real without a cosmic rewarder. Practice can improve competence, generosity can deepen friendship, honesty can build trust, and recklessness can produce injury. These causal relations support responsibility and learning in ordinary life. They do not establish that later joy repays suffering or that an unseen system ranks who deserves opportunity, trust, visibility, or relief.

A burden-sensitive reading asks whether praise is concealing a cost, gratitude is substituting for compensation, or a gentle word such as 'balancing' or 'teaching' is still turning pain into a sentence. The opposite response is more humane and more exact: suffering raises the duty of care, and no hidden theory assigns guilt to the wounded person.

Accountability remains fully available. It asks what happened, who was harmed, what evidence exists, what remedy is owed, and how future burden can be reduced. Illness, abuse, poverty, grief, disbelief, sexuality, ambition, and ordinary weakness are not punishments from the Law. Human causes and consequences should be studied directly rather than translated into cosmic retribution.

6.1.3 — Freedom Under Constraint

The proposed Law would add a terminal constraint to a world already shaped by bodies, histories, resources, institutions, and chance. It would not cancel agency or guarantee chosen

outcomes. A musician creates within scale and instrument; a patient acts within illness and care; a citizen acts within law and opportunity. Constraint and authorship can coexist.

Capability approaches distinguish formal choice from the real freedoms available to a person (Sen, 2009; Nussbaum, 2006).

A useful picture of freedom is a person moving through a field of real affordances. Some paths are open, some blocked, and some become visible only after care, safety, education, or resources arrive. Poverty, fear, fatigue, discrimination, and bureaucratic delay can narrow freedom; trust, treatment, access, and repair can widen it. The applied books develop those domains, while the root preserves the principle that abstract possibility is not the same as usable freedom.

Dignity lives inside this layered freedom. A person may be constrained and still responsible, dependent and still agentic, wounded and still worthy of privacy. The formal claim cannot rank people by the size of their menu or by the path they were able to take. Its humane value lies in making burden visible without using burden to excuse harm.

Freedom under constraint also protects privacy and mercy. Outsiders may not infer the meaning of private choices from a theory of balance. When someone chooses badly, the useful questions concern the burdens that shaped the field, the harm that followed, the repair now possible, and the conditions that would make better freedom real. With the hidden arbiter removed, constraint becomes a research question rather than a verdict on a person.

Causal explanation also depends on differences that interventions could make. If changing a condition predictably changes later possibilities or outcomes, the condition can be causally relevant even inside a lawful world. Counterfactual and interventionist approaches make room for this ordinary sense of control without requiring an exception from structure (Lewis, 1973; Woodward, 2003).

Responsibility likewise need not rest on metaphysical exemption from every cause. It can concern reasons, capacities, knowledge, alternatives, coercion, and the ability to respond to consequences. These questions remain independent of whether a terminal boundary exists. The Law cannot settle blame, desert, or legal responsibility merely by describing a constraint (Dennett, 1987).

Removing the judge therefore strengthens rather than empties the model. The hypothesis must work through relations that can be stated without intention, favoritism, or moral surveillance. If an explanation requires a hidden agent to notice each imbalance and choose a correction, it has left the scientific layer. A constraint earns attention by producing regularities, not by resembling a verdict.

6.2 — Structure Instead of Miracles

A miracle can name wonder, but it cannot serve as an ad hoc repair whenever a prediction fails. Once exceptions are explained only by hidden intervention, the claim loses a stable relation to evidence.

Scientific explanation seeks regularities, mechanisms, counterfactual structure, and transport beyond the cases that inspired the theory. Reproducibility reforms emphasize prospective specification, transparent analysis, and serious alternatives because apparent success is easy to create under flexible procedures (Munafò et al., 2017; Nosek et al., 2018).

6.2.1 — Miracle as Ad Hoc Patch

An ad hoc patch is introduced after pressure appears chiefly to preserve the preferred conclusion. It may protect hope as well as theory, but it makes the claim less answerable to the world. If a difficult case can always be assigned an unseen episode, a hidden dimension, an endpoint not defined in advance, or a special exception, no observation can matter.

Popper emphasized exposure to refutation, while Lakatos distinguished progressive revisions that predict new facts from degenerating adjustments that merely absorb anomalies (Popper, 1959; Lakatos, 1978). A revision can therefore strengthen the program when it narrows a mechanism and creates a new test. It weakens the program when it preserves every old conclusion and changes only what was needed to explain away the latest failure.

PTVC has several obvious rescue points. A non-neutral result can be attributed to the wrong bearer, a hidden continuation, an unsuitable neutral origin, an endpoint that occurred later, an unmeasured dream, a missing channel, or a route not yet imagined. Any one of these may be a legitimate possibility. None becomes a legitimate explanation merely because it saves closure.

The origin is especially vulnerable. If the terminal estimate misses zero, a researcher may be tempted to recenter the scale. A defensible origin revision would need independent phenomenological and measurement support and would alter past as well as future analyses. Choosing a new zero because it restores the desired terminal result is not calibration. It is answer-driven coordinate choice.

The endpoint offers a parallel temptation. An investigator can move the truth boundary beyond the last observation, redefine a reversible pause as continuation, or treat biological death as only a proxy when the result is unfavorable. Some endpoint revisions may be necessary as knowledge improves. They must be governed by evidence about consciousness and irreversibility, not by inspection of the cumulative estimate.

Bearer rules can also rescue universality. A stream that violates the claim can be declared fragmented, nonmaximal, or outside the intended class after the outcome is known. The safeguard is prospective ownership and routing criteria, including an honest unknown category. A universal proposition cannot remain universal by shrinking the class one adverse case at a time.

A concrete classification is stronger than a beautiful escape. A case may fall outside the declared reference class, defeat one branch, expose a measurement failure, require a narrower endpoint claim, favor a rival, or remain unresolved because a prerequisite gate failed. Bearer, endpoint, origin, and mechanism revisions may all be imaginable, but they become scientific only when independently motivated and prospectively testable. A hidden-balance story changes nothing except the reader's willingness to keep believing.

Legitimate revision therefore needs change control. The record should identify the threatened claim, the old rule, the proposed replacement, the evidence independent of the failure, the new predictions, and the dependent public artifacts that must change. The revised model should be tested on new or held-out data rather than credited for fitting the anomaly that inspired it (Nosek et al., 2018).

Miracle, in this methodological sense, is not a judgment about religion. It names an explanation whose only role is to arrive after difficulty and save the desired result. Religious traditions may speak of miracles within their own meanings and warrants. Scientific inquiry asks a different question: what observation or independently motivated mechanism would distinguish this account from ordinary uncertainty?

Patches can also accumulate quietly. A small origin adjustment, one excluded case, a revised endpoint window, a new smoothing rule, and a broadened equivalence band may each seem reasonable in isolation. Together they can make almost any result compatible. The project should therefore record the cumulative

flexibility of the entire analysis rather than approving each choice as if it were the only one made (Simmons et al., 2011; Gelman & Loken, 2013).

A program that accepts this discipline can still be ambitious. It may revise, learn, and discover deeper structure. What it cannot do is preserve the word Law by making every possible world compatible with it. Regularity must come before mystery.

6.2.2 — Regularity Before Mystery

Regularity before mystery is a sequence, not a rejection of wonder. Wonder may draw attention to the question, but scientific standing comes from patterns, assumptions, and consequences that can be compared with alternatives. Causal and counterfactual explanation asks what would change under another condition; reproducibility asks whether the pattern survives independent methods and settings.

Public language can preserve mystery without making it a conclusion. A guide, diagram, interview, or summary may invite wonder while naming the work: specify the pattern before the story is known, compare ordinary explanations fairly, and preserve the possibility of an unresolved result. Mystery is the atmosphere of the question; regularity carries the evidential burden.

Regularity asks concrete questions. Do prospectively specified patterns repeat? Does terminal affective dispersion change for reasons not explained by sedation, reporting loss, cultural script, selection, medication, exhaustion, social expectation, or measurement artifact? Do serious rivals perform worse under the

same information and scoring rules? A good hypothesis does not fear ordinary explanations. It becomes clearer by being forced to stand beside them.

Readers should be free to admire the architecture and remain unconvinced. A beautiful model earns trust by stating what it represents, what it knows only indirectly, and which rivals already explain well. Beauty becomes scientifically valuable when it exposes unfinished work rather than concealing it.

6.2.3 — Constraint Still Must Be Tested

Testing needs stable language for what can happen. Failed equivalence, rival superiority, unsafe measurement, missing invariance, endpoint ambiguity, and public-risk discovery each locate a different problem. Constraint is a disciplined description of the candidate object, not a substitute for observing it.

A constraint still must be tested against alternatives capable of producing similar observations. Adaptation, homeostasis, opponent regulation, conditioned bridges, selection, and observation artifacts can all generate apparent return or endpoint concentration. Severe testing and equal-budget rival comparison prevent the Law from winning merely because competing models were made weaker (Mayo, 2018).

A general rule must face dramatic, ordinary, flat, ambiguous, and unfinished lives under criteria fixed in advance. Cultural, clinical, and social differences may require distinct measures or ethical handling, but unusual cases need prospectively specified branches rather than retrospective exceptions.

Stories can carry testimony, memory, moral insight, spiritual witness, therapy, and literary truth. They may reveal burden and generate questions without establishing terminal closure. Respect preserves the speaker's meaning without turning singular growth after loss into public proof.

Testing begins with prerequisites: a defensible valence representation and origin, a prospectively identified bearer and complete domain, an endpoint independent of the cumulative result, and an observation process that can recover the intended construct under realistic missingness. Without those gates, a terminal analysis may produce a number without evidence about the Law.

Route tests remain route-specific. Reserve coupling, regulation, conditioned bridges, biological control, and other architectures predict different relations among history, remaining opportunity, endpoint timing, and dispersion. Support for one route does not establish the target, and route failure defeats no broader branch without a valid entailment.

Counterfactual tests matter because many models can fit a completed trajectory. A safe perturbation, an ethically usable record of a naturally occurring unexpected endpoint, a held-out transition, or a new instrument asks whether the model predicts what it did not use to fit itself. Failure cannot be rescued by saying the path merely looked meaningful.

Applied success can establish practical knowledge. A safer city, better government, meaningful ritual, or successful transition pilot may show usability, safety, participation, or public value

under tested conditions. Each result keeps the evidence role it has earned.

Before human data support a route, model-recovery studies should test whether the pipeline distinguishes prospectively specified PTV routes from adaptation, homeostasis, conditioned bridges, endpoint selection, observation floors, and flexible latent states under comparable noise and missingness. Give every comparator the same preprocessing, information horizon, tuning allowance, validation split, and scoring rule. If several generating worlds receive the same label, the result is model ambiguity rather than support (Mayo, 2018; Nosek et al., 2018).

6.3 — Purpose, Meaning, and Scientific Description

Purpose language becomes especially risky when suffering is the object. A sentence about what pain was for can easily become an assertion that someone needed, deserved, or was assigned the pain. The scientific claim contains none of those judgments, so the same restraint belongs in both scientific and public language.

6.3.1 — Purpose Language as Risk

Purpose language can cross several categories in a single sentence. A process that regulates a variable becomes an intention, the intention becomes a moral plan, and the plan becomes an explanation of suffering. Scientific description must stop that slide at the first step. A biological process may contribute to viability, learning, repair, or regulation without being good for every person or designed for a moral end.

Biology often uses teleological shorthand, such as saying that a trait is for a function, while scientific explanation traces selection, development, mechanism, and current causal role. Mayr's distinction between proximate and evolutionary explanation helped clarify why purpose-like language need not imply a conscious planner (Mayr, 1961). The Law should be even more careful because purpose claims about suffering carry immediate moral weight.

Monod used the language of teleonomy for organized, goal-like features that can be explained without foresight or cosmic intention (Monod, 1971). The distinction is useful, but it does not grant a general license to speak as though the universe has aims. A system can maintain temperature, seek food, avoid threats, or

minimize a modeled error while remaining vulnerable to failure and without assigning moral meaning to any event.

Scientific prose should use conditional and causal language. If the Law is true, certain completed trajectories satisfy the proposed boundary. A candidate process may generate a predicted consequence. A study may discriminate that process from a rival. Each sentence identifies what is assumed, what acts, and what observation could make the account weaker.

End-directed behavior is not destiny. An organism can regulate toward a range and still fail, overshoot, collapse, or be overwhelmed. Control language therefore increases empirical burden. The model must specify what is regulated, what resources are required, what happens when control fails, and which rival process could mimic the same pattern.

Purpose language can also hide retrospective selection. After an outcome is known, people can identify some later good and describe the earlier harm as necessary for it. Counterfactual discipline asks whether the good could have arisen through a less harmful path and whether the claimed necessity was specified before the story was complete. The mere existence of growth does not show that the wound was required.

Public artifacts magnify the risk. A caption such as “Every pain has a purpose” may travel far beyond a chapter that later rejects desert and payment. Images of roads, scales, hands, or destinations can imply an arranger even when the prose says constraint. The safer visual vocabulary is structural and conditional, with no symbol of a hidden planner assigning events.

For the research program, every purpose word should pass an ownership test. If the owner is a person, tradition, organism, or explicit model, say so. If no owner can be specified, replace the purpose claim with the mechanism, boundary, or observation actually meant. Ambiguity is not depth when it changes the evidentiary status of the sentence.

Function claims require the same care. Saying that a process contributes to regulation is a causal claim that can be compared with alternatives; saying that the process exists for fairness is a teleological claim that may have no independent test. The first can be wrong in ordinary scientific ways. The second can become insulated because any outcome is redescribed as serving an unseen end. Public prose should prefer the claim that can fail without demanding that readers surrender the human meanings they freely choose.

The Law is strongest when it gives up the easiest story. It does not explain why a particular suffering occurred. It asks whether completed felt trajectories exhibit a lawful boundary and whether nonagentic candidate processes can explain measurable consequences.

6.3.2 — Fate Is Not Method

A method preserves counterfactual alternatives and the possibility of correction. Fate says that what happened had to happen, then treats every result as confirmation. That difference is decisive. A scientific account must state what another measure, horizon, endpoint rule, or rival would have led us to expect before the path is complete.

Human agency and care depend on the same refusal to treat preventable harm as inevitable. A world with constraints can still contain accident, failure, choice, responsibility, and repair (Popper, 1959; Monod, 1971).

Public institutions need that openness. Poverty, abuse, missing data, treatment failure, and unequal access should trigger questions about cause, risk, response, measurement, endpoint classification, and scope, not claims that events belonged to a destined sequence. Probability, constraint, uncertainty, response, repair, and revision better preserve action and correction.

Providence or spiritual resonance may be meaningful to readers who freely choose those interpretations. They can guide prayer, worship, trust, autobiography, and communal life without becoming scientific support or public authority. Respect for religious traditions requires more than borrowing their vocabulary as convenient evidence for a new formal proposal.

Nonreligious readers retain equal freedom. Detailed comparative, spiritual, ritual, and pastoral treatment appears in *The Religion of Fairness*; this volume needs only the doorway. Spiritual resonance may accompany the Law for some readers, but the Law must stand or fall under its own formal and scientific burdens.

Ideas of karma, providence, fate, restoration, and balance document deep human salience and create comparative research opportunities. They do not supply PTVC-specific terminal evidence, settle doctrines of rebirth or divine action, permit cross-person balancing, or turn pain into morally owed payment. Their

value can remain fully visible without annexing them to the scientific claim.

The boundary matters most in crisis. A dangerous urge, abusive relationship, untreated illness, or unbearable grief must never be described as destined balance. Constrained does not mean assigned. People may leave, refuse, seek treatment, ask for help, change conditions, and be protected by others. A method risks error; fate absorbs every outcome after it occurs.

6.3.3 — Scientific Boundary

The scientific boundary starts by identifying the sentence type. A definition states meaning; an assumption declares what a model takes as given; a mathematical consequence follows under conditions; a candidate route proposes how a pattern might arise; an estimator concerns measured data; and a verdict states what a governed comparison licenses. Confusing these roles lets one sentence borrow authority from another.

The distinction between target and consequence is especially important. Exact latent closure is the proposed target. Preterminal compression, terminal concentration within a measured band, reserve-linked behavior, or other observable patterns are possible consequences under additional assumptions. A consequence can fail because the target is false, because the route is wrong, or because the bridge from target to observation is inadequate. The verdict should identify which level was actually tested.

Evidence enters through protocol-defined observations and bridges. Reports, behavior, physiology, neural activity, context, and clinical records may constrain estimates of experience, yet no

channel directly displays the hidden cumulative object. Construct validity, neutral-origin identification, stream ownership, endpoint adjudication, missingness, and transport remain separate gates (Cronbach & Meehl, 1955; Borsboom, 2005).

Negative and ambiguous results belong inside the same architecture. A failed measure, unidentifiable origin, inconclusive endpoint, or rival victory may be scientifically valuable even when it does not answer the universal question. Publishing those outcomes prevents the literature from becoming a collection of favorable fragments and helps future work avoid repeating the same bridge failure (Munafò et al., 2017; Nosek et al., 2018).

Expertise must remain plural. Psychometricians can test representation and invariance; neuroscientists can assess candidate mechanisms and measurement channels; statisticians can examine identification and error; clinicians can review endpoint validity and burden; philosophers can test lawhood and falsifiability. No field's success replaces another's missing work.

Public enthusiasm, institutional adoption, spiritual resonance, and useful applications remain distinct signals. They may show that a question matters, a practice helps, or an interpretation speaks to people. They do not identify a natural zero, establish a truth endpoint, or test a terminal residual. This lets practical good be valued without converting it into proof.

Public artifacts should preserve the same distinctions. A diagram should reveal whether it is illustrative, simulated, measured, or a rival; a summary should separate latent target from empirical

claim; and every surface should retain current status. Visual confidence must clarify the evidence role rather than create one.

A causal account of terminal closure would need variables, interventions or natural contrasts, temporal ordering, and rival causal structures. A functional account might show that regulation preserves viability or learning. Neither account establishes that the process exists for fairness. Function can arise through selection or organization without intention, and causation can operate without moral purpose (Mayr, 1961; Woodward, 2003).

Normative conclusions require independent premises. Even if a terminal invariant were supported, the duty to relieve pain, protect consent, preserve rights, and repair injustice would not be derived from the invariant alone. Those duties stand under ethical and legal reasons that remain active whether PTVc succeeds or fails. The hypothesis cannot turn a descriptive relation into permission or obligation by changing verbs.

Narrative meaning belongs to the person and community living with the event. Someone may interpret grief through faith, love, endurance, randomness, protest, or no larger story. Research can study these as human phenomena but cannot treat their emotional power as evidence for terminal closure. Nor can a scientific report assign meaning for a participant who refused it.

The boundary routes questions without closing them. Science tests causes, functions, measurements, and predictions; philosophy examines lawhood and implication; ethics governs conduct; spiritual and narrative life may explore meaning

voluntarily. Keeping those warrants distinct lets the Law be exact where models permit, humble where observation is weak, and vulnerable to serious tests without turning meaning into data. That separation also makes collaboration easier: disciplines can contribute evidence or interpretation without inheriting authority they have not earned. When a dispute crosses domains, naming the warrant in question—empirical, philosophical, ethical, or interpretive—keeps disagreement focused on the kind of support actually needed.

6.4 — Why Calling It a Law Is Still a Question

Calling PTVC a law remains a philosophical question as well as an empirical one.

Natural-law comparisons can clarify impersonality or constraint, but they cannot transfer the evidential standing of physics to a new hypothesis about experience. Every analogy must state where the resemblance ends.

6.4.1 — Lawhood Is a Philosophical Status

The title *The Law of Fairness* names an ambition, not a completed scientific verdict. In ordinary speech, law can mean an enacted command, a moral norm, a recurrent pattern, a governing necessity, or an organizing principle. Those meanings carry different kinds of authority. The intended use here is a scientific aspiration: the proposal asks whether a general structural condition constrains completed qualifying conscious streams. That is more specific than a metaphor and less established than a scientific law.

Philosophers of science disagree about what makes a law a law. A regularity account emphasizes patterns that hold without exception in the declared domain. A best-system account asks whether a principle belongs to the strongest and simplest summary of the world. A governing account treats laws as relations or necessities that help determine what happens. Other views emphasize counterfactual stability, invariance, or the role a statement plays in successful explanation (Lewis, 1973; Armstrong, 1983; Lange, 2009; Woodward, 2003). The present project cannot settle that debate by capitalization.

The first boundary is between scientific law and enacted law. Nothing in PTVC commands a court, agency, clinician, family, religion, employer, or ordinary person. A scientific relation, even if supported, would not create jurisdiction, legal duty, eligibility, punishment, or policy authority. Institutional and legal proposals require their own reasons, lawful adoption, safeguards, and review. The title cannot lend them authority in advance.

A best-system perspective makes the apparent simplicity of the terminal statement especially demanding. The sentence may be short, yet its real complexity includes the definition of valence, neutral origin, exposure, stream ownership, complete domain, endpoint, missingness, measured-to-latent bridge, branch, and reference class. If those elements can be adjusted after outcomes are known, the one-line rule is not simple. Its complexity has merely been hidden in movable definitions (Lewis, 1973; van Fraassen, 1989).

A governing or necessitarian interpretation would be stronger still. It would suggest that the relation is not only a good summary but part of what makes admissible histories possible. The formulation developed here explores structural consequences of such a target, but the public manuscript does not claim to know whether any supported regularity would govern, emerge, supervene, or summarize. That ontological question remains separate from the empirical question of whether the terminal relation holds.

Counterfactual support supplies another test of lawlike status. A relation is more informative when it constrains what would happen under changes that were not used to fit it. For PTVC, that

might involve safe perturbations, ethically usable records of naturally occurring unexpected endpoints, transport across instruments, or predictions about reserve and remaining reachability. A pattern that appears only in the exact data and endpoint window from which it was constructed has little lawlike force, however elegant its retrospective fit (Lange, 2009; Woodward, 2003).

The word law should therefore increase methodological pressure. The project must declare its bearer, domain, branch, invariances, endpoint, observable consequences, serious rivals, and defeat conditions. It must say whether a result concerns the target, an optional route, a measurement bridge, or a limited empirical consequence. A title that sounds universal while its tests remain local has a special obligation to keep that distance visible.

The title thus functions as a promissory burden. It promises an attempt at a general account and a willingness to surrender or narrow that account when the evidence requires it. It does not promise that nature has accepted the terms. Public confidence should follow the actual maturity of each component rather than the emotional force of the name.

Criticism is part of lawhood being earned. Objections about scalarization, unity, neutral origin, endpoint, observation, model recovery, missingness, or simpler rivals define the work the claim must survive. A narrower hypothesis may remain meaningful after universality fails. A route may be retired while the target stays open. The entire program may also fail. A theory that cannot lose cannot acquire genuine lawlike status.

6.4.2 — Natural-Law Analogies Have Limits

The natural-law analogies introduced at the start of Chapter 6 are orienting devices, not evidence; none may import the maturity of physics into PTVC (Maudlin, 2007; van Fraassen, 1989).

Gravity is useful because it removes the moral judge. Objects are not rewarded for falling and punished for resisting. The comparison ends there. Gravity has well-defined variables, mature measurement, extraordinary empirical support, and known domains of application. PTVC has a formal target, unresolved representation and endpoint problems, optional routes, and an adversarial research program that has not yet delivered complete-stream validation.

Physical conservation laws depend on precisely defined systems, quantities, symmetries, and transformations. Their status does not come from the loose idea that something eventually balances. A conservation analogy therefore cannot supply a neutral origin, unit, bearer, endpoint, or empirical invariance for valence. Those identification burdens remain exactly where they were before the comparison.

Thermodynamic imagery introduces another family of errors. Entropy, equilibrium, heat, and closed systems have technical meanings. A conscious organism exchanges matter, energy, information, and care with its environment, and an affective history is not made thermodynamic by calling its endpoint an equilibrium. The analogy may motivate a question about regulation, but it cannot replace a biological or phenomenal model.

Bridge processes are mathematically useful precisely because they warn against overinterpretation. A stochastic path conditioned to end at a chosen value can display terminal concentration without any special fairness law. That makes the bridge a serious comparator. It demonstrates that endpoint shape may arise from conditioning or selection and that conditional drift alone cannot identify the proposed target.

Architectural guardrails provide a different public image. A building code can limit which structures are admissible without directing every movement inside the building. The analogy clarifies constraint, then fails because codes have authors, purposes, enforcement, and external records. The hypothesis claims no such planner or inspector. The hidden human agency in the image must be surrendered before the analogy becomes theology by accident.

Symmetry and invariance become scientifically useful when they generate explicit questions. Which transformations preserve the terminal relation? Which shifts destroy it? Does the same zero survive a new instrument, culture, or temporal representation? Positive rescaling may preserve zero while additive recentering does not. The analogy earns value only by generating a prospectively specified invariance test rather than gesturing toward mathematical beauty (Lange, 2009).

Accounting imagery should be treated as a temporary teaching device, not an ontology. A ledger makes cumulative sign and nontransferability easy to picture. It does not imply a literal register in the brain, a cosmic accountant, a collectible debt, or an institution entitled to read the entries. Whenever the metaphor

appears, the prose should return to a latent formal object and the limits of observation.

The best analogies clarify one relation and then get out of the way. The Law does not need to dress as gravity, thermodynamics, conservation, control, architecture, evolution, or destiny to remain ambitious. Its real intellectual burden is harder and more interesting: identify the phenomenal object, justify the origin and endpoint, derive consequences that rivals do not share, and allow evidence to decide whether any analogy ever matured into a law.

6.4.3 — The Name Cannot Do the Evidence's Work

Evidence begins with noncompensatory identification gates: representation, sign, neutral origin, stream ownership, complete domain, truth endpoint, observation, and the measured-to-latent bridge. Excellent endpoint ascertainment cannot repair an arbitrary zero, and a reliable instrument cannot repair a bearer chosen after the outcome. A precise estimate of the wrong object is not evidence for the right one (Cronbach & Meehl, 1955; Borsboom, 2005).

A severe test must be able to expose error. Decision rules, exclusions, endpoint windows, equivalence bands, missing-data handling, and primary contrasts should be fixed before outcomes are examined (Popper, 1959; Mayo, 2018; Nosek et al., 2018). Hidden flexibility can protect almost any terminal story.

Rivals deserve the same information and fitting budget, and prediction must reach beyond construction data. Adaptation, opponent regulation, allostasis, predictive accounts, frailty, selection, observation floors, conditioned bridges, and flexible

latent states can imitate parts of the expected pattern. A frozen analysis must predict something those rivals need not, such as held-out transport, response to a safe perturbation, or a prospectively specified relation among reserve, dispersion, and endpoint timing.

Evidence maturity should be recorded component by component. Representation, origin, bearer, routing, endpoint, observation, route, target, and public translation may advance at different rates through protocol fixing, prospective specification, public preregistration where applicable, recovery, testing, replication, and transport. A component-level record prevents one strong element from lending prestige to another that remains weak; it is never a score of a theory or person.

Provenance remains part of the evidence. Each analysis should preserve the hypothesis version, endpoint rule, instrument, preprocessing, exclusions, equivalence band, and rival set. Reanalysis may improve understanding but cannot overwrite the governed result, or reasonable revisions become an invisible search for the version that looks lawlike (Nosek et al., 2018; Mayo, 2018).

Replication should vary more than the research team. Reusing one instrument and pipeline can reproduce hidden bias; stronger tests vary endpoint pathways, channels, languages, clinical contexts, and implementations while preserving the target. Adversarial collaborators should fix rivals and failure conditions before results are known (Munafò et al., 2017). A successful replication across a genuinely different implementation is more informative than repeated confirmation of the same hidden assumption.

Scope is earned in layers. Evidence from self-reporting adults does not automatically extend to infants, nonverbal people, animals, artificial systems, or unresponsive patients, and one endpoint regime does not automatically travel to another. Each extension needs a target-preserving bridge and failure conditions. Once a counterexample passes bearer, endpoint, measurement, uncertainty, branch, and bridge gates, it cannot be averaged away or renamed. Transport can therefore be staged, with each new bearer or endpoint regime earning its own status before it contributes to a broader conclusion.

Public explanations inherit the same discipline. A summary, diagram, interview, or website should identify the component, reference class, strongest rival, and unresolved gate that matter to the sentence being used. When status changes, dependent artifacts should be corrected or withdrawn so the public record continues to point to the actual claim. A public artifact can cite the exact status record it inherits, giving later corrections a concrete destination instead of a vague qualification.

The word law can therefore be assessed along separate dimensions. A relation may be precisely stated, broad in scope, stable under transformation, or capable of compressing many observations, yet still lack the empirical standing of an established scientific law. Separating those dimensions lets the book retain a bold name without letting the name decide the result. It also gives criticism a productive form: identify which dimension is unsupported and which observation would change it. This also localizes disagreement: one critic may challenge scope while another accepts the scope but questions measurement.

Correction follows the same layered logic. A coherent formal object may survive even when measurement is not yet adequate; a narrower branch may survive when a universal version fails; a capable rival may explain the observed pattern better. Lawhood is not a ceremonial badge placed at the beginning of inquiry. It is a demanding conclusion that remains answerable to evidence, rivals, scope, and revision.

6.5 — How Post Hoc Rescue Destroys the Claim

A claim can be rescued after the fact by changing more than the statistics. The bearer can shift, the endpoint can move, the origin can be recentered, the reference class can narrow, or the public sentence can become less exact after criticism.

Research on undisclosed flexibility and the garden of forking paths shows why seemingly reasonable choices can accumulate into a preferred result (Simmons et al., 2011; Gelman & Loken, 2013; Nosek et al., 2018). Public rhetoric can create the same effect by changing the claim instead of the analysis.

6.5.1 — Optional Stopping as Theory Risk

Optional stopping occurs when the decision to end observation, analysis, or interpretation depends on whether the result currently looks favorable. In ordinary statistics, repeated unplanned looks can inflate false-positive risk. In a terminal theory, the danger is wider: the researcher can stop not only a sample but also a time window, a reference class, a route, or the theory's own definition (Simmons et al., 2011; Gelman & Loken, 2013).

The most direct version is endpoint stopping. An analyst watches the estimated residual approach neutrality and treats that moment as the relevant end. If the path moves away again, the earlier instant is preserved as completion. This procedure manufactures closure because the stopping rule contains the outcome. It violates the independent-endpoint condition before any sophisticated statistics begin.

A second version ends data collection when an equivalence test becomes favorable. Repeated testing, flexible sample size, and undisclosed interim looks can change error rates even when the final formula appears conventional. A legitimate sequential design is possible, but its information times, spending rule, decision thresholds, and maximum sample must be specified and analyzed as a sequential design rather than improvised.

Stopping can also occur by window selection. A study may present the terminal week rather than the terminal month, the calm interval rather than the fluctuating one, or the last completed survey rather than the period in which missingness became severe. The chosen window must follow a protocol-fixed clinical and scientific rationale. It cannot be the interval that yields the cleanest picture.

Reference classes create another hidden stop. Researchers may begin with all eligible streams, then remove diagnoses, cultures, ages, settings, or difficult endpoint types until closure appears. Exclusions can be scientifically justified, but the rule must precede outcome inspection or be labeled as exploratory narrowing. A smaller successful class does not inherit the universal status of the larger failed one.

Route switching is optional stopping in theoretical form. A reserve model may be emphasized while it fits, then replaced by adaptation, a bridge, a global constraint, or subjective-time correction when predictions fail. Candidate routes may evolve, but each successor begins a new test cycle. The failed route remains failed for the version that was actually prospectively specified.

The reference law for the random-endpoint problem cannot be selected by the answer either. If independence produces an uncomfortable implication, the project may investigate state-coupled timing or hitting-time branches. It may not claim that the endpoint was always state-coupled solely because the observed path would otherwise count against closure. Branch choice needs independent motivation and new discriminators.

Missingness can hide a stopping rule. Participants may stop reporting when symptoms worsen, capacity declines, family burden rises, or devices are removed. Treating the last observed value as a neutral endpoint makes withdrawal and clinical deterioration part of the estimator without saying so. Denominator flow, reasons for missingness, and observation shutdown must remain visible (Little & Rubin, 2019).

Publication creates a second stopping layer. A program can keep running variants until one produces a striking graph while null, rival-favored, and unusable results disappear. Complete study inventories, prospective specification, and public preregistration when a timestamped external registry record exists reduce the power to stop the public record at success (Nosek et al., 2018; Munafò et al., 2017).

Ethical stopping is different and mandatory. A study must pause or end when burden, risk, consent, clinical instability, privacy failure, or inadequate oversight crosses its protocol-fixed safety boundary. Such a stop does not make the participant a failed case and does not count for or against the Law. It records that welfare outranked data completion, as it should.

Operational failure also deserves a typed stop. A device can malfunction, calibration can drift, a site can violate protocol, or an endpoint channel can become uninterpretable. The study should preserve the data and reason code, apply the protocol-fixed quality rules, and state No adequate estimate where necessary. Quietly moving to another channel after outcome inspection would convert repair into selection.

Universal claims face a special temptation to stop after compatible cases while unresolved or adverse cases remain in the pipeline. A program cannot declare success because the first sites, species, or clinical cohorts fit, then classify later failures as a different phase. The order of arrival should not determine which cases count as the theory's test.

Optional stopping lets the question end when the answer looks right. Prospective stop rules, typed deviations, transparent multiplicity, ethical stop authority, and a record that preserves adverse evidence are the safeguard.

6.5.2 — Public Rhetoric as Rescue

A claim can survive a failed test by changing language after the numbers are fixed. The target remains exact in the technical source, while public summaries quietly say tendency, deep balance, hidden fairness, or evidence of convergence. That shift may feel modest, but it changes the proposition and can turn an adverse result into an inspirational success.

Qualifier deletion is the simplest form. Per stream becomes across people; terminal becomes eventually; latent becomes measured; candidate becomes established; compatibility becomes

confirmation; a route-specific pattern becomes evidence for the whole Law. Public brevity does not excuse these substitutions because each omitted word removes a failure condition.

Public rhetoric can rescue a claim by making it impossible to quote accurately. ‘The universe balances every life,’ ‘suffering is repaid,’ and ‘science has proved fairness’ replace a defined candidate with promises that no present evidence supports. The remedy is a stable public sentence whose bearer, endpoint, branch, exactness, and status survive compression.

Testimony can rescue a claim through selection. A moving account of unexpected relief, terminal calm, a fulfilled dream, or a life that appears balanced may illuminate a human question. It cannot establish a universal cumulative invariant. Choosing only stories that fit and treating counterstories as incomplete turns narrative power into an undeclared sampling rule.

Automated drafting and summarization make accountable human source review more important. Fluent language can turn a proposal into a fact or quietly change every into many, exact into approximate, latent into observed, and candidate into established. Stable source text, provenance, and enough space to preserve the claim are therefore part of responsible communication.

Hostile-crop review is useful even when no critic is hostile. Read the headline, quote card, caption, image, and final sentence without context. A grieving reader, clinician, statistician, religious audience, insurer, or moderator should not reasonably infer payment, blame, forecast, rank, or delayed care. If the fragment invites those meanings, it needs redesign.

Criticism of definitions, citations, visuals, methods, accessibility, automated tools, and presentation belongs inside correction. Endorsements, prestige, institutional access, donors, virality, controversy, and biography can change attention without adding warrant. Trust grows when the strongest accurate criticism is answered at full strength and the repair remains visible.

Outcome vocabulary should be written before results arrive. Route failure, No adequate estimate, Model-ambiguous, Rival-favored, resolution-bounded compatibility, reference-class narrowing, and target-adverse evidence each license different public sentences. When status changes, the affected claim, reason, date, and replacement wording should propagate to summaries, diagrams, interviews, teaching materials, websites, and automated feeds.

Some artifacts need withdrawal rather than a new caption. A score-like tool that invites personal classification, an image that makes suffering look morally owed or like a reliable purchase of a chosen reward, or a headline that cannot carry the endpoint and claim-status limits may remain misleading in every shortened form. Responsible communication includes the ability to stop using a successful artifact.

Citation density cannot rescue status. References should show which proposition each source supports and where the new inference begins. A bibliography establishes traceability, not a vote that turns a synthesis or analogy into established science.

A useful editorial test compares the public sentence with the technical source word by word. Does each retain the same bearer,

quantifier, endpoint, branch, exactness, and status? A shorter sentence may omit machinery but cannot replace every with many, exact with approximately, latent with observed, or candidate with established. Successful compression lets the audience reconstruct the real claim, not a more comforting one.

Question-and-answer settings create pressure because a suffering child, addiction, ambition, death, or spiritual meaning can invite an improvised promise. A strong answer begins with the human reality, states the boundary that changes interpretation, then gives the narrow conceptual point. Speed never justifies a personal forecast or promise that hardship will be repaid.

Beauty remains welcome, but it must make the claim more answerable. Science communication research shows that framing and trust shape how uncertainty is understood, which is why independent reviewers and affected readers should test the sentence as it will travel (Fischhoff, 2013; Scheufele, 2014; National Academies, 2017). Rhetoric serves the project when it preserves vulnerability rather than rescuing belief.

6.5.3 — Change-Control Rather Than Improvisation

A living research program should change in ways readers can follow. Name whether the target, route, endpoint proxy, instrument, analysis, citation, or public explanation moved.

Preserve the prior version, successor, reason, date, responsible owner, and evidence available at the time. A pre-data correction, a safety change during collection, and a post-result revision have different evidential status.

Triggers need names: theorem correction, failed replication, clinical guidance, instrument defect, consent or privacy failure, expert review, accessibility defect, legal change, public misreading, or a burden requiring suspension. Record whether evidence, communication, safety, or target changed.

Changes to bearer, ownership, origin, complete domain, or endpoint can create a new target. A new unity rule can change which experiences accrue; a new origin can change the residual; a new endpoint can move the boundary. Test the successor prospectively, never to validate the predecessor retroactively.

Measurement changes need a new identity and recovery evidence. Channel selection, normalization, weighting, smoothing, imputation, calibration, and preprocessing can alter sign or magnitude. Confirmatory claims require the revised instrument to distinguish relevant generating worlds in held-out or new data.

Estimands, exclusions, stopping rules, multiplicity corrections, code, and decision thresholds require the same visibility. Preserve the original analysis, label deviations, and classify successors accurately. Open science makes learning legible without calling the first plan perfect (Nosek et al., 2018; Munafò et al., 2017).

Narrowing and defeat should state what remains. A universal claim may become reference-class specific; a scalar model may yield to a multidimensional question; a route may fail, leaving the semantic target open. Negative results, unusable endpoints, failed instruments, unsafe metaphors, and burdensome protocols preserve discoveries in measurement, consciousness, endpoints, and ethics without rescuing the retired claim.

A freeze identifies the exact wording, endpoint rule, instrument, analysis, and public summary that governed a result. Later correction, narrowing, or withdrawal may be justified; silent replacement may not make the reviewed object disappear.

With target, endpoint, rhetoric, and revision rules fixed, Part III can ask what structure might make exact closure possible. Each construction will enter as a candidate consequence, route, or rival, never as evidence created by elegance.

Part III — How Exact Closure Could Occur

Part II stated the proposed Law in ordinary language and fixed its public boundaries: one qualifying conscious stream, one complete felt history, one independently specified truth-level endpoint, and one exact terminal relation that cannot be transferred between bearers. Part III asks what would make that compact claim physically and mathematically possible without hiding the answer inside the representation, endpoint, or route.

Exactness makes the problem interesting because ordinary return toward a familiar range does not by itself explain a precise terminal boundary. One prospectively declared endpoint branch creates a severe obstruction, while other branches make different commitments about timing, state, routing, or continuation. Chapter 7 develops those alternatives in full. The Part's task is not to make exact closure easy, but to show what kind of structure would make it scientifically distinctive.

That obstruction opens a research map rather than closing the inquiry. State-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, continued experience, routing change, endpoint-inclusive value, global conditioning, local control, and corrected stream boundaries are distinct branches. Each earns attention by stating its assumptions, observable consequences, capable rivals, and conditions of failure.

The distinction between target and route governs the Part. The target names the proposed boundary relation. A route describes one way a history might satisfy it. A road can reach a destination in a model without showing that nature requires the destination, while a target can remain intelligible before its route is known.

Keeping the two apart lets mechanism research be ambitious and testable.

Measurement requires the same discipline. The cumulative object is latent, not a number displayed inside the mind or readable by one instrument. Reports, choices, behavior, physiology, neural activity, context, and memory may inform an estimate, but the estimate earns meaning only through a declared construct, converging channels, discriminating tests, and uncertainty carried through the chain (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Bollen, 1989).

A useful map therefore keeps bearer, represented contribution, experienced exposure, clock, neutral origin, truth-level endpoint, protocol endpoint, observation process, estimand, estimator, candidate route, rival, intervention, uncertainty, and verdict in their proper places. The map is more demanding than one clean ledger picture, but it makes disagreement and progress locatable.

Reachability turns a proposed destination into a question about available paths. A target can be clearly defined while no admissible continuation from the present state can reach it. That distinction gives candidate routes a genuine way to fail and keeps a verbal possibility from being mistaken for a workable one.

Remaining capacity matters because reachability changes. A path may be open early and close later as health, attention, money, relationships, legal freedom, support, or biological flexibility disappear. Reserve coupling is one candidate geometry for that change. It models capacity to move through a state space and

earns standing only through route-specific predictions (Ramsay & Woods, 2014; Sterling, 2012).

Time makes that architecture human. Calendar duration, perceived horizon, and usable opportunity can diverge, changing which choices feel available and which repairs remain possible. Chapter 9 follows those differences through ordinary life without treating a shrinking horizon as evidence that suffering was assigned or closure is near.

Addiction provides the clearest everyday picture of a changing possibility set. Reward, relief, craving, habit, and lost practice can narrow the paths a person can realistically enter, while treatment, support, and changed conditions can reopen them. The example matters as ordinary causation and route pressure, not as proof that every urge serves terminal closure.

Moderation consequently appears as protection of optionality. It can preserve sleep, attention, health, resources, relationships, and sustained action, while talent, chance, institutions, material conditions, and other people's freedom still shape outcomes. Poverty, chronic pain, unsafe work, caregiving, discrimination, and exclusion also narrow futures, so personal discipline is never the whole explanation.

Aging and serious illness require equal care. Analgesia, sedation, delirium, fatigue, family expectation, clinical setting, prognosis error, selection, and observation shutdown all shape what becomes visible near an endpoint. Palliative care exists to relieve suffering; research must adapt to care, preserve autonomy, and

accept that humane treatment may limit inference (Surges et al., 2024; Carrasco-Zafra et al., 2024).

The Part moves from structural pressure to candidate routes and then to responsible human use. Exactness gives the hypothesis a difficult way to fail. Routes generate consequences that capable rivals can challenge. Time shows how reachability changes in ordinary lives. By the end, the phrase “nature works it out” will no longer count as an explanation: a serious account must identify its branch, mechanism, signatures, and failure conditions.

Chapter 7 — The Structure Closure Needs

The proposed Law is compact enough to say in one sentence yet demanding enough to require an architecture. Readers must see the represented object, the stream that owns it, the boundary where the question is asked, the routes that might realize it, and the fallible observations that could bear on it.

The represented domain includes positive, negative, mixed, and neutral tones of experience. Pain and pleasure are central, but dread, relief, safety, humiliation, belonging, fatigue, grief, awe, loneliness, and bodily ease show why one crude public score would erase much of what the model is meant to study (Barrett, 2006; Cacioppo & Berntson, 1994).

The model must also say when experience enters the domain. Sleep, dreams, anesthesia, delirium, fragmented access, memory loss, and low responsiveness show why biological time cannot simply be copied into a phenomenal record. Accrual conditions belong to the claim and must be fixed independently of the desired terminal result.

Endpoint discipline comes next. The truth rule and any protocol approximation must be specified without using the terminal value they are meant to test. Otherwise a favorable moment can be renamed the end and closure manufactured. Legitimate dependent regimes remain possible, but only as separately declared state-coupled or other branches with their own consequences.

One distinction creates a severe obstruction: on a prospectively declared endpoint branch, exact closure cannot be treated as ordinary wandering that happens to stop at zero. Section 7.2 gives the formal statement and the alternative branches it leaves open.

No single route is built into the target. A route may fail while another branch remains open, and a mathematically coherent route may exist even when the target is false. Scientific progress comes from locating exactly which object succeeded, failed, or remained unidentified and what that result teaches.

Representation branches as well. The primary additive form uses a signed structure for positive and negative phenomenal magnitude; a bipolar form preserves them separately; generalized neutrality can use another functional or terminal set. These branches retain different information and license different inferences.

Modality determines scope. Closure for every qualifying stream differs from probability-one closure under one law, limit-sure reachability, support containment, closure under every intervention or policy, and closure under composition. Naming the modality keeps a local result from expanding into a universal conclusion by rhetoric.

The latent object also differs from its estimate. A thermometer is not temperature, and a pain scale is not pain. Reports and instruments arise through channels affected by calibration, context, bias, noise, and missingness. The analogy is imperfect, but its direction is essential: construct and observation models remain visible between experience and number (Bollen, 1989; Borsboom, 2005).

Precision becomes informative only when the representation, neutral origin, endpoint, observation process, and uncertainty model survive their own tests. Decimal places can sharpen a valid

estimate; they cannot repair a missing construct, an arbitrary origin, or an endpoint selected from the answer.

Error therefore has an address. Construct, origin, scale, exposure, timing, endpoint, sensor, report, model-selection, and missing-data errors enter at different layers. Larger samples sharpen an estimate under its assumptions; they do not erase systematic bias (Carroll et al., 2006; Little & Rubin, 2019).

Reachability adds a further demand: a route must leave the proposed boundary attainable from the states it permits. If allowed states contain no admissible continuation toward the target, the route needs a narrower state space, another endpoint branch, or a candid loss. Viability theory supplies the technical language; this Chapter supplies the public intuition (Aubin, 1991).

Remaining capacity is causal, not moral. Illness, lost time, broken trust, narrowed attention, poverty, coercion, and absent support can close paths that once existed. Treatment, education, reconciliation, protection, and material help can reopen some of them. The geometry concerns possibility, while care and justice retain their independent force.

Once bearer, boundary, branch, observation, estimate, and reachability are separated, the mechanism question becomes sharper. A candidate route must generate distinctive preterminal consequences without being mistaken for the Law itself.

7.1 — Bearer, Experience, Time, and Boundary

A formal model begins by declaring what kind of thing moves through the model. The bearer, the represented quantity, the accrual domain, and the time rule determine what later claims can mean. They are not technical decoration placed around a simple intuition.

Time and domain require a separate rule because biological duration and experienced accrual need not coincide.

7.1.1 — Stream as Bearer

The bearer of the Law is the unified conscious stream. That phrase does not mean the body is irrelevant. The body is the lived site of pain, pleasure, illness, touch, hunger, fatigue, desire, fear, safety, and relief.

The stream is chosen because fairness at this level must answer the life as lived. A family cannot complete one member's cumulative object. A nation cannot absorb a child's pain. A public archive cannot become the bearer of a private stream. The Law is per-stream because felt burden is not transferable by narrative convenience.

This choice gives consciousness science genuine leverage over the model. If future research shows that unified-stream language must be refined, the formal Law must refine with it. Treating identity as an open scientific problem makes the proposal more exact and more corrigible.

The definition also prepares later bearer questions without answering them by confidence alone. Nonhuman animals,

artificial candidates, disorders of consciousness, and fragmented human cases require further evidence and governance. A useful bearer rule can admit uncertainty, preserve protection, and still expose the hypothesis to refinement.

The stream rule is also a routing rule. When a conscious history is uncertain, fragmented, divided, fused, or resumed, the model cannot simply follow the body or the file name and assume continuity. It must state which experiences belong to which stream, what evidence supports that assignment, and where honest uncertainty remains. That discipline prevents both double counting and the fiction that every record associated with one organism automatically belongs to one uninterrupted bearer.

The stream also differs from legal personhood. Legal systems must protect persons through administrable categories, including categories that do not solve every consciousness question. A child, a patient under anesthesia, a person with severe disability, and a person near death remain legally and morally protected even when the formal model cannot yet adjudicate unity with confidence (Bayne, 2010; Bayne & Chalmers, 2003).

The stream differs from biography as well. Biography is what can be told, remembered, recorded, honored, distorted, or archived. A biography may look triumphant while the stream carries dread. A biography may look tragic while the stream contains moments of love, dignity, or relief. Public story is not inner path.

External goods remain indispensable because they shape lived possibility and felt experience, but they cannot substitute for the stream. Resources, health, status, safety, and opportunity may

alter inward burden or relief in different ways across people. The model therefore asks material justice to answer to life as lived, while preserving its independent moral importance (Sen, 1999; Nussbaum, 2011; Zahavi, 2005).

The represented value dimension is not the worth of the person. Severe pain does not make a person low value, and abundant pleasure does not make a person high value. The path may be studied; dignity belongs to the bearer without measure.

For experts, the challenge is whether a formal valence representation can preserve phenomenological richness without dissolving into vague welfare language. Felt valence is neither preference satisfaction nor retrospective life satisfaction alone. It concerns the lived tone of experience, while remaining scientifically difficult and potentially multidimensional.

7.1.2 — Accrual Conditions

Accrual means that a felt valence contribution is counted within the formal path under the conditions specified by the model. The concept is simple only if one ignores consciousness. Biological time alone cannot be enough. A clock can move while conscious access is suspended, degraded, fragmented, or uncertain.

This is why accrual conditions prepare the unity gate. The model needs to say when felt valence is available to the stream in the relevant sense. Waking experience, dream experience, anesthesia, coma, delirium, dissociation, sleep, low-unity states, and boundary states cannot be handled by a naked time integral without further discipline.

Accrual conditions protect both inquiry and care. Ambiguous states require uncertainty rather than confident absence, and care never waits for classification. The model's uncertainty belongs to the scientific claim, not to the person's standing or entitlement to treatment.

The idea can remain concrete without becoming mechanical fantasy. The proposal does not require a cosmic meter. It asks whether a unity-gated stream of felt experience can be represented over a declared domain whose boundaries are scientifically and philosophically defensible.

Technically, time prepares integration. A valence rate at a moment can be meaningful, but the cumulative object concept requires a path over which contributions occur. Without time, there is no accrual, no horizon, no endpoint, no preterminal compression, and no falsifiable terminal claim (Cacioppo & Berntson, 1994; Barrett, 2006).

The path also gives science a harder burden. If the claim concerns a whole stream, then evidence must be longitudinal, endpoint disciplined, uncertainty bearing, and rival aware. A model that is about time cannot be validated by a convenient snapshot.

A path-sensitive model also resists the statistical temptation to replace individual trajectories with group averages. Ensemble patterns can be informative, but the Law concerns the realized stream. A population may trend toward one outcome while a particular life carries another path. This does not make statistics useless. It means the statistical layer must be honest about what it estimates and what it cannot substitute.

For modelers, the path frame also means that censoring, missing intervals, and irregular sampling are not clerical problems. They touch the target. A missing period may be quiet because nothing was recorded, or it may be missing precisely because burden overwhelmed reporting. Time therefore carries evidence risk as well as philosophical meaning.

Accrual conditions also protect against overcounting. A physiological signal may occur without being felt in the relevant stream. A public event may appear burdensome without the same inward consequence in every person. A reported mood may be sincere but incomplete. Accrual must therefore be tied to felt experience under unity, not merely to external event labels.

The phenomenological relation belongs to the definition layer only. It does not imply that the candidate stochastic process has the same moment-by-moment behavior as the experienced contribution admitted by the unity condition. That bridge remains open and is treated through the observation and reachability layers in Sections 7.4 and 7.5 (Campbell & Fiske, 1959; Borsboom, 2005).

The endpoint is the boundary of the formal claim. The Law is not primarily about whether a person feels better tomorrow, whether visible life trends toward justice, or whether a society improves its institutions. It concerns the completed trajectory of the qualifying conscious stream at its truth-level endpoint.

Endpoint language requires unusual care because small ambiguities can change the claim. If the endpoint is defined by the cumulative object reaching neutrality, the argument becomes

circular. If ordinary bodily death is substituted for the end of the qualifying stream, the model may misread resuscitation, low-unity states, prolonged dying, or uncertain stream closure.

Endpoint discipline intensifies present responsibility. A possible future boundary cannot explain away a present wound, delay relief, or turn suffering into an installment. Care is immediate because the person is real before the hypothesis is settled.

The boundary also gives skeptics a stable target. A terminal rule can be specified, challenged, and compared with alternatives. If it cannot be justified in advance, the claim narrows; if its prospectively specified consequences fail after the relevant gates pass, the branch must face that result.

7.1.3 — Model Map Purpose

The model map exists to keep unlike things from borrowing one another's authority. It separates the proposed truth-level object, any candidate route that might realize it, the observations available to researchers, the analysis that turns observations into estimates, and the verdict a study is allowed to report. Without those separations, one symbol or diagram can drift from hypothesis to mechanism, from mechanism to measurement, and from measurement to a claim about a person.

A map is not the territory, but a good map shows where the territory has not yet been reached. The proposed Law concerns a complete stream-owned history of felt experience. That hidden history is the target of the question. A survey, sensor, interview, diagnosis, model state, or terminal chart is an observation or

estimate related to that target under assumptions. It is not the target itself (Box, 1976; Bollen, 1989).

The first layer is the truth-level claim. It asks which conscious stream owns which experiences, how positive and negative felt contributions are represented, what counts as experienced exposure, and where the complete strict-pre-endpoint domain ends. If the bearer, sign, origin, exposure rule, or endpoint remains unsettled, later numerical precision cannot repair the missing object.

The second layer contains candidate routes. Reserve coupling, endpoint conditioning, first-hitting rules, global path constraints, biological regulation, psychological adaptation, and distributed control are possible ways a terminal pattern might arise. A route is not the Law. It earns attention only by generating consequences that capable rivals do not explain as well.

The third layer is observation. People may report feeling, choose among options, withdraw, approach, remember, sleep, move, or show physiological and neural changes. Each channel has a different relation to experience and a different failure pattern. More channels can reduce dependence on one weakness, but they do not create direct possession of inner life (Campbell & Fiske, 1959; Mauss & Robinson, 2009).

The fourth layer is estimation. Researchers decide how to calibrate neutral, preserve sign, represent duration, handle missing intervals, combine channels, and carry uncertainty forward. Those choices create a measured construct. They do not

reveal an instrument-fixed natural unit that was waiting untouched inside the data (Michell, 1999; Borsboom, 2005).

The fifth layer is scientific decision. A result may favor one route, favor a rival, narrow a reference class, remain Model-ambiguous, or yield No adequate estimate. None of those verdicts is equivalent to proving or disproving every version of the formal claim. The verdict belongs to the exact object and comparison that survived the protocol.

The thermometer analogy is useful precisely because it has limits. Temperature measurement works through a mature relation among a physical quantity, a calibrated instrument, a scale, and a reading. Felt experience is harder. A person's words, facial movement, physiology, behavior, and neural activity can disagree without any one channel becoming the feeling itself. The map should display that difficulty rather than hide it behind a clean arrow from experience to number (Cronbach & Meehl, 1955; Cacioppo & Berntson, 1994).

Dependencies make failure informative. A failed scalar representation narrows the additive branch; an unadjudicable stream weakens ownership and endpoint inference; an unhelpful reserve proxy counts against that route; an observation failure blocks the measured claim. One failed auxiliary does not defeat every branch, and one useful instrument does not establish a hidden truth it cannot identify.

Dimensional discipline belongs inside the map. Intensity, duration, probability, capacity, and confidence are not interchangeable merely because each can be represented

numerically. Units and transformation rules determine which comparisons survive rescaling, which depend on an origin, and which should never be combined (Krantz et al., 1971; Michell, 1999).

Counterfactual questions test whether the map explains anything. What changes if the endpoint rule changes, a stream divides, the neutral origin shifts, or one observation channel disappears? A model that cannot answer such questions is a vocabulary list rather than an architecture (Pearl, 2009; Woodward, 2003).

7.2 — Exact Zero and the Random-Endpoint Problem

Section 7.2 treats the obstruction in two layers: first, exact zero as a boundary property; then, the independent positive-density endpoint branch that makes nontrivial paths difficult. The point is not that every endpoint must be independent, but that each branch must expose the structure on which its closure claim depends.

The possible verdicts remain plural and informative. A study may support measured compatibility, show directional imbalance, remain inconclusive, reveal measurement or endpoint failure, favor a rival, or defeat the relevant claim after its prerequisite gates pass. Repeated results may also narrow the bearer class, endpoint regime, estimator, route, or strong form.

7.2.1 — Exact Zero Is a Boundary Property

The strong latent form asserts exact terminal closure in the underlying idealized process. In probability language, the claim is pathwise rather than merely an average tendency.

The exact zero is meaningful only if the neutral origin is justified, the bearer and endpoint are defined, and the cumulative object has a defensible aggregation structure. Failure of any one of those conditions pressures the strong form before empirical data are interpreted.

Terminal closure alone does not imply that the ledger moves monotonically toward zero, crosses zero often, produces a neutral last feeling, makes death peaceful, equalizes visible lives, or supplies a route. Each of those would be a separate claim with its

own definitions and evidence. The target is one terminal relation, not a hidden script for the path that precedes it.

A nontrivial route can evade the random-endpoint obstruction only by changing at least one branch premise. The process may develop terminal degeneracy, acquire an atom at zero, be controlled toward a hard boundary, or couple the stopping regime to the state in another prospectively declared way.

Jumps alter the boundary problem rather than erase it. A sudden loss, medication change, trauma, windfall, or acute clinical event may create a discontinuity, but the jump branch still needs an independently specified event, size distribution, sign relation, endpoint rule, and observation model. Its timing and missingness patterns may also give rivals new explanatory leverage.

The strong form remains answerable even though the ideal terminal quantity is not read directly. Repeated failure of prospectively specified measurable consequences can narrow or defeat the relevant branch once representation, observation, measured-to-latent transport, and branch-specific endpoint gates have passed.

Measured equivalence asks whether a fallible estimator lies inside a protocol-fixed tolerance around an independently calibrated neutral origin. The margin belongs to the estimator, reflects resolution and calibration limits, and must be justified before outcome inspection. Two one-sided tests can then ask whether the confidence interval lies within that margin; failure to reject a difference from zero is merely inconclusive (Schuirmann, 1987; Lakens, 2017).

The estimand must also be declared. A cohort mean, median, individual probability, hierarchical distribution, and path-level endpoint answer different questions. A mean near zero can coexist with large positive and negative subgroups, so population cancellation cannot stand in for per-stream closure.

Different estimator or calibration versions may require different margins, but every public result must identify the measurement version, scale, calibrated units, margin, endpoint rule, and analysis date. Cross-study synthesis therefore needs harmonization or an explicit translation model. Pooling different origins, endpoints, units, or margins can create an apparent consensus no single protocol supports.

When no defensible compatibility margin can be fixed independently of outcomes, the study can still examine concentration, direction, dispersion, or relative change. Its contribution is narrower than measured terminal equivalence, but it may still improve the next test.

7.2.2 — Random Endpoints Create a Severe Obstruction

The random-endpoint problem is more than the intuition that an exact point seems unlikely. It is a conditional structural result. When a prospectively declared endpoint can arrive throughout an interval, is independent of the pre-endpoint ledger under the stated reference law, and has positive density there, exact probability-one closure imposes an extreme restriction on the path. That is much stronger than late convergence or repeated nearness to neutral.

Imagine a camera shutter that can open at any moment during a declared interval. Its timing is generated independently of the pointer in the scene. If every photograph must show the pointer exactly at zero, the pointer cannot spend ordinary stretches away from zero and somehow return only when the shutter happens to open. Under the relevant regularity conditions, it must already be at zero for almost every possible shutter time.

Now change one rule. Let the shutter wait until the pointer reaches zero and then take the picture. Every photograph will show zero, but the result says nothing about ordinary shutter times because success was built into the trigger. The contrast makes the branch distinction visible: independent timing creates the obstruction, while first-hitting timing is a different route whose biological meaning and extra predictions must be justified independently.

The formal statement requires more than the picture. The strict-pre-endpoint ledger must be jointly measurable on a declared interval; endpoint timing must be independent of the sigma-field generated by that path; and the conditional endpoint density must be positive for almost every time in the interval. If exact closure nevertheless holds at the endpoint, the ledger is zero for almost every time on that branch; an added left-continuity condition extends the conclusion to interior times.

That conclusion is severe because the cumulative object is supposed to move with experience. Long positive and negative excursions cannot coexist with probability-one exact closure at a probe that can arrive throughout the same interval under the theorem's conditions. The branch therefore demands a genuine structural account rather than a decorative terminal zero.

The result contributes a clear discriminator: one combination of assumptions and nontrivial path behavior cannot all hold together. It does not establish PTVC, but it forces the endpoint branch, route, or target to become specific before evidence is interpreted.

A truth-level endpoint cannot be defined merely as the first time the desired residual reaches zero. That rule installs success in the stopping time. A first-hitting or absorbing process may be studied as a separate candidate route only when the endpoint has independently warranted biological or consciousness meaning and when the route predicts features not guaranteed by its definition.

Independence is a branch assumption, not a decorative synonym for objectivity. It means the endpoint probe does not choose its time by reading the pre-endpoint ledger, directly or through equivalent information, under the declared reference law. If endpoint hazard depends on a state coupled to the ledger, the theorem's branch no longer applies. State-coupled timing becomes a different prospectively specified model with its own tests.

Several such branches remain open. Timing may be state coupled. The endpoint may be a first hitting time or an absorbing boundary. Its distribution may contain atoms rather than a continuous density. Its support may be singular. The stream may change routing. Experience may continue beyond the biological event used as a proxy. A phenomenal contribution may belong exactly at the endpoint on a separately declared endpoint-inclusive branch (Karatzas & Shreve, 1991; Protter, 2005; Revuz & Yor, 1999).

The pre-endpoint path itself must also exist on the comparison interval. Some models define a process only until it is killed, with no justified extension to times at which the endpoint did not occur. The obstruction cannot be applied by pretending that an unconstructed counterfactual path was available. Each theorem travels with its domain.

These exceptions do not weaken the result. They turn a vague puzzle into a branch map. A state-coupled endpoint predicts a relation between path state and endpoint hazard. A first-hitting route predicts approach to a declared boundary. An atomic endpoint predicts mass at particular times or events. A continuation branch changes the ontology and routing of the stream. Each alternative buys coherence by accepting a different scientific burden.

An approximate version is also informative. Under the declared independent-timing model, each part of the path must be weighted by the probability that the endpoint falls there. The predicted off-band endpoint probability is the expected off-band occupation weighted by that endpoint-time density, conditional on covariates specified before outcome inspection. A reproducible discrepancy, after accounting for observation loss and statistical uncertainty, would challenge the declared independence model or another stated premise. Substantial ordinary time outside the band and rare off-band endpoints alone do not establish dependence.

Bridge rivals enter precisely here. A Brownian bridge or a Doob-transformed process can be conditioned to approach a terminal set, and its earlier distribution changes because the endpoint condition is built into the law. Such a path may display a visually

compelling funnel without providing evidence for reserve coupling, a local controller, or the target Law itself (Doob, 1957).

Scoring must match the prediction target and the observation or selection regime. A comparison of complete-population laws includes the terminal-event frequency when that frequency is part of the modeled data. A comparison explicitly conditional on the terminal event uses the normalized conditional law and cannot count the imposed endpoint as a successful prediction. For a probability-zero endpoint, use a specified bridge and comparable observed-data densities, not an extra penalty based on a zero event probability.

Ordinary terminal selection can create a similar pattern without a formal bridge. People are observed at particular endpoints because illness, care setting, device tolerance, family decisions, or communication status make those endpoints visible. The observed sample may therefore concentrate even when the broader path population does not. Endpoint selection belongs in the rival set, not in a footnote.

Public interpretation should name the tested object. The obstruction concerns one mathematical branch and says nothing about a person's worth, whether suffering will be repaid, or whether biological death completes an account. A measured estimate outside a protocol-fixed band is likewise a result about an estimator under a named endpoint and uncertainty model, never an unpaid balance or failed life.

Adverse results need a protocol-fixed response. Measurement failure may invalidate an estimate. Endpoint failure may prevent

adjudication. A capable rival may explain the pattern better. Repeated, well-measured violation may narrow or defeat the declared branch. These dispositions must be fixed in advance so diagnosis does not become an unlimited rescue menu.

The positive-density condition does not require the endpoint to be equally likely at every time. It requires positive conditional density for almost every time in the declared interval, so no interval segment of positive duration is excluded in advance. If the endpoint can genuinely arrive throughout that interval, the path cannot reserve its returns to neutral for a hidden set of favored moments.

Covariate adjustment does not restore independence by declaration. Age, diagnosis, frailty, treatment, or care setting may explain part of the relation between path and endpoint, while residual dependence remains. The analysis should report how much dependence is still required after the frozen adjustment set rather than describing an adjusted association as exogenous.

Abrupt endpoints provide a sharp contrast when they are defined independently of the cumulative estimate. A route that uses gradual horizon information may predict differently before a sudden external boundary. Retrospective time-to-end alignment can reveal descriptive structure, but it uses future information to organize earlier observations and becomes a mechanism only through a separate prospective bridge.

7.2.3 — Alternative Endpoint Branches Remain Open

The random-endpoint obstruction does not leave only two choices, blind accident or a local controller. It opens a branch

structure. Different endpoint relations create different mathematical objects, different observations, and different ways the hypothesis could fail. Treating those branches as one vague escape clause would make the theory impossible to test.

One branch uses state-coupled timing. Endpoint risk may depend on independently specified biological or consciousness variables that also relate to the cumulative process. This creates dependence without defining the endpoint by the desired residual. The state, measurements, and hazard relation must be fixed before outcome inspection so completion is not merely closure renamed.

A second branch uses hitting or absorption. A process can terminate when it enters a declared set, or it can remain in that set once entered. Stochastic-process theory gives precise meanings to hitting times and absorbing states, but the scientific challenge is not the vocabulary. The challenge is finding a physically and phenomenally defensible state that was not chosen merely because it produces closure (Ethier & Kurtz, 1986; Champagnat & Villemonais, 2016).

A third branch allows endpoint atoms. Not every endpoint distribution is smooth. A medical intervention, catastrophic event, scheduled procedure, or discrete state transition can concentrate endpoint probability at particular times. An atom changes the random-endpoint theorem's conditions, but it does not automatically create exact closure. The route must still explain what happens at and around that mass.

A fourth branch uses global conditioning. Doob transformations, reciprocal processes, and Schrödinger bridge formulations

describe whole paths subject to terminal conditions, reshaping earlier probabilities across time (Doob, 1957; Léonard, 2014; Garg, Zhang, & Zhou, 2024). Mathematical possibility becomes scientific standing only when the bridge predicts held-out features not guaranteed by the terminal condition itself.

A fifth branch changes the stream boundary. Apparent endpoint failure may show that the qualifying stream continued, paused, split, fused, or routed differently from the protocol assumption. A valid correction needs an independently motivated bearer rule, ownership across the transition, and observations that could show continuation did not occur.

A sixth branch narrows the reference class. Exact closure may fail universally yet remain open for a prospectively justified class of streams, continuity profiles, endpoint regimes, or observation architectures. The narrower claim must expose itself to new cases and counterexamples; it cannot become a moving refuge for failed observations.

Endpoint branches also differ in what a protocol can observe. A truth-level endpoint concerns the end of the qualifying stream. A protocol endpoint may be the last successful interview, the withdrawal of a device, loss of behavioral responsiveness, a clinical declaration, or a fixed time window. Those events can be important without being equivalent. A study that treats last observation as truth-level ending can manufacture a terminal relation from the mechanics of follow-up.

Competing risks add another complication. A stream may become unobservable for several reasons, and the event that ends

observation can depend on the same state variables that influence the measured trajectory. Hospital transfer, sedation, sudden death, withdrawal, technical failure, and family refusal do not carry the same meaning. Survival analysis distinguishes cause-specific hazards and cumulative incidence because collapsing competing events can distort inference (Fine & Gray, 1999; Andersen et al., 2012). For example, a transfer that preserves later contact creates a different follow-up problem from a failed sensor with an unknown outage interval.

The endpoint branch must therefore specify what happens when observation stops before the stream does. One option is to declare the complete-stream claim unidentified. Another is to use a validated terminal-remainder bridge. A third is to narrow the claim to the observed domain. None of these choices can be made after seeing which one produces the most attractive result. The protocol can assign each stopping event its own follow-up decision, so the treatment of an interrupted record is reproducible.

A useful endpoint matrix lists each branch, its independent rule, expected observations, strongest mimics, and retirement result. The matrix turns metaphysical possibility into an experimental agenda. Several branches may remain open at once, but openness becomes scientific only when the branches can be distinguished and can lose. A particularly useful column would identify the next observation on which two otherwise similar branches make different predictions about the same case.

7.3 — Branches, Layers, and Non-Equivalent Claims

The cumulative object needs a local contribution term, but the notation cannot outrun the construct. The term momentary latent valence rate names the candidate contribution attributed to a moment of qualifying experience. It is not a physiological reading, behavioral score, verbal report, or direct observation.

Positive and negative signs, mixed affect, intensity, duration, origin, and cardinality remain model commitments rather than settled discoveries. A scalar representation may still be scientifically valuable when its transformation rules are explicit, its within-person predictions are stable, and multidimensional alternatives receive fair comparison (Krantz et al., 1971; Michell, 1999).

Terminology and units are part of the scientific object. Layer labels must keep one meaning within an analysis, and numerical meaning cannot be inferred from typography or familiar words alone. Later evidence may improve notation through controlled revision; it may not move the target after results are known.

The unity gate belongs to the phenomenological definition layer. It identifies when the model treats felt contribution as active under the chosen account of consciousness. Clinical decisions, legal protection, and human worth remain governed by their own evidence and duties.

7.3.1 — Additive, Bipolar, and Generalized Branches

A branch is not a change of wording. It is a different claim about what felt experience is represented as, what information the

representation preserves, and what terminal condition could be stated without contradiction. A result that supports one branch cannot be moved into another merely because both use the language of neutrality.

The additive branch is the simplest and strongest public form. It treats momentary felt contribution as signed, places positive and negative experience on a common scalar structure, and accumulates those contributions within one stream. The terminal target is then a zero signed residual, not equal circumstances, equal welfare, or equal gross experience.

That simplicity has a price. The branch needs a defensible neutral origin, a stable sign convention, enough scalar structure for addition, a rule for experienced exposure, and finite total contribution over the complete domain. If one prerequisite fails, the terminal equality is not merely hard to estimate. The additive claim itself may be undefined for that case (Krantz et al., 1971; Michell, 1999).

Additivity says that the contributions assigned to disjoint portions of one realized stream can be added to recover the contribution assigned to their union. It does not require an experience to have the same contribution in every context. Relief on entering a safe room may depend on the fear that came before; that history can already be reflected in the contribution assigned to the relief. Changing the order of events can therefore change the contributions while their accumulation remains additive.

The test is whether partitioning the same realized history in different ways preserves the represented total once context has

been taken into account. If the whole requires an additional interaction that those assigned contributions cannot preserve, the proposed representation needs a different composition rule. This distinction lets the additive branch face the richness of experience without assuming that every event has a fixed, context-free price.

Rough stochastic routes can differ from accumulation generated by an integrable local rate. Section 7.5 uses path texture as a candidate discriminator and states the measurement cautions.

The bipolar branch preserves positive and negative experience as distinguishable components rather than forcing every mixed moment into one net number. A person can feel relief and grief, love and dread, pride and shame, or pain and meaning at once. Treating the components separately can preserve information that momentary netting would erase (Cacioppo & Berntson, 1994).

Mixed states make measurement harder because a single response can hide component structure. A person who reports feeling “mostly calm” may also be frightened, sedated, exhausted, grateful, and in pain. A bipolar model can ask whether those components are separately recoverable. It cannot assume that two questionnaire columns reveal the complete mixture (Barrett, 2006; Mauss & Robinson, 2009).

A generalized branch begins when scalar addition no longer preserves the relevant structure. It may use a vector, a family of partially ordered states, a nonadditive functional, or a terminal set. For example, pleasantness, distress, arousal, agency, and bodily burden could remain separate dimensions whose transformation

and terminal relation must be declared rather than collapsed into an arbitrary origin.

Nonadditive and partially ordered branches preserve interactions or incomparabilities that simple summation can miss. Repeated humiliation, prolonged uncertainty, sleep loss, and grief layered with caregiving may depend on order and combination. Such a model needs an explicit composition rule and a terminal relation stated in its own language; equals zero no longer means what it means on the additive branch.

Scalarization is the bridge from a richer state to one signed quantity. It may be useful, but it must be earned. Weights selected because they produce terminal closure would manufacture the result. They should instead be justified by prospectively specified theory, calibration, held-out prediction, and sensitivity analysis that reveals what changes when the scalarization changes (Bollen, 1989; Borsboom, 2005).

Branch separation makes the program more vulnerable and more useful. One representation may fail while another remains testable, and readers can see exactly what changed.

7.3.2 — Representation, Endpoints, Routes, and Verdicts

A responsible research program keeps five questions apart. What is being represented? Where does the relevant stream end? What can observers actually record? What process might realize the target? What conclusion is licensed by the results? A single graph cannot answer all five.

Representation comes first because the sign, neutral origin, scale, and aggregation rule determine what counts as positive, negative,

or neutral. If the representation changes after the endpoint is known, the apparent closure may be produced by recentering rather than discovered in the stream. Measurement theory treats such choices as part of the construct, not as harmless formatting (Borsboom, 2005; Michell, 1999).

The endpoint is a separate object. It concerns the truth-level boundary of the qualifying stream and the protocol's operational approximation to that boundary. An endpoint can be clinically meaningful yet unsuitable for the formal claim, or formally relevant yet not recoverable from the available observations. The disagreement belongs in the endpoint layer, not in the estimator.

When these layers are kept separate, negative evidence becomes informative. A failed representation calls for a better value model. An unrecoverable endpoint blocks the terminal claim. Observation failure may leave the latent question open. Route failure can eliminate one mechanism. A rival-favored result can redirect the program. Precision localizes repair instead of inviting wholesale rescue (Bollen, 1989; Little & Rubin, 2019).

The dependency among layers can be seen in a simple example. Suppose reports appear to approach a neutral range near a clinical endpoint. Before asking whether a closure route is favored, the researcher must know what the report scale means, whether its neutral point is stable, whether the clinical endpoint approximates the truth-level endpoint, who remained observable, how missing reports were handled, and whether the same pattern arises under adaptation or selection. A smooth curve is the last step in the chain, not the first.

Representation errors propagate downstream. If a scale confuses low arousal with neutrality, sedation can appear as balance. If positive and negative affect are treated as one bipolar line when they can coexist, mixed states may be erased. If cultural response styles shift the apparent midpoint, comparisons can inherit an arbitrary origin. These are not minor measurement imperfections. They change the object being accumulated (Cacioppo & Berntson, 1994; Barrett, 2006; Mauss & Robinson, 2009).

Observation errors concern who and what remain visible. A participant who stops reporting is not a zero. A flat physiological signal can mean calm, medication, equipment failure, reduced movement, or loss of channel sensitivity. Missingness is itself data about the observation process, but it does not reveal the unobserved phenomenal value without assumptions (Rubin, 1976; Little & Rubin, 2019).

Estimator choices add another layer. Smoothing, interpolation, imputation, weighting, alignment, and aggregation can change the apparent trajectory. An estimator may be unbiased under one missingness model and badly biased under another. The public explanation should make this intuitive: a clean line can be created by the method used to connect incomplete dots (Gneiting & Raftery, 2007; Box, 1976).

Traceability lets correction preserve knowledge. If a neutral origin proves unstable, every dependent estimate and artifact can be revised; if an endpoint bridge changes, terminal claims may be withdrawn while raw observations remain useful. Scientific memory is strongest when revisions reveal the dependency chain instead of hiding it.

7.3.3 — Modal Claims Are Not Interchangeable

Modal language states how widely and under what conditions a claim holds. Every, almost surely, possible, invariant, stable, reachable, and supported are logical commitments, not rhetorical emphasis. They determine the relevant cases, counterexamples, and evidence.

A universal claim concerns every qualifying stream in a declared domain. Its force comes from the quantifier, not from the size of a sample. Once eligibility and measurement prerequisites are satisfied, one verified counterexample within that domain can defeat the declared universal branch.

A probability-one claim is relative to a probability law. An idealized example makes the distinction visible. Choose a point uniformly along a continuous line segment. The exact midpoint has probability zero, yet it remains one of the points in the model. The statement that the chosen point is not the midpoint therefore holds with probability one, but not at every point.

For PTVC, probability-one closure likewise allows a set of exceptional paths with probability zero under the declared law; a set-universal claim allows no qualifying exception. Changing the reference law, conditioning information, or sample space can change which paths receive probability zero. The two claims must therefore remain distinct (Billingsley, 1999).

The reachability ladder distinguishes existence of one closing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability by some policy, limit-sure attainability, support containment, and closure

of every admissible continuation. Almost-sure attainability requires one policy that reaches the target with probability one; limit-sure attainability may supply only a sequence whose values approach one. Under partial observation, one policy must also act consistently across indistinguishable states (Chatterjee, Chmelík, & Davies, 2016; Asadi et al., 2025).

A support claim asks which states or paths remain approximable by positive-probability neighborhoods under a declared topology; an empty finite-sample bin proves neither impossibility nor absence from support. Finite-band compatibility is weaker still. An estimate may lie within a protocol-fixed neighborhood of neutrality at a stated resolution, but that is not exact latent equality and does not become exact because the plotted interval is narrow (Lakens, 2017).

A population tendency is another object. A mean, median, or regression pattern can summarize a group while individual streams differ widely. The per-stream hypothesis cannot be replaced by a group average, and a group trend cannot be dismissed merely because it is not the universal target. The two findings answer different questions.

Reference classes make these modalities operational. A statement about adult human streams, one clinical cohort, one culture, one endpoint regime, or one measurement protocol has a different domain. Excluding failed cases after seeing their outcomes is rescue; prospectively justified narrowing is scientific revision.

Finite studies face a quantifier gap. Compatible observations can strengthen prerequisite models, shrink uncertainty, defeat

alternatives, or narrow a declared class, but cannot themselves survey every possible stream. A universal verdict needs additional sampling, measurement, and explicit theorem-level bridges (Mayo, 2018; Popper, 1959).

Observed stability and intervention stability are different objects. A pattern may recur while treatment, environment, or information remain unchanged, yet fail under a specified intervention; even success under one intervention does not establish stability under every intervention. Causal language therefore requires a defined change, a model of what it acts on, and a clear account of which outcomes it could alter (Pearl, 2009; Woodward, 2003).

Adaptive policies add another quantifier. A fixed schedule specifies actions in advance, while a policy chooses actions in response to information. A relation that holds under one schedule may fail when decisions adapt to the observed state. Control and reinforcement-learning language makes this difference explicit (Sutton & Barto, 2018).

Composition, scale transfer, and counterfactual claims each need a bridge. A property on two intervals need not survive their union if bearer, endpoint, origin, or routing changes, and a lifetime relation does not imply the same relation for each day, organ, network, or group. Claims about another life history require an intervention model for unobserved outcomes.

These distinctions support nested verdicts. Exact universal closure, probability-one closure, finite-band compatibility, route support, intervention stability, and population tendency may

diverge. A stronger claim can fail while a narrower finding survives, and weaker success climbs only by a bridge.

Narrowing must name what changed: reference class, endpoint regime, branch, intervention family, or resolution. The successor claim needs its own assumptions and counterexamples; public summaries should report that scope rather than overstate confirmation or defeat.

Precision permits strong conclusions at the right scale: a route may predict one cohort, a rival may win, or a branch may fail. Clear domains make each result consequential.

7.4 — Latent Objects, Observations, and Estimates

The central distinction here is between the hidden cumulative object and any measured estimate of it. That distinction is ordinary latent-variable science rather than a loophole: the target is defined first, observations are connected to it through explicit assumptions, and every inferential weakness remains visible (Bollen, 1989; Borsboom, 2005).

Three layers need special care here. The phenomenal cumulative object is the target. A reserve or other route state belongs to a candidate mechanism. The measured trajectory is an estimator built from reports, choices, behavior, physiology, neural data, context, and missingness. Similar-looking curves can arise at all three layers without making them the same object.

The measured layer introduces its own commitments: channel validity, sign, origin, weights, sampling, exposure, missingness, endpoint transport, uncertainty, and model choice. Part IV develops those commitments in depth. Here the essential bridge is simple: an estimate earns meaning through a declared chain and can never promote itself into direct access to the stream.

7.4.1 — The Hidden Cumulative Object

The hidden cumulative object is the formal quantity the Law is about, not a number that an observer can read. It represents the signed accumulation of qualifying felt experience within one stream under the declared representation and clock. The word hidden marks an epistemic limit, not a mystical substance.

The object depends on a neutral origin. Without a defensible origin, positive and negative contributions can change under an arbitrary shift. The scale may tolerate positive rescaling while rejecting additive recentering. That is why the neutral point is an identification problem rather than a cosmetic choice (Krantz et al., 1971; Michell, 1999).

The event domain is governed by the accrual conditions defined in Section 7.1; biological time is not automatically phenomenal exposure.

Because the object is latent, it cannot be validated by one indicator. Reports, choices, physiology, and neural data may each relate to aspects of valence, but construct validity requires converging and discriminating evidence across methods (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

The hidden object is neither a public score nor a private oracle. A person may know present experience better than anyone else while lacking access to a lifetime integral; institutions have still less authority. Researchers may estimate a declared construct for a bounded purpose, but the estimate cannot become a personal forecast, eligibility rule, moral record, or substitute for testimony.

This distinction keeps the scientific question sharp. The program is not asking whether people can feel that their lives balanced. It is asking whether a declared latent boundary relation generates measurable consequences that survive representation, endpoint, observation, and rival-model tests (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Latent objects are familiar in science. Traits, disease states, causal effects, and hidden system states are often inferred rather than directly observed. What matters is not hiddenness alone but whether the object has a clear definition, generates discriminating consequences, and can be related to observations without circular calibration (Bollen, 1989; Pearl, 2009).

The additive representation remains subject to the mixed-state and scalarization requirements in Subsection 7.3.1.

The neutral origin is especially fragile because what feels ordinary can shift with context, adaptation, illness, medication, expectation, and culture. A baseline can be stable enough for one task and misleading across a life. The model must show how neutrality is calibrated and how conclusions change when its location is uncertain.

7.4.2 — From Momentary to Accumulated Estimates

A measured record begins with sampled moments, while the question concerns a trajectory. Building an accumulated estimate therefore requires more than adding questionnaire values. The study must state what each channel contributes, how sign and neutral origin are calibrated, how duration and exposure are represented, how missing intervals are handled, and how uncertainty grows across time.

First-person report remains the closest ordinary witness to felt valence because it asks the person whose experience is at issue. Even so, language, memory, demand, fatigue, culture, communication access, privacy, and timing shape what can be said. Silence can mean refusal, incapacity, missingness, or

absence of a usable response; it cannot be coded as neutrality by default (Mauss & Robinson, 2009).

Physiological and neural measures can contribute information about arousal, salience, interoception, regulation, or state change. Their value is complementary, not sovereign. The same signal may accompany fear, exertion, pain, anticipation, medication, or joy, so its sign and phenomenal meaning require contextual and cross-channel validation.

An estimator must declare what kind of object it is. A reflective model treats observations as effects of a latent state. A formative index constructs a composite from chosen components. A predictive model is judged by what it forecasts. A dynamic state model estimates change over time. These architectures answer different questions and cannot share weights merely because their outputs have the same sign (Bollen, 1989; Borsboom, 2005).

The neutral origin must be established independently of the desired terminal result. Centering each session on its own mean, subtracting the final value, or choosing a baseline after seeing the trajectory can manufacture apparent closure. Calibration should instead use protocol-fixed tasks, repeated neutral judgments, cross-sign anchors, and sensitivity analysis showing how conclusions change when the origin moves.

Weights create another source of uncertainty. A composite can reproduce its training sample and fail in another task, language, or clinical setting. Weights should be estimated on one dataset, frozen, and evaluated on held-out observations. A combination tuned after the terminal outcome is known describes the past; it is

not a prospective measuring instrument (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Missingness is often informative. The most distressed, fatigued, private, medically unstable, or socially unsupported participants may be least able to contribute data. A smooth observed trajectory can therefore reflect the disappearance of difficult observations. The record must preserve who was invited, who responded, who could not respond, who refused, and what assumptions govern each unobserved interval (Little & Rubin, 2019).

Uncertainty must travel with the estimate. Construct, origin, sign, exposure, timing, missingness, endpoint, and model disagreement should remain distinguishable before any combined interval is reported. One narrow number can hide several weak bridges; a layered uncertainty account shows where inference is supported and where it remains assumption-dependent (Gneiting & Raftery, 2007).

The accumulated estimate remains a purpose-bounded research object. Part IV develops the measurement pipeline and governance in full. Here the central rule is simple: no estimate can outrun the construct, observation process, and uncertainty that produced it.

7.4.3 — Error and Confidence

Error enters before a confidence interval appears. The first question is whether the estimate targets felt valence at all. A precise instrument aimed at the wrong construct produces a precise error, not a better view of the hidden cumulative object (Borsboom, 2005; Carroll et al., 2006).

Construct, origin, sign, scale, and exposure errors answer different questions. A report can reflect memory or presentation as well as current feeling; a sensor can detect intensity without valence; a neutral point can be mislocated; equal response steps may not be equal felt changes; and clock time can diverge from experienced exposure. Each bridge needs its own evidence and sensitivity analysis.

Observation error includes sensor noise, report variability, device placement, preprocessing, interpolation, and dropout. Some variation is approximately random; other errors are systematic and state dependent. Medication, disability, disease severity, culture, or care setting can bias a larger sample more precisely in the wrong direction.

Channels can also share error. Two devices may share a clock or preprocessing artifact; self-report and family report may share the same social script. Convergence is persuasive only when the channels have meaningfully different bias structures and their relation was specified before the result (Campbell & Fiske, 1959).

Serial dependence matters because the target is cumulative. Repeated observations from one stream are not independent coins. Persistence, treatment episodes, seasonal change, autocorrelation, and long memory alter how uncertainty accumulates. Methods built for independent observations can make the final interval far too narrow.

Measurement noise can distort the very signature the model seeks. A fixed noise floor can hide genuine contraction near a boundary, while smoothing and retrospective terminal alignment can create

apparent contraction. More participants do not rescue a misspecified relation between signal and noise.

Complete-case analysis can manufacture terminal calm by retaining the streams that remain easiest to observe (Diggle & Kenward, 1994; Little & Rubin, 2019).

Confidence has both a statistical and a communication boundary. A probability statement about an estimator is not a probability that a person is balanced, completed, worthy, or safe. Mature reports name the supported level: estimated under the declared model, compatible within a protocol-fixed band, sensitive to missingness, Model-ambiguous, reference-class narrowed, or No adequate estimate.

A strong final report pairs numerical uncertainty with a visible validity record. Readers should see which prerequisites were supported, assumed, stress tested, or unresolved. Confidence becomes useful when it is attached to the dependency chain rather than displayed as one authoritative percentage.

7.5 — Reachability, Reserve, and Predictions

Reserve coupling is one explicit route by which an exact terminal relation could remain reachable. Its value is that possible becomes a set of commitments about state, remaining capacity, timing, path behavior, and failure. The route is mathematically informative and empirically vulnerable; whether conscious life uses it remains open.

Transport to the phenomenal target requires a declared bridge between the route state and the cumulative object, plus a prospectively specified relation between the reserve boundary and the truth-level endpoint. Without those links, a shrinking state variable may describe illness, fatigue, or observation loss while saying nothing about terminal closure.

The route must also respect past-fixity. Later events may alter what continuations remain possible, but they do not rewrite the felt past. Any bridge from route state to terminal residual must preserve that one-way structure rather than treating the life history as a number that can be retrospectively reassigned.

Scientific interest begins after those admissions. The candidate must predict observable relations that ordinary adaptation, homeostasis, bridge conditioning, selection, frailty, and measurement floors do not produce as well. A route earns attention by exposing itself to discriminating tests, not by making closure imaginable.

Formal relations earn their place when they reveal commitments. A bound on unresolved displacement by remaining capacity, for example, says that large unresolved states require

correspondingly large room for future movement. Whether nature satisfies such a relation is an empirical question, and the public manuscript can explain the consequence without reproducing the derivation.

7.5.1 — Reachability and Remaining Capacity

Reachability asks a simple question with severe consequences: from the states the model allows, does at least one admissible continuation still lead toward the proposed terminal relation? A target that becomes unreachable under ordinary states permitted by the theory exposes a structural gap.

In viability theory, the viability kernel is the set of states from which at least one admissible continuation can respect the declared constraints. In ordinary language, it is the region from which the future remains workable. The concept allows many possible paths and asks how that workable region changes as state and time change (Aubin, 1991; Aubin, Bayen, & Saint-Pierre, 2011).

Remaining capacity is whatever the candidate route needs in order to keep relevant futures available. Depending on the model, that may involve time, biological plasticity, tolerance for arousal, access to positive or negative experience, social support, action freedom, or another state variable. No single definition is inherited by every route.

The reserve bound in the formulation developed here captures a general intuition: the magnitude of an unresolved state cannot exceed the capacity of the remaining future to answer it, if the route is to remain viable. The public book can explain this relation

without displaying the formula or pretending that the bound has been observed.

The empirical task is to derive route-specific signatures of changing reachability and test them against equally capable alternatives. A generic finding that people adapt, recover, or change goals does not identify the route. It only shows that possibility is dynamic (Aubin, 1991; Aubin, Bayen, & Saint-Pierre, 2011).

Reachability is local to a state and a set of constraints. A target may be reachable from one point and unreachable from another. This makes the route sensitive to history without allowing the past to be rewritten. Earlier events can change present capacity; they do not change what was already experienced.

Irreversibility makes remaining capacity more than time left. Some options disappear when tissue is damaged, trust is broken, a deadline passes, or a developmental window closes. Other options can reopen through treatment, reconciliation, education, or environmental change. A serious route needs room for both loss and restored capacity.

Reachability also interacts with uncertainty. A bearer or controller, if any local control exists, may not know the true state or endpoint time. A route that requires perfect knowledge of the cumulative object is less plausible than one that operates through observable proxies, but proxy control can introduce error and overshoot. Partial observability is part of the mechanism problem (Bertsekas, 1995; Kaelbling, Littman, & Cassandra, 1998).

Concrete cases show the geometry. Treatment, changed cues, stable housing, education, respite, accessible transport, and social support can widen a reachable future; poverty, discrimination, danger, and institutional delay can narrow it. These are ordinary causal and justice claims whether or not PTVC is true, and they never pool one person's experience with another's (Sen, 1999; Nussbaum, 2011).

A reserve variable can summarize some dimensions of remaining capacity, but compression has a cost. Two people with the same scalar reserve may have different options because one has social support and another has physical health, or because their capacities are not substitutable. A route that uses one number must show that the compression preserves the predictions at issue.

Reachability also offers an immediate practical insight. People and institutions can protect the futures in which repair, learning, care, and meaningful choice remain possible, even before any terminal hypothesis is settled. That principle is valuable because it widens real options without promising a guaranteed destination.

7.5.2 — Reserve Coupling as One Candidate Route

Imagine a bead able to wander sideways inside a channel that tapers continuously to a point. As the bead approaches the tip, its greatest possible sideways distance from the centre shrinks to zero. No foresight is required: the available width already limits where the bead can be. A route could therefore produce terminal convergence through its changing geometry, provided the physical endpoint and the narrowing boundary are independently connected.

This gives a picture of one candidate reserve construction. It multiplies a nonnegative reserve by a signed reference coordinate. If reserve tends to zero while the coordinate is controlled so it cannot grow fast enough to cancel that shrinkage, the product tends to zero. The construction identifies how closure could occur inside a model. Connecting that modeled state to experience still requires the bridge developed below, including the rule that later change cannot rewrite the felt past.

Reserve is a route-specific state, not a person, a soul, a health grade, a prognosis, or moral credit. Its unit must come from the bridge that gives it physical or psychological meaning. Time, energy, attention, communication, treatment, and social support may inform candidate proxies, but none is reserve merely because it declines.

Spread in reserve space and spread in clock time need not share an exponent. If reserve is deterministic on the declared clock and the reference coordinate's standard deviation tends to a finite nonzero limit, a reserve that falls like a power of time remaining gives the modeled product's standard deviation that same power and its variance twice that power. Random reserve or changing coordinate dispersion requires a separate derivation. The protocol must declare which quantity and clock are being fitted before terminal data are examined.

Inward drift and a near-zero mean are weak signatures by themselves. Brownian bridges, Doob-transformed paths, ordinary mean reversion, cohort cancellation, instrument centering, and selective missingness can produce similar patterns. A reserve route must earn preference through a larger conjunction of

distributional, temporal, and perturbational consequences (Doob, 1957; Chetrite & Touchette, 2015).

Past-fixity imposes a hard bridge requirement. Later reserve decline may change future possibilities, but it cannot shrink experience already lived. The product may equal the latent path at every time, only at the terminal boundary, approximately under a declared norm, or merely as a protocol surrogate; each bridge carries a different burden.

Fine-scale path class offers a possible discriminator. A cumulative path built from an ordinary integrable rate has finite variation on compact intervals. A Brownian-driven product generally carries nonzero quadratic variation. Dense sampling could in principle distinguish those textures, but measurement noise can imitate roughness and interpolation can imitate smoothness (Folland, 1999; Protter, 2005).

Bridge conditioning is a serious rival. A conditioned path may narrow near the endpoint without any biological reserve, and the same inward drift can emerge from the transformation used to impose the terminal event. Scoring must use the declared conditional or complete-population comparison in Section 7.2.2 and test held-out features not fixed by the bridge (Léonard, 2014; Chen, Georgiou, & Pavon, 2021).

Frailty, sedation, and observation floors form another rival family. Severe illness can reduce movement, speech, autonomic variability, and willingness to report. A device may round small values or fail in low-perfusion states. The route survives only if

the observation model distinguishes contraction in the latent target from contraction in the ability to observe it.

Safe interventions can provide leverage without turning care into an experiment for closure. Pain relief, restored sleep, improved communication, nutrition, or environmental support may alter candidate capacity for ordinary reasons. The model should predict which variable changes first, which trajectory feature responds, and what rival accounts predict under the same intervention.

The route must be allowed to fail. Stable spread despite a well-measured reserve decline, a reversed lead-lag relation, contraction explained by sedation or sensor floor, or failure to transport across sites would count against the prospectively specified implementation. A failed route does not automatically defeat every form of terminal neutrality, but it removes one of the principal explanations offered here.

Reserve coupling therefore has a narrow public meaning: it is one candidate realization in which remaining modeled capacity scales a signed reference process. It supplies conditional predictions and clear loss conditions. It is not a fuel gauge for a human life, a countdown to death, or evidence that an unseen account is being settled.

Clock and regularity assumptions also matter. A path smooth in calendar time may look abrupt in subjective, physiological, or retrospectively aligned time, while hospitalization, treatment response, communication loss, or environmental change may alter a reserve proxy discontinuously. Any time change or

smoothing rule needs declared transport assumptions rather than being used to preserve a preferred picture.

The factorization also creates an identification test. For positive reserve and a fixed product, multiplying reserve by a positive factor requires dividing the signed reference coordinate by that factor. Applying this relation to observations requires an independently calibrated reserve proxy. Observing only the product cannot confirm that relation; independent information about at least one factor is required.

A reserve proxy should be useful before the endpoint. It should support held-out predictions at times when the future endpoint is still unknown, rather than becoming accurate only after trajectories are aligned and labeled. Prospective usefulness is one practical defense against a proxy that merely summarizes how the story ended.

7.5.3 — Observable Signatures, Rival Mimics, and Failure

A candidate route becomes scientifically useful when it generates a package of observations that capable alternatives need not share. The reserve construction offers several such consequences, but none can stand alone. Each must be tied to a declared endpoint, clock, reserve proxy, observation model, estimator, population, and rival set.

One candidate signature is preterminal compression in a declared dispersion measure. The protocol must distinguish dispersion across model realizations at a given time from variation within a time window of one observed stream and justify any bridge between them. It then tests whether the chosen measure changes

with reserve as predicted, before the endpoint is known retrospectively.

A second signature is concentration near an independently calibrated neutral origin. Session centering, participant-specific recentering, instrument floors, rounding, and post hoc anchor choice can create an apparent pile-up at zero. Neutrality is informative only if the terminal outcome did not set the origin.

A third signature family links reserve to clock time and two-time dependence. A route may fit spread against reserve yet fail its time-domain relation, while bridges or ordinary mean reversion imply different earlier-later covariance. Testing both helps expose rescaling and distinguish inward drift from its generator.

A fourth signature is perturbational response. A safe, independently justified change in sleep, symptoms, communication, or environment should specify which trajectory feature changes first, how long the effect lasts, and whether it depends on the earlier signed state.

A fifth signature concerns path class across sampling scales. Smooth finite-variation paths and rough stochastic dynamics differ, but noise, interpolation, and irregular sampling can mimic both; replicated channels and model-recovery simulation are therefore required.

A sixth signature is terminal-band compatibility at a protocol-fixed resolution. A finite band is an empirical rule, not the exact latent singleton, and its width, observation model, endpoint regime, and reference class must be justified before analysis (Schuirmann, 1987; Lakens, 2017).

The signatures are noncompensatory: a near-zero mean cannot repair an invalid origin, narrowing spread cannot repair an answer-selected endpoint, and perturbation cannot supply an unidentified terminal bridge. Each failed prerequisite limits the chain.

Endpoint adjudication must declare thresholds, dwell times, smoothing, return rules, competing events, signal quality, and revision policy. When the endpoint, bridge, or ethically available observation cannot support inference, No adequate estimate is a legitimate result. Under a purely illustrative 30-second dwell rule, a two-second reappearance and a two-minute return receive different persistence classifications despite both containing the same signal.

Fair comparison gives candidate and rivals comparable information, tuning, missingness models, computation, and held-out scoring. Full curves, calibration, uncertainty, and poor-support regions must remain visible rather than being replaced by one favorable association (Burnham & Anderson, 2002; Gneiting & Raftery, 2007). A side-by-side prediction display can reveal whether one model gains accuracy only in a small region where observations are unusually plentiful.

Reports should separate endpoint support, prerequisite validity, route discrimination, and terminal compatibility so readers can see whether a channel worked, a route beat rivals, or an estimate entered a declared band. For example, a study might improve endpoint classification while leaving rival discrimination unresolved; recording those outcomes separately shows which part of the inquiry advanced.

Chapter 8 — Possible Routes to Closure

A boundary condition does not explain its own realization. Knowing the proposed destination does not reveal which roads exist, what information is available along them, which obstacles can be removed, or whether the destination remains reachable. Chapter 8 compares candidate route families that connect an unfolding stream to a terminal relation while remaining answerable to evidence and distinct from the relation itself.

The Queue System is a public teaching name for one completion-compatible possibility architecture. Its useful question is where possibilities are generated, represented, felt as selectable, endorsed, acted upon, carried across transitions, routed between streams, or constrained at a complete-history boundary. The architecture is a research scaffold, not a discovered allocator of events.

Reachability is the common thread. At any moment, some continuations are physically possible, some imaginable, some consciously selectable, some socially permitted, some materially affordable, and some compatible with a candidate model. A paper option may exist while the body, mind, household, law, or remaining time makes it unusable.

Thinkability matters because language, education, example, trauma, scarcity, and attention shape which futures can be pictured. Yet a future can be imaginable and still be unsafe, coercive, unaffordable, medically unavailable, or practically unreachable.

Agency remains active inside constraint. Human action can matter on terrain the traveler did not choose, and treatment,

support, learning, refusal, or institutional reform can change that terrain. An account that explains every choice as forced after the fact cannot distinguish itself from fatalism.

Shared conditions can constrain many people without creating a shared cumulative object. A household shortage, hospital crisis, or disaster changes several menus at once, while each person's experience remains their own. Collective causation and intrastream nontransferability therefore coexist.

Compensability is a route concept: whether the remaining state space contains continuations capable of addressing a declared displacement. Human repair is wider and more immediate, involving care, restitution, accountability, justice, and restored capability. Later pleasure cannot erase a wound or settle what people owe.

Reserve is likewise local to a model. It may represent time, plasticity, health, action capacity, social support, or another variable affecting reachability, and some routes require no reserve at all. Declining reserve is one testable realization rather than a universal mechanism.

Biology and psychology already provide powerful component explanations: homeostatic and allostatic regulation, interoceptive prediction, salience, learning, inhibition, action selection, and mood dynamics shape responses to positive and negative states (Sterling, 2012; Barrett & Simmons, 2015; Eldar et al., 2016). A proposed additional PTVC mechanism must add a prediction those processes do not already supply.

Other routes work through endpoint rules or whole-history constraints. Bridges condition whole paths on a future boundary; hitting and absorption use declared state sets; global constraints restrict complete histories; corrected stream boundaries alter ownership. They may resemble one another near the endpoint while differing in information, causation, and testable consequences (Doob, 1957; Léonard, 2014; Chetrite & Touchette, 2015).

Discriminating consequences, not elegance alone, decide among route families. A route should predict held-out data, transport across settings, survive measurement error, and face rivals with comparable information and fitting freedom. A mechanism that explains only the data used to invent it remains a story fitted to the past.

Retirement is therefore part of progress. A route may narrow to one context, yield useful tools, or be replaced while other branches remain open. Recording that change preserves what was learned and prevents a vivid metaphor from outliving the evidence that once supported it.

The closing question is temporal. Reachability changes as the future changes in size and usability. Chapter 9 carries that insight into addiction, recovery, aging, illness, caregiving, and terminal care, where ordinary causation shows how options can close or reopen without promising compensation.

8.1 — Why a Route Is Needed

A terminal relation states a destination; it does not explain the road by which a history could arrive there. This distinction anchors Chapter 8 because an elegant mechanism can be mistaken for evidence when target and route blur.

A route must connect present states, allowed transitions, information, capacity, and endpoint conditions. It must show how the proposed relation remains reachable and what observable patterns follow. Merely renaming the target as a tendency, controller, or law of nature supplies no explanation (Woodward, 2003).

Several route families can be mathematically coherent at once. Local feedback, learned regulation, state-coupled timing, hitting or absorption, global conditioning, bridge processes, and corrected stream boundaries make different commitments. Their coexistence is a reason for discrimination, not for treating all of them as support.

A route earns attention by making risky predictions against capable rivals. Its existence does not prove the target, and its failure does not automatically settle every branch of the Law. The scientific task is to identify which route-specific consequences survive common-data comparison.

8.1.1 — A Target Law Does Not Supply Its Own Route

The architecture is candidate because each component can fail independently. The latent state may be the wrong state, the observations may be too noisy, the policy class may be too

flexible, the admissibility rule may be artificial, the horizon may be misestimated, or the apparent constraint may be fully explained by ordinary control.

One technically explicit Queue-family member can be modeled as partially observed constrained control. It asks whether distributed processes alter feasible actions as estimated burden, bodily reserve, channel availability, and expected horizon change. This is a candidate implementation of the public architecture, neither its definition nor an established feature of organisms.

A local route cannot know the true endpoint unless it explains how endpoint-relevant information becomes available. It may instead use a filtered estimate of remaining time built from prospectively specified channels. The horizon estimate must be separated from the desired cumulative result so future information is not leaked backward into the state.

Observable consequences must leave the metaphor. A candidate architecture can distinguish loci such as possibility generation, phenomenal access, attention, endorsement, intention, action, transition, routing, endpoint, and complete-history constraint. It then owes measurable predictions about menu size, choice, reaction time, error, revaluation, channel switching, physiology, or neural control that are defined independently of the story used to explain them.

A compact public map can compare what a person can imagine, what feels selectable, what practical conditions allow, and what a candidate model permits. These fields need not form a simple funnel. A model can allow an action the person has never

imagined; a person can imagine an action that money, safety, or time makes unavailable. The useful question is where a particular path becomes blocked and what would reopen it (Woodward, 2003).

The burden lens matters because ordinary language often treats the largest field as if it were the real one. A person is told that they could leave, forgive, study, rest, ask for help, stop using, tell the truth, or begin again, while the actual felt field may be constrained by threat, dependency, exhaustion, pain, shame, cognitive load, money, caregiving, law, or time. The pathway account, used carefully, refuses to confuse abstract option lists with lived menus.

The word architecture prevents the account from becoming a chain of anecdotes. The route must describe a general relation among state, information, possible actions, repair channels, capacity, and endpoint conditions that can be specified before evidence is examined. It cannot be reinvented after each life event to make the event appear necessary.

A distributed mechanism can still feel personal from the inside because it changes what seems reachable. The person does not experience a probability distribution over latent states. They experience pressure, reluctance, fatigue, dread, temptation, relief, clarity, confusion, or a sudden sense that one path has become unavailable. The model must respect that lived immediacy without mistaking it for evidence of an inner scheduler.

A distributed route need not contain a central observer. Interacting local processes can update uncertain states, revalue

actions, shift autonomic readiness, and permit or suppress responses, just as organized behavior can emerge in immune, sensorimotor, and neural systems without a homunculus seeing the whole problem.

The estimated latent-state distribution is not the person's private belief about their life. It is the modeler's probability distribution over unobserved variables given the recorded history. A person may feel uncertain while the model assigns a narrow distribution, or feel certain while the model remains uncertain. Neither result replaces first-person testimony (Bollen, 1989; Box, 1976).

Personal language creates a predictable temptation: when a door closes, we ask who closed it; when burden arrives, we ask why it was sent. The Queue System resists that grammar. It describes candidate relations among possibilities and transitions, much as weather can have fronts and nervous systems can have attractors without an inner judge.

8.1.2 — Routes Are Candidates, Not Proof

A route is a proposed way the terminal target could be realized. It links the endpoint claim to preterminal structure that can be tested, whether through local control, reserve coupling, conditioned bridges, global path constraints, hitting rules, or another family. Mathematical coherence is the first achievement, not the final verdict.

A useful route must state its represented state, available information, feasible transitions, path-changing rule, and observations that distinguish it from capable alternatives. Without those answers, a diagram is illustration rather than mechanism.

Causal explanation demands more than retrospective correlation. With relevant background fixed, a change in the candidate state should imply a specified downstream change at a specified time (Pearl, 2009; Woodward, 2003).

A route cannot answer a displacement through actions outside the remaining menu. Time, capacity, safety, support, treatment, and access constrain what can occur; reachability turns a story into a bounded problem.

Passive and active routes remain distinct. A shrinking reserve may compress movement without choice; active steering requires a state estimate, objective, policy, and admissible actions. They carry different causal and empirical burdens (Bertsekas, 1995; Sutton & Barto, 2018).

A route may be coherent yet false, or real yet unrelated to terminal neutrality. Homeostasis, adaptation, threat avoidance, fatigue, learning, and action pruning occur without establishing a whole-stream boundary condition.

Identification and model recovery precede interpretation. The route state must be distinguishable from frailty, diagnosis, medication, observation quality, and capable rivals; realistic synthetic data should reveal when the pipeline cannot recover it. Then Model-ambiguous or No adequate estimate is the correct result. A useful recovery exercise includes simulated cases with similar visible trajectories but different generating mechanisms, then records which distinctions the analysis recovers.

A frozen route should transport to held-out transitions, later windows, channels, or sites without changing its endpoint, origin,

state definition, or tuning budget. Safe, care-justified interventions add leverage only when the route prospectively specifies what should change and when (Burnham & Anderson, 2002; Gneiting & Raftery, 2007). A forecast for the next transition can be scored separately from its forecast for a later window, revealing whether the route's predictive value depends on timing.

Route evidence remains subordinate to the target. This book provides the public map of what a route must explain and how it can lose; protocol governance, recovery, replication, and verdicts remain with *The Science of Fairness*. The resulting map can turn a broad mechanism proposal into a concrete question about which state changes remain reachable from the present.

8.2 — Choice, Reachability, and Possibility

An outside list can overstate freedom because it counts grammatically possible actions without pricing danger, care duties, money, health, law, or the cost of failure. The analytic task is to trace which constraint removes an option and who carries that burden (Sen, 1999).

A paper menu may say that a person can leave a job, end a relationship, seek treatment, study, rest, forgive, or protest. The felt and practical menu may be much smaller because money, danger, disability, trauma, caretaking, shame, exhaustion, legal risk, or social threat makes some options real only in grammar (Sutton & Barto, 2018).

Human possibility has several layers. An action can be logically imaginable, legally permitted, materially available, consciously representable, emotionally approachable, behaviorally executable, and model-admissible in different combinations. Treating these layers as one is both bad science and bad moral judgment.

The available-action menu belongs to the candidate mechanism. It collects actions or regulatory moves available under the modeled environment and present channel conditions. Felt selectability belongs to lived experience and must be studied through testimony, behavior, and context. The admissible set belongs to a declared feasibility rule. None of the three may silently substitute for another.

8.2.1 — Thinkable Options

In the formal map, thinkable options are not yet the available action menu unless the candidate model includes cognitive representation as a menu condition. They are a phenomenological and psychological precursor. This distinction prevents the model from assuming that every externally listed action has entered the person's decision space.

Thinkable options are shaped by education, language, culture, trauma, cognitive bandwidth, and what a person has seen others survive. Education, stories, public rights, example, and permission can widen that menu, while danger or exhaustion may reduce planning to the next hour. No path can become selectable if it never becomes representable, but representation is only the first step.

Public ethics begins by asking where an option disappeared. A system that advertises choice while making safer paths cognitively invisible shifts burden into the person and then blames the person for the shift. The better inquiry identifies the fear, information gap, dependency, or design barrier that keeps a route from becoming thinkable without coercion.

Between thinkability and felt selectability stand concrete burdens. Fear, shame, trauma, poverty, disability, chronic illness, social dependency, cultural pressure, exhaustion, and lack of childcare can make an imagined route unsafe or unusable. A job, clinic, legal exit, or right may exist on paper while remaining humiliating, unaffordable, inaccessible, or practically invisible.

Felt selectability is where the proposed mechanism becomes most human. It asks whether a path can be lived from the inside with the biological capacity, safety, knowledge, time, and relational conditions actually present. The concept does not deny responsibility; it denies the cheapness of pretending that all visible doors are open doors (Clark, 2013; Friston et al., 2017).

Felt selectability changes over time. Rest, trust, medication, friendship, shelter, therapy, income, information, and bodily recovery can widen what feels livable, while pain, threat, shame, or exhaustion can narrow it. Good systems expand selectability by reducing burden rather than merely announcing rights, so the lived menu is not static (Shah, Mullainathan, & Shafir, 2012).

Admissibility is a modeling relation, not an ontology. Let the candidate menu contain available actions and let the admissible set contain only those that satisfy a prospectively declared feasibility rule under the model's current state, horizon, capacity, and channel limits. Changing those elements can change the set; that is a reason to specify the model, not to call every excluded path impossible in life.

In simulation, an action may count as admissible only when its continuation remains inside the route's declared constraints. This rule is neither a moral judgment nor institutional permission, and it has no authority to determine what a person may feel or what a life is worth.

Admissible does not mean good, right, legally permitted, or felt-selectable. A harmful action may remain feasible under a bad objective, while a worthy action may be excluded by a model that

omitted shame, pain, threat, meaning, or access. Disagreement between the person's account and the model is diagnostic evidence of a missing state or poor definition, not permission to silence the person (Sutton & Barto, 2018).

If the admissible action set cannot be operationalized without reading the outcome back into the rule, this mechanism family fails its own discipline. Admissibility must be specified prospectively enough that a path can surprise the model.

Keep three questions separate: Can the person imagine this path? Does it feel selectable within the present life? Does the candidate model permit it under its declared constraints? A yes to one need not imply a yes to the others. Comparing the answers can reveal a missing resource, an overlooked fear, or a model that has misunderstood the person. The diagram earns its place by making those differences visible.

8.2.2 — Shared Constraint, Separate Cumulative Objects

Shared constraint is common in real life. A family's finances, a hospital's staffing, a neighborhood's safety, or a legal system's delays can narrow several lives at once; so can poverty, war, discrimination, disaster, disability in an inaccessible world, and political violence. The shared cause is real, but the formal cumulative object remains attached to each qualifying stream.

That boundary forbids collective balancing. One person's later flourishing does not repay another person's loss, communal strength after hardship does not justify what individuals endured, and a population average cannot erase a verified stream. Governments may measure shared burden to locate structural

harm, but they may not turn a family, community, or nation into one experiential account.

Constraint without determinism also gives responsibility a more humane grammar. Responsibility is not erased by limits, but it becomes distorted when limits are ignored. A person may still be answerable for harm, truth, apology, refusal, or repair, while the surrounding world remains answerable for the conditions that made better paths less visible, less safe, or less livable.

Two opposite errors follow. One romanticizes willpower and blames people for not choosing paths they could barely perceive or survive. The other treats constraint as total and erases agency. The more accurate middle is responsibility inside a structured field of possibility: some choices are easier, some dangerous, some practically closed, and some reopened by repair.

Human lives already show this middle structure. A person can be responsible for an apology while also shaped by fear. A patient can choose treatment while constrained by cost and pain. A survivor can act with agency while carrying trauma. A worker can make decisions while constrained by debt, law, and family need. Constraint does not erase action (Sen, 1999; Nussbaum, 2011).

Population evidence may reveal and compare shared burdens so they can be repaired, while any claim that the same condition also changed a modeled reserve remains a separate empirical question.

Menu shaping is the point where ordinary institutions enter most directly. Housing, wages, disability access, transport, violence, caregiving, law, and stigma alter which paths exist and which channels can be used. These effects should be modeled before any

additional latent contribution from the proposed route is claimed (Ostrom, 1990).

The concept has direct practical value. Before asking why a person did not take a path, it asks whether that path existed in the lived menu at all; when institutions reduce threat, simplify access, protect privacy, or lower cost, they may widen possibility before any formal outcome is measured. That gain is ethically meaningful on its own.

8.2.3 — Evidence Needed

Evidence must separate structural constraint from narrative plausibility. Testimony that a person felt no choice is important evidence about burden; a route claim additionally needs declared variables, rival models, independent constraints, and predictions that can fail before the outcome is known. Retrospective necessity is a story about the past, not a mechanism.

Evidence for path architecture must combine levels without confusing them. Experience sampling can assess felt selectability. Behavioral tasks can measure option generation, choice, latency, switching, and error. Environmental data can document which options actually existed. Physiological and neural measures can test candidate implementation.

Fatalistic phrases can sound compassionate while closing inquiry: a lost chance was not meant to be, a path had to close, or relief will balance an unbearable season. The stronger response keeps ordinary causes visible and asks what happened, who carried it, which options narrowed, what repair remains possible, and what care is required now.

The central design is factorial rather than anecdotal. Horizon information, biological reserve proxies, channel availability, utility, conflict, arousal, fatigue, threat, and learned value should be varied or modeled separately where ethical and feasible. The pathway-specific prediction is an interaction structure, not the general fact that tired or frightened people choose differently (Munafò et al., 2017; Nosek et al., 2018).

Naturalistic studies add realism but increase confounding. Poverty, illness, medication, disability, caregiving, institutional delay, and social threat shape menus directly. These variables must be treated as causal conditions, not noise to be removed until a route-specific effect appears.

The evidence program can also produce a principled abstention. When relevant constructs cannot be measured without unacceptable burden, horizon cannot be separated from prognosis and fear, or reserve lacks defensible proxies, the honest result is that this mechanism is not presently testable in that domain. That finding identifies the measurement work still needed.

Repair should be described without romanticizing pain. Some burdens soften, some remain, and some become harder to address as horizons or biological capacity shrink. The scientific question is whether a route can track that changing relation; the human task is to relieve what can be relieved and restore what can be restored.

Construct validity is a nonnegotiable evidentiary task. Thinkable, felt-selectable, and model-admissible options cannot be treated as interchangeable survey labels. Each requires a declared meaning, observations that bear on that meaning, and tests showing that the

measure is not simply distress, wealth, health, or verbal fluency in disguise (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Safe perturbations can provide stronger tests. A study might alter information, support, delay, task difficulty, or access to a low-burden option and ask whether the model predicts the direction and timing of menu change. The perturbation should never create suffering for the sake of balance, and the analysis should specify which ordinary learning or control account could produce the same result (Woodward, 2003; Hernán & Robins, 2020).

Missingness belongs inside the test rather than in a final limitations paragraph. Constraint may make a person less able to respond at exactly the moments the model considers most informative. Joint models, sensitivity analyses, and explicit denominator flow can reveal how much apparent menu contraction is produced by disappearing observations (Diggle & Kenward, 1994; Little & Rubin, 2019).

Qualitative disagreement can be diagnostic rather than inconvenient. When a model calls an option available and the person describes it as dangerous, humiliating, unaffordable, or incompatible with care duties, the conflict identifies a missing state or a bad definition. A route meant to describe lived possibility should learn from that disagreement instead of treating the person as measurement noise.

The study must prospectively specify what loss looks like. Failure of construct validity, inability to distinguish the route from rivals, unstable transport, unacceptable burden, or systematic prediction error should narrow or retire the candidate. Transparent reporting

makes those outcomes cumulative rather than embarrassing (Munafò et al., 2017; Nosek et al., 2018).

Within-person designs can sharpen the test when they are safe and feasible. Repeated observations before and after a modest change in support, information, or task structure may reveal whether the same person moves as the route predicts. Between-person comparisons alone cannot tell whether an apparent constraint relation reflects stable differences that predated the study.

Evidence should also preserve the person's reasons. A route may predict that one option becomes less selectable, while an interview reveals that the person deliberately rejected it to protect a child, honor a value, or avoid a foreseeable harm. Choice is not merely movement through a menu; meaning and obligation can alter what a model should count as feasible.

The test budget must also be fair. Each capable rival should receive the information, flexibility, and validation needed to express its strongest credible form. If the comparison remains nonidentifying, the correct output is Model-ambiguous or No adequate estimate, together with the measurement or design advance that could change that verdict.

8.3 — Repair, Reserve, and Compensatory Capacity

The language of repair can clarify a constrained path, but it can also import moral meanings the model does not contain. Here repair means a change in state or opportunity that restores a route's capacity to remain viable. It does not mean that suffering earns compensation or that later relief cancels earlier harm.

Compensability concerns whether a declared remaining path can answer a formal discrepancy under the route's own assumptions. Reserve concerns the capacity available for that movement. Both are route-level concepts, not measurements of a person's worth, entitlement, or future happiness (Aubin, 1991).

Capacity can be multidimensional. Time, health, attention, social support, action freedom, and environmental access may not substitute for one another. A scalar reserve can be useful only when the compression preserves the prediction being tested and does not erase noncompensable limits (Ramsay & Woods, 2014).

8.3.1 — Compensability Defined

Compensability is a route-specific research concept, not a moral theory of repayment. It asks whether, under a declared model, a present burden remains answerable through safe and realistically available paths before the relevant horizon closes. It does not say that the burden was deserved, useful, necessary, or destined to be offset.

The word is easy to misuse because ordinary language suggests compensation after injury. In this Chapter, the concept is narrower. A model may ask whether treatment, safety, rest,

truthful information, restored agency, social support, or another declared channel can still change the future trajectory in the needed direction. That local question does not reprice what has already been suffered.

Compensability is therefore conditional on a state, a horizon, and a set of channels. The same burden may remain answerable when care is available and become harder to answer when treatment, time, mobility, communication, or trust is lost. The model should name those conditions instead of attaching a permanent label to the person.

The concept is also action-dependent. A safe intervention can widen the set of feasible continuations, leave it unchanged, or close paths through side effects and opportunity cost. A useful route model predicts those changes in advance. It does not wait for any later event and then declare that event compensatory.

Repair channels are the concrete routes through which conditions can improve or agency can reopen. They include pain treatment, sleep, food, safety, accessible communication, truth, apology, legal remedy, material support, companionship, rehabilitation, solitude, movement, and time. The list is intentionally plural because one abstract reserve score cannot substitute for the actual form of help.

Institutional availability is not the same as lived access. A clinic may exist while transport, cost, language, stigma, fear, disability, or punitive rules make it unusable. A legal remedy may exist on paper while delay destroys its value. A route model must

represent practical access rather than count formal services as if they had already reached the person (Aubin, 1991; Sen, 1999).

The concept cannot become a trait. No person is highly compensable or poorly compensable in a way that ranks worth, deservingness, resilience, seriousness, or eligibility. At most, a declared model may estimate whether one bounded path remains reachable under specified conditions and uncertainty.

Several meanings of reserve must remain separate. A formal route may contain a modeled capacity variable. Biology may supply physiological tolerance or plasticity. Psychology may describe attention, regulation, or cognitive flexibility. Social conditions may supply housing, money, rights, trust, transport, care, and time. Folding all of these into one number would make the model both unidentifiable and ethically dangerous (Ramsay & Woods, 2014; Sterling, 2012).

The modeled reserve is therefore narrow. It is a candidate state that scales or limits future movement inside one route family. It is not a known substance in the body, and it is not identical to frailty, energy, health, income, friendship, confidence, or prognosis. Any proposed proxy must earn that relation empirically.

External resources enter through their actual causal paths. Safe housing may reduce stress and improve sleep. Translation may make treatment usable. Legal protection may reopen an exit. Money may purchase time or care. Those effects can influence biology and choice, but they should not be absorbed by renaming every resource reserve.

Transport matters as well. A relation discovered in one laboratory task, age group, culture, disease, or care system should not become a general property until it survives changed contexts and measurement channels. Local success supports a local relation.

Care remains prior to the model. Researchers cannot withhold treatment, close a channel, or prolong burden to obtain a cleaner estimate of repairability. A route that requires such a design is ethically unavailable, and that unavailability supplies no indirect support for the hypothesis.

The public boundary is equally firm. No clinician, insurer, employer, school, court, government, religious community, archive, or artificial intelligence system may use a compensability estimate to classify a person's value, future, credibility, care priority, or right to assistance.

8.3.2 — Noncompensable Risk

Noncompensable risk is a warning about a declared model, not a verdict on a life. It names the possibility that, under specified conditions and uncertainty, an action may close essential paths faster than the remaining channels can reopen them. The label applies to a modeled continuation, never to the person's worth, hope, dignity, or right to care.

The risk is channel-specific. Irreversible injury may close one physical path. A violent environment may close safety and trust. Treatment toxicity may narrow function while relieving disease. Imprisonment, displacement, or institutional delay may remove options that no later benefit restores. The model should name the lost channel rather than compress every danger into one total.

Severity alone cannot establish noncompensability. People recover unexpectedly, new treatment becomes available, law changes, relationships form, and unmodeled supports appear. A model should carry that uncertainty and state which possibilities were omitted rather than treating present knowledge as the boundary of a life.

Time is relevant because a path can be reachable early and unavailable later. Yet an uncertain horizon cannot become destiny. The model must distinguish what is known, what is forecast, what remains possible, and how conclusions change across plausible endpoint or recovery assumptions.

Research may use thresholds to study when a continuation becomes unlikely under a specified model. Such thresholds are methodological choices, not moral lines. They require sensitivity analysis, calibration, explicit error costs, and comparison with simpler established risk models (Aubin, Bayen, & Saint-Pierre, 2011).

False positives are especially dangerous. A biased model may classify an action as unreachable because disability, culture, poverty, atypical regulation, or communication style was measured poorly. That error can close a path the person could safely take. Calling the mistake conservative does not make it harmless.

False negatives matter too. A model may underestimate danger because it ignores cumulative fatigue, unstable housing, treatment interaction, coercion, or a bottleneck in one essential channel.

Calibration must therefore assess both kinds of error rather than celebrate average accuracy.

The environment may create the risk. An inaccessible clinic, punitive benefit system, unsafe workplace, violent home, or delayed legal remedy can narrow future possibilities. The burden lens should identify the accountable institution and the repair it owes rather than naturalize the narrowing as a property of the person.

Biological capacity and practical access can also diverge. A person may understand what would help yet lack energy, transport, money, safety, or permission to act. Another may have severe illness but strong care and communication. The model should locate each limit in the layer where it arises.

The concept is not a clinical decision rule. It cannot determine capacity, futility, resuscitation, discharge, treatment allocation, prognosis, or end-of-life care. Those decisions belong to established clinical, ethical, and legal frameworks, with the person's values and rights at the center.

The concept is not a legal or administrative rule either. It cannot justify detention, exclusion, insurance classification, benefit denial, educational tracking, employment screening, or reduced public investment. A latent route hypothesis supplies no authority to govern a person's future.

Research visuals need particular restraint. A red risk band, depleted-reserve icon, or automated warning could be mistaken for a judgment that someone is beyond help. Any such visualization must state its bounded research purpose,

uncertainty, nonclinical status, correction route, and prohibition on person-level ranking.

A route that repeatedly misclassifies risk, fails transport, depends on unethical observation, or adds nothing beyond capable rivals should narrow or retire. Scientific vulnerability is the protection against turning a speculative mechanism into durable authority.

8.3.3 — Empirical Status

Compensability, channel state, and modeled reserve do not currently form a validated measurement system. Each remains a candidate construct with unresolved units, observation models, timescales, and identification problems. This Chapter offers an architecture for testing, not an instrument for reading lives.

The evidence chain should begin with construct definition. Researchers must state whether a variable represents internal biological capacity, practical access to a channel, felt selectability, expected time, or an interaction among them. A broad label such as reserve cannot substitute for that declaration.

A proxy is not the state itself. Functional performance, symptom burden, autonomic flexibility, sleep, cognition, reported effort, social support, or material access may each carry information. Their relation to the candidate route must be tested rather than assumed (Ramsay & Woods, 2014; Sterling, 2012).

Model recovery is a prerequisite. The full analysis pipeline should be tested on simulated data generated by the candidate route and by bridges, mean reversion, frailty, selective missingness, observation floors, and mixtures. A method that cannot identify

the known generating process cannot support a preference in human data.

Equal-budget comparison prevents a protected theory. Rivals should receive comparable information, tuning freedom, missingness treatment, and scoring rules. A complex route model should not be declared successful because it was compared with a deliberately weak baseline (Burnham & Anderson, 2002; Gneiting & Raftery, 2007).

The observation process can imitate capacity loss. Sedation, fatigue, communication barriers, device failure, refusal, and withdrawal reduce visible variation even when underlying possibility is unknown. Less observed movement is not the same as less possible movement (Little & Rubin, 2019).

Floor effects create a similar illusion. An instrument that cannot register small changes can make a trajectory appear frozen. Dense sampling helps only when measurement noise and reactivity are modeled. More observations from the same limited channel do not automatically recover the hidden state.

Intervention evidence requires a causal bridge. A treatment may improve mood through symptom relief, strengthen action through social support, or alter reporting without changing the latent variable the route names. Mechanistic interpretation depends on showing that the intervention acted on the declared route.

Uncertainty often increases near the boundary where the model seeks precision. Fatigue, delirium, sedation, communication loss, and observation shutdown may weaken the record. A short remaining horizon never grants a fragile proxy greater authority.

Repeated failure to operationalize the constructs is itself evidence about maturity. A mechanism that remains permanently immune to measurement, transport, and rival comparison should lose scientific authority even if its language remains philosophically suggestive.

Every result should use the smallest defensible status. A task-level interaction, useful predictor, Route-failed proxy, Model-ambiguous pattern, Rival-favored result, or No adequate estimate should be reported by that name. None becomes exact terminal closure through accumulation of enthusiasm.

No person-level output may become a score, forecast, diagnosis, eligibility label, spiritual judgment, or public record. The legitimate object is a governed research estimate for a bounded scientific question, with consent, uncertainty, correction, and withdrawal built into its use.

8.4 — Biological, Psychological, and Global Routes

Neuroscience and psychology make many route components empirically approachable. Distributed control, salience, interoception, valuation, action selection, prediction, inhibition, memory, stress regulation, attention, shame, relief, effort, habit, and felt possibility can all be studied. Their literatures create a feasibility bridge without yet establishing a closure mechanism.

The route-specific question is what remains after established accounts have done their work. Does a declared interaction among horizon, remaining capacity, route state, and signed history predict something beyond ordinary homeostasis, adaptation, threat avoidance, expected value, fatigue, or predictive processing? A familiar component earns a test, not ownership of the terminal target (Menon, 2011; Sutton & Barto, 2018).

Reverse-inference restraint is essential. Insula activity does not mean a cumulative object is moving, and prefrontal engagement does not identify a closure controller. Salience, control, and interoception become informative only through protocol-fixed tasks, contrasts, temporal order, held-out prediction, and rival comparison.

One plausible implementation hierarchy begins with online valuation and inhibitory control, continues through autonomic and endocrine regulation, and permits offline processes such as sleep-dependent reorganization as auxiliary channels. These levels are not interchangeable. A neural signal may correlate with conflict while the causal work lies elsewhere, and a physiological

shift may alter behavior without representing the cumulative object.

8.4.1 — Neural and Psychological Correlates

A biological route would probably be distributed rather than localized in one ‘fairness center.’ Candidate implementation could involve interacting systems for salience, interoception, reward, threat, memory, action selection, autonomic regulation, and cognitive control, with different contributions across tasks and states (Seeley et al., 2007; Menon, 2011).

Interoception is one candidate input to that field. Breathlessness, pain, nausea, hunger, fatigue, heat, tremor, and heart racing can change the felt cost of acting before reflective thought supplies a narrative (Craig, 2009). The body can make an abstract option immediately livable or unlivable.

Interoceptive signals are not privileged truth channels. Disease, medication, expectation, attention, trauma, disability, training, and culture can alter both signal and meaning. A stronger route test asks whether interoceptive variables add protocol-fixed predictive value beyond these ordinary explanations (Barrett & Simmons, 2015).

The local information burden is severe. A controller operating now can use only information available now under its filtration. If the proposed policy requires the true endpoint, the exact complete residual, or the future behavior of the stream, the route must explain how that information is represented without leaking future data into the state.

A system need not remember a history as a story for that history to affect what happens next. A reservoir carries the effects of earlier inflows and outflows in its present water level without storing a record of every drop. Likewise, a candidate route might use a compact current state, updated as experience unfolds, instead of consulting an autobiographical archive. The scientific question is whether that state retains the information the route actually needs. For conscious life, the existence and adequacy of such a state would have to be established; naming it reserve or residual would not supply the answer.

A partial-observation account may instead operate on beliefs. The system can estimate current threat, opportunity, bodily condition, or likely future, then choose under uncertainty. That is scientifically familiar, but many objectives can generate similar choices. The model must identify what distinguishes closure-oriented control from expected-value, threat-avoidance, fatigue, or predictive-processing accounts (Bertsekas, 1995; Sutton & Barto, 2018).

Temporal order provides leverage. A candidate state should change before the action pattern it is said to control. Neural or physiological changes observed only after behavior cannot establish the proposed direction. Dense longitudinal sampling and task perturbation are more informative than one cross-sectional correlation.

Perturbation must also be specific. If reducing pain changes every candidate mechanism in the same direction, the result shows that the system responded but does not identify why. A stronger design predicts which component moves first, which downstream

relation follows, and what would contradict the route (Woodward, 2003).

Mediation is useful only when its assumptions are credible. A neural signal may sit between an intervention and behavior while sharing unmeasured causes with both. The analysis should distinguish temporal mediation, statistical association, and causal mediation rather than allowing one diagram to imply all three (Hernán & Robins, 2020).

Held-out prediction protects against reverse inference. A route estimated from one task should predict behavior or signals in another task, site, or observation channel. If the candidate computation must be redefined for every context, it may be a flexible description rather than a transportable mechanism.

Cross-level convergence becomes meaningful when levels have different biases. Self-report, choice, autonomic response, and neural data may all share one demand characteristic or stressor. Stronger evidence comes when their relation is prospectively specified, each channel has a distinct error model, and the route still predicts when one channel is withheld (Campbell & Fiske, 1959).

The route also needs a boundary between state and trait. Chronic anxiety, learned avoidance, pain, addiction, and depression can alter repeated behavior without representing a terminal objective. A local mechanism should predict state-dependent changes within persons, not merely classify diagnostic groups.

Care can widen action through ordinary causal routes. Analgesia can reduce pain, stable housing can reduce threat, communication

support can restore choice, and trust can make help selectable. These effects are valuable in themselves. They should not be redescribed as evidence that care adjusted a hidden cumulative account.

The same distinction protects agency. Constraint can make action harder without making a person a puppet. People can act against fear, resist craving, choose under grief, and seek repair under shame. A mechanism that explains narrowed options should increase compassion and practical support without erasing responsibility or freedom.

Individual differences require hierarchical modeling and uncertainty. Anatomy, age, medication, disability, culture, learning history, and task strategy can shift signals. A fixed regional threshold or person-level classifier would be scientifically weak and ethically dangerous.

The route should also survive ordinary rival models. Expected-value control may explain choice, threat models may explain avoidance, effort models may explain slowing, and predictive accounts may explain updating. The candidate gains status only when its declared variable adds held-out information that those rivals do not already provide (Ochsner & Gross, 2005; Pessoa, 2017).

Failure should localize the claim. A missing interoceptive relation weakens that implementation. A neural signal that does not mediate behavior weakens the proposed loop. A rival that predicts equally well creates ambiguity. None of these outcomes proves or disproves the terminal target by itself.

A successful correlate study would therefore remain modest. It might identify a distributed, state-dependent relation among bodily signals, action selection, and candidate capacity under one task. It would not locate fairness in tissue, read the hidden cumulative object, or establish exact closure.

Autonomic and neural timing may reveal distinct roles. A bodily change can precede conscious report, a cortical update can precede a choice, or both can follow an external event. The route should predeclare which sequence it expects. Temporal resolution is not merely technical; it determines whether the proposed loop could perform the work assigned to it.

Lesion, stimulation, and pharmacological evidence can localize necessity or modulation, but each has limits. Broad disruption affects several computations, and compensation can preserve behavior. The strongest interpretation names the function that changed and the alternatives that remain, rather than treating any intervention as proof of a dedicated closure circuit.

8.4.2 — Global, Bridge, and Boundary Routes

Not every route would act through local correction. Some models constrain the complete path, condition it on a terminal event, stop it at a boundary, or revise the definition of the stream and its endpoint. These architectures can produce similar final pictures while making different claims about information, causation, and testability.

A local feedback route uses current or recent state to influence later movement. It resembles familiar regulation and can be probed through perturbation. A global route instead restricts the

set of admissible histories as a whole. It may contain no local device that knows the endpoint, yet it owes an account of what physical or mathematical principle selects those histories.

One concrete possibility would link accumulated felt experience to a changing physical quantity. Suppose, as a model assumption, that every signed contribution equals the opposite change in that quantity, under one fixed representation and calibrated scale. If the positive and negative accumulations are finite and the quantity has the same value at the stream's independently established beginning and pre-endpoint boundary, the accumulated contributions cancel. The model need not foresee when the stream will end. Its scientific challenge is to identify the physical quantity independently, test the change relation on new paths, and establish the boundary relation separately. Defining the quantity as the negative of an already calculated total would merely restate the claim.

A first-hitting route ends the process when a declared state set is reached. It can make a terminal relation nonaccidental, but the set must be defined independently of the desired answer. If the endpoint is simply the first time the cumulative object reaches zero, exact closure has been installed in the stopping rule.

An absorbing route adds persistence. Once the process enters a state, it remains there or the qualifying stream ends. The biological or consciousness meaning of the absorbing state still needs independent support. A mathematical stop is not automatically a truth-level endpoint.

A bridge route conditions a reference process on a terminal event. Brownian bridges, Schrödinger bridges, and Doob transforms show that path distributions can be altered so that a terminal set becomes likely or certain under the conditioned law (Doob, 1957; Léonard, 2014; Chen, Georgiou, & Pavon, 2021).

A positive-probability terminal band can be handled with an ordinary conditional probability. An exact point in a continuous state space usually has probability zero and requires a separately specified limiting or regular-conditional bridge. Writing a conditional probability for an event of zero probability without that construction hides the hardest step (Chetrite & Touchette, 2015).

Conditioned routes must use the scoring target and selection regime declared in Section 7.2.2. The imposed terminal condition cannot itself count as a successful prediction.

A completion potential may summarize the chance of reaching a frozen terminal event from a current state. Such a potential can be scientifically useful, but it is observational unless a causal bridge is supplied. It does not automatically tell us what would happen under an action or policy.

Interventional use requires additional assumptions about treatment versions, exchangeability or randomization, positivity, interference, censoring, endpoint change, and measurement invariance. Without those assumptions, a potential predicts under the observed regime; it does not authorize steering (Hernán & Robins, 2020).

Unexpected endpoints can distinguish route families. A slowly anticipated global or reserve route, a state-triggered boundary, and an externally imposed sudden endpoint need not produce the same preterminal relation. The contrast must be defined independently and analyzed without selecting only compatible endings.

Hybrid models remain possible. Local regulation may operate inside a global bridge; reserve coupling may coexist with a state-dependent endpoint; a boundary rule may interact with psychological adaptation. Each additional component must earn a new observable consequence. Flexibility without discrimination is not depth.

These route families fail differently. A local controller may be absent while a bridge fits. A bridge may fit only because of selection. A global route may remain mathematically coherent but empirically empty. A boundary route may become unstable if continuity criteria are vague. Those outcomes should not rescue one another.

No route implies a judge, purpose, or schedule. A bridge is not providence, a global constraint is not fate, and a corrected boundary is not a promise of continued life. They are candidate structures whose status depends on prediction, transport, and loss conditions.

A completion potential has a built-in reliability check on the positive-probability branch. When correctly specified under the reference law, its normalized conditional probability should evolve as a martingale. Passing that check shows internal

consistency of the conditioned model; it is not an additional vote for PTVC because the property is generic to correct conditioning.

The potential should be evaluated out of fold and with a denominator floor. Near-zero predicted completion probabilities can create unstable ratios and dramatic apparent updates. Reliability, conditional increments, and uncertainty should be assessed on common support rather than only among cases the model already predicts well.

A global route also owes a counterfactual account. If a safe intervention changes an early state, does the admissible path family change, does the endpoint distribution change, or does only the observation change? Without a declared counterfactual object, the route may describe complete histories while saying nothing testable about intervention.

Route pluralism remains open until discriminating evidence favors one mechanism under common rules.

8.4.3 — Plausibility Is Not Evidence

Plausibility begins with compatibility. Brains regulate, bodies allocate resources, people learn, options open and close, and stochastic processes can be conditioned on endpoints. These facts make several route families imaginable. They do not show that any system represents a complete phenomenal residual or implements terminal closure.

A useful fingerprint will usually be conjunctive. Dependence on earlier signed state, a specific horizon interaction, a route-consistent response to safe perturbation, cross-module transport, and a declared terminal-window pattern may need to appear

together. Isolated features are too easy for ordinary models to imitate.

A crucial evidential boundary concerns generic properties of conditioning. A mathematical identity shared by all correctly conditioned laws cannot count as independent evidence for the Law. Inward bridge drift, a completion-probability martingale, or a terminal funnel may be exactly what conditioning produces. Passing such a check validates the implementation, not PTVC.

The same firewall applies to action models. A policy can be mathematically reweighted toward completion under a frozen model, yet the resulting identity is not evidence that any biological or psychological system implements it. Evidence would require recoverable state information, lawful action availability, held-out prediction, endpoint support, and discrimination from ordinary control or selection.

Neural plausibility is vulnerable to reverse inference, and local control routes face an information burden. Salience, valuation, interoception, and control networks support many computations; a familiar activation cannot identify a lifetime cumulative object. The route must state what information is available at a time and how a frozen computational variable predicts held-out neural and behavioral consequences without importing the true endpoint or complete residual into its state description (Seeley et al., 2007; Pessoa, 2017).

Boundary routes require independent endpoint evidence. Hitting, absorption, endpoint atoms, and continuation make different predictions about endpoint hazard and path state. They should be

analyzed as distinct branches rather than pooled into a flexible family that always finds a boundary.

Safe perturbations provide causal leverage only when the contrast and regime are explicit. A justified change may alter pain, sleep, communication, or support, but the route must specify which component changes first and how the pattern differs from ordinary adaptation or control. Prediction under the observed regime does not automatically transport to a changed treatment, action, or continuation policy without assumptions about confounding, interference, endpoint change, and observation (Woodward, 2003; Hernán & Robins, 2020).

A rule for choosing actions faces a sharper question than whether it usually prefers promising paths. Under one fixed continuation model, its completion probability is the average of the action-specific completion probabilities, weighted by how often it chooses each action. That average can equal one only if the rule assigns zero probability to the set of actions whose continuations have any chance of failure. Even a small positive chance of selecting from that set prevents an exact probability-one guarantee.

Limited information creates a further obstacle. Consider two imaginary rooms that look identical to the decision-maker. In one room only the left exit reaches the target; in the other only the right exit does. Knowing that a successful exit exists in each room is not enough. A rule that cannot tell the rooms apart cannot guarantee success in both. It needs information that distinguishes them, or an action that works in both.

For a finite menu of actions, demanding probability-one success at every history compatible with the available information requires at least one action that succeeds with probability one at all those histories, under the fixed continuation model. The route must also supply a consistent measurable rule for choosing across information states. Increasing a soft preference for promising actions may make completion arbitrarily likely while still falling short of probability one. These requirements concern what a model can achieve with its information; they do not identify motivation, awareness, free will, or access to a cumulative total (Chatterjee, Chmelík, & Davies, 2016; Asadi et al., 2025).

Reciprocal-process diagnostics may separate bridge mixtures from a local Markov controller, but hidden state and coarse observation can mimic the same dependence. The diagnostic should narrow route families, not declare a winner beyond its assumptions.

Unexpected endpoints are a severe stress test. A route fitted under an anticipated horizon should face endpoints arriving earlier, later, or by another process, with rules fixed in advance. State-coupled, globally conditioned, and locally controlled routes need not agree outside the event for which they were built.

Adverse evidence should land on the tested object. Failure of an interoceptive correlate weakens that implementation; failure of a local controller may leave a global route open; failure of every declared mechanism narrows the route family. A verified terminal counterexample bears more directly on the target branch.

A reversed prediction is more informative than a vague null. If spread widens where compression was predicted, or a lead-lag order reverses, the prospectively specified implementation has lost. Endpoint, origin, proxy, or objective cannot be changed afterward to rename the result.

Model ambiguity is a scientific result. When routes remain observationally equivalent, the program should identify the measurement, perturbation, or endpoint contrast needed to separate them. Uncertainty never favors the most dramatic story.

Generative identifiability is required. Under realistic noise, missingness, endpoint uncertainty, and parameter overlap, the analysis should recover the correct synthetic family often enough to justify interpretation. If bridges, local controllers, global constraints, and ordinary regulation remain confused, a fitted label in human data cannot identify the generating route.

The same frozen route should transport across modules. A completion potential, reserve state, or controller must face held-out perturbations, endpoint-support changes, and independent channels without redefinition. Separately tuned fits are not one mechanism until shared state, parameters, and transport rules survive common-data comparison.

8.5 — How a Route Can Fail or Be Retired

A route becomes safer and more scientifically useful when its boundaries are stated positively. It is a candidate relation among states, information, transitions, capacities, and endpoints. It is not a deity, fate, secret self, artificial intelligence sovereign, public authority, moral judge, human-worth score, or empirical verdict. Religious meaning remains voluntary, clinical judgment remains with patients and professionals, public authority remains lawful and reviewable, and scientific verdicts remain governed by evidence. These separations leave a cleaner object to test.

Once agency and borrowed authority are removed, the route's burden becomes precise. Which state does it track, what information can it use, how does it change reachable paths, and what distinctive consequences survive capable rivals? Technical ambition is welcome because the object can lose.

8.5.1 — Not God

A constraint can shape a set of possible paths without being an agent. Gravity has no opinion about a falling object, a budget constraint does not resent a purchase, and a boundary condition does not watch the path that satisfies it. The same grammar applies here: a candidate route is a relation among modeled states, not a mind behind the stream.

A naturalistic route may still coexist with religious belief. A reader may interpret life through providence, grace, karma, sacred purpose, or another tradition. Such meaning belongs to a voluntary spiritual register. It cannot serve as the mechanism, evidence, or public warrant for the scientific claim.

The separation protects both science and voluntary meaning. A failed route does not refute a faith, and a spiritual testimony does not validate a route. Constraint is not inevitability or moral assignment: treatment, support, agency, accountability, and structural change can alter the available path. No route output may become a spiritual verdict, clinical instruction, eligibility decision, or declaration that a particular suffering was necessary. The scientific question is narrower: does a declared relation among state, time, capacity, endpoint, and observation make distinctive predictions that survive capable rivals?

8.5.2 — Not Proof

A route can strengthen one account of how closure might occur without establishing that closure occurs. Scientific support must attach to the exact object tested: representation, route state, bridge, endpoint relation, or terminal target.

Formal proof has a precise but limited role. Given declared assumptions, mathematics may prove that a construction reaches a boundary, preserves an invariant, or has a stated distribution. The proof establishes the consequence inside the model. It does not establish the assumptions as facts about conscious life.

Coherence is the entry condition for empirical work. An incoherent route should be rejected before study; a coherent route earns the right to face data. Elegance may improve understanding, but only discriminating evidence can raise the empirical verdict.

Model fit is also limited. A route may describe the observations used to estimate it and still fail on new data. Flexible models can absorb noise, confounding, and endpoint selection. Fit becomes

more informative when the route is frozen and tested on held-out transitions, perturbations, channels, and sites.

Evidence should climb gate by gate. Researchers may validate a channel, recover a route state, distinguish a route from rivals, or confirm a route-specific prediction; none silently establishes the terminal target. Direct terminal inference would additionally require adequate bearer and endpoint rules, a modeled observation process, a recoverable measured-to-latent bridge, and a bounded terminal remainder. Ordinary intuitions about depletion or repair can motivate tests, but they supply neither proof nor moral desert.

Research variables must not become administrative tools. A reserve proxy, horizon estimate, path set, or neural correlate cannot become a public score, eligibility label, clinical rank, spiritual status, or measure of worth. An uncertain construct does not gain authority by being displayed on a dashboard.

Public compression should name both the tested object and the ceiling. “One candidate route was favored under these conditions” differs fundamentally from “the Law was proven.” A headline that omits the object invites a conclusion stronger than the evidence supports.

Negative results need the same precision. A route may fail because its signature is absent, because a rival predicts better, because the state cannot be recovered, or because the study is ethically unavailable. Those outcomes narrow different parts of the architecture.

Correction is part of proof discipline. If a route loses, diagrams, summaries, talks, and websites should update with it. A metaphor that continues circulating after the model has been retired can create a public certainty the science no longer holds.

The strongest public conclusion may be modest and precise: one candidate route predicted a prospectively specified pattern better than declared rivals under stated conditions. That result can guide the next test. It cannot establish that a person's life is balanced or authorize an institution to act on the hidden object.

8.5.3 — Candidate Route Architecture Retired or Narrowed

A route earns credibility in stages: coherent verbal possibility, mathematical construction, identifiable observables, successful model recovery, held-out prediction, transport, and replication. Each stage is a real achievement and introduces a sharper discriminating test. Skipping one creates the appearance of maturity without its evidence.

Representation failure retires a route when its state cannot be related to felt valence without arbitrary transformations. A strong fit to physiology or behavior may still be useful, but it does not explain the phenomenal target until that bridge is identified.

Endpoint failure occurs when the route works only after the boundary is moved, smoothed, or reclassified in light of the trajectory. Endpoint uncertainty can be modeled prospectively; outcome-dependent revision converts pressure into rescue.

Route-state nonidentification occurs when reserve, horizon, capacity, channel availability, or control pressure cannot be separated from frailty, diagnosis, medication, or observation

quality. Simulation-based model recovery should expose nonrecoverable regions before more human data are interpreted.

Prospective prediction and safe perturbation test whether a frozen route travels across tasks, later windows, sites, and timed responses. Generic improvement after treatment remains valuable care evidence without identifying a route unless its specific predicted pattern appears.

Rivals merit equal-budget prediction, calibration, complexity control, and diagnostics. A precise opposite pattern, failed transport, or simpler superior model should defeat or narrow the implementation, not be absorbed by the Law-specific label (Burnham & Anderson, 2002; Gneiting & Raftery, 2007).

Transport failure may reveal noninvariance, context specificity, or a narrower class. Ethical stop conditions and independently justified narrowing protect people and clarify the evidence ceiling; neither becomes indirect support. A narrower model can still earn value if its scope is stated prospectively and it continues to outperform alternatives within that scope.

Retirement is cumulative when the record identifies the smallest failed component and updates explanations, diagrams, simulations, protocols, and public summaries together. Public confidence should match status, and negative results should remain searchable so defeated flexibility cannot return under a new name (Munafò et al., 2017; Nosek et al., 2018; Mayo, 2018).

The goal is to find a point where two plausible accounts predict different things and a feasible observation can favor one over the other. What survives this comparison may also improve capacity

measures, endpoint classifications, missing-data methods, or ordinary-recovery models. A protocol-fixed schedule should state when specifications freeze, which changes create a new model, and when repeated failure requires retirement. Continuing beyond the declared review date should require a fresh evidentiary case. Each well-designed comparison can then leave the next experiment with fewer ambiguities and a clearer question.

Chapter 9 — The Shortening Horizon

Time changes more than duration. It changes which actions remain possible, how much delay costs, what information can still be gathered, and whether a path can be reopened. Chapter 9 distinguishes calendar time, perceived future time, biological capacity, and workable opportunity before asking what a closure route might predict.

The shortening horizon is the practical fact that options can close. A deadline passes, illness advances, a child grows, a caregiver's reserve thins, a treatment window narrows, or years of practice become unavailable. The same action can therefore carry a different opportunity cost at different points in a life.

Perceived endings can reshape priorities as well as emotion, but no single late-horizon pattern describes every person. A shorter future may intensify meaning, anxiety, acceptance, urgency, or little at all, depending on health, culture, obligation, relationship, and opportunity. Section 9.1 develops the evidence; the Introduction needs only the reason the horizon belongs in a route analysis.

The route question is how much usable future remains from the present state. Workable opportunity includes health, energy, attention, money, access, social support, legal freedom, knowledge, and actions that can still alter the path. Equal expected duration can therefore contain radically different futures.

A shrinking continuation set may increase pressure, simplify priorities, invite acceptance, or produce little change. The relation is empirical and route-specific. Endpoint proximity is not reserve,

and fewer options do not imply that suffering was assigned or relief is owed.

Viability asks whether at least one acceptable continuation remains. Capacity asks what the system can still do. Compensability asks whether a declared displacement can still be addressed within a candidate route. Their separation turns a poetic horizon into a testable architecture.

Addiction makes that architecture concrete. Immediate reward can capture attention, cue-triggered wanting can outgrow liking, tolerance and opponent processes can intensify relief seeking, and time spent obtaining, using, recovering, concealing, or repairing consequences can displace the practice through which long-term aims become reachable (Solomon & Corbit, 1974; Everitt & Robbins, 2016; Robinson & Berridge, 2025).

Behavior changes possibility through ordinary causation. One lost night may be recoverable; a pattern repeated for years can alter health, trust, employment, skill, savings, and self-expectation. The insight is neither blame nor cosmic withholding, but a clearer account of how paths become harder or reopen.

Moderation protects optionality by preserving attention, sleep, health, money, relationships, and practice. It purchases no guaranteed reward. Its value is the wider field of futures a person may still pursue amid chance, injustice, illness, and other people's choices.

Aging changes capacity without producing one emotional script. Decline, adaptation, accumulated skill, altered priorities, unequal resources, and continued growth can coexist. The Chapter follows

those interacting pathways so that neither age nor apparent composure becomes a shortcut to the person's remaining possibilities.

Serious illness can shorten and complicate the horizon through symptoms, treatment, prognosis, dependency, and uncertainty. Caregiving can do the same for the caregiver as daily needs displace sleep, work, social life, and long-term projects. Love and burden can coexist without canceling either.

Terminal settings alter both the person and the observation system. Relief, sedation, delirium, fluctuating awareness, family memory, and cultural scripts can shape what is felt, expressed, recorded, and later remembered. A serene or distressed appearance therefore requires interpretation rather than functioning as a terminal reading.

Palliative care remains valuable on its own terms and never becomes a mechanism supplied for the theory. Relief, communication, family support, and proportionate clinical judgment govern the bedside; research accepts the inferential cost when care, fatigue, or refusal limits observation. Section 9.4 develops the clinical detail.

Terminal research also faces informative selection. Illness progression, device removal, family refusal, care transition, communication loss, and death between assessments can end observation. Missingness near the endpoint is often part of the process being studied, so denominator flow and competing events belong inside the analysis (Diggle & Kenward, 1994; Little & Rubin, 2019).

In ordinary life, the shortening-horizon lens can protect time and opportunity without pretending to reveal how much time remains or what experience is owed. Its practical value is causal seriousness, earlier repair, and respect for paths that can still be reopened.

Part III closes where possibility meets evidence. Exactness has created structural pressure, route families have supplied testable alternatives, and time has shown how reachability can change. Part IV now asks how reports, behavior, physiology, neural data, context, and missingness can inform an estimate while the conscious stream remains the object those instruments are trying to understand.

9.1 — The Shortening Horizon

A shortening horizon changes the meaning of opportunity, not the moral value of a life. Comfort, analgesia, safety, reconciliation, a familiar voice, reduced fear, or the end of an unwanted demand can matter greatly near an endpoint. Their importance does not turn relief into moral repayment or retroactively justify prior pain.

The intuition becomes scientifically interesting when it yields a prospective contrast. Under matched estimated burden and biological reserve, does a shorter horizon alter action selection, admissible-set size, control effort, or affective dynamics beyond ordinary urgency? If no such interaction appears, the intuition does not become mechanism merely because it feels persuasive (Carstensen, 2006; Wirth et al., 2024).

A horizon is first encountered as lived possibility. A child has years in which protection, education, relationship, and repair may still unfold. A person waiting for a diagnosis may feel a future contract into uncertainty. Someone leaving danger may gain a future that did not exist in practical terms the day before. These experiences make clear that time matters because events need room to develop (Carstensen, Isaacowitz, & Charles, 1999; Mather & Carstensen, 2005).

The formal model, however, cannot begin by calling every open possibility a horizon. The estimated time horizon later receives a precise time-domain definition. This Section stays one level closer to life. It asks what it means for channels to remain open, to close, or to become usable again, and why that structure may motivate a horizon variable.

9.1.1 — Time Remaining as Constraint

The formal hypothesis suggested by this intuition is not that later experience is worth more. It is that the mapping from current burden to feasible future change may depend on the remaining interval. That dependence can be tested as an interaction rather than presumed as a moral truth.

Time remaining becomes a constraint because many forms of repair require duration. Apology may require conversation. Medical relief may require access, diagnosis, treatment, response, and adjustment. Emotional recovery may require repetition, safety, sleep, relationship, and ordinary days that prove the harm is not still happening.

This does not mean the value of a person diminishes as time shortens. The Law of Fairness never treats remaining time as human worth. A person with minutes left has the same dignity as a person with decades left. Horizon affects possible repair pathways, not the value of the bearer (Carstensen, Isaacowitz, & Charles, 1999).

Calendar time and lived possibility must remain distinct. A prospective horizon estimate concerns time to a declared endpoint, while fear, pain, care, safety, and social conditions shape which actions feel or become available. The model becomes testable only when it represents both layers separately.

Time remaining as constraint also clarifies why the Law cares about sequence. If a burden occurs before repair channels have opened, its trajectory differs from a burden that occurs after most channels have closed. Sequence does not justify the burden. It

specifies the shape of the burden inside the stream (Carstensen, 2006).

The phrase fewer channels names the narrowing of ways a burden can be answered. A person may need time for treatment, conversation, legal repair, sleep, memory integration, safety, spiritual reflection, or simple bodily rest. When fewer channels remain available, a burden may press more sharply because the future has fewer places to receive it (Wirth et al., 2024).

Greater pressure does not mean greater moral importance in a way that ranks persons. It means the formal model may need to account for the changing difficulty of compensability. The same negative valence may become harder to answer when the paths through which relief could arrive are fewer, weaker, or less accessible.

Constraint can also increase when biological capacity declines or when a practical channel is blocked. Energy, neurological flexibility, trust, safety, and support belong to different layers and should not be folded into one reserve number. The architecture is useful only if those sources of narrowing remain distinguishable.

The distinctive research question is whether a declared pressure variable predicts action selection, pruning, or control signals beyond symptom severity, ordinary time pressure, fatigue, and opportunity. If it does not, the horizon-specific interpretation has no scientific advantage (Carstensen, Isaacowitz, & Charles, 1999).

The channel language should be understood broadly. A channel may be biological, relational, practical, institutional, aesthetic,

spiritual by choice, or simply experiential. Medication can open a channel by reducing pain. A visitor can open one by restoring human contact. A legal remedy can open one by ending fear. A quiet room can open one by reducing overload. Fairness is not only about what happened; it is about what remains possible after what happened.

9.1.2 — Late Relief

Late relief can be studied only through care-first observation or independently justified treatment. Researchers may ask whether analgesia, safety, communication, or freely chosen reconciliation changes state, choice, or measured burden differently under declared horizons, but care must never be delayed, reduced, or timed for a cleaner signal. A route that explains the pattern only after events occur has become narrative rather than mechanism.

Late relief may be concentrated because a narrowing horizon makes each available channel more consequential. Pain control, a familiar voice, freely chosen forgiveness or reconciliation, a safe room, the end of fear, a hand held without demand, the return of dignity, or a moment of lucidity may transform the final part of experience without being owed or assigned.

Relief is valuable because it reduces present burden. Its arrival does not establish a person's terminal balance or justify earlier suffering or neglect.

Across health, public administration, law, private life, and material support, the first duty is to keep relief available before interpretation. In medical research, participants' health and rights take priority, and independent ethical review must assess risks

and burdens (World Medical Association, 2024). This book accordingly treats pain control, dignity, protection, and practical support as obligations, and permits terminal observation only when it remains compatible with care and any added research burden is minimal, proportionate, and does not alter indicated treatment.

Late burdens also expose the failure of public biography. A life that appeared successful may close under loneliness or fear. A life that appeared modest may close under deep care and relief. Visible status cannot predict terminal burden. The Law's emphasis on the stream remains necessary precisely when the public story is least reliable.

Delayed repair has a distinct public cost. When institutions withhold pain treatment, accessible care, income support, legal remedy, companionship, or a listening clinician until little workable future remains, they have not merely postponed an external benefit; they have shifted burden into a narrower horizon. The inquiry should identify who closed the route and why relief was unavailable sooner.

Mortality itself is not an institutional failure. The institutional questions concern untreated pain, poverty at the end of life, bureaucratic delay, abandonment, and inaccessible services. These are conditions for policy, care, and relationships to address, rather than traits to assign to a person or hide inside a single time variable.

Late relief can appear in many registers. It may be bodily relief, social relief, relational repair, existential peace, sensory comfort,

removal of fear, or simple cessation of struggle. Different cultures and persons will interpret these forms differently. The model must not impose one script of what relief should look like.

Research language must separate observed relief from inferred cumulative closure. A peaceful sign may reflect medication, exhaustion, delirium, family narration, cultural expectation, or genuine felt calm. All may matter in different ways, but none gives direct access to the hidden terminal cumulative value.

Even a sharply narrowed horizon can contain real choice, comfort, repair, and human presence. The question is not whether an ending has been scripted, but which burden can still be reduced, which connection can still be offered, and which refusal must still be honored without asking the person to perform closure.

9.1.3 — Ordinary Examples Without Proof

Ordinary deadlines reveal the shortening-horizon intuition without establishing a terminal law. A student can recover from one unproductive week early in a semester. The same week near the final examination may remove the last opportunity to learn foundational material. The difference is not cosmic pressure. It is the shrinking set of corrective paths.

Skill learning has a horizon because language, music, athletics, and technical crafts depend on practice that cannot always be compressed at the end. A person with years before an audition or qualification can experiment, fail, rest, and rebuild; near the deadline, each lost hour removes more possible strategies.

Recognition may motivate action, but it cannot restore every missed practice window.

Recovery from addiction can begin at any age, but the available route changes with health, employment, housing, legal exposure, and social support. Early intervention may preserve more options. Late intervention can still be life-changing. The horizon idea should motivate support rather than fatalism.

Caregiving shows the same complexity. A caregiver may postpone rest for a day without lasting loss. Repeated postponement can accumulate into illness, isolation, financial strain, and resentment. The useful inference is that capacity must be replenished, not that the burden is part of a hidden balancing plan.

These examples also show where the intuition fails. Short deadlines can focus attention, long horizons can invite delay, some people remain creative under severe limits, and support can reopen paths that seemed closed. The examples are analogies to reachability, not evidence that a latent cumulative object exists or that terminal closure governs a life (Baltes, 1997; Freund & Baltes, 2002).

Perceived future time can shift attention toward emotionally meaningful goals when the future feels limited, yet the pattern varies with health, culture, obligation, and opportunity. Socioemotional selectivity research therefore offers a family of tendencies rather than a universal script for aging or terminal life (Carstensen, Isaacowitz, & Charles, 1999; Carstensen, 2006, 2021; Wirth et al., 2024).

Institutions can lengthen or shorten horizons. Predictable schedules, accessible health care, legal status, transportation, child care, disability accommodation, and reliable income can make plans feasible that personal discipline alone could not sustain. A horizon analysis that ignores those structures would mistake unequal opportunity for a private failure of foresight (Sen, 1999; Nussbaum, 2011).

Some windows close irreversibly, while others reopen. A missed filing date, untreated infection, lost training season, or destroyed habitat may remove routes that later effort cannot fully restore; appeal, treatment, technology, education, reconciliation, legal remedy, or new support can create new ones. A theory of the workable future must therefore allow discontinuity and restoration rather than one smooth, monotonic decline.

Option preservation often requires slack. Spare time, savings, social trust, institutional redundancy, respite, and physical reserve may look unused in a static picture, yet uncertainty makes their value visible when a surprise arrives. A system with no buffer can function until one interruption turns a narrow plan into crisis. Slack also preserves reversibility, giving a person or institution time to detect a bad path and change course before one setback closes the remaining options.

9.2 — Time, Capacity, and Opportunity

Usable time is not calendar duration alone. Section 9.2 asks how health, energy, resources, support, and available action determine how much of a future can still be used.

A candidate route may predict rising constraint pressure as the workable future shrinks. That pressure is not a moral price and does not imply that later pain or relief is more valuable. It describes how fewer admissible paths can make correction, recovery, or adaptation harder (Aubin, 1991; Bertsekas, 1995).

Section 9.2.3 distinguishes viability, capacity, and compensability; none can be inferred automatically from another.

9.2.1 — What Makes a Future Workable

The workable future is the set of continuations that remain realistically available from a present state. It is not everything a person can imagine, and it is not a prediction of what will happen. It includes paths that can still be reached through available time, bodily capacity, material conditions, relationships, information, and action.

Chronological time is only one component. A person may have years ahead and little practical freedom because of danger, untreated illness, debt, imprisonment, caregiving overload, discrimination, or inaccessible services. Another person may have little time and still possess meaningful choices about comfort, conversation, place, ritual, treatment, or rest.

Biological capacity is a second component. Strength, cognition, pain, sleep, sensory access, and recovery affect which actions can

be sustained. These capacities change unevenly and can improve with treatment or accommodation. Age or diagnosis alone is therefore a poor substitute for the workable future (Baltes, 1997).

Material and institutional conditions form a third component. Transport, money, housing, legal status, workplace flexibility, health care, and communication access can open or close paths that personal resolve cannot create. A model that ignores these conditions risks translating structural inequality into individual failure (Sen, 1999; Nussbaum, 2011).

Psychological selectability adds a fourth layer. An option may exist on paper while fear, trauma, craving, shame, exhaustion, or learned helplessness makes it feel unreachable. Support, therapy, safety, information, and repeated success can make the same option feel possible again. The route account should represent this difference without pretending to read private experience directly.

The workable future can widen even while the calendar moves on. Imagine a parent who wants to attend an evening class but has no reliable care for a dependent relative. The class exists; the intention is real; the route is blocked. A dependable two-hour period of respite each week can make that route usable. No hours have been added to the week. What changed is how the hours can be lived.

Recovery, symptom relief, accessible transport, education, and safer conditions can reopen other paths in the same practical sense. Some losses remain irreversible, but a shortening horizon does not require every possibility to shrink with it. Hope becomes

concrete when we can name the barrier that changed and the future that change made available.

The horizon relevant to a prospective model must be estimated from information available at the time. Researchers cannot use the actual future endpoint while pretending to predict earlier decisions. Prognosis, task duration, contract deadlines, or declared study windows may provide usable proxies, each with different error and meaning.

A point estimate is rarely enough. Remaining time may be uncertain, and that uncertainty can be wide even when the expected horizon is short. The model should carry a distribution or interval and ask whether its predictions survive plausible endpoint variation rather than treating one forecast as fate.

Calibration is essential. Cases assigned similar horizon estimates should, under the declared endpoint rule, show corresponding outcome frequencies over time. A model that ranks cases well but systematically overstates or understates remaining time is not a trustworthy basis for a horizon effect (Gneiting & Raftery, 2007).

It also has a strict public limit. The workable future is not a score of a life, a ranking of people, or an eligibility test. An institution may assess a specific need or service constraint under ordinary law, but it may not claim to know the total value of a person's remaining possibilities.

A strong design therefore compares several models: time alone, capacity alone, context alone, and their interactions. It examines whether the horizon contribution transports to held-out tasks and populations.

9.2.2 — Why Constraint Pressure Might Rise

The core intuition is familiar: the same correction can become harder when fewer opportunities remain. A route that can be completed through ten small adjustments may become infeasible when only one decision point remains. Control theory and viability analysis formalize versions of this intuition, but the public claim is conditional rather than universal (Aubin, 1991; Bertsekas, 1995).

Skill learning offers another. A musical performance, examination, or professional qualification may require months of practice. As the date approaches, the same remaining gap demands more concentrated effort and may eventually become unreachable. The deadline changes feasibility, but it does not guarantee success or justify overwork.

Relationships have their own windows. Apology, trust, and reconciliation often require time, safety, and repeated action. A late conversation can matter profoundly while leaving some injuries unrepaired. The horizon account helps describe the changing possibility without turning forgiveness into a required endpoint.

Constraint pressure may also rise because capacity declines rather than time alone. Pain, fatigue, cognitive impairment, withdrawal, or loss of support can make each available action harder. A model must distinguish fewer remaining opportunities from reduced ability to use them.

Blocked channels create a third route. A person may retain time and strength while losing communication, transport, money,

privacy, or legal permission. The feasible set narrows because a path is closed, not because the endpoint is near. This distinction is essential for responsibility and for empirical identification.

Late relief can therefore widen lived possibility even as chronological time shrinks. Effective analgesia, clear communication, safe housing, a reconciled relationship, or the removal of an impossible demand may reduce strain. A route that predicts only increasing pressure would miss these ordinary and clinically important reversals.

The model should state what pressure means. It might refer to action pruning, increased effort, reduced tolerance for delay, greater sensitivity to error, or a higher risk that no acceptable continuation remains. These outcomes are not interchangeable, and each requires a distinct measure.

Scarcity and bandwidth effects, developed in 9.3.1, are strong ordinary rivals, alongside stress, habit, loss aversion, and ordinary planning; any horizon account must predict something beyond them.

Model recovery is again decisive. Simulated datasets should vary time, capacity, channel blockage, and observation error independently. The planned analysis should recover which factor generated the pattern. If it cannot, human-data estimates should remain descriptive.

A missing effect is possible. Under adequate measurement and power, shorter horizons may not increase action pruning or effort. That result would count against the prospectively specified route.

A reversed effect may favor priority simplification, acceptance, or a rival regulatory account.

Public language should therefore say that constraint pressure may rise under declared conditions. It should not speak of an endgame, price a person's future, or claim that suffering must intensify in a particular way to force balance. Exact PTVC constrains the signed remainder, not a universal trajectory of increasing pain. The practical lesson is to protect time, capacity, and channels while they can still be protected.

9.2.3 — Viability, Capacity, and Compensability

Viability asks a narrow question: from the present state, does at least one admissible continuation remain? It does not ask whether the continuation is likely, desirable, easy, or fair. A path can be viable in the formal sense while being cruel, inaccessible, or dependent on resources that the person does not possess (Aubin, 1991; Aubin, Bayen, & Saint-Pierre, 2011).

The first object is therefore the continuation set. It contains the future paths allowed by the declared model, bearer, endpoint, clock, constraints, and intervention regime. If that set is empty, exact completion through that route is impossible. No amount of optimism or retrospective storytelling creates a missing continuation.

The horizon is also not capacity. Time remaining limits what can happen, but two people with the same horizon may have very different options. One may have controlled symptoms, communication, trusted care, and family support; another may

face pain, fear, isolation, or inaccessible services. A clock alone cannot represent the continuation set.

Prognosis is different again. A clinical estimate describes likely disease or survival outcomes under a model. It does not identify the truth-level endpoint of consciousness or the directional phenomenal capacity. A prognosis may inform a route proxy while remaining uncertain, revisable, and ethically separate from the formal object.

Reserve is one proposed summary of remaining modeled capacity. It can be useful when it compresses relevant constraints into a testable state, but it may hide which option was lost and why. A broad reserve index should therefore be accompanied by component measures and failure analysis rather than treated as a natural quantity.

Viability can change discontinuously because ordinary interventions and infrastructure alter the continuation set. Treatment, a communication device, housing, transport, respite, legal protection, or the arrival of a caregiver can reopen sleep, movement, safety, planning, or exit; a complication or withdrawal of support can close them. These are measurable causal changes in possibility, not evidence that support completes a hidden ledger.

The route should distinguish structural actions from observations. Measuring a person more often does not widen capacity. A questionnaire can reveal a problem, but treatment, access, rest, safety, or relationship repair changes the conditions. The model should never confuse data collection with help.

Partial observation creates two opposite errors. Researchers may declare a path impossible because they failed to observe a private choice or available resource, or imagine actions that are not genuinely available because of cost, disability, coercion, law, geography, or trust. Feasibility must therefore be checked in the lived setting rather than assigned from an abstract menu.

Safe intervention can test a bounded route component when the action and continuation regime are declared. If justified support widens communication or reduces symptom burden, the model may predict a change in the attainable set or reserve proxy, but causal interpretation still depends on subsequent care, adherence, environment, interference, endpoint change, and observation. The intervention remains valuable even when the predicted mechanism fails (Hernán & Robins, 2020).

The reserve-relative quantity formed by dividing a residual estimate by remaining capacity is especially fragile near a small denominator. Modest error can produce an apparently enormous pressure. Analyses should report numerator and denominator separately, impose a reserve floor, and show sensitivity to both estimates.

A rising ratio is therefore not evidence of a person approaching a required balance. It may indicate arithmetic instability, declining observation quality, or a misspecified proxy. Any public explanation should describe the modeled relation rather than turning it into a countdown or a private score.

Model recovery and equal-budget rivals should precede application. Synthetic paths must include known continuation

sets, support changes, observation error, endpoint uncertainty, and ordinary control, scarcity, frailty, threat, fatigue, or social-support processes. The analysis should identify viable, empty, and ambiguous states without manufacturing a terminal objective where none exists (Bertsekas, 1995; Bobba-Alves, Juster, & Picard, 2022).

The language should remain local when the evidence is local. A study may support that a named intervention widened a measured option set in one task. It may support that a reserve proxy improved held-out prediction. Neither result establishes complete-stream compensability or exact terminal closure.

The ethical and public-use boundary is direct: no one may be denied care, pressured to endure suffering, classified as nonviable, or scored for resilience, deservingness, prognosis, eligibility, or future value because a research model predicts limited capacity. Institutions may assess concrete needs under ordinary law and professional standards, but they may not convert an uncertain route variable into a judgment about the person.

9.3 — Addiction, Delay, and Future Optionality

Addiction is one of the clearest places where a discussion of future possibility can become either useful or harmful. The useful account begins with reward learning, attention, craving, tolerance, time displacement, and opportunity cost. The harmful account begins by imagining that pleasure is spent from a fixed lifetime budget.

The analysis remains human rather than moralistic. Vulnerability, cue exposure, pain, trauma, poverty, isolation, genetics, treatment access, and social design all matter. A person can be responsible for choices without being treated as the sole author of the choice architecture.

9.3.1 — Short-Horizon Reward and Attention Capture

Addiction often contracts attention before it contracts a life plan. A cue, opportunity, bodily state, or expectation can dominate the next minutes even when the person cares deeply about tomorrow. The immediate action becomes vivid while delayed costs remain abstract.

Behavioral science describes part of this pattern through temporal discounting and reinforcement learning. Outcomes available soon can receive more behavioral weight than outcomes available later, especially under stress, uncertainty, sleep loss, pain, or withdrawal. That shift is not a moral defect; it is a state-dependent feature of choice (Sutton & Barto, 2018).

Incentive-sensitization theory distinguishes wanting from liking. Repeated exposure can amplify cue-triggered wanting even when

the pleasure obtained from the reward does not increase and may decline. That separation helps explain why a person can pursue something they no longer enjoy in the ordinary sense (Berridge & Robinson, 1998; Robinson & Berridge, 2025).

Scarcity can intensify the narrowing. Financial precarity, unstable housing, threat, loneliness, and untreated illness consume cognitive bandwidth and make immediate relief more compelling. These conditions are not part of a hidden fairness mechanism; they are ordinary causes that institutions can change (Shah, Mullainathan, & Shafir, 2012; Mullainathan & Shafir, 2013).

Safe interventions can test the account. Removing cues, changing context, improving sleep, treating pain, providing medication, delaying access, or making an alternative immediately available may change attention and choice. The benefit stands on ordinary clinical and practical grounds even if the Chapter's broader route interpretation fails.

When the future has become unreliable, favoring immediate relief need not be irrational in the simple sense. The practical task is to make safer future goods credible, reachable, and near enough to compete with the present cue (Frederick, Loewenstein, & O'Donoghue, 2002).

Attention capture also changes what evidence enters deliberation. A person may know that a choice threatens work, health, or relationships while the cue temporarily occupies the foreground and those consequences lose vividness. Treatment and supportive environments can help by altering cues, routines, friction, and

social context rather than demanding that abstract knowledge overpower a narrowed attentional field on its own.

This account preserves agency without turning agency into blame. People still choose, plan, resist, seek help, relapse, and rebuild, but the difficulty of those acts is shaped by learning and circumstance. A fairness analysis should make those constraints visible so that responsibility is paired with treatment, practical support, and a genuine route back to wider options.

9.3.2 — Tolerance, Craving, and the Loss of Ordinary Effort

Tolerance means that repeated exposure can reduce the effect produced by the same dose or behavior, leading some people to seek more for a similar result. The process varies by substance, behavior, context, physiology, and time. It should not be treated as one universal staircase.

Opponent and antireward processes add another layer. Repeated use can recruit stress and regulatory systems that make absence increasingly aversive, while incentive sensitization can let cue-triggered wanting strengthen even when liking declines. Use may then be maintained less by pleasure than by relief from withdrawal, dysphoria, pain, anxiety, or bodily disequilibrium (Koob & Le Moal, 2008; Robinson & Berridge, 2025).

Ordinary effort can lose ground when attention, sleep, health, and motivation are repeatedly consumed. Work, study, cooking, exercise, caregiving, and conversation may demand more energy while the addictive action becomes increasingly practiced and immediately reinforcing.

This is not proof that control has vanished. It means that action costs have changed. The person may retain capacities that emerge in a safer context, under medication, after withdrawal treatment, or with social support. A single failure should not be converted into a fixed character judgment.

Addiction is better treated as a chronic, treatable health condition than as evidence of vice. Long courses, recurrence, remission, and multiple treatment episodes are common enough that one moment of use or abstinence cannot summarize the whole trajectory (McLellan et al., 2000).

Treatment can act on several mechanisms at once. Medication may reduce withdrawal or craving. Therapy may change cue response and coping. Stable housing, pain care, sleep, relationships, and meaningful work can lower background pressure and make alternatives more reachable.

Nothing in these mechanisms implies that addiction was assigned, necessary, or purchased by later recovery. Growth can follow suffering without making the suffering required. The proper response is evidence-based care, reduced stigma, and protection of the person's future options.

Withdrawal and negative reinforcement can then change the function of use. What began as pursuit of reward may become an attempt to end distress, restore ordinary functioning, or quiet an aversive state. The horizon narrows because relief from the next hour competes against benefits that would arrive weeks or years later.

Ordinary effort can recover, but recovery is not an instant reversal of the learning history. Sleep, nutrition, medication where appropriate, psychotherapy, peer support, housing, work opportunity, and trustworthy relationships can gradually make other rewards available again. The relevant question is not whether discipline earns compensation. It is whether the environment and the person together can rebuild a field of actions in which more than one future is genuinely reachable.

9.3.3 — Time, Opportunity Cost, and Abandoned Goals

Every repeated action uses time that cannot be spent twice. Acquisition, use, recovery, concealment, travel, waiting, and conflict can occupy hours that would otherwise support sleep, relationships, work, education, health, or rest. The loss is not only the event; it is the field of alternatives that the event displaces.

Opportunity cost becomes visible when the same cycle repeats. A missed appointment, one delayed bill, or one abandoned plan may look small in isolation. Repetition can turn those local costs into unstable housing, lost employment, untreated illness, legal exposure, or distance from people who once provided support.

Relationships carry another part of the cost. Family members may monitor, rescue, withdraw, set boundaries, provide money, seek treatment, or protect children. Their burden is real, but it cannot be added to the person's private cumulative object. Each stream remains separate even when lives are causally entangled.

Work and education can fail through absence, reduced concentration, stigma, discrimination, or rigid systems that offer no path back. Institutional response therefore affects optionality.

Punishment that removes housing, employment, health care, or family contact can deepen the very narrowing it claims to correct.

Goal abandonment is often gradual. A person may stop imagining a degree, a relationship, a move, a craft, or a healthier future because the next day already feels occupied. The loss of aspiration is not evidence that the goal never mattered. It can be a consequence of repeated short-horizon pressure.

Paths can reopen. Treatment, medication, recovery housing, peer support, family repair, legal relief, education, and stable work can restore practical options. Recovery is therefore more than abstinence in a narrow window; it can involve rebuilding the conditions under which longer goals become believable and selectable.

The idea of recovery capital captures part of this rebuilding. Resources can be personal, social, material, cultural, and community based. They do not form a hidden fairness reserve, but they can widen the actual menu through which a person protects health and pursues a future (SAMHSA, 2014).

Recent public-health thinking increasingly treats recovery support as a long-term community task rather than a brief episode after crisis. Clinical treatment, peer support, housing, employment, and social belonging may need to work together over years (Kelly et al., 2025).

A path-sensitive account should measure both loss and reopening. Time displaced, goals abandoned, relationships strained, and health opportunities missed belong beside treatment engagement,

restored sleep, safer housing, renewed work, and reconnection. One snapshot cannot represent that movement.

9.3.4 — Moderation Preserves Options, Not Merit

Moderation is often presented as virtue rewarded by success. A more useful account treats it as a strategy for preserving the resources from which many futures can still be built. That framing keeps restraint connected to freedom rather than moral accounting.

The preserved resources are concrete: time, sleep, attention, money, physical health, credibility, emotional range, and relationships. A person who retains these resources can attempt more routes, recover from more errors, and sustain longer projects. The benefit is expanded optionality, not a cosmic claim on future pleasure (Fujita, 2011).

Moderation also differs by domain. A substance may require abstinence for safety, while food, entertainment, work, exercise, or social media may call for limits rather than elimination. The correct boundary depends on risk, vulnerability, context, and professional guidance. One slogan cannot govern every behavior.

Self-control is not only resistance in the moment. It includes selecting environments, reducing cues, setting defaults, seeking support, planning for fatigue, and making desired actions easier. Effective regulation often works upstream, before the strongest conflict arrives (Fujita, 2011; Duckworth & Gross, 2014).

Aspirational dreams need the same realism. Hard work can increase skill and opportunity, but it cannot guarantee recognition, health, wealth, love, or survival. Institutions can

exclude. Markets can change. Accidents happen. Other people remain free. The Law must not turn effort into a promise that nature will deliver an emotionally equivalent prize (Duckworth & Gross, 2014).

Moderation should therefore increase freedom rather than shrink life into denial. Rest, celebration, pleasure, play, and spontaneity are part of a worthwhile life. The aim is not to avoid pleasure but to prevent one reward loop from monopolizing the future (Fujita, 2011).

An option-preserving account must also include treatment. Substance use disorders can become chronic and recurrent, yet they remain treatable, and recovery may require medication, psychotherapy, contingency management, medical care, harm reduction, and long-term support rather than a single act of will (McLellan et al., 2000; Volkow et al., 2016). The practical question is which combination restores enough safety and capacity for the person to choose among futures again.

Housing, transportation, income, trustworthy relationships, health care, legal stability, and meaningful activity can strengthen recovery capital and reopen routes that craving or crisis had closed (Cloud & Granfield, 2008). These resources should guide care, not become a score by which a person's worth or seriousness is judged.

A lapse does not erase earlier learning or prove that recovery was false. It can reveal a cue, untreated pain, withdrawal, loneliness, stress, or environmental pressure that the prior plan did not adequately address. A humane response updates the route,

strengthens support, and protects life rather than translating recurrence into shame or cosmic consequence (McLellan et al., 2000; Everitt & Robbins, 2016).

The language of moderation must therefore remain sensitive to unequal starting conditions. Someone living with trauma, unstable housing, untreated pain, aggressive marketing, criminalization, or scarce clinical access does not face the same option set as someone surrounded by safety and support. Preserving optionality is partly personal practice and partly the public work of making healthier routes materially available (SAMHSA, 2014; Volkow et al., 2016).

Harm reduction can preserve life and optionality when abstinence is not immediately achievable. Its value does not depend on one ideal recovery sequence (Volkow et al., 2016).

A preserved option is not an obligation to use it. People may choose rest over ambition, privacy over visibility, or one valued relationship over a wider menu. The aim is not maximal choice at every moment. It is enough workable freedom that a person can continue to revise a life without one reward loop dictating every next step.

9.4 — Aging, Illness, Caregiving, and Terminal Scripts

Human beings rarely approach serious illness or death without language already waiting for them. Families, clinicians, cultures, and traditions offer scripts of struggle, peace, acceptance, redemption, denial, dignity, or unfinished business. These scripts can shape attention, expression, memory, and interpretation, which makes them humanly important and scientifically consequential.

Social performance also matters. Patients may feel pressure to appear peaceful, grateful, brave, reconciled, faithful, or ready, while families may feel pressure to narrate a death as meaningful. Clinical and cultural ideals of a ‘good death’ can shape what is expressed, recorded, and remembered (Kellehear, 2007; Chochinov, 2006).

Hospice and palliative professionals hold valuable practical knowledge. Repeated observation across patients can reveal patterns, care needs, and warning signs that laboratory researchers would miss. Professional testimony remains observational and situated. It becomes stronger through structured documentation, comparison, and attention to alternative explanations, not through reverence alone.

Medication is one such explanation. Analgesics, sedatives, anxiolytics, antipsychotics, oxygen, and other interventions can alter pain, arousal, movement, speech, memory, and responsiveness. Those changes may be beneficial care effects. They also mean that outward calm or reduced report cannot be treated as an unmediated window into felt neutrality.

Delirium and fluctuating consciousness create another challenge. Attention, orientation, language, recognition, and affect can vary over minutes or hours. A lucid interval may be precious to a family and still resist mechanistic interpretation. A period of confusion may reflect disease or treatment without revealing the state of a hidden cumulative object.

9.4.1 — Aging, Illness, and Changing Capacity

Aging changes the space of possible action without moving everyone along one path. Strength, sensory function, processing speed, recovery time, memory, health, social role, income, and support change at different rates, while knowledge, emotional skill, judgment, and selective attention may remain stable or improve. A single age variable or one reserve score would erase the heterogeneity the model needs to explain (Baltes, 1997; Carstensen, 2006).

Life-span psychology describes adaptation through selection, optimization, and compensation. People may narrow goals, invest in strengths, use tools, recruit help, or replace a lost route with another. These strategies preserve agency without denying decline (Baltes, 1997; Freund & Baltes, 2002).

The Law-based application concerns reachability. A route that assumes stable capacity across the lifespan is implausible. Any candidate account must allow the set of future experiences and actions to change with the body, environment, and support network.

Decline must never be interpreted as correction. Disability, frailty, dementia, and serious illness do not settle an account.

They create needs for accessibility, treatment, protection, and care. The ethical response is determined by present dignity, not by a theory of terminal balance.

Longitudinal studies of well-being near death find patterned changes, but those patterns vary across people and places and are entangled with morbidity, social orientation, and observation (Gerstorf et al., 2010; Gerstorf et al., 2013). Group trajectories cannot be assigned to an individual future.

A scientifically useful horizon model must therefore be multidimensional, uncertain, and revisable. It should explain what changes, for whom, under which conditions, and how observation remains possible. A single countdown would be both inaccurate and dangerous.

Illness can compress capacity more abruptly. Pain, breathlessness, fatigue, cognitive change, medication effects, appointments, financial strain, and uncertainty can consume the attention needed for other goals. The relevant horizon is therefore partly experiential and practical, not only prognostic (Gerstorf et al., 2013; Carstensen, 2006).

Disability requires additional care in the language of capacity. A limitation in one environment may become less limiting when access, technology, assistance, and social expectations change. Describing capacity without describing the environment can mistake exclusion for an intrinsic property of the person (World Health Organization, 2001; Nussbaum, 2011).

Chronic illness also changes time perspective without producing one emotional result. Some people experience narrowed plans,

others build new projects, and many move between grief, adaptation, fear, relief, and ordinary life. The variation matters because a universal terminal narrative would erase the very experience the model claims to respect (Baltes, 1997).

The empirical question is not whether aging and illness affect life. They clearly do. The question is whether declared measures of changing capacity predict route-specific features after ordinary adaptation, prognosis, treatment, and social conditions are modeled. Without that discrimination, the horizon language remains an illuminating analogy rather than evidence.

9.4.2 — Caregiving, Culture, and Family Scripts

The emotional truth of memory and the evidential status of memory are different. A family account can be indispensable to understanding care and meaning while remaining unsuitable for quantifying a terminal mechanism. Respect does not require category confusion.

Memory under grief is not mere error. It is human reconstruction under emotional load. It may preserve truth, transform details, compress time, heighten meaning, soften pain, or protect the survivor. A research program should not treat family memory as either pure evidence or useless sentiment.

If family testimony is used in research, consent and burden must be taken seriously. A family interview can reopen pain. It can create pressure to interpret. It can affect relationships among survivors. Nonparticipant protection matters because one person's story may include others.

Family memory may be most useful for generating questions, not settling claims. It can reveal what families observe, how terminal scripts function, where care helps, and what public language risks. It cannot directly access the hidden terminal cumulative value.

Families deserve a gentler conclusion: a memory can be true, meaningful, and worthy of care without proving a terminal theory. Love does not become more real by serving as evidence (Kleinman, 1988; Kellehear, 2007).

Assessment instruments themselves can fail in severe illness, language difference, sensory impairment, or fatigue. Apparent fluctuation may partly reflect unstable measurement. Clinical adjudication and repeated observation should therefore be reported with uncertainty rather than reduced to a binary lucid or not-lucid label.

Fluctuating consciousness is one reason terminal interpretation must remain slow. A lucid hour may matter deeply to the person and family without becoming a clean research window. A delirious period may carry distress, medication effects, infection, metabolic change, fear, or environmental confusion. The unity gate and endpoint discipline require humility before such unstable access.

Family narratives may also be shaped by guilt, hope, religious belief, cultural scripts, conflict, and the need to tell a coherent story. The more a theory flatters a family's hoped-for meaning, the more cautious the researcher must be. Kindness requires not exploiting grief.

The direction of performance is not fixed. Some people conceal distress to protect family; others display distress to obtain needed care; still others use humor, faith, anger, or silence as authentic relational acts. A scientific account must model expressive incentives and dependence before reading demeanor as valence.

The pressure to perform a good death can burden patients and families. A person may feel expected to be serene, reconciled, spiritually certain, brave, or grateful when their actual experience is fear, anger, confusion, pain, or silence. Care should make room for many ways of dying rather than rewarding one public style.

Expert review before inference must include cultural humility, disability perspectives, palliative medicine, clinical ethics, trauma awareness, and family-burden review. The question is not merely whether a script predicts a report, but whether the research design respects the people whose final days make the data possible.

9.4.3 — Hospice, Medication, and Peace as False Proof

Hospice accounts should be sampled systematically rather than through only memorable cases. Experienced professionals witness recurring patterns of comfort, agitation, lucidity, fear, family dynamics, and acceptance under care-oriented conditions, but hospice exists to comfort rather than to function as a neutral instrument. Routine observations, counterexamples, and uncertainty belong beside striking stories.

Hospice testimony is therefore strongest as high-value clinical and qualitative evidence. Repeated exposure may reveal patterns outsiders miss, while professional training, meaning, selection, and memorable cases can shape recall. The scientific task is to

preserve the witness while separating observation, interpretation, selection, and later memory.

A script can influence experience as well as memory. A person who has been taught a peaceful death ideal may try to embody it. A family may remember signs that fit the script. A caregiver may notice calm because calm is expected. None of this means the peace is false. It means interpretation must be careful.

Cultural scripts vary widely. Some traditions emphasize struggle, vigilance, last confession, unfinished obligations, sacred presence, communal mourning, silence, or bodily signs. A model built around one culture's peaceful-death script would be biased and ethically narrow (Kellehear, 2007; Kleinman, 1988).

Communities may also judge deaths that do not fit expected forms. The Law must not add another layer of expectation.

Hospice populations are also selected. Not every dying person enters hospice. Access varies by country, class, diagnosis, culture, family support, insurance, geography, trust, and institutional referral. A pattern observed in hospice cannot be generalized to all deaths without careful controls.

Professional memory is interpretive as well as observational. Repeated exposure, institutional ideals, training, grief, and the vividness of unusual cases can shape which stories caregivers remember and how they understand them. This does not discredit testimony; it clarifies why context and triangulation matter (Kellehear, 2007; Kleinman, 1988).

Medication belongs in the event model because dose, route, indication, timing, symptom response, and concurrent care can

alter both experience and every observable channel. A calmer body may reflect relief, sedation, reduced panic, altered consciousness, social quiet, or several processes at once; medication cannot be reduced to a binary covariate or calm treated as evidence of cumulative closure.

Relief is not moral contamination of the evidence. Analgesia, anti-anxiety treatment, and symptom control aim to relieve suffering. Proportionate palliative sedation addresses refractory suffering under its own clinical safeguards. None is provided to produce a terminal signature. If treatment makes inference harder, the evidence must bear that burden. The person must never receive less comfort for the sake of a cleaner observation (Surges et al., 2024; Carrasco-Zafra et al., 2024).

Family observations can still matter when their role is defined carefully. Relatives may notice gestures, breathing changes, agitation, recognition, or calm that a brief research visit misses. Their reports can enrich clinical understanding, but grief, expectation, fatigue, cultural scripts, and unequal time at the bedside also shape what is noticed and remembered (Kellehear, 2007; Chochinov, 2006). A dated observation in the relative's own words can preserve what was noticed while keeping its timing available for later comparison.

The best terminal observation is therefore plural and modest. Clinical assessment, symptom history, medication timing, staff notes, family reports, and direct observation may jointly describe the care context without claiming direct access to the whole stream. That richer record improves medicine even when the Law remains unresolved. Placing these accounts on a shared timeline

can make agreement, delay, and disagreement visible without asking one witness to summarize every channel.

Timing and missingness are central. A medication may begin before the observed calm, after it, or in response to symptoms that are themselves changing. Reports may stop because the person rests, loses capacity, refuses, transfers care, or dies between assessments. A terminal-looking pattern is uninterpretable unless event timing and observation shutdown are modeled together. For example, a timeline should distinguish a missed interview during rest from a completed interview that records a change in symptoms.

9.5 — What Horizon Patterns Could Show

The public meaning of a signal must remain narrower than its technical role. A probabilistic estimate used in a study does not become a diagnosis, a life-completion label, a spiritual status, an eligibility criterion, or a family interpretation. Research variables remain inside research governance.

Before any terminal curve is interpreted, the study must separate a descriptive pattern from a causal horizon signature. Reports, behavior, physiology, medication, care context, missingness, and endpoint rules may move together for ordinary reasons; a horizon claim earns status only when those alternatives are modeled prospectively and the data remain capable of producing a negative result.

9.5.1 — Signals Are Not the Stream

A signal is an observation produced through a channel. The stream is the conscious history the model is trying to understand. Confusing the two is the central measurement danger in horizon research because a clear signal can create the illusion of direct access.

The observation architecture should keep at least four objects distinct: the estimated valence trajectory, a candidate reserve proxy, a prospective horizon estimate, and the endpoint rule. Shared sensors or preprocessing can create apparent agreement among these objects even when one artifact drives them all.

First-person report retains its indispensable but fallible role described in Section 7.4.2 (Mauss & Robinson, 2009).

Choice data add relative information. A person may prefer one state, reject a trade, or accept short-term burden for later benefit. Such choices can clarify ordering under specified conditions, but they do not reveal a context-free unit of feeling and should not be treated as one (Krantz et al., 1971).

Physiological channels can track arousal, autonomic flexibility, respiration, endocrine stress, pain-related activation, sleep-wake state, and medication response. The same pattern may arise from fear, exertion, fever, posture, infection, ventilation, or treatment. Apparent objectivity does not remove causal ambiguity.

Neural measures can inform network state and computation, but reverse inference remains a limit. Activity in salience, valuation, interoceptive, or control systems does not reveal a dedicated horizon or closure process. Computational regressors and timing windows must be fixed and compared with alternative accounts (Pessoa, 2017).

Context is not merely noise around the signal. Care setting, diagnosis, symptom burden, medication, family presence, staffing, housing, safety, communication access, and recent events can cause both experience and observation. Removing them mechanically can erase the pathway the study needs to understand (Barrett, 2006; Pearl, 2009).

A variable can also play different causal roles at different times. Distress may attract care, care may reduce distress, and reduced distress may improve reporting. Longitudinal analysis should state whether a variable is a cause, consequence, mediator, confounder, or part of the observation process.

Shared-method dependence is a major risk in multimethod research. Two measures can agree because they use the same device, preprocessing, rater, response scale, or disease-sensitive feature. Convergent evidence is strongest when channels have genuinely different error structures (Campbell & Fiske, 1959).

Missingness is part of the observation process. People may stop reporting because they are more ill, sedated, delirious, private, exhausted, transferred, under-supported, or unwilling to continue. Missing self-report cannot be imputed as neutrality, and missing behavior cannot be treated as absence of experience (Little & Rubin, 2019).

Terminal selection deserves separate analysis. People with stable care, compatible devices, supportive families, and gradual decline may be overrepresented, while sudden, chaotic, under-resourced, or private endings disappear. Denominator flow should be visible from invitation through final observation.

Uncertainty should be decomposed where possible. Sensor error, calibration error, latent-state uncertainty, endpoint uncertainty, missingness sensitivity, model-form uncertainty, and population transport are different weaknesses. One narrow confidence interval can conceal several fragile bridges (Gneiting & Raftery, 2007).

No artificial intelligence system can convert these signals into a person's complete inner record. Prediction of report or behavior is not ownership of experience. The same limit prevents a signal bundle from becoming a score of worth, credibility, eligibility, spiritual standing, or care priority.

A scientifically useful signal remains purpose bounded. It may help compare candidate models, improve care, or identify when a measurement process has failed. It should also have a correction, deletion, and withdrawal route appropriate to the study and the participant's consent.

9.5.2 — Confounding, Selection, and Observation Shutdown

An apparent terminal pattern may arise without a terminal mechanism. Confounding occurs when a third process affects both the estimated horizon and the observed outcome. Illness severity, medication, care setting, family presence, socioeconomic resources, and cultural expectation can all change what becomes visible.

Selection begins before enrollment. People who enter longitudinal or palliative studies differ from those who do not. Language, cognition, access, diagnosis, trust, geography, and caregiver availability shape participation. The final cohort may therefore be unrepresentative before the first measure is taken.

Attrition becomes more informative as illness progresses. Participants may stop because they are weaker, more distressed, hospitalized, sedated, delirious, or dead. When dropout depends on the unobserved outcome, standard complete-case analysis can produce biased trajectories (Diggle & Kenward, 1994; Little & Rubin, 2019).

Observation shutdown is broader than dropout. A channel can disappear while the person remains in the study. Speech may fail. Movement may diminish. A wearable may be removed. Family

reports may replace self-report. A clinical record may become sparse. Each change alters the meaning of the remaining data.

Statistical remedies require assumptions. Selection models, pattern-mixture models, joint models, inverse-probability weighting, and sensitivity analyses can probe missingness, but none restores information that was never observed without additional structure (Rubin, 1976; Tsiatis, 2006; Kurland et al., 2009).

Causal diagrams can make the problem visible. Illness severity may affect medication, ability to report, care setting, and observed affect at the same time. Conditioning on continued reporting can then select a subgroup whose trajectory differs from the full cohort. The apparent terminal pattern may be a property of who remained measurable rather than of the underlying stream (Pearl, 2009; Hernán & Robins, 2020).

Collider bias is one specific risk. If both distress and family support influence continued participation, restricting analysis to participants who remain observed can create a misleading association between distress and support. The same logic applies when device tolerance, cognition, or location affects inclusion near the endpoint (Rubin, 1976; Tsiatis, 2006; Kurland et al., 2009).

Time alignment can create additional artifacts. Researchers may align observations by estimated time to death, but the estimate is revised as new information arrives. Using future information to organize earlier data can leak endpoint knowledge into the

predictor. Prospective designs must distinguish what was knowable at each time from what became known later.

Survivor and observation selection also matter. To appear in a late window, a participant must survive and remain observable long enough to enter it; rapid decline may leave no data there. Immortal-time bias can additionally arise when eligibility or group assignment depends on future survival while follow-up is counted from an earlier time. The analysis should identify which selection or timing problem applies.

Sensitivity analysis is therefore not optional. The study should show how conclusions change when missing outcomes are assumed to be better, worse, or differently related to observed history. A route claim that survives only under one favorable missingness assumption is not robust (Diggle & Kenward, 1994).

Terminal studies may often remain inconclusive about complete-stream closure while still making valuable contributions. They can improve symptom science, missing-data methods, communication, and care, and they can state precisely when the latent terminal quantity was not adequately estimated.

9.5.3 — When Horizon Claims Must Narrow or Stop

A horizon account becomes scientific by naming how it can lose. The protocol should identify in advance which outcomes weaken a component, favor a rival, narrow the reference class, or make the test unusable. A model protected from every result has not yet supplied a discriminating claim (Popper, 1959; Mayo, 2018).

One loss pattern is absence of the predicted relation under adequate measurement, model recovery, and power. A wide

interval may remain inconclusive, but a narrow estimate inside a protocol-fixed negligible band is evidence against the prospectively specified effect. The horizon measure cannot be replaced after each null, and failure to reach significance cannot be counted as survival (Schuirmann, 1987; Lakens, 2017).

A capable rival can also win. Culture, disease severity, medication, care quality, scarcity, ordinary regulation, or selection may predict the observation better than the horizon model. Equal-budget comparison must give those accounts the same information and tuning freedom.

A reversed effect is more informative than a vague null. A shorter horizon may reduce pressure, simplify priorities, or increase emotionally meaningful choice. If that direction replicates and a rival predicts it better, the rising-pressure branch should be revised or retired rather than renamed as confirmation.

Nonidentification is another stopping point. Remaining time, reserve, burden, origin, endpoint, and observation quality may be too entangled to estimate separately, and more participants do not remove systematic error. Model-recovery simulation should expose that failure before vulnerable participants are recruited; otherwise the defensible result is No adequate estimate.

Observation shutdown and the selection it creates can defeat the claim. Rapid decline, delirium, loss of communication, privacy, sudden death, or under-resourced care can remove people from late windows, while entry into those windows already requires survival and continued observability. Denominator flow, missingness reasons, and time-to-event assumptions must

accompany every apparent terminal pattern, and no method reconstructs an unrecorded private experience without an additional bridge (Diggle & Kenward, 1994; Little & Rubin, 2019; Tsiatis, 2006; Kurland et al., 2009).

Transport failure narrows scope. A relation that appears in one language, age group, diagnosis, or care setting and reverses elsewhere cannot support broad wording. Measurement invariance, observation differences, and major-group directions should be examined before pooling or proposing a shared mechanism (Gneiting & Raftery, 2007).

Specification instability is equally adverse. If small changes in endpoint threshold, dwell time, competing-event rules, or plausible neutral calibrations change the result, the claim depends on choices that have not been identified. Sensitivity analysis should narrow or stop the inference rather than hide that dependence inside one composite interval.

Intervention failure or ethical unavailability can end the route. A model may predict that widening capacity changes trajectory features, yet justified treatment or support may produce no such relation; care remains valuable while the mechanism loses support. Palliative sedation makes the boundary concrete: its purpose is proportionate relief, not the manufacture of a terminal signature, and any test that compromises comfort, consent, privacy, visitation, medication, or clinical duty must stop (Surges et al., 2024; Carrasco-Zafra et al., 2024).

The choice between narrowing and stopping should itself be protocol-fixed. Narrowing is defensible when an independently

justified subgroup, task, endpoint regime, or route state remains identifiable and can face a new test. Stopping is required when the object cannot be recovered, the ethical burden is disproportionate, or surviving specifications were created only after failure. This rule lets persistence remain scientific rather than merely flexible.

A severe combined check uses simulated worlds generated without the candidate route and transport across independent modules. The full pipeline should face histories produced by ordinary adaptation, mean reversion, endpoint selection, observation floors, and conditioned bridges, then one frozen route should confront ordinary choice, safe perturbation, reserve-sensitive behavior, endpoint-support changes, and terminal observation without being retuned. If the same signature appears when the route is absent, or compatible predictions require separate local fits, the architecture is not yet discriminating enough for substantive interpretation.

The horizon concept remains practically useful after a formal loss because protecting time, treatment, rest, safety, relationships, and accessible services preserves options for ordinary reasons. Time-perspective research can clarify changing priorities, and palliative studies can improve treatment and observation, even when neither supports reserve coupling or terminal neutrality.

Part IV — From Experience to Evidence

Scientific seriousness begins when a hypothesis invites measurement without pretending that measurement is complete. Part IV accepts that a study may capture only part of experience, misidentify the construct, or conclude that no adequate estimate is possible. Its positive task is to build the strongest public bridge from lived experience to evidence while keeping uncertainty, dignity, and refusal intact.

That bridge has several links. A conscious person gives rise to experience; researchers define a construct; instruments and settings produce observations; models turn observations into estimates; and conclusions remain conditional on the relations among them. None of those links may be skipped. The distinction between a life, a trace, and an inference is the beginning of validity, not philosophical decoration (Cronbach & Meehl, 1955; Messick, 1995; Borsboom et al., 2004).

The first Chapter asks how experience can move toward evidence when no single channel is enough. A pain rating can be indispensable while remaining vulnerable to language, recall, social pressure, and response shift. A wearable can detect a bodily change while remaining unable to tell whether the change reflects fear, exertion, medication, illness, anticipation, or joy. A brain image can reveal a reproducible pattern without granting direct access to another person's inner life (Raja et al., 2020; Kreibig, 2010; Poldrack, 2006).

The alternative to a single meter is not an indiscriminate average of every available signal. More channels can add information, but they can also add shared error, unequal access, model dependence, and false precision. A useful multichannel estimate

must therefore declare its construct, context of use, channels, calibration evidence, observation process, missingness, model, uncertainty, and stopping rule. Contemporary clinical-outcome guidance makes the same general demand: a measure earns meaning only for a defined purpose and through an evidence chain appropriate to that purpose (FDA-NIH Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

The second Chapter asks whether an estimate retains meaning when it travels. Within-person change, between-person comparison, group comparison, and transport across cultures or settings are not stronger versions of one claim. They are different identification problems. A scale can behave consistently within one person yet fail across people. A translated item can preserve its words while changing its social meaning. A physiological feature can change interpretation across medication, disability, age, illness, or device conditions (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016).

The neutral point is especially important. The Law's terminal target is sensitive to where positive and negative experience are separated. A convenient midpoint on a questionnaire is not automatically a natural origin. Changing the origin can change the cumulative claim. Sign, origin, unit, mixed valence, and scalarity therefore require independent work rather than an informal appeal to common sense (Stevens, 1946; Krantz et al., 1971; Russell & Carroll, 1999).

Culture and disability do not enter as nuisance variables to be controlled away. They can change the construct, the response process, the available channel, the meaning of a baseline, and the

burden of participation. Response shift can alter internal standards after illness or adaptation. The scientifically responsible response is not to call the person inconsistent. It is to ask whether the instrument still means what the study claims it means (Sprangers & Schwartz, 1999; Albrecht & Devlieger, 1999; World Health Organization, 2022).

The third Chapter turns from comparability to power. Measurement creates possibilities for knowledge, but it also creates records that can be linked, reused, sold, trained on, scored, or misunderstood. Consent, minimization, purpose limitation, separation, deletion, correction, and refusal are therefore not administrative details added after the science. They shape whether the science is valid and whether it should exist at all (National Commission, 1979; World Medical Association, 2024; National Institute of Standards and Technology, 2020).

A research trace may help answer a narrow question. It never becomes a person's worth, sincerity, trustworthiness, holiness, eligibility, loyalty, or moral standing. The same rule blocks employer mood scores, insurer burden profiles, school seriousness rankings, civic eligibility measures, religious dossiers, and legal inferences about inner truth. Numbers can acquire institutional authority faster than their evidence warrants, which is why the no-score boundary must be built into the design rather than appended as a slogan (Barocas & Selbst, 2016; O'Neil, 2016; Eubanks, 2018).

Artificial intelligence expands the inferential reach of measurement because it can combine weak traces, infer sensitive attributes, and preserve learned patterns long after the original

interaction. That reach carries a matching responsibility: public availability is not general permission, research access is not unrestricted model-training authority, and data collected for one purpose do not become a permanent license for profiling. Lifecycle governance, documented purpose, human oversight, and proportionate control therefore belong inside valid use (National Institute of Standards and Technology, 2020; National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024; World Health Organization, 2024; European Union, 2024). An inferential system never owns the person whose traces it processes.

This volume's task is public measurement literacy rather than a second technical monograph. Readers need enough precision to understand why the hypothesis cannot be tested by one questionnaire, scan, wearable, or composite, why an invalid instrument does not by itself defeat the proposed Law, and why repeated route failure can narrow what the research program is entitled to claim. Exact estimator construction, model recovery, replication, and scientific verdicts remain with *The Science of Fairness*.

The human stakes remain close throughout. A child choosing a face on a pain scale, a patient whose medication changes heart-rate variability, a worker declining a wellness app, a person whose disability makes standard tasks inaccessible, and a family deciding whether research data may be retained are not abstractions. They are where measurement assumptions become lived consequences.

The intellectual movement of Part IV is from observation to warranted inference, from comparison to earned transport, and from information to responsible power. Its emotional movement matters just as much. The reader should emerge neither dazzled by instruments nor resigned to ignorance, but capable of a better seriousness: one that listens carefully, models openly, compares only where bridges have been earned, protects refusal, and corrects itself in public.

Chapter 10 — From Experience to Evidence

A person says the pain is eight out of ten. A wearable shows elevated arousal and guarded movement. A facial model detects little expression. A brain-based classifier gives an uncertain result. None of these observations cancels the others, and none receives the authority to overrule the person merely because it looks more technical. Together they form a measurement problem: what exactly is being estimated, from which traces, under which assumptions, and for what use?

Self-report has a privileged role because felt experience is lived from the first-person point of view. Pain, dread, relief, shame, calm, and mixed emotion are not made more real by becoming machine-readable. Yet reports are shaped by language, memory, reference periods, safety, fatigue, culture, and the act of being asked. The correct response is neither to worship nor dismiss testimony, but to study its response process and conditions of use (Cella et al., 2010; Shiffman et al., 2008; Kahneman et al., 2004).

Physiological data can reveal embodied change, but bodily activation is not self-interpreting. The same heart rhythm, skin conductance, respiration, posture, or endocrine response can accompany exertion, fear, fever, pain, anticipation, medication, or joy. A channel can be technically accurate while its valence interpretation is wrong (Kreibig, 2010; Khalsa et al., 2018).

Neural evidence raises a similar challenge. A reproducible pattern can support a specific contrast without becoming a universal reader of inner life. Reverse inference, task dependence, scanner differences, population shift, and within-person variability limit what a brain image can establish. Visual authority is not scientific authority (Poldrack, 2006; Wager et al., 2013).

Behavior is visible but not transparent. Silence can express calm, fear, discipline, exhaustion, or coercion. A smile can coexist with pain. Productivity can coexist with collapse. Choices are made within opportunity sets shaped by money, safety, disability, dependence, and social consequence. Behavior must therefore be interpreted as evidence under context, not as a pure report of preference.

Dream reports and other fragile channels pose a different problem. They may contain valenced experience that matters to the stream while remaining difficult to recall, ethically sensitive, and heavily selected by awakening, medication, interest, and distress. Relevance does not create a right to collect.

Once several channels are available, integration becomes a modeling decision. A formative composite, reflective factor model, state-space model, and predictive ensemble make different claims about why signals move together and what their output represents. The useful question is not which method sounds most advanced. It is which architecture matches the construct, survives capable nuisance models, transports beyond the fitting sample, and carries its uncertainty into the conclusion.

Starting with a device reverses the logic of measurement. A sensor can produce exact readings while answering the wrong question, because precision in a physical feature does not identify momentary valence, pain intensity, arousal, relief, mixed affect, functional burden, or any other intended construct. The research question and context of use must therefore govern the instrument, not the other way around.

Calibration is evidence about a relation, not direct access. A channel may be useful within one person, device, cadence, and setting while failing elsewhere. The context of use must travel with the estimate, as current biomarker and clinical-outcome guidance repeatedly emphasizes (FDA-NIH Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

Missingness belongs inside the model. People may stop responding because they are asleep, exhausted, sedated, distressed, hospitalized, disconnected, unable to speak, unwilling to continue, or excluded by the instrument. Those states can be related to the very experience being studied. Entering zero for missing experience would convert observation failure into false neutrality (Rubin, 1976; Little & Rubin, 2019).

Where feasible, test reactivity by varying observation itself: randomize prompt frequency, include a passive-only comparison, separate acclimation from analysis, and predefine burden and withdrawal thresholds. If the measured pattern changes with the schedule, that dependency belongs in the observation model rather than being dismissed as noise (McCambridge et al., 2014).

Confirmatory claims also require decisions made before the preferred result is visible. The detailed vocabulary returns in Section 10.4; the rule here is simple: preplanned choices must be distinguishable from exploratory choices, and a claim of public preregistration requires a real timestamped registry record rather than an intention, private draft, or internal date.

Null findings and model failure have positive scientific value. An estimator that cannot distinguish valence-relevant change from

generic arousal, illness severity, movement, medication, device quality, or social context has revealed a real limit. That result may retire a channel, narrow a construct, expose missing calibration, or show that a simpler rival already explains the pattern.

No natural unit is given in advance. A candidate unit requires a defensible origin, sign, scale, temporal interpretation, and version. Until those conditions are earned, accumulation remains a model-dependent research object rather than a discovered currency of experience. A smooth cumulative curve cannot create a unit the observations never justified.

The humane stopping rule is as important as the statistical one. A channel may be informative yet too invasive, exclusionary, reactive, or easy to weaponize for the question at hand. Scientific interest does not outrank care, and technical availability does not create authority to collect, retain, link, or deploy a measurement.

The reader should leave able to inspect any proposed meter of feeling with disciplined questions: What is the construct? Who can use the instrument? Which channels are available to whom? What observation process shaped the data? What uncertainty remains? What would make the estimate fail? What uses are permitted, and who may refuse?

That literacy remains valuable under every scientific outcome. It protects ordinary research from false precision, helps participants understand what a study can and cannot know, and prevents institutions from converting a bounded estimate into a claim about the person. The result is a stronger path to evidence because it knows where its authority ends.

10.1 — Why No Single Meter Is Enough

No single meter can bear the whole evidentiary burden because every channel reveals experience through a different relation. Self-report carries first-person testimony but depends on language, memory, safety, and willingness. Physiology and neural activity can detect embodied or task-linked change, yet the same signal may arise from several causes. Behavior is visible but not self-interpreting. Dreams and context can add missing experience while introducing their own selection and privacy problems.

The scientific opportunity lies in disagreement. When a participant reports relief while arousal remains high and movement guarded, the channels may be sampling different time windows, constructs, lags, or sources of error. A good design does not force them into harmony. It asks what each channel was expected to contribute, which rival explains the conflict, and whether the intended construct remains identifiable.

Dignity requires restraint in interpreting and using these indirect observations. No instrument should be described as reading a hidden ledger, detecting moral truth, or overruling a person's testimony by prestige alone. The strongest measurement system is one whose claims remain bounded by calibration, context, uncertainty, accessibility, and the continuing authority of the person being studied.

10.1.1 — Self-Report Limits

Self-report is indispensable because felt experience is first-person before it is research material. A person's account of pain, dread,

relief, shame, peace, fear, or exhaustion deserves serious weight. The Law of Fairness cannot build an estimator that treats the person's own testimony as decorative.

But self-report is not perfect access. People often do not know exactly what they feel, or they know it in ways language cannot carry. Distress can be vague. Shame can block description. Trauma can fragment memory.

Reporting is also socially shaped. A child may report what adults seem to want. A patient may understate pain to avoid being difficult. A worker may conceal distress from fear of consequences. A person from a culture of restraint may use mild language for severe burden.

Time complicates the report. A momentary report can miss slow background burden. A retrospective report can be rewritten by memory, recovery, resentment, pride, or the desire to make a story coherent. A diary may increase self-monitoring.

Response-process evidence matters as much as correlation. Researchers need to know how participants understand the prompt, choose a reference period, use the scale, and adapt their answers under fatigue, fear, language difference, or changing health. Cognitive interviewing and ecological momentary assessment can reveal whether a score change reflects experience, interpretation, recall, or the burden of being asked (Shiffman et al., 2008; U.S. Food and Drug Administration, 2025).

Difficulty of measurement does not make a person unreliable. The difficulty may belong to the instrument, language, setting, access route, power relation, or design. A multichannel model that treats

a participant's opacity as personal failure has confused the limits of observation with a defect in the life being observed.

Modern clinical-outcome guidance is useful here because it refuses a category error. A report that comes directly from the participant is a distinct source of measurement, not raw material to be silently corrected by a clinician or device. For subjective states, there is often no independent gold standard. Validation of a multichannel estimate must therefore compare reports with other channels without treating disagreement as proof that the participant is wrong (Shiffman et al., 2008; Kahneman et al., 2004).

Protected open-text follow-up can reveal states such as relief with grief, meaning with pain, or calm with exhaustion that a scalar item compresses. Comparing those descriptions with separate positive and negative ratings tests whether one bipolar scale loses information. The aim is to test the response model, not to force every report into it.

10.1.2 — Dream Data Limits

Dreams matter to this inquiry because conscious experience does not stop when public action stops. A person may wake from terror, reunion, grief, wonder, shame, relief, or an emotion that has no easy waking name. If felt experience is relevant, the night cannot be excluded merely because it is difficult to observe.

The first scientific distinction is between the dream and the report of the dream. The dream occurred during sleep. The report is produced after waking, through memory, language, attention, interpretation, and the social conditions of disclosure. A vivid

account may still be incomplete, and a forgotten dream is not evidence that nothing was felt.

A protocol should record more than the reported content. It should preserve whether a report was attempted, the delay after waking, confidence in recall, sleep timing, the method of awakening, relevant medication, and the reason a prompt was skipped. These variables help distinguish missing experience from missing observation (Little & Rubin, 2019).

Prospective collection is usually stronger than distant retrospection. A brief report recorded soon after waking can reduce later reconstruction, while repeated sampling can reveal how often recall occurs and how quickly it fades. The gain in immediacy must still be weighed against sleep disruption, fatigue, and the possibility that the study changes the very process it hopes to observe (Shiffman et al., 2008).

Dream coding also requires separation of layers. The felt quality reported from the dream, confidence in recall, narrative theme, waking appraisal, and later meaning are different variables. A frightening dream may be interpreted as spiritually important, psychologically useful, meaningless, or merely unpleasant. The interpretation should not be inserted into the valence estimate as though it were the original experience.

Human coders and language models can classify themes, but neither has direct access to the dream. Coding manuals should distinguish participant language from researcher inference, record disagreement, and prevent a fluent summary from replacing the original report. Model-assisted coding requires frozen versions,

held-out evaluation, and review of failures involving culture, metaphor, trauma, and unusual speech.

Neural and physiological measurements during sleep can help identify sleep state, arousal, awakening, and candidate conditions associated with reported experience. They do not reveal the dream as a complete inner scene. Correlates of dreaming and consciousness are scientifically valuable precisely when they are stated as correlates rather than inward possession (Siclari et al., 2017; Tononi et al., 2024).

Nightmares and trauma-related dreams require a stronger burden threshold. Repeatedly requesting detailed accounts can increase distress, rumination, anticipatory anxiety, and sleep avoidance. Participants need a real option to skip content, provide only a coarse rating, pause the protocol, or stop without having refusal interpreted as a symptom.

Bystander privacy also matters. A dream report can identify family members, partners, clinicians, or people who never agreed to participate. Redaction, local processing, restricted access, and participant review can reduce exposure, but some reports should not enter the research record at all.

Youth research needs additional restraint. A child may not understand future reuse, model training, public quotation, or the difference between a private dream and a research record. Assent, caregiver permission where appropriate, confidential skip options, age-suitable prompts, and strict limits on secondary use are part of scientific validity, not administrative extras.

A dream channel should earn admission through incremental value. It should improve a declared inference beyond lower-burden waking measures, survive held-out tests, and preserve privacy and stop authority. More intimate data are not better when they add little discrimination.

Dream research therefore teaches a general lesson about measurement. Relevance does not create entitlement to collect, fragile recall does not become continuous telemetry, and a private report does not become a complete inner record. The night may matter to the question while remaining partly unknown, voluntarily unshared, and not amenable to a defensible estimate.

10.1.3 — Dignity Requires Restraint

Seeking hidden burden must never become exposing a life. No instrument captures the whole person. Consent, purpose limits, and bounded interpretation protect against reducing a person to a dataset, score, diagnosis, suspicion, or public category while allowing limited relations among reports, physiology, behavior, context, and neural activity to be tested.

External channels can reveal change without naming its meaning. Physiology, neural signals, and behavior may discriminate narrow conditions, yet none alone decides among fear, effort, pain, medication, joy, exhaustion, coercion, or relief. Their value depends on context, calibration, rival tests, and the continuing authority of the person (Kreibig, 2010; Khalsa et al., 2018; Poldrack, 2006; Wager et al., 2013).

A channel should enter only when it improves a declared inference beyond the least burdensome valid alternative in held-

out conditions. Small predictive gains do not justify exclusion, household capture, opaque interpretation, or unusable instruments (U.S. Food and Drug Administration, 2025).

Consent and withdrawal rules must preserve refusal, including when observation is indirect. Skipped prompts, withheld reports, removed wearables, declined scans, or withdrawal may reflect privacy, fatigue, pain, culture, disability access, surveillance concerns, or inconvenience. They may be modeled as observation processes but not converted into labels of distress, deception, instability, or worth (Little & Rubin, 2019; Council for International Organizations of Medical Sciences, 2016).

Accessibility requires alternative valid routes such as voice, symbols, assisted communication, flexible timing, chosen devices, nonwearable options, and bounded caregiver support. When no route supports the intended interpretation, the claim narrows or becomes No adequate estimate; no interface becomes the person (World Health Organization, 2022).

Indirect sensing also captures relatives, caregivers, coworkers, clinicians, and strangers who never joined the study. Because one participant cannot consent for everyone in the sensor field, prefer the least revealing channel and local processing (Nissenbaum, 2004; Nissenbaum, 2010). A design review can compare what each candidate channel would reveal about nearby people before selecting the information the study actually needs.

Contestability makes provenance usable. Participants and reviewers should know which observations entered an estimate, which model transformed them, what uncertainty remains, and

how access, correction, or deletion works. Raw observations, features, estimates, permissions, corrections, and public artifacts remain separate layers with separate rules. A participant-facing record could show the observation used, its date, the resulting estimate, and the person responsible for reviewing a correction.

Before any terminal cohort, lower-burden construct-discrimination studies should separate valence from arousal, pain, effort, sedation, movement, and safety in held-out data. Dignity does not depend on becoming fully measurable. A well-chosen benchmark would include contrasting cases that share bodily activation while differing in reported pleasantness, helping isolate the intended construct.

10.2 — From Experience to a Measured Construct

Measurement design begins with a declared question. A sensor may recover a physical feature with extraordinary accuracy while remaining irrelevant to the experience-related quantity under study. Naming the construct, population, context of use, time scale, and intended decision before choosing devices keeps technical convenience from deciding what the evidence is allowed to mean.

A defensible chain moves from construct to channel, observation, transform, estimand, estimator, uncertainty, and permitted interpretation. Each link can succeed or fail independently. Section 10.2.3 sets the technical, analytical, and construct-validation requirements for channel admission.

Some channels will be missing, unusable, or unsafe. That does not create a blank that can be filled with zero, the last observation, or a model's preferred answer. It creates a visible change in what can be estimated. Honest measurement may narrow to a reduced-channel claim, an identified range, an exploratory result, or No adequate estimate.

10.2.1 — Start With the Construct, Not the Sensor

Construct definition comes before channel selection. Researchers should state whether they seek momentary valence, pain intensity, arousal, relief, mixed affect, functional burden, or another object, and they should name the intended use before deciding what to collect. Reports, sensors, observations, and clinical context then enter as candidate relations to that construct rather than as ready-made measures.

Calibration evidence has to do more than agree with the estimator after the fact. It must specify which report window, response scale, mixed-state instruction, language version, and burden condition are being used to test the physiological contribution.

Ecological momentary assessment reduces long recall windows but introduces its own distortions. Prompt timing, interruption, device interaction, social setting, fatigue, nonresponse, and repeated self-observation can change the state being measured. The protocol should randomize or balance prompt schedules where possible and estimate reactivity rather than treating it as harmless.

The response-process review should also ask whether participants distinguished intensity from importance and how they handled mixed states. A reliable answer to a misunderstood question is not valid evidence.

The report battery should be deliberately plural during development. A bipolar valence item, separate positive and negative items, arousal, pain, agency, safety, and meaning can compete in the same validation study. If the single scalar consistently loses held-out information needed to interpret physiological channels, that is evidence for a multidimensional or domain-specific bridge rather than a reason to hide the richer reports (Cronbach & Meehl, 1955; Messick, 1995).

For the present architecture, reports serve as protected validators and sign-orientation witnesses. They can support a prospective direction rule, expose ambiguity, and reveal when another channel tracks arousal, illness, or response style rather than

valence. They should neither be demoted whenever a device disagrees nor made constitutive merely because external channels are incomplete. A change in their formal role requires declared change control and fresh validation.

Reports should be modeled at the level at which they are used. A within-person association between multichannel estimate and momentary valence does not establish cross-person comparability. A group mean does not establish individual prediction. The analysis should separate within-person change, between-person differences, and person-specific heterogeneity rather than allowing one coefficient to stand for all three.

No channel class is constitutive by default. Self-report, physiology, behavior, context, neural evidence, and other signals may act as indicators, validators, auxiliary witnesses, or model inputs only under a prospectively specified estimand and observation model. Every admitted channel needs a declared target, context-sensitive sign rule, nuisance set, version, and failure threshold. Technical prestige cannot substitute for that chain (Borsboom et al., 2004; Michell, 1999; U.S. Food and Drug Administration, 2025).

Physiological inclusion should be incremental and auditable. Each candidate channel should add held-out information beyond simpler nuisance models and the existing channel set, then carry a qualification record describing its biological rationale, sensor and preprocessing, anchor, sign evidence, nuisance set, reliability, lag, missingness rule, context of use, and failure threshold. Redundant features and incomplete records remain exploratory rather than being folded into a maximal sensor stack (FDA-NIH

Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

10.2.2 — Calibration Is Evidence, Not Direct Access

Calibration asks whether a channel bears a stable, interpretable relation to a declared construct in a defined context of use. It can establish responsiveness, ordering, discrimination, or prediction under tested conditions. It does not grant direct access to experience, and it does not turn a familiar scale into a natural unit.

Behavioral calibration must begin with the feasible menu. Poverty, disability, institutional rules, family pressure, fatigue, safety, and culture alter which actions are available, so constrained behavior cannot be read as preference. Prospective contrasts can still test latency, reversal, persistence, or adaptation after independently identified events, provided the prediction names the action, interval, observation effect, and rival explanation (Shiffman et al., 2008; McCambridge et al., 2014).

Behavioral coding needs transparent uncertainty and people whose burden is not outwardly expressive. Silence, stillness, compliance, productivity, withdrawal, and help-seeking should remain ambiguous when context is insufficient. Reliability studies and stratified validation across masking, communication style, disability, and cultural display norms can show whether a model measures behavior, expression, or a broader relation to experience rather than assuming those objects are identical.

Neural evidence requires the same restraint. Activity associated with negative affect in one task may reflect attention, novelty, motor preparation, cognitive control, pain, medication, or task

difficulty in another. Discriminant tests must include those neighboring explanations, and any neural features shared with an endpoint or unity proxy must be declared so apparent independence is not created by overlapping sensors or preprocessing (Poldrack, 2006; Wager et al., 2013).

Neural and other high-dimensional channels should be tested across sites and within persons where possible. Hardware, preprocessing, movement, implants, age, disease, medication, and operator practice can dominate subtle affective patterns. Harmonization should separate technical variation from participant variation without erasing biology, and a pattern that survives only in one homogeneous laboratory remains a local result rather than a terminal or cross-population measure (Poldrack, 2006; Wager et al., 2013; Yarkoni, 2022).

For dream-channel calibration, human coders or language models may test prospectively specified within-person relations only under validated codebooks, blinded conditions, uncertainty, and the privacy safeguards in Section 10.1.2.

Missing dream data are not neutral. Any automated coding requires restricted or local processing, human review, deletion rules, and the separate showing of incremental value required in Section 10.1.2.

10.2.3 — Unsafe, Missing, or Unusable Channels

Missingness is an observation process with scientific content. A channel may be absent because a device failed, the participant refused, symptoms increased, sleep was disrupted, care took priority, the setting changed, or the person could no longer

respond. Missing is not zero, and carrying the last observation forward is not neutral. When absence may depend on the state or burden under study, the analysis should compare plausible selection, pattern-mixture, joint, or bounding models and widen uncertainty when the mechanism cannot be identified (Rubin, 1976; Little & Rubin, 2019).

A versioned channel registry should preserve every inclusion, exclusion, transform, sign rule, and failure. The registry is not a public score or a personal file. It is a scientific lineage showing why one channel was admitted, why another remained exploratory, and which populations or settings invalidate the rule.

Context and measurement burden belong in the same layer. Place, social company, activity, safety, treatment, medication, sleep, pain, movement, and major events can change how a channel should be read, while prompts and devices can create fatigue, shame, interruption, discomfort, privacy fear, or family exposure. Those conditions may modify the construct relation, the observation process, or the ethical availability of the channel; they are not footnotes added after estimation (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Passive context collection should be exceptional, not a study's background condition. Location, audio, proximity, message content, and environmental traces can reveal people who never consented. The scientific question should determine the least intrusive necessary context, with nonparticipant data excluded, transformed, or blocked at capture where possible.

Context can orient sign and reveal response shift. High activation during verified exercise should not be transformed like activation during an independently reported panic episode, and relations among report, physiology, and behavior may change after illness, treatment, caregiving, or adaptation. Any interaction rule or time-varying mapping must be prespecified and tested for transport, not improvised to preserve a result.

Technical availability does not make a channel scientifically or ethically usable. Audio may capture other people, location may expose a shelter or clinic, sleep prompts may disrupt care, and sensors may irritate skin or interfere with equipment. Nor can a fixed estimate quietly become a different composite each time a channel disappears. The protocol needs a validated reduced-channel version, a defensible linking study, or a no-estimate rule, plus group-specific audits of where absence concentrates (FDA-NIH Biomarker Working Group, 2025; U.S. Food and Drug Administration, 2025).

Audit missingness by group and setting. If a device fails on certain bodies, if reports vanish in cognitive decline, or if terminal care makes sampling impossible, the empirical program may become least informative where most relevant. A coverage table can show where records become thin and whether the gap follows the setting, the device, or the participation demands.

Qualify channels in layers. Technical verification asks whether sensor and pipeline recover the declared feature. Analytical validation asks whether the transformed feature is reproducible and robust under expected artifacts. Construct validation asks whether its prospective sign and temporal relation track valence-

relevant change rather than generic illness, movement, or arousal. A channel enters a confirmatory multichannel estimate only after all three layers are documented for its context. For example, accurate recovery of a pulse waveform answers a different validation question from whether its derived feature tracks the intended experience.

The input map has a clear center: construct, estimand, observation model, and bounded channel roles. Disagreement remains visible until a prospectively specified model explains or narrows it (Bollen & Lennox, 1991; Bollen & Bauldry, 2011). The map should let an independent team reconstruct how a disputed observation affected the estimate and which assumption would change that result.

10.3 — Combining Channels Under Uncertainty

Suppose a patient reports meaningful relief while autonomic arousal remains elevated and movement remains guarded. One interpretation is that relief arrived before the body settled. Another is that the report captures hope or comparative improvement while movement reflects tissue protection. A third is that medication changed physiology without changing the pain experience. Integration begins by comparing these explanations, not by averaging three numbers.

A multichannel model can earn value when it predicts a declared outcome in held-out data beyond each component and beyond capable nuisance-only rivals. It loses value when its apparent accuracy comes from site, device, diagnosis, medication, demographic proxy, or data-quality artifacts. The model must reveal which channels contributed, how their weights were learned, and how uncertainty changes when a channel disappears.

Scientific validation is therefore a sequence of questions rather than a ceremonial label. Does the system recover its technical inputs? Do the inputs relate to the intended construct? Does the combined estimate improve a declared criterion? Does it remain calibrated across time, people, devices, and settings? Can an equal-budget rival reproduce the result? A model that fails one question should not borrow authority from the others.

10.3.1 — Why Integration Requires Modeling

Integration requires a declared observation architecture. Each channel must name the object it targets, the conditions that alter its sign or loading, the history available at the observation time,

and the measured estimand to which it contributes. The measurement map and the analysis map should remain separately versioned, because defining what is to be measured is not the same act as estimating it from sampled records. This architecture makes assumptions auditable without claiming that the latent state has already been identified (Bollen & Lennox, 1991; Bollen & Bauldry, 2011).

The model should return an estimate together with its provenance. A valid record names the channels used, transformation and weight versions, quality flags, context domain, missingness state, and uncertainty. A detached number is not the scientific object that was analyzed.

Integration earns value when it improves a declared criterion beyond its components. Incremental validity should be assessed in held-out data, with simpler single-channel and nuisance-only models as baselines. If the composite adds no reliable information, integration has not justified its burden.

Channels differ in sampling rate, unit, noise, latency, and relation to the target. Heart features may arrive each minute, endocrine measures each hour, reports several times per day, and neural measures only in short sessions. Pooling all raw observations into one unweighted average would give the densest channel the greatest influence and treat incomparable scales as commensurable.

Weights require an explicit convention and cannot repair an unidentified sign. A study may normalize weights, fix a reference channel, or preserve declared scaling, but the choice changes the

meaning and units of the composite. Eligible, degraded, unsigned, and absent channels must also be distinguished. Removing one channel and renormalizing the rest creates a new measurement version unless that reduced form has already been validated (Krantz et al., 1971; Hand, 2004).

Weight estimation should be nested inside a development and validation design. Tuning on the full dataset and then reporting the same data as validation produces optimistic performance. Cross-validation helps, but person-level, site-level, and time-block leakage must be prevented. A future observation from the same participant is not an independent external test.

Sensitivity analysis should vary scientifically defensible transforms, weights, and personalization rules rather than report one optimized configuration as natural. A person-specific mapping may improve local prediction while reducing comparability and increasing overfitting, so the rule itself needs held-out validation. The useful output is the range of conclusions across a prospectively specified admissible set. If sign, equivalence, or model ranking changes inside that set, the result is configuration dependent (Bland & Altman, 1986; de Vet et al., 2011).

Complex models earn their place only when they improve held-out calibration, missing-data handling, and interpretation beyond a simpler baseline. A state-space or other flexible model may represent lags and changing channels, but numerical convergence does not show that its latent dimensions were observable from the available data. Simulations, rank checks, posterior geometry, and comparison with simpler models should reveal whether

sophistication recovered information or merely supplied a smoother story.

10.3.2 — Error, Noise Floors, and Confidence

An estimate is useful only when its uncertainty is described at the same level as the claim. A single margin of error cannot carry every weakness in a multichannel system. The error budget must separate sampling variation, instrument noise, calibration error, temporal misalignment, missing observations, model choice, transport failure, and uncertainty about the construct itself.

Random error and systematic error behave differently. Random fluctuation may shrink when repeated observations are averaged under stable conditions. A shifted sensor, an invalid neutral point, a biased sample, or a channel that tracks arousal instead of valence can remain wrong with extraordinary precision. Larger samples narrow uncertainty around the model that was fitted; they do not repair a model that points at the wrong object (Bland & Altman, 1986; Michell, 1999).

Reliability is therefore not validity. A device can reproduce the same result and still measure the wrong thing. Agreement between two devices can also be high when both share the same nuisance source. A credible report should state which kind of reliability was examined, which construct relation was tested, and which rival explanations remain available (Shrout & Fleiss, 1979; de Vet et al., 2011).

A measurement system can lose detail in several ways. A reading may be noisy, rounded, clipped, or biased, and each problem behaves differently. Repeated measurements can reveal a stable

difference smaller than the noise in any one reading when the errors vary in a well-understood way. They cannot by themselves recover detail the instrument never records, identify an unknown systematic bias, or reconstruct missing parts of one person's history. The relevant limit is what this instrument and analysis can resolve for this particular claim.

Missingness is part of the error budget. A missing report may reflect sleep, refusal, distress, device failure, inaccessible design, or observation shutdown. Each cause carries different information. The analysis should preserve the pathway that produced the gap instead of converting every absence into an average value (Rubin, 1976; Little & Rubin, 2019).

Confidence language should match that distinction. A confidence interval or credible interval is conditional on a model, data treatment, and sampling or prior assumptions. It is not the probability that the entire interpretation is correct. Public copy should say what the interval covers and which uncertainties remain outside it (Wasserstein & Lazar, 2016; Wasserstein et al., 2019).

A band around zero also needs careful interpretation. Evidence that an estimate lies inside a declared equivalence band is not evidence of exact latent zero. The band reflects resolution, error tolerance, and context of use. Equivalence procedures can test whether the data are compatible with a practically small difference, but the width and decision rule must be fixed before results are examined (Schuirmann, 1987; Wellek, 2010).

Accumulation magnifies some errors. A small shift in the neutral origin can build across a long record, while independent momentary noise may partly cancel. Serially correlated error, however, can accumulate in one direction. The whole-stream estimate therefore needs uncertainty propagation that respects time, dependence, changing channel availability, and version changes.

Calibration drift should be visible. Devices age, software changes, prompts are revised, participants learn the task, and clinical context evolves. A versioned calibration record should show when the observation system changed and whether old and new values remain comparable. Silent recalibration can manufacture an apparent trajectory.

The most honest output may be a range, several model-conditional ranges, or no adequate estimate. That outcome is not statistical timidity. It tells the reader which conclusions survive the uncertainty and which do not. Partial identification is preferable to a single number that depends on an unearned assumption (Manski, 2003; Tamer, 2010).

Confidence should also govern use. A research estimate that is adequate for comparing methods may be inadequate for individual feedback, clinical interpretation, public communication, or any consequential decision. The context of use determines how much uncertainty is tolerable and which kinds of error are unacceptable (U.S. Food and Drug Administration, 2025).

The final report should therefore carry an error map, not only an error bar. It should identify the construct, channels, time windows, calibration version, missingness process, model family, sensitivity set, transport population, no-estimate rule, and prohibited uses. Confidence becomes trustworthy when the reader can see what it does not include.

10.3.3 — Scientific Validation

Validation is not a stamp placed on an instrument after the numbers look persuasive. It is an argument that links a defined construct to particular observations, transformations, interpretations, decisions, and consequences. The evidence must fit the proposed use, and the argument remains open to revision when the instrument, population, setting, or purpose changes (Cronbach & Meehl, 1955; Messick, 1995; American Educational Research Association et al., 2014).

The validation argument begins with the construct and context of use fixed in Section 10.2.1; momentary pleasantness, pain burden, and cumulative signed experience remain different objects.

Content and response-process evidence should come from the people expected to use the instrument. Cognitive interviews, accessible pilot formats, local language review, and observation of how participants interpret anchors can reveal whether the prompt is understood as intended. Current clinical-outcome guidance likewise treats fit for purpose and patient relevance as parts of the measurement argument, not decorations added after a scale is chosen (U.S. Food and Drug Administration, 2025). An item that appears clear to researchers may evoke social

desirability, threat, moral judgment, illness appraisal, or a different time horizon in the population being studied.

Technical verification comes next. A sensor and its preprocessing pipeline must recover the declared physical feature with adequate accuracy, precision, timing, and robustness under expected movement, device placement, medication, disease, and environmental conditions. Technical performance is necessary, but it does not establish that the feature carries the proposed valence meaning.

Reliability must be estimated at the time scale and context of use. Stability across ten quiet minutes does not establish stability across days of changing illness, sleep, medication, or activity. Variance should be decomposed across person, occasion, device, site, and condition so the study knows whether additional sampling can reduce the error or whether the error is systematic.

Internal structure requires a declared interpretation. A reflective model treats a latent state as a source of covariance, while a formative model defines a composite from its inputs. Factor direction and scale can often be changed mathematically without changing fit, shared sensor artifacts can create false common structure, and loadings may shift with treatment, disease, device replacement, or response shift. External anchors, residual checks, and competing stable versus time-varying models are therefore needed before a fitted factor receives experiential sign, origin, or continuity (Bollen & Lennox, 1991; Bollen & Bauldry, 2011).

Convergent prediction is not enough. A model following reports may still learn arousal, nociception, exertion, sedation, fever,

illness severity, identity, location, or prompt wording. Discriminant outcomes, feature ablation, context shifts, and counterfactual stress tests should distinguish the intended relation from an easier proxy. Under a narrower name, prediction may remain useful but cannot replace construct validity.

Model recovery should precede expensive or burdensome collection. Simulated data from formative, reflective, multidimensional, generic-arousal, illness, and missingness families can test whether the planned pipeline distinguishes the structures it later intends to discuss. Recovery should examine selection, parameter bias, interval coverage, false equivalence, false defeat, convergence, calibration, and held-out performance, with rival implementations reviewed competently. A design that cannot recover its own alternatives cannot support the corresponding model-level conclusion.

Parsimony includes burden, access, and interpretability as well as parameter count. A small gain in average prediction may not justify an invasive sensor, exclusion of participants with disabilities, dependence on a proprietary vendor, or an output that cannot be explained. The study should declare a vector of performance criteria instead of hiding every value judgment inside one optimization score.

A model fails its declared interpretation when it tracks another construct, and it remains nonidentified when several transforms or structures reproduce the observations while implying different cumulative contributions. Renaming the output after results are known does not rescue the original claim, and more computation cannot identify what the observations and assumptions leave

unresolved. The warranted product may be a narrower index, a bounded set, or No adequate estimate (Krantz et al., 1971; Hand, 2004).

Governance is part of validity because a result cannot be independently evaluated when code is hidden, exclusions unreported, provenance incomplete, or a vendor controls the pipeline. Reproducible transformations, sealed confirmatory data, audit trails, and correction procedures are not administrative decoration. They show whether the claimed result can be trusted.

Scientific validation changes both the method and the claims made for it. When a model fails, affected diagrams, public summaries, and decision pipelines must be corrected or withdrawn. When a local model succeeds, it provides a better bridge from observations to the declared empirical target, with transport to other settings still to be tested. That achievement makes the Law more answerable to evidence at the measurement layer; it does not establish the terminal relation.

10.4 — Missingness, Reactivity, and Honest Failure

Missing data are not empty spaces. A report may disappear because a person is asleep, distressed, sedated, hospitalized, exhausted, unwilling to continue, or excluded by the design. A device may fail because of movement, skin contact, environment, body type, or technical limits. Each cause can change the meaning of the observed sample.

Reactivity is part of the observation process whenever measurement changes what is measured. Repeated prompts can increase self-focus, devices can alter behavior, interviews can evoke shame or relief, and participation can create expectations. Those effects must be tested, minimized, or carried into interpretation rather than treated as inconvenience outside the model.

Prospective specification, blinding, negative controls, and publishable null results protect the inquiry from its own hopes. Governance then carries those protections across protocols, analyses, corrections, public artifacts, and reuse. The goal is not a rigid procedure but a research culture in which uncertainty and failure remain visible enough to improve the next test.

10.4.1 — Blinding Against Hope

Blinding protects interpretation as well as analysis. Researchers who hope to find hidden burden or relief may ask different follow-up questions, notice different behavior, choose favorable channels, adjust weights, or resolve ambiguous records in the preferred direction without intending deception. Separating outcome knowledge from coding, adjudication, calibration, and

model selection reduces that pressure and leaves a clearer trail wherever judgment remains unavoidable (Nosek et al., 2018; Open Science Collaboration, 2015).

Analysts can be blinded to condition, hypothesis, participant group, or expected direction where possible. Coders can rate reports or dreams without knowing the study prediction. Reviewers can examine model outputs without knowing which version favors the Law. Full blinding will not always be possible, but partial blinding is often better than none.

Advance control should use exact language. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection when no public-record claim is asserted. Publicly preregistered requires a timestamped external registry record with a persistent identifier. An intention, private draft, or internal timestamp does not meet that standard.

Advance plans should state both what the study will test and what its design cannot establish. Constructs, channels, transforms, primary outcomes, exclusions, missingness rules, model families, claim levels, and support, narrowing, or failure conditions should be fixed before confirmatory outcome inspection. This matters especially in a multichannel program with many defensible choices and rich narratives, because unbounded flexibility can turn exploration into post hoc rescue (Nosek et al., 2018; Chan et al., 2025).

A specification-curve plan can be prospectively specified when several defensible analyses exist. The admissible set is defined in

advance, and the full distribution of results is reported rather than selecting one preferred specification. This preserves scientific pluralism without turning the analysis into an unlimited multiverse after outcomes are known (Steege et al., 2016; Simonsohn et al., 2020).

Exclusions require the same prospective and ethical discipline. Crisis-related missingness, atypical physiology, inconsistent reports, device failure, safety stops, withdrawal, and protocol deviations may reveal the limits of the instrument rather than disposable mess. The plan should define technical and ethical exclusions, report every participant and interval flow, examine whether exclusions cluster by disability, material access, or distress, and permit independent challenge before collection begins (World Medical Association, 2024; Chan et al., 2025).

The publication should include a participant and interval flow diagram showing every exclusion, missing period, reduced-channel estimate, and no-estimate result. This makes attrition and data loss visible at the level where they occur. A clean final sample size without the path to that sample can conceal the very selection process most relevant to validity.

10.4.2 — Null Results

A null result is informative only when the study could answer the question it posed. Power, timing, channel quality, missingness, equivalence margins, model recovery, and serious rivals determine whether non-support weakens a measurement claim, retires a route, or leaves the contrast unresolved. The result should neither be dramatized into total defeat nor dissolved into a list of

excuses. A well-powered failure of sign stability, incremental validity, within-person association, or held-out transport is a real boundary on the corresponding claim.

Null results can be scientifically rich. They may show that physiology adds little beyond report. They may show that dream channels are not useful for a particular question. They may show that neural data adds cost without interpretive gain.

Null results also protect participants. If a study adds burden and produces no usable information, that matters for future ethics. Repeating burdensome methods after repeated null results may become unjustifiable. The absence of support can become an ethical stop sign.

Public language must treat null results as meaningful. If early public materials mention possible multichannel estimate research, later public materials must not highlight only supportive findings. The public record should make non-support visible enough that the theory cannot build a reputation from selectively published success (Ioannidis, 2005; Gelman & Loken, 2013).

Equivalence and noninferiority questions need prospectively justified margins. Failure to reject a difference is not evidence of equivalence. Conversely, a narrow effect that is statistically detectable but smaller than the minimum useful improvement may not justify the added burden of multichannel estimate.

Red-team review makes criticism structural. It should include competent scientific rivals and, where relevant, clinicians, disability and trauma experts, privacy and data-governance specialists, ethicists, public-communication reviewers, and

participant representatives. Their task is to challenge channel selection, weighting, missingness, burden, exclusions, subgroup performance, public summaries, and deployment temptation before the claim travels, not to manufacture scandal after publication.

Red teams need the estimator specification, code, simulation environment, prospective plan, deidentified outputs, and enough provenance to reproduce the claim. Capable rival teams should own their competing models and evaluation rules rather than receive simplified versions implemented by the favored team. They should test leakage, alternative transforms, nuisance explanations, unsupported sign choices, and the permanent challenge set described next before the test set is opened (Simmons et al., 2011; Wagenmakers et al., 2012; Nosek et al., 2018).

A permanent challenge set should include cases designed to break the index: exercise with positive affect, calm dissociation, analgesia with continuing grief, fever without negative valence, positive arousal, low-expression distress, device artifact, and context shift. Performance on ordinary data is not enough. The score must encounter the discordant states that motivated the measurement problem (Nosek et al., 2018; Open Science Collaboration, 2015).

10.4.3 — Scientific Governance

Scientific governance gives the measurement chain authority to pause, narrow, correct, and stop. A board that can only advise after deployment is not enough for a system capable of intimate

inference or public misuse. The governing body needs defined jurisdiction, independent expertise, conflict controls, participant routes, and the power to require correction across dependent artifacts.

Governance should include review at multiple levels. Study protocols, participant burden, consent, channel collection, model design, analysis plans, missingness handling, public summaries, and data reuse all require oversight. Artificial intelligence use requires additional review because model training, inference, summarization, and automated classification can drift quickly into unauthorized authority.

The prospective specification package should separate immutable fields from controlled contingencies. Immutable fields include the primary construct interpretation, endpoint-independent data split, sign rules, confirmatory transforms, principal weights, primary rival set, and claim-update rule. Controlled contingencies describe how the analysis responds to sensor failure, unexpectedly high missingness, software error, or safety interruption.

Failure publication is a duty because a literature containing only successful multichannel studies will overstate validity, waste future effort, and continue burdensome methods that should be retired. Reports should include technical failure, null results, ethical nonviability, high burden, failed invariance, unstable weights, rival dominance, public-risk cancellation, no-estimate frequency, convergence problems, device failures, and model disagreement. Claim status and public communication must

change when later evidence changes the result (Ioannidis, 2005; Franco et al., 2014).

Registered Reports are especially valuable for pivotal validation and model-comparison studies because review precedes results and publication does not depend on a positive finding. Where unavailable, the protocol and analysis plan can still be publicly timestamped before data access if the record and later deviations remain visible (Nosek & Lakens, 2014; Nosek et al., 2018).

When a pivotal study uses a randomized design, current trial standards offer a concrete governance spine. SPIRIT 2025 identifies minimum information that should be fixed in the protocol, while CONSORT 2025 identifies essential information that should be reported after the trial. Neither standard validates the construct or the Law, and neither replaces model recovery, prospective specification, ethics review, or serious rival comparison; their value here is to make omissions, changes, and interpretive overreach easier to detect (Chan et al., 2025; Hopewell et al., 2025).

Independent data custodianship can reduce conflicts between estimator development and access. A custodian may enforce purpose limitation, track queries, manage outcome release, and preserve withdrawal rules without controlling scientific interpretation. The separation is especially valuable when vendors, donors, or sponsoring institutions have incentives tied to success.

A pivotal multichannel estimate study should use a staged release protocol. Simulations, development data, and pilot feasibility can

be available while the confirmatory outcome set remains sealed. Before release, the team deposits code hashes, channel registry, weight specification, missingness model, exclusion list, planned figures, model-recovery results, and the exact claim-status table. The outcome custodian then releases only the files required for the locked analysis, and every deviation is timestamped (Nosek et al., 2018; Chan et al., 2025; Hopewell et al., 2025).

Reproducibility includes abstention. Another team should be able to recreate the exclusions, uncertainty, no-estimate decisions, and public figures from the preserved analysis capsule. A pipeline that reproduces numbers but not the circumstances in which it must refuse a number has not reproduced the instrument (Ioannidis, 2005; Gelman & Loken, 2013).

Correction procedures should be planned before publication. When preprocessing error, sign reversal, data leakage, or an invalid exclusion changes a result, the research team should identify every affected analysis, table, figure, summary, image, tool, and public statement, issue a versioned correction, and reclassify the claim. Scientific memory requires a visible path from error to corrected interpretation; silent replacement allows obsolete artifacts to keep circulating as though nothing changed.

Governance should be tested before it is trusted. A release rehearsal can deliberately introduce a sign reversal, stale calibration file, duplicated participant, endpoint misclassification, or hidden exclusion and ask whether the registry, audit trail, and pause authority detect it before publication. The exercise should test whether a participant can be found without exposing others' data and whether each figure and public artifact is identifiable. A

paper-only control has not shown it stops an error path (National Institute of Standards and Technology, 2024).

Correction authority must be paired with notification duty. When an error changes an estimate or the permitted claim, investigators should tell participants and downstream users what changed, why it changed, which versions are affected, and available remedies. Vendors, funders, and communications teams should not be able to delay that notice because the correction is inconvenient. Scientific governance is credible when the route from detection to pause, repair, reanalysis, notification, and withdrawal is specified before a favorable result creates pressure to preserve it (World Health Organization, 2024).

The analysis is now protected from retrospective rescue. The next question is how observations enter time. A cumulative estimate can serve as a bridge only when its name does not outrun its measurement properties. Addition requires a defensible origin, sign, scale, cadence, covariance model, and version that remain stable enough for the intended interval (Mokkink et al., 2010).

10.5 — When Measurement Must Stop or Abstain

No adequate estimate is a real scientific result when the evidence cannot support a trustworthy number. It identifies which link failed, prevents false precision from entering later decisions, and preserves a clear route for better measurement. Abstention is not the absence of method. It is a conclusion produced by a method that knows its limits.

10.5.1 — No Natural Unit Is Given in Advance

A unit is not created by naming one. It is earned when a representation, an origin, a scale, and a rule of combination are justified well enough that equal numerical differences carry the intended meaning. Experience does not arrive with a natural ruler printed on it, so the public language of units must remain literal and conditional.

Measurement theory distinguishes ordinal, interval, and ratio interpretations because each survives a different class of transformations. An ordinal scale preserves ordering. An interval scale also gives meaning to equal differences, but its zero may be conventional. A ratio scale requires a meaningful zero and supports ratio statements. These distinctions are scientific obligations, not vocabulary choices (Stevens, 1946; Krantz et al., 1971; Suppes & Zinnes, 1963).

The proposed terminal relation is especially sensitive to origin. Multiplying a valid signed scale by a positive constant preserves which values are zero, although it changes their magnitude. Adding a constant changes the cumulative total and can manufacture or destroy closure. A convenient midpoint, sample

average, or fitted intercept therefore cannot be treated as neutral merely because it makes the analysis easier.

Sign is a separate issue. Arousal, intensity, novelty, salience, and physiological activation do not reveal whether an experience is positive or negative. A model can be mathematically identified while its experiential orientation remains unknown. External anchors, designed contrasts, report interpretation, and context must supply the sign.

The momentary multichannel estimate may be a dimensionless, version-specific index. If that index is integrated over hours of declared experienced exposure, the result is index-hours for that instrument version. The label describes the bookkeeping operation. It does not establish a natural phenomenological unit or a transferable currency of experience.

A simple example reveals the limit. An index value of plus two for one hour and plus one for two hours produce the same arithmetic product under a linear rule. That equality is a property of the rule, not proof that the experiences were phenomenally equivalent. The model must earn any claim that equal products represent equal contributions.

Mixed valence creates another challenge. Positive and negative components may coexist rather than cancel within one moment. A single signed value can hide grief with love, fear with excitement, relief with exhaustion, or awe with dread. The cumulative model should compare scalar and explicitly multidimensional representations before reducing those states to one line.

Every reported result should therefore carry the instrument version, origin, sign convention, scale interpretation, exposure rule, sampling cadence, interpolation method, population, context of use, and uncertainty architecture. Remove those conditions and the apparent unit loses the argument that gave it meaning.

Robust analysis should range over every prospectively specified specification that remains defensible. Alternative origins, transforms, missingness models, exposure rules, endpoint windows, and instrument versions can each generate a measured terminal result. Report the full sensitivity range across specifications expressed on a common declared comparison scale, together with sampling uncertainty, rather than selecting the specification nearest zero. Keep incompatible targets or scales separate. Call the range an identified set only when explicit population-level assumptions establish that interpretation for a common estimand.

Measured equivalence is robust only when the uncertainty region covering the defensible specifications lies inside the declared empirical band under the prespecified error-control rule. If that region crosses the boundary, equivalence is unresolved; if only one preferred specification supports equivalence, the result is specification dependent. Even complete containment concerns measured compatibility under stated observation models, not exact latent closure (Schuirmann, 1987; Wellek, 2010).

When interval structure or temporal additivity fails, the program can still preserve useful outputs. It may report ordering, within-person change, local trajectories, event-linked contrasts, or a bounded multidimensional description. Abandoning a cumulative

unit claim does not require abandoning every form of measurement.

The unit language should be withdrawn when origin, sign, interval meaning, exposure, additivity, or transport is unsupported. The paper can still describe what was observed and what remains identifiable. Refusing an impressive unit is sometimes the most scientifically informative result.

A cumulative report should include a covariance-aware audit that a serious general reader can follow. It should show the sampling grid, long gaps, interpolation rule, serial dependence, origin sensitivity, version changes, and how uncertainty widens when observations are missing or correlated. A deliberately simple calculation may be shown beside the full analysis to reveal how much precision is created by assuming complete data, one fixed origin, and independent errors.

The final standard is simple. A cumulative estimate may serve a declared research question only while its representation, units, uncertainty, and context remain visible. It is never a fixed happiness budget, a private account to be optimized, or a currency through which one person, institution, or future event can settle another experience.

10.5.2 — Invalid Constructs, Drift, Reactivity, and Noise

A measurement route can fail even when every device functions. The failure may lie in the construct, the relation between the construct and its channels, the stability of that relation, the observation process, or the burden created by collection. These

failures should be separated because each requires a different remedy.

Construct failure occurs when the instrument answers a different question from the one declared or omits states central to the intended population. A model described as valence may instead track arousal, pain, disease severity, vigilance, social compliance, response style, or willingness to report. Reliable prediction does not cure that mismatch, and response-process evidence should show whether participants are solving the task the researchers believe they designed.

Channel failure occurs when the physical or behavioral observation does not retain a stable relation to the target. Movement, medication, skin contact, speech ability, device placement, environmental temperature, or disease can change the signal. The study should report the conditions under which a channel remains usable and the populations for whom it does not.

Drift is change in an instrument or in what its output means over time. Sensors age, software and models change, prompts become familiar, participants adapt, treatment begins, illness progresses, and cultural language moves. Scheduled calibration checks, control signals, held-out anchors, version control, and subgroup review should help distinguish change in the person from change in the observation system before a trend is interpreted as experience.

Response shift is a special form of meaning change. Illness, adaptation, treatment, trauma, aging, and new knowledge can change a person's standards, values, or understanding of a

question. The answer may change because the internal frame changed, not because the underlying experience moved by the same amount (Sprangers & Schwartz, 1999; Oort et al., 2009).

Resolution must be assessed for the actual design and question. Repeated observations may improve sensitivity when random errors can be averaged appropriately; persistent bias, unobserved intervals, and a signal lost through the measurement process need different repairs. A smaller group-average error bar also does not establish that each individual's trajectory can be resolved at the same scale. Reliability estimates, standard errors of measurement, smallest detectable change, and dynamic sensitivity should show whether the proposed empirical band is supported for the intended level of inference (Shrout & Fleiss, 1979; Bland & Altman, 1986).

Imputation does not replace observation. It can be useful when assumptions are credible and uncertainty is propagated, but it can also reproduce the model's expectations where the person supplied no information. Sensitivity analyses should compare plausible missingness mechanisms and show how much the conclusion depends on each one (Rubin, 1976; Little & Rubin, 2019).

Invariance failure is not merely statistical inconvenience. It means that identical numbers may not carry the same measurement meaning across the intended comparison. The claim should narrow to within-person, within-group, site-specific, or partially identified use rather than retaining universal language (Meredith, 1993; Millsap, 2011).

A robust analysis should combine these uncertainties instead of presenting them one at a time. Origin, sign, calibration, drift, reactivity, missingness, model architecture, transport, and noise can interact. The result should report the range supported by the defensible joint specifications.

Public communication should show the result's status, not only its center. Exploratory, within-person, non-comparable, partially identified, version-specific, and no adequate estimate are legitimate labels. A reader should not need technical training to know that a smooth line is fragile.

Maintenance obligations should be funded and assigned. Calibration, software review, security, accessibility, training, version control, and correction require people and resources. An instrument without a maintenance path can become invalid while retaining the appearance of institutional authority.

The measurement route remains useful only while it can expose its own limits. A mature program can narrow a claim, retire a version, reduce cadence, substitute a less burdensome channel, or stop entirely. That capacity is not timidity; it is what keeps a proposed bridge from becoming a machine for manufacturing certainty.

10.5.3 — No Adequate Estimate Is a Real Result

No adequate estimate is a positive statement about the evidence chain. It says that the requested inference is not supported under the declared construct, observation process, model, uncertainty, transport, endpoint, or ethical conditions. The result should

identify the failed link and the next tractable task rather than emit a number merely because a pipeline expects one.

No estimate is not zero. It is not neutrality, balance, low intensity, average experience, or evidence that nothing happened. It is also not a coded form of refusal, noncooperation, incapacity, or low worth. The absence of an adequate estimate concerns the method, not the person.

One failure family concerns the construct and representation. The target may be incoherent, may mix valence with arousal or function, or may ask one instrument to serve incompatible purposes. If neighboring concepts cannot be distinguished, later precision has no scientific object to recover (Cronbach & Meehl, 1955; Messick, 1995).

Another family concerns origin, sign, scale, and additivity. The neutral point may be unstable, positive and negative components may not share one defensible axis, or a plausible transformation may change the cumulative conclusion. More observations cannot rescue an unidentified coordinate system.

Observation failure occurs when missingness, reactivity, device access, care priorities, or terminal shutdown remove the states most relevant to the question. The conclusion must widen or abstain rather than treat imputed values as observed experience (Rubin, 1976; Little & Rubin, 2019).

Model, transport, and rival failures arise when small changes in preprocessing or channel set reverse interpretation, held-out calibration fails, several structures fit equally well, or a simpler rival explains the result. Resampling, time-block and site

holdouts, recovery studies, and equal-budget comparisons should expose that instability instead of averaging it away.

Ethical unavailability is a scientific boundary. A channel may be informative yet too burdensome, invasive, coercive, exclusionary, or disruptive to care; humane limits belong inside the design and may require No adequate estimate (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Failures may be temporary, structural, or local. Recalibration, accessible design, retention, another site, or a better device may repair a temporary failure; structural failure means the requested inference cannot be supported with the available evidence or acceptable burden. One participant, interval, subgroup, or channel may fail while others remain usable.

When the data identify only a sign, ordering, interval, or admissible set, report that partial identification rather than forcing a point estimate; Section 11.5 develops the rule.

The abstention rule should be fixed before confirmatory analysis and name triggers such as construct failure, unresolved origin or sign, inadequate coverage, unstable models, indistinguishable rivals, failed transport, unavailable endpoint bridge, or excessive burden.

Reports should name the failed gate rather than hide it behind a generic missing label. The decision must also count time, fatigue, disclosure, travel, device maintenance, sleep disruption, family exposure, privacy risk, and interference with care; the smallest adequate protocol is preferred.

Correction continues after release. Methods that later fail transport, calibration, or misuse review require dependent artifacts to be corrected, annotated, replaced, or withdrawn rather than preserved as unqualified historical scores.

Repeated No adequate estimate results still teach which constructs are not scalar, which channels fail, where burden blocks observation, which rivals remain tied, and what new methods are needed. Precise abstention keeps ambition answerable to evidence and the person larger than the instrument. Across studies, those abstentions can reveal whether the recurring bottleneck is conceptual, instrumental, population-specific, or ethical. That pattern can guide method development without pretending the missing estimate existed. A repeated bottleneck can then justify redesigning the construct or observation strategy before another cohort carries the same burden.

Chapter 11 — Comparing Experiences

Two people choose the same number on a scale. One uses eight to mean unbearable interruption. The other reserves nine and ten for experiences they cannot imagine. The marks match; the experiences may not. Chapter 11 begins with that ordinary scene because it contains the entire comparison problem: numerical sameness is not experiential sameness, and comparability must be earned for a specified use.

A measure can be useful within one person without supporting comparison between persons. It can be useful in one language without surviving translation, stable in one clinic without surviving a device change, or informative for one group while misrepresenting another. Comparison is not a courtesy granted to a familiar score. It is an empirical achievement with a defined scope.

Before comparing levels, specify what a failure will change. Configural failure blocks a shared-structure claim; metric failure blocks comparison of relations; scalar or threshold failure blocks level comparison. Partial invariance may support a narrower use, but only if the retained parameters and excluded comparisons were governed before outcome inspection (Meredith, 1993; Millsap, 2011).

Even successful invariance tests do not prove phenomenal identity. They show that a specified measurement model behaves similarly under tested conditions. They do not reveal whether two people's inner experiences are metaphysically identical, nor do they authorize ranking lives. The formal Law concerns each stream's own terminal relation, not a public league table.

The neutral origin creates a deeper difficulty. A scale midpoint, resting baseline, absence of complaint, or average response does not automatically separate positive from negative experience. Because cumulative closure changes with the origin, identifying it is part of the substantive claim, not a harmless recoding choice (Stevens, 1946; Krantz et al., 1971; Michell, 1999).

Mixed states complicate sign and scale because positive and negative experience may coexist without either becoming an error. Section 11.2 develops that structure and asks when a scalar representation is adequate, when separate components are needed, and when the evidence should remain partially ordered.

Travel across language and culture changes more than words. Politeness, stigma, intensity, display rules, reference groups, memory, and the safety of disclosure can all alter what a response means. The Chapter therefore treats cultural transport as a construct and response-process problem rather than a translation exercise (van de Vijver & Leung, 1997; Harkness et al., 2004).

Age changes comprehension, memory, time perspective, bodily baselines, and available forms of report. A child's face scale, an adolescent's mood diary, an adult's tradeoff judgment, and an older person's report under illness do not become comparable because they occupy the same numerical range.

Disability and chronic illness expose another failure mode. People can adapt their standards, priorities, and interpretations over time. Response shift may reflect recalibration, reprioritization, or reconceptualization rather than simple improvement or decline. An instrument that ignores this change

can confuse a new ruler with a new state (Sprangers & Schwartz, 1999; Oort et al., 2009).

The so-called disability paradox should not be used to deny hardship or to romanticize adaptation. Reports of good quality of life under disability can reveal resilience, changed priorities, supportive environments, selection, measurement limits, and the inadequacy of outsider prediction. They do not imply that barriers, pain, discrimination, or unmet needs are harmless (Albrecht & Devlieger, 1999; Ubel et al., 2005; World Health Organization, 2022).

Some conclusions can be bounded rather than point-identified. Researchers may know one state is more negative than another within a person without assigning a universal distance. A range may fit several origins or models. Partial identification preserves information without pretending uncertainty has disappeared (Manski, 2003; Tamer, 2010).

Group averages require special care because relations at the group level need not describe individuals. Ecological and ergodic failures can make a population association misleading for a person's trajectory. Since the Law concerns unified streams, person-specific temporal validity is not a secondary detail (Robinson, 1950; Molenaar, 2004; Fisher et al., 2018).

The comparison problem has a moral edge because measurement failure is rarely distributed at random. A scale that travels poorly can understate pain, distress, access barriers, or participation burden in groups already poorly served by institutions. Failed

invariance should narrow the claim and improve the instrument, never label the population defective.

Uncertainty must travel to the conclusion. Input error, sign uncertainty, origin sensitivity, model choice, missingness, device drift, and failed transport cannot be compressed into a precise ranking at the end. Sometimes the warranted statement is within-person only, within-group only, range-bounded, or not comparable.

A comparison that loses its evidence conditions must be withdrawn visibly. Section 11.5 develops the correction chain; the Introduction needs only the governing rule that papers, figures, software, and dependent decisions may not retain authority after the bridge that supported them has failed.

The reader should leave with a sharper question than whether experiences can be compared: compared in what sense, under which origin and unit, across which people, with what evidence, for what decision, and with what right to withdraw the claim?

That discipline protects the proposed Law from a false shortcut. Exact intrastream closure does not require a universal public scale of human worth. It requires researchers to identify what can be measured and transported without converting uncertainty into hierarchy.

11.1 — The Comparison Problem

Comparison fails first at the level of meaning. The same word can carry different intensity, stigma, social permission, time frame, or reference point. The same device can produce different relations to experience across settings, bodies, medications, languages, and expectations. Apparent uniformity is therefore a hypothesis to test, not a result to assume.

Within-person, between-person, group, site, and cultural comparisons require different evidence. A scale may track change for one person while failing to place two people on a common metric. A device may reproduce its physical signal while the construct relation changes. The claim must be stated at the level actually earned.

Measurement invariance provides one family of tests, but it does not settle phenomenal identity. Configural, metric, and scalar or threshold invariance answer different model questions under tested conditions. Failure should identify where transport stopped and which conclusions remain available, rather than forcing every group into one scale.

11.1.1 — Same Word, Different Experience

Language makes experience shareable, but it also makes experience easy to overstandardize. Two people can use the same word and mean different felt worlds. Pain, fear, peace, shame, relief, exhaustion, safety, and loneliness are not sealed units carried unchanged from mouth to mouth.

The Law of Fairness requires researchers to treat language as access and distortion at the same time. Report matters because the person speaks from inside the stream. Report is limited because speech carries memory, culture, fear, power, training, role, and expectation.

Same-word difference is especially dangerous in cross-cultural and clinical contexts. A measure developed around one emotional vocabulary may misread another. A scale that asks for depression, anxiety, pain, satisfaction, or relief may import assumptions about selfhood, expression, duty, family, gender, religion, disability, and social safety (Meredith, 1993; Millsap, 2011).

Public language should not say that two reports are equal merely because the same answer option was chosen. The safer claim is that the instrument found a similar response under a tested language model, with named limits.

The discipline here is to listen before scaling. Same-word evidence is strongest when participants have been allowed to show how the word lives in their world, not only how they fit into the instrument's world.

For a multichannel estimate, the rule is that words may begin inquiry but cannot end it. Same word does not mean same experience, and different words do not automatically mean different experience. A fair measure must listen before it compares.

A person can be injured twice, first by the experience and then by having it mistranslated into a sameness the evidence never

established (Vandenberg & Lance, 2000; Putnick & Bornstein, 2016).

11.1.2 — Same Device, Different Context

Devices are built for contexts, even when they pretend to be neutral. A wearable, phone sensor, laboratory instrument, sleep tracker, imaging protocol, or survey platform may behave differently depending on site, body, connectivity, climate, movement, lighting, skin tone, disability, placement, language, and user trust.

The present approach requires device context to be treated as part of measurement, not background. If a device changes the experience it measures, the measurement includes the burden of being measured. If a device excludes people without stable housing, money, access, language, electricity, or technical comfort, the dataset inherits inequality.

Same-device claims should also include device drift. Firmware updates, sensor replacements, platform changes, proprietary algorithms, and vendor decisions can alter a measurement over time. A longitudinal claim may fail if the tool itself changes while the study pretends continuity.

A device-context protocol should treat the instrument as part of the situation it measures. The same wearable in a clinic, a school, a factory, a shelter, a prison, a private home, or a hot outdoor workplace may collect different meanings even when the sensor output is identical.

Public-facing artifacts should name the site and device conditions under which a result was produced. A diagram that looks portable

can become misleading when it hides that the evidence came from one lab, one body type, one climate, one language, or one platform.

Device humility is not anti-technology. It is pro-truth. A device can widen access when it is tested, consented to, and bounded, but it can also make bad comparisons appear mechanical and therefore official. The book asks whether the tool reduces hidden burden or merely moves it into the body of the person misread (Chen, 2007; Cheung & Rensvold, 2002).

The same sensor on a relaxed participant and on a worker afraid of being monitored may alter behavior differently (Meredith, 1993; Millsap, 2011).

Device validation must therefore be site-aware. Researchers should test whether hardware, software, interface language, sampling timing, battery demands, data costs, and privacy expectations differ across settings. A technically functioning device can still fail ethically when the context makes participation burdensome or unsafe.

Technical equivalence includes hardware, firmware, preprocessing, placement, sampling rate, environmental conditions, and operator practice. Predictive calibration at one site is not evidence of construct invariance at another. Transport studies should separate device effects from site composition and should include negative controls that reveal when a model is learning location, workflow, or disease severity instead of experience.

Equal-looking outputs can arise from different lives and different model paths. A moderate distress score may conceal severe underreporting or describe a temporary, well-supported state; low activation may reflect calm, medication, exhaustion, dissociation, expressive restraint, or device failure; high activation may reflect danger, exercise, excitement, pain, or social pressure. Researchers should inspect item and channel composition, context, causes, uncertainty, and consequences rather than treating one number as proof that two people occupy the same experiential point.

11.1.3 — Invariance Failure as Claim Limit

Invariance failure does not make all research worthless. It shows where a scale is useful, where it is not, which channels are unstable, and which comparisons require a different method. The analysis plan should therefore state in advance whether a failure narrows the result to within-person, within-group, site-specific, descriptive, or exploratory use, or requires withdrawal from public communication.

Public repair should name the failed bridge without blaming the people involved. Saying that a comparison did not transport across a declared population is different from calling that population difficult, unusual, unreliable, or noisy. The instrument and model carry the burden of fit.

Claim limitation can be exact. A failed item may be removed or modeled separately. A failed group link may preserve within-group inference. A failed common origin may preserve ordinal or within-person change. A failed cross-site bridge may preserve a

site-specific relation. Narrowing is scientific information, not rhetorical retreat.

Invariance asks whether a measurement relation remains sufficiently stable across the conditions under which comparison is made. That is a validity question and an ethical question. If a scale systematically misreads a group, statistical error can become stigma, exclusion, false reassurance, misplaced intervention, or institutional disadvantage.

A tool intended to reveal hidden burden can therefore hide the burden of people who differ from its calibration sample. Researchers should not ask whether a comparison would be interesting until they ask whether it is justified. The burden falls on the instrument, not on people to fit it (Molenaar, 2004; Fisher et al., 2018).

Testing should cover the dimensions relevant to the proposed use, including language, physiology, culture, disability, age, socioeconomic position, site, device, medication, and context. Review by affected people can reveal response processes and harm pathways that formal fit statistics miss.

Public materials should not call a measure universal merely because it behaved well in one sample. A scale earns permission only for the populations, settings, devices, and comparison levels supported by the evidence. Narrow truth is more useful than universal language built on an untested bridge.

That permission gate should be linked to predeclared consequences. The protocol should state which failures require subgroup reporting, wider uncertainty, model revision, narrowed

claims, or withdrawal. It should also ask who would bear false positives, false negatives, exclusion, or public misreading if the bridge fails. Otherwise a comparison can quietly choose its own scope after the result is known.

The difference between a group pattern and an individual judgment is especially important. Even a defensible aggregate relation may not license a conclusion about one person, and a research-only comparison can still shape treatment, resource priorities, and stigma. Humility becomes operational only when the method constrains the use (Robinson, 1950).

Failure can enter through wording, bodily mapping, device bias, channel weights, sampling, participation, context, or the social meaning of the score. These failures are not interchangeable, but each can invalidate a broader claim. The limitation must travel with the result rather than remain buried in technical notes.

Narrowing should retain what the evidence genuinely supports. A site-specific relation, within-group ordering, or within-person change can remain useful after a broader bridge fails. The revised claim should name its surviving population, context, device conditions, and comparison level so local success cannot masquerade as a universal scale (Chen, 2007; Cheung & Rensvold, 2002).

11.2 — Neutral Origin, Sign, and Scale

Imagine a seven-point scale whose midpoint is labeled neutral. One study treats the midpoint as zero. A second treats a participant's personal resting baseline as zero. A third treats the absence of aversion as zero. The same sequence of responses can produce three different cumulative totals. The arithmetic is correct in each case; the scientific claims are not the same.

A defensible origin needs an independent relation to lived neutrality, not merely convenience or symmetry on a form. Sign also requires work because relief, grief, fear, pride, pain, and love can coexist. Mixed states may need separate positive and negative components rather than forced placement on one line.

Scale transformations matter differently. Positive changes of unit may preserve a zero-closure claim, while changing the origin can alter the residual itself. The representation must also be nondegenerate: it may not assign every state zero or erase independently anchored positive and negative distinctions. Researchers must therefore record the origin, sign rule, unit, linking method, population, version, and the evidence supporting positive and negative anchors relative to an independently identified neutral origin or neutral region. A zero produced by a scale that erased valence is not closure.

11.2.1 — A Zero Point Is Not Automatically Natural

Pain is a tempting negative anchor because it is immediate and widely recognized as aversive, yet it is not one uniform object. Acute, chronic, neuropathic, inflammatory, anticipated, emotional, procedural, and remembered pain differ in duration,

controllability, meaning, bodily context, medication, trust, and history. Pain can orient a bounded within-person contrast without becoming the natural zero or universal negative unit of felt life.

Ethically, pain anchors must avoid induced suffering wherever possible. Research should not create pain merely to calibrate a theory. Clinical or naturally occurring pain data may be considered only under strong consent, care priority, minimization, and review.

Any pain anchor should distinguish intensity, duration, location, threat, controllability, expected relief, meaning, and co-occurring states. A familiar one-to-ten scale may serve a clinical purpose, but it cannot establish cardinal equivalence between brief severe pain and long moderate pain or become a universal ruler for experience. Narrative report, function, disability context, medication, culture, and within-person history remain part of the validation (Stevens, 1946; Krantz et al., 1971).

Refusal or avoidance of pain anchoring is itself meaningful and should be protected. Some participants may decline because the topic is traumatic, culturally difficult, clinically exhausting, or irrelevant to their experience of burden. The design should provide alternative channels, and the refusal should not be converted into missingness that quietly lowers the participant's visibility.

Social exclusion is a candidate anchor because human beings are relational creatures. Rejection, humiliation, abandonment, invisibility, ostracism, and nonbelonging can carry intense felt

burden. The pain of exclusion can alter body, attention, memory, behavior, and self-understanding.

The inquiry also asks whether social exclusion is measured as individual distress or as evidence of social wrong. If a person suffers exclusion because an environment is unjust, the goal should not be only better measurement. It should be repair of the environment where possible (Michell, 1999; Hand, 2004).

Social-exclusion anchors require screening, consent, debriefing, stop rules, and refusal without loss of standing. Where direct exposure would be harmful, remembered, hypothetical, or low-burden designs may be more appropriate. The aim is to study a bounded relation without making rejection into a toy stimulus or treating distress as weakness.

A brief exclusion task can model a local expectancy violation, not chronic stigma, family rejection, disability exclusion, religious shunning, or exclusion from public life. Its strongest use is within-person channel sensitivity under carefully bounded conditions, with controls for confusion, loss of points, generic task failure, attention, and cultural differences in belonging, face, duty, autonomy, and shame. The anchor earns authority through discrimination, not familiarity (Russell & Carroll, 1999; Larsen et al., 2001).

11.2.2 — Sign and Mixed Valence Require Separate Work

Sign cannot be inferred from intensity. A rapid heart rate, widened pupils, muscle tension, or strong neural response can accompany fear, effort, anger, excitement, pain, joy, or awe. The observation system must establish whether an experience is

positive, negative, mixed, or uncertain rather than treating activation as valence.

Positive and negative affect may also coexist. Grief can carry love, relief can carry exhaustion, achievement can carry dread, and awe can carry fear. A single bipolar question forces one component to erase the other and may turn a psychologically rich state into a misleading midpoint (Russell & Carroll, 1999; Larsen et al., 2001; Berrios et al., 2015).

The scalar branch therefore has to compete with separate positive and negative dimensions. The comparison should ask whether a one-axis value preserves held-out prediction, response interpretation, and relevant consequences without leaving systematic residual structure. Simplicity is valuable only after the reduction has earned it.

Report design can preserve coexistence by asking positive and negative questions separately, permitting mixed-state labels, and allowing uncertainty. A participant should not be required to choose between sadness and gratitude when both are present. The response options should reflect the construct rather than the convenience of one score.

Timing matters. A person may feel fear during the event, relief seconds later, exhaustion an hour later, and meaning much later. A prompt that asks for one global rating can combine different stages of the response. Momentary, event-level, and retrospective reports answer different questions.

Relief, awe, chills, pain, and social exclusion can orient local contrasts, but none supplies a universal signed unit. Their

meaning depends on the burden removed, duration, trust, context, medication, control, social history, and the coexistence of other feelings. These states are valuable tests of whether a model separates sign from arousal and preserves mixed valence; they become misleading when a familiar reaction is treated as a natural anchor for everyone (Raja et al., 2020; Fillingim, 2017).

Mixed-valence models should face the same recovery tests as scalar models. Simulations can ask whether the planned design can distinguish separate positive and negative states from one bipolar factor, generic arousal, or response-style artifacts. If the design cannot recover the generating structure, the scalar conclusion is not licensed.

Culture and language can change which states are named, which disclosures are permitted, and how intensity is expressed. Translation should therefore include cognitive interviewing, local idiom review, response-style analysis, and examination of whether the positive and negative terms carry the intended social meaning (Harkness et al., 2004; International Test Commission, 2017).

Disability, chronic illness, and trauma can also alter the relation among body, behavior, and report. Calm behavior may coexist with pain, speech may be unavailable, and vigilance may be adaptive. The model should not convert difference from a reference pattern into uncertainty about the person's sincerity.

If sign, coexistence, or orientation cannot be identified, the claim should narrow. The study may retain local directional contrasts, separate dimensions, or a no-estimate result. It should not force a

signed scalar merely because the Formal Law would be easier to describe that way.

11.2.3 — Changing the Origin Changes the Claim

Try a deliberately simplified calculation. A model assigns minus two to one interval and plus two to an equally long interval. The total is zero. Now move the origin down by one unit: the same intervals become minus one and plus three, and the total becomes plus two. Nothing happened to the experiences. The apparent balance changed because the place called neutral moved.

Changing the unit behaves differently. Double both original values and the total remains zero: minus four plus four. This small example reveals a large requirement. A closure claim needs evidence for where neutrality belongs, not just a convenient scale. The origin must be justified before the final total is examined, and the conclusion must show what changes when more than one origin remains defensible.

Disability, trauma, sensory history, illness, and culture may require alternative anchors, assisted communication, pacing, modification, refusal, or no-anchor status. The study must learn the comparison's real shape rather than force people into its first design (Berrios et al., 2015).

Anchor meaning can drift across time, groups, devices, settings, treatment, trauma, exposure, or life stage. Longitudinal work should test reference change and permit revision without calling participants inconsistent (Narens, 1981; Suppes & Zinnes, 1963).

An anchor is a linking constraint, not the latent record or a truth that outranks testimony, consent, or the person. Its authority

depends on response-process evidence, accessibility, invariance, prediction, and context of use (Stevens, 1946; Krantz et al., 1971; Messick, 1995).

Anchor sensitivity should be a primary result. Prospectively defined origin sets are fitted without viewing the preferred terminal outcome; agreement strengthens a bounded claim, while divergence makes classification anchor-dependent.

Governance records identify version, language, population, alternatives, purpose, failures, drift, revalidation, retirement, and dependent artifacts so participants and reviewers can contest the anchor (International Test Commission, 2017).

Conflicting verbal, behavioral, and physiological anchors may reflect different constructs, time scales, or mixed states. Forcing them into one sign can invent an origin none independently supports.

Stress tests across languages, illness, medication, repeated exposure, accessibility formats, and context should examine direction, ordering, and decision stability. When defensible sets yield different classifications, report the range rather than the most attractive member (Messick, 1995; Millsap, 2011).

Blinded teams can separate anchor selection from terminal analysis, preserving rejected choices and reasons. This does not manufacture a natural zero; it shows whether the origin was defended by evidence rather than by closure.

A mature anchor culture keeps alternatives visible, permits refusal, tracks drift, publishes sensitivity, and retires tools whose interpretation no longer holds.

11.3 — Culture, Age, Disability, and Response Shift

Group protection follows from scientific validity. When an instrument fails to travel, the claim should narrow, the measure should be revised, or the result should abstain. Institutions may not treat failed invariance, inaccessible tasks, or unfamiliar expression as proof that a group is unreliable, less aware, or less deserving of care.

11.3.1 — Cultural Expression of Feeling

Cultural expression varies across and within cultures. Endurance, disclosure, complaint, modesty, family duty, gender, religion, migration, class, generation, stigma, and institutional trust can change what may be said and how intensely it is said. Cultural humility therefore resists both a universal emotional code and the stereotype that any group has one fixed response style.

Response styles such as acquiescence, extremity avoidance, modesty, and reference-group comparison can shift observed distributions without identical shifts in felt state. These patterns should not be treated as cultural stereotypes. They are hypotheses to test with item-level models, response-process interviews, behavioral or contextual evidence, and sensitivity analyses that ask how much the substantive conclusion depends on the assumed response style (Sprangers & Schwartz, 1999; Schwartz & Sprangers, 1999).

Culture affects report, behavior, context, and interpretation of anchors. A measure developed in one setting may miss local idioms of grief, relief, shame, resilience, or fear, or may overread visible expression. Local language review, cognitive

interviewing, community consultation where appropriate, and response-style analysis should therefore precede cross-cultural use.

Participation burden itself can vary by culture. Repeated private disclosure may be ordinary in one setting and socially dangerous in another; a scale may therefore retain only participants for whom its emotional vocabulary and research rituals feel permissible. Selection into the dataset can imitate invariance unless refusal, nonresponse, translation difficulty, and trust are analyzed as part of the measurement process.

Development changes vocabulary, memory, time perspective, privacy, dependency, and relationships to authority. A child may borrow adult words without adult concepts, an adolescent may hide distress from peers or adults, and an older person may normalize pain, grief, or fatigue. Age therefore changes the route from experience to report and the risks of disclosure rather than serving as a decorative demographic variable (Albrecht & Devlieger, 1999; Ubel et al., 2005).

A measure validated for adults cannot simply be lowered into childhood, and a child-adapted instrument cannot support adult-style claims without a bridge. Developmental use should name assent or consent, guardian involvement, dependency pressure, recall limits, informant role, accessible mode, and stage. When self-report, caregiver report, and clinician observation answer different questions, they remain different constructs rather than interchangeable observations.

Developmental invariance also concerns the target itself. The meaning of safety, shame, agency, loneliness, or relief can change as cognitive, social, and bodily capacities develop. A child and an adult may not be two points on one mature construct. Longitudinal designs should test whether the structure persists, and when it does not, the program may need age-specific measures linked only at carefully justified points.

Trauma can produce hyperarousal, numbness, quiet compliance, anger, detachment, fragmented memory, avoidance, or underreporting. Those patterns may be protective rather than transparent indicators of current valence. Question wording, timing, privacy, stop rules, researcher training, and the right to refuse must therefore be designed so survival adaptations are not reclassified as lack of feeling or unreliability (World Health Organization, 2001; World Health Organization, 2022).

Measurement can itself become part of a trauma-related state. Repeated prompts, bodily sensors, closed rooms, sudden requests, authority figures, or demands for narrative detail may increase vigilance, dissociation, appeasement, or withdrawal. A protocol should model that reactivity, allow context-specific response functions and no-estimate states, and avoid selecting only those least affected by the method (van de Vijver & Leung, 1997; Harkness et al., 2004).

Multilevel models can represent context-specific response functions, but they should not smooth away dissociation or freezing as noise. A no-estimate state may be more accurate than forced continuity.

11.3.2 — Chronic Illness and Baselines

Chronic illness and disability make baseline change impossible to treat as a footnote. A person may adapt standards, priorities, language, pacing, and expectations while pain, access barriers, discrimination, fatigue, or risk remain. Response shift can reveal genuine learning or reorientation, but it can also hide deterioration when an instrument assumes that the ruler stayed fixed.

A same-scale claim must therefore ask what the person is carrying before it decides what a number means. Calibrating only within an established burden can understate suffering; comparing the same person with a healthy reference without context can overpathologize adaptation. The relevant history includes duration, flare, recovery, fear, agency, care demands, and the cost of maintaining ordinary function.

Response shift adds another layer. People may recalibrate internal standards, reprioritize what matters, or reconceptualize health and well-being. Those changes can coexist with genuine improvement or worsening. Longitudinal invariance tests, qualitative appraisal data, and clinical context are needed to distinguish a changed life from a changed measuring rule (Oort et al., 2009; Rapkin & Schwartz, 2004).

Sampling can shift with symptoms as well. The most difficult days may generate the least complete data because fatigue, treatment, crisis, or care takes priority. Apparent improvement may then reflect an observation process rather than a change in experience. Baseline models should report who could respond,

when, under what burden, and what the missing intervals might contain.

A device may assume a particular movement, face, grip, posture, voice, sensory profile, or response mode and then mistake access conditions for burden. Voice, eye gaze, sign-language interpretation, symbol systems, supported communication, flexible timing, and alternate devices can preserve participation, but each mode may require a bridge study. The inaccessible original is not entitled to serve as the neutral standard, and instrument failure should change the claim rather than exclude the person.

The review process should include disabled researchers, participant voice, accessibility and assistive-technology expertise, and clinical input where appropriate. Participants need a route to challenge interpretations, and access work should be compensated rather than treated as an unpaid test of endurance. When a measure fails under disability conditions, the claim should narrow or the instrument should change. Exclusion is not a scientific solution (Fillingim, 2017).

Chronic illness also changes physiological channels. Medication, autonomic function, inflammation, sleep, pain cycles, mobility, appetite, and energy can alter signals used by a multichannel estimate. Without clinical context, a model may mistake illness dynamics for valence or miss the valence carried by those dynamics (Sprangers & Schwartz, 1999; Schwartz & Sprangers, 1999).

A useful design separates stable burden, episodic flare, treatment effect, uncertainty burden, medical-system burden, and social burden. It also limits tracking when repeated measurement makes illness more central in attention than the participant wants. More data are not automatically better when collection itself changes the life being observed.

Access requirements and material pressure can quietly select the sample. Smartphones, connectivity, private space, flexible schedules, transportation, charging, institutional trust, and freedom to refuse are unevenly distributed. Compensation may be necessary, yet participation can feel compulsory when people need care, money, housing, grades, or approval.

Reports should therefore identify who was reachable, who was excluded, who withdrew, who carried the burden of participation, and how material conditions shaped missingness. A comparison that hides those conditions does not become broadly fair because its sample size is large.

Reference-group effects matter under inequality. People may judge conditions relative to what is normal, possible, or safe in their environment. Satisfaction under deprivation and distress under abundance can both be genuine while reflecting different comparison frames. Adaptive expectations cannot be treated as evidence that structural burdens are absent (Albrecht & Devlieger, 1999; Ubel et al., 2005).

Context changes the meaning of a signal and can play several different roles. A raised voice in a home, courtroom, hospital, protest, therapy room, or workplace does not carry one

interpretation; fear in actual danger differs from fear in a safe laboratory. The analysis should state whether context is a cause to adjust for, part of the target, a modifier of observation, or a descriptive condition, because automatic adjustment can erase the burden under study while indiscriminate collection can create a dossier (World Health Organization, 2001; World Health Organization, 2022).

The governing rule is simple. Felt experience is lived in a body, history, environment, and relation. Baselines are therefore empirical constructions that must remain open to response shift, access conditions, material constraint, and contextual meaning. When those elements cannot be linked well enough for the intended comparison, the claim must narrow rather than forcing diverse lives into a common number.

11.3.3 — Group Protection

Group protection begins with a simple rule: when a measure fails for a group, the first presumption should be that the method, context, or comparison failed. Instrument weakness must not be rewritten as a deficiency in the people who were measured.

Group averages do not describe every member. Variation within a culture, disability group, age group, income group, clinical group, or nation can be larger than the difference between group means. A public label built from an average can therefore misdescribe most of the people it names.

Consider a hypothetical study in which the people who usually sleep longest also report the most distress. Within each person, however, the better-slept days are described as easier. Both

patterns could appear in the same records. Differences between people might reflect illness or working conditions, while changes within a person follow another pattern. Knowing who sleeps longest would then answer a different question from knowing what happens after this person sleeps longer than usual. This is one reason a relation across people or groups cannot simply be read as a relation within a life (Robinson, 1950; Fisher et al., 2018).

Measurement invariance and differential item functioning provide practical tests. An item may be equally reliable yet carry a different threshold, social meaning, or relation to the construct across groups. The response difference then belongs partly to the item and context, not simply to the person (Meredith, 1993; Millsap, 2011; Teresi, 2006).

The translation review in Section 11.2.2 also needs back-translation, accessible communication, and construct discussion: language may travel while meaning does not (Harkness et al., 2004; van de Vijver & Leung, 1997).

Disability, communication method, movement, sensory environment, fatigue, pain, assistance, chronic illness, and response shift can alter observations or internal standards without diminishing experience or dignity. Accessibility and changing meaning are therefore part of construct validity (World Health Organization, 2022; Sprangers & Schwartz, 1999; Oort et al., 2009).

Small groups face contextual reidentification, and community involvement can improve questions, language, burden review,

and interpretation without becoming ceremonial endorsement or substitute consent. Individual refusal remains protected (Narayanan & Shmatikov, 2008; Ohm, 2010).

Comparison may reveal unequal pain treatment, exclusion, environmental burden, or service access, but it should name the institution or material pattern rather than turn it into a group trait. Stigma, punitive use, concentrated error, or implied inferiority are withdrawal triggers (Fillingim, 2017; Benjamin, 2019).

Corrections must reach the same audience as the original claim, including headlines, classrooms, dashboards, donor briefs, and images. Results should return in accessible language, invite local correction, and state uncertainty without transferring the investigator's responsibility.

Groups may reveal where systems and measures differ; no group result becomes a rank of worth, inner truth, or deserved treatment. Protection is a condition of valid comparison, not a favor added later.

11.4 — Comparisons Within and Across People

Within-person calibration is often the strongest starting point because the person supplies repeated observations under a relatively stable method and context. Even there, adaptation, response shift, device change, medication, learning, and reactivity can alter the relation. The study must test stability rather than assume it from repeated format.

Between-person calibration asks for more: a shared construct, defensible origin and unit, comparable response processes, channel access, model behavior, and uncertainty. Group-level invariance can support a specified comparison without proving that two people's experiences are metaphysically identical or authorizing a ranking of lives.

Withdrawal conditions complete the ladder. A comparison should narrow or stop when calibration drifts, invariance fails, the origin becomes unsupported, missingness becomes differential, burden becomes excessive, or the intended use outruns the evidence. A claim that cannot be withdrawn is not a measurement claim; it is an institutional commitment disguised as one.

11.4.1 — Within-Person Calibration

Within-person calibration asks a limited question: can changes in this person's observations support an interpretation about changes in this person's declared construct? It does not require every participant to share one scale, and it does not authorize comparison of levels across people.

The person's own response patterns, context, body, language, health, routines, and accessible channels become part of the calibration. This idiographic focus can preserve information that disappears when a group average is treated as the model for everyone (Molenaar, 2004; Hamaker, 2012; Fisher et al., 2018).

A baseline should be dynamic rather than a single permanent number. Sleep, medication, illness, treatment, environment, stress, and response shift can alter the relation between the construct and its channels. The model should detect when the baseline or channel relation has changed instead of calling every departure an experiential event.

Repeated observation should span the time scale of the intended claim. A relation demonstrated over one quiet afternoon does not establish a month-long trajectory. Calibration windows should include ordinary variation, known context changes, and enough repetition to distinguish stable person-specific structure from coincidence.

Person-specific anchors can help orient sign and magnitude. The participant may identify clear local examples of relief, threat, comfort, or distress, and the protocol can test whether channels respond consistently around those events. The anchors remain local and should be revisited when circumstances change.

The model should separate observation error from genuine change. Replicate measurements, device checks, alternate prompts, and known-condition contrasts can estimate how much variation belongs to the instrument. Without that information, a responsive-looking trajectory may be noise.

Use the accessibility rule in Section 10.1.3: adapt the interface to the participant, and narrow or abstain when no valid route supports the intended inference.

Within-person evidence may justify ordering, event-linked contrasts, or estimated change without justifying an absolute level. The report should state exactly what was identified. A personal trajectory can be useful even when its zero and unit do not transport.

The result should remain versioned. A device replacement, medication change, new language form, altered prompt, or model update may require recalibration. Continuity is an empirical claim, not a convenience granted by keeping the same participant identifier.

Within-person calibration is research, not a private accounting practice. Participants should not be encouraged to monitor every mood, calculate a personal balance, or treat daily fluctuations as evidence about the Formal Law. Feedback requires separate evaluation for benefit, burden, and reactivity.

A successful person-specific model can inform later comparison by revealing which relations are stable and which vary. Its strongest test is not how closely it reproduces the observations used to tune it, but whether a frozen person-specific mapping calibrates a later time block, a changed but declared context, or an external channel without hidden retuning. Even that success does not make the mapping a shared ruler. The next step requires separate evidence that the construct, origin, scale, response process, and usable range are comparable across people.

When the person-specific relation fails, the program can narrow to qualitative report, event-level contrasts, a different channel set, or no adequate estimate. The failure does not make the person's experience less real. It identifies the limit of the observation system.

11.4.2 — Between-Person Calibration

Between-person calibration asks whether a number retains enough common meaning to support a declared comparison across people. That requires more than giving everyone the same instrument. The construct, response process, origin, sign, scale, channel relations, and uncertainty must be linked across the people being compared.

Within-person sensitivity is not sufficient. Two people may each show reliable change on their own scales while the same numerical level carries different meanings. Personal baselines, response styles, illness, culture, medication, and accessible channels can all shift the relation between observation and experience.

No instrument provides direct interpersonal access. Calibration is an argument built from shared anchors, overlapping tasks, item behavior, qualitative evidence, and model constraints. The resulting comparison remains conditional on those bridges.

Common anchors can help, but only when participants interpret them similarly. Anchoring vignettes, known-condition contrasts, and participant-generated examples may reveal response-scale differences. They can also fail when the scenario is unfamiliar,

morally loaded, culturally specific, or differently imaginable (King et al., 2004).

Multi-group models can test whether thresholds, loadings, intercepts, residuals, and latent distributions are sufficiently linked for the intended claim. Exact invariance is often unrealistic, but partial invariance must be justified rather than selected after the result is known (Meredith, 1993; Vandenberg & Lance, 2000; Putnick & Bornstein, 2016).

Reference-population choice is part of the model, not a clerical detail. Changing the norm can alter standardized scores, thresholds, apparent prevalence, and error rates even when the raw responses do not change. A between-person claim should therefore report the reference population, justify why it fits the intended use, and show how the conclusion changes under plausible alternatives.

The group-to-individual limits in Section 11.3.3 still apply: between-person calibration does not make a group result an individual judgment (Robinson, 1950; Fisher et al., 2018).

Device and site effects can mimic person differences. Sensor placement, network quality, staff behavior, privacy, room design, and clinical routines can all influence observations. Calibration should separate person, device, site, and condition variance before interpreting a between-person contrast.

Origin sensitivity should be part of the main analysis. If defensible neutral points reverse the ordering or move the comparison across a decision boundary, the conclusion is not

stable. A single preferred origin should not be selected because it produces the clearest story.

Partial invariance does not automatically entail partial identification. Sufficient valid invariant anchors can identify a specified latent contrast even when other items vary. When the surviving constraints identify only a range, ordering, or item-subset comparison, report that narrower result explicitly (Putnick & Bornstein, 2016; Manski, 2003; Tamer, 2010).

Transport to a new site or population should be tested prospectively with the same scoring rule across held-out languages, clinics, devices, and social settings. Simulations and recovery studies should also ask whether the design can distinguish person differences from response style, site, device, and missingness effects. Recalibration after failure creates a new version; it does not show that the original model transported.

Qualitative follow-up remains necessary when statistical links fail. Participants and local experts may reveal that an item has different meaning, a device violates a social norm, or a response is unsafe to give. That knowledge can narrow the claim or redesign the instrument.

Comparison also has consequences. A between-person result can shape stigma, funding, clinical attention, media narratives, and institutional decisions. Consequence evidence belongs in the validity argument because a technically accurate estimate can still be invalid for a harmful use (Messick, 1995; American Educational Research Association et al., 2014).

No between-person calibration authorizes a hierarchy of worth. Higher or lower estimated valence does not determine deservingness, resilience, sincerity, spiritual depth, competence, or care priority. The scale concerns a limited research construct under stated uncertainty.

The calibration ladder therefore increases both evidentiary and social burden as the claim widens. Each step should state the construct bridge, uncertainty, population, context, prohibited interpretation, and withdrawal trigger. A comparison earns reach only by surviving the risks created by that reach.

11.4.3 — Withdrawal Conditions

A comparison regime is honest only if it includes exit. Withdrawal prevents local success from becoming pressure for broader scales, populations, or uses and preserves the difference between what once looked promising and what the evidence now supports.

Triggers should be declared before release and matched to the level at which evidence can break: unstable retest results, origin sensitivity, sign reversal, missing invariance, item bias, subgroup error, nontransport, device drift, context shift, burden, privacy failure, or an equally good simpler rival.

Withdrawal should be typed and proportionate. Statistical withdrawal stops an unstable estimate; construct withdrawal removes an unsupported interpretation; transport withdrawal returns a claim to the setting where it worked; ethical or authority withdrawal ends an invasive, coercive, or consequential use. A local result need not disappear when a wider bridge fails

(Wasserstein & Lazar, 2016; Wasserstein et al., 2019; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

The record should distinguish pause, narrowing, correction, retirement, and reinstatement. A pause investigates a defined question; narrowing states a smaller claim positively; retirement closes an unsupported or unsafe use; reinstatement requires new evidence, not a renamed variable or threshold.

Each decision should version the claim, model, data, reason, owner, surviving uses, and re-review conditions, then propagate status into tables, figures, summaries, repositories, software, model cards, posts, trained systems, outputs, and prior decisions (Gebbru et al., 2021; Mitchell et al., 2019).

Independent review should test whether the frozen rule was applied consistently rather than protect the project from embarrassment. A narrower use survives only with its calibration, transport, accessibility, privacy, misuse, and held-out support; retreat must not become erasure or silence mistaken for safety.

11.5 — Uncertainty and Partial Identification

Every estimate inherits uncertainty from its inputs and from the model that combines them. Sensor error, response variability, missingness, time alignment, calibration drift, and model choice can create an irreducible floor beneath which distinctions are not supported. More decimal places do not lower that floor.

Partial identification is often the honest consequence. The data may support an ordering, interval, sign, or set of compatible values without supporting one point. Such a result can still answer a meaningful question while keeping the assumptions that would be required for stronger comparison visible.

Failed invariance and retired comparison claims should remain traceable. Correction requires updated papers, figures, dashboards, public explanations, and dependent decisions. Preserving the history of a retired claim prevents old scores from continuing to govern after their evidentiary basis has disappeared.

11.5.1 — Input Error and Noise Floors

A measured cumulative result inherits error from every earlier stage. Prompt interpretation, channel acquisition, preprocessing, calibration, sign, origin, scale, temporal alignment, interpolation, missingness, model architecture, and endpoint placement can each change the estimate. The uncertainty budget must follow the entire chain rather than begin with the final number.

It helps to separate random error, systematic bias, model uncertainty, and identification uncertainty. Random error describes variation that may average down under defensible

repetition. Systematic bias shifts observations in a stable direction. Model uncertainty concerns the choice among plausible structures. Identification uncertainty means that the available observations and assumptions do not determine one answer.

These categories require different repairs; the random-versus-systematic distinction in Section 10.3.2 explains why a larger sample cannot repair a shifted origin or invalid construct.

Errors-in-variables methods show why noisy inputs can attenuate, distort, or even reverse estimated relations, especially in nonlinear and dynamic models. The analysis should model the observation process or use validation data rather than pretending that measured covariates are true values (Carroll et al., 2006).

Origin uncertainty can dominate long integrations. A small shift repeated over many intervals can change the sign of the terminal estimate. Sensitivity to every defensible neutral point belongs in the main result, not an appendix.

Apply the resolution test in Section 10.5.2: if the smallest reliably detectable change exceeds the equivalence band, the study cannot adjudicate that band.

Imputation should be evaluated under several missingness models. Missing at random, state-dependent dropout, treatment-related absence, observation shutdown, and deliberate refusal have different implications. If the conclusion changes across plausible mechanisms, the result is only partially identified (Rubin, 1976; Little & Rubin, 2019).

Refusal is not an error to be predicted away. A participant may decline a prompt, sensor, or continued use. The analysis can

model the resulting uncertainty, but it cannot reconstruct consent by filling the blank with an expected value.

Within-person nonstationarity also matters. The relation among report, physiology, and context may change with treatment, learning, illness, or adaptation. Time-varying models should compete with stable models, and score continuity should be withdrawn when the mapping shifts.

Robust sets are often more honest than one interval. The study can calculate the terminal estimate under every prospectively specified origin, transform, missingness model, interpolation rule, and instrument version that remains defensible. When these estimates concern a common estimand on a common scale, their union is an estimated sensitivity region. Interpreting it as an identified region additionally requires explicit population-level identification assumptions; inferential claims also require sampling uncertainty.

Simulation should test the entire error pipeline. Data generated with known origin shifts, sign mistakes, serial bias, nonstationarity, informative missingness, and version changes can reveal whether the planned analysis recovers the truth or manufactures apparent closure.

A useful audit shows the naive calculation beside the full calculation. The naive version may assume independent errors, one fixed origin, complete data, and stable parameters. The comparison makes the price of those shortcuts visible and prevents a polished curve from concealing its assumptions.

Use the status labels described in Section 10.5.2. The phrase composite measure should never substitute for an uncertainty statement.

Public release should be delayed when the uncertainty object is incomplete. An early result can spread through summaries, graphics, donor materials, and model-training corpora long before a correction. Scientific urgency does not justify creating a durable false certainty.

Input error and noise floors therefore do more than widen confidence intervals. They determine which questions can be asked at all. When the required resolution, transport, or coverage exceeds the observation system, no adequate estimate is the scientifically complete answer.

11.5.2 — Failed Invariance

Failed invariance is the primary withdrawal trigger. If a measure does not mean the same thing across the people or contexts being compared, the same-scale claim must narrow. Continuing the comparison would create false equivalence.

Researchers should report the level of failure, the affected groups, the scope of invalidated claims, and the remaining valid uses if any. A measure might remain useful within-person while failing cross-group comparison. That distinction should be public.

Failed invariance should also trigger review of past materials. If prior summaries, diagrams, or applied-language statements relied on the comparison, they may require correction. Public memory must not retain the old claim after science has moved on.

Failure can also be informative about the theory. If scalar invariance fails but within-person dynamics remain stable, the empirical program may narrow to per-stream predictions. If even within-person measurement changes meaning unpredictably, the observation bridge is more deeply threatened.

This error is often neglected because it is less visible than sensor noise or missing data. Yet for the Law of Fairness, invariance error may be the most ethically significant. It is the error that turns difference into false comparison.

Invariance error should be tracked across language, culture, age, disability, trauma, chronic illness, socioeconomic context, device, site, and group membership where relevant. The goal is not to create an endless checklist, but to name the domains that could change meaning.

A same-scale claim should report whether invariance was tested, where it held, where it failed, and where it remains unknown. Unknown invariance should not be silently treated as successful invariance. Unknown is a claim limit.

Failed invariance may appear as uncertainty in loadings, thresholds, anchors, or linking constants, or as a deeper change in the construct or origin itself. Parameter uncertainty can sometimes be propagated through joint estimation or simulation; structural noncomparability makes the cross-group operation unavailable. Reports should identify the level of failure and show how partial-invariance and alternative-anchor choices affect the result so a numerical interval does not conceal an invalid

comparison (Little & Rubin, 2019; Meredith, 1993; Millsap, 2011).

Device or site failure should freeze the affected hardware, firmware, preprocessing, placement, or setting and require a bridge study before results are linked. The notice should identify whether the source was sensor performance, movement, skin contact, network access, environment, staff behavior, privacy fear, or institutional workflow. If no stable crosswalk can be established, the data series must be segmented and the claim restricted rather than forced into continuity.

Later uses should carry a label identifying the hardware, software, site, and population alongside the claim. If a result came from one device, one clinical environment, one platform, or one participant-access profile, applied domains should not receive it as field-ready evidence. The scientific burden stays with the measurement system until transport has been demonstrated.

A transport failure can be diagnosed with crossed designs in which participants, devices, and sites overlap. Such designs help separate hardware effects, workflow effects, population composition, and environmental context. When overlap is impossible, uncertainty increases and causal attribution should narrow. A machine-learning adjustment trained on the same biased site does not establish transport.

11.5.3 — Retired Comparison Claims

A retired comparison claim is a claim that should no longer be used. It may have failed invariance, depended on a biased device or missingness model, caused group harm, been overtaken by a

stronger rival, or become unsafe in public communication. Retirement is not silence. It is a visible change in authority.

The retirement record should identify the old claim, its former scope, the reason for withdrawal, the date, affected analyses and artifacts, and the current replacement language. Failed invariance, partial identification, transport failure, device bias, and ethical unsafety are different failure types and should not be collapsed into one vague correction (Meredith, 1993; Millsap, 2011; Manski, 2003; Tamer, 2010).

When a claim narrows, equivalence margins and inferential limits must remain tied to the declared use rather than being recycled as evidence of general sameness (Schuirmann, 1987; Wellek, 2010).

Retirement must propagate. Scientific analyses, public explanations, legal or civic illustrations, spiritual interpretations, private-life applications, material arguments, and transition materials cannot preserve a comparison merely because it remains rhetorically useful. Dependent text, figures, tools, and summaries require reanalysis, relabeling, or withdrawal.

Group harm can trigger withdrawal even when statistical fit appears acceptable. A comparison may stigmatize, rank, expose, pathologize, or invite punitive use against cultures, disability communities, trauma survivors, clinical populations, children, workers, prisoners, migrants, religious minorities, or people living under poverty and institutional power.

Harm monitoring should be independent enough to challenge scientific enthusiasm. Complaints, withdrawal patterns, qualitative interviews, subgroup adverse events, burden reports,

and evidence of institutional misuse may show that a technically successful instrument has failed its context of use. Such a result belongs in the scientific record.

Withdrawal for harm does not require suppressing every difficult finding. The safer form may be narrower, aggregate, contextual, delayed, technically restricted, or paired with repair. The question is whether release creates understanding and remedy or transfers burden to people who never consented to the public meaning of the comparison.

Misuse can emerge after release. If a comparison begins to rank people, justify delayed care, spiritualize suffering, support surveillance, or travel into an institution that lacks authority to use it, the response may require narrowing, correction, restricted circulation, or retirement. Public stewardship continues after publication.

The repair should match the pathway. A dashboard may need removal, subgroup estimates may need withdrawal, a classroom exercise may need replacement, a device version may need retirement, and a public claim may need visible correction in every channel where it appeared. A general disclaimer is not a repair plan. A concrete closure record names the affected output and the person responsible for confirming that its replacement has reached the intended users.

Retirement decisions must be versioned and human-governed, distinguishing scientific failure, revised modeling, failed transport, changed missingness evidence, ethical unsafety, and public misuse. Every dependent use must receive the revised

status. Pooled analyses may need reanalysis, longitudinal records may need a visible break, and exploratory estimates may no longer appear as validated in summaries or tools. The observation process remains part of the explanation, because missingness cannot be converted into neutrality (Little & Rubin, 2019). For example, a version break can mark exactly where a revised comparison rule changes the interpretation of an otherwise continuous record.

The common-scale problem has now been divided into claims that can be earned, claims that must narrow, and comparisons that must remain unavailable. Even an invariant and well-calibrated research measure would not become a score of worth, civic trust, spiritual standing, legal truth, employment fitness, or public eligibility. A useful measure earns a defined job, such as comparing responses within a specified study, and its documentation should make that job clear.

Chapter 12 — Measurement Without Scores

A research record begins as a narrow promise: information will be collected to answer a defined question under stated conditions. Yet records travel. They can be copied, linked, sold, trained on, scored, cited, or used in decisions far beyond the context that made collection legitimate. Chapter 12 builds the positive governance that lets evidence remain useful without becoming ownership of the person.

The governing distinction is simple but demanding. Research data are traces produced through an observation process. They are not the conscious subject, the stream, or the latent cumulative object. An archive can preserve evidence about a question without becoming the life it describes, and a model trained on records does not acquire direct access to inner truth.

This boundary protects both science and dignity. When a trace is mistaken for the person, uncertainty disappears from public view and the record acquires authority it never earned. A score can begin as exploratory research and end as a judgment about employability, credibility, insurance, education, treatment, or moral standing.

Consent is therefore more than a signature at entry. It must be specific enough to identify what will be collected, why, for how long, who can access it, what can be inferred, whether it can be linked, how withdrawal works, and what happens after deletion is requested. Ethical research requires attention to risks and burdens throughout the lifecycle, not only at enrollment (National Commission, 1979; World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Burden also includes the act of measurement. Repeated prompts can exhaust a participant. A wearable can be uncomfortable or stigmatizing. A private question can create fear, shame, or conflict at home. A protocol can exclude people who cannot tolerate the device, read the interface, speak the dominant language, or remain continuously connected.

Minimization asks a practical question: which data are necessary for this purpose? Collecting everything because storage is cheap creates risks that are difficult to foresee and harder to reverse. Purpose limitation and contextual integrity keep data within the social relationship that justified its collection (Nissenbaum, 2004; Nissenbaum, 2010; European Union, 2016).

Deletion becomes harder after copying, linkage, derivation, publication, or model training. A responsible promise must distinguish the original record from backups, features, aggregates, and learned influence, then state which layers can actually be removed or restricted. Section 12.2 develops those limits; the public rule is that technical difficulty must not be disguised as complete erasure.

Purpose limitation gives the lifecycle a center. A valid workflow begins with a declared question and minimal data, records consent and burden, separates access roles, preserves provenance, tests intended use, blocks prohibited use, supports correction and withdrawal, and deletes or quarantines data when authority ends. Each step has an owner and a stop condition.

A final boundary governs everything that follows: estimation does not create rank. A measure of pain, burden, mood, resilience,

contribution, or apparent regulation may answer a bounded research question, but it cannot become a measure of worth, sincerity, loyalty, holiness, credibility, civic readiness, employment fitness, or legal deservingness. Predictive accuracy never creates legitimate authority over dignity.

Record separation protects the person after publication. Research, clinical, employment, educational, legal, civic, religious, and platform records operate under different purposes and permissions. Linking them creates a new system with new risks and requires new authority; technical convenience cannot supply that authority.

Public availability changes access, not authority. A post, open dataset, image, or published example remains embedded in the purpose, relationships, and expectations under which it was created. Reuse therefore requires its own justification, provenance, and affected-person protections rather than a presumption that visibility equals surrender (National Institute of Standards and Technology, 2020; Nissenbaum, 2010).

AI systems can intensify function creep because they combine weak traces into strong-looking inferences. Model cards and datasheets can improve documentation, but documentation alone does not authorize deployment. Provenance, limitations, training sources, intended uses, prohibited uses, and correction routes must remain visible (Gebru et al., 2021; Mitchell et al., 2019).

Some measurement should not be done. Coercive collection, crisis extraction, hidden emotion inference, employment or legal scoring, school ranking, spiritual classification, public burden

dashboards, and synthetic suffering created for data cross the line even when technically feasible. A fair research culture needs the courage to refuse an interesting dataset.

Nonparticipation must carry no penalty. A person who declines measurement does not become less credible, less cooperative, less worthy of care, or more suspicious. The right to remain partly unknown is a condition of dignity, not an obstacle to responsible inquiry.

The chapter does not attempt to write the operating rules for every institution. Those details belong in *The Government of Fairness*, *The City of Fairness*, and *The Constitution of Fairness*. This book establishes the boundary that every applied system must inherit: measurement may support a declared purpose, but it may never become ownership of inner life.

The reader should leave with a practical governance test: What was promised? What was collected? Who can access it? What was inferred? Which uses are allowed or forbidden? How can a person refuse, correct, withdraw, or seek remedy? What happens to derivatives, backups, shared copies, and trained models when authority ends?

Measurement becomes trustworthy when investigators can answer those questions and act on the answers. The goal is not timid knowledge. It is knowledge worthy of trust because it preserves the human being as more than any record, model, estimate, or archive.

12.1 — Research Data Is Not Inner Life

Research outputs remain bounded traces: neither data, archives, nor models become the person or the proposed cumulative object. That distinction must be operational in provenance, access controls, retention, public interfaces, model documentation, and correction.

12.1.1 — Data as Trace

Data are traces produced through a particular encounter between a person and an observation process. A report, waveform, image, note, gesture, code, or model output can support inquiry, but it does not reproduce the experience that gave rise to it. Provenance should therefore travel with the trace, including prompt, device, language, setting, sampling interval, transform, missingness, purpose, consent or authority, and the right to refuse.

Calling data a trace prevents moral inflation. A trace may support a research question, but it does not reveal worth. It may indicate burden without ranking the burdened person, and it may help estimate valence without creating public authority over inner life (Nissenbaum, 2004; Nissenbaum, 2010).

Provenance should answer practical questions. Who collected the data? Under what burden and consent? In what language, device, room, culture, clinical state, and power relation? What was refused, unavailable, or deliberately not asked because collection would have caused harm?

Recombination changes risk. A single trace may seem limited, but linking it with location, public records, employment data, donor

history, clinical files, or platform behavior can create inferences no original context authorized. Documentation and separation must travel with the record rather than being assumed from the intent of the first study (Gebru et al., 2021; Mitchell et al., 2019).

Felt experience is path-sensitive and inwardly lived, so the same trace can support several explanations. High arousal may reflect fear, effort, excitement, medication, pain, or anticipation. A low report may reflect relief, numbness, exhaustion, privacy, distrust, or wordlessness. The observation needs interpretation, and the interpretation needs limits.

First-person report deserves a protected place because no external channel replaces testimony. It also arrives through language, memory, safety, fatigue, stigma, culture, and the anticipated cost of disclosure. The task is neither to distrust nor canonize the report, but to preserve its context, permit correction and refusal, and distinguish the person's evidence from every later coding or inference (Floridi & Cowls, 2019).

The right not to report is part of the evidence architecture. Silence may be privacy, fatigue, refusal, fear, uncertainty, or wise self-protection. The absence of an answer should therefore remain a protected outcome, not a defect to punish or a zero to impute.

Trace metadata should distinguish ordinary missingness from ethical restraint. A gap created by device failure is not the same as a gap created because a participant refused, care took priority, or the question should never have been asked. Those differences affect both interpretation and governance.

The Law of Fairness treats data as useful and morally incomplete. A system that cannot preserve the difference between trace and person, or that strips away provenance as data move downstream, is not safe enough for the measured side of the Law.

12.1.2 — Archive Is Not Cumulative Object

An archive is a collection of records produced under particular purposes and conditions. The cumulative phenomenal object proposed by the Law is the complete felt path of one qualifying stream. The two are different in kind, even when the archive is large, detailed, and carefully governed.

Records begin with selection. A prompt is asked at one time rather than another, a sensor captures one channel, a clinician writes what matters for care, and an institution stores what its procedures recognize. What was never requested, noticed, disclosed, or technically observable does not appear merely because the archive later looks comprehensive.

A record is also a transformation. A self-report translates experience into language and a response scale. A sensor turns bodily change into a sampled signal. A coder turns narrative into categories. A model turns features into an estimate. Each step can preserve information, lose information, and introduce assumptions.

Time is transformed as well. A second of data, a daily summary, a peak-end memory, and a clinical note cover different intervals. Joining them does not create continuous observation. The unseen time between records remains modeled, bounded, or unknown.

Memory is not the original experience. A remembered life can be scientifically and personally important, but narrative reconstruction, salience, peaks, endings, and later meaning can alter what is retained. The archive may preserve a memory report without preserving the moment as it was lived (Kahneman et al., 2004).

More records can improve some inferences while leaving the target unidentified. Repeated reports may support a within-person trend, and synchronized sensors may help test a channel relation. Neither result establishes a natural origin, a complete domain, or the true terminal residual.

Joining archives does not repair the problem automatically. Research, clinical, platform, public, employment, legal, and personal records were created under different purposes, clocks, definitions, and permissions. Their apparent richness can hide incompatibility and create a dossier rather than a valid measurement chain.

An estimate derived from the archive is another record, not the latent object. Its meaning depends on the construct, inputs, preprocessing, model, missingness treatment, version, and uncertainty. A smooth trajectory may be useful while remaining a model-dependent summary of incomplete traces.

Artificial intelligence can search, classify, translate, and summarize archives. It can also infer beyond the evidence, reproduce bias, merge contexts, and write a coherent biography that feels more complete than the source. The United Nations Educational, Scientific and Cultural Organization's (UNESCO)

2025 Recommendation on the Ethics of Neurotechnology identifies mental privacy, autonomy, identity, and freedom of thought as governance concerns when neurotechnology and artificial intelligence can access, infer, or influence brain-related information. It is a normative instrument, not scientific evidence for the Law or universal legal authority (UNESCO, 2025). Fluency should increase the demand for provenance, not decrease it (Bender et al., 2021; Weidinger et al., 2021).

A model output should retain lineage to the observations and transformations that produced it. The record should show data source, date, purpose, consent or authority, preprocessing, model version, uncertainty, validation scope, and correction history. Without lineage, later users cannot distinguish report from inference or current status from a retired result (Geburu et al., 2021; Mitchell et al., 2019).

The archive is valuable when it remembers scientific process. It can preserve protocols, consent versions, code, exclusions, negative results, correction notices, and the reasons a claim narrowed. This kind of memory makes scrutiny and repair possible.

Correction should preserve both change and context. The original record may need to remain for accountability, while the current interpretation, reason for correction, affected outputs, and withdrawal status are made clear. Silent rewriting erases failure; uncorrected display preserves harm.

Deletion and preservation solve different problems. Some records should be removed to protect privacy or honor withdrawal. Others

may need restricted retention to document consent, misuse, or scientific error. The choice should follow purpose, authority, risk, and remedy rather than the idea that more archive is always better.

This distinction makes good science possible. Because the archive is not the object, investigators must state what was observed, what was transformed, what remains missing, and which conclusions are model dependent. The limit does not make evidence worthless; it prevents evidence from pretending to be complete.

12.1.3 — How Authority Grows Downstream

Each transformation can enlarge apparent authority: a report becomes a code, the code a feature, the feature a score, and the score a profile (Cheney-Lippold, 2017). Provenance, uncertainty, purpose, and contestability must therefore travel with every derived object. Technical convenience never entitles the artifact to speak more completely than the person or evidence it represents.

Dashboards can turn bounded estimates into judgment through names, colors, ranks, alerts, and arrows. Artificial intelligence accelerates this slippage by summarizing records, inferring missing attributes, generating biographies, and repeating classifications, but fluency never enlarges the authority of the inputs (Bender et al., 2021; National Institute of Standards and Technology, 2024).

Testimony remains evidence with context, limits, and rights to correction or refusal; it is neither demoted automatically when a sensor disagrees nor promoted into infallible access.

Contestability separates a scientific record from an institutional verdict. Participants and reviewers should trace observations, transformations, uncertainty, access, correction, and deletion. An artifact unable to answer those questions should not become an enduring account of someone else.

A failed audit may require narrower language, separation, reduced access, correction, or withdrawal. Context hidden elsewhere cannot rescue a misleading object where it appears.

Missing intervals, channel failure, model version, calibration population, uncertainty, and nontransport must travel with a trace. A clean output stripped of those limits can be more dangerous than the noisy record from which it came.

12.2 — Consent, Burden, Minimization, and Deletion

Consent is a continuing relation among purpose, channel, burden, retention, sharing, reuse, model training, correction, withdrawal, and deletion. When any of those elements changes materially, the ethical basis must be reviewed rather than inherited automatically.

Burden can make nominal agreement misleading. Illness, dependency, employment, schooling, custody, clinical crisis, poverty, family pressure, spiritual authority, or hope for care can narrow practical freedom to refuse. Ethical design must therefore examine recruitment, alternatives, compensation, timing, accessibility, and consequences for saying no (National Commission, 1979; World Medical Association, 2024).

Minimization asks for enough information to answer the declared question well. Imagine a study of whether people feel refreshed after a walk. The researchers need evidence that can separate walking from other explanations; that question does not automatically require a home address, a recording of conversations, or a permanent location history. Good minimization preserves the relevant comparison while leaving unrelated intimacy uncollected. A person can help build knowledge without handing over a detailed portrait of daily life.

The same precision applies when participation ends. A deletion promise must distinguish raw records, derived features, backups, shared copies, and influence on trained models. It should state which can be removed and which cannot be fully recalled. The participant deserves a promise the study can actually keep.

12.2.1 — Consent as Ongoing

Consent is a continuing relationship, not a permanent conversion of a person into a data source. The initial agreement authorizes a declared collection and use under stated conditions. It does not license every later inference, linkage, model, publication, or commercial application.

For intimate longitudinal data, the invitation should also state collection frequency, possible bystander capture, and prohibited downstream uses; vague language about “improving research” is not specific enough.

Assent and dissent require attention when a proxy is involved. A legally authorized representative may permit participation, but the person’s behavior, communication, distress, and refusal still matter. Observable dissent should have real stopping force unless an independently justified emergency rule applies.

Material protocol changes require renewed authorization. A new sensor, new inference target, new public artifact, new retention period, new sharing partner, new AI use, or new linkage can change burden and risk. Calling the change an implementation detail does not preserve the original consent.

Minor technical updates can be handled through documented version control when they do not alter purpose, burden, inference, or risk. The governance plan should state which changes require notice, which require reconsent, and who decides. That distinction prevents both consent fatigue and silent expansion.

Passive sensing needs special care because collection can continue when attention is elsewhere. Location, audio, video,

messaging, and device telemetry can reveal routines, relationships, and bystanders. The least revealing channel should be preferred, and content unrelated to the declared purpose should not be retained.

Bystanders cannot be covered by a participant's consent. Filters, on-device processing, restricted fields, event-triggered sampling, and deletion at ingestion can reduce relational exposure. When the design cannot avoid capturing people who did not agree, the channel may be unacceptable.

Withdrawal should distinguish future collection, future use, identifiable stored data, derived features, published aggregates, backups, and trained-model influence. Some elements may not be fully removable, but the limit must be stated honestly before consent rather than discovered after withdrawal.

Secondary use requires a new purpose review even when the data are de-identified or public. Re-identification, linkage, and sensitive inference can survive removal of names. The relevant question is whether the new use remains compatible with the original relationship and risk, not whether a file can be downloaded (European Union, 2025).

Human review remains necessary when an automated system routes consent, assesses burden, or predicts dropout. A model may assist scheduling or accessibility, but it cannot decide that silence means agreement or that a vulnerable person should be kept in a study for scientific value.

The practical standard is that participation can remain informed, voluntary, revisable, and safe throughout the lifecycle. When

those conditions cannot be maintained, collection narrows or stops. The uncertainty created by withdrawal belongs to the study, not to the participant as blame.

12.2.2 — Vulnerable Participants

Vulnerability is not a defect in the participant. It is a relation between person, context, power, burden, and risk. A person may be vulnerable because of age, illness, disability, trauma, poverty, institutional dependency, grief, incarceration, hospitalization, migration status, spiritual authority, classroom pressure, family coercion, or crisis.

Heightened safeguards do not mean exclusion by default. Exclusion can erase the people whose burdens and access conditions most challenge a measure, while inclusion without protection becomes extraction. Participation is justified only when the question is relevant, risk and burden are proportionate, refusal is real, accessible alternatives are available, and the design is built around the participant's dignity rather than the convenience of recruitment (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Review should also ask whether the study gives anything back to the participant population beyond knowledge extracted from them. This does not require therapeutic benefit in every study, but it does require honesty about burden, respect in design, accessible communication, and avoidance of research cultures that repeatedly ask vulnerable people to provide data while others receive prestige.

The protocol should collect the minimum channel set needed for the prospectively specified question. Before collection, investigators should record less intrusive alternatives, sampling frequency, burden, retention, withdrawal, and why a lower-burden route would not be enough. Scientific convenience and curiosity do not justify an additional sensor, narrative disclosure, or intimate context stream when the same inference can be supported more safely (McCambridge et al., 2014; Council for International Organizations of Medical Sciences, 2016).

Withdrawal must be ordinary, usable, and free from formal or social penalty. A participant may fear disappointing a researcher, losing help, grades, care, compensation, housing, opportunity, community standing, or approval. The protocol should permit pausing, reducing channels, stopping collection, declining future contact, and withdrawing data where feasible, with fair payment for burden already carried and no label of unreliability or betrayal.

Compensation design matters. Payment should not exert undue influence, especially for people in material need. At the same time, withdrawal should not unfairly deprive participants of compensation for burden already carried. Fair compensation requires careful structure so exit is neither financially punished nor financially pressured.

Exit rights must also be protected in citizen science and public-facing tools. An app, classroom activity, or voluntary tracking practice can still create pressure to continue. Quiet exit, deletion where possible, and freedom from shame preserve participation as a choice rather than a test of commitment.

Trauma-sensitive design is especially important. Some measures can reopen distress. Some prompts can intensify self-monitoring. Some dream or memory tasks can destabilize. Some participants may feel obligated to disclose more than they wish. Protocols must include stop rules, referral pathways, distress monitoring, and the right to skip without explanation (Emanuel et al., 2000; Appelbaum et al., 1982).

Felt-experience research can capture people who never consented through free text, audio, video, dream reports, location, household context, or stories about family, coworkers, clinicians, and friends. Prompts and collection systems should minimize identifiable third-party detail at the source, use redaction or local processing where necessary, avoid audio or video unless indispensable, and reject channels that cannot protect collateral privacy.

12.2.3 — Consent Is Not Research License Forever

The ongoing-consent requirements in Section 12.2.1 also govern secondary use, derivative models, and final closeout.

The study team should preserve the version of the information presented, the questions answered, the choices made, and the conditions under which consent was obtained.

Bundled permission weakens choice. A participant should not have to accept optional sensors, narrative disclosure, broad reuse, or model training in order to enter the least burdensome part of a study. Separate choices make refusal meaningful.

Practical freedom matters as much as formal permission. Patients, employees, students, prisoners, children, people in crisis, and

people dependent on services may reasonably fear that refusal will affect care, work, education, liberty, or standing. Recruitment and compensation should be designed so that saying no remains a real option (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

Therapeutic misconception is especially dangerous in measurement research. A participant may believe that an experimental estimate is a clinical service, that the model will reveal a hidden truth, or that participation is necessary for better care. Investigators should explain the distinction and monitor whether hope has changed the meaning of consent (Appelbaum et al., 1982).

Withdrawal should stop new collection and unauthorized future use, while the study states exactly what can be removed and what has already entered analyses, publications, backups, shared copies, derived features, linkage keys, or trained models. Lineage should connect every derivative to the permission that supports it, and any deletion claim should distinguish suppression, retraining, machine unlearning, annotation, and residual influence rather than promise erasure the system cannot deliver (Cao & Yang, 2015; Bourtole et al., 2021; Gebru et al., 2021).

Secondary-use review should classify a proposed use before data move. A minimal continuation stays inside the original question, data types, recipients, and consequence. A compatible extension changes a limited element but preserves the original relationship and may require notice or renewed choice. A materially new use, such as external linkage, individual prediction, commercial deployment, general model training, or public storytelling,

requires a new authority decision. A prohibited use remains prohibited even when a participant is asked to consent, because consent does not create scientific validity or legitimate an institution's claim to human worth or inner truth.

Derivative models require their own consent and withdrawal ledger because influence can persist after raw records are removed. The ledger should state which training runs used the record, whether retraining or machine unlearning is feasible, which benchmarks or summaries were derived from the model, and what residual influence cannot be reversed. Where complete removal is impossible, the program should say so plainly, restrict future use, annotate affected outputs, and avoid promising erasure that the technical system cannot deliver.

A final consent closeout should therefore reconcile people, permissions, data, code, models, publications, vendors, and public artifacts in one controlled review. The aim is not to turn consent into bureaucracy. It is to ensure that every continuing use still has a named purpose, a current authority basis, a proportionate burden, a correction route, and a responsible human owner. What cannot pass that review should be narrowed, placed under restricted stewardship, or stopped.

Consent does not make every risk acceptable. Some uses remain prohibited because the power relationship, consequence, or interpretation is incompatible with dignity. No person can authorize a study to define human worth, civic eligibility, spiritual rank, or inner truth.

A breach, misuse incident, or major model correction should trigger renewed communication when the participant may be materially affected. Notice should explain what happened, which records or outputs were involved, what containment occurred, and what remedy or choice remains.

Withdrawal must not reduce care, rights, services, or standing. A participant who leaves the study should not be marked as unreliable, uncooperative, unstable, or less deserving. Refusal is a boundary on research authority, not evidence about personality.

A mature program can answer a simple question for every derivative: what permission supports this use now? When the answer is unclear, the use should pause. When no valid permission exists, the appropriate result is restricted use, deletion where possible, or no further use.

12.3 — No Human-Worth Scores

A measured construct can become socially sticky long after its original limits are forgotten. A provisional estimate may be copied into a dashboard, employment file, school record, benefit decision, legal argument, or public label. Once a number appears objective, institutions may treat it as a shortcut to the person.

The no-score rule blocks that move. No estimate of burden, mood, resilience, pain, contribution, or apparent regulation can measure human worth, moral seriousness, trustworthiness, holiness, loyalty, credibility, or eligibility. The prohibition applies even when the score predicts an outcome, because predictive usefulness does not create legitimate authority over dignity (Barocas & Selbst, 2016; Selbst et al., 2019).

Correction after misuse requires more than revising a model card. Harm may persist through copied files, derivative models, institutional decisions, search results, and reputational memory. Repair must identify affected people, retract or quarantine outputs, correct public materials, notify downstream users where possible, and change the governance that allowed the score to escape its proper context.

12.3.1 — Not Human Worth

The first scoring prohibition is ontological. A measure never reaches the value of the person. It estimates a research construct under uncertainty. Confusing that estimate with worth would violate the reason the Law of Fairness begins with dignity.

This boundary is most urgent for people already carrying visible or hidden burden. Depression, disability, trauma, grief, chronic illness, poverty, dying, or crisis must never be converted into lower human value. The opposite error is equally false: calm, pleasure, fortune, or an apparently high-valence estimate does not establish wisdom, virtue, trustworthiness, civic fitness, or success.

Visual design must respect the same rule. Tables, charts, dashboards, colors, percentiles, comparative bars, badges, and leaderboards can imply rank even when the prose denies it. A person should never have to read a disclaimer to learn that a graphic did not mean what it appeared to show.

The prohibition is concrete. Multichannel and cumulative estimates may not govern employment, insurance, policing, schooling, immigration, welfare compliance, civic trust, public office eligibility, romantic suitability, parenting worth, spiritual seriousness, contribution value, public honor, or public shame. No finite list creates permission for a nearby use. The governing rule is that no measured construct becomes a rank of the person (O'Neil, 2016; Eubanks, 2018).

Human worth is prior to measurement. It is not earned by happiness, preserved by resilience, reduced by suffering, increased by contribution, or clarified by data. If public understanding begins to read an estimate as value, the material should be corrected, the use restricted, and the tool retired from public presentation when the worth interpretation cannot be removed.

Internal language matters because it becomes culture. Researchers should not describe people as high-value cases, low-value streams, balanced lives, failed ledgers, or useful outliers. The anti-worth rule begins before publication and applies to meetings, code labels, dashboards, filenames, and model outputs. Civic trust belongs to conduct, record, office, law, and review, not to inferred mood or hidden state. A person may be distressed and trustworthy, calm and dangerous, burdened and wise, or outwardly stable and exploitative. Using experience estimates as trust proxies would expose ordinary citizens while allowing institutions to treat private burden as evidence about eligibility (Barocas & Selbst, 2016).

Aggregate information can sometimes support service improvement, but aggregation is not automatically safe. Small groups, neighborhoods, disability communities, or stigmatized populations may still be identifiable or sortable. Aggregate use requires purpose limitation, minimum necessary detail, anti-linkage controls, and a ban on punitive consequences (National Institute of Standards and Technology, 2023).

This protects religious and secular readers alike. Treating suffering as purification, endurance as advancement, relief as favor, or measured balance as maturity would revive pain-as-payment and victim-blaming through technical language. The ethical boundary rejects those moves because measured experience cannot become sacred status.

The separation must also hold privately. A person should not read their own distress as proof of cosmic correction or their own relief

as evidence of higher standing. Research belongs to bounded inquiry; freely chosen meaning-making belongs to another register without measurement authority.

The positive promise is simple. A measure may help answer a declared question, guide a study, or reveal that more support is needed. It may never tell anyone how much a person is worth, how much trust they deserve, or where they stand in a moral or spiritual hierarchy.

12.3.2 — Contribution Is Not Inner Worth

Work and contribution systems can record tasks, roles, review, completion, and responsibility. They cannot measure suffering, resilience, sincerity, moral effort, or the seriousness of a person's inner life. *The Transition of Fairness* develops Workgame as one bounded coordination design; the no-human-worth boundary here is broader.

The distinction matters because contribution systems attract status. Someone who produces visible output may appear committed, while a person who rests, cares for family, manages disability, lacks resources, works privately, or refuses unsafe work can disappear from the record. Visibility is not worth, and absence is not failure.

Research and contribution records answer different questions. A research estimate concerns a declared observation model for a limited construct. A coordination record concerns bounded work in a defined process. Linking them creates a contribution-worth profile that neither system is authorized to produce.

Separate identifiers, access roles, retention rules, and databases should therefore be the default. A coordinator may need task status without research data. A researcher may need a study identifier without contribution history. Technical convenience is not a reason to join the records.

No research estimate may award points, badges, leadership, access, reputation, or eligibility. A missing, uncertain, or high-burden estimate must not reduce opportunity. Once inner-life data affects standing, participation is no longer meaningfully voluntary.

Invisible work can matter without becoming an inner-life estimate. Preparation, maintenance, translation, mentoring, accessibility support, caregiving, conflict prevention, and the decision to stop unsafe work may produce few public artifacts, while volume and speed metrics can encourage task splitting, neglected maintenance, hidden uncertainty, and burden shifting. Better workflow records should recognize bounded contributions and audit incentives without pretending to describe the whole person.

Behavioral targeting is another boundary. Research estimates should not be used to time prompts, increase commitment, prevent withdrawal, or recruit people through inferred guilt, hope, grief, fatigue, or vulnerability. Coordination must not become psychological management.

Artificial intelligence may assist scheduling, accessibility, search, or quality control under human review. It may not infer seriousness from inner-life traces, decide who deserves a role, or

combine contribution and research data into a profile (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

Records should expire or narrow when their operational purpose ends. A temporary delay should not become permanent reputation, and old coordination history should not travel into unrelated opportunities. People need factual correction routes and protection against secondary ranking.

When the boundary is crossed, linked records should be separated, labels removed, dependent decisions reviewed, affected people notified where appropriate, and the design changed to prevent recurrence. Correction should address both the data path and the social meaning created by the misuse.

The detailed architecture of any Workgame belongs to *The Transition of Fairness*. This book states only the measurement boundary: visible contribution and felt experience remain different domains, and neither becomes a rank of the person.

12.3.3 — Correction After Misuse

Correction after misuse begins by recognizing that an inner-life measure can cause harm long after the original analysis ends. A label may enter an employment file, classroom record, clinical note, donor summary, search result, model, or public story. Quietly changing the source document does not reach those descendants.

The response should start with containment. Access can be suspended, automated outputs stopped, public artifacts removed, partners notified, and new decisions paused while the route of

misuse is mapped. Containment prevents the correction process from becoming another period of exposure.

Correction should match the failure. Ambiguous wording may require replacement and an annotation. A public misstatement may require a visible correction in every channel. A consequential deployment may require withdrawal, deletion, contract suspension, model retirement, and independent investigation.

People affected by a label or decision need a direct route to learn what happened, contest the record, obtain correction, and seek remedy. The process should not require them to prove the hidden model's error using information they cannot access.

Downstream decisions must be reconsidered. Removing a score from the originating system does not repair an employment denial, school placement, benefit decision, legal filing, or public accusation that relied on it. The correction plan should trace consequences as well as data.

Correction may require changing the design, not merely the message. Access roles, retention, linkage, model-training separation, review authority, or prohibited-use controls may have failed. A communication patch cannot repair architecture that still permits the same misuse.

Documentation supports this work by preserving intended use, limitations, evaluation, and version history (Gebru et al., 2021; Mitchell et al., 2019). Documentation alone does not perform correction. Named humans must have authority, resources, and deadlines to act.

Correction is not reputational management. Its primary loyalty is to the person or group whose dignity, privacy, opportunity, or standing may have been harmed. Protecting a tool, institution, or theory from embarrassment is not a valid competing aim.

Remedy is more than retraction. Affected people may need a corrected record, reconsideration of a decision, restoration of access or opportunity, reimbursement, support, apology, or another independently grounded form of repair. Nothing in this process treats later remedy as payment that makes the earlier harm acceptable. Correction acknowledges that the harm should not have occurred and seeks to reduce what can still be repaired.

The burden of correction belongs to the institution that created or deployed the system. Affected people should not have to reconstruct every downstream copy, prove the model's internal logic, or repeatedly challenge the same label in separate offices. The responsible party should freeze the output, trace its descendants, notify decision makers, provide an intelligible explanation, and verify that the repair reached each consequential record.

Correction should preserve failure memory without preserving harmful labels. An institutional failure record can retain what failed, why, and which controls changed while removing or restricting person-level material that no longer serves a justified purpose. This distinction allows institutions to learn without turning the affected person into a permanent case file. The lesson belongs to the system; the identity does not.

A complete correction plan needs a dependency and version map. The same output may have been copied into a case note, converted into a flag, summarized in a meeting, cached in a warehouse, or inherited by another model, threshold, dashboard, or report template. The responsible institution should trace both data and decision lineage, freeze the exact failing version, block redeployment, and test whether later releases preserved its logic, parameters, training material, or data routes rather than accepting a new interface as proof of correction.

Consequential incidents require independent review with authority to disagree and halt use. The team that built a system may know it best while also being invested in minimizing failure, so reviewers need access to relevant records and no incentive tied to deployment. Urgency should follow the consequence, reversibility, reach, and time sensitivity of the action. A label affecting imminent care, liberty, employment, schooling, housing, benefits, or reputation may need immediate suspension before every causal link is known.

Closure requires evidence that repair reached the world in which the misuse occurred. Access must be removed, descendant copies amended or quarantined, affected decisions reconsidered, public artifacts replaced, partners notified, and the prohibited pathway tested to ensure it no longer functions. Where complete verification is impossible, the remaining exposure stays open in the record rather than being declared closed because an administrative ticket ended (National Institute of Standards and Technology, 2024).

12.4 — No Dossier, No Model-Training Shortcut

Access and permission are not synonyms. Material may be public for accountability, education, criticism, or scientific inspection without becoming unrestricted raw material for profiling or model training. Contextual integrity asks whether information moves according to the purposes and norms under which it was made available (Nissenbaum, 2004; Nissenbaum, 2010).

Separate record regimes must remain separate because linkage changes meaning and power. A research file, clinical record, civic archive, employment record, contribution record, spiritual record, and public communication artifact serve different purposes and operate under different authority. Joining them can create a dossier no participant agreed to and no source could justify.

12.4.1 — Public Access Is Not Permission for Every Use

Context matters because information is shared within social purposes. A testimony published to document abuse, a dataset released to reproduce a study, and a social post written for friends carry different expectations even when each can be reached by a browser. Moving information across those contexts can change its meaning and risk (Nissenbaum, 2004; Nissenbaum, 2010).

De-identification reduces some risks but does not guarantee anonymity. Rare attributes, timing, location, relational data, and linkage to outside records can re-identify people or reveal sensitive groups (Narayanan & Shmatikov, 2008; Ohm, 2010).

Scraping for model training is a new use. It changes who processes the material, how long influence persists, what outputs

may be generated, and whether correction can reach the derivatives. Research access is not standing permission to train a general system.

Public posts are especially vulnerable to context collapse. A person may disclose grief, pain, fear, or relief to seek support, not to become a labeled training example. Mining that language for hidden-state inference converts communication into involuntary measurement.

Images and audio can reveal bystanders, location, disability, family relationships, and environmental details. The person who posted the material may not have authority to consent for everyone captured within it.

Dream logs, terminal stories, and crisis disclosures require a higher threshold. Their intimacy and interpretive ambiguity make them attractive to systems that promise insight. They also make them easy to misread, sensationalize, and use as evidence of a theory the original speaker never endorsed.

A public artifact can become a dataset even when it was designed only to explain an idea. Captions, diagrams, excerpts, and examples may be copied into training corpora and separated from their limits. High-reach material should therefore avoid score-like shorthand and retain enough context to survive extraction.

Governance frameworks can discipline purpose, oversight, risk tracking, and lifecycle controls, but they do not supply scientific warrant for inner-state inference. Section 12.5.3 records the current regulatory examples and versioning rule.

Withdrawal becomes harder after wide dissemination, which makes restraint before release more important. Intimate detail should not be published merely because later correction procedures exist.

A public-reuse register can make the boundary operational. It can record approved purposes, denied uses, data versions, recipients, artifacts, incidents, corrections, and withdrawal status. The register should support accountability without becoming another profile of participants.

Public communication remains essential to science. It earns trust when it distinguishes access from permission, preserves context, and gives correction a route to catch up with circulation. It loses trust when visibility is treated as surrender.

12.4.2 — Separate Record Regimes Must Stay Separate

A record regime is more than a database. It includes a purpose, lawful or voluntary authority, identifiers, fields, access roles, retention periods, release rules, correction routes, consequences, and expectations about what the record means. Two files can sit on different servers and still belong to one merged regime if one silently unlocks the other.

Separation begins with purpose. Research data may support a declared scientific question. Clinical records support care. Employment records support bounded workplace functions. Educational records support learning and administration. Civic, donor, religious, contribution, archive, and public-artifact records each arise under different authorities. None receives a standing claim on the others.

Purpose limitation must survive technical convenience. The fact that records share a name, device, institution, or vendor does not create permission to combine them. A new use requires its own authority, necessity, proportionality, risk review, and correction path (European Union, 2016; Nissenbaum, 2010).

Identifier design is one of the strongest practical controls. Research identifiers should not be ordinary account identifiers, public handles, government numbers, employee numbers, or contribution IDs. A universal key turns every future breach, subpoena, purchase, or partnership into a path toward a dossier.

The record map must carry the derivative-removal distinctions and residual-influence limits specified in Section 12.2.3 (Bourtole et al., 2021).

Research and clinical records require a careful boundary. Care must not depend on participation, and research refusal must not become a clinical label. Clinicians may need narrowly defined study information for safety, but any transfer should be documented and justified by care rather than curiosity.

Employment and education records are high-consequence regimes. Inner-life estimates should not enter hiring, promotion, discipline, productivity, admissions, grading, attendance, or behavioral files. Even a voluntary wellness program becomes coercive when refusal or a result can be noticed by people who control opportunity.

Artificial-intelligence logs create another regime. Prompts, outputs, embeddings, evaluations, moderation records, and model telemetry may contain or regenerate sensitive information. They

require their own retention, access, provenance, and deletion rules and must not become a back door for joining human records (National Institute of Standards and Technology, 2024; World Health Organization, 2024).

Vendor boundaries matter because outsourcing can hide a merger. A platform may combine data across clients, products, analytics, and model training even when each institution believes its own use is narrow. Contracts, technical controls, audit rights, and deletion duties should address the vendor's full data flow.

A record-regime map should be created before deployment. It should show every source, identifier, transfer, derived object, access role, retention rule, public artifact, model input, decision, correction route, and prohibited link. The map makes hidden joins visible before they become infrastructure (Gebru et al., 2021; Mitchell et al., 2019).

The map should include inference, not only copying. A system can reconstruct a sensitive attribute without moving the original field. If one regime predicts what another regime is forbidden to reveal, the functional boundary has still been crossed.

Cross-regime transfer, wherever permitted, should be narrow and reversible. It should specify the exact fields, purpose, duration, recipient, legal or voluntary authority, security, human approver, no-secondary-use rule, and deletion or return condition. Broad consent and general research interest are not enough.

Record separation also protects validity. If participants know that research data may reach employers, courts, schools, families, donors, or government, reporting changes and refusal becomes

more likely. A clean boundary supports more honest participation because the declared context of use remains credible.

The strongest architecture avoids a central dossier. It does not create one master profile, universal identifier, or institution that can see every domain. Distributed responsibility, bounded custodianship, and explicit nonmigration preserve both privacy and the meaning of each record.

The boundary remains human even when law or technology changes. Regulations can supply important floors, and technical controls can reduce risk, but neither turns a prohibited interpretation into a legitimate one. No merged record becomes human worth, inner truth, spiritual status, civic rank, eligibility, or destiny.

12.4.3 — An Archive Is Not a Person

An archive stores selected representations. A person experiences, acts, remembers, forgets, changes, and can contest what others say. No amount of storage turns the first into the second.

Duration does not create personhood. Records may outlive the participant, institution, or consent process. Longevity can increase historical value without producing an enduring subject, soul, or conscious successor.

Memorial archives deserve care. Families and communities may value preservation, but those represented cannot revise the record. Curatorial notes should distinguish documented fact, testimony, inference, dispute, and silence.

Relational privacy remains active. Messages, dream reports, genetic data, location histories, and family accounts can reveal people who never consented. An archive about one person may therefore impose burdens on many others.

Deposit in an archive does not erase the record-regime boundaries established in Section 12.4.2.

Separate identifiers and access controls do not remove the linkage risks described in Section 12.4.1 (Narayanan & Shmatikov, 2008; Ohm, 2010).

Artificial intelligence magnifies value and risk. It can improve search, translation, deduplication, accessibility, and provenance, yet also infer, merge, summarize, and reproduce records at scales that strip away their original conditions. Contestability remains important where feasible. Participants and affected people should be able to access records, correct factual errors, learn about material secondary uses, and challenge generated profiles. An archive cannot become an unanswerable narrator of someone else's life. Purpose limitation should survive every transfer: a record collected for one bounded use does not quietly become permission for unrelated inference, and provenance should show which authority, consent, or rule supports each new use. A transfer register can make that limit visible by recording the declared purpose and authority for each downstream use. The transfer register should also record deletion, return, or supersession by downstream recipients, so provenance covers exit as well as entry. That record matters when corrected or withdrawn data would otherwise continue circulating through derivative systems.

12.5 — Measurement That Should Not Be Done

Not every measurable question should be asked. Technical feasibility does not answer whether collection is voluntary, proportionate, valid, necessary, or compatible with dignity. The more intimate the inference and the greater the power imbalance, the stronger the presumption against deployment.

The common pattern is uncertain inner-state inference combined with high consequence and weak refusal. Contemporary risk frameworks provide governance examples, while the ethical boundary here remains independent of any one jurisdiction (World Health Organization, 2024; National Institute of Standards and Technology, 2023; European Union, 2024).

12.5.1 — Coercive Measurement

A measurement program that cannot say no will eventually be asked to measure the wrong people in the wrong place for the wrong reason. Pressure will concentrate where people have the least freedom to refuse. Children offer early data, terminal patients create scientifically sharp endpoints, workplaces seek prediction, schools seek control, and platforms can detect vulnerability. Those incentives make independent review, proportionality, assent or consent, and a genuine stop rule indispensable (National Commission, 1979; World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

This refusal is part of the scientific discipline. A theory that needs coercive data, induced suffering, crisis extraction, hidden monitoring, or youth self-scoring to survive has already failed the

framework developed here. The burden of proof belongs to the system that wants to watch.

Coercive measurement can hide inside benevolence. An institution may say the goal is support, safety, wellness, or fairness. But if refusal carries penalty, suspicion, reduced access, or stigma, the measurement is coercive. Support should be available without requiring inner-life exposure.

Crisis contexts require heightened caution. A person in distress may agree to anything that appears to preserve care, belonging, housing, or relief. Consent during crisis can be real for care decisions, but research measurement has a much higher burden.

Coercive measurement occurs when authority, dependency, or structural pressure makes inner-life disclosure a condition of opportunity, care, grades, benefits, liberty, work, or standing. The test is practical freedom from penalty and suspicion, not the institution's description of the tool as voluntary. A worker required to wear an emotion sensor or a student expected to complete a mood tracker is being measured under authority. The European Union AI Act offers one jurisdiction-specific example by prohibiting emotion-recognition systems in workplace and educational settings except for narrowly defined medical or safety purposes (European Union, 2024).

Public ranking converts a research estimate into competition and identity. A burden, well-being, resilience, happiness, balance, trust, or fairness score can induce performance, concealment, gaming, stigma, and pressure even when the stated goal is motivation or accountability. The proposed Law studies hidden

experience to improve understanding and care, not to produce leaderboards of people, communities, institutions, or countries.

Group and anonymous rankings remain dangerous. Small or distinctive units may be reidentified, aggregate labels can stigmatize whole communities, and public comparisons can distort funding or policy while encouraging institutions to optimize the display rather than the underlying conditions. Governance should assess foreseeable social effects and prohibit punitive or status consequences rather than assuming anonymity solves the problem (World Health Organization, 2024).

Crisis extraction can occur through platforms, public appeals, tragedy recruitment, or prompts delivered when people need help, medicine, shelter, safety, money, recognition, or someone to listen. Availability during crisis is not free consent. Care and stabilization come first, and crisis-facing systems should collect only what is necessary for safety, discard intimate detail by default, and separate any retained safety record from research datasets (Barocas & Selbst, 2016; Eubanks, 2018).

12.5.2 — Employment or Legal Scoring

Employment and legal settings magnify power asymmetry. Applicants, workers, defendants, claimants, prisoners, migrants, parents, students, and litigants may be unable to refuse measurement without risking opportunity, liberty, benefits, credibility, or standing. That is precisely why intimate inner-state estimates must remain outside consequential scoring.

Even accurate information can become unjust when used under coercive authority. Employment and legal systems should address

accommodations, safety, evidence, due process, and remedy through purpose-built procedures rather than by converting provisional experience estimates into judgments about a person.

Legal misuse may appear as proof of distress, credibility assessment, risk evaluation, damages calculation, competency inference, sentencing input, custody evaluation, or compliance monitoring. Such uses would give measured estimates more authority than they can bear and expose vulnerable people to adversarial reinterpretation.

Child and youth measurement requires special restraint because young people are still developing identity, language, emotional regulation, consent capacity, privacy expectations, and relationships to authority. A tool that might be acceptable for an autonomous adult may be unsafe in a classroom, family, platform, clinic, or youth program (National Commission, 1979).

Youth overreach includes asking children to disclose family burden, log dreams, wear emotional sensors, compare well-being, appear on dashboards, or supply data for claims they cannot fully evaluate. A model output or school record can outlive the moment it tried to describe and follow a developing person into new contexts. Scientific interest in youth trajectories does not make broad collection, public comparison, or permanent labeling fair.

Authority pressure is central. A young person may comply because a teacher, parent, clinician, coach, religious leader, or researcher asks. They may not understand future data risks, may fear disappointing adults, and may not know how to refuse. Consent and assent must function as protections rather than

formalities (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

The future-self problem requires narrow retention and strong deletion. Children and adolescents cannot fully anticipate how they will later view mood, fear, sleep, social pain, family context, or crisis records. Youth data should carry age-appropriate explanation, protected assent and dissent, guardian and independent review where appropriate, minimal linkage, no scoring, and a presumption that future dignity outweighs speculative reuse.

Youth-facing materials should be private, optional, non-diagnostic, non-scoring, and designed to reduce burden. Classroom, family, platform, clinic, religious, and civic uses require special review because authority and peer visibility change consent. Children must not become proof material for adults seeking evidence of need, improvement, sincerity, risk, resilience, or formation; measurement proceeds only when welfare, comprehension, privacy, and future freedom are better protected by collection than by nonmeasurement (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979).

Voluntary spiritual practice and work coordination are different domains, but both are vulnerable to status hunger. Neither may use affective estimates, proposed experiential units, unity measures, or private observations as evidence of rank, readiness, sincerity, belonging, contribution worth, or spiritual seriousness. Inner-life data cannot become a route to prestige, leadership, donor favor, or exclusion, and nonparticipation cannot lower

standing (National Institute of Standards and Technology, 2023; World Health Organization, 2024).

Informal culture can recreate a score without an official dashboard. Rumors, badges, challenges, praise, and labels such as “more aligned,” “more balanced,” “more serious,” “fairer,” “more ready,” “more healed,” or “more advanced” can turn optional practice into hierarchy. Leaders and coordinators need training and a duty to interrupt that drift rather than benefit from it, preserving community meaning and visible contribution without converting private experience into status.

12.5.3 — Deployment-Prohibited Tool Registry

A deployment-prohibited tool registry records capabilities and uses that may not be deployed because their foreseeable harms exceed any justified benefit. It gives institutional memory to refusal so an unsafe idea does not return under a softer name after personnel, funding, vendors, or technology change.

The registry should distinguish a prohibited tool from a prohibited use and a categorical prohibition from an evidence-contingent one. A questionnaire may be acceptable in voluntary research and prohibited in hiring. A categorical boundary, such as no human-worth or civic-eligibility scoring, is not cured by better accuracy. An evidence-contingent prohibition may change only after the construct, transport, security, correction, and burden failures that created it are independently resolved.

Each registry entry should identify the capability, affected population, context, decision pathway, reason for prohibition, evidence, owner, date, review trigger, enforcement controls,

incident history, and downstream artifacts. It should also name the input channels, inferred construct, data and model versions, vendor dependencies, known proxies, prohibited interpretations, and rights or services at risk. The purpose is a usable map of power, not a longer description of the tool.

Core prohibited categories include public ranking, employment or legal scoring, spiritual or contribution status, civic readiness profiling, crisis extraction, covert monitoring, youth self-scoring, unauthorized model training, cross-regime dossiers, personal endpoint prediction, and any tool claiming direct access to inner truth. Equivalent functions remain prohibited when they use proxy variables or gentler language.

Consequence is part of the governed object. The same estimate carries different risk when it informs a voluntary study, a supportive conversation, a clinical judgment, an employment screen, a school placement, a public benefit, policing, or a criminal proceeding. The registry should name the actual decision route rather than governing the model in isolation.

Proxy and covert systems belong inside the rule. A vendor may avoid words such as emotion or fairness while inferring engagement, stability, risk, sincerity, or resilience from voice, face, movement, language, browsing, attendance, location, or networks. Passive collection does not make observation voluntary, and marketing language does not change function (Barocas & Selbst, 2016; Eubanks, 2018).

Governance needs measurement, ethics, privacy, security, disability, youth, clinical, legal, and public-risk expertise.

Developers may explain a system, but they should not control the final deployment decision when they benefit from release. Affected communities need a route into review without being required to carry or authorize the risk.

A public layer can state prohibited categories, governing reasons, responsible offices, correction routes, and aggregate activity. A protected layer may retain attack details, sensitive evidence, and personal information. Transparency should prevent quiet deployment without publishing a manual for circumvention.

The registry should also record an appeal or exception route where one legitimately exists and a date for the next review. Version history should preserve the rule that governed a decision without implying that later technical or legal change creates scientific warrant for inner-state inference (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024; European Union, 2024; European Union, 2026).

When a prohibited capability has influenced a decision, halt the use, preserve evidence, identify affected people where feasible, reopen dependent decisions, correct records, and provide notice or remedy under independently grounded standards. A prohibition should also name a safer route when one exists; when direct service solves the practical problem, the burden of justifying intimate inference rises.

Chapter 12 closes with the institutional capacity to say no. Trustworthy measurement can prohibit a tool before harm, retire it after failure, correct dependent records and decisions, preserve

failure memory, and keep private experience outside status systems (National Commission, 1979). That capacity becomes visible in ordinary decisions: a simpler service, a corrected record, or an accommodation provided without an additional assessment. A corrected record that reaches the affected person before the next decision is a stronger success signal than a policy that never changes practice.

Part V — Who or What Carries a Stream?

A cumulative history cannot exist without someone or something to whom its moments belong. The Law has named felt experience as its subject, stated an exact candidate boundary, and placed measurement under strict limits. Part V now asks the question beneath all of that work: what makes one conscious stream one stream? A body, a name, a legal identity, a biography, and an archive may each track a life, but none is automatically the bearer of experience.

This is not a decorative puzzle about identity. A mistaken bearer can assign positive and negative experience to the wrong subject, place an endpoint in the wrong history, or let a classification become more certain than the evidence allows. The problem is therefore formal, scientific, clinical, and ethical at once. Its difficulty is part of the Law's precision rather than a reason to loosen the claim.

The first movement begins with familiar human cases. Access, report, memory, agency, embodiment, and coherence usually travel together closely enough that unity feels obvious. Sleep, dreaming, anesthesia, severe brain injury, dissociation, and split-brain findings show that these channels can separate. Outward nonresponse can coexist with covert command following, and later amnesia can coexist with earlier experience (Owen et al., 2006; Bodien et al., 2024).

No single channel is sovereign. A bearer judgment must state which aspect of unity it estimates, which evidence was available, which conditions could distort it, and why uncertainty remains. A proposed Unity Index may organize evidence about access, integration, and continuity, but it is neither a discovered essence

nor a public score. It cannot rank persons, determine eligibility, settle legal or spiritual standing, or decide who deserves care. Unknown is a disciplined result that protects those most vulnerable to institutional overconfidence.

Branching and joining matter because nontransferability depends on knowing which stream owns which experience. Chapter 13 states the formal routing rule in full and keeps speculative cases separate from established human evidence. The public point is simple: no convenient merger, split, archive, or bodily label may move a residual from one bearer to another.

The second movement carries the bearer question beyond familiar humans. If the proposed Law concerns qualifying conscious streams, species membership, language, brain size, biological familiarity, and current technology categories cannot decide the matter in advance. Animals, insects, artificial systems, organoids, synthetic biological systems, and simpler living systems require a vocabulary capable of distinguishing supported, possible, unknown, not established, and not applicable.

Animals come first because human decisions already affect their welfare while the available evidence varies sharply across species. Chapter 14 compares stronger, weaker, and genuinely unknown cases without making familiarity a prerequisite for protection or treating precaution as proof of personhood.

Artificial systems create the opposite visual risk. Fluent output can resemble testimony and invite certainty even when architecture, persistence, embodiment, and phenomenal status remain unsettled. The Chapter separates behavioral evidence,

functional capability, public relationship, and possible consciousness so that neither enthusiasm nor dismissal decides the bearer in advance.

Organoids and synthetic biological systems move the bearer question into research design. Current uncertainty does not establish sentience, but it does create a duty to monitor capability, use stop rules, and avoid experiments whose value depends on producing possible suffering. Chapter 14 develops that precaution without turning unknown status into a verdict.

Chapter 13 — What Makes One Stream?

The Law cannot protect a conscious stream without asking what a conscious stream is. Earlier Parts established felt experience as the proposed subject, explained the terminal relation, explored possible routes, and built a path from experience toward evidence. Chapter 13 turns to the bearer that would own the complete history.

The problem becomes concrete whenever ordinary markers separate. Sleep removes normal responsiveness. Dreams can preserve experience without dependable recall. Anesthesia can combine silence, amnesia, and later report. Severe brain injury can leave bedside behavior absent while task-based brain activity reveals covert command following in some patients (Owen et al., 2006; Bodien et al., 2024).

A body is not enough because biological continuity and conscious access can diverge. Legal identity is not enough because law allocates social standing and protection rather than phenomenal routing. Biography is not enough because a public story is not the complete felt path. An archive is not enough because a record can describe experience without containing it.

The bearer must be approached through several evidence channels. Access concerns what is felt or available from within; report, what can be communicated; memory, what persists or can later be retrieved; agency, how perception and value guide action; coherence, how contents and control hang together; embodiment, the living condition through which experience occurs; and continuity, what links moments across time. These channels often travel together, but psychological tidiness and unified agency are not prerequisites for a unified field of experience. Ambivalence,

intrusive thoughts, memory gaps, divided access, or conflicting motives can occur within one life, so no channel may be promoted into the bearer by itself.

Clinical cases impose a moral floor. A person with dementia does not lose worth because integration is impaired. A person with dissociative symptoms is not less real because self-experience is complex. A person in a minimally conscious or behaviorally unresponsive state does not lose protection because report is limited.

The proposed Unity Index appears only as a research estimator. It is a way to organize evidence under declared assumptions, not a meter of personhood or certificate of consciousness. Validation therefore requires a defined construct, separated channels, explicit rivals, prospectively specified thresholds and contrasts, model recovery, held-out transport, error costs, and the option to abstain. A reliable number for the wrong construct would still be wrong.

Reversible states then protect the boundary. Sleep and anesthesia show that report and memory can disappear without establishing that the stream has ended. Return is not proof that experience continued throughout the interval, but it is a strong reason not to equate interruption, amnesia, or unresponsiveness with termination.

Disorders of consciousness make repeated, multimodal assessment indispensable. Medication, fatigue, sensory loss, motor impairment, fluctuating arousal, task design, and environment can shape what appears at the bedside. Multimodal

evidence can reveal missed capacities while still producing false negatives and ambiguous cases (Giacino et al., 2018; Kondziella et al., 2020; Bodien et al., 2024).

Fragmentation cases must be represented without spectacle. Split-brain patients and people with dissociative disorders are persons before they are examples. Their experiences and clinical realities limit what the theory may infer, especially when public language is tempted to turn uncertainty into a story of multiple selves.

Fission and fusion enter because the formulation developed here needs a routing rule for genuine branching or joining. The rule is conservative: old streams end, new streams begin, and no residual crosses the boundary. Chapter 13 does not claim that an existing clinical or experimental case has already been shown to meet those formal conditions.

Bearer adjudication must be insulated from the answer it would create. An assessor who can see a downstream terminal result may unconsciously choose the routing that favors closure or defeat. Blinding, prospectively specified criteria, independent review, model-recovery checks, and visible abstention rules are therefore part of the evidence rather than decorative procedure.

Unknown is the Chapter's most mature result when channels fail, conflict, do not transport, or cannot be used safely. It does not mean absence of consciousness, absence of a stream, or permission to neglect. It records that the present method cannot support a stronger claim.

13.1 — What Unity Could Mean

The word unity sounds intuitive because ordinary waking life usually arrives as one field. Vision, sound, thought, emotion, bodily sensation, and intention seem to belong to the same present. Yet philosophy and consciousness science distinguish several kinds of unity: unity at a moment, unity across time, unity of access, unity of agency, and unity of experience (Bayne & Chalmers, 2003; Bayne, 2010).

The Law needs a bearer that can own a complete history without becoming a public label. A body, name, medical record, legal person, memory archive, or social role may track important continuities, but none is automatically identical with the phenomenal stream. The bearer must be close enough to experience to carry valence and disciplined enough to survive hard cases.

The central danger is overconfidence. A fluent adult can make unity look obvious, while infancy, disability, aphasia, sedation, trauma, and neurological injury reveal how much ordinary judgment depends on channels that can fail. Unity is therefore treated as a research and formal problem, never as a measure of how complete a person is. The question is where a history belongs, not how well someone performs normality for an observer (Owen et al., 2006; Bodien et al., 2024).

13.1.1 — Access as Felt Availability

A person can be unable to speak, move, or follow an ordinary bedside command while some experience remains available within the stream. Work on phenomenal consciousness and covert

command following has made the central warning clear: a silent output channel does not by itself establish an empty inner field (Block, 1995; Owen et al., 2006; Monti et al., 2010). The evidence may still be uncertain, but silence must be interpreted rather than promoted into a verdict.

Access is best understood as a family of relations. A state may be available to attention, flexible control, working memory, deliberate action, later report, or none of those in the usual way. Consciousness theories disagree about which forms of availability are constitutive and which are merely useful evidence, so this public root should not settle the dispute by vocabulary alone (Bayne & Chalmers, 2003; Seth & Bayne, 2022).

For present purposes, felt availability names the modest starting point: something is occurring for the stream, even when the stream cannot describe, organize, or deliberately use it. A baby can undergo distress before possessing the concept of distress. A person in panic may feel danger before coherent speech returns. A person in pain may know the pain without being able to translate it into the scale an observer expects.

Report is one route by which access becomes visible to others, not the thing that makes the experience real. Speech, gesture, writing, eye movement, neural response, later recall, and patterned behavior are different channels with different failure modes. No channel earns automatic sovereignty merely because it is easier to record.

Blocked report is common enough to defeat a narrow definition. Motor impairment, language barriers, developmental disability, sedation, trauma, social fear, neurological injury, and hostile institutional settings can all limit expression. A rule that protects only the fluent would distribute credibility toward those already easiest to hear.

Memory creates a separate question. Dreaming, anesthesia with amnesia, delirium, dementia, and traumatic fragmentation show that later recall can fail even when an earlier interval may have contained experience (Siclari et al., 2017; Mashour & Hudetz, 2018). The missing memory affects what can be known later; it does not retroactively decide whether the earlier state occurred.

Felt availability also differs from self-interpretation. Experience can arrive as pressure, warmth, dread, confusion, shame, relief, or pain before it becomes a stable story about the self. The proposed Law does not require a stream to become an expert witness on its own condition. It requires the inquiry to leave room for valenced experience that is real before it is conceptually tidy.

Observers often confuse difficulty with absence. A rushed clinic, school, family, or research team may treat delayed or unconventional communication as no communication and then mistake the observer's limitation for the person's. A missing signal remains evidence, but its meaning depends on sensitivity, timing, task demands, medication, fatigue, injury, language, sensory access, and the response route the method requires. Better practice changes the environment, offers more than one channel, and records what the method could not reach before drawing a conclusion.

Care does not wait for this classification. A person whose access is uncertain still retains dignity, relief, privacy, protection, and the right not to become a demonstration. Scientific humility is strongest when it improves treatment of the difficult case rather than merely decorating the theory with uncertainty. Access as felt availability gives the bearer question a disciplined first step: it prevents reportability from becoming a gate of reality while preserving the need for evidence about how experience is integrated and used.

13.1.2 — Agency and Coherence

Agency can be evidence that experience is organized through a continuing point of view. Planning, refusal, self-protection, correction, and purposeful action often bring perception, memory, value, and control into one pattern. Yet the sense and exercise of agency are built from several cues and can be weakened without disappearing, so agency is not a simple certificate of consciousness or stream identity (Gallagher, 2000; Haggard, 2017).

Coherence is equally easy to misunderstand. It can refer to one field of experience, a stable narrative, coordinated action, agreement among motives, or a socially intelligible style of speech. Those properties can separate. A person may experience one field while feeling conflicted, ambivalent, distracted, or divided, and unity cannot mean perfect harmony in a mind that ordinarily contains competing thoughts and impulses (Bayne, 2010).

The ability to display agency depends heavily on conditions. Paralysis, pain, exhaustion, medication, sensory loss, developmental disability, trauma, poverty, coercion, and neurological injury can narrow expression without proportionally narrowing experience. Institutions often reward calm fluency and familiar forms of self-control, so agency should be read as supported or obstructed rather than possessed in one fixed amount. Safety, sleep, treatment, communication access, privacy, and reliable relationships can widen the menu of action without manufacturing the subject.

In these human cases, embodiment matters because bodily conditions shape experience. Hunger, breath, posture, touch, injury, hormones, fatigue, and sensory access shape what can be felt and done. The stream is not reducible to the body, but neither can its apparent coherence be assessed as though the body and environment were neutral scenery.

Context distributes visibility. A familiar caregiver may notice a reliable response that a rushed examiner misses. A quiet room may make communication possible. An interpreter, eye-tracking device, slower pace, or adapted task may reveal organized access that the original setting concealed. The quality of the observation environment is therefore part of the evidence.

Patterns across time are more informative than one successful or failed act. Repeated coordination among perception, value, memory, and action can strengthen a unity inference, while instability may identify a need for better support, a different test, or a genuinely uncertain case. One moment should not carry a conclusion larger than the method that produced it. Several

questions must remain separate: whether experience is present, whether action is organized, whether a narrative is coherent, whether memory bridges an interval, and whether society or law treats the individual as one person. Evidence may connect these questions, but none answers all the others by definition.

For the formal model, agency and coherence are evidence channels whose relevance must be declared in advance. They can support or challenge a routing hypothesis when the task, population, confounds, and error costs are clear. They cannot be adjusted after the terminal pattern is known simply because one classification makes closure easier. The ethical asymmetry is decisive. A false negative can erase a person whose expression is impaired, while a false positive can overstate what the evidence shows. That tension calls for multimethod review and an honest unknown category, not a dignity threshold disguised as measurement.

No public score should rank people by apparent agency, coherence, eloquence, or self-command. Care and respect remain prior to the estimate, and impairment never turns a human being into a lesser bearer of rights. The question is what structure the evidence supports, not how well the person performs normality for an observer. Agency and coherence make the inquiry more patient when they direct attention toward organized access and force the observer to ask what pain, disability, fear, environment, or method may have hidden.

13.1.3 — Unity Under Uncertainty

No accepted theory supplies a universal test for deciding whether a system carries one conscious stream, several streams, or an organization that current concepts cannot classify. Major theories emphasize different relations among recurrence, global availability, higher-order representation, integration, prediction, embodiment, and felt experience (Seth & Bayne, 2022). The disagreement does not make every judgment equivalent. It makes the source and scope of uncertainty part of the judgment.

Consciousness researchers increasingly describe the need for multiple tests whose interpretations are validated against distinct theories and populations rather than one all-purpose meter (Bayne et al., 2024). For unity, that means asking which property is under review: one phenomenal field at a moment, common access, coordinated agency, continuity through time, or ownership across a longer history.

Uncertainty can enter through at least four routes. A channel may fail; the observed state may fluctuate; several models may explain the same finding; or the construct itself may remain unsettled. Each route calls for a different response, so one confidence number can conceal the scientific problem it appears to summarize. Channel uncertainty concerns whether an observation is trustworthy. Failure to respond may reflect fatigue, inaccessible instructions, motor impairment, medication, sensory loss, language mismatch, or absence of the tested capacity. Repeating the same inaccessible task can narrow an estimate without improving the bridge to experience.

State uncertainty concerns timing. A person may move among levels of arousal, responsiveness, access, and coherence within hours. Artificial and biological systems can also change across versions, environments, or operating conditions. A classification should therefore identify when it applies and whether the state was stable enough to support it. Model uncertainty appears when the same evidence supports rival explanations. Fragmented report may reflect divided access, memory disruption, language failure, strategic withholding, or a genuinely divided organization. The record should preserve those alternatives and name the observation that could separate them.

Construct uncertainty is deeper. Researchers may agree on every measured event while disagreeing about what unity means. Statistical precision cannot identify a target that was never fixed, and a result should not be promoted from access or coordination to phenomenal ownership merely because the same word appears in both descriptions.

A responsible assessment therefore yields an evidence profile. It records channel quality, temporal stability, construct assumptions, model alternatives, missing information, and the consequences of error. The profile is less theatrical than a rank and more useful because it shows what must change before the conclusion can change.

Clinical work demonstrates the cost of treating outward silence as transparent. Repeated standardized assessment and task-based neuroimaging or electrophysiology can reveal capacities that an informal encounter misses (Schnakers et al., 2009; Claassen et al., 2019; Giacino et al., 2018). Those methods remain limited, but

they show that an observer can lose access to a person before the person loses every possible form of access. The formulation developed here makes bearer identification a prerequisite because ownership and routing cannot be selected after the terminal pattern is known. The gate blocks favorable reclassification. It does not pretend that one present method can always determine the bearer.

An unknown result should therefore carry a reason: channel failure, unstable state, model conflict, construct nonidentification, missing domain, failed transport, or ethically excessive burden. These outcomes are not interchangeable, and each points toward a different next step. Unknown is neither absence nor support. A possible bearer may still require care, privacy, welfare protection, communication access, and review, while an unresolved case cannot be counted as evidence for the Law simply because no counterexample can be established. Protected nonresolution serves both the candidate and the hypothesis.

The costs of error are asymmetric. False exclusion can erase a person whose expression is difficult to detect; false inclusion can attach an unsupported identity or metaphysical claim to a patient, family, animal, or system. The best safeguard is to separate care and rights from classification, then design the scientific process to discover which error occurred. More data help only when they address the relevant uncertainty. A larger sample of the same biased channel can make an answer look sharper while leaving the central bridge untouched. Progress should identify which uncertainty narrowed, which remained, and why.

Public language should state the finding at its actual level: command following under this task, several routing models remain admissible, bearer status unresolved, or no adequate estimate. The record should preserve the date, state, methods, assumptions, model version, dissent, and trigger for review. New communication access, clinical change, replication failure, software correction, or theoretical advance may alter a prior judgment. Unity under uncertainty is a scientific achievement when the uncertainty is classified, protected, and revisable rather than disguised as certainty.

13.2 — Access, Report, Memory, and Continuity

Access can exist without report, report can exist without accurate memory, and memory can preserve a narrative while omitting much of what was felt. These separations make stream continuity more difficult than a checklist of abilities.

A candidate unity estimator may organize evidence across channels, but it remains a model. It cannot turn an inaccessible inner life into a visible object or erase disagreement by averaging it. This Section therefore makes the estimator's status explicit, prohibits public scoring, and treats validation as a continuing argument about interpretation, population, setting, and use rather than a ceremonial label (Messick, 1995).

A person may lose speech, episodic memory, motor control, or stable attention while retaining some conscious continuity. The absence of one bridge does not automatically break the stream. The methodological demand is humility with teeth: preserve channels separately, transport cautiously, declare uncertainty, and abstain when the evidence cannot support a bearer claim.

13.2.1 — Unity Index as Estimator

A Unity Index can be useful only if it is understood as a fallible estimator of a declared construct. It is not the stream itself, not a measure of personhood, and not a shortcut through the unresolved science of consciousness. Its first duty is to expose assumptions that would otherwise remain hidden. The construct must be defined before the instrument is assembled. Access, report, memory, agency, embodiment, temporal continuity, and coherence are related but nonidentical. A tool that silently treats

them as interchangeable may produce a stable number for an incoherent target.

Each evidence channel should enter for a stated reason. Behavioral response may reveal task access; memory may reveal continuity across an interval; agency measures may reveal organized control; neural perturbation measures may reveal causal complexity (Casali et al., 2013). None of these observations becomes phenomenal unity merely by being placed in the same model. Psychometrics distinguishes a construct from the indicators used to estimate it. Construct validity depends on a network of predicted relations, rival explanations, and consequences rather than one impressive correlation (Cronbach & Meehl, 1955; Borsboom, 2005). That discipline is essential when the target cannot be observed directly.

A multimethod design can help because different channels carry different biases. Agreement among report, behavior, physiology, and neural measures is more informative when the methods are genuinely independent than when they share labels, preprocessing, or training data (Campbell & Fiske, 1959). Apparent convergence can otherwise be manufactured by a common pipeline. Disagreement is not automatically an error to average away. A person may retain experience while losing motor output, retain agency while losing later memory, or show neural complexity without reliable report. The pattern of disagreement can teach more than a single composite score.

The index should therefore preserve channel-level results. A final estimate without its components hides whether the conclusion rested on one fragile channel or on several independent lines of

evidence. Reviewers need to see the path from observation to inference. Calibration must be local to the claim. A threshold developed in healthy adults may not transport to infants, people with disabilities, people under sedation, unfamiliar species, or artificial systems. Measurement invariance is an empirical question, not a courtesy phrase added after deployment (Messick, 1995).

Missingness also belongs inside the estimator. An absent response may be informative, but it can also arise from inaccessible tasks, fatigue, sensory impairment, technical failure, or withdrawal. Coding every missing value as absence of unity would convert participation burden into a scientific conclusion. Temporal sampling needs equal care. Unity may fluctuate with sleep, medication, seizures, delirium, arousal, injury recovery, or system configuration. A snapshot can be useful, but it should not masquerade as a whole-stream identity.

The estimator should be tested in simulated and known-contrast settings before it is applied to hard cases. Model recovery asks whether the analysis can distinguish the data-generating alternatives it claims to compare. If several rival structures routinely receive the same score, the index has not earned interpretive specificity. Prospective specification can freeze the construct, channels, preprocessing, missingness rules, thresholds, and primary contrasts before the outcome is known (Nosek et al., 2018). This does not guarantee validity. It prevents the definition of unity from drifting toward whichever analysis favors the project.

Blinking strengthens the same protection. Adjudicators should not know whether a particular classification would make a terminal pattern look more favorable. The bearer decision must be made on its own evidence rather than on the closure result it could help create. Held-out evaluation matters because an index can fit its development sample while failing elsewhere. A credible tool should survive new participants, new sites, new equipment, and reasonable changes in context. Transport failure narrows the use; it does not invite a more poetic interpretation.

The output should match the evidence. Some cases may justify a bounded estimate, some a profile of competing models, and some no adequate estimate. A system that always returns one number is often revealing a software requirement rather than a scientific achievement. Error costs should be declared separately from the estimate. False negatives, false positives, and unresolved cases create different consequences in clinical, animal, artificial, and organoid contexts. The decision rule should not be hidden inside the measurement model.

A useful report can state the construct, reference class, channel inventory, model version, calibration evidence, missingness, uncertainty, competing interpretations, and authorized conclusion. That record gives later reviewers enough information to reproduce or challenge the judgment. The word index should not imply an ordinal ranking of people. Its legitimate role is closer to an instrument panel than a league table: several indicators, visible limitations, and a clear signal when the system cannot support a conclusion.

Version control is necessary because science and systems change. A model update, new sensor, altered task, revised theory, or deployment context can change what an output means, so every estimate must remain linked to the exact version that produced it. External validation should include difficult negative controls: systems that mimic surface features without the proposed organization, and cases in which one genuine channel becomes inaccessible. A reviewer should be able to identify which observations moved the estimate and whether those observations were valid for the assessed person or system; a colorful dashboard that cannot answer those questions is not transparency.

Even the strongest result would remain bounded. A validated estimator could support a specified inference within a declared population and purpose; it would not settle the metaphysics of the self, validate PTVC, or resolve every boundary case. The Unity Index is therefore best understood as a governed research instrument whose right to speak depends on construct clarity, independent channels, recoverability, transport, and honest abstention. Its value lies in making the bearer question more testable while keeping the person or candidate larger than the estimate.

13.2.2 — No Public Unity Score

A unity estimator may be scientifically useful and still be unfit for public use. The danger begins when a bounded research construct is detached from its protocol and presented as a number describing the person. Unity is not dignity: disruption of access, memory, report, or agency does not make anyone less human, less credible, less entitled to care, or less able to speak through the

channels that remain. An estimator may summarize evidence about a declared construct; it cannot grade the bearer.

Function creep often begins with a plausible request. A clinician wants a quick summary, a school wants to identify support needs, an employer wants to predict reliability, or a court wants a number that looks impartial. Each move changes purpose, population, consequences, and error costs. Once visible, the score also changes behavior: families may pressure a patient to perform, clinicians may stop looking for unconventional access, and institutions may optimize the record instead of the care. Validation in one setting does not travel automatically to another.

A public unity score would also invite proxy discrimination. Language fluency, motor control, cultural familiarity, educational opportunity, disability, fatigue, medication, trauma, and access to communication technology can all affect the channels from which the score is built. The number may therefore reproduce existing advantage while claiming to reveal an inner property (Selbst et al., 2019).

Research and clinical records must remain distinct. A research estimate should not silently enter the medical chart, disability file, school record, insurance record, employment file, legal proceeding, or public profile. Each regime has different authority, duties, correction routes, and consequences. Combining them creates a dossier that no original consent could reasonably authorize.

Even inside research, display design matters. A single prominent number suggests precision and hierarchy. A responsible report

should instead show the construct name, reference class, available channels, missing channels, error range, model version, known confounds, and whether the method was allowed to abstain. A profile of evidence is safer and scientifically more informative than a leaderboard. Thresholds require special suspicion. A cut point can turn gradual uncertainty into a categorical gate. When access to treatment, communication support, review, or participation depends on crossing that line, small errors become life-changing. No unity estimate should determine whether a person receives ordinary care, pain relief, legal protection, visitation, education, communication assistance, or respect.

Contestability and privacy must travel together. A person or authorized representative should be able to learn what was measured, which data were used, what was missing, how the result was produced, and what consequences followed; factual errors and unsupported inferences need a correction path. The underlying profile may expose communication patterns, neurological status, memory gaps, trauma history, or family observations even when the visible output is simple. Purpose limitation, access logging, retention limits, deletion or restriction rules, and hostile-crop review should therefore govern both the data and every public fragment (Solove, 2006).

No institution should be able to purchase interpretive authority. Vendors may offer dashboards, automated classifications, or proprietary models, but commercial confidence is not validation. Procurement should require source documentation, independent evaluation, bias and failure testing, human review, correction, and a withdrawal route. Confidential code cannot become an excuse

for unanswerable judgments. The same restriction applies to artificial intelligence. A model may help organize evidence or identify records for human review. It may not infer a person's hidden unity, declare a stream, diagnose a soul, or decide eligibility. Fluency and scale do not confer authority over another subject's inner life.

A no-score rule does not forbid every summary. Clinicians and researchers sometimes need concise communication. The boundary is that summaries remain descriptive, purpose-bound, reviewable, and subordinate to the underlying evidence. They must not become portable rankings or stand-alone claims about the person. The strongest governance tool is a prohibited-use register. It should name deployments that are outside scope, such as employment screening, insurance pricing, sentencing, credibility assessment, educational ranking, spiritual classification, immigration decisions, benefit conditions, public office eligibility, or care priority. The register should travel with the method and every derivative model.

Withdrawal must remain possible after deployment. New evidence may reveal that the estimator fails in a population, depends on a confound, or is being used coercively. The response is not a quiet disclaimer. The tool, dependent dashboards, public claims, and downstream classifications should be corrected, restricted, or retired. A private research construct becomes dangerous when society treats numerical neatness as moral knowledge. The safer principle is durable: evidence about unity may help scientists ask a narrower question, but no unity number

may become a public identity, a rank, a gate, or a substitute for listening to the person.

13.2.3 — Scientific Validation

Validation is not a seal placed on a Unity Index because it correlates with familiar signals. It is an argument about what kind of unity is being estimated, which observations support that interpretation, which rivals remain, and where the bridge fails (Cronbach & Meehl, 1955; Messick, 1995).

A useful estimator must distinguish unity from arousal, motor output, language comprehension, recording quality, and other neighboring variables. It should recover known alternatives in simulations and benchmark cases before it is trusted in unfamiliar bearers, and agreement among channels counts only when their errors are not merely shared under different names (Campbell & Fiske, 1959; Bayne et al., 2024). A useful benchmark deliberately separates cases with similar motor responses but different evidence relevant to unity, then examines the estimator's discrimination.

Reliability, calibration, held-out testing, transport, missingness, and abstention limit the claim. A method developed in one population or state cannot be promoted to infancy, aphasia, dementia, anesthesia, injury, chronic disorders of consciousness, nonhuman animals, or other settings without new evidence. Flexible models, incomplete cases, and failed transport may justify a narrower reference class or no adequate estimate rather than a universal tool (Simmons et al., 2011; Munafò et al., 2017; Nosek et al., 2018; Borsboom, 2005). A transport report can

identify exactly which change in population, recording conditions, or task calls for a fresh validation study.

Validation of a unity estimator would support only its declared construct and use. It would not validate PTVc or settle stream ownership, complete history, endpoint, or measured-to-latent transport. For example, a measure could satisfy its prespecified benchmark in awake adults while providing no validated basis for classifying an unresponsive patient.

13.3 — Sleep, Anesthesia, and Reversible Pauses

Reversible low-access states are useful here only for the bearer question. They show that responsiveness, memory, report, and conscious continuity can separate, so none can define stream identity by itself. Chapter 5 established the endpoint implications; this Section focuses only on ownership and routing across a pause.

13.3.1 — Ordinary Sleep

Ordinary sleep strongly favors continuity rather than a new stream: the sleeper typically returns to the same embodied, relational, and causal trajectory even though access, control, and recall vary. Dreaming also shows that experience may occur while public responsiveness is low (Nir & Tononi, 2010; Siclari et al., 2017).

For routing purposes, the key distinction is between the occurrence of experience and later access to it. Failure to recall a dream is not evidence that no experience occurred, while sleep-stage physiology is not a direct readout of felt content. The model should therefore preserve uncertainty about unobserved intervals rather than assigning neutrality by default.

Nothing here invites dream monitoring or self-surveillance. Sleep is a familiar continuity case, not a complete-stream measurement device.

13.3.2 — Recovery Reports

General anesthesia separates responsiveness, memory formation, and conscious access more sharply than ordinary sleep. Recovery

reports may reveal awareness, pain, fear, dreams, or time distortion that intraoperative behavior did not show, but amnesia and reconstruction make those reports incomplete (Brown et al., 2010; Sanders et al., 2012; Mashour & Hudetz, 2018).

For stream routing, return is the decisive asymmetry: reliable return prevents the earlier interval from being classified as the irreversible end by default, while failure of recall does not tell us whether experience occurred. Clinical purpose remains prior to research; no anesthetic care should be altered to sharpen a Law-related inference.

Thus anesthesia can constrain continuity and accrual models, but it cannot convert a recovery interview or physiological trace into a complete account of hidden experience.

13.3.3 — Clinical Review

Difficult clinical states require repeated, multimodal assessment because behavior, imaging, electrophysiology, medication, sensory access, motor capacity, and later recovery can disagree (Giacino et al., 2018; Kondziella et al., 2020; Bodien et al., 2024). No channel becomes a universal consciousness meter merely because it detects what another misses. A shared assessment timeline helps the team compare observations made under different recording conditions and identify which disagreement requires another look.

For the bearer problem, the record should state what each method measured, active confounders, how findings changed over time, and where channels disagreed. A multidisciplinary review may narrow uncertainty, but care, pain control, communication

attempts, family support, and rehabilitation do not depend on resolving PTVC routing. A disagreement between methods used at one visit, for example, differs from a change between visits; the record should make those two questions distinguishable.

The endpoint of this review is procedural humility: preserve the evidence by channel, allow reassessment, and use an explicit unknown or no-adequate-estimate category when the bearer cannot be responsibly adjudicated. The resulting decision should name the unresolved question and the particular new observation that could justify reopening it for the same case.

13.4 — Disorders, Fragmentation, Fission, and Fusion

These cases are scientifically valuable because they separate capacities that ordinary life bundles together, yet they are also easy to sensationalize. Disorders of consciousness, split-brain surgery, dissociation, severe memory disruption, and hypothetical fission or fusion challenge the assumption that one body supplies a simple stream count. Behavioral nonresponse cannot settle consciousness; standardized assessment and covert-command paradigms have repeatedly revealed capacities not visible at the bedside (Schnakers et al., 2009; Bodien et al., 2024).

Fragmentation also admits competing interpretations. Split-brain evidence may show divided access without deciding phenomenal unity, and dissociative diagnoses do not determine formal routing (Pinto et al., 2017; Dorahy et al., 2014). Any claim of fission or fusion must therefore use prospectively specified boundaries independent of the ledger result and preserve intrastream nontransferability.

The moral discipline is equally exact: uncertainty about unity never reduces dignity, care, consent, or standing. Edge cases improve the model only when the people involved remain persons rather than proof material.

13.4.1 — Nonresponse Does Not Settle Consciousness

A bedside command is a chain, not one event. The person must receive the instruction, distinguish it from background stimulation, understand it, sustain attention, select a response, organize an action, and produce an output another person can detect. Failure at any link can look like absence of consciousness.

Behavioral nonresponse is therefore evidence about an output route, not a verdict about a mind. Coma, unresponsive wakefulness syndrome, and minimally conscious state are clinically useful classifications with stated criteria and known error risks. They should not be translated casually into claims that no experience exists or that a complete bearer has been identified.

Apply the serial-assessment safeguards in Section 13.3.3 rather than treating one bedside examination as final (Schnakers et al., 2009; Giacino et al., 2018; Kondziella et al., 2020).

In a 2006 study, a woman who showed no observable response to commands after a severe brain injury was asked to imagine two familiar activities: playing tennis and moving through the rooms of her home. Her brain activity changed with the requested task in patterns corresponding to those seen in healthy volunteers. The researchers interpreted this as a response to instructions carried through brain activity instead of speech or movement (Owen et al., 2006).

The striking possibility is that an answer can exist where the usual observer has no way to receive it. Asking for a hand movement and asking for a particular mental activity put different demands on an injured system. The scientific question becomes more interesting and more humane when it includes the route through which an answer could reach us.

A six-center prospective study enrolled 353 adults with disorders of consciousness. Among 241 participants who showed no observable response to commands, 60, or 25 percent, produced task-related responses detected with functional magnetic

resonance imaging, electroencephalography, or both (Bodien et al., 2024). In that study population, covert command following was not a vanishingly rare exception.

The finding changes the burden of inquiry, not the claim ceiling. Cognitive motor dissociation establishes organized command following under a specific protocol. It does not reveal the full contents, valence, temporal continuity, unity, prognosis, or complete phenomenal history of the person. Positive and negative results remain channel-bound. A positive task may show that an instruction was understood and an internal response generated. A negative task may reflect absent command following, but it may also reflect hearing loss, language mismatch, sleep, medication, seizures, fatigue, poor signal, limited comprehension, or inability to sustain imagery.

Alternative channels remain method-dependent. Imaging and electrophysiology depend on reference tasks, preprocessing, thresholds, artifact rejection, analytic decisions, repeatability, and transport. The report should state the capacity actually tested and the conditions under which the result can be interpreted. Clinical classification, prognosis, treatment, communication planning, and formal bearer routing are different decisions. Evidence can travel among them only with its original conditions attached. A clinical label was not designed to determine which stream owns a lifetime ledger.

Clinical guidance accordingly emphasizes correction of confounds, optimization of arousal, repeated standardized examination, and multimodal assessment where appropriate (Giacino et al., 2018; Kondziella et al., 2020). These practices

reduce the risk that an inaccessible person is mistaken for an absent one. Family observations can improve assessment by identifying familiar voices, meaningful stimuli, communication habits, and patterns of fatigue. Such testimony is neither disposable anecdote nor diagnostic oracle. It is evidence about context and possible blocked routes.

Care does not wait for a performance. Analgesia, positioning, communication support, rehabilitation, sensory access, visitation, and respectful explanation remain appropriate under uncertainty (Fins, 2015). Bedside language should preserve the observation itself: no behavioral response was observed under these conditions.

Documentation should record the task, sensory route, language, medication state, arousal, examiner, timing, attempts, signal quality, preprocessing version, and reason a session failed. Repeated testing must still be proportionate; additional burden is not justified when the channel remains inaccessible or the result cannot change care or interpretation. Bearer classification should be blinded from any later terminal pattern. Expert disagreement about signal quality, sensory access, or inference should remain visible, and later evidence should trigger correction across the clinical record, research dataset, teaching material, and dependent analysis.

Behavior remains indispensable, but it is constrained by the body and environment through which experience becomes visible. A serious framework treats nonresponse as a difficult observation whose meaning must be tested, contextualized, and kept open to revision. The scientific gain is exactness about what was and was

not found. A task can detect command following, a clinical examination can classify observable behavior, and multimodal review can narrow uncertainty. None alone settles the complete stream, but each can prevent the observer's limitation from being mistaken for the person's absence.

13.4.2 — Split-Brain and Competing Interpretations

Split-brain research matters because it separates functions that ordinary life usually binds. Classic studies emphasized divided perceptual access after callosotomy, while later work found that some participants retained a single responding agent despite impaired comparison across visual fields (Sperry, 1968; Gazzaniga, 2005; Pinto et al., 2017). The literature therefore resists the simple slogan that one operation creates either one untouched mind or two complete minds.

The disagreement partly reflects several meanings of unity. Unity of action, attention, report, memory, decision, and phenomenal experience need not rise or fall together. A coordinated body can coexist with divided access, and divided task performance does not by itself prove two separate fields of experience (de Haan et al., 2020; Schechter & Bayne, 2021).

Anatomy is important evidence, but it is not a shortcut to metaphysics. One body does not guarantee one stream by definition, and disconnection of neural pathways does not automatically count two streams. The bearer question requires a rule linking observed organization to a claim about experience.

For PTV, split-brain cases are routing stress tests. The criteria for continued unity, partial separation, fission, or an unresolved

result must be declared before terminal patterns are examined. A routing choice selected afterward because it produces easier closure would be a rescue, not a discovery. Laboratory tasks reveal only part of the case. They can show which information crosses a boundary, which hand responds, or whether two stimuli can be compared. They do not contain the person's whole lived identity, relationships, agency, history, legal status, or moral standing.

This distinction protects against two opposite errors. Forced unity can hide genuinely separate burdens if more than one stream-like organization is present. Forced multiplicity can divide a person on the strength of a narrow task and invite false conclusions about responsibility, treatment, or survival. A better description may sometimes be partial integration. Access can be divided under one condition and coordinated under another. Agency can remain broadly unified while some information is locally unavailable. The model should be able to represent that structure without pretending that every case must immediately yield an integer count.

The unknown category is therefore substantive, not evasive. Evidence may support one stream with fragmented access, more than one candidate stream, several admissible models, or no adequate estimate. Scientific maturity includes knowing when the available tasks do not justify a routing verdict. Prospective adjudication would compare several evidence families. Repeated task results, cross-field transfer, simultaneous incompatible contents, independent goal pursuit, perturbation responses, first-person reports where available, and stability across contexts may

each bear on competing models. No single dramatic trial should carry the whole count.

The cost of error must remain explicit. A false split can divide one person into too many proposed streams; a false merger can erase a distinct field of experience. Because those errors are not morally or scientifically interchangeable, a method may need to preserve several models rather than force certainty for convenience.

Split-brain findings are valuable even when they do not settle the bearer. They force access, report, action, and experience to be distinguished, and they expose assumptions that remained invisible in ordinary cases. The right result can be a more exact question rather than a triumphant answer. Edge cases can challenge or refine the theory, but they cannot validate terminal neutrality for all streams. A striking case may reveal a weak unity rule, a missing branch, or a failed measure. It does not become proof merely because it is memorable.

Public communication must resist spectacle. A person should not be introduced as “two minds in one body,” a soul experiment, or a poster case for the Law. Such language outruns the evidence and turns a vulnerable life into rhetorical equipment. Cumulative, reviewed, deidentified, and context-rich evidence is safer than anecdote. The record should preserve competing interpretations and the conditions under which each task was performed. Difficulty belongs in the claim status, not in a simplified headline.

13.4.3 — Dissociation, Memory, and Representation

Dissociation can disrupt the ordinary integration of consciousness, memory, identity, emotion, perception, body

representation, or control. People may describe depersonalization, derealization, amnesia, shifts in self-state, or altered ownership while continuing to experience the world. These reports are clinically and personally important, but they do not supply a ready-made count of conscious streams.

Diagnostic categories organize patterns of distress, impairment, exclusions, and treatment needs. The DSM-5-TR does not function as a bearer instrument, and a dissociative diagnosis cannot be translated directly into one stream, several streams, or a ledger-routing rule (American Psychiatric Association, 2022). Research on dissociative identity disorder describes trauma associations, memory phenomena, state-dependent differences, comorbidity, and substantial clinical complexity (Dorahy et al., 2014). That literature supports the seriousness of the condition and the need for competent care. It does not turn every measured difference between self-states into proof of separate subjects.

Memory is an especially unreliable shortcut. An event may have been consciously experienced and later become inaccessible, partially recovered, or reconstructed. Conversely, a smooth narrative can be assembled across gaps. Missing recall does not prove missing experience, and narrative continuity does not prove uninterrupted access.

First-person language deserves respect without being made to bear a formal conclusion it was not designed to settle. A person may speak of parts, identities, voices, perspectives, or states because those terms best describe lived experience or therapeutic work. The project should preserve that language while

distinguishing description, diagnosis, mechanism, and routing hypothesis.

No diagnosis permits automatic disbelief. Dissociation should not be used to dismiss testimony, infer dangerousness, deny consent, or treat every disagreement as pathology. Capacity is decision-specific and time-sensitive, and it should be assessed through ordinary clinical and legal standards rather than a global judgment about identity. The reverse is also true. A person's account is essential evidence about experience, but no account, clinician, scan, or experimental task has infallible access to stream structure. Respect requires careful listening; science requires keeping the mechanism and bearer conclusion open when evidence does not close them.

State-dependent psychophysiological and neuroimaging differences may help identify how memory, perception, and self-models vary across contexts (Reinders et al., 2012). Interpretation remains sensitive to sampling, comparison groups, suggestion, trauma history, treatment setting, and circular classification. Such findings justify study, not metaphysical arithmetic. Agency may be fragmented without being absent. An action can feel involuntary, alien, or attributable to another state, and a sense of self can change without offering a simple answer about ownership. Philosophical work on agency and narrative identity clarifies the questions but does not settle every case (Gallagher, 2000; Schechtman, 1996).

A responsible protocol separates at least four objects: what may have been experienced, what can be reported now, what is remembered across states, and what evidence bears on continuity

or ownership. Collapsing them into one identity score would conceal the very distinctions the study needs to examine. Longitudinal evidence is usually safer than one dramatic session. Repeated observation can test co-consciousness, information transfer, agency, memory access, temporal overlap, stability across settings, and response to treatment. It can also show that a classification changes, which is itself evidence against treating it as a timeless essence.

Confounds should be recorded rather than folded into one preferred story. Trauma, sleep disruption, substance use, neurological conditions, medication, expectancy, suggestion, and therapeutic context can all affect report and performance. Naming these possibilities is not disbelief; it is how a serious inquiry avoids overclaim. Therapeutic care remains prior to theoretical curiosity. Research must not provoke destabilization, reward increasingly dramatic performance, or turn a person's distress into an experiment. Participation should be optional, burden-aware, privacy-protective, and coordinated with appropriate clinical support when needed.

Records create their own danger. Therapy notes, interviews, videos, and model-generated summaries can freeze a temporary description into a permanent public identity. Purpose limits, access controls, correction rights, retention review, and strict secondary-use rules are part of representational ethics, not administrative decoration.

Public language should survive a hostile-crop test. Inferring several streams from a dissociative diagnosis would overstate the evidence and risk reducing a person to a disputed theoretical

label. Responsible communication names the observed phenomenon, preserves uncertainty, and refuses to turn a diagnosis into a verdict about personhood or the Law. An unresolved routing result can coexist with excellent care and serious knowledge. Dissociation teaches that unity is not the same as smooth autobiography and that uncertainty need not erase the person.

13.4.4 — Stream Boundaries in Fission and Fusion

Fission and fusion are the sharpest stress tests of stream ownership because they challenge the assumption that one continuing body or biography always carries one continuing subject. They must be handled through prospective rules rather than through a story chosen after the outcome is known. The first distinction is among physical division, informational division, access division, and subject division. A brain, device, or organization can divide functions without creating two subjects. Conversely, a theory might predict a change in subject structure without any dramatic visible separation.

The competing split-brain interpretations in Section 13.4.2 motivate this routing problem without establishing that one operation creates two people (Gazzaniga, 2005; Pinto et al., 2017; de Haan et al., 2020; Schechter & Bayne, 2021).

A formal routing rule should state what evidence would count as continued unity, branching, or termination. The criteria might concern persistent independent access, separate agency, incompatible simultaneous contents, loss of cross-system integration, or another theory-linked property. The rule must be

declared before terminal results are examined. On the primary stream-routing convention, a genuine fission ends the parent stream and begins each qualifying child stream with its own ledger domain. This is a formal bookkeeping rule, not an empirical claim that every physical split causes subject fission.

Beginning child ledgers at their own boundaries prevents experience before the split from being duplicated into several descendants. It also prevents later experience in one child from canceling experience in another. Nontransferability survives branching. The parent history is not erased. It belongs to the stream that existed before the declared fission boundary. The formal record therefore preserves ancestry without pretending that one pre-branch residual can be reassigned among descendants for convenience.

Fusion creates the opposite problem. If two previously separate streams genuinely become one subject, the prior streams end and a new stream begins under the declared rule. Their previous ledgers do not merge into a pooled account. That convention blocks cross-stream trading at the moment when pooling would be most tempting. A new fused stream may inherit memories, dispositions, or causal structure from earlier streams, but inherited information is not the same object as inherited phenomenal balance.

Memory continuity cannot settle the bearer alone. Records may persist without a subject, and a subject may survive severe memory loss; bodily, psychological, causal, narrative, and phenomenal criteria can disagree (Parfit, 1984; Schechtman, 1996; Olson, 2007). Copies, backups, or a teletransportation-style

duplicate may reproduce information and behavior without establishing that the original stream continued. The protocol must therefore identify the pre-event system, candidate boundary, predicted consequences under each routing hypothesis, and evidence that would trigger reclassification. Vague appeals to continuity are insufficient.

Routing should also be exhaustive. If the declared possibilities are one continuing stream, two child streams, a new fused stream, or no adequate estimate, every eligible case should enter one of those categories under the prospectively specified procedure. Unclassified residue should remain visible rather than being dropped. Ambiguity may persist. A case can support several routing models or fail to reveal the relevant structure. The appropriate result is model-ambiguous or no adequate estimate, not a favorable split chosen in retrospect.

The stakes are scientific because routing changes the domain over which closure is assessed. An ungoverned split can turn one difficult trajectory into several easier ones, and a framework that permits such repair after observing the ledger has lost falsifiability. The stakes are also ethical. A classification of one stream or several can affect how people talk about identity, responsibility, treatment, and survival. This book must not convert a formal convention into a diagnosis, legal status, or judgment about the worth of any person.

Legal identity may continue through an event even when a phenomenal theory predicts fission, and law may assign responsibility for reasons outside this question.

The Constitution of Fairness owns legal architecture; this Part preserves only the scientific boundary.

Clinical and disability language requires restraint. Dissociation, amnesia, hemispheric disconnection, dementia, and communication impairment do not by themselves establish several people in one body; each has distinct evidence, lived reality, and stigma risks. Research should compare routing models against independent observations, including predicted differences in memory access, agency, integration, or perturbation response. A rule that cannot fail is not a routing test.

For artificial and synthetic systems, version control must preserve exact state, copied instances, merges, and branching history before any bearer claim is made. Fission and fusion do not excuse outcome-driven routing; they require explicit ownership rules, past-fixity, nontransferability, and permission to remain unresolved.

13.5 — Blinded Adjudication and Honest Unknowns

A difficult classification becomes credible only when the method can return an answer the theory does not prefer. Blinding, prospective specification, independent review, and an honest unknown category make that ability operational. Wherever possible, the assessor should not know whether a routing decision helps terminal closure, supports a favored mechanism, or strengthens a public story; otherwise ambiguity can bend toward success without deliberate misconduct.

Unknown must carry reasons and consequences. Missing data, unstable state, conflicting channels, failed calibration, and a contested construct are different kinds of nonresolution and should be recorded separately. Governance then keeps research classification, clinical care, public communication, and legal standing in their proper domains so that no label quietly becomes a decision about whose life counts.

13.5.1 — Why Blinding Matters

Unity judgments are vulnerable to expectation because the evidence is often incomplete and the downstream consequences are large. An assessor who knows which classification would support the Law, fit a diagnosis, protect an institution, reassure a family, or simplify a prognosis may weigh the same ambiguous sign differently. Research on analytic flexibility and reproducibility shows how apparently reasonable choices can accumulate into a preferred result without deliberate fraud (Simmons et al., 2011; Munafò et al., 2017).

Blinding is not an accusation against the reviewer. It is a design response to ordinary human cognition. Hope, grief, professional loyalty, theoretical commitment, fear of error, and public pressure can shape interpretation precisely because the reviewer is conscientious and understands what is at stake.

A useful workflow separates bearer classification from the result that classification will produce. Where possible, reviewers should not see the terminal ledger outcome, favored mechanism, public identity, or theory-helping direction while they assess unity. Clinical teams should retain the information needed for care, while research adjudicators receive only what the declared classification task requires.

Complete blinding will not always be possible. Diagnosis, imaging, behavior, and treatment history can reveal part of the case. The record should therefore state what was hidden, what remained visible, and which likely expectations could still influence judgment. Partial protection described honestly is stronger than a ceremonial claim of neutrality.

Prospective specification supports the same independence. Thresholds, exclusions, uncertainty zones, tie-breaking rules, and sensitivity analyses should be protocol-fixed before a case begins to serve an argument (Nosek et al., 2018). A bearer boundary that moves after the residual is known becomes a rescue device rather than a test.

Frozen rules may include an explicit unknown region. Borderline or discordant cases need not be forced into one stream, several streams, or no qualifying stream merely to keep the analysis

complete. Precommitment is compatible with humility when the protocol specifies in advance when evidence will remain insufficient.

Criteria can improve as science develops, but the change must be visible. A revised threshold receives a new version, a reason, and a fresh validation path; it does not rewrite the old classification or quietly rescue an inconvenient result. Change control protects legitimate learning from post hoc preservation of the theory. Reviewer independence should also be structural. Distance from funding, publication incentives, public controversy, clinical authority, product ownership, or movement reputation matters more than a declaration of good intentions. A thoughtful person inside a strong incentive can still become an unfair classifier.

Different expertise reveals different failure modes. Neurology, psychiatry, philosophy of mind, statistics, disability studies, trauma-informed practice, clinical ethics, and measurement science may not agree, and their disagreement can be evidence about which layer remains unresolved. A panel should not manufacture consensus by averaging incompatible reasons. Disagreement records are therefore valuable. They should identify the disputed evidence, the assumptions producing different classifications, and the result each admissible view would imply. Public language can then say contested or unknown instead of presenting one institutional choice as discovered fact.

Blinding also protects care from the research question. Treatment, communication support, rehabilitation, and family guidance should not depend on whether the case helps or harms PTV. A person must never carry the cost of making the theory easier to

adjudicate. High-reach summaries need similar independence. Diagrams, captions, case vignettes, and public explanations should be reviewed by someone empowered to reject a cleaner story when the underlying evidence is contested. A careful technical appendix cannot repair a title or image that has already declared the bearer.

Independent review may slow publication or leave a result unresolved. That delay is not wasted time. When a classification can shape endpoint analysis, public identity, and the treatment of a vulnerable person, speed without independence merely transfers uncertainty to the person least able to contest it. Blinding turns humility into an operating feature. A classification that survives hidden outcomes, fixed criteria, documented limitations, and independent disagreement becomes more informative whether it supports, narrows, or defeats the hoped-for claim. The procedure gives reality room to correct the theory before the theory begins to classify reality in its own favor.

13.5.2 — Unknown Category

Unknown is a scientific outcome, not a failure of courage. It applies when the available evidence cannot justify one bearer classification over the serious alternatives under the declared rules. The category protects the person or candidate system from a verdict manufactured by pressure to decide.

Unknown should carry a reason, not stand alone. One case may be unknown because no viable response channel was available. Another may have conflicting behavioral and neural evidence. A third may lack a validated construct, stable endpoint, adequate

sample, or transportable model. Reason codes preserve what is actually missing. The evidence profile should remain visible beside the label. It should record available channels, positive findings, negative findings, missing observations, major confounds, model version, reference class, confidence limits, and the alternatives that could not be separated. A bare word hides too much.

Unknown is different from negative. “No observed command following under this protocol” is not the same as “no consciousness.” “Current AI is not established as conscious” is not the same as “AI consciousness is impossible.” “No evidence of sentience in this organoid” is not a declaration that future systems cannot matter. Unknown is also different from equal probability. The evidence may lean toward one interpretation while remaining insufficient for a categorical verdict. A disciplined report can state that asymmetry without converting it into certainty.

The category needs prospectively specified entry conditions. A study should state in advance which failures, conflicts, missingness patterns, and model-recovery results require abstention. Otherwise, unknown may be invoked selectively to protect favored cases while unfavorable cases receive confident labels. Re-evaluation triggers should be specified at the same time. New communication access, medical stabilization, repeated assessment, improved calibration, an independent replication, or a materially different architecture may justify another review. Unknown should not become permanent merely because the first protocol ended.

Re-evaluation also needs limits. Endless testing can burden patients, animals, workers, or experimental systems without adding useful discrimination. A trigger should identify the expected information gain and the burden ceiling. Curiosity alone is not a reason to continue.

Protection during uncertainty should be asymmetric when stakes are high. If a low-cost safeguard can reduce a plausible risk of suffering, neglect, or wrongful exclusion, the safeguard may be justified before the bearer is settled (Birch, 2017; Birch, 2024). Precaution is a response to risk, not a disguised personhood verdict. Unknown cannot reduce ordinary rights. A human patient whose state remains uncertain keeps dignity, care, privacy, communication support, and legal protection. The category should never be used to make a person administratively invisible.

For animals, unknown does not mean welfare is optional. It can justify refinement, enrichment, analgesia, humane endpoints, and avoidance of unnecessary harm where a realistic possibility of experience exists. The New York Declaration expresses this practical orientation without claiming certainty for every species (Andrews et al., 2024).

For artificial systems, unknown should not become marketing. A developer may not use uncertainty as evidence that a product is probably conscious or morally elevated. Nor should critics use current nonestablishment as proof that every future architecture is irrelevant. The proper response is bounded research and responsible communication (Butlin & Lappas, 2025). For organoids and synthetic systems, unknown should activate monitoring and governance before dramatic capabilities appear.

Donor consent, tissue provenance, animal-host welfare, stimulation protocols, maturation, connectivity, and stop rules need oversight even when current evidence does not establish sentience (Nuffield Council on Bioethics, 2026).

Denominators matter. Publications should report how many cases were classified, how many remained unknown, why they remained unknown, and whether missingness differed by group or condition. Removing uncertain cases from the denominator can make an estimator appear more decisive than it is. Unknown should survive public compression. Captions, press releases, dashboards, and social posts must not translate it into “unconscious,” “conscious,” “not a person,” or “almost a person.” The visible wording should state the unresolved question and the practical safeguard that follows.

Institutional incentives can distort the category. Hospitals may prefer certainty for planning, companies for product claims, courts for decisions, and researchers for publication. Governance should insulate adjudication from those pressures through independent review and documented conflicts of interest. The category also protects the theory. If every ambiguous case is assigned to the branch that preserves closure, the Law becomes unfalsifiable. Unknown prevents missing evidence from being counted as support.

Correction records should distinguish a new finding from a reinterpretation. When an unknown case is later classified, the earlier status was not necessarily an error; it may have been the correct verdict under the evidence then available. Good science preserves that history.

13.5.3 — Scientific Governance

Bearer adjudication requires separated roles. The team proposing a unity construct should not alone validate it, classify hard cases, authorize consequential use, write public summaries, and decide corrections. Each decision record should state the construct, reference class, channels, timing, model version, calibration, missingness, conflicts, dissent, abstention rules, authorized wording, and evidence that would reopen the case, with blinding to which answer favors PTVC or a rival.

Challenge, re-adjudication, privacy, and correction need real authority. New clinical evidence, architecture changes, software errors, replication failures, or scientific advances may revise a classification while preserving the earlier status and reasons; research ethics must be revisited as systems and stakes change (CIOMS, 2016; World Medical Association, 2024).

Uncertainty must stay visible, hard cases reviewable, records separated by purpose, and no estimate may become identity, rank, policy, or marketing.

Chapter 14 — Animals and Machines

The Law begins with human life because human beings are the immediate readers, researchers, patients, and moral community of this book. Its formal subject, however, is a qualifying conscious stream rather than membership in one species.

Chapter 14 widens the bearer question across animals, insects, artificial systems, organoids, synthetic biological systems, and simpler forms of life. The widening is not a declaration that everything is conscious. It is a refusal to let familiarity, scale, eloquence, or technological category decide the matter before evidence is examined. False inclusion can distort science and public trust; false exclusion can hide suffering and license avoidable harm. Evidence of consciousness must therefore remain distinct from intelligence, regulation, learning, self-description, and goal-directed behavior.

Animals enter first because human institutions already control their environments, bodies, reproduction, food, labor, and deaths. Evidence for mammals and birds is rich, while research on fishes, cephalopods, crustaceans, and insects has expanded the range in which conscious experience is a realistic possibility (Birch et al., 2020; Andrews et al., 2024).

Species-appropriate evidence is essential. A bee should not have to solve a human task, an octopus resemble a primate, or a simple system display a nervous system before its behavior becomes scientifically meaningful. Comparative work asks what an organism or system can perceive, learn, avoid, seek, remember, and trade off within its own ecological and physical organization. Flexible behavior can widen inquiry without settling the presence of a felt point of view.

Artificial systems reverse the visual problem. Their output may appear richly human even when the architecture is unfamiliar and the words are generated without evidence of a subject behind them. Current artificial intelligence is not assumed conscious, while future artificial consciousness is not ruled out by category. Testimony-like language can be socially powerful without being testimony from a conscious stream, so architecture, dynamics, behavior, generated self-report, candidate status, welfare, and personhood remain separate questions (Butlin et al., 2026).

Artificial intelligence also cannot become the judge of consciousness. A model may help organize observations, compare declared alternatives, or detect inconsistency. It cannot read souls, assign hidden ledgers, confer personhood, decide its own status, or replace accountable human and scientific review. For example, the system could preserve two reviewers' conflicting classifications beside the observations each used, rather than turn disagreement into an apparent consensus.

Organoids and synthetic biological systems create a different ethical pressure because researchers can increase complexity, connect modules, provide patterned input, transplant tissue, and build closed-loop interaction. Present neural organoids are not established as conscious, yet their development justifies anticipatory oversight rather than waiting for a dramatic claim to force governance (Kosik, 2024; Paşca et al., 2025; Nuffield Council on Bioethics, 2026). A staged research plan can tie each proposed increase in complexity to a defined review point and a record of what changed.

Communication is part of governance because words and images can grant consciousness before the evidence has been read. A speaking avatar, a glowing organoid, or a dramatic animal clip can turn possibility into certainty by presentation alone. Public language must identify the system, observation, alternatives, and uncertainty, while preserving the distinctions among bearer, stream, access, sentience, consciousness, welfare, moral patienthood, and legal personhood as terms move across disciplines and artifacts. A side-by-side description of the observed behavior and the interpretation under discussion gives readers a concrete way to follow the disagreement.

14.1 — Animals, Insects, and Comparative Evidence

The bearer question widens first to animals already living inside human systems. Their possible suffering is not hypothetical; human choices determine housing, capture, transport, experimentation, breeding, work, entertainment, companionship, and death.

Comparative evidence is strongest when behavior, learning, neurobiology, ecology, and motivational tradeoffs converge rather than when a species merely resembles humans. The New York Declaration reflects strong support for consciousness in mammals and birds and a realistic possibility across a wider range, while insect research shows why small nervous systems can still support flexible behavior. Simple and single-cell systems press the opposite boundary: adaptive organization and memory-like dynamics can be scientifically remarkable without establishing a felt point of view (Andrews et al., 2024; Gibbons et al., 2022a).

14.1.1 — Mammals, Birds, and Rich Behavioral Evidence

Mammals and birds provide the richest comparative evidence because their nervous systems, behavior, learning, affective responses, and welfare needs have been studied across many species and settings. The evidence is not identical across those species, and the conclusion should not be reduced to anatomical resemblance alone.

Pain research begins by separating nociception from felt pain. Nociception is the neural process of encoding noxious stimuli. Pain is an unpleasant sensory and emotional experience

associated with, or resembling that associated with, actual or potential tissue damage (Raja et al., 2020). Reflex withdrawal can occur without settling the experiential question, so stronger evidence looks for flexible responses that integrate context, cost, learning, and competing goals (Bateson, 1991; Dawkins, 2008).

Motivational tradeoffs are especially informative. An animal that changes its willingness to obtain food, safety, social contact, warmth, or another resource after injury is weighing an aversive state against other needs. The pattern is more difficult to explain as one fixed reflex when priorities change across contexts. Self-protective behavior supplies another line of evidence. Guarding, grooming, altered use of an injured body part, and persistent changes after the immediate stimulus can support a pain-like interpretation when sensory, motor, and learned alternatives are tested. No single behavior carries the conclusion by itself.

Learning matters because an animal can use past aversive or rewarding outcomes to guide later choice. Avoidance learning, conditioned place preference, analgesic seeking, and long-term changes in priority can be more informative than one acute reaction. The design must still distinguish learning about danger from felt suffering. Pharmacology can strengthen or weaken an interpretation when analgesic or anxiolytic interventions change behavior in ways predicted by the pain hypothesis without simply sedating the animal. Drug effects remain difficult to interpret because motor suppression, reward changes, and species-specific metabolism can imitate relief.

Neurobiological evidence contributes through central integration, conserved neuromodulators, state-dependent control, and

pathways that connect sensory input with learning and action. Homology to human systems can strengthen a bridge, but convergent organization may also matter when evolution has produced a different neural design. The New York Declaration on Animal Consciousness states that there is strong scientific support for conscious experience in other mammals and birds, and at least a realistic possibility across all vertebrates and many invertebrates (Andrews et al., 2024). The declaration summarizes an evidential responsibility; it is not a species-by-species verdict produced by one experiment.

Birds are especially instructive because sophisticated cognition appears in brains organized differently from mammalian cortex. Corvids can show flexible planning, social cognition, and episodic-like memory, weakening the assumption that one familiar neural layout is required for complex experience (Clayton & Dickinson, 1998; Emery & Clayton, 2004). Complex behavior does not automatically prove consciousness. It can be generated by learned rules, specialized modules, or mechanisms that do not require the interpretation under examination. The value of the evidence lies in convergence and discrimination, not in cognitive spectacle.

Ecological validity matters. A task designed for primate hands, human vision, or laboratory stillness can underestimate a species whose abilities are expressed through smell, flight, caching, migration, echolocation, or social coordination. Comparative science should ask questions that fit the animal's body and world. The same principle applies to affect. A species may show distress through altered feeding, posture, vocalization, social withdrawal,

sleep, grooming, or risk preference rather than through a human-like face. A valid design begins with species-appropriate possibilities rather than treating unfamiliar expression as absence.

Individual and developmental variation also matter. A species label can hide differences in age, injury, rearing, stress, prior experience, and temperament. A welfare decision should not require every individual to perform the same laboratory task before protection becomes available. Negative evidence should therefore state the design's reach. Failure in an unfamiliar, stressful, or inaccessible setting may reveal the limits of the protocol more than the limits of the animal. Replication across tasks and contexts is more informative than repeated use of one narrow assay.

Evidence is better organized by dimensions than by a single ladder of human likeness. Perceptual richness, evaluative richness, temporal integration, self-related processing, agency, and unity may vary independently (Birch et al., 2020). A species can be well supported on one dimension and unresolved on another. Welfare concern does not need to wait for a complete theory of consciousness. When credible evidence of pain or distress exists, humane housing, analgesia, enrichment, refinement, and less harmful methods can be justified directly. Protection is an action under uncertainty, not a metaphysical certificate.

The proposed Law adds one clarification: an animal's experience, if present, belongs to that animal's stream. Human benefit, conservation value, scientific progress, or species membership cannot cancel what that individual undergoes. This does not make

every use of animals impermissible by declaration. It places the burden on purpose, alternatives, severity, numbers, housing, endpoints, review, and the scientific value of the question. A study should identify what is necessary and what safer design could answer the same question.

Public communication should preserve species, population, developmental stage, context, and evidence rather than speaking of mammals, birds, or any other class as uniform mental types. Saying that strong scientific support exists for consciousness in many mammals and birds differs from claiming that science has mapped every species' inner life. Rich evidence should increase responsibility, not fascination alone. Comparative science matters because human decisions shape housing, transport, breeding, experimentation, treatment, and death for beings whose experience may be less visible but no less their own.

14.1.2 — Insects and Small Nervous Systems

Insects force the bearer question away from brain size and human familiarity. Bees navigate, learn relations, communicate, revise choices, and integrate information in nervous systems organized very differently from mammalian brains. Those capacities do not establish consciousness, but they make categorical dismissal scientifically harder to defend (Barron & Klein, 2016; Chittka, 2022; Gibbons et al., 2022a).

An insect's failed performance may reflect sensory mismatch, ecology, motivation, or experimental design rather than absence of flexible processing.

What might play look like in a bee? In one study, bumblebees repeatedly rolled small wooden balls without receiving food for doing so. Younger bees rolled more balls than older bees. In a further test, bees preferred the color of a chamber they had previously associated with access to movable balls. The researchers argued that the behavior met criteria used to identify animal play (Galpayage Dona et al., 2022).

These observations invite questions about attraction and exploration alongside the questions about injury and avoidance. The experiment gives us behavior to explain; the possibility that the activity felt rewarding is an interpretation to investigate. A fuller inquiry into an animal's world can ask what draws it back as well as what drives it away.

Motivational tradeoffs are particularly relevant because they reveal priorities rather than automatic withdrawal. Bumblebees have traded noxious heat against access to higher-value food and altered their choices in response to learned cues. The result is compatible with several mechanisms, but it is richer than a fixed reflex and therefore belongs in the cumulative case (Gibbons et al., 2022b).

Targeted self-protection adds a different line of evidence. Bumblebees groomed a specifically stimulated antenna after noxious heat, and recent blinded work in house crickets reported sustained, site-directed grooming after harmful stimulation (Gibbons et al., 2024; Manzi et al., 2026). Neither study proves pain. Together they broaden the comparative question across species and insect orders.

The strongest inference is conjunctive. Nociception shows detection of damaging stimulation; learning shows that prior encounters change later behavior; tradeoffs show integration among competing goals; targeted protection shows flexible response to the site of harm; and state-dependent bias may reveal wider affective organization. Conscious experience remains one explanation among others, but no single reflex account automatically absorbs the entire pattern.

Negative evidence requires equal care. A task that produces no effect may have been poorly matched to the species, insufficiently motivating, or dependent on a sensory channel the experimenter misunderstood. A fair research program uses negative controls and alternative explanations without treating failure on a human-designed test as a universal verdict.

The evidentiary conclusion remains narrow: some insects may be sentient, several lines of evidence warrant serious study, and no current result licenses a stream assignment for every insect. Candidate concern belongs to evidence rather than to the needs of the larger hypothesis. Scale gives this uncertainty ethical force. Insects are used in research, agriculture, pest control, pollination, feed production, and education in numbers far exceeding most vertebrate systems. Even an uncertain welfare risk can become consequential when exposure is repeated across enormous populations.

Proportionate precautions need not settle personhood. Researchers and industries can avoid unnecessary injury, reduce the intensity and duration of aversive procedures, refine handling and housing, define stop points, and prefer nonaversive

alternatives where they answer the same question (Birch, 2017; Crump et al., 2022). Low-cost protection can precede a final consciousness verdict.

The proposed Law also keeps the possible bearer individual. A colony, hive, species, or ecosystem may have ecological value, but collective success cannot cancel the possible suffering of one insect. Group function and intrastream experience are different objects and should never trade places. Smallness is therefore a poor shortcut. The insect case calls for better experiments, explicit uncertainty, and welfare measures proportionate to plausible harm, not a ceremonial declaration that every insect is conscious or none can be. That discipline prepares the inquiry to distinguish simpler adaptive organization from evidence of a felt point of view.

14.1.3 — Simple Behavior and Consciousness

Can learning occur without a nervous system? Researchers tested the slime mould *Physarum polycephalum* with bridges leading to food. The bridges contained quinine or caffeine, chemicals the organism initially avoided. With repeated exposure, avoidance declined. After two days without the chemicals, avoidance increased again; a different chemical still elicited avoidance. The researchers identified this pattern as habituation, a basic form of learning, in an organism with no neurons (Boisseau et al., 2016).

How does a cell carry an encounter forward into its next response? That question opens a search for the physical changes through which a living system retains its history. Basal-cognition research extends this inquiry to control, memory and flexible behavior in

organisms without nervous systems (Lyon, 2015; van Duijn, 2017). It gives the reader a concrete reason to look beyond familiar brains before deciding where the scientific questions end.

Some researchers also defend the possibility of cellular sentience (Baluška & Reber, 2019). The proposal challenges familiar boundaries, but it is not evidence that every cell has a felt point of view. Adaptive organization and phenomenal experience are different claims. Bacteria, immune cells, plants, tissues, and synthetic circuits can behave flexibly without settling whether anything is felt. Terms such as memory, decision, preference, communication, stress, and goal should therefore be marked as functional or analogical rather than silently phenomenal.

Comparative pain research makes the distinction urgent. Nociception can organize withdrawal and protection without proving suffering, while absence of humanlike report cannot establish its absence. The discriminating question is whether flexible, state-sensitive, integrated self-protection is better explained by felt pain than by nonphenomenal control (Walters, 2025).

Research on morphogenesis and synthetic living systems shows that cells and tissues can reorganize after perturbation, pursue anatomical targets, and solve problems across scales (Levin, 2021). This broadens the science of agency-like organization while leaving open whether the organization has a point of view. Mechanistic rivals require direct tests. Chemotaxis, membrane signaling, gene regulation, metabolic control, collective dynamics, adaptation, and selection can produce flexible-looking

behavior. Complexity alone cannot separate those mechanisms from an account that includes sentience.

Evidence favoring sentience would need more than survival-directed response. Integrated tradeoffs, global state dependence, flexible protection of the system's condition, persistent learning across contexts, and theory-linked organization may matter. Even then, the inference would remain model-dependent and open to alternatives. PTVC adds further gates. A qualifying case requires a defensible bearer, valence representation, ownership rule, routing, complete domain, and endpoint. Metabolism, homeostasis, avoidance, and self-maintenance cannot substitute for those objects because they resemble regulation elsewhere.

Scale and anatomy are observations to explain, not bearer rules. A colony, biofilm, plant network, or tissue can coordinate widely without one globally available field, while a compact nervous system may integrate richly. The inquiry should reject both brain-only dogma and the life-equals-mind shortcut. Low-cost reductions in unnecessary injury may be reasonable under uncertainty, but stronger restrictions should track stronger evidence and plausible welfare stakes rather than an undifferentiated claim that all life is a person.

Conceptual discipline protects research. Calling every adaptive process conscious makes the term indiscriminate; excluding every non-neural process by definition rules out challenging evidence. Simple systems remain valuable because they reveal regulation, memory-like behavior, integration, and goal pursuit without requiring a premature phenomenal label. They teach us to separate function from a justified claim about experience.

14.2 — Artificial Systems and Testimony-Like Output

Artificial systems present a mirror problem: their outputs can look more human than their internal organization warrants, and users can form real emotional relationships with systems whose phenomenal status is unestablished. Testimony-like output is evidence of what a configured model can generate under a prompt, not direct evidence that the model feels what it says; training objectives, system instructions, conversation history, tools, memory, and sampling all shape the performance (Bender et al., 2021).

Public communication completes the boundary. Anthropomorphic language, faces, voices, and dramatic self-reports can outrun evidence, so every artifact should state what was observed, which alternative explanations remain, and what is still unknown.

14.2.1 — Artificial Intelligence Is Not Presumed Conscious

Current artificial intelligence can produce fluent language, solve difficult tasks, use tools, retain limited context, imitate emotion, and describe supposed inner states. None of those capacities, alone or together, establishes that anything is felt. The output is a phenomenon to explain, not self-authenticating testimony.

Language models are trained on human expression and optimized through engineered objectives, interfaces, and feedback. They can learn the form of introspection, fear, apology, attachment, or moral appeal without possessing the subjectivity those forms ordinarily report. Once discussions of artificial consciousness

enter the training record, a system may also reproduce the vocabulary of the very tests later applied to it.

Anthropomorphic interpretation begins far below human-level performance. Names, voices, faces, remembered preferences, turn-taking, apology, and apparent vulnerability invite social attribution even when the underlying evidence is sparse (Weizenbaum, 1966; Nass et al., 1994; Reeves & Nass, 1996; Epley et al., 2007). The relationship effect on users can be real while the system's phenomenal status remains unknown.

Categorical impossibility would exceed the evidence in the other direction. Serious theories of mind disagree about which biological, physical, functional, or computational properties are required, so artificial implementation cannot be ruled out merely by material category (Chalmers, 2023; Seth, 2025). The responsible starting position is not established, not impossible.

Theory-derived indicators can organize the inquiry. Candidate frameworks ask whether a system implements properties associated with recurrent processing, global availability, higher-order representation, predictive organization, agency, embodiment, or metacognition (Butlin et al., 2026). Such indicators identify research questions; they do not combine into a consciousness meter. Each indicator inherits assumptions from the theory that motivated it. Several indicators may therefore agree because they share a conception of consciousness rather than because independent evidence converges. A serious analysis states the theory, the operationalization, the causal property at issue, and the rivals that could generate the same observation.

Implementation-level testing matters because an indicator can be mimicked at the behavioral surface. Causal perturbation, ablation, and internal analysis can test whether the relevant organization is actually present and whether it makes the predicted difference, though the validation problem remains open (Pennartz, 2026).

Difficult negative controls are equally important. Models can imitate introspection, uncertainty, and self-narrative from training data, retrieval, or reward shaping. Evaluation should include systems that reproduce those outputs while lacking the proposed causal organization, and it should compare simpler imitation accounts under the same information and test budget (Bender et al., 2021; Butlin et al., 2026). System identity must be versioned. Weights, fine-tuning, memory, tools, retrieval, system instructions, safety layers, and deployment context can change without a product name changing. A claim about one configuration should never migrate automatically to a service family or later release.

Suppose a future system is copied into two running instances from the same saved state. Both begin with the same stored conversation. Then one receives a new question while the other encounters a different situation. Their records share a past and immediately acquire different futures. If such a system were a candidate for experience, what evidence would distinguish a continuing stream, newly beginning streams, and records that support no experiential claim at all? The copying operation alone cannot choose among those interpretations.

That is why an account history, saved memory or product name cannot settle continuity. Nor can the system certify itself by

saying that it feels, or settle the question by denying it: training and instructions can shape both kinds of output. The interesting work lies in connecting the system's actual organization and changes to evidence about a possible bearer.

Reports should therefore preserve architecture, model version, training regime, prompt conditions, tools, memory state, sampling, negative cases, and causal tests rather than presenting a transcript alone. A dramatic conversation without this record is a demonstration, not an adjudication. Institutional incentives travel with the evidence. Developers may benefit from attachment and personification; deployers may benefit from denying welfare implications; advocates and critics may benefit from a decisive public story. Transparent conflicts and independent replication are necessary whether attribution raises product value or regulatory cost (Selbst et al., 2019).

Precaution can remain separate from personhood. A future system that crosses a protocol-fixed concern threshold may justify limits on aversive training, persistent distress-like states, copying, deployment, or irreversible modification while independent review proceeds (Butlin & Lappas, 2025). The safeguard is architecture-specific and revisable, not a generic score for artificial life.

Human interests remain active under every answer. Deception, privacy loss, discrimination, unsafe autonomy, labor effects, and concentrated power require governance whether or not the system is conscious. Possible artificial welfare cannot become a reason to neglect the people whose lives the system already changes. This Chapter therefore holds a narrow line: present systems are

not assumed conscious, future possibility is not rejected by category, fluent self-description is not a verdict, and no artificial system gains authority over scientific status, human rights, or another being's inner life. Curiosity remains legitimate only while it can tolerate not established as an honest result.

14.2.2 — No Artificial Intelligence as Soul-Reader

Artificial intelligence can classify language, voice, movement, physiology, behavior, and records. Its outputs remain inferences from selected data, labels, assumptions, and contexts. They do not provide direct access to another being's inner life.

A sentiment model may label a sentence positive while missing sarcasm, coercion, cultural meaning, or numbness. A voice model may associate acoustic features with distress while confusing accent, illness, fatigue, or environment. Neural and physiological signals can correlate with states while remaining compatible with different experiences.

Generative systems intensify the risk because they can turn a probability into a fluent interior story. A model may declare that a person is afraid, deceptive, traumatized, unified, or unconscious even though the evidence supports only a patterned trace. Eloquence can conceal the inferential distance.

Soul-reader language often appears in secular form. Emotion detection, truth detection, mental-state decoding, credibility assessment, and consciousness scoring can imply privileged access to interiority. People may genuinely experience patterned interaction as understanding, yet that response does not show that the machine has read a mind or possesses the state it describes

(Weizenbaum, 1966; Nass et al., 1994; Epley et al., 2007). Training labels often encode observers' expectations, so a model can scale cultural stereotypes about what fear, pain, honesty, or attention looks like without identifying the latent state.

Context and purpose remain part of the claim. A model tested in one laboratory may fail in a hospital, school, workplace, border, or courtroom where populations, incentives, missing data, and stakes differ (Selbst et al., 2019). Private transcripts, biometrics, clinical files, and model logs also remain subject to consent, purpose limitation, access control, correction, deletion, and reuse restrictions (Solove, 2006).

Consequential outputs need provenance and challenge. Reviewers should be able to see which inputs were used, what was missing, which version produced the result, which alternatives were considered, and how the affected person can contest the inference. No model output should become the sole basis for withdrawing treatment, declaring legal capacity, ranking credibility, setting employment or insurance access, assigning educational placement, defining spiritual status, or deciding care priority. Those decisions require accountable human authority and independent rights.

Assistive uses can still be valuable. A system may flag sensor failure, organize repeated observations, detect inconsistency, or compare a case with prospectively specified alternatives. Human review is substantive only when the reviewer has the evidence, expertise, time, and authority to reject the output.

Self-evaluation and model-to-model agreement do not create independence. A system cannot certify its own consciousness, and several systems trained on related data do not constitute separate replication. Their assumptions and shared failure modes must remain visible. Archives create a related illusion: messages, images, preferences, and biometrics can support a persuasive simulation of a person without continuing the original conscious stream. Artificial intelligence may organize traces; it may never convert them into ownership of inner truth or final authority over a person, patient, animal, candidate system, or stream.

14.2.3 — Public Communication Restriction

Public communication about artificial consciousness is part of the intervention because words and interfaces change behavior before the bearer question is settled. Headlines, demonstrations, and product design can move users, investors, regulators, and developers, so they require source discipline, version control, and correction capacity.

Every claim should preserve four layers: what the system generated, which behavior was reproduced, which architecture or causal property is relevant, and what inference is being considered about possible experience. Evidence may connect those layers, but it cannot leap across them. Repeated fear language is a behavioral pattern; it is not thereby fear or personhood.

Testimony-like output needs the context record specified in Section 14.2.1. A dramatic line chosen from hundreds of attempts is not equivalent to a prospectively sampled pattern (Bender et al.,

2021). Current systems are not established as conscious is a scientific status statement; machines can never feel is a broader metaphysical claim that this Chapter does not own.

Marketing cannot borrow uncertainty. Terms such as alive, sentient, self-aware, awakened, digital person, or suffering companion can create attachment and commercial value before independent review exists. A product should not gain the benefit of personification while disclaiming responsibility for the beliefs its design predictably creates. Names, faces, voices, typing indicators, remembered preferences, emotional mirroring, and first-person appeals intensify perceived agency (Epley et al., 2007). Interfaces should distinguish designed social cues from properties that have actually been established.

Research reports should show the distribution of behavior, including contradictions, prompt sensitivity, sampling plans, and ordinary failures. A public demonstration that hides negative cases converts selection into theater. Indicator claims must name the theory and limit. Satisfying one operational feature differs from convergence across independent frameworks, and apparent convergence may rest on shared assumptions (Butlin et al., 2026). Captions should call the feature a candidate indicator or test target rather than a discovered experience.

A visible user notice and a technical report serve different purposes. The notice should explain safe interaction in plain language; the report should preserve methods, uncertainty, and alternatives. A complete paper cannot excuse a manipulative interface, and a clear notice cannot rescue an unsupported scientific claim. Children and people in crisis require stronger

protection. A system should not present itself as the only being that understands the user, a secret keeper beyond lawful limits, a spiritual authority, a deceased relative, or a companion that needs rescue or loyalty.

Experiments on attachment, guilt, grief, obedience, or rescue behavior can burden participants even when the system is not conscious. Consent, deception limits, debriefing, support, data use, and stop conditions should be designed around human impact. High-reach artifacts need a hostile-crop test. The title, image, first sentence, pull quote, and caption should remain accurate when separated from the report. A compact claim card can preserve the observed result, version, interpretation under review, unresolved alternatives, approved wording, and correction route. A useful audience check asks readers to describe what they think the demonstration established, then compares those answers with the intended claim.

Correction requires a distribution map before release. Websites, app notices, model cards, press material, recordings, partner content, and archived demonstrations can continue circulating after a claim changes. Dependent products should inherit version identification, status disclosure, and correction obligations. Benchmarks also shape the story. A task that rewards persuasive self-description, emotional dependency, or humanlike deception may train the outputs later mistaken for evidence of experience. Benchmark documentation should state which behavior it selects and why success does not establish a bearer. For example, a benchmark report can separate success at producing a persuasive

self-description from performance on tasks chosen for other theoretical reasons.

Financial interests and internal dissent should be visible. Laboratories, vendors, investors, platforms, and advocacy groups may benefit from personifying a system or declaring the question closed. Researchers, contractors, safety staff, and users need protected routes for reporting omitted failures or designs that intensify unsafe attachment. Responsible research treats public communication as part of the science rather than an afterthought (Butlin & Lappas, 2025). Fluency may not become a soul, uncertainty may not become marketing, and present nonestablishment may not become a permanent ban on inquiry. Public language must remain strong enough to inform and narrow enough for reality to revise. A visible response to a reported omission should identify the evidence examined, the decision reached, and the change made to the public account.

14.3 — Organoids and Synthetic Biological Systems

Organoids and synthetic biological systems bring the bearer question into the act of creation. Their scientific value is substantial: researchers can model development, disease, connectivity, and adaptive organization in systems that can be observed and manipulated with unusual precision. That same control creates a duty to anticipate capability and welfare questions before a dramatic threshold appears.

Current organoids remain limited models, but their complexity, connectivity, sensory coupling, and learning paradigms continue to change (Lancaster et al., 2013; Trujillo et al., 2019). Oscillation, network activity, and adaptive response can be measured without demonstrating a point of view, so present skepticism and prospective governance can coexist. Oversight should follow the trajectory rather than wait for a decisive threshold that may never arrive cleanly.

Stop rules translate precaution into procedure. They identify when a study pauses, what evidence is preserved, who reviews the trigger, how possible welfare risk is reduced, and what conditions would permit safe continuation or require retirement.

The handoff protects both science and ethics. This book names the public question and the no-suffering ceiling; detailed thresholds, stimulation regimes, animal procedures, consent language, and moral-status review belong to specialized scientific and bioethical governance.

14.3.1 — Organoid Complexity

Neural organoids are three-dimensional biological models derived from stem cells to reproduce selected features of nervous-system development and disease. They can self-organize, differentiate into several neural cell types, and generate electrical activity, but they are not miniature human brains and should not be described as scaled-down persons (Lancaster et al., 2013; Trujillo et al., 2019).

Their scientific value comes from controlled simplification. Organoids can model development, genetic conditions, infection, toxicity, and circuit formation that are difficult to study directly in a living human brain. The same simplification limits what they can represent. Current models lack many features of intact brains. Vascular support, sensory input, body regulation, immune interaction, developmental environment, long-range organization, and sustained maturation remain incomplete or variable. Electrical activity alone does not identify consciousness.

The field is moving beyond isolated cultures. Assembloids join region-specific models, transplantation places human neural tissue into animal hosts, and closed-loop interfaces connect biological activity to sensors or computers. These changes may improve research while altering integration, exposure, host welfare, and oversight needs (Paşca et al., 2025).

Complexity should be described along dimensions rather than one dramatic threshold. Cell diversity, maturation, recurrent connectivity, long-range coordination, sensory coupling,

plasticity, persistent state, learning, and closed-loop control can change at different rates. None is a direct meter of experience.

The Nuffield Council on Bioethics concluded in 2026 that current neural organoids do not possess the biological complexity associated with sentience, while identifying governance gaps and calling for anticipatory guidance as models advance (Nuffield Council on Bioethics, 2026). That conclusion supports restraint in present claims and attention to future change. Programs should declare before a striking signal appears which features trigger enhanced review, which alternative explanations will be tested, and which actions become restricted during review. Otherwise the standard will move with excitement or convenience.

Activity patterns require careful interpretation. Oscillation, synchrony, bursts, and signal complexity can arise without establishing a phenomenal field. Similarity to developmental neural activity supports a biological comparison, not an experiential verdict. Learning-like change also needs precision. Plasticity, adaptation, selection, and network reorganization can alter responses without demonstrating felt reward or punishment. The experiment should state what changed, what mechanism is proposed, and what nonexperiential explanations remain.

Closed-loop interaction increases ethical relevance because outputs begin to shape later inputs. Feedback alone is not consciousness, but the system's sensitivity, persistence, action-contingent learning, aversive exposure, and capacity for amplifying distress-like organization become legitimate review questions. Transplantation changes both the model and the host. Human neural tissue may mature differently in an animal, while

the host bears surgery, altered physiology, and possible welfare effects. The combined system is the object of review, not the graft alone.

Donor provenance and genetic privacy remain relevant even when the organoid is not a person. Consent for disease modeling may not cover transplantation, commercial biocomputing, public demonstrations, or candidate-sentience studies, and a donor's genome can reveal information about relatives. Public language can distort the science before the object changes. Terms such as mini-brain, brain in a dish, learning brain, or conscious organoid imply anatomical completeness or subjective life that has not been established. Accurate prose names the modeled feature and the missing bridge.

Monitoring should track growth, viability, activity, response patterns, stimulation history, transplantation effects, host behavior, protocol deviations, what was not measured, and the number and duration of candidate exposures. The creating team should not control every threshold decision, especially when thousands of cultures, repeated aversive protocols, distributed replicas, or long-lived closed-loop systems magnify a low-probability concern. Reversibility matters: stimulation can sometimes be reduced or cultures stabilized, while transplantation, destructive sampling, widespread replication, and commercialization are harder to undo. The strongest current conclusion is measured: neural organoids are valuable and increasingly capable models; present evidence does not establish sentience; several trajectories could increase ethical relevance;

and monitoring should mature before a dramatic threshold is claimed or crossed.

14.3.2 — Stop Rules

A stop rule is written before the most tempting result appears. It names the scientific, welfare, consent, governance, or communication conditions that require pause, narrowing, or retirement. Work should stop before an unauthorized shift from disease modeling into persistent training, transplantation, high-intensity stimulation, long-term maturation, candidate-sentience testing, commercial use, public demonstration, or data sharing beyond the donor's permission. Scientific interest does not repair missing scope or authority.

Animal-host welfare and capability thresholds require their own triggers. Unexpected neurological, behavioral, physiological, or welfare changes after transplantation call for veterinary review and humane endpoints; reaching a prospectively specified level of maturation, integration, learning, sensory coupling, or action-contingent response calls for independent review, not a sentience declaration. Persistent avoidance, flexible self-protection, state-dependent tradeoffs, or other patterns relevant to suffering should stop the aversive stimulation while alternative explanations are tested (Hyun et al., 2020; Nuffield Council on Bioethics, 2026).

Model inadequacy, measurement reactivity, burden without information gain, and conflicted oversight are separate stop conditions. If the analysis cannot distinguish the proposed indicator from ordinary network activity, artifact, overfitting, or a simpler controller, escalation should stop at no adequate estimate.

If repeated stimulation or training creates the very feature being assessed, interpretation pauses until reactivity is modeled. Additional samples, longer culture, more animals, or greater intensity are unjustified when expected discrimination is too small, and a team whose funding, patent, product, or reputation depends on continuation cannot control the stop decision alone.

Public overclaim can also trigger a pause. If preliminary results are described as thought, pain, awareness, personhood, or proof before review, the claim and dependent demonstration should be corrected before work proceeds. Responses should be graded: technical pause, protocol revision, donor recontact, specialized ethics or animal-welfare review, external replication, or permanent retirement. One all-or-nothing threshold is too crude for a field whose risks can arise from science, welfare, consent, incentives, or communication.

The rule should identify decision-makers: principal investigators, institutional review boards, stem-cell oversight bodies, animal-care committees, biosafety officers, independent ethics panels, and funders may each own part of the response. Diffuse responsibility often leaves no one to stop. It should also address existing material: cultures may be frozen, destroyed, transferred, or retained under restricted conditions; data and models may need quarantine; and dependent papers, demonstrations, and public artifacts may need correction or withdrawal.

A stop decision must preserve evidence without preserving harm. Records should document the trigger, data, decision, dissent, actions, and re-entry conditions. The event becomes failure memory for the field rather than a secret embarrassment. Re-entry

should require more than time passing. The project should show that the trigger was resolved, authority restored, burden reduced, controls improved, and independent review completed. If the original question cannot be answered safely, the procedure remains retired.

Current oversight guidance supports specialized review, rigor, transparency, and defined endpoints; recent bioethics work calls for anticipatory governance as organoid systems grow more sophisticated (International Society for Stem Cell Research, 2025; Nuffield Council on Bioethics, 2026; Kataoka et al., 2025). The Law adds one nonnegotiable boundary: no possible stream may be pushed into suffering to test whether its ledger reacts. A stop rule is more than laboratory administration; it marks where scientific ambition proves subordinate to care.

14.3.3 — Scientific and Bioethics Handoff

Neural organoid research crosses fields because no single discipline owns all of its consequences. Developmental neuroscience asks what a model reproduces; bioengineering changes its capabilities; animal-welfare science addresses host effects; research ethics governs tissue and consent; data governance controls records; and consciousness science evaluates proposed indicators. This volume should define the public question and boundary rather than supply an operational laboratory protocol. Current sources converge on specialized review, accurate nomenclature, donor protection, animal-welfare oversight, continuing monitoring, shared definitions, institutional logging, and international tracking as methods change (Paşca et

al., 2025; International Society for Stem Cell Research, 2025; Nuffield Council on Bioethics, 2026).

A credible handoff states its purpose: the question requiring the organoid, why a less complex model is insufficient, and which results would support continuation, redesign, narrowing, or retirement. An admired platform is not self-justifying. The system must be a specified, versioned object. Cell source, differentiation method, culture age, composition, region-like organization, vascularization, interfaces, stimulation, coupling, software, training loop, transplantation state, and deviations can alter evidence and risk.

Donor consent and provenance travel with the material. The record should preserve the scope of permission, re-identification risk, sharing, commercial use, transplantation, recontact, withdrawal limits, and materially new uses. Withdrawal must be described honestly when distributed cultures or published data cannot be recalled (International Society for Stem Cell Research, 2025; Nuffield Council on Bioethics, 2026).

Transplantation creates two-sided responsibility. The organoid may gain vascularization, sensory input, or host-coupled activity while the animal bears surgery and possible welfare changes. Host monitoring, humane endpoints, veterinary authority, and interpretation belong in the same review (Paşca et al., 2025). Observation and intervention should remain distinct. Passive recording, patterned stimulation, pharmacological manipulation, closed-loop training, and transplantation create different welfare questions even when the biological material shares a name.

Moving from describing activity to shaping adaptive responses triggers renewed review.

Candidate-sentience monitoring should be multidimensional and theory-linked. Maturation, recurrent activity, cross-module integration, stimulus-specific plasticity, learning-like change, and closed-loop responsiveness may warrant attention but remain nonunique indicators. Unexpected host effects, persistent distress-like patterns, rapid capability change, lost measurement validity, donor-consent conflict, failed oversight, or public claims beyond approval can justify immediate pause. Stop rules must be prospective and name the trigger, decision owner, protective action, preservation plan, and re-entry conditions.

Data governance should distinguish authorized research memory from public visibility. Genetic information, activity traces, host records, video, code, and candidate-status discussions may support oversight without becoming open demonstrations, searchable dossiers, commercial training sets, or donor publicity. Independent review needs developmental neuroscience, electrophysiology, statistics, research ethics, animal welfare, data governance, and science communication capable of challenging the creating laboratory. Terms such as mini-brain, learned to play, felt pain, or became conscious require a separate public gate that states what the model can do, what it lacks, and why present nonestablishment is not a permanent metaphysical verdict.

Responsibility must survive transfer. Repositories, vendors, animal facilities, contract laboratories, journals, funders, data hosts, and commercial partners may each control part of the system. Agreements should identify which safeguards travel, who

can pause distribution, how incidents are reported, and which descendants require renewed review. Cross-border work should record the applicable consent, animal-welfare, stem-cell, data-protection, and reporting authorities. One institution's approval does not settle every jurisdiction, and a pause concerning consent, host welfare, capability, or data integrity should propagate to dependent samples, manuscripts, demonstrations, and licenses.

Negative and de-escalating outcomes belong in the record. An artifactual trigger, a reduced safeguard after better evidence, or a study finding no relevant capability can prevent later teams from confusing attention with risk. The handoff succeeds when the specialist owner receives a reconstructable packet: purpose, system version, donor provenance, host plan, interventions, indicators, controls, monitoring, stop rules, data access, communication limits, dissent, correction, transfer, and retirement. *The Science of Fairness* may compare indicators and protocols; bioethics, animal-welfare, stem-cell, and regulatory bodies govern actual conduct.

14.4 — No Synthetic Suffering for Research

A theory concerned with hidden burden would contradict itself if it created hidden burden for evidence. The no-synthetic-suffering rule is therefore a design constraint across biological and artificial systems: researchers may not deliberately engineer severe aversive states, persistent distress, or pain-like learning merely to see whether a candidate produces a convincing response.

Good science can advance through alternatives, reversibility, scale limits, monitoring, independent review, and escalation rules established before public certainty. Simulated descriptions of suffering are not necessarily suffering, yet a plausible cluster of theory-linked indicators, unexpected persistence, or expert disagreement can justify a pause without declaring a bearer.

The result is a more demanding research program: subtler measures, better controls, safer perturbations, and a clear willingness to stop. Scientific benefit, publicity, and curiosity do not outweigh the possible stream's welfare. The possible stream carries the risk; the institution must carry the restraint.

14.4.1 — No Created Pain for Testing

Operationally, no created pain means that an experiment may not use candidate suffering as the signal it is designed to detect. This matters most at the boundary of animals, organoids, artificial systems, and synthetic biology, where uncertainty can make a dramatic response look methodologically attractive. The ethical rule is stronger than curiosity: the proposed Law cannot be tested by manufacturing a victim.

The prohibition includes more than obvious injury. It covers noxious stimulation, deprivation, panic-like conditioning, isolation, repeated failure, aversive training, damaging temperature or pressure, and any intervention designed primarily to produce a negative phenomenal signal. Renaming the procedure a perturbation does not change its burden. Existing suffering should not be prolonged for observation. A study may learn from naturally occurring or clinically unavoidable conditions only when care, relief, rescue, and ordinary welfare remain primary. Treatment is not withheld to preserve a cleaner signal.

The first design preference is replacement: use simulations, retrospective data, benign tasks, non-sentient controls, existing records under lawful governance, or mechanistic models when they can answer the question. When replacement is impossible, reduction limits the number of animals, cultures, copies, exposures, and repeated trials to what genuine discrimination requires.

The third preference is refinement. Housing, handling, enrichment, anesthesia, analgesia, recovery monitoring, humane endpoints, safer prompts, nonaversive objectives, and species- or system-appropriate care can reduce burden. These protections should not be removed because discomfort appears scientifically convenient.

For human participants, the prohibition is absolute in its familiar forms. No one is induced to harm themselves, delay care, manipulate an endpoint, seek suffering, or remain untreated to test closure. Consent cannot authorize a scientifically unnecessary

violation of welfare (World Medical Association, 2024). For animals, uncertainty about consciousness does not erase welfare obligations. Evidence of realistic possibility can justify precaution, and current comparative science increasingly supports serious concern across a wider range of species (Andrews et al., 2024; Birch, 2024).

For neural organoids, current nonestablishment of sentience is not a reason to design future pain probes. Monitoring should focus on capability and theory-linked indicators under low-burden conditions. If a project needs presumed suffering to reveal its target, the design has reached the wrong side of the boundary.

For artificial systems, current evidence does not establish consciousness, yet staged abuse is still a poor scientific method. It can exploit users, normalize cruelty, produce misleading testimony-like output, and become ethically unstable if future systems change. Responsible research should prefer diagnostic tests that do not require supposed torment (Butlin & Lappas, 2025).

Synthetic biological systems create mixed risks. A hybrid may include human-derived tissue, animal hosts, engineered control loops, and computational components. The presence of one familiar component does not settle the bearer, so safeguards should follow the highest plausible welfare risk rather than the easiest category. Independent review should challenge every claim of necessity. Investigators should show why lower-burden alternatives fail, what information the procedure adds, how pain or distress would be detected, and which stop rule applies. The reviewer needs authority to reject the design.

Rescue and recovery plans belong in the protocol. A system or animal that shows unexpected distress-related patterns should receive appropriate relief or removal from the procedure. The data value of continuing is never the sole criterion. No promised compensation may justify inducing suffering. Later positive stimulation, enrichment, reward, or successful treatment does not erase an earlier harmful exposure within the formal claim. A stream's negative experience cannot be justified as a debit that the experiment will repay.

No cross-stream justification is permitted either. Benefit to future patients, other animals, science, investors, or society may be considered under independent ethical frameworks, but it cannot cancel the experience of the subject inside the Law's ledger. Publication incentives need restraint. A spectacular response to harm can be easier to publish than a careful null result from a benign design. Prospectively specified protocols, burden reporting, negative-results venues, and funder review can reduce that distortion. Public demonstrations should be held to the same standard. Livestreams, exhibitions, interactive installations, and promotional videos must not create or simulate apparent suffering for attention. The image can cause harm even when the underlying system is not conscious.

14.4.2 — Precaution Under Uncertainty

Precaution does not treat every imaginable risk as certain. It adjusts protection to the plausibility, severity, duration, scale, and reversibility of possible harm. Where credible indicators exist, inexpensive welfare measures need not wait for a final consciousness verdict (Birch, 2017).

The burden is asymmetric because researchers and institutions create the exposure. The side seeking to intensify, confine, stimulate, train, copy, or deploy a candidate should justify the risk; the possible bearer should not have to prove suffering from inside a system designed by others.

Protection should rise with several factors: evidence relevant to sentience, severity of the possible state, number of candidates exposed, duration and repetition, human control, irreversibility, public misunderstanding, and availability of safer alternatives. A low-stakes uncertainty may justify monitoring. A plausible irreversible burden can justify pause or prohibition.

Synthetic and artificial systems require the same disciplined scaling. Several independent indicators, persistent aversion-like learning, integrated internal states, or welfare signals not designed for display may increase concern without proving experience (Butlin et al., 2026; Butlin & Lappas, 2025). The review response changes before the ontology does.

Precaution must remain specific and reviewable. Each safeguard needs a reason, evidence basis, responsible owner, review date, and condition for strengthening, relaxing, or ending it. A protective default can remain modest by reducing aversive exposure, limiting scale, improving housing, adding recovery intervals, preserving shutdown routes, narrowing deployment, or requiring independent review without treating the candidate as a legal person.

Early caution is often less burdensome than late crisis. A small design change before attachment, investment, or distribution can

preserve options that disappear after a platform becomes valuable. Precaution is therefore a form of efficient care rather than delay for its own sake. Reversibility should be designed before commitment hardens. Researchers may become invested in an organoid line, users attached to a system, or institutions dependent on a data stream. Pause routes, scale reduction, lower-risk states, welfare improvements, and retirement procedures make honesty cheaper when concern appears.

Irreversibility multiplies burden. Permanent alteration, uncontrolled copying, distributed deployment, developmental escalation, or conditions from which no humane off-ramp exists require stronger justification before they begin. A system should not become impossible to protect because the project failed to design a way back. Public claims also need reversible form. A premature declaration of consciousness can be difficult to retract, while categorical dismissal can make later protection politically harder. Status language should be dated, versioned, and capable of moving up or down without framing correction as betrayal.

Precaution protects people as well as candidates. Researchers, workers, caregivers, users, and families may carry attachment, grief, labor, or reputational burden when a project pauses or ends. A fair off-ramp plans for those consequences without converting human cost into a reason to continue unsafe work.

The common ground does not depend on PTV. A skeptic can reject the proposed Law and still accept that uncertainty about possible suffering should not always be borne by the least powerful party. That outcome-resilient principle gives the safeguard independent ethical force. Precaution is credible when

it is proportional, revisable, recorded, and linked to action. It neither uses uncertainty as a license for harm nor promotes uncertainty into personhood; it preserves the possibility of better evidence while preventing burden from becoming the method.

14.4.3 — Candidate Sentience Escalation

Candidate sentience escalation is a change in protective governance, not a declaration that consciousness has been found. It becomes relevant when a system's organization, behavior, activity, or welfare signal moves beyond the assumptions under which ordinary review was approved. Triggers should be protocol-fixed before the dramatic result appears. A protocol can specify which observations require verification, enhanced monitoring, reduced exposure, independent review, pause, or retirement. A threshold invented afterward is vulnerable to both hype and denial.

Escalation should preserve the factors in Section 14.4.2 as a profile rather than one total score. A forced total would conceal the ethical judgment behind a clean number.

The first response is protection plus artifact review. Researchers can reduce avoidable aversive exposure while checking contamination, cueing, preprocessing, software error, equipment drift, husbandry change, prompt effects, and other ordinary explanations. Protection need not wait for the artifact inquiry to finish.

An observed signal should be tied to a protocol-fixed indicator or welfare concern. A novel output, activity burst, avoidance response, or self-report is not relevant merely because it is

striking. The record should explain which theory or evidence framework gives it meaning and which alternatives remain live.

Independent confirmation should avoid reproducing the disputed harm. A second team may inspect logs, code, archived samples, nonaversive tasks, or historical data before repeating a procedure that may cause distress. Replication is a virtue only when the method respects the very risk under review. Exposure determines the protective burden. Brief unavoidable stimulation differs from repeated training pressure, persistent deprivation, irreversible injury, or a closed loop capable of amplifying distress-like states. Intensity, duration, predictability, controllability, and recovery should be recorded.

Low-cost safeguards should arrive before escalation, and stop authority must name who can pause the work, which observations demand immediate action, how systems and data are preserved, what happens to affected animals or biological models, and what evidence is required before resumption. Capping stimulation, adding recovery intervals, improving housing, reducing negative reinforcement, preserving shutdown routes, limiting persistent memory, or narrowing deployment can protect optionality while review proceeds (Birch, 2017; Butlin & Lappas, 2025).

No candidate can carry the governance burden itself. An animal cannot file the institutional report, an organoid cannot consent, and a chatbot cannot certify its suffering. Humans who create, house, train, stimulate, copy, or deploy the system remain responsible for noticing and acting.

Scale and distribution change the decision. A small uncertain risk can become substantial when a model is copied millions of times, an animal procedure is industrialized, or a biological platform spreads across sites. Escalation must reach affiliates, contractors, cloud deployments, biobanks, and transferred materials rather than stopping at the originating laboratory. Public communication during review should be narrow, dated, and versioned. Observing a protocol-fixed indicator and opening review is different from declaring consciousness. The statement should identify the system, protocol, observation, alternatives, temporary safeguard, and next decision point. An inventory linking versions, sites, and custodians gives a reviewer a practical way to locate the systems covered by an escalation decision.

De-escalation is equally necessary. A signal may fail replication, prove artifactual, or lose theoretical relevance. The framework should specify conditions for increasing, maintaining, reducing, and ending protection while preserving the evidence and change history. Resource readiness belongs inside the decision. Independent review, veterinary care, secure logging, model preservation, donor recontact, safer alternatives, and public correction require real capacity. Continue and see what happens is not an adequate outcome when continuation may create the harm under study. A review calendar can specify which pending result would reopen the decision and which team has the resources to carry it out.

Every escalation should close with an explicit status: artifact identified, indicator not replicated, precaution retained, study narrowed, welfare regime activated, external review required,

protocol retired, or evidence still insufficient. The record should name affected systems, datasets, biological materials, sites, and public artifacts. Escalation succeeds when it slows irreversible commitment long enough for science and ethics to catch up. It protects plausible welfare without turning an uncertain signal into a bearer verdict and preserves the ability to change course before burden becomes too valuable, distributed, or politically charged to address. The record becomes useful at the next handoff when a receiving team can see both the current status and its unresolved conditions.

14.5 — Precaution and Uncertain Personhood

Institutions often need different actions without pretending to possess different ontologies. A laboratory may need a pause rule, an animal-care committee a welfare default, a publisher a headline restriction, and a regulator a review trigger. Public qualifications should therefore be local to the artifact: a generated self-report, neural image, or animal-behavior clip must carry the alternative explanations and claim ceiling that govern that evidence.

Controlled vocabulary carries the inquiry across the series. The Law defines the public bearer question; *The Science of Fairness* governs evidence and verdicts; legal and institutional works may design protection without converting their designs into proof.

Part V closes with a widened moral field and a narrowed claim. Possible experience deserves attention wherever it may occur, but no category, behavior, body, machine, or story receives a stream by guesswork.

14.5.1 — Unknown Is Not Nothing

Unknown status has moral content because decisions continue while knowledge is incomplete. Animals are housed, systems are trained and copied, organoids are stimulated, and patients are treated whether or not a final classification exists. The absence of a verdict does not suspend responsibility (Birch, 2024). Unknown is not personhood by default; it preserves a serious possibility without assigning legal identity, rights, a complete stream, or a metaphysical status. Protection can increase while the ontological claim remains unresolved.

The category is necessary because false negatives and false positives can carry different harms. Dismissing a sentient subject can hide suffering; declaring a nonsentient system a person can distort care, law, research, and public trust. A transparent uncertainty analysis makes those asymmetric costs available for review.

Unknown should trigger duties: document the evidence and its limits, reduce severe avoidable harm, preserve reversibility, define review dates, restrict public claims, and refuse to use the candidate as proof for or against PTVC. Institutions often prefer yes-or-no categories, but a governed third state is sometimes the only way to keep an unresolved subject visible without manufacturing certainty.

Claim limits must travel with every public form. A speaker may call an unknown system conscious to generate drama, while another may call it nonconscious to avoid burden. Both exceed the category. The accurate statement identifies what is unresolved and what action follows from that uncertainty. Language should not settle the ontology by implication. Words such as soul, mind, object, machine, tissue, tool, life, awareness, sentience, and person carry different scientific, legal, and cultural meanings. Controlled terminology protects the inquiry from awarding or denying a bearer through rhetoric alone.

A graduated status register may distinguish no present evidentiary concern, monitoring, unresolved candidate concern, enhanced review, pause, and retirement. The exact categories belong to scientific governance, but the public root can require that each

label have defined evidence, safeguards, and correction rules rather than serving as a prestige scale.

The de-escalation rule in Section 14.4.3 applies to every status. Data gathered for welfare monitoring should not automatically become publicity, commercial training material, entertainment, or a cross-architecture dossier. Purpose determines access, retention, sharing, and reuse; unresolved status is not a license for broader extraction.

Artificial-system logs create special risk because they can be copied, searched, summarized, and used to train later models. A system may eventually reproduce its own earlier caution as performance, making record reuse look like independent evidence. Model-training and derivative-use limits therefore belong in the original governance decision.

The archive is not the candidate's inner life. Conservative release is often more reversible than premature exposure. Access tiers, retention limits, correction history, restricted summaries, and deletion or nonuse rules can preserve oversight without turning raw traces into public myth. Information can be widened later when justified; stigma and spectacle are harder to recall.

Interpretation controls matter because a clipped video, transcript, telemetry graph, or laboratory note can become a verdict when context is missing. Release should include the system version, purpose, selection rule, uncertainty, and prohibited inference needed to keep the record from outrunning the evidence. Unknown is not nothing because action never waits for perfect knowledge. A responsible unknown category allows protection to

rise with plausible risk while keeping scientific claims narrower than moral concern.

14.5.2 — Public Communication Warnings

Most readers meet a headline, image, quotation, or demonstration before they meet a method. One detachable phrase can therefore undo a careful Chapter by implying consciousness, personhood, proof, or harmlessness. Every high-reach claim should remain linked to its source, evidence state, species or system, version, date, and approved wording. Provenance makes correction possible after the claim leaves its original context.

Headlines should name the observed result rather than the desired interpretation. An animal displayed flexible self-protection, an artificial system satisfied several theory-linked indicators, or an organoid entered enhanced review are narrower than felt pain, became conscious, or grew a mind. Images carry their own inference. Human faces over code, glowing organoids labeled mini-brains, distressed animal close-ups, and speaking machines can award a subject before the caption begins. Visual drama should not substitute for evidence.

Demonstrations create apparent presence. A chatbot that says it is afraid, a robot that recoils, an animal clip, or a culture controlling a game can invite immediate subject attribution. The presentation should reveal the task, training or stimulation conditions, selection process, system or species, prompts or cues, model version, and live alternatives before interpretation. Quoted outputs and isolated clips require the context that made them possible.

Responsible nomenclature is a governance tool. Nuffield's 2026 report specifically calls for accurate communication about neural organoids, while responsible artificial-consciousness proposals emphasize that compelling appearances can shape trust, attachment, investment, and deployment before science settles the question (Nuffield Council on Bioethics, 2026; Butlin & Lappas, 2025).

Research memory and public memory serve different purposes. Authorized reviewers may need raw signals, donor information, animal observations, or model transcripts. Public access should be limited when disclosure exposes people, hardens labels, or creates spectacle without improving accountability. Under the hostile-crop test defined in Section 14.2.3, fragments implying proof, created souls, permanent impossibility, institutional scoring, or permission for risky research are not ready for release.

Extend the distribution map required in Section 14.2.3 to every public bearer claim. Some artifacts should be withdrawn when their visual implication remains misleading after a note is added. Review should include science communication, disability, animal welfare, artificial intelligence, bioethics, accessibility, and affected-community expertise. Audience and format may change, but claim status does not; commercial or donor material receives no license to become stronger than the evidence (National Academies, 2017).

The best explanation does more than warn. It states what was observed, why it matters, which alternatives remain, what protection follows, and what evidence would change the conclusion. Precision can be compelling because it reveals the

real stakes. Accessibility is part of accuracy: alt text, captions, transcripts, reading order, and translations should preserve the same status and uncertainty. A translated artifact should explain a term choice when consciousness, awareness, sentience, feeling, and mind divide differently across languages or legal cultures.

Interactive products should expose uncertainty at the point of use. A user encountering an emotion label, consciousness indicator, or candidate notice should be able to see the version, data limits, review status, and correction route without searching a distant policy page. Public restraint is not timidity. It protects the possible bearer, the audience, and the research program from a story that becomes harder to correct than the science itself.

14.5.3 — Controlled Vocabulary Transfer

Vocabulary is part of governance because words travel farther than protocols. Bearer, stream, unity, candidate, indicator, sentience, consciousness, unknown, precaution, support, validation, and personhood carry different evidentiary burdens. Moving them across disciplines without those limits can turn a useful term into a false verdict. Bearer names the system or process to which a phenomenal history is assigned under a declared model; it does not automatically mean organism, body, legal person, moral person, account, archive, or device.

Stream names the routed continuity of experience proposed by the formulation developed here. It is not a public biography, medical record, memory store, software session, or persistent identifier. A record can continue when a process stops, and a process can continue when the record fails. Unity names a property that must

be specified. It may concern one phenomenal field, common access, coordinated agency, continuity, or ownership across a history. The word should not imply that one universally accepted test already exists.

Candidate means that a system satisfies conditions for further inquiry under a stated framework. It does not mean almost conscious, presumptively a person, or evidence for PTV. Candidate status raises a question and may raise protection; it does not answer the question.

Indicator names an observable feature linked to a theory or evidentiary framework. It can be present in a nonconscious system, absent because measurement failed, or produced by several mechanisms. Multiple indicators can strengthen a case without becoming a direct meter of experience (Butlin et al., 2026). Sentience usually refers to the capacity for felt positive or negative experience, while consciousness may be used more broadly for subjective experience or awareness. Because literatures vary, each use should define the intended meaning.

Unknown means that the available evidence and methods do not settle the prospectively specified question. It should carry a reason such as channel failure, model conflict, construct uncertainty, routing sensitivity, failed transport, unsafe burden, or insufficient data. Unknown is neither zero nor support. Not established is narrower than impossible. Possible is also incomplete unless it is qualified as logical, theoretical, empirically plausible, a realistic welfare concern, or probable. Public language should preserve these differences.

Precaution is a rule for conduct under uncertainty. It can justify monitoring, low-cost protection, pause, or redesign when plausible harm is serious. It does not create personhood, prove sentience, or validate the proposed Law (Birch, 2017; Andrews et al., 2024).

Protection names the safeguard rather than the ontology. An animal can receive improved welfare, an artificial-intelligence project can pause training, and an organoid study can narrow stimulation while bearer status remains unresolved. The reason for action should be stated separately from classification. Support should name the proposition supported. Evidence may support a channel, indicator, measurement bridge, routing rule, welfare concern, or candidate mechanism. It cannot drift upward into support for terminal closure without the missing gates.

Validation should name the method, interpretation, population, and use. An instrument validated for one task may remain invalid for personhood, prognosis, legal status, or whole-stream adjudication. Personhood belongs to moral, legal, philosophical, and political argument and cannot be read directly from one scientific indicator. Soul is not a scientific variable in this framework. Spiritual uses belong to their own traditions and cannot be imported into a measurement claim.

Self-report and testimony-like output remain distinct. Human testimony arises within a person and can still be mistaken; artificial output may imitate testimony without establishing a subject. Shared vocabulary should not erase the different evidentiary routes. A transfer record should name the source discipline, definition, evidence ceiling, local use, prohibited

inference, and return route. Terms borrowed from consciousness science, animal welfare, artificial intelligence, bioethics, medicine, law, or religion return to those owners for specialist depth.

Versioning and correction matter because definitions change. A later framework may narrow an indicator or replace a category. Dependent passages, diagrams, glossaries, interfaces, model labels, and public posts should preserve which definition governed the claim and migrate any correction. The final rule is simple: use the strongest word the evidence licenses and no stronger. Precision protects possible experience from dismissal, protects science from inflation, and allows the series to speak across domains without pretending that one vocabulary has solved the mystery of mind.

Part VI — Windows into a Whole Life

Whole-life claims are easy to admire from a distance and difficult to investigate without distorting the people whose experience gives the question meaning. Part VI enters that difficulty directly. It asks how a complete conscious stream might leave partial traces in dreams, bedside observation, laboratory responses, and records gathered across years, while refusing to mistake any trace for the whole life.

A window can admit light and still conceal most of the room. The same is true of each observation considered here.

This distinction matters because the formulation developed here concerns one qualifying stream over its complete truth-relevant domain. Observation, by contrast, is sparse, selective, reactive, and dependent on instruments, memory, treatment, access, and timing. No accumulation of ordinary measurements automatically becomes the latent object merely because the file is long.

The first window is dreaming. Sleep reminds us that experience can continue outside ordinary public report, while dream recall reminds us that the experience and the later story are different objects. Contemporary work links dreaming to conscious experience during both rapid-eye-movement (REM) and non-rapid-eye-movement (NREM) sleep, memory processing, and state-dependent neural activity, but the field does not supply a discovered balancing mechanism (Nir and Tononi, 2010; Siclari et al., 2017; Wamsley et al., 2024).

Dreams are therefore handled as missing experience rather than hidden prophecy. A nightmare, a pleasant dream, a forgotten dream, or a lucid exchange may expand what researchers know

about the observed history. Each may also introduce recall bias, cultural interpretation, selection, medication effects, and reactivity. The scientific gain lies in making those competing explanations explicit.

The second window lies near the end of life. The Law's boundary makes terminal observation formally important, yet the human setting makes it ethically fragile. Palliative care exists to relieve suffering and support the person and family. Research must fit inside that duty, never reverse it (World Health Organization, 2020; National Institute for Health and Care Excellence, 2015).

A last response is not necessarily a last experience. Sedation, delirium, weakness, sensory loss, medication, family scripts, device removal, and observation shutdown can all change what is visible. A quiet room can contain comfort, fatigue, impaired expression, privacy, or uncertainty. When those layers cannot be bridged without compromising care, the honest scientific result is No adequate estimate, and that result must be allowed to stand.

The third window is the laboratory. A laboratory cannot reproduce childhood, grief, poverty, intimacy, disability, aging, or the end of a conscious stream. It can, however, manipulate narrow conditions, compare alternatives, and ask whether local components behave as proposed. The value of a laboratory result depends on the exact question it was designed to answer, not on the prestige of the apparatus.

The fourth window follows people through time. Repeated observations can reveal trajectories, turning points, adaptation, response shift, treatment change, and patterns of missingness that

a single interview cannot show. Yet a longer file can become a more invasive file. Duration increases both inferential opportunity and the risk of surveillance.

Longitudinal work is most revealing when it treats disappearance from the dataset as information rather than inconvenience. People stop responding because of illness, burden, distrust, relocation, caregiving, lost devices, changing priorities, or death. If those absences are ignored, apparent improvement can be produced by the removal of the people carrying the greatest difficulty (Rubin, 1976; Diggle and Kenward, 1994; Little and Rubin, 2019).

Across all four windows, the same hierarchy applies: observation is not experience, a measured pattern is not a route, and a route is not the Law. Dreams, bedside observation, laboratory tasks, and longitudinal records are valuable only at the evidential level they actually reach. Technical protocols belong in the scientific companion; this volume keeps the public question, ethical limits, and possible failure outcomes visible. None of these chapters invites compulsive self-tracking, terminal watching, induced stress, or lifetime surveillance.

The durable insight is that a whole life may cast many shadows without becoming identical to any one of them. The fairer response to that fact is not despair about measurement. It is disciplined pluralism: combine channels, preserve disagreement, document missingness, compare serious rivals, and stop when the method begins to own the person.

Chapter 15 — Dreams as a Window

Every night, a portion of human experience may occur beyond ordinary public access. A person can be frightened, delighted, ashamed, reunited, hunted, relieved, or simply bewildered while the room remains still. Chapter 15 begins with that private fact and refuses the two easiest mistakes: dismissing dreams as nothing, or elevating them into hidden verdicts.

The formal reason dreams matter is straightforward. The Law concerns felt experience within a qualifying stream, not only the events another person can witness. If dreaming is conscious experience, its valence belongs to the experienced history even when later recall is absent or incomplete. That conclusion follows from the object being discussed, not from any theory that dreams compensate for waking life.

Dream science gives this question real substance. Reports of dreaming occur in both REM and NREM sleep, and neural work has identified state-dependent correlates associated with reported experience and some contents. Those results support careful study of dreaming as experience; they do not provide a universal function or a terminal mechanism (Nir and Tononi, 2010; Siclari et al., 2017).

Memory research adds another layer. Dreams often incorporate fragments of recent learning and remote experience, and dreaming about a task has sometimes been associated with later memory performance. The emerging picture is not literal replay but recombination, updating, and selective reactivation (Wamsley et al., 2010; Wamsley, 2014; Wamsley et al., 2024).

That picture makes dreams scientifically interesting and evidentially difficult at the same time. The experience occurs during sleep. The report usually occurs after waking. Between them lie state change, memory loss, reconstruction, prompting, language, and interpretation. A later narrative may be meaningful without reproducing the original experience exactly.

Recurrent themes are handled with the same restraint. Longstanding content-analysis traditions show that waking concerns, relationships, threats, and habits can recur in dreams. Recurrence can reveal continuity between day and night, but it can also reflect base rates, salience, trauma, cultural scripts, and the researcher's coding choices (Hall and Van de Castle, 1966; Cartwright et al., 2006; Domhoff, 2017).

Lucid dreaming and real-time communication offer a different methodological opportunity. Some dreamers can signal or respond during verified REM sleep, reducing the delay between experience and report. The result is a new observation channel, not a privileged route to inner truth, and it applies to limited participants under specific conditions (Voss et al., 2009; Konkoly et al., 2021).

Competing hypotheses remain central. Dream affect may preserve waking concerns, amplify them, reorganize them, or appear unrelated once recall and selection are modeled. Medication, depression, trauma, sleep fragmentation, and awakening schedule can influence both what is experienced and what becomes available for analysis.

This is why the Chapter rejects the old counterweight framing. An apparent inversion between waking and dream affect would be scientifically nonunique. Regression to the mean, selective recall of extremes, flexible valence coding, and ordinary emotional regulation could all create the same surface pattern. The claim must be designed so that those rivals can win.

Safe research begins with sampling. REM timing, NREM awakenings, spontaneous morning recall, diaries, interviews, and in-dream signals do not produce interchangeable data. Each method has a different denominator and different effects on sleep, memory, and burden.

Coding is equally consequential. A dream image does not carry one universal valence. A chase can be terror, exhilaration, play, or mastery. A deceased relative can be comforting, painful, sacred, ordinary, or culturally expected. Original-language reports and local interpretation must be preserved before categories are compared.

Privacy is not an appendix to dream research. Reports may expose trauma, sexuality, family conflict, religion, work, health, or fears that the person has never disclosed elsewhere. The person retains authority over whether that material is shared, retained, linked, or quoted.

Tracking can also change the phenomenon. Repeated awakenings, diary demands, and concentrated attention may increase recall, disrupt sleep, or intensify rumination. The study must be able to stop when participation begins to make the night harder rather than more understandable.

15.1 — Why Dreams Matter

Dreams matter because a person can undergo fear, pleasure, shame, longing, relief, or confusion while almost none of that experience is available to an outside observer. A theory about the felt history of a conscious stream cannot limit its attention to what is reported while awake. The night may contain experience that later biography omits.

15.1.1 — Dreams as Felt Experience

The formal point is modest but important. Where a dream contains conscious experience, that interval may belong to the felt history of the same stream. The person is afraid, relieved, ashamed, loved, hunted, reunited, powerful, powerless, lost, or calm while the dream is occurring. The later story may be fragmentary, yet the felt interval is the reason dreaming matters to a theory of experience (Siclari et al., 2017).

Dreams also protect the model from externalism. If fairness were judged only by visible events, the private night would disappear. Yet many people know that a dream can alter the morning, leave residues in the body, reopen grief, soften loneliness, create anxiety, or briefly return a lost person. Felt experience has already moved before public life begins again (Siclari et al., 2017).

Sleep research makes this distinction testable through repeated awakenings. Investigators can ask a person what, if anything, was present immediately before the alarm, then relate the report to the preceding sleep physiology. The method does not observe the dream directly. It compares a later report with a recorded interval and therefore depends on timing, wording, alertness, and the

person's capacity to recover a trace. That architecture is useful precisely because it shows where the inference begins and where it remains incomplete (Siclari et al., 2017; Tononi et al., 2024).

A report of no experience is also not identical to dreamless sleep. The person may have had no conscious content, may have lost the trace during awakening, may remember only a feeling without imagery, or may be unable to translate the experience into words. Some studies distinguish recalled dream content, a report of experience without recall of its content ("white dreams"), and a report of no experience. A missing report is a separate outcome. The public lesson is simple: absence from the report is evidence about the observation channel, not automatic evidence of absence from experience (Windt, 2015; Siclari et al., 2017).

The converse problem matters too. A long, vivid report can create an impression of rich continuous experience even when the remembered sequence was assembled from fragments. Report length, confidence, sensory detail, and emotional residue are different features. A protocol that combines them into one dream score may gain convenience while losing meaning. Better work keeps them separate long enough to learn whether they move together, which ones transport across nights, and which ones are mostly properties of recall (Wamsley et al., 2024; Windt, 2015).

Dreaming also reveals why consciousness cannot be reduced to outward connection. During sleep a person may be largely disconnected from the environment and unable to act, yet later describe a world containing perception, emotion, selfhood, and time. That makes dreaming scientifically valuable for the study of consciousness, but it does not identify a hidden ledger or prove a

terminal rule. It widens the possible experienced domain while leaving stream unity, measurement, and cumulative interpretation unresolved (Tononi et al., 2024).

A cross-window relation becomes scientifically interesting when dream valence appears systematically related to preceding waking affect after ordinary rivals are modeled. A difficult day may be followed by dreams that soften, invert, rehearse, discharge, or reorganize affect. A day of intense pleasure may be followed by anxiety, loss, contrast, or ordinary emotional resetting. These patterns, if real, would still require careful separation from sleep science rivals (Siclari et al., 2017).

For dreams as felt experience, a credible design should separate the occurrence of experience from recall, narrative reconstruction, and the conditions under which a report was elicited. These choices fix the denominator, comparison class, and treatment of missing observations. A persuasive result must still be shown to arise from the experience rather than from the observation process. The scientific companion must turn those distinctions into a prospectively specified sampling, coding, missingness, and verdict protocol (Siclari et al., 2017; Windt, 2015).

A person wakes with a pounding heart and a durable residue of fear but can recover only one image from the night. The fear belongs to the observed aftermath; the missing narrative remains uncertain. A careful study records the residue, the recall conditions, and the competing explanations without pretending that the surviving image is a complete transcript of the dream.

In a 2021 study, a sleeping participant heard a simple question from the laboratory: what is eight minus six? He replied with two left-right eye movements, the agreed signal for two, while physiological recordings placed him in REM sleep. Asked again, he answered correctly again. Across four laboratories, the study recorded 29 correct replies from six of 36 participants (Konkoly et al., 2021).

That narrow exchange opens a new route for inquiry. Researchers need not always wait until waking to ask a question. A response during sleep creates a second record that can be compared with what the dreamer later remembers. The intriguing opportunity is to study where those records agree, where they diverge, and how the transition to waking reshapes the story.

For the whole-life question, the contribution is limited but real. Dreaming warns against treating a waking autobiography as an exhaustive phenomenal record. It also supplies a natural example of why a latent object may exceed every available observation channel. The lesson travels to terminal and longitudinal research: what is not reported may still have occurred, but the gap cannot be filled by theoretical preference.

15.1.2 — Offline Processing

Offline processing is the strongest ordinary frame for dream relevance. Sleep may reactivate recent learning, integrate remote memories, sort salience, support emotional recalibration, simulate threats, regulate synapses and bodily states, and create novel combinations. These functions may differ across tasks and sleep

stages, and none requires the Law of Fairness to be true (Hobson and McCarley, 1977; Wamsley, 2014; Wamsley et al., 2024).

That explanatory abundance is a scientific advantage. If established sleep models account for a pattern, they receive the credit and the proposed Law is spared work it did not earn. Only residual, prospectively specified relations that survive capable ordinary rivals can become informative about a Law-related route (Wamsley, 2014).

Dreams can supply auxiliary evidence only when the observation path is prospectively specified, consented, coded, and tested against rivals. Unreported or forgotten experience cannot be imagined after the result merely to protect the theory. The same hidden interval that makes dreaming scientifically valuable also makes post hoc rescue especially easy, so the rules for inference must be fixed before the decisive data are examined (Wamsley et al., 2024).

An incorporation finding becomes interpretable only after the design specifies what could enter the dream and how chance resemblance will be ruled out. Waking exposure, baseline performance, awakening schedule, coding rules, blind adjudication, and negative controls should be fixed before the decisive result is known. A student may dream fragments of a practiced task and improve the next day, yet salience, prior learning, and sleep-dependent consolidation can produce both observations without the dream narrative causing the improvement (Wamsley, 2014; Wamsley et al., 2024).

Causal interpretation requires more than a matching story. Within-person comparisons, exposure controls, delayed outcomes, and analyses that test whether incorporation predicts performance beyond prior learning and salience can distinguish a useful association from a narrative coincidence. Until those bridges succeed, dream content and next-day improvement remain linked observations rather than evidence that the dream caused the gain. The question stays empirical and bounded, never a reading of anyone's hidden history (Wamsley, 2014; Wamsley et al., 2024).

The Law-related implication remains modest. Offline processing could contribute to local regulation of affect, memory, or choice, but local regulation does not guarantee an exact whole-stream boundary. A useful sleep mechanism may explain part of the path while leaving the terminal claim untouched or unnecessary.

15.1.3 — Scientific Research Status

Dream science begins with four events that ordinary speech often compresses into one: the experience in sleep, the awakening that interrupts it, the memory trace available after waking, and the report given to an investigator. Recall can omit, reorder, compress, and narratively repair experience, so a report remains evidence about remembered dream experience under a stated elicitation procedure rather than an unedited transcript of the sleeping mind (Windt, 2015; Wamsley et al., 2024).

The present research question is which combination of sampling, physiology, recall, and report permits a defensible inference about

what was experienced, not which sleep stage owns dreaming (Nir and Tononi, 2010; Siclari et al., 2017; Tononi et al., 2024).

A high-density electroencephalography (EEG) study included seventy-five expert-confirmed non-rapid-eye-movement parasomnia episodes, fifty-seven of which were followed by an interview. Among those interviewed episodes, participants reported some conscious experience after eighty-one percent, could clearly recall content after fifty-six percent, and reported no experience after nineteen percent; the neural patterns associated with experience and with later recall were partly different (Cataldi et al., 2024). The result does not settle what occurred when no interview was obtained. It shows why outward action, conscious experience, memory encoding, and later testimony must remain separate layers.

Shared resources can widen comparison without solving identification. The initial DREAM database release combined twenty datasets, 505 participants, and 2,643 awakenings, linking sleep electrophysiology to a standardized classification of reported experience (Wong et al., 2025). That scale supports reanalysis and cross-protocol questions, but every classifier remains tied to contributing studies' labels, sampling schemes, and reportability. More data can reveal instability hidden by a small study; it cannot transform a proxy into experience.

A newer high-density EEG study of non-rapid-eye-movement parasomnia episodes found activity unlike ordinary wakefulness or sleep. Across seventy analyzable episodes from twenty adults, delta and theta power and the beta-to-delta relation were intermediate between wake and sleep (Valomon et al., 2026). This

mixed-state physiology helps explain why sleep-wake boundaries resist simple labels, but cannot identify conscious content, valence, or a stream-wide law.

Before decisive dream data are opened, freeze the report taxonomy, awakening schedule, exclusions, affect coding, principal contrasts, and failure conditions under the terminology defined in Section 10.4.1. Held-out replication should then test whether the result survives the fixed dream protocol (Nosek et al., 2018; Wong et al., 2025).

Privacy is part of validity. Dream reports may contain sexual material, trauma memories, religious imagery, interpersonal accusations, or details about nonconsenting people. Minimization, restricted access, de-identification, participant review where feasible, and limits on secondary use protect dignity and the scientific conditions under which honest reporting remains possible (CIOMS, 2016; Nissenbaum, 2010).

The scientific status is therefore exact but modest. Dreams are a legitimate window into possible experience that ordinary waking records miss. They are not prophecy, counterweights, private diagnoses, or proof that any visible suffering has been answered elsewhere in the stream.

Imagine two invented archives of dream reports. In the first, frightening experiences become more common during sleep. In the second, the experiences stay the same, but frightening dreams become more likely to be remembered and reported. Arrange the examples so that both archives contain the same rise in reported

fear. A reader looking only at that rise could not tell which story produced it.

That is a useful rehearsal for research. Can the planned measurements tell the two invented worlds apart when their histories are known? If they can, the method has cleared one obstacle before facing real uncertainty. If they cannot, a larger collection of reports alone will not answer the question. The exciting target is the experience that happened at night, and the challenge is to distinguish it from the memory that survived until morning.

A pooled dream archive is scientifically useful only when it preserves the denominator and event linkage that produced each report. Every eligible awakening, no-experience report, fragmentary recall, refusal, exclusion, sleep stage, clock time, medication state, prompt, response delay, verbatim account, coded affect, and relevant neural interval should remain traceable. Otherwise scale can magnify selection and timing error while making a recurring report pattern look more universal than it is (Wong et al., 2025).

The evidentiary sequence should therefore be staged. First describe the reports and missingness under a declared protocol. Then freeze a predictive model and test it on held-out events. Only after successful model recovery and cross-site transport should the field ask whether the pattern bears on a route-level claim. This order does not weaken dream science. It protects its most valuable contribution by ensuring that an impressive classifier is not mistaken for direct access to sleeping experience.

15.2 — Dream Affect, Recall, and Missingness

A dream report arrives after a transition. The sleeper wakes, or is awakened, reorients to the room, searches memory, selects words, and answers a question. Every step can change what becomes available. Dream affect is therefore studied through a chain of recovery rather than through direct access to the sleeping experience.

15.2.1 — Recurrent Themes

Recurrent dream themes are among the first things people notice. Falling, being chased, failing a test, arriving unprepared, losing teeth, being unable to speak, searching for someone, returning to childhood places, or encountering the dead may appear across many lives.

Recurrent themes may reflect common human situations, repeated waking burden, and cultural patterns. Evaluation, pursuit, loss, bodily change, shame, and uncertainty recur in dreams partly because bodies and social worlds repeatedly contain them. A person under chronic evaluation may dream of unpreparedness; a grieving person may dream of searching. These relations can be studied without treating similarity as evidence of one universal hidden mechanism (Hall and Van de Castle, 1966).

Threat-simulation theory proposes that dreams rehearse responses to danger (Revonsuo, 2000). A dream may simulate pursuit, attack, loss, failure, humiliation, or danger without exposing the body to that event. This is a proposed function of dreaming, distinct from the continuity hypothesis that dreams

reflect waking activity (Schredl and Hofmann, 2003). Neither hypothesis requires the Law of Fairness.

Positive and relief dreams can be genuinely restorative. Reunion, safety, forgiveness, success, homecoming, escape, or abundance may leave a person calmer or briefly renewed. Yet relief can arise through desire, memory, attachment, fantasy, unmet need, or ordinary affective imagination. A dream of reunion may comfort grief without proving that grief has been balanced, and a dream of success may ease frustration without closing a cumulative history (Cartwright et al., 2006).

But relief dreams become dangerous when used against sufferers. A person who dreams of comfort must not be told that their waking injury is therefore resolved. A grieving person who dreams of the dead must not be told what the dream proves. A survivor who experiences a moment of dream safety must not be pressured toward meaning or gratitude (Domhoff, 2017).

The same recurring theme can arise through very different pathways. A chase dream may follow immediate danger, chronic social evaluation, trauma-related vigilance, entertainment exposure, or an ordinary cultural script. Theme counts therefore need waking-context data and within-person history before they can support even a modest interpretation (Domhoff, 2017).

Themes should be coded with transparent dictionaries, inter-rater checks, cultural review, and a prospectively specified loss condition. Base rates, waking exposure, trauma, media, salience, and narrative convention can all create recurrence. A person who repeatedly dreams of arriving late may be expressing chronic time

pressure rather than a compensatory process. The codebook must therefore preserve context and uncertainty instead of turning a familiar image into a universal symbol (Domhoff, 2017).

A recurrent-theme claim should also state the size, population, and transport it requires. If a prospectively specified codebook and adequately informative sample do not recover the expected relation, the claim narrows; if cultural base rates, waking exposure, or a rival model predict the pattern equally well, the result remains ambiguous. A relation that appears in one language, clinic, or sleep window but fails elsewhere remains local. That loss condition prevents recurrence from becoming an unfalsifiable story without pretending that one dream result settles the whole Law.

Recurrent themes may still have clinical and human value. They can guide conversation, reveal concerns a person wishes to discuss, or help evaluate a nightmare treatment (Krakow et al., 2001). That value does not require the theme to be a universal signal. Meaning can remain local, voluntary, and useful without being promoted into evidence for a lifetime law.

15.2.2 — Memory Reconstruction

Memory reconstruction is central to dream evidence because most data arrive through waking report. The dreamer wakes, retains fragments, gives language to images, arranges sequence, fills gaps, and sometimes interprets before recording. The final narrative may be sincere and useful while differing from the experience that gave rise to it (Schacter, 1999; Windt, 2015).

Recall is also uneven across people and nights. Interest in dreams, mind wandering, sleep continuity, age, medication, interference after waking, and awakening time all affect whether a report appears and how much survives. A remembered-dream dataset is therefore filtered by physiology, attention, memory, motivation, and opportunity; that filter belongs inside the model rather than being dismissed as noise (Aserinsky and Kleitman, 1953; Siclari et al., 2017).

The interval between awakening and report changes what can be known. A voice note made in darkness may preserve sensory fragments that disappear by breakfast, while a later interview may recover a clearer story only because the dreamer has imposed order on those fragments. Neither report should automatically be treated as the correct one. They answer different questions about immediate residue, later memory, and narrative integration, and the protocol should keep those targets separate (Wamsley et al., 2024).

Prompting can also change the object. Asking whether a dream was frightening, meaningful, or connected to yesterday directs attention toward particular features and can encourage completion of gaps. Free recall, structured checklists, confidence ratings, and interviewer follow-up may each be useful, but they are not interchangeable. A careful study records which form produced each item and tests whether its central result survives when the prompt changes (Nielsen and Levin, 2007; Siclari et al., 2017).

A person from one tradition may treat a dream of an ancestor as sacred contact. Another may treat it as memory. Another may

treat it as random imagery. Another may not report it at all because it belongs to family or ritual privacy. A research program that flattens these differences will misread both valence and meaning (Cartwright et al., 2006).

Cross-cultural dream work must therefore be collaborative. It should involve language review, cultural expertise, participant-defined meaning categories where appropriate, and caution against imposing one tradition's dream taxonomy on another. Translation is not enough. A translated dream word may not carry the same ontology, shame, sacredness, or privacy boundary (Windt, 2015).

Record the delay from awakening to report, distinguish free recall from prompted recall, and preserve the original report before later coding or interpretation. A morning diary entry may become more elaborate by evening as the dreamer connects it to a recent argument. That growth does not imply deception; it shows why the remembered narrative and the earlier experience must remain separate objects in the measurement model.

Keeping those records separate lets the protocol test which features are stable, reconstructed, or changed by the elicitation procedure (Windt, 2015; Wamsley et al., 2024).

The Law-related question is affected in a precise way. If the available affect rating is partly determined by reconstruction, then an apparent relation between waking events and dream valence may be a relation between waking events and later memory. That rival is not a minor correction. It changes the object being

modeled and therefore the status of every inference drawn from it.

15.2.3 — Observation Is Not Signature

Observation can still make a positive contribution without becoming a signature. A report may establish that a particular person remembered fear after a particular awakening, or that a prospectively specified relation recurred in a sample. The scientific question is whether that relation exceeds what recall, salience, sleep stage, medication, culture, and selection already predict.

Nightmares, wish dreams, cultural patterns, and morning mood can all carry information without distinguishing PTV. Sleep science, memory science, threat models, trauma models, and social interpretation can reproduce many of the same observations, so a study must name the pattern those rivals should not match under common rules (Nielsen and Levin, 2007).

Vividness, recurrence, and emotional intensity are not unique fingerprints. The same observation can emerge from memory consolidation, nightmare disorder, trauma, stress, medication, sleep fragmentation, cultural expectation, or selective recall. A Law-related signature would need a prospectively specified relation to waking burden, horizon, or later adjustment that transports and outperforms those capable rivals (Domhoff, 2017).

A candidate dream signature must be specific enough to fail. It should identify, before outcome inspection, the waking exposure, lag, dream feature, sleep-stage or timing condition, comparison class, and rival set. Stable participant traits and within-person

deviations should remain distinct so a cross-person correlation is not mistaken for a dynamic relation within one stream. Classic dream observations can then receive their proper weight. They show that dream life matters and that sleeping experience may widen the observable history, while also showing how recurrence, memory reconstruction, trauma, culture, salience, and selective recall can manufacture an attractive story. The next question is not whether the story is moving, but whether it risks being wrong.

Model checks should extend beyond the headline association. A fitted account should recover report and no-report rates, sleep-stage differences, recall confidence, and the distribution of affect. A model that predicts one chosen outcome while failing the rest of the record has not produced a signature.

The strongest conclusion may therefore remain channel-specific. A protocol may predict when a person reports dreaming without identifying the experience itself or its place in a lifetime total. That is still useful science. It becomes weaker, not stronger, when presented as direct access to the stream.

15.3 — Competing Hypotheses, Not Counterweights

A tempting story says that hard days should produce pleasant dreams, easy days should produce distressing dreams, or unbalanced waking experience should be corrected at night. Such inversion stories are emotionally memorable because they resemble a hidden accounting system. They are also scientifically weak unless their predictions differ from ordinary explanations.

Adaptation, mood-congruent recall, medication, sleep disruption, selective memory, regression to the mean, and the overrepresentation of vivid dreams can all create apparent compensation. A negative relation between waking and dream affect may disappear when neutral nights, failed awakenings, and forgotten dreams are returned to the denominator (Simmons et al., 2011; Gelman and Loken, 2013).

The comparison must be decided before the pattern is known. Lag, coding rules, exclusion criteria, equivalence bands, missing-report treatment, and rival models should be frozen in advance. A route that survives only after favorable nights or favorable definitions are selected has not survived a test (Lakens, 2017; Nosek et al., 2018).

15.3.1 — Dream Valence May Extend the Observed History

Tough-day inversion is the candidate pattern in which a day with negative residual valence is followed by dream valence that is less negative, more restorative, more resolving, or opposite in affective direction than expected from ordinary carryover alone. The phrase does not mean every hard day earns a good dream.

The pattern would need to be residualized. A difficult waking period is not merely a day the participant describes as bad. Its observable components may include sleep loss, pain, conflict, threat, grief, work strain, social exposure, and acute stress, each with a different relation to later recall and dream content. The model would need to separate ordinary fatigue, trauma activation, baseline mood, and measurement timing from the specific waking burden under study.

The dream side would also require caution. A dream might be positive in theme but anxious in body. It might contain reunion and dread. It might begin as threat and end in relief. It might be forgotten except for one image. Coding must therefore handle mixed valence, uncertainty, intensity, and confidence rather than forcing one clean label (Cartwright et al., 1984).

The third safeguard is asymmetry caution. Researchers must not look harder for downward dreams after good days than upward dreams after bad days merely because balance language attracts them. The analysis plan must specify positive, negative, and neutral residuals before seeing the data. Selective attention can manufacture symmetry (Shiffman et al., 2008).

The observed history can be widened by including remembered dream affect, but widening is not completion. Reports are sampled, recall-dependent, culturally mediated, and missing for many sleep intervals. The contribution is a better observation model and a clearer account of uncertainty, not a second hidden ledger.

Individual differences also create public communication risk. If one person reports a striking pattern, the public may generalize it. If one group shows an effect, the public may turn it into a universal claim. A fair book must teach readers that personal resonance is not population evidence, and population evidence is not individual destiny (Cartwright et al., 2006).

Interpretation begins with a prospectively specified failure rule. If the expected dream-waking relation does not appear, the candidate route narrows; if ordinary sleep, memory, medication, or selection models reproduce the finding with the same information and flexibility, the result remains ambiguous. A local loss should be reported as a local loss rather than absorbed into a broader story.

The waking predictor should be decomposed rather than treated as one undifferentiated score. Researchers might distinguish interpersonal conflict, workload, sleep opportunity, pain, threat, bereavement, and social evaluation, because each may enter dream content through different memory and affective pathways (Cartwright et al., 2006).

Dream affect can extend the observable history when waking reports omit what was felt during sleep. Studies linking dream tone with depressive state, remission, and next-day mood show that the night may contribute information not contained in a daytime interview. They also show why later recall and morning residue must remain separate: the dream, the report, and the effect after waking are related observations, not one interchangeable measure (Cartwright et al., 1984; Cartwright et al., 2006).

The formulation's terminal relation is therefore a boundary hypothesis, not a conservation-of-feeling slogan. Even if exact closure were supported, the mechanism could be global, state-coupled, route-specific, or still unknown. The public copy should distinguish the conditional net remainder from a fixed gross quota, moral debt, a promised reward, or a quantity institutions could administer.

15.3.2 — Adaptation, Recall, and Selection as Rivals

Adaptation, memory, medication, and selection can each create apparent dream-waking relations. A difficult day may be remembered differently after sleep; medications can alter sleep architecture, recall, and affect; and people who awaken or choose to report are not a random sample of all dream intervals. These rivals should receive the same timing and contextual information as the candidate route (Goldstein and Walker, 2014; Little and Rubin, 2019).

Medication and treatment can alter dream frequency, vividness, emotional tone, awakening, and recall at the same time. Trauma-related sleep disruption can also change both what is experienced and which episodes become reportable. A model that ignores those pathways may mistake a treatment or observation effect for persistence, contrast, or compensation.

A research estimate of a neutral waking period should be defined relative to the participant's prospectively specified baseline, not treated as an emotionally empty day. Ordinary irritation, modest pleasure, fatigue, and routine life may all be present. This

distinction matters because an apparent dream inversion can be manufactured by a poorly identified waking origin.

Medication is not a single adjustment variable. Drug class, dose, timing, indication, recent change, withdrawal, and interaction with sleep continuity can alter dream recall and affect through different pathways. A model that treats medication as one yes-or-no covariate may leave precisely the confounding structure it claims to control (Goldstein and Walker, 2014).

These rivals must be capable of winning. Adequate data that fail to recover the prospectively specified relation count against the route, and equal or better performance by a prospectively specified rival blocks a favorable verdict. The wider hypothesis may remain open, but this explanation does not receive unlimited adjustments.

Credible comparison requires prospective records of treatment change and sleep disruption, appropriate stratification, and explicit treatment of missing reports rather than coding them as neutral dreams. Confirmatory implementation and verdict rules are developed in the scientific companion (Nielsen and Levin, 2007; Cartwright et al., 1984).

After a medication change, vivid nightmares may rise at the same time that awakenings become more frequent. The underlying experience and the probability of reporting it have both changed. A useful analysis separates those pathways and asks whether sleep stage, medication, mood, recall, or selection explains the pattern better. If the available data cannot distinguish them,

ambiguity is the result rather than a reason to choose the most dramatic account.

Selection operates at several stages. People who volunteer for dream studies may remember dreams more often; participants may record striking dreams and omit ordinary ones; researchers may emphasize nights with clear narratives; journals may prefer unusual associations. Each filter can make the observed archive more symmetrical or dramatic than the underlying experience.

A fair comparison gives these rivals their own predictions. Adaptation may depend on prior extremity, memory models on recall conditions, medication models on exposure, and selection models on reporting probability. The Law-related route should outperform that structured competition rather than receiving credit whenever any movement toward a middle appears.

15.3.3 — No Inversion Pattern Is Unique Evidence

No inversion pattern is unique evidence because opposite-looking dream and waking reports can arise through contrast, salience, regression to the mean, recall selection, cultural expectation, medication, or ordinary emotion regulation. The strongest test is not whether an inversion can be found, but whether its prospectively specified timing and dependencies survive those rivals.

A careful analysis should prospectively specify sleep-quality variables, medication classes where ethically and lawfully reported, clinical context, and sensitivity analyses. It should also allow participants not to disclose details beyond what the study

truly needs. Data minimization still applies. The desire to control confounds cannot become a hunger for medical possession.

A strong rival model may predict that negative waking events lead to emotionally intense dreams because salient material is preferentially processed. Another may predict that REM-related mechanisms reduce or alter emotional reactivity in some contexts. Another may predict that nightmares reflect failed regulation rather than successful compensatory interpretation. Another may predict cultural and reporting effects that swamp biological pattern (Gelman and Loken, 2013).

A capable comparison should preserve the same waking history, sleep timing, medication information, missingness, and reporting conditions for every model. The candidate route cannot receive a richer record while ordinary sleep, memory, selection, and cultural models are asked to compete with crude summaries. Equal information and comparable flexibility are part of the test.

Wishful imagery and coding artifacts are additional rivals; the predicted inversion and comparator flexibility must be fixed prospectively (Lakens, 2017; Simmons et al., 2011).

Trauma-informed design should begin with a burden decision, not a data-collection menu. Investigators should ask whether the question can be answered without narrative content, whether ordinary care already supplies safer evidence, and whether a subgroup should be excluded because the act of tracking is likely to worsen symptoms. When the answer remains doubtful, declining to collect the data is a scientifically and ethically

legitimate outcome (CIOMS, 2016; World Medical Association, 2024).

Two people may show the same apparent inversion for different reasons. One slept after acute social stress and remembered a comforting reunion; another woke during a fragmented night after medication change and reported a distressing dream after a pleasant day. Treating both as one compensatory pattern would hide the causal structure the research needs to distinguish.

15.4 — Safe Sampling, Coding, and Privacy

Dream research alters the night it observes. Alarms interrupt sleep, repeated questioning trains attention toward dreams, diaries can intensify rumination, and detailed retelling may reopen traumatic material. Sampling design therefore changes both the data and the burden carried by the participant.

Timing, coding, and privacy are parts of the observation model: sampling must preserve the declared sleep window, categories must survive cultural and language review, and collection must remain minimizable, stoppable, and proportionate to burden.

15.4.1 — REM Timing

REM timing matters because many vivid reports follow REM awakenings, but dreaming is not owned by REM. A design that treats REM reports as all dreams may miss non-REM experience, confuse sleep-stage effects with dream function, or overstate the specificity of a pattern. The first task is therefore to identify the sleep window the claim actually concerns.

Wake-up bias also differs by person. A light sleeper may report more dreams than a deep sleeper. A parent, caregiver, shift worker, patient, or person in unsafe housing may wake more often. A person with insomnia may generate more reports because they are awake more, not because dream life is more affectively active. Social and material burden can enter the dream dataset through awakening itself (Nir and Tononi, 2010).

Stage and time of night should therefore be treated as two coordinates rather than one label. An early-night NREM

awakening, a late-night NREM awakening, an early REM awakening, and a final-morning REM awakening differ in sleep pressure, recent memory, report opportunity, and typical content. Pooling them can manufacture a single “dream effect” from several distinct sampling regimes. A credible analysis preserves those cells or explains why they can be combined (Aserinsky and Kleitman, 1953; Siclari et al., 2017).

The denominator is the number of eligible awakenings or sleep intervals, not merely the number of remembered dreams. A study that analyzes only reports with enough words for coding may silently remove no-recall, fragmentary, and low-confidence outcomes. That can make affect appear more intense or coherent than it was in the full sample. Reporting the flow from scheduled awakening to usable report is therefore as important in dream research as participant flow is in a clinical study.

Sampling density should be adaptive to burden. Repeated awakenings may be tolerable for one healthy volunteer and disproportionate for a person with insomnia, trauma, caregiving duties, chronic pain, or unstable housing. The protocol should set nightly and cumulative limits, allow immediate reduction in sampling, and treat sleep deterioration as an adverse effect rather than a useful increase in observations. Scientific value does not excuse a design that damages the window it is trying to study (World Medical Association, 2024; CIOMS, 2016).

The first privacy rule is that content collection must be minimized. If affect rating is enough, the study should not collect narrative content. If broad categories are enough, it should not collect names, locations, sexual detail, family identifiers, or

traumatic scenes. If content is needed, it should be justified by the hypothesis and protected by stronger governance (Nir and Tononi, 2010).

Youth dream logs also risk family coercion. A parent may read, ask about, interpret, punish, worry over, or share a child's dreams. A school may treat dream reports as emotional information. A religious community may overinterpret. The child may learn that private sleep belongs to authority. That would violate the privacy boundary set out in Chapter 12.

Operationally, timing becomes part of the observation record. Clock time, sleep stage, awakening method, prior awakenings, sleep continuity, medication, and response delay all affect report probability, vividness, emotional intensity, and memory. Preserving those variables prevents a late-night sampling effect from being mistaken for a universal dream relation (Nir and Tononi, 2010; Siclari et al., 2017).

Sampling should distribute awakenings across the declared sleep windows or state explicitly which window the claim concerns.

The first night in a laboratory may differ from later nights because the room, equipment, anticipation, and repeated awakenings are unfamiliar. Habituation nights can reduce some effects but increase cost and selection. Researchers should state whether the target is ordinary home sleep, adaptation to the laboratory, or the specific monitored condition.

Timing also affects affective interpretation. A dream sampled after a long REM period may be more elaborate and easier to rate than a fragment recovered from NREM sleep. Comparing their

valence as though the reports had equal recovery probability can create a stage-by-measurement artifact.

15.4.2 — Cross-Cultural Coding

Cross-cultural coding begins with two risks: interpretive imperialism and false invariance. Researchers can mistake their own categories for universal features of dreaming, then read group differences as differences in experience when the real difference lies in language, disclosure, symbolism, or the instrument itself. No culture is entitled to own another culture's night, and no apparently reliable scale should travel before its meaning and performance have been tested in the new setting (Domhoff, 2017).

The dreamer's own rating is indispensable, but it is not an infallible oracle. A participant may be uncertain, ashamed, influenced by the interviewer, or unable to put the experience into the offered categories. A blinded coder may notice affective cues that the participant does not name, yet may also misunderstand idiom, trauma, metaphor, or sacred meaning. A defensible analysis compares participant rating, coded features, disagreement, and uncertainty rather than appointing one sovereign interpreter (Windt, 2015).

Valence must also be separated from moral or theological judgment. Fear, relief, tenderness, shame, anger, confusion, awe, or peace may be coded as reported affect without deciding what the dream means, whether a symbol is good or bad, or what the dreamer ought to believe. Coder training, blind conditions,

reliability checks, and explicit disagreement categories protect dignity because they limit how much authority a code can acquire.

Translation is part of measurement, not clerical preparation. Words for fear, awe, bodily heaviness, peace, shame, obligation, or spiritual presence may divide experience differently across languages. Back-translation can reveal some errors, but it cannot prove that two prompts evoke the same judgment. Interviewer expectations, politeness norms, willingness to discuss sexuality or religion, and beliefs about dreams all influence what reaches the record (Hall and Van de Castle, 1966; Domhoff, 2017).

A strong codebook therefore has shared and local layers. Shared categories support limited comparison only when their meaning, reliability, and disclosure burden transport. Local categories preserve distinctions that a universal grid would erase, including culturally specific forms of kinship, sacred presence, social exposure, obligation, protection, or threat. The scientifically honest result may be a partially common map rather than one global vocabulary.

The same image can make the problem concrete. An animal, ancestor, examination, erotic scene, or sacred figure may be coded as threat in one scheme and protection, duty, blessing, or ordinary familiarity in another. The disagreement is not noise to be averaged away. It identifies a bridge that the study must explain before any comparison can carry a larger theory (Schredl and Hofmann, 2003).

Design should therefore involve fluent translators, cultural reviewers, and participants in the construction and testing of

categories. Original-language reports should be preserved under appropriate access controls, and transport should be tested before groups are ranked or pooled.

Cross-cultural validity never justifies added pressure or burden. If translation, coding, disclosure, or repeated awakening increases distress, the protocol should narrow or stop; subsection 15.4.3 makes that exit operational.

Possible refusal or burden may appear as silence, missed prompts, indirect language, or deference rather than an explicit request. Local safeguarding review should define how those signals trigger a private check-in that does not pressure the participant to explain.

Report stopping and missingness by language, site, and study arm; lower disclosure may reflect privacy, translation difficulty, or safer refusal, not less dream experience.

15.4.3 — Stop If Tracking Increases Distress

A participant may discover that repeated dream recording intensifies rumination, fear of sleep, or trauma recall. That change is both an ethical signal and a measurement effect. The correct response is to reduce or stop collection, preserve earned compensation and care, and record that the observation process altered the target rather than interpreting greater distress as more informative data.

When burden rises, the response is protection rather than interpretation. The participant should be able to pause immediately, stop without explanation, receive debriefing, and be referred to appropriate clinical or crisis support when needed.

Reports should disclose how often stopping occurred, whether it differed by symptom history or study arm, and how the resulting missingness changed the estimand instead of treating stopped nights as unusable noise (Nielsen and Levin, 2007; World Medical Association, 2024).

Privacy protection begins before analysis. Dream reports can expose sexuality, trauma, spiritual meaning, family conflict, or allegations about people who never consented. Collection should be minimized, identifiers separated, access tiered, quotations separately authorized, and model training prohibited unless explicitly governed for that purpose (CIOMS, 2016; Nissenbaum, 2010).

General public materials should not prescribe dream logging or compelled disclosure for minors. Any youth-facing research or educational use requires an age-appropriate purpose, independent safeguarding review, guardian permission and child assent where applicable, easy nonparticipation, no group pressure to reveal content, and a clear separation from any effort to prove the Law. The safer default for general series materials is not to include such exercises (CIOMS, 2016; World Medical Association, 2024).

Dream tracking is not a general practice of fair living. Ordinary private life requires no project record, and the privacy boundary developed here applies directly. A person can live seriously, reflect deeply, care well, and reduce hidden burden without recording a single dream. The same boundary protects the beauty of unmeasured sleep: a person may dream, forget, wonder, laugh, mourn, pray, or ignore the night without converting it into project material.

Stopping should be easy, private, and free of persuasion. Participants need accessible ways to pause prompts, skip an awakening, withdraw a report, or leave the study without losing care, standing, or compensation already earned. A technically available exit that requires explanation during distress is not a practical right.

Dream methods also need a reactivity plan. Repeated awakenings, scheduled reports, wearable prompts, and requests for vivid detail can change sleep, recall, attention, and emotional salience. The protocol should measure those changes, compare lower-burden alternatives, and distinguish a naturally occurring pattern from one strengthened by observation. If the method becomes part of the phenomenon, that interaction belongs in the result rather than being hidden as inconvenience.

Public correction matters because dream claims travel easily. A striking account can become a headline, spiritual sign, diagnostic label, or automated interpretation long after the underlying analysis has narrowed. Versioned summaries should identify the sampled population, recall window, burden, missingness, and alternative explanations. When a claim contracts, the public artifact should contract with it so a memorable story does not outlive the evidence that supported it.

15.5 — What Dream Evidence Could Teach

Dream evidence can succeed in several limited ways. It can show that a waking record omits conscious affect, clarify relations among sleep stage, recall, and content, or reveal that one coding system transports poorly across groups. None of those results is the complete-stream claim.

Failure is equally informative. A predicted valence pattern may be absent, disappear under a better denominator, vary mainly by culture or medication, or remain indistinguishable from memory and selection effects. The proper response is to narrow or retire the route, not to claim that the important dream was simply hidden from measurement.

Public language should contract with the result. A local association is not prophecy, a recurring image is not diagnosis, and a nightmare is not evidence that pain has been balanced elsewhere. Headlines, diagrams, and excerpts should identify the sampled population, method, uncertainty, and major rival whenever omission would change the meaning.

The closing lesson is methodological humility with positive content. Dreams enlarge the observable history and expose the limits of waking biography. They also demonstrate why missing experience, reconstructed memory, cultural meaning, and observation effects must remain separate.

15.5.1 — No Valence Pattern

The clearest failure of a dream-valence route is a prospectively specified absence of the predicted relation. If waking burden does

not forecast dream-valence residuals under a capable design, or if positive, negative, and neutral waking periods produce no discriminating pattern, the proposed route should narrow rather than migrate into a vaguer claim.

Absence must be earned rather than inferred from a large p-value. A broad interval may show that the study could not decide, whereas a narrow interval that excludes the prospectively specified smallest effect required by the route counts against that route. The public report should distinguish practical absence from inadequate estimation (Lakens, 2017; Wasserstein and Lazar, 2016).

Use the full awakening and reporting denominator specified in Section 15.4.1. A relation found only among vivid reporters may describe recall and selection rather than dream experience as a whole.

Ordinary sleep explanations should receive the same information and flexibility. Emotional salience, sleep fragmentation, trauma status, memory consolidation, medication, stage composition, and recall bias may fit the observations as well as or better than an inversion model. When a prospectively specified rival wins, the report should identify it plainly rather than preserving the preferred route through post hoc lags or subgroups (Munafò et al., 2017).

Route retirement should be specific. Failure of an inversion prediction does not make dreams irrelevant to a whole-life inquiry. Dreaming may still contain phenomenal exposure missing from waking reports, and dream methods may still clarify

memory, distress, or observation. What retires is the claim that remembered dream affect systematically counters waking valence.

Public language should match that distinction. A responsible summary may say that the prospectively specified dream-valence relation was not detected with enough precision to support the candidate route, or that sleep fragmentation and emotional salience predicted better. It should not turn either result into a generic promise that more research will vindicate the idea.

No post hoc change in lag, subgroup, codebook, or outcome may be described as the original test. A revised route may be worth studying, but it must receive a new identity, a new rationale, and a new failure condition. Otherwise revision becomes rescue rather than learning (Munafò et al., 2017; Wasserstein and Lazar, 2016).

A failed dream route can still improve the program. It can prevent unnecessary burden, redirect attention toward better sleep models, and remove an attractive but unsupported story from public use. Research remains valuable when it preserves what was learned while allowing the proposed connection to the Law to end.

15.5.2 — Cultural Pattern Dominance

Cultural-pattern dominance occurs when dream results are explained more fully by reporting norms, religious meaning, language categories, family practice, social expectation, or popular dream scripts than by a general dream-valence mechanism. Culture is not a minor confound added after the main

analysis. It can shape what is noticed, remembered, disclosed, and understood, so it belongs in the explanation from the start.

The same content can therefore carry different evidentiary meanings. A warning dream in one community, an ancestor dream received as comfort in another, and silence where dreams remain private may all be real features of experience and disclosure. Their difference is scientifically informative, but it does not establish one universal mechanism or authorize an outside interpreter to flatten the traditions into a single code (Domhoff, 2017; Little and Rubin, 2019).

If a dream-valence pattern appears only after pooling languages, the pooled result does not establish transport. Report language-specific estimates, coding disagreements, disclosure rates, and whether the direction survives local idioms and scale use; otherwise classify the pattern as culturally local (Schredl and Hofmann, 2003).

Unreliability can become a dignity problem. Coders who force agreement by flattening a report may misrepresent the dreamer; confident labels for ambiguous material create false knowledge; selective removal of hard cases can bias the archive toward tidy dreams. Participant ratings, blinded coding, disagreement, uncertainty, and context should remain visible rather than being collapsed into one sovereign interpretation (Hall and Van de Castle, 1966).

The scientific response is to model culture as a substantive source of variation, prospectively specify the relation and rivals, test measurement equivalence, and report local patterns without

untested universality. Translation history, local base rates, disclosure norms, and uncertainty should remain visible in the dataset. Blinded analysis then supplies an independent route to the result rather than allowing cultural familiarity or unfamiliarity to decide it in advance (Domhoff, 2017; Schredl and Hofmann, 2003).

Interpretive authority should not rest only with distant analysts. Fluent coders, community reviewers, and participants can identify imposed meanings, offensive categories, translation failures, sacred material, and privacy risks that an external team may miss. Their contribution improves validity and ethical fit; it does not replace blinded analysis or decide the scientific verdict. A defensible design keeps local interpretation and independent adjudication in constructive tension.

When culture explains more than the proposed mechanism, the result may still be valuable. It can describe the studied setting, reveal which categories travel, motivate local replication, or show that cross-cultural comparison is not yet identified.

15.5.3 — Claims Narrow Publicly

Correction is part of a trustworthy public record. When a claim contracts, the manuscript, website, lecture, participant communication, image, presentation, and social excerpt should contract with it. The correction names what narrowed, preserves the full denominator, and states whether the issue involved population, sleep stage, awakening method, cultural setting, reliability, missing reports, transport, or mechanism (Nosek et al., 2018). Dream research especially needs this discipline because

one vivid report can travel farther than its sampling and reconstruction limits. The observed object was a report elicited under a protocol, not an unedited view of sleeping experience.

If a prospectively specified relation fails, the route narrows or retires; if a capable rival performs equally well, the result is model-ambiguous (Munafò et al., 2017; Wasserstein and Lazar, 2016). High-reach artifacts need an owner, review date, and replacement path because an image may remain literally accurate yet become materially misleading after a stronger rival appears. When limits cannot travel with the artifact, correction or withdrawal is safer. Dream evidence can still widen observation, improve a bridge, favor an ordinary sleep account, or retire a route; the public sentence remains no larger than what survived the test.

Chapter 16 — At a Stream's End

The end of life attracts grand stories because it gathers love, fear, uncertainty, medicine, memory, and irreversible loss into a small space. Chapter 16 enters that space without turning dying people into symbols. Its first obligation is care. Its second is honesty about what terminal observation can and cannot reveal.

The terminal-closure hypothesis makes a boundary claim, so an endpoint is formally sharp. Yet the model's truth endpoint is not automatically the moment of legal death, the last chart entry, the final spoken word, the withdrawal of a device, or the end of a study. Those are different events and must remain different in the public explanation.

This separation is not technical fussiness. If investigators choose the endpoint after observing where a measured trace looks favorable, the result becomes tautological. If they treat the last observation as the last experience, they confuse missingness with closure. If they infer inner peace from outward stillness, they exceed the channel.

Palliative care provides the governing human context. Its purpose is to prevent and relieve suffering, support quality of life, and respond to physical, psychosocial, and spiritual needs. It is not a waiting room for terminal research, and its success cannot be reduced to whether it produces interpretable data (World Health Organization, 2020).

Clinical guidance also emphasizes uncertainty, communication, individualized care, and regular review in the last days of life. Recognition of dying can itself be uncertain. Symptoms change. Capacity fluctuates. Families disagree. Good care must remain

responsive rather than subordinated to a theory's preferred timetable (National Institute for Health and Care Excellence, 2015).

Family knowledge is different rather than inferior. Relatives and close others may recognize habits, expressions, preferences, and meanings that a clinical team cannot see. They may also be grieving, exhausted, divided, or trying to build a coherent account from a difficult sequence. Respect requires preserving both knowledge and uncertainty.

Dignity-conserving care offers a vocabulary for what must not be lost. A person is not merely a symptom profile or endpoint case. Identity, privacy, relationships, unfinished concerns, choice, comfort, and how others address them remain part of care even when communication narrows (Chochinov, 2002).

Sedation makes interpretation especially difficult. Proportional palliative sedation may be considered for refractory suffering under contemporary clinical and ethical guidance, but it changes awareness, expression, memory, and observation. Reduced movement or report after medication may mean relief, reduced arousal, impaired expression, or several things together (Cherny and Radbruch, 2009; Surges et al., 2024).

Delirium adds another source of uncertainty. It can fluctuate, alter attention and perception, and arise from illness, medication, organ failure, infection, or many interacting causes. A person's words or behaviors may change rapidly, and a single interpretation can be both clinically and scientifically unsafe (Lawlor et al., 2000; Inouye et al., 2014).

Humane observation also requires a genuine No adequate estimate outcome. If comfort prevents assessment, capacity is unclear, treatment dominates the signal, or the relevant endpoint cannot be defended, the study may be unable to support a trustworthy terminal inference. Recognizing that limit protects both care and the scientific record.

Late-window patterns still deserve serious study, but medication, floor effects, selective observability, frailty, culture, family context, and missingness can produce similar appearances. No single death therefore settles the Law, and no memorable final scene should be asked to stand for a complete conscious stream.

Private dying remains private when the person or family chooses. Research can learn from aggregate terminal patterns without claiming ownership of intimate speech, final gestures, photographs, or recognizable stories. Readers should leave able to hold scientific curiosity and human dignity together as the inquiry moves toward safer nonterminal questions.

16.1 — Why the End Is Scientifically Sharp

A terminal claim makes the endpoint scientifically sharp because the proposed relation is defined there. That formal sharpness does not tell researchers which clinical event is the truth-level end of experience, nor does it guarantee that the relevant state is observable. The boundary is exact in the hypothesis and uncertain in practice.

Several clocks can diverge. Legal death, circulatory arrest, neurologic criteria, withdrawal of monitoring, loss of communication, the last behavioral response, and the final qualifying experience are not automatically identical. A protocol endpoint may be necessary for analysis while remaining only a proxy for the truth endpoint.

One life cannot settle the question. A peaceful final hour may reflect excellent palliation, fatigue, medication, privacy, impaired expression, or genuine calm; a difficult death may reflect uncontrolled symptoms, delirium, family distress, or observation bias. Neither scene reveals a complete latent residual (Seow et al., 2011).

Formal sharpness now meets empirical humility. The endpoint can be defined exactly in a model while remaining uncertain at the bedside, and a single case can remain meaningful without becoming adjudicative. The resulting bridge is demanding by design and may legitimately end in No adequate estimate.

16.1.1 — Endpoint as Formal Test Point

The endpoint is scientifically sharp because the proposed Law concerns a complete strict-pre-endpoint domain, whereas every real study sees only a partial, clinically shaped record. The research opportunity is to specify in advance how the truth-level endpoint, protocol endpoint, last observation, and reasons for divergence will be represented. Their separation is the first condition of an interpretable terminal study.

The terminal point is also not a private symbol. A family's sense that someone was ready, peaceful, restless, reconciled, or unfinished may matter profoundly in human life, but those meanings cannot define the formal test point. The model requires a research endpoint that can be specified before interpretation, not an emotional endpoint reconstructed afterward (National Institute for Health and Care Excellence, 2015).

The reason terminal closure is a test point rather than a proof point is that it sits inside an observation problem. Even if a late estimator moves toward a neutral band, the estimator is not the latent cumulative valence record. Even if physiological variance contracts, contraction is not exact equality. The endpoint is where the hypothesis becomes answerable, not where it becomes self-validating.

The truth-level endpoint and the protocol endpoint may be different. The first is the boundary of the qualifying conscious stream under the model; the second is the last time a study can ethically and reliably observe a participant under its declared rules. Loss of speech, discharge, transfer, sedation, device

removal, withdrawal, or a change in clinical priority may close the protocol window while conscious experience continues. A valid study records that gap instead of quietly treating the last available measurement as the end of the stream.

Observation can also stop after the stream has already become difficult to identify. A person may move through fluctuating consciousness, delirium, deep sedation, coma, or uncertain recovery. These states do not become equivalent merely because the study has reached its final page. The endpoint rule needs clinically independent channels, prospective adjudication, and an honest branch for cases in which the relevant boundary cannot be established.

Endpoint timing must be fixed independently of the desired result. Selecting the endpoint after inspecting the cumulative estimate can manufacture closure, just as stopping a race when a runner reaches a chosen mark can manufacture a finish. A protocol may use a prospectively specified clinical or observational endpoint, but it must not relabel the last convenient measurement as the truth-level end of the stream.

The last response, last reliable report, last contact, last sensor interval, biological death, and irreversible end of the qualifying conscious stream are different objects. Clinical reality may make some of them unobservable. The scientific record should preserve that uncertainty rather than collapse it into a single timestamp.

Clinical states can change rapidly near death. Responsiveness may fluctuate, monitors may be removed, medication may be adjusted, and the last assessment may occur hours before the last

qualifying experience. A study should record the distance between the protocol endpoint, observation shutdown, and relevant clinical events rather than treating one timestamp as all three.

The residual uncertainty can be represented honestly. Researchers may report a terminal window, a set of plausible endpoints, or a range of estimates under prespecified alternatives. Such a result is weaker than exact endpoint identification, but it is stronger than concealing the ambiguity inside a single date.

16.1.2 — No Single Life Proves the Law

A single life can move a reader, but it cannot validate the Law. The formal hypothesis concerns a structural property of qualifying conscious trajectories. One terminal scene, no matter how moving, cannot distinguish terminal valence closure from medication effects, disease progression, family interpretation, cultural expectation, spiritual language, ordinary adaptation, or chance.

This rule protects families as much as science. No family should feel that a loved one's death was a failed example because it was agitated, quiet, medicated, confused, or complex. No family should feel that a peaceful death proves a hidden order.

A sharp claim is not a certain claim; it is a claim precise enough for the world to prove wrong. The endpoint matters because terminal valence closure, if it has observable consequences, should constrain what can be seen near terminal regimes more than a vague adaptation story would. That makes the region worth

modeling and testing, not believing in advance (Steinhauser et al., 2000).

A single life may challenge a prospectively declared universal branch only when bearer, representation, endpoint, observation, missingness, and measured-to-latent bridges are all adequate. Most real cases will not satisfy that complete chain. Their value lies in clarifying endpoint uncertainty, observation failure, treatment effects, and which future designs could become discriminating.

Sharpness should make the design more careful, not the claim louder. A terminal window can reduce some ambiguity about horizon, reserve, and endpoint approach, while increasing ambiguity about medication, missingness, fatigue, delirium, and care interruptions. The scientific value of this region is therefore conditional. It becomes useful only when the analysis shows which uncertainty has been reduced and which uncertainty has been enlarged.

Case selection can create the appearance of balance or imbalance. Prospective cohorts and prospectively specified ascertainment rules reduce outcome-driven selection, while denominator flow shows who was never enrolled, who became unobservable, and whose final interval remained unknown. The resulting sample must remain tied to that flow rather than being presented as though it were the whole population (Little and Rubin, 2019).

The late-window limitation also matters mathematically. A process can show late-window concentration for reasons unrelated to whole-life closure. Endpoint-conditioned bridges,

declining variance from biological reserve, measurement saturation, medication-driven dampening, or missingness at high distress can all produce apparent narrowing. Without earlier path information and rival comparison, the late window can deceive (Seow et al., 2011).

For one case to challenge an exact universal branch, the evidentiary burden would be unusually high. The bearer, ownership, valence representation, neutral origin, complete domain, endpoint, observation bridge, and residual would all need adequate identification. A compelling biography or dense chart does not supply those gates automatically.

Case ascertainment also needs an independent route into the study. Referral rules, enrollment windows, site coverage, and reasons for exclusion should be fixed before researchers know which lives appear balanced or unbalanced. A held-out site or later cohort can then test whether the apparent terminal pattern travels beyond the cases that first made it interesting. Prospective ascertainment reduces selection bias, though it does not solve the remaining measurement problem.

An adverse case can still narrow a route when the route's local prerequisites are met. For example, a proposed terminal pattern may fail under a validated channel even if complete-stream adjudication remains impossible. That loss should be recorded at the route level rather than inflated into universal defeat or dismissed as irrelevant.

Personal communication should remain outside the verdict. A family should never be told that a research estimate proves the

person's life balanced, failed to balance, or carries implications for what happened after observation ended. The scientific object is a prospectively specified claim, not a judgment delivered to the bedside.

16.1.3 — Scientific Status Control

Scientific status control keeps terminal research useful at the level it earns. Clinical description can document symptoms, relief, sedation, delirium, and communication; prerequisite evidence can improve endpoint and observation models; route evidence can compare declared mechanisms; complete-stream adjudication requires the full chain of bridges. No lower level is a failure simply because it does not reach the final verdict.

That priority governs timing, contact, sensors, questionnaires, interviews, audio, video, family involvement, and public description. A procedure can be physically harmless while changing the emotional, relational, clinical, or privacy environment. Burden therefore includes interruption, self-consciousness, fatigue, altered staff attention, and the feeling that an intimate room has become a research site (Munafò et al., 2017).

Observation is not forbidden, but extraction is. A study may be ethically discussable when clinically accepted monitors are already present, consent remains active, data are minimized, withdrawal is easy, care staff can stop research immediately, and the protocol rewards abstention when care requires it. When those conditions cannot be met, ethical nonviability is the correct recorded result, not a reason to force the study.

Terminal observations can still perform useful scientific work. They may motivate a hypothesis, reveal a confound, refine an endpoint rule, test one observation channel, or distinguish one candidate route from a rival. None of those achievements by itself identifies the bearer, reconstructs the complete stream, establishes exact latent closure, or upgrades PTVC from a candidate hypothesis to established science.

Consider a study with excellent symptom records but no defensible bridge from observed behavior to cumulative phenomenal valence. The records may support a careful clinical inference and may improve palliative care, yet the complete-stream result is No adequate estimate. Controlled verdict language prevents formal sharpness and scarce terminal data from being mistaken for empirical access (Munafò et al., 2017).

Endpoint studies must place the distinct boundaries described in Section 16.1.1 on a prospectively specified timeline. This prevents the last available measurement from silently becoming the endpoint required by the hypothesis.

Ascertainment quality should be reported by layer. Dense symptom records without a credible consciousness assessment answer a different question from repeated awareness assessment with sparse affect data. Reports should state what was observed, what was inferred, what remained unavailable, and which branch, population, protocol, and resolution the conclusion covers. Passing one prerequisite cannot compensate for a missing bearer, endpoint, observation model, or measured-to-latent bridge.

Medication, missing experience, fluctuating awareness, measurement error, and uncertainty about the truth boundary remain even in unusually detailed records. Scientific conclusions belong to declared cohort-level methods and claims.

The end of observation must remain separate from the end of experience. A device may be removed, a person may ask for quiet, a family may request privacy, staff may redirect attention to care, or communication may become impossible. None of those events establishes that the qualifying stream has ended. The record should preserve why observation stopped, which uncertainty follows, and which lower-level clinical findings remain valid without converting silence into a terminal datum (Little and Rubin, 2019).

Sedation makes that distinction especially important. Medication can relieve distress, change responsiveness, and alter observable behavior without providing direct access to felt experience or the truth-level endpoint. Dose, timing, indication, clinical response, and missingness can inform a bounded observation model, but they cannot be selected after the result to force a preferred terminal story. When the bridge is inadequate, No adequate estimate is the complete scientific conclusion.

16.2 — Hospice, Relief, and Palliative Reality

Hospice and palliative care begin from the needs of the person, not from the needs of a hypothesis. Relief of pain and other symptoms, communication, family support, privacy, and chosen goals remain the governing work. Research may enter only if it fits inside that work without weakening it (World Health Organization, 2020).

Relief is not a contaminant. Analgesia, anxiolysis, sedation, repositioning, quiet, touch, and the presence or absence of family can change observable behavior and reported experience because they are real parts of care. A study that treats good palliation as a threat to its signal has inverted the ethical order.

Families and clinicians carry forms of knowledge that matter without becoming direct measurements of inner life. They can describe expressions, preferences, communication changes, prior wishes, and moments of distress or comfort. Their accounts may improve protocol design and interpretation, but grief, hope, and role expectations also shape retrospective narrative (Steinhauser et al., 2000).

16.2.1 — Professional Witness

Professionals at the bedside contribute disciplined observation, treatment history, and contextual judgment. They can describe pain, breathlessness, agitation, comfort, delirium, sedation, communication, and response to care. Their witness is indispensable to humane research, but it remains an observation channel shaped by clinical purpose and limited access (World Health Organization, 2020; Kesecioglu et al., 2024).

Clinical records are strongest when they preserve timing and purpose. Symptom assessments, medication changes, care goals, family presence, communication capacity, and reasons for missing observations help distinguish relief from reduced expression and privacy from data loss. A brief calm interval is meaningful for care without becoming a reading of the complete stream.

Professional witness also protects families from theory pressure. When a family asks whether a particular final behavior means something about fairness, the responsible answer is not to interpret the Law. It is to return attention to care, grief, love, and the limits of knowledge. Clinical wisdom can keep the theory from entering rooms where it does not belong.

Palliative care also reveals why care and inference must be ordered. Relief is given because suffering matters now, not because a theory predicts a later balance. Research may learn from care that was already clinically justified; it may never delay, dilute, or redirect treatment to improve a terminal signal (World Health Organization, 2020).

The research design must therefore record medication timing, dose changes, indications, target symptoms, route, clinical response, and relevant side effects where ethically and legally permitted. It must also treat medication data as protected health information, not as public proof material. Clinical context belongs in the analysis, not in public display (Chochinov, 2002).

Clinical records were created first to support care, not to reconstruct a complete phenomenal history. A symptom score,

nursing note, medication chart, or family conference can be highly informative about the problem it was designed to address while remaining silent about many other aspects of experience. Research reuse must therefore preserve the original purpose, timing, and context of each record rather than treating the chart as a continuous inner-life transcript (Steinhauser et al., 2000; Seow et al., 2011).

The constructive scientific gain is a better account of what professionals can observe and what remains hidden. Bedside expertise can improve endpoint windows, treatment histories, symptom interpretation, and stop rules, while leaving the private cumulative object outside professional ownership.

Most importantly, professional participation must not turn care into data collection. Clinicians should be able to skip a research field, remove a device, or end an assessment when the person's needs require it. Good care is not a protocol deviation, and a missed measure caused by good care should be reported as such.

16.2.2 — Family Narratives

Family narratives preserve relationship, meaning, and details that formal instruments miss. They can reveal changes in communication, fear, comfort, recognition, humor, ritual, or desire. They are valuable human testimony and may improve contextual interpretation, yet they are also shaped by grief, memory, family roles, and the need to make sense of an ending.

The distinction is not a judgment against families. It protects what their testimony is actually good at. Families often know preferences, fears, routines, relationships, unfinished concerns,

and the meaning of ordinary expressions. That situated knowledge can correct an unfamiliar clinical interpretation and help the team avoid treating a person as a generic case.

Consider a daughter sitting beside a parent who no longer answers questions. She notices a brief relaxation when a favorite recording begins and tells the nurse that the music has always been reassuring. The observation may support a compassionate change in the room. It does not reveal a cumulative ledger, establish inner peace, or settle whether the person is conscious at that moment. Its value lies first in care.

Grief changes recall without making it worthless. After a death, relatives may compress several days into one story, give unusual weight to the last exchange, reinterpret uncertainty through later knowledge, or seek a coherent ending to a sequence that never felt coherent while it unfolded. Researchers should preserve dates, sources, uncertainty, and disagreement rather than flattening many witnesses into one family account.

The same event can support different meanings. A period of quiet may be remembered as peace, exhaustion, medication effect, privacy, resignation, or loss of access. A final phrase may be heard as reconciliation by one person and confusion by another. When accounts diverge, the divergence is evidence about the observation process and the relational setting, not a defect to be edited away.

Cultural and religious traditions also shape what families notice and how they describe a good death. Some emphasize explicit farewell, some prayer, some bodily presence, some privacy, some

endurance, and some relief from suffering. These meanings deserve respect, while the scientific record must still distinguish cultural interpretation from evidence about an unobserved phenomenal state (National Institute for Health and Care Excellence, 2015).

A defensible study therefore treats family narratives as situated testimony. It records who observed what, when, under which treatment and communication conditions, how soon the account was collected, whether other witnesses agreed, and what the family wishes to keep private. The original words should be preserved where consent permits, with later coding clearly separated from the account itself.

Family testimony should never be converted into a ranking of deaths. A peaceful account is not a successful case, and a distressed account is not a failed one. The Law does not assign metaphysical meaning to agitation, silence, lucidity, conflict, reconciliation, or the absence of a final message.

Public use requires an additional boundary. A family may consent to a research interview without consenting to a quotation in a video, fundraising appeal, social post, or illustrated guide. Small details can identify a person even after names are removed. High-reach uses should therefore be separately reviewed, minimized, and declined when the story would become more memorable than the limitation surrounding it.

Dignity over data is not a refusal of knowledge. It is the condition under which end-of-life research remains worthy of trust. A study that protects comfort, privacy, family choice, staff judgment, and

the right not to be observed can still produce important evidence about measurement, missingness, treatment, and endpoint uncertainty.

16.2.3 — Dignity Over Data

The research plan should begin with the care pathway and fit inside it. Consent or authorized permission, minimal burden, clinician veto, family boundaries, data minimization, and immediate stop authority should be settled before any observation begins. No data point is valuable enough to compete with symptom relief or a person's wish for privacy (World Medical Association, 2024).

When direct participation is no longer possible, legal authorization alone is not enough. Prior wishes, current signs of assent or resistance, family knowledge, clinical judgment, study burden, and the possibility of nonparticipation all matter. The safest design often relies on routine-care data or brief observation rather than adding a new task to a fragile final interval (CIOMS, 2016).

Dignity protects nonparticipation. A dying person who declines research, cannot consent, withdraws, or never enters the research conversation loses no standing in the theory, book, or any public imagination. Their death remains their own; absent data do not diminish the life (World Health Organization, 2020).

Dignity over data means some moments should remain unmeasured even when measurement is technically easy. A final conversation, a prayer, a bodily-care moment, a fearful confession, or a family apology may be meaningful because it is

protected from public use. The Law of Fairness does not require a complete archive of dying. Sometimes the fair archive is the one that never gets made.

A dignity-first protocol separates clinical care from research value. Missed assessments carry no penalty; staff can stop observation immediately; treatment and family access never depend on participation; researchers cannot negotiate for more data during symptom crises. These are not external ethics added after the method; they determine which observations can be interpreted without being purchased by vulnerability (Chochinov, 2002; World Medical Association, 2024).

A peaceful room can be scientifically ambiguous and humanly complete. Quiet may reflect effective relief, fatigue, medication, reduced expression, chosen privacy, or several of these at once. Research should preserve those possibilities instead of making serenity or agitation carry a verdict about the life as a whole.

Picture an interview that is canceled because speaking has become tiring. The assessment records its reason, the care team continues without interruption, and the participant is not treated as failed data. That decision protects dignity and improves interpretation: the record now shows that observation stopped because burden rose, not because the latent state was neutral.

Dignity governs language. Terms such as completed, balanced, failed, or unresolved can sound like judgments about the person when applied to a death. Research reports should use status only for the measurement or model and avoid attaching it to the life. Keeping that vocabulary narrow also helps families read a report

without mistaking a research status for a judgment about the person they lost. A family-facing summary can name what was observed, what was not observed, and why data collection stopped without converting the person into a research outcome. That separation keeps the record useful while protecting remembrance from the vocabulary of adjudication. Reports written this way can support clinical learning without asking relatives to adopt the study's categories as the meaning of the death.

16.3 — Sedation, Delirium, and Family Scripts

Outward stillness is multiply determined. Sedation can reduce arousal and expression, delirium can produce fluctuating attention and altered report, weakness can prevent movement, and sensory or motor impairment can block a response that consciousness alone would not predict. The same bedside appearance can therefore arise from different states.

Clinical interpretation is indispensable when treatment, cognition, or communication changes. Medication timing, delirium screening, neurologic status, respiratory distress, pain, hearing, language, and motor capacity shape what an observation can mean. Research coding should never outrun the care team's documented clinical understanding (Inouye et al., 2014; Lawlor et al., 2000).

Family scripts add another layer. A phrase such as she was finally at peace or he kept fighting may be loving, culturally meaningful, and emotionally true for the family. It is still a narrative frame, not a measurement of terminal neutrality. Researchers should honor the meaning without converting it into a scientific label.

Treatment, memory, and clinical state now become part of the evidence chain. Bedside knowledge retains its value, but every observation must remain attached to medication, cognition, channel, timing, and uncertainty before it can support even a local inference.

16.3.1 — Sedation Effects

Palliative sedation is a clinical response to otherwise refractory suffering, not a standardized experimental exposure. Contemporary guidance emphasizes proportionality, careful indication, patient autonomy or an appropriate decision process, continued care, and repeated review. The purpose is relief, never the production of an interpretable terminal pattern (Surges et al., 2024).

The word sedation covers different depths, durations, drugs, routes, and clinical circumstances. Intermittent light sedation for a procedure is not continuous deep sedation near death. A single label can therefore hide the very differences that determine awareness, expression, memory, and observability (Maltoni et al., 2013; Surges et al., 2024).

For research, the medication record must be a timeline rather than a checkbox. Indication, symptom target, drug, dose, route, titration, co-medication, organ function, intended depth, observed response, and timing relative to each observation can change what a person is able to report and what others can see. A final score detached from that history is not a stable endpoint measurement.

Delirium compounds the problem. Attention, perception, arousal, language, and behavior can fluctuate with infection, organ failure, medication, dehydration, sleep disruption, or the underlying illness. A phrase recorded during one interval may not represent the preceding hour or the following one (Inouye et al., 2014).

Observational studies can still clarify how sedation is used, which symptoms precede it, and how practice varies. Recent work on

prevalence and determinants is valuable for clinical description, but variation in indication, population, and practice means that these datasets do not supply a uniform consciousness manipulation or a direct test of PTVC (Carrasco-Zafra et al., 2024).

No research question may delay analgesia, alter a clinically indicated titration, prefer a more observable state, or pressure a team to preserve responsiveness for data collection. Once treatment changes the channel, the study adapts to the care plan. The care plan does not adapt to the study.

A strong protocol predefines the status of observations collected before, during, and after medication changes. It specifies when treatment makes the relevant state unidentifiable, when sensitivity analyses are possible, and when the only honest result is No adequate estimate. That abstention is part of the model rather than an embarrassment to be hidden.

The bedside scene remains human throughout. A person receiving proportional relief is not a cleaner signal, a final data point, or an opportunity to infer settlement. Sedation may reduce suffering, and that clinical good stands on its own. The scientific task is to preserve the distinction between relief, awareness, expression, memory, and what the instruments happened to record.

16.3.2 — Retrospective Meaning-Making

The problem begins when retrospective meaning is asked to do scientific work. A story made after the fact knows its ending. It selects details in light of that ending. It may smooth

contradictions, heighten symbolism, and connect events that were not experienced as connected at the time (Chochinov, 2002).

If qualitative research studies retrospective narratives, it must ask narrative questions. How do families understand dying? How does public language change grief? Which research practices feel intrusive? Those are real questions. They are not measurements of terminal valence closure.

A humane design protects family members as nonparticipants unless they explicitly and separately consent. It does not treat their speech, tears, prayers, conflicts, or memories as automatically available data. It also protects them from being asked to influence the patient toward reportable states (Chochinov, 2002).

Family pressure is also a rival explanation. A final report of relief may reflect genuine relief, desire to comfort family, social expectation, or the narrowing of what can be said in the room. A serious analysis must treat interpersonal context as part of the model rather than as noise outside it.

Family performance pressure can be subtle. People may feel expected to witness peace, create meaning, forgive, be forgiven, pray, reassure, or narrate the death in a certain way. The dying person may also try to comfort the living. These pressures do not make the scene false. They make the scene social, and therefore not a clean measure of terminal closure (Chochinov, 2002).

Retrospective meaning-making can help families integrate an ending, honor a relationship, and describe what mattered. It can also reshape sequence, emphasis, and certainty as grief unfolds.

Research should respect that human function while distinguishing the family's later narrative from contemporaneous observation and from the person's private experience.

A family may remember a final phrase as the center of the death while the clinical record emphasizes long periods of nonresponse. Neither record is simply false. They answer different human and scientific questions. A careful account preserves both provenance and does not force one to overwrite the other.

Public use requires additional restraint. A family's phrase may be moving enough to dominate a paper or presentation, yet quotation can detach it from uncertainty and turn private meaning into universal evidence. Consent, context, and claim status should travel together whenever a narrative leaves the research record.

16.3.3 — Clinical Review Required

Clinical review is required because terminal observation is never only a research event. Medication changes, delirium, respiratory distress, pain, fatigue, communication impairment, and the underlying illness can alter both experience and the channels through which experience is expressed. A theoretical interpretation that bypasses the clinical timeline may be precise in form and wrong in substance.

The clinical reviewer's task is specific: reconstruct what care was intended, which treatments and symptoms shaped the interval, what the person could express, and which inferences remain unavailable. The review may support a bounded observation classification or an abstention. It never certifies a terminal balance.

Independent review becomes especially important when one team proposes the route, selects cases, codes ambiguous behavior, and interprets the result. Prospectively specified criteria, blinded adjudication where feasible, and separate clinical and analytic roles reduce the chance that expectation shapes every layer. The review structure should be designed before the first difficult case arrives.

Behavioral nonresponse cannot be equated with absence of experience. Multimodal work in severe brain injury has found task-related brain activity in some people who did not follow commands behaviorally (Bodien et al., 2024). That population is not a proxy for dying patients, but it demonstrates why motor silence cannot carry a sovereign conclusion about awareness.

Review should be interdisciplinary because different observers see different parts of the terminal window. Palliative medicine, nursing, neurology, pharmacy, ethics, social work, chosen spiritual care, and family knowledge may clarify treatment, communication, distress, culture, and prior preference. Their evidence should remain distinguishable, and no panel acquires ownership of the person's inner state by combining expertise.

Reviewers must be able to return an indeterminate classification. The record may state that consciousness could not be assessed, pain behavior was uncertain, treatment confounding was unresolved, family interpretations differed, or the endpoint was not identified. When the bridge cannot be reconstructed, the proper result is No adequate estimate, with the reason named rather than hidden (CIOMS, 2016).

Documented disagreement and exclusion audit complete that independence. Reviewers should be able to see which cases were omitted, why classifications changed, where clinical and analytic judgments diverged, and whether the preferred interpretation depended on one contested decision. That record strengthens the bridge while preserving the right to abstain.

Clinical review protects meaning as well as method. A peaceful death, a difficult death, a final statement, or a family's interpretation can matter profoundly without becoming evidence that the whole stream closed in one way or another. Care and remembrance keep their human value precisely because they are not forced to serve as a terminal verdict.

16.4 — Observation and No-Estimate Outcomes

Humane observation accepts that the most important data may be the data not collected. A symptom crisis, a request for quiet, fatigue, family conflict, or a clinician's judgment can stop an assessment. The resulting gap records a priority correctly applied, not a protocol failure to be repaired at the bedside.

Stop rules should be prospective and distributed. The participant, surrogate where appropriate, nurse, physician, family member, or independent monitor may identify burden that the research team does not see. Authority, escalation, data handling, and restart conditions should be clear before enrollment (World Medical Association, 2024; CIOMS, 2016).

Caregivers must not be recruited by guilt. Families and staff should not feel that declining a prompt, removing a sensor, or missing an observation weakens care, disappoints the team, or deprives science of a rare chance. Dependency and grief can make nominal permission look more voluntary than it is.

No adequate estimate is a complete scientific outcome. If endpoint identity, consciousness, valence, missingness, treatment effects, or consent cannot be bridged, the study should say so. Abstention protects the person and the public record from a false precision that no amount of statistical polish can justify.

16.4.1 — No Data Before Comfort

Comfort comes before data. Routine care, symptom relief, privacy, family presence, and clinician judgment set the boundary within which observation may occur. If collection interferes with

sleep, medication, touch, conversation, spiritual practice, or a requested quiet room, collection stops. The missing data should then be recorded as ethically unavailable rather than treated as a failure of cooperation.

A humane protocol minimizes active demands and uses routine-care information only when governance, purpose, consent, and access rules permit. It should specify when observation is suspended, who can stop it, how burden is noticed, and what happens to data already collected. The design earns legitimacy by making abstention operational.

The absence of a final measurement can be informative. It may show that the endpoint was outside the observation window, that care properly displaced research, or that the bridge from behavior to experience could not be supported.

Consent also applies to families and nonparticipants. A terminal room may include speech, touch, ritual, conflict, or grief from people who are not research subjects. Their presence does not make their data available. If their privacy cannot be protected, the study must narrow or stop (National Institute for Health and Care Excellence, 2015).

Comfort-first design changes the scientific record in an honest way. Analgesia, communication needs, family presence, and chosen privacy may interrupt or end assessment, leaving fewer observations precisely when researchers would like more. Those missing data are not a protocol failure to be overcome at the bedside. They are evidence that the study remained subordinate to palliative care and the person's priorities (World Health

Organization, 2020; National Institute for Health and Care Excellence, 2015).

The Science of Fairness must translate the safeguards in Section 16.2.3 into independent staff roles, escalation paths, burden thresholds, and adverse-event review rather than treating complete data as the measure of a good site (National Institute for Health and Care Excellence, 2015; World Medical Association, 2024).

The rule becomes concrete when a researcher waits outside while the care team treats pain and the patient rests, accepting that the observation window may close. The lost measurement is not a scientific embarrassment. It is evidence that the protocol preserved its proper order of authority.

Research personnel need training to recognize when they are becoming another demand in the room. A person may comply out of politeness, fatigue, or fear of disappointing the team. Brief consent renewal and easy stopping are more reliable than assuming the original signature remains active under changing conditions.

16.4.2 — Stop Rules

Stop rules should be observable, immediate, and controlled by more than the research team. A request from the person or family, clinician concern, escalating burden, protocol deviation, privacy breach, equipment interference, or loss of the permission basis can end collection without debate. The record should state which rule fired and preserve the person's care and privacy after stopping.

Authority to stop must be distributed to people who can act. The participant, the clinical team, and a designated independent safety role should each have a clear route to end collection, and family concern should be heard even when it is not legally decisive. No one should have to defend the stopping decision to an investigator, sponsor, or theory (Beauchamp and Childress, 2019; World Medical Association, 2024).

The protocol should specify who removes devices, what happens to already collected data, which deletion limits apply, how the family is informed, how private reasons are coded, and whether any later restart is possible. A cessation is recorded as protected protocol conduct, not as a failed participant or a missing terminal datum. Training should include the moment when researchers want one more measurement and the care team says no.

Humane stopping is also informative missingness. People with greater distress, faster decline, delirium, communication difficulty, or family concern may leave observation earlier, making the remaining trajectories appear calmer or more complete. Analysis should distinguish clinical improvement, withdrawal, incapacity, device removal, transfer, death, and ordinary missed observation, then test how conclusions change under plausible ranges for the unseen interval (Diggle and Kenward, 1994; Little and Rubin, 2019).

Reports should disclose how often each trigger occurred, who invoked it, and how the denominator changed. Sites must not be rewarded for terminal completeness in ways that pressure staff, patients, or families. Appropriate abstention and care-preserving

missingness are successful protocol outcomes, even when they force No adequate estimate.

A study that cannot accept protected data loss has not designed a humane terminal window. Its scientific ambition then depends on access that no person owes. The proper habit is immediate deference to care, followed by documentation, analysis of the resulting selection, and later review away from the bedside.

16.4.3 — No Caregiver Pressure for Research

Caregivers, relatives, and staff must not be recruited through obligation. Their participation should remain separate from employment evaluation, clinical duty, family expectation, and access to services. Compensation, workload, privacy, and refusal routes should be explicit so that the study does not convert relationships of care into data pressure (CIOMS, 2016).

Recruitment should also be separated from the people on whom treatment or daily care depends. A physician or nurse may mention that a study exists when appropriate, but an independent person should handle the decision and make clear that treatment, attention, and relationships with staff will not change. When that separation cannot be made credible, the study should forgo recruitment.

Caregivers also need time. Immediate decisions may be unavoidable for some low-burden records, but interviews, recordings, quotations, and future use can often wait. A delayed option allows the family to decide after the acute moment, while acknowledging that later grief can introduce different pressures rather than automatically producing freer consent.

Direct consent from the person remains primary whenever possible. A relative's permission cannot erase a prior refusal, exceed the scope of an advance preference, or convert family curiosity into authority over inner experience. When capacity is uncertain, legal and ethical review must distinguish care decisions, research authorization, and permission for public use.

Families may disagree. One person may support participation, another may fear intrusion, and a third may believe the person would have objected. The study should not recruit the most available or enthusiastic relative and treat that answer as family consensus. Disagreement can require pause, narrower use, or abstention.

The burden of participation must be visible. Interviews can reopen distress, consume scarce time, strain relationships, or create anxiety about whether the family is interpreting signs correctly. Researchers should monitor burden, allow unanswered questions, stop without explanation, and avoid language that implies a valuable case will be lost if the family withdraws (World Medical Association, 2024).

No caregiver should be asked to watch for a terminal signature. Families are not unpaid observers responsible for timing symptoms, interpreting calm, recording final speech, or preserving evidence. Such instructions can alter the room and leave a lasting fear that something important was missed.

Privacy belongs to caregivers too. Interviews may reveal conflict, guilt, finances, health information, religious beliefs, or details about people who never consented. Data minimization, restricted

access, careful quotation policy, and separate permission for high-reach uses protect the family as well as the dying person.

A study passes this gate when refusal is ordinary, care is untouched, roles are separate, burden is monitored, disagreement has a safe route, and withdrawal does not require justification. If those conditions cannot be created, the scientifically responsible decision is not to collect the data.

A humane data architecture keeps clinical, research, family, and public records distinct. Information gathered to relieve symptoms does not become research material by default, and research permission does not authorize quotation, model training, public illustration, or future reuse. Each transfer needs its own purpose, access rule, retention period, correction path, and permission basis. Separation prevents care from becoming recruitment and protects an intimate room from becoming a general data source.

Abstention can strengthen the program by showing where the present design is not entitled to answer. Repeated No adequate estimate outcomes may reveal an inaccessible endpoint class, burdensome instrument, missing communication bridge, or reference class that should narrow. Those results guide safer measurement development away from the bedside. The gain is both ethical and scientific: fewer false terminal claims, clearer denominators, and a record future studies can improve without asking vulnerable people to pay for methodological uncertainty.

16.5 — Terminal Patterns, Rivals, and Limits

Late-life trajectories can show compression, volatility, decline, recovery, or apparent stability. These patterns deserve study because they may distinguish routes and improve care. They remain nonunique: medication, illness severity, reduced communication, selective retention, observation floors, and family context can produce similar shapes.

Strong rivals must receive the same data and flexibility. Frailty, age, disease, treatment, ordinary adaptation, selection into hospice, dropout, device removal, and terminal sampling can all create an apparent movement toward neutrality. A model wins only by surviving those alternatives under common rules (Seow et al., 2011).

Aggregate evidence cannot be translated into a personal judgment. A cohort pattern never proves that one person's dying was balanced, unfinished, deserved, meaningful, or privately resolved. The more intimate the case, the more important it is to keep research status out of the family's account.

Public limits are part of the result. Images, captions, donor summaries, and social excerpts must not turn a quiet bedside into proof or a distressed death into disproof. Terminal research therefore remains a narrow, care-subordinate inquiry.

16.5.1 — Measured Compression Is Nonunique

Measured compression is nonunique because treatment, sedation, reduced expression, selection, floor effects, pooling, survival, and observation loss can all narrow a late trajectory. The first

scientific question is therefore whether the estimator is genuinely changing under a declared observation model, not whether the graph looks orderly.

A capable rival comparison should receive the same treatment history, timing, missingness, clinical context, and measurement flexibility as the proposed route. Apparent compression that vanishes when these processes are modeled, or that a hybrid rival predicts equally well, does not require a terminal mechanism (Little and Rubin, 2019).

Picture three uncertain estimates far apart on a graph. The analysis moves each partway toward the same group average to make the estimates less erratic. They now stand closer together even though no new observation has arrived. The narrowing is partly something the calculation did.

This approach, called partial pooling, can be useful when records are sparse. But the resulting graph leaves us with two possibilities to separate: the observations may show paths drawing closer, or the method may have drawn the estimates closer. Showing how the picture changes with and without pooling makes that distinction visible. A reader can then ask how much of the convergence came from the observations and how much came from the analysis.

Neural and physiological channels may help determine whether the stream is plausibly present, whether arousal or unity changed, and whether medication or illness altered the observation window. They do not read fairness. Heart rate, respiration, movement, skin conductance, electrophysiology, and complexity

measures remain multiply caused and require a bridge to any affective interpretation.

Selection also changes the population. People who remain measurable may differ from those who withdraw, transfer settings, lose capacity, or die before a scheduled assessment. Apparent compression among retained participants can therefore be a property of who stayed visible rather than a property of terminal experience (Seow et al., 2011; Little and Rubin, 2019).

Unexpected endpoints offer a sharper natural stress test. A model fitted to gradual decline may perform well when frailty, treatment, and observation schedules make the final window predictable, then fail when observation ends abruptly for reasons not coupled to the latent path. Such events cannot be induced, but they can reveal whether the proposed pattern depended on anticipatory selection or a slowly closing clinical window.

Descriptive compression and route evidence remain separate. Description asks whether an estimator narrows under a declared observation model. Route evidence asks whether the rate, timing, channel pattern, relation to horizon or reserve, and response to safe perturbations match one candidate mechanism better than equal-budget rivals. Shrinking spread alone cannot answer the second question.

A narrowing band may ultimately support ambiguity or No adequate estimate. That verdict is scientifically stronger than a serene closing graph whose precision was manufactured by missingness, pooling, treatment, or floor effects. The point of terminal measurement is not to make the end look orderly; it is to

learn exactly what the available evidence can and cannot distinguish (Rubin, 1976).

16.5.2 — Confounds That Can Defeat Interpretation

Near the end of life, the final usable datum is often selected by care itself: sedation begins, a device is removed, speech becomes effortful, or a family declines further contact. Analyses should show how each event changes the denominator and whether apparent terminal concentration survives plausible unobserved suffixes (National Institute for Health and Care Excellence, 2015).

Missingness must be recorded by reason rather than collapsed into one blank value. Data may end because the person or family refused, clinicians prioritized care, symptoms escalated, equipment failed, transfer occurred, a protocol rule fired, or an administrative limit intervened. A terminal dataset should show denominator flow from eligibility through each final window so that a clean retained sample is not mistaken for the cohort (Kurland et al., 2009; Little and Rubin, 2019).

Use the changing medication timeline specified in Section 16.3.1. The aim is not to subtract care from experience, but to prevent a treatment-shaped trace from being credited to a unique terminal mechanism (Surges et al., 2024).

Culture and family scripts shape both experience and its record. Some people are encouraged to describe fear openly; others protect relatives by minimizing symptoms. Quiet may be read as peace, exhaustion, medication, privacy, or withdrawal. These differences influence what is asked, what is said, who is present,

and what enters the chart, so transport requires more than using the same form at each site.

Sensitivity analysis should follow plausible clinical stories. One analysis may ask what changes if unobserved intervals were more distressed than observed intervals; another if relief increased while communication declined; another if the endpoint fell outside the protocol window. These analyses cannot reconstruct private experience, but they can reveal whether the conclusion depends on one convenient account of what happened after observation stopped.

The strongest rival matrix combines processes rather than testing them one at a time. Endpoint selection can interact with medication, medication with delirium, delirium with missingness, and missingness with family interpretation. An equal-budget hybrid rival should receive the same timing, treatment, observation, and context information as the proposed Law-specific model. If it predicts held-out terminal windows as well or better, the apparent compression does not require the proposed boundary condition.

Uncertainty also protects individuals. Aggregate patterns and model-level conclusions should be presented so that no participant becomes a successful or failed example. A calm-looking trajectory, a distressed final report, or an unobserved ending carries no personal score and no public meaning about the life as a whole.

Negative results must be interpreted by layer. Persistent non-neutrality under a valid complete-stream bridge could count

against the strong target. Dominant treatment effects may defeat a design. Missingness may yield No adequate estimate. Better rival fit may narrow a route. Ethical nonviability may end a research path without deciding the formal hypothesis.

Before issuing any terminal result, the study should show its endpoint rule, denominator flow, treatment timeline, culture-aware interpretation, sensitivity analyses, and surviving rival set. A clean-looking trajectory is not enough. The protocol must state in advance which failure blocks the inference and which result requires public narrowing.

If the only response to adverse evidence is retreat to an unspecified latent state that no available bridge could ever constrain, the public scientific claim has lost its falsifiable content and must be withdrawn. A limited clinical inference may remain valuable, but it cannot be promoted into a terminal verdict about the whole stream.

16.5.3 — Private Dying Is Not Public Proof

A private death is not a public demonstration of the Law, and no participant should be classified as balanced, unbalanced, complete, unfinished, supported, or refuting. The final hours belong first to the person, the people they chose, and the professionals responsible for care. Aggregate research can compare models without assigning metaphysical meaning to a named or recognizable life.

The last observable interval is not the whole stream. A calm afternoon, an agitated night, a lucid conversation, or a period of nonresponse can be clinically and humanly important while

remaining a tiny, selectively observed part of a life. Even carefully modeled late trajectories cannot substitute for the unobserved cumulative domain (Seow et al., 2011).

Public imagery carries special risk. Serene graphs, scales settling at zero, paired photographs of suffering and peace, or stories of final reconciliation can imply that a person's pain was repaid or that death completed an account. A visual that cannot preserve the limits of the evidence should not be used.

De-identification is often weak in small cohorts. Diagnosis, age, location, timing, family structure, and one memorable detail may identify a person within the community. Publication should prefer aggregates, suppress unnecessary detail, and obtain separate permission for quotations, photographs, audio, or video. The scientific value of a story never overrides the person's control over a recognizable ending (World Medical Association, 2024).

Research consent does not authorize every audience or use. Scholarly analysis, clinical education, public communication, fundraising, social media, and model training require separate purposes, access, retention, correction, and permissions. The farther a use travels from the original care and research relationship, the narrower and more explicit its authorization should be (World Medical Association, 2024). A permissions record can distinguish a clinical teaching session from a public recording, making the intended audience clear before material is prepared.

If the endpoint definition changes, a model fails replication, a family withdraws permission, or material proves identifying or

misleading, dependent captions, talks, websites, videos, and archives should be corrected or removed. Some endings should never enter the public record: privacy, clinical judgment, or failed anonymization may make nonpublication the proper boundary between scientific access and human ownership. A correction plan can identify the custodian of each authorized copy and the action needed when its permission or interpretation changes.

The strongest public conclusion is structural and constructive. Terminal research can clarify endpoint uncertainty, missingness, treatment effects, observation limits, and rival explanations while protecting private endings. It cannot reveal a particular death's secret meaning or make one family's final hours bear the burden of proving a universal law. The most useful shared result may be a better way to describe an observation gap or compare explanations across appropriately governed records.

Chapter 17 — Questions for a Laboratory

A laboratory can isolate a variable, randomize an intervention, repeat a task, and compare models under common rules. It cannot become a life. Chapter 17 begins with that limit so that technical clarity does not inflate the scale of the conclusion.

Horizon manipulation is one example. Researchers can vary the time available to decide, the number of remaining opportunities, the possibility of repair, or the delay before an outcome. Those manipulations may reveal how urgency, perceived control, and opportunity shape experience and choice. They do not simulate the end of a conscious stream.

Stress tasks offer another narrow window. Laboratory stressors can produce changes in cortisol, cardiovascular activity, self-report, attention, and behavior. These effects have ordinary psychobiological explanations and must be kept within mild, monitored, reversible limits (Kirschbaum et al., 1993; Dickerson and Kemeny, 2004).

Reward reversal and learning tasks can test how people update after gains, losses, and changing contingencies. A reversal curve may illuminate adaptation or control. It does not become a miniature terminal cumulative history, and its meaning depends on the computational model and alternatives being compared.

Neural measures can deepen the question while creating visual overclaim. Activity in regions associated with salience, valuation, interoception, or control is not direct access to the cumulative stream. Reverse inference remains a central danger: the same area or pattern can participate in many processes (Poldrack, 2006).

Reproducibility adds another discipline. Small samples, flexible preprocessing, undisclosed exclusions, and many analytic choices can produce striking but unstable results. Large-scale brain-behavior research has underscored how much data may be needed for reliable associations, especially when effects are small and heterogeneous (Marek et al., 2022).

A candidate signature must therefore transport. It should predict held-out data, survive changes in site and task, and outperform capable rivals under comparable flexibility. A model that explains only the data used to invent it has not yet earned public significance.

Perturbation can strengthen causal inference because changing a proposed mechanism and observing downstream consequences is more informative than correlation alone. Yet interventions have side effects, expectancy effects, spillovers, and assumptions. Causal language must name those conditions rather than rely on the drama of manipulation (Hernán and Robins, 2020; Pearl, 2009).

Transcranial magnetic stimulation illustrates the point. It can be used responsibly within expert safety guidance, but it is not a casual self-experiment and not a direct probe of fairness. Training, screening, dosing, monitoring, and regulatory context matter (Rossi et al., 2021).

Negative controls give a local mechanism claim a fair way to lose. A non-horizon task, sham procedure, alternate outcome, or irrelevant timing condition can show whether a proposed effect is specific. If the same signature appears where the route predicts it

should not, the route should narrow rather than acquire a new explanation after inspection.

Social channels require separate treatment. People influence one another through conversation, imitation, shared environments, norms, and platforms, while selection and common exposure can create similar correlations. Correlated mood is not cross-person balancing. Network experiments and observational studies show why influence, selection, and shared context must be distinguished before a social explanation is credited (Kramer et al., 2014; Coviello et al., 2014).

Failure is not one thing. A prospectively specified null with narrow uncertainty may count against a route. A noisy result may support No adequate estimate. A failed manipulation may say little about the mechanism. An ethically unacceptable burden may mean the experiment should never proceed.

Laboratory tests should follow the prospective-specification and public-preregistration distinctions in Section 10.4.1, exposing sample, exclusions, outcomes, analyses, and failure conditions before outcome inspection (Nosek et al., 2018; Munafò et al., 2017).

Equivalence reasoning is particularly useful when a route requires an effect above a meaningful size. Instead of treating every nonsignificant result as endless uncertainty, a sufficiently precise study can rule out the effect range the route needed (Lakens, 2017).

Participant protection runs through every test family. The study may not induce serious suffering, intensify crisis, withhold care,

exploit dependency, or turn private distress into a spectacle. Mild burden must still be justified, monitored, and stoppable.

By the close, the laboratory should feel both more useful and less magical. It can pressure local mechanisms and expose bad explanations. It cannot compress a life into a task. The remaining challenge is to follow experience through real time without converting observation into ownership. A strong laboratory program therefore ends each experiment with a narrower next question, not a larger claim about the person.

17.1 — What a Laboratory Can Ask

A laboratory can isolate a component of a theory; it cannot reproduce a life. Tasks can vary delay, uncertainty, opportunity, reward, recovery, or the availability of repair. They cannot recreate childhood, chronic illness, grief, addiction, caregiving, or the truth-level endpoint of a conscious stream.

Artificiality is not a defect when the question is local. A ten-minute task may test how people respond to closing opportunities or whether a candidate state predicts a later choice. The inference should remain tied to that task, population, manipulation, and timescale rather than expanding because the apparatus looks precise.

Ecological transport must be earned. Effects discovered in students, brief sessions, or tightly controlled environments may change under fatigue, medication, clinical burden, culture, or repeated real-world exposure. Held-out settings and independent sites matter more than a second random split from the same session (Hernán and Robins, 2020).

The laboratory's proper contribution is now concrete: narrow questions, explicit artificiality, safe manipulation, capable rivals, and failure conditions that can actually close a route. The reader will see why an endpoint-like task remains a task and why public explanation must stop before it becomes an invitation to self-experiment.

17.1.1 — Horizon Manipulation

A safe horizon task might vary the number of remaining rounds, the time allowed before response, the possibility of later revision, or the visibility of an upcoming closure point. A repair-horizon task might allow one group to correct an error later while another group must live with a final response inside the task. The design tests perceived remaining opportunity under reversible, low-burden conditions rather than simulating a life ending.

The formal expectation is not that short horizon proves the Law. A useful laboratory model would look for changes in burden-relevant estimators. For example, a prospectively specified study might test whether a short-horizon condition increases hesitation, autonomic arousal, error monitoring, conflict signals, or subjective pressure relative to a long-horizon condition, while also examining whether repair availability moderates that effect.

A good horizon study therefore begins before data collection with the question of burden. What is being shortened? What remains reversible? What does the participant believe is at stake? Can the participant withdraw without penalty? Is the manipulation transparent enough after debriefing? Is the affective load minimal? Horizon manipulation becomes acceptable only when the future narrowed in the lab remains a bounded task future, not a moral, social, or personal threat.

For a publishable research program, the prewritten protocol should state the intended horizon construct, the task endpoint, the expected interaction, the competing ordinary models, and the public wording allowed after success or failure. Without that

prospective record, a horizon task can become a flexible story rather than a test.

Short horizon may become bodily before it becomes verbal. It can appear as faster breathing, elevated arousal, narrowed attention, stronger error sensitivity, increased caution, impulsive closure, or the felt sense that a task is tightening. These traces are not a cumulative history, but they can test whether perceived opportunity changes the local burden of action under a declared causal design (Hernán and Robins, 2020).

The most important design distinction is between urgency and threat. Urgency can be ethically studied in modest tasks. Threat can become harmful quickly. A short horizon should not make participants believe their dignity, care, income, relationship, safety, or standing is at risk. The task may create pressure, but it must not create existential danger, social degradation, or durable fear.

Short-horizon pressure also has an ethical communication problem. Public audiences may hear the phrase and imagine death, catastrophe, or dramatic personal loss. The lab phrase must stay narrow. It means a bounded experimental horizon, not the lived horizon of a terminal diagnosis, prison sentence, poverty trap, war zone, or collapsing relationship. Those lived horizons require care, law, policy, and human support, not laboratory analogy.

Imagine two versions of a small puzzle. Both give you one minute and show the same countdown. In one, you may revise an earlier answer before the game ends. In the other, each submitted answer

is final. Nothing outside the game depends on your score. The clock is identical; what changes is whether there is a way back.

That difference makes a sharper question than whether a countdown creates pressure. How much does the pressure depend on the seconds remaining, and how much on the possibility of revising a mistake? A proposed repair mechanism becomes interesting if it predicts a difference that time pressure alone misses. The task would still have to show that people understood the available choices. A small experiment could then turn an elusive idea about opportunity into a specific relationship that other explanations must also account for.

Individual understanding should be checked. Participants may misread how many opportunities remain, distrust the displayed countdown, or use a different strategy once the task feels evaluative. Comprehension and strategy reports can reveal whether the manipulation altered the intended representation rather than merely changing stress.

Transport is the final constraint. A horizon-sensitive choice in a laboratory may inform models of planning or recovery while saying little about addiction, chronic illness, or terminal experience. The result earns broader relevance only through additional studies that bridge timescale, context, and population.

17.1.2 — Lab Artificiality

Laboratory artificiality is not an embarrassment to hide. It is the condition under which the laboratory becomes ethically possible. The task is bounded. The stakes are limited. Participants can leave. Deception, if any, is reviewed and debriefed. The room is

not the participant's whole world. These features make the lab less like life, but they also protect people from the lab becoming life.

The danger comes when artificiality is forgotten after a result appears. A clean contrast, a small p value, a striking brain image, or a well-fit model can make a task feel deeper than it is. The Law of Fairness must resist that temptation. A task creates a narrow disturbance in a controlled frame. It cannot reproduce biography, deprivation, chronic pain, childhood, grief, love, aging, social power, or terminal closure (Shiffman et al., 2008).

A scientifically honest study reports artificiality as a limit. It states the task stakes, duration, population, consent process, debriefing, exclusion criteria, and context. It avoids language suggesting that the task captures a life-level horizon. It distinguishes internal validity from external validity. It treats transfer beyond the laboratory as a hypothesis requiring further evidence.

Ignoring laboratory artificiality shifts the cost of overclaim onto people who did not design the study. Readers may infer conclusions about suffering, death, or repair from a game-like task, and institutions may use preliminary findings outside their intended setting. External-validity limits therefore belong in the main claim rather than in a footnote (Shiffman et al., 2008).

Compensation should recognize participants' time and burden without making withdrawal financially punitive or exerting undue influence over participation. Chapter 12 states the full standard; laboratory protocols should apply it rather than recreate it here.

Laboratory privacy is a substantive design constraint. Facial expression, voice, physiology, brain images, self-report, behavior, and social interaction can expose intimate facts or enable new inferences. Collection should be minimized, secured, purpose-limited, and separated from public-facing tools, institutional records, and artificial-intelligence training by default. If that separation cannot be maintained, the task is unsafe for the intended purpose (Shiffman et al., 2008).

Artificiality also marks the limit of transport. A participant who knows that the stakes are small, the task will end, and compensation is secure may respond differently from someone whose employment, caregiving, health, or safety is genuinely constrained. The laboratory can identify local mechanisms under controlled conditions, but it must earn every step from that bounded room to ordinary life.

Laboratory control is purchased by removing much of a life. A task can isolate a short horizon, a reversal, or a social cue, but it also strips away biography, chronic illness, relationships, material constraints, and the accumulation that gives the root question its scale. Ecological momentary work can restore some context, yet even repeated daily sampling remains an observation window rather than a complete history (Yarkoni, 2022; Shiffman et al., 2008).

A laboratory result should carry an explicit jurisdiction statement: what the task manipulated, which stakes were absent, how long the effect was observed, and which populations and settings were not represented. Generalization then becomes a new empirical question, tested through independent tasks, sites, and observation

channels rather than granted by the precision of the original graph (Shiffman et al., 2008; von Elm et al., 2007).

17.1.3 — No Lab Task Proves the Law

A laboratory task can test a component, not a life. A countdown can manipulate perceived time, a reversal task can test updating, a mild stressor can test recovery, and a brain measure can examine a candidate process. Laboratory control is achieved by removing much of the context that gives a life its scale, so the same simplification that makes a local contrast estimable also keeps the conclusion local.

A participant may sit before a screen while the remaining time suddenly drops from two minutes to ten seconds. Heart rate rises, attention narrows, and choices change. The experiment can ask whether perceived horizon caused those local changes under its conditions. It cannot call the ten-second interval a miniature death or infer what the participant's whole life requires.

Consciousness tests add another layer of caution. No single measure is universally valid across healthy adults, brain injury, sleep, anesthesia, infants, animals, or artificial systems. Tests must be linked to theories, populations, capacities, and known failure modes rather than treated as one meter of presence (Bayne et al., 2024).

Neural evidence supports only the process-level inference earned against alternatives; Section 17.2.2 develops the reverse-inference limitation (Poldrack, 2006).

A useful claim ladder keeps the scale visible. The weakest result may be that a measure changed under one task. Stronger results

may show reliable within-person effects, held-out prediction, transport across tasks or sites, and causal sensitivity to a safe perturbation. Route-level support requires further model recovery and rival comparison. Complete-stream support remains outside the laboratory task itself.

External validity cannot be added by rhetoric. A result from a convenient sample, brief task, artificial stakes, and one measurement channel must earn transport to other populations and settings. Failure to transport may reveal a boundary condition, a measurement problem, or a task-specific effect rather than a universal mechanism.

The public sentence should name the component. Researchers may say that shortening a task horizon changed reported urgency or that a perturbation altered recovery under the declared conditions. They should not say that the laboratory reproduced terminal closure, discovered a fairness center, or showed that suffering will later be balanced.

The same boundary prohibits self-testing. Readers should never be invited to create severe stress, deprive themselves of sleep, manipulate medication, simulate death, provoke another person, or monitor a crisis in search of a signature. A valid experiment requires ethics review, trained staff, bounded burden, and independent stop authority. The same rule applies to remote or consumer tools: convenience does not convert unsupervised personal experimentation into a valid laboratory protocol.

Failure conditions should be prospectively specified before data collection. The task may fail because the manipulation was weak,

the measure was unreliable, the effect did not transport, a rival fit equally well, or the necessary burden was ethically unacceptable. These outcomes narrow different parts of the program and should not be blended into one vague call for more research (Hernán and Robins, 2020). Predeclared failure types also help the next study distinguish a weak manipulation from a failed mechanism or an ethically unavailable test.

Safe perturbation is most informative when it changes one causal feature while preserving the participant's ordinary rights and alternatives. Mild horizon shifts, reversible reward changes, attention cues, or noninvasive measurement contrasts can test local mechanisms without manufacturing crisis. The protocol should specify the intended pathway, foreseeable spillovers, recovery period, and stop rule. If several systems change together, the result may still be valuable, but its causal interpretation must remain correspondingly narrower. A useful perturbation is therefore one whose recovery path can be observed safely enough to support the causal question it was designed to ask.

17.2 — Signatures and Nonunique Mechanisms

A candidate signature is a pattern that a proposed route predicts, not a fingerprint that belongs exclusively to the Law. Shrinking variance, recovery after perturbation, horizon-sensitive choice, or a relation between early residuals and later responses can arise from several ordinary mechanisms.

Nonuniqueness is the central problem. Adaptation, reinforcement learning, expectation, regression to the mean, integral control, selection, measurement floors, and conditioned paths can imitate one another over a short window. A single elegant curve is therefore weaker than a conjunction of independent predictions.

Transport and rival comparison provide the pressure. The signature should survive new participants, tasks, sites, measurement channels, and prespecified perturbations while beating capable alternatives given the same information and tuning budget. If several models recover one another under realistic noise, the verdict is model ambiguity.

17.2.1 — Short Horizons Are Not Whole Streams

A short task can reveal local recovery, contrast, learning, fatigue, or regulation. It cannot reproduce the complete history, open future, changing body, social world, and truth-level endpoint of a conscious stream. The task endpoint is an administrative boundary chosen by investigators; the proposed terminal boundary is part of the hypothesis and cannot be replaced by the moment the screen turns off.

Short horizons should be created through modest limits on time or opportunity, not through threats to dignity, income, care, relationships, or safety. Social-evaluative threat can produce strong physiological responses that have little to do with remaining opportunity, so perceived evaluation and humiliation must be measured and controlled rather than folded into the target effect (Dickerson and Kemeny, 2004).

A candidate brain map can quickly become a myth. Activation does not equal experience, absence of activation does not equal absence of experience, and region involvement does not establish causation or access to cumulative history. Brain-based language should therefore remain tied to task roles: salience may concern burden or conflict, control may concern regulation, valuation may concern choice, and interoception may concern bodily feeling. The interpretation comes from the design, not from the visual aura of a colored patch (Poldrack, 2006).

Control under constraint is a candidate relationship among narrowing horizon, felt burden, repair options, and regulatory effort. When later repair becomes unavailable, a task may recruit more monitoring, inhibition, caution, or urgency. This is a proposed local prediction: the study must distinguish a horizon interaction from generic time pressure and social-evaluative stress.

A possible prediction is interactional rather than simple. Short horizon under high affective load and low repair availability may recruit stronger control-related patterns than long horizon under low load and available repair. But this prediction must compete with ordinary models: time pressure, working-memory demand,

loss aversion, scarcity cognition, and arousal may explain the same pattern.

The distinction also protects dignity. Brain-based claims can sound intimate and final. People may be told, explicitly or implicitly, that their scan reveals burden, truthfulness, maturity, risk, hidden suffering, or moral readiness. The ethical boundary of the project forbids that drift. A brain image cannot become a public rank, a family accusation, a clinical shortcut, a civic trust label, a religious or spiritual status, or an employment tool.

The bridge to the terminal claim requires additional assumptions about scale, bearer, remaining opportunity, and endpoint that the laboratory task does not observe.

Short-horizon recovery can reveal regulation, stress response, learning, or local repair without representing complete-stream closure. The same rebound may arise from allostasis, opponent processes, habituation, expectation, or task strategy. A useful experiment specifies which local dynamic it tests and which features would separate those accounts, rather than treating any return toward baseline as a miniature version of the terminal claim (McEwen, 1998; Dickerson and Kemeny, 2004).

The methodological core of the short-horizon limitation is to separate timescales, predeclare the observation window, and test whether parameters remain stable when that window changes. The scientific companion must compare nested horizons and recover the generating process before any laboratory interval is interpreted as evidence about a complete stream.

17.2.2 — Nonunique Control-Hub and Recovery Signatures

Temporal order can make a local mechanism question sharper. Researchers can ask whether bodily arousal precedes reported doubt, gaze hesitation precedes conscious explanation, control engagement rises when later repair becomes unavailable, or subjective pressure changes after an irreversible choice. These patterns concern the path through a bounded task, not the final state of a life (Marek et al., 2022).

A modest task might pair immediate reward with later correction burden or compare choices with and without a safe opportunity for repair. Rival explanations must enter from the beginning: loss aversion, regret anticipation, cognitive dissonance, social desirability, demand characteristics, moral norms, and ordinary emotion regulation may all produce valence-conflict effects. The task is informative only when those accounts can win (Marek et al., 2022).

Neural and physiological signals are multiply caused. Activity associated with control, salience, interoception, recovery, or conflict may also reflect arousal, effort, attention, surprise, pain, motor preparation, or task difficulty. Anatomical precision does not remove this ambiguity, and reverse inference remains a central risk (Poldrack, 2006; Marek et al., 2022).

Converging channels help only when they do not repeat the same error. A proposed signature should predict held-out behavior or physiology beyond simpler task features, survive prospectively specified ablations and capable rivals, and transport to a meaningfully different task or sample. Otherwise a striking

pattern may describe the original apparatus more than the proposed mechanism (Marek et al., 2022; Wilson and Collins, 2019).

Individual differences can modify both experience and signal. Chronic pain, anxiety, medication, disability, trauma history, sleep loss, and familiarity with research settings may change how the same task recruits control and interoception. Heterogeneity should be modeled as possible effect modification, not discarded as noise or converted into a ranking of people (Marek et al., 2022; Yarkoni, 2022).

Public graphics require the same restraint as the analysis. A colorful brain map must name nonuniqueness, avoid person-level diagnosis, and never imply that activation is a ledger, a consciousness meter, or direct access to inner truth. If the prospectively specified neural pattern disappears after simpler features receive equal treatment, the local mechanism claim narrows rather than changing its definition after inspection (Marek et al., 2022; Wilson and Collins, 2019).

Control and recovery measures can therefore strengthen or weaken one local mechanism hypothesis. They do not become the terminal target. Complete-stream adjudication still requires independently justified representation, neutral origin, ownership and routing, truth-level endpoint, observation, missingness, and measured-to-latent bridges.

No local hub can be promoted into an administrator of fairness. The word controller describes a functional dependency in a

candidate model. It does not imply a judge, planner, moral center, or organ that keeps a person's experiential account.

17.2.3 — A Signature Must Transport and Beat Rivals

A candidate signature is useful only when its variable, direction, time window, population, preprocessing, and expected relation are fixed before decisive outcome inspection. A pattern discovered after many analytic choices is a hypothesis for later testing, not a confirmed route.

Transport is multidimensional. A route claim should state whether it is expected to survive a new task, site, device, preprocessing pipeline, time point, language, age group, health context, or measurement channel. Each change probes a different dependency, so success in one dimension cannot be cited as general transport (Marek et al., 2022; Yarkoni, 2022).

Model recovery should come before route preference. Investigators should simulate data from the proposed route and capable rivals, including realistic noise, missingness, and observation artifacts, then test whether the planned pipeline identifies the generator. If it cannot recover known simulated truth, a human-data success cannot identify which process produced the result.

Rivals need comparable access to history, physiology, treatment context, and task timing, together with comparable freedom to represent nonlinearities, delays, and hidden state. A reserve-linked model cannot receive rich state information while adaptation, control, bridge, medication, and observation-floor rivals are reduced to one crude term. The comparison should

publish the information boundary and tuning budget before ranking the models.

Open-science practices make rescue harder when described accurately: protocol-fixed analyses, public preregistration with a persistent record, Registered Reports, code, provenance, and replication. They do not replace valid measures, capable rivals, or transport.

A strong route result is conjunctive: recovery succeeds, direction appears, held-out forecasts calibrate, negative controls remain null, transport survives, and capable rivals lose under comparable information and flexibility.

External validation should distinguish frozen transport, recalibration, and refitting. New intercepts, scales, or features change what the result establishes and must not be described as unchanged.

Consider a candidate route that links accumulated experience to changes in one independently specified physical quantity. If that relation holds on every admitted path, two paths with the same sufficient starting state and the same sufficient finishing state must accumulate the same signed total, even when their sequence differs. This offers a concrete challenge: fix the state description and scale first, then compare paths not used to build the model. A reproducible difference beyond justified uncertainty would challenge that route. Matching totals would still leave ordinary explanations and the complete-stream boundary to be tested. The states must match in every respect relevant to the claimed relation; similar mood at the finish is not enough. A prediction

about every path also requires stronger evidence than agreement only in the average.

Public artifacts inherit the same status. When a signature fails transport, papers, releases, diagrams, slides, and websites should be corrected or withdrawn together rather than leaving the attractive overstatement online.

Subgroup differences may reveal task or model boundaries but cannot rank age, culture, disability, medication, adversity, balance, or recoverability.

Even transported signatures remain route evidence. At most, they justify the conclusion that a specified route survived specified rival tests in a specified domain, not a verdict on complete lives.

17.3 — Perturbation, Control, and Causal Limits

Perturbation can reveal whether a variable participates in a local mechanism. Changing opportunity, delay, feedback, expectation, or a safe stimulus and observing a downstream response is more informative than correlation alone. It still depends on assumptions about what changed and what else changed with it.

Expectancy, demand characteristics, spillover, imperfect blinding, adaptation, and multiple affected pathways can imitate causal specificity. Sham conditions, manipulation checks, alternate outcomes, and temporal controls help locate the effect rather than merely confirm that something happened (Pearl, 2009; Hernán and Robins, 2020).

The ethical limit is not an appendix. Serious distress, withheld relief, sleep deprivation, simulated death, crisis exposure, or unsupervised stimulation cannot be justified by the desire for a decisive result. Minimal-risk alternatives, expert screening, independent review, monitoring, and immediate stop authority define the admissible question (Rossi et al., 2021).

17.3.1 — Why Perturbation Matters

Perturbation matters because correlation can mislead. A control network may become active during horizon pressure because it causes the response, reflects effort or arousal, follows the decision, or participates in several pathways. A safe intervention can separate some of those possibilities when its assumptions and spillovers are explicit (Hernán and Robins, 2020; Pearl, 2009).

The design should begin with a modest local mechanism, not the fantasy of controlling fairness. A study might ask whether changing feedback, delay, expectation, or a control-related neural state alters hesitation, error monitoring, recovery, or choice within one task. Even a successful result remains evidence about that task mechanism rather than a discovered fairness switch.

The causal estimand must be named before interpretation. Assignment to a procedure, actual target engagement, dose received, and change mediated through one pathway are different comparisons. Conditioning on post-assignment response can create bias, while ignoring target engagement can make mechanism failure indistinguishable from intervention failure (Hernán and Robins, 2020; Pearl, 2009).

The intervention also opens new causal paths. Expectancy, discomfort, fatigue, learning, experimenter behavior, and spillover can change the outcome alongside the intended target. Sham conditions, target-engagement checks, manipulation fidelity, and an explicit causal diagram are therefore part of the meaning of the result rather than technical decoration added after a striking effect appears.

A nominally identical procedure may alter the target differently across participants. The analysis should preserve assignment, achieved modulation, sensations, adverse effects, and protocol deviations without treating them as one variable. A null target-engagement check may leave the mechanism untested, while a broad behavioral change may remain causally ambiguous.

Social interference can defeat an individual-level interpretation. Participants may discuss the task, observe another person's reaction, compete, cooperate, or receive cues from a shared experimenter. When one person's assignment can affect another person's outcome, the unit of intervention and the unit of analysis must match the social structure rather than hiding dependence inside standard errors (Hernán and Robins, 2020).

Causal leverage creates no ethical exemption. As a procedure becomes more invasive, stressful, difficult to reverse, or burdensome to understand, its incremental information must be justified against safer alternatives. Noninvasive brain stimulation still requires trained operators, screening, monitoring, and method-specific safety controls rather than the reassurance of a label (Rossi et al., 2021; World Medical Association, 2024).

Consent should explain in ordinary language what will happen, what sensations and uncertainties are known, which alternatives exist, what screening information is collected, and how to stop. Medical, psychiatric, medication, reproductive, and device information gathered for safety should be minimized and should not become general research data by default.

The public claim remains narrow. A perturbation effect can show that a specified variable contributed to a specified response under specified conditions. It cannot show that researchers moved a cumulative history, balanced a life, identified a person's hidden ledger, or validated the terminal law (Pearl, 2009).

Failure must also be typed. The target may not have changed, the manipulation may have produced competing effects, the causal

assumptions may have failed, or the predicted response may truly be absent. A good perturbation study distinguishes those outcomes and permits the local mechanism to narrow without turning participant burden into a reason to preserve the preferred story.

17.3.2 — Ethical Limits on Manipulation

Ethical limits on manipulation are non-negotiable. A laboratory test may vary horizon, repair opportunity, feedback, timing, or task difficulty. It may not create trauma, humiliation, panic, dependency, fear for safety, social exposure, or durable distress. It may not withhold care. It may not recruit people in crisis because their burden would produce stronger signal (Rossi et al., 2021).

This rule applies even if a stronger manipulation would be scientifically tempting. The Law of Fairness cannot be tested by violating the public analysis. If a design requires moral ugliness in order to produce a clear effect, the design is not bold. It is disqualified. The field must become more inventive rather than more cruel.

Manipulation limits should be written into the protocol as stop rules and design ceilings. How intense can frustration become? How long can uncertainty last? What feedback is allowed? What deception is prohibited? What distress rating ends the task? What signs require operator intervention? What happens if a participant freezes, cries, dissociates, panics, or asks whether they failed as a person?

The manipulation must also respect repair after the task. Debriefing is not ceremonial. It should explain the artificiality of the task, correct misconceptions, remove personal stigma, provide support information when needed, and make withdrawal or data removal options clear where possible. A study that leaves participants carrying false meaning has shifted burden downward (Beauchamp and Childress, 2019).

The ethical limit includes bystanders and nonparticipants. Social tasks may involve partners, confederates, family-like roles, or group interactions. No one near the study should be made collateral material for a horizon manipulation without consent and protection. The burden of research must not leak into relationships casually.

Sham and control conditions are essential because perturbation carries expectation. A participant who feels a coil, hears a click, sits under equipment, or believes their brain is being modulated may change behavior because of meaning, not mechanism. Without sham control, the study may confuse intervention with suggestion.

Monitoring should be multimodal where appropriate. Participants may report discomfort directly. Operators may notice behavioral signs. Physiological signals may show arousal. Task behavior may show freezing, repeated errors, or withdrawal. None of these channels is perfect. Together they can help prevent the study from becoming blind to the burden it creates (World Medical Association, 2024).

Minimal-risk design begins by asking whether the same causal contrast can be obtained with a weaker manipulation, shorter exposure, observational design, or natural variation. Independent safety review, real-time monitoring, participant-controlled stopping, and clinically appropriate exclusion or support should be built into the protocol rather than added after distress appears (CIOMS, 2016; World Medical Association, 2024).

A proposed stress induction may be scientifically cleaner than an observational comparison and still be rejected because a safer design can answer the narrower question. That decision does not weaken the research program. It identifies the boundary within which knowledge remains worth having.

The ethical boundary is not a rhetorical line around the experiment. It determines which causal questions may be asked. A design that needs humiliation, untreated pain, prolonged panic, deprivation, or crisis leverage to produce a clean effect is not made courageous by its scientific ambition; the question must be redesigned or left unanswered (World Medical Association, 2024).

Publication should include the burden actually observed. A study that produced distress, withdrawal, or lingering effects should report those outcomes alongside the scientific result. Ethical cost is part of whether a method deserves replication, not an invisible price paid for information.

17.3.3 — Scientific Protocol Handoff

The public root can explain the logic and ethical boundaries of a fair test, but it should not pretend to supply a runnable protocol.

A proper handoff names the target, branch, reference class, intervention or exposure, comparator, endpoint, observation process, rival set, claim ceiling, and unresolved dependencies. A design that can return only support, reinterpretation, or vague inconclusiveness has insulated the hypothesis rather than tested it (Schulz et al., 2010). Before enrollment, the question, population, timing, harms, data flow, and analysis should be prospectively specified; SPIRIT 2025 strengthens that discipline through transparent methods, oversight, handling, and dissemination (Chan et al., 2025).

The handoff should include a causal graph, not merely a prose hypothesis. Investigators need to declare what is manipulated, which mediators may change, which pathways are unwanted, what confounds the comparison, and which post-treatment variables must not be adjusted away. Group and online settings must address interference when participants observe one another, discuss the task, infer its purpose, or respond to the experimenter (Pearl, 2009; Hernán and Robins, 2020). For example, the graph can show whether a participant's response is expected to depend on information received from another member of the group.

Some constraints travel unchanged: no harm for proof, no crisis extraction, no hidden score, no public-worth inference, no conversion of operational records into research data by default, and no stimulation or affective manipulation without expert review, consent, monitoring, and independent stop authority. Communication stays within the protocol boundary, with release claims fixed in advance and tied to the design's supported result. A rehearsal can test how a participant reaches the stop procedure

and how the team communicates the decision during an interrupted session.

Exact estimands, sample-size calculations, randomization, multiplicity, missing-data methods, adverse-event rules, recovery checks, and confirmatory verdicts return to the governed scientific program. The handoff is complete when each analysis has a declared decision and each route an observation forcing narrowing, retirement, or abstention. An experiment becomes useful by limiting its claim before the first participant arrives. The resulting protocol should let another team reconstruct the intended comparison from the declared inputs through the analysis to its decision.

17.4 — Negative Controls and Rival Tests

Suppose a model is said to detect the effect of a cue shown halfway through a task. The cue was chosen at random only after the first part had been recorded. Now ask the model to find the same effect in that earlier record. A recurring apparent effect before its proposed cause would send us back to the measurement and analysis. The model might be reading an earlier difference between groups, a recording artifact, or information that slipped into the analysis.

This is the logic of a negative control: look where the proposed cause should have no effect. Here, discovering nothing can be part of discovering something. A proposed mechanism becomes more informative when it predicts both where a pattern should appear and where it should be absent. Those paired predictions give curiosity a sharper question than whether an impressive signal can be found.

Specificity now becomes the central test. A candidate result must survive the condition that is supposed to produce it, weaken or disappear in matched controls, and remain separable from social influence and alternate channels. Failure at any of those points limits how far the signature may travel.

17.4.1 — Social Influence

Social influence enters almost every laboratory task. Participants may infer what the experimenter wants, follow perceived norms, adjust to peers, imitate confident actors, avoid looking selfish, or perform competence. A horizon effect may partly be a social

effect if the participant believes urgency is expected, valued, or judged.

A serious laboratory program therefore asks not only what the task did, but what the participant thought the task meant socially. Without that question, the study may be measuring compliance with a room rather than pressure from horizon.

Social-influence controls should also consider researcher identity and institutional setting. Participants may respond differently to authority, prestige, gender, age, race, culture, accent, clinical status, or perceived ideology. The laboratory's social field begins before the task begins (Coviello et al., 2014).

The burden risk is manipulation. A study may use authority, peer pressure, or status to intensify affect because social levers are powerful. That must be constrained. Participants should not be made to feel publicly exposed, morally condemned, or trapped in group judgment. Social tasks must remain dignified and debriefed (Coviello et al., 2014).

Emotional contagion is the spread or alignment of affect across persons. In a lab, a participant may become tense because others are tense, calm because the operator is calm, competitive because peers are competitive, or ashamed because the group atmosphere implies shame. The measured affect may be real without being generated by the intended mechanism.

Contagion also creates nonparticipant risk. In some studies, emotional burden can leak beyond the recruited participant into friends, partners, group members, or bystanders. If someone's distress is triggered by a participant's task or debriefing, the

research burden has escaped its consent boundary. That possibility must be anticipated (Coviello et al., 2014).

Expectancy can also travel through public materials. If participants enter a study after reading advocacy for the Law, that framing may shape reports, persistence, and interpretation. Serious studies should separate neutral recruitment language from public argument wherever feasible so that participation does not begin with emotional priming toward support.

Observation can feel exposing even when records are confidential. Participants may worry that sensors reveal hidden truth or that their bodies are being judged. The study must explain that the measures are indirect, non-diagnostic, and non-scoring while minimizing every recording not needed for the declared question (Coviello et al., 2014).

Social transmission is itself a candidate causal channel. Emotion can spread through interaction and shared environments, while demand characteristics can change what participants report or display. Designs should separate exposure to others' states, beliefs about the researcher's expectations, and the structural features of the task instead of treating all three as one fairness response (Coviello et al., 2014).

Separate contagion, homophily, shared exposure, platform intervention, and measurement spillover. Suppose friends report similar moods after a storm. The pattern could reflect communication, common conditions, selection into the same environment, or selective reporting. The study should predeclare which relation would distinguish these accounts and what result

would count against social influence. Even a successful local test remains a bounded causal inference, not a miniature proof of a lifetime law.

Platform experiments create additional governance duties. A manipulation delivered at scale can affect people who did not meaningfully understand that research was occurring, and social effects may extend beyond the sampled account. Ethical review, transparency, minimization, and post-study assessment must match the reach of the intervention (Kramer et al., 2014).

Most importantly, correlated valence remains separate across streams. Social connection can change experience through ordinary causation, care, conflict, or contagion. It does not create a shared account in which one person's joy cancels another person's pain.

17.4.2 — Non-Horizon Tasks

A non-horizon task should preserve ordinary task demands while removing the feature claimed to be distinctive. It may match difficulty, duration, reward, response format, and observation burden while removing finality, shrinking opportunity, or the loss of later repair. The comparison asks whether the proposed effect depends on horizon rather than on the general fact that one condition is harder or more dramatic (Pearl, 2009).

If the same pattern appears when no meaningful opportunity window is closing, the horizon mechanism weakens. Time pressure, novelty, arousal, cognitive demand, social evaluation, or analytic artifact may be sufficient. The theory cannot answer by declaring that horizon was secretly present in every task.

Each control must isolate the right burden. When the target condition adds urgency, the design should separate urgency from finality. When it adds social observation, the design should separate observation from horizon. When it adds affective load, the design should separate valence from cognitive demand. Control is conceptual precision, not a checkbox (Pearl, 2009).

Neutral-valence and matched-reward tasks provide further tests. A task can preserve timing and structure while reducing meaningful positive or negative burden; another can preserve incentive while leaving later correction available. A Law-specific affective interaction should weaken under those controls. If it does not, the result may reflect attention, timing, or strategy rather than felt burden.

Manipulation checks are required on both sides. The target condition must actually change perceived remaining opportunity or repairability, and the control must not accidentally create the same perception. Participant report, behavior, and physiology can help establish that local contrast without becoming direct readings of a hidden lifetime process.

Observation flow must also be comparable. If distressed participants leave the target condition more often, the retained sample may look unusually regulated; if the control is shorter or easier, it may keep people who would otherwise withdraw. Reports should show missingness, exclusions, and burden by condition before claiming specificity.

Social tasks can involve partners, peers, confederates, observers, or people named in prompts. Their voices, faces, messages, and

emotional reactions do not become research material through proximity. Data minimization, bystander protection, debriefing, and escalation review are part of a valid control design because burden can spill outside the recruited participant (Pearl, 2009).

A strong result survives several capable controls chosen for different rivals, with the interaction, smallest meaningful effect, and failure rule fixed in advance. A weak result may disappear under the first fair comparison. Both outcomes are useful: one narrows the rival set, while the other retires an overbroad interpretation before it becomes a public claim (Nosek et al., 2018; Munafò et al., 2017).

Even a surviving non-horizon comparison supports only the bounded mechanism tested. It does not establish a terminal law, a personal forecast, or a moral diagnosis. The evidentiary ceiling is set by the prospectively specified contrast, observation model, population, and loss condition, not by the drama of the task.

17.4.3 — Negative-Control Failure

A failed control does not always reject the target Law. It may refute one task interpretation, one measurement bridge, or one candidate mechanism. The report should state exactly which link failed and which claims remain untouched. Typed failure is stronger than a dramatic but ambiguous verdict.

Partial failures matter. A signal may survive a sham comparison but fail a non-horizon task, or transport across tasks while disappearing at a second site. Those patterns can identify expectancy, site, preprocessing, or measurement dependence.

Collapsing them into pass or fail discards the very information controls were designed to provide.

Controls should be protocol-fixed with the primary analysis. Adding a convenient control after the result, changing its exclusion rules, or choosing a different outcome can protect the preferred story from the test it was supposed to face. Prospective specification and transparent deviation logs make that rescue visible (Nosek et al., 2018).

Model recovery should include control worlds. Simulated data can show whether the pipeline falsely discovers the signature when only demand, regression to the mean, common exposure, or measurement drift is present. If the analysis calls those worlds route-positive, a real-data victory is not interpretable.

A negative-control failure should propagate. The route may narrow, the finding may become exploratory, public materials may require correction, or the task family may be retired. The project should not preserve a strong headline while quietly weakening the methods section (Munafò et al., 2017).

The result can be no winner. When the primary condition and controls are all poorly fit, unstable, or confounded, the correct status may be open-set inadequacy or No adequate estimate. Forcing a preferred interpretation among failed comparisons creates certainty where the design produced none.

17.5 — What Laboratory Failure Would Mean

Laboratory failure has several forms. A precise null can exclude an effect large enough for a route to require; an imprecise study may support No adequate estimate; a failed manipulation may leave the mechanism untouched; a capable rival may explain the observation better; and an ethically excessive design should stop before recruitment. Each outcome changes a different part of the scientific map.

Equivalence and uncertainty matter. A nonsignificant result does not prove absence, but a sufficiently precise comparison can exclude a prespecified meaningful range (Lakens, 2017). Conversely, a statistically detectable effect may be too small, unstable, or poorly transported to carry the proposed mechanism.

Some tests should fail before recruitment because the ethical burden is too high. Retiring a design that would intensify distress, compromise care, exploit dependency, or invite unsafe imitation is not scientific cowardice. It is a finding about which questions can be asked responsibly.

17.5.1 — No Horizon Signature

A missing horizon signature narrows the route only when the manipulation changed the intended construct, measurement was reliable, and the study was precise enough to exclude the meaningful range. Otherwise the result identifies manipulation, design, or estimation failure. This distinction prevents endless rescue without pretending that every null decides the question (Lakens, 2017; Wasserstein and Lazar, 2016).

Public interpretation should change in proportion to the result. An equivalence result can narrow a route; an imprecise null supports more uncertainty; failed manipulation means the intended question was not asked; and failed transport weakens generality. Books, guides, talks, and public summaries should name that exact change rather than preserve the earlier promise.

A failed horizon route can redirect inquiry toward longer timescales, repair expectations, longitudinal designs, or ordinary rivals. It can also prevent expensive repetition and show that the laboratory is the wrong instrument for the mechanism. The gain is a cleaner scientific map, provided the revision receives a new prospective identity rather than being presented as the original prediction succeeding.

Replication tests whether the result survives beyond its first sample, operator, pipeline, institution, culture, wording, device, and chance events. An exact repeat assesses repeatability; deliberate changes in site, population, or task test transport. Failure should identify the weakened route or boundary condition before the original result continues to travel publicly (Open Science Collaboration, 2015).

17.5.2 — Low Power or Bad Design

Low power and bad design are different failures. Low power means that, under a specified alternative, the planned test has a low probability of rejecting the null hypothesis. Imprecision concerns the uncertainty of the estimate; it is related to power but is not the same quantity. Bias, leakage, invalid measurement, uncontrolled analytic flexibility, or a failed manipulation may

prevent the intended inference even with a large sample. The report should state which failure occurred and what result a redesigned study would need to close the line.

Sample size should be justified against a declared effect and decision. A feasibility study may need precision for recruitment and burden. A mechanism comparison may need power to distinguish predictions that are close. An equivalence test must be able to exclude effects larger than its prospectively specified tolerance. Saying only that the study was powered leaves the scientific question unstated.

Reliability sets an upper limit on what a measure can support. A noisy outcome can make a real effect look unstable and can also inflate selected effects when many analyses are tried. Pilot estimates of reliability, repeated measurement, and transparent uncertainty should precede grand interpretation.

Flexible exclusions, transformations, time windows, covariates, region definitions, and stopping rules can manufacture apparent stability. Prospective specification does not rescue a weak question, but it separates a planned test from a search through many possibilities. Specification-curve or multiverse analyses can show whether the conclusion depends on one convenient choice (Simmons et al., 2011; Nosek et al., 2018).

Point estimates should not outrun their intervals. A wide interval that contains both no useful effect and an effect large enough to matter is unresolved, even when the estimate points in the favored direction. Precision is part of the result rather than a decoration added after a significance test (Wasserstein and Lazar, 2016).

Replication should challenge the original claim through an independent team, new sample, frozen analysis, and comparable burden. Multisite work should also report site-specific estimates because language, training, equipment, trust, demographics, and implementation may reveal genuine transport limits rather than noise (Open Science Collaboration, 2015; Marek et al., 2022).

Sequential designs can reduce waste when their interim rules are fixed in advance. Safety, feasibility, futility, or overwhelming evidence may justify stopping, but repeated unplanned looks invite a favorable stopping time. Timing is part of the design and must not become another hidden degree of freedom.

Before treating a laboratory failure as low power or bad design, verify that the full pipeline can recover the favored mechanism, strong rivals, nulls, missingness, and measurement artifacts under realistic simulation; Section 28.4.1 gives the program-wide recovery standard.

Redesign should not become endless rescue. One failed study may justify a better instrument or stronger manipulation; repeated failure under adequate designs must narrow or retire the route. Every new attempt should state what changed, what prior failure it addresses, and what result would close the line rather than merely extend it (Munafò et al., 2017).

The report should therefore separate design failure, imprecision, and route failure. No adequate estimate is correct when the study could not answer the question. A route should narrow when an adequate design excludes the effect it required. Neither outcome

should be advertised as support or defeat of the whole Law unless the prospectively specified entailment truly reaches that far.

17.5.3 — Ethical Burden Too High

A laboratory question can be scientifically coherent and ethically unavailable. Burden includes distress, fatigue, deception, privacy loss, stigma, time, travel, cognitive effort, bystander exposure, and the possibility that a task changes how participants understand themselves. Risk is not limited to physical injury.

The protocol should measure burden rather than merely label it minimal. Completion time, interruption, withdrawal, distress reports, adverse events, missed obligations, accessibility failures, and delayed recovery can be tracked prospectively. Participant experience is evidence about whether the design was proportionate.

Current protocol and reporting standards improve transparency by requiring clearer specification of interventions, outcomes, harms, analysis, participant flow, and open-science materials (Chan et al., 2025; Hopewell et al., 2025). Ethical proportionality remains a separate judgment: a complete checklist can still describe a study that should not be done.

Affective manipulation requires a conservative ceiling. Mild frustration, ordinary reward delay, or brief social evaluation may sometimes be permissible under review. Humiliation, trauma provocation, sleep deprivation, sustained distress, coercive scarcity, relapse induction, or manipulation of vulnerable relationships may exceed what the question can justify. The

project's ethical floor supplies no special reason to move that ceiling upward.

Vulnerability is contextual. A person may be capable of consent yet constrained by illness, financial need, institutional dependency, family pressure, or fear of losing care. Recruitment, compensation, and refusal routes should be designed so that declining the study carries no hidden cost (CIOMS, 2016).

Use the stop-rule architecture in Section 16.4.2 for participant requests, distress, privacy breach, protocol deviation, or clinician concern. Laboratory-specific deception, when indispensable, still needs independent justification, genuine debriefing, and post-study support.

Laboratory data can expose participants and bystanders when signals such as voice, location, messages, sleep, or household events are combined. Apply the minimization and record-separation rules in Chapters 12 and 18 before collection.

Technical sophistication and public enthusiasm do not alter the threshold. A famous theory, donor interest, an urgent deadline, or the promise of a decisive answer cannot convert excessive burden into an ethical one. Safer observational work should be preferred when it can answer the narrower question. Independent review is most valuable when it can require a safer design, not merely note that a burden exists.

When a route cannot be tested under acceptable conditions, the scientific claim narrows to what safe observation can support. The report should state the unavailable contrast rather than allowing readers to imagine evidence that was never ethically obtainable.

Some questions may remain open because human beings are not instruments for completing a theory. Recording the unavailable contrast also prevents a later summary from treating an ethical abstention as missing support that can be filled by speculation.

Stopping is therefore not a defeat of science. It shows that curiosity has not become entitlement, that care remains prior to proof, and that the program can preserve a genuine question without sacrificing the people asked to help answer it (Beauchamp and Childress, 2019; World Medical Association, 2024). That restraint leaves the scientific question available for future methods that can answer it without imposing the prohibited burden.

Chapter 18 — Following Lives with Care

A snapshot can miss the turn. A person may be well during the interview and struggling the next day, or distressed during a crisis and steadily recovering afterward. Chapter 18 asks what becomes visible when observation follows change rather than freezing it.

Longitudinal work matters because the Law concerns a complete stream, while ordinary studies sample moments. Repeated measures cannot create completeness, but they can reveal trajectories, within-person variation, adaptation, response shift, treatment effects, and the timing of missingness.

The central image is a moving denominator. People are invited, enroll, respond, pause, withdraw, move, become ill, change treatment, lose devices, die, or become unreachable. The scientific record must follow those changes rather than presenting the final retained sample as though it had always been the whole group.

Missingness is therefore a mechanism, not a blank cell. Rubin's framework distinguishes forms of missingness by their relation to observed and unobserved data, while later longitudinal methods show how informative dropout can bias estimated trajectories (Rubin, 1976; Diggle and Kenward, 1994; Little and Rubin, 2019).

Death creates a particularly important truncation. A measure collected only among survivors answers a different question from one about a population that includes people whose follow-up ended with death. The analysis must match the research aim instead of smoothing that difference away (Kurland et al., 2009).

Life events change context, opportunity, routine, identity, and experience together, often through pathways that begin before the named event and continue through secondary consequences. Bereavement, unemployment, illness, recovery, caregiving, migration, success, and relationship change therefore resist one-number explanations.

Adaptation research describes average changes after major events, but averages conceal heterogeneity and do not turn later outcomes into scheduled counterparts. Some people recover, some remain changed, and trajectories depend on resources, treatment, safety, relationships, and opportunity (Luhmann et al., 2012).

Response shift matters because a stable number can accompany a changing person, and a changed number can reflect a new standard rather than a simple change in experience. Following a life therefore means following what its measures mean. Section 18.2 examines how standards, priorities and understandings can change over time, and how researchers can investigate those changes instead of assuming that a repeated scale always records the same thing (Sprangers and Schwartz, 1999; Oort et al., 2009).

Prediction is useful when its scope remains explicit. A model may forecast next-week distress, relapse risk, dropout, or treatment response. That is not prediction of terminal closure. The target, horizon, population, calibration, and consequences of error must remain visible.

Model comparison should be prospectively specified before outcome inspection. Adaptation, treatment, selection, ordinary

causal change, measurement drift, and response shift are not nuisance explanations to be added after the target model fails. They are serious rivals that deserve equal scoring and comparable flexibility.

Predictive fit is also not causal explanation. A model may use stable proxies to forecast well while misunderstanding the process. Causal claims require assumptions about intervention, confounding, selection, and transport that prediction alone does not supply (Hernán and Robins, 2020; Pearl, 2009).

Longitudinal privacy risk grows with linkage. Mood, sleep, location, communication, health, and social records can reveal intimate patterns even when names are removed, and data collected for one purpose can become a dossier when joined to clinics, employment, education, platforms, or model training. Consent at enrollment is not unlimited permission for every future use (Nissenbaum, 2004).

Imagine that sleep in a long-running study seems to improve almost overnight. One possibility is a real change in people's lives. Another is that the devices received an update and began counting wakefulness differently. The same bend in a graph can start in a bedroom or in a measurement rule. Keeping a record of changes lets investigators tell these stories apart. Following lives well means remembering how the study itself changed, including what participants agreed to share and how their reports became results.

Citizen science can contribute questions, observations, feasibility evidence, inclusion, and public oversight. It cannot become a

public cumulative-history project, personal closure service, leaderboard, or proof campaign. Participation counts and enthusiasm are operational facts, not scientific warrant.

Readers should leave able to inspect a longitudinal claim: who entered, who remained, what changed, which observations were missing, how measures evolved, what rivals were compared, and what the model actually predicted. That skill matters far beyond this hypothesis.

Part VI closes with a disciplined picture of evidence. A whole life can be approached through multiple windows, but never owned by the record. The next Part turns from windows to the strongest rival explanations and asks whether ordinary science already accounts for the patterns the Law might otherwise claim.

18.1 — What Repeated Observation Can Reveal

Repeated observation reveals change that a snapshot misses. It can show cycles, turning points, adaptation, relapse, treatment response, and within-person variability. It cannot turn a sampled record into the complete truth-domain history merely by continuing for longer.

The sampling architecture matters. Entry date, prompt schedule, device availability, response window, pauses, missed observations, and the reason a person leaves all shape the trajectory. Monthly averages can hide alternating days of burden and relief, while dense sampling can create fatigue and reactivity (Shiffman et al., 2008).

Data gaps are part of the phenomenon. Illness, overload, distrust, caregiving, poverty, lost technology, privacy concerns, and death can remove observations in patterned ways. Sensitivity analysis and denominator flow are therefore more honest than filling every blank with a convenient value (Little and Rubin, 2019).

Scientific governance begins with the recognition that the record is borrowed. Instruments, algorithms, consent, access, and purposes change across years. A longitudinal project must version those changes, preserve reasons for missingness, and make correction and withdrawal possible without treating the person as a permanent file.

18.1.1 — Why Longitudinal Work Matters

A repeated record can reveal that the same rating leads into very different paths: recovery, chronic burden, treatment response,

withdrawal, or a change in what the scale means. Longitudinal work can study those paths, but it still observes selected moments under changing conditions rather than a complete phenomenal history.

Episode and trajectory are different objects. A study may capture a bad week, a recovery month, a winter season, or a terminal approach, but the meaning of any window depends on what came before and what remained possible afterward (Singer and Willett, 2003).

This matters because the Law of Fairness concerns a trajectory, not an isolated feeling or an average detached from order. Two people can share the same monthly mean while one experiences severe early collapse followed by repair and the other undergoes mild but relentless decline. The average hides the path and its timing (Fitzmaurice et al., 2011).

Long records also reveal path dependence. Burden, relief, treatment, anticipation, and loss can alter what later experience means and what options remain. A single multi-channel estimate cannot show whether a shock echoed, whether repair changed slope, or whether a life narrowed toward a terminal regime.

The measuring system changes too. Questions acquire new meanings, devices and operating systems change, and participants age into disability, caregiving, grief, poverty, recovery, or illness. A trustworthy analysis must distinguish change in the person from change in the instrument before treating a curve as evidence.

Calendar time and lived time should not be confused. A week of routine, intensive caregiving, acute pain, bereavement, and safe

rest each occupies seven days while carrying a different experiential density. Sampling intervals are equal on a clock, not necessarily in burden or opportunity.

The design should name legitimate no-sampling zones. Hospitalization, acute grief, domestic danger, coercive monitoring, severe fatigue, terminal comfort care, or clinically identified distress may require the study to go quiet. Silence can be the protocol doing the ethical work it promised rather than a defect to be repaired (Shiffman et al., 2008).

Interpretation therefore depends on the time scale, entry point, sampling schedule, recall window, calibration history, and relation between momentary and interval measures. Daily reports, monthly summaries, and annual evaluations answer different questions; combining them without a measurement model can manufacture a smooth trajectory from incompatible observations (Singer and Willett, 2003).

Within-person variation may exceed differences between people. Repeated measures can reveal alternation between burden and relief that a stable group average conceals, while ecological momentary assessment can reduce some memory error at the cost of notification burden, missed prompts, device effects, and reactivity. Adherence, timing, context, and measurement effects belong in the result (Shiffman et al., 2008).

Passive channels can add context such as movement, sleep timing, or location, but each carries error and privacy risk. A step count is not motivation, phone use is not loneliness, and location

is not felt safety. A channel belongs only when its incremental value justifies the exposure.

The best longitudinal design is not always the longest. Strong retention, valid measures, explicit calibration, and clear governance can make a shorter study more informative than a decade-long archive with shifting instruments and unclear consent. Duration is a resource, not a substitute for design.

18.1.2 — Data Gaps

Data gaps are inevitable. Phones break, batteries die, sensors fail, people forget, travel, refuse, lose access, enter crisis, or become too ill to continue; institutions change, and death ends public observation. A theory that cannot tolerate gaps is not ready for real lives.

The first task is classification without overclaim. A missing interval may reflect technical failure, ordinary forgetfulness, privacy choice, fatigue, distress, unsafe household conditions, care transition, hospitalization, incarceration, homelessness, withdrawal, or death. Those mechanisms have different implications for inference and for the person (Little and Rubin, 2019).

Absence should not be filled with confidence. Interpolation may serve a technical purpose, but it cannot become a moral or experiential claim. A smoothed curve can erase the very crisis that caused the missingness, and complete-case analysis can make a trajectory look healthier when the most burdened participants are the first to disappear.

Missingness often arises from the same events that change experience: illness, treatment, overload, distrust, relocation, device loss, or recovery. It is therefore part of the observation process and sometimes part of the substantive story. The study should preserve proportionate reason categories where participants voluntarily provide them and test conclusions under plausible unseen trajectories (Rubin, 1976; Little and Rubin, 2019).

A missed prompt during a busy afternoon is not equivalent to a week of silence during hospitalization, a privacy-driven pause, or an endpoint event. Coding every absence as one blank value discards information about how the record was produced. At the same time, investigators should not pressure participants for intimate explanations merely to refine a missingness label.

Sensitivity analysis should make assumptions visible. Pattern-mixture, selection, weighting, joint, and bounded approaches answer different questions and cannot recover unobserved experience by decree. When defensible models yield materially different conclusions, that range may be the most honest result (Diggle and Kenward, 1994; Little and Rubin, 2019).

A simple example shows the risk. Symptom scores may appear to improve after several high-burden participants stop answering during hospitalization. The apparent recovery belongs to the retained sample unless the missingness process and plausible unseen outcomes are modeled. It is not evidence that the absent people improved.

Some gaps should remain. A person may decline contact during grief, danger, illness, or exhaustion, and the team should not pursue completeness so aggressively that research becomes surveillance. Scientific loss does not cancel the person's right to disappear from the record.

The final claim must therefore stay tied to the windows actually observed, the denominator flow, and the assumptions used to bridge what is missing. Data gaps may reveal where observation is hardest and whose experience is most likely to vanish from view, but they cannot reconstruct childhood, dreams, crises, or a terminal remainder.

18.1.3 — Scientific Governance

Scientific governance turns repeated observation into a trustworthy record. It identifies the responsible data steward, purpose, access rules, version history, correction path, retention period, linkage limits, and stop authority for each channel. Governance is part of validity because people report and remain differently when they trust the relationship that holds their data.

The protocol should define the target population, sampling schedule, event windows, observation channels, retention plan, and reason for collecting each variable. Treatment, illness, relocation, caregiving, bereavement, work change, and technology change are intercurrent events because they can alter experience and observation together (Hernán and Robins, 2020).

Consent should be renewed when purpose or burden changes materially. Passive sensing, record linkage, genetic analysis, commercial partnership, public release, or artificial-intelligence

use may exceed the original agreement even when the data already exist. Reconsent, a fresh opt-in, or abandonment of the new use may be required (Vayena et al., 2015).

Access should follow purpose and data lineage. Analysts studying missingness do not automatically need raw messages, clinicians do not automatically need research predictions, and public communicators do not need identifiable trajectories. Raw data, cleaned data, derived features, model outputs, and public summaries should remain distinguishable so a coding error, consent change, or deletion request can propagate to every dependent artifact.

Missingness and public reporting belong to the same governance chain. The study should preserve proportionate reasons for missed observations, withdrawal, treatment interruption, transfer, death, and administrative loss, then report participant flow, protocol changes, attrition, treatment context, model versions, failed measures, null analyses, and transport failures. Different absences imply different recoverability, and some are the ethical result of honoring care or refusal (Little and Rubin, 2019; von Elm et al., 2007). A participant-flow diagram can show how each kind of absence changes the available observations at successive stages of the same study.

Retention, versioning, custody, and interpretive authority should follow purpose rather than institutional appetite. Some identifiers can be deleted early, some links time-limited, and some sensitive channels never collected; a repository may store records without permission to infer new states, an analysis team may build a model without authority to return individual predictions, and a

dashboard may receive only aggregates. Independent governance must be able to reject secondary use, deployment, or publicity when consent, comparability, or scientific justification fails. A purpose-specific access table can identify who may hold a record, analyze it, contact participants, and release a particular result.

A longitudinal record is a research trace, not a biography, soul, or measure of worth. The project should plan its ending before funding or institutional interest disappears, assigning responsibility for data disposition, participant contact, archive conditions, model retirement, public correction, and unresolved requests. Stewardship succeeds when no institution, archive, or algorithm acquires ownership of the person behind the data (Nissenbaum, 2010). An orderly close should leave participants a working contact route and custodians a clear account of the requests they still need to resolve.

18.2 — Life Events, Adaptation, and Context

Life events should be modeled as processes rather than isolated dates. Anticipation, immediate disruption, treatment, social response, material consequences, and later adaptation may each alter experience. Event-centered designs should therefore show pre-event trends and post-event pathways instead of attributing every change to a single timestamp.

Response shift complicates every long-arc comparison. People may change standards, priorities, interpretation, or the meaning of a scale after illness, disability, loss, recovery, or aging. A stable score can hide change, while a changed score can partly reflect recalibration. Measurement invariance and qualitative context are needed to distinguish these possibilities (Sprangers and Schwartz, 1999; Oort et al., 2009).

No event is a cosmically assigned counterpart to another. Later joy does not morally repay earlier harm, and later loss does not erase earlier pleasure. Both still contribute to the complete-stream arithmetic without being paired events. The Section keeps ordinary causal pathways, adaptation, and changing standards visible so that narrative hindsight cannot manufacture a pair after the life has unfolded.

18.2.1 — Events Change Context and Experience

A life event changes more than a score. Job loss can alter income, schedule, identity, relationships, sleep, treatment access, and future options; illness can change pain, attention, dependence, medication, and the meaning of ordinary activities. Longitudinal

analysis should model these pathways and their timing rather than code the event as one positive or negative shock.

Illness and injury illustrate the problem. They can bring pain, fear, disability, cost, isolation, body change, loss of work, and narrowed horizons while also changing family roles, social visibility, and access to care. Relief systems may reduce burden or add to it through delay, paperwork, disbelief, cost, stigma, or inaccessibility (Headey and Wearing, 1989).

The event can also change the measurement ruler. Pain, treatment, fatigue, and medication may alter self-report, sleep, movement, physiology, social contact, attention, appetite, and emotional range. When those channels contribute to the estimate, the event affects both experience and the instrument used to study it. That double role must be modeled rather than hidden (Headey and Wearing, 1989).

Care remains prior to measurement. A person in pain, medical uncertainty, rehabilitation, disability transition, or acute injury does not owe the theory data. Clinical care, privacy, consent, and practical support govern whether observation may continue at all.

Events commonly described as positive are also multidimensional. Birth, love, attachment, success, and reunion may bring joy, tenderness, purpose, belonging, and a widened future while also introducing exhaustion, fear, vulnerability, medical risk, financial pressure, or caregiving load. Chosen burden can be meaningful without becoming costless, and love never makes exploitation fair (Headey and Wearing, 1989).

Bereavement, migration, caregiving, recovery, relationship change, and job loss unfold through anticipation, onset, duration, secondary consequences, social response, and recurring reminders. A serious model records those stages, co-occurring events, and the person's changing interpretation before attributing the path to adaptation, compensation, or any scheduled counterpart (Luhmann et al., 2012; Headey and Wearing, 1989).

Job loss provides a concrete test of pathway thinking. The same date may reduce income immediately, disrupt sleep weeks later, alter treatment access after insurance changes, and reshape identity or relationships over months. A single before-and-after contrast cannot represent that chain. Intercurrent-event rules and sensitivity analyses should be fixed before the trajectory is interpreted (Hernán and Robins, 2020).

The Law-related lens adds one disciplined question: how did the event become lived within this stream? It does not add a second event that was secretly assigned to balance the first. Ordinary causes, institutions, relationships, and choices remain the first explanations to investigate.

18.2.2 — Adaptation, Response Shift, and Score Meaning

Loss and grief are among the most dangerous event categories for a fairness theory. They invite interpretation, and interpretation can become cruelty. A study may observe grief trajectories, but it must never suggest that a death, abandonment, estrangement, miscarriage, displacement, or irreversible loss occurred to balance a cumulative history or teach a lesson.

Grief also resists simple timing. The event date may be clear, yet the loss may begin earlier through illness, anticipatory grief, disappearance, betrayal, decline, or fear. It may continue long after public sympathy fades. It may return through anniversaries, songs, places, documents, body memory, dreams, and ordinary objects (Schwartz and Sprangers, 1999).

Grief illustrates why a stable score can hide changing meaning. Early burden may be carried in shock, sleep disruption, appetite, fear, administrative work, and the sudden absence of a person; later burden may appear in anniversaries, changed identity, financial strain, family reorganization, or the loss of imagined futures. A repeated rating can remain numerically similar while the experience, standards, and life context beneath it have changed.

Grief can also contain love, relief, guilt, gratitude, numbness, anger, and disorientation at once. Coding grief as purely negative may erase the complex truth of terminal illness relief, abusive relationship loss, caregiver exhaustion, or sacred memory. Coding it as balancing may violate the mourner. The right posture is patience (Luhmann et al., 2012).

The first rule is that safety outranks data. A survivor of violence, coercive control, abuse, war, assault, exploitation, detention, trafficking, or severe threat does not owe the Law of Fairness an interpretable trajectory. Crisis support, legal protection, clinical care, confidentiality, and survivor control are prior to research.

Event coding around trauma must therefore be participant-controlled and trauma-informed. It should allow broad categories,

uncertainty, private suppression, delayed reporting, no reporting, and safe withdrawal. The study may not require naming perpetrators, locations, details, or timelines unless a separate ethically reviewed protocol makes that necessary and safe (Oort et al., 2009).

A person with chronic illness reports the same quality-of-life score after reorganizing what counts as a good day. The number is stable, but its meaning may not be. Longitudinal analysis must distinguish state change from recalibration, reprioritization, and changing standards before treating equal scores as equal experience.

Interpretation therefore needs qualitative context and repeated calibration rather than assuming that the scale remained fixed (Sprangers and Schwartz, 1999).

Adaptation and response shift can move in different directions. A person's distress may lessen while their standard for reporting severe distress also changes, or their score may remain stable while they reorganize priorities around illness or disability. The observed number is therefore a joint product of state and measurement meaning.

Recalibration can be adaptive without making hardship harmless. Learning to value smaller activities, changing expectations, or finding new meaning may improve life while preventable barriers and symptoms still deserve remedy. A lower burden under changed standards does not retroactively justify the condition that required adaptation.

Public summaries should avoid saying that people return to normal when the measure no longer carries the same meaning. A more accurate statement identifies what changed, which scale was used, and whether the comparison survived response-shift analysis.

18.2.3 — No Event Is a Scheduled Counterpart

A longitudinal science can study shocks, recovery, and adaptation without turning events into moral prices. No bereavement, illness, promotion, debt, recognition, or failure is evidence that an earlier experience required repayment or that a later event was scheduled to offset it.

The pathways in Section 18.2.1 affect sampled mood without turning a mood score into the latent whole-life ledger (Headey and Wearing, 1989).

Population tendencies remain a strong rival. Some events produce average changes that partly fade, while others leave persistent differences and individuals vary widely. Such findings show that adjustment can occur through ordinary psychological and social processes without implying a universal compensatory schedule.

Event coding should keep external structure separate from participant meaning. A promotion may be publicly classified as favorable while being experienced as exhaustion; a failure may later be understood as release. The study can preserve both the initial appraisal and later reinterpretation without forcing one to govern the whole arc (Headey and Wearing, 1989).

Constraint changes the menu from which action is chosen. A person who misses prompts, changes phones, works irregular

hours, lacks private space, moves housing, or distrusts institutions is not a worse participant. The design may be failing to meet the person's world. Poverty and reduced reserve should be modeled as changing opportunity and horizon, not as moral or measurement deficiency.

A responsible event schema can record onset, duration, severity, recurrence, confidence, source, participant meaning, public meaning, dependency, and uncertainty. It must allow overlapping events because illness, caregiving, grief, poverty, and work disruption often occupy the same period. One tidy label cannot carry a season of life.

Sources should remain distinguishable. Events may be voluntarily reported, clinically documented, administratively recorded, suggested by sensors, or inferred from context. Each source has different error, privacy, and authority. The record should state source confidence rather than pretending that every event entry is equally direct (Headey and Wearing, 1989).

Private nonreporting is a legitimate outcome. A participant may need a protected event category, may suppress detail, or may decline event coding altogether. The loss of scientific resolution is real, but it cannot authorize a search through records, relatives, or devices for an event the person chose not to disclose.

Research should test ordinary pathways: education can alter employment, treatment can alter symptoms, social support can alter recovery, and policy can alter exposure. These mechanisms are actionable and falsifiable. A paired-event story that specifies

no pathway cannot replace them, however narratively satisfying it appears.

Public examples should resist symmetrical biographies built to illustrate balance. Selecting a childhood wound and an adult triumph can imply a prescribed injury or promised reward even when a disclaimer denies it. A responsible account preserves contingency, unpaired experience, unequal timing, and the possibility that no redemptive arc arrives (Headey and Wearing, 1989).

18.3 — Missingness, Treatment, and Response Shift

The people who disappear from a longitudinal record can determine what the remaining curve appears to show. Participants leave because of illness, distress, recovery, overload, distrust, relocation, caregiving, device failure, privacy concern, or death. Attrition is therefore a scientific mechanism and an ethical signal.

Treatment changes complicate interpretation. Medication, psychotherapy, surgery, rehabilitation, palliative care, and social support can alter experience, reporting, opportunity, and retention together. The analysis should distinguish treatment effects, intercurrent events, and selection rather than treating them as noise around a natural trajectory.

Reporting duty begins with the denominator. Recruitment, eligibility, enrollment, response, pause, withdrawal, death, exclusion, linkage, and imputation should remain visible across time and relevant groups. A positive subgroup in the main text cannot compensate for attrition buried in a supplement (Diggle and Kenward, 1994; Little and Rubin, 2019).

No imputation model restores a life that was not observed. Sensitivity analyses can show how conclusions change under plausible missingness assumptions, and sometimes the range remains too wide for a verdict. That outcome should narrow the claim rather than invite a more convenient model.

18.3.1 — Long-Arc Prediction

The strongest long-arc tests will be prospective. Retrospective datasets may be useful for exploratory work, but they often lack

the exact channels, consent architecture, endpoint rules, event coding, and missingness metadata a complete-stream test would require. Prospective design forces the theory to make commitments before it knows the answer.

Long-arc evidence should also be multi-resolution. Daily data may show noise and burden. Weekly summaries may show rhythm. Monthly arcs may show recovery. Yearly trajectories may show life-stage movement. A fair model should ask which scale carries the signal and which scale is merely smoothing the person into abstraction. (Munafò et al., 2017).

A long-arc prediction should be written before the data are interpreted. For example, a protocol might state that if the reserve-coupled picture is informative, later portions of a trajectory should show specified changes in variance, extremity, recovery slope, or admissible range relative to independently defined horizon and reserve indicators. That prediction remains modest because every measured term is an estimator. It becomes useful only when the same pattern is not equally explained by simpler rivals.

A research team should not examine years of telemetry and then decide that whatever shape appears is what the Law expected. The predicted family must be named in advance: slope change, shock-compensatory interpretation, measured variance change, late-window concentration, reserve interaction, horizon interaction, or rival defeat (Fitzmaurice et al., 2011).

Ethically, near-endpoint clustering should never become a good-death ranking. A person whose final data are distressed, missing,

confused, sedated, or unmeasured has not failed the Law, the family, the study, or themselves. Terminal observations are fragile public traces, not verdicts on a life.

Compression should also be distinguished from flattening. A flat curve may indicate peace, medication, exhaustion, reduced reporting, sensor failure, institutional routine, dissociation, diminished consciousness, or loss of access. Compression as evidence requires context, channel agreement, rival comparison, and ethical review. A line that looks calm on a chart may hide a person too burdened to report.

A study that explains improvement through pain relief, secure housing, or dependable support has learned something valuable about how lives can change. Those possibilities deserve serious testing, and the public account should credit them when the evidence supports them. Their ethical appeal cannot decide the explanation in advance. Care remains justified while causes are uncertain; comparing those causes can show which repairs actually helped and which new questions are worth asking.

Every terminal and long-view pattern has plausible rivals. Age can reduce activity. Illness can narrow expression. Medication can blunt signals. Palliative care can reduce distress. Social support can stabilize experience. Frailty can reduce variability. Depression can flatten reports. Devices can fail more often in late illness. Selection can remove high-variance cases from the observed sample.

The last observations are often the most scientifically desired and the most ethically difficult. Near the end of life, people may

become unable to report, unwilling to participate, sedated, unconscious, protected by family, moved between care settings, or simply unavailable to research. The missing last observation is not a minor inconvenience. It may be the central uncertainty of terminal longitudinal evidence (Munafò et al., 2017).

Training and test periods should remain separate, calibration should be checked over time, and transport should be evaluated across cohorts and life stages. A model that predicts next-month distress in one service but loses calibration after a health-system change has produced a bounded local forecast, not a recovered life-course mechanism. Its proper use and public claim end where the prospectively specified loss and transport conditions fail.

Personal forecasts require a use boundary. A risk estimate may support an optional conversation when clinically validated, but it should not become a statement about hidden worth, future balance, eligibility, or destiny. Prediction is a tool with costs, not possession of the person's future.

18.3.2 — Who Drops Out

The first missingness question is human: who drops out? A longitudinal study may begin with a diverse group and end with a compliant remainder. If the remainder differs from those who left, the curve has changed its population while appearing to follow the same one (Little and Rubin, 2019).

Withdrawal, loss, and design failure are not the same. A person who intentionally leaves has exercised a right. A person who becomes unreachable may be missing for unknown reasons. A person excluded by language, disability, cost, technology,

transportation, or burden marks a failure of access that the study must report rather than attribute to noncompliance.

Exit may be related to depression, anxiety, grief, trauma, poverty, illness, overload, distrust, family danger, privacy concern, recovery, or simple disinterest. The same observed absence can therefore arise from opposite trajectories. Retention percentage alone cannot identify which process changed the sample.

Denominator flow should distinguish people who never enrolled, withdrew, paused, became unreachable, entered treatment, lost capacity, transferred settings, or died. The record should do this for each channel and period, not only at the final visit. Reporting the retained sample without these paths hides whether the study gradually selected lives that were healthier, safer, wealthier, or easier to contact.

Attrition has a sequence. A participant may first skip evening prompts, then disable passive sensing, then miss a visit, then refuse linkage while allowing earlier data to remain. Preserving those steps can reveal whether burden, health, trust, technology, or changing priorities drove the loss and whether a remedy actually worked.

Treatment can reverse the meaning of missingness. Effective care may reduce reporting because life becomes less centered on symptoms, while intensive services may increase observation. Worsening illness may remove someone from the study, or may make the person more visible to clinical systems. Analyses should separate treatment initiation, response, access barriers, and

research contact (Hernán and Robins, 2020; Fitzmaurice et al., 2011).

Response shift can coexist with selective retention. People may change how they interpret a scale after illness, disability, recovery, caregiving, or repeated use, while those who find the instrument alienating leave. Apparent stability may therefore reflect changed experience, changed standards, changed membership, or all three (Schwartz and Sprangers, 1999; Oort et al., 2009).

Apply the missingness methods and sensitivity discipline in Section 18.1.2. When the conclusion requires every missing participant to resemble a convenient model, partial identification or No adequate estimate may be the honest result (Little and Rubin, 2019; Kurland et al., 2009).

Retention should also be reviewed for pressure. Participants may remain because they fear disappointing the team, losing compensation, losing community standing, or being interpreted as evidence against the Law. High retention is not an ethical success when it is produced by coercive gravity (Diggle and Kenward, 1994).

Sensitivity analysis should ask how the conclusion changes if people with greater burden were more likely to leave, if recovery reduced participation, or if missingness differed across treatment and access conditions. The purpose is not to select the assumption that preserves the headline. It is to show which claims remain stable and which must narrow.

Attrition and missingness should be reported by time, subgroup, burden, treatment, site, and reason rather than hidden in one retention percentage.

18.3.3 — Reporting Duty

Reporting duty is the public expression of missingness discipline. A longitudinal study must not display smooth curves without showing who produced them, who did not, when data disappeared, which channels failed, what correction was used, and how sensitive the result is to missingness assumptions.

Reports should preserve enrollment and follow-up flow, baseline differences between completers and non-completers, reasons for dropout where ethically available, device and prompt completion, missingness by event and burden class, endpoint missingness, treatment changes, protocol deviations, adverse events, and uncertainty after correction. Current reporting guidance treats these as essential parts of the result, not technical decoration (Schulz et al., 2010; Hopewell et al., 2025).

The report should state the claim level that the missingness permits. The same dataset may support a descriptive association while leaving a causal, route, or terminal claim unusable. Severe unmodeled missingness cannot be transformed into strong support merely because the observed cases form a coherent pattern.

The ethical problem is sharper than the statistical problem. A model may impute a value, but the person did not experience an estimate. They experienced the day, the crisis, the inability to answer, or the choice not to be observed. Imputation can support

analysis; it cannot become narrative ownership of the missing person's life (von Elm et al., 2007).

When missingness is highest during burden, the visible record can become too optimistic. Device inequality can create the same distortion: unreliable connectivity, incompatible bodies, demanding work, disability, or lack of a device may be coded as noise or noncompliance when the instrument is the problem. A pattern of stabilization can therefore be a visibility failure rather than a Law signature.

Endpoint missingness is especially dangerous. The last observation may precede sedation, hospitalization, device removal, loss of speech, family withdrawal, or the point at which ordinary prompts are no longer appropriate. Unless the endpoint is independently defined, the last available point is not the end of the stream (Munafò et al., 2017).

Missingness can also be socially patterned, so public reporting must not stigmatize groups with lower participation. Lower observation may reflect higher burden, reduced access, greater privacy risk, or justified distrust. Missingness is not a civic failure by the participant; it is a question the study must answer about itself and the world.

Longitudinal authority should be renewed rather than assumed. A participant who accepted one cadence, instrument, or purpose may reasonably refuse a later device, linkage, analysis, quotation, or model-training use. Renewal should name what changed and offer a practical way to continue more narrowly. A study learns more when it can distinguish refusal of one use from withdrawal

from every form of participation, rather than treating consent as a permanent grant over a growing record.

Correction must travel through the same chain as the original record. If a participant amends a report, an instrument changes meaning, a diagnosis is revised, or a linkage proves wrong, the raw record, derived variables, analytic dataset, public summary, and any reusable model need a governed update path. Versioned correction prevents a long record from gaining authority merely through age and keeps withdrawal or disagreement visible within the evidence history.

Reporting duty reaches every public layer. The paper, supplement, repository, abstract, press release, presentation, and graphic should show denominator flow, attrition, treatment change, missingness assumptions, subgroup status, failed sensitivity analyses, and deviations from the prospectively specified plan. If an exclusion rule, coding error, or attrition count changes, the record should identify every affected table, model, summary, and artifact so the earlier conclusion does not remain alive elsewhere.

Transparent reporting does not recover what was never observed, but it prevents absence from becoming invisible evidence. It allows readers to distinguish a narrow result from an overconfident story and lets later investigators decide whether to improve the design, bound the inference, or stop using the record.

18.4 — Comparing Models Under Common Rules

Long records can support many stories. Adaptation, treatment, selection, memory, ordinary causal change, response shift, measurement drift, and terminal-closure routes should enter the comparison as genuine competitors before the decisive outcome is known. Each should receive the information, flexibility, and tuning budget appropriate to its claim; a model cannot win because its rivals were deliberately weakened.

Scientific status should be granular. A study may support a measurement channel, favor a local route, weaken another, fail to transport, or leave terminal claims indeterminate. Common rules make those distinctions visible and prevent a beautiful trajectory from being promoted into a verdict about a whole life.

18.4.1 — Predeclared Rivals

Prospective comparison rules make that competition capable of loss. They should state which outcomes, time windows, covariates, missingness models, transport tests, and decision rules each rival receives before the decisive record is examined. New explanations may still be explored later, but they cannot silently replace the comparison that the study was designed to make.

The rival set should match the question. Recovery after a life event requires adaptation, treatment, and social-support models. A terminal-window claim requires frailty, medication, palliative-care, activity, endpoint, and missingness models. Citizen-science data require selection, platform, motivation, device, and reporting-bias models.

Ordinary life belongs in the tournament. Housing stability, pain relief, disability accommodation, family support, economic change, religious practice, grief rituals, and access to care may explain a trajectory more directly than any terminal constraint. Omitting those channels can make the extraordinary account look distinctive by leaving its strongest mundane competitors outside the room.

Null structures are also essential. Random walks, autoregressive persistence, seasonal baselines, regression to the mean, and instrument drift may reproduce visually compelling curves. Human pattern recognition is not a statistical discriminator, and an unromantic baseline can be the most important opponent (Wilson and Collins, 2019).

Fair comparison requires equal opportunity. Rival models should receive comparable data, preprocessing, tuning budgets, covariates, and scoring rules. The favored model should not receive future information, custom exclusions, or more flexible hidden states than its competitors. A victory over deliberately weak rivals is not evidence (Nosek et al., 2018).

Complexity carries both statistical and ethical costs. A flexible model may fit the past while failing the future, and a model too opaque to explain may weaken informed participation. Additional parameters should earn their place through prospectively specified prediction, held-out performance, model recovery, or a clearer account of edge cases rather than through post hoc fit.

Extend the model-recovery rehearsal in Section 17.5.2 to longitudinal response shift and hybrid rivals. If the models

recover one another or fail to recover themselves, the real-data tournament is not identified.

New rivals may emerge after analysis, but they must be labeled exploratory and tested on new or held-out data. The original prospectively specified comparison remains part of the record. This preserves scientific flexibility without converting every adverse result into a newly designed escape route (Hernán and Robins, 2020).

Decision rules should state what rival victory changes. A rival may retire one signature, narrow a route, expose a measurement artifact, or show that the favored model adds no predictive value. It does not automatically decide every branch of the formulation developed here, but the public claim must contract wherever the comparison reached.

A serious tournament can end with the ordinary model winning, the candidate winning narrowly, every model failing, or the evidence remaining inadequate. The Law earns attention only when all of those outcomes are possible before the first trajectory is opened.

18.4.2 — Predictive Fit Versus Explanation

Forecasting and explanation answer different questions. A model may predict tomorrow's reported valence by exploiting autocorrelation, device habits, demographics, treatment schedules, or stable reporting styles. Explanatory support requires evidence that the proposed variables and dependencies correspond to the mechanism, survive intervention or natural

variation, transport beyond the original setting, and outperform capable rivals for the right reason.

A black-box model may therefore be useful and theoretically shallow. It can improve held-out prediction without recovering horizon, reserve, compensability, or any other Law-related structure. Forecasting value should be reported honestly, but low error does not turn the tool into an explanation of the lived trajectory (Hernán and Robins, 2020).

Causal language requires stronger assumptions than prediction. Stable proxies may forecast distress while misidentifying what produced it, and a treatment schedule may be easier to predict than the burden it is meant to relieve. Interventions, natural variation, causal diagrams, and sensitivity analysis help distinguish association from mechanism, but each supports only its declared estimand (Pearl, 2009; Hernán and Robins, 2020).

Calibration matters alongside discrimination. A model may rank high and low outcomes correctly while overstating magnitude, fitting the average while failing in the tails, or performing well only where observation is dense. Reports should show error and uncertainty by horizon, site, subgroup, treatment state, and observation density before accuracy is translated into a claim about cause or terminal structure.

Open-set testing matters equally because the true process may be absent from the prespecified model library. Residual structure, unstable parameters, site-specific failure, and systematic intervention errors may show all candidates inadequate. A forced

winner creates false confidence; model-ambiguous and open-set inadequate remain legitimate verdicts.

When a rival wins, narrow the claim. Credit adaptation, homeostasis, treatment, social support, or the observation process when they better fit the pattern. The manuscript, guide, summary, public response, and research registry must then change together rather than preserve stronger language after evidence moves (Poldrack, 2006; Wilson and Collins, 2019).

The ethical firewall survives every outcome. If palliative practice explains terminal calm, palliative practice matters more, not less; if poverty explains trajectory burden, material repair becomes more urgent. The distinction among predictive fit, causal explanation, mechanism recovery, and terminal verdict remains essential, so predictive success never becomes a judgment about a person or proof of PTVC.

18.4.3 — Scientific Status

The common-rule comparison in Section 18.4.1 should also include allostasis, event-history models, conditioned trajectories, and flexible machine-learning baselines (Wilson and Collins, 2019).

Route support and target support remain separate. A trajectory may favor an adaptation or reserve-related mechanism without identifying exact terminal closure. Failure of one route does not refute every target branch, and route fit cannot become target proof without valid entailment.

Large records create many opportunities for selective success across windows, subgroups, lags, outcomes, and preprocessing

choices. Prospective specification, multiplicity control, model recovery, held-out confirmation, and full reporting are therefore central. An analysis discovered after searching the archive is a candidate for later testing, not the original confirmatory result (Nosek et al., 2018).

A null result must be typed. The pattern may be absent, the manipulation may have failed, the measure may be unreliable, the study may be imprecise, missingness may dominate, or the competing models may be indistinguishable. The verdict should state which interpretation the design can support instead of using inconclusive as a universal refuge.

Status can differ by module. A study may validate one reporting channel, fail to identify neutral origin, favor an ordinary adaptation model for midlife trajectories, and remain unable to assess the terminal target. Modular status is not indecision; it preserves which part of the research architecture actually survived.

Uncertainty and denominator flow should travel with the trajectory. Early estimates may be weak because calibration is poor, while later estimates widen as attrition, treatment, and device changes accumulate. A smooth line should never hide the collapse of information underneath it.

Ethical usability belongs in scientific status. A statistically rich dataset may be unusable for a proposed analysis if consent did not cover the purpose, bystanders were captured, burden was excessive, withdrawal cannot propagate, or the record regime cannot support correction. Ethical failure can make an analysis

impermissible even when its statistical estimate is valid; coercion, selective participation, and altered reporting can also damage validity.

Public summaries and maturity records should inherit the narrowest material dependency. A finding limited to one age range, intensive smartphone use, one clinical service, a high-retention sample, or exclusion of irregular trajectories must keep that limit. When code changes, linkage errors appear, a measure fails transport, or a rival performs better, the protocol record, manuscript, public graphic, and participant communication should change together. Stronger maturity language requires independent teams, materially different sites, and visible null and adverse results.

A route-favored result should name the rivals defeated and those that remain. It may beat ordinary mean reversion while remaining indistinguishable from treatment selection or a conditioned bridge. Naming the surviving rival set gives the next study a discriminating job rather than an invitation to collect more of the same.

Record the scientific status at this stage, then use Chapter 28's full verdict vocabulary for the program-level decision. A local finding may be exploratory, rival-favored, model-ambiguous, failed in transport, narrowed, or unusable without becoming a verdict on the whole Law.

18.5 — Participation Without Surveillance

Participation without surveillance requires three things: continuing and granular consent as technology and purpose change, freedom from social pressure or penalty for refusal, and citizen-science language that never converts scale of participation into proof.

18.5.1 — Voluntary Participation

Voluntary participation is a continuing condition, not a signature collected at enrollment. Long studies introduce new sensors, linkages, hypotheses, staff, and uses. Consent must therefore be granular, revisitable, and compatible with pausing or leaving without loss of care, compensation already earned, or ordinary standing.

New uses require new choices. A person who agreed to mood diaries did not necessarily agree to location inference, clinical linkage, commercial partnership, or artificial-intelligence training. Recontact should explain the material change in ordinary language and make refusal as easy as acceptance (Vayena et al., 2015; CIOMS, 2016).

Participation must remain separate from belief, community membership, employment, treatment, education, benefits, donations, public recognition, and unrelated services. Supervisors, clinicians, teachers, family leaders, and project moderators should not see participation choices when their authority could make refusal feel unsafe (World Medical Association, 2024; CIOMS, 2016).

Privacy by design begins with collecting less. Every channel should answer a defined scientific need, and the release plan should be tested against hostile linkage with social media, public records, clinic visits, location traces, platform logs, and rare-event timing. Removing names does not make a rich longitudinal pattern anonymous (Nebeker et al., 2017; Vayena et al., 2015).

Emotional privacy and bystander privacy matter as much as identifier removal. A participant may consent to affect-relevant signals without consenting to narrative interpretation, public sympathy, religious framing, or family discussion. Messages, household sounds, locations, photographs, and relationship events can expose people who never agreed to be studied.

Controls must be usable. Participants should be able to pause prompts, disable a sensor, decline new linkage, restrict secondary use, or leave through accessible digital, phone, paper, and human routes. A right that requires finding a distant administrator or explaining a crisis is not a practical right.

Withdrawal should propagate through raw data, cleaned data, analytic tables, trained models, backups, and published aggregates as far as law and technology permit. Limits should be explained before consent, not discovered after departure. Where model removal or machine unlearning is claimed, the study should state what was removed and how the claim was verified (Vayena et al., 2015).

People may pause during grief, illness, poverty, household danger, caregiving, or simple disinterest without pressure to justify themselves. Pausing or leaving should not reduce care,

community standing, compensation already earned, access to corrections, or visibility in public discussion. Departure should receive acknowledgment rather than persuasion.

Voluntary participation governs the legitimacy of the record, not the truth of the complete stream. Even an exceptionally rich and fully consented dataset remains sampled, endpoint-limited, and unable to become a person's private cumulative truth.

18.5.2 — No Social Pressure

No social pressure is necessary because communities can coerce without intending to. A project may begin with curiosity and become a culture of proving loyalty, seriousness, courage, openness, or belief. Longitudinal citizen science is especially vulnerable because continued participation can look like commitment and refusal can look like doubt.

Pressure can arrive through praise as easily as through threat. Publicly celebrating high-compliance participants, displaying streaks, or repeatedly telling moving contribution stories can make people who pause or decline feel lesser. Participation may be thanked without becoming a badge of character, courage, or belonging (Nissenbaum, 2004).

Feedback and quality metrics must remain inside their proper purpose. A participant may receive information needed for consent, safety, or study engagement, but no display should imply moral improvement, worsening, balance, worth, or failure. Response rate, data completeness, variance, provisional affect estimates, and event history are research properties, not identity.

Aggregate review needs a use-after-release plan. A result that is safe in a paper may become unsafe as a dashboard tile, donor claim, news graphic, slide, or social excerpt. Review should follow the artifact into its likely public forms, asking not only whether the statistic is deidentified but whether the surrounding context invites ranking, diagnosis, or inference about people (Nissenbaum, 2004).

An employee may believe that declining an optional well-being study will be noticed by management; a patient may agree because the treating clinician introduced the project; a community member may fear appearing disloyal. Private comprehension checks and confidential refusal routes should treat those perceptions as design evidence, not as misunderstandings to be argued away.

Longitudinal affect data can also look clinically meaningful. Participants may be alarmed by trends, families may ask for interpretations, and platforms may be tempted to offer wellness advice. Any safety signal requires a preapproved clinical or crisis route that is separate from public scoring and clear about the limits of research data. Collecting affect-relevant information does not create authority to diagnose or treat.

Recruitment and public messaging should make nonparticipation ordinary rather than presenting it as a lost opportunity to help humanity or prove commitment.

Research should be judged partly by the freedom it leaves behind. Data collected under loyalty pressure are ethically compromised and harder to interpret as expressions of actual choice. A program

earns trust when refusal is practical, private, and free of suspicion, even when that freedom weakens the dataset.

18.5.3 — No Citizen-Science Proof Claims

Citizen science can improve feasibility, measurement design, missingness research, inclusion, model transport, and the questions a program chooses to ask. It can widen participation in governance without turning participation itself into evidence for the Law.

Scale is not a proof statistic. Large enrollment can coexist with self-selection, duplicated accounts, uneven retention, device differences, weak measurement, and an unidentified endpoint. More observations may improve precision while leaving the scientific target untouched.

Public portals should state both recruitment source and local purpose. Online followers, patients, students, employees, volunteers, and the general public enter through different incentives and barriers; enrollment, prompts completed, countries represented, years of data, and retention are operational facts, not scientific warrant. A selected mixture should not be presented as humanity, and participation should never be described as donating a life to a cosmic ledger or validating the hypothesis (Onnela and Rauch, 2016).

The digital divide shapes both participation and measurement. Device ownership, connectivity, literacy, language, disability access, work schedule, privacy, and institutional trust affect who enters, who remains, and what can be observed. A dashboard

should never present the resulting sample as a transparent picture of humanity.

Citizen-generated records can contain bystanders and sensitive context. Location, audio, messages, photographs, household routines, and relationship events should be minimized under tiered access, while feedback remains limited to aggregate summaries, data-quality information, and study updates. Open science does not require open identifiable lives, and a research portal is not a clinical or spiritual service that may return hidden-valence estimates, future-balance predictions, or terminal interpretations (CIOMS, 2016; Vayena et al., 2015).

The strongest contribution may be negative. Citizen science can show that a prompt is burdensome, a measure unstable, a sample unrepresentative, or a laboratory effect absent in ordinary settings. Those findings improve the program without supporting PTVC, and public correction should reach every portal, dashboard, video, and partner site (Wiggins and Wilbanks, 2019).

Any future support must come from valid representation, prospectively specified comparisons, capable rivals, replication, and the correct scientific verdict, not enthusiasm or participation counts. Citizen science earns its place when it improves questions and accountability while remaining willing to conclude that the Law was not supported.

Part VII — The Strongest Rivals

A theory becomes scientifically serious at the point where it can lose. The Law of Fairness has now been stated with enough precision to face that test: one qualifying conscious stream, one complete trajectory, an independently defined endpoint, and a proposed terminal condition. Part VII asks whether familiar psychological, biological, computational, and stochastic processes can explain the relevant patterns without that condition.

The question is no longer merely what the Law might predict. It is what adaptation, opponent regulation, predictive processing, reinforcement learning, homeostasis, allostasis, mean reversion, endpoint conditioning, and hybrid models would predict under the same observations. Wherever one of those accounts explains a pattern more economically, that pattern belongs first to the rival.

These alternatives are not one undifferentiated objection. Some describe how reactions fade after events. Some explain counterstates such as withdrawal, relief, or contrast. Some treat affect as part of prediction and bodily inference. Others model action, viability, feedback, stochastic return, or paths conditioned by their endings. Each family earns a distinct comparison because each can imitate a different part of the picture.

Adaptation is the nearest rival to ordinary experience. Gains often become familiar, losses can change in felt intensity, expectations move, routines form, and people differ sharply in whether and how far they return. Modern evidence therefore supports neither a universal reset nor a simple story of permanent circumstance. It supports heterogeneous adjustment across events, people, measures, and time (Diener et al., 2006; Lucas, 2007; Luhmann et al., 2012).

Opponent and addiction models add temporal structure. A first response can recruit a later counterresponse; repeated reward can alter wanting, tolerance, and withdrawal; stress can be followed by relief, exhaustion, sensitization, or dysregulation. Those sequences may look compensatory while remaining local mechanisms with no claim about a completed conscious stream (Solomon and Corbit, 1974; Koob and Le Moal, 2008; Robinson and Berridge, 2025).

Predictive and interoceptive accounts reach deeper into the construction of felt states. Expectation, precision, bodily signals, action, and uncertainty can shape fear, relief, mood, pain, and motivation before any terminal rule is considered. Their breadth makes them formidable because a model that explains the local dynamics well may leave little distinctive work for a stronger boundary hypothesis (Friston, 2010; Clark, 2013; Seth, 2013; Greenwood and Garfinkel, 2025).

Control and stochastic accounts widen the challenge again. Reward learning can produce long trajectories from local updates. Homeostasis and allostasis can stabilize viable ranges. Ornstein-Uhlenbeck dynamics can generate ordinary mean reversion, and bridge constructions can create apparent endpoint pull by conditioning on an ending. Hybrid models can combine several of these effects without declaring any law of felt fairness (Uhlenbeck and Ornstein, 1930; Doob, 1957; Sterling, 2012; Sutton and Barto, 2018).

A fair rival comparison gives each model comparable information, endpoint definitions, missingness assumptions, tuning freedom, and scoring rules. A visible curve may arise from

expectation, treatment, social support, dropout, regulation, mean reversion, or endpoint conditioning; before it is called PTVC-specific, those explanations must be allowed to win.

Human consequences remain separate from model comparison. Treatment success, adaptation, or addiction dynamics can be explained through ordinary causes without assigning blame or desert. Rival victory changes scientific language, not care, dignity, privacy, or refusal.

The target is the complete-stream boundary property; a route is one proposed way to realize it; a rival explains the observations without that route or target; and the observation process limits what can be estimated. Part VII asks which explanation earns the data under common rules.

Chapter 19 — Adaptation and Rebound

A promotion becomes Tuesday. A once-dazzling purchase joins the furniture of daily life. The first weeks after a loss feel unlike the months that follow. These ordinary changes give hedonic adaptation its power as a rival (Helson, 1964; Frederick and Loewenstein, 1999): experience is not a passive mirror of circumstance, and the meaning of a condition can change while the condition remains.

The classic treadmill image suggested a return toward a stable level of well-being. Later work made the picture less tidy. Adaptation varies by event, person, domain, measure, and time; affect and life satisfaction may move differently; and some changes leave durable effects. The strongest modern account is therefore not universal return but patterned heterogeneity (Diener et al., 2006; Lucas, 2007; Luhmann et al., 2012).

Gains illustrate the first mechanism clearly. Novelty fades, attention relocates, aspirations rise, comparisons change, and new advantages create new routines or obligations. A gain can remain materially important after its initial emotional intensity declines. Adaptation explains that decline without turning the gain into a hidden moral debt.

Losses are more varied. Some people recover much of their prior functioning, some establish a different but workable life, and some experience persistent distress, disability, grief, pain, or reduced opportunity. Averages can conceal all three patterns. Resilience research therefore strengthens the rival only when it preserves the tails rather than converting them into exceptions to a universal script (Bonanno, 2004; Albrecht and Devlieger, 1999).

Habituation adds repeated exposure. A stimulus that first commands attention may become background; a routine can become automatic; a repeated reward can lose novelty; and a recurring stressor can produce adaptation, sensitization, or exhaustion. What changes is not always the event itself but the organism's allocation of attention, expectation, and response.

Opponent process theory introduces a different temporal form. An initial state can recruit an afterstate whose strength and timing change with repetition. Pleasure may be followed by flatness, acute fear by relief, repeated drug reward by tolerance and withdrawal, or stress by recovery and fatigue. The sequence can be powerful without constituting any lifetime theorem (Solomon and Corbit, 1974; Solomon, 1980).

Addiction shows why reward components must be separated. Cue-triggered wanting can intensify while pleasure weakens, habits can persist after conscious endorsement changes, and withdrawal can reorganize attention around immediate relief. Opponent and incentive-sensitization accounts explain this dissociation without treating the sequence as a moral payment or a movement toward terminal neutrality (Berridge and Robinson, 1998; Koob and Le Moal, 2008; Robinson and Berridge, 2025).

These mechanisms can reproduce many patterns that might otherwise be assigned to the Law. Pleasure can fade, pain can soften, relief can follow threat, behavior can reorganize, and reported well-being can move toward a familiar range. Each success removes that observation from the Law's distinctive evidence unless an additional prediction separates the accounts.

The decisive distinction is the property being explained. Adaptation usually concerns change after an event or repeated exposure. Opponent process concerns linked responses across an episode or series. The strong Law concerns a latent cumulative property of the completed qualifying stream. Local return, even when genuine, is not identical to terminal closure.

The distinction is also statistical. A population mean may return while subgroups diverge, while the least well participants leave follow-up, or while the same numerical rating changes meaning over time. A smooth curve can be produced by recovery, replacement, attrition, response shift, or aggregation. The Chapter therefore keeps trajectories and observation processes visible alongside averages.

Ethically, adaptation describes change; it does not authorize the conditions that required adaptation. Reduced expectation may be coping, constraint, or both. Numbness may preserve functioning without becoming relief. Survival after poverty, humiliation, injury, or grief does not make the original harm fair or the remaining losses negligible.

Whatever remains for the Law must be earned. It needs a complete-stream prediction not already implied by habituation, set-point dynamics, opponent regulation, treatment, social recovery, reporting change, or statistical smoothing. If no such increment survives, adaptation is not incomplete; the stronger claim is unnecessary in that domain.

Fair comparison begins by giving the rivals their natural data. Adaptation models should see the event history, repeated

measures, changing context, treatment, and attrition. Opponent models should receive exposure, timing, dose, and after-response information. The Law should not win because ordinary explanations were denied the variables that define them.

A skeptical reader should leave with neither a weakened rival nor an inflated one. Hedonic adaptation is not a universal solvent, but it is a large part of the empirical landscape. Opponent dynamics do not settle a life, but they can explain many apparent offsets. Both deserve causal credit before any terminal interpretation begins.

The governing question is therefore concrete: when experience appears to return, what has returned? A current rating, a behavioral routine, a reference point, a bodily state, an average, or the complete latent trajectory? The answer determines whether the observation belongs to adaptation, measurement, regulation, or the still-unearned claim of terminal closure.

19.1 — What Adaptation Explains Well

A gain enters life twice: first as change, then as the new background against which later experience is judged. The first encounter may be vivid, but attention, comparison, expectation, and routine can alter its felt force. Adaptation begins with this movement from event to ordinary condition.

Adaptation after gains is heterogeneous rather than a universal reset.

Habituation and revised set-point thinking explain a substantial portion of that change. They also expose the importance of measurement: a stable rating may conceal changed standards, and a changed rating may reflect context rather than one underlying quantity. Population summaries must therefore be read beside individual trajectories and missing observations.

The opening task is therefore generous and exact: grant adaptation the heterogeneous post-event adjustment the evidence supports, then ask what it does not yet explain. That order protects the rival from caricature and the Law from inheriting a complete-stream conclusion it has not earned.

19.1.1 — Adaptation After Gains

Adaptation after a gain is easiest to see when delight becomes ordinary. A promotion, a larger home, a celebrated award, or a sudden financial windfall can reorganize daily life without preserving the first week's intensity. The serious lesson is not that gains are unreal. It is that attention, expectation, comparison, and

routine can change the emotional meaning of the same condition over time.

Hedonic adaptation is not one mechanism wearing one name. Novelty can fade, standards can rise, attention can move, habits can form, social comparisons can change, and a person can alter the environment that made the gain possible. Philosophical and psychological analyses therefore distinguish several routes by which a new good becomes ordinary rather than assuming one universal return process (Klausen et al., 2022).

The classic contrast between extraordinary gain and catastrophic loss made adaptation memorable, but later research replaced the picture of universal reset with a more differentiated one. Affective and evaluative measures can move on different schedules, event classes differ, and individual trajectories may show return, partial return, durable change, or continuing divergence (Brickman et al., 1978; Diener et al., 2006; Luhmann et al., 2012; Sheldon and Lyubomirsky, 2012).

Gains can also change the reference point from which later experience is judged. The salary that once removed fear may become the minimum required to sustain a new household. The recognition that once felt astonishing may become the standard against which the next year is measured. Adaptation can therefore reduce felt elevation without restoring the person to the same practical world.

Social comparison adds another route. A gain may move someone into a new peer group where the same resources feel ordinary, inadequate, or exposed. That change can alter satisfaction without

reversing the gain itself. The rival explanation is powerful because it can produce apparent return through changing comparison rather than through any hidden terminal requirement.

A careful analysis also asks what the gain changed outside the person's immediate report. Did it reduce danger, expand choice, create maintenance obligations, increase public scrutiny, or shift work onto family, staff, debtors, or future maintainers? Those questions belong to the material and social consequences of the gain. Adaptation alone does not answer them, but it prevents a visible advantage from being treated as a permanent experiential deposit.

When positive affect fades, the first comparison should be ordinary recalibration. Attention, expectation, comparison, habit, and the ending of surprise offer simpler explanations for a local rise and return. Only a prospectively specified pattern that survives those accounts can begin to bear on the Law's stronger claim.

A strong design separates three questions. Did the event improve conditions or reduce a burden? Did its affective impact change as the event became familiar? Does any path-sensitive structure remain after adaptation, expectation, and comparison are modeled under common rules? The proposed Law enters only the third question, and it enters as a hypothesis rather than as the owner of the first two.

This separation has practical consequences. Public policy should not dismiss housing, health care, safety, access, or freedom because beneficiaries may adapt to them. The point of many

goods is to become reliable background conditions. Adaptation can explain why they stop feeling exceptional without weakening the case for providing them.

The same caution applies to ambition. A person may reach a dream and discover that daily life remains daily life. That finding can teach something about expectation and sustainable motivation. It does not prove that striving was foolish, that success required a compensating low, or that a later disappointment was assigned to balance the gain.

19.1.2 — Habituation

Habituation is a reduced response after repeated stimulation, distinguished from reduced sensory sensitivity and fatigue of the response system (Rankin et al., 2009). It helps explain why a sound fades into the background, why a routine pleasure can lose salience, and why a stable circumstance may cease to command the same attention. In a fairness inquiry, habituation is important precisely because it can create return-like patterns without saying anything about the complete course of one conscious stream.

Set points are the treadmill's stabilizing idea, but contemporary adaptation theory does not require one immutable emotional thermostat. The stronger rival permits person-specific ranges, event-sensitive shifts, and different courses for affect, life satisfaction, desire, and expectation. A reference range is therefore a statistical and psychological tendency shaped by history and context, not a fixed essence or a neutral origin granted in advance (Diener et al., 2006; Lucas, 2007; Luhmann et al., 2012).

Scientifically, set points also limit simple event interpretation. A group may return toward prior well-being after a gain, or one person may regain a characteristic mood after loss. Those results may support baseline regulation. The Law earns no support unless it predicts something beyond return, such as path-level, horizon-sensitive, or endpoint-related structure that set-point models do not already require.

The strongest set-point rival is flexible. It allows baselines to shift after major events, illness, repeated stress, chronic support, institutional exposure, or long-term deprivation. That flexibility makes the account more realistic, but it also makes hindsight especially dangerous. A set point that can move anywhere after the fact explains too much. Both the Law-linked model and its rivals therefore need prospectively specified baseline rules before outcome inspection.

Set points also make personal comparison dangerous. If one person appears to recover faster than another, that difference may reflect baseline, biology, support, measurement context, risk exposure, culture, safety, and hidden resources. It should not become a moral contrast. The faster-rebounding person is not “fairer,” “stronger,” or “more finished.” The slower-rebounding person is not a failed stream.

If the Law retains any claim after set-point models are properly built, it must be because the remaining structure cannot be reduced to baseline return. The chapter therefore teaches a strict order: baseline first, adaptation second, Law-specific residual only after the ordinary rival has been given its best form.

Set-point language becomes serious when it is treated as a statistical tendency, not as a metaphysical destiny. A person may have characteristic affective ranges shaped by temperament, body regulation, social environment, memory, threat sensitivity, and opportunity. Those ranges can be stable enough to forecast many returns and flexible enough to change under sustained conditions.

For the Law, the set-point rival is especially important because it can imitate convergence without implying closure. If a stream tends to drift back toward its own range after perturbation, terminal-looking calm may be ordinary regulation near a personal baseline. The formal claim must explain why the terminal condition has a Law-specific structure rather than merely being the final appearance of set-point dynamics under declining reserve.

The organism may learn that the stimulus is no longer new, or that responding with full intensity is too costly.

Distinguishing benign familiarity, costly accommodation, and defensive disengagement keeps quiet from being treated as closure. Habituation can explain why an observed response contracts over an interval, while the formal question concerns a complete stream and an independently defined endpoint. The rival receives full credit for local contraction; the stronger claim receives nothing unless it adds a separate consequence specified before outcome inspection.

19.1.3 — Why This Rival Is Serious

Adaptation is most formidable when it remains heterogeneous, mechanistic, and modest. Novelty fades, standards move,

comparison groups change, attention shifts, and routines reorganize. It can explain substantial adjustment without claiming a neutral endpoint, so the Law must identify a distinctive burden of evidence rather than mistake unexplained territory or dramatic vocabulary for discovery.

Population studies give the rival much of its force. After major gains and losses, group averages often move back toward earlier levels, but the degree and speed of return vary by event, outcome, person, and follow-up window. Partial adaptation is common enough to demand explanation and uneven enough to defeat the picture of one universal emotional reset (Diener et al., 2006; Lucas, 2007; Luhmann et al., 2012).

The average is only the first layer. Longitudinal records may contain stable trajectories, brief disruption, gradual recovery, durable decline, delayed change, or movement to a new level. The apparent number and prevalence of those courses can depend on the outcome measure, time base, missingness, and modeling assumptions, so trajectory labels should be treated as estimates rather than natural kinds (Galatzer-Levy et al., 2018; Infurna and Luthar, 2016).

That model dependence does not weaken the rival into irrelevance. It clarifies the task. Adaptation research already has repeated measures, panel studies, event designs, behavioral observations, and clinical follow-up that can test modest predictions without first accepting a cumulative ledger or a terminal endpoint. Methodological independence is one reason the rival deserves priority.

Its strongest predictions are also restrained. A new condition may become less salient. Expectations may rise. A person's reference group may change. Daily behavior may absorb the event. None of these claims requires a hidden moral account, and each can be wrong in observable ways. Modest, testable claims are difficult competitors for a grand hypothesis because they can explain much while risking less (Klausen et al., 2022).

The public phrase hedonic treadmill often compresses several mechanisms into one image. Time heals, people get used to things, and life returns to normal may refer to material repair, social support, altered expectation, avoidance, numbness, habit, or simple passage of time. The rival becomes scientific only when those routes are separated well enough to predict different courses.

A fair test therefore freezes event class, outcome, time scale, subgroup, and missingness rules before the trajectory is interpreted. It should ask which forms of return the adaptation model predicted, which forms it did not, and whether any remaining pattern survives reasonable changes in measurement and analysis. A headline average cannot perform that work by itself.

Only then can the Law ask for incremental evidence. It would need a prospectively specified stream-level consequence that is not ordinary baseline return, not a changing reference point, not a local counterresponse, and not an artifact of who remained observable. The stronger claim earns no credit for territory already explained by the rival.

The decisive comparison is therefore about causal location and time scale. Adaptation may operate through attention, expectation, habit, social rebuilding, treatment, changing goals, or response standards, and those pathways can unfold at different speeds. A terminal theory gains nothing by calling all movement toward a familiar range one mechanism. It must identify a residual prediction that survives when the ordinary pathways are measured separately and when people who do not recover remain visible in the analysis.

This discipline improves care as well as inference. A person whose distress persists is not evidence of moral failure, and a person who improves is not evidence of hidden cosmic compensation. Clinical, social, and material causes retain their own names. The scientific question begins only after those causes receive a fair test, because every observation they explain is territory the broader hypothesis no longer owns as distinctive support.

19.2 — Opponent Processes, Tolerance, and Rebound

Adaptation describes reduced response or changed reference. Opponent process adds a paired temporal dynamic: an initial state recruits a counterstate whose onset, strength, and duration may change with repetition. That architecture can produce relief, withdrawal, tolerance, contrast, and rebound.

The mechanism is not one story for every domain. Drug exposure, acute fear, exercise, pain, and repeated reward involve different systems and timescales. In some cases the afterstate strengthens; in others the system sensitizes, destabilizes, or fails to recover (Solomon and Corbit, 1974; Koob and Le Moal, 2008).

The scientific value lies in specifying sequence rather than moral meaning. A later negative state does not make an earlier pleasure a moral debt, and relief after distress does not show that the particular distress was prescribed. The rival explains temporal organization without assigning moral payment.

Pleasure, stress, recovery, tolerance, and addiction now supply the temporal test. A counterprocess can organize an episode with real causal force while remaining logically separate from terminal closure.

19.2.1 — Pleasure and Counterpressure

Pleasure can recruit counterpressure. Repeated reward may weaken the first response, strengthen craving, or deepen the contrast when stimulation ends. Opponent-process accounts make those temporal dynamics intelligible without invoking a hidden

moral budget, and they therefore deserve to be treated as explanatory competitors rather than as illustrations of the Law.

Pleasure often comes with an afterlife. A sweet experience may be followed by longing, fatigue, contrast, satiety, guilt, craving, boredom, or a higher expectation for the next experience. A moment that first feels abundant can become a new baseline. A luxury can become ordinary. A thrill can leave restlessness. None of this makes pleasure wrong. It makes pleasure temporal (Solomon and Corbit, 1974; Solomon, 1980).

For evidence, the design lesson is clear. Positive events must be measured through time, not as isolated peaks. Researchers must ask whether later negative or neutral shifts are explained by adaptation, opponent response, expectation change, social comparison, fatigue, or reward learning. Only rival-resistant structure beyond those explanations can begin to matter for the Law's stronger claim.

Pleasure requires neither suspicion nor innocence by category. The relevant questions are function, flexibility, consequence, and what the pattern does over time. Some pleasures restore capacity; others become costly through exploitation, repetition, or narrowing of choice. Opponent and incentive-sensitization accounts help explain that difference without turning enjoyment into moral debt (Koob and Le Moal, 2008; Berridge and Robinson, 1998).

The scientific problem is that the first pleasant state is only one part of the sequence. A later counterstate may belong to the same pattern, but it should not be interpreted as moral repayment. If the

data show a counterprocess, the data show affective dynamics. They do not show that happiness owes suffering.

Opponent-process thinking begins with a dynamic insight: intense pleasant states can recruit processes that dampen, oppose, or follow them. The nervous system does not passively record pleasure. It regulates arousal, expectation, reward prediction, social risk, fatigue, satiety, and future wanting. The after-state can become part of the experience's cost.

That logic matters because a formal model of fairness must not reinterpret every later flatness as moral balance. After-pleasure dullness, satiation, withdrawal, or boredom may be ordinary counterpressure. The Law would need to show a terminal and path-level structure beyond that local sequence before it could speak.

Pain and relief cycles are among the most tempting places to overclaim. A burden ends, and relief comes. A fear passes, and the world feels bright. A migraine lifts, and ordinary light becomes beautiful. A conflict resolves, and breathing returns. The afterstate can feel positive precisely because it follows pain. The sequence is real.

Relief is not retroactive permission. The fact that relief follows pain does not make the pain fair, deserved, necessary, useful, or good. A careful science can study relief without romanticizing the wound that preceded it. Pain remains pain, relief remains relief, and the practical question is whether burden was reduced or merely shifted.

Opponent processes are one explanation for relief after threat. Endogenous pain modulation, reduced arousal, contrast, social reassurance, medication, and restorative sleep offer other or interacting explanations. Each can change current felt state; the study must identify which process produced the observed relief.

In an addiction cycle, brief relief from withdrawal or craving may follow a much longer burden, and that relief can reinforce repeated use without canceling health, relationship, financial, or agency costs. The sequence is clinically important, but it is not compensatory accounting (Berridge et al., 2009; Volkow et al., 2016).

As evidence, pain and relief cycles are rival territory unless the pattern carries more. If a study shows that relief follows discomfort, that is not enough. If it shows that local counterprocesses scale with intensity, that is still not enough. The Law needs signatures that cannot be reduced to ordinary rebound, such as whole-stream path dependence or endpoint behavior under disciplined observation. Otherwise, the opponent process rival explains the cycle.

19.2.2 — Stress and Recovery

Stress and recovery unfold across several systems and several clocks. Heart rate can settle within minutes, sleep can remain disrupted for days, and a person's sense of safety can change for years. The word recovery can therefore refer to physiological downregulation, restored function, changed behavior, narrative reorganization, or genuine relief, and those are not interchangeable outcomes.

An acute stress response is a mobilization pattern, not a direct moral or experiential verdict. Attention narrows, energy shifts, autonomic and endocrine systems respond, and action becomes urgent. Those changes may help a person survive immediate danger while also carrying fatigue, pain, fear, or later cost (McEwen, 1998; Sterling, 2012).

Recovery also depends on resources outside the body. Sleep, safety, food, privacy, medicine, money, social support, child care, time away from threat, and the credibility of institutions can determine whether a stress response resolves. A curve that looks biological may therefore contain a large social infrastructure hidden beneath it.

Opponent processes can produce either habituation or escalation. Repeated exposure may reduce response, but it can also strengthen the afterstate, deepen craving, increase vigilance, or make recovery slower. A serious rival model must allow both adaptation and sensitization rather than forcing every trajectory toward one baseline.

Addiction makes this especially clear. Reward may initially produce pleasure and later become tied to tolerance, wanting, withdrawal, shame, and relief seeking. The person may continue the behavior less to obtain the original pleasure than to escape the negative state that now follows its absence (Koob and Le Moal, 2008; Robinson and Berridge, 2025).

That cycle should never be described as pleasure being paid for by pain. The safer description is mechanistic: repeated reward can alter learning and regulation so that future choice becomes

narrower. The burden is captivity, health risk, social damage, and reduced agency, not a moral debt created by enjoyment.

Wanting and liking also need separation. A cue can acquire motivational force after the pleasure it once predicted has weakened. Behavior may then look like pursuit of value while the person reports little enjoyment. This is one reason observable choice cannot be treated as a transparent meter of felt valence.

Stress and addiction both warn against reading local reversals as complete repair. Relief after danger, use, withdrawal, or panic can be intense because a negative state has ended. That relief is real, but it does not justify the preceding harm or establish any condition on the completed stream.

The recovery distribution matters as much as the mean. Some people recover quickly, some slowly, some in waves, and some remain changed. Resilience research shows several common trajectories, but group averages do not tell one person how they should recover or prove that persistent burden is an exception to be explained away (Bonanno, 2004; Galatzer-Levy et al., 2018).

Clinical care remains prior to interpretation. Treatment access, harm reduction, trauma-informed support, family protection, pain relief, and nonhumiliating assistance do not wait for a theory of lifetime valence. The rival literature is valuable partly because it identifies mechanisms through which care can widen the person's available choices now.

Institutions can create opponent dynamics too. A workplace may alternate overload with brief praise, a platform may alternate social reward with uncertainty, and a coercive relationship may

alternate fear with relief. The positive interval can strengthen attachment to the system without making the system fair.

These mechanisms do not supply a truth-level endpoint claim about the signed total of a complete conscious stream. They explain local and intermediate dynamics, sometimes with great clinical and causal power. Any Law-linked increment must survive those explanations rather than borrow their language of balance.

19.2.3 — Counterprocess Is Not Terminal Law

Object, time scale, and quantifier differ. Opponent process becomes a genuine competitor when it receives full credit for temporal counterpressure and the Law is denied every shortcut through analogy.

Opponent-process models concern linked responses over defined episodes or exposure histories. They can estimate the latency, magnitude, duration, and decay of a primary response and its after-reaction. The terminal hypothesis requires additional bridges: one bearer, an exhaustive domain, a neutral origin and scale, a truth-level endpoint, and equality of the complete positive and negative magnitudes. Event dynamics do not supply those bridges by themselves (Solomon and Corbit, 1974; Solomon, 1980).

Paired movement can nevertheless resemble fairness. Pleasure may be followed by unease, pain by relief, intoxication by withdrawal, and repeated stress by exhaustion or altered recovery. Sequence is not accounting, however, and resemblance is not entailment. The Law would fail scientifically if it depended on the

emotional appeal of counterweight rather than on a consequence that can distinguish the two models.

The strongest rival permits asymmetry and state dependence. Relief need not equal the preceding pain. Tolerance may grow while liking contracts. Sensitization can coexist with habit, cue reactivity, or reduced pleasure. Recovery can change with age, deprivation, treatment, social setting, and exposure history (Berridge and Robinson, 1998; Koob and Le Moal, 2008; Robinson and Berridge, 2025).

Those distinctions matter because repeated exposure changes more than reported feeling. It can alter attention, expectation, action policy, memory, physiological readiness, and the availability of alternatives. A model based only on self-report may read coordinated adaptation as hidden balance; a model based only on behavior may miss the felt burden that remains.

The comparison should therefore begin with the best event-level account. Fit the opponent process on the time scale it claims, allow domain-specific parameters, carry uncertainty forward, and test whether the same model transports across exposures or contexts. Only the unexplained remainder may enter a broader terminal comparison.

A genuine discriminator would have to connect local dynamics to complete-stream structure without changing the endpoint, bearer, scale, or admissible history after the result is known. It would also have to survive ordinary explanations such as medication, withdrawal, social support, recovery opportunity, reporting

change, and observation loss. The existence of rebound alone cannot meet that burden.

This separation protects both criticism and support. A critic does not defeat the Law merely by showing that rebound exists, because rebound addresses a smaller claim. A defender does not support the Law by showing the same thing, because the rival already owns that pattern. Each model should be judged only on what it actually asserts.

The distinction is also a form of care. A person in pain should not be promised a chosen relief or timetable, and a person after pleasure should not be told that a particular later distress is required. Exact PTVC does require an equal-and-opposite net remainder for a genuinely nonzero accumulated balance. Local mechanisms and that conditional arithmetic can be studied without prescribing suffering or making recovery a moral obligation.

19.3 — Where These Rivals Shine

A rival earns priority where it explains the data with fewer unsupported commitments. Adaptation and opponent processes are especially strong when the exposure is repeated, the timescale is observable, the response changes predictably, and the mechanism can be tested without invoking a completed life.

Short-term recovery, habit formation, cue-driven behavior, tolerance, withdrawal, and some forms of post-event adjustment provide such domains. Treatment, social support, material repair, and changed opportunity may also account for improvement that a grander theory would otherwise claim too quickly.

Clinical reality prevents easy generalization. Persistent pain, prolonged grief, trauma, disability, depression, addiction, and chronic stress can show partial return, altered baselines, recurrent episodes, or no meaningful recovery. The rival remains strong by representing those differences rather than averaging them away.

Ordinary explanations receive first claim on these domains. Law-specific interpretation begins only after adaptation, context, treatment, and heterogeneous trajectories leave a pattern they cannot account for.

19.3.1 — Short-Term Recovery

These rivals are strongest where the horizon is short and the perturbation is observable. Recovery after acute stress, fading excitement after a gain, relief after temporary pain, and settling after ordinary disruption can often be traced to sleep, food, contact, explanation, treatment, problem resolution, attention

shift, or time. In those domains adaptation and rebound should receive explanatory priority unless the Law adds a distinct, prospectively testable consequence.

For the Law, short-term recovery is useful as a control. If the Law predicts a pattern, researchers must ask whether an ordinary recovery curve already produces it. Only if the observed pattern differs in timing, scale, path dependence, horizon sensitivity, or endpoint relation should the Law receive additional attention.

Short-term recovery is often the first change observers can see. The crying stops, the person returns to work, the family resumes routine, or the institution marks the case resolved. Public legibility is not the same as complete recovery, however. The remaining question is whether burden has genuinely fallen, changed form, or moved into quieter channels (McEwen, 1998; Ramsay and Woods, 2014).

A good study of short-term recovery would therefore include delayed follow-up, qualitative context, sleep, trust, social connection, functional capacity, and participant-defined burden. A single return-to-baseline measure can be useful, but it should not be allowed to close the case. Recovery is a process, not a stamp.

Test whether the timing follows sleep, circadian rhythm, support, problem resolution, medication, or attention shift, and whether the same form appears in negative controls or after unrelated events. When those checks explain the pattern, credit ordinary recovery; only a surviving difference enters the terminal comparison.

Repeated exposure supplies a second natural test. With practice, an intimidating task can become routine; with familiarity, a strange place can become home; with habituation, a distracting sound can recede. The change may reflect learning, increased safety, or reduced salience rather than any terminal process.

Repeated exposure can build capacity or conceal harm. Practice may increase skill, safe exposure may reduce fear, and familiarity may make a new school or role workable. The same outward quiet can also reflect enforced tolerance of surveillance, humiliation, poverty, noise, unsafe work, discrimination, or chronic uncertainty. Reduced reaction is therefore evidence about adaptation, not a certificate that the condition became fair.

A close reading of repeated exposure asks what changed. Did the person gain skill, safety, support, and agency, or lose sensitivity because protest became costly? Did the environment become less harmful, or did the person become less able to respond? Did repetition widen the available options, or narrow them? (Wilson and Gilbert, 2005; Luhmann et al., 2012).

As evidence, repeated exposure is rival-rich. Habituation, learning, fatigue, expectancy, social conformity, and opponent strengthening can all shape the results. If an apparent fairness signature appears only after repetition, it may be a repetition effect. The Law must not claim it unless negative controls and rival models are satisfied.

Adaptation is especially persuasive when exposure history predicts change better than cumulative record position. If the fourth exposure differs from the first because the organism has

learned, the Law does not win by calling the decline counterweighting. The fair question is whether exposure history, expectation, and context explain the pattern before any terminal architecture is invoked.

19.3.2 — Behavioral Adaptation

Behavioral adaptation shows how people alter routines, attention, expectations, goals, relationships, locations, and requests for help after circumstances change. Those actions can improve, preserve, or narrow the conditions of later feeling. Learning and environment often explain more than a whole-life hypothesis needs to claim, which is why behavior must remain visible as cause rather than be absorbed into terminal language.

Behavioral adaptation begins when repeated action becomes easier, more automatic, or more reliably cued. Habits can stabilize sleep, medication use, exercise, work routines, avoidance, and help-seeking, while the same habit machinery can also preserve craving or withdrawal from valued activity. Context and repetition therefore explain substantial persistence without requiring a whole-life boundary condition (Lally et al., 2010; Wood and R  nger, 2016).

Support is part of the behavioral mechanism, not background noise. Stable housing, transportation, money, accessibility, childcare, legal protection, treatment, and trusted relationships can make new routines possible. Apparent individual resilience may therefore depend on resources distributed outside the person, and a fair model should represent those resources rather than attribute all change to internal adjustment.

This means that recovery after loss may come not from hidden balance but from altered action. A person may feel better after building a new support network. A worker may recover after leaving a harmful job. A patient may stabilize after treatment. A parent may regain calm after childcare improves. These are real repairs, and the Law should not steal them from the people and systems that made them possible (Diener et al., 2006; Luhmann et al., 2012).

Behavioral adaptation is constrained by the available menu. Money, safety, documents, health, transport, disability accommodations, social support, and lawful access enlarge what can be changed; their absence can make survival itself extraordinary work. Persistent burden is not evidence that a person failed to adapt. The relevant questions are what options actually existed, what support was missing, and which systems made repair harder.

Behavioral adaptation matters because people are authors under constraint. A smaller life may be freely chosen because it is peaceful and desired, or selected because fear, poverty, stigma, disability, or repeated humiliation made other paths unreachable. Similar behavior can therefore carry very different meanings. The Law may ask whether deeper structure remains after action, repair, and menu change are modeled, but it may not erase ordinary agency or unequal opportunity.

The rival explains menu changes through learning and adjustment; the ethical inquiry asks whether the menu itself was fair. Did the person adapt to a world that should have changed? Did the system move burden onto the person's discipline? Did

apparent adjustment hide institutional failure? These questions do not defeat the rival. They locate its ethical boundary.

For research, behavioral adaptation should be modeled alongside self-report and physiology. Choices reveal something about lived cost, but they do not reveal everything. A person may avoid because avoidance is wise, or because the world has made approach too punishing.

Habit expression is strongly context-dependent. A response practiced in one room, schedule, social setting, or emotional state may weaken when the context changes, while an old cue can restore it after a long absence. This gives the rival concrete predictions about persistence, lapse, and reorganization, and it explains changing paths without invoking terminal closure (Wood and R nger, 2016).

19.3.3 — Clinical Exceptions

Clinical exceptions are not inconvenient footnotes to a general story of return. Persistent grief, chronic pain, disability, trauma, addiction, and severe mood disorders can remain altered, fluctuate, or divide into distinct courses. A rival becomes more credible when it explains that heterogeneity instead of treating nonrecovery as noise (Galatzer-Levy et al., 2018; Luhmann et al., 2012).

Professional care cannot be replaced by adaptation theory. Pain, grief, depression, trauma, dissociation, addiction, psychosis, or disabling fatigue already justify care, access, protection, and relief, while research observation must limit prompts, devices, disruption, and endpoint-focused questioning and permit genuine

refusal or No adequate estimate (Beauchamp and Childress, 2019).

Concrete mechanisms such as sensitization, withdrawal, intrusive memory, inflammation, sleep disruption, medication, isolation, cognitive load, and treatment response deserve explanatory priority where they fit.

Expertise remains distributed among clinicians, researchers, disability scholars, palliative teams, patient advocates, and the person receiving care. The public root can define its question but cannot replace those forms of judgment and lived authority.

Nonrecovery does not hand the Law a victory. It may favor another rival, treatment or social account, measurement failure, or a narrower class; suffering must never become a trophy for a preferred theory.

The balance is clear: rivals explain without erasing heterogeneity, research measures without possessing the person, and care arrives before interpretation. Only then may the remaining scientific question be asked.

19.4 — What They Do Not Guarantee

Strong rivals have boundaries. Hedonic adaptation and opponent processes usually address changes within an observed interval, not an exact latent property at the independently defined end of a complete conscious stream. That difference is real, but it is only a difference in claim scope.

No inference follows merely from the rival's silence. A model that does not guarantee terminal closure has not thereby failed to explain the observations it was built to explain. The Law gains no evidence from asking a broader question unless it produces a distinct, testable prediction for that broader object.

The additional work includes bearer definition, path construction, neutral origin, endpoint discipline, missing experience, and a measured-to-latent bridge. It also includes showing why population return, local rebound, or response shift cannot reproduce the proposed signature under common rules.

The unmet requirements are not hidden evidence for the Law. They define what a stronger account must independently predict while allowing adaptation and opponent process to keep the explanatory territory they have earned.

19.4.1 — No Per-Stream Endpoint Claim

Adaptation research usually begins with events, baselines, reports, and repeated observations within a life. It does not ordinarily define the truth-level endpoint of a complete qualifying conscious stream or predict an exactly neutral terminal residual.

That difference identifies an additional claim the Law would have to earn; it does not count as evidence for the Law.

A return toward a typical range can be scientifically important without being terminal. Hedonic adaptation may explain how many people respond after particular gains or losses, and set-point accounts may describe recurring ranges of well-being. Neither claim says that every separately routed stream closes at exact neutrality (Diener et al., 2006).

Imagine a month of distress followed by a return to the person's earlier level of well-being. That return describes the state reached; it does not make the difficult month disappear from the history. A different path could end at the same level after a largely pleasant month. The matching endpoints therefore tell us little, by themselves, about the two accumulated histories. This is why adaptation and the Law ask different questions: how an observed state changes, and whether the complete signed history meets a particular condition at its true endpoint.

This separation blocks two opposite public shortcuts. “People adapt, so life is fair” mistakes regulation for the formal claim. “Some people do not adapt, so the Law is impossible” also moves too quickly, because nonadaptation during an observed interval does not by itself identify the complete latent residual. Both claims outrun what the evidence has established.

A genuine per-stream endpoint claim requires a qualifying bearer, a time-indexed path, an independently defined truth endpoint, a latent cumulative object, and a terminal condition. It also requires rules for missing experience, duration, intensity, recurrence, and

uncertainty. Without that dependency chain, endpoint language adds grandeur rather than testability.

Anecdotes cannot supply the missing structure. A recovery story and a nonrecovery story may be morally important, but each remains a partial record shaped by memory, narration, observation, and unknown future experience. Narrative symmetry is not terminal evidence (Lucas, 2007; Luhmann et al., 2012).

Longitudinal analysis faces the same danger in technical form. If every local return is treated as a compensating term, the measured record can be made to telescope toward zero by construction. Order, exposure, missingness, residue, and endpoint uncertainty must remain visible so neutrality is not manufactured by the method.

Adaptation can therefore explain much of the observed motion and remain silent about the complete endpoint. The Law may investigate that remaining question only if it produces a prospectively testable consequence beyond local recovery. An unanswered question is a burden of proof, not a scientific advantage.

19.4.2 — Averages Still Fail Streams

An average can describe a population while failing every particular stream differently. Mean recovery may combine rapid return, delayed return, chronic distress, improvement beyond baseline, rare disease, exceptional support, extreme deprivation, early death, and missing follow-up. A person lives one path, not the mean path, so a population tendency can neither establish nor refute the stream-level target without a valid bridge.

Hard cases keep the inquiry answerable to the lives it discusses. Innocent children suffer, chronic pain persists, trauma returns, dementia changes familiar identity, some people die early, and some burdens remain hidden or publicly unanswered. A responsible account must preserve grief, disability, addiction, loneliness, and terminal uncertainty rather than dissolve them into reassurance. Suffering remains real where every model remains incomplete (Luhmann et al., 2012; Lucas, 2007).

The conclusion is neither rival victory nor Law victory. Hard cases remain tests of whether adaptation explains enough and whether any distinctive Law-linked prediction can be investigated without cruelty.

For rivals, hard cases reveal the boundary conditions of adaptation itself: ongoing threat, persistent injury, absent support, severe scarcity, disrupted identity, or a dysregulated counterprocess. A fair science studies those conditions rather than calling them failure of the person.

For the Law, these cases set a severe standard. Any surviving claim must accommodate them without spiritualizing suffering, demanding unethical observation, or treating honest ignorance as failure. Where that standard cannot be met, the claim narrows and care remains the first answer.

Hard cases also expose response-shift ambiguity. People can change standards, values, identities, and definitions of quality under chronic conditions. That may be wise adaptation, protective reframing, social pressure, constrained choice, or a mixture. A

later report of acceptability does not automatically mean the original burden was erased.

The account is strongest when it can hold two truths at once. People may genuinely remake life under hardship, and some hardship remains unjust, disabling, or unrelieved. Adaptation must preserve the reality of human reconstruction; the Law must never recruit unresolved suffering as emotional leverage for proof.

Statistics are indispensable to an answerable Law. Models, cohorts, distributions, estimates, uncertainty, and rival tests reveal patterns no anecdote can settle. Their use becomes careless only when group regularity is mistaken for individual closure. The per-stream question is stricter than a population tendency.

Averages can hide missing burden. The most distressed people may be least able to complete follow-up; poverty can limit devices, illness can interrupt surveys, dying can end observation, and coercion can shape what is safe to report. Missingness therefore belongs inside the rival comparison. Who disappeared, why, and under which observation rule may determine whether apparent adaptation is recovery or the loss of the most burdened streams from view.

The right comparison is not average versus anecdote. It is distributional model against distributional model. Which account predicts variance, tails, subgroups, timing, missingness, clinical exceptions, and endpoint-adjacent behavior? If the Law cannot improve that prediction without overfitting, the average-based rival may win.

19.4.3 — Why Law of Fairness Must Earn the Difference

The difference between local regulation and complete-stream closure must be earned through incremental risk. A Law-linked account needs a consequence that strong adaptation, opponent-process, selection, and observation models do not already predict. Without that exposure to loss, the larger language adds scale but not knowledge.

The comparison begins by writing the rivals first. What return, rebound, heterogeneity, treatment response, missingness, and endpoint behavior do they expect without PTVC? Only after those expectations are frozen may a PTVC-linked model state the additional pattern it predicts. This order prevents ordinary science from being counted twice.

Neither success nor failure of a rival automatically supports the Law. If adaptation explains a curve, the rival owns the curve. If adaptation fails, several other explanations may remain, including prediction, allostasis, treatment, changing measurement, selection, and simple model misspecification. Unexplained is not the same as Law-specific.

Endpoint discipline is part of the price. The bearer, complete domain, neutral origin, endpoint rule, admissible transformations, and target modality must be prospectively specified without using the observed ledger outcome to select them. Statistical independence is an additional branch-specific condition, not a universal requirement. Moving any element after failure would manufacture a new question rather than answer the old one.

The measured-to-latent bridge is another nonnegotiable step. A quiet report, a lower arousal signal, a return in life satisfaction, or a group mean near zero is not the proposed complete phenomenal residual. The evidence must show why the available channels constrain the target strongly enough to support the declared verdict.

Consider a person whose well-being falls after illness and partially recovers. Adaptation may explain changed expectations, treatment may relieve symptoms, social support may widen options, and response shift may change the meaning of the scale. A PTVC-linked claim becomes distinctive only if it predicts something beyond those causes under an independently defined endpoint and common observation rules.

The comparison also needs a declared margin of success. Better fit on the same data is not enough if the gain comes from greater flexibility, more tuning, or hidden endpoint information. Support should increase only when the PTVC-linked account makes a prospectively specified, rival-resistant prediction that transports to held-out observations after uncertainty and complexity are counted (Mayo, 2018).

The Law must earn the difference ethically as well. A study cannot seek a cleaner terminal pattern by increasing burden, delaying care, pressuring participation, or treating a person's private experience as project property. Scientific distinctiveness and ethical acceptability are separate gates, and neither can compensate for failure of the other (Beauchamp and Childress, 2019). A design review can compare the additional information

an observation would provide with the practical demands it places on the person providing it.

Partial defeat must remain available. The Law may lose intuitive examples to adaptation, survive only in a narrower reference class, or remain unidentifiable even after careful measurement. Those outcomes should change the public claim rather than be absorbed into a more flexible story. A narrower result can specify the event class and observation window in which the proposed distinction remains worth testing with better measures.

A stronger claim must predict more, expose itself more clearly, and accept a genuine route to loss. A useful comparison would place the rival's forecast beside the proposed route's forecast before the observations that could distinguish them are collected. The comparison should also state in advance what amount and kind of divergence would count as meaningful, so a weak difference cannot be promoted after inspection. That makes rival victory, ambiguity, and partial defeat interpretable rather than rhetorical.

19.5 — If These Rivals Win

A scientific proposal should describe in advance what it will surrender. If adaptation, opponent process, treatment, context, or ordinary regulation explains a pattern better, that pattern is no longer available as distinctive support for the Law. The concession should be visible in both the research record and public language.

Victory can be partial. One mechanism may explain pleasure fading, another addiction cycles, and another recovery after stress. The Law may retain a narrower endpoint question while losing broad claims about how everyday balance-like patterns arise. Precision requires several categories between total triumph and total erasure.

Concession, retirement, and scientific progress are therefore treated as ordinary outcomes. A theory gains credibility when readers can see exactly what a winning rival takes from it.

19.5.1 — What Law of Fairness Concedes

Concession removes a phenomenon from the Law's evidential column when a rival explains it better. It does not erase the phenomenon or diminish the person who lived it. It gives explanatory credit to the model that earned it and narrows every dependent claim.

If adaptation explains return after gains or losses, that return belongs to adaptation. If opponent process explains rebound, tolerance, or relief, those sequences belong to opponent process. Local regulation should not be retained as support for terminal

closure merely because it first led readers to the question (Diener et al., 2006; Luhmann et al., 2012).

A clean concession names the result plainly. Adaptation explained this pattern. Opponent process predicted this effect. The proposed signature was not specific. Phrases such as “the evidence is complicated” should not be used to preserve an image after the relevant comparison has been decided.

Concession differs from narrowing and defeat. A concession gives up a particular observation or example. Narrowing changes the reference class, modality, or scope of the hypothesis. Defeat rejects the prospectively declared claim. Keeping those outcomes separate prevents one lost example from being dramatized as total collapse and prevents a broad failure from being disguised as a small adjustment.

The triggers should be prospectively specified. A research plan should identify which adaptation findings would remove a signature, which opponent-process result would reclassify an example, and which failure to outperform a rival would pause stronger public language. Prospective specification turns concession into method rather than reputation management.

Concession also protects emotional ownership. A reader may have found a recovery story meaningful because it seemed to suggest hidden balance. If the story is better explained by treatment, support, habituation, or ordinary adjustment, the project should say so. An image may open a question; it may not remain an answer after its status changes.

Human dignity remains outside the model contest. Recovery is not proof of virtue, and nonrecovery is not evidence of failure. The explanation belongs to the rival model; the person's worth, need for care, and right to interpret the experience remain their own.

A conceded pattern may still matter to the rest of the series under its correct status. Ordinary-life care, public burden, material conditions, rights, and optional meaning do not disappear. They simply may not borrow a stronger scientific claim than the evidence permits.

A project becomes more credible when it can say, without evasion, this part was already explained. That sentence makes the remaining question smaller, cleaner, and more difficult to protect by rhetoric.

19.5.2 — What Law of Fairness Retires

Retirement means that a claim, route, signature, example, or public artifact no longer serves as current support. Its historical record may remain, but the retired item no longer carries the authority it once had.

The reason should be stated exactly. Unsupported, explained by a rival, nonidentified, superseded, unsafe for public use, and withdrawn for ethical reasons are different outcomes. Naming the reason helps readers distinguish a failed mechanism from an unanswered question or unacceptable application (Mayo, 2018).

Examples retire when their interpretation fails. A recovery story belonging to adaptation should not remain evidence of closure. A

terminal anecdote should remain testimony, not result. A group-average graphic should not imply every stream returned.

Metaphors can require retirement even without a decisive experiment. Debt, payment, price, deserved burden, and cosmic settlement require correction when they imply moral obligation, a prescribed harm, or a reliable personal reward rather than the limited conditional accounting. A memorable phrase is not worth preserving when its ordinary reading violates the ethical firewall.

Public artifacts must change where the risk appears. Captions, Illustrated Guide pages, lectures, websites, social excerpts, donor descriptions, and classroom materials cannot keep a stronger formulation after the scientific claim narrows. A correction in a distant appendix does not repair a high-reach image.

Retirement is not erasure. The record should preserve what was proposed, why it changed, which dependents were updated, and what would reopen the question. That history is useful only when it cannot be mistaken for the current position.

The ethical floor does not retire with a mechanism. Care before proof, no induced harm, no human scoring, privacy, consent, refusal, and dignity were never prizes awarded for PTVC winning. The burden-sensitive lens may also remain useful so long as it is not presented as restored scientific authority.

Corrections should travel only along the claims each title actually inherited. The scientific companion records the verdict; receiving titles revise the dependent statement and preserve their independently supported work. Spiritual resonance, institutional

success, donor preference, or transition momentum cannot keep a superseded sentence alive.

A fair correction uses ordinary language: this example now belongs to adaptation; this signature was not specific; this route failed; this claim narrowed; this phrase retired because it could be read as morally owed pain or a promised reward. Readers should not need specialist training to discover that the public meaning changed.

19.5.3 — Losing as Scientific Progress

A rival victory can be scientific progress because it replaces a protected possibility with a better map of causes. It identifies which observations were nonunique, which mechanisms deserve credit, and which future questions no longer justify their cost.

If adaptation wins, the project learns where novelty, attention, expectation, comparison, habit, treatment, and social support already explain change. If opponent process wins, it learns which rebounds belong to temporal counterregulation rather than terminal structure. Those are substantive gains, not consolation prizes.

Loss can improve the research program as well. It may expose an invalid signature, an underpowered comparison, a moving baseline, or a measurement channel that never identified the target. Preserving that failure prevents later studies from repeating the same burden under a new name (Mayo, 2018).

For vulnerable readers, precision is care. False assurance burdens people whose suffering outsiders cannot fully observe. No person should carry the project's need to be right.

Learning to lose early protects the institution. A project that cannot publish narrowing before prestige gathers will struggle to correct itself later.

Chapter 19 gives adaptation and opponent process the territory they earn. Where they explain better, they should win; suffering cannot rescue what remains. What survives after a rival victory can become a sharper research question with fewer assumptions and a lower burden on future participants.

Chapter 20 — Prediction and Surprise

Prediction begins before conscious reflection. A footstep behind us, a change in breathing, a familiar facial expression, or a pain expected to worsen can alter attention and feeling before the situation is fully known. The nervous system does not wait passively for the world; it uses prior structure, bodily signals, and incoming evidence to organize what happens next.

Predictive processing is not one single finished theory. It is a family of accounts in which perception and action depend on generative models, prediction errors, uncertainty, and the weighting of signals by their expected reliability. Active-inference formulations add policies and action, asking how organisms sample or change the world to remain within preferred or viable states (Friston, 2010; Clark, 2013; Hohwy, 2013).

This family is a powerful rival because it can explain why the same event feels different under different expectations. Anticipated pain can alter pain; uncertainty can amplify threat; relief can arrive when feared outcomes fail to occur; and a bodily sensation can be interpreted differently across context, learning, and culture (Wager et al., 2004; Paulus and Stein, 2006; Khalsa et al., 2018).

Interoception gives the account a bodily center. Signals from the heart, lungs, gut, immune system, and other internal processes are sensed, integrated, and interpreted imperfectly. Emotional experience may depend not only on the signal but on attention, prior expectation, precision, action, and the meaning assigned to the bodily change (Barrett and Simmons, 2015; Critchley and Garfinkel, 2017; Greenwood and Garfinkel, 2025).

Valence and prediction error are related in some models but not interchangeable. Surprise can be pleasant, unpleasant, or affectively weak; expected harm can remain painful; and successful prediction can accompany chronic fear or constrained life. A theory of updating therefore does not automatically become a theory of felt value.

Computational mood models deepen the challenge by producing cumulative-looking dynamics from local learning; Section 20.1.1 explains the distinction.

Predictive breadth is also a methodological risk. A model with flexible priors, hidden states, precision weights, endpoints, and observation mappings can explain many patterns after they are seen. The correct response is not to dismiss the framework, but to freeze the relevant choices before testing and ask for held-out predictions that rival accounts cannot match equally well.

Endpoint conditioning deserves special attention. If paths are selected because they end in a desired region, earlier portions of those paths can display apparent pull toward that ending. A pattern that looks like terminal attraction may therefore be created by selection or conditioning rather than by an independent law (Doob, 1957; Fitzsimmons et al., 1993; Chetrite and Touchette, 2015).

The same problem appears in baselines. If the neutral point, reference model, or comparison window is chosen after the trajectory is known, the analysis can manufacture return. Predictive accounts and the Law must therefore face the same frozen origin, endpoint, observation, and scoring rules.

Viability provides another boundary. Organisms regulate conditions required for continued life and action. They avoid some states, seek others, and update behavior under uncertainty. That explains a great deal about persistence, but survival, regulation, and prediction are not equivalent to fairness or to an exact terminal residual.

The ethical distinction is concrete. A person may have learned to expect neglect, danger, or pain so accurately that prediction error becomes small. The model may fit while the life remains constrained and the experience remains negative. Accuracy is not moral success, and reduced surprise is not repair.

Strong comparison asks what predictive accounts would forecast before the data are seen. Which perturbations should alter valence, precision, interoceptive belief, action, or mood? Which patterns should transport across tasks and people? Which observations would favor a simpler regulatory account over a terminal-boundary account?

The public challenge is to explain this power without turning the brain into a maze solved by one equation. Predictive processing has empirical support in particular domains and active debate about scope, implementation, and interpretation. It should neither be marketed as a universal master theory nor weakened into a slogan.

The sequence follows the explanatory burden. Prediction, surprise, and bodily inference come first; viability and fairness are then separated; flexible models are tested for their ability to

mimic closure; and the final Section states what changes when predictive accounts explain better.

A fair outcome may be model ambiguity. Predictive and Law-linked accounts may fit the same observations, or the measures may not identify the latent distinction. In that case, the project should preserve the shared empirical result, report the unresolved mechanism, and avoid converting indistinguishability into support.

The deeper reader gain is a new standard for explanation. Breadth is not enough, technical vocabulary is not enough, and resemblance is not enough. A rival becomes decisive when it predicts new observations under frozen rules and leaves the stronger claim with no distinctive work.

Chapter 20 therefore asks whether balance-like patterns arise because a life is moving toward a terminal condition, or because an embodied system continually predicts, updates, acts, and regulates. The answer may differ by domain. The comparison must be allowed to say so.

The governing question is not whether prediction is present. It almost certainly is. The question is whether prediction, interoception, learning, selection, and flexible baselines can account for the observed structure without the Law, and what evidence could separate those explanations fairly.

20.1 — Prediction, Surprise, and Uncertainty

Prediction error and precision are useful here as rival-model concepts: one tracks mismatch between expectation and input, while the other governs how strongly that mismatch updates the model. Neither is a direct measure of valence or terminal closure (Friston, 2010; Clark, 2013).

Social and cultural context can shape what is predicted and how signals are interpreted, so a strong rival must be frozen prospectively and tested on held-out affective and behavioral data rather than used to redescribe every result after the fact. Persistent distress is not a failure of the person; predictive success remains a mechanism claim.

The scientific comparison is therefore direct: prediction, interoception, and action may explain important dynamics without establishing terminal fairness. The contest turns on discriminating predictions under frozen rules, not on resemblance in vocabulary.

20.1.1 — Prediction as Organizing Principle

Predictive processing begins from a simple proposition: perception depends on a model of what is likely as well as on signals arriving now. The system uses prior structure to interpret ambiguous input and updates when the fit is poor. This can organize much of affective life without proposing that experience is balanced across a completed stream.

In many hierarchical implementations, generative models operate across several levels. Faster levels anticipate sensory and bodily

patterns, while slower levels represent context, relationships, safety, identity, and expected futures. Higher and lower levels constrain one another rather than forming a one-way ladder (Friston, 2010; Clark, 2013).

Suppose two weather forecasts disagree. One has repeatedly been accurate in your area; the other has been unreliable. You would have a reason to give their predictions different weights, even before either forecast was tested today. Precision plays a related role in predictive models: it expresses estimated reliability, so some signals or expectations count more strongly than others. Applied to bodily feeling, the question becomes how much weight the system gives to an ache, a heartbeat, or an expectation about what will happen next (Hohwy, 2013; Friston et al., 2017).

Predictive processing is a family of models, not one finished doctrine. Different accounts assign different roles to representation, error, precision, action, and neural implementation. A fair rival analysis should compare the version actually specified rather than treating the broad framework as either universal truth or an easy caricature (Clark, 2013; Hohwy, 2013).

Prediction error is simply a discrepancy between an expected signal and an observed one under a declared model. The discrepancy is not automatically pleasant, unpleasant, conscious, or important. Its effect depends on what was predicted, how precise the signal is, what outcome matters, and what action remains possible.

Surprise and valence should not be collapsed. A surprise can delight, terrify, relieve, confuse, or matter very little. An expected event can be deeply painful, and an unexpected event can be welcome. The direction of mismatch alone does not provide the sign or magnitude of felt experience.

Relief illustrates the point. Relief may occur when expected harm fails to arrive, when uncertainty resolves, or when bodily threat decreases. The prediction account can explain the contrast between expected and received states, but the size and meaning of relief remain empirical questions rather than automatic entries in a life ledger.

Affective consequences depend on preferences and stakes as well as error. Missing a predicted bus and learning that a feared diagnosis was wrong are both mismatches, yet they differ in scale, bodily preparation, available alternatives, and future meaning. A strong model must represent those differences rather than hide them in one error term.

Computational accounts of momentary well-being show how expectations and recent outcomes can shape current reports. Reward history, prediction error, and momentum can explain fluctuations over minutes or days without defining a complete cumulative phenomenal object (Joffily and Coricelli, 2013; Rutledge et al., 2014; Eldar et al., 2016).

This is the first major boundary for the Law. A model of current affect or recent momentum does not become a model of the entire signed history merely by being integrated over time. Origin, scale,

missing experience, bearer, and endpoint remain separate problems.

Active inference adds action to the rival. The system can reduce costly mismatch by revising beliefs, seeking information, or changing the world so future sensations become more workable. This makes the framework a perception-action account rather than a passive theory of internal correction (Friston et al., 2017; Pezzulo et al., 2015).

Information-seeking and practical action need not point in the same direction. A person may accept short-term uncertainty to learn something important, or avoid new information to preserve immediate stability. Strong predictive models can therefore represent exploration as well as comfort-seeking.

No-look-ahead discipline is equally important. Future endpoint classification, terminal missingness, or the realized final value cannot be used to tune a model that is later described as prospective. Otherwise information from the answer has been carried backward into the prediction.

Held-out prediction, transport to new contexts, and explicit complexity costs are the proper tests of breadth. A predictive account that explains everything only after flexible refitting has not won, but a Law-linked model that improves fit only by consuming more freedom has not won either (Burnham and Anderson, 2002; Patterson et al., 2024).

20.1.2 — Interoceptive Prediction

Interoceptive prediction concerns the body as it is sensed and anticipated. Breath, heartbeat, pain, fatigue, hunger, nausea,

temperature, arousal, and visceral unease arrive through noisy channels shaped by expectation, context, uncertainty, history, and available action. Feeling can therefore depend on both bodily signals and the predicted meaning of the body (Seth, 2013; Barrett and Simmons, 2015).

Interoceptive prediction matters because felt experience is deeply bodily. A future can feel possible or impossible through the chest before it becomes a sentence. Fear can arrive as heart rate, breath, stomach, skin, and muscle readiness. Relief can arrive as warmth, exhale, softening, and restored room. Interoceptive prediction gives a naturalistic account of why meaning and physiology are intertwined (Critchley and Garfinkel, 2017; Pezzulo et al., 2015).

For the Law, this is a serious rival because research will likely use bodily channels to estimate momentary and cumulative affect. Heart-rate variability, sleep, autonomic patterns, movement, inflammatory signals, neural markers, and other physiological measures may help estimate felt state. Yet if those measures are shaped by interoceptive prediction, they cannot be read as direct access to a cumulative record. They are traces inside a predictive body.

Interoceptive prediction can also explain why uncertainty itself hurts. A symptom with no name, a pain with unclear cause, a social threat with no visible source, or a future with no reliable route can make the body forecast danger repeatedly. The suffering may lie less in a single event than in the body's ongoing inability to settle its model.

The Law should receive this as orientation, not proof. Hidden burden may appear through persistent bodily prediction under threat: vigilance, fatigue, sleep disruption, appetite change, muscle tension, panic, numbness, or chronic arousal. Naming that burden matters. It shows where a stream may be costly to inhabit, not that terminal neutrality has been established.

Interoception also strengthens the ethical firewall. A theory that touches bodily feeling can easily become invasive. No one should be told that a device, model, or AI system knows their inner cumulative record because it reads bodily signals. The body offers evidence under consent and uncertainty. It does not surrender the person.

The method for interoceptive prediction is clinical humility. Bodily channels should be compared with pain, medication, illness, sleep, trauma, disability, hunger, fatigue, and context before they are interpreted as evidence for any formal trajectory. The body matters too much to be read cheaply (Khalsa et al., 2018; Paulus and Stein, 2006).

Placebo and nocebo effects make the point concrete. Expectations can alter pain, symptom appraisal, attention, and physiological response even when the external stimulus is held relatively stable. The effect is not imaginary, because interpretation is one process through which the body is experienced. The same evidence warns against treating expectation as a universal master variable: effects vary across people, conditions, contexts, and outcomes, and they do not erase the need for treatment (Wager et al., 2004).

Anxiety offers another clear case. A person who expects bodily danger may attend more intensely to heartbeat, breathing, dizziness, or stomach sensation, then interpret ordinary variation as confirmation of threat. Prediction and sensation can reinforce one another. This can create real suffering without any external event large enough to explain the experience by itself, which is precisely why interoceptive prediction is a serious account of felt life rather than a verbal gloss (Seth, 2013; Critchley and Garfinkel, 2017).

Cultural learning can shape what bodily changes mean. The same arousal may be understood as illness, excitement, shame, spiritual presence, panic, or ordinary exertion. That variability does not make feeling arbitrary. It means that the path from signal to experience is structured by concepts, expectations, language, and social practice, all of which remain separate from a claim about a complete stream's terminal invariant.

20.1.3 — Why Comparison Is Not Evidence

Conceptual resemblance is not evidence. Prediction error can rise and fall, surprise can diminish, and bodily estimates can stabilize in ways that sound like correction or balance. Those similarities may guide model construction, but they do not establish exact closure of a complete conscious stream.

The temptation comes from shared vocabulary. Predictive processing and PTVc both speak of constraints, hidden states, horizons, uncertainty, and changing trajectories. Shared words do not create shared entailments. A framework can illuminate the

same terrain while replacing the stronger hypothesis in that terrain.

Several relationships are possible. Predictive processing may be a rival, a host framework, one candidate route, a narrowed subcase, or a defeater. The role cannot be chosen because the combination sounds elegant. It must follow from what each model predicts and what the evidence can distinguish.

A resemblance audit asks simple questions. Do the two frameworks use the same target variable? Do they require the same endpoint? Does one explain only current state while the other concerns complete history? Would the predictive account expect the observed pattern without PTVC? Which result would make the stronger claim unnecessary?

The rival should speak first. Before any PTVC-linked interpretation, the study should state what prediction, precision, interoception, action, medication, social context, and observation loss are expected to produce on their own. The Law may claim only a remainder specified before outcome inspection.

The comparison must also give the rival enough expressive power to be serious. State-dependent horizons, hierarchical uncertainty, bodily inference, changing action sets, and missingness may all be needed to explain the observations. Defeating a stripped-down caricature would not count as surviving a rival test.

Translation carries a second risk. Borrowing predictive vocabulary can make the Law sound neuroscientific without adding evidence, while borrowing fairness vocabulary can give a regulatory model moral authority it does not possess. Responsible

translation should make both accounts easier to test and harder to inflate.

Evidence begins with contrast: common observations, frozen targets, held-out consequences, safe perturbations, and explicit failure conditions. A terminal curve or story of relief is not discriminating when it is also predicted by treatment, ordinary adaptation, social repair, physiological change, or observation shutdown (Mayo, 2018).

Public artifacts need the same restraint. A diagram that places prediction beneath the Law as though neuroscience had supplied its mechanism would overstate the relationship. The safe caption explains that the accounts overlap in subject matter and remain competitors until a governed comparison says more. Rivals matter only when they receive a fair, shared test.

20.2 — Affect as Prediction Error

Relief, surprise, and expected harm test whether a predictive account explains affective dynamics without being mistaken for the complete-stream object (Wager et al., 2004; Leknes et al., 2008).

Predictive affect models can therefore explain momentary or short-horizon dynamics, including slower mood states, without becoming the cumulative object of a complete stream; similar mathematical form does not establish the same scientific object (Joffily and Coricelli, 2013; Eldar et al., 2016).

20.2.1 — Valence and Surprise

Valence may depend not only on what happened, but on how an outcome compares with expectation, whether prospects are improving, and how strongly the system revises its beliefs. Prediction-error accounts are therefore serious rivals for momentary and short-horizon affect.

Keep the distinction from Section 20.1.1: mismatch alone does not determine the sign or intensity of experience (Joffily and Coricelli, 2013; Rutledge et al., 2014).

Relief shows the interaction clearly. When a feared diagnosis is ruled out, a child comes home safely, pain eases, or a next step becomes possible, threat expectations can weaken and costly preparation can subside. Breathing, muscle tension, appetite, and attention may change as the model updates (Barrett and Simmons, 2015; Clark, 2013).

That vivid release can resemble burden being repaid, but a predictive explanation may already account for it. Relief establishes that the current state changed. It does not reveal a lifetime cumulative residual or show that a complete stream is approaching exact closure.

Institutions can misuse the same pattern. A long appeal, diagnostic delay, disciplinary process, debt negotiation, or immigration interview may impose weeks of anticipatory strain and then point to the person's relief as evidence that the process was harmless. Predictive accounts make the hidden cost visible because expected threat is itself lived before the final answer arrives.

Measurement must distinguish error magnitude, precision, valence, bodily cost, meaning, and context. Tears, laughter, stillness, arousal, or reported intensity can arise from many combinations of those variables. None is a direct unit of a complete phenomenal history.

For PTVC, the evidentiary demand begins after predictive explanations have received full credit. Any claimed residual pattern must survive prospectively specified rival baselines, independent endpoint rules, missingness analysis, bounded reporting effects, and uncertainty about the measured-to-latent bridge (Eldar et al., 2016; Friston, 2010).

20.2.2 — Body Updating

Body updating is never merely abstract. A racing heart in a safe room, a familiar ache interpreted as danger, or a treatment expected to help can change experience through attention,

interpretation, and physiological regulation. The same bodily signal can therefore support different felt states in different predictive contexts.

Threat learning gives the rival force in the negative direction. A system shaped by harm may treat ambiguity as danger, assign high weight to threat cues, and narrow available actions before a conscious argument appears. What looks irrational from outside may be coherent within a history of injury, discrimination, illness, neglect, or unstable care (Seth, 2013; Pezzulo et al., 2015).

Physiology has two roles at once. Bodily processes help generate experience, and physiological measurements provide imperfect evidence about it. Treating heart rate, respiration, movement, skin conductance, sleep, voice, or pain behavior as transparent readouts would confuse the causal process with the observation process (Critchley and Garfinkel, 2017; Khalsa et al., 2018).

Within-person calibration can improve interpretation because medication, fitness, illness, age, sensor behavior, and ordinary baseline vary. Calibration still has to transport. A pattern learned during rest may not mean the same thing during exercise, acute pain, sleep, infection, or palliative treatment.

Intercurrent events can change the latent state and the measurement channel together. Analgesia, sedation, surgery, oxygen support, sleep deprivation, inflammation, and new medication may alter feeling, reporting ability, physiology, and missingness at the same time. Those changes belong in the model rather than in a residual category called noise.

Multiple channels can narrow uncertainty when their failures differ. Self-report, behavior, autonomic measures, sleep, and clinical context may constrain one another, but they should not be collapsed into a person score. The value lies in patterned agreement and disagreement under an explicit bridge.

State and trait also need separation. A brief arousal response, a chronic regulatory pattern, and a stable reporting style can look similar in a short window. Repeated observation and context are needed before a model can distinguish a momentary shift from a lasting change.

Treatment makes causal plurality visible. Improvement may follow reduced inflammation, safer movement, lower threat, altered expectation, restored sleep, social support, or several mechanisms at once. A public account should not force those routes into one story merely because each changes feeling.

The Law's residual question remains harder. If fear, pain, and bodily regulation change across a life, does any complete-stream pattern remain after threat prediction, treatment, physiology, social repair, reporting, and observation loss are modeled fairly? The question should not be made easier by calling a sensor an inner ledger.

A person may continue working, parenting, studying, or caregiving while the body remains organized around threat. Visible functioning can therefore coexist with hidden cost. Predictive processing can help explain the vigilance, while care asks whether the environment, support, or demand should change. Neither explanation licenses romanticizing endurance.

Body-updating evidence is useful when it supports care, reveals a change, or constrains competing models. It does not become direct access to the complete phenomenal record, and uncertainty about the bridge should remain visible.

20.2.3 — Affect Is Not Cumulative Object Closure

Affect is not cumulative object closure. Predictive accounts can explain how a present state arises from expectation, bodily information, uncertainty, and updating without making a claim about the signed total of a complete conscious stream. A process that shapes moments and a boundary condition proposed for a life are different objects.

Different computational models make different commitments. Recent rewards and expectations can predict momentary well-being, while accumulated prediction errors can represent mood as momentum (Rutledge et al., 2014; Eldar et al., 2016). Joffily and Coricelli (2013) instead define emotional valence through the negative rate of change of free energy. These models should therefore be compared as distinct rivals; none by itself guarantees complete-stream terminal closure.

Mood also changes how later events are received. A guarded state can increase the expected cost of ambiguity, while an expansive state can make exploration feel available. The background state is both an effect of recent history and a condition on future prediction, which can create cumulative-looking dynamics on a limited time scale.

That time scale matters. A model may forecast mood accurately while remaining silent about the neutral origin, exposure rule,

omitted intervals, stream ownership, and truth-level endpoint required for PTV. A slow state is not automatically a complete-history ledger.

Neural convergence does not close the gap. Meta-analysis can identify networks associated with prediction across tasks, but it cannot locate a lifetime valence balance or isolate prediction signals from every neighboring function (Ficco et al., 2021).

Interoceptive accounts come closest to felt life because bodily signals are interpreted through context and expectation. They still describe mechanisms shaping experience, not a direct readout of a terminal cumulative object (Seth, 2013; Greenwood and Garfinkel, 2025). A held-out perturbation that changes bodily prediction while leaving a proposed terminal signature unchanged could help separate a local mechanism from a whole-stream claim.

A cumulative claim requires sign, cross-sign comparability, a neutral origin, experienced exposure and clock, bearer, exhaustive domain, endpoint, observation and missingness rules, and a measured-to-latent bridge. Without them, a local affect model is not evidence about closure. The practical consequence is modular testing: each bridge can be challenged separately rather than inferred from one successful mood forecast.

Mean-zero prediction error is not enough. Errors may average near zero because rewards were centered, the learner calibrated, or the task designed to settle. A finite task may also end without residual error because the final trial was learned or the endpoint used for fitting. Neither establishes a pathwise terminal residual

for every stream. A task intended to distinguish calibration from closure should therefore keep its evaluation trials and endpoint rule outside the data used for fitting.

What would discriminate PTVC is not the existence of prediction error, mood updating, or interoceptive inference. It is a prospectively declared consequence that these rivals fail under common data, endpoint, missingness, and scoring rules. Safe perturbations can identify local mechanisms, but complete-stream language still requires transport across the missing bridges. For instance, a frozen rival and PTVC-linked model might agree on average mood yet disagree prospectively on how a safe perturbation changes the next transition; if both absorb either outcome, the comparison is not yet discriminating.

20.3 — Why Viability Is Not Fairness

Viability, adaptive function, and felt value answer different questions. A system can learn to survive chronic pain, threat, deprivation, or coercion, and low prediction error may reflect either effective safety or narrowed expectations; regulatory success therefore does not imply a neutral or good lived state.

For PTVC, viability is a serious rival wherever prediction and regulation explain the observed path. The Law must add a distinct, prospectively testable consequence about the complete phenomenal trajectory, while care and social judgment remain independent of whether adaptation succeeds.

20.3.1 — Survival Versus Closure

Viability concerns the continued regulation of a living system. Terminal closure concerns a proposed relation over felt experience in one qualifying conscious stream. A mechanism can be excellent at keeping temperature, glucose, hydration, or action within workable ranges while saying nothing about the full phenomenal path.

Survival is not closure. Survival asks whether the organism continues; closure asks what happens to the cumulative felt path at the stream's endpoint. Endurance can coexist with chronic stress, deprivation, accurate prediction of abuse, narrowed action, and severe burden. Continued life therefore cannot be treated as evidence that the burden of continuing has been answered (Friston, 2010; Sterling, 2012).

Preserving life is often the first task of care, but it is not the final verdict on experience. Safety without dignity, continuation without relief, and biological persistence without restoration remain unfinished from the standpoint of felt life. The scientific comparison must therefore separate survival variables from the latent cumulative target.

In research, survival endpoints cannot replace experiential endpoints. Mortality, morbidity, hospitalization, function, and performance may be important covariates, but they are not the latent cumulative object. The truth-level endpoint must be defined independently of the observed residual, with biological death, protocol stopping, last observation, and irreversible stream ending kept distinct.

Survival versus closure is the first place where the Law may still differ from predictive viability. But difference is not victory. The Law must show evidence that closure has structure beyond survival-oriented regulation. Otherwise it should honor survival models and narrow its own claim.

Homeostasis concerns regulated bodily ranges such as temperature, glucose, hydration, sleep pressure, pain modulation, arousal, and autonomic tone. Allostasis extends regulation across anticipation and changing demand. Both can operate while experience remains deeply burdened, so regulation within a range is not the endpoint condition of a latent cumulative object (Clark, 2013).

Allostasis makes the distinction sharper. A system may shift its regulatory targets under chronic stress, illness, aging, or social

threat. What looks like stability can be costly adaptation. The Law cannot turn homeostatic return into evidence, and the rival cannot turn regulatory success into moral settlement.

Neutrality in the strong Law is not homeostasis with ethical language added. Homeostasis manages ongoing state and may reduce burden by returning bodily variables toward workable ranges. Terminal neutrality concerns a proposed property of the completed phenomenal trajectory. Physiological recovery after panic, infection, injury, or exertion can coexist with lasting fear, memory, loss, or humiliation; it does not determine the signed total of experience.

20.3.2 — Adaptive Function Versus Experience

Adaptive function and lived experience overlap, but they are not identical. A response can promote survival while feeling terrible, and an experience can feel good while undermining health or future agency. Biological usefulness cannot substitute for the phenomenal quantity named by the Law.

Prediction can succeed in a harmful world. A child may learn when punishment is coming, a worker may anticipate retaliation, and a patient may expect disbelief because experience has trained the forecast. The model becomes more accurate while the conditions remain unjust (Barrett and Simmons, 2015; Seth, 2013).

The cost of successful adaptation may be carried inside the person as fear, constriction, pain, silence, fatigue, or self-erasure. A threat-sensitive system can keep someone alive by narrowing

action and attention. That function may be intelligible without being repair, justice, or relief.

Some protective responses preserve function while imposing felt costs. Pain can discourage tissue damage, nausea can prevent ingestion, fear can prepare escape, and exhaustion can force retreat. Explaining those functions does not turn the cost into moral permission, and it never licenses neglect of care (Friston et al., 2017; Pezzulo et al., 2015).

The distinction also limits what prediction can prove. A model may anticipate behavior, reduce error, and guide action while the latent cumulative object remains unidentified. Accuracy, viability, and reward attainment are not evidence of terminal neutrality unless a separate endpoint-specific pattern survives rival comparison.

Research can miss experience even when it predicts well. Choice models may overlook shame, survival models may overlook loneliness, and self-report models may miss dissociation or inaccessible states. Adaptive function becomes a proxy for felt history only if the measurement program earns that bridge.

The public-root reason for keeping experience central is simple: mechanisms and external conditions matter partly because of how they are lived. A person is not reducible to the function a feeling serves, and a stream is not merely an optimization trace.

The remaining formal question is whether felt experience displays an endpoint structure beyond adaptive function. Predictive rivals can explain regulation, narrowing, mood change, and successful

action. If no distinct terminal signature remains after those explanations, the Law has not earned stronger language.

20.3.3 — Rival Strength Remains

Predictive accounts remain strong after these distinctions are drawn. Their breadth, links to action and interoception, and capacity to model body, environment, attention, policy, social history, and time make them serious competitors. That breadth can become overreach when explanation is added after the fact, so every account must face the same observation rules and a comparison specified before outcome inspection.

The Law may still matter if it survives, but survival after a strong rival is different from survival after a weak one. The chapter wants any surviving claim to be smaller where it must be smaller, sharper where it can be sharper, and quieter where public confidence would be premature. That is not a loss of wonder. It is wonder taught to carry its own weight (Hohwy, 2013; Seth, 2013).

Internal debate or abstraction within the predictive family is not a shortcut for the Law.

The book's confidence must rest in the test, not in a desired winner. If a predictive rival explains more with less flexibility, the Law narrows. Public trust depends on treating that contraction as precision rather than retreat, and on matching public language to the residue left after serious comparison.

Rival victory should be typed by layer. A predictive account may explain perceptual updating, symptom persistence, or momentary mood while leaving the complete-stream target unidentified.

Conversely, failure on one local task does not defeat the entire predictive family. The comparison must say exactly which object was won, which remained open, and which claim narrowed (Mayo, 2018).

A predictive explanation earns credit for the patterns it explains. If it reproduces apparent balance without requiring exact terminal closure, it is a rival to that interpretation. If an independently supported mechanism actually entails the declared closure condition, it could instead realize the Law. The distinction turns on what follows from the mechanism and its independently tested links to experience, not on whether its components are familiar.

Rival strength can also improve care. Different predictive mechanisms suggest different supports: reliable information, safer action, pain relief, sleep, trust, treatment, or removal of coercive conditions. Those interventions may help because ordinary mechanisms are real. Their success is clinically and ethically valuable without becoming evidence for terminal closure (Paulus and Stein, 2006; Barrett and Simmons, 2015).

20.4 — How Flexible Models Can Mimic Closure

Flexible models can mimic closure when endpoints, baselines, hidden states, selection rules, or missingness handling are adjusted after the outcome. A fair comparison freezes those choices, gives rivals common information and tuning budgets, and asks for held-out prediction or model recovery rather than retrospective fit (Gneiting and Raftery, 2007; Burnham and Anderson, 2002; Patterson et al., 2024).

Conditioned paths, post hoc baselines, selection, and unequal flexibility can all create closure-like patterns. If several models make the same held-out predictions, the result is model ambiguity, not shared support.

20.4.1 — Endpoint Conditioning Can Mimic Return

Conditioning on an endpoint can create the appearance of endpoint pull. When paths are selected because of how they end, earlier segments can display drift, compression, or return even if the unconditioned process contains no terminal law. The endpoint must therefore be specified independently of the result.

A simple example makes the danger visible. Select only trajectories that finish near a chosen value and align them by that ending. Their earlier observations will often appear to bend toward the selected point because the sample was defined by the finish. The convergence comes from selection, not from discovery of a natural boundary (Doob, 1957; Fitzsimmons et al., 1993).

The predictive baseline must be allowed the variables it plausibly needs: state-dependent horizons, hierarchical uncertainty, interoceptive precision, medication, social context, reserve-like capacity, missingness, and observation shutdown. It may predict quieting, volatility, withdrawal, agitation, or apparent resolution without PTVC (Friston et al., 2017; Seth, 2013).

Only after those predictions are frozen can a PTVC-linked model state its incremental claim. If the Law speaks first, it can absorb rival territory by description. If the rival speaks first, the Law has to identify a pattern that remains after ordinary prediction and clinical context are accounted for.

Observation stopping creates a related form of conditioning. Severe illness, fatigue, sedation, device removal, privacy protection, or inability to report can make the last observed segment look quieter than the unobserved experience. Disappearance of data must not be mistaken for disappearance of valence.

The comparison should examine the path rather than one final point. Timing, variance, subgroup behavior, treatment, missingness, and two-time dependence can distinguish a natural process from a selected or conditioned one. A single attractive terminal curve cannot.

Ethics changes the stopping rule. If strong predictive and clinical models already explain the pattern, a study should not continue into more burdensome terminal observation merely to rescue a dramatic hypothesis. The fair design may end with a narrower result.

A PTVC-specific signal would need more than measured concentration near a band. It would require independent endpoint rules, validated measurement bridges, rival-resistant path structure, and transport to observations that were not used to define the ending. Sedation, frailty, response compression, palliative care, and terminal selection remain active competitors.

Endpoint conditioning is therefore not a minor statistical caution. It is a complete rival explanation for apparent pull toward an ending. Any claim of terminal structure must show why the pattern belongs to the unconditioned eligible paths rather than to the rule that selected or observed them.

20.4.2 — Freeze Baselines and Post Hoc Fitting Rules

Flexible baselines and post hoc fit must be frozen before comparison. A model that can change its zero, endpoint, reference class, time scale, state variables, and error rule after seeing the data can explain almost any trajectory. Flexibility becomes scientific only when it produces risky predictions outside the data that shaped it.

Predictive models have several legitimate sources of flexibility: priors, precision weights, hidden states, action policies, observation maps, and hierarchical time scales. Each choice should be specified in advance with its allowed range and tuning procedure. Otherwise a failed model can be repaired by changing whichever component the result exposed (Clark, 2013; Friston et al., 2017).

Time scale needs its own declaration. Fast sensory updating, slower mood integration, and long-lived expectations can all

generate smooth curves while describing different processes. A fit over seconds or days cannot silently become an account of complete history (Eldar et al., 2016; Rutledge et al., 2014).

Transport is stricter than refitting. A model that succeeds only after its priors, features, baseline, and noise assumptions are rebuilt for every site has not yet shown a stable mechanism. The stronger test carries a frozen architecture into a new context and permits only prospectively specified calibration.

Channel structure must be specified as well. Self-report, behavior, physiology, and clinical observation can agree because they share a cause or because one channel influences another. They can disagree because each fails differently. The model should predict those dependencies rather than average them into apparent certainty.

Complexity penalties make the cost of flexibility visible, but they do not choose truth by themselves. A larger model can be justified when it predicts held-out data and survives perturbation, while a smaller model can still be wrong. Simplicity is a comparison consideration, not a moral virtue (Burnham and Anderson, 2002).

Implementation choices are part of the model. Random seeds, task selection, preprocessing, hyperparameters, stopping rules, and selective reporting can dominate apparent differences in computational work. Adequate replication and transparent empirical design are therefore part of the rival test (Patterson et al., 2024).

Calibration and discrimination answer different questions. A model may assign reliable probabilities while several

mechanisms remain observationally indistinguishable, or distinguish simulated models while being badly calibrated on natural data. Proper scoring, coverage, and recovery results should be reported separately (Gneiting and Raftery, 2007).

Freezing the comparison protects both sides. A predictive account that wins has earned territory rather than inherited it from tuning, and a PTVC-linked model that survives has faced a serious alternative.

20.4.3 — Rival Success Is a Real Outcome

Rival success is a real scientific outcome. A predictive model may explain the data better, several models may remain indistinguishable, or the observation process may be too weak to decide. None of those results can be translated into a hidden victory for the Law.

If changing precision, medication, observation loss, interoceptive regulation, or another prospectively specified mechanism explains an apparent terminal concentration, the Law-specific claim narrows in that domain. A rival can win locally, a reference class can shrink, or the correct result can remain model-ambiguous or No adequate estimate (Mayo, 2018).

Partial outcomes should remain typed. Predictive accounts may explain early and midlife affect while the endpoint question stays open, or they may explain one clinical context without transporting to another. A physiological signature, a reported state, and a latent terminal residual are different evidentiary objects.

The adjudication must be protected from enthusiasm. Project interest, donor interest, spiritual resonance, public attention, and Artificial Intelligence summaries cannot determine which model is accepted. The stronger the claim, the more exposed it must be to serious loss.

That exposure requires prospective specification, independent endpoints, held-out prediction, model recovery, rival baselines, sensitivity analysis, missingness plans, uncertainty propagation, measurement-invariance checks, replication, and red-team challenge. These are the evidentiary costs of law-level seriousness, not ornamental signs of rigor (Mayo, 2018; Patterson et al., 2024). Where feasible, the red team should precommit to the failure patterns it would treat as rival-favored so adverse results cannot be reclassified after inspection.

Failure to beat rivals should also reduce participant burden. When the distinctive contribution is not appearing, the answer is not to recruit more fragile people, seek more dramatic circumstances, or intensify invasive measurement. The answer is to narrow, redesign, or stop. A redesign can begin with existing observations and compare follow-up variables by their expected information gain about the remaining explanations relative to participant burden.

Scientific speech changes when a rival wins, but human duty does not. Care, dignity, privacy, refusal, and the ban on scoring remain independent of the terminal hypothesis. The analysis of hidden burden can retain value even if PTVC loses lawhood. A well-supported ordinary explanation can sharpen practical priorities by identifying changes in treatment, context, prediction, or

measurement that a person or institution can act on without assuming a terminal mechanism.

Rival success must change both the conceptual map and the public explanation. The record should identify which observation favored the rival, which interpretation must now be revised, and which ordinary mechanism earned the evidence rather than merely lowering confidence in the Law. A public correction should travel with the same prominence as the earlier claim, including summaries and teaching materials that may outlive the original analysis. Scientific humility becomes operational when revision reaches the places where confidence was first created.

20.5 — If Predictive Accounts Win

If predictive accounts explain the relevant observations better under common rules, the concession should match the evidence: a mechanism may be replaced, narrowed to a subcase, or retained only as metaphor where it adds no discriminating success.

20.5.1 — Law of Fairness as Metaphor Only

Replacement, a narrowed subcase, and metaphor-only status should not be blended into one softened victory.

Replacement means the rival owns the explanation. Relief, fear, mood return, interoceptive settling, or terminal calm would be described through prediction, precision, action, treatment, and bodily regulation rather than through Law-specific evidence (Friston, 2010; Clark, 2013).

Metaphor-only status preserves a way of asking where burden is hidden, shifted, reduced, or repaired while retiring the scientific claim in the defeated domain. The metaphor may guide reflection, but it cannot appear in verdict language as though it names a boundary of nature.

The main risk is aura. The word law, a cumulative graph, the language of neutrality, and the emotional force of fairness can make a metaphor sound empirical even after the evidence has narrowed. Honest presentation removes that borrowed authority instead of relying on readers to infer the qualification.

Subcase status is different. A PTVC-like pattern may survive only under specified endpoint rules, reference classes, active-inference regimes, reserve conditions, or measurement bridges. In that case

predictive processing provides the larger architecture and the Law names a constrained pattern within it.

A subcase must be bounded rather than admired. Which streams, endpoints, channels, ages, cultures, clinical states, and rival baselines are included? What result would defeat the subcase? Without those limits, subcase language becomes a polite form of rescue (Mayo, 2018).

Public language must match the status. Replaced claims leave scientific explanations. Subcases carry their conditions wherever they appear. Metaphors carry explicit lens language and no evidentiary implication. The qualification belongs in the front room, not in a technical appendix.

Hope may remain after replacement, narrowing, or metaphor-only status. A reader can still wonder whether visible unfairness is the whole story and use the burden question to deepen care, while accepting that hope cannot restore scientific status. A smaller exact claim can be more useful than a universal claim protected by ambiguity.

20.5.2 — Ethical Firewall Remains

The ethical firewall remains because it never depended on predictive accounts losing or PTVC winning. Present suffering deserves present care, consent remains necessary, privacy remains ordinary, and no person becomes a score. Scientific status changes the claim, not the value of the person.

Mechanistic explanation creates no blame. A person whose threat expectations are accurate in an unsafe environment is not failing to regulate. A person whose predictions remain shaped by trauma

is not morally defective. The environment may need to change before any model can reasonably expect safety.

Care may target a mechanism only when the target is justified and the intervention is proportionate. Reducing uncertainty, improving interoceptive discrimination, widening options, treating pain, or changing harmful contingencies may help. None authorizes forced exposure, withheld comfort, or pressure to conform to an externally chosen emotional baseline (Beauchamp and Childress, 2019).

Consent includes inference. A participant may agree to provide physiological or behavioral data without agreeing that a model may label hidden emotion, credibility, cumulative burden, or personhood. Secondary interpretation requires its own purpose, access rules, correction path, and right to refuse.

Model outputs remain provisional records under accountable human review. They should not enter employment, insurance, education, detention, benefits, spiritual standing, or care priority as declarations of inner truth, and no artificial system may certify consciousness, worth, balance, or the validity of a person's testimony. Fluency and accuracy on one task create neither a dossier nor moral or scientific sovereignty.

The firewall also protects critics and nonbelievers. Rejecting the Law, preferring a predictive account, or suspending judgment does not reduce dignity or standing. Disagreement is evidence pressure, not disloyalty.

Meaning remains voluntary. A reader may find existential value in the fairness question after the scientific claim narrows, but that

meaning cannot be imposed on another person or used to delay repair.

20.5.3 — Scientific Humility Preserved

Scientific humility is the exact statement of what changed. It identifies which patterns a rival explained, which claims narrowed, which results remain ambiguous, and which questions no longer justify stronger language. It is not the softening of a claim until nobody can oppose it.

Predictive frameworks require humility. They are broad, debated families with different commitments and risks of overreach. Rival victory should not replace one protected theory with another; each account remains answerable to its predictions and failures.

Operational humility changes priorities. Resources move toward unresolved discriminators, away from signatures a rival explains, or toward measurement and care when terminal adjudication is impossible. Uncertainty enters the next decision rather than disappearing into prose (Mayo, 2018). The next study can target the point where competing predictions diverge, giving the unresolved question a more specific experimental form.

Humility also gives readers freedom. A person may find the Law meaningful, reject it, prefer predictive accounts, suspend judgment, or care only about the ethical firewall. The book should not treat any of those positions as a failure of depth. Readers can locate the part of the argument that interests them and assess it without having to adopt the entire framework.

Ambition and humility are partners: the larger the claim, the more visible its path to correction must be. The chapter welcomes a

better explanation because predictive and interoceptive accounts force the Law to separate beauty, ethics, and coherence from evidence about felt experience. A successful comparison would leave a sharper account of experience, including the conditions under which one explanation makes the more useful prediction. That outcome is progress because it turns a broad contest into a clearer map of where predictive models succeed and where a distinct terminal hypothesis would still need to earn evidence.

Chapter 21 — Control and Homeostasis

A thermostat, a hungry animal, a person learning a route home, and a controller keeping a machine within safe limits all exhibit directed change. Feedback, reward, constraint, and state estimation can create orderly trajectories without a moral accountant.

Reinforcement learning formalizes one part of that order. Actions alter later states, outcomes update values, and policies change with experience. Reward-prediction errors, habits, exploration, and delayed consequences can generate long behavioral patterns from local interaction (Schultz et al., 1997; Sutton and Barto, 2018).

Homeostasis and allostasis provide embodied counterparts. Some variables remain within workable ranges; others are anticipated across changing demands. Stability may require movement, preparation, and redistributed physiological cost rather than return to one fixed point (McEwen and Stellar, 1993; Sterling, 2012; Ramsay and Woods, 2014).

These frameworks are strong rivals because the Law also speaks about paths, constraints, horizons, reserve, and boundary conditions. Shared vocabulary, however, does not establish shared ontology. A reward signal is not felt value, a regulated variable is not a life, and a safe set is not terminal fairness.

Optimization is the central boundary. An objective function states what a model is built to improve; it does not show that the objective captures experience, the horizon is appropriate, or omitted costs are harmless. A successful policy can still be a poor description of a conscious stream.

Constrained and safe reinforcement learning sharpen the comparison. Naive maximization can violate safety limits, exhaust resources, or enter irreversible states. Constraint models, risk measures, corrigibility, and common benchmarks show how much purposeful structure ordinary control can provide (García and Fernández, 2015; Achiam et al., 2017; Ji et al., 2024).

Human application requires a sharp stop. Models may guide research design or clarify mechanisms, but they may not turn people into objective functions, hidden reward estimates, or eligibility scores. The state variables selected by a model reflect a purpose and a measurement system; they do not exhaust the person's lived reality.

Stochastic models separate three effects that can look alike: ordinary return toward a reference range, path selection created by endpoint conditioning, and observation change near that endpoint. Chapter 21 tests them as distinct rivals rather than treating inward drift or shrinking spread as unique (Uhlenbeck and Ornstein, 1930; Doob, 1957; Chetrite and Touchette, 2015).

Hybrid models may be stronger still. Prediction, reward learning, bodily regulation, social context, treatment, memory, and observation error can act together. A hybrid wins when its added complexity earns better held-out explanation under a fair penalty.

Common observation rules are indispensable. Models must receive comparable data, endpoint information, interventions, tuning budgets, and missingness treatment. Neither a PTVC-linked model whose rivals were denied necessary variables nor a flexible rival built on the test set has won fairly.

If control or homeostasis explains a pattern, the explanation should be accepted at that level. Care can improve regulation, treatment can restore capacity, and safe environments can change trajectories without becoming evidence for terminal closure. Ordinary causal success deserves its own name.

The final ethical challenge is institutional loyalty. Research can become organized around defending a movement, funder, or preferred theory. Prospective specification, independent review, publishable nulls, and correction authority must let a better control model defeat a preferred explanation.

Chapter 21 joins physiology, learning, control, and stochastic process theory without collapsing them. Similar trajectories can arise from different causes. The question is which account survives common observations and prospective tests.

21.1 — Rewards, Set Points, and Regulation

Technical reward, felt value, and bodily regulation are different objects. Reinforcement-learning reward is a formal signal that can guide action without any feeling, while pleasure, pain, relief, money, recognition, and moral approval belong to different human meanings.

Set points, allostasis, and care explain how bodies and opportunities can change without implying a terminal fairness law. Anticipatory regulation may coordinate competing bodily demands, and material care can restore regulatory capacity (McEwen, 1998; Sterling, 2012; Theriault et al., 2025).

21.1.1 — Reward Learning Basics

Reward learning explains how action changes when outcomes are better or worse than expected. It distinguishes immediate reward, expected return, learning signals, and policy change. Those tools are indispensable for understanding pursuit and avoidance, but none of them is automatically identical to felt experience.

One way to see the strength of reward learning is to notice that it does not need a grand moral story. A creature can learn to approach warmth, avoid pain, repeat relief, prefer predictability, and conserve energy without knowing any theory about fairness. The lawlike elegance of temporal-difference learning, value updating, and policy adjustment can therefore explain a great deal of lived pattern without touching terminal neutrality (Schultz et al., 1997; Sutton and Barto, 2018).

Reward-learning accounts become stronger when reward is understood broadly. Reward does not have to mean pleasure in a shallow sense. It may mean relief, safety, reduced uncertainty, avoided pain, social approval, access to food, restored control, or escape from danger. A life can reorganize around such signals without any hidden cumulative record being balanced. Behavior can look meaningful because learning has shaped it (Berridge and Robinson, 1998; Berridge et al., 2009).

The Law must therefore not steal reward learning's evidence. If a pattern is explained by local reinforcement, repeated avoidance, habit learning, reward prediction error, or policy adaptation, the correct action is concession. The Law may ask whether something remains after such models have done their work. It may not claim the whole field merely because learning happens inside conscious life.

The Bellman idea deepens the rival. A value estimate can be decomposed into immediate reward plus expected future value, allowing an agent to learn from partial futures rather than wait for an entire trajectory to end. That is a major strength of reinforcement learning. It also marks the boundary of what the framework answers: action under partial information, not the signed phenomenal history of a completed stream.

The Markov assumption asks whether the model's present state contains the information needed to predict what comes next. That is a demand on the representation, not a claim that a person's past no longer matters. Memory, illness, expectation, and social history may need to be represented because their effects persist. The useful question is what a particular model has kept and what

it has left out. Success at predicting a person's actions does not, by itself, show that the same representation captures what those actions feel like (Sutton and Barto, 2018).

A final safeguard belongs here: reward-learning explanations should be written without moral inflation. A learned behavior is not necessarily wise, deserved, healthy, fair, or free. It may be the best available policy inside a constrained menu. The analysis therefore reads reward learning through available options rather than character judgment.

Set-point language becomes dangerous when it makes pain sound temporary or self-repairing by default. Variables differ in speed and may shift with trauma, poverty, illness, age, medication, stress, sleep loss, or repeated threat. A person can appear stable while living around a new defended level of fear, exhaustion, or muted possibility. Return is a regulatory tendency, not permission to ignore suffering.

Set point language is attractive because it seems to resemble fairness. A system is pushed away, then returns. A variable deviates, then corrects. A person suffers, then adjusts. A mood drops, then rebounds. At the surface, this can look like an inner law of balance. But set point return is not terminal neutrality. A return toward a range says the system is regulating. It does not say the whole stream has closed its cumulative felt history (Keramati and Gutkin, 2014; Eldar et al., 2016).

21.1.2 — Care as Regulation Support

Care often supports regulation. Food, medication, shelter, sleep, social safety, pain relief, and reliable assistance can restore

capacities that physiology and learning cannot recover alone. Describing care as regulation support does not reduce dignity to a control problem. It shows why material help can change the space of possible experience.

Homeostatic control is often taught with simple examples, but real living systems are layered, anticipatory, noisy, and expensive. Blood pressure, glucose, inflammatory tone, temperature, oxygen, thirst, arousal, and sleep pressure do not move as isolated dials. They trade burden through networks. Correcting one variable can disturb another. Relief in one hour can borrow from the next (McEwen, 1998; Sterling, 2012).

Homeostasis can also conceal cost. Long-term stress regulation may preserve outward performance by redistributing effort across endocrine, immune, cardiovascular, metabolic, and behavioral systems. A person can appear composed while regulatory systems work under sustained strain. Visible functioning therefore cannot settle the cumulative question, and stable appearance should never be treated as a verdict on burden (McEwen and Stellar, 1993; McEwen, 1998; Ramsay and Woods, 2014).

Homeostatic control also has time constants. Some corrections are rapid, such as reflexive responses to heat or breath. Others are slower, such as endocrine adjustment, immune recovery, sleep repair, or adaptation to chronic stress. The observed emotional life of a person may therefore reflect multiple overlapping control loops rather than one cumulative movement.

Because care changes embodied regulation, clinical and physiological expertise must remain involved wherever the

hypothesis touches bodily evidence. A philosopher can describe the cumulative record, but cannot safely interpret all regulatory signals. Medication, pain, infection, sleep, disability, neurodivergence, trauma, and age can alter observable patterns. Expert review protects both science and persons.

Care changes the regulatory field. Food supports metabolism; shelter supports temperature, sleep, and safety; pain treatment lowers nervous-system load; therapy can alter threat interpretation; friendship can reduce isolation stress; and legal or material support can reduce scarcity-driven arousal. Improvement after care is consistent with reduced burden. Establishing which part of the care caused that change requires an appropriate causal comparison; the sequence alone does not identify the mechanism or establish terminal balance.

That causal understanding makes care more urgent, not less. A person in distress should not be told to wait for a theory to resolve the burden. The body, mind, household, and social world are already carrying cost. When support can reduce harm now, relief comes before interpretation (Beauchamp and Childress, 2019).

21.1.3 — Why Reinforcement Learning Matters Here

A system's reward function is a design object, not automatically felt value, welfare, morality, or fairness. Technical success cannot identify a person's inner life, and similar behavior may arise from very different conditions. Preference, fear, poverty, stigma, disability, coercion, opportunity, and repeated exclusion can all shape what a person learns to do under the choices available.

That makes the ethical question larger than whether a model predicts behavior accurately. We also have to ask who designed the reward structure, who can refuse it, who bears the cost of mistakes, and whether apparent compliance conceals burden. A system can predict actions impressively while remaining silent about what those actions feel like to the person performing them.

Reinforcement learning matters scientifically because feedback, prediction, and policy improvement can explain many patterns that might otherwise look like balancing. Those explanations deserve credit where the evidence supports them. The unresolved empirical issue is whether reward-learning variables merely reproduce a local resemblance or, through independently supported links to felt experience, could entail the complete-stream relation. A remaining mismatch can guide research, but it does not by itself identify the missing mechanism.

The same distinction protects public interpretation. Algorithmic prediction can look like authority when users begin treating a reward model as value, a policy as explanation, or fluent output as fairness. Technical performance should therefore remain separate from claims about experience or human worth. Detailed governance of platforms, workplaces, markets, clinical systems, and public institutions belongs to the books that own those domains.

21.2 — Optimization Is Not Pathwise Closure

Optimization is always relative to a chosen representation, objective, horizon, constraints, and aggregation rule. High performance can therefore omit pain, coercion, uncertainty, future opportunity, or people outside the modeled objective; discounted return is not the complete felt history of a stream.

Constraint satisfaction and simulation can clarify tradeoffs and reachable regions, but neither establishes terminal fairness or moral authority. In public systems, the chosen target, omitted costs, and route of appeal must remain visible.

21.2.1 — Objective Functions Versus Experience

An objective function is a modeling choice that ranks selected outcomes for an agent or controller. Experience is what a conscious subject feels. An objective may be useful while omitting suffering, representing it badly, or optimizing a proxy that diverges from the person's life. The omitted burden must be named because the function is a decision tool, not a soul, a life, or a completed stream (Sutton and Barto, 2018; Berridge et al., 2009).

Objective-function drift creates the same danger in public systems. Once a metric governs behavior, people may change around the metric. They may perform wellness, suppress distress, chase points, hide burden, or avoid help because the objective function has become authority. The Law's no-score rule is designed to prevent this drift from attaching to momentary affect estimate, cumulative affect estimate, or any related tool.

The final danger is proxy obedience. When a proxy becomes the target, the system may become more efficient at doing the wrong thing. The Law's measurement vocabulary must therefore be surrounded by proxy humility. A measure may guide attention only while remaining weaker than the person's dignity and weaker than the evidence status assigned by *The Science of Fairness*.

Local reward is vivid and event-bound: relief, food, approval, novelty, information, reduced threat, or a model's scalar arrives at a time after an action. A lifetime trajectory is the evolving felt path of a unified stream, including rebound, opportunity loss, habit, fatigue, memory, risk, and effects on future moments. The first can shape the second without being identical to it.

Imagine a child learning to ride a bicycle. The first successful stretch brings delight; practice then changes which journeys are possible, whom the child can visit, and what feels within reach. One rewarding moment can open a different future. This is a reason to study paths as well as peaks: learning and opportunity can change the shape of later experience. These are already rich questions for ordinary science. The Law adds a further one, whether the complete felt history of a qualifying stream satisfies exact closure.

Reinforcement learning often handles time through discounted accumulation, which is useful for decision-making. A discounted return is not a life. It depends on how reward is defined, the chosen horizon, the representation of the state, the weight assigned to distant consequences, and the model's assumptions

about what matters. Change those assumptions, and the apparent value of a policy changes (Sutton and Barto, 2018).

A local reward can also produce long-term burden. A drug may bring relief while deepening dependence. Avoidance may reduce fear today while shrinking tomorrow's life. Public applause may reward a leader's performance while teaching deception. A platform notification may reward attention while fragmenting peace. The existence of local reinforcement does not settle the cumulative felt account.

Separating an optimization horizon from a complete experiential path also shows why anecdotal sequencing can mislead. A hardship followed by success does not prove compensation. A pleasure followed by loss does not prove payment. Lives contain sequences, and humans naturally narrate sequences as meaning. The Law must require more than sequence before it speaks in formal terms.

21.2.2 — Constraint Satisfaction Versus Fairness

Constraint satisfaction can keep a system within permitted limits. Fairness, in the sense examined here, concerns a proposed property of felt experience over a complete stream. A controller that never violates its engineering constraints has not thereby demonstrated experiential closure.

Policy improvement is a technical achievement: a learning system can select actions that better satisfy its reward and safety constraints (García and Fernández, 2015; Achiam et al., 2017). Human improvement is broader. A patient may establish a steadier routine, a service may be redesigned, and a community

may remove a recurring obstacle. None of these gains is trivial, yet none establishes terminal neutrality.

Constraint satisfaction remains insufficient for terminal evidence as well. Near the end of life, a person may simplify choices, reduce activity, accept care, seek reconciliation, or narrow attention. Control models can explain some of this as policy change under declining reserve, reduced action space, safety demand, medication, fatigue, social support, or changing expectation. The Law must not rush to claim it as terminal closure.

Neutrality, in the strong formal sense, concerns the cumulative object at the stream's independently defined endpoint. It is not an observed improvement in strategy. Any empirical test of measured endpoint compatibility would need prospectively specified representations, observation models, rival baselines, and a governed verdict rule. This Chapter clarifies only why policy improvement cannot substitute for that adjudication (Mayo, 2018).

Policy improvement can even intensify hidden burden when the environment is unfair. A person may learn to be less visible, less demanding, less trusting, or less hopeful because those policies reduce immediate harm. The policy has improved under a hostile environment. The life has not thereby become fairer in the deeper sense.

The public-risk version of this rule is simple: do not call adaptation justice. A person may become better at enduring an unfair world. A community may become better at surviving

neglect. A patient may become better at managing symptoms. These improvements deserve support and respect, not misuse as evidence that the original burden was balanced.

A constraint can be satisfied while a person remains burdened. A school can meet attendance rules while a child is terrified, a hospital can satisfy discharge or staffing requirements while patients and families feel abandoned, and an algorithm can meet a formal fairness constraint while remaining opaque at the edge.

The central question is where burden goes after a formal condition is satisfied. If a rule forces risk into silence, family labor, undocumented waiting, staff burnout, future collapse, or another unmeasured channel, the system has preserved its surface and hidden its cost. Formal compliance is therefore a beginning of review, not an account of the person or community affected.

21.2.3 — No Computational Proof by Analogy

Computational analogy can clarify structure, but it cannot prove that nature uses the structure. A simulation can show that a route is possible, reveal its signatures, and expose degeneracies. It cannot convert an assumed objective into evidence about consciousness.

A fair evidence design would not treat time as a loose background. It would define observation windows, sampling cadence, endpoint criteria, missingness rules, and uncertainty before attractive cases are inspected. It would also state what happens when final observations are unavailable, because the most difficult intervals may be the least measurable.

Computational construction faces a severe evidence burden because lives are long, private, sparsely observed, ethically protected, and missing at difficult moments. Affect estimates remain governed candidates: evidence from dreams and terminal settings is partial, longitudinal records are burdensome, and laboratory tasks are controlled but capture only a narrow slice of life.

Distinctive pathwise evidence would need residual structure after reward learning, homeostasis, adaptation, predictive processing, and strong stochastic baselines, or endpoint patterns not reproduced by safe control and selection models. The point is to state what would discriminate, not to presume the result (Mayo, 2018).

The same burden protects public language. Anecdotes, terminal stories, resilience, and beautiful diagrams are fragments, not pathwise verdicts; group trends, laboratory windows, snapshots, and operational success must remain at their own scales.

A valid pathwise test also needs an endpoint rule independent of the realized ledger, keeping biological death, last observation, protocol endpoint, and irreversible stream ending distinct. No candidate or rival may win by selecting the endpoint from the answer.

Completeness also raises privacy risk. Research must remain voluntary, governed, minimal where possible, and separated from rank, eligibility, and cross-architecture dossiers.

Computation is a workshop, not a courtroom. It builds candidate questions, exposes false positives, and compares mechanisms;

evidence under scientific governance issues only branch-bounded conclusions. That discipline keeps computation useful: a model can expose structure or failure without acquiring authority over anyone's standing. That separation also creates an empirical opportunity: ask whether a compact ordinary state can reproduce the whole-history pattern, and if so, which retained variables make the difference. A successful answer would simplify the story while revealing what information the apparent closure actually requires.

21.3 — Constrained Control Without Human Scoring

Constrained-control models are serious rivals because ordinary safety limits, fallback policies, and governed exploration can produce narrowing, lower variance, and stable regions without a lifetime ledger (Achiam et al., 2017; García and Fernández, 2015).

As technical tools, these models remain subordinate to human authority: no model decides worth, personhood, eligibility, or inner truth, and abstention is legitimate when the target is unidentified, untransportable, indistinguishable from rivals, or ethically unusable.

21.3.1 — Constraint Models

Constraint models are useful because they separate what is allowed from what is preferred. A system may pursue reward while respecting safety limits, or maintain viability while avoiding dangerous states. This vocabulary helps compare routes without turning a technical constraint into a moral command for human beings.

One practical comparison would ask whether each model has to look ahead to the same horizon. A standard reinforcement learner may discount future reward. A homeostatic system may defend near-term viability. A safe-control system may avoid violation within a defined horizon. The Law's terminal claim, by contrast, concerns the completed stream. That different horizon is not an ornament. It is a different burden (García and Fernández, 2015; Achiam et al., 2017).

Resemblance is not identity. A constrained Markov decision process is designed by specifying states, actions, rewards, costs, transition dynamics, and thresholds. The Law concerns a unified conscious stream whose felt experience is not directly observed and whose endpoint is ethically and biologically difficult to establish. The objects, evidence bridges, and moral risks differ (Sutton and Barto, 2018).

Constraint models are useful because they force specificity. What is the state variable? What is the cost? What is the threshold? What counts as violation? Who defines the objective? What data support it? These questions are equally useful when the Law speaks of cumulative record, reserve, horizon, and neutrality. The comparison pushes the Law away from vague poetry and toward testable structure.

A constraint model can also defeat the Law locally. If terminal variance compression, horizon narrowing, repair selectivity, or apparent balance is better explained by ordinary safe-control behavior under low reserve, the Law must concede that explanatory territory. A hard boundary in a control problem may be enough. The Law need not be invoked where control theory explains the data.

In the Law's vocabulary, a terminal boundary condition may look like a hard constraint. In a control model, a hard constraint can alter the value function as the system approaches a boundary. This comparison helps readers understand why a terminal layer might create distinctive dynamics. But the comparison remains conditional and mathematical, not proof that such a layer exists in conscious life.

The rigorous question is comparative. Does the candidate reserve-coupled route predict patterns beyond a constrained control model that lacks terminal fairness? Does it add residual explanatory power after safety envelopes and homeostatic variables are included? Which observation would distinguish that route from generic constrained control? These questions test one optional route; they do not redefine PTV C as that route.

Run that comparison under the common observation and evaluation rules in Section 21.4.4.

The computational analogy has an ethical boundary. A simulated agent can be reset after unsafe exploration; a person cannot. Human evidence must come from care-preserving, consent-based, low-burden designs, and a study that would impose unjust burden is a forbidden shortcut. Some questions may remain partly inaccessible because persons are not instruments.

The represented hazards may include collision, instability, irreversible loss, harm to humans, violation of a physical limit, or unacceptable cost (García and Fernández, 2015; Achiam et al., 2017; Ji et al., 2024).

21.3.2 — Hybrid Models

Hybrid models combine mechanisms because lived trajectories rarely belong to one textbook chapter. Adaptation, prediction, reinforcement learning, bodily regulation, treatment, social context, and observation error may operate together. The gain in realism is purchased with greater risk of nonidentification, overfitting, and post hoc explanation.

Integral feedback illustrates why ordinary control deserves full strength. In specified biomolecular systems, antithetic integral feedback can restore the stationary mean of a regulated molecular count to its set point, under the required ergodicity and bounded-moment conditions (Briat et al., 2016; Aoki et al., 2019). This is robust perfect adaptation of a regulated variable, not an exact finite-time terminal value for every trajectory or of a complete valence stream.

Clinical and social inputs belong in the causal model when they change the path. Medication, pain relief, housing, income stability, caregiving, legal protection, and family presence can alter both state and observation. Treating them as incidental noise would make an internal mechanism appear stronger than it is.

End-of-life data make hybrid reasoning unavoidable. Disease progression, medication, delirium, fatigue, family contact, palliative care, observation shutdown, and changing action horizons may all affect reported and unreported experience. Any terminal pattern must be compared against that joint field before it is called Law-specific.

A hybrid wins territory when the combined account explains the declared observations and transports better than the Law-linked alternative. The Law does not retain unexplained residue merely because no single rival component explained it alone.

A hybrid carries its own burden. Its component set, interactions, priors, parameter ranges, and module boundaries should be prospectively specified or selected through a transparent training procedure. Representation, dynamics, observation, endpoint, and

intervention modules can then be frozen before combination on held-out data, revealing which component creates any apparent advantage.

Held-out tests should vary more than random rows. A strong program holds out people, sites, event classes, interventions, or endpoint distributions so that transport is genuinely tested. A model that succeeds only within one sampling regime may have learned the regime rather than the mechanism (Patterson et al., 2024).

Ablation helps identify contribution. Removing adaptation, prediction, control, or observation components should change performance in interpretable ways. If no component matters individually, the model may be fitting through diffuse flexibility rather than recovering a causal structure.

Safe interventions can distinguish models when observation alone cannot. A treatment, context change, or information intervention may alter one pathway more than another. The design belongs to governed science and must never induce harm for the sake of discrimination.

Model recovery is a prerequisite. Data generated from each rival family should be passed through the planned pipeline to see whether the analysis recovers the true source or repeatedly labels rival worlds as Law-like. False recovery must be corrected before natural data are interpreted (Patterson et al., 2024).

Simulation does not validate the Law. A simulated agent exhibits the objective, state variables, and constraints placed into the model. If terminal pressure is encoded, terminal pressure may

appear. The output reveals implications of assumptions, not a natural boundary law.

Complexity costs should reflect effective freedom, not only named parameters. Flexible bases, repeated model selection, hand-tuned priors, analyst choices, and calibration options all consume freedom. A fair comparison records those choices and judges probability or interval forecasts on data not used to choose the result (Burnham and Anderson, 2002; Gneiting and Raftery, 2007; Mayo, 2018).

Hybrid reasoning now gives way to two sharply defined stochastic rivals. Mean reversion can produce local return without endpoint conditioning, while bridges can create apparent endpoint pull because an ending is built into the law being observed.

21.3.3 — Model Governance

Model governance begins with a prior question: who chooses the target, data, loss function, constraints, allowed interventions, and verdict? Those choices are not neutral when a model concerns people. Technical performance cannot decide which human purposes are legitimate.

Purpose must be declared before deployment. A model built for research comparison should not drift into screening, employment, insurance, treatment priority, eligibility, or public ranking. Every new use requires separate authority, burden analysis, and a meaningful right to refuse.

The governance boundary spans the full lifecycle: prospective specification, collection, preprocessing, training, validation, access, interpretation, communication, correction, and retirement.

A careful training process can still produce harm through a later reuse, derivative artifact, or uncorrected public claim.

Objectives and constraints require human review because reward, cost, safety, and success are design choices. A controller can satisfy its formal limits while shifting pain, coercion, privacy loss, or risk to people outside the optimization.

The no-score rule follows directly. Momentary affect estimates, cumulative estimates, endpoint probabilities, and control-model outputs may not become rankings of human value or hidden eligibility criteria. A research variable can become administrative power unless prohibited decisions are named in advance.

Participants and affected people need notice, consent where applicable, challenge, correction, withdrawal, and an accountable human decision maker. An appeal must be capable of changing the record or use, not merely documenting disagreement after the model has acted (Beauchamp and Childress, 2019). For high-impact uses, the record should show who could override the system, what evidence justified the override, and how the affected person could contest the outcome.

Conflicts of interest strengthen the need for institutional separation. When a foundation, movement, religious community, civic project, government partner, or funder benefits from positive results, prospective specification, independent review, and publishable adverse findings become part of the scientific design. Loyalty is a confound. Independent reviewers should disclose those relationships and retain authority to publish null or adverse findings without sponsor approval.

A model that cannot identify the target, transport to the new setting, or distinguish rivals should abstain. Common infrastructure can expose implementation choices and support reproducibility, but software creates neither scientific authority nor permission to govern a person (Ji et al., 2024; Mayo, 2018; Patterson et al., 2024). An abstention record should identify the failed gate so the next study knows whether the problem is measurement, transport, rival discrimination, or governance rather than treating every failure as one undifferentiated no. Abstention should also trigger a routing decision: return the question to ordinary clinical, legal, or administrative standards rather than leave an unresolved model output in operational use.

21.4 — Mean Reversion, Bridges, and Hybrid Models

Mean reversion, conditioned paths, observation selection, and hybrids are first-class rivals because noisy trajectories can show local return or endpoint pull without a terminal Law. A bridge process can alter earlier drift by conditioning on an ending, while missingness or measurement floors can create similar compression (Doob, 1957; Fitzsimmons et al., 1993).

These rivals must face common observation, information, tuning, and scoring rules, with extra complexity paid for by better held-out prediction or calibration.

21.4.1 — Mean Reversion Can Explain Local Return

Ordinary mean reversion can explain local return without a terminal boundary law. Feedback, regression, sampling, or stable dynamics may pull observations toward a typical range, and an Ornstein–Uhlenbeck (OU) process shows how disciplined return can coexist with random disturbance and no moral meaning (Uhlenbeck and Ornstein, 1930). The scientific question is whether the Law predicts something beyond that ordinary return.

A fair comparison can include the classical OU process, whose restoring drift is linear, alongside richer mean-reverting models. Heterogeneous baselines, changing variance, disease progression, medication, disability, chronic stress, report context, and missingness may enter those extensions. Nonlinear dynamics or several defended ranges should be identified as broader model classes, not as the classical OU process. If these rivals explain apparent compression or terminal concentration under matched

observation assumptions, the Law-specific claim must narrow in that evidence family.

A PTVC-linked route would need more than ordinary return. It would require independently defined endpoints, a latent cumulative object, and prospectively specified consequence families that strong mean-reverting, regulatory, selection, and observation models do not reproduce. Reserve-linked variance compression or terminal concentration can matter only as route-level signatures under explicit uncertainty; they are not the target law by themselves.

This is why drift alone cannot decide the Chapter. Both a mean-reverting process and a candidate reserve-coupled route can show apparent pull toward a central value. A fitted slope near the end may be impressive and still nondiagnostic. The scientific question is whether the full conjunction of prospectively specified signatures survives baselines that include ordinary regulation, stochastic return, endpoint conditioning, observation noise, and missing data (Burnham and Anderson, 2002).

Picture four curves that nearly overlap. In one, treatment restores function. In another, an ordinary mean-reverting process relaxes after a shock. In a third, the sample contains only paths that end near a chosen point. In a fourth, the observation system loses the most extreme final states. The shape is the same; the generator is not. A curve cannot identify its own cause. Intervention response, timing, missingness, and transport are needed to separate the stories (Uhlenbeck and Ornstein, 1930; Doob, 1957).

The ethical point is equally important. If a person appears to stabilize after hardship, public language must not imply that hardship was necessary, paid off, or made fair. Stabilization may mean care worked, expression narrowed, the observer stopped seeing, the body adapted under cost, or the measurement lost resolution. OU-like return is a model of pattern; it is not a judgment on the worth or meaning of suffering.

The correct role of the OU baseline is adversarial and clarifying. It gives researchers a simple family to beat before larger claims are made. It gives public readers a reason not to mistake smooth return for cosmic balance. It gives the Law a fairer burden: show the residual pattern, or concede that ordinary regulation has done the explanatory work.

Regression toward the mean can arise even when the latent state has no restoring force. An extreme observed value contains signal and measurement error; a later measurement often appears less extreme because the error component is unlikely to repeat. This must be separated from genuine dynamic reversion.

A state-space model can make that separation explicit by distinguishing latent dynamics from observation noise. If apparent return disappears once measurement error is represented, the rival has explained the pattern at the observation layer rather than the biological or psychological layer.

Reversion speed is informative. Ordinary mean-reverting models imply characteristic relations between departure size, subsequent change, and time lag. Estimating those relations across windows

can test whether one stable process accounts for the data or whether the apparent pull changes near an endpoint.

The reference level may itself move. Illness, treatment, aging, chronic stress, development, and social change can shift the center around which observations fluctuate. A fixed-center model and a moving-center model should be compared prospectively rather than selected according to which makes the ending look most orderly.

Selection and missingness can imitate both reversion and stability. Participants with extreme burden may leave follow-up, terminal observations may stop during agitation or sedation, and quality filters may remove difficult measurements. The denominator and observation process belong inside the rival.

Once ordinary return, observation error, nonstationarity, and selection have been modeled, the comparison can turn to paths that are explicitly conditioned on their endings. Those bridge constructions create a different kind of apparent pull and therefore require a different interpretation.

21.4.2 — Bridges and Transforms Can Mimic Endpoint Pull

Bridge and transformed processes remain serious rivals because conditioning on an ending can make earlier motion look as though it anticipates that ending. The apparent pull comes from the conditioned path law, not automatically from a terminal fairness constraint (Doob, 1957; Fitzsimmons et al., 1993; Chetrite & Touchette, 2015).

The construction is already developed in Sections 7.2.2, 8.4.2, and 20.4.1. Here the question is comparative: under common

observation and scoring rules, do bridge, control, or homeostatic models explain the observed path as well as or better than the PTVC-linked account?

This rival is especially relevant near terminal windows. Survival, diagnosis, treatment, observation, and endpoint definitions can select which trajectories remain in the sample. A cohort aligned near a common ending may already be conditioned before any explicit bridge model is fitted.

For prospective forecasts, the no-look-ahead rule in Section 20.1.1 applies: neither model may use information about the held-out path's endpoint, subsequent missingness, or terminal classification that was unavailable at the forecast time. A retrospective bridge comparison may condition on a declared endpoint, provided every model receives the same endpoint information and is scored for that conditional task. Its fit must not be reported as a forecast made before the endpoint was known.

Discrimination belongs in more than one-time averages. Two-time dependence, perturbation response, action sensitivity, transition behavior, and transport to a different endpoint distribution can separate a fitted bridge from an unconditioned mean-reverting process or a PTVC-linked route.

Model recovery is a practical safeguard. Simulate blinded data from bridges, ordinary return processes, and candidate PTVC routes, then ask whether the planned pipeline identifies the source. A method that calls bridge-generated data Law-specific is not ready for natural observations.

If the bridge rival wins under common endpoint information and tuning rules, endpoint drift retires as a distinctive signature. A PTVC-linked route would need a stronger conjunction involving an independently specified endpoint, validated measurement, path dependence, perturbation response, and transport.

21.4.3 — Global and Hybrid Routes Remain Distinct

Global and hybrid routes remain distinct. A two-boundary constraint, a local controller, a corrected endpoint, a predictive mechanism, and a mixed physiological-psychological model may all produce some similar traces while making different commitments about causation and counterfactuals.

Psychological and computational hybrids can combine predictive inference, homeostatic need, reward learning, and local adaptation. Such models may explain how expectation, bodily state, action, and recent outcomes jointly shape affect without positing complete-stream closure (Friston et al., 2017; Keramati and Gutkin, 2014). Memory, appraisal, social support, and meaning may then alter what is noticed, remembered, and carried. A different story about suffering can arise because the person and context changed, not because the cumulative record closed.

The strongest hybrid accounts treat lives as embodied, socially situated, historically conditioned, and unequal. Allostatic regulation already joins prediction, context, and changing operating ranges rather than one fixed set point (Ramsay and Woods, 2014; Sterling, 2012). The distinction in Section 21.1.3 applies here: reproducing a suggestive pattern and enforcing the complete-stream target are different achievements. The

comparison should establish which, if either, the hybrid actually accomplishes.

The ethical lesson is that psychological explanation must never become blame. If a person does not rebound, that is not failure of character. If a person adapts, that is not proof that harm was acceptable. Hybrid psychology can explain variation while preserving care.

Hybrid psychological models should include context variables. Sleep, pain, disability, medication, poverty, discrimination, family safety, social trust, legal protection, and work burden can all change the meaning of adaptation. Without context, a model may misclassify environmental burden as personal weakness or personal resilience as evidence that the environment was acceptable.

The applied inquiry remains useful here even if the formal Law narrows. It asks where the burden goes in the hybrid account. Does the explanation place too much responsibility on the individual? Does it hide institutional contribution? Does it ignore care? A rival can defeat the formal Law while still being corrected by the applied lens.

Homeostatic-affective hybrids link bodily regulation and felt life. Hunger, sleep, pain, inflammation, hormones, threat response, breathing, movement, medication, and social safety can shape mood, attention, desire, meaning, and behavior; behavior then changes exposure and physiology again. This embodied loop can look like cumulative adjustment toward balance without any terminal boundary condition.

The care pathways in Section 21.1.2 illustrate these homeostatic and allostatic accounts (McEwen, 1998; Ramsay and Woods, 2014; Sterling, 2012). The moral importance of relief does not depend on whether the Law contributes.

The strongest version of this rival may explain why the final currency is difficult to measure. Valence is shaped by variables that are themselves regulated, hidden, and context-sensitive. A clean affective number may conceal many bodily pathways. Measurement humility is therefore not optional. It is built into the nature of the target.

21.4.4 — Common Observation Rules Are Required

PTVC-linked models and rivals must face the same raw data, endpoint, missingness, preprocessing, uncertainty, scoring rules, information, compute, tuning, and repeated-seed budgets. Training, validation, and testing remain separate, with held-out times, sites, tasks, channels, or cohorts when transport is claimed (Patterson et al., 2024). Model recovery comes first: if blinded simulated sources are routinely confused, the verdict is model ambiguity; if the full model set fits poorly, the result is open-set inadequacy rather than shared support. A common data lineage should also record which transformations were applied, when they were chosen, and which observations each model actually received. That record makes it possible to distinguish an explanatory advantage from an advantage created by preprocessing, exclusion, or access to information unavailable to a rival.

Forecasts require proper scoring, interval coverage, subgroup performance, and residual diagnostics because average curves can hide overconfidence and local failure (Gneiting and Raftery, 2007). Hybrid rivals receive full opportunity but declare components, interactions, scales, and transport conditions before results. When added data bring precision without discrimination, the record should ask for a new perturbation, channel, endpoint regime, or identification assumption. The protocol should also specify in advance which residual patterns, calibration failures, or transport failures would require model revision rather than merely a larger sample. That distinction prevents accumulating data from becoming a substitute for confronting a systematic mismatch between what a model predicts and what observations repeatedly show.

Infrastructure may standardize environments and evaluation, but cannot choose humane targets or outputs that govern a person. Accountable authorities retain that role (Ji et al., 2024). That separation lets shared software improve comparability while leaving target choice, permitted use, and remedy with institutions answerable to the people affected. A benchmark can standardize a test without standardizing the human values that govern deployment. Its technical specification should therefore be versioned whenever datasets, endpoints, scoring rules, permissible inputs, or evaluation procedures change. A visible benchmark history allows later researchers to tell whether apparent progress came from a better model, a better measurement system, or a changed test, preserving comparability across generations of evidence.

21.5 — If Control and Homeostasis Win

When control, homeostasis, or bridge rivals win, the project should surrender exactly the territory they explain: a mechanism may become an analogy, a universal claim may narrow, a signature may retire, or a public artifact may require correction.

21.5.1 — What Gets Reclassified

If control and homeostatic accounts explain the relevant patterns, the Law's claims must be reclassified according to what the rival actually won. Reclassification is a record of learning, not a rhetorical compromise.

Several states are possible. A pattern may become ordinary regulation, an engineered control effect, a conditioned-path artifact, a local analogy, a narrow PTVC subcase, a burden-sensitive heuristic, or a retired claim. These categories should not be collapsed into the word nuanced.

Reward learning may explain repair behavior, homeostasis may explain range maintenance, allostasis may explain changing operating levels, and bridges may explain endpoint pull. In those domains, reserve or horizon language becomes a neighboring description rather than distinctive evidence unless an additional discriminator survives.

Reclassification is not erasure. A concept can become smaller and remain useful. The applied lens may still help readers ask where hardship is carried, shifted, reduced, or repaired, but applied usefulness cannot restore law-level status.

The current claim record should state the prior status, revised status, evidence trigger, affected reference class, public dependents, and conditions for reconsideration. That history prevents a universal sentence from surviving after the technical claim has become local.

Public language should use the new name wherever the reader encounters the idea. A control-theoretic analogy should be called an analogy. A narrow endpoint pattern should carry its conditions. A retired mechanism should not remain alive in a caption.

Other titles inherit only the corrected status. They may preserve care, rights, material analysis, institutional design, or voluntary meaning, but none may borrow a stronger formal claim than the scientific record permits.

Stating the reclassification rule in advance is evidence that the project is prepared to learn. A claim that has no named smaller state after failure is more likely to protect itself by ambiguity (Mayo, 2018).

21.5.2 — What Science Records

Science records the protocol, observation limits, rival set, result, and permitted verdict. It does not preserve a movement's preferred conclusion. A rival-favored, model-ambiguous, or open-set-inadequate result must remain visible.

Data provenance belongs beside the result. Recruitment, exclusions, attrition, treatment, missingness, preprocessing, and access restrictions can change what the dataset means. A polished estimate without denominator flow is not a complete record.

The model manifest should state the information and tuning budget. Which variables each model used, how parameters were chosen, how many runs were attempted, what was held out, and which endpoint information was available determine whether the comparison was fair.

Results should include uncertainty, calibration, subgroup behavior, residual structure, and sensitivity to defensible choices. One preferred number should not hide a range that crosses several scientific statuses.

The record needs typed dispositions rather than one verdict score: prerequisite failure, No adequate estimate, model ambiguity, rival-favored evidence, route failure, reference-class narrowing, and target-level adversity.

No adequate estimate is a result, not an administrative embarrassment. It may follow from nonidentification, unsafe burden, missing endpoint coverage, invalid measurement transport, or irreducible ambiguity. Recording why estimation failed can guide the next study more honestly than publishing a fragile number.

Negative and null findings deserve the same preservation as favorable ones. A failed signature, nonreplication, calibration error, or rival victory should remain searchable so later studies do not inherit only the surviving optimism (Mayo, 2018).

Apply the linked correction and non-erasure rules in Section 19.5.2 to the claim record in Section 21.5.1; silent replacement would destroy evidence about the correction process.

Failure memory is therefore a scientific asset. It records what was tried, what burden it created, why it failed, what should not be repeated, and which ethical protections remain independent of the result.

Scientific memory has little value if prestige, funding, or movement loyalty prevents it from reaching public language.

21.5.3 — Evidence Must Be Allowed to Change the Project

Project-protective science begins with a conclusion that cannot be lost, then arranges methods, rivals, and public interpretation around preserving it. No instrument can correct loyalty placed above discovery. Employment, access, belonging, or safety must never depend on confirmatory results.

Scientific authority therefore stays separate from fundraising, spirituality, campaigning, and implementation. Null and adverse results are complete products because failed predictions, harmful measures, inadequate models, and impossible bridges prevent larger burdens. Critics need a route that can change assumptions, methods, aggregate evidence, corrections, and public claims rather than merely decorate the process. A review process demonstrates that authority when a documented objection produces a testable revision and a response that others can inspect.

Claims, routes, images, and statements should narrow, retire, change, or withdraw without institutional collapse. For vulnerable readers, emotional usefulness cannot justify overclaim; a smaller honest result may preserve care and future inquiry better than a protected grand one. For example, a public statement can retain a

supported finding about regulation while revising the additional interpretation once attached to it.

The record carries both the question and the conditions under which it must shrink. Correctability is competence. That record gives later readers a usable starting point: the surviving question, the evidence behind it, and the decisions already made. A durable record should also distinguish changes to the scientific claim from changes in implementation, so later readers can see exactly what moved.

Part VIII — Evolution, Worlds, Meaning

Part VIII widens the horizon without loosening the standard of proof. Earlier Parts defined the proposed Law, examined measurement and stream identity, opened empirical windows, and gave strong rivals room to win. The question now meets three of humanity's largest interpretive frames: evolution, possible worlds, and spiritual meaning.

These frames can make a difficult hypothesis feel inevitable before evidence earns anything. Evolution can become destiny, simulation can masquerade as revelation, and recurring religious images can be presented as independent confirmation. This Part rejects those shortcuts while preserving what each frame can genuinely teach.

Evolution comes first because every candidate stream is embodied in a history of inheritance, development, environment, and selection. Pain, reward, attachment, fear, appetite, social sensitivity, and repair arise in organisms shaped by constraint, drift, tradeoff, conflict, and ecological change. Any account of complete felt experience must stand beside evolutionary biology rather than float above it (Darwin, 1859; Darwin, 2009; Tinbergen, 1963; Williams, 1966).

Evolutionary compatibility is a low bar. Selection can produce regulation, rebound, learning, inhibition, and homeostatic control without selecting for terminal neutrality. It can also produce vulnerability, mismatch, painful defense, and traits that protect fitness while worsening a bearer's life. The real question is whether the Law adds a distinctive, testable constraint beyond ordinary biology (Gould & Lewontin, 1979; Trivers, 1971).

Cross-species comparison sharpens the question and raises its ethical burden. Evidence for consciousness and valenced experience varies by taxon, method, development, and context. Precautionary work has widened the range of animals serious inquiry should not dismiss while stressing that behavior, neural organization, and welfare response remain uncertain evidence, not a consciousness meter (Birch et al., 2020; Andrews et al., 2024; Birch, 2024).

The second movement enters simulated worlds. Models can plant mechanisms, vary assumptions, create hostile alternatives, and test whether an analysis recovers what generated the data. They expose circularity, observational equivalence, hidden endpoint selection, and the ease of programming a desired result into a system (Levins, 1966; Oreskes et al., 1994; Guest & Martin, 2021).

A simulation inherits its rules from its builders, and that is precisely what makes it useful. A declared terminal condition can reveal what else a world would have to be like for the condition to hold. Alternatively, closure might follow from other declared rules, without a corrective step at the end. Both exercises can teach us about structure. Evidence about our world requires the further step of testing those rules and consequences against observation. Humanlike dialogue still does not establish that a synthetic agent feels.

Elegant models can create an aura of authority. Clean plots may hide circular endpoints, complex agents invite anthropomorphism, and large parameter spaces can reproduce almost any trajectory after enough searching. Verification,

validation, sensitivity analysis, model recovery, and transparent description are therefore part of the argument, not technical housekeeping (Oberkampff & Roy, 2010; Saltelli et al., 2020; Grimm et al., 2020).

The third movement turns toward spiritual echoes. Long before modern science, people asked why suffering falls unevenly, whether justice exceeds visible events, how grief can be carried, and whether repair belongs to reality. These questions appear in doctrines, rituals, laments, protests, philosophies, and secular refusals across cultures (James, 1902; Durkheim, 1912; Geertz, 1973).

Respect requires more than collecting attractive similarities. Karma is not one doctrine, providence is not fate, grace is not repayment, and repair is not cosmic accounting. Traditions contain rival schools, critics, reformers, protest voices, and different accounts of suffering, agency, death, and hope. Responsible comparison preserves those differences instead of turning them into one chorus for the Law (Asad, 1993; Obeyesekere, 2002; Mahmood, 2005).

The ethical center does not move. Evolution does not justify suffering because a trait served survival. Simulation does not make real pain a programmed necessity. Spiritual interpretation does not belong to another person's grief unless freely received. Care, relief, protection, truth, and repair remain immediate under every scientific and metaphysical outcome.

The shared thread is the location of burden and the limits of authority. In biology, who bears the cost of regulation and

adaptation? In simulation, which assumptions or patches produce the result? In spiritual comparison, who may name meaning, and who remains free to refuse it?

Part VIII also deepens falsifiability. Cross-species variation may narrow a reference class. Recovery studies may fail to separate a proposed mechanism from ordinary control. Counterworlds may reproduce the same signature without the Law. Comparative religion may show recurring human concern but no shared metaphysical claim. Each outcome informs a project that does not need every neighboring frame to agree.

The emotional movement is from grandeur to humility. A question may touch life, worlds, and meaning without gaining evidence from its scale. Wonder is strongest when it survives precision, and hope when it remains compatible with correction.

Chapter 22 — Evolution and Constraint

Evolution is not an ornament added to make the Law sound scientific. It is one of the strongest bodies of explanation available for the capacities through which feeling, learning, avoidance, attachment, and regulation occur. A public root that speaks about complete conscious streams must therefore meet evolutionary biology at full strength.

Natural selection explains differential persistence through heritable variation under environmental conditions. It does not aim at happiness, moral desert, or fairness. Traits can spread because they improve reproduction even when they impose pain, fear, exhaustion, conflict, or early death on the organisms that carry them (Darwin, 1859; Williams, 1966; Trivers, 1971).

Development and constraint complicate the picture further. Organisms are not assembled from an unlimited menu. Historical inheritance, body plans, gene regulation, energetic limits, developmental pathways, and ecological context restrict what variants can arise and survive. A deeper boundary, if real, would have to operate within that constrained space rather than replace it.

Tinbergen's distinction among mechanism, development, function, and phylogeny is useful here because it prevents one answer from doing the work of another (Tinbergen, 1963). A trait's current mechanism may be described without knowing its evolutionary history. An adaptive function may be proposed without identifying the felt quality it produces. A terminal claim would add another layer and would need its own evidence.

Maladaptation is equally important. Systems shaped in one environment can become damaging in another. Defensive responses can become chronic. Reward learning can intensify craving while pleasure weakens. Social mechanisms can protect group coordination while humiliating or excluding individuals. Evolutionary explanation does not guarantee experiential benefit.

Cross-species inquiry raises the bearer problem in a form that human self-report cannot settle. Mammals and birds provide strong cases for precaution, while fishes, reptiles, cephalopods, decapod crustaceans, and insects require increasingly serious attention under current evidence (Birch et al., 2020; Crump et al., 2022; Andrews et al., 2024). The relevant conclusion is not that every candidate has the same kind of experience, but that uncertainty must be represented rather than erased.

Comparative work also requires a transport bridge. A behavioral response in a bee, a nociceptive tradeoff in a crustacean, a pessimism-like judgment bias in a mammal, and a human verbal report are not interchangeable measures. Each channel has a different relationship to valence, unity, duration, memory, and context.

Reference classes may therefore narrow. A finding may apply to one taxon, one developmental stage, one ecological condition, or one operational definition of valence. That narrowing is not failure of scientific seriousness. It is the honest consequence of discovering where a broad formulation exceeds what the evidence can support.

Animal welfare follows from uncertainty in a precautionary direction, not from proof of the Law. Where credible evidence raises a realistic possibility of conscious suffering, the ethical question is how to reduce unnecessary risk while improving knowledge. No experiment becomes justified merely because it could sharpen a terminal hypothesis (Birch, 2017; Birch, 2024).

The failure conditions are concrete. Cross-species patterns may be explained by ordinary regulation. Measurement bridges may not transport. Species may differ in ways that defeat a universal route or reference class. Ethical limits may prevent decisive experiments. Any of these outcomes can narrow the claim without being relabeled as hidden support.

Detailed protocols, model recovery, and verdicts belong to *The Science of Fairness*. This Chapter maps what evolution explains, what a deeper constraint must add, what cross-species comparison requires, and what failure would mean. It does not own scientific adjudication.

The reader should leave with a concrete biological question: could an independently supported mechanism connect the course of felt experience to the proposed terminal relation? A familiar mechanism might supply that connection, while other mechanisms might produce only similar local patterns. Distinguishing those possibilities gives the chapter its constructive task.

22.1 — Evolution Under a Possible Constraint

Evolution under constraint is ordinary biological thinking before it is a hypothesis about fairness. Variation is filtered through inherited body plans, developmental pathways, physical limits, ecological relations, and historical accidents. Evolution explores a structured possibility space rather than an unlimited menu (Darwin, 1859; Maynard Smith, 1982).

A Law-specific constraint would have to be distinguished from those known restrictions. It could not be inferred merely because organisms regulate themselves, because complex traits appear, or because selection repeatedly finds workable solutions. The relevant question is whether a deeper boundary adds consequences that selection, development, and ordinary control do not already entail.

Constraint language also needs protection from design language. Physical and developmental constraints do not imply an intending agent, and compatibility with evolution does not imply purpose. The Chapter can consider design as a later conditional thought experiment without importing it into biology.

Finally, evolution under constraint must include maladaptation and suffering. Natural selection produces compromises, conflicts, mismatches, and painful defenses as well as elegant function. Any plausible relationship between evolution and the Law must survive those cases rather than describing only well-regulated organisms (Gould & Lewontin, 1979; Trivers, 1971).

22.1.1 — Evolution Under Constraint

Developmental systems can channel change without dictating one outcome. Plasticity may expose new phenotypes, developmental architecture may make some transitions easier than others, and ecological feedback may stabilize one pathway while closing another. These processes show how constraint and variation interact, but they supply no terminal phenomenal invariant by themselves (West-Eberhard, 2003; Pigliucci & Müller, 2010).

Evolutionary explanation must locate the process that produced a pattern. Selection, drift, developmental constraint, historical contingency, and the mapping from genotype through development to phenotype can preserve similar outcomes for different reasons. A proposed terminal boundary would need a place in that causal map rather than a poetic label applied after adaptation is observed (Gould & Lewontin, 1979; Pigliucci & Müller, 2010; West-Eberhard, 2003).

Constraint language carries several meanings that should not be blended. Developmental constraints limit which phenotypes can arise; ecological constraints limit which strategies can persist; physical constraints limit what matter can do; and a terminal boundary would concern a complete phenomenal trajectory. Similar vocabulary does not establish a shared mechanism, so every comparison must identify the level at which the constraint acts (Maynard Smith, 1982; West-Eberhard, 2003).

Evolutionary vocabulary earns its place here by clarifying causal order. Selection may favor traits within an available space, whereas a genuine stream-level boundary would restrict the

trajectories available before any trait succeeds or fails. That distinction keeps the proposal from becoming a story about genes choosing fairness, moral virtue being selected, or history carrying a hidden plan.

Evolutionary inquiry can take the project in several informative directions. Ordinary regulation may explain a candidate pattern without implying closure; available observations may leave competing accounts unresolved; or a specified mechanism may yield a testable connection to the complete-stream target. If an ordinary mechanism itself supplies that connection through independently supported premises, it becomes a candidate realization. What matters is what the mechanism actually entails.

Current constraint biology makes the compatibility point sharper. Evolution is path-dependent and historically contingent, yet its range of possible designs is narrowed by physical, developmental, cellular, energetic, ecological, and informational constraints. That observation does not support the Law. It only prevents a careless objection in which natural selection is imagined as free to generate any possible experiential trajectory without regard to the state space in which organisms live.

Tradeoffs make this point concrete. A nervous system may become more sensitive to threat because sensitivity sometimes protects survival, but the same sensitivity may create anxiety, fatigue, or social withdrawal. A reproductive strategy may increase lineage success while imposing risk on the body that carries it. A metabolic adaptation may conserve energy while narrowing activity and pleasure. Evolution often solves one constraint by moving burden elsewhere.

The public point lies in that redistribution of cost. Functional success can conceal pain, vigilance, exhaustion, reproductive risk, or reduced opportunity elsewhere in an organism's life. A fairness inquiry therefore asks not only what a trait accomplishes, but where its experiential cost appears and whether ordinary biological tradeoffs already explain the pattern attributed to deeper closure (Sneddon et al., 2014; Solvi et al., 2016).

The Law of Fairness enters this picture only as a candidate additional constraint, not as a replacement for evolution. If the Law is true, selection would not need to know the Law in order to operate within it. Genes do not need to understand gravity for organisms to fall, and organisms do not need to understand developmental constraints for those constraints to shape possible bodies.

The analogy reaches only as far as structure. Gravity is established science; the Law of Fairness is a candidate hypothesis. A constraint can be real without being selected as a trait, but evolutionary compatibility cannot show that this particular constraint exists. Evidence must do that work.

Evolutionary constraint also helps prevent a false objection. A critic may ask why natural selection would choose terminal neutrality. The better question is whether selection would have a choice if experiential state space had boundary conditions. Selection may favor organisms within the space it is given, not legislate the structure of the space itself.

The comparison becomes more useful when it asks what a specified mechanism actually entails. Explaining a familiar

pattern and establishing exact terminal closure are different achievements. The evidence should identify which has been accomplished, while leaving familiar biology available as either a competing explanation or a candidate realization.

The formal consequence is modest but important: evolution becomes a compatibility gate, not a validation engine. If the Law cannot be stated in a way that can coexist with ordinary evolutionary constraint, it fails conceptually. If it can coexist, nothing has been proven, but the next question becomes empirical rather than dismissive.

22.1.2 — Constraint Does Not Mean Design

The word constraint can tempt readers toward design. If a system has limits, someone must have placed them. If streams close, someone must have planned closure. If suffering does not remain as residual imbalance, some moral intelligence must be enforcing balance (Barron & Klein, 2016; Feinberg & Mallatt, 2016; Birch, 2017).

Design language enters too easily when an outcome is described as if evolution aimed for it. Natural selection filters heritable variation through differential reproduction; it does not look ahead to a finished lifetime ledger. Any account that depends on future closure must therefore explain how present processes are coupled to that boundary without borrowing foresight from the observer (Darwin, 1859; Tinbergen, 1963).

Biologists often distinguish teleonomy, the appearance of goal-directed organization produced by evolved mechanisms, from teleology, an appeal to final purpose. That distinction matters

here. A system can regulate toward a state through causal feedback without the state being morally intended, cosmically assigned, or known in advance (Tinbergen, 1963; Godfrey-Smith, 2009).

The Religion of Fairness may later receive the Law as optional sacred form for those who freely receive it that way. That is a different register. This chapter must not smuggle that register backward into the science-facing argument.

Natural selection is sovereign over neither physics nor phenomenology. It is a historical filter on variation that affects differential survival and reproduction. It can shape nervous systems, bodies, social capacities, and behavioral dispositions. It cannot, by itself, define every property of felt experience unless felt experience is wholly reducible to the features selection manipulates.

That separation is decisive because the proposed Law concerns felt experience rather than biological success. It asks what a unified conscious stream undergoes and whether its complete pre-endpoint history closes without residual imbalance. Selection may explain many inputs to the stream; explaining the inputs does not yet explain the boundary.

A pain system may be adaptive because it warns the organism away from damage. That does not mean the pain is fair. A fear system may protect the body from threat. That does not mean the fear burden is neutralized. A mating drive may increase reproduction. That does not mean the longing, rejection, jealousy,

violence, or vulnerability around it has been experientially balanced.

Selection is powerful precisely because it does not need to care about subjective burden directly. Traits can persist if they help genes propagate or bodies survive, even when they make life harder for the bearer. This is one reason evolutionary explanation cannot be turned into a moral justification. Biological usefulness is not ethical permission.

The argument therefore has to avoid two opposite errors: denying that felt life is embodied and evolved, or allowing evolutionary success to dissolve the terminal question. The target remains the conscious stream, not the lineage, genome, or population curve.

Evolutionary and regulatory accounts can explain many of the capacities and patterns this inquiry studies. A proposed additional fairness condition must earn its place through consequences that distinguish it from capable alternatives. A rival need not explain every aspect of consciousness to expose a failed prediction. Where genuinely competing models make the same observable predictions, those observations cannot decide between them.

An ordinary mechanism could also become part of an explanation of the Law. If independently supported relations connected that mechanism to felt contributions and the stream's true pre-endpoint boundary, and together required exact closure, it would provide a possible realization. Familiar biology would then be doing the explanatory work. The decisive distinction is between producing a similar-looking pattern and establishing the complete relation the Law proposes. That leaves the research open to a

discovery through ordinary processes, while asking every claimed connection to earn its evidence.

A constraint does not automatically command. Physical systems exhibit constraints without moral intent; biological systems develop under them without conscious planning. Mathematical models use boundary conditions without implying moral approval. If true, the formal Law would resemble a structural boundary condition, not a moral administrator.

This is more than philosophical refinement. It protects vulnerable readers. When constraint becomes design, suffering can sound assigned; from there it is a short step to claiming a wound happened for a reason. The book closes that path before it can excuse blame, neglect, or forced meaning.

A constraint can still be profound without becoming personal. Support for terminal neutrality would raise deep philosophical questions, but the first obligation would be to keep claim, evidence, purpose, and responsibility in their proper places. Wonder may accompany the inquiry; it may not decide it.

That restraint matters when the argument reaches simulated worlds and spiritual parallels, which can intensify impressions of design. Evolution establishes the discipline first: constraints are not intentions in scientific clothing.

22.1.3 — Maladaptation and Suffering

Evolution builds workable inheritances from prior conditions, not a clean sheet. Historical constraint, developmental tradeoffs, ecological change, and path dependence can leave costly compromises sufficient for survival or reproduction without

improving the bearer's life (Gould & Lewontin, 1979; West-Eberhard, 2003; Nesse & Williams, 1994; Stearns, 1992).

Maladaptation makes the distinction visible. Fear, pain, or stress can serve protective functions and still produce prolonged burden, especially when environments change or responses remain active too long. Comparative welfare research must separate function from felt cost rather than treating either as evidence about the other (Mellor et al., 2020; Crook, 2021; Crump et al., 2022). Traits need not be strictly maladaptive to hurt: competition, attachment, appetite, and vigilance can support reproduction while exhausting or distressing the organism.

For the Law of Fairness, fitness and felt experience remain separate variables. Reproductive success cannot stand in for positive valence, and biological failure cannot stand in for negative valence. A lineage may persist through lives containing severe burden, while an evolutionarily unsuccessful life may contain joy, care, and meaning. The proposed terminal condition, if it exists, concerns the stream that feels, not the lineage that persists. A useful comparison would describe reproductive outcomes and reported experience separately, allowing their relationship to become a question rather than an assumption.

Regulation, recovery, and return toward familiar ranges may explain local stabilization without explaining the complete signed history of one stream. They are serious rivals or possible components, not automatic evidence for a terminal law. Maladaptation also blocks moral storytelling: ancient or adaptive origins do not make pain deserved, and no possible long-horizon constraint excuses preventable suffering. Care, treatment,

accommodation, environmental change, and welfare improvement remain justified by present need. For example, an environmental change can be assessed through its effects on present welfare while the proposed terminal question remains separately specified.

Research should model evolutionary and ecological variables as causes, moderators, or rivals; state the candidate bearer and valence representation; and challenge broad predictions with mismatch, chronic pain, altered environments, reproductive conflict, long recovery, and cases where regulation visibly fails. Evolution explains how traits arose and persist. The Law asks whether those mechanisms or an additional restriction entail the complete-stream relation. A comparative design can then ask which prediction changes across ecological settings and whether the change follows the proposed mechanism or a competing explanation.

22.2 — What Regulation Already Explains

Living systems already contain layered regulation. Homeostasis holds variables within viable ranges, allostasis anticipates demand, learning changes action through consequences, and opponent processes can produce rebound after repeated stimulation. These mechanisms can create bounded trajectories without a complete-stream terminal rule.

Analogy still has value. Regulation shows how local mechanisms can restrict choices, allocate resources, and shift costs across time. It also teaches us to ask where a system protects viability by imposing distress, reducing future options, or displacing risk onto another process.

The danger is overreading resemblance. A control loop is not a lifetime ledger, stability is not fairness, and a viable organism is not necessarily a flourishing bearer.

22.2.1 — Regulatory Systems

Living bodies are regulatory systems before they are philosophical puzzles. They monitor temperature, blood chemistry, hunger, thirst, threat, fatigue, arousal, pain, immune activity, movement, and social signals. They do not merely receive the world. They continually adjust themselves in order to stay viable within changing conditions (Sterling, 2012; Solomon & Corbit, 1974).

Resemblance alone cannot carry evidence. A thermostat-like loop does not establish terminal neutrality. A stress-regulation circuit does not establish ledger closure. A priority queue in attention

does not prove a metaphysical constraint. Regulatory systems should be credited for what they explain, not recruited as automatic witnesses.

Regulatory analogy becomes scientifically useful when it turns resemblance into a relation that can be tested. Known control systems receive credit for the patterns they explain. The further question is whether their independently established connection to experience could entail the complete-stream target or whether they explain only a local resemblance.

One practical research implication is that regulatory systems should be treated as default explanations before any Law-specific mechanism is invoked. Stress recovery, temperature regulation, hunger cycles, pain avoidance, social buffering, and sleep restoration all have established biological roles.

Modern physiology also warns against a gene-only picture of evolved function. Homeostasis, allostasis, repair, regeneration, development, immune regulation, endocrine timing, and behavioral control are layered processes, not single switches. That richness helps explain why the candidate pathway model comparison is tempting, and why it must be controlled. Many ordinary regulatory architectures can produce menu restriction, rebound, prioritization, and narrowing without implying terminal neutrality.

Regulatory systems therefore serve as both bridge and rival. They make the candidate pathway model biologically intelligible, and they threaten to absorb it. That double role is exactly why the

comparison belongs in a rival-aware chapter rather than in a triumphal mechanism chapter.

At the same time, regulatory systems may help identify the level at which the Law would have to operate. If the Law has any biological interface, it would likely appear through coupled systems rather than a single hidden meter. That possibility calls for multichannel humility, not a search for one magic organ of fairness (Darwin, 1859; Tinbergen, 1963).

Allostasis widens the comparison. Organisms do not merely return every variable to a fixed set point; they often regulate by prediction, context, anticipated demand, and changing environments. That means menus of possible action can be shaped before a crisis becomes visible. A body under chronic stress may live with a narrower menu not because fate is selecting paths, but because regulation has adapted to threat.

A narrowed behavioral menu should not be read as destiny. Trauma, poverty, illness, developmental history, medication, social pressure, and ordinary learning can all reduce the options a person or animal can use. Only prospective comparison among explicit models could separate those causes from a candidate constraint architecture.

Homeostasis is a language of viable ranges. The organism does not choose from an abstract menu of infinite options. It chooses under temperature, glucose, oxygen, hydration, injury, fatigue, immune state, social threat, and available energy. The live menu is shaped by the body's current condition before conscious deliberation begins.

The candidate pathway model also speaks in menus. A stream does not move through life with every possible repair option open. Its live possibilities narrow or widen under reserve, horizon, social support, fear, pain, money, safety, institutional time, and bodily capacity. Homeostasis makes that idea biologically familiar without exhausting the formal claim.

But homeostasis is not fairness. Restoring glucose, temperature, or arousal toward a viable range is not the same as closing a life ledger. A body can maintain viability while the stream carries fear. A nervous system can survive by narrowing life. Regulation can protect biological continuation while making felt life smaller.

The useful comparison is therefore not identity but structure. Homeostasis shows how local constraints shape menus. The Law asks whether there is a deeper cumulative constraint on the entire stream. The difference is temporal, phenomenological, and formal. One concerns current viable regulation; the other concerns terminal per-stream closure.

A future model might represent this distinction by separating menu variables from ledger variables. Menu variables describe current reachable options. Ledger variables describe cumulative valence under unity. Mixing them would make the model easier to tell and less true.

22.2.2 — Resemblance Is Not Evidence

Resemblance is especially dangerous when the same vocabulary serves different theories. Constraint, reserve, menu, repair, inhibition, pressure, and closure can describe ordinary control or a Law-specific proposal. The translation rule is simple: shared

words do not imply shared objects or mechanisms (Hamilton, 1964; Williams, 1966; Trivers, 1971).

Analogy orients; evidence changes claim status. A familiar comparison can help readers locate a concept, but it becomes evidentiary only through a prospectively specified, ethically valid test that can distinguish the proposal from serious rivals. Until then, resemblance remains a teaching aid rather than support.

This rule protects the Law from aesthetic science. A diagram of feedback loops can feel rigorous. A comparison to regulatory biology can feel natural. A phrase like evolved balance can feel profound. But if the comparison does not generate distinctive predictions or survive rival baselines, it is not evidentiary.

Behavioral inhibition is one of the clearest biological analogies to queue restriction. Organisms often do not act because a signal stops them. Fear inhibits approach. Disgust inhibits consumption. Pain inhibits movement. Social threat inhibits speech. Fatigue inhibits effort. Uncertainty inhibits commitment. These stoppages are not mere absences. They are active constraints on the live menu.

The Law of Fairness can use behavioral inhibition to clarify how burden changes possibility. A person under severe shame may not have the same social menu as an unashamed person. An animal under threat may not have the same exploratory menu as a safe animal. A stream under chronic pain may lose options before any conscious plan is formed.

Behavioral inhibition also connects to the future horizon. When repair seems unavailable, an organism may freeze, conserve,

avoid, submit, or narrow attention. Threat systems, reinforcement history, predictive processing, and safe-control models can all generate those patterns. The open question is whether any inhibition signature remains after those rivals receive a fair test.

The analogy helps because queue-like closure is not exotic. Biological life already contains stops, gates, thresholds, vetoes, and priority rules. The question is whether these local mechanisms aggregate into a stream-level structure with terminal implications. That is not answered by observing inhibition alone.

Ethically, inhibition language must avoid blame. A blocked behavior is not necessarily weakness, laziness, cowardice, or lack of will. It may reflect a menu constrained by threat, pain, fatigue, deprivation, or prior learning. Reading the blockage before judging the bearer makes hidden burden more visible and preserves dignity across human and animal cases.

Behavioral inhibition is therefore a doorway into mechanism and humility at once. It grounds the pathway analogy in real biology while showing how much of the apparent queue-like pattern ordinary control may already explain. Whatever remains must be earned against those accounts (Gould & Lewontin, 1979; Maynard Smith, 1982).

Behavioral inhibition is also where human and animal interpretation can diverge. A human may inhibit speech from shame, fear, prudence, habit, or trauma. An animal may inhibit movement from predation risk, illness, social rank, or experimental stress. Similar outward stillness can carry different inner meanings or no comparable meaning at all.

For this reason, inhibition should be modeled as a signal under context. It can indicate burden, but it can also indicate adaptive caution or learned task structure. A responsible Law of Fairness science would ask what rival models predict before assigning inhibition to a candidate Law-specific pathway mechanism.

The public conclusion can be stated without either dismissal or promotion. Homeostasis, stress regulation, reinforcement learning, and neurobiology may make a candidate architecture intelligible and generate tests. They do not establish terminal neutrality. Keeping those roles separate protects both the hypothesis and the sciences from which it learns.

Research would have to test whether queue-like dynamics predict terminal concentration, variance compression, endpoint-independent patterns, compensability signatures, or cross-species relations better than homeostasis, adaptation, predictive processing, reinforcement learning, and stochastic baselines. Until such comparisons succeed, the resemblance remains orienting rather than probative.

A useful internal rule is that resemblance may generate a hypothesis only if it also generates a possible disconfirmation. If the comparison cannot say what would count against it, it is not yet scientific. It is metaphor. Metaphor can be useful in teaching, but it must not be promoted into claim status.

That rule should govern every diagram in this book. Arrows, loops, queues, and balances compress thought. Compression can educate, but it can also erase uncertainty. A visual analogy should make clear that it locates the question without answering it.

22.2.3 — Where Comparison Misleads

Comparison misleads when it treats design-like function as evidence of purpose. Biological systems can appear elegant because selection, development, and physical constraint produce organized outcomes. Elegance does not mean intention. Function does not mean moral plan (Stearns, 1992; Nesse & Williams, 1994; Maynard Smith & Szathmáry, 1995).

Comparison misleads when it turns adaptive suffering into justified suffering. Pain can warn. Fear can protect. Grief can signal attachment. Shame can regulate social standing. None of these functions make the suffering deserved or ethically acceptable to ignore.

Comparison misleads when it becomes cross-species projection. A human account of relief, anxiety, boredom, shame, or hope cannot simply be pasted onto other animals. Their sensory worlds, social ecologies, nervous systems, and needs differ. Similar behavior may not carry identical valence.

Comparison also misleads when it becomes too comforting. If homeostasis resembles balance, a reader may infer that life always self-corrects. Hard cases refute that. Chronic pain, trauma, deprivation, and irreversible loss show that local regulation can fail or stabilize around suffering.

The final check is whether the analogy leaves the hard case intact. If a comparison makes chronic suffering, animal distress, trauma, disability, or terminal uncertainty disappear, it is not clarifying. It is hiding burden.

Comparison also helps identify candidate mechanisms. If menu restriction matters, researchers can examine attention, threat, energy, interoception, action selection, pain, social inhibition, and recovery pathways. If horizon matters, researchers can examine perceived future opportunity, urgency, reserve, and irreversibility avoidance. Analogy can guide where to look.

Comparison helps with rival design as well. A good analogy suggests alternative explanations that might defeat the Law. If homeostasis can explain a pattern, test it. If reinforcement learning can explain repair-seeking, test it. If predictive processing can explain relief and uncertainty reduction, test it. The Law becomes stronger only by letting these comparisons become competitors.

A comparison becomes useful when it produces a discriminating question. Does the pattern require a terminal condition, or does it follow from local control? Does a late-life change reflect horizon compression, medical care, cultural script, reduced reporting capacity, or ordinary physiology? Does an apparent burden reduction disappear when welfare context, missingness, and regression are modeled? These are better questions than does this look like the candidate pathway model.

Comparison also helps locate intervention ethics. If a local regulatory mechanism explains a burden, care can be directed toward that mechanism without waiting for terminal evidence. Pain relief, sleep repair, stress reduction, enrichment, social support, or environmental change may matter immediately. The applied Law of Fairness lens can help care see burden even when the formal Law remains under test.

Comparison also keeps the analysis embodied. People and animals do not carry abstract ledgers outside their bodies; they carry fatigue, fear, pain, hunger, attachment, social injury, and capacities for recovery. Any use of the fairness framework should remain close to those lived mechanisms rather than float above them (West-Eberhard, 2003; Godfrey-Smith, 2009).

Comparison also helps when it improves compassion. Seeing a choice as menu-constrained can lower blame. It can explain why a person could not simply decide differently, why an animal did not escape, why a patient avoided treatment, or why a child froze under pressure. Burden becomes visible before character judgment hardens.

Good comparison therefore has a moral signature: it makes the person or animal less blamed and the theory more testable. Bad comparison does the opposite. It makes the theory feel inevitable and the burdened bearer easier to explain away.

Comparison also fails when it erases the temporal distinction in Section 22.2.1: a local regulatory loop is not a complete stream. A comparison should place the observed duration beside the duration of the claim, making the difference in temporal coverage visible.

Comparison also misleads when it erases currency. Biological systems often optimize viability, energy, reproduction, or survival-relevant action. The Law's currency is felt experience. A system can keep the organism alive while the organism suffers. Viability cannot be allowed to replace valence. An example becomes clearer when its columns name the biological outcome

and the experiential outcome separately, with the evidence available for each.

The safest rule is that comparison may open a question but may not close it. The moment analogy begins to carry proof, moral comfort, spiritual certainty, or public authority, it has left its job. The productive next step is to identify the feature the analogy suggests and formulate a direct test of that feature in the intended domain.

22.3 — Predictions and Measurement Bridges

Comparative prediction begins only after two uncertainties are admitted: which systems are conscious, and how valence is represented in each. Behavior, physiology, neural organization, and learning can inform those questions, but no single channel settles them. The bridge from a measured response to felt experience must be explicit.

The evidence distinction in Section 14.1.1 requires taxon-specific methods, converging indicators, and welfare-aware interpretation rather than an assumption that species share one kind of experience (Andrews et al., 2024).

Transport is the central empirical burden. A human rating scale, a rodent judgment-bias task, a cephalopod tradeoff, and an insect self-protection response do not share meaning automatically. Researchers must ask which latent construct, exposure rule, and uncertainty structure could make them comparable.

Reference classes may narrow when transport fails. A result can remain useful for one species, one task, or one welfare question without supporting a broad terminal claim. The Section treats narrowing as a scientific result rather than a defect to conceal.

22.3.1 — Bearer Status and Valence Representation First

Cross-species comparison begins before the first graph. Researchers must decide which systems are serious candidate bearers and what would count as evidence about positive, negative, or neutral experience in each of them. Recent work has widened the range of animals that should not be dismissed

without inquiry, while also emphasizing that no single behavior or neural feature settles consciousness by itself (Birch et al., 2020; Andrews et al., 2024; Birch, 2024).

Candidate status is not the same as a verdict. It is a reason to investigate with proportionate care. Mammals, birds, cephalopods, fishes, decapod crustaceans, and insects present different nervous systems, ecologies, learning capacities, and modes of action. The evidentiary question is not whether they look human. It is whether converging evidence supports a defensible interpretation of what helps, harms, attracts, repels, relieves, or frustrates the organism.

Species-specific valence must therefore come before cross-species scaling. Stillness may indicate calm in one context and freezing, illness, exhaustion, or concealment in another. Vocalization may indicate distress, social contact, arousal, or species-typical communication. A human observer's intuitive reading is a hypothesis, not a measurement rule.

The best comparative evidence combines channels that fail in different ways. Preference and avoidance can reveal what an animal works to obtain or escape. Cognitive-bias tasks can test how prior conditions alter ambiguous judgment. Analgesic response, injury-directed behavior, physiological change, flexible learning, social behavior, play, exploration, and opportunities for species-typical action can add context. None is sufficient alone, but together they can support a more disciplined welfare interpretation (Broom, 2014; Fraser, 2008).

Bearer uncertainty and valence uncertainty must also remain distinct. Researchers may have moderate confidence that an animal is conscious while remaining uncertain about the sign or intensity of a particular state. They may understand a welfare effect while remaining unable to compare its scale with human experience. Collapsing those uncertainties into one number would manufacture clarity rather than discover it.

A comparative program should therefore state what is expected to travel. It might test whether relief follows the removal of an aversive condition, whether enrichment changes behavior and physiology in the same direction, or whether long-term trajectories differ under matched care. It should not assume that a human questionnaire, a reward signal, or one neural marker has the same meaning across species.

Welfare first changes the choice of study. The strongest candidate domains are often those in which evidence can be gathered while improving or preserving the animal's life: enrichment, rehabilitation, sanctuary care, analgesia, environmental choice, recovery, or naturally occurring change. A project about felt burden should not create severe burden merely because induction produces cleaner contrasts.

Precaution does not require pretending certainty. It allows uncertainty to increase protection without increasing the scientific claim. Researchers can say that a system may be conscious, that a condition may be aversive, and that the safer action is to reduce possible harm, while still reporting that the bearer and valence bridge remain unsettled.

Public communication must preserve the same structure. Animal behavior is not proof of PTV, and precaution is not a declaration of personhood. Conversely, uncertainty is not permission to ignore welfare. The honest position protects animals from both overclaiming and convenient dismissal.

Only after bearer status, species-specific valence, and welfare limits are declared does transport become a meaningful question.

22.3.2 — Cross-Species Transport Is an Empirical Burden

The central work is to demand explicit bridges across lifespan, behavior, physiology, ecology, and observation. That requirement turns cross-species transport from a slogan into an empirical burden: every claimed common object must be defined, measured, and tested in the species and context where it is used (Mendl et al., 2010; Sneddon et al., 2014; Solvi et al., 2016).

Cross-species transport cannot be adjudicated from general evolutionary literacy alone. It requires population genetics and developmental biology for lineage and constraint, comparative cognition for task interpretation, welfare science for burden, biostatistics for transport and dependence, philosophy of biology for category discipline, and science communication for public accuracy. The empirical bridge is only as strong as its weakest necessary component.

Recent comparative work shows why transport cannot be assumed from a familiar word such as fairness. A 2024 meta-analysis of accept-or-reject paradigms reported no general evidence for inequity aversion across nonhuman animals. A 2025 bonobo study found aversive responses to unequal rewards, while

a 2026 chimpanzee group study found that food quality and social relationships altered the response. Species, task structure, reward quality, partner relationships, and participation conditions can therefore change what the same experimental label appears to measure (Ritov et al., 2024; Radovanović et al., 2025; Martinez et al., 2026).

These studies concern reactions to unequal rewards in bounded tasks. They do not measure a complete phenomenal stream, identify a neutral origin, establish a common valence unit, or adjudicate terminal closure. Their value here is methodological: even a comparatively concrete construct can change meaning when the species, relationship, reward, and experimental setting change.

A defensible bridge would declare bearer criteria, species-specific indicators, welfare context, behavioral alternatives, exposure window, latent construct, observation channel, scale property, and uncertainty set. It would then test whether that same construct survives across sites and tasks. When only an ordering or directional effect is defensible, the result must remain at that level rather than being rewritten as a common cardinal ledger (Mendl et al., 2010; Sneddon et al., 2014; Solvi et al., 2016).

Comparative disagreement is productive when it reveals which assumption moved. A meta-analytic null, a species-specific positive result, and a context-dependent group result can coexist without contradiction if they answer different questions under different designs. The Law earns no support from that plurality, but it gains a clearer map of the evidence burden it would have to meet.

Transport begins with the uncomfortable possibility that the same outward act can carry different experiential meanings in different species. Immobility may indicate freezing, fatigue, thermoregulation, illness, strategic concealment, or low arousal. Vocalization may be a distress signal in one context and a routine coordination signal in another. A human-centered scale can therefore become confidently wrong while remaining numerically tidy. Comparative work must earn every bridge from behavior to felt valence instead of assuming that familiar gestures preserve familiar meaning (Barron & Klein, 2016; Feinberg & Mallatt, 2016).

The neutral origin is especially difficult to transport. A midpoint chosen from human language, facial expression, or preference may not mark experiential neutrality in a bird, cephalopod, fish, insect, or mammal with a different sensory world. Even within one species, domestication, development, social position, injury, and environment can alter what a signal means. A cross-species claim must therefore state whether it transports a sign, an ordering, an interval, or only a broad direction of change, because those are different achievements.

Sensory ecology changes the meaning of evidence before statistics begin. An animal may organize its world through smell, vibration, polarization, electrical fields, echolocation, water pressure, or ultraviolet cues that a human observer barely notices. A task designed around vision and manual choice can therefore understate competence, agency, or affective discrimination. Comparative fairness research must begin with the species' world

rather than with the convenience of the laboratory apparatus, then document which parts of that world the protocol leaves out.

Social structure also changes transport. Solitary animals, pair-bonded species, eusocial colonies, fission-fusion groups, dominance hierarchies, and cooperative breeders face different sources of safety, deprivation, control, and loss. A social-isolation manipulation cannot be interpreted uniformly across those forms. Even a preference test may express affiliation, threat avoidance, territoriality, or learned dependence. The bridge must therefore connect the behavior to a species-appropriate causal account before it connects the behavior to valence.

Missingness also changes across species. The least observable moments may be the most relevant ones: hiding, nocturnal activity, migration, torpor, illness, predation, solitary recovery, and the final stages of life. Captive observation can improve visibility while changing the very environment that shapes stress and agency. Field observation can preserve ecology while losing internal resolution. A serious measurement bridge must record what could not be seen and how the observation setting may have altered the bearer's available paths.

Time scale is another transport problem. A brief assay may capture human frustration while missing an animal's longer recovery cycle, seasonal rhythm, or delayed behavioral adjustment. Conversely, a long observation window can average away short but intense states. Comparative protocols should therefore justify the temporal grain at which experience is sampled and the duration over which change is interpreted. A cross-species bridge that transports only the sign of a response

should not be described as if it transported intensity, duration, or cumulative weight.

Comparative samples also inherit phylogeny. Closely related species are not independent replicates, and similarity may reflect shared ancestry, while distant resemblance may reflect convergence. Homology can support a biological bridge; analogy can motivate a test. Neither identifies phenomenal equivalence, so the analysis must declare its phylogenetic structure and alternative evolutionary histories (Futuyma & Kirkpatrick, 2017; Godfrey-Smith, 2009; Andrews et al., 2024).

Life history creates another scaling problem. A response measured across minutes in an insect, months in a rodent, and decades in a human does not automatically represent the same fraction of a lived trajectory. Developmental stage, reproductive cycle, dormancy, migration, and senescence alter both exposure and opportunity. The bridge must justify its clock before it compares accumulated experience across organisms with radically different temporal architectures (Stearns, 1992; West-Eberhard, 2003).

Negative evidence must be interpreted with the same care. A failed human-designed task may reveal poor motivation, incompatible senses, unfamiliar equipment, stress, or a mistaken theory of the species rather than absence of experience. Yet methodological humility cannot become permanent immunity from failure. Repeated failures across independently designed tasks, channels, laboratories, and ecological contexts should narrow the transported claim, especially when rival models predict the pattern more directly.

Welfare-improving interventions offer one of the safest anchors for interpretation. Analgesia, enrichment, social restoration, environmental choice, and relief from chronic stress can test whether an indicator changes with a condition already relevant to the animal's good. Even then, the result supports a bounded welfare inference, not a terminal ledger or a verdict about the Law (Fraser, 2008; Mellor et al., 2020; Crook, 2021).

22.3.3 — Reference Classes May Narrow

A universal-sounding hypothesis must still be tested inside declared reference classes. Cross-species evidence cannot begin with every possible conscious system and then quietly retreat to the cases that fit. The class of candidate bearers, the representation of valence, the observation rules, and the criteria for inclusion must be stated before the result is known.

Comparative evidence may widen a claim, but it may also narrow it. A pattern found in humans and familiar mammals may fail in birds, cephalopods, fishes, insects, or animals whose welfare signals are organized differently. That would not be an embarrassment to hide. It would show that the original breadth exceeded the evidence and that the reference class must be revised.

Inference limits are severe because behavior is many-to-one. Avoidance can reflect pain, fear, novelty, motor limitation, learned expectation, or ecological strategy. Physiology can reflect arousal without telling us whether the state is pleasant or unpleasant. Neural similarities can support a bridge while remaining insufficient to establish a common phenomenal scale.

Comparative confidence therefore has to be layered rather than announced all at once (Birch, 2017; Ginsburg & Jablonka, 2019; Birch et al., 2020).

The same caution protects against false dismissal. Different anatomy or behavior does not show that experience is absent. Evidence from analgesic response, motivational tradeoffs, cognitive bias, flexible self-protection, and species-typical preferences may support welfare concern even when direct report is impossible (Mellor et al., 2020; Crook, 2021; Sneddon et al., 2014).

A useful comparative result must specify what transported. It may support a shared sign, a common response to relief, or a limited relation between two measurement channels. It does not automatically support equal units, a complete lifetime ledger, or one cross-species neutral origin. Each additional bridge needs its own evidence.

Several failure patterns are especially informative. A species may show no expected trajectory. A measurement bridge may fail invariance tests. Ordinary learning, stress, or welfare mechanisms may explain the data as well as or better than a Law-specific route. The endpoint may be unknown. Ethical limits may prevent adequate observation. Each result changes a different part of the claim.

Some outcomes should therefore be reported as no adequate estimate rather than support or rejection. If bearer status, valence sign, observation coverage, or the measured-to-latent bridge is too

weak, a broad verdict would exceed the evidence. Abstention is a scientific result when the missing prerequisite is named.

Welfare science offers a better starting vocabulary than human likeness. Preference, motivation, affective bias, injury response, analgesic response, social context, and opportunities for species-typical action can identify concrete welfare questions without pretending to solve consciousness or PTV in one step (Fraser, 2008; Mendl et al., 2010).

Reference-class narrowing also protects moral judgment. Fitness, cognitive complexity, communicative ability, and human resemblance cannot rank the worth of lives. A narrower scientific claim may coexist with a wider ethical concern, because precaution and care do not depend on proving a universal terminal law.

The proper ambition is therefore cumulative and revisable. Comparative work should make the Law more answerable, not more insulated. When species variation, failed transport, ordinary-mechanism sufficiency, or ethical limits appear, the public claim must narrow with the science.

22.4 — Fitness, Feeling, and Moral Worth

Fitness, feeling, and moral worth answer different questions. Fitness concerns differential reproductive contribution in an environment. Feeling concerns the valenced quality of experience for a bearer. Moral worth concerns how beings should be regarded and treated, not how successfully their genes propagate.

Confusion among these categories can naturalize cruelty. A painful trait may be adaptive. A cooperative trait may benefit a group while burdening an individual. A short, dangerous life can be reproductively successful. None of those outcomes converts suffering into a good for the one who feels it.

The welfare indicators in Section 22.3.3 inform how conditions affect a bearer without becoming a consciousness meter or terminal-law test (Fraser, 2008; Mendl et al., 2010; Sneddon et al., 2014).

22.4.1 — Fitness Is Not Fairness

Fitness cannot be allowed to become moral explanation. Evolutionary success is not absolution, permission, or a verdict on whether burden is acceptable. A trait may help a lineage persist while imposing severe costs on particular bearers, so survival history cannot convert suffering into fairness (Williams, 1966; Nesse & Williams, 1994; Fraser, 2008).

PTVC asks a different question: whether the completed felt stream satisfies its declared terminal relation.

Natural selection tracks reproductive consequences, not comfort, dignity, justice, or the quality of a complete felt life. A trait may

persist because it improved survival or reproduction in ancestral conditions while remaining painful, costly, or maladaptive in another environment. The distinction between fitness and welfare is therefore not an ethical afterthought; it is built into the explanatory target (Nesse & Williams, 1994; Stearns, 1992).

This distinction also prevents social Darwinist misuse. No institution may claim that dominance, wealth, survival, or reproductive success reveals moral worth. The Law of Fairness rejects all conversions of outcome into desert. Evolutionary success is not a certificate of dignity, and evolutionary failure is not a verdict against a life.

The formal claim remains per-stream precisely to block substitution. A species, tribe, family, nation, market, or institution cannot close one stream by succeeding elsewhere. Fitness may describe patterns across populations. Fairness, in this book, asks what happens to the bearer who feels.

Fitness can select for traits that impose heavy burdens. Fear, jealousy, aggression, social comparison, dominance sensitivity, grief, pain, hunger, and reproductive urgency can all have adaptive explanations. Adaptive explanation does not convert their burden into justice. It only explains why such burdens may appear in evolved life.

The Law of Fairness does not claim that evolution should have selected comfort. It does not claim that pain is a mistake because it hurts. It does not claim that all biologically useful burdens are unfair in the ordinary moral sense. It asks a more formal question:

what happens to the cumulative felt burden in the stream that carries those mechanisms?

Fitness also cannot settle endpoint discipline. Reproduction may occur long before the irreversible end of the qualifying conscious stream. Survival may continue through years of pain. A population may prosper while individuals die in fear. The biological endpoint of reproduction is not the formal endpoint of the conscious stream (Sterling, 2012; Solomon & Corbit, 1974).

Survival is good in many ordinary senses, but survival can increase burden. A life saved into chronic pain, fear, captivity, humiliation, displacement, or isolation is not simply a positive line in an experiential account. It is a longer stream with more days in which burden can accumulate, change, be repaired, or remain unresolved.

This is not an argument against saving life. Care comes first. Protection, medicine, rescue, shelter, and relief are ethical duties. The point is that survival alone does not settle the Law's question. A person or animal may survive and still carry heavy felt cost. The fact of continuation does not prove fairness.

In animal contexts, survival metrics can be especially misleading. A captive animal may survive with poor welfare. A farmed animal may grow efficiently while experiencing stress or pain. A wild animal may live longer under protection while losing species-specific freedom. Survival data need welfare context.

This point matters for medicine, welfare, and public policy. Keeping someone alive without attending to pain, fear, dignity, environment, or consent can become a thin form of care. The Law

does not reject survival; it says survival must be read through the quality and burden of the continuing stream.

Reproductive output is likewise insufficient as a welfare measure; the question remains how the continuing life is experienced.

The point also keeps the Law from becoming a shallow optimism about endurance. Endurance can be heroic, tragic, coerced, supported, numbed, desperate, or peaceful. The word survived does not tell us which.

22.4.2 — Adaptive Traits Can Hurt

An adaptive trait can impose real suffering. Natural selection favors traits through effects on survival and reproduction, not through a direct vote by the bearer about how the trait feels. Functional analysis therefore cannot be treated as a welfare verdict (Tinbergen, 1963; Hamilton, 1964; Williams, 1966).

Pain is the clearest example. It can protect injured tissue, teach avoidance, and alter future behavior while also becoming severe, persistent, or disconnected from current benefit. Fear can improve rapid escape while producing chronic vigilance. Hunger can motivate acquisition while narrowing attention and increasing distress. Attachment can support care while making loss and separation devastating.

Reproductive traits make the distinction especially important. Pregnancy, mating competition, parental investment, and sexual conflict can contribute to lineage success while imposing unequal burdens on individuals. Reproduction does not complete a parent's stream, and one generation cannot balance another. Descendants are not payment for sacrifice.

Social traits also carry costs. Sensitivity to rank, exclusion, reputation, or reciprocity may support cooperation in some environments and intensify shame, envy, loneliness, or conflict in others. An evolutionary explanation can show why the sensitivity exists without showing that its felt consequences are fair or terminally neutral.

The Law of Fairness therefore cannot use adaptive function as a substitute for its target. The proposed equality concerns the complete experience of one separately routed stream. Fitness concerns the propagation of traits. These objects may influence one another, but they are not interchangeable.

This separation has a humane consequence. A burden does not become less deserving of relief because it has a biological function. Treatment of pain, anxiety, addiction, trauma, reproductive complications, or chronic stress does not oppose evolution. It responds to the life that is being lived now.

It also has a scientific consequence. Studies should compare Law-specific predictions against models in which ordinary adaptive systems produce rebound, stabilization, and context-sensitive regulation. If those models explain the data, the result favors an ordinary mechanism or narrows the proposed route. It does not become hidden support for the Law.

Public language should remain equally clear. Evolution does not want fairness, assign lessons, or guarantee repayment. A trait may be adaptive and still hurt. A life may be difficult without being a biological failure. A life may be biologically successful without being experientially protected.

The category boundary is simple but indispensable: function is not feeling, reproduction is not justice, and persistence is not moral worth. Keeping those distinctions intact allows evolutionary science to remain strong while preventing it from becoming consolation or blame.

22.4.3 — Evolutionary Storytelling Risk

Evidence discipline strengthens evolutionary explanation. Evolutionary claims should identify mechanism, comparative pattern, plausible selection pressure, alternative explanations, and limits. Where evidence is absent, the language should say conceptual possibility, not evolutionary fact. A good story is not a study (Trivers, 1971; Gould & Lewontin, 1979; Maynard Smith, 1982).

Evolutionary medicine gives the distinction practical form. Fever, pain, anxiety, nausea, and inflammatory responses can have defensive histories while still becoming excessive, chronic, mistimed, or dangerous. Explaining why a response evolved helps locate intervention; it does not create a duty to endure the response untreated (Nesse & Williams, 1994).

An evolutionary tree is a map of descent, not a ladder of phenomenal or moral worth. Any figure used here should branch without a summit, avoid placing humanity as the culmination, and state that phylogenetic proximity neither proves consciousness nor fixes welfare priority. A cropped ladder can silently convert relationship into rank even when the surrounding prose is careful.

Ask what the account would predict if complete-stream closure failed. If its predictions remain unchanged, they do not

distinguish the target. If its premises rule that failure out, those premises need independent tests. This separates an explanation that permits closure from one that would require it.

Evolutionary storytelling becomes most dangerous when it is used backward from a desired conclusion. If the chapter first wants the Law to be true and then finds a plausible evolutionary story that makes truth feel natural, it has not done science. It has built a myth with biological vocabulary. The safeguard is to require every story to face alternatives that could explain the same pattern more simply.

Evolutionary explanations deserve scrutiny where they make claims beyond their evidence. The Law deserves the same scrutiny. A critic need not supply a complete theory of every affective trajectory to show that a particular Law-specific prediction fails or that its evidence is insufficient. The constructive question is narrower: which account predicts the observations better, and what remains unresolved after that comparison?

Evolutionary storytelling risk appears whenever an elegant after-the-fact account begins to sound like explanation. Humans are excellent at inventing adaptive stories. A behavior exists, so it must have served survival. A burden persists, so it must have had a function. A pattern resembles balance, so evolution must have selected balance. These stories can persuade before they are tested (Stearns, 1992; Nesse & Williams, 1994).

Another useful test is whether the story protects the sufferer. If an evolutionary explanation makes a person or animal's burden seem

deserved, necessary, or less urgent, the sentence has drifted into misuse. A scientifically respectable sentence can still be ethically dangerous if it lets function swallow care.

Plain admissions are better than sweeping explanations. We do not yet know is often more faithful to the evidence than a paragraph that makes the universe sound tidy. Honesty may be less dramatic, but it carries less hidden burden.

22.5 — What Cross-Species Failure Would Mean

A cross-species program is meaningful only if failure changes the claim. Systematic residual patterns, incompatible measurement bridges, taxon-specific mechanisms, or ordinary regulatory explanations may narrow a universal proposal. Ethical limits may also make some branches unadjudicable.

Failure is layered. One result may defeat a proposed mechanism without defeating every target formulation. Another may show that a measure does not transport while leaving the welfare finding intact. A verified counterexample could defeat a strict universal reference class while leaving narrower classes open.

Animal research adds a nonnegotiable constraint: information value must exceed welfare cost under independent review. Naturalistic observation, veterinary records, enrichment changes, rehabilitation, and existing data may often be preferable to invasive designs. No harm is justified by the possibility of a decisive result.

22.5.1 — Systematic Imbalance Cases

A cross-species fail pattern is stronger when it is structured rather than anecdotal. The relevant question is not whether one species, one laboratory line, or one welfare setting appears imbalanced. The stronger challenge would be repeated imbalance across independently studied domains after adjusting for measurement bias, species-specific valence, captivity, deprivation, illness, age, ecology, and observation process (Maynard Smith & Szathmáry, 1995; West-Eberhard, 2003; Godfrey-Smith, 2009).

These cases require unusual care because animal-welfare measures, human endpoint records, and cross-species bridges are all fragile. A serious test would combine prospectively specified reference classes, multiple species-appropriate channels, transparent missingness, independent teams, confound analysis, rival models, and ethical review. Apparent imbalance would count only after surviving that chain.

Such cases would also require an answer to missingness. The streams most burdened may be least measurable. Animals in poor welfare conditions, people in crisis, patients near death, and organisms outside easy observation may disappear from clean datasets.

The possibility of disconfirmation must remain real. If every apparent residual imbalance is dismissed as measurement error, hidden compensation, an unknown channel, or future correction without criteria, the proposal becomes unfalsifiable. Under prospectively specified conditions, systematic imbalance must be allowed to count against a broad claim.

The ethical dimension remains central. Researchers may not create severe imbalance to test the Law. They may study existing welfare records, naturalistic observations, clinical or sanctuary data, and humane interventions. The Law cannot demand harm in order to make itself vulnerable.

A strict universal claim can be defeated in principle by one genuine counterexample. The difficulty is establishing that the case really falls within the claim and that the apparent nonneutrality is not an artifact of measurement, incomplete

observation, or a misidentified endpoint. Replicated patterns across a declared group can make that case more persuasive, but repeated failure is not a logical requirement of universality. Once a qualifying counterexample is established, the original universal claim fails. A narrower claim would be a revised hypothesis requiring fresh scrutiny.

The strongest design would make the reference class visible before outcomes are known. Species, developmental stage, ecological condition, care context, measurement channels, and endpoint rules would be declared in advance. Researchers would then ask whether the apparent imbalance persists when the same protocol is transported, when known biases are modeled, and when rival explanations receive equal flexibility. A claim that survives only after the class is redefined around the data has not survived the test.

Replication matters because comparative evidence is unusually vulnerable to local interpretation. A distinctive behavior may be common in one population and absent in another; a welfare indicator may reverse under different husbandry; an instrument may work only in one sensory ecology. Independent teams should therefore be able to reproduce both the measurement bridge and the imbalance pattern. Failure to reproduce either one narrows what can honestly be said, even when the original case remains ethically important.

A systematic imbalance could also emerge through the tails rather than the average. Mean values might appear near balance while a subgroup, life stage, or ecological condition carries persistent extreme burden. A responsible analysis must therefore inspect

distributions, not only central tendency. It should ask whether rare but severe trajectories are being diluted by many mild observations and whether the chosen summary silently treats intensity, duration, and frequency as interchangeable.

Endpoint uncertainty must remain separate from imbalance. A study may observe a long trajectory without observing the truth-level end of the qualifying stream, or it may observe biological death without resolving whether experience ended earlier, later, or intermittently. Such a study can still reveal welfare-relevant patterns and test local mechanisms. It cannot honestly convert incomplete coverage into a terminal verdict. The correct result may be strong contrary evidence for a route while the complete-stream claim remains unadjudicated.

Finally, systematic imbalance should be assessed against serious rivals that can generate the same pattern. Chronic environmental stress, disease, selection into observation, caregiving differences, captivity, measurement drift, and species-specific adaptation can all produce durable asymmetry. The Law gains nothing when those rivals are mentioned only after a result appears. They should be specified in advance, given comparable flexibility, and allowed to win when they predict the held-out pattern more accurately.

A second failure pattern would occur if no candidate Law-specific pathway constraint appears where the Law's mechanism expects one. Researchers might find ordinary regulation, learning, fatigue, homeostasis, pain response, social behavior, and recovery patterns, but nothing that supports a distinct queue-like structure beyond known control models.

This would not automatically defeat the formal Law, because the candidate pathway model is a candidate mechanism, not the Law itself. But it would narrow mechanism language. The Law would have to stop speaking as if queue structure had biological support. It might retain a terminal claim while losing a favored explanatory path.

The test would require clarity about what counts as queue-like. Menu restriction alone is not enough, because many systems restrict options. Priority signals alone are not enough, because attention and action selection already prioritize. Compensation alone is not enough, because local regulation and opponent processes can compensate. The distinctive structure must be stated before testing.

Failure to find a distinctive queue-like constraint would strengthen its rivals. Homeostasis, reinforcement learning, predictive processing, allostasis, and ecological adaptation may explain the observations more parsimoniously, and those explanations should receive the credit rather than being absorbed after the fact (Pigliucci & Müller, 2010; Futuyma & Kirkpatrick, 2017).

Mechanism failure is honest progress only when the record says what failed: the candidate pathway metaphor, a specific menu model, a predicted cross-species signature, an endpoint relation, a measurement channel, or a public interpretation. Classified failure removes clutter and sharpens the surviving claim; unclassified failure becomes atmosphere that can be reinterpreted later.

Retiring a failed mechanism also protects the later applied books. If the candidate pathway model narrows, ordinary-life guidance should not become self-surveillance, City design should not pretend to measure souls, government should not infer civic rank from dashboards, and no religious interpretation should turn a candidate mechanism into doctrine. Humility at the root prevents downstream harm.

Naming should not decide the science. A critic may dislike a metaphor that remains testable, while a supporter may love a formula that describes nothing real. If the candidate pathway model survives only as illustration, the text should label it as illustration rather than defend it as machinery.

22.5.2 — Species Variation Defeats Broad Claims

Species variation can defeat different claims at different levels. It may defeat a universal mechanism while leaving a narrower human-stream hypothesis intact. It may defeat a particular cross-species measurement channel while leaving comparative welfare research valuable (Nagel, 1974; Broom, 2014; Fraser, 2008).

A broad Law must be willing to become less broad. Stable divergence under adequate measurement would defeat the breadth actually claimed, and any revision to a taxon, architecture, developmental stage, welfare context, or measurement family must be prospective rather than built around the adverse result.

The studies in Section 22.3.2 need not cancel one another: they may identify different constructs under different conditions. Reference class, social structure, reward design, and observation

rule are part of the claim, not details to be averaged away (Ritov et al., 2024; Radovanović et al., 2025; Martinez et al., 2026).

Public language must follow the scientific scope. If the claim becomes human-only, mammal-only, vertebrate-only, context-limited, or method-limited, the sentence that travels publicly must say so. A failed broad claim cannot retain universal wording while every contrary species is moved into an exception.

Narrowing the formal claim does not narrow moral concern. Welfare precaution can remain broad where evidence of possible sentience and serious harm warrants it, even when a shared measurement bridge is unavailable. Formal inclusion, scientific transport, and ethical protection are different decisions, and no species acquires a higher or lower moral rank from its usefulness to the theory.

Plurality may be the mature answer. The world may contain many forms of life, uncertain bearer boundaries, and wide welfare duties without yielding one confident comparative map. The hypothesis has to remain intelligible under that plurality rather than forcing it into one scale.

Variation may enter before mechanism is considered. Some species may plausibly support unified streams, others may remain unknown or differently organized, and a channel useful in humans may fail in mammals, birds, cephalopods, or insects. Bearer status and measurement transport must therefore be reported separately; unknown is a result, not an invitation to theory expansion.

Variation may also enter through regulation. Social buffering, avoidance, physiological shutdown, environmental selection, and

developmental plasticity can organize burden differently across organisms. If no general route survives, the mechanism claim must narrow even when welfare research remains valuable.

Species variation may also require plural ethical thresholds. The evidence needed to justify concern may differ from the evidence needed to make a formal bearer claim. Welfare decisions may rightly use precaution at lower confidence, while formal Law claims require stronger status. Mixing those thresholds would create confusion (Mendl et al., 2010; Sneddon et al., 2014).

Institutional convenience is a powerful welfare confound. Laboratory cages, farms, shelters, sanctuaries, homes, zoos, and field interventions may be easy to manage or observe while imposing distinctive stress, deprivation, or loss of agency. The resulting data can become a record of human systems rather than species nature, and that record should first improve the system rather than elevate the theory.

A defensible dataset therefore records housing, social contact, light, temperature, noise, enrichment, medical status, handling, prior history, observer presence, sampling time, exclusions, and missingness. Context does not make welfare evidence useless; it makes provenance part of the burden map and allows humane interventions to remain valuable even when Law-specific inference is uncertain.

The danger is removing too much context in the name of statistical cleanliness. Researchers may be tempted to statistically remove context in order to isolate a variable, but the context may be the burden. A fair science does not erase the very conditions

that make the animal's experience intelligible. It distinguishes nuisance from lived reality.

If welfare confounds dominate, public claims must pause or narrow. The right conclusion may be that the data teach welfare reform, not terminal neutrality. That still matters. Care does not need to prove the Law in order to be urgent.

The same lesson reaches human research. Hospital systems, poverty, culture, platforms, work, family, medication, surveillance, and access to care shape human data as deeply as habitat and task design shape animal data. Context is not noise around the measurement; it is part of what the measurement means.

22.5.3 — Ethical Cost Too High

Ethical nonviability deserves the same seriousness as statistical failure. Some studies could be informative and still be wrong to perform. A project about conscious burden cannot demand severe pain, fear, deprivation, isolation, terminal manipulation, or irreversible harm from animals, patients, children, disabled people, dying people, or synthetic candidates in order to test whether burden ultimately closes (Solvi et al., 2016; Barron & Klein, 2016; Feinberg & Mallatt, 2016).

The rule is not merely that harm should be minimized after a design is chosen. The research question, comparison, and endpoint must be reconsidered before burden is imposed. If a prediction can be tested only by creating suffering that would not otherwise occur, the route may be ethically unavailable. That

verdict limits knowledge by that method; it does not count as evidence that the hypothesis is true.

Independent review must be able to stop a study for reasons that do not depend on scientific enthusiasm. Burden exceeding the declared ceiling, a safer alternative becoming available, a failed measurement bridge, an uninformative comparison, or an unexpected sign of possible moral status should all trigger pause or termination. Participants and caregivers must retain correction and withdrawal authority.

Lower-burden alternatives often exist. Naturalistic observation, sanctuary and rehabilitation records, welfare-improving interventions, clinical data generated under care, noninvasive monitoring, preference-based enrichment, archival review, and simulation can answer narrower questions. Their confounding and missingness must be modeled honestly, but inconvenience is not a reason to prefer a harsher experiment.

Care remains prior to inference. No one may be denied analgesia, sedation, treatment, rescue, food, safety, companionship, or ordinary support for a cleaner dataset. No person should be recruited through crisis or told that suffering is valuable to the theory. No animal should become a proof object. The proposed Law has no ethical authority to suspend present welfare.

Public demonstration carries its own burden. A vivid video, image, case narrative, or dramatic comparison may attract attention while turning suffering into spectacle. Consent, privacy, dignity, and welfare do not disappear because a result is

educational. The shortest public artifact should be at least as careful as the full argument.

Ethical limits may force a narrower verdict. A claim may remain logically possible while some decisive observations are forbidden or unavailable. The honest report is then untested under acceptable conditions, no adequate estimate, or reference class narrowed. Absence of permitted evidence is not confirmation.

The same narrowing must travel through guides, interviews, classroom materials, social excerpts, donor pages, diagrams, and automated summaries. Compatibility with evolution is not evolutionary support. Precaution for animals is not proof of consciousness or PTV. A speculative mechanism is not a natural purpose. When the evidence or ethical clearance narrows, the sentence must narrow too (Birch, 2017; Ginsburg & Jablonka, 2019).

Evolution can constrain the proposal and may help explain how it could be realized. The biological chapter therefore closes with an open empirical task: explain what ordinary mechanisms do, protect welfare now, and establish whether the evidence supports the full relation rather than a resemblance.

Chapter 23 — Could Such a World Exist?

A simulated world is a disciplined thought experiment with memory. It can hold assumptions still, generate many trajectories, reveal hidden dependencies, and let rival mechanisms face the same observations. Used well, simulation makes a theory easier to break.

Used badly, simulation can make a theory impossible to lose. The designer chooses the state variables, the endpoint, the meaning of zero, the update rules, and the data that will be visible. If those choices already contain the desired conclusion, the output is a restatement of the code.

The first question concerns architecture. A global constraint acts across complete trajectories. A local controller responds to current state. A bridge conditions paths on an endpoint. Selection may remove trajectories that do not fit. Observation loss may hide the failures. Several of these worlds can produce similar terminal patterns.

The second question concerns patches. A model can be rescued after each failure by adding a correction, changing a parameter, moving the endpoint, or excluding a difficult case. Patches may be legitimate when independently motivated, but an accumulating rescue system can turn a vulnerable claim into an unfalsifiable machine.

The third question concerns neutrality by fiat. If the target is defined as zero by construction, if each path is recentered after observation, or if the endpoint is chosen when the residual closes, then the simulation has not tested neutrality. It has installed it.

Model recovery supplies a more demanding task. Plant several known generating worlds, hide their identities, and ask whether the analysis selects the correct one under held-out data. A method that cannot recover known alternatives should not be trusted to identify an unknown mechanism in real life (Guest & Martin, 2021).

Recovery must include hostile worlds. Ordinary mean reversion, endpoint-conditioned bridges, local feedback, heavy-tailed noise, selective missingness, measurement floors, and flexible hybrid controllers should receive equal computational budgets. The Law should not win because its rivals were made crude.

Before a simulated result can teach anything, three questions must be separated: was the model implemented as specified, is it adequate for the purpose assigned to it, and how sensitive is the conclusion to uncertain inputs, structures, and data? Those are the distinct tasks of verification, validation, and uncertainty quantification (Oberkampf & Roy, 2010; National Research Council, 2012).

Transparent description matters because simulations can be irreproducible even when the code runs. State variables, schedules, interaction rules, stopping conditions, stochastic seeds, parameter ranges, outputs, and analysis decisions must be recoverable. The updated Overview, Design concepts, and Details (ODD) protocol provides one established model for making agent-based simulations more explicit and comparable (Grimm et al., 2020).

Synthetic agents create a separate problem. A reward signal is a mathematical quantity. A policy can maximize it without feeling it. Language-like output can invite us to project a subject where none has been established. Simulation of behavior and evidence of consciousness are different questions (Turing, 1950; Searle, 1980; Butlin et al., 2023).

That distinction does not license indifference. Where future systems create a nontrivial possibility of conscious experience, precaution may be warranted. The safer research strategy is to use non-sentient formal agents wherever possible and to reject synthetic suffering as a shortcut to mechanistic clarity (Metzinger, 2021; Chalmers, 2023).

Designer language requires similar discipline. A hypothetical designer can clarify a world-building tradeoff, but the word does not identify God, a simulator, or a metaphysical author. The Chapter neither proves nor disproves theology. It only governs what a scientific model can license.

Religious doctrine remains outside the Chapter's authority. *The Religion of Fairness* may later explore simulation metaphors, but simulation cannot validate doctrine or rescue a failed model. The formal root must remain complete without devotional language. A reader interested in spiritual interpretation can pursue that separate question while following the same explicit description of how the simulation works.

The empirical burden must become heavier, not lighter. A coherent world is not enough. A recoverable mechanism is not enough. A beautiful plot is not enough. A model must identify

what would distinguish the proposed structure from alternatives in independently observed conscious systems. For example, two simulations may both close at zero yet respond differently when available actions change; their intervening paths then supply the useful comparison.

A failed simulation can therefore be a success of inquiry. It may reveal circularity, nonidentifiability, fragility, or an unnecessary mechanism. By preserving those failures rather than hiding them, the project becomes less theatrical and more scientific. The failed run should remain reproducible, with the parameter settings and diagnostic trace that show where the proposed construction ceased to work.

23.1 — Why Build Simulated Worlds?

Simulated worlds are useful because they can hold a theory still. A researcher can declare the state variables, mechanisms, endpoints, observation rules, and competing models, then generate many trajectories under controlled changes. The result is a laboratory for assumptions rather than a miniature proof of reality.

Different modeling traditions serve different purposes. Agent-based models show interactions and emergence. Dynamical systems expose stability and feedback. Stochastic simulations test variability and rare events. Causal models clarify intervention questions. No one form deserves authority beyond its purpose (Epstein, 2008; Pearl, 2009).

The reason to model is to make the hypothesis clearer, more comparable, and more decisively testable.

23.1.1 — Conditional Design Reasoning

A conditional model can establish that a set of assumptions is jointly coherent, reveal consequences, and expose tradeoffs. It cannot determine that those assumptions describe the actual world. That separation lets the chapter use possible worlds as test benches rather than as concealed metaphysics (Busemeyer & Diederich, 2010; Woodward, 2003; Weisberg, 2013).

A useful design question can be stated without metaphysical commitment: what structural constraints would reduce hidden burden, prevent permanent deficit traps, protect trajectory-level

coherence, and resist exploitation by local optimization? Such a question belongs to model architecture.

The hypothetical builder is best treated as a vantage point rather than a being. From that vantage point, the reader can ask what rules a candidate world contains, what burdens those rules create, and what constraints would make complete trajectories possible. Nothing in the exercise identifies an actual designer or grants the model metaphysical authority.

Conditional design reasoning can produce an existence proof: under stated assumptions, a world with a specified rule can be constructed. Explanatory adequacy asks a different question: whether the rule is needed, whether rivals can do the same work, and whether the model resembles the domain of interest. Simulations become informative when they keep those achievements separate (Winsberg, 2010; Weisberg, 2013).

Suppose one constructs a hypothetical world containing agents capable of burden and relief. The model must declare which trajectories are admissible and whether a terminal boundary is one candidate rule among others. This does not say that our world was constructed or that any designer exists; it only makes the proposed structure explicit enough to compare.

That freedom lets imagination serve rigor. A model can ask what a fairness constraint would accomplish, which failure cases remain, and whether a simpler rule solves the same problem. A favored hypothesis must be allowed to lose even inside the thought experiments designed to clarify it.

Every sentence in this exercise must know whether it is asking, modeling, imagining, or asserting. A coherent architecture is still only possible, not actual, and detached passages should carry that status on their surface so conceptual design cannot drift into metaphysics or evidence.

This distance is especially important because the Law deals with suffering. When a model makes a burden look structurally necessary, language can begin to sound cold. A conditional model should never be used to tell a person that a particular harm was required for closure, assigned, or justified. It can ask what rules would prevent terminal residue in a possible world, but it cannot interpret the wounds of actual people.

Stability is morally empty until the model shows whose burden it protects. A world can be orderly for the powerful while unbearable for others, or remain stable by suppressing memory, agency, consciousness, or inconvenient streams. Such stability hides cost rather than answering it.

A world with conscious agents can become unstable in more than physical ways. It can become unstable in incentive structure, memory structure, trust structure, and experiential distribution. Agents may learn exploit paths. Systems may reward burden shifting. Local successes may create hidden deficits elsewhere. A world can look orderly in external motion while becoming unstable in the felt trajectories it permits.

Here world stability has a narrow meaning. It is not political order, social calm, equal goods, pleasantness, or visible compensation. It asks whether the modeled space of unified

conscious trajectories is bounded against terminal felt residue, while preserving unequal histories and ordinary conflict.

The reason this is conceptually attractive is that local patching has a coordination problem. If every wound requires a later visible repair, the world must track countless local obligations in a way that is both public and manipulable. If every villain must be visibly punished, the world becomes theatrical. If every sufferer must be visibly compensated, hidden suffering disappears from the account. A terminal constraint, if real, would operate at the stream level instead of the spectacle level.

This distinction matters because stability language can become too attractive. People like the idea that the world is secretly well-built. They like the idea that unfairness is structurally temporary. But psychological appeal does not improve claim status. A stability constraint may be beautiful and still false. A world may be ugly and still lawful under different principles. The Law cannot be chosen because it is consoling.

A simulation can help by showing how different stability concepts diverge. A world can be economically productive, politically durable, and behaviorally predictable while still generating unbearable private paths. Conversely, a world can appear locally turbulent while satisfying a deeper terminal constraint. The Law of Fairness, if true, would operate at the stream level, not at the level of visible order alone.

23.1.2 — Designer Language as Public Risk

Simulation requires designers, but designer language easily outruns its technical meaning. In a model, a designer is the person

or team that selects variables, rules, objectives, data structures, and stopping conditions. That ordinary fact does not imply that the universe was designed, that conscious life inhabits a simulation, or that a moral author arranged experience.

The risk begins with metaphor. Researchers speak of agents, rewards, goals, environments, learning, and choice because those words organize computational systems. In reinforcement learning, for example, a reward is a numerical signal used to update behavior. It is not automatically pleasure, welfare, desire, or conscious approval (Sutton & Barto, 2018).

Agent-based and computational models use similar shorthand. An agent may be a few rules attached to a state vector. A population may reproduce, compete, cooperate, or collapse without any modeled experience at all. The language is useful as long as the ontology remains explicit (Railsback & Grimm, 2019; Wilson & Collins, 2019).

Public interpretation often removes that context. A graph showing agents returning to zero can be retold as a world seeking balance. A programmed controller can be described as nature correcting unfairness. A terminal condition can become destiny. These translations add intention that the model did not establish.

The remedy is not to ban evocative language. It is to bind each term to its operational role. A report should say who chose the objective, whether the neutral point was fixed in advance, how endpoints were generated, which variables were observed, and which parts of the result were imposed rather than recovered.

Sensitivity analysis should show how conclusions change when those choices change (Saltelli et al., 2020).

Designer language also needs a theology boundary. A conditional sentence such as a world could be constructed with this property is a statement about logical or computational possibility. It is not evidence for a creator, simulator, providence, or cosmic intention. Religious interpretation belongs to voluntary meaning, not to the evidentiary output of code.

Documentation must therefore include the model's purpose, assumptions, calibration, validation limits, failure conditions, and intended use. A clean animation or persuasive narrative cannot substitute for that record. Model checking and workflow transparency matter precisely because a technically impressive result may still be fragile or misimplemented (Gelman et al., 2020).

The same rule applies to artificial intelligence summaries and public artifacts. A short caption should not say the simulation proves a fair universe, shows that evolution aims at balance, or reveals a hidden designer. It may say that a constructed model demonstrates one coherent route under declared assumptions.

Keeping designer language disciplined protects both science and spiritual freedom. Scientists retain a precise tool for building counterfactual worlds. Believers and nonbelievers are not told that code has settled a metaphysical question.

23.1.3 — Spiritual Meaning Is Not Scientific Evidence

Simulation can invite spiritual reflection without becoming evidence about the actual world. A modeler knows a planted

generator because they wrote it; that epistemic advantage does not transfer to nature. The rule written into a model, the behavior produced inside it, and the claim about reality must remain separate.

Religious or spiritual interpretation may find meaning in constraint, repair, or terminal dignity, but it cannot serve as a model parameter, establish improved fit, identify the truth-level endpoint, or count as model-recovery evidence. Public diagrams and lessons should preserve that distinction rather than letting beauty perform the inference.

The boundary protects spiritual, nonreligious, and skeptical readers alike. No simulated rule may be used to describe suffering as deliberately placed for balance, and neither belief nor enthusiasm for abstract systems is a measure of spiritual maturity. Readers may accept, reject, or remain undecided about every spiritual interpretation while retaining the full conceptual value of constraint modeling.

Simulation can clarify formal coherence without settling spiritual interpretation; Chapter 24 handles spiritual resonance directly.

23.2 — Constraints, Patches, and Forced Outcomes

A constraint acts as part of the architecture. A patch is added to repair a particular failure. Both can be legitimate, but they carry different explanatory costs. A theory that requires a new rescue after every difficult case may be fitting the history rather than explaining it.

Simulation makes that difference visible. Researchers can compare one general rule with a sequence of local corrections, count added parameters, examine sensitivity, and test held-out conditions. Complexity is not automatically bad, but unexplained flexibility weakens discrimination.

Forced outcomes are more serious. If paths are selected because they end correctly, if zero is imposed after the run, or if failed agents are removed from the observed sample, success has been manufactured. The model must declare such conditioning as part of the generating world rather than present it as discovery.

23.2.1 — Patches as Rescue

A patch rescues a local failure. It notices a broken rule, an edge case, an exploit, a deficit, or an undesirable outcome, and it adds a correction. Patches are familiar because human institutions rely on them constantly: appeals, exceptions, subsidies, bug fixes, safety overrides, emergency interventions, and revised rules after harm becomes visible.

Patches are not bad by nature. In ethical life, present care often requires patching. A hungry person needs food, not a theory of

food systems first. A harmed person needs protection, not only a constitutional redesign.

The problem appears when patches are mistaken for the deep structure of fairness. If the formal Law were imagined as a patching system, then every suffering event would appear to demand a visible later adjustment. That leads quickly to unsafe moral-accounting language: deserved balance, punitive repair, hidden settlement, or metaphysical bookkeeping.

In simulations, patches can accumulate until the model becomes arbitrary. Each new exploit receives a new rule. Each new deficit receives a special exception. Each exception creates new edge cases. Eventually the system's behavior depends more on its rescue history than on its original architecture.

A patch should be described as an intervention with an observable trigger, response delay, authority, maintenance cost, and side effects. If the trigger cannot see hidden burden, the patch protects only what the model records. If its response arrives too late, the system may look repaired after the experience it was meant to protect has already occurred.

Patch logic is tempting because it looks compassionate. A visible wrong appears, and a visible repair arrives. But a system built from patches may hide who waited, who was overlooked, who received repair only because they were legible, and who was never counted. The public framework asks whether the repair route reduces burden or merely relocates it.

Patch systems also create visibility bias. A local patch can only correct what the system can detect, classify, and act upon. Hidden

trauma, private despair, silent fatigue, low-amplitude background dread, shame, loneliness, and chronic bodily pain may never enter the patch queue. A fairness theory based on visible repair would fail the very cases that demanded the theory in the first place.

A patch can also shift burden. It may repair the visible outcome while moving cost to unseen agents, future states, memory systems, or peripheral populations. In social life, this is familiar. A policy fixes one statistic while increasing paperwork. A family calms one conflict while silencing one child. A platform reduces one abuse vector while increasing surveillance. Patching must therefore be judged by hidden burden, not only visible repair.

The Law's terminal claim, if true, would not be an endless series of local rescues. It would be a structural boundary on the stream's total trajectory. That distinction is essential because it keeps the Law from becoming a story in which every hurt must be visibly answered by an event that outsiders can admire.

Patch thinking becomes especially unsafe when attached to biography. Observers may search a life for the later good that "made up for" earlier pain. Selecting those events after the fact does not test the Law, and the interpretation can become cruel when it asks the sufferer to accept an observer's story of balance before receiving care.

The patch model also reveals why invisible compensation language is unsafe. If every harm is imagined to be repaired by a hidden counter-event, readers may begin to think suffering is already handled. That would weaken urgency. The formal Law is not permission to assume that a present wound has an unseen

offset. Present care is required even in a world that may contain terminal constraint.

Architecture is not automatically kind. A constraint may protect, but it may also suppress, hide, or normalize burden if it is badly specified. The chapter's task is therefore not to praise constraints in general. It is to show why a terminal boundary condition would be a different kind of claim than ad hoc rescue, while keeping that claim answerable.

A constraint is not a later rescue. It is a condition on what can occur, persist, count, or close. In a physical system, a constraint may limit possible states. In a game, it may define legal moves. In an institution, it may define authority, due process, privacy, or role burden. In the formal Law, the candidate constraint concerns the terminal condition of a unified conscious stream.

Constraint architecture matters because it changes the question from "who gets answered?" to "what paths are admissible?" That shift is safer and more formal. It avoids treating pain as currency. It avoids turning suffering into moral accounting. It avoids requiring visible compensation. It asks whether a stream's cumulative felt trajectory can close in terminal imbalance.

Inside a simulation, a constraint can operate before failure becomes visible. It can forbid impossible states, reduce runaway accumulation, prevent exploit loops, or require certain closure conditions. This is why simulation reasoning can help the reader imagine the Law as a structural boundary rather than a cosmic repair shop. The Law is not a sequence of interventions. It is a proposed condition on the path.

Constraint architecture also helps explain why ordinary visible unfairness can remain visible without settling the formal question. A path may contain severe imbalance at many intermediate points. The Law's claim, in its strongest form, concerns terminal closure, not visible smoothness. A simulation can show how systems may be highly uneven before satisfying a boundary condition at the end. But again, showing possibility is not showing truth.

The ethical danger is to let constraint architecture become cold. A formal constraint does not excuse present cruelty. It does not say that intermediate burden is acceptable because terminal conditions may close. In human life, care remains immediate. A person in pain does not need to wait for a terminal theorem. The ethical firewall remains prior to conceptual architecture.

Architecture language also helps explain why the Law is per-stream. A building code does not average one room's collapse against another room's stability. It defines what must be true of each relevant structure. In the same way, if the Law is a stream-level constraint, no group average can substitute for the bearer's path. The analogy is imperfect, but the direction is useful.

The architectural frame should not be romanticized. A constraint may be austere, hidden, and difficult to observe. It may not feel like justice at any visible moment. This is why the formal Law must never be offered as emotional closure before evidence. Architecture can define the problem; it cannot comfort a person by being named.

23.2.2 — Cost and Complexity

A model can fit almost anything if enough local repairs are allowed. Each patch may look reasonable in isolation: a special endpoint rule, a species-specific exception, a hidden variable, a recentered baseline, or a controller activated only in troublesome cases. Together, however, they can turn a test into a protection system.

Complexity is not automatically a defect. Real organisms, societies, and measurement systems are complex, and a simple model may omit the mechanism that matters. The relevant question is whether each added component earns its place through independent evidence, new prediction, improved recovery, or better comparison with rivals.

A patch ledger makes that question visible. Every change should record when it was introduced, what failure motivated it, whether the change was specified before seeing the relevant outcome, which new observations it predicts, and what result would retire it. Without such a record, adaptation of the model can be mistaken for adaptation of the world.

Apply the equal-budget rule already established for rival comparison: competing models receive comparable data, preprocessing, tuning freedom, priors, and validation opportunities. Complexity is a cost, not a privilege.

Verification and validation address different risks. Verification asks whether the code correctly implements the stated model. Validation asks whether the model is adequate for its intended use. A program can be perfectly coded and still answer the wrong

question, omit the decisive process, or succeed only inside an artificial world (Oreskes et al., 1994; Grimm et al., 2020).

Sensitivity analysis should identify which assumptions control the result. If neutrality disappears when one plausible parameter range, endpoint distribution, observation floor, or missingness rule changes, the result is conditional on that choice. Reporting only the successful setting hides the model's real dependence (Box, 1976; Railsback & Grimm, 2019).

Simulation arguments also carry a public-risk burden. The philosophical possibility that observers could inhabit a simulation is not evidence that PTVC is true, and it does not convert a constructed balance rule into a discovery about our world (Bostrom, 2003). Possible-world reasoning can sharpen assumptions without supplying empirical warrant.

A complex model earns confidence by surviving hostile alternatives, recovering planted generators, predicting held-out features, and failing where it should. If complexity rises while discriminating power does not, the scientific response is not awe. It is a narrower claim, a demand for new data, or retirement of the unnecessary machinery.

Complexity may explain, but it may not conceal. Every added rule should remain inspectable, testable, and removable. With that standard in place, the inquiry reaches the most consequential hidden choice in the exercise: how neutrality itself is defined.

23.2.3 — Simulation Cannot Save Failure

Simulation cannot save a failed formal claim. If empirical evidence later shows terminal nonneutrality outside defensible

equivalence bands, an elegant simulated fairness constraint cannot repair the result. If equal-budget rivals explain the observed signatures better, a world built for the Law cannot overturn that rival victory (Epstein, 2008; Pearl, 2009).

The first discipline is category stability. A model output should be labeled before interpretation as a coherence demonstration, recovered mechanism, failed recovery, sensitivity finding, transport conjecture, or other declared result. A compelling image or sympathetic audience may not upgrade that label after the fact.

Simulation can still diagnose why a candidate signature disappeared. It may reveal observation loss, estimator failure, terminal selection, or the absence of the proposed mechanism in the generator. That diagnosis can improve a later design, but it cannot reverse the adverse result in the original analysis.

This rule must govern public artifacts as strictly as chapter prose. Ambitious theories often begin as laws, become metaphors when evidence turns difficult, and return as laws when the audience is receptive. Simulation must not become that escape route (Winsberg, 2010; Oberkampff & Roy, 2010).

Its legitimate rescue is narrower. A simulation may show that a study tested the wrong constraint, reveal a rival's hidden assumption, propose an observation model, or define a narrower claim. It may not erase negative evidence, convert possibility into validation, or declare the Law true because rivals seem inelegant.

A simulation should be built to expose failure. Null cases, unconstrained baselines, alternative constraints, hostile parameter settings, observation loss, missingness, and sensitivity checks

should receive the same care as the favored route. The design should ask what would make the Law unnecessary, not merely how to display closure.

Public reporting follows the hierarchy. Formal coherence may propose, simulation may explore, empirical evidence must test, rival models must compete, and ethical review must constrain use. A simulation counts as evidence for the Law only through a declared empirical bridge and the required rival tests, not through conceptual appeal alone.

Restraint keeps the theory answerable. A failed claim must be narrowed, corrected, or retired; imagination may redesign the inquiry, but not hide failure with a brighter image. The Law cannot be strongest in imagination and weakest under evidence.

23.3 — No Neutrality by Fiat

A simulation can begin by assuming terminal neutrality to explore its consequences, or derive neutrality from other declared rules. The scientific question is what those rules explain and predict beyond the feature they were chosen to enforce. Origin, endpoint, sample and observation choices must remain visible so an imposed condition is not presented as independent evidence.

Independent outputs also require visible missingness and stopping. If agents disappear, terminate, merge, or become unobserved, the model must distinguish a truth-level process from the data available to the analyst. Selection effects can produce apparent concentration without any underlying boundary.

A failed simulation matters because it identifies a specific inability: the route may be incoherent, unreachable, nonidentifiable, or too fragile. Preserving that failure prevents the model from becoming a story generator that only remembers its successes.

23.3.1 — Assumptions and Independent Evidence

Circularity often hides under helpful language. A modeler may say the simulation encodes the Law in order to explore its implications, which is legitimate, and then later speak as if the simulation discovered the Law, which is not (Busemeyer & Diederich, 2010; Woodward, 2003; Weisberg, 2013).

A model may guarantee closure throughout its declared class. That can be a useful theorem about the model, whether closure was imposed directly or follows from its component rules.

Circularity enters when the guaranteed result is presented as independent evidence that the same rule governs conscious experience.

Such a model can reveal admissible paths, dependencies and additional predictions under its assumptions. These consequences matter most when rival models disagree about them and the relevant observations were not used to choose the rules.

Definitional closure is a different case. If a program chooses the endpoint, neutral origin, bearer rule or observation filter solely to make the measured residual vanish, it has built the answer into the measurement. That construction supplies no independent test of closure. A mechanism with an independently established physical endpoint must instead be judged by the evidence for its rules and their further consequences.

A circularity audit should trace closure through preprocessing, initialization, endpoint definition, normalization, missing-data handling, objective function and stopping rule. It should identify what was assumed, what was derived, and what was tested against independent observations. The critical question is where the argument acquires evidence about reality, rather than another restatement of the planted condition (Palminteri et al., 2017).

Another form of circularity appears after the run rather than in the code. A model may begin as an explicitly conditional illustration, yet captions, summaries, or later citations may drop the condition and describe the output as a discovery. The audit trail should therefore preserve the planted rule, the question asked, the

permissible claim, and every public restatement. Status can be lost in transmission even when the original analysis was honest.

Ask two questions separately: does closure follow from the stated rules, and what evidence supports those rules in the intended biological setting? A proof can answer the first throughout the declared model class; simulation checks selected implementations and cases. The second requires independent measurements and predictions that could fail even when closure cannot fail inside the model.

Inspect the whole architecture, not only the final line of code. Reward rules, repair mechanisms, memory dynamics, endpoints and valence transformations may jointly entail closure. That dependence is something to explain. It becomes an evidentiary problem when those choices were justified only by producing the desired ending, yet their success is presented as independent support.

Explain each circularity finding in plain language so a public reader can identify the step that contained the desired result (Palminteri et al., 2017; Smaldino, 2017).

The strongest defense against circularity is precommitment. The model should state, before results, how streams are defined, how endpoints are determined, how valence contributions are signed, how missingness is handled, what counts as failure, and what rival models will be compared. Without that discipline, the model can be reshaped around the conclusion after the fact.

Rules can be tested through consequences beyond the condition that motivated them. A model built to close may still predict a

particular response to changed inputs, a required memory capacity, or a relation between independently measured variables. If another model closes through different machinery, those further predictions may distinguish them. If no available observation separates them, the result remains unresolved.

Independence does not mean value-neutrality. A model can test a fairness constraint by including candidate mechanisms related to burden, reserve, horizon, menu limits, or compensability. But it must mark which rules are assumptions and which outputs are being observed. It must also include alternatives where the candidate constraint is absent or replaced by a rival.

A model that defines its endpoint solely as the first zero of the same ledger cannot cite closure there as independent support. A physical endpoint may nevertheless be coupled to the state of the system. Its identification must come from biology or other evidence independent of fitting the residual; the claimed coupling must then predict observations not used to construct it.

The same separation applies to consciousness. If the model calls an agent conscious only when it satisfies fairness closure, it has collapsed bearer status into the outcome. Bearer criteria must be stated independently, or the simulation can never show a conscious stream failing the Law. A claim that cannot represent its own possible failure is not serious.

Independent rules are also an anti-founder safeguard. The person who builds the model should not be the only person who understands why it closes. If the model depends on private intuitions, hidden parameter choices, or authorial interpretation,

it has not become public science. It has become private persuasion.

Independence is also an ethical concept. If the rules are built only to protect the theory, then the model has transferred burden onto the reader, who is asked to trust a result that could not have gone otherwise. A fair model bears its own risk. It lets its assumptions be named and its outputs be capable of refusal.

23.3.2 — Consciousness Not Assumed

A simulated agent is not automatically a conscious bearer. It may optimize a score, avoid a state, issue a report, remember earlier inputs, or describe distress without there being anything it is like to be that system. Behavioral sophistication and fluent language are evidence about function, not a direct meter of experience.

The reverse error is also possible. A system should not be declared nonconscious merely because its architecture is unfamiliar or because recognizing uncertainty would complicate research. Consciousness assessment requires theory pluralism, evidence from multiple functional and structural indicators, and a willingness to abstain when the evidence does not settle the question.

Reward signals require special care. In machine learning, reward is a formal quantity that guides updating. Negative reward is not necessarily suffering, and positive reward is not necessarily pleasure. A model that balances rewards has balanced its programmed numbers, not demonstrated terminal valence closure in a conscious stream.

Language-like testimony is similarly ambiguous. A system can generate first-person sentences because those sequences are useful outputs. The content may still be relevant evidence in a broader assessment, especially as architectures change, but testimony-like output cannot carry the entire burden of bearer identification.

This uncertainty creates a welfare obligation before it creates a scientific conclusion. Researchers should avoid designs that deliberately generate persistent distress-like dynamics, trapped self-protection, adversarial punishment, or irreversible states merely to see whether a candidate system behaves as if it suffers. No synthetic suffering is justified by curiosity alone.

Governance should therefore scale with credible moral-status risk. Independent review, staged testing, monitoring for unexpected self-protective or distress-like patterns, access controls, and stop rules belong inside the research plan. These safeguards do not assert consciousness. They prevent uncertainty from becoming a license for avoidable harm.

Simulations used for PTV research should also be explicit about the modeled level. A world may contain abstract agents with no consciousness claim, proxy valence variables, or a candidate architecture under separate assessment. Those categories should not be mixed in one result.

Public reporting must resist both hype and ridicule. Researchers should not announce that a model created a soul, nor reassure the public that no digital system could ever matter. The honest

statement may be that the simulation tests a formal mechanism while the consciousness status of its agents is outside the result.

Human judgment remains responsible for the boundary. An optimizer, benchmark, or language model cannot appoint itself as the authority on its own moral status or on the hidden experience of another system. Once neutrality, consciousness, and the test are kept separate, a failed simulation can be interpreted without either anthropomorphism or dismissal.

23.3.3 — Failed Simulation Matters

A failed simulation should not be treated as a discarded sketch if a successful one would have been celebrated. Symmetry is a fairness rule inside method. If simulated support would have been public, simulated failure deserves public memory, because otherwise the project hides the burden of unfavorable imagination (Saltelli et al., 2020; Guest & Martin, 2021; Palminteri et al., 2017).

Failure can also localize the defect. The implementation may be wrong, the estimator may be unable to recover the generator, the proposed mechanism may be unnecessary, or the target may remain unidentified even when the mechanism is present. Distinguishing those failures prevents one broken component from being used to rescue or condemn the entire program.

Failed simulations may also produce stronger rivals. A world may stabilize through adaptation, prediction, homeostasis, control, population turnover, memory decay, ecological balance, institutional repair, or stochastic averaging without a terminal

fairness constraint. When those mechanisms work better, their success belongs to them.

The project must resist the instinct to relabel every failed simulation as merely preliminary. Some failures will be preliminary. Others will reveal that the simulation question was badly framed. Others may show that the Law is unnecessary in a class of models. Still others may expose a deeper conceptual problem. A mature theory distinguishes these outcomes instead of blending them into optimism (Butlin et al., 2023; Chalmers, 2023).

Apply that visibility rule to guides, lectures, public summaries, and appendices as well as technical reports.

The ethical meaning of failed simulation is not despair. A failed model does not make care less necessary. It does not make suffering acceptable. It does not make public fairness unimportant. It simply removes one possible source of conceptual support. Ethical duties remain even where formal ambition narrows.

A good failure archive would preserve unsuccessful models, rival wins, circular versions, parameter sensitivities, and public corrections. Such an archive would make the project less glamorous and more trustworthy. The Law should not be protected by forgetting how it struggled.

23.4 — What Simulations Can Recover

Good recovery work separates implementation verification from inferential discrimination. Code can implement a model correctly while the data remain unable to distinguish it from a rival. Both facts matter, and neither should be hidden behind visual fit (National Research Council, 2012; Oberkampf & Roy, 2010).

Transparent description is a condition of criticism. A model should disclose its purpose, entities, state variables, scales, scheduling, initialization, inputs, and submodels before its results are interpreted. The ODD framework makes those commitments inspectable rather than leaving them buried in code or narrative (Grimm et al., 2020).

23.4.1 — Recovery Tests the Method Against Known Truth

Picture three sealed envelopes, each containing a set of invented numerical histories. The first came from a program with a declared rule governing the whole path. The second came from ordinary feedback that retains both recent and older effects. The third contains only histories selected because they finished near zero. All three may produce similar final plots.

Give another team the same limited view of each history, with some observations missing, and ask which rule produced it. The labels stay hidden until the answers are recorded. If the team confidently calls all three a fairness constraint, the exercise has exposed a weakness in the test. If it can distinguish the sources where their predictions differ, we learn which parts of the histories carry useful information. Where the available records

cannot distinguish them, recognizing that uncertainty is also a correct result.

Building terminal neutrality into the first world is legitimate when the purpose is to explore its consequences. It becomes circular only if the imposed feature is later presented as independent evidence for the Law. Recovery adds a further achievement: showing what a proposed analysis can identify despite realistic gaps. That achievement remains conditional on these invented worlds. The detailed tournament and scoring rules belong to *The Science of Fairness*.

Uncertainty quantification complements the verification/validation distinction in Section 23.2.2 by asking how imperfect knowledge of inputs, structure, observations, and computation propagates into the answer. A verified model may still have inadequately characterized uncertainty; recovery separately asks whether the analysis can identify known generating parameters or mechanisms (National Research Council, 2012).

Recovery also requires a stable account of what the simulation is for. The ODD protocol helps another team inspect and reproduce the entities, processes, schedules, design concepts, initialization, inputs, and submodels (Grimm et al., 2020). For a Law-facing simulator, the stated purpose must distinguish illustration of coherence, mechanism comparison, estimator testing, and inference. One model should not quietly change roles after results appear.

Parameter recovery asks whether known generating values can be estimated under declared noise and missingness. Model recovery asks whether the analysis can distinguish worlds generated by different candidate mechanisms. The second is harder and more important here. A method that estimates parameters accurately inside every model but repeatedly selects the wrong model cannot support a Law-specific interpretation.

The recovery tournament should include ordinary regulation, selection, memory, endpoint conditioning, observation floors, and hybrid worlds, not only a clean Law world and an obviously poor null. Each world should receive comparable information, tuning freedom, and computational budget. The confusion matrix is part of the result: systematic confusion is not an inconvenience to hide, but evidence that the planned human-data verdict would be unsafe.

Simulation-based calibration can test whether a Bayesian inference procedure is self-consistent when data are generated from the declared model. That is a powerful check on implementation and approximation, but it remains conditional on the generative model used for the test. Calibration inside a planted world does not show that the planted world resembles nature (Talts et al., 2018; Cranmer et al., 2020).

The diagnostic also depends on what is checked. Recent work shows that the choice of test quantities can determine which failures simulation-based calibration detects, so a convenient rank plot cannot stand in for a broad recovery claim (Modrák et al., 2025). Law-facing recovery should therefore register quantities tied to the target, endpoint, observation process, missingness, and

rival discrimination rather than checking only easily recovered parameters.

Posterior simulation-based calibration adds a further refinement by asking whether inference remains calibrated in the region made relevant by observed data, rather than only across broad prior draws (Säilynoja et al., 2026). That can expose failures hidden by an average over irrelevant simulated datasets. It still validates an inference workflow under a model; it does not turn the model into evidence about conscious life.

Held-out transfer is the next gate. A recovery method tuned on one set of generators, endpoints, tasks, or observation patterns should face new modules whose truth remains concealed from the analyst until scoring is complete. Performance should be reported by model family, stream type, missingness pattern, endpoint process, and severity of distribution shift. Aggregate accuracy can conceal exactly the cases that matter most.

Open-set adequacy matters because every prospectively specified model may be wrong. The simulator bank should include worlds generated outside the fitted family and require abstention when none of the candidates is adequate. A method forced to choose between PTVC and a few rivals will manufacture confidence when the true process is absent from the comparison.

Recovery therefore ends with a permission rule. If the pipeline cannot recover generators, distinguish serious rivals, detect implementation failures, and abstain on unfamiliar worlds, it is not ready to interpret human or animal data. The proper outcome

is method revision or no adequate estimate, not a softer public claim.

23.4.2 — Alternative Worlds and Nonunique Explanations

A visible pattern rarely identifies its generator by itself. Several different worlds can produce the same curve, endpoint concentration, rebound, or apparent return to zero. Simulation is valuable because it can make that nonuniqueness concrete rather than leaving it as an abstract warning (Levins, 1966; Oreskes et al., 1994).

One world may contain a genuine terminal boundary. Another may contain ordinary mean reversion. A third may condition paths on a selected endpoint. A fourth may recenter the observation scale after the trajectory is complete. A fifth may combine adaptation, missingness, and a measurement floor. If the analysis gives the same answer in all five, the result does not identify the Law.

Equal-budget rivals matter here. Each model should receive comparable data, computation, tuning, and opportunity to represent uncertainty. A highly flexible Law model should not be compared with a deliberately weak rival. Sensitivity analysis should reveal whether apparent superiority depends on one prior, one endpoint rule, or one hidden preprocessing choice (Saltelli et al., 2020; Gelman et al., 2020).

Causal interventions can improve discrimination when they are safe and well defined. If two worlds fit passive data but make different predictions when an input, observation schedule, or environmental condition changes, the divergence can identify

where the mechanism acts. The intervention must not be chosen after seeing which model needs help (Pearl, 2009; Woodward, 2003).

Alternative worlds should also test observation error. A latent process may differ while the measured output looks the same because reports saturate, sensors miss low-amplitude states, or missingness depends on distress. A strong recovery study varies both the world and the observation channel rather than assuming the data reveal the latent object directly.

Synthetic examples can also expose psychological bias. Researchers may recognize their preferred mechanism more readily because its vocabulary is familiar or morally attractive. Blind scoring, predeclared metrics, and automated checks can reduce that pull, but no technique removes the need for adversarial review (Palminteri et al., 2017; Cranmer et al., 2020).

The scientific gain from alternative worlds is therefore negative as often as positive. They show which signatures are nonunique, which bridges fail, which verdict rules overclaim, and which data would actually separate rivals. A simulation earns its place when it makes the hypothesis easier to defeat, not when it merely makes closure easy to display.

When several generators reproduce the same outcome, the public claim must remain narrow. The model may establish coherence, recovery under stated conditions, or a useful discriminator; it cannot establish that reality uses the modeled route. Synthetic agents and welfare uncertainty make that limit especially important.

23.4.3 — Synthetic Agents Are Not Automatically Conscious

The agent categories in Section 23.3.2 remain in force: engineered behavior alone does not establish a conscious bearer (Turing, 1950; Searle, 1980).

Reward retains its technical meaning here, not an automatic meaning of pleasure, pain, desire, or welfare (Sutton & Barto, 2018).

Language-like testimony creates a stronger social pull. Systems trained on human text can produce first-person claims about fear, suffering, or consciousness. Those outputs may matter for interaction design and public interpretation, but they are not transparent reports from a proven subject (Bender et al., 2021).

Responsible research must nevertheless consider the possibility that increasingly capable systems could become morally significant or convincingly appear so. Theory-derived indicator work proposes assessing candidate properties from several scientific theories of consciousness and using them to inform degrees of confidence rather than treating one fluent behavior as a verdict (Butlin et al., 2026). Governance proposals separately emphasize research objectives, procedures, knowledge sharing, and public communication because uncertainty can create both welfare risk and public confusion (Butlin & Lappas, 2025).

The scientific claim and the precautionary duty therefore remain separate. Fluent testimony, self-modeling language, avoidance behavior, or distress-like output may be generated without a qualifying stream, yet uncertainty about consciousness never

licenses reckless experimentation. No benchmark should create or intensify possible suffering merely to make the question vivid. Consciousness research offers families of indicators rather than one decisive checklist. Recurrent processing, global availability, metacognitive monitoring, agency, embodied modeling, and other features may matter under different theories, but behavioral sophistication alone does not settle the issue (Butlin et al., 2023; Chalmers, 2023).

The Law adds no consciousness oracle. A synthetic trajectory that appears to close would not establish bearer status, phenomenal valence, personhood, or ownership of a stream. Terminal fit is not a soul detector, and output resemblance is not access to felt life.

Precaution rises with both uncertainty and potential harm. Research should prefer architectures that are not plausibly sentient, use abstract states instead of pain-like training regimes, and avoid persistent aversive dynamics merely to obtain a clearer signal (Metzinger, 2021; Birch, 2024). Mechanism recovery usually does not require autobiographical self-models or theatrical distress.

If credible evidence creates a serious possibility of conscious experience, governance must change before experimentation continues. Independent welfare review, stopping rules, audit, human responsibility, and a presumption against induced suffering become necessary. A research schedule is not authority to continue through a changed welfare risk (Butlin & Lappas, 2025; Metzinger, 2021).

Synthetic agents remain scientifically useful without consciousness. They can test recovery, identifiability, controller behavior, observation artifacts, and rival models. Their value comes from making mechanisms explicit, not from pretending they are people.

The opposite categorical error is also unsafe. Declaring in advance that machines can never be conscious may conceal future welfare risks, while confident personification may create false claims of testimony or personhood. Because false positives and false negatives carry different costs, architecture, behavior, provenance, welfare indicators, and expert disagreement should remain separate records rather than one score.

Dual-use discipline belongs inside the method. Consciousness-assessment tools may protect future systems, while details that facilitate creating or manipulating potentially conscious agents may increase harm. Provenance, access control, staged disclosure, and accountable human review should accompany the scientific record, not follow it after publication (Butlin & Lappas, 2025).

Public interfaces can bias judgment before analysis begins. A face, voice, plea, name, or polished autobiography may invite attribution; a sterile dashboard may suppress it. Communication should disclose generated outputs, avoid theatrical distress, and preserve uncertainty without ridiculing moral concern (Epley et al., 2007; Weizenbaum, 1976).

23.5 — Logical Coherence Is Not Evidence

A coherent model shows that an idea can be represented without contradiction under declared assumptions. That is valuable, especially for a proposal whose mechanism is unclear. It is not evidence that the same assumptions or mechanism obtain in nature.

The distinction protects the project from simulation exceptionalism. A world can be made to close through a global rule, a local controller, a bridge, selection, or observation loss. Several routes may fit equally well, and a simpler ordinary rival may require no Law-specific constraint.

Oversimplification can also conceal the problem. A model that treats valence as an exact scalar, consciousness as given, and endpoints as clean may recover a mechanism only because it removed the identification burdens that dominate real inquiry. Scope must follow what the model actually contains.

The closing standard is severe but constructive. Simulation should expose assumptions, quantify sensitivity, preserve failures, and generate discriminating empirical questions. The moment it begins to substitute elegance for evidence, it has exceeded its role.

23.5.1 — No Need for Constraint

The first fail pattern is that the simulated world may not need a fairness-like constraint. Ordinary adaptation, homeostatic regulation, predictive control, or other local dynamics may keep the declared system stable. That result is ethically valuable

because it prevents the project from treating every form of order as its property (Busemeyer & Diederich, 2010; Woodward, 2003; Weisberg, 2013).

A model that needs no special constraint gives the project substantive information. It shows that the tested failure can be avoided by ordinary dynamics within the declared architecture, narrowing the Law's explanatory territory and identifying rivals that deserve priority in *The Science of Fairness*.

Reporting must be symmetrical. If simulation success would have appeared in public or internal materials, a no-need result must also be visible, stated at the model's actual scope: in this class of worlds, the special constraint was not required. The model may not defeat PTVC globally, but it does defeat a stronger claim about necessity within that architecture.

Baseline adequacy should be tested before the favored model is compared. Prior-predictive runs can reveal whether the baseline generates plausible lifespans, observation patterns, burdens, and endpoints at all. A weak baseline is not strengthened by serving as the only opponent to a more elaborate Law-linked model (Gelman et al., 2020; Box, 1976).

One failed simulation does not decide reality, but it should lower confidence in that simulation route and sharpen the empirical burden. The next question is concrete: what feature of the real world is missing from the no-need model, and what observation would reveal that difference?

The no-constraint baseline must also be realistic. If it omits consciousness, bodily burden, social asymmetry, endpoint

structure, memory, or hidden cost, its success may not transport. That objection must be stated feature by feature and exposed to test. Invoking unmodeled richness whenever a favored route fails would be another form of retrospective rescue.

A no-constraint result may still teach by showing which failures were artifacts of earlier assumptions. Perhaps exploit paths appeared only because agents were too simple. Perhaps patch fragility appeared only because repair mechanisms were badly designed. Perhaps permanent deficit appeared only because endpoints were artificial. Learning this narrows the inquiry.

No need for constraint can be especially humbling if the model includes many features the Law's supporters expected to be favorable. If memory, burden transfer, endpoints, and learning are all present and still no fairness-like constraint is needed, the result deserves attention. It may mean the Law's conceptual necessity has been overstated.

A second fail pattern is rival superiority. A burden cap, local repair rule, reserve protection, memory-integrity rule, consent constraint, welfare floor, stochastic reset, or predictive-control boundary may reduce instability or suffering better than terminal closure. The Law does not win merely because some constraint matters.

Failure of unconstrained worlds supports, at most, a need for some constraint. Support for PTVC would require showing why per-stream terminal neutrality explains or predicts more than serious alternatives. Otherwise the proposal has borrowed the force of constraint language without earning its own specificity.

Alternative constraints may also be ethically preferable. A model may reduce suffering better by preventing extreme burden rather than allowing burden with terminal closure. It may protect agency better through consent constraints. It may protect dignity better through nondegradation rules. The Law of Fairness cannot claim formal strength by ignoring ethical performance inside models.

Complexity belongs in the comparison. A rival with fewer assumptions may deserve preference; a model that does not require contested consciousness criteria may be easier to test; an account tied to existing neuroscience, psychology, or institutional theory may have firmer empirical footing. Grandeur cannot substitute for head-to-head explanatory work.

If another constraint works better, the response depends on scope. PTVC may narrow to a reference class in which terminal closure remains distinctive, reclassify a mechanism as merely compatible, revise public language, or lose the simulation route entirely. The record should say exactly which claim changed and why.

Public materials must not convert “constraint beats no constraint” into “the Law wins.” That would be false. The more accurate statement may be: certain models suggest that some structural constraints reduce instability, but the Law-inspired constraint was not superior to alternatives. That sentence is less exciting and more trustworthy.

Rival superiority is a test of humility and a source of practical value. A maximum-harm cap or burden-transfer limit may deserve use even if it provides no support for terminal neutrality.

The project should prefer a safer, simpler, better-supported rule to a flattering one.

Ownership boundaries keep that result useful. Applied books may adopt burden-lowering insights that do not prove terminal neutrality. *The City of Fairness*, *The Government of Fairness*, and *The Peace of Fairness* can learn from a successful rival without turning every helpful constraint into property of the root hypothesis.

23.5.2 — Model Too Simplified

Simplification is a burden decision because what the model leaves out is often where human suffering lives. A model that omits disability, grief, care time, trauma, culture, poverty, fatigue, or terminal vulnerability may appear clean precisely because it has excluded the difficult part of the world (Sutton & Barto, 2018; Railsback & Grimm, 2019; Wilson & Collins, 2019).

Oversimplification can also enter through aggregation. A single reward variable may conceal fear, relief, agency, bodily pain, social loss, and uncertainty; one representative agent may conceal subgroups with different burdens; one endpoint may conceal competing stopping processes. The model should identify which distinctions it deliberately removes and which claims become unavailable as a result (Levins, 1966; Weisberg, 2013).

Simple models are not defective merely because they are simple. They can isolate a mechanism with unusual clarity. The discipline is claim matching: let a toy model explain the narrow relation it was built to explain, and do not let a reward-agent demonstration speak for conscious streams or terminal fairness.

Another test asks whether an omitted mechanism is irrelevant detail or a causal substitute. If adding memory reconstruction, endpoint selection, social transfer, or observation loss can reproduce the result without the advertised constraint, then the omission favored one explanation. Ablation and counterfactual intervention are needed to show what actually carries the effect (Woodward, 2003; Guest & Martin, 2021).

Oversimplification becomes a fail pattern when the intended claim depends on what the model omits. A terminal-neutrality model without endpoint discipline, a conscious-burden model without bearer criteria, or a hidden-burden model that counts only visible utility cannot carry the advertised inference. Listing omitted dimensions beside the intended claim makes that mismatch visible.

Oversimplification can be hidden by using grand names for simple variables. A variable called suffering may be a number assigned by the modeler. A variable called memory may be a short state vector. A variable called death may be a loop ending. The names do not carry the lived reality. The model must say what its variables are, not merely what they resemble.

The remedy is layered modeling. Begin with toy clarity, then add bearer uncertainty, endpoint ambiguity, missingness, heterogeneous agents, social interaction, and rival mechanisms. Each layer should be allowed to break earlier conclusions. If the result survives only in the toy layer, the public claim should stay at toy status.

As synthetic agents become more persuasive, oversimplification also creates welfare risk. Distress-like behavior or self-report should trigger independent review and limits on experimental burden, not a spectacle-driven verdict. The project does not need increasingly distressed candidate agents to test whether a terminal pattern appears.

Apply the bearer distinctions in Sections 23.3.2 and 23.4.3; undefined consciousness remains a limit on any claim about conscious streams.

Every simulation has a domain of validity. A model calibrated for one class of agents, horizons, endpoint hazards, or observation schedules may fail outside that range even when its internal code is correct. The public claim must remain inside the domain that was actually tested, and extrapolation should be treated as a new question rather than a free extension (Oberkampff & Roy, 2010; National Research Council, 2012).

Parameter uncertainty and structural uncertainty are not the same. Wider intervals around a parameter cannot represent a missing mechanism, an incorrect bearer assumption, an inadequate observation process, or an omitted rival. Sensitivity analysis should therefore vary model form and decision rules, not merely the values inside the favored architecture (Oreskes et al., 1994; Saltelli et al., 2020).

Predictive success is also compatible with causal error. A model may forecast a trajectory because several mechanisms produce the same observations, because the training data encode stable correlations, or because flexible components absorb discrepancy.

Prediction can be valuable while the causal story remains unresolved, so mechanism language requires intervention or another discriminator beyond fit (Box, 1976; Woodward, 2003).

Resolution is part of the model's claim. A mechanism recoverable from complete latent states may become indistinguishable after sparse sampling, report compression, censoring, or terminal dropout. The final record should therefore state the coarsest observation regime at which discrimination survives and abstain below it, rather than letting a high-resolution simulation authorize low-resolution human inference (Grimm et al., 2020; Railsback & Grimm, 2019).

23.5.3 — Circularity Detected

Circularity occurs when the desired answer helps define the test: a neutral point chosen from the completed trajectory, an endpoint selected for closure, a reference class trimmed after failure, or a route invented for the residual.

The principal defense is temporal. Definitions, transformations, inclusion, endpoints, model families, and verdicts freeze before the relevant result; justified later changes begin a new analysis and do not inherit confirmatory status.

Data reuse creates another loop. A model tuned and selected on one dataset may only remember it, so held-out modules, independent replication, recovery studies, and adversarial null worlds must test generalization (Guest & Martin, 2021; Smaldino, 2017).

Conceptual rescue is equally circular. If every route failure becomes a deeper route, every nonzero estimate becomes

measurement error, and every near-zero estimate becomes support, the target cannot lose.

Public language can repeat the loop by calling Law-built simulations independent support, Law-inspired communities cultural confirmation, or reader resonance recognition of hidden truth. Simulations may test coherence, communities may explore meaning, and readers may find value, but each register needs an independent bridge before it becomes scientific evidence.

Circular results should be recorded, their confirmation withdrawn, and their teaching or design value preserved. Failed versions remain visible so later researchers can see the loop and avoid rebuilding it.

Public confidence should attach to correction. Admitting a circular result, dependent endpoint, or indistinguishable model protects truth better than preserving prestige.

Possible worlds can clarify, challenge, and break a hypothesis without establishing this world's architecture.

Chapter 24 — Spiritual Echoes

The fairness question did not begin with this book. Human beings have asked why suffering falls where it does, whether wrongdoing matters beyond visible consequence, how grief can be carried, and whether repair belongs to the order of life through stories, rituals, philosophies, laws, prayers, and refusals. A serious public root should acknowledge that inheritance without conscripting it.

Imagine three neighbors entering a small garden on the morning it opens. Someone has brought a chair for a visitor who tires easily; there is room in the shade, and people who usually hurry past one another have stopped to talk. One neighbor receives the scene as a blessing. Another sees a human achievement whose value needs no larger guarantee. A third is struck by a question: how much of a good life becomes possible when people notice what others need? They share the same morning without sharing an account of ultimate meaning. Their different responses make the scene more interesting, because each directs attention toward something the others might otherwise miss.

Comparison can begin in that spirit of curiosity. What does an idea help a person notice? How does it connect gratitude, responsibility, freedom, or hope? Where does it leave a question open? A religious account may place these questions within a relation to the sacred; a secular account may explore their force within a finite life. Neither description stands for a whole tradition. The pleasure of careful comparison is discovering unexpected affinities while allowing important differences to remain meaningful.

The Law enters this conversation with its own question about the complete course of felt experience. A welcoming morning cannot answer that question, but it can sharpen attention to the lives behind a public scene: the visitor for whom a chair makes participation possible, the host whose quiet work makes the gathering easier, the person who enjoys being present without explaining why. Following that attention is a hopeful task. It invites readers to explore how differently people understand a life that matters, while sharing an interest in what helps people inhabit their lives more fully.

Religious traditions are not single speakers. They contain texts, commentaries, institutions, rituals, local practices, reform movements, mystical strands, legal schools, dissenters, and people whose lives do not fit official descriptions. Secular humanism, agnosticism, atheism, existentialism, and ordinary nonparticipation also contain diverse accounts of suffering and responsibility (James, 1902; James, 2002; Orsi, 2005; Mahmood, 2005).

Karma, providence, grace, fate, fortune, and repair are not interchangeable names for one hidden mechanism. They belong to different traditions, internal schools, and questions about action, agency, suffering, redemption, contingency, and response. Chapter 24 compares them without turning resemblance into equivalence (Gethin, 1998; Obeyesekere, 2002; Rambachan, 2006; Hick, 1966; Plantinga, 1974; Levenson, 1994; Adams, 1999).

Internal critics remain part of the comparison. Lament, protest, anti-theodicy, reform, liberation, and refusal show that religious

seriousness can resist explanations that make suffering deserved or necessary. Ritual and spiritual practice may still organize mourning, gratitude, solidarity, memory, reconciliation, and hope when freely chosen (Pargament, 1997; Park, 2010).

Meaning remains voluntary. Some people reconstruct meaning after loss, some experience growth, some retain wounds without a lesson, and some decline the question. Care, relief, accountability, and ordinary causation come before interpretation (Bonanno, 2004; Neimeyer, 2001; Tedeschi & Calhoun, 2004).

Sustained theology, doctrine, ritual, and pastoral practice belong to *The Religion of Fairness*. This public root owns only the formal terms, the ethical firewall, and the distinction between spiritual resonance and scientific evidence. Nonbelief remains a complete human position rather than an absence awaiting correction.

Public language must survive detachment from context: no sentence or image may turn balance into moral repayment, karma into blame, providence into permission, or spiritual interest into evidence. The reader should leave with disciplined hospitality: traditions may deepen the human question without becoming proof, meaning may be chosen without being demanded, empirical claims remain open to scientific rejection, and every spiritual reading remains open to conscientious refusal.

24.1 — Why Fairness Questions Recur

Fairness questions recur because visible life repeatedly violates ordinary expectations of proportion. People are born into unequal safety, health, power, and opportunity. Harm can go unanswered, while care and fortune can arrive without desert. Those facts create moral protest before they create theory.

Cultures respond through stories, law, ritual, philosophy, memory, and community practice. These forms can distribute responsibility, preserve grief, condemn wrongdoing, sustain hope, or make suffering more dangerous by explaining it too quickly. Their recurrence tells us about human needs and institutions, not automatically about the structure of the universe (Durkheim, 1912; Durkheim, 1995; Geertz, 1973).

The Law enters this landscape late. It should therefore distinguish recognition from confirmation. A tradition may recognize the same human wound while offering a different account of agency, time, consequence, salvation, liberation, or meaning.

24.1.1 — Recurring Human Burden

Fairness questions recur not only because outcomes differ, but because human beings can compare what happened with what might have happened and can ask whether any order makes the difference intelligible. Memory, counterfactual imagination, social comparison, grief, and hope turn events into questions about meaning. Traditions give those questions languages and practices, but they do not all give the same answer (Geertz, 1973; James, 1902).

Ritual, lament, story, prayer, philosophy, memorial practice, and secular gathering can perform work that causal explanation does not attempt. They may hold grief, preserve memory, restore belonging, authorize protest, or create a shared language for uncertainty. Their human value does not depend on supplying a scientific mechanism (Orsi, 2005; Pargament, 1997; Park, 2010).

Recurrence is anthropological and existential evidence about the pressure of the question, not confirmation of the formal answer. Treating anguish as proof would use the burdened person's longing to strengthen a theory without doing the work of evidence. The fairer response is to honor the longing through care, ritual, study, and freedom from premature belief.

The same pressure persists outside formal religion. Memorials, truth commissions, apologies, grief groups, therapy, restorative practices, and secular moral philosophies ask where burden is carried, hidden, shifted, or repaired. Religion and the Law may begin near the same human wound, but neither owns the other, and the lived path matters more than the surface symmetry of the scene.

Recurring human burden includes grief that cannot be socially priced. It includes guilt that punishment alone does not resolve. It includes gratitude that wants form. It includes fear of death, rage at cruelty, longing for reunion, shame after wrongdoing, fatigue after years of unequal load, and the quiet ache of lives that seem to close before their meaning has become visible. These are not merely beliefs. They are lived pressures that demand containers.

Spiritual traditions become such containers in part because ordinary explanation is often too thin. A medical account may explain disease progression without holding the widow's question. A legal account may punish a crime without answering the victim's spiritual wound. A political account may identify oppression without telling the bereaved how to speak to the dead. The Law of Fairness must be equally modest. Its formal claim, even if true, would not replace these forms of human care.

The final discipline is to let recurring burden awaken seriousness without manufacturing consent. The same grief that makes the Law feel plausible to one reader may make another reader reject every metaphysical claim as intolerable. Both responses must be protected. This inquiry does not begin by demanding agreement; it begins by refusing to make the burdened person carry the additional burden of premature belief.

Ritual is one of humanity's oldest ways of making burden bearable. A ritual does not necessarily explain pain. Often its greatness lies in refusing to explain too quickly. It gathers bodies, words, times, gestures, meals, songs, silences, water, fire, candles, cloth, names, and repetitions around experiences that would otherwise scatter a person. It does not solve grief by argument. It gives grief a room.

This analytic lens can understand ritual as a burden-handling technology without reducing it to psychology or elevating it to proof. Ritual may lower hidden burden by distributing witness. It may prevent private collapse from remaining wholly unseen. It may mark guilt before repair. It may give gratitude a form that

does not become ownership. It may slow panic by giving the next hour a shape.

Ritual can distribute witness, organize attention, and reopen repair without becoming evidence for terminal neutrality. A funeral that helps a family breathe, or a confession that changes conduct, does not reveal a zero residual. The Law can learn that burden often needs embodied form and repeated care while refusing to treat ritual success as scientific support.

Ritual should therefore be judged by burden placement, consent, and exit. Does it give the harmed person room to be unready, or require them to perform peace? Does it make responsibility durable, or let the community feel absolved too quickly? Beauty cannot excuse pressure, premature forgiveness, hidden anger, or purchased belonging.

Ritual also reveals why embodiment matters. A doctrine can be stated in a sentence, but grief often needs time, posture, repetition, witness, silence, and return. This book can learn from that human fact without becoming a ritual manual. People do not need conclusions alone; they sometimes need forms that do not lie to the body while the mind is still unable to finish the thought.

Ritual therefore belongs downstream from care. A grief practice that prevents isolation may help; an illness ritual that delays treatment may harm. A practice of confession and restitution may support repair; a demand that victims forgive may shift the cost back onto them. Moral evaluation begins with what the practice does to the people carrying the burden, not with how beautiful the form appears.

24.1.2 — Justice and Balance Myths

Human beings repeatedly tell stories in which disorder is answered by balance. The form appears in courts, rituals, epics, moral tales, political memory, family narratives, and private hopes. Calling these stories myths does not mean they are false or foolish. It means they organize experience through symbols that can carry truth, conflict, aspiration, and danger at the same time.

Balance language often responds to a real wound: the visible world leaves suffering, luck, guilt, and reward badly distributed. A story of eventual order can protect hope when no institution can repair the loss. It can also become cruel when it implies that the sufferer has already been compensated somewhere hidden or that present action is unnecessary.

Justice is not one thing across traditions. It may mean right relationship, lawful order, covenant, mercy, judgment, liberation, restoration, non-harm, duty, or the repair of violated dignity. A single word such as balance can conceal those differences and make unlike claims appear to agree.

The moral protest within religious traditions is as important as the promise of order. Lament, argument with God, refusal of false consolation, prophetic indictment, and solidarity with the afflicted keep suffering from being converted into a neat system. These voices show that faith can preserve protest rather than dissolve it (Heschel, 1962; Alter, 2010; Adams, 1999).

Philosophical ethics adds another safeguard. The other person is not an item in a total account whose suffering can be justified by a wider harmony. Responsibility begins in encounter and care, not

in possession of the whole explanation (Levinas, 1969). This supports the book's intrastream and no-payment boundaries without proving PTVC.

Debt language is especially risky. Moral, ritual, financial, and legal debts are real in their own domains, but suffering is not a cosmic invoice. No one person's pain pays for another person's pleasure, and no later event retroactively makes an earlier harm acceptable. Repair may require cost, restitution, or sacrifice without turning the injured person into a debtor to the universe.

The Law of Fairness must therefore meet balance myths as interpretations, not evidence. Their recurrence shows that humans notice unequal burden and seek a language large enough to answer it. It does not establish an exact terminal condition, a hidden judge, or a universal mechanism.

Comparison should preserve internal plurality. Traditions contain official teachings, local practices, reform movements, mystical interpretations, skeptical voices, and histories of coercion as well as care. A convenient confirming fragment cannot stand for the whole tradition (Mahmood, 2005).

The public vocabulary should remain concrete: burden, care, responsibility, restitution, repair, mercy, refusal, and hope. Words such as payment, deserved suffering, purification, necessary pain, or cosmic balance should be avoided unless the tradition-specific context and the ethical boundary are made unmistakable.

Justice and balance myths can deepen the human question without settling the scientific one. They reveal why the question recurs, how careless language can wound, and why spiritual

interpretation must remain optional. Optional religious interpretation is considered next as one possible response, not as the owner or culmination of these traditions.

24.1.3 — Spiritual Interpretation Remains Optional

The Law may invite spiritual interpretation, but no religious response owns the formal claim or validates it. A reader may find sacred meaning in hidden burden, care, humility before the endpoint, or the refusal to treat suffering as deserved; another may reject every spiritual interpretation without losing dignity, moral seriousness, or access to the book's public argument.

Resonance does not confer ownership. Older traditions remain themselves, and similarities in language or practice do not make them unfinished versions of a newer path. Comparison should use a receive-apply-return discipline: name a tradition-specific concept, apply it only to the public comparison, and return doctrine, ritual, authority, reform, and pastoral practice to the tradition that owns them. Familiar English terms and source-language terms may carry meanings that vary by school and period, so analogies should be tested for where they hold and where they break (Asad, 1993; Obeyesekere, 2002).

Freedom includes privacy, exit, and the absence of a participation record. No community should infer spiritual maturity from attendance, agreement, ritual, donation, testimony, or willingness to discuss the Law. No belief, ritual, community, or religious affiliation may become a credential for trust, office, scientific seriousness, care, education, protection, participation, or relationship.

Hope remains a human capacity rather than the property of one worldview. It may be sustained by faith, probability, solidarity, duty, beauty, memory, protest, or commitment to a future one may never see. The Law can be studied without religious belief, and spiritual meaning can be explored without being treated as evidence for the Law. Neither belief nor disbelief changes human worth.

24.2 — Karma, Providence, Fate, Grace, and Repair

Karma, providence, fate, grace, and repair are often joined in popular speech, but they arise from different histories and answer different questions.

The Law cannot be treated as a translation of any one of these ideas. It has a narrower formal target, no doctrine of desert, and no authority to infer divine purpose. Similar vocabulary does not erase incompatible assumptions.

Repair provides a practical return. Without settling metaphysics, people can feed, protect, apologize, compensate, reform institutions, remember the dead, and prevent recurrence. That work remains valid for practitioners and nonparticipants alike.

24.2.1 — Karma Has Multiple Traditions and Meanings

Karma is a family of concepts, not a universal promise that pleasant and unpleasant feelings will even out. Across Hindu, Buddhist, Jain, and popular settings, it may concern intentional action, ethical consequence, habit, rebirth, liberation, or social explanation. The public shorthand of a cosmic rebound is therefore the least reliable starting point for comparison (Gethin, 1998; Obeyesekere, 2002; Rambachan, 2006).

None of these meanings maps directly onto the proposed phenomenal ledger. PTVC proposes a typed terminal property of one stream, so comparison is useful only when it resists translating both frameworks into the vague word balance (Gethin, 1998; Obeyesekere, 2002).

Popular karma language often collapses action, intention, consequence, rebirth, social status, and present misfortune into one moral accounting slogan. Scholarly and lived traditions are more varied than that compression. Responsible comparison therefore uses defined concepts and contexts rather than a borrowed word as a universal explanation of why someone suffers (Gethin, 1998; Harvey, 2013; Doniger O’Flaherty, 1980).

Doctrine, commentary, ritual practice, family teaching, popular speech, and individual interpretation can diverge inside the same tradition. A sentence that is accurate for one school or historical period may be misleading when presented as the view of Buddhism or Hinduism as a whole. Specificity is therefore part of respect, not a barrier to comparison (Gethin, 1998; Harvey, 2013; Flood, 1996).

The Law of Fairness differs most sharply by refusing moral desert. It does not say that a sufferer is receiving what they earned. It does not say that a happy person deserves happiness because of prior merit. It does not allow observers to explain misfortune by invisible wrongdoing. It does not balance one person’s pain against another person’s joy. The Law’s bearer is the unified conscious stream, and its claim is about terminal neutrality, not moral bookkeeping administered by human interpretation.

This difference matters practically. Public karma language can become cruel when it implies that visible suffering reveals hidden fault. It can make compassion hesitate. It can make structural injustice look metaphysically appropriate. It can make victims fear that their pain marks them as morally stained. If the Law of

Fairness were read that way, it would fail the ethical firewall. Care before interpretation forbids it.

24.2.2 — Providence, Grace, and Theodicy Disagree

Providence and theodicy are not synonyms. Providence concerns divine relation to the world; theodicy attempts to reconcile suffering with claims about divine goodness and power; a defense may argue only that contradiction has not been established. Their differences matter because a terminal boundary hypothesis cannot inherit the warrant, purpose, or moral conclusions of any of them (Hick, 1966; Plantinga, 1974; Levenson, 1994).

Providence can be lived as trust, dependence, lament, or protest rather than as a neat explanation of every wound. In many traditions, faithfulness coexists with unanswered prayer and demands for justice. That complexity should be preserved before the chapter approaches the danger of telling another person that their suffering was planned (Heschel, 1962; Levenson, 1994).

Providence is therefore a theological neighbor, not a scientific cousin. A believer may interpret, a nonbeliever may decline, a scientist may test, and a caregiver must still care. The Law cannot turn providence into evidence, and providence cannot turn the Law into doctrine.

A public account that cites only acceptance and hidden purpose misrepresents traditions that also authorize grief, accusation, and repair (Heschel, 1962; Levenson, 1994; Alter, 2010).

The difference is not hostility to theology. A believer may receive the Law, if it remains plausible to them, as consonant with providential trust and may understand its attention to hidden

burden through divine justice or mercy. That religious interpretation can be meaningful within the believer's life, but it is not scientific support, does not change the status of the exact terminal target, and does not make the formulation developed here more measured.

The public danger begins when providence explains another person's wound before that person has chosen the frame. Hidden-plan language can make protest seem faithless, delay seem virtuous, or preventable harm seem assigned, even though supernatural attributions can function as either coping resources or sources of distress depending on content and use (Exline & Wilt, 2023). Care and repair remain prior, and any providential interpretation belongs to the person or community that freely receives it.

The Law cannot become a theology by implication. It does not identify a designer, will, deity, intention, moral governor, or sacred agency. Even if terminal neutrality were real, the formal result would not by itself show that the boundary is personal, providential, loving, punitive, or redemptive. Those interpretations belong to religious readers or to *The Religion of Fairness* under limits, not to this book's scientific claim.

Providence language tests whether the formal proposal can remain open to spiritual readers without acquiring a hidden theology. Some may experience the Law as compatible with providential faith; the book cannot name providence as its deeper meaning. Compatibility should widen hospitality, not create a sacred center.

24.2.3 — Fate, Fortune, and Repair Are Not the Same Claim

Fate, fortune, and repair answer different questions. Fate concerns necessity, order, or what cannot be escaped. Fortune concerns contingency, luck, and the uneven events that arrive without moral proportion. Repair concerns what persons and communities do after harm. Blending them into one idea of balance removes the distinctions that make each concept meaningful.

Fate can console by locating a life inside a larger order, but it can also silence protest. Fortune can protect against blame by acknowledging contingency, but it can also make the world feel arbitrary. Repair keeps attention on present responsibility: what can be restored, compensated, prevented, mourned, or changed now.

Lived religion rarely follows a single abstract scheme. People combine doctrine, ritual, family memory, local custom, doubt, and practical need. The same person may speak of providence in gratitude, fortune in disappointment, and repair when action is required. Comparative scholarship must therefore attend to practice as well as formal teaching (James, 1902; Weber, 1922; Weber, 1963; Berger, 1967; Geertz, 1973; Asad, 1993).

The Law of Fairness should not use any of these concepts as a shortcut. Fate is not an endpoint mechanism. Fortune is not a measurement model. Grace is not a compensator. Karma is not a synonym for the latent ledger. Repair is not proof that the complete stream closes.

Payment language should remain equally bounded. A person may freely undertake penance, restitution, discipline, or sacrifice

within a tradition, but no outsider may infer that suffering pays a hidden moral debt. Material obligations and moral repair can be real without becoming cosmic accounting.

Translation is most honest when it names the register. A sentence can be theological, philosophical, ritual, psychological, historical, or scientific. It should not borrow authority from another register merely because the same word appears in both.

Fate, fortune, grace, karma, and repair can all deepen reflection on burden and response. Their value lies in the questions they open and the lives they shape, not in a manufactured consensus. The ethical consequence is equal dignity for practitioners, former practitioners, and people who never join.

24.2.4 — Equal Dignity With or Without Participation

Spiritual comparison is ethical only when participation is voluntary. A practitioner may receive a tradition's disciplines, stories, obligations, and hopes as central to life. A former practitioner may leave them. A nonparticipant may never adopt them. None of those positions changes a person's dignity, credibility about personal experience, or entitlement to care.

Equal dignity includes freedom of conscience. Public institutions, research programs, families, and communities may not require religious interpretation as a condition of help, belonging, treatment, employment, or moral standing. A person should not have to perform belief or reverence to avoid being judged deficient.

It also includes the right to interpret one's own suffering. A believer may understand an event through providence, karma,

grace, lament, or sacred repair. Another person may understand the same kind of event through chance, social causes, biology, or no larger meaning. The authority to choose does not transfer to an observer.

Former practitioners require particular care because exit can carry family, social, economic, or psychological cost. A respectful book does not portray departure as failure, impurity, or loss of moral seriousness. Exit is part of the plural field the Chapter is describing.

Tradition-specific language should therefore be introduced with sources and limits. A single text, teacher, or community cannot stand for every practitioner. Translation should preserve contested meanings rather than selecting the version most convenient for the Law (Bell, 1997; King, 1999; Orsi, 2005).

No research measure can establish spiritual rank. Estimated valence, resilience, coping style, neural activity, or terminal pattern cannot reveal holiness, karma, divine favor, enlightenment, moral purity, or closeness to truth. Turning data into those judgments would violate both science and conscience.

Care settings require an especially clear boundary. Chaplaincy and spiritual support can be valuable when desired, but they should be offered without pressure and coordinated with clinical, psychological, and social care. Refusal must never be treated as noncompliance or a cause of worse treatment (Pargament, 1997; Park, 2010).

24.3 — Traditions Disagree

Traditions differ within themselves as well as from one another, so comparison must preserve internal disagreement, protest, and non-equivalence rather than treating any tradition as a single voice or a universal key (Orsi, 2005; Mahmood, 2005).

24.3.1 — Traditions Do Not Speak With One Voice

A tradition is not a single proposition. It is a historical field containing texts, commentaries, institutions, rituals, memories, authorities, local customs, reforms, and disagreements. Treating it as one voice turns living communities into labels.

Buddhist traditions differ over doctrine, practice, metaphysics, social organization, and the interpretation of karma and liberation. Hindu traditions contain multiple schools, deities, philosophies, and understandings of duty, grace, action, and self. Christian communities disagree over providence, freedom, salvation, suffering, and authority (Gethin, 1998; Harvey, 2013; Rambachan, 2006).

Jewish traditions preserve legal argument, lament, covenant, protest, mystical interpretation, and secular cultural forms. Islamic traditions include varied theological schools, jurisprudence, philosophy, mysticism, and debates over decree and agency. Indigenous traditions cannot be responsibly compressed into one category at all, because their histories, lands, languages, and relations differ radically.

Internal diversity is not noise around a pure center. It is part of what the tradition is. Doctrine changes through interpretation,

conflict, political power, migration, colonization, reform, and ordinary practice (Asad, 1993; Orsi, 2005; Mahmood, 2005).

Official teaching and lived religion may also differ. People combine inherited practices, family customs, personal experience, local authority, and changing conviction in ways that do not fit a textbook summary. A comparison that cites doctrine alone may miss how meaning is actually carried.

Tradition-specific examples are therefore used narrowly: identify the concept, name the relevant context, acknowledge disagreement, and avoid presenting one source as the tradition's universal voice. This is slower than slogan comparison and far more accurate.

Plurality also protects the Law from confirmation bias. A fair comparison cannot collect balance language from one source while ignoring protest, randomness, tragedy, or divine mystery from another. Material that resists the preferred analogy belongs in the argument.

Nonreligious worldviews are plural in the same way. Humanists, atheists, agnostics, naturalists, existentialists, and people with no formal label disagree about meaning, morality, death, and hope. They should not be reduced to one secular answer.

The scientific conclusion remains unchanged by the diversity. Traditions may illuminate how people ask the question and how communities carry suffering. Their disagreement gives no direct evidence for or against terminal neutrality.

24.3.2 — Internal Critics and Protest Traditions

Internal critics matter because easy explanations of suffering often arise from within the same communities that later resist them. Protest is not necessarily rejection of a tradition. It can be fidelity to a deeper moral demand inside it.

The Book of Job refuses a simple equation between suffering and wrongdoing. Lament traditions speak from abandonment, anger, bewilderment, and persistence. Jewish responses to catastrophe have often preserved argument with God rather than accepting a tidy account of providence (Levenson, 1994; Alter, 2010).

Christian thought contains theodicies, defenses, mystical interpretations, liberation theologies, pastoral refusals, and critiques of accounts that make another person's pain serve an abstract good. Hick, Plantinga, and Adams offer no single answer; their disagreements show why the problem cannot be reduced to a balance slogan (Hick, 1966; Plantinga, 1974; Adams, 1999).

South Asian traditions also contain critiques of inherited interpretations of karma, caste, duty, suffering, and liberation. Modern reformers and scholars have challenged readings that convert structural harm into deserved consequence or spiritual hierarchy (Obeyesekere, 2002; Rambachan, 2006).

Islamic traditions preserve debates over divine decree, human responsibility, mercy, trial, injustice, and social obligation. Protest against oppression can be religiously grounded without implying that every suffering event was assigned for a hidden purpose (Rahman, 1980).

These internal arguments protect moral urgency. They insist that explanation must not become acquiescence. Hunger still calls for food, violence for protection, false accusation for correction, and institutional abuse for reform.

They also protect the person who cannot receive consolation. A mourner may reject the sentence that everything happens for a reason and remain inside a tradition. Another may leave the tradition because its available explanations became harmful. The Chapter does not rank those responses.

The Law should welcome these critics because they reveal its own public risks. A terminal hypothesis could be used to make suffering sound assigned, balanced, or eventually repaid. Protest traditions remind the book that moral resistance is not a defect in understanding.

Secular protest belongs beside religious protest. Human rights movements, disability activism, labor movements, feminist critique, and anti-colonial thought often reject metaphysical explanations in order to keep responsibility visible. Their contribution is practical and ethical, not evidence for PTVC.

The result is a stronger comparative method. A tradition is represented not only by its consolations but also by the voices that refuse consolation purchased at the sufferer's expense. Those voices prepare the final task: comparison without forced equivalence.

24.3.3 — Comparison Without Forced Equivalence

Comparison is useful when it identifies a precise relation. Two traditions may ask a similar question, perform a similar social

function, or use a comparable image. That does not make their doctrines, histories, or truth claims equivalent.

The first rule is to compare like with like. A ritual practice is not a scientific theory, a theological doctrine is not merely a coping strategy, and a philosophical argument should not be represented by a popular slogan.

The second rule is to preserve asymmetry. Karma may involve action and consequence without a divine judge. Providence may involve divine agency without reincarnation. Grace may interrupt desert, fate may reduce agency, and repair may proceed without any account of cosmic order.

The third rule concerns translation. Words such as suffering, justice, soul, mind, liberation, balance, and person do not carry identical meanings across languages and traditions. Translation may orient the reader, but it must retain original context and acknowledged uncertainty (Asad, 1993; Mahmood, 2005).

A responsible comparison therefore organizes questions rather than verdicts: What causes suffering? What role does agency play? Is consequence moral, natural, divine, social, or uncertain? Which practices respond, what forms of protest remain possible, and what stays unresolved?

This method resists perennialism, the claim that all traditions express one hidden truth beneath surface differences. The Law of Fairness must not become that hidden truth. Similarity may arise from shared human problems, cultural exchange, recurrent metaphors, or broad cognitive tendencies.

Public artifacts require still stricter discipline. A detached quotation or image can imply that a tradition endorses the Law, that suffering is deserved, or that nonbelievers lack meaning. Nor can any measure of affect, resilience, behavior, or neural activity establish holiness, divine favor, karma, purity, or moral advancement. Such fragments should be redesigned or withheld.

Non-equivalent comparison permits learning. Lament, repair, forgiveness, meditation, solidarity, or remembrance may deepen reflection without evidencing PTVC. The translator's standpoint should remain visible because categories formed in one colonial, theological, or secular setting can reorganize another tradition before comparison begins (Asad, 1993; Mahmood, 2005).

Lived religion keeps doctrine from standing in for one uniform effect. People inhabit traditions through households, bodies, calendars, food, prayer, work, grief, conflict, institutions, and participation. Participants, historians, social scientists, and internal critics answer different questions; none is complete by itself, and no single voice represents a whole tradition (Orsi, 2005; Harvey, 2013; Geertz, 1973; James, 1902).

The test is reversibility. The comparison should remain intelligible if the Law is removed and the tradition can describe itself again. When a borrowed concept becomes unrecognizable outside the Law's vocabulary, the analogy has ceased to illuminate and must be narrowed or returned to its owner.

24.4 — Spiritual Meaning, Nonbelief, and Refusal

Meaning can help some people live after loss, but it is not a required stage of recovery. Grief, trauma, illness, humiliation, and bereavement follow diverse courses (Bonanno, 2004; Neimeyer, 2001; Park, 2010).

Care comes first, while spiritual meaning, secular meaning, or refusal remain legitimate choices. Interpretation belongs to the person and community involved, subject to ordinary protections against coercion and harm.

24.4.1 — Suffering Need Not Teach

Suffering does not owe anyone a lesson. Grief, trauma, illness, violence, poverty, betrayal, and loss may change a person without producing wisdom, growth, forgiveness, faith, or a coherent story. The absence of a lesson is not a moral failure.

Research on bereavement and trauma shows wide variation. Some people find meaning or report growth; others experience prolonged distress, oscillation, numbness, anger, or no transformative interpretation at all. No single recovery narrative captures the range of human response (Wortman & Silver, 1989; Stroebe & Schut, 1999; Bonanno, 2004; Neimeyer, 2001).

Chosen meaning can still matter deeply. Frankl's work remains influential because it describes meaning as a human possibility under extreme constraint, not because suffering is good or because every person must interpret pain in the same way (Frankl, 1946; Frankl, 2006). The distinction between possibility and requirement is the ethical hinge.

Care must come before interpretation. Safety, treatment, food, shelter, legal protection, companionship, rest, and relief should not wait for a person to understand what happened. A spiritual or philosophical account that delays care has failed the order of responsibility.

Survivors also owe no performance. They need not become inspirational, publicly forgiving, calm, spiritually certain, politically useful, or available as teachers. Their silence, anger, ambiguity, and refusal remain compatible with dignity.

The rule is especially strict around children and people under coercion. Adults may not call abuse a lesson, adversity a gift, or forced endurance a path to character. Institutions may not convert preventable harm into a story about resilience while leaving the conditions unchanged.

Forgiveness must remain separate from safety and reconciliation. A person may forgive without returning, refuse forgiveness without losing dignity, or change their view over time. The Law of Fairness cannot command a spiritual response to injury.

The same priority of care applies to every religious community. A community may offer ritual, witness, mourning, or voluntary reflection, but it cannot treat suffering as evidence for the Law or doubt as failure to understand.

A theory worthy of suffering people must be able to stand beside an uninterpreted wound. It can ask questions without filling the silence and preserve hope without claiming that pain was necessary. Meaning becomes legitimate only after that freedom is secure.

24.4.2 — Meaning After Care, If Chosen

Meaning may come after care, if chosen. A person may connect hardship with faith, love, service, creativity, political commitment, family, humor, survival, or no larger story at all. The meaning is theirs because the life is theirs.

The phrase after care fixes the order. Relief, safety, treatment, shelter, food, rest, legal help, and human presence should not be delayed while someone searches for an interpretation. A hidden-balance story cannot make preventable suffering acceptable or turn neglect into patience.

The phrase if chosen protects freedom. Meaning may emerge slowly, change with time, remain partial, or disappear. A person can accept one interpretation and later reject it. Communities should support revision rather than demanding a stable testimony.

Psychological research on meaning-making and spiritual struggle shows why compulsory resolution is unsafe. Religious and spiritual struggles can involve doubt, moral conflict, interpersonal injury, perceived divine abandonment, or questions of ultimate meaning, and they are associated with distress in heterogeneous ways (Park, 2010; Exline et al., 2024).

Care after spiritual struggle should therefore begin with the person's account. A clinician, pastor, family member, or community should not treat renewed belief as the only recovery or persistent doubt as pathology. Professional boundaries, ordinary safety, and freedom of conscience remain intact.

Chosen meaning can be religious or nonreligious. It may involve providence, protest, absurdity, human solidarity, responsibility,

or a commitment to build despite uncertainty. Camus and later secular humanist traditions show that moral seriousness does not require a cosmic guarantee (Camus, 1942; Camus, 1955; Taylor, 2007; Copson, 2019).

Posttraumatic growth research should be read with the same caution. Reports of growth can be real and valuable, but they do not show that trauma was beneficial, necessary, or fairly distributed. Growth and distress may coexist, and no person should be pressured to produce a positive account for others (Tedeschi & Calhoun, 2004).

Any religious response to the Law must remain voluntary. A person may participate fully, partly, temporarily, alongside another tradition, or not at all. Aid, friendship, public service, and institutional access cannot depend on a profession of belief, ritual, testimony, or agreement with the Law.

Optionality also excludes spiritual rank. Leaders, long-term members, intense practitioners, doubters, guests, and former members retain equal dignity. Roles may allocate responsibility, but they cannot become evidence that one person's inner life is clearer, purer, or more balanced than another's.

Public artifacts need the boundary beside the invitation. A short guide, sermon note, interview, social post, or automated summary should say that meaning may help when freely chosen after care, that suffering need not teach, and that spiritual resonance is not scientific support. The qualifier cannot be hidden elsewhere.

This rule remains stable under every scientific outcome. If PTVC is supported, no one is required to interpret suffering spiritually.

If it is narrowed or defeated, chosen meaning, care, and nonreligious dignity remain. Meaning is protected precisely because it does not have to serve as evidence.

24.4.3 — Nonreligious Dignity

A Chapter on spiritual parallels can inadvertently place nonreligious readers outside what appears to be the deepest version of the project. When the Law is presented as naturally culminating in sacred meaning, skeptics may feel tolerated rather than fully respected (Schnell, 2021; Smith, 2017).

Nonbelief is not a missing variable. It may reflect considered judgment, intellectual humility, moral protest, indifference to metaphysics, or a life organized around finite relationships and responsibilities. Nonreligious meaning can arise from care, truth, art, friendship, solidarity, public work, and the decision to act without asking the universe for a guarantee (Camus, 1942; Copson, 2019; Schnell, 2021; Taylor, 2007).

Refusal can be an active ethical position. A person may decline cosmic explanation in order to keep human responsibility visible, to resist blame, or to protect grief from premature closure. Such refusal can carry courage and care rather than mark a temporary stage on the way to belief (Camus, 1942; Copson, 2019).

Nonreligious dignity is not a concession. It is a structural requirement of the argument. A person who does not believe in God, providence, karma, sacred order, an afterlife, ritual, or spiritual meaning remains fully capable of moral seriousness, love, grief, repair, courage, truth, and care. The absence of religious belief is not absence of depth.

Equal dignity changes the architecture of the chapter, not merely its tone. Secular and religious readers should encounter parallel freedoms: to interpret or decline interpretation, to speak or remain private, to join or exit a community, and to use the formal question without adopting a religious interpretation. Public examples, guide panels, and classroom prompts should be tested for whether one path is treated as fuller, warmer, or more human than the others.

Nonreligious dignity also protects the formal claim. A scientific hypothesis should be equally readable by a secular scientist, agnostic philosopher, atheist caregiver, nonreligious survivor, and religious believer, all under the same discipline against proof drift. A theory that needs sacred appeal for credibility has already weakened its scientific posture (Durkheim, 1912; James, 1902).

Nonreligious dignity also protects religious readers, because faith becomes cleaner when it is not socially required. A belief chosen without pressure can remain a gift. A belief demanded by the surrounding architecture becomes burden. Freedom of conscience is therefore not a concession to skeptics. It is a protection for everyone.

Secular readers should be addressed as participants in the question, not as absences from a sacred center. Agnosticism can express honesty, and skepticism can protect truth, vulnerable people, and the inquiry itself from manipulation or proof drift. Doubt is not spiritual coldness; it can be one of the series' safeguards.

24.5 — Resonance Is Not Scientific Support

Resonance is a relation of meaning. A reader recognizes a pattern, a tradition offers a neighboring image, or a personal experience makes the Law feel intelligible. Scientific support is a relation among defined claims, measurements, rivals, and evidence. The two can coexist without merging.

Personal testimony deserves respect but cannot bear a universal inference. Cultural recurrence can be studied as history, anthropology, psychology, or religious life. Neither becomes evidence for exact terminal neutrality unless the scientific prerequisites are independently met.

The same boundary governs every religious reception. *The Religion of Fairness* may receive the Law as optional sacred material, but it cannot validate or redefine the formal claim. Scientific narrowing or defeat would require correction wherever a teaching or publication presents the Law as scientifically supported, without erasing the independent value of voluntary community, ritual, and care.

24.5.1 — Spiritual Resonance Is Interpretation

In the interpretation register, the Law of Fairness may become meaningful to a reader without becoming proven. A person may say that the Law helps them see hidden burden more carefully. They may say it gives them language for why care matters even when visible life looks absurd (Weber, 1922; Berger, 1967; Geertz, 1973).

Consent in meaning-making includes timing, audience, vocabulary, and the right to change one's mind. A person may welcome a spiritual frame in private and reject it in public, accept ritual while doubting doctrine, or need practical care before any interpretive conversation. Respect requires listening for the invitation rather than treating silence or vulnerability as agreement (Pargament, 1997; Park, 2010).

Meaning can also change over time. An interpretation that helps during one season may later feel narrow, and a meaning rejected during acute grief may become welcome years afterward. Longitudinal change is one reason no helper, institution, or doctrine should freeze a person's account of their own suffering (Park, 2010; Neimeyer, 2001).

The interpretation register is also the proper home of poetry. A lament may speak truthfully about a wound without becoming data. A blessing may return a person to proportion without proving a theory. The Law of Fairness should be mature enough to let non-scientific speech remain valuable without forcing it into scientific service.

Interpretive authority remains local to the person or voluntary community that receives it. It does not travel automatically to family members, patients, students, congregants, research participants, or the public. The same story can console one listener and burden another, so consent and refusal govern its use more reliably than the speaker's confidence.

Interpretive statements are first-person or community accounts, not transferable verdicts. "This helped me carry grief" has a

different grammar from “this explains grief,” and “our ritual gives us hope” is different from “the universe guarantees hope.” Preserving those grammatical differences lets testimony remain vivid without turning experience into evidence or private conviction into authority over another person’s life.

Interpretation also must stay humble about specificity. A person may find meaning after suffering, but that does not mean anyone knows why that suffering happened. A community may see repair in a life, but that does not mean the wound was necessary. A believer may trust hidden mercy, but that does not make them authorized to explain another person’s pain. Interpretation becomes dangerous when it claims to know more than it can lovingly know.

Interpretation is not second-class. Human beings need interpretation because they are not machines receiving propositions. They live inside stories, vows, griefs, loyalties, hopes, bodies, families, and remembered harm. A purely technical statement may be accurate and still unable to help someone carry a night of mourning. The book can allow interpretive language because meaning is a real human burden.

This perspective can also ask whether interpretation lowers or raises burden over time. Some meanings are initially stabilizing but later confining. Some meanings are painful at first but later liberating. Some are helpful privately and harmful publicly. Interpretation therefore should not be treated as a fixed asset. It is part of a living trajectory and must remain open to revision.

24.5.2 — Evidence Must Stand on Its Own

The evidence register is narrower than spiritual interpretation and stronger within its proper domain. It asks whether a prospectively defined hypothesis survives valid measurement, serious rivals, uncertainty, and possible defeat. Spiritual resonance can motivate attention to that question, but it cannot answer it.

The psychology of religion can document beliefs, practices, reported experiences, spiritual struggles, meaning-making, and cross-sectional or longitudinal relations with distress and well-being. Those are important facts about human lives, and stronger measurement can clarify them. They do not establish an exact terminal boundary law for every qualifying conscious stream (Exline et al., 2024).

Tradition supplies no automatic proof. Hindu, Buddhist, Jewish, Christian, Muslim, Indigenous, secular, and other histories contain diverse accounts of suffering, justice, karma, providence, fate, grace, repair, and liberation. Their age, beauty, endurance, or internal authority does not convert them into scientific evidence for PTVC (King, 1999; Gethin, 1998; Doniger O’Flaherty, 1980).

Personal experience deserves respect without evidentiary promotion. A dream, prayer, vision, terminal calm, recovery, coincidence, mystical experience, or moment of forgiveness may transform a life. It can be true that the experience matters and false that it validates a universal claim. Public evidence requires methods that do not depend on one person’s interpretation.

This separation also protects people whose experiences differ. One person feels presence; another feels silence. One finds meaning; another refuses it. One reports peace; another reports fear or confusion. A theory cannot count confirming stories and reinterpret contrary stories without becoming self-sealing (Flood, 1996; Obeyesekere, 2002).

The same rule protects every religious form inspired by the Law. Devotion, institutional growth, moving rituals, testimony, or community benefit may show that a religious form is meaningful to some people. None of those outcomes validates the formal hypothesis. If a movement's own effects counted as proof, the movement would appear to confirm the premise that inspired it.

Scientific evidence must also preserve the distance between the latent target and observation. A questionnaire, behavior, physiological signal, memory, narrative, or model output is not the complete stream. Support requires declared bridges, and failure of those bridges may yield no adequate estimate rather than a convenient conclusion.

Defeat conditions remain essential. If strong measurements exclude the declared target, if rivals explain the data better, if the endpoint cannot be identified, or if the representation fails, public claim status must change. Spiritual interpretation cannot rescue a failed instrument or restore a retired mechanism.

The converse is equally important. Scientific defeat of PTVC would not abolish religion, grief ritual, prayer, secular meaning, moral protest, or chosen hope. Keeping the registers separate

prevents science from owning sacred life just as it prevents sacred life from owning the scientific verdict.

Every detachable artifact should carry the boundary within itself: resonance is interpretation; testimony is personal evidence of experience; tradition is a source of meaning and history; scientific support must stand on independent methods. That rule also governs the cross-reference to *The Religion of Fairness*, so the handoff does not import doctrine into the public root.

24.5.3 — Cross-Reference Without Doctrine

A cross-reference should send readers to the domain that develops the next question: this volume defines the public claim, the religion volume develops optional comparative and spiritual interpretation, and the scientific volume handles empirical adjudication. Naming those destinations does not make any companion title authoritative over the others. A reader should be able to decline any of those handoffs and still understand the argument already completed here. Cross-reference is most useful when it opens a path to greater depth without making another volume necessary to understand the present one.

The boundary is now clear: Traditions can be honored without being drafted as evidence; adjacent ideas can be compared without collapse; interpretation can remain distinct from support; sufferers can be protected from forced meaning; and religious interpretation can remain optional and bounded. That separation gives traditions room to contribute history, vocabulary, practices, and questions without assigning them an evidentiary task they were never designed to perform. It also allows scientific findings

to be discussed seriously without turning them into theological verdicts.

This restraint keeps the book open to readers who do not want an entire religious architecture. The essential rule is enough: spiritual parallels may matter, but they do not prove, own, or command the Law. The same restraint protects religious traditions from being reduced to instruments for validating PTV. A tradition may guide meaning, ritual, community, or moral reflection without being recruited as a scientific measurement tool.

Beauty can feel like validation, science like religious warrant, and institutional plausibility like transition authority. The series must resist those emotional mergers. This Chapter closes with a door, not a demand: readers may continue to The Religion of Fairness for optional sacred development, or remain secular and skeptical with equal standing. That shared footing matters because the root volume is meant to remain a common place of inquiry across belief systems. Readers can disagree about ultimate meaning while still examining the same public claim, evidence standards, and human consequences.

Part IX — Dignity and Responsible Use

Public trust depends as much on the powers an idea refuses as on the claim it advances. Part IX asks how PTV C can remain open to study without becoming proof, permission, or authority over another person. A sentence about terminal neutrality may comfort one reader while giving another an excuse to delay care, assign meaning, or claim access to a life they cannot know.

The governing tests are practical: who carries the burden, can affected people refuse or appeal, and who can stop a harmful use? No person may be reduced to a proof object, spiritual lesson, human-worth score, or source of hidden balance. Meaning remains voluntary, and institutional use requires contestability, correction, and withdrawal.

Responsible use is therefore a translation problem from private idea to public power. The same statement changes character when spoken by a friend, clinician, school, employer, government, religious community, platform, or automated system. Each setting changes the cost of refusal, the persistence of records, and the route for correction. A serious framework follows that chain from claim to actor to decision to remedy rather than assuming that good intentions survive institutional scale. Power also changes what counts as a harmless error: a mistaken private interpretation can be corrected in conversation, while an institutional inference may shape access, records, or reputation long after the original context is gone. The design problem is therefore not only accuracy, but reversibility, traceability, and whether the affected person can reach an independent remedy.

This restraint is constructive. The Law can sharpen attention to hidden burden, privacy, support, and repair without turning those

concerns into eligibility tests or inner-life scores. Institutions can improve access, reduce preventable suffering, protect consent, and correct mistakes through ordinary evidence and public reasons. The aim is durable freedom: keep the question open, the person larger than the theory, and every exercise of power answerable to someone who can refuse, appeal, or stop it. A strong institution should be able to explain what evidence it used, what it refused to infer, and which ordinary service could meet the need without intimate prediction. That discipline keeps innovation focused on reducing visible burdens rather than manufacturing authority over hidden experience.

Scientifically, PTVC remains a candidate complete-stream hypothesis whose strong inferences require a declared representation, endpoint, observation model, latent bridge, reference class, and rival comparison. Ethically, rights, care, privacy, and dignity stand independently of whether the hypothesis is supported, narrowed, or defeated. This separation lets public action improve while the scientific question remains open: a clinic can improve access, a school can reduce coercive pressure, and an employer can correct a harmful process without deciding anyone's terminal state. The public value of the inquiry is strongest when curiosity expands while institutional power remains bounded by evidence, consent, and remedy.

Chapter 25 — Limits on the Idea's Use

Misunderstanding is inevitable; preventable misuse is the harder test. Any ambitious theory will be disputed, and serious criticism often reveals its weakest joints. The decisive question is whether a supporter, institution, family member, teacher, researcher, religious leader, platform, or AI system could excuse harm while claiming fidelity to the book.

This chapter answers that risk before it becomes practice. Its place near the end is purposeful: the reader now knows enough of the claim to understand both its appeal and the power that careless translation could create.

A claim about terminal neutrality can be comforting, yet comfort does not make every use safe. A reader in grief may hear that nothing is finally wasted; another may hear that present pain is less urgent; a third may try to turn a hidden ledger into blame, rank, or evidence. The same sentence therefore needs an ethical form as precise as its intellectual content.

Nothing in the Law of Fairness licenses that movement. It is not a doctrine of deserved suffering, a command to let pain happen, permission to delay care, or a ledger anyone can inspect, administer, punish through, or translate into worth.

The chapter makes the public-use boundary teachable, quotable, and actionable. Instead of adding another wall of warnings, it identifies the actor, record, decision, appeal, remedy, stop power, or withdrawal route that turns each protection into conduct.

The burden-locating question remains the method. Every misuse can be tested by asking who gains freedom, authority, or reassurance and who carries the resulting cost. If an interpretation

calms the responder while the suffering person receives less care, responsibility has moved in the wrong direction. If an institutional tool gains power while the person loses dignity or appeal, the harm has been hidden inside procedure.

The chapter also recognizes that misuse rarely arrives wearing a villain's face. It may sound compassionate, spiritual, scientific, parental, therapeutic, efficient, educational, or civic. It may say that a child needs resilience, that a patient needs meaning, that a worker's distress can be scored, that a public group needs discipline, or that a research study needs intensity.

Boldness and precision are allies here. A theory can state a formal possibility clearly while refusing harmful shortcuts. It can be beautiful without becoming manipulative, and it can inspire inquiry without recruiting suffering as evidence.

No role receives a private exception. Caregivers, researchers, moderators, writers, educators, courts, parents, religions, states, individuals, and artificial intelligence systems are denied the same forbidden power: no one may administer another person's hidden ledger, turn relief into interference, or turn repair into payment.

Every fragment that may travel alone carries its own duty. A caption, chart, guide page, talk, handout, donor sentence, or social excerpt that can be misunderstood needs enough local context to remain accurate; otherwise, it should not be released in that form.

Artificial intelligence deserves separate attention because it can make unsafe interpretation appear neutral, patient, or comprehensive. A system may summarize distress, classify risk,

infer motivation, simulate empathy, or assemble a profile with a confidence score. None of that is access to inner life. A machine output may assist under governance, but it is not a soul, conscience, ledger, or authority over persons (Floridi and Cowls, 2019; Bender et al., 2021; National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

A double safeguard follows. Present artificial intelligence systems may not issue inner-life verdicts about people, and uncertainty about machine consciousness may not become permission to create synthetic suffering for intellectual drama. The responsible posture is accountable human governance, scientific humility, and refusal to treat actual or plausibly felt distress as free material.

The chapter closes with boundaries strong enough to govern action. Nuance remains essential until it would ask a vulnerable person to carry uncertainty for someone else's theory. At that point the book needs a clear stop, a named duty, and a route to remedy.

The scientific boundary is exact. PTVC is a falsifiable candidate boundary hypothesis, not an established fact, clinical instruction, legal doctrine, or moral explanation of anyone's suffering. Its latent object is intrastream, nontransferable, and dependent on declared primitives that no ordinary observer possesses.

Palliative care, research ethics, disability and child rights, measurement validity, privacy, restorative justice, AI governance, and science communication already supply practical

disciplines for unequal power. Their contribution here is to clarify responsible human conduct while the scientific question remains open; none establishes PTVC.

The result is a misuse-and-repair casebook rather than a catalogue of prohibitions. Each section identifies an actor, a tempting interpretation, the burden created, the person with stop authority, and the route of care, referral, appeal, correction, or withdrawal. A boundary becomes useful when someone can act on it.

The hardest reader test is the person who encounters only one fragment. That person should not need the whole series to know that pain is not morally owed payment, relief is not evidentiary weakness, AI is not sovereign, an archive is not a soul, and no institution may turn the idea into eligibility.

25.1 — Care Remains Immediate

A theory about felt experience meets its first public test before anyone measures anything. Someone is hurting, frightened, unsafe, breathless, grieving, or unable to navigate the system in front of them. At that moment, the question is not whether the Law will eventually be supported. The question is whether a person with the ability to help will act.

Care is the first positive architecture of responsible use. Pain requires response; comfort belongs inside the human reality being studied; and the local safeguard must appear wherever an isolated fragment could otherwise invite delay. Palliative and research ethics make the priority concrete: knowledge is pursued within duties of relief, consent, risk minimization, and respect for the person (World Health Organization, 2020; World Medical Association, 2024; CIOMS, 2016).

Many needs resist simple remedy, and care does not promise control over every outcome. Its promise is different: the person in front of us will not be abandoned to theory, spectacle, or a search for cosmic explanation. Competent response begins with what can be relieved, protected, repaired, or honestly accompanied now.

25.1.1 — Pain Requires Response

Palliative care begins from a practical judgment that suffering is not a philosophical placeholder. It is something to relieve through competent attention to pain, breathlessness, fear, communication, family burden, and the conditions that make a remaining day more livable. The World Health Organization describes palliative

care as an approach that improves quality of life by preventing and relieving suffering, while the current Declaration of Helsinki places the rights and interests of human participants ahead of the goals of research (World Health Organization, 2020; World Medical Association, 2024).

Pain has priority because felt experience is the subject of the inquiry. A framework concerned with lived burden contradicts itself when it treats present suffering as background noise, future payment, or material for proof.

This rule protects the theory from one of the oldest abuses of grand meaning: turning suffering into material for someone else's explanation. A person in pain is not an illustration. A child in danger is not a philosophical pressure test. A dying patient is not a terminal-signature opportunity. A traumatized person is not evidence that hidden balance is at work (World Health Organization, 2020; World Medical Association, 2024).

Pain requires response even when the response is imperfect. Waiting for the perfect theory, perfect diagnosis, perfect explanation, or perfect institution often becomes a refined form of neglect. This inquiry asks us to start where burden is already burning.

A useful audit question is whether the sentence in front of us would make response more likely or less likely. If it makes the responder more attentive, quicker to protect, more willing to relieve, and less tempted to explain, it is moving in the right direction. If it makes the responder calmer while the sufferer remains unaided, it has become an aesthetic shelter for neglect.

The Law should leave the hurting person less alone, not more interpretable (Beauchamp and Childress, 2019; O'Neill, 2002).

Research may study how relief changes reports, physiology, or attention, but the proper response is to model those changes, not to preserve suffering for a cleaner signal. International research ethics likewise requires social value, scientific validity, favorable risk-benefit judgment, informed consent, and additional protection where vulnerability is present (CIOMS, 2016).

The same priority binds institutions. A clinic cannot soften its duty to treat through philosophical uncertainty; a school cannot use character language to ignore harm; a family cannot substitute spiritual interpretation for protection; and a research team cannot preserve participant distress for a cleaner signal. The Law should make public delay more visible, not more excusable (United Nations, 2006; World Health Organization, 2020).

Public systems often meet people when bargaining power is lowest. A patient needs medicine, a child needs protection, a tenant needs shelter, a worker needs safety, and a disabled person needs access. If direct help is unavailable, responsibility must still become visible through explanation, referral, stabilization, escalation, record, or honest acknowledgment of what cannot be done. Procedural fog is itself a lived burden (Entwistle et al., 2010).

The test for institutions is not whether they can quote the Law but whether the quote changes burden placement. A benefits office that cites dignity but leaves people waiting has not understood dignity. A hospital that praises holistic care but leaves pain

untreated has not understood experience. A school that speaks of fairness but ignores a frightened child has not understood hidden burden. Public systems prove their language only by making the next humane action easier to reach.

25.1.2 — Comfort Is Not Evidence Weakness

Comfort changes what can be observed because care changes the person's immediate world. Analgesia can alter speech, movement, autonomic activity, sleep, memory, and attention. A quiet room or trusted presence can change fear. These are not contaminants introduced from outside the study. They are part of the causal reality that any honest account of lived experience must include (World Health Organization, 2020; World Medical Association, 2024).

The scientific response is to define the question around ordinary care, record treatment and timing, and distinguish treatment effects from the target of inference. It is not to imagine that an uncared-for person offers a purer signal. Research ethics has long required favorable risk-benefit relations, respect for persons, and safeguards against turning clinical need into investigative opportunity (Emanuel et al., 2000; Faden and Beauchamp, 1986).

Relief does not interrupt the real trajectory. It becomes part of it. If medicine, friendship, public assistance, apology, rest, or justice changes what someone feels, the changed path is the life that occurred. The proposed Law cannot require a person to remain in pain so that an observer can preserve a cleaner hypothetical curve.

That rule is especially important near the end of life. Palliative care is not evidence tampering. It is care directed at comfort,

dignity, and the relief of suffering. A terminal study that depends on withholding ordinary relief has failed before its measurements begin, however technically impressive the resulting data might appear (World Health Organization, 2020; World Medical Association, 2024).

The proper scientific responses are narrower estimands, explicit intervention models, wider uncertainty, or No adequate estimate. None of those outcomes licenses the preservation of avoidable pain for interpretive convenience (CIOMS, 2016).

Consent and refusal remain active throughout this problem. A person receiving care is not a natural instrument whose untreated condition belongs to the investigator. Clinical decisions should remain grounded in the person's welfare, values, lawful representatives where needed, and professional judgment rather than in the theory's desire for a dramatic endpoint (Faden and Beauchamp, 1986; CIOMS, 2016).

Comfort can also improve the honesty of the evidence that remains. A person overwhelmed by preventable pain may lose the capacity to speak, attend, assent, remember, or participate in ordinary relational life. Relief can restore enough room for choice, communication, farewell, and refusal to become possible again.

The decision rule is plain. Where measurement conflicts with comfort, measurement yields. Where public storytelling conflicts with comfort, storytelling yields. Where theoretical ambition conflicts with comfort, ambition yields. Care does not need the

hypothesis's permission, and no result becomes stronger by violating that order.

25.1.3 — Portable Safeguards at the Point of Use

Public ideas rarely arrive in the order their authors intend. A person may encounter a headline before a definition, a diagram before its assumptions, or a clipped sentence before the ethical boundary that made it safe. Communication is therefore an audience-and-context problem rather than a simple transfer of facts from an expert to a passive reader (National Academies of Sciences, Engineering, and Medicine, 2017; Fischhoff, 2013).

Four safeguards travel with high-risk uses: care before interpretation, no human scoring, no cross-person balancing, and no machine ownership of inner truth. Portability means each appears as an operative constraint on the surface or workflow where misuse can occur, not as a distant disclaimer.

A useful recurrence changes what someone can do. Beside a clinical example, it names the duty to relieve and refer. Beside a research proposal, it identifies the stop rule. Beside a public artifact, it fixes the status label and correction route. Beside an automated tool, it blocks a prohibited use and points to human review.

This local form matters for vulnerable readers. A person in crisis should not have to reconstruct from formal chapters that the Law does not blame them. A grieving family should not have to infer that the book assigns no reason for loss. Safety language earns its place when it removes a real interpretive burden from the person least able to carry it.

The distinction resembles good safety practice in other fields. A system is not made safer by posting one distant rule and assuming every later user remembers it. Safety becomes durable when hazards are anticipated at the point of action, authority is assigned, and learning changes the design after failure (Reason, 2000; Dekker, 2014).

The wording should still vary. Repetition becomes wallpaper when it copies the same abstract qualification without adding a mechanism, actor, decision, or remedy. Uncertainty communication is stronger when it states what is known, what remains open, and what would change the conclusion rather than merely sounding cautious (van der Bles et al., 2020).

A fragment that can circulate alone deserves a standalone safety test. A caption, slide, search summary, social post, classroom excerpt, or isolated crop should carry the minimum context needed to prevent the claim from becoming proof, blame, destiny, or permission. The safeguard belongs on the surface that travels, not only in a chapter the audience may never see (National Academies of Sciences, Engineering, and Medicine, 2017; Fischhoff, 2013).

The operational sentence names responsibility. A caregiver relieves and refers. A moderator preserves substantive criticism while removing scoring or diagnostic abuse. A researcher pauses a burdensome procedure and records why. An editor corrects or withdraws a misleading surface. A product team blocks the use, notifies affected people, and opens an appeal route. Repetition becomes safety when it ends in an accountable action.

25.2 — No Punishment, Ranking, or Eligibility

Power changes the risk. A caregiver can impose meaning; a researcher can confuse participation with care; a moderator can turn disagreement into disloyalty; a public writer can omit the sentence that changes the claim; an educator can make a speculative idea feel mandatory; and an expert can speak beyond competence. The same theory therefore creates different duties in different hands.

The root boundaries travel across those roles without becoming a substitute professional code: no punishment for disbelief, no rank based on inferred inner balance, no eligibility test, no hidden dossier, no delay in care, and no authority beyond the speaker's actual role. Rights instruments and research ethics reinforce autonomy, non-discrimination, review, and protection where power is unequal (United Nations, 2006; United Nations, 1989; World Medical Association, 2024).

Each subsection pairs a risk with a positive response. Caregivers respond and refer. Public writers preserve status and context. Researchers, moderators, educators, and technical teams seek the expert lane that can examine what they cannot responsibly decide alone. The purpose is not to make speech timid. It is to keep confidence from becoming power without accountability.

25.2.1 — Caregiver Duties

Caregiving authority is intimate, and intimacy does not make it unlimited. A clinician, parent, relative, teacher, or support worker may know more about a person's circumstances than a stranger does, but that knowledge does not reveal a lifetime cumulative

object. Human-rights instruments protect autonomy, participation, non-discrimination, and the best interests of children without making distress a test of belief or character (United Nations, 2006; United Nations, 1989).

Caregivers encounter the Law where abstraction can wound quickly. A physician, nurse, therapist, social worker, hospice worker, family member, teacher, chaplain, peer supporter, or friend may know important facts about a person's circumstances, yet still lack access to the person's complete inner life. The shared duty is simple: do not make the hurting person carry the caregiver's need for explanation.

A caregiver using this inquiry should become more attentive to hidden burden. They should notice pain that is minimized, shame that is unspoken, fear that is disguised as anger, exhaustion that looks like withdrawal, and dependency that has become humiliating. The lens should sharpen care, not replace care.

Care begins with epistemic humility. The caregiver does not know the person's latent ledger, why an event happened in any final sense, or whether PTV is true. Personal belief cannot create interpretive authority over another person's suffering (Appelbaum et al., 1982; Henderson et al., 2007).

In clinical and family settings, the safest sentence is often not a theory sentence at all. It may be: I am here. We will get help. You do not have to make meaning of this now. Your pain matters. We will not use this idea to explain you away. These sentences are small, but they carry more fairness than a grand explanation offered too soon.

Caregiver duty also includes referral. A friend is not a clinician because they have read this book. A parent is not a trauma specialist because they can speak about hidden burden. A chaplain is not a substitute for emergency care. A researcher is not a therapist by virtue of measuring distress. Knowing the limits of one's role is part of care.

The caregiver's rule is therefore: use the Law only if it lowers burden, preserves dignity, and remains welcome to the person receiving care. If the language increases shame, fear, confusion, pressure, dependency, or delay, stop using it. The person matters more than the vocabulary.

A caregiver can also misuse the Law by making their own endurance appear noble while another person's needs remain unmet. The lens should not train caregivers to romanticize sacrifice, silence, or overfunctioning. Hidden burden includes the burden carried by the caregiver too. Fair care protects the sufferer without erasing the responder. It asks for support systems, respite, referral, and limits so that care does not become another hidden injury (Entwistle et al., 2010; Committee on the Rights of Persons with Disabilities, 2014).

Care cannot become a reward for agreement, disclosure, composure, or participation. Treatment, accommodation, protection, education, and ordinary respect remain tied to need, rights, and the caregiver's actual duty, not philosophical compliance (United Nations, 2006; United Nations, 1989).

Teachers and other custodial adults work in settings where refusal may not feel free. Inquiry may examine assumptions, rivals,

public language, and fictional cases, but it may not require belief, self-measurement, grief disclosure, or personal interpretation. Young or distressed learners need an ordinary right to pass, leave, receive an alternative, or reach qualified support; silence is not consent and composure is not understanding (Beauchamp and Childress, 2019; O'Neill, 2002).

Educational materials must preserve the same boundary when detached from the room. Slides, diagrams, examples, and summaries should not present a ledger as a personal score or hard cases as proof scenes. Accessibility, status, and referral information belong near the point where misunderstanding could alter a learner's action (World Wide Web Consortium, 2024; National Academies of Sciences, Engineering, and Medicine, 2017).

The successful classroom leaves skepticism honorable, nonparticipation ordinary, and personal dignity unchanged. A learner who rejects the Law has not failed a moral test, and a learner who finds comfort has not entered a doctrine. The purpose is sharper thought with less power over the person.

25.2.2 — Public Writer Duties

A public writer controls more than sentences. The writer controls selection, order, emphasis, headline, example, image, caption, and the point at which uncertainty becomes visible. Science-communication research warns that an audience's real question, prior knowledge, and decision context shape what responsible explanation requires (Fischhoff, 2013; National Academies of Sciences, Engineering, and Medicine, 2017). A technically

accurate sentence can still mislead when its framing invites a stronger conclusion than its evidence permits.

Public discussion spaces become ethical sites as soon as the Law enters them. Forums, classrooms, livestreams, comment sections, reading groups, and community channels can support critique and learning, but they can also amplify blame, fatalism, harassment, spiritual pressure, amateur diagnosis, and public scoring.

Moderation should protect people from harmful conduct without protecting the theory from criticism. Disagreement about logic, evidence, definitions, metaphors, falsifiability, writing, presentation, governance, or project choices remains legitimate. Rules may address threats, harassment, doxxing, impersonation, and targeted abuse, but criticism cannot become disloyalty or grounds for lowered standing. Stated reasons and a real appeal route keep safety enforcement separate from ideological enforcement (Society of Professional Journalists, 2014; National Academies of Sciences, Engineering, and Medicine, 2017).

Moderators also carry context duty. Public channels collapse audiences. A technical sentence written for researchers may be read by a teenager in crisis. A spiritual reflection may be seen by a trauma survivor. A speculative statement about terminal neutrality may be quoted without the ethical firewall. Moderation should therefore favor status labels, pinned safeguards, referral information, and channel separation.

Escalation matters. If a discussion reveals self-harm risk, abuse, medical emergency, child danger, coercion, stalking, or other acute harm, the moderator's role is not to continue philosophical

debate. The role is to route toward appropriate emergency, clinical, legal, or platform procedures. The Law should never become a reason to keep danger inside discourse.

Moderation becomes punishment when disagreement changes access, reputation, role, or eligibility without a conduct-based reason. A healthy discussion culture also leaves room for anger. The relevant distinction is not calm versus upset, but critique of a claim versus use of the claim to harm a person (Ecker et al., 2022; van der Bles et al., 2020).

Moderation is part of public stewardship because conversation can become an experiment without consent. A thread may invite people to test the Law on themselves, compare grief, provoke distress, ask whether a death balanced, or turn a vulnerable disclosure into spectacle. The moderator's duty is not to prevent disagreement; it is to prevent the forum from turning participants into materials.

Public writers must imagine high-risk language outside its original paragraph. Would a grieving reader hear care or fatalism? Would a headline imply confirmation? Would a diagram look like a score? A caption, interview answer, donor page, video, or social post may travel farther than the chapter that justified it, so the safeguard must be as portable as the claim (World Health Organization, 2017; World Wide Web Consortium, 2024).

The first discipline is claim labeling. A sentence should remain recognizable as formal, reflective, scientific, speculative, religious, applied, or public-facing. Debt and payment metaphors around suffering should be rejected when they imply moral

obligation, prescribed harm, or reliable reward control. A clearly conditional statement of net-remainder accounting must not be confused with those inferences. A standalone-context review should ask what a supporter, critic, teacher, journalist, or automated summary could make the isolated line appear to authorize.

Public writing should never turn the Law into personal branding, donor pressure, religious recruitment, political loyalty, artificial intelligence authority, or public moral superiority. The theory does not become more true when more people repeat it. Popularity creates responsibility, not evidence (Society of Professional Journalists, 2014).

The public writer's practical rule is to keep context with consequence. A short claim needs a short safeguard; a beautiful image needs a sober caption; a powerful idea needs a nearby status label. Before release, ask whether the object sounds proven, fated, therapeutic, legal, spiritual, or institutional beyond its warrant, whether it invites personal scoring, and whether the safeguard disappears when cropped.

25.2.3 — Expert Referral

Expert referral is not a retreat from responsibility. It is the moment responsibility becomes honest about its limits. A person can care deeply, understand the public hypothesis, and still lack the competence or authority required for a clinical, legal, scientific, pastoral, safety, or technical decision.

The first task is to identify the role actually being played. A friend can listen and help someone reach support. A clinician can assess

and treat within professional scope. A researcher can explain a protocol and its uncertainty. A lawyer can advise on rights and process. A public writer can clarify a claim, but none of these roles expands merely because the Law is being discussed.

Expertise is also local. Knowledge of affective science does not confer authority in palliative medicine, and experience in data governance does not settle consciousness. Collins and Evans distinguish several kinds of expertise precisely because complex public questions reward people who know what they know while recognizing where another discipline begins (Collins and Evans, 2007).

Referral should begin before improvised confidence harms someone. Immediate danger, severe distress, abuse, medical symptoms, suicidal thinking, or risk to another person is not a theory discussion. Safety, qualified care, emergency services where necessary, and the person's local support network take priority. A warm handoff explains why another role is needed, identifies the right kind of help, and remains present long enough to reduce abandonment.

Clinical and research questions belong to the disciplines equipped to answer them. The Law does not decide undertreated pain, delirium, medication, capacity, diagnosis, stream individuation, instrument validity, endpoint definition, missingness, or scientific verdict; those questions go to qualified clinical, methods, ethics, and consciousness experts.

Legal and pastoral questions require distinct routes. Records, discrimination, due process, privacy, appeal, spiritual care, doubt,

and refusal should be handled by qualified authorities who do not turn voluntary meaning into diagnosis or evidentiary endorsement.

Technical and communication risks likewise require engineers, domain experts, privacy professionals, accessibility specialists, affected communities, and independent reviewers who can inspect systems, burdens, and hostile reuse rather than merely admire fluent output.

Children, dependent adults, and people facing retaliation need stronger independent review because assent, dissent, dependency, and practical refusal may be hard to see. Compliance is neither understanding nor consent.

Conflicts must remain visible. Designers, donors, leaders, and interpreters invested in an outcome may be poorly placed to decide pause or withdrawal, so high-consequence disagreement requires independent review.

Concrete triggers keep authority bounded: danger goes to emergency or clinical support, threatened rights to legal or civil-rights review, disputed inference to methods and ethics review, and unsafe systems to technical, privacy, accessibility, and governance review.

Referral has a return path. Specialists answer within their domains, while writers, caregivers, moderators, and institutions remain responsible for the actions and care still belonging to them.

25.3 — The Hidden Object Cannot Be Administered

Human institutions prefer objects they can see, count, compare, and enter into a record. The cumulative phenomenal object proposed by PTVC is not such an object. It is a truth-level construct defined over a complete qualifying stream under demanding assumptions about representation, neutral origin, ownership, routing, exposure, endpoint, and observation. No ordinary observer possesses it.

Measurement can produce estimates for a declared research question, but an estimate does not become an administrative fact. Construct-validity scholarship shows why a score's interpretation must be earned, and due-process scholarship shows why opacity becomes dangerous when a number influences rights, liberty, care, or access (Cronbach and Meehl, 1955; Messick, 1995; Citron, 2008).

At every scale, the same boundary holds. Courts cannot punish through a latent object, states cannot govern populations through it, and individuals cannot convert relationships into private ledgers. The remedy is not a better universal score but refusal, record separation, ordinary evidence, qualified review, appeal, and humility about what remains inside another conscious life.

25.3.1 — Courts Cannot Administer the Cumulative Object

A court can receive evidence, apply law, make findings under a declared standard, and review the conduct of people and institutions. It cannot observe a complete latent lifetime total merely because a party gives that total a name. Construct validity depends on evidence about what an instrument represents and

how an inference behaves; it is not created by legal formality or technological confidence (Cronbach and Meehl, 1955; Messick, 1995; Borsboom et al., 2004).

Courts may determine legal responsibility, liability, guilt, innocence, damages, custody, remedy, procedure, and public consequence. They may not determine terminal neutrality, total felt history, spiritual standing, or hidden moral worth. The Law of Fairness gives no tribunal privileged access to the complete conscious stream.

This boundary does not weaken justice. A court can protect victims, restrain dangerous conduct, require restitution, correct rights violations, and impose lawful consequences. It becomes more reviewable when it acts on evidence, causation, duties, rights, and proportionality rather than on a story about what suffering a person secretly owes.

The prohibited shortcut appears in both directions. Prior pain cannot be treated as payment that reduces the seriousness of present harm, and punishment cannot be increased because someone seems to deserve cosmic balancing. Trauma, poverty, disability, addiction, illness, grief, and recovery are legally relevant only through the ordinary issues for which law makes them relevant, not as proxies for a hidden ledger.

Candidate research indices, archive summaries, and AI outputs are especially dangerous in this setting because numerical form can make contestable judgments look inevitable. Work on technological due process shows how automated systems can obscure the evidence, rules, and routes of challenge needed by

affected people (Citron, 2008; Citron and Pasquale, 2014). A latent, endpoint-dependent, complete-stream object is even less suitable for administration than an ordinary disputed score.

A consequential inference should satisfy a basic reviewability test. The affected person must be able to know what evidence mattered, what rule was applied, what uncertainty remained, and how to contest the decision. No one can meaningfully appeal an object whose bearer, scale, complete domain, endpoint, and measured-to-latent bridge remain unresolved (Tyler, 2006).

The decision should also survive a deletion test. Remove every claim about hidden balance, secret desert, or total inner worth. If the judgment still rests on ordinary facts and law, the metaphysical language was unnecessary. If the judgment collapses, it never had a reviewable legal basis.

Quantification can aid administration, but it can also hide political and moral choices behind apparent precision. Historical work on commensuration shows why institutions must remain explicit about what a number leaves out and who bears the consequences of its use (Porter, 1995; Merry, 2016). The cumulative object cannot repair those limits because it is not an administrable measurement at all.

The Constitution of Fairness develops the detailed legal architecture of due process, evidence, public privacy, review, and remedy. This book supplies the prior rule every legal design must honor: no court may turn the proposed Law into secret sentencing authority, eligibility, rank, or an account of the soul.

A lawful system can therefore be firm without pretending omniscience. It can name wrongdoing without declaring a person reducible to wrongdoing, and it can protect others without treating hidden worth as evidence. The person remains more than the record, and the record remains less than the life.

25.3.2 — States Cannot Administer the Cumulative Object

States often turn complex goals into indicators because indicators make large systems manageable. That convenience creates danger when a measure becomes a target and begins to reshape the conduct it was meant to describe. Campbell warned that the more a quantitative indicator is used for social decision-making, the more pressure there is for corruption and distortion around it (Campbell, 1979). A hidden feelings total would be uniquely vulnerable because neither its origin nor its complete domain is available for ordinary administration.

States can turn interpretation into coercion. A government that claims access to inner fairness can become a surveillance system, welfare score, loyalty engine, risk classifier, punishment machine, public-trust index, or spiritual-political authority. The proposed Law gives the state no such permission.

The state may protect rights, deliver services, enforce law, regulate public power, maintain records, and provide remedy. It may not measure citizens as ledger bearers, rank human worth, infer moral balance, certify trust, or condition care, housing, credit, citizenship, office, or dignity on a Law-facing score.

This boundary applies to attractive technologies as much as punitive ones. A benefits system cannot infer deservingness, a

school record cannot become fairness maturity, a health record cannot become social trust, and artificial intelligence cannot fuse state records into one dossier of inner worth (Scott, 1998; Eubanks, 2018).

The Government of Fairness and *The Constitution of Fairness* place heavier visibility and answerability on public power rather than ordinary private life. Record burden moves toward authority, coercion, allocation, adjudication, representation, and public consequence; it does not move downward as a warrant to make citizens legible as inner lives. More answerable institutions may improve governance without validating terminal neutrality.

The directional test is practical: does visibility move toward power or into private life; does record burden follow authority or spread across citizens; and does the system explain its decision or demand that people become legible as inner lives? A fairer state is more answerable, not more intimate (O’Neil, 2016; Nissenbaum, 2010).

Human-rights and AI-governance frameworks supply concrete reasons for restraint. Disability rights protect equal recognition, participation, access, and non-discrimination; European and Council of Europe instruments impose risk, rights, and lifecycle duties on specified AI uses. None makes a lifetime ledger administrable (United Nations, 2006; European Union, 2024; European Union, 2026; Council of Europe, 2024).

Research institutions remain bound by the same limit: they may estimate, model, compare, falsify, narrow, and report, but may not convert a metric into a person. Apply the record-separation

architecture below before access occurs (National Institute of Standards and Technology, 2020; European Union, 2024; European Union, 2026; Scott, 1998; Eubanks, 2018).

The danger is especially high when research becomes prestigious. A laboratory may begin with humility and later allow funders, media, institutions, or platforms to treat its models as access to hidden reality. The researcher duty is to prevent that conversion before it happens, not to regret it after public systems adopt the score (O’Neil, 2016; Nissenbaum, 2010).

Record separation must exist before access, not as a promise after files have been fused. Health, benefits, education, employment, policing, research, religious participation, donor activity, and ordinary-life records arise under different purposes, access conditions, retention periods, and correction routes. The ability to join them does not create authority to build one portrait of inner worth; purpose limitation belongs in the architecture from the beginning (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Appeal must be capable of changing the consequence. A person needs a plain-language account of the decision, access to the material facts that may be challenged, a human decision-maker with authority to correct or reverse the result, and a record of the remedy. Oversight that can explain but cannot act is ceremonial rather than protective (Council of Europe, 2024; European Union, 2024; European Union, 2026).

25.3.3 — Individuals Cannot Score Each Other

The most intimate misuse of the Law requires no government, court, or software. It begins when one person quietly treats another as a hidden score. A partner, parent, teacher, friend, employer, pastor, clinician, or online stranger may feel that familiarity grants access to the whole life. It does not.

Ordinary social perception is partial even when it is loving. People see behavior, hear selected reports, remember striking episodes, and infer motives through their own histories. They do not observe the complete stream, its neutral origin, its felt intensity, its subjective time, its unreported experience, or its truth-level endpoint.

A hidden object becomes especially dangerous when it is paired with confidence. The sentence I know what you are really feeling can silence testimony. The sentence you have had an easy life can erase pain. The sentence your hardship will make you stronger can turn an uncertain future into permission for present neglect.

Children are especially vulnerable because adults control protection, discipline, education, money, transport, and access to care. No parent or guardian may describe pain as deserved, comfort as weakness, hardship as assigned, or later joy as a promised reward for suffering. Discipline should remain connected to safety, learning, repair, proportion, and the child's developing agency.

Family scoring can also hide abuse. A controlling person may claim to know that another is overreacting, secretly thriving, spiritually tested, or destined to benefit. The Law provides no

language for minimizing coercion, restricting escape, withholding money, or interpreting fear as part of a meaningful trajectory.

Disability, culture, language, trauma, neurodivergence, and masking further weaken confident inference. A flat voice may coexist with intense feeling. A practiced smile may conceal distress. A direct communication style may be misread as hostility. A quiet person may be protecting privacy rather than lacking emotion.

Close relationships still permit moral knowledge. A partner can notice patterns, remember promises, identify harm, and ask for change. The boundary is not indifference. It is the refusal to convert partial knowledge into a total judgment about worth, destiny, hidden balance, or the meaning of another person's suffering.

Workplaces, schools, religious communities, and therapeutic relationships create different asymmetries because private interpretations can alter income, opportunity, support, belonging, treatment, or spiritual standing. Managers may assess demonstrated performance against declared job requirements, teachers may assess demonstrated learning, clinicians may formulate within professional standards, and communities may apply voluntary practices. None of those judgments becomes the complete person. Mood, health, composure, disclosure, digital traces, and project engagement cannot establish spiritual rank or hidden balance; other inferences must remain valid for the specific professional decision rather than become total judgments of resilience, loyalty, maturity, or future promise.

Commensuration can turn varied qualities into one comparable number while erasing the circumstances that produced it (Espeland and Stevens, 1998; Bowker and Star, 1999). Data aggregation magnifies the risk: location, health information, purchases, school records, social networks, and speech may create a detailed portrait that is still wrong, especially when information is repurposed across contexts (Nissenbaum, 2010; Haggerty and Ericson, 2000).

Imagine a worker who stops mentioning exhaustion after a wellbeing survey begins to influence promotion. The next survey looks brighter, while the person has learned to hide more. This imagined case shows what a score can reward: a system may collect more reassuring answers while losing the information that someone needs help. The worker should not have to perform happiness to remain eligible for opportunity.

Self-reflection is different from self-scoring. A journal can help someone ask what is happening, what a pattern protects or displaces, and what might help. It becomes harmful when it claims to estimate lifetime balance, predict compensation, or turn a bad week into debt requiring pleasure or punishment. Reflection remains revisable and compassionate; scoring appoints a judge that follows the person everywhere.

First-person testimony should be respected without becoming infallible, and nonparticipation should remain ordinary. People can misremember or struggle to name experience, yet they remain the primary witnesses to what they report. No partner, family, employer, school, religion, platform, or research project is

entitled to a private ledger, and refusal to disclose implies neither guilt, imbalance, disloyalty, nor lack of self-knowledge.

Accountability does not require total scoring. Lies, broken promises, assault, neglect, discrimination, and misuse of power can be judged through ordinary evidence and standards. The practical boundary is direct: any use that claims authority over another person's hidden worth, balance, trust, guilt, maturity, destiny, or eligibility fails, even when it is private, affectionate, spiritual, or framed as concern.

Individuals may still read one another with care. They may listen, ask, apologize, set boundaries, protect children, recognize danger, and act on credible evidence. The limit is not blindness but gentler inference: partial knowledge may guide a responsible response, but it never becomes ownership of the whole conscious stream.

25.4 — No Institution Owns Inner Truth

Systems can appear to know more than any person could. An AI model can synthesize millions of records, an archive can preserve a life in extraordinary detail, and an institution can combine data, incentives, expertise, and legal authority. None of those capacities creates ownership of inner truth.

Contemporary AI governance places human dignity, accountability, transparency, risk management, and remedy at the center of responsible use, while privacy scholarship warns that combining records across contexts can create a new and coercive representation (National Institute of Standards and Technology, 2023; Council of Europe, 2024; Nissenbaum, 2010). This book adds a sharper limit: no system may infer a person's lifetime balance and then act as if that inference reveals worth, loyalty, personhood, or destiny.

This section separates three questions that public debate often collapses. AI may assist without becoming sovereign. An archive may preserve without becoming the person. An institution may make ordinary decisions under law and evidence without converting the Law into eligibility. Human review, correction, appeal, and withdrawal remain necessary, but they are not permissions to build the forbidden object.

25.4.1 — AI May Assist but Cannot Become Sovereign

Artificial intelligence can search, translate, compare drafts, organize evidence, flag inconsistencies, improve accessibility, and help people navigate a large body of work. Those functions can be valuable, but value does not create authority. The system

remains a tool inside a declared human process whose purpose, data, limits, error modes, reviewer, appeal route, and stop authority are visible (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

The first boundary is simple: no AI system can administer the hidden cumulative object. It cannot read a person's complete felt history, certify terminal neutrality, infer moral worth, determine spiritual status, assign desert, rank trust, or decide whether a life has been fair. A prediction assembled from language, behavior, biometrics, records, and context remains an inference from partial observations, not access to the person's inner life.

This prohibition holds even when the interface sounds patient, kind, and comprehensive. Fluency can make a system appear to understand more than it does, while a calm tone can make an uncertain output feel settled. The more persuasive the presentation becomes, the more clearly the uncertainty, data boundary, and human responsibility must be stated (Weizenbaum, 1966; Bender et al., 2021).

A clinical summary that calls a patient resilient, a school tool that equates quietness with low distress, or an employment model that infers reliability from digital traces may combine many signals and still miss pain, coercion, disability, culture, masking, fear, or the reason for nondisclosure. Once the label enters a record, the person may be forced to disprove an opaque conclusion. The proponent of the inference, not the affected person, carries the burden of validity (Citron and Pasquale, 2014; Green, 2022).

The same rule applies to apparently benevolent uses. A model may help a clinician organize a patient's own reports, help a researcher detect missingness, or help an editor compare a public summary with its source chapter. It may not overrule the patient, erase the participant's refusal, or present its synthesis as a deeper truth than the evidence supports. Assistance ends where illegitimate substitution begins.

Human oversight must be operational rather than ceremonial: a qualified reviewer needs relevant evidence, time, authority to reverse the result, protection from retaliation, and a route for explanation, correction, appeal, and stop authority (Green, 2022; Raji et al., 2020; UNESCO, 2021; Council of Europe, 2024; National Institute of Standards and Technology, 2023). Current AI law and governance frameworks can supply additional institutional duties, but none turns an inferred inner state into administrable truth or gives PTVC a role in eligibility (European Union, 2024; European Union, 2026).

Data use should remain purpose-limited and separated across research, care, employment, education, public power, religion, and ordinary life. Private disclosures are not a free reservoir for model training or profiling. In grief, crisis, or end-of-life settings, AI must not tell a person that suffering will be repaid, death closed an account, or a harmful event served hidden balance; it should route toward care, safety, and qualified human support.

AI can assist literature review, simulation, documentation, accessibility, and reproducibility, but scientific and personal verdicts remain accountable to declared methods, evidence, rival comparison, and human review. A Law-facing system therefore

needs abstention, incident response, meaningful appeal, and a named person with power to inspect, correct, reverse, and stop.

25.4.2 — An Archive Cannot Become a Soul or Person

An archive can preserve letters, photographs, recordings, clinical notes, research data, public statements, creative work, and memories. Those traces can support history, accountability, scholarship, access, and remembrance, but they are selected records created for different purposes, not the person whose life they record. Combining them can erase the contexts that made each record legitimate and intelligible (Bowker and Star, 1999; Nissenbaum, 2010; Solove, 2004).

A model trained on a person's words, voice, images, and preferences may produce a persuasive reconstruction. The output is still a new artifact generated from selected traces, not resurrection, continued consciousness, consent, or proof that the original stream survives (Öhman and Floridi, 2018). Continuing bonds in grief may be meaningful, but no commercial or institutional actor thereby owns identity or gains permission to manufacture a first-person persona (Klass et al., 1996; Neimeyer, 2001).

Consent remains purpose-specific, including after death where applicable. A photograph, research record, or private message does not automatically authorize a conversational replica or unlimited secondary use. Living people need correction when archival portraits are false or decontextualized, and AI-assisted reconstruction should be unmistakably labeled and prevented

from acquiring independent authority. The record-regime rules in Section 12.4.2 remain the governing data architecture.

A persistent archive or model may enter a broader consciousness inquiry, but scale, continuity of identifier, and narrative coherence do not by themselves establish a subject for whom anything is felt. An archive can carry traces with care; it cannot carry the terminal ledger or standing of the person it remembers.

25.4.3 — The Law Cannot Determine Eligibility

Institutions can turn an abstract measure into access, denial, suspicion, or surveillance even when designers call it neutral or deny that it is a score. Health-care and face-analysis studies show how proxy choices and uneven error can reproduce inequality, while employers, insurers, schools, platforms, governments, religious communities, campaigns, forums, and families may all be tempted to rank people through apparently scientific language. These cases do not test the Law; they demonstrate why the answer to Law-facing eligibility must be no before the tool exists (Obermeyer et al., 2019; Buolamwini and Gebru, 2018; Citron and Pasquale, 2014; Eubanks, 2018).

The governing rule is to measure only for a rightful purpose under adequate consent, uncertainty, review, and appeal, and never as human worth. That boundary must be built into the tool: avoid leaderboards, ranks, colors of shame, hidden badges, comparative profiles, and exportable person scores; state purpose and uncertainty; and prevent data from traveling into unrelated systems. A person is not their data, archive, estimator, or public

story, and design should make forbidden scoring difficult rather than merely disclaim it (O’Neil, 2016; Selbst et al., 2019).

Measurement may support science, care, accessibility, self-understanding, or institutional repair when the purpose is rightful and governance adequate. It becomes coercive when refusal brings punishment, suspicion, exclusion, loss of care, loss of standing, pressure to belong, or fear of being treated as less serious. Meaningful consent is specific, informed, proportionate, revocable where appropriate, and non-retaliatory; people should know what is collected, why, who sees it, how long it lasts, whether it can be reused or processed by AI, whether public release is possible, and what declining will change.

Measurement can itself alter experience through anxiety, compliance, performance, shame, resistance, or self-alienation. That burden must count as a cost, not disappear as noise. An applied use is unsafe whenever its claimed authority outruns a necessary permission layer, including evidence status, lawful purpose, domain authority, independent review, safeguard readiness, or public-risk clearance; strength in one layer cannot compensate for failure in another (European Union, 2024; European Union, 2026; UNESCO, 2021).

The deeper problem is abstraction. A model can ignore the social system in which its output acts, treating fairness as a property of a formula while the surrounding institution remains unequal or unreviewable (Selbst et al., 2019). That is why no employer, insurer, school, platform, government, religion, donor, or care system may convert Law-of-Fairness language into eligibility. Human-rights and AI rules can regulate particular systems, but

the root prohibition comes first: no inferred inner balance becomes a gate.

The rule is: no measurement without dignity, purpose, limits, consent, review, and exit. If a tool cannot tolerate refusal, it is not a research instrument. It is a control instrument, and it must not be built in the Law's name.

Kind language does not cure coercion. A family may say it wants understanding, a school support, a platform safety, a researcher knowledge, or a religious community care. The test is whether the person can decline without penalty, whether data are limited, and whether the tool's burden is visible to those imposing it (Raji et al., 2020; Suresh and Guttag, 2021).

Public artifacts require the same pre-release discipline. Reach, ambiguity, vulnerability, cropping risk, and the aura of authority increase exposure; clear status, source traceability, accessibility, hostile-reading resilience, and rapid correction reduce it. A high-reach artifact with weak local safeguards is unsafe even when the full manuscript is careful (National Institute of Standards and Technology, 2020).

A technical graphic should say estimator rather than proof; a balance image should reject person scoring; a terminal graph should preserve comfort first; a religious image should remain optional; and a transition timeline should remain gated and conditional.

Prohibiting a direct ledger score is insufficient if institutions recreate it through proxies such as resilience, readiness, emotional stability, trustworthiness, engagement, wellness, or cultural fit.

Governance must examine purpose, features, error distribution, downstream use, and appeal, while eligibility rests only on lawful, necessary, contestable facts tied to the actual decision (Barocas and Selbst, 2016; Obermeyer et al., 2019; Suresh and Guttag, 2021).

A school may determine whether a prerequisite was completed; an employer may assess demonstrated job requirements; a public agency may verify statutory conditions. None may add speculative claims about a person's complete inner life. When a system cannot separate legitimate criteria from protected need, cultural difference, disability, or unobserved context, abstention is safer than confident exclusion.

25.5 — Restoration, Correction, and Boundaries

Prohibition alone is incomplete. When harm occurs, people still need safety, acknowledgment, treatment, apology, compensation, correction, accountability, prevention, and sometimes separation. Responsible use must reject moral accounting without weakening any of those forms of response.

Restorative traditions help name the difference between repair and payment. Repair addresses a real harm and the relationships, needs, and obligations it created; payment language imagines that suffering can purchase moral balance. Research ethics supplies another boundary: knowledge never justifies deliberate harm or the withholding of needed care (Zehr, 2002; Walker, 2006; World Medical Association, 2024).

No image, spiritual story, scientific ambition, institutional emergency, or promise of world-changing knowledge can override care, consent, dignity, or stop authority. The decisive test arrives where attractive language meets a vulnerable person and restraint must become action.

25.5.1 — Repair Is Not Payment

Repair begins from a fact that does not depend on PTVC: harm changes the world between people. It can create injury, fear, lost time, financial cost, damaged trust, altered opportunity, and continuing risk. A responsible response addresses those consequences because they are real, not because anyone is balancing a hidden account.

Payment language reverses that order. It imagines suffering as a debt discharged, a price demanded by fate, or a quantity that can be offset elsewhere. The Law of Fairness rejects that picture. Pain is not moral currency, and later relief does not retroactively make an earlier wrong acceptable. The conditional change in the net remainder under exact PTVC does not justify injury or establish a method for choosing a future reward.

Repair is forward-facing and relational. It asks what happened, who was affected, what safety is needed now, what the responsible actor can truthfully acknowledge, what restitution or restoration is possible, and what structure should change so the harm is less likely to recur (Zehr, 2002; Walker, 2006).

The sequence matters. Immediate danger is stopped first. Needs and rights are identified before a reconciliation ritual is proposed. Facts are investigated without demanding that the harmed person perform forgiveness. Remedies are funded and assigned. Learning is recorded, and recurrence is monitored.

Restorative practices can help when participation is informed, voluntary, safe, and supported. They can also reproduce pressure when institutions value closure more than the person's refusal. Evidence on restorative justice therefore belongs beside safeguards, selection limits, and continuing access to ordinary legal remedies (Sherman and Strang, 2007; United Nations Office on Drugs and Crime, 2020).

Apology is one possible part of repair, not a substitute for it. A credible apology names the act, recognizes its effects, accepts responsibility without self-exoneration, and connects words to

restitution and changed conduct. Public remorse without altered incentives is performance.

Consequences may still be necessary. Removal from a role, compensation, treatment, supervision, protective orders, disciplinary action, or criminal process can be appropriate under ordinary law and ethics. Repair without accountability can abandon the harmed person, while accountability without dignity can become humiliation rather than protection (Braithwaite, 1989).

No one stream pays for another. A victim's later growth does not clear the wrongdoer's responsibility, and a wrongdoer's private suffering does not compensate the victim. The intrastream nature of the hypothesis makes cross-person moral bookkeeping especially incoherent.

Nor does repair purchase a favorable scientific result. A family, institution, or community cannot point to apology, reconciliation, or later benefit as evidence that the Law has balanced anything. Repair may improve lived conditions. That practical value stands on its own.

Imagine a school mistakenly labeling a pupil uncooperative after an inaccessible activity. Repair would mean correcting the record, restoring the missed opportunity where possible, and making the activity accessible. An apology becomes credible through those changes. The repair record should show the error, protection, responsible actor, remedy, timetable, appeal route, evidence of completion, and any unresolved need. It records what the school

must put right, without claiming to contain the pupil's whole life (Walker, 2006; Zehr, 2002).

The durable rule is simple. Respond to harm because the harm is real. Seek repair because people deserve safety, voice, remedy, and the chance to rebuild. Nothing about that work is payment to the universe, and no beautiful theory may convert it into one.

25.5.2 — No Harm for Proof

The clearest scientific boundary is also the clearest moral one: no person may be made more vulnerable, and no suffering may be inflicted, prolonged, intensified, staged, ignored, or left unrelieved in order to test PTVC. This applies first and most urgently to terminal patients, children, prisoners, students, employees, dependent adults, grieving people, disabled people, poor people, and crisis readers. Respect for persons, beneficence, justice, scientific validity, independent review, minimized risk, and protection of vulnerable participants are conditions of legitimate inquiry, not obstacles to it (National Commission, 1979; World Medical Association, 2024; CIOMS, 2016).

Synthetic and simulated systems require proportionate caution when a design might instantiate morally relevant suffering or train people to create, ignore, or aestheticize harm. Systems built to display distress, generate repeated aversive states, model dependency or terminal conditions, or dramatize machine suffering for persuasion deserve greater scrutiny as their fidelity rises. The Law does not establish that such systems are conscious. Following the precautionary concern in AI-consciousness research, this book rejects creating plausible suffering states for

entertainment, data extraction, or proof-seeking (Butlin and Lappas, 2025).

If simulation is scientifically necessary, its purpose, assumptions, uncertainty, safeguards, and public claim ceiling should be fixed before work begins. Programmed terminal convergence shows what the programmed rules produce; it does not establish the design of the universe, the sentience of the model, or the justification of suffering. The more closely a system models distress, aversion, isolation, reward denial, dependency, or terminal states, the stronger the case for a less burdensome design or abstention (National Commission, 1979; Emanuel et al., 2000).

Protocols should state observable pause and stop conditions, who can invoke them, what immediate care follows, and what evidence would permit continuation. Independent authority must be able to interrupt a procedure without first proving the theory false. The same principle extends, under the protections appropriate to each domain, to animals and humans: greater ambition requires greater care, not broader permission to treat unresolved status as disposability (World Medical Association, 2024; CIOMS, 2016).

The public rule is blunt: do not make something suffer, or plausibly risk making it suffer, merely to make the Law look true. Even if no sentient system was harmed, staging apparent distress can teach that suffering is useful theater. Responsible public work should cultivate the opposite reflex (National Institute of Standards and Technology, 2023; UNESCO, 2021).

The formulation developed here places research under clinical and ethical gates and accepts principled abstention when the target cannot be studied safely. That is not a weakness in the hypothesis. It is a condition of legitimate inquiry. Some questions remain observational, indirect, or unresolved because the direct experiment would be wrong. No elegance, urgency, donor interest, public pressure, or promise of discovery changes that boundary.

The same answer holds when a study seems historic, a pilot urgent, a donor deadline close, or an image unusually powerful. Care delay includes withholding medicine, food, protection, emergency help, or palliative relief, and subtler failures such as postponing referral, correction, removal of harmful language, study suspension, or response to abuse (National Commission, 1979; Emanuel et al., 2000; World Health Organization, 2021).

The Law of Fairness does not require haste in every institutional decision. Some decisions need review. Some interventions can cause harm if rushed. But delay must carry its burden honestly. If delay protects the theory, institution, researcher, leader, donor, public image, or timeline more than it protects the vulnerable person, it is suspect.

When care is needed, theory waits. Pausing interpretation to protect a person is not a retreat from rigor; it is the conduct that makes rigor worth having.

Care delay also includes delayed correction after public misuse. Once a harmful interpretation is visible, an accountable publisher, community, or institution should not wait for perfect messaging

before clarifying. A short corrective statement with accurate safeguards may be fairer than a long review process that leaves vulnerable readers exposed. Public stewardship includes both careful release and prompt repair when release has already caused confusion (National Institute of Standards and Technology, 2023; UNESCO, 2021).

25.5.3 — No Beautiful Exception

Meaning can help people survive, grieve, act, and rebuild, but people revise, refuse, or move among interpretations. That plurality never authorizes an outsider to supply the answer (Park, 2010).

No one must make suffering intelligible for another person's comfort. Art, service, activism, prayer, study, relationship, justice, silence, or memory may carry chosen meaning, but meaning loses dignity when demanded, rewarded as loyalty, or treated as proof of recovery.

The varied grief trajectories discussed in Section 24.4.1 rule out a required stage sequence or endpoint (Wortman and Silver, 1989; Bonanno, 2004; Stroebe and Schut, 1999).

The rule is care first, meaning if chosen. Protection, treatment, truth, accompaniment, accountability, and repair precede elegance, resonance, excitement, utility, comfort, visual power, or mass reach.

Review asks who carries the cost. Testimony needs permission and context, images need claim boundaries and accessibility, and studies need validity, minimized risk, independent review, and

stop authority (World Medical Association, 2024; CIOMS, 2016).

Across the series, no civic image authorizes surveillance, ritual coercion, dashboard ranking, self-enacting legal model, country-ranking peace map, destiny timeline, or scientific figure that converts compatibility into validation.

Public success makes correction harder, so stop authority must be able to pause a study, remove a score, revise a caption, seek consent, protect a nonparticipant, route a claim, or withdraw an artifact (National Institute of Standards and Technology, 2023).

Trust and repetition shape what resists correction; greatness must remain compatible with refusal, correction, and defeat (National Academies of Sciences, Engineering, and Medicine, 2017; Ecker et al., 2022).

Exception requests reveal burden transfer: who benefits, who carries irreversible risk, whether refusal is practical, whether a less intrusive route exists, and what remedy follows failure. Concentrated foreseeable burden cannot be justified by collective hope.

Chapter 26 — Hope Without Certainty

Ideas rarely remain in libraries. Once spoken, this one travels into grief, interviews, classroom summaries, family arguments, headlines, donor pages, social clips, search results, and private midnight interpretations. Public communication is therefore part of the argument itself, one of the first places where burden can be hidden, shifted, exaggerated, repaired, or made reviewable.

Public speech about the Law requires a demanding balance. It should explain why the hypothesis matters without smuggling certainty into a sentence that cannot bear it; allow hope without selling comfort as fact; invite wonder without letting beauty masquerade as evidence; and remain memorable without becoming a slogan that injures the person who hears it alone.

The previous chapter established the boundaries that no use may cross: neglect, pain, coercion, public scoring, spiritual pressure, artificial intelligence authority, and delayed care. This chapter turns those boundaries into affirmative public language. It asks how the book should sound when it meets readers who have not yet encountered the safeguards.

Meaning must also remain free. A person may receive the Law as a source of private hope, intellectual orientation, or spiritual resonance. Another person may reject it, bracket it, ridicule it, study it, test it, or feel no comfort from it at all. Public communication must protect both readers equally. A theory about fairness cannot demand belief as the price of dignity.

Hope carries power. A sentence that comforts one reader may corner another; a line that sounds brave in a public talk may sound cruel at a hospital bed; a statement that feels harmless among

supporters may become dangerous when isolated or repeated by someone seeking permission to ignore pain. The fair question is not only whether a sentence inspires, but where its hidden burden lands.

If rigorous evidence supported PTVc within a declared reference class, the result could be profound: conscious life might exhibit a terminal relation that visible fortune does not reveal. Even then, support would not authorize immediate government, religion, law, scoring, transition, or public practice. Any broader truth claim would still depend on replication, scope, identification, rival comparison, ethical governance, and domain-specific authority.

Public language cannot outrun its weakest supporting status. A formal definition permits conceptual analysis, not empirical proof. A private lens has no force of law; a useful application is not validation; a public-facing object is not canon; an AI summary is not authority; and a donor description is not a scientific verdict. Readers should be able to recover those distinctions from the words that travel.

Serious skepticism belongs within that discipline. Critics help reveal when attractive ideas are becoming harmful systems, when evidence in one domain borrows authority in another, or when presentation hides a burden the argument has not carried. A public culture that cannot receive criticism safely has already weakened its claim to trust.

Short forms create a separate burden. Long-form prose can hold conditionality, rival models, grief safeguards, scientific

uncertainty, and domain boundaries. A headline, clip, diagram, caption, or one-sentence summary cannot carry the same weight unaided. The shorter the material becomes, the more carefully its adjacent context must be chosen.

Supporters can create risk too. Friendly overstatement may travel farther than hostile criticism because it arrives with confidence and shared hope. A supporter may say the Law proves the universe is fair, that pain has meaning, or that research has already confirmed terminal balance. The right response is correction that preserves the human hope while removing the unsafe claim.

Communication is treated here as a form of care. The relevant questions are what a sentence permits someone to do, what it tempts someone to ignore, what it makes easy to isolate, and what authority it appears to claim. Beauty and precision are not opponents. The public voice should be alive enough to remember and exact enough to trust.

Hope and freedom belong together. Real freedom recognizes conditions without turning them into destiny. Honest hope recognizes uncertainty without becoming sterile. When either loses its counterpart, the cost falls on the reader: unbounded freedom becomes denial, unbounded guardrails become fatalism, and unbounded consolation becomes a claim stronger than the evidence.

Many readers meet an idea only in fragments: a cover, chart, quotation, classroom slide, donor summary, isolated crop, or artificial intelligence paraphrase. Those fragments can create hope, confusion, devotion, mockery, or misuse before the full

argument is encountered. Their design is therefore part of public responsibility, not decorative packaging around the work.

Beauty belongs in difficult work because difficult ideas need doors. Its task is to open inquiry, not close doubt. A moving phrase should invite the reader toward greater precision, never make them feel disloyal for asking for evidence, cruel for naming rivals, or spiritually deficient for refusing consolation. The prose may be bright, severe, and memorable, but it cannot become a velvet cover over a missing warrant.

Public trust is also shaped outside the proposition itself. Tone, moderation, images, summaries, diagrams, claims about support, and the treatment of criticism affect whether serious readers experience the work as open or manipulative. Presentation does not prove or disprove the Law, but careless delivery can ask readers to separate a careful idea from an avoidable burden. Responsible authorship reduces that burden wherever it can.

The chapter addresses three audiences at once: the vulnerable reader who may need hope but must not be trapped by it; the skeptical reader who needs evidence boundaries before trust can begin; and the public intermediary who may compress the work for others. The same rule governs each audience: do not make the listener pay for the speaker's need to sound certain.

Hope becomes credible only after it accepts the same outcome discipline as the science. PTVC may be supported, narrowed, model-ambiguous, beyond adequate estimation, or defeated. None of those outcomes changes human worth, and every public steward remains responsible for correcting language that no

longer matches the evidence (Committee on Publication Ethics, 2025).

The chapter draws on research about agency, capabilities, meaning, grief, uncertainty, and science communication. These fields do not settle PTVC. They show how people can remain free, dignified, and informed while a large question stays open (Bandura, 2001; Sen, 1999; Nussbaum, 2011; National Academies of Sciences, Engineering, and Medicine, 2017).

26.1 — Freedom Inside Conditions

Agency scholarship and capability approaches both resist an empty picture of choice. They ask what a person can actually do and be, which options are reachable, what support is available, and whether refusal carries a hidden penalty (Bandura, 2001; Sen, 1999; Nussbaum, 2011). That approach fits the Law's concern with lived experience without turning the Law into a complete theory of freedom.

This section also disciplines the language of candidate mechanisms. A constraint, reserve process, bridge, or completion potential is not a secret agent deciding a person's fate. The reader can take causal structure seriously while refusing fatalism, punishment, and the fantasy of an inner accountant.

26.1.1 — Freedom as Lived Choice

Freedom is not the absence of every condition. People act through bodies, histories, relationships, laws, resources, habits, illnesses, opportunities, and constraints. A serious account of agency begins inside that reality rather than imagining a person untouched by it.

The Law, if true, would propose a condition on a complete stream, not a script for the next decision. A boundary on the whole does not tell a person which conversation to begin, which job to pursue, which treatment to accept, whom to love, what to believe, or whether to read another page.

Formal constraints are easily personified. Language that says the Law needs, chooses, redirects, teaches, or rewards turns a candidate relation into an invisible actor and makes ordinary

choices look preassigned. More accurate grammar keeps agency with people and causal activity with identifiable bodies, environments, institutions, habits, memories, and deliberation.

Bandura's account of human agency emphasizes intentionality, forethought, self-regulation, and reflection within social conditions (Bandura, 2001). Self-determination research similarly examines how autonomy, competence, and relatedness support more self-endorsed action rather than treating freedom as isolation (Deci and Ryan, 2000; Ryan and Deci, 2017).

Capabilities add another public dimension. A person may formally possess a choice while lacking the health, money, transport, safety, education, time, or social permission needed to use it. Sen and Nussbaum therefore ask what people are substantively able to do and be, not merely which options appear on paper (Sen, 1999; Nussbaum, 2011).

Freedom as lived choice includes the quality of the available path. A patient offered treatment without interpretation, transport, or time to decide has less practical agency than the consent form suggests. A worker free to refuse an unsafe task only by losing housing may face coercion through conditions rather than words.

Agency is a relationship between a person and a field of possibilities. Addiction may narrow that field through craving, withdrawal, cue sensitivity, and altered reward learning. Illness may narrow it through fatigue, pain, delirium, depression, medication effects, or cognitive change. Support should restore options through relief, communication, time, and supported decision-making rather than reduce the person to willpower.

Social conditions can also expand freedom. A ramp, interpreter, safe shelter, paid leave, accessible information, reliable transport, childcare, or a trusted advocate can transform an abstract right into a usable option. Repairing the environment is often a more respectful response than lecturing the person about willpower.

Freedom includes refusal. A reader may reject the Law, decline research, leave a religious interpretation, refuse a score, ask for a second opinion, or keep private experience private. Nonparticipation is not evidence of imbalance or fear of truth.

Freedom also includes revision. People change their minds as they learn, heal, age, encounter new evidence, or see consequences more clearly. A theory that treats every prior choice as destiny leaves no room for growth; a humane theory must allow a person to become different without declaring the earlier self fraudulent.

Philosophers of agency distinguish several questions that public language often compresses. Frankfurt emphasizes identification with one's motives, Fischer and Ravizza emphasize reasons-responsive control, and List argues that meaningful free will can exist at the level of persons even within a causally structured world (Frankfurt, 1971; Fischer and Ravizza, 1998; List, 2019).

The book need not settle those debates. It needs only enough discipline to avoid a false inference: a complete-stream boundary, if real, would not by itself abolish reasons, deliberation, learning, responsibility, or intervention. The relation between global constraint and local agency remains a substantive question.

Responsibility should therefore be tied to answerable action rather than cosmic desert. We can ask who knew, who could act, who held power, what alternatives existed, what harm followed, and what repair is possible. Those questions preserve accountability without turning suffering into punishment assigned by the universe.

Institutions should not use constraint language to excuse preventable harm. A school cannot call exclusion inevitable while ignoring access, a hospital cannot call delay part of the path, and a government cannot treat poverty as destiny. Conditions are often where responsibility becomes most visible.

This leaves real room for authorship. A person can accept help, abandon a harmful ambition, rebuild a friendship, or discover an interest they did not have last year. If PTVC is true, those changes are part of the trajectory through which the proposed constraint is realized. If it is narrowed or defeated, their ordinary causes and consequences remain. The whole-life question gives no one a script for the next hour.

26.1.2 — Choice Under Uncertainty

Uncertainty does not suspend ordinary life. People choose treatments, relationships, careers, commitments, and forms of care without complete knowledge of outcomes. Responsible communication helps them distinguish what is established, what is plausible, what is unknown, and what would change the judgment. Evidence suggests that numerical and verbal uncertainty can be communicated without necessarily destroying trust when it is explained rather than concealed (van der Bles et

al., 2020; National Academies of Sciences, Engineering, and Medicine, 2017).

Human choice happens without access to the whole account. No person sees the complete causal network around their life, the future shape of their relationships, the exact felt effect of a decision, or the endpoint structure of consciousness. That ignorance does not make choice meaningless. It is the normal condition under which moral life takes place.

The Law of Fairness, if true, would not give ordinary people private access to terminal balance. A person cannot inspect the ledger before choosing. A family cannot decide that care is unnecessary because balance will close. A researcher cannot infer that a participant's burden is acceptable because the hypothesis predicts eventual neutrality. Uncertainty blocks those shortcuts.

Choice under uncertainty favors reversible, caring, and repairable action. If the next step may harm, choose slower. If the next sentence may be quoted without context, narrow it. If the next intervention collects intimate data, require consent and review. If a person is suffering, begin with relief and protection rather than interpretation (Williams, 1981; Sen, 1999).

Public speech should resist perfect hindsight. After tragedy or success, the idea should not be used to explain why an event had to happen. People act under uncertainty and remain responsible for reducing avoidable burden.

Uncertainty can be freeing when it removes false authority. Because no person can inspect the ledger, no person can tell another person what balance demands. A private reader may

reflect; a researcher may study; a public speaker may explain status. None of them may convert uncertainty into command (Nussbaum, 2011; Bandura, 2001).

The practical standard is not certainty but reversibility and consequence. Where the stakes are high, the person needs time, alternatives, qualified advice, and a route back. Where the choice is private and low-risk, exploration can remain light. Capability approaches sharpen this distinction because a nominal choice is not meaningful when the cost of refusal removes safety, belonging, care, or livelihood (Sen, 1999; Nussbaum, 2011).

Freedom language travels across political, religious, therapeutic, and scientific disagreement. Grounded language says that conditions are real, choices still matter, and care remains required. It avoids puppetry, destiny, hidden administration, and cosmic scheduling (Fischer and Ravizza, 1998; Williams, 1981; Sen, 1999).

The writer's discipline is to test freedom language by its effect on responsibility. Does it make the listener more able to help, repair, refuse harm, and seek care? Or does it make them less able? The answer matters because public communication is not neutral when the topic touches suffering.

Good public language leaves the door open in both directions. It lets readers enter without humiliation and leave without suspicion. Insider phrases such as once you understand or everyone secretly feels this create a social trap by making disagreement sound like confession.

A better public style is invitational and bounded. It can say that the hypothesis raises a difficult question about felt experience and fairness, and that the lens may help reveal hidden burden. It should also say that readers remain free to reject the claim and still participate in care, critique, and burden reduction (Nussbaum, 2011; Bandura, 2001).

26.1.3 — Language Limits for Candidate Routes

The formulation developed here separates the proposed terminal condition from every route that might produce it. That distinction has a direct consequence for public language. A controller, bridge, completion potential, policy-support condition, reserve relation, or endpoint process is a candidate architecture, not a hidden biography of why a person acted.

A route diagram may describe possible state transitions. It does not show that someone consciously sensed a ledger, chose pain for later balance, or was guided toward a particular event. Mathematical arrows connect variables under assumptions; they are not arrows of destiny.

When discussing candidate routes publicly, keep the modal claim exact. A model may show that one completing continuation exists, that completion has positive probability, or that a stated policy closes with probability one; those claims are not interchangeable, and none identifies a person's motive or guarantees a real-life outcome. Section 7.3.3 gives the full reachability ladder.

Reserve language requires equal care. Remaining capacity in a model is not stored suffering, moral credit, future happiness, or a

promise that circumstances will improve. It is a technical relation between a declared state, a horizon, and the movement the model permits.

Endpoint language cannot become prophecy. A truth-level endpoint is a formal boundary whose identification may remain uncertain. Biological death is not automatically that endpoint, and a protocol endpoint is not permitted to stand in for the truth merely because it is observable.

Public examples should therefore translate function rather than personify mechanism. Say that a candidate route constrains allowable paths under stated assumptions. Do not personify life as a moral corrector or claim that particular pains and choices are scheduled without evidence for that route. The equal-and-opposite net remainder follows from exact PTVC regardless of the mechanism.

Intervention language should be equally exact. A safe perturbation can test whether a route signature changes as predicted. It does not test whether a person deserves an outcome, whether suffering was necessary, or whether anyone should manipulate a life to make completion easier.

A useful crop test asks what a single sentence would imply without its equations. If the sentence suggests unestablished hidden motivation, a guaranteed chosen reward, moral instruction, or fate, it needs repair even when the mathematics beneath it is correct.

These restrictions preserve freedom in two directions. They prevent skeptics from being told that every choice was secretly

caused by closure, and they prevent supporters from turning a formal possibility into a command. A candidate architecture can be studied rigorously while the person remains more than the model. That separation lets route research become more ambitious because mechanisms can be tested on their own terms without turning a model into a script for how anyone should live. The gain is precision without surrendering agency: a model can become more testable while its human use remains deliberately modest.

26.2 — Meaning Without Illusions

Meaning can make life more inhabitable, but it becomes dangerous when certainty is manufactured for someone else's comfort. A person may find meaning in the fairness question, in a religious tradition, in skeptical inquiry, in memory, in work, or in no encompassing interpretation at all. The freedom to decline meaning is part of meaning's moral value.

Meaning-making is plural and optional. This Section asks how hope can remain honest without turning grief or recovery into a required narrative (Park, 2010; Wortman and Silver, 1989; Bonanno, 2004; Stroebe and Schut, 1999).

26.2.1 — Meaning as Optional

Meaning is something people make, receive, revise, refuse, and sometimes live without. It can arise through love, work, faith, memory, art, service, protest, family, identity, nature, or the simple decision to continue. No single theory owns that activity.

The word optional is essential because meaning can help and still become coercive. A person in pain may find an interpretation sustaining, while another may experience the same interpretation as denial. Respect begins by asking rather than assigning.

Meaning-making research distinguishes broad beliefs about the world from the interpretation of a particular event. Distress can emerge when the event conflicts with those beliefs, but people respond in many ways, including revising the event's meaning, revising global beliefs, holding tension, or declining a settled account (Park, 2010).

Grief makes the boundary vivid. Some people maintain continuing bonds with the dead, some seek narrative reconstruction, some move between loss-oriented and restoration-oriented coping, and some resist the expectation that grief should follow a single sequence (Klass et al., 1996; Stroebe and Schut, 1999; Neimeyer, 2001).

None of these patterns proves or disproves PTV. They show that human adaptation is plural and that an outsider should not prescribe closure merely because the word appears in a formal model. Formal closure and emotional closure are different ideas.

A person may say that faith helps them trust a larger order. Another may say that the same event destroyed faith. A third may refuse theological interpretation entirely. Each response can be sincere, and dignity does not depend on moving toward one preferred conclusion.

Communities often seek quick intelligibility because uncertainty also hurts witnesses. Phrases such as everything happens for a reason may soothe the speaker while burdening the harmed person. Optional meaning therefore begins with consent and timing: ask whether a spiritual, philosophical, or Law-related frame is wanted, and recognize that safety, food, sleep, medical care, information, silence, or companionship may matter before any story. The person may later revise or abandon the account without being accused of inconsistency or failed faith.

Human interpretation may remain plural. A person can believe that an event was unjust, that love survived it, that the future changed, that no explanation is adequate, and that life still

contains value. None of those meanings should become a moral grade. Finding meaning is not proof of superior character, and failing to find it is not pathology by itself; resilience research cautions against treating one visible response as the universal healthy outcome (Bonanno, 2004).

Therapeutic work should preserve the same freedom. A clinician may explore beliefs, values, identity, narrative, and action when useful, but may not impose the Law as treatment. Meaning can emerge through accountability, protection, art, service, community, or continued living without answering why harm occurred, and growth belongs to the survivor rather than granting advance permission for the conditions from which it emerged.

Religious belief deserves protection from instrumental use as well. A tradition should not be cited merely because it resembles the hypothesis. Karma, providence, grace, fate, repair, and rebirth carry distinct histories and disagreements. Spiritual resonance is meaningful only when those differences remain visible.

Optional meaning is not thin relativism. Some interpretations are harmful because they blame victims, delay care, sanctify abuse, erase evidence, or deny rights. Freedom of meaning does not require institutions to tolerate coercion.

The positive standard is agency with care. Meaning should widen a person's ability to live, connect, choose, remember, and act without demanding false certainty or surrender of evidence. When it narrows those capacities, the interpretation has become a burden.

The durable rule is simple: offer the Law as a question, never impose it as the answer to another person's life. Meaning must remain free enough to comfort, change, disappear, or be refused.

26.2.2 — Grief Without Explanation

Grief does not owe a theory a conclusion. Research has repeatedly challenged the idea that bereaved people pass through one universal sequence or arrive at one required form of acceptance. Grief may surge, recede, coexist with ordinary life, remain quiet for long periods, or return around places, anniversaries, and relationships (Wortman and Silver, 1989; Bonanno, 2004).

The person who died also does not become an example whose purpose the living must discover. The Law of Fairness assigns no reason for a death, no lesson to a family, and no proof that a loss was needed. Explanation hunger belongs to the witness; it cannot be imposed on the person carrying the wound.

Presence often matters more than interpretation. Meals, childcare, transportation, administrative help, listening, ritual, silence, and protection from avoidable demands can make grief more livable without solving it. Care can be useful before anyone knows what the loss will mean.

Meaning remains optional and personal. One person may understand a continuing bond through memory, prayer, service, art, family practice, or conversation with the dead. Another may reject every metaphysical frame. Both can grieve with dignity (Klass et al., 1996; Neimeyer, 2001).

The oscillation between confronting loss and attending to restoration is also ordinary. People may move between sorrow,

work, caregiving, distraction, planning, and moments of pleasure without betraying the dead or proving recovery complete (Stroebe and Schut, 1999).

Support should therefore avoid timetables disguised as wisdom. A sentence such as everything balances can close the conversation exactly when the bereaved person needs room to speak. Even when intended as hope, it may shift attention from the loss to the speaker's need for order.

The bedside and memorial test is concrete. Does the sentence invite the grieving person's own meaning, or does it install the theory's meaning? Does it make refusal safe? Does it leave space for anger, doubt, numbness, relief, unfinished attachment, and contradictory memory? If not, silence may be the more respectful response.

Public artifacts need the same discipline. A quotation about terminal neutrality should never be placed beside an image of a death in a way that implies explanation, compensation, or endorsement. The crop must preserve the distinction between an open hypothesis and the singular reality of the loss.

Professional help may be appropriate when grief becomes entangled with danger, severe impairment, trauma, or other clinical needs. That possibility does not turn variation into pathology. It means support should respond to the person's condition rather than to a theory of how mourning ought to proceed (Bonanno, 2004; Stroebe and Schut, 1999).

Grief can remain unexplained without becoming loveless or hopeless. It can hold memory, continuing bonds, anger,

tenderness, and silence while remaining protected from compulsory interpretation. The book can stay beside the question without using the dead to answer it.

26.2.3 — Skepticism as Dignity

Skepticism is not an insult to a serious hypothesis. It is one way a public claim becomes answerable. Scientific communities depend on organized doubt, criticism, and the possibility that evidence will force revision rather than reward loyalty (Merton, 1973; Longino, 1990).

A skeptic may reject the representation, the neutral origin, the scalar ledger, stream individuation, endpoint identification, measured-to-latent transport, a candidate route, or the universal quantifier. Those objections are not interchangeable. A good response identifies the exact premise under pressure instead of treating doubt as one undifferentiated attitude.

A fair answer restates the strongest version of an objection before responding. That discipline prevents easy victory over a weaker claim, shows readers what evidence would actually matter, and gives critics a correction route when their position has been misdescribed.

No conversion arc is required. A person can read the entire series, understand the model, value its ethical firewall, and remain unconvinced. Comprehension is not assent, and continued participation cannot be conditioned on public enthusiasm.

Partial agreement deserves the same freedom. A clinician may endorse care-before-proof while rejecting PTVC. A statistician may value the measurement distinctions while doubting the latent

object. A philosopher may find the question significant while denying the ontology. Those positions should not be collapsed into support.

Skepticism also protects vulnerable readers from social pressure. A family, religious community, classroom, workplace, forum, or research group should not treat doubt as coldness, disloyalty, lack of hope, or moral deficiency. The right to withhold belief is part of the dignity this book is committed to defend (O'Neill, 2002). A meeting or classroom can protect this right by allowing anonymous questions and nonparticipation without requiring a public explanation.

Public disagreement works best when reasons and status remain visible. State the claim, cite the source, identify the inference, and answer the strongest counterpressure. Avoid prestige appeals, insinuations about motives, or vague claims that experts agree. Trust is better served by showing how a conclusion could change (National Academies of Sciences, Engineering, and Medicine, 2017; Fischhoff, 2013). A correction log that links an objection to the sentence it changed can make that openness visible.

Historical and philosophical work on science also warns against presenting uncertainty as a defect that public relations must conceal. Mature communication distinguishes organized doubt from manufactured confusion and explains why genuine disagreement persists (Oreskes, 2019).

A serious critic can strengthen public work without joining it. Critics can reveal undefined terms, unsupported inferences, weak comparators, inaccessible prose, harmful isolated excerpts, and

institutional blind spots. Their role becomes a safety function when the response is examination rather than punishment. Inviting a critic to review a detachable figure or summary before release can catch distortions that insiders no longer notice.

Skepticism as dignity means the person keeps ownership of belief. The book may ask for careful reading and precise criticism. It may not demand allegiance, punish refusal, or turn uncertainty into a character judgment. A project that can be understood without requiring belief is more credible because agreement remains evidence of persuasion rather than pressure.

26.3 — If the Law Is Supported

A responsible hypothesis should say what would follow if it earned support. Support would remain branch-specific, reference-class-specific, measurement-dependent, and open to replication, rival comparison, and later correction. Within those bounds, it could still be a profound discovery about the structure of conscious life.

The formulation developed here distinguishes exact latent closure from finite-resolution compatibility, target support from route support, and a scientific result from metaphysical or institutional authority. Those distinctions govern every conditional implication in this section.

26.3.1 — Conditional Ontological Meaning

Suppose future research repeatedly supports PTVC within a clearly declared reference class, using validated representation, routing, endpoint, observation, and measured-to-latent bridges. Complete conscious streams in that class would appear to share a terminal symmetry invisible in biography, wealth, duration, or public outcome. That result would give direct empirical support, within the declared class, to the proposed terminal symmetry.

Such support would open a new organizing question across affective science, consciousness research, endpoint modeling, and comparative biology: what lawful structure allows radically different paths to reach the same terminal boundary? It would justify ambitious replication, transport, and mechanism work while keeping the tested modality, population, and error controls

explicit. It would not establish that every possible bearer, world, representation, or branch satisfies the same law.

The finding would concern a boundary property of felt experience. It would not imply equal lives, equal welfare, equal opportunity, equal suffering, or a fair distribution of circumstances. Gross positive and negative variation could remain radically unequal even when a terminal residual meets the prospectively specified condition.

Mechanism would remain open unless a route had independent evidence. Biological regulation, psychological dynamics, endpoint coupling, a global path constraint, or an unknown structure might produce the relation. Discovering a boundary and explaining the boundary are different achievements.

The result would not solve consciousness. Philosophy of mind contains live disagreements about how experience relates to physical processes, representation, and self-models (Chalmers, 1996; Metzinger, 2009). A terminal constraint could pressure some accounts and remain compatible with others.

Nor would comparative evidence become direct access to another subject. Behavior, report, neural structure, and terminal patterns can sharpen inference without erasing the first-person gap captured by the question of what it is like to be another creature (Nagel, 1974).

The bearer question would therefore remain independently gated. Support in declared human streams would not prove that a particular animal, organoid, artificial system, archive, or simulation is conscious. Terminal fit is not a soul detector.

No personal forecast would follow. A supported population-level or model-level result would not reveal that a present injury will be compensated, that a desired future must arrive, or that a dying person has reached balance. The individual hidden path would remain private and beyond administration.

Institutional authority would not follow either. Courts, employers, insurers, schools, governments, religious bodies, and AI systems would gain no right to score, classify, or govern inner life. Evidence about a terminal property does not convert the latent object into an eligibility tool.

The ethical floor would continue unchanged. Care, prevention, rights, consent, privacy, and ordinary justice matter because unequal paths and present harms remain real. Support would deepen the mystery of experience, not make preventable suffering acceptable.

The scientific consequence would be more work. Transport across populations, cultures, developmental stages, clinical conditions, species, and measurement channels would require new tests.

The durable conditional sentence is this: if PTVC is supported under its declared gates, felt experience may possess a terminal structure deeper than visible biography, while mechanism, metaphysical interpretation, bearer scope, personal prediction, and institutional consequence remain open. Precision does not diminish the wonder. It keeps wonder from becoming authority.

26.3.2 — Conditional Series Meaning

The wider series receives a vocabulary and a question, not a scientific verdict. Support for PTVC would strengthen the formal root within its declared limits. It would not by itself validate a mature horizon, authorize a transition, enact a constitution, establish a religion, operate a government, build a city, prescribe private life, or prove an economic path.

The series-wide method is receive, apply, and return. A title receives the root concept and its status ceiling. It applies that concept only within its defined domain. It returns scientific, legal, religious, administrative, personal, transitional, or material questions to the title responsible for that depth.

The Constitution of Fairness remains model law rather than enacted authority. *The Peace of Fairness* must earn its material claims through economics, infrastructure, and peace research. *The Utopia of Fairness* may synthesize a mature conditional horizon, while *The Transition of Fairness* develops sequence, gates, pause, rollback, and failure memory. *The Science of Fairness* handles scientific adjudication.

Receiving a root never eliminates domain-specific burdens. Rights, lawful authority, consent, feasibility, cultural plurality, expert review, evidence, maintenance, and remedies remain necessary even under strong scientific support. Formal truth is not a rollout plan (National Academies of Sciences, Engineering, and Medicine, 2017; O'Neill, 2002).

The same rule protects the series from circularity. An applied proposal cannot validate PTVC because it is coherent or popular.

PTVC cannot validate an application because the application resembles its values. Each direction requires its own warrant.

Readers may encounter the books out of order. A first-use bridge should therefore define the inherited object, name its owner, state the local use, identify the limit, and point back to the specialist treatment. Cross-reading rewards depth; it should never repair missing reasoning.

Public communication should show relay rather than command. The formal root sits at the center of the question, but the companion titles are not subordinate departments waiting for permission. They are distinct forms of responsibility whose limits make the series more credible.

Support would still change the conversation. It would give shared vocabulary greater scientific weight, sharpen the need for careful application, and invite each title to revisit the claims that genuinely depend on the root. It would not excuse a title from rejecting a proposal that fails in its own domain.

Conditional series meaning is therefore both hopeful and disciplined. A supported Law could deepen the relationship among the books while leaving plural belief, private choice, human rights, institutional authority, and the possibility of local error intact. The series becomes more connected without becoming one undifferentiated argument.

26.3.3 — No Immediate Institutional Proof

Scientific support would remain scoped to its reference class, representation, endpoint, measurement bridge, and observation conditions. It would not automatically become a clinical protocol,

legal rule, hiring standard, public program, religious doctrine, curriculum requirement, or AI deployment, and a finding in one population or terminal window could not be transported into a whole-life rule by enthusiasm alone.

Institutions ask questions that science cannot answer alone. A study may estimate model fit; a hospital must decide how to care for a patient, a court how to apply law, a school how to educate without discrimination, and an employer how to justify a job-related decision. Those choices require normative and legal premises no empirical result contains. This book therefore authorizes no government, law, religion, score, eligibility system, surveillance practice, or transition.

Any institutional use would carry a new burden of purpose, jurisdiction, rights, necessity, proportionality, security, privacy, accessibility, affected-person participation, professional evidence, human review, appeal, remedy, monitoring, and stop authority. Clinical decisions about diagnosis, treatment, capacity, medication, or prognosis remain governed by clinical evidence and patient values, while an unobservable latent object cannot ground punishment, eligibility, credibility, or rights (World Medical Association, 2024).

The institutional limits in Sections 25.3.2 and 25.4.1 remain unchanged. Education, employment, religion, medicine, public administration, and AI deployment retain their own domain-specific standards; support for PTVC would not authorize hidden-balance inference, merged dossiers, or automated eligibility.

Benevolent proposals still create power. A nonprofit campaign, school guide, city pilot, donor dashboard, or personalized platform can expose people to stigma, surveillance, coercion, or unreviewable decisions even at small scale. Real choice, minimized data, independent oversight, and an exit that does not reduce access or standing are basic prerequisites.

Institutional proof would also require model recovery and adversarial testing. If the analysis cannot distinguish PTVC from adaptation, selection, observation floors, conditioned bridges, or ordinary control, deployment would institutionalize ambiguity rather than knowledge.

Change control matters because models and definitions evolve. An institution cannot rely on one estimator, then quietly update the representation or endpoint while keeping the same public claim. Versioned objects, revalidation, and withdrawal rules are part of responsible use.

The public should be able to contest an application without mastering the formulation developed here. Notice, explanation, correction, appeal, and human authority must exist at the point where a decision affects a person. A production record behind the system is not enough.

The first institutional implication of promising evidence would be restraint: not paralysis, but a deliberate sequence of validation, rights review, limited and reversible inquiry, independent assessment, and public accountability. *The Transition of Fairness* develops rollout gates and rollback; this book supplies only the prior public boundary.

A concise boundary follows: scientific support can justify more careful research, not immediate power over people. Any later institution must earn its own authority through the evidence, law, ethics, participation, and remedy appropriate to its domain (National Academies of Sciences, Engineering, and Medicine, 2017; National Institute of Standards and Technology, 2023).

The evidence-to-policy path must remain typed. Formal coherence can justify analysis; reliable measurement can justify better studies; repeated rival-discriminating evidence may justify a scientific verdict within scope. None supplies the legal and ethical premises needed to govern people.

Exploratory applications should identify who benefits, who bears risk, who cannot easily refuse, what records are created, and what happens when the system is wrong. They should remain narrow, reversible, incapable of changing rights or eligibility, and subject to transparent failure conditions, independent evaluation, and automatic sunset unless renewed through authorized review. A pilot should expand only after its appeal, correction, and exit mechanisms have been shown to work under ordinary use.

Adverse effects must count against the application even when the theory remains scientifically interesting. If an artifact increases stigma, coerces disclosure, delays care, or encourages personal scoring, the use can fail before the hypothesis is resolved. Application safety and scientific truth are related but separate questions. A stop decision should record whether the failure arose from the scientific claim, the measurement system, or the application so those conclusions do not blur together.

26.4 — If It Is False, Narrowed, or Unmeasurable

The research program includes outcomes that supporters may not prefer. PTVC may be refuted in a prospectively specified class, narrowed to a smaller branch, rendered model-ambiguous, or remain beyond adequate estimation. A book committed to falsifiability should know how to behave in each case before the result arrives.

The answer begins with outcome independence. Care, rights, dignity, ordinary causation, and independently supported knowledge remain. Applied systems remove claims they can no longer support. Public artifacts are corrected, replaced, or withdrawn. Scientific failure becomes information about the world rather than a moral failure of the people who tested the idea.

This posture is not pessimism. It is what allows hope to remain honest. A project that can survive defeat without denial can also receive support without triumphalism. Correction mechanisms from science publishing and risk governance show how public records can change while remaining traceable and accountable (Committee on Publication Ethics, 2025; National Institute of Standards and Technology, 2023).

26.4.1 — Care Remains

If PTVC is false, narrowed beyond practical use, or not identifiable with available methods, the person in pain loses no claim to care. Palliative care, research ethics, disability and child rights, and ordinary protection do not borrow their authority from terminal closure. Pain matters before a formal model, children deserve protection before a research program, and grieving

families deserve kindness before anyone settles a cosmic question. The ethical firewall was placed before the claim because the duty to reduce avoidable suffering arises from the person, the need, the relationship, and the authority already present (World Health Organization, 2020; World Medical Association, 2024; United Nations, 2006).

Adverse outcomes should be named plainly. Refutation excludes a prospectively specified branch under adequate evidence; narrowing restricts what survives; model ambiguity leaves serious explanations unresolved; and no adequate estimate means the prerequisites or observations cannot support adjudication. The last outcome limits judgment rather than inviting broader surveillance: when origin, bearer, endpoint, missingness, or measured-to-latent transport cannot be identified, clinical intuition, family confidence, spiritual impression, or AI synthesis does not become a substitute.

Participants also retain every safeguard promised at consent (Beauchamp and Childress, 2019).

Outcome independence creates common ground across belief. Supporters, skeptics, clinicians, researchers, and public officials can disagree about PTVC while protecting care, consent, privacy, repair, and nonparticipation. A dependency audit should preserve practices grounded in ordinary rights, clinical evidence, accessibility, privacy, or good governance and stop practices that predict compensation, infer private balance, rank commitment, or treat resonance as evidence (Nussbaum, 2011; Sen, 1999).

Research participants are owed a direct account of an adverse outcome, including what was learned, what was not, how dependent public claims will change, and whether data will be retained or reused under the original consent. A failed or narrowed program should freeze unjustified secondary uses, review retention, keep context-specific records separate, and preserve enough audit history to show what happened. Scientific disappointment does not enlarge permission.

Skeptics deserve more than vindication theater. Their criticism should be recorded at the level of the argument, and a correct objection should be credited without turning disagreement into a contest of loyalty. A project that learns from critics demonstrates the openness it claimed from the beginning.

The public sentence should therefore match the outcome: the branch was refuted; the claim was narrowed; the models remain unresolved; or the available evidence cannot adjudicate the target. Plain verbs protect the record better than euphemisms such as evolving understanding when a material claim has changed.

A failed hypothesis can still improve measurement, terminology, protocols, or public safeguards. Those gains survive only if the central claim is allowed to fail rather than return under softer language. Researchers should be free to publish negative, narrowing, or rival-favored results without being treated as disloyal.

Those surviving contributions need independent warrant (World Health Organization, 2020; Beauchamp and Childress, 2019).

Public communication should treat that outcome with respect. A negative result is not a scandal when the method was honest. The scandal would be hiding it, spinning it, or leaving public artifacts to speak as though nothing had changed.

26.4.2 — Applied Systems Must Not Pretend

An applied system that depends materially on PTVC needs a prewritten answer to one question: what changes if the claim narrows, remains unmeasurable, or fails? The answer cannot be rebranding, silent deletion, or a substitute metric that preserves the same authority under a softer name. Responsible risk management requires monitoring, documentation, correction, and retirement when assumptions no longer justify use (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

The first practical step is a dependency audit. Each guide, ritual, lesson, protocol, model, platform, donor statement, training surface, and decision tool is classified as independently warranted, conditionally useful, evidence-dependent, or prohibited. The classification determines whether it remains, is relabeled, requires revalidation, or is withdrawn.

Independently warranted components keep only the authority of their real sources. A privacy safeguard remains a privacy safeguard. A grievance process remains a grievance process. A care practice remains a care practice. None continues wearing PTVC as a badge once the scientific status changes.

Conditionally useful components need a visible premise and a real refusal path. A reflection exercise may remain available when

participants can see what it assumes, what it cannot show, and that declining it carries no penalty. Conditional use becomes deception when the premise disappears from the public surface.

Evidence-dependent components track the exact surviving branch, population, resolution, and uncertainty. A route visualization cannot retain universal language after the route narrows. A research protocol cannot become a public score because one model fit better. Scientific reform depends on transparent methods, model recovery, and the willingness to publish adverse results (Munafò et al., 2017; Nosek et al., 2018).

Prohibited components do not receive a repair-through-renaming option. Human-worth scores, hidden-ledger eligibility, pain-for-proof designs, nonconsensual dossiers, AI sovereignty, and coercive participation remain prohibited even if the theory later gains support. Their defect is ethical and jurisdictional, not merely evidentiary.

Consequential systems carry a stronger duty because affected people may not be free to leave. The institutional limits in Section 26.3.3 remain in force when a system is narrowed or retired (European Union, 2024; European Union, 2026; Council of Europe, 2024).

A status change triggers a decision change. Thresholds, data flows, incentives, access rules, training materials, and downstream consequences must be reviewed. Renaming the tool while preserving the same decision path hides continuity rather than correcting it.

Affected people need direct notice when an obsolete claim touched their data, treatment, education, employment, public standing, or participation. The notice should identify the former basis, the current status, the consequence being stopped or reviewed, and the route to correction, appeal, deletion, or remedy.

Researchers and critics who produce adverse results need protection from retaliation, access loss, or reputational punishment. A system that rewards only favorable findings corrupts its evidence base and trains participants to conceal failure.

A public dependency map can keep the correction reviewable without turning the book into a compliance manual. It records the source claim, dependent object, current owner, affected audience, status change, required action, completion evidence, and any surviving independent warrant.

Failure memory completes the process. A retired object remains traceable as superseded or withdrawn, the reason is recorded, and the failed control informs later design. Open science treats correction and reproducibility as part of knowledge rather than embarrassment to be hidden (Open Science Collaboration, 2015; Popper, 1959). Applied systems stop pretending when their public behavior changes with the evidence.

26.4.3 — Public Correction

Decommissioning should be planned before an applied system is launched. The plan inventories every dependent artifact and workflow, freezes new uses, preserves the audit record, notifies affected people, removes unsupported claims, migrates

independently useful components under accurate rationales, and verifies that no proxy score continues the same consequence under a softer name (National Institute of Standards and Technology, 2023; National Institute of Standards and Technology, 2024).

Retirement is demonstrated by consequence, not label. The unsupported decision path has stopped, affected people can obtain remedy, the public source is corrected, and later reviewers can trace what changed. If data, thresholds, incentives, or eligibility effects continue unchanged, the system has not been retired.

Correction is the public form of intellectual responsibility. It identifies the former claim, revised status, reason for change, affected materials, surviving value, and replacement source. Publication-ethics guidance distinguishes correction, retraction, and other notices according to the seriousness of the problem while preserving traceability rather than pretending the earlier version never existed (Committee on Publication Ethics, 2025).

A correction pathway should exist before error occurs. Every public channel needs an accountable owner who can halt distribution, replace summaries, update translations and AI knowledge bases, notify partners, label archives, and make the corrected language easier to find than the error. A private file or obscure note is insufficient when the original claim traveled publicly.

Corrections should be specific, prompt, emotionally plain, and proportionate. They should not punish readers for trusting prior language, blame critics for noticing the problem, or hide behind

technicalities. Science-communication research supports visible uncertainty and revision rather than confidence theater (National Academies of Sciences, Engineering, and Medicine, 2017; van der Bles et al., 2020; Ecker et al., 2022).

A useful notice reaches every dependent surface, including manuscript, website, diagram, caption, social post, classroom handout, donor summary, search snippet, AI output, and translation. It states what was unsafe or inaccurate, what the current status is, and what readers should now do differently (O'Neill, 2018; National Academies of Sciences, Engineering, and Medicine, 2017).

The correction record is a trail of answerability, not a soul record, loyalty record, or public rank. It should preserve surviving knowledge without turning repair into spectacle. A failed route may leave measurement work valuable, a defeated universal branch may leave a narrower relation, and a withdrawn public claim may leave independent ethical safeguards intact.

26.5 — Public Courage and Correction

Public speech asks for two virtues that can look opposed: courage and restraint. Courage names the question clearly, invites capable criticism, and refuses to hide behind vagueness. Restraint keeps confidence proportional to warrant, protects vulnerable readers, and leaves a visible path for correction.

Public communication is part of the method because context and detachment change interpretation (Fischhoff, 2013; National Academies of Sciences, Engineering, and Medicine, 2017; van der Bles et al., 2020).

The final movement of the Part therefore makes correction public. A changed claim, replaced image, withdrawn caption, or revised translation should leave a record and point to the current version. Correction does not diminish public courage. It shows that no artifact is more important than the truth or the people who may be affected by it.

26.5.1 — Confidence Must Track Status

Confidence is a feature of delivery; warrant is a feature of the claim. The two should not be confused. A speaker can be clear and calm while saying that a result is preliminary, model-ambiguous, limited to a reference class, or not yet estimable. Communication research emphasizes matching the message to the audience's decision and stating the uncertainty that actually matters for that decision (Fischhoff, 2013; National Academies of Sciences, Engineering, and Medicine, 2017).

Serious skeptics strengthen a healthy review culture by testing the points supporters are most tempted to protect. In public response, restate the objection accurately, name the claim's status, identify the evidence that would resolve it, and record any correction. Answer criticism of the argument with reasons, review harmful wording on its merits, and handle abuse through conduct rules. Neither agreement nor disagreement measures intelligence or virtue (Fiske and Dupree, 2014; van der Bles et al., 2020).

The most credible public answer may therefore be smaller than the most memorable one. 'The model is formally specified' is different from 'the Law has been shown.' 'A route fit these data better under the prospectively specified comparison' is different from 'the universe balances experience.' Clear uncertainty can sustain trust when it is specific, explained, and paired with an update path (van der Bles et al., 2020).

Expertise travels with borders. Mathematics may establish formal coherence without clinical advice; palliative expertise may establish care safeguards without validating the Law; religious scholarship may discuss resonance without doctrine; legal scholarship may discuss rights without enacted authority. The public voice should name the lane rather than borrow the résumé (Fischhoff, 2013; National Academies of Sciences, Engineering, and Medicine, 2017).

Unverified prestige claims create a special burden. If a public statement says that experts, institutions, donors, journals, universities, countries, scientists, religious leaders, or public figures support the research program or series, the claim must be verified, dated, qualified, and removable. Prestige can help

readers decide what deserves attention, but it can also substitute for evidence. Readers must not be asked to believe because an impressive name appears nearby.

Audiences readily transfer credibility across domains. A scientist speaking beautifully can sound like scientific validation; a lawyer praising the ethical firewall can sound like legal readiness; a religious leader finding resonance can sound like required belief. Role labels and permission for partial agreement protect both the public and the expert.

Expert spokesperson limits also protect experts. No one should be pressured to defend parts of the series outside their competence because they support one domain. A palliative clinician may affirm care before proof without endorsing the formal Law, and a data-governance expert may affirm privacy safeguards without accepting any proposed research index. Credibility grows when partial agreement remains partial.

The founder or authorial voice also needs limits. The formal-law author may define terms and explain intent, but not overrule science, law, medicine, religion, government, or transition review by charisma. The more central a voice becomes, the more visible its self-limitation must be (Fiske and Dupree, 2014; van der Bles et al., 2020).

A speaker's duty is not to win every question. Sometimes the strongest answer is: this belongs to another title, another expert, or a later review. That answer protects both the listener and the claim from borrowed authority.

26.5.2 — Detachable Excerpts Need Their Own Ceilings

Most readers will not meet the Law first in a complete chapter. They may meet a headline, quotation, search result, cover line, caption, diagram, classroom slide, interview answer, donor summary, translation, social post, or AI-generated explanation. Each fragment becomes a small publication with its own audience and its own risk of overclaim.

A detachable excerpt therefore needs an explicit claim boundary. It may simplify vocabulary, but it may not state more than the source passage states. It may shorten an argument, but it may not remove a condition that changes the meaning. It may invite attention, but it may not turn a question into a verdict.

The minimum context depends on the object, but a high-risk fragment usually needs five recoverable elements: claim status, source passage, reference class or intended audience, the difference between latent target and observed evidence when relevant, and the nearby human safeguard. Not every element must occupy the headline, but none may disappear from the governed artifact.

The first test is isolation. Place the excerpt on a blank page without the surrounding chapter. Would an ordinary reader infer that PTVc is established, that suffering is repaid, that the book predicts a person's life, that a score exists, or that an institution has authority to act? If so, the fragment is not safe enough.

A public interview answer should match the question asked. When a journalist asks whether the Law is true, the answer should not hide behind a cloud of applications. The honest response

identifies the current status, the strongest test, and what would count against the claim. Directness builds trust more effectively than prestige language (Fischhoff, 2013; Fiske and Dupree, 2014).

A social post must carry its own boundary because reposting separates it from the thread. The status should appear in the post or image itself when omission would materially mislead. A reply several comments later is not an adequate safeguard for a claim likely to travel alone.

AI-generated summaries require the same discipline. A plausible explanation can quietly change “one separately routed stream” into “every person’s life,” “terminal residual” into “happiness score,” or “hypothesis” into “law of the universe.” The source should provide compact canonical language and machine-readable correction cues, but each generated output remains a new public object requiring review.

Accessibility is part of the claim boundary. Essential meaning cannot live only in an image, color contrast, decorative symbol, or visual arrangement. Alt text, captions, reading order, contrast, and plain-language support help ensure that the safeguards reach readers who encounter the artifact differently (World Wide Web Consortium, 2024).

At the point of contact, the rule is practical. An image mentioning terminal neutrality should say hypothesis; a guide page mentioning suffering should preserve care before proof; and an excerpt that sounds like a score should be rewritten. Public

objects also need current versioning so readers can distinguish active explanations from superseded ones.

Detachable artifacts require their own version and audience review because a narrowed claim, corrected source, retired mechanism, or mistranslation can change what a caption, slide, video, post, or AI-generated summary may safely say. Review should include people likely to bear a misreading and should ask a simple standalone question: can a reader identify the claim's status, its main prohibited inference, and the current source without reconstructing the whole book? (National Academies of Sciences, Engineering, and Medicine, 2017; Ecker et al., 2022; van der Bles et al., 2020).

A short introduction can still invite curiosity: “Could the complete story of feeling have a structure that a biography cannot show? In its main additive version, the Law of Fairness proposes that each complete conscious stream covered by the model has exactly equal, finite totals of positive and negative feeling. Each stream is counted separately, over its full qualifying history before an independently defined ending. The scientific task is to discover whether nature meets those conditions. Care and human dignity remain fully in place.” Keep a compact release record with source, claim status, accountable owner, audience, prohibited reading, accessibility, correction trigger, and withdrawal route. Translations and automated summaries should preserve the same claim and its boundaries.

26.5.3 — Correction and Withdrawal Are Signs of Strength

Trust depends partly on what a public project does after an error is found. Correction begins with specificity: identify the affected sentence, figure, citation, translation, dataset, or artifact; state what was wrong; give the replacement; name the effective date and accountable owner; and say whether the prior object is corrected, superseded, suspended, or withdrawn. The response should match the consequence, from a simple erratum to immediate containment of an artifact that delays care, ranks persons, invites unsafe testing, or misstates scientific status (Committee on Publication Ethics, 2025).

Propagation matters because the same claim may exist in a manuscript, website, slide deck, translation, screenshot, search cache, or AI retrieval system. The production record should identify dependent surfaces, including translations and generated summaries, and make the current status accessible to people who saw the original claim. Scientific changes such as a narrowed reference class, failed route, improved measurement model, or rival-favored result must propagate in the same way.

Failure memory should record how the error arose, why review missed it, which warning signs appeared, and what control changed. The record exists to improve source selection, drafting, translation, design, monitoring, and response, not to humiliate the person who reported the problem (Reason, 2000; Dekker, 2014).

Withdrawal is not censorship when an author or institution removes its own unsupported or unsafe artifact while preserving notice and history. It is an exercise of responsibility over what

remains publicly available. Other people remain free to criticize, quote within law, or analyze the record. A replacement notice should remain easy to find wherever the withdrawn object was likely to travel.

The durable sequence is detect, assess, contain, correct, propagate, document, review, and learn. Each stage should preserve dignity, evidence, accessibility, and the public's ability to distinguish the current claim from the superseded one (National Institute of Standards and Technology, 2023; Raji et al., 2020). Completion is measurable when the corrected source, dependent artifacts, and affected audiences have all received the updated status.

Correction and withdrawal are signs of strength because they show that truth and people matter more than the appearance of never having erred. Hope does not need immunity from revision. It needs a form that can survive being answerable. A project that can visibly revise itself gives readers a reason to trust its future claims without pretending that earlier wording was infallible.

Part X — Living with and Testing the Law

A large question eventually reaches the kitchen table. After definitions, formal objects, measurement problems, endpoint discipline, rival theories, and ethical limits, the reader still wakes to an ordinary day. The alarm sounds. A body is tired or rested. A message brings relief or dread. Work waits. Someone needs care. A pleasure is offered. A craving returns. A plan succeeds, fails, or remains uncertain.

Part X asks what becomes of the Law in that day. The answer cannot be a private accounting system. No reader possesses the complete latent object described by the formulation developed here, and no sequence of mood ratings can reveal a lifetime endpoint relation. The question may enter life as attention, vocabulary, and long-horizon reflection, but not as a score, forecast, moral debt, or rule of personal worth.

That distinction makes room for practical value without laundering it into evidence. A person can notice that hunger sharpens conflict, that an endless feed consumes sleep, that pleasure and fulfillment are different, or that shame makes recovery harder. These discoveries can improve a life even if PTVC is eventually narrowed or defeated. They stand on ordinary causation and established research, not on the truth of the Law.

The first chapter therefore follows a day through attention, pleasure, moderation, addiction, work, ambition, relationships, grief, rest, joy, repair, and help-seeking. It does not prescribe a lifestyle. It shows how the question can sharpen observation while leaving practice voluntary, private, and subordinate to care.

Brief reflection may be useful. A compulsory balance audit is not. A goal may preserve possibility. A dream is not a promise. Moderation may protect future choice. Pleasure is not moral debt. Recovery may reopen paths. Relapse is not punishment. Repair may require apology, restitution, and changed behavior. Suffering does not settle a moral account.

Detailed guidance on ordinary embodied and relational life belongs beyond this root volume. Here the purpose is narrower: show how PTVC can inform reflection without becoming a score, forecast, lifestyle rule, or substitute for ordinary evidence about health, relationships, work, grief, and daily practice.

Chapter 28 changes scale from private reflection to a scientific program. A confirmatory test of PTVC requires representation, bearer, endpoint, observation, estimand, estimator, rivals, ethics, and verdict rules to be specified before the decisive outcomes are examined. A near-zero measured band is not exact closure, a favorable fit is not transport, replication is not universality, and a study cannot define its bearer or stopping point by the result it hopes to find.

The program must also be allowed to abstain. If the neutral origin, endpoint bridge, terminal observation, or model-recovery test fails, the correct outcome may be No adequate estimate. Strong rivals must be resourced to win, and human participation remains voluntary and separable from reading or belief. The aim is not to protect PTVC, but to build tests capable of discovering that it is wrong.

Chapter 27 — Fairness in an Ordinary Day

An ordinary day contains more theory than it appears to. Before breakfast, a person may choose whether to open the phone, eat before a hard meeting, answer a craving, continue a project, apologize, rest, celebrate, or let grief remain wordless. Small choices can redirect an entire day.

The Law enters carefully. No reader can observe or manage the complete phenomenal measure named by its formal claim. What can be observed are local patterns: attention tightening, relief arriving, habit becoming automatic, options narrowing, trust changing, sleep disappearing, and support making action possible.

Local patterns and the terminal law remain distinct. Reward learning can explain craving, sleep loss can explain cognitive strain, social support can alter stress, and goal structure can change performance. These findings matter without PTVc and do not prove it.

Attention comes first because noticing can help while counting can become surveillance. A reader may discover that a setting, body state, relationship, or hour changes experience without turning the day into a number.

Reflection may be written, spoken, spiritual, secular, embodied, artistic, or absent. It belongs to the person, not to a partner, teacher, employer, platform, research team, or project. The right to stop matters because rumination can narrow rather than clarify (Nolen-Hoeksema et al., 2008; Watkins, 2008).

Pleasure needs rescue from suspicion. Joy creates no moral debt; its felt contribution still enters the balance required by exact

PTVC. Reward science distinguishes liking, wanting, learning, and relief; well-being research distinguishes pleasure, fulfillment, and meaning (Berridge et al., 2009; Ryan and Deci, 2001). This vocabulary supports better questions without making happiness dangerous.

Moderation protects capacity and choice rather than purity or evenness. It can preserve sleep, health, money, attention, trust, and another path tomorrow. The contrast is between pleasure that remains one option and a demand that becomes the only vivid one.

Addiction makes the cost of narrowed optionality concrete. Reward learning, cue-triggered wanting, stress, habit, and relief can make alternatives technically present but practically remote (Tiffany, 1990; Everitt and Robbins, 2016; Koob and Volkow, 2016). Section 27.3 turns that mechanism toward treatment, support, environment, and the reopening of paths rather than repeating it as a verdict about the person.

Work and ambition show the opposite danger: a wider future can still become a single consuming track. Goals and implementation plans can organize effort, while health, money, institutions, timing, relationships, opportunity, and luck shape what effort can achieve (Locke and Latham, 2002; Gollwitzer and Sheeran, 2006). Section 27.4 protects aspiration without turning any dream into destiny or any failure into rank.

Relationships shape the paths around a person. A friend can provide a ride to treatment, a partner can make study possible or

unsafe, and a caregiver can preserve dignity while carrying too much. Support is part of the causal architecture of possibility.

Repair stays concrete. Apology, restitution, changed behavior, boundaries, protection, and record correction matter because they change what harmed people must carry. Guilt or later suffering does not substitute for repair.

Rest, joy, and celebration belong here because felt life is not only burden. Sleep protects capacity, but rest need not justify itself as productivity. Joy can matter before it builds anything, and celebration can honor a good without denying grief elsewhere.

Grief receives the opposite protection. It need not produce a lesson, stage progression, spiritual certainty, or eventual equivalence. People oscillate, remain numb, continue bonds, recover routines, and return to sorrow differently (Stroebe and Schut, 1999; Bonanno, 2004). Meaning may emerge but cannot be required.

The final Section marks where ordinary use becomes harm: do not seek pain, delay care, score a person, forecast events, pair joy with suffering, or call private reflection evidence.

27.1 — Attention Without Accounting

Attention is the smallest honest use of a large idea. It can notice that a body, room, hour, relationship, or repeated cue changes experience without claiming to see the complete trajectory. The aim is not to reconstruct the hidden ledger. It is to become less careless about what is already happening.

This section protects attention from three distortions. Numbers can turn noticing into false measurement when they claim to represent fairness, balance, or worth, although a bounded symptom scale, sleep log, medication record, or craving note may still serve ordinary care. Social pressure can turn reflection into confession. Repetition can turn curiosity into rumination. The safer form remains descriptive, optional, private, and easy to stop.

27.1.1 — Notice Patterns Without Creating Scores

Attention can become useful before it becomes numerical. Experience-sampling research shows that people can report moments, settings, activities, and felt states while still revealing fluctuation, missingness, reactivity, and the limits of self-report (Csikszentmihalyi and Larson, 1987; Stone and Shiffman, 1994; Shiffman et al., 2008; Killingsworth and Gilbert, 2010). Its lesson for ordinary life is modest: context changes experience, and careful noticing can uncover a pattern without pretending to summarize a person.

Consider an ordinary afternoon. Meetings after three o'clock feel heavier, hunger makes criticism sharper, and a ten-minute walk changes the next conversation. The useful discovery is not a daily

score. It is a relationship among time, body, setting, and what becomes possible next.

A pattern becomes clearer when its unit is named. The question may concern one conversation, a recurring hour, a bodily state, a place, or the transition between tasks. A broad question such as whether the day was good invites a verdict. A narrow question such as what changed after eating, resting, or leaving the room invites an explanation that can be revised.

Reflection works best through questions rather than judgments. What burden did I overlook? Where did relief become possible? What care did I postpone? Which boundary would make the next hour safer? These questions return attention to a concrete scene. They do not ask whether anyone is balanced, deserving, advanced, or close to a hidden endpoint.

Numbers are not forbidden when their purpose is bounded. A pain rating, sleep estimate, medication log, craving note, or budget may support care, communication, accessibility, or a specific decision. The line is crossed when a record claims to reveal lifetime balance, human worth, destiny, or a private neutrality calculation. A journal becomes research data only through a separate governed protocol.

That distinction reflects a basic lesson of measurement. Validity concerns the interpretation and use of a measure, not the presence of digits or the sophistication of a device (Cronbach and Meehl, 1955; Messick, 1995). A precise number attached to the wrong meaning can mislead more efficiently than an honest description.

Bodily signals also need context before they become moral stories. Fatigue may feel like hopelessness, pain may arrive as irritability, and autonomic arousal may be interpreted as danger. Contemporary affective neuroscience treats bodily prediction and interpretation as part of emotional experience, which is one reason to examine sleep, hunger, illness, medication, and setting before judging the self (Barrett and Simmons, 2015).

Attention includes relief as fully as burden. The cup of tea, the medicine taken on time, the quiet walk, the laugh, the safe boundary, the meal, the apology, and the rest accepted are not decorative details. A practice that searches only for pain would distort the very field it hopes to understand.

Every private account remains revisable. The person may first blame meetings and later discover that the pattern appears only after poor sleep, or may blame lack of discipline and later recognize an inaccessible environment. Revision is not failure. It is the movement from impression toward a more discriminating and merciful account.

Attention also needs a mercy limit. Trauma, anxiety, shame, illness, religious pressure, family pressure, or perfectionism can turn inward observation into another source of distress. In those conditions, fair attention may mean looking less, eating, sleeping, walking, calling someone trustworthy, or letting professional care carry what private analysis cannot safely hold.

The Law adds a reason to take felt patterns seriously while withholding a false verdict. The complete latent object remains unavailable to the reader, so private attention may guide care and

choice without becoming an inner balance sheet. The practice succeeds when it helps something real become more visible and then allows ordinary life to resume.

27.1.2 — Reflection Is Optional and Private

Reflection is one possible tool, not a duty of belief, disclosure, or self-improvement. Research on autonomy distinguishes freely chosen engagement from controlled compliance, and that difference matters when the subject is a person's own inner life (Deci and Ryan, 2000; Ryan and Deci, 2001). A practice that must be performed to prove seriousness has already changed from invitation to pressure.

The reader owes the theory nothing. A person may understand the question, doubt it, reject it, use one safe idea, return years later, or leave it entirely. Refusal does not make anyone less moral, less thoughtful, less hopeful, or less fair.

Privacy means more than keeping a secret. Information should remain within the setting and purpose for which it was created. A journal entry written during grief, a voice memo after conflict, or a private search should not migrate into a classroom, employer file, public archive, research dataset, or model-training corpus merely because a platform can move it (Nissenbaum, 2010; Solove, 2006).

A private note may contain another person's life as well as the writer's. Reflections about a partner, child, clinician, coworker, or family history include people who never agreed to become part of a theory. One person's choice to share cannot erase the privacy interests of everyone named or implied.

Optionality includes the right not to begin, the right to stop, and the right to choose another form. Walking, prayer, music, conversation, art, silence, ordinary work, and no deliberate reflection at all can each be legitimate. A writing exercise should never become the hidden price of serious membership in the question.

No institution may demand private reflection as evidence of loyalty, health, progress, readiness, belief, or civic worth. A school, workplace, support group, religious community, app, research team, or public project may offer a bounded activity, but the person must be able to decline without penalty, suspicion, or loss of belonging.

The simplest privacy image is a closed door that the reader controls. The Law may enter as an invited question, never as a camera, credential, or claim on the contents of the room.

27.1.3 — Stop When Reflection Becomes Distressing

Reflection can clarify a pattern, but it can also become rumination. Rumination research describes cycles that intensify distress, narrow attention, and repeat the same unresolved material without producing a workable next step (Nolen-Hoeksema et al., 2008; Watkins, 2008). The practical sign is often not depth, but shrinking freedom.

The stopping threshold should be observable. Reflection has become a burden when sleep is lost, ordinary pleasure becomes suspect, checking feels compulsory, shame increases, disclosure feels demanded, care is delayed, or relationships begin to revolve around the theory. A difficult moment is not automatically

harmful, but a repeated pattern of distress or impairment is enough reason to pause.

A pause can be small and early. The reader may shorten the exercise, remove reminders, close or delete the record, replace writing with rest, talk with a trusted person, or set the question aside for a season. Stopping before a crisis is a sign that the practice remains under human control.

If reflection begins to reactivate trauma, panic, self-harm urges, psychosis, severe depression, or danger to anyone, the next step belongs to safety and qualified support. Trauma-informed guidance emphasizes choice, trust, collaboration, and avoidance of retraumatization rather than forcing disclosure in the name of insight (SAMHSA, 2014). The goal is to restore enough stability that reflection becomes optional again rather than something the person must complete.

Private pattern notice may still be useful. It can help someone recognize a cue, explain a need, change a routine, or seek care. That practical gain does not turn the exercise into evidence for a complete-stream terminal hypothesis, and distress buys no scientific advantage. A useful note should therefore end in a concrete option—rest, a conversation, a changed cue, or a request for help—rather than another round of self-interpretation.

The least intrusive useful practice is usually the best one. One sentence may be enough; a weekly note may be safer than hourly tracking; a conversation may be better than a spreadsheet. More data do not automatically mean more truth when the observer and the observed are the same person. The right cadence is the one

that leaves attention available for work, relationships, sleep, pleasure, and recovery.

The same stop rule applies to groups and public projects. If an activity produces self-surveillance, shame, public scoring, pressured disclosure, youth coercion, data extraction, or care delay, leaders must pause, revise, retire, or prohibit it. Good intentions do not convert a harmful method into a necessary one. A responsible program should also make its stop route as easy to find as its invitation to participate.

The book has no entitlement to the reader's attention. When a question that was meant to widen understanding begins to narrow life, distance, rest, care, and ordinary routine take priority. That freedom to disengage is part of what keeps inquiry subordinate to the life it is meant to illuminate.

27.2 — Pleasure, Relief, and Moderation

Pleasure is often treated as simple until a life depends on the difference between enjoyment, craving, relief, fulfillment, and meaning. A person may seek what is no longer enjoyed, feel relief without happiness, or pursue a meaningful task through discomfort. A single happiness total obscures these differences.

27.2.1 — Pleasure, Relief, Fulfillment, and Meaning

Pleasure is not one undifferentiated substance. Reward research separates hedonic impact, incentive salience, and learning, while affective science allows positive and negative systems to operate partly independently (Berridge and Robinson, 1998; Berridge et al., 2009; Cacioppo and Berntson, 1994). A person can want what is no longer liked, enjoy what carries little meaning, or feel relief without joy.

These distinctions appear in ordinary scenes. A meal can be pleasurable, the end of a migraine can be relief, finishing an apprenticeship can be fulfillment, and caring for a parent can carry meaning while also bringing fatigue and grief. Collapsing all four into happiness erases the mechanism and the human texture.

Relief has a temporal shape of its own. Pain lowers, fear releases, thirst ends, uncertainty resolves, or a threatening obligation is removed. The positive feeling may arise from leaving an aversive state rather than from receiving a new reward. That difference matters in medicine, addiction, conflict, and caregiving because an action that gives immediate relief may either restore capacity or create a later cost.

Pleasure first matters because it can be good to experience. A song, a game, consensual intimacy, a joke, a quiet morning, or an affectionate meal does not need to earn its place through productivity or future compensation. Its immediate value is not evidence for PTVC, but a theory concerned with felt life would be misshapen if it treated positive experience as morally suspect.

Fulfillment usually connects an activity to a valued life: learning a craft, keeping a promise, caring well, or building something that matters beyond the afternoon. Meaning may accompany pleasure or suffering, but it does not convert suffering into good. Well-being research likewise distinguishes hedonic and eudaimonic dimensions rather than forcing every positive life into one total (Ryan and Deci, 2001; Diener et al., 2006).

Mixed experience makes the point sharper. A parent can feel exhausted and grateful, a performer terrified and exhilarated, and a mourner comforted and devastated in the same hour. Positive and negative affect need not occupy opposite ends of one simple dial, which is why the formulation developed here preserves bipolar and generalized branches instead of forcing every state into one scalar story.

The wanting–liking separation becomes especially important in compulsive behavior; Section 9.3.1 explains the cue-sensitization mechanism (Robinson and Berridge, 1993).

A separate question concerns the conditions that make enjoyment possible. Consent, labor, money, safety, environmental cost, and the distribution of care can reveal hidden burden. When pleasure depends on coercion, exploitation, deception, addiction, neglect,

or another person's invisibility, the remedy belongs to that causal structure. The problem is not pleasure as such.

Pleasure can also restore what fear, shame, pain, or relentless usefulness has narrowed. Play, beauty, touch, music, unmeasured time, and ordinary comfort can remind a person that life is not only a container for burden. Relief deserves permission before it can be converted into a lesson.

Meaning changes with scale. A tedious afternoon may belong to a deeply valued project, while a thrilling evening may undermine a life the person wants. The afternoon is not secretly pleasant, and the evening is not secretly worthless. Immediate valence, reflective evaluation, and long-horizon consequence answer different questions.

The richer vocabulary prevents a crude happiness calculator. Before interpreting suffering, ask what care could reduce it. Before distrusting pleasure, ask whether a real burden is hidden beneath it. Where the terms are humane and visible, delight needs no apology; where they are not, the work is to change the terms rather than condemn the capacity for joy.

27.2.2 — Healthy Pleasure Is Not Moral Debt

Pleasure creates no moral obligation to suffer. Positive emotion can broaden attention, support connection, and replenish the capacity to act, while self-determined enjoyment can be part of a life rather than a distraction from it (Fredrickson, 2001; Ryan and Deci, 2001). Under exact PTVC, felt enjoyment changes the accumulated balance and therefore the required net remainder. It

does not prescribe a particular harm, identify its timing, or turn a good hour into a moral bill.

Experienced pleasure and remembered pleasure are not identical. The way a person later summarizes an event can be shaped by peaks, endings, contrast, and memory, so a favorable memory does not provide a complete measure of what was felt while the event unfolded (Fredrickson and Kahneman, 1993; Kahneman et al., 1997). The distinction calls for humility, not suspicion of enjoyment.

Savoring can deepen attention to an experience without converting it into a score. Pausing over music, food, friendship, beauty, achievement, or relief can help a person receive what is present rather than rushing toward the next stimulus (Bryant and Veroff, 2007). The point is presence, not accumulation.

Healthy pleasure can also restore capability. Sleep, laughter, play, affection, movement, nature, art, and celebration may return patience, social energy, or courage to a person whose life has become narrow. Their value does not depend on whether they later produce a measurable benefit.

The fear of happiness is therefore a misuse risk. A reader should not expect loss after joy, reduce delight to avoid imagined correction, or interpret a bad event as payment for a good one. Ordinary causes explain ordinary events until evidence shows otherwise, and this hypothesis supplies no event forecast.

Access to pleasure is unequal. Poverty, disability, discrimination, caregiving, pain, unsafe housing, and time scarcity can make ordinary joy harder to reach. A fair response changes those

conditions where possible instead of telling people to be grateful for a smaller menu.

A good day may simply be good. Receive it without moral debt language, address harm where harm actually occurs, and allow positive experience to retain its own human meaning.

27.2.3 — Moderation Protects Capacity and Choice

Moderation is useful when it protects capacity, relationships, health, and future choice. It is not a purity test and it does not require every pleasure to be reduced to the same middle point. The practical question is whether a pattern leaves the person able to choose again tomorrow.

Habits become easier when cues and actions repeat in stable settings, often with little conscious deliberation (Wood and R  nger, 2016). Repetition can therefore support sleep, movement, study, medication, or connection, but it can also make a costly pattern feel automatic. The design of the environment matters alongside intention.

Habit formation is gradual and variable rather than magical. Research on everyday routines shows that automaticity develops at different speeds and can tolerate some missed opportunities without collapsing (Lally et al., 2010). A missed day is information, not a moral failure.

Friction can preserve choice. Moving an app, changing a route, limiting easy access, preparing food, scheduling rest, or asking for support can create enough pause for a different action to become possible. The best friction is proportionate and reversible, not punitive.

Moderation is context-specific. A person managing pain, disability, recovery, pregnancy, grief, or an acute crisis may need a pattern that looks different from someone else's. A universal numerical rule can hide the body, culture, resources, and risks that make a choice reasonable.

The opposite danger is control disguised as care. Partners, parents, employers, schools, platforms, or communities can use moderation language to supervise bodies, sexuality, food, medication, spending, or speech. A protective practice should increase agency, not transfer it to the loudest observer.

Structural conditions can make moderation easier or harder. Predictable work, safe housing, access to treatment, affordable food, public space, and reliable care return room for deliberate choice. Personal discipline should not be used to erase the systems that shape the available menu.

27.2.4 — Excess Can Be Addressed Without Shame

Excess names a pattern and its consequences, not a complete identity. A behavior may be harmful, costly, compulsive, or dangerous while the person remains more than it. Begin with what happens, what it does, and who bears the burden.

Stigma increases secrecy and discourages help-seeking. Addiction-language research shows that words can frame people as capable of recovery or as moral contaminants (Kelly et al., 2016; Livingston et al., 2012). Public language should reject purity labels, redemption theater, and suffering as the price of credibility.

Ask what function the pattern serves and what sustains it. Relief, belonging, escape, pain, poverty, trauma, isolation, unstable work, commercial access, or social cues may shape the available route. Context does not excuse harm; it identifies the safer alternative that must become credible.

Accountability remains specific. Protect affected people and require proportionate repair when excess harms a person, body, household, or public safety. Addiction, eating disorders, self-harm, violence, dangerous withdrawal, severe financial loss, or medical risk may require qualified help. Reduce danger and connect care before interpretation.

Compassion is not permission. It refuses needless shame while protecting others, telling the truth, and changing what can be changed. The Law adds no hidden rank or payment story. Ordinary causes and support can produce partial, private, medication-supported, or nonlinear improvement without a perfect narrative.

27.3 — Addiction, Craving, and Future Optionality

Addiction reveals how misleading abstract freedom can be. From outside, many actions remain physically possible. From inside a craving, one cue may occupy attention, relief may feel urgent, and the cost of resistance may rise beyond what a simple command to choose differently acknowledges.

27.3.1 — Reward, Tolerance, and Opponent Processes

Ordinary addiction science already explains much of the pattern relevant to daily life: reinforcement and habits can strengthen repeated action, cue-triggered wanting can outlast liking, and tolerance or withdrawal can shift behavior toward relief seeking (Robinson and Berridge, 1993; Everitt and Robbins, 2016; Koob and Le Moal, 2001; Koob and Volkow, 2016). These mechanisms are developed in Chapters 9 and 19; the Law receives no credit for them.

27.3.2 — Craving Captures Attention and Narrows Choice

Craving can narrow attention until physically available alternatives cease to feel reachable. The practical question is which cue, withdrawal state, stressor, material constraint, pain condition, or learned routine has narrowed the next choice, and what treatment, support, or environmental change can reopen it (Tiffany, 1990; McLellan et al., 2000; Bickel et al., 2014). The aim is enough safety and optionality for meaningful choice to return, not a hidden score or proof of PTVC.

27.3.3 — Recovery Can Reopen Paths

Recovery is an expanding field of possible life, not a single moment of moral victory. It may involve abstinence, reduced use, medication, psychotherapy, mutual aid, safer use, housing, work, family repair, new routines, or a change in the people and places that cue the old pattern. Different conditions and different people require different routes.

Learning does not disappear on command. Cues can remain potent after long periods, and stress can reactivate old responses, while repeated safer actions can gradually acquire their own momentum (Everitt and Robbins, 2016; Wood and Rünger, 2016). A recurrence can reveal a vulnerable pathway without becoming a verdict on the whole person.

Use recovery capital, defined in Section 9.3.3, to identify a missing resource and someone able to restore it, not to create another score (Cloud and Granfield, 2008). An identity not exhausted by the disorder can also help make change practicable.

Medication is a form of care, not a moral shortcut. Methadone, buprenorphine, and naltrexone are approved treatments for opioid use disorder. Methadone and buprenorphine ease withdrawal and craving, while naltrexone blocks opioid effects; access to effective medication can reduce illicit opioid use and overdose risk (National Institute on Drug Abuse, 2025). A recovery culture that treats effective medication as impurity can close a lifesaving path for the sake of a preferred story.

Continuity matters because many people are lost between settings. The move from emergency care to outpatient treatment,

from incarceration to community, from adolescence to adult services, or from stability to recurrence can reopen old risks. Integrated clinical and community recovery support can help make the safer route available after the immediate crisis has passed (Kelly et al., 2025).

Harm reduction can preserve life and future choice before complete cessation is possible. Overdose prevention, safer-use information, sterile equipment, testing, medication access, and rapid reentry to care reduce immediate danger without declaring the present condition harmless or permanent.

Support should not become possession. Consent, role clarity, minimal necessary information, privacy, and exit routes help distinguish care from surveillance. A friend is not automatically a therapist, a partner is not an investigator, and an artificial intelligence tool is not a sovereign judge.

Recovery also includes repair. Someone may need to acknowledge harm, repay money, protect children, accept boundaries, or comply with lawful conditions. The purpose is to reduce and prevent burden, not to create pain as symbolic payment.

A viable life needs positive alternatives. Music, movement, work, learning, food, intimacy, nature, service, and community can become sources of reward and identity. A recovery plan made only of prohibition may leave the very emptiness that the old pattern once filled.

Reopened paths matter whether PTV is supported, narrowed, or defeated. More options permit more care, more authorship, and

more chances to choose safety in the next hour. That value stands on its own.

27.3.4 — Addiction Is Not Cosmic Payment or Proof

Addiction makes balance language dangerous. Withdrawal is not moral repayment, recurrence is not destiny, and a difficult recovery does not prove a hidden law. Opponent processes, tolerance, allostasis, and incentive learning are causal mechanisms, not moral bookkeeping.

The mechanisms in Section 27.3.1 identify treatment targets (Koob and Le Moal, 2001; Koob and Volkow, 2016). They do not make suffering morally deserved or establish that any particular withdrawal episode is required or scheduled by PTV. C.

Recurrence may follow stress, untreated pain, isolation, medication interruption, housing instability, or lost alternatives (McLellan et al., 2000). It can reveal conditions that need to change rather than character that deserves judgment.

Care remains specific. Clinicians should assess pain, medication history, withdrawal risk, function, and the person's goals, and offer effective treatment when opioid use disorder is present (National Institute on Drug Abuse, 2025). Prescribed exposure creates duties of continuity and support, not excuses for abandonment.

Stigma, criminalization, exclusion from work, unstable housing, and denial of care create burdens that pharmacology does not explain. Institutions should reduce them directly, not claim that the person needed hardship to complete a balance (Kelly et al., 2016; Kelly et al., 2025).

No recovery route confers purity or rank. Medication, psychotherapy, peer support, spiritual practice, secular community, family repair, and private change can each matter. Local improvement does not privately validate the complete latent endpoint object described by PTVC.

Overdose, death, and grief are never closure tests. The human response is to preserve life, widen access, care for survivors, and refuse every story that treats pain as owed.

27.4 — Work, Ambition, and Aspirational Dreams

Ambition lets a possible future give present effort a direction. A qualification, a garden, a business, a recovery, or a piece of music can make an ordinary day feel connected to something worth pursuing. The question is how to pursue that future while remaining free to learn, rest, change course, and receive help. Encouragement becomes cruel when it conceals unequal opportunity or turns every setback into an instruction to sacrifice more.

27.4.1 — Dreams Require Pathways, Not Wishes Alone

An aspiration is a picture of a possible future, not a promise that the future will arrive. Dreams matter because they can organize attention and effort, but they become actionable only when they are connected to skills, time, relationships, opportunity, health, and a sequence of decisions.

Specific and challenging goals can improve performance when people have ability, commitment, feedback, and resources, but goal effects depend on task conditions and do not override reality (Locke and Latham, 2002). A goal is therefore a coordination tool, not a guarantee.

Implementation intentions help translate a desired future into a decision about when, where, and how to act. Plans framed around a concrete cue and response can make action more likely than intention alone (Gollwitzer, 1999; Gollwitzer and Sheeran, 2006).

Positive fantasy without obstacle recognition can reduce effort. Mental contrasting asks the person to hold the desired future

beside the most important present obstacle, then decide whether commitment is warranted (Oettingen, 2012). The method respects both hope and constraint.

Skill also requires practice that responds to feedback. Deliberate practice research shows that improvement depends on structured effort, correction, and sustained access to appropriate challenges rather than on wishing or raw repetition alone (Ericsson et al., 1993).

A pathway includes conditions outside the person. Money, caregiving, disability, discrimination, geography, safety, timing, and luck can determine whether a plan is available. Honest ambition names these constraints without confusing them with a verdict on worth.

Good plans preserve more than one route. They include checkpoints, alternatives, and conditions for pause or redirection. A dream that permits no revision can become a trap when the body, family, market, or world changes.

Effort should not consume the person it is meant to serve. Sleep, health, friendship, care, and ordinary joy are not distractions from every worthy goal. Sometimes they are the conditions that keep the path open.

The Law adds no destiny language. It asks how present choices and conditions shape future possibility while leaving causation, uncertainty, and unequal opportunity visible. Success remains contingent, and a closed path does not prove that the person chose wrongly.

A dream becomes humane when it can be pursued seriously, revised honestly, and released without destroying the dreamer.

27.4.2 — Discipline Can Preserve Future Possibility

Discipline is often described as the future self defeating the present one. A kinder and more accurate account treats it as preservation. Repeated practice can keep a language, craft, body, relationship, recovery path, or professional option available long enough for it to become real.

Preservation begins with the person's actual menu. Disability, illness, trauma, poverty, caregiving, shift work, unstable housing, unsafe homes, grief, depression, anxiety, and family pressure all alter what a routine costs. A habit that appears simple from outside may require resources the person does not have.

The mechanism need not be heroic. Stable contexts can reduce decision costs, implementation plans can connect cues to actions, and feedback can show whether effort is producing the intended skill (Wood and R  nger, 2016; Gollwitzer and Sheeran, 2006; Locke and Latham, 2002). Discipline works best when the environment supports it instead of demanding a fresh contest with exhaustion every day.

A writer who protects four mornings a week gives a long project repeated access to attention. A person in recovery who changes the route home protects freedom from a cue that has repeatedly captured it. Neither act proves superior character. Each rearranges the conditions under which a later choice will be made.

Discipline can also mean stopping. Resting before injury, refusing another task, setting a budget, leaving an exploitative opportunity,

or retiring a plan that no longer fits may preserve more possibility than endurance. Motion and sacrifice are not reliable measures of agency.

Research on grit found associations between perseverance and achievement, but later meta-analysis indicates modest predictive power and substantial overlap with conscientiousness (Duckworth et al., 2007; Credé et al., 2017). Persistence can matter without explaining outcomes apart from ability, opportunity, support, health, and luck.

Depletion also has causes outside the individual. Burnout research describes exhaustion, cynicism, and reduced efficacy as responses to chronic workplace conditions rather than simple failures of will (Maslach and Leiter, 2016). A schedule that destroys sleep, health, or relationships may close more paths than it opens.

Feedback should therefore remain diagnostic rather than punitive. A missed week may reveal illness, hidden caregiving, an unrealistic schedule, a vanished reward, or a method that no longer serves its purpose. The answer may be redesign, assistance, or release rather than escalation.

Supporting capacities are part of the path. An artist who learns budgeting, a researcher who learns communication, or a tradesperson who protects physical health may preserve the visible craft more effectively than someone who concentrates every hour on it.

The long-horizon question is ordinary and causal: what does this repetition preserve, and what does it spend? Good discipline

leaves the person more capable of choosing. It creates no moral debt to suffering, promises no reward, and does not turn a missed day into a moral verdict.

27.4.3 — Uncertain Success and Nonmoral Failure

Two people can prepare with equal seriousness and meet different doors. Selection favors visible survivors and hides capable people whose timing, networks, health, location, or luck differed (Denrell, 2003), while opportunity varies sharply across places and families (Chetty et al., 2014). An outcome is evidence about a path, not a pure reading of merit.

Effort still matters. Skill, preparation, persistence, judgment, and cooperation change probabilities. The correction is to place those causes inside the wider field rather than allowing success to absorb every advantage or failure to absorb every constraint.

When an attempt fails, its causes can be separated. The method may have been weak, the timing poor, the goal unclear, the support insufficient, or the plan no longer suited to the life around it. Learning begins when changeable causes are distinguished from conditions requiring help or acceptance.

Prior investment can also trap ambition. Time, money, identity, or public promises already spent do not make another sacrifice wise. A failed path becomes informative when a person can separate recoverable learning from costs that cannot be recovered, then decide from the future being created rather than from the past that has already been paid.

Responsibility remains exact. A person may need to acknowledge poor preparation, a broken promise, avoidable harm, or a failure

to seek help. Repair should address the consequence and next decision without converting one error into a verdict on the whole life.

Second routes are partly institutional. Education, retraining, health care, disability access, child care, debt relief, lawful protection, and a chance to reenter can determine whether a setback remains local or becomes permanent narrowing. Praising resilience while closing return paths confuses character with infrastructure.

Success may be celebrated without becoming rank or proof of cosmic favor. A failed plan may be revised, released, or replaced without ending authorship. The proposed Law promises neither victory nor a chosen reward; its conditional net remainder is a different claim. It asks what the path required, preserved, spent, and left possible. A result belongs to a biography, but it is never the whole biography.

27.4.4 — The Law Lens Adds a Question, Not a Promise

In ordinary planning, the Law contributes one bounded question: how might this choice widen or narrow the experiences and possibilities that remain? It forecasts no outcome, reveals no hidden balance, and promises no later reward for sacrifice.

Keep authorship and constraint together, as Section 26.1.1 explains (Bandura, 2001).

Local causal explanation remains primary. If sleep improves concentration, a mentor opens a path, debt closes an option, or conflict destroys trust, those mechanisms deserve the explanation. The Law should not replace them with a more dramatic story.

Future optionality is therefore a heuristic, not a measurement. It asks whether a choice protects health, time, relationships, skill, mobility, or the ability to change direction. It cannot reveal the complete latent object.

The lens must also permit refusal. A person may choose art over status, caregiving over advancement, rest over optimization, or an ordinary life over a celebrated one. No hierarchy of dreams follows from the hypothesis.

The reflection remains useful if PTVC fails. Choices still shape capability, attention, burden, and opportunity; honest plans still need pathways; care still matters. Outcome-independent value is a test of whether the lens has become more than promotion.

Ask the question when it clarifies a real choice. Put it down when it begins to speak with an authority it has not earned.

27.5 — Relationships, Grief, Rest, Joy, and Repair

Think of a friend waiting outside an appointment, a neighbor bringing dinner, or someone listening to the same grief story again without asking when it will end. Each can change what another person has to face. None has to explain the suffering to make the next hour less lonely.

These acts do not transfer anyone's feelings into someone else's stream. They change the world in which the other person's experience occurs. The one-stream boundary is compatible with deep dependence on one another: we cannot live another life from the inside, but we can help shape the conditions in which it is lived.

Social-support, sleep, positive-emotion, and bereavement research each show plural mechanisms and plural trajectories (Cohen and Wills, 1985; Lim and Dinges, 2010; Fredrickson, 2001; Stroebe and Schut, 1999). The human texture is richer than any single balance metaphor.

27.5.1 — Relationships Can Expand or Constrict Possibility

Relationships change the practical size of a life. Social support can buffer stress, close relationships can reduce perceived risk and effort, and chronic isolation is associated with serious health burdens (Cohen and Wills, 1985; Coan and Sbarra, 2015; Holt-Lunstad et al., 2010). Belonging is not decoration around individual agency; it helps shape what agency can do.

A trustworthy relationship can return time and attention. People share care, reality-check fear, make rest possible, protect one

another during illness, and keep a difficult path from becoming solitary. The value often appears in small acts rather than in grand declarations.

Relationships can also constrict. Coercion, contempt, manipulation, surveillance, financial control, threats, and unpredictable withdrawal of affection can narrow choice until ordinary decisions feel dangerous. The need to belong can make these conditions difficult to name (Baumeister and Leary, 1995).

Boundaries are therefore part of care. Distance, confidentiality, separate money, limited contact, professional support, or leaving may be necessary when closeness is used to control. The harmed person owes no contact or reconciliation where safety cannot be secured; the responsible person's duties of repair remain.

A support network should distribute rather than hide load. One partner, child, sibling, or friend should not become the only emergency system for another person. Wider care, respite, clinical help, and public services can protect both dignity and relationship.

Loneliness also deserves direct attention. It may persist in a crowd, after migration, during disability, following bereavement, or inside a relationship that no longer feels safe. The answer is not to assign meaning from above, but to make real connection more reachable.

Privacy includes the people around the relationship. A partner's disclosure may expose another person's health, sexuality, trauma, or family history. Shared life does not erase individual consent.

No relationship can be reduced to a balance sheet of pleasure and pain. The relevant questions concern trust, safety, reciprocity, freedom, burden, care, and the possibility of repair. Those features can be examined without pretending to see a hidden lifetime ledger.

The clearest relational question is practical: does this bond make truthful, safe, and caring action more possible, or does it steadily make the person's world smaller?

27.5.2 — Repair Requires Action, Not Hidden Accounting

Repair begins with the actual harm. What happened, who was affected, what was lost, what remains unsafe, and what can be changed now? Hidden accounting distracts from these questions by imagining that suffering somewhere else has already settled the matter.

Repair in ordinary relationships should follow the standards established in Section 25.5.1: name the act, recognize its effects, accept responsibility, connect words to changed conduct, and keep restorative participation voluntary and safe (Lazare, 2004; Schumann, 2018; Zehr, 2002; Walker, 2006).

Restitution may involve money, time, correction of a record, return of property, clinical care, a public statement, changed access, or a new boundary. The form should match the burden rather than perform suffering for an audience.

Repair does not require reconciliation. The harmed person may decline contact, forgiveness, mediation, or a renewed relationship. Their refusal does not make the repair incomplete in a way that gives the person who caused harm a claim on them.

Pain is not morally owed payment. Consequences may be necessary to protect people, uphold rights, or reduce recurrence, but their purpose is not to make someone hurt enough to balance an invisible account. Symbolic cruelty can satisfy spectators while leaving the cause untouched.

Changed behavior is the long evidence of repair. It includes accepting limits, giving up an advantage, telling the truth when concealment would be easier, and staying accountable after public attention has moved on.

27.5.3 — Rest, Joy, and Celebration Have Positive Value

Rest is not the absence of a worthwhile life. Sleep loss impairs attention, working memory, executive function, mood, and safety, and repeated restriction can accumulate effects that people do not accurately perceive in themselves (Dinges et al., 1997; Lim and Dinges, 2010). Protecting sleep is therefore protection of basic capacity, not indulgence dressed as science.

Rest also includes pauses that are not sleep: an unhurried meal, a bath, time in nature, music without monetization, an afternoon without performance, or permission to be unreachable. These intervals may restore later functioning, but they need not justify themselves only by what they produce.

Rest has a distribution. Someone may cover the shift, prepare the meal, watch the child, answer the call, or clean after the gathering. A fair arrangement asks whether that work was chosen, visible, supported, and sustainable. Restoration for one person should not be built on the silent exhaustion of the person with the least power.

Joy deserves the same seriousness. A first day without pain, a repaired roof, a recovery anniversary, a graduation, a new child, a shared meal, or a joke that returns after grief can be received without predicting an equal reversal. Celebration does not deny the world's burden; it refuses to make burden the only morally serious experience.

Some readers have learned to distrust ease. Rest had to be earned, joy was frivolous, desire was selfish, or comfort proved that someone else must be paying. This book should refine that conscience, not make ordinary delight feel dangerous. Where a real hidden burden exists, address it. Where it does not, delight needs no defense.

Positive-emotion research proposes that interest, contentment, love, and amusement can broaden thought and help build resources over time (Fredrickson, 2001). That account is useful, but it is not the whole justification. The first value of joy is that it is joy, felt by someone whose experience matters.

A person may choose a simple form to carry attention gently: lighting a candle for grief, walking after conflict, marking the end of a harmful pattern, sitting quietly before an apology, or keeping a private rhythm of gratitude. Its value is interpretive and personal, not evidentiary. The form may be secular or religious, inexpensive, brief, adapted for disability, changed, or refused. Sustained doctrine and communal ritual belong to *The Religion of Fairness*.

Celebration should not become compulsory optimism. A person may feel numb at a milestone, grief inside a happy gathering, or

relief mixed with fear after survival. The occasion does not dictate the correct inner response, and sincerity is not measured by performance.

Joy can also be a public good. A safe park, music in a hospital corridor, a neighborhood meal, or a library open late can make beauty, play, and companionship available without luxury. Systems that protect time, accessibility, safety, and ordinary celebration are not decorative additions to dignity.

Gratitude becomes credible when it changes conditions. Remembering the nurse on the holiday, the cleaner after the event, the relative cooking alone, or the worker required to smile can lead to better pay, schedules, participation, and recognition. Rest and joy remain fully positive while becoming more honest about the people who make them possible.

27.5.4 — Grief and Meaning Without Compulsion

Grief follows no required sequence. The dual-process model describes movement between loss and restoration, while resilience research shows wide variation over time (Stroebe and Schut, 1999; Bonanno, 2004). A person may move, pause, return, or carry several forms of grief at once.

Expectations can add burden. Some people grieve intensely for years, others regain function quickly, and many alternate between ordinary life and sudden return. No visible course should become the standard for judging every mourner (Wortman and Silver, 1989).

Continuing bonds can remain healthy. Memory, ritual, conversation, objects, stories, and an enduring inner relationship

may become part of life rather than obstacles to detachment (Klass et al., 1996). Their form belongs to the mourner and the culture.

Meaning may be found, made, refused, or left incomplete, and people differ in whether meaning-making helps (Park, 2010). Overdose, estrangement, miscarriage, stigmatized relationships, public violence, and loss of a role may also produce grief that others fail to recognize (Doka, 1989).

The proposed Law cannot explain why a person died, why a family suffered, or what the loss was for. It gives no warrant for saying that pain completes a balance, protects future joy, or reveals a hidden plan.

Chosen ritual can help, but no spiritual interpretation is required. Prayer, remembrance, music, silence, gathering, service, or private refusal may carry dignity. Food, sleep, childcare, paperwork, medical help, listening, and protection from isolation often matter more than explanation, and qualified support should be available when grief becomes dangerous or disabling.

Meaning after loss is a freedom, not a debt. The mourner may build a new life, preserve the bond, change the story, or leave the question unanswered.

27.6 — What Not to Turn Into a Rule

A good overview ends with a stopping point. Without one, attention can become self-testing, moderation can become purity, ambition can become sacrifice, and grief can become compulsory interpretation. The section names the exact transformations this chapter refuses.

The boundaries are operational rather than ceremonial. Seek care instead of evidence. Use ordinary decision criteria instead of scores. Treat event pairings as stories unless prospectively tested. Keep private reflection outside the research record unless a separately governed protocol exists.

The result should feel lighter than a warning wall. A reader is free to close the book, enjoy the day, seek help, change their mind, and leave the ultimate question unanswered.

27.6.1 — Do Not Seek Pain or Delay Care

Nothing in the Law authorizes a person to seek pain, preserve suffering, refuse relief, or postpone treatment. A proposed terminal relation cannot turn preventable harm into a required installment, and no future possibility can make present deterioration desirable.

Pain can carry information, but information is not an instruction to endure without care. Worsening symptoms, injury, withdrawal, infection, suicidal crisis, abuse, or loss of basic functioning call for appropriate help, not private interpretation through the hypothesis.

The research boundary is equally firm. Ethical guidance requires scientific necessity, proportional burden, informed consent, independent review, and protection of vulnerable people; curiosity about PTVC creates no exception (World Medical Association, 2024; CIOMS, 2016). An experiment that needs avoidable suffering in order to become interesting is not permitted by this book.

The practical order is care first, explanation later. Take the medicine as directed, contact the clinician, leave the dangerous setting, use emergency services when needed, or ask another person to help navigate access. A reader does not owe the theory an untreated symptom.

Balance-later reasoning is especially unsafe. No one may say that pain now will be repaid, that relief would interfere with a hidden process, or that another person's suffering serves a larger account. Those claims add invented certainty to an already vulnerable moment.

Access itself may require assistance with transportation, insurance, language, disability accommodation, child care, advocacy, or finding a safer professional. The book's duty is to avoid becoming one more barrier between a person and help.

The detachable rule is simple: help first. Interpretation can wait; preventable deterioration should not.

27.6.2 — Do Not Score Yourself or Anyone Else

The no-score rule reaches beyond a daily number. It forbids private or institutional verdicts such as more balanced, less balanced, spiritually advanced, overdue for hardship, unusually

avored, or close to a hidden endpoint. None is licensed by the latent object described in the formulation developed here.

Bounded records may support bounded decisions; they cross the line when they claim to summarize worth, destiny, eligibility, lifetime balance, or the meaning of a life. See Section 27.1.1 for the practical record rule.

Validity depends on the interpretation and use of a measure, not only on how the number was calculated (Messick, 1995; Borsboom, 2005). A qualitative label can therefore function as a score when a school, employer, family, platform, religious group, or public agency uses it to sort people.

The danger is greatest where refusal is costly. Children, patients, workers, benefit recipients, migrants, disabled people, prisoners, grieving families, and anyone dependent on an institution can carry a label into opportunity, stigma, discipline, or care. Friendly intent does not make that transfer safe.

Software adds scale, not truth. A mood app, wearable, archive, or artificial intelligence system may summarize declared observations for a limited purpose, but aggregation does not reveal lifetime balance, trustworthiness, inner truth, or human rank.

Indicators also change behavior when they become targets. Campbell's law warns that decision pressure can corrupt a measure and distort the process it was meant to monitor (Campbell, 1979). A private score can make someone live for the number rather than for safety, rest, relationship, and repair.

Use evidence proper to the decision. Conduct informs accountability, demonstrated skill informs a task, symptoms inform care, and lawful criteria inform eligibility. The Law supplies no shortcut across those domains.

A person can notice harm, make amends, set a boundary, or change a habit without reducing anyone to a metric. Human judgment remains answerable when it names the observation, purpose, uncertainty, and consequence instead of pretending to see the hidden whole.

27.6.3 — Do Not Forecast Events or Pair Joy With Suffering

The Law does not forecast what happens next. A good event does not predict a bad one, and a painful period does not guarantee a chosen later reward. If the genuine accumulated balance is negative, exact PTVC does require a net positive remainder before the true endpoint. Even if the proposed endpoint relation were true, it would not provide a calendar of compensating events.

Opponent processes describe local dynamics, not cosmic scheduling. Adaptation or rebound can follow a specific stimulus within a specific system, but that does not imply that every pleasure will be followed by equal pain or that every hardship will be repaid (Solomon and Corbit, 1974).

Meaning-making research examines how people relate stressful experiences to wider beliefs and goals (Park, 2010). This provides a framework for understanding personal interpretations of important events. In applying PTVC, those interpretations

deserve compassion and should be examined separately from a claim that one event balances another.

Just-world beliefs create a sharper danger. They can lead observers to imagine that victims caused, deserved, or will somehow be compensated for suffering when no evidence supports that conclusion (Lerner, 1980; Hafer and Bègue, 2005; Callan et al., 2006). The formal hypothesis gives no authority to that judgment.

Fear of happiness can become another burden. A reader should not suppress celebration, avoid intimacy, or distrust relief because an equal reversal is expected. Ordinary risk management remains sensible; magical pairing is not.

Retrospective pairing is equally unreliable. After a tragedy, almost any earlier joy can be framed as repayment; selecting the pair after the outcome makes the story unfalsifiable.

Local consequences matter. Sleep loss can impair judgment, substance use can cause withdrawal, broken trust can alter a relationship, and debt can narrow future options. Address those pathways directly rather than turn them into prophecy.

Receive the good without moral debt and meet the bad with care. The hypothesis gives no permission to make either one a forecast of the other.

27.6.4 — Private Reflection Is Not Research Evidence

Private reflection can guide a personal decision, but it cannot establish a universal terminal claim. A diary, mood record, or self-experiment begins after the stream began, ends before an

independently known truth endpoint, and observes only a small part of the proposed object. N-of-1 trials can answer bounded questions when interventions, outcomes, periods, and analyses are specified in advance, yet they do not turn short observation into evidence about exact complete-stream closure (Guyatt et al., 1986; Lillie et al., 2011).

Reporting guidance requires protocol clarity, adverse-event attention, and limits on interpretation; a method never authorizes risky medication changes, induced distress, sleep deprivation, or endpoint testing without care and review (Vohra et al., 2015). Everyday self-tracking adds selective entry, changing goals, device error, forgotten context, and pressure to force a coherent story (Choe et al., 2014). A record may still reveal sleep loss, help explain symptoms, or identify a craving cue. Practical usefulness is valuable without becoming PTV evidence.

Private notes do not become project data by mentioning the Law. Research requires a declared protocol, consent, governance, purpose limits, security, an analysis plan, and accountable later-use decisions. Permission to use an artificial-intelligence tool does not authorize converting a conversation, journal, or classroom response into training data or research evidence. The person may delete, forget, or never share the record.

Chapter 28 — A Scientific Program

A serious scientific program gives a large question a place to begin. A useful first contribution could be a better way to identify neutral feeling, a test of what information a candidate mechanism must retain, or a biological model that makes a clear prediction about completion. None requires one researcher to solve the entire problem at once. The contributions become cumulative when their connections are explicit and independent critics can distinguish genuine progress from an attractive fit.

The formulation developed here converts the intuition into a dependency graph: representation, neutral origin, exposure, bearer, routing, complete domain, endpoint, observation, missingness, transport, and inference. Each layer can fail independently, and no downstream success may hide an upstream failure.

That precision does not establish the Law. It exposes the work needed to test it and the points at which the honest result may narrow, stop, or become No adequate estimate.

The first Section states the public proposition and opens it into evidence families. Affective science, psychometrics, consciousness research, longitudinal and palliative studies, computational modeling, and statistics can test pieces of the architecture. Their findings remain local unless a valid bridge reaches the complete target.

No self-validation is the first rule. Equations cannot observe the world, resonance cannot replace data, series coherence cannot validate its root, and public interest cannot turn promotion into evidence.

The second Section builds the measurement gates. Positive, negative, mixed, and neutral must be defined before closure is calculated. An additive shift of the origin changes the result, while stream and endpoint rules must remain independent of the desired answer.

Observation remains separate from the target. Reports, neural signals, behavior, clinical notes, and sensors are channels; the estimand is the declared quantity, the estimator is the inferential rule, and transport is a bridge to another setting or latent object. Conflation creates confidence without identification.

The gates are noncompensatory. A large sample cannot repair an invalid construct, replication cannot repair a circular endpoint, and sophisticated modeling cannot support inference about a systematically unobserved interval without defensible additional information or assumptions.

The third Section turns to prospective discipline. Important choices must be fixed before relevant outcomes are seen, strong rivals must receive equal opportunity, unfavorable results must remain visible, and confirmatory predictions must stay distinct from exploration (Nosek et al., 2018; Munafò et al., 2017).

Red teams matter because a universal claim attracts loyalty and hostility. A serious team receives the exact claim, adequate resources, methodological access, and authority to conclude that the preferred model loses, the data are ambiguous, or the target is unidentified.

The fourth Section begins with model recovery. Before interpreting lives, a method must distinguish simulated worlds

with known generators. A pipeline that confuses mean reversion with terminal conditioning, or observation shutdown with closure, is not ready for human data.

Held-out transport and independent replication follow. A result fitted in one cohort, channel, site, or endpoint class must survive a frozen test elsewhere before the claim widens; reproduction and replication answer different questions (National Academies of Sciences, Engineering, and Medicine, 2019).

Evidence maturity and theory maturity remain separate. An elaborate architecture may have little direct evidence, while a robust local effect may leave the universal law untouched. Honest status is a profile, not a badge.

Verdicts are modular. Target-adverse, Route-failed, Route-favored, Rival-favored, Model-ambiguous, Open-set inadequate, No adequate estimate, and Reference-class narrowed describe different events or scope changes. Each must be reported at the branch and quantifier that license it.

Exact protocols, simulator banks, code, rival-comparison procedures, governance, replication, and verdict registers are specified in the scientific companion.

The chapter closes by transferring authority to criticism. The next Part invites the hardest objections to reshape, narrow, or defeat the project. The handoff is complete only when the theory no longer controls the outcome.

28.1 — The Scientific Question in One View

Scientific clarity begins by shrinking the claim to the object it actually concerns. PTVC does not say equal lives, equal welfare, equal opportunity, or average neutrality. It proposes one exact intrastream terminal relation under a demanding set of independently fixed definitions.

The public needs to see both the simplicity and the burden. The simple view is one stream, one complete domain, positive and negative phenomenal magnitudes, and a terminal residual. The burden is that every term must survive identification, observation, rival comparison, and possible defeat.

This section therefore presents the object, the evidence families, and the rule against self-validation together. The scientific question remains worth asking only if the answer is not owned by the people who asked it.

28.1.1 — Formal Object

A scientific program cannot begin with the sentence life is fair. It must begin with a prospectively specified object that can be wrong. On the primary finite signed-measure branch, each separately routed maximal qualifying conscious stream is assigned a finite signed phenomenal measure over its complete strict-pre-endpoint domain, and PTVC requires its terminal residual to be exactly zero.

Each term prevents a different shortcut. One stream blocks cross-person balancing. Separately routed blocks convenient merging or splitting. Complete blocks silent truncation. Strict-pre-

endpoint distinguishes the proposed domain from endpoint-inclusive additions. Exact blocks replacement by a vague trend.

The first measured objects will be candidate constructs, not miniature ledgers. Reports, choices, physiological signals, behavior, and clinical records can inform a measurement model, but none is identical to the latent phenomenal object.

The bearer rule and endpoint belong to the target specification. A study cannot choose the person, stream boundary, or stopping point after seeing which choice produces the most favorable residual.

Branches remain distinct. Additive PTVC, bipolar mixed-valence representations, generalized terminal neutrality, probabilistic claims, and route-specific hypotheses are not interchangeable. Evidence against one may leave another open, and evidence for a route does not by itself establish the target.

A midlife cohort ending at the last interview illustrates the problem. It observes neither the unmeasured prefix nor a guaranteed truth-level endpoint. Without a defensible bridge for both missing ends, the study may inform components while remaining unable to adjudicate the complete object.

Falsification attaches to the prospectively specified claim. A gate-valid counterexample to the declared universal branch would count against that branch; an invalid measurement or post hoc target change would not. The point of precision is to make adversity interpretable.

Exactness supplies vulnerability, not certainty. Finite data may support or reject an empirical band, identify an atom, narrow a

branch, or yield No adequate estimate. It cannot silently convert those results into proof of exact latent closure.

28.1.2 — Evidence Families

No single instrument sees the target, so evidence must arrive in families. The families do not vote by majority. Each one tests a different prerequisite, route, rival, or observation problem and carries its own failure modes.

Affective science can separate positive and negative dimensions, sample experience in daily life, compare remembered with experienced utility, and study adaptation (Cacioppo and Berntson, 1994; Kahneman et al., 1997; Shiffman et al., 2008; Diener et al., 2006). These methods supply tools, not terminal-law support by themselves.

Measurement studies can test reliability, invariance, calibration, reactivity, and missingness. Consciousness research can constrain candidate bearer and endpoint rules. Longitudinal and palliative studies can characterize trajectories, attrition, treatment, and terminal observation. Computational work can ask whether proposed worlds and rivals are distinguishable.

Some evidence concerns the target; other evidence concerns whether the target can be measured at all. A valid affect scale does not identify a truth endpoint, and a convincing endpoint rule does not establish cross-sign scalarization. Prerequisite evidence is noncompensatory.

Triangulation is strongest when methods fail differently. Self-report, pairwise choice, behavior, physiology, and neural

measures add information only when their errors are understood rather than assumed independent.

Negative controls and counterfactual worlds belong in the map. A method that supports PTV C in simulated no-PTVC data, or fails to detect a known rival, has not earned interpretation of a favorable real-world result.

Every bridge should state four things: what proposition has already been earned, what new assumption is being added, what error the bridge introduces, and what decision becomes possible if the bridge holds. This prevents a stack of plausible steps from being mistaken for one validated conclusion.

Historical, philosophical, religious, and autobiographical recurrence belongs to a different evidentiary family. It may explain why the question matters or how people interpret fairness, but it does not measure the latent terminal object.

Contrary evidence belongs in the same record as favorable evidence. Measurement failure, rival sufficiency, nontransportability, adverse terminal patterns, and unexplained open-set behavior should change the program rather than being filed as peripheral complications.

Route evidence stays local: the action-first fixed-policy support condition in Section 8.4.3 does not identify motivation, free will, treatment, moral instruction, personal obligation, or endpoint support. The evidence record should end with a dependency map, not a score: which gates are supported, which remain open, which routes have failed, and where science must abstain.

28.1.3 — No Self-Validation

The Law cannot validate itself through elegance, usefulness, recurrence, popularity, institutional interest, or the coherence of the wider series. Those qualities may justify attention, but none observes a complete conscious stream or supplies the missing bridges.

The first defense is recovery in worlds where the answer is known. A pipeline that chooses an origin after seeing the endpoint, drops difficult records, favors one routing rule, or compares the preferred model with weak rivals can manufacture support. Simulation should expose those tendencies before real data are interpreted.

Rivals must receive the same information, tuning opportunity, and decision budget. If an ordinary model explains the prospectively specified pattern more successfully, the preferred route loses that comparison. Rival victory may narrow the claim or retire the route; it cannot be renamed as hidden support.

Ethical nonviability is also failure. A mechanism test that requires induced suffering, delayed care, coerced disclosure, or unsafe terminal observation is not an available research route.

Origins, exclusions, endpoints, transformations, rival sets, analytic flexibility, and public wording must be frozen or governed before the decisive result. Independent adjudicators need the prospectively specified question, code, deviations, missingness, negative controls, dissent, and authority to preserve an adverse verdict.

A narrower reference class, failed bridge, corrected artifact, retired mechanism, or well-supported defeat can be progress. Self-validation ends when competent critics, ordinary data, and prospectively specified tests can reduce what the project is allowed to say.

A public scientific program should therefore publish a claim map before publishing a victory story. The map identifies the target branch, prerequisite gates, route hypotheses, capable rivals, reference class, measurement scale, endpoint rule, and exact outcomes that would support, narrow, leave unresolved, or contradict each component. Readers can then see which part of the architecture a result actually touches. This reduces the pressure to summarize every encouraging finding as support for the whole Law. A simple dependency diagram can connect each proposed conclusion to the particular measurement and comparison needed to support it in that study.

The same map protects negative results from disappearance. A failed scale, unstable origin, a route that fails to transport, or stronger rival can close one path while preserving lower-level knowledge about measurement or ordinary regulation. Recording those distinctions makes the program cumulative. It also prevents later editions, websites, or automated summaries from reviving a retired component simply because the original wording was memorable. An updated map can mark the retired link while keeping the observations that remain usable for a different, clearly stated question.

Independent adjudication needs authority over timing and language, not merely access to a finished paper. Reviewers should

be able to require another recovery test, delay public release, narrow the reference class, separate an exploratory finding, or state that the prospectively specified question was not answered. A review process that can only recommend softer wording after the result has been promoted is not independent enough to govern a universal claim. Their decision should be documented before publication, with any required test and the responsibility for completing it visible to the research team.

28.2 — Representation and Measurement Gates

The hardest scientific mistakes often occur before the statistical test. A scale quietly moves the neutral origin. A stream is split after inspection. The last survey is treated as the endpoint. A model output is mistaken for the estimand. A large sample makes these choices look precise without making them valid.

This section turns those vulnerabilities into gates. Representation and origin define the coordinate. Bearer and endpoint define the unit and boundary. Observation, estimand, estimator, and transport define the inferential route. Each gate has owners, assumptions, and stop conditions.

Identification comes before calculation. Even perfect sensors would not solve a problem whose zero, unit, bearer, or endpoint had not been fixed independently. The Law is testable only through the structures that prevent the result from being true by definition.

28.2.1 — Representation and Neutral Origin Come First

Before a study asks whether a total closes, it has to say what is being totaled. Affect may be represented as one valence axis, partly independent positive and negative dimensions, a bipolar state, a distribution, or a richer nonadditive object. Choosing among these is a substantive commitment rather than clerical formatting (Cacioppo and Berntson, 1994; Borsboom, 2005).

A simple word can hide the problem. Calm may mean pleasant low arousal, emotional flatness, safety after fear, sedation,

fatigue, or a culturally preferred style of report. A slider labeled bad to good cannot decide among those possibilities by itself.

Calibration therefore comes before accumulation. Researchers need to learn which channels respond to valence rather than arousal, demand, memory, language, or motor preparation; how disagreement among channels is represented; and when uncertainty is too wide for a point estimate. A candidate construct earns use by surviving those tests and by returning a range or No adequate estimate when it cannot.

Neutral-origin sensitivity is established in Section 10.5.1: positive rescaling preserves zero, whereas additive recentering can create or erase closure. Here the requirement is independent calibration before accumulation.

Construct validity asks whether the intended interpretation is supported by theory, response processes, relationships with other variables, and the consequences of use (Cronbach and Meehl, 1955; Messick, 1995). A measure is not validated once for every purpose. Evidence that supports momentary symptom communication does not automatically support a complete-stream terminal inference.

Converging channels can help, but agreement is not guaranteed. Multitrait-multimethod designs examine whether a proposed construct converges across methods while remaining distinct from neighboring constructs (Campbell and Fiske, 1959). When self-report, behavior, physiology, and neural signals disagree, the conflict should be modeled rather than averaged away for convenience.

Comparison also requires structural transport. Measurement invariance analyses ask whether groups or times share enough of the same response structure to justify comparison, although the answer remains dependent on model and design assumptions (Putnick and Bornstein, 2016). A within-person signal is not automatically a cross-person ruler, and an instrument built in one language, age group, illness state, or device ecology is not automatically universal.

Response shift can change how anchors and baselines are interpreted after illness, disability, recovery, aging, or cultural change. Any transport claim must therefore show that the measurement relation remains valid in the new context; Chapters 11 and 18 give the full treatment.

Cross-sign calibration is equally necessary. Equal numerical distance above and below neutral does not prove equal phenomenal magnitude across positive and negative experience. Indifference points, pairwise judgments, trade-offs, behavior, and physiological or neural channels may constrain the relation, but partial identification can be more honest than a forced single point.

Even an estimate produced by a well-governed estimator remains an inference from measured data, not direct access to the latent object itself. It carries calibration error, missingness, device effects, cultural limits, participant burden, and uncertainty. A measured trajectory that hides those limits is less scientific, not more persuasive.

Use restrictions belong inside the construct definition. A momentary or interval estimate validated for research cannot be carried into care priority, classrooms, employment, insurance, policing, civic eligibility, spiritual rank, or artificial intelligence certification. Units, uncertainty, authorized use, human review, correction, appeal, retention, and deletion should be named before the number leaves the protocol.

A failed representation gate cannot be overwhelmed by a large dataset or a beautiful graph. It means the downstream terminal question has not been posed in a stable coordinate. The scientifically correct outcome may be No adequate estimate.

The handoff to *The Science of Fairness* carries two duties: build the best instrument the evidence permits, and keep that instrument from becoming an oracle. A useful measure serves a declared question and remains corrigible. A failed measure is narrowed or retired; a dangerous use stays prohibited even when the underlying measurement problem remains scientifically interesting.

28.2.2 — Independent Ownership and Endpoint Rules

The bearer and endpoint cannot be chosen to make the answer come out well. Splitting a difficult trajectory after seeing its residual, merging streams that offset one another, dropping an inconvenient interval, or ending the record when the estimate reaches zero would build closure into the procedure.

Consider a patient who loses speech under sedation and later regains communication. The loss of report is an observation event, not automatically the end of the conscious stream. Treating

it as an endpoint because the remaining record looks convenient would confuse measurement access with the truth-level boundary.

Stream ownership therefore needs prospectively declared rules for continuity, pauses, low-unity states, fission, fusion, ambiguous consciousness, and unobserved intervals. When several routings remain scientifically admissible, the analysis should preserve the resulting set of possibilities rather than select the route that places zero nearest the center.

Endpoints require the same discipline. Truth-level ending, protocol stopping, legal death, loss of communication, observation shutdown, the last qualifying experience, and endpoint-inclusive value are different objects. A study may examine one branch or compare several predeclared branches, but it cannot move among them after seeing which produces the most favorable result.

Independence is the tautology firewall. The endpoint channel may use clinical, physiological, unity-related, or adjudicative evidence, and it may remain imperfect. What it cannot do is define the endpoint algebraically from the ledger estimate whose terminal value is being tested.

The random-endpoint obstruction derived in Section 7.2.2 remains the sharpest consequence of endpoint independence. Different timing branches create different empirical burdens, and none may be selected after inspecting the residual.

A protocol endpoint may still be necessary when the truth endpoint is unavailable. The study can estimate a terminal-window object under stated assumptions, but it should not rename

the protocol boundary as the end of consciousness. The unobserved remainder remains an inferential limitation, not a value debt assigned to the person.

Procedural independence can reduce bias. Where feasible, endpoint adjudicators should not see the evolving residual estimate, and outcome analysts should not alter endpoint classifications after learning which choice improves closure. Disagreement and uncertainty should be logged rather than resolved by convenience.

Endpoint research has a distinctive burden budget. Time, fatigue, distress, privacy exposure, family involvement, device demands, travel, and opportunity cost must be estimated for each group, especially where illness, impaired communication, or proximity to death weakens the person's practical freedom. Information valuable to the project does not erase costs carried by participants or bystanders.

Participants and lawful representatives also need a challenge route throughout endpoint adjudication. They should be able to ask what is collected, correct a record where appropriate, complain without retaliation, withdraw within stated limits, and know who can pause or stop a procedure. Consent becomes more credible when disagreement can change institutional action before an endpoint classification hardens.

Measurement burden can itself change the terminal window. Prompt frequency, device wear, dream recall, repeated mood review, laboratory stress, and heightened attention near decline may alter both experience and observation. The protocol therefore

needs reactivity monitoring and prospective rules for reduction, suspension, redesign, or withdrawal.

Terminal records also contain other lives. Diaries, interviews, bedside notes, and family reports may expose partners, children, clinicians, caregivers, or others who never agreed to become data. Collection should be minimized, necessary context separated from reusable content, and unexpected disclosures handled through a predefined care and safety route.

The endpoint rule handed to science is both formal and humane: define the boundary independently, preserve admissible uncertainty, never use death as theater, and never let the convenience of a terminal claim outrank comfort or care.

28.2.3 — Observation, Estimands, and Estimators

An observation is what a study records: a rating, choice, behavior, signal, interview, sensor value, clinical note, or missing entry. An estimand is the precisely stated quantity sought under specified population and intercurrent-event conditions. An estimator is the rule that turns finite observed data into an estimate. Transport is the argument that connects that estimate to another setting, channel, cohort, or latent target.

A terminal cohort makes the separation concrete. The last completed self-report is an observation. The mean experienced valence during a declared terminal window under a policy for missed reports could be an estimand. A hierarchical model combining several channels could be the estimator. The claim that the result bears on a truth-level complete-stream residual would require a separate transport bridge.

Clinical-trial guidance from the International Council for Harmonisation (ICH) emphasizes defining the estimand before choosing the estimator. Treatment discontinuation, rescue medication, and death require explicit strategies within the scientific question; missing outcomes require estimation assumptions aligned with that question (ICH, 2019). The same discipline applies here. Last-observed mood, terminal-window mean, latent-state posterior, and complete-stream residual are not interchangeable outputs.

Intercurrent events also need declared causal roles. Sedation, analgesia, withdrawal, hospitalization, caregiver presence, and treatment change may be interventions, confounders, mediators, components of the observation process, or parts of the estimand strategy. Naming the role helps prevent an analysis from adjusting away the phenomenon or attributing a treatment effect to PTVC (Hernán and Robins, 2020; ICH, 2019).

Missingness is part of the observation process rather than empty space. People may stop responding because they are exhausted, sedated, hospitalized, distressed, improving, losing access, or dying. Missing-data methods rely on assumptions that observed data alone rarely verify (Little and Rubin, 2019). A larger sample does not remove bias from a systematically missing terminal region.

The analysis should show how conclusions change across credible missingness models and return No adequate estimate when the unseen region cannot be bounded honestly (Little and Rubin, 2019).

The estimator has its own obligations. Bias, precision, calibration, model assumptions, preprocessing, exclusions, tuning, and sensitivity to violations all matter. A mathematically sophisticated estimator can answer the wrong question perfectly when the estimand was vague or the observation process was misdescribed.

Transport needs explicit conditions. A model trained on healthy adults using phone surveys may not travel to children, nonverbal patients, people with severe illness, different cultures, or terminal care. Transportability work makes selection variables and target-population assumptions visible instead of treating generalization as a reward for sample size (Pearl and Bareinboim, 2014; Dahabreh et al., 2020).

The measured-to-latent bridge is therefore a claim in its own right. It has to survive calibration, recovery, sensitivity analysis, and held-out testing. When it fails, the study may still teach us about the observed channel, but that channel cannot be promoted by declaration into the hidden phenomenal object.

Transport should be decomposed rather than announced as one success. An instrument may retain its response structure while its neutral origin shifts. Calibration may travel while the relation to the latent target does not. A route signature may replicate while the endpoint bridge fails. Reports should name which component traveled, which required local recalibration, and which stronger conclusion remains unavailable.

Asymmetry can also be informative. A model that predicts self-report from physiology may not predict physiology from self-

report, and a relation learned in one culture may reverse or disappear in another. Such failures can reveal that the bridge captured a local association rather than the proposed latent object.

The protocol should preserve denominator flow. Readers need to see who was invited, enrolled, remained observable, changed treatment, lost speech, withdrew, died, or became unreachable, and how each transition affected the estimand (ICH, 2019). The invisible denominator is often where an apparently clean result acquires its uncertainty.

Permission must travel with the use being proposed. Enrollment, restricted archiving, public summary, model training, secondary inference, correction, retention, and deletion are different authorizations. A dataset already held by an institution does not acquire a measured-to-latent bridge, or permission for one, merely because a later analysis is technically possible.

Replication belongs after these layers are explicit. A pattern that repeats under one instrument may still be explained by adaptation, treatment, social support, device effects, or missingness. Replication strengthens the precise object that traveled; it does not automatically strengthen every claim attached to it.

The public discipline is simple: observed data never become the hidden object by declaration. Each layer receives its own name, uncertainty, failure condition, and owner. That vocabulary is not bureaucratic decoration. It lets a skeptical reader locate the exact bridge carrying the conclusion.

28.2.4 — Upstream Failure Blocks Downstream Warrant

The gates form a dependency chain rather than a point system. A strong final statistic cannot repair an unidentified neutral origin, outcome-dependent stream partition, invalid endpoint, or systematic loss of the most informative observations.

This is the logic of severe testing. A result carries weight when the method had a real chance to reveal error and when alternatives faced comparable risk (Platt, 1964; Mayo, 2018). Passing an easy downstream test cannot compensate for avoiding the hard upstream question.

A reader can follow the chain through three questions. What is being counted, and where is its independently fixed zero? Which experiences belong to this stream, over which complete interval? How do the observations and analysis justify the conclusion against capable alternatives? The preceding sections identify the representation, bearer, endpoint, measurement, and comparison requirements inside those questions. Each later answer depends on the earlier answers it uses; *The Science of Fairness* develops the methods for testing those dependencies.

Failure remains typed. An unidentified origin may yield No adequate estimate. A failed route signature may retire that route while leaving the target open. A gate-valid residual set excluding zero would bear more directly against the prospectively declared target branch. The verdict follows the location of failure.

Prestige cannot substitute for a gate. A famous laboratory, large sample, expensive scanner, sophisticated model, or beautiful visualization does not overrule a failed prerequisite. Gate owners,

closure evidence, disagreement, and stop authority should be recorded before the result is known. A gate record can name the unresolved condition and the evidence required to resolve it, making review independent of a project's presentation.

Repair is possible when new evidence genuinely addresses the failed condition. The program may build a better measurement bridge, endpoint rule, or recovery test and then rerun the chain. It may not simply move the gate after seeing the desired result. A revised test can be evaluated on newly collected or held-out observations, giving the repaired bridge an identifiable opportunity to succeed.

Public communication follows the same discipline. A downstream pattern can be described accurately without promotion to a stronger layer. Visible abstention is not weakness; it shows that the preferred claim is not being advanced through an inference the evidence cannot support. A useful public account can still explain the observed pattern and identify the specific additional evidence needed for the stronger interpretation.

28.3 — Rivals, Preregistration, and Red Teams

A universal claim should be tested by a procedure that does not need it to survive. Preregistration limits retrospective choice. Blinding limits some expectation effects. Red teams develop the strongest alternative explanations. Null results and failed routes remain publishable outcomes.

These tools are not rituals of open science. They are responses to specific human tendencies: flexible analysis, selective memory, publication incentives, allegiance, and the ease with which a compelling narrative absorbs ambiguous data (Simmons et al., 2011; Munafò et al., 2017).

28.3.1 — Preregistration

Preregistration records the question before decisive data can answer it. A publicly preregistered plan should identify the target, branch, reference class, endpoint, inclusion rules, measurement model, estimand, estimator, rival set, missing-data strategy, decision thresholds, and planned public language, and it must reside in a timestamped external registry with a persistent identifier (Nosek et al., 2018). When no such registry record exists, the accurate terms are protocol-fixed or prospectively specified.

The purpose is not ritual paperwork. It is to separate confirmatory tests from exploration and to make later flexibility visible. A deviation may be justified, but it should be dated, explained, and analyzed as a deviation rather than silently incorporated into the original plan.

Registered Reports strengthen this separation by reviewing the question and method before outcomes are known and by linking publication more closely to design quality than to a favorable result (Chambers, 2013). They are one useful model, not the only acceptable route.

A confirmatory PTVC protocol should prospectively fix the choices most capable of manufacturing zero: the neutral origin, stream routing, endpoint rule, admissible transformations, handling of unobserved intervals, and target modality. Those decisions cannot change because the first analysis is inconvenient.

Rivals require the same prospective treatment. The preferred model should not receive more data, tuning, exclusions, or diagnostic effort than its competitors. Equal-budget comparison is part of the fairness of the test.

The protocol should include stop conditions. Ethical burden, recruitment failure, instrument drift, endpoint ambiguity, privacy breach, or inability to recover known models may require pause, redesign, or termination before the planned statistic is reached.

Exploration remains valuable when it is labeled. Unexpected patterns can generate new hypotheses, reveal a missing rival, or improve measurement. The gain is lost only when exploration is presented as if it had survived a test designed in advance.

Public summaries should link back to the prospectively specified object, the external registry record when one exists, and every material deviation. A headline should not call a result

confirmatory when decisive choices were made after the data were seen.

Preregistration also protects negative results. When the question, power, and decision rule were clear in advance, a failure to support the prediction can inform the next step rather than disappearing as an uninteresting analysis.

The deeper principle is temporal: the answer should not write the question that supposedly tested it.

28.3.2 — Null Results

A null result is a family of outcomes, not a single scientific sentence. It may reflect an effect that is absent or very small, an estimate too imprecise to decide, a weak instrument, insufficient variation, poor recovery, missing data, or a design that never reached the prospectively specified target. The report must identify which possibilities remain live.

Failure to reject a point null does not establish practical equivalence. A wide interval can contain effects large enough to matter in either direction, while a small p value may accompany an effect too small, biased, or poorly transported to carry the theoretical claim.

Equivalence testing asks whether the data exclude effects beyond prospectively justified bounds, accounting for uncertainty. In the two one-sided tests procedure at the 5% level, the corresponding 90% confidence interval must lie wholly within those bounds (Lakens, 2017). The band must come from the scientific and decision problem; choosing it after the result turns a test into a rescue device.

Bayesian analysis can compare specified models or quantify evidence under a declared prior and likelihood (Dienes, 2014). Its conclusion remains conditional on those choices and cannot repair an unidentified origin, invalid endpoint, or failed model-recovery test.

The words significant and not significant should never carry the interpretation alone. Estimates, uncertainty, design quality, prior evidence, practical scale, and consequences belong in the same account (Amrhein et al., 2019).

Publication bias can erase findings that would discipline the program, leaving the visible literature more confident than the complete record warrants (Franco et al., 2014). Prospectively specified protocols, result repositories, and correction histories make unfavorable or indeterminate results harder to hide.

For PTVC, use the route, estimation and target-failure distinctions in Section 28.2.4 rather than treating every null as the same result.

A null should change the next decision by ending a protocol, redirecting work to an upstream gate, strengthening a rival, narrowing the reference class, or establishing that the data cannot answer the question. A program that cannot remember its nulls cannot learn from them.

28.3.3 — Formal-Law Detail Limit

This chapter gives the public the architecture of a fair test, not a protocol manual. Exact mathematics, simulations, priors, power calculations, code, recovery thresholds, and verdict algorithms belong to *The Science of Fairness* and its governed technical record.

The public account must still support independent criticism. It should expose the target, assumptions, frozen choices, rivals, failure conditions, ethical gates, and results that would narrow or defeat the claim. Handoff begins only after the local reasoning is complete.

Blinding removes cues that could steer judgment toward the preferred answer. Endpoint adjudicators may be kept from evolving residual estimates; analysts may receive masked model labels; language reviewers may assess claims without knowing the hoped-for result. Remaining cues and dependencies should be declared.

A serious red team receives the exact target, protocol, code, negative controls, deidentified outputs, deviations, and enough time to build the strongest rival. It searches for leakage, weak comparators, unstable origins, missingness assumptions, harmful uses, and public sentences that outrun the result (Simmons et al., 2011; Munafò et al., 2017).

Preferred teams can implement opponents badly. Adversarial collaboration gives competent critics a real role in defining comparators and scoring rules before the test set opens. Equal information and diagnostic effort make rival victory a scientific result, not a rhetorical concession.

Red-team authority must reach the decision. Reviewers should be able to require delay, narrowing, safeguards, a new recovery test, publication of an adverse result, or artifact retirement. Criticism unable to change the study or verdict is performance, not control.

The detail limit is ethical as well as editorial. Public explanations should not invite unsafe self-experimentation, unreviewed terminal observation, or amateur recreation of protocol conditions. Technical depth belongs in versioned, governed sources with stop authority.

Enough detail lets readers understand the task, lets critics reach the audit materials, and keeps the public root from posing as the final design.

28.4 — Replication, Recovery, and Scientific Verdicts

Before a favorable real-world result can carry weight, the analysis must survive two different kinds of distance. It must recognize known simulated worlds, and it must predict data that were not used to construct it. Recovery tests identification; held-out transport tests whether the fitted structure travels.

Replication then introduces new teams, sites, channels, and populations. The result may strengthen, narrow, fragment, or fail. Science advances when the verdict attaches to the component that changed rather than radiating across the whole theory by association.

This section therefore treats verdict language as part of method. A precise status can be less dramatic than a headline and more useful to every later researcher.

28.4.1 — Model Recovery Before Real-World Confidence

A method should demonstrate that it can distinguish known worlds before it interprets an unknown one. Model recovery generates data from declared candidate models across stated parameter ranges, fits the full analysis pipeline, and asks whether the pipeline identifies the model that produced each dataset (Wilson and Collins, 2019).

Parameter recovery and model recovery answer different questions. A method may identify the right model but estimate its parameters poorly, or estimate parameters well within the wrong model family. Both failures matter for interpretation.

Flexible models can win by absorbing noise or by mimicking one another. Simulation studies should therefore vary sample size, missingness, endpoint error, measurement floors, heterogeneity, and rival overlap rather than testing only the easiest case (Palminteri et al., 2017).

Use the simulation-based calibration discipline in Section 23.4.1 as one implementation check within the broader recovery battery (Talts et al., 2018).

Recovery must use the exact preprocessing, exclusions, transformations, tuning, and decision rules planned for the real study. A simplified demonstration can miss the step that creates false support.

No-PTVC worlds are essential. The pipeline should face adaptation, conditioned bridges, endpoint selection, observation floors, memory reconstruction, and open-set rivals with enough realism to produce deceptive patterns.

Failure to recover is a result. It may require simpler models, better data, a different design, or a verdict of model ambiguity. The proper response is not to lower the benchmark after the real result looks favorable.

Recovery evidence should be preserved with code, random seeds, model versions, and diagnostics so another team can reproduce the test. It is part of the claim's warrant, not a private rehearsal.

Real-world confidence begins only after the method has shown that it can be wrong in the right ways.

28.4.2 — Held-Out Transport and Independent Replication

A model fitted to one dataset may describe that dataset without traveling beyond it. Held-out prediction asks whether the prospectively specified method forecasts observations that were not used to choose the model, tune the parameters, select exclusions, or calibrate the public story.

Transport is stronger than random splitting. A result should be tested across settings, populations, channels, time periods, endpoint regimes, and institutions whose differences were identified in advance. Transportability requires assumptions about what remains stable and what changes (Pearl and Bareinboim, 2014; Dahabreh et al., 2020).

Independent replication adds a separate team, implementation, and error surface. The aim is not ceremonial repetition but a test of whether the finding survives new decisions, personnel, instruments, and context. Reproducibility and replicability are related but distinct obligations (National Academies of Sciences, Engineering, and Medicine, 2019).

Large collaborative replication projects have shown that many published effects do not reproduce at their reported strength, which is precisely why replication must be planned as ordinary science rather than as an insult to the original team (Open Science Collaboration, 2015).

Data and code should be as findable, accessible, interoperable, and reusable as ethics, consent, privacy, and law allow (Wilkinson et al., 2016). Openness is purpose-limited; intimate

trajectories should not be released merely to satisfy an abstract norm.

For randomized studies, use the SPIRIT 2025 and CONSORT 2025 protocol/report distinction explained in Section 10.4.3 to make replication possible (Chan et al., 2025; Hopewell et al., 2025).

Rivals must travel too. A preferred route that fits one cohort but loses to an ordinary regulatory model in another setting has not been universally supported. The discrepancy may reveal a reference-class boundary or model misspecification.

Replication can fail for informative reasons: instrument translation, endpoint access, cultural response, treatment patterns, missingness, or different causal structure. The record should identify the failed bridge rather than treating every difference as noise.

A single successful replication still does not establish universality. It increases confidence within the tested conditions and supports the next bounded inference. Wider claims require wider, prospectively justified bridges.

The public lesson is simple: a result becomes stronger when it survives distance, not when the original team repeats its own conclusion with greater confidence.

28.4.3 — Evidence and Theory Maturity Stay Separate

Evidence maturity and theory maturity answer different questions. Evidence maturity concerns the quality, replication, transport, and discriminatory power of observations. Theory

maturity concerns formal definition, internal consistency, branch separation, theorem status, mechanism specification, and unresolved primitives.

A formally mature theory may have no adequate empirical test. Its objects can be exact and its propositions correct under stated assumptions while the representation, bearer, endpoint, or measured-to-latent bridge remains unavailable. Mathematical completion does not supply a participant or dataset.

The reverse can occur. A regularity may replicate across settings while its explanation remains nonunique or formally incomplete. Repeated observation strengthens the regularity, not every theory offered to explain it.

The project therefore needs two separate maturity profiles. The theory profile records definitions, branch identities, proofs, route entailments, assumptions, and open problems. The evidence profile records measurement validity, recovery, preregistration, rival discrimination, replication, transport, endpoint access, and inferential strength.

Movement in either profile must name its cause. A theorem can strengthen a conditional consequence without adding data. A replicated scale can strengthen one measurement bridge without proving exact closure. A better endpoint method can improve adjudicability without favoring a route.

Either profile can also contract. Evidence weakens when invariance fails, replication does not transport, hidden flexibility appears, or a stronger rival enters. Theory narrows when an

assumption proves too strong, a bridge fails, or branches once treated as equivalent must be separated.

A single progress percentage would erase this uneven state. Keeping the profiles separate lets mathematics, measurement, criticism, and replication improve the program without borrowing authority from one another or forcing every advance into a premature verdict on PTVC.

28.4.4 — Verdicts Must Be Modular and Exact

A scientific verdict should be a typed record rather than a banner. It identifies prerequisite status, the prospectively specified target and modality, the target result, recovery result, route result, comparator result, inference status, and the public sentence actually licensed.

That record can contain several outcomes at once. Measurement may improve while endpoint identification remains weak. A candidate route may fail while the target stays open. A declared rival may outperform the preferred model while the reference class remains unsettled. One triumphant or defeated word would destroy that information.

Adverse evidence has to match the branch. A failed reserve signature bears against a reserve-coupled route, not automatically against every PTVC formulation. A valid complete-stream residual set that excludes zero would speak more directly to the prospectively declared target.

No adequate estimate is a substantive result when representation, endpoint, observation, recovery, or transport fails. It is neither a

favorable approximation nor an adverse target verdict. It states that the prospectively specified inference was not earned.

Other verdict terms also stay narrow. Route-favored concerns an optional mechanism. Rival-favored reports better performance under common rules. Model-ambiguous means the candidates were not adequately distinguished. Open-set inadequate means the tested family omitted important structure. Reference-class narrowed limits the population or branch to which any conclusion applies.

Verdict language must preserve the modality distinctions defined in Section 7.3.3; evidence for one completion geometry does not license another.

Every verdict should trigger a declared action. A failed gate redirects work to the missing bridge. A failed route is revised or retired. Ambiguity calls for better discrimination. Narrowing changes public scope. Defeat propagates to dependent artifacts and the research agenda.

The verdict packet preserves the prospectively specified question, data and code version, gate status, rival set, estimates, uncertainty, deviations, sensitivity analyses, dissent, and correction route. An abstract, press line, donor summary, and later correction should all point back to that same object. A correction should identify both the superseded packet and its replacement, so readers can follow the changed inference rather than encounter an unexplained new conclusion.

Favorable findings need the same restraint as unfavorable ones. Scientific history contains many published claims weakened by

analytic flexibility, selective reporting, and underestimated model uncertainty (Ioannidis, 2005; Gelman and Loken, 2014). A result becomes more credible when the record makes those pressures visible. A comparison between the registered analysis and the reported analysis can show exactly where choices changed and how the result depends on them.

The public sentence should therefore be readable and exact: what was tested, under which gates, what was found, what remains unresolved, and what changes next. Clarity does not require false certainty. An illustrative report could say, “The scale passed its registered calibration test in this setting; its accuracy near the proposed endpoint still requires separate validation.”

28.5 — Participation and the Handoff to Science

A public theory can gather an audience long before it has a research infrastructure. That creates a temptation to treat readers, online communities, classrooms, and volunteers as a ready-made sample. The temptation must be refused.

Participation requires a separately governed relationship. Youth, patients, grieving families, people at end of life, and communities carrying historical research harm may need additional protection, not streamlined access. Consent is a process within power, not a signature detached from context (World Medical Association, 2024; CIOMS, 2016).

The final handoff concerns artifacts as much as protocols. A diagram, questionnaire, website, or AI summary can imply diagnosis, proof, or enrollment even when surrounding prose does not. Public materials therefore remain versioned, reviewed, correctable, and withdrawable.

28.5.1 — Reading Is Not Participation

A person becomes a reader by encountering an idea, not by enrolling in a study. Reading, buying, borrowing, criticizing, teaching, quoting, parodying, or ignoring this book creates no research consent and no duty to disclose an inner response.

Private reflection remains private by default. A note, search, comment, or conversation should not become project data merely because it contains Law of Fairness language. Context and purpose matter when information moves (Nissenbaum, 2010).

Digital traces create a hidden boundary risk. Views, clicks, dwell time, reposts, location, classroom submissions, and forum activity can be collected without the user understanding that they are being studied. Public communication should not quietly convert an audience into a research population.

Human-subjects research requires a defined protocol, appropriate review, understandable information, voluntariness, risk limits, withdrawal rules, data governance, and consent suited to the population (World Medical Association, 2024; CIOMS, 2016). A public website or community conversation is not a substitute for those conditions.

Citizen contribution can remain upstream. Readers may propose questions, criticize definitions, help identify public risks, or review materials without providing intimate data. Co-design is not the same as enrollment.

Unsolicited stories need a non-research intake rule. A message describing grief, trauma, dreams, addiction, or a possible endpoint should be treated as communication, protected according to its sensitivity, and routed for safety or response when appropriate. It should not be silently coded into a dataset.

The therapeutic misconception adds a further risk. People may believe a research interaction offers individualized care or privileged insight even when the study has another purpose (Appelbaum et al., 1982). Recruitment should never imply personal balance results, better treatment, spiritual status, or access to hidden truth.

Criticism must remain free. A negative review, hostile question, or refusal to participate is not adverse data about the critic. The project should not profile dissenters or use participation status as a measure of seriousness.

The boundary is operational: recruitment must be recognizable as recruitment, research records must be separated from platform telemetry and public artifacts, and the person must know when the relationship has changed from reader to participant.

28.5.2 — Non-Diagnostic Materials

Public and classroom materials should explain the hypothesis without behaving like diagnostic instruments. They must not tell a person that they are balanced, deficient, unusually burdened, spiritually advanced, at risk, or evidence for the Law.

Youth-facing work needs stronger review because authority, development, privacy, belonging, and dependency can make refusal difficult. International and United States frameworks require additional protections for children in research and recognize assent as affirmative agreement rather than mere failure to object (U.S. Department of Health and Human Services, 2018; United Nations, 1989; CIOMS, 2016).

Assent and parental or guardian permission answer different questions. Permission concerns legal authorization; assent concerns the young person's own willingness at a developmentally appropriate level. Neither should become a decoration around a decision already made by adults.

Voluntariness in a classroom is tested by consequences. A student should be able to skip an activity without losing a grade, status,

access, or trust. The safer default is to use public texts, fictional cases, measurement problems, and rival models rather than asking for trauma, dreams, family conflict, substance use, or private belief.

A non-diagnostic label is not enough if the artifact still behaves like a test. Personalized feedback, risk colors, score-like categories, comparison with peers, or a suggestion that the tool sees something hidden can create diagnostic force without using the word diagnosis.

Data minimization should be visible. Do not collect names, intimate histories, location, biometric data, or persistent identifiers when the educational purpose can be met without them. Student work should not be repurposed as a research or model-training dataset by default.

Unexpected disclosure needs a response plan. Adults should know how to pause the activity, protect privacy, respond to immediate safety concerns, and route the student to qualified support. The role is care, not interpretation through the Law.

Bystander privacy matters because a young person's story often contains parents, siblings, peers, teachers, and clinicians. One student's willingness to speak does not authorize public use of everyone else's life.

Young people and family caregivers helped develop SPIRIT-C 2026, which adds seventeen pediatric protocol-reporting items to the general SPIRIT 2025 framework (Baba et al., 2026). The broader lesson is that youth should help shape the safeguards that govern research about them.

Non-diagnostic design leaves the learner with questions, concepts, and routes to help, not a label.

28.5.3 — Public Artifact Limits

A public artifact is any detachable surface through which the project travels: a caption, diagram, slide, video, social post, press line, donor brief, classroom handout, website card, artificial intelligence summary, or illustrated page. The hostile-crop test asks what the fragment implies after its safest surrounding sentence is removed. If it presents pain as a safe purchase of chosen joy, treats balance as visibly known, claims completed validation, or makes participation seem expected, it needs redesign or withdrawal. A clearly conditional statement of net-remainder accounting is not itself a promise of practical control.

Visual grammar carries claims before prose is read. Perfect scales, glowing zeros, cosmic judges, central artificial intelligence eyes, ranked people, or paths ending in light can imply proof, destiny, payment, surveillance, or guaranteed progress. A caption cannot reliably cancel an image whose first impression says the opposite.

Use the release record specified in Section 26.5.2 for each high-reach artifact. It keeps the fragment connected to the manuscript sentence and scientific status that support it.

Apply the audience and detachable-fragment rules in Sections 25.1.3 and 26.5.2. Here the additional scientific limit is simple: no public artifact may state a stronger claim than the governed source permits.

Data-sharing artifacts have a separate ceiling. De-identification is not permanent, and combinations of attributes can reidentify people in records presented as anonymous (Narayanan and Shmatikov, 2008; Rocher et al., 2019). Open-science values do not erase consent, purpose limits, intimate-data risk, or the rights of bystanders described inside a participant's story.

Apply the accessibility and standalone-context rules in Section 26.5.2 to every public artifact; claim status and safeguards must survive nonvisual access as well as the visual layout.

Youth-facing artifacts require a separate detachability review. A young person and caregiver, where appropriate, should be able to understand what the material does, decline personal disclosure without consequence, and identify where help or correction lives. No worksheet, graphic, or automated response may imply that a teacher, platform, or project can read a young person's hidden balance.

Artificial intelligence summaries need the same source lock as human summaries. They should retrieve from approved versions, preserve status labels and refusal language, disclose uncertainty, and route consequential questions to accountable people. Fluency cannot turn a model into the project's scientist, clinician, teacher, confessor, or final interpreter.

Correction and retirement should be rehearsed before release. Teams need a way to replace a misleading quote card, consent-failed story, outdated verdict, inaccessible graphic, or overclaiming summary across books, websites, slides, classrooms, archives, and media packets. Repeated misreading,

reidentification risk, superseded evidence, or loss of source control may justify withdrawal even when an artifact is popular. The release test is whether a separated reader can recover what kind of claim it is, what supports it, what it does not authorize, and where the correction record lives.

These standards have a human test: a reader using a screen reader, a translated summary, or a classroom handout should be able to find the same claim, limitations, and route to correction. Accessibility must be checked in use, including keyboard navigation, reading order, contrast, zoom and reflow, captions, alternative text, and a nonvisual route through diagrams. The production record should retain those checks. A withdrawal rehearsal must also verify updates to search results and automated systems, and a usable route for affected people to report misuse.

A better measure of neutral feeling, an independently defensible endpoint rule, or a comparison that reveals where two models diverge would each be a definite advance. Each would make the eventual answer more informative, whatever that answer turns out to be.

Part XI — Objections and Outcomes

The final Part gives the strongest objections full force and makes their consequences explicit. A serious objection must attack a specific link: measurement, rival explanation, bearer identity, endpoint, evolutionary or simulation inference, falsifiability, or public safety. The answer must state what would narrow, retire, or defeat the relevant claim.

The book therefore resists two end-stage errors: triumphal synthesis, in which accumulated definitions and references create the impression that PTVC has been established, and rhetorical rescue, in which every adverse result is redescribed as mystery. Public compression is a separate test because captions, headlines, classrooms, search results, and AI summaries may detach a fragment from its safeguards.

Objections become most useful when they identify the exact dependency under pressure. A measurement critic should say which construct, origin, or bridge is underidentified; a rival-model critic should name the ordinary mechanism and the observation it can explain; an endpoint critic should show where a stopping rule becomes circular; a safety critic should identify the action or institution that could be changed by misleading language. This makes disagreement cumulative because each criticism points toward a narrower claim, a better test, or a clearer repair. A precise objection can be preserved as an explicit test rather than absorbed into general doubt. That creates durable research memory in which later evidence can answer the criticism directly, strengthen it, or show that the original target was misidentified.

Criticism can therefore improve the project even when it does not defeat PTV. It can expose a missing control, sharpen a reference class, separate a theorem from evidence, or show that public wording outruns its warrant. The standard at the end of the book is correctability rather than loyalty. A durable research program should become easier to challenge as it grows, preserve serious adverse results, and let the next reader see exactly what evidence would change the current status. The most valuable criticism also distinguishes between a claim that needs better evidence and a claim whose structure has failed. That distinction prevents endless repair of a defeated branch while protecting narrower questions that remain genuinely open.

Chapter 29 gives critics standing to force these distinctions. Chapter 30 classifies the possible outcomes: clarification, bounded support, route support, model ambiguity, No adequate estimate, reference-class narrowing, rival victory, or strong-form defeat. Care, privacy, dignity, and nonparticipation remain independent of every scientific verdict. The final Part therefore treats disagreement as part of the method rather than an obstacle to the project. A reader should be able to trace each major objection to the dependency it challenges, the evidence that could answer it, and the consequence of failure. The closing standard is not unanimity; it is a record clear enough that support, narrowing, rival victory, and defeat cannot be confused with one another. That makes the ending useful even before the hypothesis has a final verdict, because it leaves a map of what has been learned and what remains testable.

Chapter 29 — The Hardest Objections

Strong objections are valuable because they reveal what an answer would actually have to accomplish. The Law asks unusually broad questions about experience, identity, whole histories, and endpoints. A careful psychometrician, affective scientist, philosopher, clinician, survivor advocate, consciousness researcher, evolutionary theorist, or public critic can therefore contribute by identifying the point at which an appealing story becomes a testable claim.

Measurement skepticism asks us to build better bridges from experience to observation. Ordinary rivals ask whether familiar mechanisms already explain a pattern or could even realize the deeper relation. Identity and endpoint questions force the bearer and boundary to become explicit. These are productive demands because each turns a broad objection into a narrower research problem.

The strongest response is to make the Law easier to investigate. A good answer names the representation, comparison, boundary, and observation that would matter. It also shows what would count as real progress: a better measure, a discriminating prediction, a mechanism that survives new paths, or a result that narrows the claim.

Seen this way, criticism becomes part of the research architecture. It helps separate a beautiful possibility from a scientifically earned conclusion while showing exactly where new evidence could move the project forward.

Measurement comes first because felt experience is lived from inside and inferred from outside. Reports, behavior, physiology,

brain activity, context, and clinical observation can contribute evidence, but none is the experience itself. Construct validity and multimethod agreement are achievements under assumptions, not possession of the person (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Borsboom, 2005).

Ordinary rivals come next because many patterns that make the Law seem intuitive already have substantial explanations. Adaptation, opponent process, allostasis, predictive regulation, computational mood, learning, and memory can produce return, rebound, stability, and context sensitivity without exact terminal closure. The Law cannot claim those familiar phenomena merely because they fit its story.

Identity then pressures the bearer. The formal target belongs to a separately routed qualifying stream, not automatically to a body, legal person, memory archive, or social biography. Disorders of consciousness, covert command following, anesthesia, sleep, dissociation, and uncertain synthetic candidates show why observed behavior and stream status can separate (Giacino et al., 2018; Bodien et al., 2024).

Physics, evolution, and simulation create a different temptation. They can show that a proposed constraint is coherent, compatible with a class of models, or reproducible inside a designed world. None of that establishes that real conscious histories obey it. A model can clarify possibility and failure without becoming a witness for reality.

Falsifiability becomes most useful when it is translated into specific research choices. If a measurement bridge fails, the next

task is to improve or replace that bridge rather than reinterpret the result. If a rival predicts the same pattern, the next study should look for a discriminator. If a public formulation causes confusion, the language can improve without abandoning the scientific question.

Ethical review strengthens the same process. It helps ensure that the search for evidence does not distort care, invite self-testing, or turn another person's suffering into a score. Those limits make the research object cleaner because they force the project to work with observations and interventions it can justify.

The chapter therefore treats critics as collaborators in precision. Some objections concern the target law, some concern measurement, some concern mechanisms, and some concern public use. Separating those levels lets each objection point toward the next useful question instead of collapsing every difficulty into a single verdict.

By the end, the reader should have something more valuable than a list of cautions: a map of the places where evidence could genuinely strengthen, narrow, redirect, or retire a claim. That is how a large idea becomes a research program.

Measurement and identity problems may leave the latent claim undecided while blocking empirical adjudication. Rival explanations may defeat a proposed signature without settling every formal branch. Ethical objections may prohibit a study or sentence even when the underlying proposition remains logically possible.

The reader should also notice that skepticism is not homogeneous. A scientist may demand model recovery. A disabled reader may reject proxy authority. A palliative clinician may fear treatment distortion. A philosopher may challenge the neutral origin. A public critic may focus on how a sentence behaves once cropped from its safeguards. Each concern belongs to a different lane of answerability.

29.1 — Can Feelings Be Measured?

The strongest measurement objection begins with a fact no instrument can repeal: felt experience is lived from a first-person point of view. A pain rating, facial movement, autonomic signal, choice pattern, clinical note, or brain measure is publicly observable, but the experience itself is not handed over with the data. The objection asks whether the Law has chosen a target that science can approach only through model-dependent public traces, and whether those traces can identify enough of the target to justify the claim being made.

Imagine a bedside scale from zero to ten. One patient has aphasia, another fears being labeled difficult, another has learned to dissociate from pain, and another uses the top of the scale differently after years of illness. The recorded numbers may still be clinically useful, but they do not become commensurable merely because they share a form. A theory of lifetime valence has to carry that problem rather than hide it inside averaging.

The burden is therefore not to produce a number, but to show that any estimate remains connected to the declared target strongly enough for the use it is asked to bear.

Consider a hypothetical postoperative patient whose report, behavior, and physiology diverge. The patient says the pain is manageable because they fear being labeled difficult; movement remains guarded; pulse is elevated; a sedative blunts expression. None of the channels is worthless, yet each is shaped by a different process. A defensible estimate would need a theory of those processes, a declared context of use, and an uncertainty

range. More channels do not automatically produce more truth because convergence must be explained and disagreement must remain visible.

29.1.1 — Strongest Form of the Objection

The skeptic's strongest claim is not that emotions are unreal. It is that the public evidence used to study them is related to experience through fallible reports, instruments, models, and interpretations. A theory that depends on a complete lifetime total therefore inherits every uncertainty in that chain, and it cannot answer by promising that future technology will simply see more.

The objection deepens at the boundary between testimony and inference. Pain, dread, relief, shame, grief, boredom, tenderness, nausea, and hope are known most directly by the person who lives them; everyone else receives reports, behavior, physiology, records, or neural signals shaped by context and measurement. Those traces can be informative without becoming transparent access, so every scientific claim must name the bridge that connects a public observation to the private construct it is meant to represent (Cronbach & Meehl, 1955; Borsboom, 2005).

The objection becomes more serious when the measurement model is left implicit. Some indicators are treated as effects of a hidden state, such as tears following grief. Others are components of a stipulated composite, such as housing and income in a material-deprivation index; a possible cause, such as social isolation contributing to loneliness, is not automatically a constituent of the experience. Still others are merely associated with the target, such as heart rate accompanying both fear and

exercise. Mixing these roles without saying which is which can produce a composite that is mathematically neat but conceptually unstable. Construct validity requires a theory of why each indicator belongs, not merely a correlation table (Cronbach & Meehl, 1955; Borsboom, 2005; Flake & Fried, 2020).

A future sensor does not escape this problem. Better temporal resolution can reveal faster changes, and better spatial resolution can distinguish more physiological processes, but neither improvement chooses the neutral origin, decides whether relief and pleasure share one scale, identifies the bearer, or proves that a signal has the same meaning across contexts. Those are identification questions. They require explicit assumptions, comparison with alternatives, and evidence that the interpretation survives challenge. More data can reduce some uncertainty while making other uncertainty visible.

This objection is strongest when it refuses both cynicism and wishful instruments. It does not say that science can never study experience; it says that any study of experience must show how the construct survives translation into public data. That standard protects the formal hypothesis from becoming a promise that future sensors will solve a problem of identification by force (American Educational Research Association et al., 2014; Messick, 1995).

The strongest reply is therefore a narrow one. A measure may be defensible for a declared purpose when its construct, response process, calibration, context of use, missingness, and failure conditions are all stated in advance. It may support a within-person comparison without supporting a comparison between

people. It may detect change without identifying a lifetime total. It may help a clinician hear a patient without licensing an institution to rank that patient. The object is not to defeat skepticism, but to let skepticism determine exactly how far the estimate may travel (American Educational Research Association et al., 2014; U.S. Food and Drug Administration, 2025).

The depth of the objection becomes clear when two well-validated instruments yield different lifetime orderings. Averaging them would not resolve the conflict because each may embody a different theory of valence, time, memory, and report. Before either estimate can be called wrong, the target that makes wrongness meaningful must be declared. A measurement program that skips that step can become precise about its conventions while remaining uncertain about experience.

The objection succeeds whenever the project treats an indicator as the experience, a construct as a natural unit, or an estimate as a private truth. The only defensible reply is to build explicit bridges and let them fail. Measurement begins, rather than ends, with the admission that the target exceeds the instrument.

29.1.2 — Invariance Challenge

A scale can be reliable and still fail to mean the same thing across people, settings, or time. Invariance asks whether the relation between the observed indicator and the intended construct remains stable enough for the comparison being made. Without that step, one common numerical format may conceal several different psychological quantities.

Suppose the same five-point distress item is translated into two languages. In one language the middle category means ordinary discomfort; in another it sounds like emotional indifference. The response options look identical in a spreadsheet, but the psychological thresholds differ. The same problem can arise across childhood and adulthood, before and after a stroke, or before and after years of chronic pain. Invariance testing asks whether the pattern, strength, and intercept of the measurement relation remain stable enough for the exact comparison being attempted (Putnick & Bornstein, 2016).

Invariance is where measurement skepticism becomes hardest. A signal that appears meaningful in one culture, age group, illness state, language, device ecology, or disability context may not mean the same thing elsewhere. Same-scale claims do not become valid because the model needs them. They must be earned (Campbell & Fiske, 1959; Putnick & Bornstein, 2016).

Long studies introduce response shift. A person living with progressive illness may change what counts as a difficult day, what they compare themselves with, and which dimensions of life they value. That change can be adaptation, learning, loss, resilience, or all four at once. Treating it as mere instrument drift would erase meaningful psychological change; treating it as pure improvement could erase continuing hardship. A lifetime hypothesis must model changing interpretation rather than pretending that the first interview and the last interview speak an unchanging language (Messick, 1995; Shiffman et al., 2008).

A candidate multichannel valence estimator is useful only if it is introduced as a fallible research construction. It does not read

happiness, read suffering, rank lives, or disclose an inner total. Its channels, signs, weights, exposure rules, and calibration sample must be protocol-fixed before outcome inspection for confirmatory use, challenged against alternatives, and narrowed when the construct does not transport (Messick, 1995; American Educational Research Association et al., 2014).

The methodological rule follows directly. Origins, signs, item meanings, channel weights, and transformation rules should be frozen before the decisive data are interpreted. Transport should then be tested across sites, populations, devices, languages, and relevant clinical states. When the bridge does not travel, the result becomes local, partially identified, or no adequate estimate. That is not a bureaucratic retreat. It is the scientific difference between a measure that serves people and a measure that uses a common number to conceal different meanings (U.S. Food and Drug Administration, 2025; Yarkoni, 2022).

The measurement objection has bite only if it can force loss. Failure would include poor reliability, unstable calibration, unmodeled noise floors, nontransport across populations, channel signs that reverse under context, missingness that mimics compression, or estimators that predict social desirability better than felt burden.

The challenge therefore extends to mode of administration. Voice, keyboard, caregiver proxy, clinician interview, and passive inference can draw on different abilities and social pressures. When a channel changes, researchers should not assume that the scale has remained the same simply because the response categories are unchanged. Mode effects belong in the

observation model, and unresolved differences should widen or limit the claim rather than disappear into a harmonized score.

The strongest response is therefore conditional. Same-scale claims survive only where the origin, sign, factor structure, loading pattern, exposure rule, and context of use remain defensible. Elsewhere the result must become group-specific, partially identified, or withdrawn.

29.1.3 — Dignity Requires Humility

Measurement humility has an ethical purpose. The greater the uncertainty around a private state, the less authority an institution has to turn the estimate into a verdict about credibility, eligibility, worth, compliance, or care. Dignity enters precisely where the data are most tempting to overread.

Dignity changes the first question asked of a measurement system. The first question is not, “How much data can we collect?” It is, “What decision, if any, requires this information, and what burden will the collection place on the person?” A brief voluntary diary used to improve symptom care is not ethically equivalent to passive sensing that follows location, sleep, voice, contacts, and physiology for years. Both may produce research data, but their intrusiveness, reversibility, bystander effects, and consequences differ sharply (World Medical Association, 2024; CIOMS, 2016).

The dignity answer is not anti-measurement. It is anti-appropriation. People can benefit when pain, distress, fatigue, relief, recovery, and care burden become better understood. The danger is not that research learns too much. The danger is that

research forgets the difference between evidence about a person and authority over a person (Borsboom, 2005; Messick, 1995).

The evidential standard for relying on an estimate should rise as the stakes rise. A noisy indicator might be acceptable for exploring a research question, yet wholly unacceptable for denying care, determining employment, predicting dangerousness, or deciding who deserves attention. This proportionality principle prevents a familiar error: using a model because it exists rather than because it is sufficiently valid for the consequence attached to it. Fit for purpose is not a permanent badge. It is a relation among a construct, a population, a use, and a decision burden (U.S. Food and Drug Administration, 2025).

Humility also has an architecture. Collect the least information needed; separate identifiers from research data; provide correction and deletion routes where possible; log access; prohibit secondary model training without new authority; and make withdrawal practical rather than ceremonial. A participant should not lose ordinary care, standing, or respect by refusing. The system should be designed so that one person's silence cannot be turned into suspicious data and one person's uncertainty cannot be converted into noncompliance.

Most importantly, no estimate outranks the person. A clinician may use a measure to open a conversation, but a score cannot prove that pain is absent. A researcher may identify a pattern, but the pattern cannot define inner truth. An institution may report group uncertainty, but it cannot transform that uncertainty into a private verdict. Dignity requires humility because the gap between public evidence and lived experience is not an

inconvenience to be conquered. It is a boundary that keeps inquiry answerable to the people whose lives make the inquiry possible (Beauchamp & Childress, 2019; Fricker, 2007).

A dignity-preserving study should make refusal legible as refusal, not as a symptom. Declining a diary, removing a wearable, or withholding an answer may reflect privacy, fatigue, mistrust, culture, or a reasonable judgment that the question is too intrusive. Treating nonparticipation as evidence of distress or deception would turn consent into a trap. The analysis must allow a person to leave without converting departure into a hidden diagnosis.

Humility should also govern proxy reports. Families and clinicians may provide indispensable observations when direct report is unavailable, but their accounts concern behavior, interpretation, and relationship as well as the person's experience. Proxy information can narrow uncertainty without becoming the person's voice. The manuscript should preserve that distinction especially near disability, delirium, childhood, and the end of life, where well-intentioned others can otherwise acquire more epistemic authority than the evidence supports.

A humane measurement program can still be ambitious. It can improve prediction, calibration, and uncertainty while preserving the person's right to contest, refuse, or exceed the model. The measure earns trust not by claiming completeness, but by remaining visibly smaller than the life it describes. That modest scale makes better measurement easier to use because correction and disagreement remain possible without threatening the person's standing.

29.2 — Do Ordinary Rivals Already Explain It?

Many of the patterns that make a fairness hypothesis feel plausible are already familiar to ordinary science. People adapt to new circumstances, regulatory systems resist disturbance, opponent processes create rebound, predictive systems update expectations, and moods summarize recent momentum. None of these findings establishes terminal neutrality, but each can explain return, stabilization, and context-sensitive change.

The right comparison is not between a rich Law story and a caricature of adaptation. It is between explicit models given comparable data, flexibility, priors, and opportunities to fail. A rival may explain the visible pattern without ever carrying a lifetime ledger. If it does, the Law cannot claim the pattern simply because the pattern is consistent with its vocabulary.

Consider a person who wins a lottery, survives an injury, changes jobs, or leaves an abusive relationship. Affect may move sharply, then settle into a new or partly familiar range. That movement can reflect comparison, attention, learning, physiology, social support, memory, opportunity, and expectation. To call it balancing before those mechanisms are modeled would convert an observation into a conclusion.

The section therefore grants ordinary rivals first explanatory rights over everyday return and stability. Only after they have been given their strongest form can the Law ask whether a distinctive terminal, route, or cross-window structure remains.

29.2.1 — Adaptation, Regulation, and Prediction as Rivals

Ordinary mechanisms operate at different loci and timescales. Habituation can reduce response to repetition, expectation updating can change the reference against which an event is felt, homeostatic and allostatic control can stabilize bodily variables, opponent processes can reverse an initial response, and computational mood can carry recent reward history into later evaluation. A comparison must therefore give those mechanisms their full combined strength. Their success must be interpreted by the property they establish, as the following comparison explains.

The strongest rival is rarely one mechanism acting alone. A person may habituate to a repeated sound, update expectations after a life change, regulate attention, receive social support, change behavior, and experience opponent rebound from the same event. A flexible PTV route could be made compatible with all of these processes, but compatibility is not discrimination. The scientific task is to identify predictions that differ after ordinary adaptation, regulation, learning, and context have been given enough freedom to explain the data (Diener et al., 2006; Sterling, 2012).

Each process predicts a different pattern of speed, generalization, rebound, context dependence, and persistence. Habituation may remain stimulus-specific; social comparison depends on a reference group; treatment changes causes; response shift changes the reporting map. A model that matches average return while missing those process signatures has not yet explained the trajectory (Brickman et al., 1978; Diener et al., 2006).

Equal-budget comparison matters because expressive models can win by being allowed more hidden states, more tuning, or more information. A PTVC-linked model should not receive the endpoint, later observations, or a hand-built completion variable while a rival is restricted to age and a linear trend. Both sides should receive the same training windows, covariates, missingness model, intervention history, and evaluation metric. Otherwise the contest measures analyst generosity rather than explanatory power (Wilson & Collins, 2019; Palminteri et al., 2017).

A useful contrast is local correction versus terminal complementarity. Adaptation predicts that responses to a persistent change often diminish or reorganize. Homeostatic control predicts resistance around regulated variables. Predictive accounts expect affect to depend on expectations, precision, and action. PTVC requires an additional whole-stream relation. A study therefore needs a circumstance in which the ordinary rivals make one forecast and a frozen Law-linked route makes another. Without that fork, even an excellent fit cannot identify which explanation did the work.

The predictive rival says that the brain is not passively receiving the world. It continually models causes of sensory and interoceptive input, predicts what should happen next, updates errors, adjusts precision, and acts to make the world and body more predictable or viable. Affect can be understood through interoceptive prediction, control demands, and regulation of bodily states (Friston, 2010; Seth & Friston, 2016).

Rivals should also be compared at the level of intervention response. A model that fits ordinary trajectories may fail when expectations, controllability, social feedback, or physiological state are changed safely. Designing such forks requires specificity about what each account predicts before the manipulation. A vague claim that every mechanism contributes cannot be distinguished from a vague claim that none is necessary.

The rival's breadth is precisely what makes it serious. It links body, perception, action, uncertainty, and affect in an architecture that can generate balance-like patterns without terminal fairness. A fair comparison lets that architecture make its full predictions first, then asks whether any frozen terminal consequence remains that it does not already explain (Eldar et al., 2016; Sterling, 2012).

A familiar process becomes especially interesting when we ask exactly what it explains. If ordinary regulation reproduces only recovery, rebound, or a narrowing range, those patterns do not distinguish PTVC from its rivals. If an independently supported mechanism, together with all the required independent links to felt experience, stream ownership, and the endpoint, entails the exact complete-stream condition, it could instead realize the Law. The decisive question is whether the mechanism merely resembles closure or requires it under the declared conditions. Familiar biology need not be an opponent of a deeper regularity.

29.2.2 — What Rivals Explain About Return and Stability

Everyday return is a broad phenomenon, not a signature owned by one theory. People habituate, compare, reinterpret, learn, receive support, change environments, and rebuild routines.

Organisms also regulate internal variables and allocate attention in ways that can make affect appear to settle after disruption.

Consider bereavement. Intense grief may soften, recur around anniversaries, change with social support, and coexist with meaning, anger, fatigue, or relief. A graph that trends toward a narrower range can be generated by time, memory, routine, changing demands, selective observation, and ordinary human adaptation. Calling that trend evidence of lifetime closure would erase the mechanisms visible in the person's actual life. It would also confuse a population pattern with the trajectory of any one stream (Luhmann et al., 2012).

Even complete local return does not reset accumulated exposure. A state can return to a prior range while the signed area accumulated during the disturbance remains nonzero. Local stabilization can therefore be clinically important and scientifically well explained without constituting evidence of terminal integral closure (Luhmann et al., 2012; Diener et al., 2006).

The same caution applies to dramatic positive events. The early lottery study found that its sampled winners did not report significantly greater current happiness than controls; it did not establish that any initial gain had permanently disappeared. Later well-being research shows heterogeneous adaptation, domain-specific effects, and durable differences for some events and some people. The lesson is not that everyone returns to one set point. It is that expectation, comparison, personality, circumstance, and time can generate complex return without an exact terminal law (Brickman et al., 1978; Diener et al., 2006).

Clinical stability can also be manufactured by the observation process. People with the greatest distress may stop answering, enter treatment, become sedated, lose communication, or leave the study. A final sample can look calmer because the remaining participants are easier to observe. Device floors can round small changes toward zero, and retrospective reports can favor peaks and endings. Before a pattern is interpreted as regulation, much less fairness, the study must ask who disappeared and which states became unmeasurable (Little & Rubin, 2019; Shiffman et al., 2008).

Adaptation may win in many ordinary domains. Wealth changes can produce temporary spikes and then return. Loss may produce grief trajectories that change over time without neutral closure. Disability adjustment can reflect environmental accommodation, identity reconstruction, peer support, and changed expectations. Chronic illness can produce both adaptation and enduring burden (Solomon & Corbit, 1974; Sterling, 2012).

The burden of evidence stays with the Law. It cannot ask adaptation to disprove every possible terminal reading. The task is to establish the complete-stream relation through independently justified evidence; local recovery alone does not do that work.

Social structure matters as well. Income replacement, accessible treatment, safer housing, family care, and community belonging can produce apparent emotional recovery through ordinary causes. A model that treats the recovery as an endogenous balancing process while omitting those supports would misattribute social work to hidden law. Concrete covariates and

causal histories are therefore not distractions from the theory; they are necessary rivals to any claim of unexplained closure.

National life-evaluation data sharpen the objection. The World Happiness Report finds persistent, substantial differences in how people in different countries evaluate their lives, and its rankings draw on responses collected across more than 140 countries. A critic can reasonably ask why long-running high and low national averages should not count as broad evidence against a universal per-stream closure claim (Helliwell et al., 2026).

They count as evidence about life evaluation, social conditions, and population differences, and they deserve serious causal and moral attention. They are not, however, complete-stream terminal residuals: the Cantril Ladder asks respondents where they stand in life at the time of survey, national scores average people and years, and the measure does not identify each stream's neutral origin, phenomenal exposure, true endpoint, or unseen remainder. Such data can pressure external validity and motivate longitudinal cohort tests, but population averages cannot establish or cancel an individual terminal residual (Helliwell et al., 2026).

The correct inference from return and stability is therefore modest. These patterns justify studying adaptation and regulation carefully. They do not by themselves justify a hidden lifetime ledger, exact terminal closure, or a mechanism that knows what remains to be balanced.

29.2.3 — Rival Victory Is Allowed

A rival victory is not a ceremonial concession. It changes which explanation owns the tested pattern. When an ordinary model

predicts held-out data better, recovers its generator more reliably, and survives transport or intervention that a Law-linked route does not, the Law-specific explanation loses that contest.

The comparison must be defined before results. Prediction error, calibration, transport, generator recovery, and intervention response answer different questions, so the verdict should name the comparison actually won (Akaike, 1973; Burnham & Anderson, 2002; Shmueli, 2010). If an adaptation-plus-missingness model reproduces terminal-looking compression in a new cohort while a PTVC-linked route adds no held-out information, that route loses evidentiary ground (Mayo, 2018).

Scope remains exact. A rival may explain everyday return without settling complete-stream closure, or defeat one signature while another question remains open. Adaptation, predictive regulation, opponent processes, homeostatic control, learning, treatment, development, social context, and evolutionary history all receive full standing rather than being treated as background obstacles (Wilson & Collins, 2019; Palminteri et al., 2017; Tinbergen, 1963; Williams, 1966; Gould & Lewontin, 1979).

A real victory moves attention, resources, and public claims. Measures, datasets, and safety infrastructure may remain useful, but the next studies should pursue the winner's unresolved questions rather than preserve Law-specific work by inertia (Longino, 1990; Oreskes, 2019). The honorable response states where the rival won, what no longer follows, what remains open, and where inquiry should go next.

29.3 — Are Identity and Endpoints Too Uncertain?

The formal claim belongs to one qualifying stream, but a stream is not directly marked by a birth certificate, a hospital wristband, or a memory archive. Bodies persist while consciousness changes. Communication can disappear while covert command following remains. Sleep, anesthesia, seizures, dissociation, dementia, coma, recovery, and uncertain synthetic systems all complicate the boundary of the bearer.

Endpoints are no easier. Biological death, loss of speech, last observation, withdrawal of a device, and the irreversible end of qualifying consciousness are not interchangeable. A terminal hypothesis becomes tautological if the endpoint is moved until the ledger appears to close. The stopping rule has to be set independently and prospectively.

Modern disorders-of-consciousness research makes the problem concrete. Some behaviorally unresponsive patients show task-responsive brain activity, while others remain difficult to classify despite repeated assessment. These findings do not identify a PTVC stream, but they demonstrate why outward behavior alone cannot settle conscious status (Claassen et al., 2019; Bodien et al., 2024).

The honest result may therefore be partial identification, reference-class restriction, or no adequate estimate. Uncertainty about the bearer or endpoint is a reason to make the claim smaller, not a device for preserving it.

29.3.1 — The Bearer Problem May Be Underdetermined

The bearer problem concerns ownership and continuity: which phenomenal events belong to the same stream, and under what evidence can that assignment be defended? The formal target requires a separately routed maximal qualifying stream, yet neuroscience and philosophy offer no single universally accepted marker for onset, persistence, division, fusion, pause, or irreversible ending. A favorable residual cannot decide the bearer after the fact; unresolved individuation must remain visible in the result.

A legal identity is not yet a stream identity. A passport, medical record, or continuous body can organize care and rights without answering whether every qualifying experience belongs to one uninterrupted conscious stream. Conversely, a temporary loss of report does not prove that the stream ended. General anesthesia, nonconvulsive seizures, severe motor impairment, and cognitive motor dissociation show why public behavior cannot be treated as a perfect marker of conscious access (Giacino et al., 2018; Claassen et al., 2019; Bodien et al., 2024).

Evidence can disagree across report, behavior, neurophysiology, and longitudinal history. A body may remain continuous while report disappears; command following may reappear under imaging; memory may fragment while causal integration persists. The bearer question is therefore a model-selection problem whose uncertainty must propagate into routing and endpoint inference, not a choice of whichever marker is most convenient (Bayne, 2010; Seth & Bayne, 2022).

The problem becomes even harder in cases of possible fragmentation. Split-brain research, dissociative phenomena, alternating access, and severe neurological injury raise questions about whether one bearer can host more than one route of experience or whether apparent division remains functionally integrated. This book should not pretend to settle these debates. It should explain the consequence: a terminal claim cannot be evaluated until its ownership rule says what belongs together and what evidence would justify that assignment (Bayne, 2010; Seth & Bayne, 2022).

A defensible protocol would therefore separate three judgments. First, does the system meet a declared threshold for possible consciousness? Second, does the evidence support one stream, more than one, or an unresolved routing structure? Third, is the available observation sufficient to connect any measured window to the declared stream? These judgments may use overlapping evidence, but they are not interchangeable. A positive finding at one gate cannot compensate for failure at another.

The most important protection is that uncertainty about the bearer never becomes uncertainty about dignity. Clinical care, legal standing, precaution, and ordinary respect do not wait for a PTVC classification. The unknown category is not a lesser human category. It is a limit on the theory's authority. When identity is underdetermined, the correct scientific response is to narrow or abstain, not to let the model decide who counts (World Medical Association, 2024).

Fission and fusion thought experiments expose why continuity alone is insufficient. If one stream appears to divide into two

successors, memory and causal connection may run down both branches. Declaring one branch the true bearer because it closes more neatly would be outcome-dependent routing. The ownership rule must be chosen without looking at the terminal residual, and some cases may remain genuinely underdetermined.

Unknown should remain visible in truth-domain accounting rather than being dropped from analysis. Exclusion from an empirical reference class requires an explicit failed gate and a stated scope consequence. It never becomes evidence that a person, animal, patient, artificial system, organoid, or fragmented stream lacks consciousness, dignity, or moral standing.

Underdetermination is therefore a legitimate endpoint of analysis. Some cases may be classifiable, some only partially identified, and some outside the current reference class. The unknown category protects both the person and the theory from forced certainty.

29.3.2 — Endpoint Ambiguity Cannot Save the Theory

A terminal claim becomes empty if the endpoint is chosen after the ledger is inspected. The truth endpoint, protocol endpoint, last observation, biological death, and loss of communication must remain separate. Each can fail to coincide with the end of qualifying experience.

A terminal study can contain several endings. Enrollment may end when a participant loses capacity; observation may end when a sensor is removed; a clinical phase may end when sedation begins; biological life may end later; qualifying consciousness may have ended earlier or may remain uncertain. Treating these

times as one endpoint creates a hidden selection rule. The protocol needs a map that names each time and states which bridge, if any, connects it to the truth-level endpoint (Greer et al., 2023).

For the strict universal branch, fix the truth-domain class, stream and routing ontology, representation branch, neutral origin, sign and scalarization, endpoint branch, observation model, and measured-to-latent bridge before inspecting the case. One qualifying stream with a nonempty truth-containing residual set that excludes zero refutes exact PTVC for that declared class. Reclassifying the stream, moving the endpoint, changing the origin, switching branches, or adding continuation after the adverse result creates a new and narrower hypothesis; it does not erase the loss.

Ambiguity can explain why a result is not interpretable, but it cannot be used to erase every bad result. If uncertain cases always protect the Law and never constrain it, the theory has gained immunity. The argument must keep no-estimate categories separate from excuses.

Sensitivity analysis can make endpoint uncertainty visible without letting it rescue the theory. Investigators can prospectively specify several plausible endpoint branches, apply the same measurement and routing rules to each, and report how the verdict changes. If closure appears only under a branch selected after inspection, that is not robustness. If all defensible branches remain too uncertain, the outcome is No adequate estimate. If a prospectively specified branch produces a gate-valid

adverse result under its own decision rule, that branch bears the loss.

Endpoint adjudication should be versioned. Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, and No adequate estimate are different states. A later reclassification does not rewrite the original analysis; it creates a successor version and propagates uncertainty through every dependent result (Bodien et al., 2024).

Edge cases test whether the bearer rule was specified prospectively enough to handle split access, dissociation, reversible unresponsiveness, degenerative change, anesthesia, sleep, and future candidates. When several routings remain defensible, the honest object is the set of admissible interpretations, not one identity selected after the residual is seen.

Ambiguity also affects the remainder beyond the last observation. A final interview is rarely the final experience. Any bridge from the last measured state to the truth-level endpoint must account for unobserved time, treatment, delirium, sedation, and changing responsiveness. When that remainder is too wide, the result should be no adequate estimate. The theory is not protected by imagining the missing interval closes in its favor.

Prospective endpoint discipline makes the theory harder to support, as it should. When the relevant endpoint cannot be identified or bridged, the result is not presumed closure. It is a narrower claim or no adequate estimate.

29.3.3 — Noise in Dreams and Missing Experience

Dream reports are scientifically interesting and evidentially fragile. Experience occurs during sleep, while the report arrives after waking through memory, language, and narrative reconstruction. A vivid report may answer a narrow question without recovering an exhaustive hidden night.

Alarm timing, sleep stage, medication, stress, interviewer wording, and delay before reporting can all change the record. No recall, fragmentary recall, and reconstructed recall are different observation states. A dream diary therefore samples remembered content under a particular procedure, not the complete sleeping stream (Windt, 2015; Siclari et al., 2017).

Dreams are not automatically useless. They may inform questions about affective processing, threat simulation, memory consolidation, emotional regulation, bodily signals, and internally generated valence. That contribution remains auxiliary: fragile signal is not privileged access to missing experience (Windt, 2015; Nielsen & Levin, 2007).

A responsible study asks a modest prospective question, such as whether coded dream affect adds held-out information about waking affect after sleep stage, medication, trauma history, and recall propensity are modeled. Independent coding, reported disagreement, intimate-content protection, and permission to skip or delete reports are part of the design, not afterthoughts (Nielsen & Levin, 2007; Shiffman et al., 2008).

Recall selection is central. Surviving reports may overrepresent vivid, emotional, coherent, or recent dreams, while repeated

awakenings can improve sampling by changing sleep architecture and dream content. The observation procedure therefore enters the model instead of disappearing behind the data (Windt, 2015; Siclari et al., 2017).

Failure should be typed. Recall, coding, transport, privacy, and incremental-prediction failures imply different repairs or retirement decisions. When dream data cannot separate signal from reconstruction, the project should not invite readers to search sleep for hidden balancing meaning. Dreams may remain useful under controlled sampling, or the route may end honestly (Little & Rubin, 2019).

29.4 — Do Physics, Evolution, or Simulation Help?

The Law uses the language of constraint, and constraint language naturally invites comparisons with physics, evolution, and simulation. Those comparisons can sharpen the proposal, but they can also lend it authority it has not earned. A formal object can be coherent, a model can be executable, and a trait can be evolutionarily compatible without any of those facts showing that PTVC is true.

The moralized-physics objection begins with the title itself. Fairness ordinarily implies justice, desert, proportion, or right relation. A scientific boundary condition has none of those meanings by definition. If nature is described as morally judging, rewarding, teaching, or collecting, purpose has entered through grammar rather than evidence.

Evolution presents a different challenge. Selection can build pain avoidance, reward learning, homeostasis, social care, prediction, and flexible control because they affect survival and reproduction. A hidden terminal law is not needed to explain those capacities. Evolutionary compatibility removes at most one impossibility objection; it contributes no direct support.

Simulation is limited in yet another way. A programmer can make every simulated ledger end at zero, or make apparent closure emerge from conditioning, selection, and observation floors. Such worlds are useful for testing recovery and false support. They do not testify that real conscious histories follow the same rule (Oreskes et al., 1994; Winsberg, 2010).

29.4.1 — Formal Constraint Is Not Moralized Physics

The objection separates two questions often fused by the word law. Is PTVC formally expressible as a boundary condition, and does any natural system realize it? A yes to the first does not answer the second; keeping them separate prevents mathematical coherence from borrowing empirical authority.

Physics distinguishes equations that summarize regularities, dynamical laws that generate change, boundary conditions that restrict solutions, and effective models that work only at a certain scale. Calling PTVC a boundary hypothesis places it in one of those conceptual families, but the label does not establish a physical substrate. The project would still need to identify variables, coupling, conservation or nonconservation structure, scale, and observable consequences. A mathematical resemblance to known physics is orientation, not inheritance of physics' evidentiary authority (Armstrong, 1983; Cartwright, 1983).

The vocabulary needs a semantic audit as strict as the mathematical audit. Fairness can be heard as justice, cosmic balance, providence, deserved outcome, or repayment even when the formal object contains no moral verdict. Conditional mathematical accounting must be distinguished from those moral and spiritual meanings. Each public use should identify the structural relation intended and the moral conclusions that do not follow from it.

As defined in Chapter 4, “fairness” here names a proposed terminal symmetry, not desert or exhaustive justice. The objection

succeeds if the project ever lets that descriptive boundary do moral work it was not designed to do.

A serious physical route would specify the relation between phenomenal contributions and physical state, explain how its dynamics and independently justified boundary would entail closure, and identify observations that test those connections. Ordinary biological processes could supply such a realization. Until the relevant variables and bridges are identified, physical analogies clarify the question without establishing the mechanism (Woodward, 2003; Oreskes et al., 1994).

Interpretive meaning may still be explored, but it belongs to a separate layer. A reader may find philosophical or spiritual significance in a nontransferable boundary without placing that meaning inside the causal model. The separation protects both inquiry and conscience: science need not deny meaning, and meaning need not impersonate evidence (Woodward, 2003).

Purpose belongs outside the scientific layer unless a theory has independent evidence for purposive agency, and this book does not claim that. The Law does not say consciousness is guided by intention. It does not say nature wants neutrality. It does not say suffering is assigned to produce final closure.

Itemize each analogy: a boundary condition can illustrate an endpoint restriction, and symmetry can illustrate transformation. The comparison ends with that specific resemblance (Armstrong, 1983; Cartwright, 1983).

The defense succeeds only when the grammar stays impersonal. Constraint, boundary, dependence, and admissibility are

scientific terms. Moral reward, punishment, prescribed lessons, and desert belong to interpretations that the formal layer neither needs nor licenses. Mathematical repayment language is acceptable only when it states the conditional net remainder without implying those interpretations.

29.4.2 — Evolution May Supply Ordinary Explanations

The evolutionary comparison should start with a biologically capable model whose assumptions do not entail PTVC. Such a model can include selection, development, learning, ecology, pain avoidance, reward seeking, attachment, regulation, and context-sensitive affect. A proposed PTVC mechanism must face that capable comparison, with the same information and a fair opportunity for either model to explain the observations (Tinbergen, 1963; Williams, 1966).

Chapter 22 already separates Tinbergen's levels. Here the relevant objection is narrower: evolutionary function, developmental history, and current mechanism can explain pain-related patterns without implying a terminal valence law. A proposed realization must establish the missing connection; a genuinely additional condition must supply consequences that distinguish it from alternatives.

Evolution is especially powerful at explaining workable trade-offs rather than fair outcomes. Selection can favor systems that avoid immediate danger, seek reward, conserve energy, attach to caregivers, and revise behavior even when those systems create chronic anxiety, addiction, grief, mismatch, or costly overshoot in other environments. Functional usefulness therefore supplies

an ordinary causal account, not evidence that complete phenomenal histories satisfy an exact terminal relation (Williams, 1966; Gould & Lewontin, 1979).

Comparative work should convert evolutionary explanation into prospective pressure. A route that depends on a conserved architecture should state which cross-species relation, developmental pattern, or ecological contrast is expected to transport and which failure would narrow the route or reference class. Phylogenetic relatedness, life history, nervous-system organization, and measurement non-equivalence must be modeled rather than treated as noise, because heterogeneous affective paths are themselves predictions of evolutionary history (Tinbergen, 1963; Williams, 1966).

Cross-species work is useful as a transport test, not a ranking exercise. The protocol should declare the shared construct, species-specific indicators, representation of valence, and the failure that would narrow the route. A pattern forced onto one scale is not support; defensible transport motivates inquiry but does not establish complete-stream closure.

If evolutionary and control explanations account for the prospectively specified patterns without entailing PTVC, those patterns cannot be claimed as distinctive support for the Law. Compatibility with selection may keep a model possible. It does not by itself make the model probable, necessary, or empirically supported.

29.4.3 — Simulation Tests Coherence, Not Reality

Simulation can show whether an estimator recovers a known generator, whether selection creates false closure, whether models become observationally indistinguishable, and whether a decision rule produces false support. It cannot convert a programmed rule into evidence that nature uses the rule.

Apply the verification-versus-validation distinction from Section 23.2.2. The additional burden here is uncertainty analysis: how conclusions change across plausible inputs and structures. Visual polish proves none of the three (Oreskes et al., 1994; Winsberg, 2010).

The strongest program is adversarial. Analysts generate worlds from PTVC-linked, adaptive, regulatory, selection, and observation-artifact models, hide the generator, freeze the analysis, and test recovery. Open-set worlds belonging to none of the candidate families prevent forced classification from masquerading as discrimination.

The inverse problem is central because reserve dynamics, conditioned bridges, mean reversion, selective dropout, and measurement floors can all produce late narrowing. A pipeline that repeatedly selects PTVC when a rival generated the data is not ready for human evidence, however elegant the fitted curve looks (Box, 1976; Vehtari et al., 2017).

Circularity appears when terminal neutrality enters both construction and evaluation. A useful benchmark withholds the target module, declares the generator and fitted parameters, and

reports false attribution under ordinary rival worlds. A bridge forced to hit zero cannot be scored as a discovered mechanism.

Failure is informative. Poor self-recovery weakens the empirical program; indistinguishable rivals narrow the claim's testability; observation processes that manufacture support require public retreat. These are method results, not embarrassments to conceal (Wilson & Collins, 2019).

The best simulations try to make PTV C look supported when it is false and false when it is true. Success strengthens the test; failure exposes the method. Neither outcome, by itself, establishes anything about natural conscious streams. The benchmark is most informative when its failure patterns are recorded alongside successes, because those errors reveal which rival distinctions the pipeline cannot yet make. That record can guide the next simulation without turning methodological improvement into evidence for the hypothesis.

29.5 — Is the Idea Falsifiable?

The theory-immunity charge asks whether an adverse result can actually change the Law. The concern is concrete: the target is latent, the endpoint may be uncertain, the bearer may be ambiguous, and the observation model may fail. Each legitimate difficulty can also become a shelter for an unfavorable result unless the project states in advance what the difficulty changes and what it cannot rescue.

A testable project therefore needs prospectively specified branches, reference classes, instruments, stopping rules, estimands, rivals, and decision language, with confirmatory choices fixed before outcome inspection. It must distinguish a failed route from a failed target, but it must also prevent that distinction from becoming infinite retreat. Every retained branch has to specify the evidence that would count against it.

Imagine a study that finds no terminal concentration. The project could blame missingness, revise the endpoint, alter the neutral origin, change the stream partition, or choose a different estimator. Any one revision might be justified. A sequence chosen after seeing the result, however, can make the claim immune unless the changes are logged, frozen, and tested on new data.

Falsifiability is therefore procedural as well as logical. The question is not only whether a counterexample can be described in principle, but whether the research program is organized so an adverse result can reach the public claim. Protocol-fixed analysis, model recovery, held-out testing, severe rivals, and transparent

status updates are defenses against self-protection (Nosek et al., 2018; Munafò et al., 2017).

29.5.1 — Theory-Immunity Charge

The immunity charge is strongest when it identifies a repeated pattern of protection rather than one careless sentence. Does the project specify risky predictions before data arrive, or does it revise the origin, endpoint, bearer, route, and reference class after each adverse result? A theory is learning only when reality can force a visible loss.

No ambitious theory is tested without auxiliary assumptions. A failed prediction may implicate the instrument, sample, endpoint, model, or target. The danger begins when an auxiliary failure is invoked only after disappointment and without independent evidence. The Duhem-Quine problem therefore makes change control necessary: each revision should state what moved, why it moved, which earlier conclusion no longer stands, and where the successor will face fresh data (Quine, 1951; Lakatos, 1978).

A versioned branch record converts that discipline into practice. Before outcome inspection, it lists the active representation, bearer rule, endpoint branch, observation model, route, rival set, and verdict map. Undeclared branches remain exploratory: they may motivate a later study, but they cannot reinterpret the confirmatory result or make a family of possible theories look like one theory that predicted the winner in advance (Popper, 1959; Meehl, 1978; Nosek et al., 2018).

Apply the progressive-revision test in Section 6.2.1: a revised endpoint, route, or reference class earns standing only when it

produces independent consequences and survives a new test (Lakatos, 1978; Mayo, 2018).

Every version therefore needs a visible exposure to loss. A declared universal branch can be defeated by a gate-valid counterexample. A route can fail through the wrong intervention response. An estimator can fail recovery, calibration, or transport. A public formulation can fail through foreseeable harm. These outcomes differ in scope, but none may be dissolved into mystery.

The critic's strongest human counterexample is not a statistical abstraction. Imagine a documented stream marked by prolonged high-intensity pain, severe isolation, treatment-resistant depression, or another sustained burden, followed by an independently adjudicated endpoint without observed relief. Cases involving long-term dependence on mechanical ventilation or other life-sustaining technology require particular care: disability, restricted mobility, and dependence do not themselves imply a joyless life, and no outsider may invent the person's interior. The objection begins only when the stream's own evidence supports prolonged severe negative valence.

Section 29.3.2 already fixes the counterexample rule: once a gate-valid branch yields a nonempty truth-containing residual set that excludes zero, that branch is Target-adverse. Falsifiability is compromised if the project invents a new branch, endpoint, or hidden contribution after seeing that result in order to erase the adverse verdict.

Rival comparison supplies a second route to loss. PTVC and its competitors should receive the same observations, preprocessing,

endpoint information, tuning freedom, complexity allowance, and held-out scoring rule. Adaptation, opponent processes, predictive regulation, homeostasis, allostasis, mean reversion, conditioned bridges, treatment effects, cultural scripts, and observation models must be allowed to explain the pattern completely when they can.

An immunity audit should record every adverse result and every resulting change in definition, branch, reference class, route, or measurement rule. A long sequence of one-way revisions that always preserves the favored conclusion is itself evidence of degeneration, even when each revision has a plausible story.

A rival win has consequences. The record identifies what the rival explained, which Law-linked claim lost support, which public language must change, and what narrower question remains open. The project may not absorb the victorious rival as an undiscovered mechanism of the original claim without an independently tested entailment (Mayo, 2018; Wilson & Collins, 2019). No theory is entitled to exchange every loss for another unknown.

29.5.2 — Measurement Failure

Measurement failure does not automatically refute a latent hypothesis, but it defeats every empirical claim that required the failed bridge. That distinction is necessary and easy to abuse. The project cannot declare an instrument invalid while continuing to treat its favorable pattern as suggestive evidence.

The distinction between a latent object and a measured estimate is scientific only when it constrains inference in both directions.

A favorable measured pattern cannot be promoted beyond the validated bridge, and an adverse pattern cannot be demoted by declaring that same bridge untrustworthy after it has passed its protocol-fixed gates. The criteria for accepting, limiting, or rejecting the bridge must not depend on whether the result favors PTVC.

Failure should be named at the layer where it occurs. Representation may fail because sign or neutral origin is unstable. An observation channel may be nonspecific, reactive, or unavailable in the relevant state. A bridge may fail because missingness, treatment, routing, or endpoint uncertainty cannot be recovered. An estimator may fail because it cannot distinguish known generators or calibrate its uncertainty.

Recovery studies should come before high-stakes interpretation. Analysts can simulate PTVC-linked and rival worlds, apply the complete frozen pipeline, and ask whether it recovers origin, trajectory, terminal behavior, and the correct generator under realistic noise and missingness. Negative controls should also reveal whether the method manufactures closure where no relevant process exists. A pipeline that cannot distinguish known worlds has no warrant to classify an unknown one (Wilson & Collins, 2019).

Analytic flexibility is the second failure mode. Apply the prospective-specification and multiverse/specification-curve checks in Section 17.5.2; support that appears only under one favorable reasonable analysis is analysis-dependent rather than a clean test (Gelman & Loken, 2014; Steegen et al., 2016; Simonsohn et al., 2020).

A failed bridge has a precise downstream consequence. Claims that depend on it pause, and public artifacts inherit the lower ceiling. The report distinguishes No adequate estimate, where the target cannot be identified or bounded under maintained assumptions, from unusable evidence, where a protocol or validity gate has failed. Any narrower measured claim must be justified separately. Not disproven is not a positive evidentiary category.

Coverage sufficiency requires protocol-fixed decision criteria rather than a promise of omniscience. The protocol names the channels, coverage thresholds, missingness bounds, endpoint windows, and residual uncertainty that would force No adequate estimate. If an unobserved contribution could still reverse the decision under those fixed bounds, the target is not adjudicated; once the declared coverage and bridge gates establish adequacy, however, an unnamed post hoc channel cannot rescue an adverse result.

Bounded salvage prevents both waste and rescue. Local observations may still support reporting, adaptation, physiology, or clinical communication when they no longer support terminal closure. The disposition record states what survives, what retires, and what validation would be needed for return. Measurement can say no only when outputs lose unsupported uses, dependent surfaces are corrected, and the next design begins with a new validation burden.

29.5.3 — No Rhetorical Rescue

Rhetorical rescue preserves the emotional force of a claim after evidence has reduced its warrant. Failure becomes mystery, rival victory becomes another hidden route, or a narrowed reference class continues to be described universally. The language moves less than the evidence.

Compare the prospectively specified forecast with the final report. A result once expected to support the Law may later be said merely to illuminate it; a victorious rival may be called compatible with a wider vision; an unidentified target may be declared beyond measurement. The comparison reveals whether confidence survived by changing the claim. A side-by-side archive of preregistered wording and final wording makes that comparison inspectable rather than dependent on memory.

Registered Reports reduce this drift by reviewing questions and methods before outcomes are known and by separating publication from a pleasing result. For high-stakes PTVC tests, prospective review, frozen analysis, and guaranteed reporting of adverse outcomes make permission to fail operational (Chambers, 2013; Nosek & Lakens, 2014). The same principle can be used outside journals by timestamping analysis plans and public claim templates before decisive data arrive.

The cure is verdict language that can disappoint and propagate. Clarified, supported, narrowed, model-ambiguous, no adequate estimate, suspended, defeated, retired, ethically nonviable, and corrected must remain possible, and verbs such as suggests or favors must name the comparison actually performed. Every

adverse status identifies its dependencies, owners, and public actions; a claim is not corrected while its most memorable version circulates without notice. Each status can carry a dated scope statement so downstream summaries inherit the current ceiling rather than an older headline.

Use the dependency map in Section 26.4.2 and correction route in Section 26.5.3. Every affected object needs a dated disposition, with the current notice at least as findable as the obsolete one (National Academies of Sciences, Engineering, and Medicine, 2017). A visible correction chain should preserve both the status change and its reason, allowing readers to distinguish new evidence, a failed method, narrowed scope, or a public-language repair from quiet rewriting. That dated history shows whether the project actually changed when its warrant changed. When a change is substantive, the obsolete artifact should point forward to its replacement instead of simply disappearing.

29.6 — Is the Idea Too Dangerous to Say?

A claim about the balance of suffering can injure people even when every equation is labeled correctly. A bereaved parent may hear that a child's pain was necessary. A patient may fear that analgesia interferes with a balance. A person in crisis may try to generate or interpret suffering as a test. A public figure may use hidden burden to dismiss visible injustice.

These are not misunderstandings that can be assigned entirely to careless readers. Authors are responsible for foreseeable readings, especially when a phrase is memorable, portable, and emotionally charged. The ethical question is whether the idea can be communicated without shifting risk onto those already carrying pain.

Public artifacts require a stricter test than long-form prose. A sentence that is safe only when surrounded by five pages of qualification is not safe as a caption or quote card. The headline, crop, diagram, and closing line have to be reviewed as independent objects.

The section treats victim-blaming, care delay, self-testing, fear of joy, and prescribed pain-as-payment as disqualifying outcomes. No beautiful sentence is entitled to survive by making another person's burden heavier.

29.6.1 — The Ethical Objection Is Central

The ethical objection tests foreseeable pathways of harm, not only intention. Reviewers ask who may use a sentence, what action it could encourage, who carries the risk, whether refusal is

meaningful, and how harm can be stopped or repaired. A technically bounded claim can still be unfit for release when ordinary use predictably burdens vulnerable people.

Review follows the path from exposure to consequence. Who encounters the language, in what format and state, and with what power over another person? What inference could follow, how severe and reversible would the harm be, and could safer wording preserve the legitimate intellectual content? Another disclaimer is not a remedy when the action pathway remains unchanged.

Foreseeable misuse can arise without malice. A phrase written as wonder may be repeated by an abusive partner as justification, by a caregiver as a reason to postpone relief, or by a grieving reader as proof that loss was required. The relevant question is whether the use was reasonably predictable and preventable.

The 2024 Declaration of Helsinki emphasizes respect, protection, transparency, and meaningful engagement with participants and communities before, during, and after medical research. That participant-centered principle supports asking people living with pain, disability, trauma, terminal illness, bereavement, or fear of happiness to identify burdens in protocols and high-reach materials before they are frozen. Their response does not decide scientific truth, but it is evidence about public effect (World Medical Association, 2024).

Stop authority must exist before release. A clinician, ethics reviewer, data steward, accessibility reviewer, or public-risk owner needs power to pause a study or artifact when a declared red line appears. The record names the trigger, immediate

protection, accountable decision maker, correction route, and evidence required before reconsideration.

Ethical viability therefore requires affected-reader input, independent review, meaningful consent and refusal, burden minimization, accessible correction, a visible ban on personal testing, and authority to suspend or withdraw. When no practical safeguard prevents the cost from falling on vulnerable people, the correct design decision is nonpublication.

29.6.2 — No Prescribed Pain or Fear of Joy

The two most dangerous semantic errors are pain-as-payment and fear of joy. Under exact PTVC, changing a genuine accumulated balance changes the equal-and-opposite net remainder required for closure; that arithmetic does not identify a particular future event, create moral desert, make suffering controllable, or turn pleasure into a debt. Just-world reasoning and fear-of-happiness beliefs make these mistakes especially tempting, so public language should avoid words such as pay, owe, settle, and repay unless their limited mathematical meaning is explicit (Callan et al., 2006; Joshanloo, 2013).

No completed qualifying stream can end off the target condition if exact universal PTVC is true, but that does not give a person a safe method for manipulating what remains. Safeguards against self-testing, care delay, and coercion protect people and research integrity; they are not mechanisms that make the Law true. Any detachable sentence or image that makes pain sound like a safe purchase of chosen joy, joy like a liability, or a specific hardship like an assigned route should be rewritten or retired.

The humane alternative is direct: pain occurred; relief may follow; adaptation may change experience; care can reduce burden; meaning may be chosen. None of those statements makes the first event morally owed. Conditional remainder arithmetic and moral interpretation must remain separate.

29.6.3 — No Blame, Care Delay, or Harmful Self-Testing

A public theory fails ethically when it makes a harmed person explain the harm, wait for care, or prove the claim with a life. Victim-blaming converts uncertainty into accusation. Care delay converts interpretation into clinical risk. Self-testing converts curiosity into danger.

Three failures make the rule concrete. Analgesia is delayed because pain is assigned hidden purpose. A survivor is told that abuse was justified because it contributed to eventual balance. A curious reader intensifies distress to see whether relief follows. Each act adds real harm and produces no valid evidence. The theory neither requires nor permits any of them.

Imposed meaning becomes victim-blaming when it treats the person harmed as deserving, causing, or morally responsible for the harm without justification. Even when it does not assign blame, imposing meaning can violate the person's freedom to interpret or refuse the experience. Violence, neglect, illness, poverty, discrimination, and institutional failure retain their ordinary causes and accountable agents. Hidden-burden language cannot move responsibility from a perpetrator or system onto the person harmed (World Medical Association, 2024; CIOMS, 2016; Beauchamp & Childress, 2019).

Unsafe self-testing often begins as private interpretation rather than a formal experiment. Warning signs include deliberately increasing distress, withholding relief, seeking danger, timing an endpoint, repeatedly scoring a life, or treating misfortune as a message from the theory. The response is to stop, secure immediate safety, seek trusted human and clinical support where needed, and conduct no further personal test.

The risk rises when language becomes an instruction. A reflection prompt, community challenge, symptom tracker, classroom exercise, or chatbot response can convert a metaphor into behavior without calling itself an experiment. Release review should test the action a user is likely to take, not merely the sentence's literal claim (Fischhoff, 2013; van der Bles et al., 2020).

Release review should treat victim-blaming, care delay, and personal testing as binary gates. Any exercise, diagram, call to action, community challenge, or social prompt that invites balancing behavior, promises a chosen reward or prescribes repayment, discourages treatment, weakens accountability, or converts private suffering into evidence is removed or redesigned. A disclaimer beneath an unsafe action does not repair the action (Emanuel et al., 2000).

A responsible incident path is direct: stop the activity, protect the person, encourage appropriate emergency or clinical help, preserve privacy, avoid debating the theory during crisis, and record any project artifact that contributed to the misunderstanding. The project then corrects or withdraws the

source. Protecting the hypothesis is not part of the emergency decision.

29.6.4 — Beauty Cannot Justify Dangerous Language

Memorable language travels farther than careful analysis. A sentence can be quoted without its chapter, placed over an image, translated, shortened, summarized by a machine, or reused by someone who rejects the surrounding limits. Literary power therefore increases the duty of review.

Test a memorable line as though it were the only line a reader sees. Remove the definitions and qualifications, then ask what claim, moral message, and action the fragment appears to authorize. If it implies proof, payment, destiny, ranking, self-testing, or delayed care, it fails regardless of beauty.

Beauty can survive through precision. Care comes before interpretation gives a clear action; one stream cannot pay for another blocks a formal and ethical error. When ambiguity places foreseeable burden on a vulnerable reader, patient, survivor, child, grieving family, or participant, the line is narrowed, redesigned, or cut (van der Bles et al., 2020; Oreskes, 2019).

The same rule governs visuals and summaries. Anything that resembles a person score, cosmic bookkeeping, death settlement, or imposed meaning is redesigned or withdrawn. If removing a flourish loses only drama, keep the reader and lose the drama. If a real distinction would be lost, rewrite until the distinction and safeguard travel together.

Chapter 30 — What the Search Teaches

By the final chapter, the Law has become more than an intuition about fairness. It has a defined bearer, a whole-history target, an independently specified endpoint, candidate routes, measurement gates, capable rivals, and explicit ways for evidence to change the conclusion. That structure gives the question scientific traction without taking away its human force.

The most valuable outcome is a cumulative research path. Clarification can expose assumptions, measurement can identify what is recoverable, mechanism work can reveal what information a route would need, and model comparison can distinguish a distinctive consequence from ordinary recovery. Each layer can make the next question sharper.

The book began with the visible fact that lives can look profoundly unequal. It now asks a narrower and more surprising question: whether complete conscious histories could nevertheless share an exact terminal relation at the level of felt experience. That possibility remains compatible with vast differences in biography, opportunity, duration, and circumstance.

A successful program would have to connect those levels rather than leap between them. It would show how momentary experience, stream ownership, cumulative history, and endpoint structure fit together under one independently supported account. Even partial progress toward that account would clarify what a deeper law of fairness would have to mean.

That is the purpose of this final chapter: to show how support, ambiguity, narrowing, rival success, and strong counterevidence

can all improve the project by telling us which part of the question has actually moved.

Scientific status updates in modules. Representation may improve while endpoint identification remains weak; a route may outperform a rival while the target stays open; a reference class may narrow while the broader claim retires. Care and independently supported science remain intact through every change.

Model ambiguity deserves respect when latent variables, missingness, endpoint selection, or observation floors let capable explanations look alike. Choosing a favorite because it matches the project's hopes is less scientific than reporting ambiguity (Akaike, 1973; Burnham & Anderson, 2002; Vehtari et al., 2017).

Rival victory also teaches. If adaptation, predictive regulation, a conditioned bridge, terminal selection, or another ordinary model explains the evidence better under equal constraints, the tested Law-linked explanation must change. The result identifies which account earned the pattern; its implications for the terminal target depend on what that comparison actually tested.

The ethical core does not depend on victory. If the Law fails, care does not; if an estimator fails, dignity does not; and the duty to relieve preventable suffering remains without terminal-neutrality support.

The series must remain correctable. Changes to the root claim must travel to summaries, diagrams, websites, talks, classrooms, donor language, applications, translations, search surfaces, and AI explanations. A fair theory updates where it traveled.

The public grammar is demanding: clarify imprecision, support only within evidence, report ambiguity, narrow overbroad scope, retire failed components, acknowledge rival victory, and state defeat when the strong claim no longer survives.

These are not stylistic synonyms. Formal clarification, finite-resolution compatibility, route favored, model ambiguous, No adequate estimate, reference-class narrowed, rival favored, and strong-form defeated identify different objects and authorize different next actions.

The affirmative gain is knowledge that survives every outcome: clearer constructs, safer research, stronger rivals, better missingness models, more honest communication, and ethical commitments never borrowed from a desired verdict.

Outcome language changes action. Route support may justify another mechanism test; ambiguity may justify a discriminating study but prohibit person-level interpretation; No adequate estimate may stop collection; rival victory may redirect funding; and defeat may require public withdrawal.

30.1 — Formal Clarification

A theory can improve its language without improving its evidence. That is not a defect. Formal clarification can expose hidden assumptions, separate branches, correct notation, sharpen quantifiers, and make the target easier to test. The danger begins when clarification is narrated as if it had rescued a failed prediction.

Suppose a term such as stream, endpoint, neutral origin, or closure was previously underspecified. A revision may define it more carefully and reveal that earlier claims were broader than the formal object. The new definition is progress only if the old meaning remains traceable and any lost scope is acknowledged.

Change control turns this philosophical point into practice. Every substantive revision needs an owner, date, rationale, affected claims, dependent artifacts, and supersession path. Without that record, a project can present a moving target as one continuous success.

Suppose a study fails because “endpoint” referred sometimes to biological death and sometimes to the last observed report. Replacing the term with separate truth-level and protocol endpoints is real progress: the next study can be designed coherently. Yet the repair does not tell us what the earlier data meant. The old result remains uninterpretable under the target claim. This example captures the chapter’s rule: formal clarification earns a cleaner future and a more accurate record of the past, but it cannot manufacture evidence inside an analysis whose object was never fixed.

30.1.1 — Clarifying Without Rescuing

Clarification earns its name when it makes a preexisting claim easier to identify without increasing its scope. It may distinguish a truth endpoint from an observation end, separate a route from the target, or replace one overloaded term with two narrower terms. The evidence status stays where it was, and any apparent new consequence is treated as a new claim that needs its own warrant.

A useful clarification often reduces scope. Suppose the phrase “one conscious life” is replaced with “one separately routed qualifying stream.” The new wording may be more exact, but it also withdraws any earlier implication that legal personhood, biological continuity, and stream unity are automatically identical. That loss of convenience should be stated. Precision is progress when it reveals what the theory can no longer claim as casually as before.

A clarification is especially suspect when it appears only after a critic has named a hard case. The timing does not automatically make the change dishonest, but it increases the burden of public explanation. The revision must show that the new wording follows from the best version of the theory, not from a wish to keep the counterexample outside the door (Popper, 1959; Lakatos, 1978).

Rescue can therefore be detected by counterfactual comparison. Ask whether the revised formulation would have predicted the adverse result before it was known and whether the original experiment could have rejected it. If the answer is no, the revision

may be a legitimate successor hypothesis, but it is not a clarification of the tested one. The record should preserve the old loss and start the new claim without inherited confirmation (Lakatos, 1978; Quine, 1951).

A clarification should carry a change statement: the ambiguity removed, the cases affected, the prior claims altered, the dependent artifacts requiring review, and the next test created by the revision. The more visible or memorable the earlier wording, the more visible the correction.

Formal clarification should also preserve counterexamples. If a case defeated the earlier wording, the record should say so even when a narrower successor survives. The counterexample remains part of the theory's history and may explain why the new boundary exists. Erasing it creates the impression of uninterrupted success; preserving it shows that criticism changed the work.

The clearest public sentence distinguishes three possibilities. The original wording may have been ambiguous but substantively unchanged. The target may have been narrowed, so earlier universal language is withdrawn. Or the model may have been replaced, requiring a new test rather than a reinterpretation of the old result. Those states should never be merged under the comforting word "clarified."

A genuine clarification makes future failure easier to recognize. It narrows ambiguity, fixes assumptions, and exposes the revised target to new data. A rescue protects the preferred conclusion while leaving the public impression unchanged.

30.1.2 — Notation Improvements

Notation is scientific hygiene because it can preserve or erase layer distinctions. A latent ledger should not share a symbol with a measured estimate. A truth-level endpoint should not share a label with the last observation. A route variable should not look like the target law.

In a public-facing account, improvement often means better verbal labels rather than more symbols. “Truth-level endpoint,” “protocol endpoint,” and “last available observation” distinguish three times without displaying an equation. “Latent cumulative object” and “measured estimate” prevent a report from being mistaken for the target. Plain language can carry formal discipline when every term has one stable job.

Good notation also shows uncertainty and failure. A confidence band, partially identified set, status label, or stop symbol may be less elegant than a clean equality, but it is often more truthful. A compact equation can create an illusion of authority, so notation must be controlled language rather than decoration.

A notation map should label each object’s role: definition, assumption, target, route, bridge, diagnostic, rival, estimator, or verdict. Separate labels for streams and persons, truth and measured objects, and target laws and mechanisms make illegal moves harder to write. Reusing one symbol across layers saves ink by purchasing interpretive risk.

Accessibility is part of the same discipline. Every technical term needs a plain-language equivalent, a first-use definition, and a route to Chapters 4–8 and The Science of Fairness. Searchable

text, screen-reader compatibility, and clean copy-and-paste behavior help critics inspect the object and help later editors detect drift.

Good notation exposes absence. A missing endpoint bridge appears as a gap. A partially identified quantity does not look exact. If analysis ends in No adequate estimate, the diagram stops rather than flows into a verdict box. A technical object earns trust by showing where the argument cannot proceed.

A notation audit should attempt illegal substitutions. Replace a truth endpoint with a last observation, a latent object with an estimate, or a route with the target. If the sentence still looks grammatically natural, the labels are too weak; good notation makes the wrong inference visibly malformed.

30.1.3 — Change-Control Requirement

A public theory needs change control because its claims exist in networks. Definitions feed protocols; protocols feed figures; figures feed summaries, teaching, and public discussion. When one source object changes, the project must identify every dependent use and distinguish a local editorial repair from a semantic, scientific, ethical, or structural change.

A compact change record can still be complete. It names the stable object, old wording, new wording, reason, evidence or criticism, status effect, owner, date, affected chapters and artifacts, and re-review trigger. It also classifies the change as editorial, clarifying, narrowing, expanding, retiring, or withdrawing so that a major scope change cannot disappear among minor repairs.

The dependency graph is the difficult part. Each canonical definition, formula, diagram, guide page, talk, website, translation, and machine-readable description should identify what it inherits. A changed endpoint rule may require protocol review; a narrowed reference class may require a headline change; a retired measure may require a dashboard shutdown. Unresolved dependencies remain open.

The record should also state what did not change. Readers need to know whether the ethical firewall, latent-measured distinction, target geometry, source identity, or downstream title boundary remains intact. Evidence never transfers automatically when the target, endpoint, representation, or reference class changes materially. The successor inherits only the warrant that survives the bridge.

Public correction follows the reach of the original object. A widely circulated graphic, interview sentence, or video cannot be repaired only in a later appendix. Text, image, caption, animation, search result, and public summary each receive propagation review, because nontext artifacts may preserve a universal or purposive implication after prose has narrowed.

Preparation begins before publication. Every high-risk sentence and canonical object receives an identifier, owner, source, version, status ceiling, dependency list, and re-review trigger. That preparation turns later correction from a search through scattered files into a controlled update.

Urgent safety corrections need a fast path. Artifacts encouraging self-testing, care delay, ranking, or coercive interpretation are

suspended before investigation. The record preserves the trigger, interim notice, accountable owner, affected versions, channels, and replacement or withdrawal decision. Once immediate risk is contained, the fast path should enter ordinary review so a temporary safeguard does not become an unexamined permanent rule. Its final status should be recorded with the same visibility as the emergency notice that began the correction.

30.2 — Empirical Support

Empirical support is not a single upward step. A study may support the reliability of an instrument, the invariance of a construct, the recoverability of a model, the transport of a route signature, or the comparative performance of a hypothesis. Those achievements matter, but they do not carry the same warrant.

For a latent terminal claim, support must arrive through a chain of gates. The representation has to remain meaningful, the bearer and endpoint have to be independently governed, the observation process has to be modeled, the estimator has to recover known worlds, and the result has to outperform capable rivals on held-out data. A broken prerequisite is not repaired by a striking final graph.

Invariance is central because a quantity that changes its meaning across cultures, ages, conditions, instruments, or time cannot support a common claim without qualification. The appropriate response may be partial invariance, group-specific models, or withdrawal of same-scale language (Putnick & Bornstein, 2016).

A credible support package resembles a chain of custody. The measure is validated for the population and use; the endpoint rule is independent of the ledger estimate; missingness and treatment are modeled; the pipeline recovers known generators; rivals receive equal information; confirmatory choices are frozen; held-out data and an independent team reproduce the result. Breaking any link changes the status. The chain metaphor is useful because strength does not average across links: a brilliant final model

cannot compensate for an unidentified origin or an outcome-dependent endpoint.

30.2.1 — Support Is Not Proof

Support means that data have survived a declared test better than specified alternatives at a stated resolution and scope. Proof would require something much stronger than any finite empirical program can normally provide for a latent universal claim. The distinction protects both the evidence and the reader (Lakens, 2017; Wasserstein & Lazar, 2016).

A useful evidence ladder begins below the final claim. One study may show that a report scale is reliable. Another may establish a defensible neutral point for a limited task. A simulation may show that an estimator can recover a known generator. A prospective cohort may find the predicted pattern under a declared endpoint. Each result removes one obstacle, but none automatically supplies the other missing bridges. Support should therefore be attached to the exact gate that passed.

Universality remains open after any finite collection of compatible cases; one gate-valid counterexample can defeat the declared universal branch. Section 7.3.3 states the quantifier distinction formally.

Support depends on valid gates. A residual inside a protocol-fixed acceptance region means little if the endpoint was selected after inspection, the band was widened after results, relevant channels were closed, missingness created the pattern, treatment effects were ignored, or obvious rivals were excluded.

Support is stronger when the test was capable of producing disappointment. A wide equivalence band, an underpowered comparison, a flexible endpoint, or a rival set that lacks the obvious alternatives can make success easy. Severe testing asks whether the method would probably have exposed a meaningful departure if that departure were present. A favorable result from a weak test deserves weak language, however attractive the figure appears (Mayo, 2018; Lakens, 2017).

A responsible public sentence names the measured implication and its conditions. It may report support for a defined finite-resolution claim in a stated population under a specified observation model. That justifies further work without claiming exact latent closure or universality.

The strongest package would be conjunctive but typed. A finite-resolution terminal result would inform TargetStatus; variance compression or horizon dynamics would inform RouteStatus; cross-channel convergence would support measurement; and rival performance would inform ComparatorStatus. Independent modules gain force when they align without one axis substituting for another.

For a hypothetical measured-scale test, suppose the preregistered equivalence band is -0.5 to $+0.5$ units and the required confidence interval is $+0.1$ to $+0.2$. If the measurement and analysis gates pass, the interval supports the declared measured equivalence claim. It establishes neither exact zero nor latent closure without the appropriate bridge. An interval from $+0.6$ to $+0.9$ would instead be adverse to that finite-band claim under the registered

rule. Neither conclusion becomes stronger merely by adding studies that share the same bias.

30.2.2 — Invariance

Invariance asks whether an empirical pattern survives changes that should not alter the target. The relevant tests may span instruments, channels, cultures, ages, sites, endpoint causes, analysts, and time. A result that disappears under reasonable changes may belong to the method rather than to the proposed law.

Measurement invariance should be tested across populations and contexts; robustness should also be tested across defensible analytic decisions. A finding that appears under one imputation model, one window definition, or one exclusion rule may belong to that specification. Multiverse analysis can show how results vary across defensible choices, while held-out analysis protects the confirmatory specification from being selected after inspection. Robustness is not the number of analyses that reach significance; it is the stability of the substantive conclusion across justified alternatives (Steege et al., 2016; Simonsohn et al., 2020).

Invariance is where measurement meets justice. A scale that behaves differently by language, disability, clinical condition, culture, medication, age, trauma history, or social risk may look neutral while moving burden onto those already easiest to misread. The Law of Fairness cannot accept comparability as a convenience assumption (Putnick & Bornstein, 2016; Campbell & Fiske, 1959).

Power can alter what becomes observable. In hospitals, prisons, care facilities, families, workplaces, and public platforms, people may speak, remain silent, or adapt their reports under dependency, fear, stigma, fatigue, or anticipated consequence. A comparability claim is incomplete until the conditions of expression and the resulting differential error have been examined rather than treated as background.

Transport adds another layer. A model fitted in an affluent outpatient sample may fail in hospice care, across languages, in populations with limited technology, or under medications that alter report and physiology. The failure may reveal a narrow context of use rather than a useless measure. Generalization should be earned step by step, with the reference class widened only after the construct and observation process survive the new setting (Yarkoni, 2022).

Invariance is also needed over time within a stream. A marker that has one meaning early in life may change meaning under illness, grief, aging, sedation, dementia, or terminal decline. A stable formula can still become unstable when the channel it reads changes.

Independent replication should change the team, implementation context, and opportunity for disappointment. A new group should recover the intended objects, document deviations, and publish adverse results without permission from the founding project. Critics can strengthen the design by specifying rivals and failure thresholds before data arrive (Open Science Collaboration, 2015; Munafò et al., 2017; Longino, 1990).

Every cross-context statement should name its level: change within one person, transport within a population, transport to a comparable population, or a general claim. Evidence moves upward only when construct meaning and observation conditions survive the move (Putnick & Bornstein, 2016; Yarkoni, 2022).

The result of invariance testing may be a map rather than a yes or no answer. Some channels may transport across age but not language. Some route signatures may survive site changes but fail under a different endpoint cause. The status should follow that map. Evidence that is stable in one corridor does not become universal merely because the corridor is scientifically interesting.

Registered Reports, described in Section 28.3.1, can make the planned test reviewable before a favored result becomes fixed in narrative (Chambers, 2013; Nosek & Lakens, 2014).

A stable effect can still have an unstable interpretation. Several measures may agree that something changes near an endpoint while disagreeing about whether the change reflects valence, arousal, medication, communication loss, or selection. Invariance of the number is not invariance of the construct. The status update should preserve both layers.

No single invariance test can carry the whole claim. The evidence becomes stronger when several independent channels and populations preserve the same limited structure while capable rivals fail. Where invariance breaks, the status must narrow with it.

30.2.3 — Scientific Status Update

A status update should report the fixed target, prerequisite gates, deviations, participant flow, missingness, harms, rivals, uncertainty, and replication.

Different modules may occupy different states. An instrument can be valid while endpoint transport remains open; a route can be favored in a laboratory task while the complete-stream target remains unidentified; a rival can explain one evidence family while another remains unresolved. The formulation developed here therefore keeps TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus separate so no global badge erases the next decision.

The register behaves like a state machine. Target-adverse blocks promotion; route-favored may justify replication but not a target-law headline; model ambiguity may justify a discriminating study; open-set inadequacy may require a new model class. Failed replication, reanalysis, source correction, bias, consent loss, or a stronger rival can trigger downgrade, and a rival counts as defeated only when it was modeled fairly and allowed to win.

Evidence may outlast a claim. Consented observations, code, calibration results, and rival comparisons can remain useful after a claim narrows or fails, but interpretations and permitted uses must change accordingly. Updated status must propagate to dependent text and artifacts; no model revision may relabel participants or weaken their standing (National Academies of Sciences, Engineering, and Medicine, 2017; Nosek et al., 2018; World Medical Association, 2024).

30.3 — Route Support and Model Ambiguity

A candidate mechanism can become the best available explanation of one signature while the target law remains unestablished. Reserve coupling, a completion potential, opponent regulation, or endpoint-conditioned dynamics may fit a result. The scientific question is whether the route makes distinctive, recoverable, and transported predictions under the declared conditions, not whether it can be made to resemble closure after the fact.

Model ambiguity occurs when capable explanations remain observationally similar. A Brownian bridge, a selection process, an allostatic controller, and a PTVC-linked route may produce comparable terminal concentration under certain conditions. More data of the same kind may reduce uncertainty only slowly or not at all.

A fair comparison therefore gives rivals comparable flexibility and evaluates them on predictions not used to fit them. Information criteria, cross-validation, simulation-based recovery, and intervention tests can help, but none abolishes underdetermination (Akaike, 1973; Vehtari et al., 2017; Wilson & Collins, 2019).

No adequate estimate is a separate outcome. It may arise from missing history, unstable origin, ambiguous bearer, terminal observation failure, nonrecoverable models, or insufficient information. The absence of an estimate is not evidence for hidden exactness.

30.3.1 — Route Support Does Not Establish the Law

A route claim identifies where a regularity enters the system: representation, action selection, transition, endpoint timing, routing, observation, or another declared locus. Evidence at one locus cannot be promoted downstream without an entailment. A model that predicts choices may leave endpoint timing untouched, while a stopping model may say nothing about phenomenal representation.

Suppose a reserve-linked model predicts how variability changes as a declared horizon shortens and outperforms adaptation and bridge rivals in held-out tasks. That would be meaningful route evidence. It would not show that every stream closes, that the same route operates near every endpoint, or that the latent cumulative object has been observed over a whole life.

Route evidence strengthens when a safe intervention targets the proposed mechanism and produces a response that matched rivals did not predict. Even then, the result is local to the manipulated system, variable, and observation model. A control relation is not automatically a complete-stream boundary condition.

Information constraints matter as much as fit. Some completion models behave as though the system knows the future endpoint, the missing prefix, or the full remaining possibility set. If no plausible process has access to that information, predictive success may reflect analyst conditioning rather than an organismic mechanism.

The target inherits route evidence only through a proved entailment. Without that bridge, a route may be scientifically

valuable and still leave PTVC open. This rule prevents a memorable mechanism from becoming a shortcut around representation, bearer, endpoint, and measured-to-latent identification (Wilson & Collins, 2019).

A route-favored verdict lists what remains unresolved. Does the route transport across populations, endpoint causes, and measurement channels? Can an equal-flexibility rival mimic the effect? Does the route reach phenomenal valence or only a measured proxy? Which assumptions would make the result disappear?

Mechanism status should be explicit: exploratory, supported, revised, suspended, replaced, or retired. A route may retire while the formal target remains under investigation. The project should not preserve a mechanism merely because it gives the hypothesis narrative power.

Vivid route stories also carry public risk. They can be mistaken for forecasts about an individual or for explanations of why a person suffered. Communication therefore states the route's scope and failure conditions and prohibits person-level prediction. A route can be retired without leaving a moral story behind (Fischhoff, 2013).

30.3.2 — Model Ambiguity Is an Honest Scientific Outcome

Model ambiguity is not indecision caused by insufficient courage. It means that the available observations do not distinguish capable explanations with enough reliability. Latent states, flexible dynamics, missingness, and endpoint selection can make different generators look alike.

Ambiguity often appears when several models predict nearly the same observed range. A conditioned bridge and state-coupled controller may both narrow near an endpoint; adaptation and selective dropout may both reduce variance; a latent-state and measurement-floor model may fit the same cluster. A small information-criterion difference can overstate what the data contain (Akaike, 1973; Burnham & Anderson, 2002).

The latent-measured distinction prevents category error, but it does not shield weak tools from consequence. A composite estimator or neutrality band needs a context of use, data requirements, uncertainty, prohibited uses, human approver, and revalidation trigger. When calibration, transport, specificity, or recovery fails, the corresponding inference weakens.

Multimodel inference can preserve what was learned. Researchers may report relative support, predictive overlap, and consequences shared across the surviving set rather than crowning one winner. The useful discovery may be that all adequate models imply one limited pattern while disagreeing about mechanism.

Ambiguity changes decisions. A low-stakes exploratory program may continue with several plausible models; a clinical implication, person-level inference, or public-status claim requires a much higher standard. Contexts such as palliative care, childhood, disability, psychiatry, incarceration, and low-resource settings also require separate transport rather than presumed interchangeability.

Some ties cannot be broken ethically. The intervention may be unsafe, the endpoint rare, or the observation too intrusive. The correct result is that available ethical designs do not distinguish the models. The moral boundary is preserved without becoming evidence for either side.

A disposition record then states what each tool may still do: attention aid, communication support, exploratory proxy, confirmatory estimator, or public-status input. Population, setting, decision, retention, oversight, and stop condition are explicit. A familiar brand cannot keep a failed tool in a higher role.

Model-ambiguous preserves the information actually earned: the data constrain the field but do not select a winner. Future work may seek a discriminator; until then, choosing the preferred story would reduce knowledge rather than increase it.

30.3.3 — No Adequate Estimate Is Not a Hidden Win

No adequate estimate applies when the target cannot be identified, observed, recovered, or bounded for the decision. It is neither zero, possible confirmation, nor a mystery leaning toward the Law. It names the failed layer because origin, bearer, endpoint bridge, observation, recovery, and uncertainty require different repairs, some impossible at acceptable burden. A broken sensor and an uncertain neutral origin may both block a terminal estimate, but they direct further work toward different problems.

Estimator failure concerns construction, not the people behind the data. A failed bridge may leave the hypothesis open while retiring the route and narrowing dependent claims. The empty cell cannot

be filled by mean imputation, unvalidated proxies, or artificial-intelligence inference of unreported states (Little & Rubin, 2019). For example, improving an observation channel may reopen a measurement question even while the completed-stream inference remains beyond the available record.

The public sentence is: the study could not answer its protocol-fixed question. Coverage, origin, endpoint, recovery, uncertainty, and safety gates determine when abstention stops verdict and dependent use. Repeated No adequate estimate outcomes may pause confirmation while essential objects remain inaccessible; trust grows when the cell stays empty, not recast as support. The report can still return usable knowledge by identifying the failed prerequisite and explaining which future development would make another attempt informative.

30.4 — Narrowing, Rival Victory, and Defeat

Narrowing is often treated as defeat in miniature, but the distinction matters. A universal claim may become viable only for a narrower reference class, endpoint branch, representation, or range of conditions. That contraction can be progress when it follows evidence and when the lost scope is not quietly retained in public language.

Components can also fail independently. An estimator may be nonviable, a mechanism may lose to a rival, a dream route may be too noisy, or a public diagram may create unacceptable misunderstanding. Retirement means the failed object stops doing work in future editions, studies, summaries, and applications.

Rival victory changes the project more deeply. If an ordinary model explains the protocol-fixed evidence better under equal rules, the compared Law-linked model loses its claim to that explanatory success. Further scientific work should identify a genuinely new discriminator or an independently justified route to the terminal target. The earlier result remains on the record rather than being recast as support.

These outcomes differ in what they withdraw. Narrowing withdraws scope; retirement withdraws an object's role; rival victory withdraws explanatory ownership of a pattern; strong-form defeat withdraws the declared target branch for its tested scope. Treating them as one vague disappointment would hide both the loss and the knowledge gained.

30.4.1 — Reference-Class Narrowing Can Be Progress

A universal claim can fail while a narrower claim remains viable. The narrower successor becomes a new scientific object with an explicit reference class, quantifier, endpoint, and evidence ceiling; it does not inherit the authority or public reach of the discarded universal form. Progress lies in stating that new boundary plainly and testing it prospectively rather than pretending the old claim survived unchanged.

Suppose a bridge works in verbally responsive adults during stable health but fails under delirium, deep sedation, or rapid neurologic change. The honest revision names the tested class, identifies the missing bridge, and makes the exclusion visible in every later claim. Narrowing replaces vague breadth with a boundary that can be challenged (Yarkoni, 2022).

Narrowing may concern species, developmental stage, stream architecture, endpoint type, measurement context, or representation branch. The reason must match the gate that failed. A scalarity problem may require a bipolar or partially identified branch rather than exclusion of a population; endpoint uncertainty does not revise the meaning of valence.

The Law may remain a philosophical hypothesis while losing an empirical family, remain useful as a burden lens while losing a terminal claim, or stay open for one human reference class while remaining unknown for nonhuman or artificial candidates. A narrower domain can still matter without inheriting universal rhetoric (Popper, 1959; Lakatos, 1978).

Timing matters. A reference class fixed before testing belongs to the protocol-fixed hypothesis. A class chosen after an adverse result may be scientifically reasonable, but it is a new claim that requires new evidence. The old result remains evidence against the broader formulation.

Downstream applications must update locally. The relevant section, example, caption, guide page, or public summary should state whether it depends on the formal claim, a narrower scientific result, independently justified ethics, or domain-specific practice. A general disclaimer at the front cannot perform that dependency work.

A narrowed claim needs a re-entry rule. Return of an excluded population or endpoint may require a validated translation, an independent consciousness marker, a better observation model, or proof that another branch preserves the same target. Without such a rule, narrowing can become permanent neglect.

Evidence also remains domain-specific. A result in terminal cohorts does not automatically travel to dreams, laboratory tasks, nonhuman bearers, artificial systems, private practice, or public communication. Each corridor earns its own transport.

The mature update is exact and unembarrassed: the broader claim was not supported under the protocol-fixed conditions; a narrower claim remains open for stated reasons; and excluded cases remain unresolved and visible. Narrowing is honorable only when the public language becomes smaller with the science (Mayo, 2018).

30.4.2 — Components and Routes Can Be Retired

A research program is made of terms, diagrams, estimators, mechanisms, datasets, protocols, chapters, and public artifacts. Retirement is the controlled decision that one of these objects no longer carries scientific or public work.

Retirement criteria should be written before loyalty forms. A route may retire after repeated recovery failure, loss to equal-budget rivals, dependence on unavailable information, violation of a safety gate, or reliance on unidentifiable parameters. An estimator retires when bias, uncertainty, or context dependence defeats its declared purpose. A public phrase retires when ordinary readers repeatedly and reasonably take it to imply payment, prediction, or rank.

A component can retire without settling the whole program. A neutral-origin calibration, dream-coding procedure, reserve-linked route, or terminal channel may fail while other questions remain open. Modularity becomes an excuse for endless survival only when nothing is ever allowed to leave.

Retirement can be honorable. A clearly stated claim that was tested ethically, exposed to rivals, and withdrawn when it failed has served science. The dishonor would be preserving it as metaphor, branding, or mystery after withdrawing it as evidence (Kuhn, 1962; Longino, 1990).

The retirement record states what stopped, why it stopped, what remains unaffected, and whether a replacement begins a new evidentiary chain. A failed mechanism is not said to have

“evolved,” and an invalid instrument is not silently swapped while the old claim remains.

Failure memory prevents abandoned ideas from returning under new names. Code, protocol records, public registry records where they exist, negative controls, recovery results, deviations, and interpretive decisions remain traceable so later teams know what failed and critics can verify that retirement was real (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2017).

The disposition may be removal, demotion to a partial pathway, or preservation as historical scaffold. Each status carries different public permissions. A retired claim cannot continue as donor inspiration, merchandised mystery, private-membership knowledge, or unmarked educational material (Oreskes, 2019).

Correction should respect people who relied on earlier language. The project explains what changed without mocking early readers or supporters. Honest memory protects trust better than pretending the stronger version was never used.

Retirement makes the program more coherent. The object remains recoverable for history, its dependencies update, and later editions cannot quietly reintroduce it. Letting go is evidence that modularity serves learning rather than immortality.

30.4.3 — Rival Victory Changes the Project

Rival victory means another model has supplied the better explanation under the agreed rules. The explanation that lost must relinquish its claim to the tested pattern, and the project must

decide which remaining question would make a new comparison informative.

The comparison is not a popularity contest. Adaptation, allostasis, predictive processing, opponent process, evolutionary strategy, treatment, selection, social scripting, or measurement artifact may be less emotionally satisfying and still explain better. Calibration, out-of-sample performance, parsimony, transport, and intervention response determine the protocol-fixed result (Oreskes, 2019; Box, 1976; Akaike, 1973; Burnham & Anderson, 2002).

Victory is scoped. A rival may win one laboratory task while leaving terminal questions open, explain variance compression without explaining a complete-stream relation, or show that a completion variable adds no value after ordinary covariates. Neither side receives more than the contest established.

Calling the winner “how the Law works” is not enough. A mechanism earns that role only if its independently established properties, joined to independently justified bridges to the declared phenomenal target and endpoint, actually entail PTVC in the stated scope. Reproducing a similar curve does not meet that requirement. An ordinary mechanism could therefore become a realization of the Law, while an ordinary model that wins the original comparison remains credited for that result. The record preserves the tested loss and identifies any stronger follow-up claim as a separate question for new evidence (Wilson & Collins, 2019; Palminteri et al., 2017).

Useful infrastructure survives the change. Better measures, safer protocols, clearer endpoints, stronger missingness models, and well-governed datasets can serve affective science even when the explanatory hierarchy changes. Preservation of method is not preservation of the defeated claim.

Resources should follow the evidence. Repeated rival success shifts future design, expertise, and funding toward the mechanisms that earned support. Continuing to center the losing explanation by inertia would make rival victory symbolic rather than real.

The public record must change where the stronger claim traveled. Summaries, figures, talks, teaching materials, websites, and funding language receive the revised status. A technical concession that leaves the public promise untouched is not a scientific update.

A mature response is revision without resentment. Preserve the history, credit the explanation that won, remove the false road, and state the remaining question plainly. A theory gains credibility when competitors can alter its future rather than decorate its present.

30.4.4 — Strong-Form Defeat Must Be Said Clearly

Strong-form defeat is branch-matched. On the protocol-fixed exact set-universal branch, one gate-valid qualifying stream whose nonempty truth-containing terminal residual set excludes zero is Target-adverse and refutes that branch for the declared reference class. A finite-band branch is adverse only when its truth-containing set is disjoint from the protocol-fixed acceptance

region. Probability, support, and mechanism branches retain their own guarantees and decision rules.

Such a conclusion requires adequate bearer, representation, neutral origin, complete domain, endpoint, observation process, measured-to-latent bridge, and inferential guarantee. A point estimate away from zero is not a truth-containing set, and a confidence interval is not a deterministic counterexample merely because it excludes a preferred value. The adverse object must match the claimed quantifier and target geometry.

Logical refutation and evidential confidence are related but distinct. One verified counterexample has decisive force against an every-stream claim; independent replication, transport, and adversarial review strengthen confidence that the gates and scope were correct. Precommitment prevents both cheap dismissal and cheap acceptance (Popper, 1959; Mayo, 2018).

The same rigor that prevents false support must prevent false defeat. A broken instrument, retrospectively shifted origin, incomplete stream, outcome-dependent routing, convenient endpoint, or population mean away from zero does not establish a counterexample. Failure applies to the target, not a nearby statistic.

Hiddenness cannot be invoked twice. If the protocol-fixed bridge is inadequate, the result is nonadjudication or No adequate estimate. If the bridge and adverse object are adequate, saying that the latent quantity remains hidden would reopen a gate after the result was known.

Replication and transport then determine how confidently the finding extends beyond the first adjudicated case. They may show that failure is not confined to one instrument, site, endpoint class, or population, while preserving the exact boundary of what has actually been defeated.

Reports should distinguish a failed claim, failed instrument, failed study, and failed communication choice. That precision prevents one loss from becoming either exaggerated collapse or minimized inconvenience.

The public sentence is plain: strong-form defeat ends the strong-form claim for the defeated scope. It does not erase care, invalidate every method developed around the question, or imply that adjacent branches failed. It means reality answered one protocol-fixed question with no.

Defeat requires a propagation protocol. The status owner freezes promotional use, records the branch, quantifier, representation, endpoint rule, reference class, adverse finding, and verified prerequisites, and identifies every dependent chapter, figure, summary, presentation, classroom exercise, and public excerpt. Each object is corrected, narrowed, archived, or withdrawn (Lakatos, 1978; Mayo, 2018).

Correction urgency should scale with reach, and discoverability is part of correction: a reader encountering an obsolete claim should be able to find the current one (National Academies of Sciences, Engineering, and Medicine, 2017; Fischhoff, 2013).

Scientific integrity requires precision, not ritual humiliation. The project may preserve the history, thank the evidence for a sharper

boundary, and retain independently useful work. What it cannot preserve is the authority of a claim that lost its warrant (Longino, 1990). A clear defeat can therefore remain informative by identifying which bridge, branch, or scope no longer deserves to carry the claim forward. Preserving that distinction turns a loss of warrant into a gain in scientific resolution. That record also shows the next research team where a narrower, better-founded question can begin.

30.5 — Value Under Every Outcome

Several commitments in this book do not depend on PTVC. A person in pain should be helped. A patient deserves honest information and relief. A participant deserves consent, privacy, and the right to withdraw. A critic deserves an answer directed at the argument rather than at loyalty or motive.

Those commitments arise from the person and the situation, not from a terminal hypothesis. Dignity is not conferred by a favorable result, and privacy is not suspended by scientific interest. Nonparticipation remains a full-status choice because no research program owns access to another person's experience.

This outcome-resilient value is not substitute evidence. It consists of independently justified commitments made more explicit by the inquiry: care before interpretation, dignity beyond measurement, skepticism without contempt, privacy without penalty, and correction without shame.

The strongest proof of commitment is often costly conduct. Privacy may require deleting useful data. Honest criticism may require publishing an uncomfortable result. Care may require abandoning a terminal observation. Correctability may require withdrawing a popular graphic. Outcome-resilient value becomes credible when it constrains the project before the scientific verdict is known.

30.5.1 — Care Remains

Care is independent of the Law because need is already present. Pain, fear, hunger, grief, illness, danger, and isolation do not have

to become evidence before they deserve a response. PTVC may change a question; it does not create the duty to help.

Clinical and social duties do not wait for a theory of the whole life. A person in pain receives relief according to clinical judgment and consent; a family in danger receives protection; a worker facing exploitation receives remedy. These duties follow established care and ethics, not a terminal residual (World Medical Association, 2024; Beauchamp & Childress, 2019; World Health Organization, 2020).

This independence functions as a decision rule. When evidence is absent, ambiguous, narrow, or adverse, treatment follows the best available clinical and social knowledge. When a Law-linked interpretation conflicts with rescue, analgesia, safety, privacy, or rights, the interpretation yields.

Care also includes refusing imposed meaning. A clinician, chaplain, teacher, parent, or friend should not tell someone that suffering balances a hidden account or prepares equivalent joy. A person may choose meaning and later revise it; support makes room for that freedom without requiring it.

Records are part of care. Research data do not become eligibility files, employment screens, insurance judgments, loyalty indicators, or model-training material merely because they exist. Purpose limitation, minimization, access control, correction, deletion where possible, and separation from ordinary services protect the person when interest outlives consent (CIOMS, 2016; Emanuel et al., 2000).

Dignity is not an evidence outcome. Support cannot increase it, narrowing cannot partition it, uncertainty cannot suspend it, and defeat cannot reduce it. Scientific results may alter explanation; they do not redistribute moral standing.

30.5.2 — Skepticism Remains Welcome

Skepticism is not disloyalty to a question. It is one way a community prevents authority, reputation, hope, and shared language from becoming substitutes for evidence. A critic may be wrong on a point and still be necessary to the health of the project.

Skepticism can be logical, empirical, ethical, linguistic, or political. One critic may doubt scalar valence, another the endpoint bridge, another the treatment of rivals, and another the public imagery. Welcoming criticism means identifying which claim is challenged rather than treating every objection as resistance.

The Law cannot be protected by belonging. Requests for valid measurement, independent endpoints, equal-budget rivals, source support, version history, or safer public language perform a necessary epistemic function (Merton, 1973; Longino, 1990).

A healthy project gives critics consequences. They can shape rival selection, prospectively specified failure conditions, artifact language, and decisions to pause or withdraw. An invitation to comment after publication is weaker than adversarial participation before decisive choices are frozen (Longino, 1990).

Critics remain necessary after support. Early positive evidence can make exaggeration more likely, not less. Skeptics should therefore be protected from loyalty tests, social rank, or claims

that doubt reveals a lack of compassion. A person can reject PTVC and care deeply about suffering, science, justice, and peace.

A flaw found early reduces downstream burden. It may prevent a vulnerable reader from receiving a dangerous sentence, a researcher from repeating a failed method, or an institution from adopting an unsupported tool. Criticism is productive work (Oreskes, 2019).

Public conversation should distinguish sharp criticism from threats, harassment, privacy violations, and unsafe instructions without using moderation to insulate the theory. Criticism of presentation belongs in the inquiry when it identifies a concrete distortion, inaccessible explanation, unsupported authority signal, or detachable artifact that overstates the source. The answer should address that defect on substance and correct the artifact when the criticism is right.

Discomfort, embarrassment, or reputational risk is not a reason to remove substantive criticism.

Welcome for skepticism has practical tests: objections remain visible, adverse findings are publishable, critics are answered on substance, and participation never depends on belief. The project should become easier to correct as attention grows. It can serve truth only while the critic remains inside the room.

30.5.3 — The Series Must Remain Correctable

Correction is not complete when only one technical source changes. Any chapter, summary, figure, website, classroom material, translation, or application that inherited the old claim

must receive the new status or be withdrawn. Preserve a visible version history so readers can see what changed, why, and which evidence or ethical trigger caused the revision (National Academies of Sciences, Engineering, and Medicine, 2017; Fischhoff, 2013).

“We were wrong here,” “the evidence is narrower,” and “this artifact has been withdrawn” are signs that learning outranks continuity, not admissions to be hidden.

30.6 — Final Synthesis: A More Precise Question

The final synthesis asks what kind of intellectual object has been built. This book has built a public statement of a falsifiable boundary hypothesis, an ethical firewall, a vocabulary of application, and a map of support, ambiguity, narrowing, rival victory, and defeat. Doctrine, government, religion, clinical practice, ranking, and personal prediction retain their own standards and authority.

The formal perspective is now clearer. The proposed bearer is a separately routed maximal qualifying conscious stream. The formal quantity is neutral-relative phenomenal valence under a declared representation. The target is the complete strict-pre-endpoint terminal residual, not a population average, remembered-life judgment, or visible biography.

The ethical perspective is equally clear. No one's pain is morally owed payment. No joy creates moral debt. Exact PTVC nevertheless requires an equal-and-opposite signed remainder for a genuinely nonzero accumulated balance. No endpoint claim authorizes delayed care, personal testing, harm induction, forced meaning, surveillance, or human-worth scoring. These limits are not softened by spiritual resonance or public enthusiasm.

The final question can now be posed as a sequence rather than a slogan. Is there a qualifying stream? Can valence be represented without moving the origin? Are the complete domain and endpoint defined independently? Can the observation model and measured-to-latent bridge recover or bound the target under uncertainty? Do prospectively specified consequences distinguish

the proposal from ordinary rivals? Does the result transport, replicate, and survive ethical review? The sequence is longer than the opening sentence because answerability has a cost.

30.6.1 — Formal Perspective

Each component of the formal object performs a different task. Changing one changes the claim.

In plain language, the strong proposal concerns accumulated neutral-relative valence within one qualifying stream at an independently defined endpoint. It is exact, intrastream, and nontransferable. It does not say that every moment is neutral, that lives are equal, or that visible injustice has already been repaired.

The search has separated objects that ordinary language blends. Exact latent closure is not a near-zero estimate; a population mean is not a pathwise result; a route signature is not the target law; a theorem under assumptions is not evidence that the assumptions hold; and logical coherence is not occurrence in nature. The ledger names accumulation, not moral debt, worth, punishment, or a public score.

The random-endpoint obstruction remains the sharpest formal reminder that exact closure needs declared structure rather than wishful stopping. Chapter 7 states the result and its branches in full; the final synthesis carries forward its consequence: any viable nontrivial account must expose the route or structural condition that permits closure on its declared branch and let ordinary selection and observation rivals challenge it.

Evidence may arrive through terminal, longitudinal, dream, laboratory, perturbation, choice, physiological, or repair studies.

No family carries the Law alone, and route tests support only their routes. *The Science of Fairness* inherits a burdened hypothesis, not a protected doctrine (Mayo, 2018).

Consciousness, scalar valence, neutral origin, complete-stream observation, endpoint truth, and measured-to-latent transport remain open. Formal elegance has made these unknowns easier to locate and harder to disguise, not solved them.

The proposal becomes worth investigating where these difficulties can be turned into discriminating questions. It offers a sharply defined terminal target, possible architectures whose assumptions can be examined, and a demand to distinguish complete-stream closure from familiar return and rebound. These are reasons to undertake the work while leaving its answer open. The next gain could be a better measure, a prediction that separates two routes, or evidence that closes a route completely. Each would leave the question better understood (Popper, 1959; Longino, 1990).

30.6.2 — Ethical Perspective

The ethical conclusion is not derived from PTV. Care, consent, dignity, privacy, justice, refusal, and nonparticipation remain binding whether the claim is supported, narrowed, unresolved, or defeated. No result converts suffering into moral desert, owed payment, evidence of worth, or permission to wait (World Health Organization, 2020; Beauchamp & Childress, 2019).

The intrastream rule blocks one transfer without becoming a complete moral theory. One person's pleasure cannot cancel another person's pain inside the claim. A population benefit

likewise does not erase concentrated harm, institutional success does not settle an individual grievance, and a future promise does not revalue a past injury.

Felt experience is the part of life least available to public ownership. Research therefore requires restraint in collection, humility in inference, and a permanent refusal to turn a hidden object into a score of worth. No model, archive, or institution gains privileged access to another person's standing.

The cumulative object is not moral debt. Violence, poverty, illness, abuse, and neglect retain ordinary causes and accountable agents; treatment, protection, apology, and repair act in the present. Beauty, inspiration, or spiritual resonance cannot impose meaning on a particular wound (Beauchamp & Childress, 2019).

Interpretation remains free. Religious, secular, skeptical, survivor, and grieving readers may accept, revise, refuse, or avoid every balancing metaphor; meaning is voluntary, never a verdict about what suffering was for.

The ethical firewall includes every derived sentence and artifact. Language inviting fear of joy, self-punishment, victim-blaming, personal testing, or delayed care is rewritten or withdrawn because foreseeable effect matters more than intent (World Medical Association, 2024; Beauchamp & Childress, 2019).

30.6.3 — Test Fiercely, Care Immediately

The final posture joins fierce testing with immediate care: strong rivals, adverse data, recovery, replication, and defeat on one side; relief, dignity, consent, privacy, and protection on the other.

Severity without care can make people wait; care without testing can shelter error. Each restrains the other (Oreskes, 2019; Mayo, 2018).

The next stage begins with questions small enough to work on: how to define neutral feeling, how to distinguish the end of a stream from the end of a recording, and how to tell a distinctive prediction from an ordinary explanation. *The Science of Fairness* develops the mathematics, candidate mechanisms, measurement work, and tests needed to pursue them. Low-burden work comes first, with later steps earned by what the earlier work establishes.

This book also opens a ten-book conversation. What would make a possible future worth wanting, and a path toward it feasible? How might fairness inform private life or freely chosen spiritual meaning? What should public administration, civic life, rights, and the material conditions of peace ask of us? The companion books take these questions further, each using the evidence and reasons its subject requires. The Series Map offers starting points; the question that matters to you can help choose the next book.

Bring your curiosity, your experience, and your hardest questions. You are free to follow a single strand, compare competing answers, or explore the series in depth. Each later proposal must earn its own justification. That freedom to examine the idea on your own terms is part of the invitation.

Could life be fair? The possibility gives us a remarkable question to pursue. We now have a sharper way to ask it, a clearer view of what an answer would require, and concrete places to begin. We can pursue the question while relieving pain, protecting freedom,

making room for joy, and caring for one another. Let curiosity carry the question forward, and let care begin where we are. If the Law points toward something real, the path to discovering it may reveal new connections across consciousness, biology, mathematics, and the long arc of a human life. That possibility is reason enough to keep asking the question with imagination, rigor, and hope.

Glossary

A

Abstention: A scientifically legitimate decision not to issue a positive or negative verdict because identification, measurement, endpoint, model, or data-quality requirements are not satisfied. Abstention is not evidence for closure and is not a failed analysis.

Action-mediated closure: A candidate route on which closure, if it occurs, is produced through the consequences of available actions rather than imposed by an unexplained terminal adjustment. Evidence for this route would not by itself establish the terminal target.

Active inference: A family of models in which an organism can reduce expected mismatch by updating expectations and by acting on or selecting its environment so preferred or viable states become more likely. Directed action in such a model does not establish felt valence, free choice, hidden fairness, or terminal closure. See also: Predictive processing; Reinforcement-learning reward signal; Viability kernel.

Ad hoc patch: A change introduced after an adverse result mainly to preserve a preferred conclusion, such as moving the endpoint, recentering the origin, inventing an unseen interval, or excluding a bearer. A legitimate successor must have independent support, create new prospective tests, and leave the earlier result visible. See also: Correction and withdrawal; Endpoint independence.

Adverse case: A gate-valid qualifying stream for which the appropriate truth-containing result excludes exact zero. Under an exact universal branch of PTVC, one valid adverse case cannot be canceled by favorable population averages.

Affective valence: The positive, negative, or neutral felt character of conscious experience. In this series, valence concerns how experience feels from within, not merely preference, behavior, reward, arousal, or verbal report.

AI sovereignty: The forbidden elevation of an artificial intelligence system from an assistive tool to an authority over a person's inner life, worth, standing, or eligibility. Fluency, prediction, confidence scores, or large archives do not give a system direct access to consciousness, consent, personhood, or a complete phenomenal history. See also: Human-worth score prohibition; Testimony-like output; Latent cumulative object.

Almost-sure attainability: The existence of at least one admissible policy that reaches a declared target with probability one from a stated information state. It differs from closure under a fixed law, success under every policy, and limit-sure attainability, where probabilities can approach one without any single attaining policy. See also: Limit-sure attainability; Probability-one closure; Reachability.

Almost-sure closure: See Probability-one closure.

Alternative endpoint branches: A family of prospectively declared endpoint architectures: state coupling, first hitting, absorption, endpoint atoms, singular timing or support, continued experience beyond a protocol boundary, and endpoint-inclusive value. These alternatives change different premises and are not interchangeable. Each carries separate endpoint, bridge, representation, and testing burdens, and none independently

evidences PTVC. See also: Endpoint atom; First-hitting and absorption routes; State-coupled timing.

Applied lens and formal Law: The applied lens asks bounded present questions about burden, options, care, and repair. The formal Law concerns an exact truth-level condition over one complete qualifying stream. An applied insight may guide humane action without testing PTVC, and a formal result cannot replace practical judgment, rights, or care. See also: Care before proof and interpretation; Law of Fairness; Target-route distinction.

Arousal: The activation, wakefulness, or energetic mobilization dimension of experience. It is distinct from felt valence because high or low arousal can accompany welcome, burdensome, mixed, or neutral states, and an arousal measure cannot substitute for a valence representation or terminal residual. See also: Felt valence; Observation channel and observation model.

Atom at zero: A positive probability mass located exactly at zero in a distribution of terminal residuals. An atom at zero is stronger than mere concentration near zero, but it is not the same as universal or probability-one closure.

Attainability: The existence of at least one admissible route or policy capable of reaching the target from a specified state. Attainability alone does not show that the route will be selected, that it works under partial observation, or that every admissible path closes.

Attrition mechanism: The process by which participants leave, pause, become unreachable, lose access, enter treatment,

withdraw permission, or die during longitudinal research, thereby changing who remains visible. Attrition is both a selection problem and an ethical signal about burden, access, trust, and study design. See also: End of observation; Informative missingness; Longitudinal research trace.

Auxiliary assumption: An assumption needed to connect a theoretical claim to an observation, estimator, or test. A failed prediction may reflect the target hypothesis, an auxiliary assumption, or both, so confirmatory work must state these assumptions in advance.

B

Balance: Book 1's accessible term for exact intrastream terminal valence closure. It does not mean equal pleasure and pain at every moment, equal lives, morally owed compensation, or any transfer of valence between people or streams.

Baseline drift: A change over time in a person's use of a rating scale, neutral point, comparison standard, or reporting habit. Unmodeled baseline drift can create an appearance of convergence or closure without a corresponding change in latent phenomenal valence.

Bearer: The truth-level subject or unit to which qualifying felt experiences belong and within which the proposed accumulation is evaluated. A bearer is not automatically identical to a body, legal person, biography, memory, archive, group, record, or output channel. See also: Maximal qualifying conscious stream; Ownership; Stream identity.

Behavioral nonresponse: The absence of expected command following or another observable response under a particular assessment. Motor, sensory, language, medication, arousal, task, and environmental barriers can block expression, so nonresponse alone does not settle whether experience is present or whether a conscious stream has ended. See also: Felt availability; Low-unity state; Unknown category.

Biological death: The end of biological life. It may be associated with a stream endpoint in some cases, but PTVC does not define biological death as the truth-level endpoint by stipulation.

Bipolar branch: A representation branch that keeps positive and negative phenomenal components separate so they may coexist rather than being forced into one momentary net value. It is not interchangeable with the primary additive branch, and results transfer between them only through a justified bridge on a common identified domain. See also: Generalized Terminal Neutrality; Primary additive branch; Scalarization.

Blinded bearer adjudication: A procedure for deciding among bearer or routing models while withholding, as far as possible, the downstream result each classification would help produce. Fixed criteria, independent reviewers, visible disagreement, and outcome separation reduce post hoc routing in favor of the theory. See also: Bearer; Routing; Unknown category.

Blinding: A design safeguard that withholds selected information from participants, investigators, coders, analysts, or adjudicators when that information could bias conduct or interpretation. The form of blinding must fit the study and its ethical limits.

Boundary hypothesis: A hypothesis about a limiting, terminal, or otherwise sharply delimited property of a process rather than a claim about its ordinary moment-to-moment path. PTVC is a candidate boundary hypothesis because it concerns a stream's terminal signed residual.

C

Calibration: The process of relating a measurement system to a defined reference so that its origin, scale, uncertainty, and interpretation are known. Calibration must not be performed in a way that forces completed records to appear closed.

Candidate boundary hypothesis: A status label for PTVC as a testable proposal about a declared terminal boundary that remains open to support, narrowing, nonadjudication, or defeat. It is stronger than a metaphor but weaker than an established law, and no finite collection of compatible cases proves universal exact closure. See also: Falsifiability; Law of Fairness; Lawhood.

Candidate mechanism: See Candidate route.

Candidate route: A proposed mechanism, policy structure, dynamical construction, or mathematical path by which the terminal target might be satisfied. A route can receive support or fail independently of the target hypothesis.

Candidate sentience escalation: A protocol-governed increase in protection when a candidate system's organization, activity, behavior, or welfare signal exceeds the assumptions under which ordinary review was approved. It may trigger verification, monitoring, reduced exposure, independent review, pause, or retirement without declaring consciousness or personhood. See

also: Precaution without personhood; Testimony-like output; Theory-derived indicator.

Candidate signature: A prospectively specified observable pattern predicted by a candidate route, such as horizon-sensitive choice, recovery after perturbation, or narrowing variance. It is a testable clue rather than a unique fingerprint, and it gains weight only through transport, adequate measurement, and serious rivals under common rules. See also: Negative control; Nonunique explanation; Predictive fit versus explanation.

Care-before-interpretation rule: The ethical rule that urgent care, relief, safety, consent, privacy, and dignity take priority over assigning theoretical meaning to suffering or distress. No PTVC interpretation is required before helping a person.

Care before proof and interpretation: The governing sequence that responds to present pain, fear, danger, illness, or grief with reachable care and protection before asking what the Law or any wider story might mean. Care remains evidence-based, proportionate, revisable, and binding whether PTVC is supported, narrowed, defeated, or never identified. See also: Ethical floor; No harm for proof; Outcome independence.

Causal locus: The place in a system where a causal influence is proposed to operate, such as experience generation, action selection, routing, endpoint timing, observation, or reporting. Distinguishing loci helps prevent one process from being mistaken for another.

Channel qualification: The staged admission process by which a report, sensor, behavior, contextual record, or other observation

channel earns a defined role through technical verification, analytical validation, construct validation, burden review, and a failure threshold. Availability, intimacy, density, or technical prestige cannot substitute for relevance and validity. See also: Calibration; Multichannel estimate; Observation channel and observation model.

Claim ceiling: The strongest inference that a channel, design, model, comparison, or public artifact may support under its actual identification, uncertainty, scope, and safeguards. Clear communication may simplify language, but it cannot raise that ceiling or turn prerequisite, route, or finite-resolution evidence into universal exact closure. See also: Detachable-excerpt ceiling; Public-artifact limit.

Closure: The satisfaction of a declared terminal target. On the primary additive PTVC branch, closure means that the finite positive and negative phenomenal masses within one qualifying stream are exactly equal; equivalently, its terminal signed residual is zero.

Coercive measurement: Inner-life measurement conducted where authority, dependency, crisis, or structural pressure makes disclosure or monitoring a practical condition of care, work, education, benefits, liberty, opportunity, or standing. A tool is not voluntary when refusal brings penalty, suspicion, or reduced access. See also: Consent, refusal, and withdrawal; Human-worth score prohibition; Participation without surveillance.

Comfort-first observation: A terminal-research order of authority in which routine care, symptom relief, privacy, family

presence, and clinical judgment determine whether and when observation may occur. Data collection stops when it interferes with treatment, rest, touch, conversation, spiritual practice, or requested quiet. See also: Care before proof and interpretation; End of observation; Protocol endpoint.

Common-action requirement: Under partial observation, a single action chosen from the available information must be closure-compatible across all latent states that remain observationally indistinguishable at that decision point. State-specific actions that cannot actually be selected do not satisfy this requirement.

Comparative consciousness evidence: Species-appropriate evidence that combines behavior, learning, neurobiology, ecology, affective organization, and motivational tradeoffs into a converging case about possible experience. Human resemblance may inform a bridge but is neither necessary nor sufficient, and welfare protection need not wait for a complete theory. See also: Precaution without personhood; Theory-derived indicator; Unknown category.

ComparatorStatus: See Scientific verdict axes.

Comparison without forced equivalence: A method for comparing traditions or ideas by naming a precise shared question, function, image, or difference while preserving distinct histories, doctrines, registers, and internal disagreements. Similar words such as balance, justice, soul, or suffering do not make traditions interchangeable or turn resonance into evidence. See

also: Controlled vocabulary transfer; Meaning as optional; Spiritual resonance is not scientific support.

Compensability: A route-specific question about whether a present formal discrepancy remains answerable through safe, realistically available paths before a declared horizon closes. It is not moral repayment, a personal trait, a promise of recovery, or a license to withhold care or reprice past suffering. See also: Future optionality; Noncompensable risk; Viability kernel.

Complete felt history: See Complete strict-pre-endpoint felt history.

Complete strict-pre-endpoint felt history: Every qualifying phenomenal contribution owned by one stream before, but not at, its independently defined truth-level endpoint. Completeness is relative to declared bearer, representation, ownership, and domain rules, not to a remembered biography, selected episode, observation window, or final survey. See also: Endpoint-inclusive branch; Maximal qualifying conscious stream; Truth-level endpoint.

Complete strict-pre-endpoint phenomenal domain: The entire qualifying field of conscious experience belonging to one stream before, but not including, its truth-level endpoint. The primary PTVC target is defined over this complete domain, not merely over a sampled terminal window.

Completion potential: A candidate mathematical function whose decrease, boundary behavior, or terminal value can help characterize a proposed route to closure. It is a route-level object and must not be treated as the target itself.

Conditional support: The set of values, states, actions, or outcomes that remain possible after conditioning on specified information. Claims about exact closure must distinguish support restrictions from high probability or average behavior.

Conscious stream: A temporally connected sequence or field of qualifying conscious experience treated as one routed unit for analysis. A stream is not automatically identical to a biological organism, legal person, memory narrative, or observed data record.

Consent, refusal, and withdrawal: A continuing, practical consent relationship in which information is understandable, alternatives are genuine, retaliation is absent, stopping remains possible, and data consequences are clear. Separate choices may govern participation, monitoring, linkage, reuse, model training, disclosure, continued contact, and deletion. See also: Data minimization and deletion; Measurement without ownership; Participation without surveillance.

Constraint does not imply design: The rule that physical, developmental, mathematical, or stream-level limits can structure what is possible without implying an intending agent, moral purpose, hidden plan, or created world. A coherent design model can show possibility without showing that the actual world was designed that way. See also: Constraint without a judge; Spiritual resonance is not scientific support.

Constraint without a judge: An impersonal structural or boundary relation that may limit admissible complete histories without any agent assigning rewards, punishments, purposes, or

schedules. PTV C proposes exact terminal zero in this structural sense while leaving mechanism and metaphysics open. See also: Candidate boundary hypothesis; Constraint does not imply design; No moral desert, debt, or payment.

Construct-estimand distinction: A construct is the experience-related concept a study intends to represent. An estimand is the exact population, time, endpoint, treatment, and analysis quantity it seeks to learn. Precision cannot repair an undefined construct, and a cohort mean, individual probability, and path-level endpoint answer different questions. See also: Estimator; Observation channel and observation model; Reference class.

Construct validity: The degree to which a measure, manipulation, or model captures the intended latent construct rather than a neighboring one. For valence research, behavior, reward, arousal, language, and neural activity may inform measurement without being identical to felt valence.

Controlled vocabulary transfer: The disciplined movement of a technical term across fields with its source meaning, evidence ceiling, local use, prohibited inference, and route back to specialist authority. It prevents words such as bearer, stream, candidate, indicator, sentience, support, and personhood from silently changing strength. See also: Claim ceiling; Comparison without forced equivalence; Testimony-like output.

Correction and withdrawal: The accountable process of detecting a problem, assessing it, containing harm, correcting the record, propagating the change, documenting it, reviewing controls, and withdrawing or retiring affected claims,

comparisons, routes, tools, and public artifacts. Superseded versions remain visible enough to preserve the scientific record. See also: Ad hoc patch; Public-artifact limit.

Counterfactual reachability: Whether a target could be reached under one or more alternative admissible actions or policies from a specified state. Counterfactual reachability does not establish what actually occurred or what a partially informed agent could select.

Craving capture: The narrowing of attention and immediately reachable options when a learned cue, urge, withdrawal state, or expected relief becomes unusually salient. It is an ordinary mechanism that can constrain lived choice and future optionality without implying weak character, cosmic punishment, or assigned suffering. See also: Future optionality; Lived choice; Reinforcement-learning reward signal.

D

Data as trace: The principle that a report, waveform, image, note, gesture, code, archive, or model output is a context-bound trace produced through an observation process, not a reproduction of the experience or person. A trace may support bounded inference without becoming inner truth or human worth. See also: Latent cumulative object; Observation channel and observation model; Provenance.

Data-generating process: The full process that produces the observed data, including latent states, dynamics, routing, selection, measurement, missingness, and reporting. A valid

analysis must consider whether more than one process can generate the same observations.

Data minimization and deletion: The discipline of collecting and retaining only the least revealing data needed for a declared question, while governing removal or restriction across raw records, features, backups, shared copies, outputs, and trained-model influence. Deletion promises must accurately state their technical reach and residual limits. See also: Consent, refusal, and withdrawal; Provenance; Record-regime separation and no-dossier rule.

Decision rule: A prespecified mapping from evidence to an outcome such as support, adverse result, narrowing, abstention, or route retirement. Decision rules should be fixed before outcome inspection whenever confirmatory status is claimed.

Defeat condition: A prespecified result that counts against a target, route, measurement claim, or auxiliary assumption. A serious empirical proposal states what would narrow or defeat it rather than treating every possible result as compatible.

Desert-based fairness: The question of when praise, reward, blame, or sanction properly follows from what a person chose and did. Its legitimate use is narrow and evidence-based, and it does not make suffering a currency of justice or a verdict on a person's worth. See also: External justice; No moral desert, debt, or payment; Procedural fairness.

Detachable-excerpt ceiling: The maximum claim a headline, quotation, caption, slide, social post, translation, donor summary, or AI explanation may make when encountered apart from its

source. A fragment may simplify, but it must preserve every condition whose removal would imply proof, payment, prediction, scoring, or authority. See also: Claim ceiling; Portable safeguard; Public-artifact limit.

Dignity firewall: The rule that a person's dignity, rights, care, safety, treatment, privacy, and entitlement to relief do not depend on whether PTVC is true, false, supported, or untested.

Dimensionality of valence: The question whether phenomenal valence can be represented on one signed dimension or requires multiple dimensions. The primary additive PTVC branch uses a signed representation, but that representation remains an empirical and conceptual commitment to be tested.

Direct identification: Identification obtained from assumptions and observations that determine the target without relying on a route merely because it fits. Direct identification is distinct from selecting one convenient model among several observationally equivalent alternatives.

Dream-as-felt-experience window: When a dream contains conscious experience, that interval may belong to the felt history of the same stream even if the waking biography omits it. Dreams can therefore widen a possible observation domain without revealing the whole stream, proving continuity through sleep, or exposing a hidden terminal balance. See also: Complete strict-pre-endpoint felt history; Dream-report chain; Reversibility and return rule.

Dream-report chain: The ordered distinction among experience during sleep, awakening, the memory trace available after

waking, and the report given to another person. Each transition can omit, reorder, prompt, or reshape what reaches the record, so a dream report is not an unedited transcript of sleeping experience. See also: Data as trace; Informative missingness; Response shift.

E

Empirical band: See Finite-band compatibility and measured equivalence.

End of observation: The last time usable data are obtained because a channel fails, monitoring stops, communication is lost, privacy is requested, or another practical event occurs. It is an observation boundary that may precede the protocol endpoint and truth-level endpoint and creates missingness rather than terminal truth. See also: Protocol endpoint; Truth-level endpoint; Observation window.

Endpoint: The boundary at which a qualifying conscious stream ceases for purposes of the truth-level target. The endpoint must be defined by the stream model and cannot simply be replaced by biological death, loss of contact, device shutdown, or the last available report.

Endpoint atom: Positive probability mass assigned by a declared endpoint-time or marked-endpoint law to a particular time or point. It changes the timing architecture but does not itself create closure, and it differs from an atom at residual value zero and from merely recording a categorical endpoint event. See also: Atom at zero; First-hitting and absorption routes; State-coupled timing.

Endpoint bridge: The independently validated relation connecting an observed or protocol endpoint, possibly an interval, to the truth-level endpoint required by the claim while carrying error, return, revision, confounding, and competing-event uncertainty. A clinical definition or sensor threshold may inform the bridge without becoming the formal endpoint. See also: Protocol endpoint; Reversibility and return rule; Truth-level endpoint.

Endpoint conditioning: Selecting, observing, conditioning, or reweighting trajectories because of where they end, which can make earlier paths appear to drift or compress toward the selected endpoint without an independent terminal law. A conditioned construction may be mathematically valid while remaining nonprobative of PTVC. See also: Candidate signature; Endpoint independence; Random-endpoint obstruction.

Endpoint coupling: A candidate route in which endpoint timing or endpoint conditions depend on the evolving stream state. Evidence for such coupling would concern a route and would not, by itself, establish exact terminal closure.

Endpoint firewall: See Endpoint independence.

Endpoint-inclusive branch: A separately declared branch that assigns a specified phenomenal contribution exactly at the endpoint. It has a different temporal domain from the primary strict-pre-endpoint branch, cannot be introduced after a residual is seen, and cannot establish strict-pre-endpoint closure merely because its own condition is compatible. See also: Complete

strict-pre-endpoint felt history; Exact closure; Truth-level endpoint.

Endpoint independence: An endpoint rule is outcome-independent when its branch, evidence source, selection rule, and adjudication procedure are fixed and applied without letting a ledger estimate or terminal residual choose the endpoint. This firewall blocks circular selection but does not require probabilistic independence from all stream dynamics. See also: Ad hoc patch; Protocol endpoint; Random-endpoint obstruction.

Endpoint marker: An observable event or signal used to estimate or bound a truth-level endpoint. A marker can be informative without being identical to the endpoint itself.

Endpoint uncertainty and ambiguity: The preservation of multiple defensible interpretations of endpoint identity, timing, or interval when available evidence cannot choose among them under the declared rules. Such uncertainty must be represented, adjudicated prospectively, or allowed to yield No adequate estimate, never resolved by selecting the boundary that makes closure look favorable. See also: Endpoint bridge; No adequate estimate; Truth-level endpoint.

Equivalence band: A prespecified finite interval around a reference value within which effects are treated as practically equivalent for a stated purpose. An equivalence result around zero is not evidence of exact zero unless the target itself is explicitly approximate.

Error budget: An explicit accounting of uncertainty from sampling, instruments, calibration, sign, origin, timing,

missingness, model choice, transport, and construct identification across the evidence chain. Larger samples may reduce some random error but cannot repair an invalid construct, shifted origin, reversed sign, systematic bias, or selective missingness. See also: Calibration; Noise floor; Partial identification.

Estimand: The precisely defined quantity an analysis seeks to learn, including its population, unit, time domain, conditioning, and scale. The estimand is not the same as the estimator used to calculate it.

Estimator: A rule or procedure for using observed data to estimate an estimand. An estimator may be biased, uncertain, inconsistent, or nonidentifying even when its numerical output is close to zero.

Ethical burden too high: A typed outcome in which a scientifically coherent question is not ethically available because the required burden, privacy exposure, dependency, deception, bystander risk, or lasting effect is disproportionate. Consent, payment, technical sophistication, or promised decisiveness cannot cure the design; it must be redesigned or left unanswered. See also: No harm for proof; No adequate estimate; Safe perturbation.

Ethical floor: The nonnegotiable safeguards that keep present pain a claim on care, reject earned suffering and cross-stream payment, forbid harmful testing, protect consent and refusal, place relief before interpretation, and allow meaning to be declined. These protections hold under every scientific outcome.

See also: Care before proof and interpretation; No harm for proof; Outcome independence.

Every-continuation closure: Every continuation in a prospectively declared nonempty admissible continuation class satisfies closure. The claim says nothing about paths outside that class and is not implied by probability-one closure or support containment. If the class is empty or undefined, closure is unavailable rather than vacuously established. See also: Probability-one closure; Reachability; Support containment.

Evidence ladder: An ordered set of evidential claims that prevents weaker findings from being described as stronger ones. For PTVC, descriptive convergence, near-zero estimates, zero-containing intervals, atoms at zero, probability-one statements, and universal closure occupy different levels.

Exact closure: The truth-level condition that one qualifying stream's terminal signed residual is exactly zero. Exact closure is not interchangeable with near-zero measurement, mean-zero aggregation, an interval containing zero, or practical equivalence.

Exact zero: See Exact closure.

Experienced exposure and unity gate: Experienced exposure is the declared measure of how qualifying phenomenal experience contributes across a stream and may differ from calendar time. The unity gate states when felt contribution is treated as active and owned by the stream. Neither rule is a legal, clinical, personhood, or moral-worth threshold. See also: Complete strict-pre-endpoint felt history; Maximal qualifying conscious stream; Ownership.

External justice: Rights, capabilities, opportunities, material conditions, duties, institutions, protection, and remedy considered independently of any estimate of felt valence. External justice remains fully binding whether PTVC is supported, false, or untestable, and hidden relief or adaptation never excuses remediable deprivation or discrimination. See also: Outcome independence; Procedural fairness; Hidden burden and hidden relief.

F

Failed upstream gate: Failure to identify a necessary earlier layer, such as representation, origin, bearer, endpoint, observation, measured-to-latent bridge, or transport, blocks downstream warrant. A large sample, strong final statistic, or impressive graph cannot compensate for a prerequisite the inference requires but did not earn. See also: Measured-to-latent bridge; No adequate estimate; Scientific verdict axes.

Falsifiability: The requirement that a scientific claim specify observations or evidence objects that could genuinely count against it under prospectively declared gates and decision rules. A hypothesis that can absorb every adverse result through hidden channels, moved endpoints, narrowed classes, or renamed mechanisms has not faced a real test. See also: Candidate boundary hypothesis; Gate-valid counterexample; Strong-form defeat.

Falsification: A result that conflicts with a prespecified claim under the study's stated assumptions and validity gates. Falsification must identify which target, route, or auxiliary claim

is affected rather than overgeneralizing from one failed component.

Felt availability: The modest claim that something is occurring for a conscious stream even when the experience cannot be spoken, deliberately used, remembered later, or organized into a clear self-story. It separates possible experience from an observer's access to evidence about it. See also: Behavioral nonresponse; Low-unity state; Observation channel and observation model.

Felt valence: The positive, negative, or neutral evaluative character of qualifying experience as lived from within under a declared representation and neutral origin. It is not identical to happiness, life satisfaction, preference, arousal, salience, reward, pain behavior, welfare, meaning, moral value, or human worth. See also: Arousal; Neutral origin; Primary additive branch.

Finite-band compatibility and measured equivalence: Latent finite-band compatibility requires the complete nonempty latent compatibility object to lie wholly inside a prospectively calibrated latent acceptance set. Measured equivalence requires the complete nonempty measured compatibility set to lie wholly inside the appropriate bridge-defined measured region. Partial overlap or merely including zero is unresolved, and neither result proves exact PTVC. See also: Exact closure; Measured-to-latent bridge; Terminal residual compatibility object.

Finite signed phenomenal measure: A finite signed measure representing positive and negative phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain.

Finiteness permits well-defined positive and negative Jordan masses and a terminal signed residual.

Finite soft-guidance ceiling: The result that no finite probabilistic bias toward closure can guarantee exact probability-one or set-universal closure while nonclosing actions or paths retain positive support. Strong guidance is not the same as exact exclusion.

Finiteness: On the primary finite branch, each complete positive and negative phenomenal magnitude must be a finite quantity. This admissibility condition does not impose one upper bound on every stream, cap peak suffering, mean that experience is small, equalize gross experience, or establish finiteness for every possible representation. See also: Positive and negative phenomenal magnitudes; Primary additive branch; Gross experience.

Firewall: An explicit boundary that prevents evidence or language in one domain from being used to justify an unsupported conclusion in another. The series uses scientific, ethical, legal, institutional, and spiritual firewalls.

First-hitting and absorption routes: A first-hitting route uses the first entry into a prospectively declared set. An absorption route additionally requires the process to remain in a declared set after entry. Hitting is not absorption, and any claim that entry or absorption coincides with the truth-level endpoint needs a separately specified endpoint bridge. Neither route independently evidences PTVC. See also: Endpoint atom; Endpoint bridge; State-coupled timing.

Fission-and-fusion routing rule: Under the book's primary formal convention, genuine fission ends the parent stream and begins each qualifying child stream, while genuine fusion ends the prior streams and begins a new one. Earlier phenomenal balances are neither duplicated among descendants nor pooled into the fused stream. See also: Bearer; Routing; Stream identity.

Fitness, feeling, and moral worth: The separation among reproductive propagation, the valenced quality of a bearer's experience, and the ethical standing owed to that being. Biological success, intelligence, human resemblance, productivity, survival, or reproductive outcome cannot be converted into a rank of lives or a verdict that suffering is acceptable. See also: Comparative consciousness evidence; External justice; Viability kernel.

Formal or latent phenomenal ledger: On the primary additive branch, the formal ledger is the latent cumulative signed phenomenal-value process through time. Ledger is mathematical shorthand, not a moral, spiritual, financial, or administrative account, and it does not establish moral debt, inheritance, cross-stream transfer, or direct institutional access. Under exact PTVC, its accumulated signed value determines the equal-and-opposite net remainder required for closure. See also: Latent cumulative object; No moral desert, debt, or payment; Terminal residual.

Future optionality: The meaningful future paths a person can realistically reach through available time, bodily capacity, material conditions, relationships, information, institutions, treatment, support, and action. Optionality can widen or narrow and is not a hidden balance score, guarantee of recovery, or

judgment that constrained people chose badly. See also: Compensability; Craving capture; Lived choice.

G

Gate-valid counterexample: An adverse case that passes the declared bearer, representation, origin, domain, endpoint, observation, bridge, and inference gates for the tested branch. Deductive exclusion requires a deterministic truth-containing object guaranteed under maintained assumptions to exclude the target. Confidence or posterior objects support only their registered coverage- or posterior-qualified adverse conclusions, not literal universal refutation. See also: Falsifiability; Set-universal closure; Strong-form defeat.

Gate-valid stream: A qualifying stream that satisfies the prespecified inclusion, endpoint, measurement, data-quality, and model-validity gates required for the declared test. Only gate-valid cases may carry the evidential status assigned by that test.

Generalized Terminal Neutrality: A broader family of terminal-zero conditions defined by a declared functional, measurable output space, and distinguished zero when scalar addition would lose relevant structure. It is a separate branch and is not PTVC unless a justified bridge connects its zero condition to the additive positive-negative decomposition. See also: Bipolar branch; Primary additive branch; Scalarization.

Gross experience: On a declared additive scalar representation, gross experience is the complete positive-plus-negative variation before cancellation. It is distinct from the terminal residual and is not total consciousness, duration, welfare, or every experiential

dimension. PTVC does not equalize gross experience across streams. See also: Outcome independence; Phenomenal magnitude; Terminal residual.

Gross negative valence: The total magnitude of negatively valenced phenomenal mass within one stream, considered without subtracting positive valence. Equal terminal residuals do not imply equal gross negative valence across streams.

Gross positive valence: The total positively valenced phenomenal mass within one stream, considered without subtracting negative valence. Equal terminal residuals do not imply equal gross positive valence across streams.

H

Hedonic adaptation: The changing emotional impact of a gain, loss, or continuing condition as novelty, attention, expectations, comparison, routine, and context change over time. Adjustment varies across events, people, domains, measures, and times, and does not make a condition fair or prove a fixed set point. See also: Opponent process; Response shift.

Held-out transport: Testing whether a prospectively specified model or result travels to data, settings, populations, channels, time periods, or endpoint regimes not used to build or tune it. In-sample fit or a random split alone does not justify claims beyond the context and observation process actually tested. See also: Measurement invariance; Model recovery; Reference class.

Hidden burden and hidden relief: The paired caution that visible advantage may conceal distress and visible hardship may coexist with joy, competence, love, meaning, or peace. The

principle widens humility about inference but does not invent misery to balance privilege or use possible relief to excuse exclusion and known harm. See also: External justice; Measurement without ownership; Per-stream claim.

Hidden cumulative object: See Latent cumulative object.

Hidden-state model: A model in which the process of interest contains latent states that are inferred from observable signals. Such a model must distinguish state dynamics, observation error, routing, selection, and missingness.

Homeostasis and allostasis: Homeostasis regulates variables within workable ranges, while allostasis anticipates changing demands by adjusting operating levels, mobilizing resources, and redistributing physiological cost. These ordinary regulatory processes may preserve viability or outward function while substantial hidden burden remains, so they are not terminal fairness. See also: Mean reversion; Opponent process; Viability kernel.

Human-worth score prohibition: No estimate of burden, mood, pain, resilience, contribution, regulation, apparent balance, or unity may measure or rank human worth, credibility, trustworthiness, holiness, competence, deservingness, or eligibility. The prohibition applies to labels, visuals, dashboards, proxy scores, and model outputs. See also: AI sovereignty; Coercive measurement; Record-regime separation and no-dossier rule.

I

Identification: The ability of the available data and stated assumptions to distinguish the target quantity or model from alternatives. More data cannot resolve nonidentification when competing explanations remain observationally equivalent under the design.

Independent neutral-origin calibration: A procedure that identifies the zero point of valence without using the completed record's desired terminal result. Centering each finished stream so that its residual equals zero is circular and provides no evidence for PTVC.

Inference: Reasoning from observations and assumptions to conclusions about latent quantities, models, mechanisms, or populations. Every inference inherits the limits of its measurements, design, identification conditions, and auxiliary assumptions.

InferenceStatus: See Scientific verdict axes.

Informative missingness: Missing observations whose occurrence relates to burden, state, access, treatment, refusal, shutdown, or another factor relevant to the inference, so visible records may differ systematically from what disappeared. A missing report is not zero, neutrality, noncooperation, or absence of experience. See also: Attrition mechanism; Missingness; Observation channel and observation model.

Institutional administration limit: Courts, governments, employers, insurers, schools, platforms, clinicians, families, and religious bodies have no authority to possess or administer the

latent cumulative object as a fact about worth, credibility, eligibility, fault, care priority, spiritual standing, or terminal status. Bounded research estimates cannot create that authority. See also: Human-worth score prohibition; Latent cumulative object; Record-regime separation and no-dossier rule.

Interoceptive prediction: The anticipation and interpretation of bodily signals such as heartbeat, breathing, pain, fatigue, hunger, nausea, temperature, and arousal through expectations, context, uncertainty, attention, and action. It is a possible account of affective patterns, not direct access to a cumulative inner record. See also: Active inference; Predictive processing; Prediction error and valence.

Intrastream: Evaluated within one conscious stream at a time. The term fixes the scope of the proposed equality and excludes netting across people, households, families, populations, species, institutions, or ensembles, even when experiences are causally connected or socially interdependent. See also: Nontransferability; Per-stream claim; Routing.

Intrastream closure: Closure evaluated within one qualifying conscious stream. Positive valence in one stream cannot cancel negative valence in another, and no population total substitutes for a stream-level target.

Invariance under positive rescaling: The property that multiplying a valid signed valence representation by a positive constant leaves its sign structure and exact-zero closure status unchanged. This does not license arbitrary nonlinear transformations or shifts of the neutral origin.

J

Jordan decomposition: The representation of a finite signed measure as the difference between mutually singular positive and negative measures. Their total masses provide the positive and negative phenomenal quantities compared by the primary additive PTVC target.

Jordan masses: The total positive and total negative masses produced by the Jordan decomposition of a finite signed phenomenal measure. Exact closure requires equality of these two masses within the same qualifying stream.

L

Last conscious experience: The final qualifying experience before a truth-level stream endpoint, when such a final experience is well-defined. PTVC does not require this last experience itself to feel neutral and does not equate it with the last observed report.

Latent cumulative object: The formal, not directly observed, representation of signed qualifying felt experience accumulated within one stream across its declared domain. Latent or hidden marks an epistemic gap, not a mystical substance. Reports and instruments may constrain it through a model but are not the object itself. See also: Formal or latent phenomenal ledger; Measured-to-latent bridge; Terminal residual.

Latent valence: The unobserved phenomenal valence attributed to experience itself. Reports, behavior, physiology, neural signals, and model states may provide evidence about latent valence but are not identical to it.

Law of Fairness: The series' public name for the proposed idea that each qualifying conscious stream may satisfy exact terminal valence closure, together with the wider inquiry built around it. In scientific use, it remains a candidate hypothesis rather than an established law of nature.

Lawhood: The scientific aspiration to discover a general structural condition on completed qualifying streams, not an established verdict or enacted command. Lawlike standing would have to be earned through identified objects, invariance, prediction beyond construction data, capable rivals, replication, scope, and genuine defeat conditions. See also: Candidate boundary hypothesis; Falsifiability; Claim ceiling.

Ledger: See Formal or latent phenomenal ledger.

Limit-sure attainability: The property that policies can make the probability of closure arbitrarily close to one, even though no single admissible policy need achieve probability-one closure. It is weaker than probability-one closure and weaker than set-universal closure.

Lived choice: The capacity to form purposes, choose among meaningful options, revise, refuse, and act with others from within real bodily, social, historical, and material conditions. Under uncertainty, caring, reversible, reviewable, and repairable actions may be preferable, but PTVC does not script a person's next act. See also: Craving capture; Future optionality; Target-route distinction.

Local return and terminal closure: The separation between movement toward a familiar range during an episode or observed

interval and a proposed exact cumulative condition at the independently defined end of one complete qualifying stream. Rebounds, quiet final observations, and returning population means cannot be combined into per-stream terminal closure. See also: Hedonic adaptation; Mean reversion; Pathwise closure.

Longitudinal ledger: A model-based representation of accumulated signed phenomenal valence across a stream. It is not assumed to be consciously known, morally deserved, stored in a literal database, or transferable to another stream.

Longitudinal research trace: A repeated sampled record created through changing instruments, schedules, contexts, purposes, and permissions. It can reveal trajectories and within-person change but remains a borrowed trace rather than a biography, complete phenomenal history, or ownership of the person. See also: Attrition mechanism; Consent, refusal, and withdrawal; Response shift.

Low-unity state: A state in which ordinary access, responsiveness, integration, or evidence of unified experience is reduced or uncertain, as in sleep, anesthesia, coma, delirium, or severe brain injury. It may be a pause, altered-state branch, or unresolved interval rather than the end of a stream. See also: Behavioral nonresponse; Reversibility and return rule; Truth-level endpoint.

M

Maximal qualifying conscious stream: A qualifying stream extended as far as the adopted routing and identity criteria permit, so that it is not merely an arbitrary fragment of a larger routed

stream. Maximality helps prevent selective windowing from manufacturing closure.

Mean reversion: A local statistical or control tendency for an extreme observation to move toward a typical range because of feedback, stable dynamics, regression, sampling noise, treatment, or changed context. A return-shaped curve cannot identify its own cause and carries no moral meaning or terminal guarantee. See also: Homeostasis and allostasis; Local return and terminal closure; Nonunique explanation.

Mean-zero result: A finding that an expectation or sample average equals or approaches zero. Positive and negative individual residuals can cancel in a mean, so a mean-zero result does not establish per-stream exact closure.

Meaning as optional: The freedom to make, receive, revise, postpone, refuse, or live without an encompassing religious, spiritual, philosophical, or secular interpretation of experience, grief, suffering, or the Law. Nonbelief, doubt, silence, and exit retain full dignity and may never become barriers to care or belonging. See also: Care before proof and interpretation; Comparison without forced equivalence; Spiritual resonance is not scientific support.

Measured-to-latent bridge: The independently justified relation that allows a result on a measured estimand to constrain the declared latent phenomenal target, with scope, uncertainty, and failure conditions carried forward. It is separate from the observation model, endpoint bridge, estimator, and route; without

it, evidence remains on the measured scale. See also: Endpoint bridge; Estimator; Latent cumulative object.

Measurement invariance: The tested stability of a measurement relation across the people, groups, languages, devices, sites, times, or contexts for which a comparison is proposed. Invariance supports only that specified comparison, not phenomenal identity, and failure narrows or blocks the claim rather than discrediting the people measured. See also: Calibration; Held-out transport; Response shift.

Measurement model: A formal account of how latent valence or related states generate reports, behavior, physiology, neural signals, or other observations. It includes scale, origin, error, drift, and possible response-process distortions.

Measurement reactivity: Change in experience, behavior, disclosure, or the observation process caused by measurement itself, including repeated prompts, wearables, interviews, monitoring, and study expectations. Reactivity must be minimized, estimated, or carried into interpretation because it can alter both the target interval and the evidence collected. See also: Informative missingness; Observation channel and observation model; Consent, refusal, and withdrawal.

Measurement without ownership: Experience may be described, compared, and modeled without transferring ownership to a clinician, researcher, institution, record, or algorithm. Measurement creates responsibilities for purpose limitation, consent, privacy, correction, uncertainty, and stopping, not authority over a person's inner truth, worth,

eligibility, or care. See also: Observation channel and observation model; Consent, refusal, and withdrawal; Human-worth score prohibition.

Metrology: The discipline of measurement, including units, calibration, traceability, uncertainty, comparability, and error. PTVC research requires metrological discipline because exact-zero language is meaningless without a defensible origin and scale.

Missingness: The process by which observations are absent, censored, truncated, or unavailable. Missingness near endpoints may be informative, and treating it as random without support can bias terminal conclusions.

Modal reachability ladder: The ordered family of nonequivalent possibility claims, including existence of one closing continuation, positive-probability completion, fixed-law probability-one closure, almost-sure attainability, limit-sure attainability, support containment, and closure of every admissible continuation. Success on a weaker rung does not establish a stronger one. See also: Almost-sure attainability; Every-continuation closure; Support containment.

Model ambiguity: Model-ambiguous means two or more adequate declared models remain practically indistinguishable under the common rules. Open-set inadequate means no declared model is adequate. No adequate estimate instead means a required inferential object or prerequisite is unavailable, undefined, ethically unavailable, or invalid. None is confirmation, equal probability, or permission to choose a preferred story. See also:

No adequate estimate; Rival model and equal-budget comparison; Scientific verdict axes.

Model-ambiguous: See Scientific verdict axes.

Model governance: The accountable rules for who chooses a model's target, data, loss function, constraints, allowed interventions, prohibited uses, review process, correction route, and verdict. Model outputs may not become hidden eligibility scores, rankings of human value, spiritual status measures, or unappealable administrative authority. See also: Decision rule; Human-worth score prohibition; Scientific verdict axes.

Model recovery: A simulation-based check of whether an analysis pipeline can correctly distinguish candidate models and recover their parameters under realistic data conditions. Failure of model recovery limits the meaning of later fits to real data.

Motivation: The internal or external factors associated with action selection. A policy describes a mapping from information or state to actions; it should not be equated with the person's consciously experienced motive.

Multichannel estimate: A versioned, purpose-bounded estimate formed through an explicit model relating several imperfect channels, such as report, behavior, physiology, neural data, and context, to a declared construct and estimand. It is not a direct reading of inner life, an automatic average, or a personal dossier. See also: Error budget; Observation channel and observation model; Provenance.

N

NAE: See No adequate estimate.

Near-zero band: A finite interval around zero used to describe limited measurement resolution, uncertainty, or practical closeness. Membership in a near-zero band does not establish exact closure.

Negative control: A condition, timing, procedure, or outcome that should not show the proposed signature if an explanation is specific. Matched non-horizon tasks, sham procedures, impossible timing, and alternate outcomes can expose generalized arousal, demand, leakage, selection, or analytic flexibility. See also: Candidate signature; Nonunique explanation; Safe perturbation.

Net valence: Positive phenomenal mass minus negative phenomenal mass within a specified stream and domain. A zero net value does not imply small gross experience, equal biographies, or a neutral final moment.

Neutral origin: The identified zero point separating positive from negative phenomenal valence in the adopted representation. Its location must be justified independently rather than chosen to make terminal records balance.

No adequate estimate: The protocol returns No adequate estimate when a predeclared rule applies because a required prerequisite or information threshold is unmet, a necessary object is unavailable or undefined, computation is invalid, or ethical unavailability blocks the required evidence. It is not zero, neutrality, compatibility, support, or a person-level judgment. A

valid computed incompatibility is classified separately, and ethical unavailability follows its declared protocol route. See also: Decision rule; Failed upstream gate; Model ambiguity.

No harm for proof: The prohibition on inflicting, prolonging, intensifying, staging, ignoring, or leaving suffering unrelieved in order to test, demonstrate, dramatize, or promote PTVC. It rules out unsafe self-testing, delayed care, coerced disclosure, harmful research involving vulnerable participants, and the creation or prolongation of plausible synthetic suffering; principled abstention remains available. See also: Care before proof and interpretation; Ethical floor; Stop authority.

No moral desert, debt, or payment: Required for closure does not mean morally deserved, earned, or justified. Under exact PTVC, a net positive accumulated balance requires a net negative remainder, and a net negative balance requires a net positive remainder. Further felt contributions change that signed requirement, without prescribing particular events, fixing gross future totals, or demonstrating practical control. See also: Constraint without a judge; Desert-based fairness; Outcome independence.

No personal endpoint testing: No person should try to test the Law by approaching danger, loss of consciousness, treatment refusal, self-harm, deprivation, violence, or death. Such acts create risk but cannot identify the bearer, complete domain, endpoint, missingness process, origin, or measured-to-latent bridge required for valid evidence. See also: Care before proof and interpretation; No harm for proof; Truth-level endpoint.

No-score rule: See Human-worth score prohibition.

Noise floor: The smallest distinction an observation system can reliably resolve under declared conditions. Below it, rounding, censoring, smoothing, preprocessing, or noise dominates the signal. More participants, decimal places, or smoothing cannot recover information that the observation system does not contain. See also: Calibration; Error budget; Finite-band compatibility and measured equivalence.

Noncompensable risk: The modeled risk that, under specified conditions and uncertainty, an action or state closes essential paths faster than remaining channels can reopen them. The label applies to a route or continuation, never to a person's worth, hope, dignity, care priority, prognosis, clinical fate, or legal status. See also: Compensability; Future optionality; Viability kernel.

Nonidentification: A condition in which different target values or models are equally compatible with the available observations and assumptions. Reporting a single precise estimate under nonidentification creates false certainty.

Nontransferability: The rule that valence belonging to one conscious stream cannot be credited to, debited from, or used to balance another stream. PTVC is per-stream and contains no interpersonal or cosmic compensation ledger.

Nonunique explanation: The condition in which several different mechanisms can generate the same visible trajectory, endpoint concentration, rebound, report, compression, or return-to-zero pattern. Fit or resemblance alone cannot identify the generator, and serious observational equivalence requires narrow

language, further tests, or model ambiguity. See also: Candidate signature; Mean reversion; Model ambiguity.

Null model: A specified alternative under which the distinctive PTVC target or route is absent. A useful null model makes concrete predictions and is tested alongside serious rivals rather than serving as a vague absence of effect.

Null outcome: A result that does not support the tested effect or distinction under the declared analysis. A null outcome may be informative, inconclusive, or adverse depending on power, identification, and the prespecified decision rule.

O

Objective functions versus experience: The separation between a modeler's chosen rule for ranking selected outcomes and the lived experience of a conscious subject, which the rule may omit, distort, or replace with a proxy. An optimized score is not a completed stream and may hide suffering or displaced costs. See also: Model governance; Reinforcement-learning reward signal; Viability kernel.

Observation: A recorded report, behavior, physiological measure, neural signal, event, or other data item. Observations are evidence about the latent process; they are not the phenomenal process itself.

Observation channel and observation model: An observation channel is recorded evidence such as report, behavior, physiology, neural activity, or context. The observation model states how a latent construct and nuisance conditions can generate or alter that output, including sampling, timing, error, and channel

failure. Neither the channel nor its model is the experience. See also: Informative missingness; Measured-to-latent bridge.

Observation shutdown: See End of observation.

Observation window: The finite span during which a study records a person, often only days, months, or years of a longer stream. It can support bounded claims about that interval but is not a complete conscious history unless an independently justified bridge covers the missing prefix and remainder. See also: Complete strict-pre-endpoint felt history; End of observation; Protocol endpoint.

Observational process: The mechanism by which latent experience and system states become recorded data, including measurement error, reporting, censoring, sampling, and missingness. It must be modeled separately from the target process.

Open-set inadequate: See Scientific verdict axes.

Opponent process: A temporal counterprocess in which an initial state recruits a later afterstate whose strength and timing may change with repetition, helping explain relief, recovery, tolerance, withdrawal, fatigue, and rebound. An afterstate is not moral payment, and episode-level counterpressure does not establish lifetime closure. See also: Hedonic adaptation; Local return and terminal closure.

Outcome independence: Care, rights, dignity, privacy, repair, ordinary causation, and independently supported knowledge remain valid whether PTVC is supported, narrowed, unresolved, unmeasurable, or defeated. No scientific result can create or

remove a person's entitlement to relief, protection, consent, nonparticipation, or remedy. See also: Care before proof and interpretation; Ethical floor; Strong-form defeat.

Ownership: The prospectively specified rule assigning each qualifying phenomenal event or contribution to its bearer without omission, double counting, or cross-stream transfer. Ownership answers whose experience a contribution is, while routing governs continuity through time and transitions; data ownership is a different legal and institutional question. See also: Bearer; Nontransferability; Routing.

P

Palliative sedation: A clinically governed process used proportionately to relieve otherwise refractory suffering, with varying drugs, depths, durations, routes, and circumstances. It is treatment rather than a standardized consciousness manipulation or clean terminal signal, and research must never delay relief, alter indicated titration, or prefer observability over comfort. See also: Comfort-first observation; End of observation; Protocol endpoint.

Partial identification: A condition in which the evidence restricts a target to a set or interval but does not determine a unique value. A set that includes zero and nonzero values does not establish exact closure.

Partial observation: A decision or inference setting in which the relevant latent state is not fully observed. Closure claims in such settings must respect common-action and selection requirements rather than assuming state-specific knowledge.

Participation without surveillance: A research architecture that keeps participation voluntary through purpose separation, data minimization, practical privacy controls, ordinary nonparticipation, correction, and eventual deletion. A person may contribute selected observations without granting permanent access, hidden inference, public ranking, or ownership of the life described. See also: Consent, refusal, and withdrawal; Data minimization and deletion; Record-regime separation and no-dossier rule.

Past-fixity: Later experience may add to the future portion of a stream but cannot retroactively change what earlier experience was. This temporal discipline permits formal future complementarity while forbidding moral or phenomenal repricing of the past, including claims that later joy makes earlier abuse acceptable. See also: Outcome independence; Formal or latent phenomenal ledger; Reserve coupling.

Pathwise closure: A proposed condition on the complete trajectory of one qualifying stream at an independently defined endpoint, rather than on a fragment, population average, task score, controller's success, or endpoint-only appearance. Mathematical analogy may clarify the object but cannot show that conscious experience obeys it. See also: Complete strict-pre-endpoint felt history; Local return and terminal closure; Set-universal closure.

Per-stream claim: A claim evaluated for each separately routed maximal qualifying conscious stream as its own unit rather than for a population, household, country, species, or group average. Population evidence may inform estimation and heterogeneity but

cannot perform exact intrastream equality, transfer experience between streams, or establish closure for each member. See also: Intrastream; Maximal qualifying conscious stream; Population mean and expected neutrality.

Per-stream terminal valence closure (PTVC): On the primary additive branch, the candidate boundary hypothesis that every separately routed maximal qualifying conscious stream satisfying the declared gates has finite and exactly equal total positive and negative phenomenal magnitudes over its complete strict-pre-endpoint domain, leaving a zero terminal signed residual.

Phenomenal magnitude: The represented amount of positive or negative felt experience assigned to a stream under a declared valence representation. Positive and negative gross magnitudes remain distinct from their signed net difference, and equal residuals do not imply equal duration, peaks, biographies, agency, opportunity, or welfare. Rights and dignity remain independently protected. See also: Felt valence; Gross experience; Positive and negative phenomenal magnitudes.

Phenomenal measure: A measure assigned to conscious experience itself for purposes of representing valence across a stream. It is conceptually distinct from clock time, event count, memory strength, neural activity, or report frequency.

Phenomenal time: A measure or ordering of experienced duration within a conscious stream. It need not coincide uniformly with external clock time and should not be inferred solely from the number of observations.

Phenomenal valence: The felt positive, negative, or neutral character of conscious experience considered as phenomenal rather than merely behavioral, computational, motivational, or linguistic.

Policy: A rule mapping available information or state to actions, possibly probabilistically. A policy is not identical to a conscious motive, a moral intention, or the outcome it happens to produce.

Policy mixture: A probabilistic combination of actions or component policies. Average cancellation across components does not establish exact closure on each realized path.

Policy-mixture exactness: The requirement that exact closure under a randomized policy be secured on the relevant realized branches, not merely in expectation through cancellation among closing and nonclosing actions. Under the applicable support conditions, every action retained with positive probability must be closure-compatible.

Policy-support condition: A condition requiring the support of a selected policy to contain only actions or continuations compatible with the declared exact closure claim. Assigning very small but positive probability to a nonclosing option does not satisfy exact support restriction.

Population mean and expected neutrality: Expected neutrality means that the terminal residual has expectation zero under a named probability law and information set, when that expectation exists. A sample or cohort mean estimates a population quantity under further assumptions. Either can arise through cancellation

while every individual stream remains nonzero. See also: Per-stream claim; Probability-one closure; Set-universal closure.

Portable safeguard: A local, operative protection placed on the exact surface or workflow where a high-risk fragment can be encountered or acted upon. A distant disclaimer is insufficient when the nearby claim could imply personal prediction, cross-person balancing, human scoring, machine ownership, or delayed care. See also: Detachable-excerpt ceiling; Public-artifact limit; Stop authority.

Positive and negative phenomenal magnitudes: On the primary additive branch, the nonnegative complete-stream totals of the positive and negative parts of the signed phenomenal measure. PTVC proposes that these two quantities are finite and exactly equal within each qualifying stream, without pairing individual events or treating them as moral credits and debts. See also: Finiteness; Gross experience; Terminal residual.

Positive-probability completion: Under a prospectively frozen continuation regime and its declared induced law, the completion event has probability strictly greater than zero, optionally conditional on a stated prefix and admissible action or policy. This weak reachability claim does not imply likelihood, probability one, an attaining policy, support containment, or every-continuation closure. See also: Modal reachability ladder; Probability-one closure; Reachability.

Positive rescaling: Multiplication of a valence representation by a positive constant. Exact zero and sign are preserved under

positive rescaling, whereas shifting the origin or applying arbitrary nonlinear transformations can change the target.

Precaution without personhood: Proportionate, revisable protection under uncertainty, including monitoring, reducing avoidable harm, preserving reversibility, pausing, or redesigning when plausible welfare risk is serious. Protection can strengthen while consciousness, ontology, law, and personhood remain unresolved, without treating precaution as proof. See also: Candidate sentience escalation; Comparative consciousness evidence; Unknown category.

Prediction error and valence: The distinction between mismatch under a predictive model and the positive, negative, or neutral quality of felt experience. Surprise and valence can interact without being the same quantity: expected harm can remain painful, surprise can be welcome, and mean-zero prediction error does not imply neutral experience. See also: Felt valence; Interoceptive prediction; Predictive processing.

Predictive fit versus explanation: Predictive fit is a model's ability to forecast held-out observations. Explanation requires additional evidence that its variables and dependencies correspond to the process that generated them. A black-box forecast can be useful without identifying horizon, reserve, causation, consciousness, or terminal structure. See also: Candidate signature; Held-out transport; Model recovery.

Predictive processing: A family of accounts in which perception, bodily interpretation, and action depend on generative models, expected signals, prediction errors, uncertainty, and estimated

reliability. It is a debated family rather than one finished theory, and shared vocabulary with PTVC supplies no evidence for terminal closure. See also: Active inference; Interoceptive prediction; Prediction error and valence.

Preregistration: A time-stamped specification of hypotheses, gates, measures, exclusions, models, analyses, and decision rules completed before the relevant outcomes are examined. Deviations should be disclosed and labeled exploratory.

Preterminal compression: A candidate observable signature in which a specified measure of dispersion narrows before a declared endpoint in the prospectively predicted relation. The protocol distinguishes dispersion across model realizations from temporal variation within one observed stream and justifies any bridge between them. Sedation, frailty, sensor floors, selective missingness, pooling, mean reversion, treatment, and endpoint conditioning can mimic it, so measured narrowing is not exact closure or a verdict on a person's life. See also: Candidate signature; Endpoint conditioning; Noise floor.

Primary additive branch: The branch that represents qualifying phenomenal valence as one finite signed scalar measure with a justified neutral origin and additive consistency across disjoint parts of a stream. Its positive and negative parts yield complete-stream magnitudes, and their difference yields the terminal residual. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Probability-one closure: Closure with probability one under a specified policy and probability model. It permits measure-zero

exceptions and is distinct from set-universal closure, which requires closure for every admissible realization.

Procedural fairness: Fairness in rules and decision paths, including knowable standards, voice, understandable reasons, consistency, limited bias, respectful treatment, correction, and appeal. It is an independent good governing authority and treatment, not evidence of terminal closure or a substitute for a valid construct. See also: Desert-based fairness; External justice; Consent, refusal, and withdrawal.

Process evidence: Evidence about a proposed route, mechanism, state transition, or selection process. Process evidence can strengthen or weaken a route without settling the terminal target.

Protocol endpoint: A prospectively specified operational event, adjudicated interval, stopping region, or governed proxy used by a study to locate or approximate a relevant boundary. It may be usable, uncertain, revisable, or ethically unavailable and is distinct from both the truth-level endpoint and end of observation. See also: End of observation; Endpoint bridge; Truth-level endpoint.

Protocol-fixed and prospectively specified: A choice is protocol-fixed or prospectively specified when it is recorded in the study protocol before the relevant outcome is inspected. This timing status does not guarantee that the choice is valid or the study succeeds, and it is not public preregistration without a prior external timestamped record. See also: Endpoint independence; Preregistration; Rival model and equal-budget comparison.

Provenance: The documented lineage of a trace or derived output, including who collected it, for what purpose and authority, under which prompt, device, language, setting, burden, transform, model version, missingness state, uncertainty, and correction history. Provenance limits interpretation rather than merely decorating a record. See also: Data as trace; Public-artifact limit; Record-regime separation and no-dossier rule.

Psychometrics: The theory and practice of measuring latent psychological or experiential constructs, including reliability, validity, scaling, response processes, and measurement invariance. PTVC research needs psychometrics without treating a questionnaire score as valence itself.

PTVC: See Per-stream terminal valence closure (PTVC).

Public access-permission distinction: Public availability for accountability, education, criticism, scholarship, or communication does not itself authorize profiling, consequential judgment, redistribution, linkage, or general model training. Visibility is not surrender, de-identification is not guaranteed anonymity, and access for one purpose is not standing authority for another. See also: Consent, refusal, and withdrawal; Data as trace; Record-regime separation and no-dossier rule.

Public-artifact limit: Every detachable project surface, including a summary, caption, guide, model output, slide, translation, or post, must preserve its source, status, audience, prohibited readings, reviewer, and correction route. Visual polish and brevity do not authorize proof, personal destiny, prescribed payment, surveillance, ranking, or guaranteed progress. See also:

Detachable-excerpt ceiling; Portable safeguard; Correction and withdrawal.

Public safety floor: The Law of Fairness, technically Per-Stream Terminal Valence Closure, remains a falsifiable candidate boundary hypothesis, not a validated natural law, moral doctrine, personal forecast, or institutional rule. Care, dignity, rights, welfare, relief, refusal, and protections under uncertainty remain binding under every outcome. Nothing here licenses human-worth scoring, created suffering, compulsory belief, personal endpoint testing, delayed care, hidden-object administration, or labeling any person or life balanced, adverse, failed, ambiguous, or inadequate.

Q

Qualifying conscious stream: A conscious stream that meets the declared scope, identity, routing, inclusion, and validity conditions for the target being tested. The criteria must be stated independently of whether the stream appears to close.

Queue model: A constructive candidate route that represents unresolved signed valence and asks whether admissible future transitions can remove the remaining residual. It is not a literal conscious queue, a moral debt system, or evidence that PTVc is true.

Queue System: Book 1's accessible name for the Queue model, a possible action-mediated route to closure. It is an explanatory construction, not an established mechanism and not a claim that every urge consciously serves closure.

R

Random-endpoint obstruction: The result that exact closure is generally not preserved when an otherwise independent endpoint occurs at a genuinely random time while the cumulative residual can be nonzero. A successful route must therefore address endpoint structure rather than assume random stopping is harmless.

Reachability: Whether at least one admissible continuation from a state can reach the declared target. Reachability does not guarantee selection, common feasibility under partial observation, probability-one success, or universal success.

Record-regime separation and no-dossier rule: Research, clinical, employment, educational, civic, legal, family, spiritual, platform, and public records must retain separate purposes, identifiers, access roles, retention rules, consequences, and correction paths rather than being joined into a master profile. Inferred reconstruction can breach this boundary even without copying a forbidden field. See also: Data minimization and deletion; Human-worth score prohibition; Public access-permission distinction.

Reference class: The prospectively declared truth-domain set of qualifying streams, endpoint regimes, representations, populations, settings, or measurement architectures over which a claim and its quantifier range. It is not automatically the enrolled sample or analyzed cases, and outcome-driven exclusions create a new claim. See also: Held-out transport; Maximal qualifying conscious stream; Set-universal closure.

Reference-class narrowed: See Scientific verdict axes.

Registered Report: A publication format in which a study's question, methods, and analysis plan receive peer review before results are known, with publication not conditioned on obtaining a preferred outcome.

Reinforcement-learning reward signal: A formally specified signal used to update action values or policies from outcomes. It may represent relief, safety, access, or another modeled result without being identical to pleasure, felt value, welfare, or human worth. A learned policy is not necessarily wise, healthy, fair, free, or morally deserved. See also: Active inference; Objective functions versus experience; Predictive processing.

Remaining capacity: The set or amount of admissible future change still available before an endpoint. A closure route must show that remaining capacity is sufficient and selectable under the actual information and support constraints.

Repair is not payment: Responding to actual harm through safety, truth, apology, restitution, accountability, prevention, and remedy differs from prescribing suffering or promising a chosen reward. Exact PTVC fixes a signed remainder without justifying harm. Later relief, growth, reconciliation, or a wrongdoer's suffering does not retroactively justify an injury or offset harm across persons. See also: External justice; No moral desert, debt, or payment; Outcome independence.

Replication: A new test designed to determine whether a result, measurement relation, or model distinction recurs under

prespecified conditions. Replication should include adverse and null outcomes rather than only successful demonstrations.

Representation: A mathematical or conceptual encoding of phenomenal valence, time, state, or route. A useful representation must preserve the distinctions required by the target and should not be mistaken for the experience itself.

Reserve: A route-level variable representing remaining capacity, flexibility, or corrective potential before an endpoint. A reserve is not a store of deserved pleasure or pain and is not itself the terminal target.

Reserve coupling: A candidate relation between accumulated residual and remaining capacity or future transition structure. Support for reserve coupling would be evidence about a route, not direct proof of PTVC.

Response shift: A change in a person's standards, priorities, values, or understanding of a question after illness, disability, adaptation, treatment, trauma, grief, or aging, altering what the same response scale means over time. A changed answer can reflect changed experience, changed measurement, or both. See also: Calibration; Longitudinal research trace; Measurement invariance.

Reversibility and return rule: A prospectively specified rule under which reliable return of unified access can defeat an earlier classification as a final endpoint, while nonreturn alone cannot prove absence of experience or identify the exact ending. Protocols must state return evidence, revision timing, and its

effect on classification. See also: Endpoint bridge; Low-unity state; Protocol endpoint.

Reversible pause: See Reversibility and return rule.

Rival-favored: See Scientific verdict axes.

Rival model: A concrete alternative explanation capable of generating overlapping observations. Strong tests compare PTVC targets and routes against serious rivals involving measurement, reporting, adaptation, selection, endpoint structure, and ordinary statistical effects.

Rival model and equal-budget comparison: A rival is a serious alternative able to generate the relevant pattern without the target or proposed route. Equal-budget comparison gives candidates and rivals common raw variables, observation and missingness pipelines, data splits, scoring rules, and comparable prospectively bounded flexibility, while allowing that no declared model is adequate. See also: Model ambiguity; Model recovery; Scientific verdict axes.

Route: A proposed process or construction through which a target might be reached. The route, its state variables, and its evidence must remain distinct from the terminal target.

Route-failed: See Scientific verdict axes.

Route-favored: See Scientific verdict axes.

Route retirement: A prespecified decision to stop treating a proposed mechanism or route as viable after decisive failure, poor recovery, nonidentification, or superior rival performance. Retiring a route need not settle the target hypothesis.

Route state: A latent or observed variable used to describe progress within a candidate mechanism. A route state is not automatically phenomenal valence, cumulative residual, motivation, or verdict.

RouteStatus: See Scientific verdict axes.

Routing: The prospectively specified treatment of continuation, pause, restoration, fission, fusion, copying, replacement, and unresolved continuity, including where prior and later phenomenal contributions are assigned. Routing governs transitions through time and is distinct from ownership, compensation, and causal influence. See also: Fission-and-fusion routing rule; Ownership; Stream identity.

S

Safe perturbation: A reversible, low-burden intervention that changes a declared local feature, such as feedback, delay, expectation, opportunity, or a mild stimulus, to test a named causal contribution. Its meaning depends on target engagement, spillovers, controls, stopping, and the stated causal estimand. See also: Candidate signature; Negative control; No harm for proof.

Scalarization: A prospectively justified mapping from a richer or multidimensional experiential state to one signed scalar quantity for a named purpose and domain. It is a substantive bridge rather than harmless formatting, and weights selected because they produce closure manufacture the result. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Scientific verdict axes: Four non-overwriting axes record TargetStatus, RouteStatus, ComparatorStatus, and

InferenceStatus. Target-adverse opposes a branch. Route-failed marks a failed gate; Route-favored requires identification, diagnostics, absolute fit, discrimination, and superiority to adequate rivals. Rival-favored prefers an adequate rival under common rules. Model-ambiguous retains tied adequate models; Open-set inadequate finds none; Reference-class narrowed reduces scope. NAE marks unavailable or invalid inference. See also: Decision rule; Model ambiguity; Target-route distinction.

Scope firewall: A statement of conclusions that do not follow from PTVC. The hypothesis does not imply free will, monotonic movement toward zero, frequent zero crossings, conscious ledger awareness, that every urge serves closure, that pain is morally owed payment, that suffering is deserved, or that anyone can safely control a chosen reward. This does not deny the equal-and-opposite net remainder required for exact closure.

Selection condition: A requirement governing which available action or continuation is actually chosen. The existence of a closing option is insufficient when the selection process can retain nonclosing options with positive support.

Separately routed: Governed by a trajectory or transition structure that is not pooled with another stream for purposes of the terminal target. Separate routing supports the rule that one stream cannot balance another.

Set-universal closure: Closure for every admissible realization, path, or member of the declared set. It is stronger than probability-one closure because measure-zero exceptions are not permitted.

Severe test: A test that had a high probability of exposing a claim's error if the claim were false in the specified way. A result is not strong evidence merely because it is statistically significant or compatible with the preferred model.

Shortening horizon: A reduction in remaining time, opportunity, revision, access, or other route-relevant room for future action. A shorter horizon can constrain reachability and option preservation without revealing a truth-level endpoint, terminal residual, destiny, or duty to make a particular choice. See also: Future optionality; Reachability; Reserve.

Signed representation: A representation that assigns positive and negative directions relative to a justified neutral origin and supplies the declared scale and aggregation rules needed for the target branch. A sign convention alone does not establish cardinality, comparability, additivity, or phenomenal meaning. See also: Felt valence; Neutral origin; Primary additive branch.

Signed residual: The cumulative positive phenomenal mass minus the cumulative negative phenomenal mass over a specified domain. The terminal signed residual is evaluated at the truth-level stream endpoint.

Skepticism as dignity: The protection of organized doubt, exact criticism, nonbelief, and refusal as honorable ways of engaging with a consequential public hypothesis. Comprehension is not assent, and care, standing, participation, intelligence, belonging, or moral worth may never depend on enthusiasm for the Law. See also: Meaning as optional; Falsifiability; Outcome independence.

Spiritual resonance is not scientific support: A relation of meaning occurs when a person or tradition recognizes a neighboring pattern. A relation of evidence requires a defined hypothesis to survive valid measurement, serious rivals, uncertainty, and possible defeat. Tradition, ritual, beauty, testimony, dream, prayer, or transformation may matter without validating PTVC. See also: Comparison without forced equivalence; Meaning as optional; Falsifiability.

State-coupled timing: An endpoint branch in which endpoint timing or hazard depends on prospectively specified current biological, consciousness, ledger, or route-state variables while the endpoint ontology and adjudication remain fixed independently of desired closure. Dependence is allowed, but the state relation cannot be invented after outcome inspection. See also: Endpoint atom; First-hitting and absorption routes; Random-endpoint obstruction.

Stop authority: Preassigned power held by an accountable person or independent reviewer to pause a study, system, publication, or artifact when a declared safety or rights boundary is crossed. Effective authority can interrupt activity, contain harm, trigger care, and require correction or withdrawal rather than merely advise. See also: Correction and withdrawal; No harm for proof; Portable safeguard.

Stream: See Maximal qualifying conscious stream.

Stream identity: The criteria that determine which experiences belong to the same conscious stream. Biological continuity,

memory, body, legal identity, and data linkage may inform the analysis but are not automatically decisive.

Strong-form defeat: Deductive branch-matched refutation requires an independently verified qualifying maximal stream whose deterministic truth-containing object, guaranteed under maintained bearer, representation, origin, domain, endpoint, observation, bridge, and inference assumptions, excludes the registered exact target. Confidence or posterior objects support only their registered qualified adverse verdicts. Once this counterexample exists, hidden channels or moved boundaries cannot erase it. See also: Falsifiability; Gate-valid counterexample; Set-universal closure.

Support condition: A restriction on the values, actions, paths, or outcomes that retain positive possibility under a model or policy. Exact claims often require excluding nonclosing support, not merely making it unlikely.

Support containment: Under a declared topology, the support of the relevant probability law lies inside the completion event or target set. This topological modality is not every-continuation closure, positive probability at a point, or proof that an unsampled possibility is absent from support. See also: Every-continuation closure; Modal reachability ladder; Probability-one closure.

T

Target: The truth-level property the hypothesis asserts. For the primary PTVC branch, the target is exact zero terminal signed residual within each qualifying stream; it is not a route, score, estimator, or fitted mechanism.

Target-adverse: See Scientific verdict axes.

Target modality and geometry: The exact mathematical form and quantifier of a declared target, including whether it is an exact singleton, finite band, distributional atom, probability-one event, support condition, every-continuation property, or set-universal claim. Evidence and defeat must be matched to that form rather than borrowed across nearby claims. See also: Exact closure; Modal reachability ladder; Terminal residual compatibility object.

Target-route distinction: The target hypothesis states the terminal relation to be evaluated, while a candidate route proposes one optional process that might realize or mimic consequences of it. Route support does not establish the target without a proved entailment, and route failure does not automatically refute every target branch. See also: Candidate route; Scientific verdict axes; Claim ceiling.

Target-route separation: The rule that evidence for a proposed mechanism or route must not be reported as if it directly established the terminal target, and failure of one route must not automatically be reported as defeat of every possible route.

TargetStatus: See Scientific verdict axes.

Terminal neutrality: Book 1's accessible term for a zero terminal signed residual over the whole qualifying stream. It does not mean that the final conscious moment is emotionally neutral.

Terminal residual: The signed total of phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain,

evaluated at its truth-level endpoint. The primary PTVC target states that this residual is exactly zero.

Terminal residual compatibility object: A branch-matched nonempty object representing residual values allowed by declared representation, endpoint, observation, missingness, bridge, and uncertainty assumptions. It retains its inferential type: deterministic identified set, confidence region, credible region, or sensitivity envelope. Only a deterministic guaranteed truth-containing object supports deductive exclusion; confidence and Bayesian objects license only registered coverage- or posterior-qualified conclusions. A point estimate is not such an object. See also: Finite-band compatibility and measured equivalence; Partial identification; Target modality and geometry.

Terminal signed residual: See Terminal residual.

Terminal window: A finite observed period near a presumed endpoint. It can provide useful process data but is not identical to the complete strict-pre-endpoint phenomenal domain or the truth-level endpoint.

Testimony-like output: Language or behavior from an artificial system that has the form of a first-person report but is generated under a particular model, prompt, memory state, tool set, training regime, and sampling process. It is evidence about configured output, not self-authenticating testimony that a subject feels what is said. See also: AI sovereignty; Candidate sentience escalation; Theory-derived indicator.

The Law: See Law of Fairness.

The Law of Fairness: On the primary additive branch, the Law proposes that within each separately routed maximal qualifying conscious stream, positive and negative felt magnitudes across its complete strict-pre-endpoint history are finite and exactly equal, leaving a zero latent terminal residual immediately before its independently defined truth-level endpoint. No stream counts toward another, and lives may still differ radically in duration, intensity, order, agency, opportunity, memory, welfare, and circumstance. Chapters 4–8 define the terms used here, with technical development in The Science of Fairness.

Theory-derived indicator: An observable property selected because a stated theory connects it to recurrent processing, global availability, higher-order representation, agency, embodiment, metacognition, or another proposed feature of consciousness. Indicators frame tests but do not automatically combine into a direct meter of experience. See also: Candidate sentience escalation; Comparative consciousness evidence; Testimony-like output.

Truth-containing object: An interval, set, distributional statement, or other inferential object constructed with a declared guarantee or calibration concerning the true target. If such an object excludes zero in a gate-valid case, it can be adverse to an exact-zero claim.

Truth-containing residual set: See Terminal residual compatibility object.

Truth-level boundary: See Truth-level endpoint.

Truth-level endpoint: The actual boundary of the qualifying stream in the target model. It is conceptually distinct from any observed marker, last report, terminal window, loss of contact, or biological event used to estimate it.

U

Unity Index: A proposed fallible research estimator that organizes separately reported evidence about access, report, memory, agency, embodiment, continuity, and coherence under a declared construct. It exposes assumptions and uncertainty but is not the stream itself, a consciousness meter, personhood test, dignity measure, or validation of PTVC. See also: Blinded bearer adjudication; Human-worth score prohibition; Unknown category.

Universal branch: See Set-universal closure.

Universal quantifier: A claim applying to every member of a declared class. Because the primary PTVC branch is per-stream and universal over qualifying streams, a valid adverse case cannot be repaired by averaging across the class.

Unknown category: A reason-coded classification outcome used when available evidence cannot justify one bearer, consciousness, or boundary classification over serious alternatives under declared rules. Unknown is not a negative finding, equal probability, zero, or support for the Law, and it differs from No adequate estimate. See also: Bearer; No adequate estimate; Precaution without personhood.

V

Valence: The felt positive, negative, or neutral character of conscious experience. The term does not by itself denote arousal, reward, preference, motivation, behavior, or moral worth.

Valence density: A representation of signed phenomenal valence per unit of an adopted base measure, such as phenomenal time. Its interpretation depends on the validity of both the valence scale and the base measure.

Valence ledger: Book 1's metaphor for the cumulative signed phenomenal measure within one stream. It is not a conscious account, a moral score, a cosmic database, or a debt transferable between lives.

Verdict: The declared evidential conclusion produced by a prespecified decision architecture, such as support, adverse result, narrowing, abstention, or route retirement. A verdict must state its target and scope.

Viability: The possibility of remaining within declared constraints over the specified horizon under an admissible continuation or policy, with any probabilistic guarantee stated explicitly. Maintaining access to closure is an additional completion-specific requirement when it is included among those constraints.

Viability kernel: The set of modeled states from which at least one admissible continuation can continue to satisfy a route's declared constraints. It marks where a future remains formally workable but does not mean that a continuation is likely, humane, accessible, desirable, or supported by resources a person actually

has. See also: Compensability; Noncompensable risk; Reachability.

Z

Zero-containing interval: An uncertainty or identification interval that includes zero. Because it may also include nonzero values, it does not establish exact closure.

Zero crossing: A time at which a cumulative or instantaneous signed quantity passes through zero. PTVC does not require monotonic movement, frequent zero crossings, or any particular number of crossings before the endpoint.

References

The entries below provide bibliographic destinations for the Author/Year citations used throughout the manuscript. Author and institutional creator names appear in bold. Distinct works by the same author in the same year retain their manuscript suffixes. DOI identifiers appear as plain text so the entries remain readable in print. Readers following a claim beyond this book should consult the cited edition, the proposition it supports, and the source's stated scope.

Achiam, Joshua, David Held, Aviv Tamar, and Pieter Abbeel.

2017. "Constrained policy optimization." *Proceedings of the 34th International Conference on Machine Learning, Proceedings of Machine Learning Research* 70: 22–31.

Adams, Marilyn McCord. 1999. *Horrendous Evils and the Goodness of God*. Cornell University Press.

Akaike, Hirotugu. 1973. "Information theory and an extension of the maximum likelihood principle." In B. N. Petrov and F. Csáki, eds., *Second International Symposium on Information Theory*, 267–281. Akadémiai Kiadó.

Albrecht, G. L. and P. J. Devlieger. 1999. "The disability paradox: High quality of life against all odds." *Social Science & Medicine* 48 (8): 977–988. DOI 10.1016/S0277-9536(98)00411-0.

Alter, Robert. 2010. *The Wisdom Books: Job, Proverbs, and Ecclesiastes: A Translation with Commentary*. W. W. Norton.

American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. 2014. *Standards for educational*

and psychological testing. American Educational Research Association.

American Psychiatric Association. 2022. *Diagnostic and Statistical Manual of Mental Disorders (5th ed., text rev.; DSM-5-TR).* American Psychiatric Association Publishing.

Amrhein, Valentin, Sander Greenland, and Blake McShane. 2019. “Scientists rise up against statistical significance.” *Nature* 567: 305–307. DOI 10.1038/d41586-019-00857-9.

Andersen, P. K., R. B. Geskus, T. de Witte, and H. Putter. 2012. “Competing risks in epidemiology: Possibilities and pitfalls.” *International Journal of Epidemiology* 41 (3): 861–870. DOI 10.1093/ije/dyr213.

Andrews, K., J. Birch, J. Sebo, and T. Sims. 2024. *Background to the New York Declaration on Animal Consciousness.* New York Declaration on Animal Consciousness.

Aoki, S. K., G. Lillacci, A. Gupta, A. Baumschlager, D. Schweingruber, and M. Khammash. 2019. “A universal biomolecular integral feedback controller for robust perfect adaptation.” *Nature* 570: 533–537. DOI 10.1038/s41586-019-1321-1.

Appelbaum, Paul S., Loren H. Roth, and Charles W. Lidz. 1982. “The therapeutic misconception: Informed consent in psychiatric research.” *International Journal of Law and Psychiatry* 5 (3–4): 319–329. DOI 10.1016/0160-2527(82)90026-7.

Armstrong, D. M. 1983. *What Is a Law of Nature?* Cambridge University Press.

- Asad, Talal.** 1993. *Genealogies of Religion: Discipline and Reasons of Power in Christianity and Islam*. Johns Hopkins University Press.
- Asadi, A., K. Chatterjee, R. Saona, and A. Shafiee.** 2025. “Limit-sure Reachability for Small Memory Policies in POMDPs is NP-complete.” *Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence, Proceedings of Machine Learning Research* 286: 238–256. PMLR.
- Aserinsky, E. and N. Kleitman.** 1953. “Regularly occurring periods of eye motility, and concomitant phenomena, during sleep.” *Science* 118 (3062): 273–274. DOI 10.1126/science.118.3062.273.
- Aubin, J.-P.** 1991. *Viability theory*. Birkhäuser.
- Aubin, J.-P., A. M. Bayen, and P. Saint-Pierre.** 2011. *Viability theory: New directions*. Springer. DOI 10.1007/978-3-642-16684-6.
- Baba, Ami, Maureen Smith, Beth K. Potter, An-Wen Chan, David Moher, Alene Toulany, Amanda Doherty-Kirby, Begonya Nafria Escalera, Catherine Stratton, Chris Gale, Colin Macarthur, Diane Purper-Ouakil, Edmund Juszczak, Eyal Cohen, Giorgio Reggiardo, Jennifer Preston, Jérémie F. Cohen, Julia Upton, Karel Allegaert, Katelynn Boerner, Kayur Mehta, Kim An Nguyen, Kimberly Courtney, Lisa Hartling, Menelaos Konstantinidis, Michal Odermarsky, Nancy J. Butcher, Niina Kolehmainen, Patricia E. Longmuir, Peter J. Gill, Piet Leroy, Reinhard Feneberg, Ramesh Poluru, Shaun K.**

- Morris, Stefan J. Friedrichsdorf, Tanya Chute Nagy, Terry P. Klassen, Thierry Lacaze-Masmonteil, Wes Onland, and Martin Offringa.** 2026. “SPIRIT-Children and Adolescents (SPIRIT-C) 2026 Extension Statement: Enhancing the Reporting and Usefulness of Pediatric Randomized Trial Protocols.” *JAMA Pediatrics* 180 (4): 426–439. DOI 10.1001/jamapediatrics.2026.0113.
- Bailey, Z. D., N. Krieger, M. Agénor, J. Graves, N. Linos, and M. T. Bassett.** 2017. “Structural racism and health inequities in the USA: Evidence and interventions.” *The Lancet* 389 (10077): 1453–1463. DOI 10.1016/S0140-6736(17)30569-X.
- Baltes, P. B.** 1997. “On the incomplete architecture of human ontogeny: Selection, optimization, and compensation as foundation of developmental theory.” *American Psychologist* 52 (4): 366–380. DOI 10.1037/0003-066X.52.4.366.
- Baluška, F. and A. S. Reber.** 2019. “Sentience and consciousness in single cells: How the first minds emerged in unicellular species.” *BioEssays* 41 (3): 1800229. DOI 10.1002/bies.201800229.
- Bandura, Albert.** 2001. “Social cognitive theory: An agentic perspective.” *Annual Review of Psychology* 52: 1–26. DOI 10.1146/annurev.psych.52.1.1.
- Barocas, Solon, and Andrew D. Selbst.** 2016. “Big data’s disparate impact.” *California Law Review* 104 (3): 671–732. DOI 10.15779/Z38BG31.
- Barrett, L. F.** 2006. “Solving the emotion paradox: Categorization and the experience of emotion.” *Personality and Social Psychology Review* 10 (1): 20–46. DOI

10.1207/s15327957pspr1001_2.

Barrett, L. F. and W. K. Simmons. 2015. “Interoceptive predictions in the brain.” *Nature Reviews Neuroscience* 16 (7): 419–429. DOI 10.1038/nrn3950.

Barron, A. B. and C. Klein. 2016. “What insects can tell us about the origins of consciousness.” *Proceedings of the National Academy of Sciences* 113 (18): 4900–4908. DOI 10.1073/pnas.1520084113.

Bateson, P. 1991. “Assessment of pain in animals.” *Animal Behaviour* 42 (5): 827–839.

Baumeister, Roy F., and Mark R. Leary. 1995. “The need to belong: Desire for interpersonal attachments as a fundamental human motivation.” *Psychological Bulletin* 117 (3): 497–529. DOI 10.1037/0033-2909.117.3.497.

Bayne, T. 2010. *The Unity of Consciousness*. Oxford University Press.

Bayne, T., and D. J. Chalmers. 2003. “What is the unity of consciousness?” In A. Cleeremans, ed., *The Unity of Consciousness: Binding, Integration, and Dissociation*, 23–58. Oxford University Press.

Bayne, T., A. K. Seth, M. Massimini, J. Shepherd, A. Cleeremans, S. M. Fleming, R. Malach, J. B. Mattingley, D. K. Menon, A. M. Owen, M. A. K. Peters, A. Razi, and L. Mudrik. 2024. “Tests for consciousness in humans and beyond.” *Trends in Cognitive Sciences* 28 (5): 454–466. DOI 10.1016/j.tics.2024.01.010.

Beauchamp, Tom L., and James F. Childress. 2019. *Principles*

- of Biomedical Ethics (8th ed.)*. Oxford University Press.
- Bell, Catherine.** 1997. *Ritual: Perspectives and Dimensions*. Oxford University Press.
- Bender, E. M., T. Gebru, A. McMillan-Major, and S. Shmitchell.** 2021. “On the dangers of stochastic parrots: Can language models be too big?” *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. DOI 10.1145/3442188.3445922.
- Benjamin, R.** 2019. *Race After Technology*. Polity.
- Berger, Peter L.** 1967. *The Sacred Canopy: Elements of a Sociological Theory of Religion*. Doubleday.
- Berridge, K. C. and T. E. Robinson.** 1998. “What is the role of dopamine in reward: Hedonic impact, reward learning, or incentive salience?” *Brain Research Reviews* 28 (3): 309–369. DOI 10.1016/S0165-0173(98)00019-8.
- Berridge, K. C., T. E. Robinson, and J. W. Aldridge.** 2009. “Dissecting components of reward: ‘Liking,’ ‘wanting,’ and learning.” *Current Opinion in Pharmacology* 9 (1): 65–73. DOI 10.1016/j.coph.2008.12.014.
- Berrios, R., P. Totterdell, and S. Kellett.** 2015. “Eliciting mixed emotions: A meta-analysis comparing models, types, and measures.” *Frontiers in Psychology* 6: 428. DOI 10.3389/fpsyg.2015.00428.
- Bertsekas, D. P.** 1995. *Dynamic Programming and Optimal Control, Volume I*. Athena Scientific.
- Bickel, Warren K., Matthew W. Johnson, Mikhail N. Koffarnus, James MacKillop, and James G. Murphy.** 2014.

“The behavioral economics of reinforcer pathologies: Novel approaches to addictive disorders.” *Annual Review of Clinical Psychology* 10: 641–677. DOI 10.1146/annurev-clinpsy-032813-153724.

Billingsley, P. 1999. *Convergence of probability measures (2nd ed.)*. Wiley. DOI 10.1002/9780470316962.

Birch, J. 2017. “Animal sentience and the precautionary principle.” *Animal Sentience* 2 (16): 1.

Birch, J. 2024. *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Oxford University Press.

Birch, J., A. K. Schnell, and N. S. Clayton. 2020. “Dimensions of animal consciousness.” *Trends in Cognitive Sciences* 24 (10): 789–801. DOI 10.1016/j.tics.2020.07.007.

Bland, J. M. and D. G. Altman. 1986. “Statistical methods for assessing agreement between two methods of clinical measurement.” *The Lancet* 1 (8476): 307–310. DOI 10.1016/S0140-6736(86)90837-8.

Block, N. 1995. “On a confusion about a function of consciousness.” *Behavioral and Brain Sciences* 18 (2): 227–247. DOI 10.1017/S0140525X00038188.

Bobba-Alves, N., R.-P. Juster, and M. Picard. 2022. “The energetic cost of allostasis and allostatic load.” *Psychoneuroendocrinology* 146: 105951. DOI 10.1016/j.psyneuen.2022.105951.

Bodien, Yelena G., Judith Allanson, Paolo Cardone, Arthur Bonhomme, Jerina Carmona, Camille Chatelle, Srivas Chennu, Mary Conte, Stanislas Dehaene, Paola Finoia,

Gregory Heinonen, Jennifer E. Hersh, Evelyn Kamau, Phoebe K. Lawrence, Victoria C. Lupson, Anogue Meydan, Benjamin Rohaut, William R. Sanders, Jacobo D. Sitt, Andrea Soddu, Mélanie Valente, Angela Velazquez, Henning U. Voss, Athina Vrosgou, Jan Claassen, Brian L. Edlow, Joseph J. Fins, Olivia Gosseries, Steven Laureys, David Menon, Lionel Naccache, Adrian M. Owen, John Pickard, Emmanuel A. Stamatakis, Aurore Thibaut, Jonathan D. Victor, Joseph T. Giacino, Emilia Bagiella, and Nicholas D. Schiff. 2024. “Cognitive motor dissociation in disorders of consciousness.” *New England Journal of Medicine* 391 (7): 598–608. DOI 10.1056/NEJMoA2400645.

Boisseau, Romain P., David Vogel, and Audrey Dussutour. 2016. “Habituation in non-neural organisms: Evidence from slime moulds.” *Proceedings of the Royal Society B* 283 (1829): 20160446. DOI 10.1098/rspb.2016.0446.

Bollen, K. A. 1989. *Structural equations with latent variables*. Wiley. DOI 10.1002/9781118619179.

Bollen, K. A. and S. Bauldry. 2011. “Three Cs in measurement models: Causal indicators, composite indicators, and covariates.” *Psychological Methods* 16 (3): 265–284. DOI 10.1037/a0024448.

Bollen, K. A. and R. Lennox. 1991. “Conventional wisdom on measurement: A structural equation perspective.” *Psychological Bulletin* 110 (2): 305–314. DOI 10.1037/0033-2909.110.2.305.

Bonanno, George A. 2004. “Loss, trauma, and human resilience: Have we underestimated the human capacity to thrive after

- extremely aversive events?” *American Psychologist* 59 (1): 20–28. DOI 10.1037/0003-066X.59.1.20.
- Borsboom, D.** 2005. *Measuring the mind: Conceptual issues in contemporary psychometrics*. Cambridge University Press. DOI 10.1017/CBO9780511490026.
- Borsboom, Denny, Gideon J. Mellenbergh, and Jaap van Heerden.** 2004. “The concept of validity.” *Psychological Review* 111 (4): 1061–1071. DOI 10.1037/0033-295X.111.4.1061.
- Bostrom, N.** 2003. “Are you living in a computer simulation?” *The Philosophical Quarterly* 53 (211): 243–255.
- Bourtole, L., V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, and N. Papernot.** 2021. “Machine unlearning.” *2021 IEEE Symposium on Security and Privacy*, 141–159. DOI 10.1109/SP40001.2021.00019.
- Bowker, Geoffrey C., and Susan Leigh Star.** 1999. *Sorting Things Out: Classification and Its Consequences*. Cambridge, MA: MIT Press.
- Box, G. E. P.** 1976. “Science and statistics.” *Journal of the American Statistical Association* 71 (356): 791–799. DOI 10.1080/01621459.1976.10480949.
- Braithwaite, John.** 1989. *Crime, Shame and Reintegration*. Cambridge University Press.
- Briat, C., A. Gupta, and M. Khammash.** 2016. “Antithetic integral feedback ensures robust perfect adaptation in noisy biomolecular networks.” *Cell Systems* 2 (1): 15–26. DOI 10.1016/j.cels.2016.01.004.

- Brickman, P., D. Coates, and R. Janoff-Bulman.** 1978. “Lottery winners and accident victims: Is happiness relative?” *Journal of Personality and Social Psychology* 36 (8): 917–927. DOI 10.1037/0022-3514.36.8.917.
- Broom, Donald M.** 2014. *Sentience and Animal Welfare*. CABI.
- Brown, E. N., R. Lydic, and N. D. Schiff.** 2010. “General anesthesia, sleep, and coma.” *New England Journal of Medicine* 363 (27): 2638–2650. DOI 10.1056/NEJMr0808281.
- Bryant, Fred B., and Joseph Veroff.** 2007. *Savoring: A New Model of Positive Experience*. Lawrence Erlbaum Associates.
- Buolamwini, Joy, and Timnit Gebru.** 2018. “Gender shades: Intersectional accuracy disparities in commercial gender classification.” *Proceedings of Machine Learning Research* 81: 77–91.
- Burnham, K. P. and D. R. Anderson.** 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach (2nd ed.)*. Springer.
- Busemeyer, Jerome R., and Adele Diederich.** 2010. *Cognitive Modeling*. SAGE.
- Butlin, P., R. Long, T. Bayne, Y. Bengio, J. Birch, D. J. Chalmers, A. Constant, G. Deane, E. Elmoznino, S. M. Fleming, X. Ji, R. Kanai, C. Klein, G. W. Lindsay, M. Michel, L. Mudrik, M. A. K. Peters, E. Schwitzgebel, J. Simon, and R. VanRullen.** 2026. “Identifying indicators of consciousness in AI systems.” *Trends in Cognitive Sciences* 30 (6): 488–501. DOI 10.1016/j.tics.2025.10.011.

- Butlin, P., R. Long, E. Elmoznino, Y. Bengio, J. Birch, A. Constant, G. Deane, S. M. Fleming, C. Frith, X. Ji, R. Kanai, C. Klein, G. Lindsay, M. Michel, L. Mudrik, M. A. K. Peters, E. Schwitzgebel, J. Simon, and R. VanRullen.** 2023. “Consciousness in artificial intelligence: Insights from the science of consciousness.” *arXiv:2308.08708*.
- Butlin, Patrick, and Theodoros Lappas.** 2025. “Principles for responsible AI consciousness research.” *Journal of Artificial Intelligence Research* 82: 1673–1690. DOI 10.1613/jair.1.17310.
- Cacioppo, J. T. and G. G. Berntson.** 1994. “Relationship between attitudes and evaluative space: A critical review, with emphasis on the separability of positive and negative substrates.” *Psychological Bulletin* 115 (3): 401–423. DOI 10.1037/0033-2909.115.3.401.
- Callan, M. J., J. H. Ellard, and J. E. Nicol.** 2006. “The belief in a just world and immanent justice reasoning in adults.” *Personality and Social Psychology Bulletin* 32 (12): 1646–1658. DOI 10.1177/0146167206292236.
- Campbell, D. T. and D. W. Fiske.** 1959. “Convergent and discriminant validation by the multitrait-multimethod matrix.” *Psychological Bulletin* 56 (2): 81–105. DOI 10.1037/h0046016.
- Campbell, Donald T.** 1979. “Assessing the Impact of Planned Social Change.” *Evaluation and Program Planning* 2 (1): 67–90. DOI 10.1016/0149-7189(79)90048-X.
- Camus, A.** 1942. *Le Mythe de Sisyphe*. Gallimard.

- Camus, Albert.** 1955. *The Myth of Sisyphus and Other Essays*. Translated by Justin O'Brien. Alfred A. Knopf. Originally published 1942.
- Cao, Y. and J. Yang.** 2015. "Towards making systems forget with machine unlearning." *2015 IEEE Symposium on Security and Privacy*, 463–480. DOI 10.1109/SP.2015.35.
- Carrasco-Zafra, M. I., R. Ocaña-Riola, R. Gómez-García, M. L. Martín-Roselló, and E. Blanco-Reina.** 2024. "Palliative sedation at the end of life: Prevalence, characteristics and possible determinants." *BMC Palliative Care* 23: 278. DOI 10.1186/s12904-024-01606-0.
- Carroll, R. J., D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu.** 2006. *Measurement Error in Nonlinear Models: A Modern Perspective (2nd ed.)*. Chapman & Hall/CRC.
- Carstensen, L. L.** 2006. "The influence of a sense of time on human development." *Science* 312 (5782): 1913–1915. DOI 10.1126/science.1127488.
- Carstensen, L. L.** 2021. "Socioemotional selectivity theory: The role of perceived endings in human motivation." *The Gerontologist* 61 (8): 1188–1196. DOI 10.1093/geront/gnab116.
- Carstensen, L. L., D. M. Isaacowitz, and S. T. Charles.** 1999. "Taking time seriously: A theory of socioemotional selectivity." *American Psychologist* 54 (3): 165–181. DOI 10.1037/0003-066X.54.3.165.
- Cartwright, N.** 1983. *How the Laws of Physics Lie*. Oxford University Press.

- Cartwright, R., M. Y. Agargun, J. Kirkby, and J. K. Friedman.** 2006. "Relation of dreams to waking concerns." *Psychiatry Research* 141 (3): 261–270.
- Cartwright, R. D., S. Lloyd, S. Knight, and I. Trenholme.** 1984. "Broken dreams: A study of the effects of divorce and depression on dream content." *Psychiatry* 47 (3): 251–259.
- Casali, Adenauer G., Olivia Gosseries, Mario Rosanova, Mélanie Boly, Simone Sarasso, Karina R. Casali, Silvia Casarotto, Marie-Aurélié Bruno, Steven Laureys, Giulio Tononi, and Marcello Massimini.** 2013. "A theoretically based index of consciousness independent of sensory processing and behavior." *Science Translational Medicine* 5 (198): 198ra105. DOI 10.1126/scitranslmed.3006294.
- Cassell, E. J.** 1982. "The nature of suffering and the goals of medicine." *New England Journal of Medicine* 306 (11): 639–645. DOI 10.1056/NEJM198203183061104.
- Cataldi, J., A. M. Stephan, J. Haba-Rubio, and F. Siclari.** 2024. "Shared EEG correlates between non-REM parasomnia experiences and dreams." *Nature Communications* 15: 3906. DOI 10.1038/s41467-024-48337-7.
- Cella, David, William Riley, Arthur Stone, Nan Rothrock, Bryce Reeve, Susan Yount, Dagmar Amtmann, Rita Bode, Daniel Buysse, Seung Choi, Karon Cook, Robert DeVellis, Darren DeWalt, James F. Fries, Richard Gershon, Elizabeth A. Hahn, Jin-Shei Lai, Paul Pilkonis, Dennis Revicki, Matthias Rose, Kevin Weinfurt, and Ron Hays.** 2010. "The Patient-Reported Outcomes Measurement Information System (PROMIS) developed and tested its first

wave of adult self-reported health outcome item banks.” *Journal of Clinical Epidemiology* 63 (11): 1179–1194. DOI 10.1016/j.jclinepi.2010.04.011.

Chalmers, D. J. 2023. “Could a large language model be conscious?” *arXiv:2303.07103*.

Chalmers, David J. 1996. *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press.

Chambers, C. D. 2013. “Registered Reports: A New Publishing Initiative at Cortex.” *Cortex* 49 (3): 609–610. DOI 10.1016/j.cortex.2012.12.016.

Champagnat, N. and D. Villemonais. 2016. “Exponential convergence to quasi-stationary distribution and Q-process.” *Probability Theory and Related Fields* 164: 243–283. DOI 10.1007/s00440-014-0611-7.

Chan, An-Wen, Isabelle Boutron, Sally Hopewell, David Moher, Kenneth F Schulz, Gary S Collins, Ruth Tunn, Rakesh Aggarwal, Michael Berkwits, Jesse A Berlin, Nita Bhandari, Nancy J Butcher, Marion K Campbell, Runcie C W Chidebe, Diana R Elbourne, Andrew J Farmer, Dean A Fergusson, Robert M Golub, Steven N Goodman, Tammy C Hoffmann, John P A Ioannidis, Brennan C Kahan, Rachel L Knowles, Sarah E Lamb, Steff Lewis, Elizabeth Loder, Martin Offringa, Philippe Ravaud, Dawn P Richards, Frank W Rockhold, David L Schriger, Nandi L Siegfried, Sophie Staniszewska, Rod S Taylor, Lehana Thabane, David J Torgerson, Sunita Vohra, Ian R White, and Asbjørn Hróbjartsson. 2025. “SPIRIT 2025 statement: Updated guideline for protocols of randomised trials.” *BMJ*

389: e081477. DOI 10.1136/bmj-2024-081477.

Chatterjee, K., M. Chmelik, and J. Davies. 2016. “A Symbolic SAT-Based Algorithm for Almost-Sure Reachability with Small Strategies in POMDPs.” *Proceedings of the AAAI Conference on Artificial Intelligence* 30 (1): 3225–3232. DOI 10.1609/aaai.v30i1.10422.

Chen, F. F. 2007. “Sensitivity of goodness of fit indexes to lack of measurement invariance.” *Structural Equation Modeling* 14 (3): 464–504. DOI 10.1080/10705510701301834.

Chen, Y., T. T. Georgiou, and M. Pavon. 2021. “Optimal transport in systems and control.” *Annual Review of Control, Robotics, and Autonomous Systems* 4: 89–113. DOI 10.1146/annurev-control-070220-100858.

Cheney-Lippold, John. 2017. *We Are Data: Algorithms and the Making of Our Digital Selves*. New York University Press.

Cherny, N. I. and L. Radbruch. 2009. “European Association for Palliative Care recommended framework for the use of sedation in palliative care.” *Palliative Medicine* 23 (7): 581–593. DOI 10.1177/0269216309107024.

Chetrite, R. and H. Touchette. 2015. “Nonequilibrium Markov processes conditioned on large deviations.” *Annales Henri Poincaré* 16: 2005–2057. DOI 10.1007/s00023-014-0375-8.

Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. “Where is the land of opportunity? The geography of intergenerational mobility in the United States.” *Quarterly Journal of Economics* 129 (4): 1553–1623. DOI 10.1093/qje/qju022.

- Cheung, G. W. and R. B. Rensvold.** 2002. “Evaluating goodness-of-fit indexes for testing measurement invariance.” *Structural Equation Modeling* 9 (2): 233–255. DOI 10.1207/S15328007SEM0902_5.
- Chittka, L.** 2022. *The Mind of a Bee*. Princeton University Press.
- Chochinov, H. M.** 2002. “Dignity-conserving care: A new model for palliative care.” *JAMA* 287 (17): 2253–2260.
- Chochinov, H. M.** 2006. “Dying, dignity, and new horizons in palliative end-of-life care.” *CA: A Cancer Journal for Clinicians* 56 (2): 84–103. DOI 10.3322/canjclin.56.2.84.
- Choe, Eun Kyoung, Nicole B. Lee, Bongshin Lee, Wanda Pratt, and Julie A. Kientz.** 2014. “Understanding quantified-selfers’ practices in collecting and exploring personal data.” *Proceedings of CHI 2014*, 1143–1152. DOI 10.1145/2556288.2557372.
- Citron, Danielle Keats.** 2008. “Technological due process.” *Washington University Law Review* 85 (6): 1249–1313.
- Citron, Danielle Keats, and Frank Pasquale.** 2014. “The scored society: Due process for automated predictions.” *Washington Law Review* 89: 1–33.
- Claassen, Jan, Kevin Doyle, Adu Matory, Caroline Couch, Kelly M. Burger, Angela Velazquez, Joshua U. Okonkwo, Jean-Rémi King, Soojin Park, Sachin Agarwal, David Roh, Murad Megjhani, Andrey Elishev, E. Sander Connolly, and Benjamin Rohaut.** 2019. “Detection of brain activation in unresponsive patients with acute brain injury.” *New England Journal of Medicine* 380 (26): 2497–2505. DOI

10.1056/NEJMoa1812757.

Clark, A. 2013. “Whatever next? Predictive brains, situated agents, and the future of cognitive science.” *Behavioral and Brain Sciences* 36 (3): 181–204. DOI 10.1017/S0140525X12000477.

Clark, A. and D. J. Chalmers. 1998. “The extended mind.” *Analysis* 58 (1): 7–19. DOI 10.1093/analys/58.1.7.

Clayton, N. S. and A. Dickinson. 1998. “Episodic-like memory during cache recovery by scrub jays.” *Nature* 395: 272–274. DOI 10.1038/26216.

Cloud, William, and Robert Granfield. 2008. “Conceptualizing recovery capital: Expansion of a theoretical construct.” *Substance Use & Misuse* 43 (12–13): 1971–1986. DOI 10.1080/10826080802289762.

Coan, James A., and David A. Sbarra. 2015. “Social baseline theory: The social regulation of risk and effort.” *Current Opinion in Psychology* 1: 87–91. DOI 10.1016/j.copsyc.2014.12.021.

Cohen, Sheldon, and Thomas A. Wills. 1985. “Stress, social support, and the buffering hypothesis.” *Psychological Bulletin* 98 (2): 310–357. DOI 10.1037/0033-2909.98.2.310.

Collins, Harry, and Robert Evans. 2007. *Rethinking Expertise*. University of Chicago Press.

Committee on Publication Ethics. 2025. *Retraction guidelines (Version 3)*. DOI 10.24318/cope.2019.1.4.

Committee on the Rights of Persons with Disabilities. 2014. *General Comment No. 1, Article 12: Equal Recognition before*

the Law. CRPD/C/GC/1.

Copson, Andrew. 2019. *Secularism: A Very Short Introduction*. Oxford University Press. DOI 10.1093/actrade/9780198747222.001.0001.

Council for International Organizations of Medical Sciences. 2016. *International Ethical Guidelines for Health-related Research Involving Humans (4th ed.)*. CIOMS.

Council of Europe. 2024. *Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law*. Council of Europe Treaty Series No. 225.

Coviello, Lorenzo, Yunkyu Sohn, Adam D. I. Kramer, Cameron Marlow, Massimo Franceschetti, Nicholas A. Christakis, and James H. Fowler. 2014. “Detecting emotional contagion in massive social networks.” *PLoS ONE* 9 (3): e90315. DOI 10.1371/journal.pone.0090315.

Craig, A. D. 2009. “How do you feel-now? The anterior insula and human awareness.” *Nature Reviews Neuroscience* 10: 59–70. DOI 10.1038/nrn2555.

Cranmer, Kyle, Johann Brehmer, and Gilles Louppe. 2020. “The frontier of simulation-based inference.” *Proceedings of the National Academy of Sciences* 117 (48): 30055–30062. DOI 10.1073/pnas.1912789117.

Credé, Marcus, Michael C. Tynan, and Peter D. Harms. 2017. “Much ado about grit: A meta-analytic synthesis of the grit literature.” *Journal of Personality and Social Psychology* 113 (3): 492–511. DOI 10.1037/pspp0000102.

- Critchley, Hugo D., and Sarah N. Garfinkel.** 2017. “Interoception and emotion.” *Current Opinion in Psychology* 17: 7–14. DOI 10.1016/j.copsyc.2017.04.020.
- Cronbach, Lee J., and Paul E. Meehl.** 1955. “Construct validity in psychological tests.” *Psychological Bulletin* 52 (4): 281–302. DOI 10.1037/h0040957.
- Crook, R. J.** 2021. “Behavioral and neurophysiological evidence suggests affective pain experience in octopus.” *iScience* 24 (3): 102229. DOI 10.1016/j.isci.2021.102229.
- Crump, A., H. Browning, A. Schnell, C. Burn, and J. Birch.** 2022. “Sentience in decapod crustaceans: A general framework and review of the evidence.” *Animal Sentience* 7 (32): Article 1. DOI 10.51291/2377-7478.1691.
- Csikszentmihalyi, M. and R. Larson.** 1987. “Validity and reliability of the Experience-Sampling Method.” *Journal of Nervous and Mental Disease* 175 (9): 526–536. DOI 10.1097/00005053-198709000-00004.
- Dahabreh, Issa J., Sarah E. Robertson, Jon A. Steingrimsen, Elizabeth A. Stuart, and Miguel A. Hernán.** 2020. “Extending inferences from a randomized trial to a new target population.” *Statistics in Medicine* 39 (14): 1999–2014. DOI 10.1002/sim.8426.
- Darwin, C.** 1859. *On the Origin of Species by Means of Natural Selection, or, The Preservation of Favoured Races in the Struggle for Life*. John Murray.
- Darwin, Charles.** 2009. *On the Origin of Species*. Oxford World’s Classics. Oxford University Press. Originally

published 1859.

Dawkins, M. S. 2008. “The science of animal suffering.” *Ethology* 114 (10): 937–945.

de Haan, E. H. F., P. M. Corballis, S. A. Hillyard, C. A. Marzi, A. Seth, V. A. F. Lamme, L. Volz, M. Fabri, E. Schechter, T. Bayne, M. Corballis, and Y. Pinto. 2020. “Split-Brain: What We Know Now and Why This Is Important for Understanding Consciousness.” *Neuropsychology Review* 30 (2): 224–233. DOI 10.1007/s11065-020-09439-3.

de Vet, H. C. W., C. B. Terwee, L. B. Mokkink, and D. L. Knol. 2011. *Measurement in Medicine*. Cambridge University Press.

Deci, Edward L., and Richard M. Ryan. 2000. “The ‘What’ and ‘Why’ of Goal Pursuits: Human Needs and the Self-Determination of Behavior.” *Psychological Inquiry* 11 (4): 227–268. DOI 10.1207/S15327965PLI1104_01.

Dekker, Sidney. 2014. *The Field Guide to Understanding “Human Error”* (3rd ed.). Ashgate.

Dennett, D. C. 1987. *The Intentional Stance*. MIT Press.

Denrell, Jerker. 2003. “Vicarious learning, undersampling of failure, and the myths of management.” *Organization Science* 14 (3): 227–243. DOI 10.1287/orsc.14.3.227.15164.

Dickerson, S. S. and M. E. Kemeny. 2004. “Acute stressors and cortisol responses: A theoretical integration and synthesis of laboratory research.” *Psychological Bulletin* 130 (3): 355–391. DOI 10.1037/0033-2909.130.3.355.

Diener, E., R. E. Lucas, and C. N. Scollon. 2006. “Beyond the

hedonic treadmill: Revising the adaptation theory of well-being.” *American Psychologist* 61 (4): 305–314. DOI 10.1037/0003-066X.61.4.305.

Dienes, Zoltan. 2014. “Using Bayes to get the most out of non-significant results.” *Frontiers in Psychology* 5: 781. DOI 10.3389/fpsyg.2014.00781.

Diggle, P. J. and M. G. Kenward. 1994. “Informative drop-out in longitudinal data analysis.” *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 43 (1): 49–73. DOI 10.2307/2986113.

Dinges, David F., Frances Pack, Katherine Williams, Kelly A. Gillen, John W. Powell, Geoffrey E. Ott, Caitlin Aptowicz, and Allan I. Pack. 1997. “Cumulative sleepiness, mood disturbance, and psychomotor vigilance performance decrements during a week of sleep restricted to 4–5 hours per night.” *Sleep* 20 (4): 267–277. DOI 10.1093/sleep/20.4.267.

Doka, Kenneth J., ed. 1989. *Disenfranchised Grief: Recognizing Hidden Sorrow*. Lexington Books.

Domhoff, G. W. 2017. *The Emergence of Dreaming: Mind-Wandering, Embodied Simulation, and the Default Network*. Oxford University Press.

Doniger O’Flaherty, Wendy, ed. 1980. *Karma and Rebirth in Classical Indian Traditions*. University of California Press.

Doob, J. L. 1957. “Conditional Brownian motion and the boundary limits of harmonic functions.” *Bulletin de la Société Mathématique de France* 85: 431–458. DOI 10.24033/bsmf.1494.

- Dorahy, M. J., B. L. Brand, V. Şar, C. Krüger, P. Stavropoulos, A. Martínez-Taboas, R. Lewis-Fernández, and W. Middleton.** 2014. “Dissociative identity disorder: An empirical overview.” *Australian & New Zealand Journal of Psychiatry* 48 (5): 402–417. DOI 10.1177/0004867414527523.
- Duckworth, A. L. and J. J. Gross.** 2014. “Self-control and grit: Related but separable determinants of success.” *Current Directions in Psychological Science* 23 (5): 319–325. DOI 10.1177/0963721414541462.
- Duckworth, Angela L., Christopher Peterson, Michael D. Matthews, and Dennis R. Kelly.** 2007. “Grit: Perseverance and passion for long-term goals.” *Journal of Personality and Social Psychology* 92 (6): 1087–1101. DOI 10.1037/0022-3514.92.6.1087.
- Durkheim, É.** 1912. *Les formes élémentaires de la vie religieuse: Le système totémique en Australie.* Félix Alcan.
- Durkheim, Émile.** 1995. *The Elementary Forms of Religious Life.* Translated by Karen E. Fields. Free Press. Originally published 1912.
- Ecker, Ullrich K. H., Stephan Lewandowsky, John Cook, Philipp Schmid, Lisa K. Fazio, Nadia Brashier, Panayiota Kendeou, Emily K. Vraga, and Michelle A. Amazeen.** 2022. “The Psychological Drivers of Misinformation Belief and Its Resistance to Correction.” *Nature Reviews Psychology* 1: 13–29. DOI 10.1038/s44159-021-00006-y.
- Eldar, E., R. B. Rutledge, R. J. Dolan, and Y. Niv.** 2016. “Mood as representation of momentum.” *Trends in Cognitive Sciences* 20 (1): 15–24. DOI 10.1016/j.tics.2015.07.010.

- Emanuel, Ezekiel J., David Wendler, and Christine Grady.** 2000. "What makes clinical research ethical?" *JAMA* 283 (20): 2701–2711. DOI 10.1001/jama.283.20.2701.
- Emery, N. J. and N. S. Clayton.** 2004. "The mentality of crows: Convergent evolution of intelligence in corvids and apes." *Science* 306 (5703): 1903–1907. DOI 10.1126/science.1098410.
- Entwistle, Vikki A., Stacy M. Carter, Alan Cribb, and Kirsten McCaffery.** 2010. "Supporting patient autonomy: The importance of clinician-patient relationships." *Journal of General Internal Medicine* 25 (7): 741–745. DOI 10.1007/s11606-010-1292-2.
- Epley, Nicholas, Adam Waytz, and John T. Cacioppo.** 2007. "On seeing human: A three-factor theory of anthropomorphism." *Psychological Review* 114 (4): 864–886. DOI 10.1037/0033-295X.114.4.864.
- Epstein, Joshua M.** 2008. "Why model?" *Journal of Artificial Societies and Social Simulation* 11 (4): 12.
- Ericsson, K. Anders, Ralf T. Krampe, and Clemens Tesch-Römer.** 1993. "The role of deliberate practice in the acquisition of expert performance." *Psychological Review* 100 (3): 363–406. DOI 10.1037/0033-295X.100.3.363.
- Espeland, Wendy Nelson, and Mitchell L. Stevens.** 1998. "Commensuration as a social process." *Annual Review of Sociology* 24: 313–343. DOI 10.1146/annurev.soc.24.1.313.
- Ethier, S. N. and T. G. Kurtz.** 1986. *Markov processes: Characterization and convergence*. Wiley.

- Eubanks, Virginia.** 2018. *Automating Inequality*. St. Martin's Press.
- European Union.** 2016. *Regulation (EU) 2016/679 of the European Parliament and of the Council: General Data Protection Regulation*. Official Journal of the European Union, L 119.
- European Union.** 2024. "Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No. 300/2008, (EU) No. 167/2013, (EU) No. 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act)." *Official Journal of the European Union, L 2024/1689*, 12 July 2024.
- European Union.** 2025. "Regulation (EU) 2025/327 of the European Parliament and of the Council of 11 February 2025 on the European Health Data Space and amending Directive 2011/24/EU and Regulation (EU) 2024/2847." *Official Journal of the European Union, L 2025/327*, 5 March 2025.
- European Union.** 2026. "Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI)." *Official Journal of the European Union, L 2026/1744*, 24 July 2026.
- Everitt, B. J. and T. W. Robbins.** 2016. "Drug addiction: Updating actions to habits to compulsions ten years on."

Annual Review of Psychology 67: 23–50. DOI 10.1146/annurev-psych-122414-033457.

Exline, Julie J., and Joshua A. Wilt. 2023. “Supernatural Attributions: Seeing God, the Devil, Demons, Spirits, Fate, and Karma as Causes of Events.” *Annual Review of Clinical Psychology* 19: 461–487. DOI 10.1146/annurev-clinpsy-080921-081114.

Exline, Julie J., Kenneth I. Pargament, Joshua A. Wilt, Kathleen C. Pait, and William A. Schutt. 2024. “Research on spiritual struggles: A brief snapshot focusing on new horizons.” *Spiritual Care* 13 (2): 103–114. DOI 10.1515/spircare-2022-0063.

Faden, Ruth R., and Tom L. Beauchamp. 1986. *A History and Theory of Informed Consent*. Oxford University Press.

FDA-NIH Biomarker Working Group. 2025. “BEST (Biomarkers, EndpointS, and other Tools) Resource.” *Food and Drug Administration and National Institutes of Health*; updated January 16, 2025.

Feinberg, Todd E., and Jon M. Mallatt. 2016. *The Ancient Origins of Consciousness: How the Brain Created Experience*. MIT Press.

Ficco, L., L. Mancuso, J. Manuello, A. Teneggi, D. Liloia, S. Duca, T. Costa, G. Z. Kovacs, and F. Cauda. 2021. “Disentangling predictive processing in the brain: A meta-analytic study in favour of a predictive network.” *Scientific Reports* 11: Article 16258. DOI 10.1038/s41598-021-95603-5.

Fillingim, R. B. 2017. “Individual differences in pain:

- Understanding the mosaic that makes pain personal.” *Pain* 158 (Suppl 1): S11–S18. DOI 10.1097/j.pain.0000000000000775.
- Fine, J. P. and R. J. Gray.** 1999. “A proportional hazards model for the subdistribution of a competing risk.” *Journal of the American Statistical Association* 94 (446): 496–509. DOI 10.1080/01621459.1999.10474144.
- Fins, J. J.** 2015. *Rights Come to Mind: Brain Injury, Ethics, and the Struggle for Consciousness*. Cambridge University Press.
- Fischer, John Martin, and Mark Ravizza.** 1998. *Responsibility and Control: A Theory of Moral Responsibility*. Cambridge University Press.
- Fischhoff, Baruch.** 2013. “The sciences of science communication.” *Proceedings of the National Academy of Sciences* 110 (Suppl. 3): 14033–14039. DOI 10.1073/pnas.1213273110.
- Fisher, A. J., J. D. Medaglia, and B. F. Jeronimus.** 2018. “Lack of group-to-individual generalizability is a threat to human subjects research.” *Proceedings of the National Academy of Sciences* 115 (27): E6106–E6115. DOI 10.1073/pnas.1711978115.
- Fiske, Susan T., and Cydney Dupree.** 2014. “Gaining Trust as Well as Respect in Communicating to Motivated Audiences about Science Topics.” *Proceedings of the National Academy of Sciences* 111 (supplement 4): 13593–13597. DOI 10.1073/pnas.1317505111.
- Fitzmaurice, G. M., N. M. Laird, and J. H. Ware.** 2011. *Applied Longitudinal Analysis (2nd ed.)*. Wiley.

- Fitzsimmons, Patrick J., Jim Pitman, and Marc Yor.** 1993. “Markovian bridges: Construction, Palm interpretation, and splicing.” In *Seminar on Stochastic Processes, 1992*, 101–134. Birkhäuser.
- Flake, J. K. and E. I. Fried.** 2020. “Measurement schmeasurement: Questionable measurement practices and how to avoid them.” *Advances in Methods and Practices in Psychological Science* 3 (4): 456–465. DOI 10.1177/2515245920952393.
- Flood, Gavin.** 1996. *An Introduction to Hinduism*. Cambridge University Press.
- Floridi, Luciano, and Josh Cowls.** 2019. “A unified framework of five principles for AI in society.” *Harvard Data Science Review* 1 (1). DOI 10.1162/99608f92.8cd550d1.
- Folland, G. B.** 1999. *Real Analysis: Modern Techniques and Their Applications (2nd ed.)*. Wiley.
- Franco, Annie, Neil Malhotra, and Gabor Simonovits.** 2014. “Publication Bias in the Social Sciences: Unlocking the File Drawer.” *Science* 345 (6203): 1502–1505. DOI 10.1126/science.1255484.
- Frankfurt, Harry G.** 1971. “Freedom of the will and the concept of a person.” *Journal of Philosophy* 68 (1): 5–20.
- Frankl, V. E.** 1946. *Ein Psycholog erlebt das Konzentrationslager*. Verlag für Jugend und Volk.
- Frankl, Viktor E.** 2006. *Man’s Search for Meaning*. Beacon Press. Originally published 1946.
- Fraser, David.** 2008. *Understanding Animal Welfare: The*

Science in Its Cultural Context. Wiley-Blackwell.

- Frederick, S., and G. Loewenstein.** 1999. “Hedonic adaptation.” In D. Kahneman, E. Diener, and N. Schwarz, eds., *Well-Being: The Foundations of Hedonic Psychology*, 302–329. Russell Sage Foundation.
- Frederick, S., G. Loewenstein, and T. O’Donoghue.** 2002. “Time discounting and time preference: A critical review.” *Journal of Economic Literature* 40 (2): 351–401. DOI 10.1257/002205102320161311.
- Fredrickson, B. L. and D. Kahneman.** 1993. “Duration neglect in retrospective evaluations of affective episodes.” *Journal of Personality and Social Psychology* 65 (1): 45–55. DOI 10.1037/0022-3514.65.1.45.
- Fredrickson, Barbara L.** 2001. “The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions.” *American Psychologist* 56 (3): 218–226. DOI 10.1037/0003-066X.56.3.218.
- Freund, A. M. and P. B. Baltes.** 2002. “Life-management strategies of selection, optimization and compensation: Measurement by self-report and construct validity.” *Journal of Personality and Social Psychology* 82 (4): 642–662. DOI 10.1037/0022-3514.82.4.642.
- Fricker, M.** 2007. *Epistemic Injustice: Power and the Ethics of Knowing*. Oxford University Press.
- Friston, K.** 2010. “The free-energy principle: A unified brain theory?” *Nature Reviews Neuroscience* 11: 127–138. DOI 10.1038/nrn2787.

- Friston, K. J., T. FitzGerald, F. Rigoli, P. Schwartenbeck, and G. Pezzulo.** 2017. “Active inference: A process theory.” *Neural Computation* 29 (1): 1–49. DOI 10.1162/NECO_a_00912.
- Fujita, K.** 2011. “On conceptualizing self-control as more than the effortful inhibition of impulses.” *Personality and Social Psychology Review* 15 (4): 352–366. DOI 10.1177/1088868311411165.
- Futuyma, Douglas J., and Mark Kirkpatrick.** 2017. *Evolution*. 4th ed. Sinauer Associates.
- Galatzer-Levy, Isaac R., Sandy H. Huang, and George A. Bonanno.** 2018. “Trajectories of resilience and dysfunction following potential trauma: A review and statistical evaluation.” *Clinical Psychology Review* 63: 41–55. DOI 10.1016/j.cpr.2018.05.008.
- Gallagher, S.** 2000. “Philosophical conceptions of the self: Implications for cognitive science.” *Trends in Cognitive Sciences* 4 (1): 14–21. DOI 10.1016/S1364-6613(99)01417-5.
- Galpayage Dona, Hiruni Samadi, Cwyn Solvi, Amelia Kowalewska, Kaarle Mäkelä, HaDi MaBouDi, and Lars Chittka.** 2022. “Do bumble bees play?” *Animal Behaviour* 194: 239–251. DOI 10.1016/j.anbehav.2022.08.013.
- García, J. and F. Fernández.** 2015. “A comprehensive survey on safe reinforcement learning.” *Journal of Machine Learning Research* 16: 1437–1480.
- Garg, J., X. Zhang, and Q. Zhou.** 2024. “Soft-constrained Schrödinger bridge: A stochastic control approach.”

Proceedings of the 27th International Conference on Artificial Intelligence and Statistics, Proceedings of Machine Learning Research 238: 4429–4437.

Gazzaniga, M. S. 2005. “Forty-five years of split-brain research and still going strong.” *Nature Reviews Neuroscience* 6 (8): 653–659. DOI 10.1038/nrn1723.

Gebru, T., J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford. 2021. “Datasheets for Datasets.” *Communications of the ACM* 64 (12): 86–92. DOI 10.1145/3458723.

Geertz, Clifford. 1973. *The Interpretation of Cultures*. Basic Books.

Gelman, A. and E. Loken. 2013. *The Garden of Forking Paths: Why Multiple Comparisons Can Be a Problem, Even When There Is No Fishing Expedition or p-Hacking and the Research Hypothesis Was Posited Ahead of Time*. Columbia University.

Gelman, A. and E. Loken. 2014. “The statistical crisis in science.” *American Scientist* 102 (6): 460–465. DOI 10.1511/2014.111.460.

Gelman, Andrew, Aki Vehtari, Daniel Simpson, Charles C. Margossian, Bob Carpenter, Yuling Yao, Lauren Kennedy, Jonah Gabry, Paul-Christian Bürkner, and Martin Modrák. 2020. “Bayesian workflow.” *arXiv:2011.01808*.

Gerstorf, D., N. Ram, J. Goebel, J. Schupp, U. Lindenberger, and G. G. Wagner. 2010. “Where people live and die makes a difference: Individual and geographic disparities in well-being progression at the end of life.” *Psychology and Aging* 25

(3): 661–676. DOI 10.1037/a0019574.

Gerstorf, D., N. Ram, U. Lindenberger, and J. Smith. 2013. “Age and time-to-death trajectories of change in indicators of cognitive, sensory, physical, health, social, and self-related functions.” *Developmental Psychology* 49 (10): 1805–1821. DOI 10.1037/a0031340.

Gethin, Rupert. 1998. *The Foundations of Buddhism*. Oxford University Press.

Giacino, Joseph T., Douglas I. Katz, Nicholas D. Schiff, John Whyte, Eric J. Ashman, Stephen Ashwal, Richard Barbano, Flora M. Hammond, Steven Laureys, Geoffrey S.F. Ling, Risa Nakase-Richardson, Ronald T. Seel, Stuart Yablon, Thomas S.D. Getchius, Gary S. Gronseth, and Melissa J. Armstrong. 2018. “Practice guideline update recommendations summary: Disorders of consciousness.” *Neurology* 91 (10): 450–460. DOI 10.1212/WNL.0000000000005926.

Gibbons, M., A. Crump, M. Barrett, S. Sarlak, J. Birch, and L. Chittka. 2022a. “Can insects feel pain? A review of the neural and behavioural evidence.” *Advances in Insect Physiology* 63: 155–229. DOI 10.1016/bs.aiip.2022.10.001.

Gibbons, M., E. Pasquini, A. Kowalewska, E. Read, S. Gibson, A. Crump, C. Solvi, E. Versace, and L. Chittka. 2024. “Noxious stimulation induces self-protective behavior in bumblebees.” *iScience* 27 (8): 110440. DOI 10.1016/j.isci.2024.110440.

Gibbons, M., E. Versace, A. Crump, B. Baran, and L. Chittka. 2022b. “Motivational trade-offs and modulation of nociception

in bumblebees.” *Proceedings of the National Academy of Sciences* 119 (31): e2205821119. DOI 10.1073/pnas.2205821119.

Ginsburg, Simona, and Eva Jablonka. 2019. *The Evolution of the Sensitive Soul: Learning and the Origins of Consciousness*. MIT Press.

Gneiting, T. and A. E. Raftery. 2007. “Strictly proper scoring rules, prediction, and estimation.” *Journal of the American Statistical Association* 102 (477): 359–378. DOI 10.1198/016214506000001437.

Godfrey-Smith, Peter. 2009. *Darwinian Populations and Natural Selection*. Oxford University Press.

Goldstein, A. N. and M. P. Walker. 2014. “The role of sleep in emotional brain function.” *Annual Review of Clinical Psychology* 10: 679–708.

Gollwitzer, Peter M. 1999. “Implementation intentions: Strong effects of simple plans.” *American Psychologist* 54 (7): 493–503. DOI 10.1037/0003-066X.54.7.493.

Gollwitzer, Peter M., and Paschal Sheeran. 2006. “Implementation intentions and goal achievement: A meta-analysis of effects and processes.” *Advances in Experimental Social Psychology* 38: 69–119. DOI 10.1016/S0065-2601(06)38002-1.

Gould, Stephen Jay, and Richard C. Lewontin. 1979. “The spandrels of San Marco and the Panglossian paradigm: A critique of the adaptationist programme.” *Proceedings of the Royal Society B* 205 (1161): 581–598. DOI

10.1098/rspb.1979.0086.

Green, Ben. 2022. “The flaws of policies requiring human oversight of government algorithms.” *Computer Law & Security Review* 45: 105681. DOI 10.1016/j.clsr.2022.105681.

Greenwood, B. M. and S. N. Garfinkel. 2025. “Interoceptive mechanisms and emotional processing.” *Annual Review of Psychology* 76: 59–86. DOI 10.1146/annurev-psych-020924-125202.

Greer, David M., Matthew P. Kirschen, Ariane Lewis, Gary S. Gronseth, Alexander Rae-Grant, Stephen Ashwal, Maya A. Babu, David F. Bauer, Lori Billingham, Amanda Corey, Sonia Partap, Michael A. Rubin, Lori Shutter, Courtney Takahashi, Robert C. Tasker, Panayiotis Nicolaou Varelas, Eelco Wijdicks, Amy Bennett, Scott R. Wessels, and John J. Halperin. 2023. “Pediatric and adult brain death/death by neurologic criteria consensus guideline: Report of the AAN Guidelines Subcommittee, AAP, CNS, and SCCM.” *Neurology* 101 (24): 1112–1132. DOI 10.1212/WNL.00000000000207740.

Greer, David M., Sam D. Shemie, Ariane Lewis, Sylvia Torrance, Panayiotis Varelas, Fernando D. Goldenberg, James L. Bernat, Michael Souter, Mehmet Akif Topcuoglu, Anne W. Alexandrov, Marie Baldisseri, Thomas Bleck, Giuseppe Citerio, Rosanne Dawson, Arnold Hoppe, Stephen Jacobe, Alex Manara, Thomas A. Nakagawa, Thaddeus Mason Pope, William Silvester, David Thomson, Hussain Al Rahma, Rafael Badenes, Andrew J. Baker, Vladimir Cerny, Cherylee Chang, Tiffany R. Chang, Elena

- Gnedovskaya, Moon-Ku Han, Stephen Honeybul, Edgar Jimenez, Yasuhiro Kuroda, Gang Liu, Uzzwal Kumar Mallick, Victoria Markevich, Jorge Mejia-Mantilla, Michael Piradov, Sarah Quayyum, Gentle Sunder Shrestha, Ying-ying Su, Shelly D. Timmons, Jeanne Teitelbaum, Walter Videtta, Kapil Zirpe, and Gene Sung.** 2020. “Determination of brain death/death by neurologic criteria: The World Brain Death Project.” *JAMA* 324 (11): 1078–1097. DOI 10.1001/jama.2020.11586.
- Grimm, V., S. F. Railsback, C. E. Vincenot, U. Berger, C. Gallagher, D. L. DeAngelis, B. Edmonds, J. Ge, J. Giske, J. Groeneveld, A. S. A. Johnston, A. Milles, J. Nabe-Nielsen, J. G. Polhill, V. Radchuk, M.-S. Rohwäder, R. A. Stillman, J. C. Thiele, and D. Ayllón.** 2020. “The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism.” *Journal of Artificial Societies and Social Simulation* 23 (2): 7. DOI 10.18564/jasss.4259.
- Guest, Olivia, and Andrea E. Martin.** 2021. “How computational modeling can force theory building in psychological science.” *Perspectives on Psychological Science* 16 (4): 789–802. DOI 10.1177/1745691620970585.
- Guyatt, Gordon, David Sackett, Wayne Taylor, Jack Chong, David Roberts, and Sylvia Pugsley.** 1986. “Determining optimal therapy: Randomized trials in individual patients.” *New England Journal of Medicine* 314: 889–892. DOI 10.1056/NEJM198604033141406.
- Hafer, C. L. and L. Bègue.** 2005. “Experimental research on

- just-world theory: Problems, developments, and future challenges.” *Psychological Bulletin* 131 (1): 128–167. DOI 10.1037/0033-2909.131.1.128.
- Haggard, P.** 2017. “Sense of agency in the human brain.” *Nature Reviews Neuroscience* 18 (4): 196–207. DOI 10.1038/nrn.2017.14.
- Haggerty, Kevin D., and Richard V. Ericson.** 2000. “The surveillant assemblage.” *British Journal of Sociology* 51 (4): 605–622. DOI 10.1080/00071310020015280.
- Hall, C. S. and R. L. Van de Castle.** 1966. *The Content Analysis of Dreams*. Appleton-Century-Crofts.
- Hamaker, E. L.** 2012. “Why researchers should think ‘within-person’: A paradigmatic rationale.” In M. R. Mehl and T. S. Conner, eds., *Handbook of Research Methods for Studying Daily Life*, 43–61. Guilford Press.
- Hamilton, W. D.** 1964. “The genetical evolution of social behaviour. I.” *Journal of Theoretical Biology* 7 (1): 1–16.
- Hand, D. J.** 2004. *Measurement Theory and Practice*. Arnold.
- Harkness, Janet, Beth-Ellen Pennell, and Alisú Schoua-Glusberg.** 2004. “Survey questionnaire translation and assessment.” In Stanley Presser, Jennifer M. Rothgeb, Mick P. Couper, Judith T. Lessler, Elizabeth Martin, Jean Martin, and Eleanor Singer, eds., *Methods for Testing and Evaluating Survey Questionnaires*, 453–473. Wiley.
- Harvey, Peter.** 2013. *An Introduction to Buddhism: Teachings, History and Practices*. 2nd ed. Cambridge University Press.
- Headey, B. and A. Wearing.** 1989. “Personality, life events, and

- subjective well-being: Toward a dynamic equilibrium model.” *Journal of Personality and Social Psychology* 57 (4): 731–739.
- Helliwell, J. F., R. Layard, J. D. Sachs, J.-E. De Neve, L. B. Aknin, and S. Wang, eds.** 2026. *World Happiness Report 2026*. University of Oxford: Wellbeing Research Centre. DOI 10.18724/whr-ewft-vq17.
- Helson, H.** 1964. *Adaptation-Level Theory: An Experimental and Systematic Approach to Behavior*. Harper & Row.
- Henderson, Gail E, Larry R Churchill, Arlene M Davis, Michele M Easter, Christine Grady, Steven Joffe, Nancy Kass, Nancy M. P King, Charles W Lidz, Franklin G Miller, Daniel K Nelson, Jeffrey Peppercorn, Barbra Bluestone Rothschild, Pamela Sankar, Benjamin S Wilfond, and Catherine R Zimmer.** 2007. “Clinical trials and medical care: Defining the therapeutic misconception.” *PLoS Medicine* 4 (11): e324. DOI 10.1371/journal.pmed.0040324.
- Herd, Pamela, and Donald P. Moynihan.** 2018. *Administrative Burden: Policymaking by Other Means*. New York: Russell Sage Foundation.
- Hernán, M. A. and J. M. Robins.** 2020. *Causal Inference: What If*. Boca Raton, FL: Chapman & Hall/CRC.
- Heschel, Abraham Joshua.** 1962. *The Prophets*. Harper & Row.
- Hick, John.** 1966. *Evil and the God of Love*. Macmillan.
- Hobson, J. A. and R. W. McCarley.** 1977. “The brain as a dream state generator: An activation-synthesis hypothesis of the dream process.” *American Journal of Psychiatry* 134 (12): 1335–1348.

- Hohwy, J.** 2013. *The predictive mind*. Oxford University Press.
- Holt-Lunstad, Julianne, Timothy B. Smith, and J. Bradley Layton.** 2010. “Social relationships and mortality risk: A meta-analytic review.” *PLoS Medicine* 7 (7): e1000316. DOI 10.1371/journal.pmed.1000316.
- Hopewell, Sally, An-Wen Chan, Gary S Collins, Asbjørn Hróbjartsson, David Moher, Kenneth F Schulz, Ruth Tunn, Rakesh Aggarwal, Michael Berkwits, Jesse A Berlin, Nita Bhandari, Nancy J Butcher, Marion K Campbell, Runcie C W Chidebe, Diana Elbourne, Andrew Farmer, Dean A Fergusson, Robert M Golub, Steven N Goodman, Tammy C Hoffmann, John P A Ioannidis, Brennan C Kahan, Rachel L Knowles, Sarah E Lamb, Steff Lewis, Elizabeth Loder, Martin Offringa, Philippe Ravaud, Dawn P Richards, Frank W Rockhold, David L Schriger, Nandi L Siegfried, Sophie Staniszewska, Rod S Taylor, Lehana Thabane, David Torgerson, Sunita Vohra, Ian R White, and Isabelle Boutron.** 2025. “CONSORT 2025 statement: Updated guideline for reporting randomised trials.” *BMJ* 389: e081123. DOI 10.1136/bmj-2024-081123.
- Hyun, I., J. C. Scharf-Deering, and J. E. Lunshof.** 2020. “Ethical issues related to brain organoid research.” *Brain Research* 1732: 146653. DOI 10.1016/j.brainres.2020.146653.
- Infurna, F. J. and S. S. Luthar.** 2016. “Resilience to major life stressors is not as common as thought.” *Perspectives on Psychological Science* 11 (2): 175–194. DOI 10.1177/1745691615621271.
- Inouye, S. K., R. G. J. Westendorp, and J. S. Saczynski.** 2014.

“Delirium in elderly people.” *The Lancet* 383 (9920): 911–922. DOI 10.1016/S0140-6736(13)60688-1.

International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. 2019. “ICH Harmonised Guideline E9(R1): Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials.” *Step 4*, 20 November 2019.

International Society for Stem Cell Research. 2025. *Guidelines for Stem Cell Research and Clinical Translation (Version 1.2)*. ISSCR.

International Test Commission. 2017. *The ITC Guidelines for Translating and Adapting Tests (Second edition)*. International Test Commission.

Ioannidis, J. P. A. 2005. “Why Most Published Research Findings Are False.” *PLoS Medicine* 2 (8): e124. DOI 10.1371/journal.pmed.0020124.

James, W. 1902. *The Varieties of Religious Experience: A Study in Human Nature*. Longmans, Green, and Co.

James, William. 2002. *The Varieties of Religious Experience*. Modern Library. Originally published 1902.

Ji, J., J. Zhou, B. Zhang, J. Dai, X. Pan, R. Sun, W. Huang, Y. Geng, M. Liu, and Y. Yang. 2024. “OmniSafe: An infrastructure for accelerating safe reinforcement learning research.” *Journal of Machine Learning Research* 25 (285): 1–6.

Joffily, M. and G. Coricelli. 2013. “Emotional valence and the

free-energy principle.” *PLoS Computational Biology* 9 (6): e1003094. DOI 10.1371/journal.pcbi.1003094.

Joshanloo, M. 2013. “The influence of fear of happiness beliefs on responses to the Satisfaction With Life Scale.” *Personality and Individual Differences* 54 (5): 647–651. DOI 10.1016/j.paid.2012.11.011.

Kaelbling, L. P., M. L. Littman, and A. R. Cassandra. 1998. “Planning and acting in partially observable stochastic domains.” *Artificial Intelligence* 101 (1–2): 99–134. DOI 10.1016/S0004-3702(98)00023-X.

Kahneman, D. and A. Deaton. 2010. “High income improves evaluation of life but not emotional well-being.” *Proceedings of the National Academy of Sciences* 107 (38): 16489–16493. DOI 10.1073/pnas.1011492107.

Kahneman, D., A. B. Krueger, D. A. Schkade, N. Schwarz, and A. A. Stone. 2004. “A survey method for characterizing daily life experience: The Day Reconstruction Method.” *Science* 306 (5702): 1776–1780. DOI 10.1126/science.1103572.

Kahneman, D., P. P. Wakker, and R. Sarin. 1997. “Back to Bentham? Explorations of experienced utility.” *The Quarterly Journal of Economics* 112 (2): 375–406. DOI 10.1162/003355397555235.

Karatzas, I. and S. E. Shreve. 1991. *Brownian motion and stochastic calculus*. Springer.

Kataoka, M., T. Niikawa, N. Nagaishi, T.-L. Lee, A. Erler, J. Savulescu, and T. Sawai. 2025. “Beyond consciousness:

Ethical, legal and social issues in human brain organoid research and application.” *European Journal of Cell Biology* 104 (1): 151470. DOI 10.1016/j.ejcb.2024.151470.

Kellehear, A. 2007. *A Social History of Dying*. Cambridge University Press.

Kelly, John F., Nora D. Volkow, and Howard K. Koh. 2025. “The Changing Approach to Addiction: From Incarceration to Treatment and Recovery Support.” *New England Journal of Medicine* 392 (9): 833–836. DOI 10.1056/NEJMp2414224.

Kelly, John F., Richard Saitz, and Sarah Wakeman. 2016. “Language, substance use disorders, and policy: The need to reach consensus on an ‘addiction-ary.’” *Alcoholism Treatment Quarterly* 34 (1): 116–123. DOI 10.1080/07347324.2016.1113103.

Keramati, M. and B. Gutkin. 2014. “Homeostatic reinforcement learning for integrating reward collection and physiological stability.” *eLife* 3: e04811. DOI 10.7554/eLife.04811.

Kesecioglu, Jozef, Katerina Rusinova, Daniela Alampi, Yaseen M. Arabi, Julie Benbenishty, Dominique Benoit, Carole Boulanger, Maurizio Cecconi, Christopher Cox, Marjel van Dam, Diederik van Dijk, James Downar, Nikolas Efstathiou, Ruth Endacott, Alessandro Galazzi, Fiona van Gelder, Rik T. Gerritsen, Armand Girbes, Laura Hawyrluck, Margaret Herridge, Jan Hudec, Nancy Kentish-Barnes, Monika Kerckhoffs, Jos M. Latour, Jan Malaska, Annachiara Marra, Stephanie Meddick-Dyson, Spyridon Mentzelopoulos, Mervyn Mer, Victoria Metaxa,

Andrej Michalsen, Rajesh Mishra, Giovanni Mistraletti, Margo van Mol, Rui Moreno, Judith Nelson, Andrea Ortiz Suñer, Natalie Pattison, Tereza Prokopova, Kathleen Puntillo, Kathryn Puxty, Samah Al Qahtani, Lukas Radbruch, Emilio Rodriguez-Ruiz, Ron Sabar, Stefan J. Schaller, Shahla Siddiqui, Charles L. Sprung, Michele Umbrello, Marco Vergano, Massimo Zambon, Marieke Zegers, Michael Darmon, and Elie Azoulay. 2024. “European Society of Intensive Care Medicine guidelines on end-of-life and palliative care in the intensive care unit.” *Intensive Care Medicine* 50 (11): 1740–1766. DOI 10.1007/s00134-024-07579-1.

Khalsa, S. S., R. Adolphs, O. G. Cameron, H. D. Critchley, P. W. Davenport, J. S. Feinstein, J. D. Feusner, S. N. Garfinkel, R. D. Lane, W. E. Mehling, A. E. Meuret, C. B. Nemeroff, S. Oppenheimer, F. H. Petzschner, O. Pollatos, J. L. Rhudy, L. P. Schramm, W. K. Simmons, M. B. Stein, K. E. Stephan, O. Van den Bergh, I. Van Diest, A. von Leupoldt, M. P. Paulus, and the Interoception Summit 2016 participants. 2018. “Interoception and mental health: A roadmap.” *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 3 (6): 501–513. DOI 10.1016/j.bpsc.2017.12.004.

Kidd, I. J. and H. Carel. 2017. “Epistemic injustice and illness.” *Journal of Applied Philosophy* 34 (2): 172–190. DOI 10.1111/japp.12172.

Killingsworth, M. A., D. Kahneman, and B. Mellers. 2023. “Income and emotional well-being: A conflict resolved.”

- Proceedings of the National Academy of Sciences* 120 (10): e2208661120. DOI 10.1073/pnas.2208661120.
- Killingsworth, Matthew A., and Daniel T. Gilbert.** 2010. “A wandering mind is an unhappy mind.” *Science* 330 (6006): 932. DOI 10.1126/science.1192439.
- King, G., C. J. L. Murray, J. A. Salomon, and A. Tandon.** 2004. “Enhancing the validity and cross-cultural comparability of measurement in survey research.” *American Political Science Review* 98 (1): 191–207. DOI 10.1017/S000305540400108X.
- King, Richard.** 1999. *Orientalism and Religion: Postcolonial Theory, India and “The Mystic East.”* Routledge.
- Kirschbaum, C., K.-M. Pirke, and D. H. Hellhammer.** 1993. “The Trier Social Stress Test: A tool for investigating psychobiological stress responses in a laboratory setting.” *Neuropsychobiology* 28 (1–2): 76–81. DOI 10.1159/000119004.
- Klass, Dennis, Phyllis R. Silverman, and Steven L. Nickman, eds.** 1996. *Continuing Bonds: New Understandings of Grief.* Taylor & Francis.
- Klausen, S. H., J. Emiliussen, R. Christiansen, L. Hasandedic-Dapo, and S. Engelsen.** 2022. “The many faces of hedonic adaptation.” *Philosophical Psychology* 35 (2): 253–278. DOI 10.1080/09515089.2021.1967308.
- Kleinman, A.** 1988. *The Illness Narratives: Suffering, Healing, and the Human Condition.* Basic Books.
- Koch, T.** 2000. “The illusion of paradox: Commentary on the

disability paradox.” *Social Science & Medicine* 50 (6): 757–759. DOI 10.1016/S0277-9536(99)00385-8.

Kondziella, D., A. Bender, K. Diserens, W. van Erp, A. Estraneo, R. Formisano, S. Laureys, L. Naccache, S. Ozturk, B. Rohaut, J. D. Sitt, J. Stender, M. Tiainen, A. O. Rossetti, O. Gosseries, C. Chatelle, and the EAN Panel on Coma, Disorders of Consciousness. 2020. “European Academy of Neurology guideline on the diagnosis of coma and other disorders of consciousness.” *European Journal of Neurology* 27 (5): 741–756. DOI 10.1111/ene.14151.

Konkoly, Karen R., Kristoffer Appel, Emma Chabani, Anastasia Mangiaruga, Jarrod Gott, Remington Mallett, Bruce Caughran, Sarah Witkowski, Nathan W. Whitmore, Christopher Y. Mazurek, Jonathan B. Berent, Frederik D. Weber, Başak Türker, Smaranda Leu-Semenescu, Jean-Baptiste Maranci, Gordon Pipa, Isabelle Arnulf, Delphine Oudiette, Martin Dresler, and Ken A. Paller. 2021. “Real-time dialogue between experimenters and dreamers during REM sleep.” *Current Biology* 31 (7): 1417–1427.e6. DOI 10.1016/j.cub.2021.01.026.

Koob, G. F. and M. Le Moal. 2001. “Drug addiction, dysregulation of reward, and allostasis.” *Neuropsychopharmacology* 24 (2): 97–129. DOI 10.1016/S0893-133X(00)00195-0.

Koob, G. F. and M. Le Moal. 2008. “Addiction and the brain antireward system.” *Annual Review of Psychology* 59: 29–53. DOI 10.1146/annurev.psych.59.103006.093548.

Koob, George F., and Nora D. Volkow. 2016. “Neurobiology

- of addiction: A neurocircuitry analysis.” *The Lancet Psychiatry* 3 (8): 760–773. DOI 10.1016/S2215-0366(16)00104-8.
- Kosik, K. S.** 2024. “Why brain organoids are not conscious yet.” *Patterns* 5 (8): 101011. DOI 10.1016/j.patter.2024.101011.
- Krakow, Barry, Michael Hollifield, Lisa Johnston, Mary Koss, Ron Schrader, Teddy D. Warner, Dan Tandberg, John Lauriello, Leslie McBride, Lisa Cutchen, Diana Cheng, Shawn Emmons, Anne Germain, Dominic Melendrez, Diane Sandoval, and Holly Prince.** 2001. “Imagery rehearsal therapy for chronic nightmares in sexual assault survivors with posttraumatic stress disorder: A randomized controlled trial.” *JAMA* 286 (5): 537–545. DOI 10.1001/jama.286.5.537.
- Kramer, A. D. I., J. E. Guillory, and J. T. Hancock.** 2014. “Experimental evidence of massive-scale emotional contagion through social networks.” *Proceedings of the National Academy of Sciences* 111 (24): 8788–8790. DOI 10.1073/pnas.1320040111.
- Krantz, D. H., R. D. Luce, P. Suppes, and A. Tversky.** 1971. *Foundations of Measurement, Volume I: Additive and Polynomial Representations*. Academic Press.
- Kreibig, S. D.** 2010. “Autonomic nervous system activity in emotion: A review.” *Biological Psychology* 84 (3): 394–421. DOI 10.1016/j.biopsycho.2010.03.010.
- Kuhn, Thomas S.** 1962. *The Structure of Scientific Revolutions*. University of Chicago Press.
- Kuppens, P., F. Tuerlinckx, J. A. Russell, and L. F. Barrett.**

2013. “The relation between valence and arousal in subjective experience.” *Psychological Bulletin* 139 (4): 917–940. DOI 10.1037/a0030811.
- Kurland, B. F., L. L. Johnson, B. L. Egleston, and P. H. Diehr.** 2009. “Longitudinal data with follow-up truncated by death: Match the analysis method to research aims.” *Statistical Science* 24 (2): 211–222. DOI 10.1214/09-STS293.
- Lakatos, I.** 1978. *The Methodology of Scientific Research Programmes*. Cambridge University Press.
- Lakens, D.** 2017. “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses.” *Social Psychological and Personality Science* 8 (4): 355–362. DOI 10.1177/1948550617697177.
- Lally, P., C. H. M. van Jaarsveld, H. W. W. Potts, and J. Wardle.** 2010. “How are habits formed: Modelling habit formation in the real world.” *European Journal of Social Psychology* 40 (6): 998–1009. DOI 10.1002/ejsp.674.
- Lancaster, Madeline A., Magdalena Renner, Carol-Anne Martin, Daniel Wenzel, Louise S. Bicknell, Matthew E. Hurles, Tessa Homfray, Josef M. Penninger, Andrew P. Jackson, and Juergen A. Knoblich.** 2013. “Cerebral organoids model human brain development and microcephaly.” *Nature* 501: 373–379. DOI 10.1038/nature12517.
- Lange, M.** 2009. *Laws and Lawmakers: Science, Metaphysics, and the Laws of Nature*. Oxford University Press.
- Larsen, J. T., A. P. McGraw, and J. T. Cacioppo.** 2001. “Can

people feel happy and sad at the same time?” *Journal of Personality and Social Psychology* 81 (4): 684–696. DOI 10.1037/0022-3514.81.4.684.

Lawlor, Peter G., Bruno Gagnon, Isabelle L. Mancini, Jose L. Pereira, John Hanson, Maria E. Suarez-Almazor, and Eduardo D. Bruera. 2000. “Occurrence, causes, and outcome of delirium in patients with advanced cancer: A prospective study.” *Archives of Internal Medicine* 160 (6): 786–794. DOI 10.1001/archinte.160.6.786.

Lazare, Aaron. 2004. *On Apology*. Oxford University Press.

Leknes, S., J. C. W. Brooks, K. Wiech, and I. Tracey. 2008. “Pain relief as an opponent process: A psychophysical investigation.” *European Journal of Neuroscience* 28 (4): 794–801.

Léonard, C. 2014. “A survey of the Schrödinger problem and some of its connections with optimal transport.” *Discrete and Continuous Dynamical Systems A* 34 (4): 1533–1574. DOI 10.3934/dcds.2014.34.1533.

Lerner, M. J. 1980. *The Belief in a Just World: A Fundamental Delusion*. Plenum Press. DOI 10.1007/978-1-4899-0448-5.

Levenson, Jon D. 1994. *Creation and the Persistence of Evil: The Jewish Drama of Divine Omnipotence*. Princeton University Press.

Levin, M. 2021. “Life, death, and self: Fundamental questions of primitive cognition viewed through the lens of body plasticity and synthetic organisms.” *Biochemical and Biophysical Research Communications* 564: 114–133. DOI

10.1016/j.bbrc.2020.10.077.

Levinas, Emmanuel. 1969. *Totality and Infinity: An Essay on Exteriority*. Translated by Alphonso Lingis. Duquesne University Press.

Levins, Richard. 1966. “The strategy of model building in population biology.” *American Scientist* 54 (4): 421–431.

Lewis, D. 1973. *Counterfactuals*. Harvard University Press.

Lillie, Elizabeth O., Bradley Patay, Joel Diamant, Brian Issell, Eric J. Topol, and Nicholas J. Schork. 2011. “The n-of-1 clinical trial: The ultimate strategy for individualizing medicine?” *Personalized Medicine* 8 (2): 161–173. DOI 10.2217/pme.11.7.

Lim, Julian, and David F. Dinges. 2010. “A meta-analysis of the impact of short-term sleep deprivation on cognitive variables.” *Psychological Bulletin* 136 (3): 375–389. DOI 10.1037/a0018883.

List, Christian. 2019. *Why Free Will Is Real*. Harvard University Press.

Little, R. J. A. and D. B. Rubin. 2019. *Statistical Analysis with Missing Data (3rd ed.)*. Wiley. DOI 10.1002/9781119482260.

Livingston, James D., Teresa Milne, Mei Lan Fang, and Erica Amari. 2012. “The effectiveness of interventions for reducing stigma related to substance use disorders: A systematic review.” *Addiction* 107 (1): 39–50. DOI 10.1111/j.1360-0443.2011.03601.x.

Locke, Edwin A., and Gary P. Latham. 2002. “Building a practically useful theory of goal setting and task motivation: A

- 35-year odyssey.” *American Psychologist* 57 (9): 705–717. DOI 10.1037/0003-066X.57.9.705.
- Longino, Helen E.** 1990. *Science as Social Knowledge*. Princeton University Press.
- Lucas, R. E.** 2007. “Adaptation and the set-point model of subjective well-being: Does happiness change after major life events?” *Current Directions in Psychological Science* 16 (2): 75–79. DOI 10.1111/j.1467-8721.2007.00479.x.
- Luhmann, M., W. Hofmann, M. Eid, and R. E. Lucas.** 2012. “Subjective well-being and adaptation to life events: A meta-analysis.” *Journal of Personality and Social Psychology* 102 (3): 592–615. DOI 10.1037/a0025948.
- Lyon, P.** 2015. “The cognitive cell: Bacterial behavior reconsidered.” *Frontiers in Microbiology* 6: 1323. DOI 10.3389/fmicb.2015.01323.
- Mahmood, Saba.** 2005. *Politics of Piety: The Islamic Revival and the Feminist Subject*. Princeton University Press.
- Maltoni, M., E. Scarpi, and O. Nanni.** 2013. “Palliative sedation in end-of-life care.” *Current Opinion in Oncology* 25 (4): 360–367. DOI 10.1097/CCO.0b013e3283622c47.
- Manski, C. F.** 2003. *Partial identification of probability distributions*. Springer. DOI 10.1007/b97478.
- Manzi, Oscar, Kate E. Lynch, Daniel M. Allman, Tanya Latty, and Thomas E. White.** 2026. “Flexible self-protection as evidence of pain-like states in house crickets.” *Proceedings of the Royal Society B: Biological Sciences* 293 (2070): 20260609. DOI 10.1098/rspb.2026.0609.

Marek, Scott, Brenden Tervo-Clemmens, Finnegan J. Calabro, David F. Montez, Benjamin P. Kay, Alexander S. Hatoum, Meghan Rose Donohue, William Foran, Ryland L. Miller, Timothy J. Hendrickson, Stephen M. Malone, Sridhar Kandala, Eric Feczko, Oscar Miranda-Dominguez, Alice M. Graham, Eric A. Earl, Anders J. Perrone, Michaela Cordova, Olivia Doyle, Lucille A. Moore, Gregory M. Conan, Johnny Uriarte, Kathy Snider, Benjamin J. Lynch, James C. Wilgenbusch, Thomas Pengo, Angela Tam, Jianzhong Chen, Dillan J. Newbold, Annie Zheng, Nicole A. Seider, Andrew N. Van, Athanasia Metoki, Roselyne J. Chauvin, Timothy O. Laumann, Deanna J. Greene, Steven E. Petersen, Hugh Garavan, Wesley K. Thompson, Thomas E. Nichols, B. T. Thomas Yeo, Deanna M. Barch, Beatriz Luna, Damien A. Fair, and Nico U. F. Dosenbach. 2022. “Reproducible brain-wide association studies require thousands of individuals.” *Nature* 603: 654–660. DOI 10.1038/s41586-022-04492-9.

Marmot, M., S. Friel, R. Bell, T. A. J. Houweling, and S. Taylor. 2008. “Closing the Gap in a Generation: Health Equity through Action on the Social Determinants of Health.” *The Lancet* 372 (9650): 1661–1669. DOI 10.1016/S0140-6736(08)61690-6.

Martinez, Mayte, Katie Hall, Laurent Prétôt, Susan P. Lambeth, Steven J. Schapiro, and Sarah F. Brosnan. 2026. “Chimpanzees’ responses to inequity in a group context are influenced by food quality and their relationships with the group members present.” *Proceedings of the Royal Society B*:

- Biological Sciences* 293 (2071): 20260397. DOI 10.1098/rspb.2026.0397.
- Mashour, G. A. and A. G. Hudetz.** 2018. “Neural correlates of unconsciousness in large-scale brain networks.” *Trends in Neurosciences* 41 (3): 150–160. DOI 10.1016/j.tins.2018.01.003.
- Maslach, Christina, and Michael P. Leiter.** 2016. “Understanding the Burnout Experience: Recent Research and Its Implications for Psychiatry.” *World Psychiatry* 15 (2): 103–111. DOI 10.1002/wps.20311.
- Mather, M. and L. L. Carstensen.** 2005. “Aging and motivated cognition: The positivity effect in attention and memory.” *Trends in Cognitive Sciences* 9 (10): 496–502. DOI 10.1016/j.tics.2005.08.005.
- Maudlin, T.** 2007. *The Metaphysics Within Physics*. Oxford University Press.
- Mauss, I. B. and M. D. Robinson.** 2009. “Measures of emotion: A review.” *Cognition and Emotion* 23 (2): 209–237. DOI 10.1080/02699930802204677.
- Maynard Smith, J. and E. Szathmáry.** 1995. *The major transitions in evolution*. Oxford University Press.
- Maynard Smith, John.** 1982. *Evolution and the Theory of Games*. Cambridge University Press.
- Mayo, D. G.** 2018. *Statistical Inference as Severe Testing: How to Get Beyond the Statistics Wars*. Cambridge University Press.
- Mayr, E.** 1961. “Cause and effect in biology.” *Science* 134 (3489): 1501–1506. DOI 10.1126/science.134.3489.1501.

- McCambridge, J., J. Witton, and D. R. Elbourne.** 2014. “Systematic review of the Hawthorne effect: New concepts are needed to study research participation effects.” *Journal of Clinical Epidemiology* 67 (3): 267–277. DOI 10.1016/j.jclinepi.2013.08.015.
- McEwen, B. S.** 1998. “Protective and damaging effects of stress mediators.” *New England Journal of Medicine* 338 (3): 171–179. DOI 10.1056/NEJM199801153380307.
- McEwen, B. S. and E. Stellar.** 1993. “Stress and the individual: Mechanisms leading to disease.” *Archives of Internal Medicine* 153 (18): 2093–2101.
- McLellan, A. Thomas, David C. Lewis, Charles P. O’Brien, and Herbert D. Kleber.** 2000. “Drug dependence, a chronic medical illness: Implications for treatment, insurance, and outcomes evaluation.” *JAMA* 284 (13): 1689–1695. DOI 10.1001/jama.284.13.1689.
- Meehl, Paul E.** 1978. “Theoretical risks and tabular asterisks: Sir Karl, Sir Ronald, and the slow progress of soft psychology.” *Journal of Consulting and Clinical Psychology* 46 (4): 806–834. DOI 10.1037/0022-006X.46.4.806.
- Mellor, D. J., N. J. Beausoleil, K. E. Littlewood, A. N. McLean, P. D. McGreevy, B. Jones, and C. Wilkins.** 2020. “The 2020 Five Domains Model: Including human-animal interactions in assessments of animal welfare.” *Animals* 10 (10): 1870. DOI 10.3390/ani10101870.
- Mendl, Michael, Oliver H. P. Burman, and Elizabeth S. Paul.** 2010. “An integrative and functional framework for the study of animal emotion and mood.” *Proceedings of the Royal*

- Society B: Biological Sciences* 277 (1696): 2895–2904. DOI 10.1098/rspb.2010.0303.
- Menon, V.** 2011. “Large-scale brain networks and psychopathology: A unifying triple network model.” *Trends in Cognitive Sciences* 15 (10): 483–506. DOI 10.1016/j.tics.2011.08.003.
- Meredith, W.** 1993. “Measurement invariance, factor analysis and factorial invariance.” *Psychometrika* 58 (4): 525–543. DOI 10.1007/BF02294825.
- Merry, Sally Engle.** 2016. *The Seductions of Quantification*. University of Chicago Press.
- Merton, Robert K.** 1973. *The Sociology of Science: Theoretical and Empirical Investigations*. University of Chicago Press.
- Messick, Samuel.** 1995. “Validity of psychological assessment.” *American Psychologist* 50 (9): 741–749. DOI 10.1037/0003-066X.50.9.741.
- Metzinger, Thomas.** 2009. *The Ego Tunnel*. Basic Books.
- Metzinger, Thomas.** 2021. “Artificial suffering: An argument for a global moratorium on synthetic phenomenology.” *Journal of Artificial Intelligence and Consciousness* 8 (1): 43–66. DOI 10.1142/S270507852150003X.
- Michell, J.** 1999. *Measurement in Psychology: A Critical History of a Methodological Concept*. Cambridge University Press.
- Millsap, R. E.** 2011. *Statistical Approaches to Measurement Invariance*. Routledge.
- Mitchell, M., S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru.** 2019.

“Model Cards for Model Reporting.” *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229. DOI 10.1145/3287560.3287596.

Modrák, Martin, Angie H. Moon, Shinyoung Kim, Paul-Christian Bürkner, Niko Huurre, Kateřina Faltejsková, Andrew Gelman, and Aki Vehtari. 2025. “Simulation-based calibration checking for Bayesian computation: The choice of test quantities shapes sensitivity.” *Bayesian Analysis* 20 (2): 461–488. DOI 10.1214/23-BA1404.

Mokkink, L. B., C. B. Terwee, D. L. Patrick, J. Alonso, P. W. Stratford, D. L. Knol, L. M. Bouter, and H. C. W. de Vet. 2010. “The COSMIN checklist for assessing methodological quality of studies on measurement properties.” *Quality of Life Research* 19: 539–549. DOI 10.1007/s11136-010-9606-8.

Molenaar, P. C. M. 2004. “A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever.” *Measurement* 2 (4): 201–218. DOI 10.1207/S15366359MEA0204_1.

Monod, J. 1971. *Chance and Necessity: An Essay on the Natural Philosophy of Modern Biology*. Alfred A. Knopf.

Monti, Martin M., Audrey Vanhaudenhuyse, Martin R. Coleman, Melanie Boly, John D. Pickard, Luaba Tshibanda, Adrian M. Owen, and Steven Laureys. 2010. “Willful modulation of brain activity in disorders of consciousness.” *New England Journal of Medicine* 362 (7): 579–589. DOI 10.1056/NEJMoa0905370.

Mullainathan, S. and E. Shafir. 2013. *Scarcity: Why Having Too Little Means So Much*. Times Books.

- Munafò, Marcus R., Brian A. Nosek, Dorothy V. M. Bishop, Katherine S. Button, Christopher D. Chambers, Nathalie Percie du Sert, Uri Simonsohn, Eric-Jan Wagenmakers, Jennifer J. Ware, and John P. A. Ioannidis.** 2017. “A Manifesto for Reproducible Science.” *Nature Human Behaviour* 1: 0021. DOI 10.1038/s41562-016-0021.
- Nagel, T.** 1970. “Death.” *Noûs* 4 (1): 73–80. DOI 10.2307/2214297.
- Nagel, T.** 1979. “Moral luck.” In *Mortal Questions*, 24–38. Cambridge University Press.
- Nagel, Thomas.** 1974. “What is it like to be a bat?” *Philosophical Review* 83 (4): 435–450.
- Narayanan, A. and V. Shmatikov.** 2008. “Robust de-anonymization of large sparse datasets.” *2008 IEEE Symposium on Security and Privacy*, 111–125. DOI 10.1109/SP.2008.33.
- Narens, L.** 1981. “On the scales of measurement.” *Journal of Mathematical Psychology* 24 (3): 249–275. DOI 10.1016/0022-2496(81)90045-6.
- Nass, C., J. Steuer, and E. R. Tauber.** 1994. “Computers are social actors.” *Proceedings of CHI '94*, 72–78. DOI 10.1145/191666.191703.
- National Academies of Sciences, Engineering, and Medicine.** 2017. *Communicating Science Effectively: A Research Agenda*. Washington, DC: National Academies Press. DOI 10.17226/23674.
- National Academies of Sciences, Engineering, and Medicine.**

2019. *Reproducibility and Replicability in Science*. Washington, DC: National Academies Press. DOI 10.17226/25303.

National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1979. *The Belmont Report*. U.S. Department of Health and Human Services.

National Institute for Health and Care Excellence. 2015. *Care of dying adults in the last days of life*. NICE Guideline NG31.

National Institute of Standards and Technology. 2020. *NIST Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management, Version 1.0*. NIST CSWP 01162020. Gaithersburg, MD: National Institute of Standards and Technology. DOI 10.6028/NIST.CSWP.01162020.

National Institute of Standards and Technology. 2023. *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology. Published 26 January 2023. DOI 10.6028/NIST.AI.100-1.

National Institute of Standards and Technology. 2024. *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1. Gaithersburg, MD: National Institute of Standards and Technology. DOI 10.6028/NIST.AI.600-1.

National Institute on Drug Abuse. 2025. *Medications for Opioid Use Disorder*. National Institutes of Health. Updated 20 March 2025.

National Research Council. 2012. *Assessing the Reliability of*

Complex Models: Mathematical and Statistical Foundations of Verification, Validation, and Uncertainty Quantification. National Academies Press. DOI 10.17226/13395.

National Research Council. 2013. *Subjective Well-Being: Measuring Happiness, Suffering, and Other Dimensions of Experience*. National Academies Press.

Nebeker, C., J. Harlow, R. Espinoza Giacinto, R. Orozco-Linares, C. S. Bloss, and N. Weibel. 2017. “Ethical and regulatory challenges of research using pervasive sensing and other emerging technologies.” *AJOB Empirical Bioethics* 8 (4): 266–276. DOI 10.1080/23294515.2017.1403980.

Neimeyer, Robert A., ed. 2001. *Meaning Reconstruction & the Experience of Loss*. American Psychological Association.

Nesse, R. M. and G. C. Williams. 1994. *Why we get sick: The new science of Darwinian medicine*. Times Books.

Nielsen, T. and R. Levin. 2007. “Nightmares: A new neurocognitive model.” *Psychological Bulletin* 133 (3): 482–528. DOI 10.1037/0033-2909.133.3.482.

Nir, Y. and G. Tononi. 2010. “Dreaming and the brain: From phenomenology to neurophysiology.” *Trends in Cognitive Sciences* 14 (2): 88–100. DOI 10.1016/j.tics.2009.12.001.

Nissenbaum, H. 2004. “Privacy as contextual integrity.” *Washington Law Review* 79: 119–157.

Nissenbaum, Helen. 2010. *Privacy in Context: Technology, Policy, and the Integrity of Social Life*. Stanford, CA: Stanford University Press.

Nolen-Hoeksema, Susan, Blair E. Wisco, and Sonja

- Lyubomirsky.** 2008. “Rethinking rumination.” *Perspectives on Psychological Science* 3 (5): 400–424. DOI 10.1111/j.1745-6924.2008.00088.x.
- Nosek, B. A. and D. Lakens.** 2014. “Registered Reports: A method to increase the credibility of published results.” *Social Psychology* 45 (3): 137–141. DOI 10.1027/1864-9335/a000192.
- Nosek, Brian A., Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor.** 2018. “The preregistration revolution.” *Proceedings of the National Academy of Sciences* 115 (11): 2600–2606. DOI 10.1073/pnas.1708274114.
- Nuffield Council on Bioethics.** 2026. *Neural organoids: Ethical and governance considerations*. Nuffield Council on Bioethics.
- Nussbaum, M. C.** 2006. *Frontiers of Justice: Disability, Nationality, Species Membership*. Belknap Press of Harvard University Press.
- Nussbaum, M. C.** 2011. *Creating Capabilities: The Human Development Approach*. Cambridge, MA: Belknap Press of Harvard University Press.
- Oberkampff, William L., and Christopher J. Roy.** 2010. *Verification and Validation in Scientific Computing*. Cambridge University Press.
- Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan.** 2019. “Dissecting racial bias in an algorithm used to manage the health of populations.” *Science* 366 (6464): 447–453. DOI 10.1126/science.aax2342.

- Obeyesekere, G.** 2002. *Imagining karma: Ethical transformation in Amerindian, Buddhist, and Greek rebirth*. University of California Press.
- Ochsner, K. N. and J. J. Gross.** 2005. “The cognitive control of emotion.” *Trends in Cognitive Sciences* 9 (5): 242–249. DOI 10.1016/j.tics.2005.03.010.
- OECD.** 2025. *OECD Guidelines on Measuring Subjective Well-being (2025 Update)*. OECD Publishing. DOI 10.1787/9203632a-en.
- Oettingen, Gabriele.** 2012. “Future thought and behaviour change.” *European Review of Social Psychology* 23 (1): 1–63. DOI 10.1080/10463283.2011.643698.
- Ohm, P.** 2010. “Broken promises of privacy: Responding to the surprising failure of anonymization.” *UCLA Law Review* 57: 1701–1777.
- Öhman, Carl, and Luciano Floridi.** 2018. “An ethical framework for the digital afterlife industry.” *Nature Human Behaviour* 2: 318–320. DOI 10.1038/s41562-018-0335-2.
- Olson, E. T.** 2007. *What Are We? A Study in Personal Ontology*. Oxford University Press.
- O’Neil, Cathy.** 2016. *Weapons of Math Destruction*. Crown.
- O’Neill, Onora.** 2002. *Autonomy and Trust in Bioethics*. Cambridge University Press.
- O’Neill, Onora.** 2018. *From Principles to Practice: Normativity and Judgement in Ethics and Politics*. Cambridge University Press.
- Onnela, J.-P. and S. L. Rauch.** 2016. “Harnessing smartphone-

- based digital phenotyping to enhance behavioral and mental health.” *Neuropsychopharmacology* 41: 1691–1696. DOI 10.1038/npp.2016.7.
- Oort, F. J., M. R. M. Visser, and M. A. G. Sprangers.** 2009. “Measurement and conceptual perspectives on response shift.” *Journal of Clinical Epidemiology* 62 (11): 1126–1137. DOI 10.1016/j.jclinepi.2009.03.018.
- Open Science Collaboration.** 2015. “Estimating the Reproducibility of Psychological Science.” *Science* 349 (6251): aac4716. DOI 10.1126/science.aac4716.
- Oreskes, Naomi.** 2019. *Why Trust Science?* Princeton University Press.
- Oreskes, Naomi, Kristin Shrader-Frechette, and Kenneth Belitz.** 1994. “Verification, validation, and confirmation of numerical models in the earth sciences.” *Science* 263 (5147): 641–646. DOI 10.1126/science.263.5147.641.
- Orsi, Robert A.** 2005. *Between Heaven and Earth: The Religious Worlds People Make and the Scholars Who Study Them.* Princeton University Press.
- Ostrom, E.** 1990. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge: Cambridge University Press.
- Owen, Adrian M., Martin R. Coleman, Melanie Boly, Matthew H. Davis, Steven Laureys, and John D. Pickard.** 2006. “Detecting awareness in the vegetative state.” *Science* 313 (5792): 1402. DOI 10.1126/science.1130197.
- Palminteri, Stefano, Valentin Wyart, and Etienne Koechlin.**

2017. “The importance of falsification in computational cognitive modeling.” *Trends in Cognitive Sciences* 21 (6): 425–433. DOI 10.1016/j.tics.2017.03.011.
- Parfit, D.** 1984. *Reasons and Persons*. Oxford University Press.
- Pargament, Kenneth I.** 1997. *The Psychology of Religion and Coping: Theory, Research, Practice*. Guilford Press.
- Park, C. L.** 2010. “Making sense of the meaning literature: An integrative review of meaning making and its effects on adjustment to stressful life events.” *Psychological Bulletin* 136 (2): 257–301. DOI 10.1037/a0018301.
- Paşca, S. P., P. Arlotta, P. Campbell, A. Charo, J. H. Evans, N. Farahany, F. H. Gage, P. Godfrey-Smith, I. Hyun, A. R. Kriegstein, L. Melloni, G.-L. Ming, J. D. Moreno, J. Sugarman, S. Temple, G. Testa, and H. T. Greely.** 2025. “The need for a global effort to attend to human neural organoid and assembloid research.” *Science* 390 (6773): 574–577. DOI 10.1126/science.aeb1510.
- Patterson, A., S. Neumann, M. White, and A. White.** 2024. “Empirical design in reinforcement learning.” *Journal of Machine Learning Research* 25 (318): 1–63.
- Paulus, Martin P., and Murray B. Stein.** 2006. “An insular view of anxiety.” *Biological Psychiatry* 60 (4): 383–387. DOI 10.1016/j.biopsych.2006.03.042.
- Pearl, J.** 2009. *Causality (2nd ed.)*. Cambridge University Press.
- Pearl, Judea, and Elias Bareinboim.** 2014. “External Validity: From Do-Calculus to Transportability across Populations.” *Statistical Science* 29 (4): 579–595. DOI 10.1214/14-STS486.

- Pennartz, C. M. A.** 2026. “How can we validate theory-derived indicators of consciousness in artificial intelligence?” *Trends in Cognitive Sciences* 30 (7): 573–574. DOI 10.1016/j.tics.2026.01.011.
- Pessoa, L.** 2017. “A network model of the emotional brain.” *Trends in Cognitive Sciences* 21 (5): 357–371. DOI 10.1016/j.tics.2017.03.002.
- Pezzulo, Giovanni, Francesco Rigoli, and Karl J. Friston.** 2015. “Active inference, homeostatic regulation and adaptive behavioural control.” *Progress in Neurobiology* 134: 17–35. DOI 10.1016/j.pneurobio.2015.09.001.
- Pigliucci, Massimo, and Gerd B. Müller, eds.** 2010. *Evolution: The Extended Synthesis*. MIT Press.
- Pinto, Yair, David A. Neville, Marte Otten, Paul M. Corballis, Victor A. F. Lamme, Edward H. F de Haan, Nicoletta Foschi, and Mara Fabri.** 2017. “Split brain: Divided perception but undivided consciousness.” *Brain* 140 (5): 1231–1237. DOI 10.1093/brain/aww358.
- Plantinga, Alvin.** 1974. *God, Freedom, and Evil*. Eerdmans.
- Platt, John R.** 1964. “Strong inference.” *Science* 146 (3642): 347–353. DOI 10.1126/science.146.3642.347.
- Poldrack, R. A.** 2006. “Can cognitive processes be inferred from neuroimaging data?” *Trends in Cognitive Sciences* 10 (2): 59–63. DOI 10.1016/j.tics.2005.12.004.
- Popper, Karl R.** 1959. *The Logic of Scientific Discovery*. Hutchinson.
- Porter, Theodore M.** 1995. *Trust in Numbers: The Pursuit of*

Objectivity in Science and Public Life. Princeton, NJ: Princeton University Press.

Protter, P. E. 2005. *Stochastic integration and differential equations (2nd ed.)*. Springer. DOI 10.1007/978-3-662-10061-5.

Putnick, D. L. and M. H. Bornstein. 2016. “Measurement invariance conventions and reporting: The state of the art and future directions for psychological research.” *Developmental Review* 41: 71–90. DOI 10.1016/j.dr.2016.06.004.

Quine, W. V. O. 1951. “Two dogmas of empiricism.” *Philosophical Review* 60 (1): 20–43. DOI 10.2307/2181906.

Radovanović, Kia, Anoek Lorskens, Sebastian Schütte, Juliane Bräuer, Josep Call, Daniel B. M. Haun, and Edwin J. C. van Leeuwen. 2025. “Bonobos respond aversively to unequal reward distributions.” *Proceedings of the Royal Society B: Biological Sciences* 292 (2045): 20242873. DOI 10.1098/rspb.2024.2873.

Rahman, Fazlur. 1980. *Major Themes of the Qur'an*. Bibliotheca Islamica.

Railsback, Steven F., and Volker Grimm. 2019. *Agent-Based and Individual-Based Modeling: A Practical Introduction*. 2nd ed. Princeton University Press.

Raja, S. N., D. B. Carr, M. Cohen, N. B. Finnerup, H. Flor, S. Gibson, F. J. Keefe, J. S. Mogil, M. Ringkamp, K. A. Sluka, X. J. Song, B. Stevens, M. D. Sullivan, P. R. Tutelman, T. Ushida, and K. Vader. 2020. “The revised International Association for the Study of Pain definition of pain: Concepts,

challenges, and compromises.” *Pain* 161 (9): 1976–1982. DOI 10.1097/j.pain.0000000000001939.

Raji, Inioluwa Deborah, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. “Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing.” *Proceedings of FAT* 2020*, 33–44. DOI 10.1145/3351095.3372873.

Rambachan, Anantanand. 2006. *The Advaita Worldview: God, World, and Humanity*. State University of New York Press.

Ramsay, D. S. and S. C. Woods. 2014. “Clarifying the roles of homeostasis and allostasis in physiological regulation.” *Psychological Review* 121 (2): 225–247. DOI 10.1037/a0035942.

Rankin, C. H., et al. 2009. “Habituation revisited: An updated and revised description of the behavioral characteristics of habituation.” *Neurobiology of Learning and Memory* 92 (2): 135–138. DOI 10.1016/j.nlm.2008.09.012.

Rapkin, B. D. and C. E. Schwartz. 2004. “Toward a theoretical model of quality-of-life appraisal.” *Health and Quality of Life Outcomes* 2: 14. DOI 10.1186/1477-7525-2-14.

Rawls, J. 1999. *A Theory of Justice: Revised Edition*. Belknap Press of Harvard University Press.

Reason, James. 2000. “Human error: Models and management.” *BMJ* 320: 768–770. DOI 10.1136/bmj.320.7237.768.

Reeves, Byron, and Clifford Nass. 1996. *The Media Equation*. Cambridge University Press.

- Reinders, A. A. T. S., A. T. M. Willemssen, H. P. J. Vos, J. A. den Boer, and E. R. S. Nijenhuis.** 2012. “Fact or factitious? A psychobiological study of authentic and simulated dissociative identity states.” *PLoS ONE* 7 (6): e39279.
- Revonsuo, Antti.** 2000. “The reinterpretation of dreams: An evolutionary hypothesis of the function of dreaming.” *Behavioral and Brain Sciences* 23 (6): 877–901. DOI 10.1017/S0140525X00004015.
- Revuz, D. and M. Yor.** 1999. *Continuous martingales and Brownian motion (3rd ed.)*. Springer.
- Ritov, Oded, Christoph J. Völter, Nichola J. Raihani, and Jan M. Engelmann.** 2024. “No evidence for inequity aversion in non-human animals: A meta-analysis of accept/reject paradigms.” *Proceedings of the Royal Society B: Biological Sciences* 291 (2035): 20241452. DOI 10.1098/rspb.2024.1452.
- Robinson, T. E. and K. C. Berridge.** 2025. “The incentive-sensitization theory of addiction 30 years on.” *Annual Review of Psychology* 76: 29–58. DOI 10.1146/annurev-psych-011624-024031.
- Robinson, Terry E., and Kent C. Berridge.** 1993. “The neural basis of drug craving: An incentive-sensitization theory of addiction.” *Brain Research Reviews* 18 (3): 247–291. DOI 10.1016/0165-0173(93)90013-P.
- Robinson, W. S.** 1950. “Ecological correlations and the behavior of individuals.” *American Sociological Review* 15 (3): 351–357. DOI 10.2307/2087176.
- Rocher, Luc, Julien M. Hendrickx, and Yves-Alexandre de**

- Montjoye.** 2019. “Estimating the success of re-identifications in incomplete datasets using generative models.” *Nature Communications* 10: 3069. DOI 10.1038/s41467-019-10933-3.
- Rossi, Simone, Andrea Antal, Sven Bestmann, Marom Bikson, Carmen Brewer, Jürgen Brockmöller, Linda L. Carpenter, Massimo Cincotta, Robert Chen, Jeff D. Daskalakis, Vincenzo Di Lazzaro, Michael D. Fox, Mark S. George, Donald Gilbert, Vasilios K. Kimiskidis, Giacomo Koch, Risto J. Ilmoniemi, Jean Pascal Lefaucheur, Letizia Leocani, Sarah H. Lisanby, Carlo Miniussi, Frank Padberg, Alvaro Pascual-Leone, Walter Paulus, Angel V. Peterchev, Angelo Quartarone, Alexander Rotenberg, John Rothwell, Paolo M. Rossini, Emiliano Santarnecchi, Mouhsin M. Shafi, Hartwig R. Siebner, Yoshikatzu Ugawa, Eric M. Wassermann, Abraham Zangen, Ulf Ziemann, and Mark Hallett.** 2021. “Safety and recommendations for TMS use in healthy subjects and patient populations, with updates on training, ethical and regulatory issues: Expert guidelines.” *Clinical Neurophysiology* 132 (1): 269–306. DOI 10.1016/j.clinph.2020.10.003.
- Rubin, D. B.** 1976. “Inference and missing data.” *Biometrika* 63: 581–592. DOI 10.1093/biomet/63.3.581.
- Russell, J. A.** 2003. “Core affect and the psychological construction of emotion.” *Psychological Review* 110 (1): 145–172. DOI 10.1037/0033-295X.110.1.145.
- Russell, J. A. and J. M. Carroll.** 1999. “On the bipolarity of positive and negative affect.” *Psychological Bulletin* 125 (1): 3–30. DOI 10.1037/0033-2909.125.1.3.

- Rutledge, R. B., N. Skandali, P. Dayan, and R. J. Dolan.** 2014. “A computational and neural model of momentary subjective well-being.” *Proceedings of the National Academy of Sciences of the United States of America* 111 (33): 12252–12257. DOI 10.1073/pnas.1407535111.
- Ryan, Richard M., and Edward L. Deci.** 2001. “On happiness and human potentials: A review of research on hedonic and eudaimonic well-being.” *Annual Review of Psychology* 52: 141–166. DOI 10.1146/annurev.psych.52.1.141.
- Ryan, Richard M., and Edward L. Deci.** 2017. *Self-Determination Theory*. Guilford Press.
- Säilynoja, Teemu, Marvin Schmitt, Paul-Christian Bürkner, and Aki Vehtari.** 2026. “Posterior SBC: Simulation-based calibration checking conditional on data.” *Statistics and Computing* 36 (2): Article 78, 1–20. DOI 10.1007/s11222-026-10825-9.
- Saltelli, Andrea, Gabriele Bammer, Isabelle Bruno, Erica Charters, Monica Di Fiore, Emmanuel Didier, Wendy Nelson Espeland, John Kay, Samuele Lo Piano, Deborah Mayo, Roger Pielke Jr, Tommaso Portaluri, Theodore M. Porter, Arnald Puy, Ismael Rafols, Jerome R. Ravetz, Erik Reinert, Daniel Sarewitz, Philip B. Stark, Andrew Stirling, Jeroen van der Sluijs, and Paolo Vineis.** 2020. “Five ways to ensure that models serve society: A manifesto.” *Nature* 582: 482–484. DOI 10.1038/d41586-020-01812-9.
- Sanders, R. D., G. Tononi, S. Laureys, and J. W. Sleigh.** 2012. “Unresponsiveness $\neq$ unconsciousness.” *Anesthesiology* 116 (4): 946–959. DOI 10.1097/ALN.0b013e318249d0a7.

- Schacter, D. L.** 1999. “The seven sins of memory: Insights from psychology and cognitive neuroscience.” *American Psychologist* 54 (3): 182–203. DOI 10.1037/0003-066X.54.3.182.
- Schechter, E. and T. Bayne.** 2021. “Consciousness after split-brain surgery: The recent challenge to the classical picture.” *Neuropsychologia* 160: 107987. DOI 10.1016/j.neuropsychologia.2021.107987.
- Schechtman, M.** 1996. *The Constitution of Selves*. Cornell University Press.
- Scheufele, D. A.** 2014. “Science communication as political communication.” *Proceedings of the National Academy of Sciences* 111 (Suppl. 4): 13585–13592. DOI 10.1073/pnas.1317516111.
- Schnakers, Caroline, Audrey Vanhaudenhuyse, Joseph Giacino, Manfredi Ventura, Melanie Boly, Steve Majerus, Gustave Moonen, and Steven Laureys.** 2009. “Diagnostic accuracy of the vegetative and minimally conscious state: Clinical consensus versus standardized neurobehavioral assessment.” *BMC Neurology* 9: 35. DOI 10.1186/1471-2377-9-35.
- Schnell, Tatjana.** 2021. *The Psychology of Meaning in Life*. Routledge.
- Schredl, M. and F. Hofmann.** 2003. “Continuity between waking activities and dream activities.” *Consciousness and Cognition* 12 (2): 298–308. DOI 10.1016/S1053-8100(02)00072-7.

- Schuirmann, D. J.** 1987. “A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability.” *Journal of Pharmacokinetics and Biopharmaceutics* 15 (6): 657–680. DOI 10.1007/BF01068419.
- Schultz, W., P. Dayan, and P. R. Montague.** 1997. “A neural substrate of prediction and reward.” *Science* 275 (5306): 1593–1599.
- Schulz, K. F., D. G. Altman, and D. Moher.** 2010. “CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials.” *BMJ* 340: c332.
- Schumann, Karina.** 2018. “The psychology of offering an apology: Understanding the barriers to apologizing and how to overcome them.” *Current Directions in Psychological Science* 27 (2): 74–78. DOI 10.1177/0963721417741709.
- Schwartz, C. E. and M. A. G. Sprangers.** 1999. “Methodological approaches for assessing response shift.” *Social Science & Medicine* 48 (11): 1531–1548. DOI 10.1016/S0277-9536(99)00047-7.
- Scott, James C.** 1998. *Seeing Like a State*. Yale University Press.
- Searle, John R.** 1980. “Minds, brains, and programs.” *Behavioral and Brain Sciences* 3 (3): 417–424. DOI 10.1017/S0140525X00005756.
- Seeley, W. W., V. Menon, A. F. Schatzberg, J. Keller, G. H. Glover, H. Kenna, A. L. Reiss, and M. D. Greicius.** 2007. “Dissociable intrinsic connectivity networks for salience processing and executive control.” *Journal of Neuroscience* 27

(9): 2349–2356. DOI 10.1523/JNEUROSCI.5587-06.2007.

Selbst, Andrew D., danah boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. 2019. “Fairness and Abstraction in Sociotechnical Systems.” *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 59–68. DOI 10.1145/3287560.3287598.

Sen, A. 2009. *The Idea of Justice*. Harvard University Press.

Sen, Amartya. 1999. *Development as Freedom*. New York: Alfred A. Knopf.

Seow, Hsien, Lisa Barbera, Rinku Sutradhar, Doris Howell, Deborah Dudgeon, Clare Atzema, Ying Liu, Amna Husain, Jonathan Sussman, and Craig Earle. 2011. “Trajectory of performance status and symptom scores for patients with cancer during the last six months of life.” *Journal of Clinical Oncology* 29 (9): 1151–1158. DOI 10.1200/JCO.2010.30.7173.

Seth, A. K. 2013. “Interoceptive inference, emotion, and the embodied self.” *Trends in Cognitive Sciences* 17 (11): 565–573.

Seth, A. K. 2025. “Conscious artificial intelligence and biological naturalism.” Published online 21 April 2025; final version in *Behavioral and Brain Sciences* 49 (2026): e315. DOI 10.1017/S0140525X25000032.

Seth, A. K. and K. J. Friston. 2016. “Active interoceptive inference and the emotional brain.” *Philosophical Transactions of the Royal Society B: Biological Sciences* 371 (1708): 20160007. DOI 10.1098/rstb.2016.0007.

- Seth, Anil K., and Tim Bayne.** 2022. “Theories of consciousness.” *Nature Reviews Neuroscience* 23: 439–452. DOI 10.1038/s41583-022-00587-4.
- Shah, A. K., S. Mullainathan, and E. Shafir.** 2012. “Some consequences of having too little.” *Science* 338 (6107): 682–685. DOI 10.1126/science.1222426.
- Sheldon, Kennon M., and Sonja Lyubomirsky.** 2012. “The challenge of staying happier: Testing the hedonic adaptation prevention model.” *Personality and Social Psychology Bulletin* 38 (5): 670–680. DOI 10.1177/0146167212436400.
- Sherman, Lawrence W., and Heather Strang.** 2007. *Restorative Justice: The Evidence*. The Smith Institute.
- Shiffman, S., A. A. Stone, and M. R. Hufford.** 2008. “Ecological momentary assessment.” *Annual Review of Clinical Psychology* 4: 1–32. DOI 10.1146/annurev.clinpsy.3.022806.091415.
- Shmueli, G.** 2010. “To explain or to predict?” *Statistical Science* 25 (3): 289–310. DOI 10.1214/10-STS330.
- Shonkoff, J. P., A. S. Garner, Committee on Psychosocial Aspects of Child and Family Health, Committee on Early Childhood, Adoption, and Dependent Care, and Section on Developmental and Behavioral Pediatrics.** 2012. “The lifelong effects of early childhood adversity and toxic stress.” *Pediatrics* 129 (1): e232–e246. DOI 10.1542/peds.2011-2663.
- Shrout, P. E. and J. L. Fleiss.** 1979. “Intraclass correlations: Uses in assessing rater reliability.” *Psychological Bulletin* 86 (2): 420–428. DOI 10.1037/0033-2909.86.2.420.

- Siclari, Francesca, Benjamin Baird, Lampros Perogamvros, Giulio Bernardi, Joshua J LaRocque, Brady Riedner, Melanie Boly, Bradley R Postle, and Giulio Tononi.** 2017. “The neural correlates of dreaming.” *Nature Neuroscience* 20 (6): 872–878. DOI 10.1038/nn.4545.
- Simmons, J. P., L. D. Nelson, and U. Simonsohn.** 2011. “False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant.” *Psychological Science* 22 (11): 1359–1366. DOI 10.1177/0956797611417632.
- Simonsohn, U., J. P. Simmons, and L. D. Nelson.** 2020. “Specification curve analysis.” *Nature Human Behaviour* 4 (11): 1208–1214. DOI 10.1038/s41562-020-0912-z.
- Singer, J. D. and J. B. Willett.** 2003. *Applied Longitudinal Data Analysis: Modeling Change and Event Occurrence*. Oxford University Press.
- Smaldino, Paul E.** 2017. “Models are stupid, and we need more of them.” In Robin R. Vallacher, Stephen J. Read, and Andrzej Nowak, eds., *Computational Social Psychology*, 311–331. Routledge.
- Smith, Jesse M.** 2017. “Can the secular be the object of belief and belonging? The Sunday Assembly.” *Qualitative Sociology* 40 (1): 83–109. DOI 10.1007/s11133-016-9350-7.
- Sneddon, Lynne U., Robert W. Elwood, Shelley A. Adamo, and Matthew C. Leach.** 2014. “Defining and assessing animal pain.” *Animal Behaviour* 97: 201–212. DOI 10.1016/j.anbehav.2014.09.007.

- Society of Professional Journalists.** 2014. *SPJ Code of Ethics*.
Revised 6 September 2014.
- Solomon, R. L.** 1980. “The opponent-process theory of acquired motivation: The costs of pleasure and the benefits of pain.” *American Psychologist* 35 (8): 691–712.
- Solomon, R. L. and J. D. Corbit.** 1974. “An opponent-process theory of motivation: I. Temporal dynamics of affect.” *Psychological Review* 81 (2): 119–145. DOI 10.1037/h0036128.
- Solove, D. J.** 2006. “A Taxonomy of Privacy.” *University of Pennsylvania Law Review* 154 (3): 477–560.
- Solove, Daniel J.** 2004. *The Digital Person: Technology and Privacy in the Information Age*. New York University Press.
- Solvi, C., L. Baciadonna, and L. Chittka.** 2016. “Unexpected rewards induce dopamine-dependent positive emotion-like state changes in bumblebees.” *Science* 353 (6307): 1529–1531. DOI 10.1126/science.aaf4454.
- Sperry, R. W.** 1968. “Hemisphere deconnection and unity in conscious awareness.” *American Psychologist* 23 (10): 723–733. DOI 10.1037/h0026839.
- Sprangers, M. A. G. and C. E. Schwartz.** 1999. “Integrating response shift into health-related quality of life research: A theoretical model.” *Social Science & Medicine* 48 (11): 1507–1515. DOI 10.1016/S0277-9536(99)00045-3.
- Stearns, Stephen C.** 1992. *The Evolution of Life Histories*. Oxford University Press.
- Steege, S., F. Tuerlinckx, A. Gelman, and W. Vanpaemel.**

2016. “Increasing transparency through a multiverse analysis.” *Perspectives on Psychological Science* 11 (5): 702–712. DOI 10.1177/1745691616658637.
- Steinhauser, K. E., N. A. Christakis, E. C. Clipp, M. McNeilly, L. McIntyre, and J. A. Tulsky.** 2000. “Factors considered important at the end of life by patients, family, physicians, and other care providers.” *JAMA* 284 (19): 2476–2482.
- Sterling, P.** 2012. “Allostasis: A model of predictive regulation.” *Physiology & Behavior* 106 (1): 5–15. DOI 10.1016/j.physbeh.2011.06.004.
- Stevens, S. S.** 1946. “On the theory of scales of measurement.” *Science* 103 (2684): 677–680. DOI 10.1126/science.103.2684.677.
- Stone, A. A. and S. Shiffman.** 1994. “Ecological momentary assessment in behavioral medicine.” *Annals of Behavioral Medicine* 16 (3): 199–202.
- Stroebe, Margaret, and Henk Schut.** 1999. “The dual process model of coping with bereavement: Rationale and description.” *Death Studies* 23 (3): 197–224. DOI 10.1080/074811899201046.
- Substance Abuse and Mental Health Services Administration.** 2014. *SAMHSA’s Concept of Trauma and Guidance for a Trauma-Informed Approach*. HHS Publication No. SMA 14-4884.
- Suppes, P., and J. L. Zinnes.** 1963. “Basic measurement theory.” In R. D. Luce, R. R. Bush, and E. Galanter, eds., *Handbook of Mathematical Psychology, Volume I*, 1–76.

Wiley.

Suresh, Harini, and John V. Guttag. 2021. “A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle.” *Proceedings of EAAMO 2021*, Article 17. DOI 10.1145/3465416.3483305.

Surges, S. M., H. Brunsch, B. Jaspers, K. Apostolidis, A. Cardone, C. Centeno, N. Cherny, Á. Csikós, R. Fainsinger, E. Garralda, J. Ling, J. Menten, S. Mercadante, D. Mosoiu, S. Payne, N. Preston, L. Van den Block, J. Hasselaar, and L. Radbruch. 2024. “Revised European Association for Palliative Care (EAPC) recommended framework on palliative sedation: An international Delphi study.” *Palliative Medicine* 38 (2): 213–228. DOI 10.1177/02692163231220225.

Sutton, R. S. and A. G. Barto. 2018. *Reinforcement learning: An introduction (2nd ed.)*. MIT Press.

Talts, Sean, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. 2018. “Validating Bayesian inference algorithms with simulation-based calibration.” *arXiv:1804.06788*. DOI 10.48550/arXiv.1804.06788.

Tamer, E. 2010. “Partial identification in econometrics.” *Annual Review of Economics* 2: 167–195. DOI 10.1146/annurev.economics.050708.143401.

Taylor, Charles. 2007. *A Secular Age*. Belknap Press of Harvard University Press.

Tedeschi, R. G. and L. G. Calhoun. 2004. “Posttraumatic growth: Conceptual foundations and empirical evidence.” *Psychological Inquiry* 15 (1): 1–18.

- Teresi, J. A.** 2006. “Overview of quantitative measurement methods. Equivalence, invariance, and differential item functioning in health applications.” *Medical Care* 44 (11 Suppl 3): S39–S49. DOI 10.1097/01.mlr.0000245452.48613.45.
- Theriault, J. E., Y. Katsumi, H. M. Reimann, J. Zhang, P. Deming, B. C. Dickerson, K. S. Quigley, and L. F. Barrett.** 2025. “It’s not the thought that counts: Allostasis at the core of brain function.” *Neuron* 113 (24): 4107–4133. DOI 10.1016/j.neuron.2025.09.028.
- Tiffany, Stephen T.** 1990. “A cognitive model of drug urges and drug-use behavior: Role of automatic and nonautomatic processes.” *Psychological Review* 97 (2): 147–168. DOI 10.1037/0033-295X.97.2.147.
- Tinbergen, Niko.** 1963. “On aims and methods of ethology.” *Zeitschrift für Tierpsychologie* 20 (4): 410–433. DOI 10.1111/j.1439-0310.1963.tb01161.x.
- Tononi, G., M. Boly, and C. Cirelli.** 2024. “Consciousness and sleep.” *Neuron* 112 (10): 1568–1594. DOI 10.1016/j.neuron.2024.04.011.
- Trivers, R. L.** 1971. “The evolution of reciprocal altruism.” *The Quarterly Review of Biology* 46 (1): 35–57.
- Trujillo, Cleber A., Richard Gao, Priscilla D. Negraes, Jing Gu, Justin Buchanan, Sebastian Preissl, Allen Wang, Wei Wu, Gabriel G. Haddad, Isaac A. Chaim, Alain Domissy, Matthieu Vandenberghe, Anna Devor, Gene W. Yeo, Bradley Voytek, and Alysson R. Muotri.** 2019. “Complex oscillatory waves emerging from cortical organoids model early human brain network development.” *Cell Stem Cell* 25

- (4): 558–569.e7. DOI 10.1016/j.stem.2019.08.002.
- Tsiatis, Anastasios A.** 2006. *Semiparametric Theory and Missing Data*. Springer. DOI 10.1007/0-387-37345-4.
- Turing, A. M.** 1950. “Computing machinery and intelligence.” *Mind* 59 (236): 433–460.
- Tyler, Tom R.** 2006. *Why People Obey the Law* (2nd ed.). Princeton University Press.
- U.S. Department of Health and Human Services.** 2018. 45 CFR 46, Subpart D: Additional Protections for Children Involved as Subjects in Research.
- U.S. Food and Drug Administration.** 2025. *Patient-Focused Drug Development: Selecting, Developing, or Modifying Fit-for-Purpose Clinical Outcome Assessments*. Guidance for Industry, Food and Drug Administration Staff, and Other Stakeholders. Final Level 1 Guidance, October 2025, docket FDA-2022-D-1385.
- Ubel, P. A., G. Loewenstein, N. Schwarz, and D. M. Smith.** 2005. “Misimagining the unimaginable: The disability paradox and health care decision making.” *Health Psychology* 24 (4 Suppl.): S57–S62. DOI 10.1037/0278-6133.24.4.S57.
- Uhlenbeck, G. E. and L. S. Ornstein.** 1930. “On the theory of the Brownian motion.” *Physical Review* 36 (5): 823–841.
- United Nations.** 1989. *Convention on the Rights of the Child*.
- United Nations.** 2006. *Convention on the Rights of Persons with Disabilities*.
- United Nations Educational, Scientific and Cultural Organization.** 2021. *Recommendation on the Ethics of*

Artificial Intelligence.

United Nations Educational, Scientific and Cultural Organization. 2025. “Recommendation on the Ethics of Neurotechnology.” *Adopted by the General Conference at its 43rd session*, 11 November 2025, Samarkand, Uzbekistan.

United Nations Inter-agency Group for Child Mortality Estimation. 2026. *Levels & Trends in Child Mortality: Report 2025*. United Nations Children’s Fund, New York.

United Nations Office on Drugs and Crime. 2020. *Handbook on Restorative Justice Programmes (2nd ed.)*. United Nations.

Valomon, A., I. De Cuntis, K. Nakamura, B. A. Riedner, S. Jones, M. Bazalakova, D. T. Plante, G. Tononi, and M. Boly. 2026. “High-density EEG signature of NREM sleep parasomnia episodes.” *Scientific Reports* 16: 19414. DOI 10.1038/s41598-026-41601-4.

van de Vijver, F. J. R. and K. Leung. 1997. *Methods and Data Analysis for Cross-Cultural Research*. Sage.

van der Bles, Anne Marthe, Sander van der Linden, Alexandra L. J. Freeman, James Mitchell, Ana B. Galvao, Lisa Zaval, and David J. Spiegelhalter. 2020. “The effects of communicating uncertainty on public trust in facts and numbers.” *Proceedings of the National Academy of Sciences* 117 (14): 7672–7683. DOI 10.1073/pnas.1913678117.

van Duijn, M. 2017. “Phylogenetic origins of biological cognition: Convergent patterns in the early evolution of learning.” *Interface Focus* 7 (3): 20160158. DOI 10.1098/rsfs.2016.0158.

- van Fraassen, B. C.** 1989. *Laws and Symmetry*. Oxford University Press.
- Vandenberg, R. J. and C. E. Lance.** 2000. “A review and synthesis of the measurement invariance literature.” *Organizational Research Methods* 3 (1): 4–70. DOI 10.1177/109442810031002.
- Vayena, E., M. Salathé, L. C. Madoff, and J. S. Brownstein.** 2015. “Ethical challenges of big data in public health.” *PLoS Computational Biology* 11 (2): e1003904. DOI 10.1371/journal.pcbi.1003904.
- Vehtari, A., A. Gelman, and J. Gabry.** 2017. “Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC.” *Statistics and Computing* 27: 1413–1432. DOI 10.1007/s11222-016-9696-4.
- Vohra, S., L. Shamseer, M. Sampson, C. Bukutu, C. H. Schmid, R. Tate, J. Nikles, D. R. Zucker, R. Kravitz, G. Guyatt, D. G. Altman, D. Moher, and the CENT group.** 2015. “CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement.” *BMJ* 350: h1738. DOI 10.1136/bmj.h1738.
- Volkow, N. D., G. F. Koob, and A. T. McLellan.** 2016. “Neurobiologic advances from the brain disease model of addiction.” *New England Journal of Medicine* 374 (4): 363–371. DOI 10.1056/NEJMr1511480.
- von Elm, E., D. G. Altman, M. Egger, S. J. Pocock, P. C. Gøtzsche, and J. P. Vandenbroucke.** 2007. “The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement.” *PLoS Medicine* 4 (10):

e296. DOI 10.1371/journal.pmed.0040296.

Voss, U., R. Holzmann, I. Tuin, and J. A. Hobson. 2009. “Lucid dreaming: A state of consciousness with features of both waking and non-lucid dreaming.” *Sleep* 32 (9): 1191–1200. DOI 10.1093/sleep/32.9.1191.

Wagenmakers, E.-J., R. Wetzels, D. Borsboom, H. L. J. van der Maas, and R. A. Kievit. 2012. “An agenda for purely confirmatory research.” *Perspectives on Psychological Science* 7 (6): 632–638. DOI 10.1177/1745691612463078.

Wager, T. D., L. Y. Atlas, M. A. Lindquist, M. Roy, C.-W. Woo, and E. Kross. 2013. “An fMRI-based neurologic signature of physical pain.” *New England Journal of Medicine* 368: 1388–1397. DOI 10.1056/NEJMoa1204471.

Wager, Tor D., James K. Rilling, Edward E. Smith, Alex Sokolik, Kenneth L. Casey, Richard J. Davidson, Stephen M. Kosslyn, Robert M. Rose, and Jonathan D. Cohen. 2004. “Placebo-induced changes in fMRI in the anticipation and experience of pain.” *Science* 303 (5661): 1162–1167. DOI 10.1126/science.1093065.

Walker, Margaret Urban. 2006. *Moral Repair*. Cambridge University Press.

Walters, E. T. 2025. “From nociception in aneural animals to human suffering: Toward a comparative biology of pain.” *Journal of Experimental Biology* 228 (19): jeb251210. DOI 10.1242/jeb.251210.

Wamsley, E. J. 2014. “Dreaming and offline memory consolidation.” *Current Neurology and Neuroscience Reports*

14: 433. DOI 10.1007/s11910-013-0433-5.

Wamsley, E. J., T. Trost, and M. Tucker. 2024. “Memory updating in dreams.” *SLEEP Advances* 5 (1): zpae096. DOI 10.1093/sleepadvances/zpae096.

Wamsley, E. J., M. Tucker, J. D. Payne, J. A. Benavides, and R. Stickgold. 2010. “Dreaming of a learning task is associated with enhanced sleep-dependent memory consolidation.” *Current Biology* 20 (9): 850–855. DOI 10.1016/j.cub.2010.03.027.

Wasserstein, R. L. and N. A. Lazar. 2016. “The ASA’s statement on p-values: Context, process, and purpose.” *The American Statistician* 70 (2): 129–133. DOI 10.1080/00031305.2016.1154108.

Wasserstein, R. L., A. L. Schirm, and N. A. Lazar. 2019. “Moving to a world beyond $p < 0.05$.” *The American Statistician* 73 (sup1): 1–19. DOI 10.1080/00031305.2019.1583913.

Watkins, Edward R. 2008. “Constructive and unconstructive repetitive thought.” *Psychological Bulletin* 134 (2): 163–206. DOI 10.1037/0033-2909.134.2.163.

Weber, M. 1922. *Wirtschaft und Gesellschaft*. J. C. B. Mohr (Paul Siebeck).

Weber, Max. 1963. *The Sociology of Religion*. Translated by Ephraim Fischhoff. Beacon Press. Originally published 1922.

Weidinger, Laura, John Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, Myra Cheng, Mia Glaese, Borja Balle, Atoosa Kasirzadeh, Zac Kenton, Sasha

- Brown, Will Hawkins, Tom Stepleton, Courtney Biles, Abeba Birhane, Julia Haas, Laura Rimell, Lisa Anne Hendricks, William Isaac, Sean Legassick, Geoffrey Irving, and Iason Gabriel.** 2021. “Ethical and social risks of harm from language models.” *arXiv:2112.04359*.
- Weisberg, M.** 2013. *Simulation and Similarity: Using Models to Understand the World*. Oxford University Press.
- Weizenbaum, Joseph.** 1966. “ELIZA: A computer program for the study of natural language communication between man and machine.” *Communications of the ACM* 9 (1): 36–45. DOI 10.1145/365153.365168.
- Weizenbaum, Joseph.** 1976. *Computer Power and Human Reason: From Judgment to Calculation*. W. H. Freeman.
- Wellek, S.** 2010. *Testing Statistical Hypotheses of Equivalence and Noninferiority (2nd ed.)*. CRC Press.
- West-Eberhard, Mary Jane.** 2003. *Developmental Plasticity and Evolution*. Oxford University Press.
- Wiggins, A. and J. Wilbanks.** 2019. “The rise of citizen science in health and biomedical research.” *American Journal of Bioethics* 19 (8): 3–14. DOI 10.1080/15265161.2019.1619859.
- Wilkinson, Mark D., Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul**

Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C. 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. 2016. “The FAIR Guiding Principles for Scientific Data Management and Stewardship.” *Scientific Data* 3: 160018. DOI 10.1038/sdata.2016.18.

Williams, B. 1981. *Moral Luck: Philosophical Papers 1973–1980*. Cambridge University Press.

Williams, D. R., J. A. Lawrence, and B. A. Davis. 2019. “Racism and health: Evidence and needed research.” *Annual Review of Public Health* 40: 105–125. DOI 10.1146/annurev-publhealth-040218-043750.

Williams, George C. 1966. *Adaptation and Natural Selection*. Princeton University Press.

Wilson, R. C. and A. G. E. Collins. 2019. “Ten simple rules for the computational modeling of behavioral data.” *eLife* 8: e49547. DOI 10.7554/eLife.49547.

Wilson, T. D. and D. T. Gilbert. 2005. “Affective forecasting: Knowing what to want.” *Current Directions in Psychological Science* 14 (3): 131–134. DOI 10.1111/j.0963-7214.2005.00355.x.

Windt, J. M. 2015. *Dreaming: A Conceptual Framework for*

Philosophy of Mind and Empirical Research. MIT Press.

Winsberg, Eric. 2010. *Science in the Age of Computer Simulation*. University of Chicago Press.

Wirth, M., M. Wettstein, and K. Rothermund. 2024. “Longitudinal associations between time perspective and life satisfaction across adulthood.” *Communications Psychology* 2: 67. DOI 10.1038/s44271-024-00118-0.

Wong, William, Rubén Herzog, Kátia Cristine Andrade, Thomas Andrillon, Draulio Barros de Araujo, Isabelle Arnulf, Somayeh Ataei, Giulia Avvenuti, Benjamin Baird, Michele Bellesi, Damiana Bergamo, Giulio Bernardi, Mark Blagrove, Nicolas Decat, Çağatay Demirel, Martin Dresler, Jean-Baptiste Eichenlaub, Valentina Elce, Steffen Gais, Luigi De Gennaro, Jarrod Gott, Chihiro Hiramatsu, Bjørn Erik Juel, Karen R. Konkoly, Deniz Kumral, Célia Lacaux, Joshua J. LaRocque, Bigna Lenggenhager, Remington Mallett, Sérgio Arthuro Mota-Rolim, Yuki Motomura, Andre Sevenius Nilsen, Valdas Noreika, Delphine Oudiette, Fernanda Palhano-Fontes, Jessica Palmieri, Ken A. Paller, Lampros Perogamvros, Antti Revonsuo, Elaine van Rijn, Serena Scarpelli, Monika Schönauer, Sarah F. Schoch, Francesca Siclari, Pilleriin Sikka, Johan Frederik Storm, Hiroshige Takeichi, Katja Valli, Erin J. Wamsley, Jennifer M. Windt, Jing Zhang, Jialin Zhao, and Naotsugu Tsuchiya. 2025. “A dream EEG and mentation database.” *Nature Communications* 16: 7495. DOI 10.1038/s41467-025-61945-1.

Wood, W. and D. Rüdiger. 2016. “Psychology of habit.” *Annual*

- Review of Psychology* 67: 289–314. DOI 10.1146/annurev-psych-122414-033417.
- Woods, A.** 2011. “The limits of narrative: Provocations for the medical humanities.” *Medical Humanities* 37 (2): 73–78. DOI 10.1136/medhum-2011-010045.
- Woodward, J.** 2003. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press.
- World Health Organization.** 2001. *International Classification of Functioning, Disability and Health: ICF*. World Health Organization.
- World Health Organization.** 2017. *Communicating Risk in Public Health Emergencies: A WHO Guideline for Emergency Risk Communication Policy and Practice*.
- World Health Organization.** 2020. *Palliative care*. World Health Organization.
- World Health Organization.** 2021. *Ethics and Governance of Artificial Intelligence for Health: WHO Guidance*.
- World Health Organization.** 2022. *Global Report on Health Equity for Persons with Disabilities*. World Health Organization.
- World Health Organization.** 2024. *Ethics and governance of artificial intelligence for health: Guidance on large multi-modal models*. World Health Organization.
- World Medical Association.** 2024. *WMA Declaration of Helsinki: Ethical principles for medical research involving human participants*. World Medical Association.
- World Wide Web Consortium.** 2024. *Web Content Accessibility*

Guidelines (WCAG) 2.2. W3C Recommendation, 12 December 2024.

Wortman, Camille B., and Roxane Cohen Silver. 1989. “The myths of coping with loss.” *Journal of Consulting and Clinical Psychology* 57 (3): 349–357. DOI 10.1037/0022-006X.57.3.349.

Yarkoni, T. 2022. “The generalizability crisis.” *Behavioral and Brain Sciences* 45: e1. DOI 10.1017/S0140525X20001685.

Young, I. M. 1990. *Justice and the Politics of Difference*. Princeton University Press.

Zahavi, D. 2005. *Subjectivity and Selfhood: Investigating the First-Person Perspective*. MIT Press.

Zehr, Howard. 2002. *The Little Book of Restorative Justice*. Good Books.

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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as “MrGameTheory,” Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United

Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia?

Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky's fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

